ALL INTERVIEW TRANSCRIPTS – COMBINED MASTER FILE

Akshay-spacequant-srv-first

Unknown Speaker 00:03

What we can do,

Speaker 1 00:05

could you please disconnect your experience latest company? What was your daily task? What probably some challengeable task, probably some forward you're proud of. After that, I will describe our company and our precision project, and if you don't need to solve expectations from both sides, we can continue, and also, I will describe all steps

Unknown Speaker 00:35

for the interview process.

Speaker 2 00:42

I Okay. So hi, my name is accessing and I've been working as a full stack tower developer for almost five years now, and I've been working mostly on our best technologies, utilizing frameworks such as Spring, Hibernate and industrial services. And my current client is that old dominant pipeline where I have line working as a senior software engineer involved mostly in migration of, you know, monolithical applications to microservices. And some of the services that I recently migrated encode Experience API and maintenance API. And you know the Experience API, primarily used for user sentiments, and maintenance API, usually any changes related to user profile preferences. And basically, you know, primarily the tech stack that involves users, includes like Java Spring Boot, Spring Data, JP and everything is, you know, deployed on AWS cloud. And I have been involved in, you know, promising these services as well. And some of the services that we use include like AWS ECD, ESCs and SQS and SS and lambdas. And I have also been involved in, you know, writing unit testing to unit and Mockito and little bit of, you know, automating using cucumber. So, yeah, that's, you know, bit background about me and my current experience at aluminum flight line.

Speaker 1 02:38

My question is to me,

Speaker 2 02:46

to currently, you know, working on a freightline, but my, you know, contract is coming to an end.

Unknown Speaker 03:00

Sorry, what's that you should

Unknown Speaker 03:06

inform company?

Speaker 2 03:13

Yeah, like, I don't have, like, any kind of obligations. I just need, you know. And my project is also coming and aim. So anyways, you know, I'll be leaving soon, so that's one of the reasons I'm looking for new opportunities.

Unknown Speaker 03:41

Like in more detail.

Unknown Speaker 03:43

So what is especially, particularly what

Unknown Speaker 03:47

AWS, AWS, services,

Unknown Speaker 03:52

database,

Unknown Speaker 03:59

Postgres,

Unknown Speaker 04:02

OpenSearch, source or maybe,

Speaker 2 04:04

sure, so our back end code is, you know, written in Java, and we use Spring Boot for all our microservices, and some of the microservices are deployed on AWS lambda, But, you know, the team is primarily Java, heavy applications.

Unknown Speaker 04:31

So sorry, what's that?

Speaker 2 04:38

Oh, the dollar version, the law version we have been using is $17.17 right now. So, you know, recently, you know, we migrated some of the cloud services that our team use, included, like AWS ECS to for our legacy applications, and then AWS ECS for continuations as well. And we also use like, you know, lambda for like, event based architecture, and then we use AWS s3 for, you know, storing static files, and then AWS RDS for relational database. And all those databases that we use is Postgres, and we also have our dynamic dB, and you know, in one of our Use cases as well. Okay,

Speaker 3 05:31

so thank you for now.

Unknown Speaker 05:50

Okay, describe

Unknown Speaker 05:58

our company in

Speaker 1 06:00

commercial real estate, Application for customers from different

Unknown Speaker 06:15

sources,

Unknown Speaker 06:18

transaction for properties for

Unknown Speaker 06:26

example,

Speaker 1 06:29

if this property is okay to invest or something some risk analysis, also for business customers who provide

Unknown Speaker 06:40

DOCUMENT MANAGEMENT scope, can

Unknown Speaker 06:45

be other systems customers

Speaker 1 06:48

is using. Also, we provide portfolio management to be done by the can combine different properties in portfolio, create reports. Also, there is functionality related with Loan request. So this end to end process, user can request one particular property, and there is different stages of this loan, different assignments, documents, review and so on. So the application is kind of a combination between document management system, CRM system can create contacts, teams, ticket system, application contains a lot of table information, a lot of data tables, micronutrient information, for example, we can show full traffic in this area where properties Located,

Unknown Speaker 07:46

technologies,

Speaker 1 07:49

AWS, primary and all spring in and all spring related libraries, also for some microservices. Reason, Python elements is

Unknown Speaker 08:11

Python based, and some microservices

Speaker 1 08:15

written in TypeScript for modules and oftentimes databases indexing, like

Unknown Speaker 08:32

for data databases,

Unknown Speaker 08:36

that's pretty much about the

Speaker 1 08:39

project. About the project, about the process, methodology after each iteration, CICD, UI, specialist,

Speaker 1 09:02

using specialist

Unknown Speaker 09:10

business, probably for some questions,

Unknown Speaker 09:17

yeah, I think for now, yeah,

Speaker 2 09:21

I should be good. But I think, Yeah, I'm good for now, perfect.

Speaker 1 09:28

So my question, is it interesting for you? Can you continue? Yes,

Speaker 2 09:33

I think, like all the technologies that you have is kind of I've been using, and I think that's what I'm looking for, you know, in the new opportunities, and it also write out my skill set as well. And I think I'd be excited to, you know, continue further with you guys.

Unknown Speaker 09:53

So the

Speaker 1 09:57

next steps. So today we'll have short technical interview with Java, some development the next step interview will be causing so you will use your pte or any Mumbai. Pte will

Unknown Speaker 10:12

be a

Speaker 1 10:15

task for Java about seven lines the certain steps of interview will be introduced. It's more technical. It's about your motivation, your experience, and after that, after this last step, we will make a decision about before. Before before we start, I would like to provide some disclaimer that you know, company during interview, we are not allowed to use, for example, voice recognition and other stuff that can answer situation on behalf of you. So just remind you about that.

Unknown Speaker 11:05

Okay, yeah, so,

Speaker 1 11:09

architectural question, I have an application one and have another application, and this application

Unknown Speaker 11:15

invoke an estimate point application

Speaker 1 11:22

and receive a response. It's okay, but,

Speaker 3 11:25

like, it takes quite a lot of time

Speaker 1 11:31

to do some logic and doing the response. For example, five min, which is not good.

Unknown Speaker 11:40

Probably you can suggest any ideas how we

Unknown Speaker 11:44

can make this communication more reliable.

Speaker 2 11:56

Does the application making the call, or does it need to rely on the response coming from back from the server.

Speaker 2 12:14

Well, so I guess the first thing I was suggest is, you know, to do event based systems, you know, introduce some kind of messaging queue so, you know, you can send messages to the topic, and then, you know, the server would basically act as a consumer that will, you know, put This and process data, second roster, and during this I think we should just do a fallback response, like an accepted or something like that, to say that request is being processed, and after the request is completed. You know, we just send a notification back to the user that the process that they initiated is completed.

Unknown Speaker 13:08

Also, if you talk about

Unknown Speaker 13:14

software development day to date,

Speaker 2 13:20

well, so, you know, we have, you know, pilot invaded in our intel is a,

Unknown Speaker 13:27

you know, co pilot, like we

Speaker 2 13:35

think that's It, like things are no other areas that that's it.

Unknown Speaker 13:48

Annotation

Unknown Speaker 13:50

one more time. Can you repeat

Unknown Speaker 13:53

that or annotation

Unknown Speaker 13:59

I can't understand? Can you

Speaker 1 14:04

talking about any property, how we can relate the complexity, how we can compare bottlenecks?

Speaker 1 14:17

For example, I have Bishop is more

Speaker 2 14:26

efficient well, so the Sony algorithm, you know, that is like, most efficient, right? I think,

Unknown Speaker 14:41

you know, like

Speaker 2 14:45

the most, you know, the most sorting is like, should be like the, you know, most efficient way. So usually, because it has, you know, a pretty good runtime complexity,

Speaker 2 15:10

basically like the way it works. It like it basically, you know, splits your list and then the splitted things, you know, because like, it usually, like divides, like the list into, like a multiple chance, and then, you know, the recursive, which shorts theme, and then, you know, combines the result. I

Unknown Speaker 15:51

bunch of ideas out implement, for example,

Unknown Speaker 15:57

only one cycle but another cycle, Team

Speaker 1 16:01

cycle. Inner cycle, inner cycle. What is more more efficient?

Speaker 2 16:12

So usually, like that is, you know, compared using the runtime complexity with the goal notations, right? So if you have embedded, or, you know, listed loop, it would be like a quadratic runtime complexity, which is not good, right? And so if you like have for you just mentioned that, you just mentioned

Unknown Speaker 16:42

he, yes,

Speaker 2 16:47

so you want to so give example about that, right?

Unknown Speaker 16:58

Sure, so,

Speaker 2 17:01

like I mentioned, you know, the merge sort like it has a big own addition of and log n, and, you know, this bubble short, like it has, like, a big O of n squared, because, like, it loops and then each element twice. So these are the, you know, two ones that you know, like, top of my head,

Unknown Speaker 17:42

one more.

Speaker 2 17:52

Well, I have, you know, worked with executor, you know, service on the history command, and then the his command, basically, you know, is wrapped on a, you know, executed service. And

Unknown Speaker 18:14

we basically question, so if

Speaker 1 18:17

you ask questions, so the question is described

Unknown Speaker 18:21

situation in court

Speaker 1 18:25

because to stress, and there is Very kind of ability to favor that law.

Speaker 2 18:40

So basically, you know, the deadlock, or when they are like, you know, multiple threats, they are waiting for each other to, you know, release some kind of resources, right? And so if, let's say, you know, we have a service that is like holding a lock on another service, and then we have that basically holding on a log and on the one that is basically locks the services, and you basically, know, wait for each other for the release that lock, and which is basically leads to a deadlock.

Unknown Speaker 19:32

So this is more fun questions,

Unknown Speaker 19:38

but I was just

Speaker 3 19:42

so you definitely use it. Yeah, okay, okay, so could you please explain what this operator does? Can

Speaker 4 20:03

you

Speaker 2 20:09

repeat the last question? Because I cannot hear from

Unknown Speaker 20:18

operation.

Unknown Speaker 20:20

So what is intent? How does it work?

Speaker 2 20:24

Okay? So, basically, you know, the stream operations, you know, was introduced in Java eight, right? And then you know, stream is used for collections processing. And basically, you know, we can convert, like any collections into a stream and then process it using intermediary operations, like using filter and map and flaps. And then we can also use, you know, terminal operations like collect or reduce or so on. And it basically, you know, makes working with collections, easier

Speaker 3 21:08

question is, what is distinct?

Speaker 2 21:17

Oh, yeah, yeah. So basically, the string operator. In String, it is, you know, basically used to collect, like, distinct elements, right? And what I mean by that is that it just removes the duplicates, basically. And I think the equal methods to compare, you know, the duplications, but

Speaker 3 21:48

Okay, and what is the difference between this team and collect?

Unknown Speaker 21:57

You can actually remove duplicates

Speaker 2 22:10

so, yeah, basically, you know distinctly intermediate operation, right? So what I mean by this is, whenever you know, we use this thing you can do, like a further processing of that stream, like you can filter it, or you can map it and convert into another, updates types, or something like that. And whenever you do this thing right, like using collections to a state, you can, you know, keep that order that input data is in, because, whenever you do a collect to set, it, basically, you know, converts data into a set where We call order is, you know, not guaranteed.

Speaker 3 23:01

Okay, correct, we can guarantee that

Unknown Speaker 23:06

order

Speaker 3 23:08

if we would like to collect, to set. So is it possible

Speaker 2 23:17

to create a service? So there is, you know, set implementation that, you know, preserve the order. And I think it's called preset and,

Unknown Speaker 23:41

and, you know,

Unknown Speaker 23:44

it might be even,

Unknown Speaker 23:47

you know, Link lay link headset,

Unknown Speaker 23:53

because, you know, because it's like A Link To

Speaker 3 23:58

explain, how does that actually work so and what is the complexity of gradient elements, and why

Speaker 2 24:13

so? Well, basically, you know, Link has a set which has a node like structure, right? And what I mean by that is it basically like each element of that, you know, say, has a pointer to the next element. And so, you know, you have have your element and it will basically have a reference to the next element you know, which is in the order of the insulations. And whenever you know, you're talking about complexity, the you know in like insulation complexity is like a big old notations for it, and it's constant because you just need to update the pointers, right? And so you do not have to go through the whole set, but whenever you are trying to read it, it's like a big old notation. And because you don't have the, you know, direct access to the element at the end,

Speaker 3 25:28

you say that it's before, do you know if, now, values is allowed? Now,

Speaker 4 25:44

value is

Unknown Speaker 25:49

allowed in so i

Speaker 2 26:03

So, so in a links, Link has said, you know, it has a null value, just like you're, you know, simple, and it has a right where it can store one null value. And, okay, okay,

Speaker 3 26:24

to database using a Postgres right to explain what is truncate table operation? Sure.

Speaker 2 26:39

So, yeah, basically the truncate table, you know, it like removes, like all the columns from the you know, and then, like it removes all the rows. Sorry,

Speaker 3 27:01

you also can remove all the rows using the Delete, right? You can write delete from table, and do not provide any clarification. So what is the difference between gate table and delete?

Speaker 2 27:21

So, yeah, basically, whenever you, you know, you do a delete, it removes like, row by row, right? And then delete can be like, roll back. So if you, you know, if you delete inside a transaction, then you can roll you know, roll that index and back. But like, truncate can't roll, you know, can be rolled back, okay.

Unknown Speaker 27:51

And VG vase

Speaker 2 27:55

faster. So truncate is much faster because, you know, it deletes row by row, you know, because delete, you know, deletes like row by row and truncate does it faster and efficiency.

Unknown Speaker 28:10

There is a

Speaker 3 28:14

very common problem, and you have a big database right, and many indexes in different tables, and those indexes consume much space at some point of time. So how would you determine which indexes are really used right now by your game or some indexes are so important, we can draw

Unknown Speaker 28:48

your indexes Postgres,

Speaker 2 28:53

yeah, so, like, like you mentioned, you know, you know, whenever you're working with big applications, there is going to be, like a big indexes that might have been created for like, performance reasons, but you can, you know, check the user indexes using Postgres internal tables, and I think it's called user index table, but you know, if that is not available, then you can also just look for the EXPLAIN execution plan, you know, using explain keyword, and see if you are really need to, you know, indexes for solving queries, and then Determine like, what can be deleted. Okay,

Speaker 3 29:41

any experience with open search or Elastic Search,

Speaker 2 29:48

so like, I haven't worked directly with elastic church.

Unknown Speaker 30:02

But, yeah, I haven't worked directly with

Speaker 3 30:06

those, mainly experience data frameworks like Yes.

Speaker 2 30:21

So I have been a work with Apache Spark in the past. You know, we basically, you know, had a file lab load process where, you know, we would get a file from a client, and then, you know, we have to spin up about the Spark clusters to populate the our internal database. And it was basically when I was working during our United Health Group Project,

Speaker 3 30:53

use architecture works.

Unknown Speaker 31:02

Yeah,

Unknown Speaker 31:03

so basically, you know,

Speaker 2 31:10

you know, we can use map, produce architecture, basically when we handle, like, a large amount of data across nodes. So there is a two phases, and it is usually divide and conquer, where you know, you have your mapping phases and where your input data is, you know, basically mapped into smaller chunk using, you know, some kind of Mapper tool, right? And each mapper, you know, basically processes are part of the data. And there is a you know, reader faith, which basically uses to generate those values from the mapper. And you can also, you know, sometimes have internally failed, basically to stop all those data. But you know, those are these, you know, the two primary phases. And there is also a layer called STFs, you know, which is used for storing the data.

Unknown Speaker 32:19

Basic question,

Speaker 3 32:22

it's but we have multiple calculations in our pipelines, different forms, indexes, some not information and so on. So this is advantage if you have some mass

Unknown Speaker 32:47

preparation. So the question is, you have

Unknown Speaker 32:53

least with

Speaker 3 32:54

integer values, let's say

Unknown Speaker 33:00

values. So how would you calculate,

Unknown Speaker 33:04

averaging processes and media offices?

Speaker 2 33:15

So basically, you know, whenever I are,

Speaker 2 33:25

you know, basically, whenever you want to, you know, calculate, like, average of the list. I think we can still use the stream, right,

Unknown Speaker 33:37

and then, you know, you can. It's

Unknown Speaker 33:40

not not it's not about three

Unknown Speaker 33:46

average.

Speaker 2 33:56

So in order to do that, you know we can send a map radius, right?

Unknown Speaker 34:09

Just

Unknown Speaker 34:12

okay, what is the difference between average,

Unknown Speaker 34:17

media, average,

Speaker 2 34:18

and median? So basically, average is like, you know, some of all the data divided by like the number of data, and median is like the value that occurs in the middle of that list.

Unknown Speaker 34:33

It's like the same

Unknown Speaker 34:35

middle, middle of that list.

Unknown Speaker 34:40

So

Speaker 2 34:42

basically, if there are, like, you know, 100 elements in it, then the 51st element, but the list has to be sorted for that.

Unknown Speaker 34:55

Okay, thank you.

Unknown Speaker 34:57

On my side, I

Unknown Speaker 35:07

BBB, some

Unknown Speaker 35:19

questions.

Speaker 2 35:22

I mean, from my side, so, yeah. So basically, first I'd like to, you know, thank you for, you know, for giving me this opportunity. And the first question, you know, that I have is, like, what are like, you know, some day to day activities for developers that get involved in it like, you know, they get involved in designs along with implementations.

Speaker 1 35:49

So we started with daily standard we discussed between planning so each developer received a bunch of tasks depending on the velocity, then

Unknown Speaker 36:01

the bending velocity.

Unknown Speaker 36:03

Now we discuss progress,

Unknown Speaker 36:04

workers,

Unknown Speaker 36:12

architecture as

Speaker 1 36:18

well, technically, quite what

Unknown Speaker 36:26

changes

Speaker 2 36:36

implementation? Are we like working on some kind of, you know, migration project. Or it is like fresh grass, you know, fresh project on the spreads. So this is the

Speaker 1 36:51

product connection, because the application applications to get the new features.

Speaker 4 37:01

So yeah,

Speaker 2 37:07

I think yeah, that's the questions that I have for now.

Unknown Speaker 37:13

Okay, so let us few days.

Unknown Speaker 37:18

Thank you. Have a nice

Speaker 2 37:22

day in mind. Thank you. Thank you. Have a good nice day as Well. You

Unknown Speaker 37:59

damn right

Unknown Speaker 38:01

exit. Crazy.

Unknown Speaker 38:22

Crazy,

Unknown Speaker 38:34

I

Speaker 4 38:41

I give? One hour plastic. Thank

Unknown Speaker 38:52

you. What's your

Unknown Speaker 39:01

email? LinkedIn, live follow ups, it went well,

Unknown Speaker 39:05

nevertheless understand

Unknown Speaker 39:08

when he was having difficulty,

Unknown Speaker 39:21

like feedback as like reach out and Follow

Unknown Speaker 39:29

let's see what goes up. Thank You,

Unknown Speaker 39:38

Steve, i

Unknown Speaker 40:26

i kitchen. Roo,

Aman Maharjan-JPMC- SRV-Final-offered

Speaker 1 00:00

Role and your architecture of your product.

Speaker 2 00:04

Yeah, surely. So as I mentioned, I was like, basically, was like working as a full time developer on James over there. I was like, primarily working and engineering, focusing mainly on, like, Spring Boot applications. Also, I was like, involved in ADM too. So that project was, like, already deployed on, like AWS cloud. I think you might know that. So, yeah, the application was, like, primarily the architecture service designed for, like, managing the calls from like different clients to like downstream services, talking about, like Alien services. I was, like, working there. Our team was, like, heavily emerald and like implementing the GraphQL

Unknown Speaker 00:56

so

Speaker 1 01:01

you were done with Chase in May of 2024, right? That's where you went back to your home country or your emergency, right? But then that was your last project. You were, you're not on any current project this part

Speaker 2 01:18

right now, yeah, I'm nothing like in the current project. That was my last

Unknown Speaker 01:26

project, sorry, sorry, go ahead, yeah, sorry, yes.

Speaker 2 01:29

So what I was saying, Yeah, I was like, working over there as an AUM developer over there, we, our team were, like, heavily involved in, like, some of the experimental project. Basically we had to, like, develop different like content for content for like different producers, and we were, like implementing different APIs for them so that they can reservoir. So also, like we had, like, some of the work in Spring Boot application stream, managing it in the AWS team. So, yeah, these are the like, was it that I was, like, heavily involved? So

Speaker 1 02:10

how was, how was your architecture? Was it a monolithic application on your

Speaker 2 02:16

side? Moreover, it was, like, previously, it was like a monolithic architecture, but we use envelopes into the micro services as of now.

Unknown Speaker 02:28

So it was

Unknown Speaker 02:30

monolithic, but it got changed over to

Speaker 1 02:34

it, right, yeah, so you were part of the process where they split it up into micro services.

Unknown Speaker 02:40

Yeah, we were like, involved on

Unknown Speaker 02:44

those on the project. So, so

Speaker 1 02:49

how did you decide the endpoints or micro services when you're splitting up from that

Unknown Speaker 02:56

monolithic to individual

Unknown Speaker 03:00

micro services? That was your design approach.

Speaker 2 03:04

So, yeah. So basically what we had, like, we want to, like, go with a microservice route easily, like, think of microservices as in, like a single dome. So basically had, like, their responsibility. Like, for example, if we want a service that was like, responsible for like, for example, we I have done like in image extraction job. We have like, those services to those services were like, into a monolithic architectures. So we change it into like separate services. For example, we had like, as I told you, image extraction job, what we did was like, we created a network where all the logic of like,

Unknown Speaker 03:55

for example, if we have like, 200

Unknown Speaker 03:59

we will, like, convert it

Unknown Speaker 04:02

into like more, if it was like

Unknown Speaker 04:07

last, and

Speaker 2 04:11

validate it and store it in our storage, in Our database. So,

Unknown Speaker 04:20

so in this way, we like

Speaker 2 04:23

separate the different services and make it as and like,

Speaker 1 04:28

how do you identify the boundary? How do you define the boundary of the vertical endpoint, right

Unknown Speaker 04:35

for that one,

Speaker 2 04:38

those are really responsible in order to like, define the boundaries. You know, you basically first we had like P meetings. We had like different product engineers who were accounted like this and that. So there's like something that we found it within a certain domain. We have like context for like that conceptual aspect, right? So basically, using this domain model that applies within this like individual service, and that's how we like define it. So basically, we define the four business functions, and then we give to the penalty of this like functions, and then make sure that we have a single responsibility of like his service. So that's how we define those boundaries. Okay,

Unknown Speaker 05:31

so how many endpoints are there

Unknown Speaker 05:33

applications

Unknown Speaker 05:36

like multiple endpoints of that one? So

Unknown Speaker 05:41

I don't know, like boundary link as the

Unknown Speaker 05:44

I don't want exact number, right, but

Speaker 1 05:48

some say just four or five points, but it

Unknown Speaker 05:53

seems right, yeah.

Speaker 1 05:57

Okay, so you have multiple endpoints, and you're putting up all these endpoints on AC two, yeah.

Unknown Speaker 06:06

So one endpoint, one EC

Speaker 2 06:09

two, these were deployed as like ECS so, like, basically one instance was like, deployed on like ECS cluster. So each service were like compromised and deployed to like ECS cluster. So, yeah, not like one easy to write, so only easy to point like endpoints will be like too much.

Unknown Speaker 06:36

So how did you manage security on that part?

Speaker 2 06:40

Yeah, for the security aspect of implementing it to it, we had like token base, so we basically

Unknown Speaker 06:51

once, like, there is like,

Speaker 2 06:54

providers who inject the JWT tokens and our service would like, basically make sure the validity of the token by doing the signature matching and also looking at the expired of the payload and then, like having a role based assets control across those endpoints.

Unknown Speaker 07:16

And how was your How was your end

Unknown Speaker 07:19

point to end point communication? How did you do that

Unknown Speaker 07:24

into input to Input

Speaker 1 07:29

Connections? You may be having to do APIs talking to each other, right? So, how was the communication out there? How did you buy that?

Speaker 2 07:39

So, yeah, so we were, like, still using the REST APIs, using the API keys

Unknown Speaker 07:49

and so, yeah. So

Speaker 2 07:53

you mean to say we have to, like, join the API schemes for, like, other vendors, APIs?

Speaker 1 08:01

No, I mean, you're still into the same cluster, right? You said you deployed in ECS, right? So you're the same cluster, but then definitely there will be a point when we'll have the top two other endpoints.

Speaker 2 08:18

So yeah, on that context, basically based on those, like API keys and JWT tokens. We basically make sure we have like role based across those services. So if you are referring to like a separate service, or I think you're referring in terms of like networking security, because for like microservice communication, we just made that we have like, proper inbound rules on this services, so that once I was like, so that like, once the services would like communicate with the other one. And after that, we made sure that we had, like, passing proper authorizations header

Speaker 1 09:03

Okay, okay, okay, so now I know your application.

Unknown Speaker 09:10

Did you talk to any external APIs?

Speaker 2 09:13

We had couple of like from the Google studio. We had like using and all the also vendors we had it so we did like use external API.

Speaker 1 09:26

So let's take a scenario the similar ways, right, that both API, which is no on your side, right? Doesn't matter if it is on prem or cloud application, right? We need to talk to multiple of those. API, write external API, get the data from there, aggregate it into one and then throw it back to the UI. Right?

Unknown Speaker 09:53

How will you design that system?

Speaker 2 09:57

So basically, the security is the main process when we are, like, using the external APIs. So basically, let's say we have like a service that needs to talk with like external clients then basically aggregate the data, right? So basically, first you need to make sure you have like external bank communications done properly using the HTTP protocols, and having a proper API key and or taken token is needed for data, right? So basically, first you need to need to make sure that you have those external client communication done properly using the SDK protocols and having a proper like API key and OAuth token added into your request. So once you have that, you should be able to call this like APIs. But again, it depends on the APIs that they expose and the documentations that they provided. So yeah, for authentications and authorizations, we do that one we provided in the header on your request. And once you have those, we will, like, basically integrate a data aggregation layer once which is once done your external like data space. So you would like basically continue with this data source and those maybe like use like parameter or filters to look at the data. So security

Speaker 1 11:36

token, I get it right. That part is there, but that's the only thing. What we need to do, right, to speed up the process. You're calling multiple APIs, right, to get the data and then aggregate it. Do you think of any other better way to speed up the process?

Unknown Speaker 12:00

Yeah, for like, speeding up. So during

Speaker 2 12:04

the external, like API's call, you will basically do parallel processing and doing an astronomers communications, right? So if you utilize like something like spring wave blocks that we can do, that can do, like, super nice communications with this APIs. So, yeah, we definitely want to make sure that you have like analog processing and then do like data managing or aggregations. After all this like excellent calls have been, like, complicated. Okay,

Speaker 1 12:43

okay, so now, I mean just expanding the current scenario right, right now, the same scenario to call it 1234, data, aggregate, push it to back to the UI right now, getting it work, making it more complicated out there, right? You're getting the data from API one, giving it to or pushing it, posting it to API two, right? And getting it from API three, and posting it to API four, kind of thing. Now, how would you maintain a transaction

Unknown Speaker 13:21

with all this scenario,

Unknown Speaker 13:24

yeah, that's like, really

Speaker 2 13:29

managing all those transactions is, like, really difficult. So if you want to, like, make sure that your data is like, consistent across like different services, whenever you have, like, sequential API calls, you first need to like some kind of, like distributed transactions, maybe using, like, either design patterns, something like that, which is, like very popular in like Microsoft service, You know, making transaction into series of like local transitions, and then handling each of those, like individual can be so, and then also come over, like the compensation actions, which is a rollback stays, Wherever there is failure. So and the way it works like is you start a transaction by invoking the first service, and if the APIs once it's like successful, then you proceed to the API too. And if it's successful, then you proceed to the disk API and so on. And if any of like those service fails, then user, if you could be compensating actions to undo the changes.

Speaker 1 14:52

Okay, have you used any design patterns in your your project?

Speaker 2 14:57

Yeah, we had like, couple of like design patterns that I was like, heavily involved with. Moreover, we use like singleton pattern. Also like, try to condense patterns we were like, using it. For example, when we were like, using the singleton pattern, get like, a single instance of a class which will, like, provide a double point of access. So we were using those for electronic ships with the databases and so on, yeah. So we had like, couple of windows patterns that we have used in our Have you ever heard

Unknown Speaker 15:37

of circuit breaker pattern?

Speaker 2 15:41

Yeah. I have heard about it. So basically, I haven't, like designed, how much involved into it, but definitely I have, like, a conceptual knowledge about it. So basically, circuit breaker pattern is, like for resiliency. So it's basically making sure that you do not like pressure on the downstream services, right? So basically, let's say if our like, APIs wants it like hanging continuously, and then it does not get time to recover, because you keep, like, constantly sending the request to it. But if you need to get a certain paper, I think, from our service right, then what it does is like, whenever there is like certain persons, that there is a failure, instead of calling the downstream services, we just go to the callback method and return that data so that it gives like service one to like recover it so it just acts like a circuit as it does on like electrical circuit breaker.

Unknown Speaker 16:57

Okay? It, yeah, sure.

Unknown Speaker 17:07

My questions do want to say

Unknown Speaker 17:11

anything? Yeah,

Speaker 1 17:14

I feel like you have a lot of experience in financial tech, yeah, my questions will be more conceptual and short answer questions, have you worked with Angular two at all on the UI side?

Unknown Speaker 17:30

Oh yeah, I had like, involved in Angular Yeah.

Speaker 1 17:36

Let's go over this. Go over some basics with Angular components, we use hooks, life cycle hooks, you know, what are they and what are they used for? It was a summary.

Speaker 2 17:54

So basically Angular like life cycles hooks to the first group that gets executed after the constructor in the NG on changes right. So basically it is called an input on property changes. And then there is a ng on a which is called like, once the component other properties are like initiated. And then there is like ng after content in it, where it is called after, you know your view content is like initialized, so. And then there is like ng on display, which is called paper, your Angular destroys the company. So, yeah, basically, those are the

Speaker 1 18:48

that's really good. And then in Angular, we have different components, and then you want to share data in between them, obviously, data sharing. Could you explain it to us? Pass data from one component onto another.

Unknown Speaker 19:05

How do we do that?

Speaker 2 19:08

Yeah, so basically, when you are like, transferring data or sharing data in Angular, we have like, we can use like parent style components, data sharing, using like input and outdoor output decoder, so basically on the data you want like,

Speaker 1 19:36

That was the first part, and then they use services to, you know, share data between components.

Speaker 2 19:48

Oh, yeah. So we have shared services that we can use to share data across like components, which we can use across like multiple like components. So that's the other way of like, sharing the app.

Speaker 1 20:04

Yeah, and then those services, for example, you want that dependency inject to be, you know, unique across all the components. What should you do? How should you include that service to different components that dependency,

Speaker 2 20:23

the dependency that we are using. So you mean to say we have to, like, show those dependencies.

Speaker 1 20:28

For example, service a have injected to component one and component two. Are they the same? Necessarily, same

Unknown Speaker 20:39

thing? Instance, are they the same?

Unknown Speaker 20:43

Any idea about that?

Speaker 2 20:47

Not exactly I have, like, it's been a long time I've been using both. We can

Speaker 1 20:55

just skip it, and then, let's say data directly, you know, we subscribe to observe variables. And we found some data synchronous or synchronous data.

Unknown Speaker 21:12

So, yeah, also, variables are, like, more

Speaker 1 21:16

asynchronous, okay, okay. And then we got the data from the, you know, asynchronous call, you know, observable. Do we have a way to manipulate that data make changes on it? How do we do that?

Speaker 2 21:33

So, yeah, if we want to, like, make changes. So basically, whenever. So observers, whenever the observer is provides like your what you call the subscriptions, right? So you can like, subscribe to the observer and then to like add to transform the data operation to ventilate those so you can maybe do a map or filter or maybe use like windows.

Unknown Speaker 22:14

Have you

Unknown Speaker 22:16

worked with redox stores and

Unknown Speaker 22:20

NGRX at all. Have you set up a store?

Speaker 2 22:26

Yeah, I have done, but it's not like, I have, like, walked into it, like, directly, but yeah, doesn't matter. What's the purpose of it.

Speaker 1 22:35

That's what I'm asking. What's the purpose of setting up a store? How do we set up a store?

Speaker 2 22:41

So, yeah, so it's mainly for, like, managing the state of the applications, right? So basically, it specifically designed mainly to, like, handle the states it uses. It uses the stores, which is like a container that owns, like the complete state of the application itself. I'd

Unknown Speaker 23:07

say, let's

Unknown Speaker 23:12

say you release the service,

Speaker 1 23:17

you know, Kubernetes, and then it's not spinning in a crash. Bucha,

Speaker 2 23:25

yeah. So probably whenever there is when we are like using cells, if you're soliciting like equipments in pass loop. So basically, there could be like multiple reasons for the one of the main reasons that might be the application errors, right? So if your applications like, doesn't start up, then you might need to, like, check the input endpoints. And also you might like need also need to like, check the environment animals that you might have to maybe like in bird to some

Speaker 1 24:06

higher region variable, specifically, you know, there's a whole bunch of environment variables, but you know, it's not starting up at all. You know, it's just crash those. You know, it's not coming out live, spinning

Unknown Speaker 24:24

on that context, if the

Speaker 2 24:28

if the application is not like running at all, if you have like, those like problem, it might be the container image that might not have been like incorrectly used, or maybe like corrupted. So that might be the result.

Speaker 1 24:45

Okay, and then let's say to our services, we get an influx of requests. Is there a way to, you know, adjust our side pot number with respect to the request number you're getting,

Speaker 2 25:02

yeah, if you have, like, multiple numbers, we can define it.

Speaker 1 25:10

For example, I have two to three plus and then I'm in the rush hour, for example, JPMC during the day, and maybe we want to make it five. Is there a way to do that like intelligently?

Speaker 2 25:27

We can like, defend the ethnicity rule if I'm not like, wrong. But I think there's a concept of like definitely. Is the concept of like ethnicity rule where you can like, apply like, specific rules to the specific which partial circuit service.

Speaker 1 25:47

Okay, okay. And then, for example, we use databases. Eventually we get the databases to be retraining some data. What do you know about, you know, optimizing the database is healthy. How do you optimize queries? How do we make them faster?

Speaker 2 26:07

Yeah, that's great questions, because optimizations is, like, really important part of, like, any businesses households we have. So basically, if you are, like, doing really heavy queries, then on column where you might have like specific process, then you would need to add like indexes also. You also need to like, add indexes on like, normally you drains. So this will help you in like, speaking of the data retrievals, you might also want to, like, interpret some kind of like data casting, so you do not like all the data stability for, like, heavy

Unknown Speaker 26:50

casting. What do you mean

Unknown Speaker 26:52

by that caching?

Speaker 1 26:54

So, caching, okay, okay. Then, let's say you test some data, and then, you know, change data in the cache is failed right now, you're serving the wrong data. How do you overcome that? How do you deal with that?

Speaker 2 27:15

Yeah, caching, when we do like, cachings So on the like, you know, we can, like, educate some kind of like trigger to basically, like, invalidate the cash and restore it again. But

Speaker 1 27:32

all together, all of the cash, you're just purging all the entries in there and then repopulating. Or how they how they invalidate

Speaker 2 27:43

them. So, yeah, in like casing like, we are, like storing CAS based on like certain identifiers, and we can, like, invalidate, does that identify? Like in validations needed, if the data is like change, we will basically direct the entire guest system.

Speaker 1 28:07

Okay, okay, and then, yeah, let's continue with the database. Databases. You patch a number of columns from a table, let's say, and then you fetch the same, very same columns for another from another table, and then you want to merge them into one whole step. How do you do that?

Speaker 2 28:31

So for like, joining the tables, you need to say, right,

Speaker 1 28:34

no, no, no. You fetch a number of, like, five columns from table A and then the same column you just, you know, retreat from the table B, let's say, and then you want to combine it into a one hole that those two,

Speaker 2 28:57

yeah, we can use the like The union operations, all right, so the union operator combines the results of two or more, like select difference between

Unknown Speaker 29:06

the union and union all.

Speaker 2 29:12

So basically, union is, moreover, like what you say, what you call that one which does not like remove the duplicate type, so union removes the duplicate, but union all does not remove the duplicate. For example,

Speaker 1 29:36

same data says you want to exclude the big you know, intersect him from the first step, from the other body.

Unknown Speaker 29:48

If we want to do that, we will.

Speaker 2 29:52

So that will be like outer zone if I'm not done. So basically, if you want to exclude from the

Unknown Speaker 29:59

same

Speaker 1 30:00

example, you use union and union. Combine them this time I want you to, you know, find the find the, exclude the intersection of those two from the first set the operator.

Unknown Speaker 30:21

I will definitely,

Speaker 1 30:26

yeah, that's right, that's right, yeah, that's correct. And then Could you summarize drop, delete and truncate,

Speaker 2 30:38

so on the context of like, village and chunk it. So basically drop delete

Unknown Speaker 30:44

from the database itself.

Speaker 2 30:47

If you want to, like, drop the database itself. Delete removes a row from the table. So, like, we use the drop.

Unknown Speaker 30:57

Can you roll back a drop? Can you roll back a drop? Do

Speaker 2 31:00

if I'm not wrong, it cannot,

Unknown Speaker 31:06

okay, okay, continue and then delete.

Unknown Speaker 31:10

Yeah. So for the

Speaker 2 31:13

delays are like, single role, right? So basically you can, like, specify where clause, and then it may be multiple rows based on that.

Unknown Speaker 31:25

Can you roll it back?

Speaker 2 31:26

Yeah, so where plus, so just to, like, remove those from the database.

Unknown Speaker 31:33

And then can you roll it back? We can use Delete.

Unknown Speaker 31:38

Can you roll back and delete?

Speaker 2 31:40

You should be able to roll back on, I

Unknown Speaker 31:48

think so

Unknown Speaker 31:54

you are coming.

Speaker 1 31:56

You explain the lead and then truncate. Have you heard of a truncate?

Speaker 2 32:03

Yeah, don't get it basically, remove all rows from table weekly,

Unknown Speaker 32:08

and then can you roll it back?

Unknown Speaker 32:11

We cannot. Yeah,

Unknown Speaker 32:17

you got it right? Okay, on the same. Sorry,

Speaker 1 32:23

since you since you said truncate, delete, and everything right? What happens to the sequence? Because if I have a sequence at the table,

Speaker 2 32:30

the data sequence that you are storing, Oracle sequence, right?

Unknown Speaker 32:37

Sequence.

Unknown Speaker 32:37

So

Speaker 1 32:39

what happens to the sequence? When you truncate a table, what happens to a sequence? Then we delete the record to the table.

Speaker 2 32:51

So basically deleting a row, if you are deleting right, it doesn't affect the sequence. If you use the trunk do it does not affect the sequence at all, but when you use the sequence, it will remove as well. Okay. And

Speaker 1 33:15

then in Spring Boot, for example, there is dependent speaking Jackie. Could you summarize

Unknown Speaker 33:22

what part that is?

Speaker 2 33:25

Yeah, sure. So in like springboard, basically Dependency injection is just a principles where the dependencies that you need for your classes is like provided by external source, right? So in terms of, like Spring Boot, it provides from the spring container to the inversion of control IOC. So basically you can, like, auto wire the dependencies that we need from those spring containers. And the dependency injection in spring can be gone through like constructors, setters and failing systems.

Unknown Speaker 34:08

Define the dependency yourself.

Speaker 2 34:12

You can do that and for the same object, let's

Speaker 1 34:17

say you have two separate dependencies, but they're functionally different. You want to pick at your wall from it and inject it into your variable.

Speaker 2 34:25

How do you achieve that? So if you have like depend, so if you have like multiple dependencies, and for that, you just make sure that you have your pin with the name, and then you can use quantifier to provide the name of that build during the injection.

Speaker 1 34:50

Let's say you have a private method. Then you want to provide test coverage for it. How do you achieve that? How do you do that

Speaker 2 34:58

for the private network. So basically, typically, it involves like calling the public network. That inbox is like private network. So

Unknown Speaker 35:11

whenever you are like calling it, whenever you go

Speaker 2 35:16

for like testing approach, you do not have the ability to like test it in like public network directly. You have an options to use like reflections, to invoke like this, private networks directly, but those are not like user connected in like job.

Unknown Speaker 35:38

Why can't you use the reflection?

Speaker 2 35:42

I mean, we can use it, but it's not like recommended for Okay,

Speaker 1 35:49

let's, let's pass the new point, and then you, let's say you have a variable like a database string, whatever, in your manifest file for application, and you want to inject it into a variable. How do you do that? You have your database being defined in your manifest file, and then you want to inject it into your variable. How do you do that? What kind of annotation do you use?

Speaker 2 36:18

We can use like value annotations on that team provide the key to that value.

Unknown Speaker 36:27

And what

Speaker 1 36:30

kind of an annotation do you use in your database

Unknown Speaker 36:35

when you invoke a Data database

Speaker 1 36:37

transaction to you know, ensure the integrity of the transaction. What's the annotation for that?

Speaker 2 36:45

So, yeah, Spring Boot provide transition annotations that we can use to make sure the

Speaker 1 36:53

What's that? All of that.

Speaker 2 37:02

So we can use that to make sure the operation within the network in like single transactions. When

Speaker 1 37:10

does it roll back that transactional

Unknown Speaker 37:15

annotation?

Speaker 1 37:15

And does it roll back the transaction?

Speaker 2 37:20

So, yeah, basically, when you are using the transitionals the springboard, what it does do is that Spring Boot automatically helps to, like, know whether the transaction is like, successful. For example, if you are, like, sending money to someone. But in the meantime, it has, like, complications and it does not. So rollback will be implemented by the what's

Unknown Speaker 37:46

the measure of success or failure?

Speaker 1 37:49

How does it know the success or failure? It just passes off from one line to another to another, like,

Unknown Speaker 37:56

what? What's the barrier? So,

Speaker 2 37:58

yeah, basically, we can define a way to roll back and when not to usually, if you don't provide like anything, an exception will like positive to roll back. But if

Speaker 1 38:11

Yeah, an exception defines a failure and but if you use tricad around that you know,

Unknown Speaker 38:20

does it roll back? Or what does it

Unknown Speaker 38:25

so if

Speaker 2 38:26

there's exceptions, as I already told you, it will be rolled back. The method will need to, like, throw an exception. So within crank as block, you will need to throw an exception if you want the pin to be I

Speaker 1 38:41

didn't throw an exception, I just caught it and then I logged out an error. That's it.

Unknown Speaker 38:47

I love it. Roll back.

Unknown Speaker 38:50

So in that case, no,

Unknown Speaker 38:53

okay. Back to you. Kieran, thank you.

Unknown Speaker 38:57

So I I know

Unknown Speaker 39:05

did a lot of

Speaker 1 39:07

database things, but few questions from my side, right on the database, right? What is the difference between virtualized view and a normal view? Yeah,

Speaker 2 39:20

so when we are like talking about like views and Spring Boot itself, so the basic difference between virtual and personalized view and the normal view is that your normal view essentially stores the SQL queries that present the data from one or more tables, but virtualized view refers to the view that is like, mainly designed to improve the performance by storing the result. Data may be like caching or like computing you normally use, like storing SQL, building without like physical storage, but your visualized view may be stored physically or in like gas pressure.

Unknown Speaker 40:15

So I have a scenario, right?

Speaker 1 40:18

I have one table with a 25 column. I columns. Just say, an employee, first name, last name, city, social, all this, biographics and everything. Five columns out there, right? And we have a UI right, based on that particular table, right? And the UI also has all the 25 you know, fields out there as such criteria, right? The users are complaining about, you know, the performance of a particular query, right? Like, I say, female employees with name Arman and it starts to clock, kind of thing, right? What could be your approach to improve the performance of such situation?

Speaker 2 41:18

Yeah, I think this one was, like already, I mentioned it earlier, too. So if you have like table with like large number of columns, and then you have like

Unknown Speaker 41:29

complex, like different queries.

Speaker 2 41:33

So in that cases, and you can have like different searches, you may also be okay with just like using indexes like,

Speaker 1 41:42

so you're going to put credit indexes on that table.

Speaker 2 41:49

I think that will, like, really affect the operation if you are like, using like indexes, right? Exactly. So then

Speaker 1 42:00

adding credit for indexes is not approach. Do to speed up

Unknown Speaker 42:11

the search for

Unknown Speaker 42:15

like doing that if

Speaker 2 42:17

we are not like using the indexes, I guess we, instead of doing select all like to we could do like select particular fields or coding and then maybe, like, implement those, like a limit on how many results we get back, better than, Like, getting everything and getting all those requests,

Unknown Speaker 42:44

if it's worked,

Speaker 1 42:48

right? I mean, you're talking about not getting 100,000 records in one go, but all the 100 records. But to get those 100 records, you still need to speed up the query. Right? The

Unknown Speaker 43:06

in that cases,

Unknown Speaker 43:12

definitely what?

Unknown Speaker 43:16

Maybe I'm not like, I can only

Speaker 2 43:21

change any of that right now on top of my head. Okay,

Unknown Speaker 43:28

good, from my side, right? You

Unknown Speaker 43:29

have any questions for us right now,

Speaker 2 43:36

I like to know a little more about like individuals like working on the team, how, what is the responsibility of the team, and what the team like actually is like doing the work or like involved in like projects.

Speaker 1 43:54

So we are part of the Business, Enterprise, business resiliency plans, right? So essentially, you know the volume of taste, right? It's spread across

Unknown Speaker 44:09

the globe, right? So

Speaker 1 44:14

business, let's say plans for each and every right from jbmc

Unknown Speaker 44:22

from the team perspective, right?

Speaker 1 44:26

Our team is spread across multiple locations in the US.

Unknown Speaker 44:31

Have teams in UK and also in India,

Unknown Speaker 44:34

about

Unknown Speaker 44:38

eight or nine Scrum teams

Speaker 1 44:42

in all now, and we are looking to expand our team in Columbus.

Unknown Speaker 44:50

So

Speaker 1 44:52

Arman and I, we are part of the Columbus team out here, right? So that's where we are interviewing candidates to expand our team. Are

Speaker 2 45:03

also based on, like, not in Columbus, but in others. In

Unknown Speaker 45:08

Columbus. I'm in Columbus, so

Unknown Speaker 45:13

this position is in Columbus.

Unknown Speaker 45:14

We're just expanding

Unknown Speaker 45:18

Columbus, expanding in Columbus region.

Unknown Speaker 45:21

I look

Unknown Speaker 45:22

after the Columbus side kind of thing.

Unknown Speaker 45:27

Other teams, they are also expanding, but they are different ideas, because

Speaker 2 45:33

I know there's like the team in Delaware also that works on this, isn't it? No, we do not have delivery. No, Texas, Texas, okay,

Unknown Speaker 45:55

any more questions?

Unknown Speaker 45:55

I'm good right now.

Unknown Speaker 45:56

No, I think that will be

Speaker 1 46:00

fine. I Thank you, thank you for taking some.

Aman Maharjhan- Bofa- SRV- final- Offered

Speaker 1 00:00

Experience. Dr pans on development, right? Yeah, almost

Unknown Speaker 00:03

like the seven years.

Speaker 1 00:05

Okay, seven years. And then what you what did you do? I think you graduated about seven years ago, right? Let's see here, looking at your resume. So you graduated with a Bachelor of Science in Computer Science, okay? And you sort of COVID in 2017

Unknown Speaker 00:24

so you consider yourself a full stack developer?

Speaker 2 00:27

Yeah, I was, like, in my last role was, I was like, appointed as, like full stack developer, but I was like working more over like 80 persons as in back end and like 20% as in front end. So, yeah, moreover, like heavy on like back end rather than like front end.

Unknown Speaker 00:50

So 80% back end. Did you say? Ah, yeah, okay,

Speaker 1 00:57

so you're more back end focus, but obviously leverage the UI and, yeah, just once you open it up, you know, just kind of give us, give us a little bit detail of your hands on experience over the last six years, right? So you do have web service integration experience too. Yes, yes,

Unknown Speaker 01:24

I do like working

Speaker 1 01:28

with Go ahead, kind of walk me through your the back end, your experiences there, and some of the detail, what you were able to accomplish on the back end from a job, a middleware perspective, kind of touch on the UI as well and your kind of day to day activities, right?

Unknown Speaker 01:44

Yeah, sure,

Speaker 2 01:45

no problem. So as I talk about my name is Alan Morrison. I've been like a software engineer for about like seven years and running, working mainly primarily on my Java based web applications, mainly like utilizing like framework, for example, spring hibernate and other like rest COVID services for like API development recently been like doing a lot of like micro service work, primarily mainly migrating like monolithic to like micro services, and then also like designing them, implementing them, and also like developing them, like AWS and in like, some of the cloud services that I have like work with include like AWS, ECS, EQs and so on, on the like front end side I have, like, done with Angular and can work with, like JavaScript and TypeScript on the testing side, framework, if I have to, like, say, I have been working on, like, doing it, and mokito for, like, developing different test cases. So like, yeah. So yeah. These are, like, tools and technologies that I have been like working on my recent projects. Okay, all right.

Speaker 1 03:18

You know your debugging experience, how do you go go about detecting issues from the front end, from the UI side to the back end?

Speaker 2 03:32

Yeah. So debugging is, like, a crucial part in like software development. So talking more about like from the US side, you know, whenever there's like, any kind of like issues that I need to figure out, I usually like, look the look into the like developer tools. So mainly like console in is a really nice place to, like, see what errors you have when, like starting JavaScript on your front end. But apart from the apart from like console tab, also, I also been using the network app, that's where a lot of like informations about like, what API has been like calling or has been like invoke you can see on the network cap. So, yeah, these are, like, the two primary places that I look for whenever there's like issue that is, like, detachable in the front side, right? So, but like, if your issue is not like, cannot be like, fine on the front end, and it's strictly back in then we have to look into like monitoring tools that we have in like places. I've been like, primarily working with like new reli for like response code, and also like possibility database, but talk also I've been using for like different blocks. So I usually look into more, over into Splunk and sprung dashboard, like dress down, like decode, showing like any kind of like exceptions or like what heroes might be, like ordering. And based on those like arrows, I usually track them down into like, different workflows, and also like, try to, like, figure out what the exceptions can be, and then also recreate to the lower environments to try and tackle those issues and then fix also,

Unknown Speaker 05:41

worked with an agile development environment, yes. Okay,

Speaker 1 05:47

what's your what's your definition of add to the agile methodology, what you've done with the Agile development world?

Speaker 2 06:00

Yeah, so working like, I've been like working like agile team, so basically just a process that like team follows, right? We basically, like emphasize on like more flexibility and collaborations, and we basically put like small batches of like, coding iterative cycles so that we can, like, follow a sprint, which is worth, like two weeks. And every script, like we usually have a release or, like every other sprints we have maybe a release date that is like different. So we basically add up to like change for like continuous improvement. And that's, I think, the definitions for like as, so it's more like what you call, like flexible and collaborative work, if I have to, like explain the team.

Speaker 1 07:01

Okay, I see that TJ is joined. So TJ was just quick background with Anon, and he's got about 80 80% Java. Nick Mulk of his development experience on the Java side, but as UI side as well, Debugging perspective and that kind of thing. So, what I like to do next is kind of turn over the line of questioning. TJ, getting a little more in depth, but I'm a more technical q&a; with you there and you know maybe.

Unknown Speaker 08:16

Pretty much how

Speaker 3 08:17

someone answers some entry question and move forward to the

Unknown Speaker 08:24

intermediate questions

Unknown Speaker 08:27

and some advanced questions,

Speaker 3 08:30

COVID questions. So in your opinion, right

Unknown Speaker 08:41

Oh, yeah. So

Speaker 2 08:43

in like object oriented programming languages, so basically in like object, for me, is like representations of like something, right? So let's say it's a class which has been like initiated or so anything that can have like in Java, represent a cat class. If you use a new keyword before, like that constructor, you create an object. So it's just that like instance of a class that represents, like certain data and moreover, like its behavior.

Speaker 3 09:23

Okay. So what is like the default constructor? Yeah.

Speaker 2 09:30

So whenever you are, like defining class itself, so default constructor is a constructor without like any

Unknown Speaker 09:40

parameters.

Speaker 3 09:44

So let's assume that we create like an object, and we have the object have a constructor that has parameter is there a way, or can you have a default constructor once you create the class that have constructed with

Unknown Speaker 10:12

parameters? So

Speaker 2 10:18

if you like, yeah, you can have like default constructor, if you have, like, already called the if you have, like, already have these like parameters. So, yeah, you can call like parameterized constructor from the default constructor using like this keyword. Yeah.

Speaker 3 10:46

Like JPH, and we've had our class like this class already have a constructor on JPH, instantiate the object of our class is a company, because now that we've created a class and we added our constructor to it, that is not the default constructor. So how is JP going to instantiate our class? Well, yeah.

Speaker 2 11:17

So in that case, when like, JPA is like around and you can like, call constructor, which is your like, default constructor, or you can, like, call the parameterized, mainly the constructor, right? So basically you just need to, like, pass the values or as the parameters of like, what constructor you want to like call. So it is just like call like constructor overload, where you define like constructor that may have, like parameters or like no parameters at all,

Speaker 3 12:02

right? In this case, right? We only had

Unknown Speaker 12:08

a constructor that we created that has the list is on

Speaker 3 12:11

one perimeter, and we now we want JPA to interact with this constructor, right? So what you said earlier is that like we can have a class right that has our constructors, that has this one parameter, that we can also call our default constructor. So structure. Do you still think that's true?

Speaker 2 12:45

Yeah, in the case, I guess like, in order to like, create an object, it depends on like, which like parameters like we have to which diameter like constructor you are like calling, right? So you can call like any constructor, maybe in order to like create that object, it's just like constructor just contains like information about how your methods and like fields are initially set up. So if you can call like, default constructor for like one field value, it will be initialized as like small but if you want to, like, have an initial property, you can

Speaker 3 13:37

get to the point where I'm saying that up once you have a constructor right, and you created your constructor to have a perimeter right, and you don't have the constructor, I'm trying to actually Like, are you still able to call a default construct. And when you say that was this, right? If

Speaker 2 14:06

you have like, the constructor that you have like defined, then you cannot, like, actually invoke it. I think you will have to, like define it with that within that, like within the classes,

Speaker 3 14:21

so you cannot call it default constructor if you create a constructor that, so you spoke about Like overloaded constructor. Can you tell me what overloaded is, or what overloading is, generalizes. Let's say we have unsafe, right? Can you tell me what is like? I asked you to overload, override this method, what is equity? Is like overriding,

Speaker 2 15:00

yeah. So in Java itself, we have like network overloading and network over lighting. So basically, method overloading is like a way of like achieving polymorphism, right? So network overloading is just like basically defining, like multiple networks with the same name, but you either have like different return type so or different like parameters, or different types of like data, types of like parameters, so it's done within the same class, and you Just like have multiple networks with the same name method overriding is also like another way of like polymorphism, but this is done in like the time class, where you have the exact same method definitions, but overriding the parent method by giving like separate implementations.

Speaker 2 16:06

So yeah, so you can achieve like polymorphism, as I told you, so the Yeah. So the main benefit, like of Metro is like, it makes it more flexible, right? It allows you to like, define multiple methods with the same class, with the same name, but like different parameters, which makes it like, able to handle various inputs and like scenarios. So basically, it will improve like the growth of like, predictability of the code itself. So for an instance, right, if you you can have like a YouTube guest that has like that adds like you may have like integers or double so you can define like add methods that takes two integers or but also can define to add like methods that takes like two doubles, right? So, yeah, it will like basically be doing the same functionality, but like with different data types, making it like more flexible. Can you

Unknown Speaker 17:25

override a static method? The static

Speaker 2 17:35

method? So, yeah, unlike technical top, yes, you can you tag, you can give like, different implementations of like static method, but it does not like, technically means like, you can override that method, right? You can like, hide like static method, which happens when you give like meta definitions in our sub class. But it is not like technical definitions of like static overloading or like overriding, right? Because mainly static network does not belong to an object. It belongs to, mainly the class difference

Speaker 3 18:25

between a class. So

Unknown Speaker 18:33

class is basically

Unknown Speaker 18:37

when you are like,

Unknown Speaker 18:40

what you call that one

Speaker 2 18:43

a template, we can say that that gives like details, like about the structures and behavior, right? So you can have like class, and also an object is an instance of that class, so, right? So it represents the particular entity, right? So, like, I can, like, have a car object which represents the car, and then I have, like, another car objects that represents your car. So basically, is like, definitions and object is like, basically created from that. Definitions.

Unknown Speaker 19:27

Have you done anything with concurrency,

Unknown Speaker 19:33

conference, bps, I have worked with Katie COVID.

Unknown Speaker 19:40

Race condition,

Unknown Speaker 19:43

you mean to say risk condition.

Speaker 2 19:48

So basically, race conditions in situations when there are when there are like two or more threats, which are like simultaneously working, and the outcome depends on the order of like executions, right? So basically it is an input system, okay, so, yeah, basically it is in like inconsistency and then unpredictable, like results because of like

Unknown Speaker 20:27

bad synchronizations.

Unknown Speaker 20:35

So yeah, in order to like, namely, fix

Speaker 2 20:40

it, fix the race conditions itself, we have to like, make sure we use like, proper synchronizations across the shared resources, making like operations more like thread set you can, like, use synchronized methods and blocks. Also use like, maybe like handling those synchronizations. Also you can use it. You may also be able to use like executors service package to provide more highly concurrency features.

Unknown Speaker 21:23

How can

Unknown Speaker 21:25

you fix situation?

Unknown Speaker 21:33

Yeah, sure,

Speaker 2 21:34

so deadlocks. So basically, dead loss is like situations where like two or like more trades are like blocked forever, right? So basically, what is mean that one threat is like waiting for like resources held by another, and that threat is like waiting for a trade or a resource held by like the first one, right? So in situations like where multiple threats are, like basically waiting forever, and in order to, like, fix the depth locks, mainly we can, like, usually need to make sure there is, like, no listed blocking, and you should have like, proper time offset for like, those

Speaker 3 22:38

shoes. You to make the class immutable.

Speaker 2 22:47

Yeah, so to meet like immutable, we can have like, we can do like a couple of steps. Like, the first thing is to like, make sure the class is like Final. Just use the final keyword on the class definitions. Then you might need to like, make all fields ID and final class final, so that so that the class won't be able to replicate itself, means that it won't be instantiated. If you don't make the class like final, you will be able to sub class that class, which means that the behavior may be changed, right? So just like final, prevent subclassing of the utable class that you are like defining

Speaker 3 23:57

is want to meet the class to be able to be

Unknown Speaker 24:07

like the

Speaker 2 24:14

Yeah, so then, like, you need to, like, make all the fields private and final, so that you don't have like, direct access to those like fields, and they cannot, like, be changed once it's set. And then you can. You do not like provide any set of methods, right? So you only, like, expose the gator networks, and then you use the constructor to, like, initialize all fields.

Speaker 3 24:46

One of my field is like a limit strike. Do you think that there's a problem? There's a problem with that?

Unknown Speaker 25:02

If I have one,

Speaker 3 25:08

and from your from your definition, so another person can call this class, get this list, and them into it. So then that's not surely immutable, because now I can see the list that this class, right? So how can you get up around that, and how can you fix that?

Speaker 2 25:37

Yeah, so there will be a problem, because you will be like changing it. So if you have a list, then you need to make sure the definitions or the installations of that list needs to be on modifiable list in the constructor or the other way I can, like, think of this right now is like returning a copy of a list whenever you are, like, doing Gator network, or whenever you are calling. So, yeah, okay, sounds good

Unknown Speaker 26:15

to your game,

Unknown Speaker 26:18

some COVID question.

Speaker 3 26:27

You are going to solve these particular problems. So let's say I asked you to write a method, six, eight. Takes an integer array and then it also has

Unknown Speaker 26:45

another parameter called Target, right?

Speaker 3 26:50

What I want you to do is for you to find any two number that is inside of the array that has up to the target number, how would you how'd you go about

Speaker 2 27:05

you want me to like code or just like,

Unknown Speaker 27:09

just sort of, what sort of code, just

Unknown Speaker 27:14

how you think it helps me

Speaker 3 27:18

creating? This particular method that will print out the two number, any two number to

Unknown Speaker 27:27

add to the target number.

Speaker 3 27:30

So I'm going to go ahead and add the method signature inside of inside of the chat, just so that it gave

Unknown Speaker 27:40

clarity to it.

Unknown Speaker 27:51

So I just added the message signature

Speaker 3 27:57

inside of like your own. So if you look at your at your own WebEx, you should see the

Unknown Speaker 28:05

census had

Unknown Speaker 28:06

got the message signature.

Unknown Speaker 28:12

Do you see it?

Speaker 3 28:15

Okay? So talking about creating this particular like method that prints out in a suit number that's part of the array that adds up to the target number. So, yeah,

Speaker 2 28:34

if I like really need to find like, two numbers that, like add up to current target. Then I will try to, like, find a compliment to that like Target. So basically, I'm like, moreover, thinking of like, iterating over, over, it like numbers right? And then like, I will find like compliment, which is like difference between the target and the number. And then I will like, basically add that number into the including like asset

Unknown Speaker 29:22

and right? So if like

Speaker 2 29:26

number like, if this like compliment already exists on the asset, then I'll basically be returning that like actual number and the compliment, but if not, and just keep on, like adding. So I think that will, like, take care of, like, finding, like, the two numbers and then add up to the target

Speaker 3 29:55

you were saying that you that you're creating a set right,

Unknown Speaker 30:02

putting inside of the set.

Speaker 2 30:05

So in the set, I will be like, putting the main, mainly numbers that like I'm iterating over,

Unknown Speaker 30:15

so you're adding the number that you're

Speaker 3 30:24

saying, that you're adding that you're adding the number well, that you're putting the number that you're iterating over inside of,

Speaker 2 30:37

mainly like inserting. I mean adding means like inserting the number and iterating over and like before insertion. I would like, if like the compliments, like edges or not.

Unknown Speaker 30:56

If that makes sense,

Speaker 3 30:59

when, when you sit compliment, are you saying like the Delta, or you just adding the number in it? That's that's a reflect, but

Unknown Speaker 31:13

delta into slightly

Speaker 2 31:22

so mainly the compliment is the delta, right? So which is like a difference between the target and the current number, right? So it's like a compliment, like IGS, then I will return my current number and the compliment. But if not, then I will add, like, my current number

Unknown Speaker 31:50

sounds good part where you said,

Speaker 3 31:57

let's say this question is a little tricky,

Unknown Speaker 32:04

still doable.

Unknown Speaker 32:07

The main thing about this question

Speaker 3 32:12

is just the concept, right? So if I was to ask you to write a method reverse string, right? But you cannot, but you cannot use any loops, any built in method part of JDK, right, from the String class, right? So any

Unknown Speaker 32:42

method of screen class, right?

Speaker 3 32:46

Anything else, sort of like iteration, you cannot do that, how to go by running a method that would refresh a string without using an in loop.

Speaker 2 33:05

Okay, but you can use like methods, right?

Speaker 3 33:10

You can use methods only part of the strain class, so like for example, let's just say that you can use,

Unknown Speaker 33:19

that you could use

Speaker 3 33:22

writing a method that will reverse a list with a method that's

Unknown Speaker 33:32

inside of the string class, apart from

Speaker 2 33:38

reversion. Yeah. So I'm like thinking more over like recursion in this case, because I would like need to go like, through each characters. So yeah, if more of a like regression will be the best this one. So

Unknown Speaker 34:03

good. That's That's how you would

Speaker 3 34:07

do it back to Mark, if he has any question for you to continue. Thanks. Thank you. Thank you, Mark, I think you're on mute if you're talking

Unknown Speaker 34:28

anybody Thanks. Thanks.

Speaker 1 34:31

All right. Any questions for me? Thanks again to Jay for the question. I

Speaker 2 34:38

have, like, right now, I think I like to first, like, thank you all, all of you guys like, for giving me a chance to come here and give me some time. It was, like, really nice talking with you all guys. And also like one more questions, like, what are the stacks that you guys like it seems like working on, and how much like developers get involved, involved

Speaker 1 35:10

working within the merchant services base. I'm not sure what kind of explains a little bit of what that is. But the small business side, we have merchants that onboard terminals to take financial transactions with credit cards and so forth. So we enable those processes. So the full end to end as it relates to procurement, boarding those relationships. So So we work with primarily a specialized within the merchant services space. So that's our primary products and services. And so a lot of service integration, we also interact with the business process management suite, BPM, if you're not below that, there's a vendor called app yet that has the framework where those processes reside as well. So you'll be kind of working from a services integration perspective on that side. And then there's also other side of the house as well, a BPA and application. So again, services, Java and integration. But as always, when we, you know, we develop, we get letters to UI, we do our testing. So having that front end experience, hands on experience, is important as well, so that we are able to quickly diagnose issues and get to the root of the issue quickly, and that kind of thing. So, yeah, so, so it's good to have a kind of a be able to wear that full stack hat and continue to learn.

Unknown Speaker 36:54

But

Unknown Speaker 36:58

all right, thank you.

Unknown Speaker 36:59

Yeah, thanks guys, thanks.

Speaker 1 37:02

Appreciate it. Thanks a lot. Have a great, great adversity. Thanks guys. Thank You.

Unknown Speaker 38:08

Hello. I

Aman-deloitte-citi-srv-final-Offered

Speaker 1 00:00

Am I just gonna make you the host so that you know I can leave.

Unknown Speaker 00:10

JD should also be joining shortly.

Speaker 1 00:12

He's running late, but why don't you guys get connected first,

Unknown Speaker 00:18

and I'll leave. I can connect with you later.

Unknown Speaker 00:22

Thank you. Thank you.

Unknown Speaker 00:28

Just give me

Unknown Speaker 00:29

one minute. I'm offering my it's

Unknown Speaker 00:40

mainly going to be

Unknown Speaker 00:44

technical for coding exercise sort of around,

Speaker 2 00:48

what we would expect is, I'll send you the exercise, hopefully in maybe another 510 minutes. Just wanted to understand from you. I mean, I think generally we get the CVS. I don't think so CV is going to actually, but if you could maybe just quickly run past your sort of the technical experience which you have, considering what our technology tool sets you have used, and I will also there, by the time exercise is sent out to you,

Speaker 3 01:26

I'll have your email address as 043, gmail.com,

Speaker 2 01:40

let's do one thing. I send out emails. Send out email experiences. What

Speaker 4 01:49

you currently do? Thank you for giving me a chance. Yeah. My name is Oman Murzin. We have been I've been, like, working as, like, software engineer for about like six years or so, working primarily on like Java based web applications, mainly utilizing the frameworks like Spring, Hibernate and rest web services for the APIs. Been like recently working on like, lot of like micro services, and been involved in like mainly designing those micro services, developing them and also deploying them to like AWS cloud. I have a pretty good like experience working with AWS using like services, like to ECLS, lambda s2, buckets, SQS, SNS and so on. Recently, like, unlike my scrum team, my team, like, consist of like five papers. We have like product owner, and also we have like Scrum Masters, and we follow, like, as I'll met Rosie, my daily responsibility, like, working on, like back end, helping like COVID and also like products and opportunities. So yeah, these are like work that I have been, like involved in my recent projects.

Unknown Speaker 03:22

Sorry. Give me one

Unknown Speaker 03:30

minute. I think we also have JD on call now.

Speaker 2 03:35

Joined in from my you should be able to see me as well. Hopefully. Thank you. Have you received the exercise? Which I sent you? Amar sent you on your treatment address. Can you just see whether you have The exercise? Yes, just give me say.

Unknown Speaker 04:24

I think I'm logged out. I'm just like logged into mine.

Speaker 2 04:31

It's on your Gmail address, so once you have it, let us know, and I can maybe just quickly go through what we plan to sort of understand, right? So the idea is to

Unknown Speaker 04:48

sorry, I think we lost you on your video.

Unknown Speaker 04:53

Yeah, just some,

Unknown Speaker 05:01

I think we lost among the

Speaker 4 05:19

back. Hello, can you hear me? Guys,

Unknown Speaker 05:28

I can we can see you now. What

Unknown Speaker 05:29

I wanted to give you was just

Speaker 2 05:34

to give a once you have the email, you can obviously open the email.

Unknown Speaker 05:41

The idea is,

Unknown Speaker 05:43

is for you to sort of share your screen.

Speaker 2 05:47

Obviously, we expect you to use any IDE you want to use or an issue. And you should try to treat this exercise as you would do any real world coding, which you have to do on a day to day basis. So if you need to occasionally use Google just to find an API or something, that's perfectly fine. Not an issue, because we all do. But as long as your screen is shared, we can see what you're doing. That should be fine. Any questions to you have to sort of understand the exercise or etc, please feel free to ask them five minutes for you to maybe just go through that exercise and share your screen. Online ID will be great. I mean, it's up to you. If you don't have any, I think what you would find it easier is to if you have your IDE but if you want to use online ID, that's fine. I mean, it's up to you.

Speaker 4 06:46

Okay, let me just like start my screen and I'll like online ID.

Unknown Speaker 06:52

That's fine.

Unknown Speaker 07:09

Can you see the screen? Yes, yes. We can see i

Speaker 4 07:26

I'll just use this, I think probably a good compiler. I can Use that. I

Speaker 5 08:54

question, would this Id be able to allow you by unit test, along with your

Unknown Speaker 09:00

implementation?

Speaker 4 09:05

I don't think it will allow me to write unit tests, but we can just like, make use of like equal methods to like assert equals, okay.

Speaker 4 09:31

But if you want me, I can like write like the general structure of the English test that will be like writing if I were to like work on the like real project.

Speaker 5 09:47

It's up to you. I mean, if you are able to demonstrate, because on the day to day working scenario at depauci management, this is just we follow TDD. So we were hoping that we are able to demonstrate that as well. In this exercise,

Speaker 4 10:06

I'll demonstrate how I will, like, write a unit test after I'm done with, like, the actual, maybe implementation.

Speaker 4 10:19

So, like, basically, like, we want to, like, support his take operations. And what I'm thinking like is maybe, let's say we look through this length of like, this string, and maybe then like, if the character is like itself is a number, and then we start Like,

Unknown Speaker 10:50

if it's like that, then let me think I

Speaker 2 11:04

we have given a basic tummy better signature signatures right in that exercise. So Feel free to use both.

Speaker 4 12:14

So we have like three cases at the character of like index assignment, right? So it can be like a digit, can be an opening bracelet, and it can be like closing bracket. So our it can just be a character. So we have to do like some kind of checks if it's a digit, or like actual character, or like opening bracket or a closing bracket.

Unknown Speaker 12:53

So just let me start here.

Speaker 4 12:58

I'll just gonna basically keep track of like, all those like indexes.

Speaker 4 13:08

So yeah, this index will just be used to, like, loop around the eye, and we basically do not want to like increment the index unless we are done with that like particular character, right? So we'll stay at that index until all like repetitions of that character is done. So for an example, I

Speaker 4 13:56

so here for like ABC, I will on that, like ABC, I'll stay like on that, like ABC blog, or The remainder, like of my number indexes, number, right? I

Speaker 4 14:23

so let me, like, start with, like, a string builder, and basically, to you, there's a placeholder for, like, attending that main value.

Speaker 4 14:41

And no, I'll just one basically loop until I get to the end of the link. I

Unknown Speaker 15:07

The input is just a string, right? Are

Unknown Speaker 15:12

you going to tokenize

Speaker 4 15:16

that? Yes, no, no. Will this be like playing with the strings, we are not going to like converting that into like anything. It will just get the character at like that at an index here.

Speaker 4 15:51

So yeah, this year we can get the back at that index. I

Speaker 4 16:06

so yeah, for our like first example, we will have like three, I

Speaker 4 16:30

so if character is like digit right, Then I need to like kind of I think I

Speaker 4 16:47

Yeah, so we basically need to keep track of like, How many times the looping needs to be happen. I

Speaker 4 17:52

Yeah, so if the character is the visit, then we

Unknown Speaker 17:59

basically, we

Speaker 4 18:06

I find what this soul, there will be a case where the digit is like better than 10, like, it's like multi digit number. It can be like 15 or like, basically 20?

Unknown Speaker 18:27

Is that valid? I

Unknown Speaker 18:32

think that's valid.

Speaker 4 18:39

So if in that like case, then we have to, like, basically, look again until, like, we reach to the, you know that, right? So it will be, like, another loop maybe in, like, maybe loop inside, is there

Speaker 2 19:00

a factor you can observe in the string? Right? I mean, I understand what you're trying to do. You're going through character by character. But is there a better way,

Speaker 2 19:15

repeating blocks? Repeating blocks are always within the bracket. There is a number before that bracket.

Unknown Speaker 19:24

So is there something

Unknown Speaker 19:27

that can help you?

Unknown Speaker 19:29

Because if you keep on going this character by character

Speaker 2 19:33

to have 100 and then ABC, what are you going to do? You're just going to say, oh, 100, right? So for

Unknown Speaker 19:40

that case,

Speaker 4 19:44

so we definitely can use like pattern and seen using like regex to like, basically like, extract the duplicate numbers and also like duplicate visits, right, and try to, like, play with it, so to see if I can

Unknown Speaker 20:10

think if there's like, any specific way to do

Unknown Speaker 20:19

that. So yeah, I'll basically use

Speaker 4 20:23

regex to, like, extract the values if that I

Speaker 2 20:40

do you want to try doing that, or do you want to understanding if you feel free, if you want to Google for The syntax, operating, etc, right? Not an issue.

Unknown Speaker 20:57

Thanks. Do

Speaker 4 21:06

I'm just like, thinking out loud, like, so with the red X, I'll get the number sign. I used to be like, repeat, so, and then like, if I get like that, I will have like split up into

Unknown Speaker 21:29

blocks to use.

Speaker 4 21:40

And then, like, if once I had that, Then I should be okay. I

Speaker 4 22:04

h, i, so for pattern matching, I think we can get natural. I

Speaker 4 22:44

so let me do An online regex generation. I

Speaker 4 23:36

process so the regex should something be like that reject where it can be anything after the square break it, and I think I should just be a blood star.

Unknown Speaker 24:01

And once

Unknown Speaker 24:04

I'm dead, I think I probably can do that. I

Speaker 4 24:41

BBB, so we can, like, create a New Vegas pattern using the pattern and match it. I think it's like, okay, if I google the pattern, right? That's

Unknown Speaker 24:55

fine. I'm very sure

Unknown Speaker 24:58

people don't remember. People

Unknown Speaker 25:00

don't remember that on site you can Google. Thanks. You

Unknown Speaker 26:02

So, I think this is a syntax I need to use, like

Unknown Speaker 26:07

regex, get The pattern and get the matches.

Speaker 4 26:21

For us the regex to be what I like generated earlier,

Unknown Speaker 26:36

and generates the code here i

Unknown Speaker 26:43

Oh, Actually, I think

Unknown Speaker 26:56

my compellates like frozen. Okay, it's back.

Speaker 4 27:08

So yeah, instead of like, let's see that the gigs gave me I

Unknown Speaker 27:13

need, I think,

Unknown Speaker 27:18

anything i

Speaker 4 27:27

So, yeah, so once I have that, so I have a digit, I have anything inside the bracket, and then I still have like, Like, like, everything that is needed. I

Speaker 4 28:06

then I should be extracting it and retrieve it's

Speaker 4 28:46

this will be a fine method, and I think I'll need to use that, putting it

Speaker 4 29:02

in so as long as I match or like Find something, we basically need to

Unknown Speaker 29:14

populate it like I

Speaker 4 29:26

to find what you call and then find it and then store it In like a list of

Unknown Speaker 29:38

me just like to string area.

Unknown Speaker 29:58

So when we have the value, we

Unknown Speaker 30:02

basically like create a

Speaker 4 30:05

new entry on the matches layers,

Speaker 4 30:17

and the string audio will contain the first place, which is like integer value, and The second value will just be the actual value. So

Speaker 4 30:48

alright, so let me see if this program runs and above this i

Unknown Speaker 31:13

illegal character,

Unknown Speaker 31:18

You need two practices. You Bishop,

Unknown Speaker 31:43

it should be like two plexuses

Unknown Speaker 31:47

instead of 23

Speaker 4 32:02

actually. Need to import the pattern. Let me see where I can Like, import.

Unknown Speaker 32:40

I thought some

Speaker 5 33:14

group does not know. Just return a stream does the return? I

Speaker 4 33:24

No, I think I will match it like I was, like, using a wrong keyword. I

Unknown Speaker 34:10

let me call the mirrors and See like what it prints Now I

Unknown Speaker 35:02

This And then,

Unknown Speaker 35:48

debug.

Speaker 4 35:55

I don't think I can like debug in like online compiler, but I can just, like, try printing it maybe

Unknown Speaker 36:18

don't have any

Unknown Speaker 36:21

biology machine.

Speaker 4 36:24

Actually, my PC was, like broken, so I'm using, like my laptop to Okay, do

Unknown Speaker 37:29

I think maybe we can try with

Unknown Speaker 37:32

now we're using regular

Speaker 5 37:35

expression to for this problem. Maybe we can try with a simpler case first, maybe just a simple number with bracket with a few letters in there, as if we can use that to at least build our initial version of the solution, then we can carry on building On top of that to refine our implementation. If that makes sense, okay,

Speaker 4 38:09

don't I think want the pattern matching right? So you said, with the pattern maximum without,

Speaker 5 38:19

let's carry on with the pattern matching. And what I'm saying was, let's start with a simpler I know currently you are trying to test your code using the input three, bracket, ABC, four, A, B and then C. Right at the moment, looks like we didn't capture the last letter C in there, which might be a problem, but at least we capture the first two groups. So if we were just to have a simple input just with three bracket ABC, how would you proceed with the rest of The code?

Unknown Speaker 38:54

In that case? Yeah,

Unknown Speaker 39:04

let me Thank you.

Speaker 4 40:27

I think I just realized something like the metric group like starts with like, first index. I think that's who be like, taking care of my initial flow.

Speaker 4 40:42

So So I find it, and then maybe, Like I get this, I

Speaker 4 41:20

Okay. Man, I think I just still have to, like, play with the regular regex, but I'll see what I can like. But I fixed that later, so once I had, like, that strings grouped out, then all I need to do is, like, need to loop. I

Speaker 4 42:00

so for like each of the matter, right? So

Unknown Speaker 42:05

firstly, I need a string builder.

Speaker 4 42:13

Once I have like string builder set up for like each of the matches, the first part will give me the number of times I need to append it. And second will give what character, right? So I

Speaker 4 42:56

so basically I will look like from until the account and then just obtain it to my building. I

Unknown Speaker 43:24

and then, like, once that is done, I'll

Speaker 4 43:27

be like, kind of done, but I think this pattern, like matching, only gets up to the end, so I'll need to, like, obtain the rest of the items. Do.

Speaker 4 43:44

So let's see, like an example, I can, like, have multiple digits. So I think the research I basically probably need to like progress to, yeah,

Speaker 4 44:10

and then I'll just do any character like inside there, so it will give me like everything I

Speaker 4 44:25

Yeah, that's exactly what I wanted so as like three ABCs and AB.

Speaker 4 44:42

Um, I think the only like problem here, it's gonna be like missing character, so see which is like at the end. Now this is like, simple to like extract. I just need to get, like, all the characters after like, closing all the parentheses then which I can like get to using the last index of right?

Unknown Speaker 45:16

So I'll get the like last index of the bracket. So

Unknown Speaker 45:38

so basically I get the like soft screen of my

Unknown Speaker 45:43

given input from the last index of the lane I

Unknown Speaker 46:12

and then, like, I'll just written this.

Speaker 4 46:21

I Can like attend The remainder and I

Unknown Speaker 47:14

we change the number Three maybe to 10.

Speaker 4 47:21

Okay, let's see. Okay, this should work because we are using the matter, right?

Unknown Speaker 47:38

Okay, it did not work. So

Speaker 4 47:42

So because we are just, like extracting single digit here, right? So we basically need to account for, like, multiple digits. So to just be a

Unknown Speaker 47:54

plus sign on,

Unknown Speaker 48:07

yeah, I think that.

Speaker 2 48:10

Will it work if there is an estate characters,

Unknown Speaker 48:16

if you see the second example,

Unknown Speaker 48:20

I'm sure I didn't get that one which one.

Speaker 2 48:24

So there's a second example. If you see what it has done is it is saying that repeating blocks, basically they want a to be repeated thrice, and then that block in itself should be repeated, points, three, A's then B, and then again, three, A's then

Speaker 4 48:49

B, right? This should work. Let me, like, check it out. Like, real quick. I

Unknown Speaker 49:03

also, that doesn't work. Interesting.

Unknown Speaker 49:06

Let's see. I

Speaker 4 49:26

I need to, I think, account for, like nested one, which is, I

Unknown Speaker 49:44

needs to Be like some

Unknown Speaker 49:47

kind of Like, recursive call, maybe I

Speaker 4 50:22

so if I have like understood character, then I

Unknown Speaker 51:04

so fine here on that, and

Speaker 4 51:15

We probably need to, like recursively call the method itself. I

Unknown Speaker 51:56

for them, see I

Unknown Speaker 52:36

I mean, what we can do is

Speaker 2 52:40

give You maybe another couple of minutes if you can fix it out right. If not, can we just continue with because I think you're almost there. Since I thought maybe it's more like a strange exercise, if you can do it right, let's give a couple of minutes if you're able to do it good, good. Otherwise, we obviously want the next Round of the process Do chew

Speaker 4 54:45

basically, I think that we need to do here is like, we need to look outside the matcher to, like, check if the matching is found. Again, right? So basically, we find that pattern, then again, we do this like whole operation inside that loop.

Speaker 2 55:08

Okay? I think what we can do is we can maybe cover some of the Java questions, right? So I think just wanted to understand,

Unknown Speaker 55:19

if I say TDD approach. What do you

Unknown Speaker 55:24

understand when I say TDD

Speaker 4 55:27

generally? Yeah, so generally TDD approach is, like, basically the first things that comes around when we are, like, developing any software at all. So I think of like, maybe red, green approach, basically, you start with, like, something that might be a little backward from your normal development process, right? So you start by writing a test which, like, basically defines some kind of like functions, or like, some kind of like improvement that you are working on, and then, like, you move backward, right? So firstly, since you do not have the code at all, the base will fail, which is like, kind of like a red state. Now you write the minimum amount of code that is required to basically make that code pass. Then you build of that right so you basically range and like, repeat the same process over and over again until, like, all the functionality required on your test is like, done. So yeah, that's that kind of like test driven development process approach will be. So you basically write a test, write a code, refactor the code, and then build on like top of that. So that's what like test driven development, I think is okay.

Speaker 2 56:58

So that's the mechanical aspect of how will you implement the test driven development, but what are the positives of that

Speaker 4 57:07

approach? The main advantage of the positive approach of like test driven development is that, like, when I was like involved in like in test driven development, I found that it's like very good documentations approach, where you have like this on like different cases and different scenarios that you have, right? So it does, like, provide anyone, like, trying to, like, understand the code itself, a good understanding of, like, how the code behaves like different cases. And you also, like, after the deployment, also it has like less debugging time, because it encourages mainly to write those small blocks of like this. So you basically end up, like, having lesser box, because you can, like, start with like failing stages, which is like, kind of like bug states that you are trying to like fix, right? So just reduce that regarding time, whenever you encounter like, any kind of like effect, which is very like minimal

Speaker 2 58:16

in terms of your understanding generally. So you, I think you obviously are a back end developer. Do you have an experience on your front end technologies, react,

Unknown Speaker 58:27

or anything with that?

Speaker 4 58:30

Yeah, I have worked with like, both react, mostly React and Angular, but like, mostly in, like Angular, like work in my previous project

Speaker 2 58:42

and it most of your experiences non banking experiences that

Unknown Speaker 58:49

we work in the banking environment.

Speaker 4 58:52

Oh, yeah. The latest project that I was like, worked in was on like, JP Morgan, so yeah, Julie,

Speaker 5 59:04

okay, cool. So maybe I'll go for a quick sort of system design question. So let's picture a scenario. Let's say we have to develop an API. So first of we, let's say we have a database, which large amount of historical data, let's say they are trade data which accumulated over 10 years. And now there's a requirement to be able to provide an API to allow the user to go through those records at

Unknown Speaker 59:40

their own time.

Speaker 5 59:42

And now, once we build the API, we encounter two issues, mainly the first one is the performance aspect, because the API read the data from the database, the implementation read the data from a database, and it's pretty slow. And also, given the large amount, large volume nature of these data sets, we frequently encounter out of memory issues in our services, because sometimes user pooled data for the past five years, maybe with 100,000 records. So with it, with these two particular problems in hand. What are some things that you would consider doing to improve the scalability as well as the sort of improve the performance as well as the stability of that API?

Speaker 4 1:00:34

Yeah. So basically, when we are working with the APIs, or whenever you are dealing with the application that will deals with the back end services, and you work with like, large volume of data, there are like, couple of like things that you can like, basically help in, like, improving the performance of the overall the applications. So in that case, the first thing will be the database optimizations, right? So database optimization basically means how you like optimize the query execution plan, which can be viewed using the EXPLAIN keywords, and here like explain gives you the query executions plan where you can analyze what the bottleneck for like certain inputs are, if you are missing, like certain indexes and so on, once you have like that in hand, then what you can like, basically say, like, add indexes or like, add any kind of like query enhancement that will help in like, improvement of this, like particular database query, if you do not have like that particular issues, or like, if your issue is like something, yes, then I will say like we would indicate some kind of like change in like API designs, like, maybe For long running queries are, if you have like complex complexity, like data processing, we have, like some asylum processes in hand, and we probably make like some kind of great linking to that. If we do not have, like multiple users processing the same new servers, and have it like, probably originally, like, scale to different nodes of like, different APIs, mainly for like interactions. We also like, need to think about the data processing as well as like memory management by, you know, having like some kind of workers in the background to process and handles and to offload the work from the API server itself, because it may become like unresponsive whenever there is, like a process hanging. But apart from that, your infrastructure should be horizontally, like scaled. So you have like, multiple instances of the same server behind, like load balancers, and sometimes right it might be also helpful if you have like, some kind of, like pre aggregation sponsors, like the sum and addresses in some kind of, like pre aggregated data. So, yeah, I think database optimization using like indexing and query optimizations and API optimization using like synchronous processing rate limiting and also like data processing using the background jobs should be like helpful in like restoring the stability of your application.

Speaker 5 1:03:44

So you mentioned database indexes, right? So I just wanted to a bit more. So let's say we have someone, a junior in our team, and then who saw this optimization approach and decided to put the index on every single column in that table. Do you see any issues with that approach?

Speaker 4 1:04:06

Yeah, so in that cases, if like, adding index is like, really good for like retrieval, right So, but it also like comes with, like, a hit, like, what happens is that index is really good for, like, optimizing the mainly the read operations, but whenever, like, you do write operations, that means that every single index is that you wrote will need to be updated once there is like, any kind of, like, insert, update or delete in a row. So basically it makes you write the operations very slowly. And indexes are like, basically the objects. So once you create a index, it increases storage. And if you have like indexes in like, all of those like columns, and then that means that you have like, unnecessary objects created and stored. So that is like you something that you can see right away, but also having like, too many queries and like indexes can, like, actually be bad to like, having some kind of like analysis on the query plans, right? So these are like disadvantages of like adding columns

Unknown Speaker 1:05:25

with the indexes.

Speaker 5 1:05:28

Okay, cool. My other part of the question was this out of memory exception with frequent BC on a server you mentioned you could potentially using a synchronized approach with Walker thread to doing the retrieval. Can you just elaborate a little bit more on that approach? How does the workers thread solve this memory issue?

Speaker 4 1:05:54

Yeah, so definitely, memory optimizations is, like really unnecessary thing. So basically, what I was like mentioning was, you know, whenever you have, like, synchronous processing with these threads, right? So it basically, what it does is, like a block certain part of your application, so that this process, or this mainly, the trade is like using that part of your CPU memory, so and it is not like available for like others to use, right? So. And when you have like heavy tasks, by offloading this CPUs intensive tasks to work or like trades, you prevent the main like trade, right? The main application threads to become overwhelmed and consumes like access in memory, right? So this is like basically offloading that heavy tasks into those, like worker threads. And you know, once you do that, you will have like other worker threats and level to solve those. And once you do that, you will have, like other workers, as I mentioned. So it will basically help in, like, improving the responsiveness, because worker threats will be handling all those, like heavy processes, rather than the main thread.

Speaker 5 1:07:25

I agree. I think with the walker thread approach, you definitely can help to increase your CPU efficiencies. But does the worker threat is just going to take the same amount of memory anyway, and does it help to resolve the out of memory issue that we are currently facing.

Speaker 4 1:07:44

So yeah, for like, out of memory for the main applications,

Unknown Speaker 1:07:52

I don't think it will be like, really like,

Speaker 4 1:07:59

let me think if like, you like I mentioned the worker

Unknown Speaker 1:08:05

said, will still use the same memory.

Speaker 4 1:08:09

I think once you choose the worker fits you probably like, need to do some like data processing in like trunks. So instead of like, loading the entire like data set into the memory serves, then you can, like, rather like, basically divide the data into like certain chunks, like limiting the certain amount of data and then losing like streams to basically process those data. And like, instead of like, having everything done in like, one shot. So yeah, the advantage of like this is that you have, like, worker trade with does not, like, really rely on the main thread, and it basically be working until you reach the end of the data that you need to, like process, and then it accumulates the data right? So you will, like, basically be able to like, reuse that worker, to like, efficiently retrieve all the data and process but in jobs, come back to you. I think

Speaker 2 1:09:21

we covered all the questions which we have, any questions you have for myself,

Speaker 4 1:09:29

and G Yeah, first of all, like, Thanks for, like, giving me the opportunity to talk with you guys today. I'm like, sorry about my details here, because my issue was, like, I still, I had to use my friends species to use it, I will have, like, done a little bit like there if I had, like, my own PC. So yeah, some of the questions that like included, what activities to developers, like, do I need to, like, learn more about, like, what stack like you are using in your companies. So, yeah,

Speaker 2 1:10:06

so maybe I can maybe give a quick so obviously, you're kind of coming back through. I think the

Unknown Speaker 1:10:15

role is, obviously, as you know, is already Tampa based

Speaker 2 1:10:21

teams, generally, the Tampa Bay teams, are focused both on the UI aspect and the server aspect, so that so the out of the dev leads we are looking needs to have a knowledge on both aspects of third of the applications. Right? The team in itself is quite a global in nature. We have a presence

Unknown Speaker 1:10:45

in Nam itself. We have a presence in York

Speaker 2 1:10:49

desk, generally traders, if they promise. We are based in New York. But then we have a presence of developers in thari and China. We have myself, JD and few other developers in the AVR region, London, very fast, etc, right? So the person who comes in as a dev lead is expected to obviously be working with other developers across regions and deliver solutions for our interviewer, solutions for the business, right? So there will be specific projects a person will be working in you will obviously have some help and handover with the permanent member of the staff in the city who will sort of help you with that you're kind of getting onboarding, understanding the applications, but the devil lead responsibility also means the person is more proactive enough to figure things out, ask the correct questions, navigate the correct things and etc, etc. So that's also sort of a part of the role. So we're not just looking as a of course, technical ability has to be there. That's a prerequisite, but there are certain other qualities a person needs to have under the technical abilities as well, which is what the job leads are, we are looking

Speaker 4 1:12:08

at. So like, how much like management role, like that I have to like play on this, like specific team.

Speaker 2 1:12:17

So, I mean, I can't provide a specific Coronavirus as a dev lead, I think the point here is you are going to have a team of junior developers working with you from the delight itself. So yes, I will call it as a management role, but there will be a mentorship you will have to do, manage their work, code review their work, etc. So it's not like a project management work, but it will be more of a mentorship, senior developer, kind of a work where, yes, you are, you will be sort of, I won't call it as a company responsible, but in a way, responsible for your the team's work

Speaker 4 1:13:02

as because I don't want to, like, really work as like in management, because I want to, like, have, like, some kind of work on the back end and big front end too, on the big sides too. So that's why I just, like, ask the questions, like, how it is not

Speaker 2 1:13:22

much? But you will be still responsible for ensuring that the quality delivery for other members of the team is also met. That's probably appropriate. Any more

Unknown Speaker 1:13:38

questions? Sorry,

Unknown Speaker 1:13:41

no, I think that's it. I have one

Unknown Speaker 1:13:43

final question

Unknown Speaker 1:13:48

before we wrap

Unknown Speaker 1:13:50

up, or what is the motivation to make a change?

Speaker 4 1:13:55

I think the better opportunity to like grow myself as as they take like industries, like going to so i want to, like, get more knowledge about like the industries and like, also, like, I'm like, thinking about, like, moving to the southern part, because I've been, like, eastern part of like, four months. So that's why, too, I think,

Unknown Speaker 1:14:23

yeah, you can say that,

Unknown Speaker 1:14:27

Ohio, Columbus,

Speaker 4 1:14:31

Columbus, yeah, so much wonderful. I think because of that, I'm like, trying to, like, move into the southern part. So that's

Speaker 5 1:14:40

enough said. No idea. Okay.

Unknown Speaker 1:14:57

Thanks so much.

Aman-deloitte-srv-Second-lead-position

Unknown Speaker 00:00

That again. I

Unknown Speaker 00:40

cologne Interview,

Unknown Speaker 01:14

sidekick, you do it, my loyal sidekick. Have you met my sidekick? Oh, I didn't. Oh, oh.

Speaker 1 02:17

I'm going to make you the host, because some people want me to leave. So let me just meet you the host, and then so what? Sandeep joins you will get the notification that he's waiting in the waiting room.

Unknown Speaker 02:34

Just admit him.

Unknown Speaker 02:39

Thank you. Thank you. Internet,

Speaker 1 02:42

yeah, he should be joining soon, because somebody was paying much on time.

Unknown Speaker 02:57

Hopefully your doctor's appointment will went around everyone

Speaker 2 03:02

on the just like in last second. So that's only okay. No worries. Thank you so much.

Speaker 1 03:12

Just keep a watch to see if he's in the meeting room. Yeah, I'll keep him up.

Unknown Speaker 03:33

I think I admitted him, yeah,

Unknown Speaker 03:36

okay, great,

Unknown Speaker 03:41

hey,

Speaker 1 03:44

thank you for your flexibility. I know we had to move this around a couple of times, so appreciate your flexibility and leave and let you and I'm in have a good discussion. Thank you. Thank you.

Unknown Speaker 04:02

How are you

Unknown Speaker 04:04

not too bad. Can you hear me?

Unknown Speaker 04:07

Okay, I can hear

Speaker 3 04:10

you. So my name is Sandy Bucha. I work in the equity finance team as part of the London thing. So good to meet you, if you can start off today's interview with a brief introduction to yourself and some of your recent experiences.

Speaker 2 04:30

Oh, yeah, sure, yeah. My name is Aman Martin, and I've been like, working as a like, full stack developer for more than about, like, six years now, working primarily on, like Java based web applications, utilizing mainly like frameworks, especially with like spring cabinet and rest wave services. For like ABI development, we primarily like involving, like more into like Microsoft development, where I have been like involving in like designing, developing and also developing them to like AWS clouds to some of my cloud services that I have worked with include like, more about like AWS ECS to ECS, SQS, SNS and lambda. So yeah, I do have, like, pretty good experience working with Angular and react on the front end side for also, like testing side. I work with like, say, JUnit and mokito. Currently I'm on like, Scrum team. The team mainly consists of like five developers which have, like, product owner and Scrum Masters and on the daily basis, I mostly worked on like back in Java code, helping the team with like code reviews, and also on like some products and support duties.

Speaker 3 05:55

Thank you very much. So if you can tell me, first of all, you mentioned code review, right? So what do you typically look for in a code review when conducting

Speaker 2 06:07

Oh, yeah. So in our team, where I, like mostly work is like for code reviews, there are, like, couple of things that I look for. Firstly, I just look at how the Navy conditions is like, if something that are like named widely, or something doesn't make sense, like the meta naming, the variable meanings, that's something that I look for right away. But apart from like that, I do keep in mind how much of the functional requirements are there, right? So if there are like, certain changes that like actually makes no sense, or might have like some scientific to order flows, I make sure that I point out and I point if the court meets the specified like requirements, or if the business logic is met. And then, apart from that, I look like, as if there's like edge cases that I've had to, like, told it if that might be handled right, if cases are like handled, maybe like normal or empty values, or anything like that, as then I look for like code structures. And then after that, I look for, moreover, into like unit tests. I personally feel that unit test acts as like documentation for the code itself. So, yeah, I I do like all those things. So mainly, these are the things that I look for, like during like code reviews.

Speaker 3 07:42

Okay, you talked about the fact that you do unit tests, obviously, one strategy is TDD. Can you explain TDD and how you have applied that in your previous projects?

Speaker 2 07:54

Yeah, mostly like test driven development. That's right, yeah. Test Driven Development, like is a kind of, like a strategy where you have, like, red and green phase, right? So basically the main idea is you do a red phase and then you do a green phase, then you refactor those phases. And what I mean that is like you start with failing test first. So you before you even write any test, I'm sorry, like before you even write a code, you start with a unit test that defines the what the functions will behave. Since you do have like the functionality implemented, it will fail, right? Filling is called the red face. And then once you have, like, identified that you write a minimum amount of code required to make the particular test pass. So you just have that it of code to make it pass, which is like, kind of like green, as quickly as possible. And once that is done, you just, like, refactor it, right? And then you print and repeat the same process over and over again until you have, like, a well tested all around solution for it. So for our projects, we adopted for like, six months, and then we had to keep it up because of how our like team did not feel too productive over there. I feel that it was like, well, way of like getting defects earlier in the process. But for one of our like, micro service development, we had, like, a peer programming session for, like, creating a RESTful API using those GDD approach.

Speaker 3 09:51

Okay, thank you so. How would you instill a behavior of TDD in a team that you're leading that historically has not been talking about practice.

Speaker 2 10:04

Oh yeah, I have, like, ministry didn't have like, those experienced programmer with, like, TTD. So I would say that TV is like, kind of like, gets a little bit difficult to initially, like, standardize right away, right? So, because it is a very gradual and incremental approach. So basically, first thing that I would like to do is, like, kind of like to start with, like, some really very small example, right? Like, maybe utility class, or maybe, like a utility function events. So I will try, like, for example, if I want to, like, develop a calculator functions, then I will start by, like, giving an example of how I will have, like, add method that should, like return certain value, and then I would like, just explain, and then build on top of that, I'll say I will lead, like, basically by example for very, like, first knowledge transfer sessions. And then we know we will, like, basically have, like, a spread out option on how the expectation from the team needs will be. And even after this knowledge transfer session, if things don't clear up, I will suggest to have like a peer programming sessions possible where there is like a driver who basically tells the other teammate what to do, and then there's like other teammates who access who have like those quotes, right? And then they will switch towards implementing those certain parts of the code. And then I make sure, like to give like constructive questions when it is needed, for the base, for the team itself, or also basically celebrate and kind of like win that the team lines have like followed for like encouragement. So I think that's the strategy I would like basically approach with, Okay,

Speaker 3 12:15

what's your understanding of behavior driven development?

Speaker 2 12:20

Oh, yeah. So much more over like. Behavior Driven Development is a little bit different from the case driven development. So basically you have like different behaviors defined so you have like, let's say, business logic and user requirements that like will return on your like tickets that may be in like cheetah board story you like our flow. So what happens is like, there will be given statements, and there will be a win statements, and then there will be like, like these statements for which give you a clear outline of what the user expect and what the business like expects. And then after that, I think you basically develop on top of that, so that kind of like, how approach with like behavior driven, approach

Speaker 3 13:21

understood Are you guys, any particular frameworks that you're aware of within BDD?

Speaker 2 13:30

Yeah, we can, I think, implement, I've been like, I haven't, like, directly used the framework, but I think for the testing of the BDD approach we have like cucumber, I think, yeah, that's the cucumber, and we're getting like syntax to, Excuse me, I

Unknown Speaker 14:49

I do apologize. Just have to deal with something at home.

Unknown Speaker 14:51

No, no problem. I cannot stand so

Unknown Speaker 14:56

sorry. You're in the middle of your

Speaker 2 14:58

office. Oh, yeah. So what I was like telling is that, like, for the testing of the GDG approach, we have like cucumber with like guidance syntax, and I think built in like syntax is like what we use In our like projects.

Speaker 3 15:18

You also talking about like using cloud moving into containerization, just as a basic concept, what is containerization, and what are the motivations moving into a containerization environment?

Speaker 2 15:36

So, like, for example. So basically, what I will say is like contrary, containerization is like just like with like virtualizations, right? What I mean by that is like everything that you need for a program so it basically package into like one container, right? So basically your container contains all your code. That library is like required, the dependencies and the system to and the configurations that you have. So Docker is like a containerization tool, which the main NRT is like of the using like the system is like, all your code or applications are, like, more portable and lightweight. So you won't have the problem of saying that it works in my computer and it doesn't work on your computer, right? So basically, Douglas provide all those, like, available settings and everything so and working with it, you can specifically, like, have, like you have, like, your S in it, and have the simple Docker file that gives you what you need to do then, like, order to, like, set that applications, in order to really build that applications and from that container. So, yeah, that's like, what fundraisation is used, and Docker is like, one example of it.

Speaker 3 17:11

You also occurred by the tech lead. What do you think are the key characteristics that you need to be doing. Let me rephrase, as a tech lead, what do you think your key responsibilities are on a day to day basis?

Speaker 2 17:31

So yeah, basically, as a tech lead, I think Moreover, what you need is like, there are, like, certain activities or like characteristics that you need a part of, like having like technical ability, right? It's like important to have the technical abilities too, but communications, it's like the other things that is like much needed when you are on like tech lead road, and I have like quality of black code that you are like prompting, also, you need to check on those things, which is like a very hard skills like you gain that with like experience. Also, like mentoring to the team members is like kind of the thing that the technique should like have it, and you have to be some kind of like mentoring to other people's and if there is any kind of like knowledge sharing or providing any feedbacks or giving directions, mainly on how to look those like ideas up. So yeah, you should be more of a responsible as a tech lead and basically collaborating and communicating with the team, and then take that idea to like stakeholders, or like businesses, or, you know, other people who might be like interested in the things that you are working on. So it also an important for me as technique to be like driving the team for some innovations and improvements as well. And I think techniques will also be leader and take like ownerships and mostly like, be accountable for all the team actions also in

Speaker 3 19:28

your recent experiences, what's been your most, proudest achievement as a tech leader?

Unknown Speaker 19:35

Oh, yeah. So

Speaker 2 19:38

I'll say, like, as a click, I will be like, one of like, my progress at shipment as a tech link will be like, work with like, top of my engineers and then like, solve enterprise like, which are like in the backlog, not only for my particular team, but also for like, other things. Right? What I mean by that is, like, our company has enterprise wide tech backlog, right? And this is basically, is your replicating between to stay, like, compliant with the company's policies, and that we had to, like, basically incorporate like SDK into like applications, and, you know, configuring it to, like, send some data to like streaming workflows, we basically, like owned a lot of like micro services, and just like, doing the same thing over and over Again, and all the services same redundant. So I and my couple of like engineers and came with my team, we started strategizing or how we can, like, achieve those services and active this in and like efficient manner. And I think what I advised was to, like, basically create a service that will basically handle the metadata, and then, based on the metadata, take, like RESTful API call and, you know, pick that call and then forward it to the team or the service that can like, handle this, like stga integrations to an API call directly, without having to like particle reporting, the STK integrations and all the other things that come with it. So, you know, leading a team to provide solutions that only that not only help my particular team, but allowed other teams to decrease their backlog and also, like clearing efforts time was, like, something that I'm proud of, because I was like, able to lead like, a couple of like my engineers to provide, like, enterprise wide solutions. So yeah, thank you.

Speaker 3 22:00

What do you think is, let's say you were successful in this role, what would be your strongest asset that you bring to the role?

Speaker 2 22:10

Oh, yeah. So thinking about, like, my skills too. I have, like, got, like, six years, and I've experienced also, right? You know, I have, I think I have a pretty good technical background. You like modern technologies, and especially working with, like microservices and so on. There are, like, fun skills, apart from this, like hard skills. I think I'm, like, more of a very effective communicator that I can like voice up what I think will be better for the team and for the company itself, as I like to be more like, transparent with the teammates and provide them like, really positive mindsets and feedbacks when needed, and then be very constructive when needed to so yeah, I think I will also bring collaborative environment to the team, because greater minds working together leads to very good work cultures that everybody wants to be a part of it. And apart from that, I think I will also mainly help the Team Drive to our success in a way. On

Speaker 3 23:29

the flip side of it, what's your what's one characteristic you feel that you need to improve or to further yourself? Sorry I didn't get Can you what's one weakness and one feel that you need to work more to improve yourself. Oh,

Speaker 2 23:42

yeah. I think every persons are like, different from each others. They have like, strong assets, as well as they have, like, some characteristics that are not like, really good. So I think one of the characteristics that I need to like improve on, what I think is like I would be some kind of, like delegations, right? Because whenever, like, there's any challenge, whenever there's any challenging tasks, I basically picked that all upon myself, right? So sometimes I think I do that in order to basically find the solutions. Thinking like help might not be needed. But you know, whenever the like smaller research actions or small research that may help that I'm like reaching on that needs to be delegated. So not only my like experience increases, my team's experience increases on that like particular manner as well. So I think I will work on like, delegating like smaller tasks to my teammates. And you know, work like, from the

Unknown Speaker 24:56

from the will to like plan manaway,

Speaker 3 25:01

what would you describe your proficiency level

Speaker 2 25:06

in Java? So, yeah, Java. I've been, like, working with Java for like, more than like, four or five years, so I think I'm pretty, like, proficient. I would read myself somewhere around eight and eight and half out of like 10.

Speaker 3 25:33

Can you explain to me? I'm going to give you a scenario. Let's say that you've got a Java process which is being accessed via front end. Front end is showing that your request was initiated from the front end and it's in a stuck state. What would be the series? Obviously, the front end calls the back end, which is a Java process. What would be the series of events that you would follow to work out what went wrong? Oh, grow.

Speaker 2 26:03

So, yeah, working itself in the front end, the first mine that will come in, like, do the console log, right? So whenever there's like something, so first thing you will do the console log. Then I'll look out for like network cab or cons to of the developers, tools of the front end, then whatever, like browsers that I'm using it right? So, you know, does that like helping that will help me identify what endpoints are, whatever it's calling it. And then after that, like particular one, I will look for, like the application logs on the back end as well. So if it's like in a production environment, or like the higher environment, I've been like, primarily working with the new real dashboards, and also with like Splunk for like logging out. So I'll basically query and like our applications against, like Splunk against the indexes and the arrows, or any other like codes that might be available for us. And once like that is done, I will have like, some kind of like idea what process is like failing and the work process is like, stuck into it. So once I identify that, I will see, to see if I can, like, restart the process to restore the responsiveness of the UI, and after that, I will basically go about like debugging mode. The way I developed is like, I tried to like, recreate the issue in the first lower environments are the in the locals to have, like, better controls of the system itself. You can, I think you know, the log into, like the box and then run like some Linux commands, like, see what Java process is like running and get the keep dump of the applications, like using J stack to like, exit, extract the dump, and then use a visual virtual machines to see what the process livestock on and like. Then go from there, if there are like database query exceptions, I analyze the query to if there are like sockets timer exceptions, I see what kind of like sockets are like open if, mainly like, if there's like deadlocks that I will look for like trade states that are blocked, and then I think those are like series of like steps that I will mainly follow to, like, find out what is like going on and what causing the error.

Speaker 3 28:52

What does observability mean? Oh,

Speaker 2 29:00

so yeah, absolutely, I think basically means a way of, like, mainly how the representing the system health and responsiveness, basically it contains, like, a couple of things to mainly help detect, like to investigate, and also, like resolves the issues, right? So, for an example, if you have neural tool, it helps you detect issues, like only you know, because, based on like, certain thresholds, it will alert systems. So it is a tool for, like, detecting any anapolis. It also helps in like investigation, in a sense of like, on those like dashboards, you can visualize what part of like application is failing, like if it is not working for like, certain APIs or like, if certain like process are failing, or threats are stopped, or anything like that in case, so that that's what like observability is like, is like helpful and banking blocks metrics and different process of the application itself understood. I

Unknown Speaker 30:23

Are you familiar with the term Dora?

Unknown Speaker 30:27

Oh yeah, I have heard about it,

Unknown Speaker 30:31

but I'm not, like, really

Unknown Speaker 30:35

into it. I have. Do you

Unknown Speaker 30:37

know what it stands for?

Unknown Speaker 30:41

Philosophy is I I'm

Speaker 2 30:42

not sure about it. That's fine.

Speaker 3 30:52

How do you stay appraised of current market standards in technology?

Unknown Speaker 31:00

So, yeah, we do have like

Unknown Speaker 31:04

resistance and everything.

Speaker 2 31:07

So basically, I would say, like, I like to be up to date, also with like resistance to our like, interact with them. What are like? The things that is like needed for our applications, I'll basically object myself with like, with like new technologies. Go for like udev courses to if needed, and work, walk around and try to like, solve the problems, read more blogs and yeah, and also, like, if there's like, some projects that is like going on in the outside of the projects to out, like involved into it. So yeah, these are the steps that I took, mainly to, like, gain the knowledge. So

Speaker 3 32:03

it you say that you're obviously a full stack developer. What do you think are there? But just basically what makes good UI design?

Speaker 2 32:19

Oh, yeah, I think the this is like a vague I think it can contest, like, big answers for, like, the good UI design. I think it needs to be mainly like the users can, like, interact with easier, simple and clear, and also, like visually appealing, right? So basically, it will consist across, like the web pages, so that you don't have a button of like one color here and then another, like color over there, or like different size of the button itself, or anything like so. So it needs to be, like, persistent, the word I think, yeah, consistence will be like the useful one over here, and have like, certain standards for it. So it needs to be clear that, basically, it gives you an idea of what you are doing. It needs to be intuitive, so that you know your users do not need. Don't have to, like think too much what something things to do. So it should be like, responsive to should be flexible as well. And last but not least, it needs to be visually appealing, right? So it needs to be looked I think those are like things that makes good UI. And as I said, it's a big, big questions, because the design itself might be like different from one user to another user. So basically, the consistency is like what I think it should be in, like UX design or the UI itself.

Speaker 3 34:06

Open source libraries, have you used in recent years? And how do you come about those libraries?

Unknown Speaker 34:15

Yeah, so

Speaker 2 34:17

really, are you talking about like UI, or especially for UI,

Unknown Speaker 34:24

okay, contributions

Unknown Speaker 34:26

to open source or interested?

Unknown Speaker 34:32

Are you interested open source?

Speaker 2 34:36

Not exactly, but some of the like open source libraries or like Angular that I have worked with, like, includes natural UI for like, different components, like buttons. I have, like, worked with ng x for the forms itself. I've also worked with ngi X for, like, state management a little bit in like, if you are, like, talking, especially for, like, back into, mainly, like Java packages I have, like, worked with, so, yeah, I think I will also, like, say some of like ones that includes the Spring Framework also itself, right? So the spring with like JPA hibernate in everything around the spring ecosystems I have, like, worked with fronters for, like generating mainly like dashboards. So, yeah, these are, like, the libraries that I have worked with, and also, like, I have also worked with, like, recently with like, cloud gateway, yeah, I think those are, like, the libraries that are happening, like understood. Those are

Speaker 3 35:55

the key questions I had for you today. Do you have anything for me?

Speaker 2 36:02

She so this one is like, for like back end, or like front end. It does it like on this of like border,

Speaker 3 36:12

looking to do, we're looking to onboard a scrum team in where we have a scrum master that's very strong in the teams, they engage with requirements as well as managing these Scrum teams. But then also the tech lead has to be quite strong. So we're looking for a combination of both scrum based scrum master and a tech lead to run a team of 10 people, essentially managing delivery for our area of business, we work in the equity finance states. So we have traders in New York, London, Hong Kong, Japan and Australia. We also have middle Office users in Tampa and in Belfast and Hong Kong. What we aim to do is build the trading application and the supportive functions to support trading directly. In house, we have a front end that's built existing in WPF, C, sharp, and strategically, we're moving towards react. A back end is a composite of Java services operating in a micro service environment. So we're looking for an individual that can lead that team

Unknown Speaker 37:25

and

Speaker 3 37:27

provide accountability for the delivery of that team or items we're working now book of the work.

Speaker 2 37:38

Any further questions? I think that's it. He explained it quite well. So thank you for your time. And it was, like, really nice talking to you. Also, like, sometimes, like, yeah,

Unknown Speaker 37:51

like, I would like, like to ask, like,

Speaker 2 37:56

also, what like will have, like, maybe successful on this role, like, what like the stack that you are using it and what will be like a roadmap for that

Speaker 3 38:11

back end is in mentioned, the front end. Back end is Java

Unknown Speaker 38:16

services. We use the Windows processing

Unknown Speaker 38:19

so using

Speaker 3 38:22

things like, how capitalize and communication, I do have a couple of other questions. I'm going to ask you those as well today. But then we use Oracle as a data persistence layer, capital as a messaging layer. We have caches with produce and ignite. We also, I mean, we have an in house framework they also use for serialization. But we moving more towards things like restable services for Jason,

Unknown Speaker 38:58

that's pretty much our tech stack.

Speaker 2 39:02

We other first questions? Yeah, I think that's it like. Thank you so much for your time. Thank you.

Unknown Speaker 39:10

One question I had for you.

Unknown Speaker 39:13

Just remember I was gonna ask you this, but I thought,

Unknown Speaker 39:16

well, obviously

Speaker 3 39:18

Kafka is a key thing for us, what's your understanding of Kafka, and how would a concept known as partitioning be used to scale up?

Speaker 2 39:29

Yeah. So we have, like integrated Kafka in our projects too. So basically, what am I understanding? Kafka is like model, right? So basically you have your publishers that like write, call your topics, and then you have like consumers, which mainly reads from the topics. So publishers, like publishers, the consumers, subscribers. Now how it works is that you have like topic, or is like category where misses are like published, and then with the topic, there are like multiple there are like multiples, like messages that are like published, and within a topic, we have like partitions. So a topic is like split into multiple partitions. So basically, people like parallel processing of those, like messages. So as you increase the partitions itself, it might include the parallelism, but depends on like different factors, and there are, like other concepts within where you need, like brokers, which is like the service, and then Zookeeper, which is like the managers and so on. But how it likes, how partitions helps in, like scaling up, is that your purchase usually publish those messages to the partitions mainly in like a round robin manner, right? So it will, like, basically go around the partitions, or if you want, like, specific ordering of, like certain messages that you use, like key based manners, and then consume consumers, mainly in a consumer group reads from like different partitions in like parallel. So if you have multiple consumers in a group itself, and there's partitions on those like topics, then it will basically enable parallel processing of those so as partisans like distribute the load, allowing like multiple consumers to like process the data simultaneously, which you know will help you scale up and like processing mainly those data so you just like horizontally scale by adding more brokers or like consumers

Speaker 3 41:55

understood. But thank you. That's no more questions for me. Thank you very much for your time and best of luck. Thank

Unknown Speaker 42:04

you so much. Have a good Day. Bye. Day.

Aman-deloitte-srv-first

Unknown Speaker 00:00

So my name is a

Speaker 1 00:04

senior software engineer. I'm working for the like for two years. I have 15 years of experience as a software engineer, and pretty much I have been working with Java microservices in cloud and using Kafka, mostly database. And other kind of shows. So if you can give me your history,

Speaker 2 00:30

so yeah, sure. My name is Aman. I've been working as an like software developer for about, like, more than six years, mainly working on, like, Java based web applications. I like mainly frameworks like Spring, Hibernate and web services for the API development I've been, like, primarily involved in, like developing micro services, where I've been designing them, deploying them, and also developing game. Some of the cloud services that I have worked with to like deploy applications include AWS, ECq, ECS and lambda. I'm also involved in, like testing using like JUnit and Mok. Currently I'm on like, strong team. The team like consist of like active lovers. We have product owner and Scrum Masters and on daily Basics, I'm mostly like, involved working on, like banking in like, mainly in Java, but helping the team with the code reviews, and also on like products and support many duties.

Speaker 1 01:46

So right now I'm going to introduce you some questions that I have here with me that we are looking for for the project, and we can keep going during this time. And if you have any questions, anything to me, you can ask. Feel free and don't, don't worry about having time to think you can take your time. We have a lot of time. And for sure you can, you know, like, wish that time well and without any rush, you know. So, yeah, so I'm gonna start with you said that review will change microservices, right? Can you explain me, why did you guys choose to use the microservices architecturally? Or do you have any idea why you guys choose work with member services instead of using a monolith, for example? Yes,

Speaker 2 02:47

yeah. So basically, in Microsoft DISA architecture, it is like very distinct in a sense that it is like more flexible, right? So basically, you have this small services that serve small business logic or some kind of domain that it handles. So for an example, you have had a service that is like, responsible for mainly handling changes to the profile preferences. And then you have like another services that is like, responsible for handling your request management, or you have, like another service that is like responsible for mainly experience APIs, which basically handles the entitlements. So you have this different kinds of like services that are independent, and they are like, developed independently, with its like on scalability and policies attached to it, mainly. So basically, what this does, it allows for like, faster development and then adapting to like different technology stats. So for our like maintenance in maintenance API, for our profile references, we use like back in database of like cross case SQL for Experience API, since it actually leverages and JSON based data structures, we use like DynamoDB as its backend. So disadvantages like make it more easier to maintain an update and do nearly the sequential update of each of those services individually. So that's the main reasons we went with like microservice architectures.

Unknown Speaker 04:40

So did you have any experience with tech,

Speaker 1 04:45

whether you can kill some advanced system view your career?

Speaker 2 04:50

Oh, yeah. So yeah, I do have, like, a lot of experience with the Kafka in my recent position, I have, like, implemented that one. But I work with like weekly messaging through SNS and SQS. Basically, we publish like messages to SNS topics, and then we have like SQS view, mainly like subscribe to that topics, and then there is a lambda that gets triggered once there is like message on SQ these are offshore models like Kafka, and it's like something similar we do for like notification service stream. But I am, like, familiar with like, how Kafka works and the overall like architectures, about his topics for producers and consumers and offsets and so on.

Speaker 1 05:51

So do you like to describe, like, how you guys are using STS? Or work like, which kind of services using that? Like, do you mind? So just give me, like, a small example how you guys are using message queue.

Speaker 2 06:12

So on our like, mainly legacy code base, we have this, like, email notification service, which was like, put it into our own applications. This notifications, like services, was a downstream partner team which handling which handles mainly like sending emails out to the users. So what will happen was like, sometime missing a surveillance will go down. And basically, because of this failure, and any like other workflow that was like, outside of like transactions will be mainly marked as like failure. So now this causes a lot of like manual work, basically go back and hit some data fixes and also COVID like sending the email manually now to basically resolve this like issue, what we did was created SPS two to mainly like subscribe to SNS, top topics that we had and with our applications, we'll do our modern framework. We'll take that messages in and they forward it to like SNS topic. Now, once like, once like our back in Java code, the publishers like to SNS topics. SQS will subscribe to that and then one there is message on that, SQS, AWS lambda. Will that get triggered? This lambda responsibility for the notification flow. Now, the advantage of using mainly the this decouple like system is that the main application, it's only responsible for, like, publishing the messages to the SNS topics, they are, like, backed by rates and so on. So mainly in case of like Fabio, we don't feel the whole applications flow. And now, since it's an SQS queue, the message will be like Phil said, because we have three retreats with exponential block so, and also our date letter queue. So in case of like, notification system basically fails for some reasons. It works for a certain period of time, and retries until like three times. And even if it is like sales after that, it goes to the date later key that can be like, manually a bit triggered for processing at like, later. Okay, cool. Okay, yeah. Other Yeah.

Speaker 1 09:03

So, let me ask you something. How would you handle like distributed transactions, like rollbacks, transaction managing microservices environment when, for example, each service has your own database. How would you handle like drawbacks transactions that are related distributed transactions out of kind of things.

Speaker 2 09:29

So, yeah, this is, like, basically one of the advantages that Microsoft has right. So in like distributed systems, you have like different services interacting with its own database, handling its own like many transactions, but to like basically overcome this, there is a different pattern called the saga Pattern Type. Inside the saga pattern, we basically have like different ways of like handling our main projects, which uses an orchestration based saga, which is basically a central controller that handles like the transactions of ecosystems, right? So basically what happens is that you initiate an actions into microservice. I mean the mind service, mean like succeed or fail. If it succeeds, if push it forward, and in any like chain if it fails, then what happens is that there will be like a compensating action present in those like services. So then this orchestrations will perform those like compensation actions. Okay?

Speaker 1 10:42

So basically you have, like, third service that's going to be the one that's going to be responsible for those transactions, right? Orchestrator service.

Speaker 1 10:59

So assuming that you have, like, I'm gonna give you a scenario right now, okay, so assuming that you have three services, service A, B and C, right? And service a has a customer table with the customer information, and service D and service C needs to interact with that table, but service B and C has your own database. How would you handle that? How would you do to for service B and C interactive service

Unknown Speaker 11:29

that belongs to service a?

Speaker 2 11:33

So if I'm not wrong, you need to say, service B and C has to interact with a, right? Okay, so if they have like service data needed in service EOC, basically you should use like service to service communications, right? Basically, it will be a RESTful API that exposes those endpoints. So based on like certain URI, which may be like customer ID or all customers, and then this can be like passed by those services to basically not be hitting the customer service over and over

Speaker 1 12:16

again. So you mean, we could have, like a real eye on the service a that we just have something like a gap right expected from service B and service C to be concerned? Yes, exactly. Yeah. Okay, any other kind of approach that you do,

Unknown Speaker 12:43

right now, I think we were like,

Speaker 2 12:48

if it's just been my thought, like, maybe service can, like, probably certain, certain kind of events when customer data changes, and then BFC can, like, live with some kind of event to basically consume those that will be like one way. Or you can also have like, some kind of, like distributed CAs that basically has their like shared data across some table. But I think the best way, I think will be like, like, risk, API integrated. It will be the best office.

Unknown Speaker 13:32

So

Speaker 1 13:36

let me go to the next one. So assuming that we have, like, a performance issue on the database. I want to give you examples to help you. I suppose that we have like one query, and that query, it's a big one. It's giving us like performance issue. Which steps do you think like to try to fix that variation.

Speaker 2 14:05

So it's just like a performance issue. So if you're like a

Speaker 1 14:12

performance issue that you you are monitoring, and you just over that a particular query was giving you performance issues. You know, it's the bottleneck of the service. So how would you which kind of steps do it take, like to fix that query to make sure that the service is not going to have those performance issues anymore?

Speaker 2 14:36

So Right? So basically, if you want to look at the query execution plan you can like simply add the explain, analyze keyword before your query, which gives you a step by step, of where there is like stable scan, or what kind of like indexes are like used on those tables. And so what kind of execution strategies too. So right when you took at this, you will be able to see where the bottlenecks are, and usually it indicates that you are doing like cable scans instead of colon scans. So you can basically see if you need to add like indexes or not. You can also see what kind of like fluents You are like using it, and it might lead to some kind of like issue if there is like indirect fluence that you are using. So basically, the solution will be to like add in excess whenever it's possible, to basically speed up the lookup based on the filtering that you are doing, or mainly like joints that you are doing, that you can maybe have the query modified to the designated response by using limit and offset, And also like modify the query to only really like Selected Columns better than selecting everything if you are using that. So will be the best way, I think.

Speaker 1 16:15

So I can change the subject a little bit and talk about reading files. So let's suppose that you have to read a file for your computer using Java, but you have like, let's suppose, as a geeks, file like, very validate

Unknown Speaker 16:36

memory issues.

Speaker 1 16:40

How would you do to read that file in a way that you not having those issues?

Speaker 2 16:46

Yeah, so in that case, mainly like reading a file. If you are, like, reading like, really large files that have like lot of rows, it will definitely read, but you will definitely have like, some kind of like error. So basically, if you are running into that issue, you read all the lines to write. So basically the way we can handle is you basically chop up the data and then handle it right. So basically, instead of, like loading everything, you will convert files into some kind of like Buffer reader, and then you will read it line by line. Or you can, like, basically do some kind of like allocations after much buffer you want, ended with that buffer until there is, like, some buffer remaining. So I think that will be like how you do basically, okay,

Speaker 1 17:57

okay, so which can you choose? Perfect,

Unknown Speaker 18:02

Java. 17.

Speaker 1 18:05

Okay, can you describe me some features that comes after Java date,

Speaker 2 18:12

some features we have, like records, also in Java 70, that we mostly use, which will have, like pattern matching, also users,

Speaker 1 18:31

anything that you remind that after Java eight release that comes, which was really important,

Speaker 2 18:41

yeah, we widely use the optional classes, which is like rampart or no office, that have like functional interface and also like lambda expressions. They have, like stream APIs, also data and APIs and yeah. So these are like features that was in others.

Unknown Speaker 19:08

Okay, cool. So,

Unknown Speaker 19:17

so do you know what is the function interface? Yeah.

Speaker 2 19:22

So basically, functional interface is an interface that contains, like, only one abstract method. There is such a way to, like represent mainly the Lambda expressions.

Speaker 1 19:36

What about the format in an interface?

Speaker 2 19:42

In an interface, right? So, basically, default method was introduced as a way to like add backward compatibility on like interfaces, and it is a method that has like implementations

Unknown Speaker 20:00

within like implementation interface.

Unknown Speaker 20:07

So okay, so now

Unknown Speaker 20:10

I'm gonna give you,

Speaker 1 20:13

like a little exercise, something like very simple, and you don't need to code using any ID, anything like that. You can just give me like instructions on the chat, or you can just tell me if you prefer how to handle that. So I don't want to see like I don't need like Java codes that, anything like that, you know, because we don't have the appropriate tools here for work on this, so it's not going to be shared. So let me just give you and just tell me, how would you fix that?

Unknown Speaker 20:58

Actually, I'm sorry, not fixed but so

Unknown Speaker 21:03

can you

Unknown Speaker 21:08

take a time just

Speaker 2 21:19

do so, so we will we always have like, sorted, added,

Speaker 1 21:35

actually, now That's not sorted, which you see like

Unknown Speaker 21:40

have like, 11126460,

Unknown Speaker 21:45

yeah, I see the issues.

Unknown Speaker 21:48

So it's okay.

Speaker 2 22:02

So, in this case, I think basically we need to, like, travel to the list and then like, find the difference. And I think the best way we can do here is like, we need to, like, basically populate a set, right? This set will then keep track of like, what elements I have already looked through, and then, with my iterations, I find the difference from my target to the number I'm in now, if that number with the difference. Exits on the set. Then it means that I have a pair. And then I will like, create this pair, and then add it to the add a list. And after the executions like complex, it will like, basically return the value. Both. Yeah,

Speaker 1 23:07

that works. Okay, so let me ask you about best student development

Speaker 2 23:16

we have, like the music test driven. Do?

Speaker 2 23:29

So basically, like test driven development is, like I would say, a practice or like pattern where you approach by writing the test first, right? So basically, you write a failing test, which is like, usually known as the Red State, and then you write the minimum amount of like both, just to make that this pass, like, for example, like if you want to create a service, then you start by like, failing this, and then, just like, write the list code to make pass in your service, and then you like increment on those like tests until you with the functionality. And yeah, so that way, you basically have all that test cases covered

Unknown Speaker 24:19

as you needed for your application.

Unknown Speaker 24:30

I don't have any more questions for you

Unknown Speaker 24:34

to give your feedback. You did very well, like

Unknown Speaker 24:38

prior to the other things that we have.

Speaker 1 24:45

So now I'm gonna send my feedback to Kelly. I think you were talking with Kelly, right? Yeah, she's gonna exchange the process with you. From my end, you did very well, so it was a real pleasure to talk with you. Sorry to, like, give you so much questions, because usually, that's not the way that we usually interview here at the

Unknown Speaker 25:16

Library clients. We had to

Speaker 1 25:18

be sure that all those topics have been covered. You know, yeah,

Unknown Speaker 25:22

I think so often.

Anish Ghimire- Infosys- SRV-Final

Speaker 1 00:00

The like, customer support, or, you know, support team, basically, like, it's an issue with the back end server, right? You cannot do anything you know from the client side. See, there is, like, something called, you know, locally cast, you know, that might be, you know, you might want to, like, you know, clear, clear, the case. But that's basically it, right? Because, you know, that's an issue in the back end solver, right? And you cannot do anything you know, rather than the you know, client side.

Speaker 2 00:33

Okay, I have just shared any API do?

Unknown Speaker 00:43

Are you able to see it in the team shot?

Unknown Speaker 00:45

Yeah.

Speaker 2 00:47

Can you write a pseudo code if you have to consume This get web service, how about you do it? I

Unknown Speaker 01:19

BBB.

Speaker 1 01:33

So we might like, probably be, you know, using some kind of, you know, client, right? So, like, I work with, like rest templates, I know, along along with, like apartheid closable clients. So I think if you are not talking about like pseudo code, I can write it as

Speaker 2 01:56

well. Yeah, technology choice,

Speaker 1 02:10

sorry, did you want the like actual code or the pseudo code?

Unknown Speaker 02:16

If you can write actual code, that would be great, not the step by step

Unknown Speaker 02:24

descriptions, right? Like algorithm, not Like,

Unknown Speaker 02:29

not like, What, Not

Speaker 3 04:19

so we know you Want him to write the REST client right, basically, because if I see his logic right, I don't think is going on the same ground

Speaker 2 04:32

is basically right. Yeah, he's trying to add the dependency Rest Template and see If he can your Okay,

Unknown Speaker 04:47

Okay, yeah, whatever, Okay. I

Speaker 1 10:19

Yeah, so I would like to be like, do something like this, but, you know, it's not, like, accurate, but because there might be, you know, some system syntactical errors, but basically, like, you define your client using, you know, using the rest template as a dependency, And then, you know, have a set of headers that that I needed to add, and I have like query patterns added as a hash map. And well, then, like, I'm exchanging, you know, for HTTP, like method to get with the headers and query, you know, parenthesis for the particular URL. So I would say, like, it would be a rough way to, you know, consume an API. But you know, then you know, value of like exchange operation of the like waste template.

Speaker 2 11:19

Do you know what does the exchange method contains, in general, as what does it do with this URL, Get adder, QP,

Unknown Speaker 11:34

so, in general. So

Unknown Speaker 11:40

basically, like

Unknown Speaker 11:44

the exchange, exchange method, like,

Speaker 1 11:48

you know, like, basically like ticks, like URL looks at the you know, headers, and then, you know, have a like query parameter that you know, that, like, based on the request verb that, like you are passing. And well, internally, like it makes the, you know, HTTP call. And so the flexibility of, like, what kind of data you want to, you know, like variables, like, what kind of no you want to add, and then so on. Yeah questions.

Unknown Speaker 12:28

Okay, good with this. Okay,

Speaker 3 12:31

so you mentioned about lot of interactions which you are using with Kafka in your application, right?

Unknown Speaker 12:39

So can it explain like, where you are using Kafka.

Speaker 1 12:44

So, thinking about a previous project, I like, basically, you know, used Kafka, like, sometimes to handle a lot of data, and whenever, like, initially we had in, you know, email notification, like service, you know, embedded in our application, right? And when the email like server was down, or our in our application, like would go to, you know, down as well. So basically, like that was a separate concern. So we started, you know, publishing like messages to Kafka topic. And basically, like created, like Kafka, or consumer, you know, that would like, read the messages and then, like, you know, power process these messages, asyncrotant, like, asynchronously, and also send, you know, an email out to the users. So, yeah, that's, that is the basic like, use cases you know, that, that, you know, I used in Kafka, the project that I particularly worked on. So what

Speaker 3 13:46

I understood is like you basically were posting that when email need to be triggered, you were posting it at Kafka topic, and then your email server was listening to that topic, and then it was triggering that email Correct. Yeah. Okay, so, can you tell me, like your role, some consumers and all right, can you just explain me, what are partitions in like, what you're different written partition and a topic, basically.

Speaker 1 14:17

So differentiating between like partition and a topic. So like topic and partisans are like two, you know, basic blocks. And I think like topic is like a channel and, you know, it's just like a feed name where, you know, messages can be like, published, and we, you know, very like, loosely, you can write, like, think of topic as, you know, like a table in your database, and, you know, it basically, like organizes, you know, those records in, you know, some kind of, like logic, but you know, part like partition is a very small unit of data. And so partition, like, basically, like, you know, contents and ordered, you know, immutable, like, sequence of, you know, those Kafka messages, right? And image, like, record length, you know, is a partition and has some, you know, kind of offset, just like, which, which is, like the position or an integer, right? So, yeah, that's the basic difference between, you know, topic and like, topic is just a logical channel, but you know, partisan is just a, you know, physical unit that stores these topics.

Unknown Speaker 15:30

Okay, can you

Unknown Speaker 15:32

just write a pseudo code? How you can write

Unknown Speaker 15:35

consumer for a topic? Do

Unknown Speaker 15:43

uh, well, let me think

Speaker 1 15:52

so like in order to write a consumer like you need, you know, a couple of things well,

Unknown Speaker 16:08

thinking about those.

Speaker 1 16:12

So, like, you need to, like, define, like, the bootstraps are right. And, you know, like, right? You know, digitalizer and class and and like, then, you know, subscription to like, particular, you know, topic. And I think, yeah, you can do that in this way. Yeah. So

Speaker 3 16:33

let's say, like, do you want to write a consumer Java class? Okay, for that. So how your consumer Java class will look like, at least like, from a high level perspective, which method you will be instantiating, or which method you will be overriding, anything like that. I just need some idea, like this high level idea about it. I got like you were trying to you will try to push up a server, and then you will create some you will give some name to topic name, and you will create some consumer group ID and everything, right? And then you are mentioning about serializer and decentralize. That is fine, but I'm that is like the properties of your consumer group, right? But how you will be writing the consumer code like when you want to read the message right from a topic, How it Looks Like I

Unknown Speaker 22:17

Yeah, so I think this should be able to do this. Okay,

Speaker 3 22:24

why do you need this while loop here? Why you're in this while loop here?

Speaker 1 22:31

Well, so, like, basically, you know, to constantly, like, keep on, you know, pulling. You know, I don't want to just, like, pull one message and you know, be done with it. And like it will, but you know it will keep on between forever. And while you know, because you know, like, you need to, like, be looking for the messages, right? It just like, constantly, like continuously, like consuming, consumes the messages from the topic and, well, then the you know, consumer, like, basically, you know, looks for whatever message that I have in, you know, an internal like, in an interval, and every second I will pull And then do a certain operation. But, you know, this can be like basic change, you know, something by using some kind of flag, like a feature flag, which can be done, like on and off, you know, based on certain like, based API, or something like that.

Speaker 3 23:37

So one more question. So, have you ever came across any issues while consuming or producing any message to a Kafka topic, let's say like you are trying to consume some message, but some message are lost in between. Have you faced any such issues? So like

Speaker 1 24:00

from my past experience, basically, like, whenever, you know we have, like, any processing failure, like, we send the message, you know, to our, like, deleted queue, and you know that we are, like, easily able to recover those messages, like, even though, like, there might be, you know, some, like, processing, you're right. So we, like, we have added, you know, retry mechanism, and along with like, exponential, like, back off, and, you know, even if it like falls off, then we have data queue. So usually, yeah, don't, like, have any loss of loss in messages. Because, you know, the main point of like Kafka is that it like, acknowledges the, you know, messages, and then also like guarantees, you know at most, like one or like, like processing, and so, like, personally, I like, haven't, you know, have any message loss, because Kafka is, like, very resilient in terms of, you know, messages as it is like, supposed to, like, handle, like, multiple numbers of messages at a time.

Speaker 3 25:14

Okay, so do you know there is an concept of an Offset?

Unknown Speaker 25:20

Offset? Offset,

Unknown Speaker 25:22

what is the use of it?

Speaker 1 25:27

So, basically, like Kafka, offset is an identifier, you know, like like it is just a unique like integer, you know, for each like messages within within a like particular parties, and like it represents, right, like, what position like messages, you know, consumes in like it represents the inner position of the messages. And like in the party sun and like allows, you know, our like Kafka Consumer to, you know, keep track of the like messages that have the like that are like being consumed and that are left to be like, consumed. Okay, sorry.

Speaker 2 26:15

Is it offset? Is it like a string? Where is it a UID, usually,

Unknown Speaker 26:22

usually, offsets

Speaker 1 26:25

should, should be like a number, just, just a number.

Speaker 2 26:33

What is the key for a message in Kafka? What is the importance of a key? So

Unknown Speaker 26:44

could you like,

Unknown Speaker 26:46

ask the question again? Yeah,

Speaker 2 26:48

for a Kafka message, right? That is dropped on the topic is something called key.

Unknown Speaker 26:57

What is it and why is it important? I

Speaker 1 27:03

so basically, like key in Kafka, like messages, I think it's like optional tier, like it is like I think it is, you know, it is used to, maybe, like ordering the messages, or like partitioning, because you know your key like fields are not mandatory. But

Speaker 2 27:31

sorry, sorry. Can you explain? So how does it help in ordering? You said it will help in the ordering, right? What is that mean?

Speaker 1 27:41

I so ordering of like, it means like, if you need to, like, ensure that, you know, the messages are in like, particularly ordered, you know, like ordered way, like, you know, certain kind of, you know, messages like that are, you know, related to like, certain you know entity, and you need to, like process them, you know, in a particular order, so then like it will be placed and on that particular like partition, like Which, you know, guarantees like ordering within a partition. Go ahead.

Speaker 3 28:28

Yeah, okay, so let's say like, right now you were created a consumer, okay, okay, and maybe it's, let's say like it's listening to Topic. Topic name is topic one, okay? And let's say, like, do you know pods, right? Like, what pod like, correct, servers. Like, built, like, I'm less like, I'm having two servers where

Unknown Speaker 28:54

we have created this consumer group. Okay?

Speaker 3 28:58

So will both, and if you are given the same consumer group name, okay for the topic, so will that message will be consumed in both the pods, or it will be consumed in single part. And how that does it work? Basically.

Speaker 1 29:16

So we have two more parts, yes,

Speaker 3 29:19

which are having the same consumer group name, listening to the same topic? Okay, so whenever you send a message, will the message be consumed in both the both the part, or it will be consumed in single part.

Speaker 1 29:37

So basically, like, we have a horizontally, like scale system, where, like, you have, you know, multiple instances of the like, same, like, code running, you know, like, against different pods or clusters. The consumer group, like, basically, is, you know, just a consumer like that, you know, work together to, you know, consume those messages, right? Like your Kafka will, like, differentiate by, you know these topics, partisans, and you know these like partisans, and like these partisans can can become consumed, like by, you know, only one consumer, you know, within a particle, like consumer group. And so like, you know, if you have like, two consumers in the same group, Kafka will, like, assign partition to consumers like, such that partitions, you know, is consumed by like, only one consumer, and there is, like, no overlapping. And like, for example, you know, if you like, have a topic, like, a topic with like four partisan and then like on consumer group with like two consumers. Then, you know, like consumer like will only, you know, listen to the first two partition, and then you know, the other consumer like will be assigned to the rest of the two participants.

Unknown Speaker 31:07

Okay, okay, vida, do you have any questions?

Speaker 3 31:16

Okay, yeah, I think I am also done from my

Speaker 2 31:25

Okay. Anish, do you have any questions for us? Anish, yeah, like,

Speaker 1 31:29

what, what would be the rest of the processes, you know, and I would also like to know more about the team structure, and I, like, wasn't told about the location as well. I was given two locations.

Speaker 2 31:49

So usually the location here in Atlanta, here in

Speaker 1 31:54

Texas, I was given Atlanta and bought through the location

Unknown Speaker 31:59

I'm currently in Indianapolis, currently

Unknown Speaker 32:03

in Indianapolis,

Speaker 2 32:06

yeah, I think probably we don't know exact team that you might be hired into, right? So depending on that, probably they will decide on the team. They will give the feedback to the organizer

Unknown Speaker 32:25

interview and then where they will take it to the next level. So

Speaker 2 32:36

with holidays place, right? I'm not exactly sure of the time.

Unknown Speaker 32:42

most of the books are on vacation.

Speaker 1 32:48

Yeah, sounds good. And what about, like you mentioned.

Anup Basnet-Optum-SRV-Final-Offered

Speaker 1 00:00

Will. We will end with you, where you can then share some of your background, professional backgrounds, with your experience with us. If anything else that you're comfortable with sharing, feel free to share during your introduction. Give you a quick overview of the organization and the role itself after the introduction, and then we'll jump right into some questions. We'll start with some behavior style questions, and then we'll jump into some more technical questions. And then we try to leave about 10 to 15 minutes on the clock to allow you to ask us any questions that you may have. And then we'll, I will kick things off. So I am Kelly Holland, and I am a talent advocate here and also serve and what, what my role is today. What I do today is I'm a people manager. So my primary focus is, they're on individuals and their overall performance. I incorporate a focus on development plans and performance goals, and then simply provide day to day support the team members. I've been with Optum serve since July of 2024, so not quite a year yet, but I come from another area within the organization called optim insight, where I was a business systems analyst for many years and then led the IC operations and production support group in my prior role, I moved over here to The talent advocate role, where I support the teams, and I am based in Pensacola, Florida. That is a little bit about myself. I will turn it over to Scott, because you're next in line.

Speaker 2 01:54

Thank you. Kelly, hi. My name is Scott cordick. I'm an engineering lead here at optumserve. I joined Optum serve last February for this particular project that we're seeking to add some some engineers to my background is in software engineering. I did about six, seven years ago, start to move over a little bit more into the cloud, specifically AWS. And then I, you know, recently, kind of took a position within Optum server. I'm kind of doing a little bit of both, a little bit of software management and cloud management as well. So I based out of the suburbs of Chicago.

Speaker 3 02:49

My name is desol. I'm a lead software engineer in opt south, and I joined Optus south for maybe a little few months over two years, two and seven months. I'm not already working on this contract, BWS, you know. I'm happy in the team, you know, for interview. And my background is Microsoft, shelf with CSF, you know. And I also work with Angular. And being, you know, my experience total will be 20 plus years, 2425 plus. I mean, 2023 24 years. And you know, work on variety of projects I based off in

Unknown Speaker 03:37

San Francisco, Bay Area,

Speaker 3 03:40

East Bay. Okay, that's about it.

Speaker 1 03:48

All right, if you'd like to jump in and then give us a little bit about your background and experience,

Speaker 4 03:53

sure. So first of all, nice to meet you all, and thank you for the person that for a couple minutes late. And as you know, my name is Anup, and you know, I have been working as a software engineer, you know, for like, about five years. You know, working primarily on Java based with application utilizing the framework like Spring Boot, you know, Hibernate, and you know, RESTful web services for API development. And, you know, pretty good experience working on the services, also involved in, you know, designing and developing and developing, then some of the cloud service based application and that, you know, I had more with AWS and, you know, ECS, SPS, SNS, lambda. And, you know, cloud formation template, you know, involved, you know, creating the Jenkins pipeline for CICD tool and leveraging, you know, the script for history, like the strays. I know, definition and cloud formation template for opposing, you know, the the for the like the exploit, like the provision the cloud, the infrastructure for the like the the cloud based application, you know, and for the testing, you know, I have been over with like the J unit and you know, so you know, and Mockito for the Unit, writing the test cases and, you know, and the like the I have the work with the team. That team consists of, like the five developer, like the product owners and Scrum Master. And, you know, on the daily basis, I have involved working on the both, you know, Java coding and helping the team with the code review. And also, you know, production support duty as well. So, you know, this is my kind of rundown for the last five years. Background, excellent.

Speaker 1 05:54

Yeah, we will definitely have some follow up questions for you relating to your background and experience there. So thank you for the information you shared. We will give you a quick overview of the organization and role, so you know who you're interviewing with, and more about the role that you're interviewing for. So Office serve, who are we and kind of what we do? So here at office serve. We were dedicated to serving our nation's military and veterans. We have the honor to support federal agencies and their efforts to advance the United States healthcare system and improve the overall health and well being of those who serve and who have served our country. That's just briefly a little bit about officer serve and what our focus is. I am going to hand it off to Scott to do a role overview.

Unknown Speaker 06:51

Thank you, Kelly. So

Speaker 2 06:55

we are looking to expand our like our little AWS engineering team here by adding a few different software engineers and data engineers to help us take a solution that we started building last spring and summer and kind of had to put on the shelf for a little bit, and, you know, kind of taking it from where it was left off and bringing it to completion, to like, production level. We have a pretty tight timeline in which to work, you know, roughly about three months to take it from where it is, get it to production ready. You know, we're it's unique to Optum serve, and the fact that it's the first time we're building anything in AWS, that's why we're kind of looking for folks with experience in building within the AWS cloud realm. Because of we're primarily an Azure shop. Typically, we run an agile fashion, but we're going to kind of try and run a little bit later, just because of the timeline that's that's forwarded to us. We're going to try and, you know, kind of forego some of the traditional ceremonies that come along with running in an agile format, and just really, just trying to hit the ground and just just build. We're definitely looking for folks that you know, have experience in building software applications three tier like, you know, three Tier solutions, you know, back end, front end, middle API layer, you know, kind of have a good, good figure on the pulse. So there are different resources that are out there, and really kind of trying to pull this solution together, as we're, you know, attempting to prepare for the lottery for this project, okay, trying to think if there's anything else that that's, you know, worthwhile to know, I mean, I guess we have, we have some, some four real big components that we're trying to stand up as part of the solution. The first one is kind of like a contact center type solution. We're going to be using a couple of different AWS products to try and build out this contact center, Amazon Connect, which does phone and chat, Amazon chime, SDK, which is going to bring a video component and then pinpoint, which is going to allow us to Do like marketing outreach via SMS, text. We have a restricted web presence, basically just a content based website that we have to stand up. We actually already got that part of the solution pretty much done and ready to go. We're using an off the shelf content management system, Doc CMS to provide that functionality that, for all intents and purposes, is ready to go. We're not going to have to do too much on that. There's still quite a bit to build out on the contact center side. And then we're going to have, like, a data, data and analytics center, slash platform thing. We're going to bring in a bunch of data into redshift. You know, make it available for use and analysis in Amazon, QuickSight. And then we also are going to be using ServiceNow as our case management tool. But even though it's going to be hosted in AWS, you know, there is a ServiceNow workstream that's going to be responsible for building out the ServiceNow solution, and we just may have to lend a hand with actually bringing me into AWS, but

Unknown Speaker 10:50

that's kind of it in a nutshell.

Speaker 1 10:58

All right, so we're going to jump into the interview, maybe the questions now, Interview Questions section, I will kick us off with with two questions, and then I'll hand it over to Scott and bet to jump in with the more technical side. Okay, so this first question here, can you tell us about a recent project you were part of. What parts of that project did you own, and then What value did that project raise the business,

Unknown Speaker 11:31

okay, so

Speaker 4 11:34

the recent project that I worked so, you know, so recently, you know, I was working on the project that that was a fulfillment of the enterprise, like the take backlog. So, you know, nothing that directly helped the business, the teams, but, you know, it is something that had that helps our you know, auditing team. So we had Amazon with some, sometimes, like we have the you know, process that, you know, interact with the the, you know, PR, data, payment system and you know, so and we being in the orchestration for layer for that, you know, man, like the manages, the routine of the, you know, all of those requested. So, you know, have the audit need, where we need to, like, aggregate all of the processing in the form of the information and the stream those logs, you know. So is it common? You know, snowflake tables, so, so for me, this particular process, you know, we had a like the, like the company wide, you know, backlog where, you know, we had, like the, the integrated SDK to our project. And, you know, so the process, like those logs and then send it forward. And, you know, but that our team had, like, ended just a lot of micro service and adding the SDK, like the multiple service, and in the little bit, you know, duplicate of the logic and just, you know, just having like, the different configuration. So, you know what I actually like, the proposal was to basically come up with the, like, the, you know, API that will be used, like the throughout the, you know, for the micro service, by just making the API call, and it will be deployed as a separate microservice, and, you know, as a standalone service, like service like, where the service could, like, interact with each other. And you know, often the first time, you know, the like the onboard, and then they send the API call to use the particle API and so, so that was the most recent one that I have, no it brought to our team. And, you know, partner with the team. So that is the kind of, like the reduce the food for onboarding to the platform and streaming, like the logs, so, so, you know, they'll that we like them, like the with their printer 25 like Microsoft became really small, and then also like the like partner teams that were, like, able to leverage the same, you know, API. So, you know, I was particularly Warner of the actual design of those service and also implemented. So, you know, that was the project that I have done. It

Speaker 1 14:49

okay, and you said you as you're part of the process, you were the actual you were the owner of the design. So the

Speaker 4 14:56

actual Yeah, design was mine, yes on that one. And the Yes,

Unknown Speaker 15:13

right.

Speaker 1 15:16

Okay, so when you say, just for a follow up on that, when you say owner of the actual design. So you engineer the design, did you? Did you go in and do the actual development work, or, more or less just

Unknown Speaker 15:29

designing it, and

Speaker 4 15:30

some type of so on that one like, so I actually know the owner design and owner and the design and implementation also. So actually the design documentation and approval and also work on it.

Speaker 1 15:51

Great. All right, okay, for that, we'll ask one more question here. All right, can you tell us about a time when you were not able to meet a work commitment? But did you learn what steps today do you take to ensure you meet your commitments?

Speaker 4 16:15

Okay? So, so you know talking about the not meeting the the timeline there. So, you know, someone of the times, you know, when I was not, you know, able to meet the deadline. And, you know, was whenever I was working with a new, you know, listener. And then the listener was supposed to be, basically, you know, listening to the event on our SPS queue. And the process was like the incoming message to the, you know, loaded into our persistent layer. So, you know, it seems like they really like the simple on that face value, but with the different finding throughout, you know, the way you know that this actually, like, kind of took longer than, you know, usual, like us. And so what was the like, the expected, right? So it was just like, this seems like the, you know, seems like they listen to a message and process it out and then put into into the database so, but often starting of the development, we found, found out that the actual persistent model was very complex out and in. It contained, you know, a lot of embedded relation shape and, you know, get those listen details. We need to leverage a like the a lot of API calls and, you know, and also, there were some field that within the persistent layer that you know, exists, like the form of the data, and not, you know, like the for these kind of, and so, but we never need to populate this. So this was one of the time when I was unable to, like the kind of meet the deadline. And, you know, kind of,

Unknown Speaker 18:15

kind of, I know,

Speaker 4 18:17

like the this feature was a couple of, like the sprints, and because of it was the critical aspect of one of the workflow, you know, like the four corresponding and their events, like, related to it. So, you know, like the for this particular scenario, I would say I did not meet the deadline, but it was great, you know, learning opportunity, because of, you know, our team was kind of moving to a new, like, the modern platform, like the for, like, retrieving the data and aggregate those together. So I'll say, I know, I was able to basically create, like, the create like a, like, kind of a checklist, and, you know, and from there, like the, you know, I was able to go and, you know, be needed and to the next time to have it, like something similar. And you know what? We all the information we needed to have before, and like the we even proceed with, like implementation. So, you know, that was the time, you know, I was not able to meet the deadline, and the attitude of the project,

Speaker 1 19:29

all right? And some, one of the things in my office, right that you do today to ensure you meeting commitments you said, You created some type of checklist to help you ensure you

Speaker 4 19:41

Yes. So, yes. So basically, you know, I come up with the documentation, like the for our internal team to basically, you know, follow in order to, like, be, you know, like the kind of new workflow and like the, like the listener on to the queue and for the next complex kind of project, right? So we need to have a proper detail on like, the model that we are, like persisting and like down the relationship that had the external elements, and, you know, any other API call that was involved. So it was kind of like the rough take list that, you know, teammate could use whenever they were working with something similar, like, so what need to be checked and what need to be done, like those kind of checklists. Okay,

Speaker 1 20:31

all right, okay, so that is all for me. I am going to turn it over now to the technical portion of our questions here, so I will head it off to Scott.

Unknown Speaker 20:52

I said, Thank

Speaker 2 20:53

you, Kelly. I will go ahead and jump in right now. Okay, so I don't want to switch into some AWS background and experience here. So can you tell me a little bit about like, when you're designing a scalable or fault scalable and fault tolerant solution in AWS like, what? What considerations do you take into account?

Speaker 4 21:21

Okay, so talking about the designing kind of fault tolerance and scalable solution here. So, you know, in order to make the ice level and, like, a fault tolerant solution for any service, like, you need to keep the track of the number of things, right. So I'll say, like, the first thing I would say, like, would be auto scalable, right? And so it would be, like, automated through the, like, the coordinate cluster ECS, or, you know, just like the auto scaling from, like the, like the foreign ACS, like the, you know, she do. So, you know, like, like, it's just already Scaling group for easy. Then it's basically make sure that application can be like, you know, can handle increased load, maybe, like, through like, peak traffic, or increasing the number of customer, like the customer, or something like that. So basically, it sits and, you know, on the on the top of, you know, your load balancer. And then basically you have to make sure that you have the healthy instances available to serve the load and, you know, and the second thing that I mentioned that was the load balancing, basically is, you know, like the making sure that the incoming traffic get routed to the instances that are able to take those loads and fulfill them, right? So, you know, rather than sitting queue it in the same instances to So, basically slow your service down the, you know, the other other like, the thing is, I'd like to, you know, address the that is, you know, we may have you using kind of stressless orchestras, like the architecture, like ECS or, you know, fargate or AWS lambda for, like, the process and so, so that, that, I mean, like, you don't, you know, have to keep the track of those session or, Like, you know, so on. So, you know, for like the currents, you need to like the, you know, have to like they have your application deployed across the different various region. So, you know, multiple region deployment with the multiple availability zone using kind of like the route, you know, like the SD to route the traffic. Or, you know, failover route routing, and then you have your data, you know, I know, like the you have to have the primary in this, then the secondary and read, like, replica, like these, some other reason, right? So, you know, then, you like, they probably implement a lot of resilience patterns like history, try and, you know, block off the pattern. And then, you know, I would say, like, you have the proper cloud that's all, like the COVID alarm, I will track, and, you know, tracing the like, the, basically, you know, then have some kind of, like, the, you know, monitoring from, like, the cloud was alarmed for the monitor, the instances. And then, you know, if they go down, you get the alert. And, you know, you get the alert on the like the basically supporting great dollar, and act on it. What is going on?

Unknown Speaker 24:35

Cool. Thank you.

Speaker 2 24:40

Let's talk a little bit about containers, a little bit. So do you know what the difference is between containerization and virtualization?

Speaker 4 24:49

Okay, so different between the containerization and the virtualization. So

Unknown Speaker 24:57

let me think about it. So,

Speaker 4 24:59

so basically, I think the controllation and the virtualization, so is likely it is using the some kind of like the virtual machine to, you know, it your hardware, right? And, and, you know, it is kind of basically allow you to run the multiple virtual machine on, like in the physical machine, some or somewhere, right? And so you can have a multiple operating system or libraries or dependency in our virtual machine, and then basically it is isolated, and, you know, runs independently, right? And, and I know, computerized is basically using your host operating system to run on isolated process, rather than having a voice, right, the whole BM put up into a new operating system. So, you know, container basically, are like kind of a lightweight and, you know, fast, and that includes just the, you know, application code and you know and dependency, rather than the whole operating system. So, so I think that's a different between the kind of configuration and personalization.

Speaker 2 26:17

Um, so kind of sticking with with containers. A little bit like, tell me a little bit about your experience with, with building on containers and AWS.

Unknown Speaker 26:32

Okay, so

Speaker 4 26:36

I'm talking about the experience you know, on that. So you know, all of our, you know, microservices are, you know, containerized. And so basically, you know, have a Docker files that, you know as the step on how to basically run the application. So, you know, building our application, you know, we build, you know, like, kind of our Docker image, and then, you know, published to the, you know, Artifactory. And then these, you know, Docker images are, you know, deployed on the AWS ECS, or, you know, target instances. And, you know, during the, you know, in, like, the diploma, diploma phase, it like the kind of post, I know, like, kind of a publicity like, and then, you know, Docker image are deployed on the AWS and Institute and and during the deployment phase that it will pull, like, those kind of container image and then build it and run it, right? So, you know, I have been involved in writing these Dockers files. Okay? Okay,

Speaker 2 27:46

what did you use for doing the deployments, for the

Speaker 4 27:54

like, for the deployment, like the, you know, it's just the AWS CloudFormation template. And, you know, like the backbone, and we have, like, the pipelines, you know, where, like, there was like, once we go to the pipeline, it goes to the kind of different stages, right from the there was like the, like, the different main system. There was like two, you know, preview, like the pre deployment and the post like that and and from there, like the, you know, you have the package command, and then it stays to the like the Docker file. So, and we, like the user, runs the Docker file. Then the next stage, you know, is actually, I know, deploy the application using the like the cloud permission template, so, you know, deploying through the cloud permission template using the existing like the Jenkins as our, you know, orchestra.

Speaker 2 28:53

One last real quick question, and I'll hand it over to do you have any experience you mentioned using cloud formation. Do you have any experience using TerraForm at all, writing TerraForm for your infrastructure as code so?

Speaker 4 29:09

So talking about the, you know, TerraForm, mostly we did in the cloud formation. But, you know, I had seen the TerraForm script. And, you know, I haven't directly dealt with the TerraForm, but you know, I'm kind of familiar with, like, the concept. It's just like the the cloud formation and the TerraForm is like, I think it's kind of like the single, do the same thing, but it's on the different and it's on the different, different system, right? So, you know, so I'm familiar with the concept that I know I have done the minor changes on the TerraForm and, you know, script that already in the kind of existence. So, you know, I have some experience on it, but not all extensive, alrighty. All

Unknown Speaker 29:58

right, cool. Thank you. I will

Unknown Speaker 30:00

hand it over to you now.

Speaker 3 30:02

Thanks, Scott, thanks. So, speaking about Docker, can you, you know, describe how you, you know, write the Docker file. Okay, so steps

Unknown Speaker 30:19

already by the Docker file, you know, yeah.

Speaker 4 30:22

So, so basically, you know, when you like the Docker file, you start with your busyness, right. So, so then, you know, first thing is the way you want the busyness to be pulled from. So that is kind of, you know, excuse me, kind of first time. And then the Docker file, can, you know, can like the contents that you basically create like, you know, like the working directory you want, you want your best directory to be. And then, you know, you would basically install your dependencies. Maybe, you know, copy package from part to other, and, you know, then build your application and, and then I can you do, expose, you know, you know, a part that then you like, the starting the command on how, now the application is start and, and that's basically, it is, right. So it is a very, you know, high level definition of step by step. You know, you like a basic needs, you know that is working directly. You know, any copy certain file system sorted like the dependency, and, you know, you expose a port so, so, you know, that is kind of like your like the Docker file, it would be, you know, used basically build on and deploy the Docker container.

Speaker 3 31:55

So this Docker file to create the Docker file, what about the, you know, how can we optimize the Dockerfile image?

Unknown Speaker 32:04

Okay, so,

Speaker 4 32:07

talking about optimizing the Dockerfile image, so you know. So you know, in order to optimize the Docker file image, so you basically do some kind of lightweight version and reduce the any kind of layers, or you know, kind of combination of commands, right? So basically, instead of doing kind of multiple commands, so you just kind of, you know, clean up those command and just like the do, like the simple runs. So you may be able to like, then, kind of, like, they removed any kind of care within it, right? And then they use some kind of, like the normalized, like the drill. And then I think you can also use a Docker compose to handle these. You know, multi container application.

Speaker 3 33:08

What about the security, though? How about we screw the Docker image the vulnerabilities? How will we prevent that, avoid the vulnerabilities that reduce those?

Speaker 4 33:17

Okay, so talking about the security here, so, you know, so basically, you know, in order to, you know, talk about the vulnerability, and you can have your static scan, at least the base image, then that you have, and, you know, we usually have a kind of, you know, three months rotation of the business to have a, you know, proper, like the Milton. And you can also, you know, can kind of incorporate those, you know, as package and build and and everything is scanning and to make sure that you are not using the vulnerable business. So, yeah, this is I will do it.

Speaker 3 34:10

So, so talk how you use the talk Are you familiar with talkers? What about the properties? Have you worked with the properties? Use it before?

Speaker 4 34:19

So talking about the Kubernetes, yeah, I have used that before. I have used it.

Speaker 3 34:26

Okay, how do you deploy the Docker image to the Kubernetes? Can you

Unknown Speaker 34:32

explain all that? Yeah,

Speaker 4 34:33

so, so talking about the deploying the Docker image on the Kubernetes. So you know. So basically, in order to like the deploy your Docker to Kubernetes, you basically have your Docker image build and push to like a like a registry. And once you have done with that, I know you basically create the deployment, YMO, like the configuration file here, and you would have some kind of, like the metadata and the specification like for that, know, where you'd have and what kind of, like, the container, what, you know, what is the name of the image, and, you know, the portal that it need to be, you know, actually exposed. And then you would have a like the service, like dyml file and like for like, accessing kind of these containers, I know. And you'd basically use a like the cubicle command like to actually apply this, like the deployment, like the yml and the service, like yml files, and then you can, like, just see the status of the the parts and the end service, right? So this is, like the this is kind of like how you do Kubernetes deployment or the Docker image.

Speaker 3 36:02

So if your application have configurations, right? Some some configurations is sensitive configuration data, and also your application have sensitive data, like password by token. And when you deploy the Kubernetes, where do you store them? One is config, you know, config, data, configuration, data, settings, data, one another, one is sensitive data, as they are still configuration by sensitive data,

Unknown Speaker 36:32

okay, so, so

Speaker 4 36:36

you know, basically you have, like the configuration that you know you can just use like, the config maps. And these are like, the use for the configuration values, right? And so, so you would have like, they have like, like yml file for config map type when you like. They have like, you can have like, what environment, what URL, etc, right? It gets stored, and that would be the like the conflict map for the password and the token and API keys. And you would use, like the secret that you can mount or into the app itself, right? So, so you know, you would use the config map for the configuration and the second, like the for the sensitive data, and you would have like, like, an encryption, like, you know, cloud, and you like, in your cluster for like, you know, and if you're like the your application, like it used there, like the head score vault, in order to like the manage this service. Basically, you know, during the bootstrapping of the application, the path of the particular circuit get, you know, like, kind of resolved and get injected into the application.

Speaker 3 37:55

So last question for the cover these stuff so we have, like, multiple environments to deploy the Docker image right on it so Dev, UAT, SA, staging and production for environments. How do you manage that? How do you set up that? Where make it like, you know, efficient, more efficient way to deploy individual, you know, environments. Of enrollment.

Speaker 5 38:31

So make it small. I use my helms heart and basically have some kind of main space for those particular environments.

Speaker 4 38:52

So you know. So basically, you know the space for those you know particle environment. You can have those deployment, right so, you know, and a party, you know, in a parameter those deployment and, you know, using your, you know, Helm chart, and basically have some kind of, you know, namespace for those particular environment.

Unknown Speaker 39:20

And, that's particular

Speaker 3 39:22

name, speed and particular environments,

Speaker 4 39:27

kind of, I have some same kind of like the some kind of name is for those particular environment,

Unknown Speaker 39:34

alright, so,

Speaker 3 39:38

Scott, do you have any I haven't talked with that programming programming yet, but I think I go to you so you have a chance to ask more questions. Yeah,

Speaker 2 39:46

let's before, before we if there's anything we want to ask, see if you has any any additional questions for us.

Speaker 1 39:57

For about 15 minutes left. So if you have any questions, or more than happy to answer those. Okay,

Speaker 4 40:09

so, you know, I did it when it was, like the very beginning of the brief. I did a little bit so, you know, so even just like, kind of, you know, it's just like the cloud engineering position, but you know, a lot of more like the site liability engineer, or strict like influence structure, or like the, you know, kind of a like, it's a position, it's like, kind of return, like structure related. Or it is just like the more like the hybrid, or like the cloud and the influence price structure. So kind of clear on trying to get clear on that one, yeah, this

Unknown Speaker 40:46

is it is,

Speaker 2 40:50

I would, I would probably say hybrid, a little bit we are going to have cloud slash infrastructure engineers that are going to be supporting our software team here to, you know, help ensure that we're, you know, following best practices. You know, doing our deployments properly. You know, you know, peer reviewing any any kind of TerraForm that we need to write, because we will be using TerraForm for infrastructure as code but, but I would still say 70 price, 75% of our focus is going to be setting up, developing and integrating Amazon Web Services. We're gonna we're gonna have some like, for example, we have a chat client that we need to create and deploy on AWS services. So, so that's like, kind of an example. We've got some front end kind of development work that we're gonna have to do as well. So,

Speaker 4 41:59

okay, so it's kind of front end, back end, and the for the AWS infrastructure, okay,

Unknown Speaker 42:14

okay, I think

Unknown Speaker 42:16

so. Thank you for the information

Speaker 4 42:19

otherwise you're looking for. And I think

Unknown Speaker 42:24

that will be like just one question for Now,

Speaker 2 42:30

having said that we do have a few extra minutes if you wanted to jump into some stuff.

Speaker 3 42:36

Yeah, so can you tell me about his play, about solid principle.

Unknown Speaker 42:45

Can you repeat the question again? Please? Solid

Unknown Speaker 42:47

principle.

Speaker 4 42:50

Okay, so talking about the solid principle here, so you know, basically, you know, its principle is like the, you know, five years for the application development, like, so you have, like, the kind of a single responsibility, kind of principle, which basically mean you should, you know, have a class that has the only one region to I can change, right? So it's like, the module or class only, like, like they do the one thing right? And then kind of, you have your, you know, open, close, kind of principle where you have to say, you know, you have the, like, the software entities, and it should be like the open for the extension, and, you know, and it, but it should be not closed for the modification. So, for example, you need to, like, you know, like discounting model, and then kind of like you have, like the discount interface, so that then you have the different kinds of, you know, discount that are the attached to it. And then, you know, and then, like the you have, like kind of, your kind of list of the substitution principle and where, like the, which is basically mean that that kind of subtypes that must be substitutable, and for these kind of, you know, best types. So, I know so, so if you have like the like the printer and the scanner, right? You can have all of these. Those are like the printer, and you know, like as a single object. And then you have like, kind of have like kind of implementation of those printer that you know only prints, and kind of, and then you have like a dependency, in words, in principle, which you know, basically mean that you like that you have a high level module that should not be depend on, like, kind of like, like the low level muscle, but you know, it depends on the section, so try to like the injector dependency as interface. And then, you know, interface, like the interface, like the segregation so likely, like the Wizard basically says that, you know, clients should not be enforced like to like depend on the interface, and there's like the kind of they do not use, the right on this plate, or like the kind of out of Your interface, as much as like you like before this principle. So basically, it helps to make your code more maintainable and scalable.

Speaker 3 45:33

So have you apply any of the principle in your code?

Speaker 4 45:39

Okay? So, you know, like, I have done it before. Yes, I had applied those, you know, you know, I regularly follow the, you know, single responsibility, interface, like, you know, segregation, dependency, like the inversion and, you know, open, close principle, you know. So, you know, I do follow these principles.

Speaker 3 46:01

Yes, give me a sample example, like when you refer to the code, like, Which one, which principle you use for them?

Speaker 4 46:12

Okay, yeah. So, you know, basically, you know, like one of our, you know, classes like the like the word calculator, so, you know, basically calculate the weight, or, you know, the like, the different kinds of like, the accounts and and then, you know, allocate between the users. So we have, like, the neutral weight, or which basically is like, the like, like, this is the like, the weight based value. And you know, then we have, you know, another kind of weight that is, like the function of those neutral weight so both of the, you know, like the calculator that we work with depending on the multiple, you know, abstraction. So we have a base with the, like, the the kind of weight calculator that has, like the, you know, calculate method then, depending on what kind of the weight we need. So, you know, we can inject either the like, the benchmark like, to do the neutral weight, right? So we have the two different kind of, like these things, single recipe class kind of, and you know, then, you know, we will have those, like, the two different types of module on the benchmark. So, so, you know, like, kind of, we actually need to, like the leverage then, you know, external interface, rather than just the wage calculators. So the benchmark, which kind of calculate the implement on an actual Weight Calculator. And then, you know, and we like the kind of hierarchy map, or based on what kind of hierarchy mapping it and it relates to So, so you know, from there, like that is your interface exaggeration principle as well. And you know, and you know, these are like the kind of injected as a dependency using the interface rather than the kind of object itself. So I know it is like it.

Speaker 3 48:15

What about the design patterns? Any design pattern you know, you know, you use in your project,

Speaker 4 48:23

okay, so talking about the design pattern like, you know, you know, I have, you know, work with, I can know, a lot of design pattern, and some of it like the, some of the like them is being a singleton pattern, You know, factory pattern, Director pattern, and, you know,

Unknown Speaker 48:43

those pattern I have worked with.

Unknown Speaker 48:45

Okay, so Okay,

Speaker 3 48:50

for the Geo, for US sales, your Java developer. I'm a C sharp, so I don't have any questions for the Java so I see your resume that you use the you know, one of your project like enough pitch radiance, you mentioned about used indexes on my secret to make the service provider ecosystem select queries, queries 75% more efficient. What do you mean by that? What do you use indexes? Can you explain why? We mean, how you use indexes, how you make sure that you hit the indexes, or, you know, or you create indices. I'm not sure that you know. Can you explain? Okay,

Speaker 4 49:33

so you know, so you know. So basically, like, the indexes are just on the database object. I know that helps in improving the like, the database improvements, right? So, like, if you have the complex query that libraries on a lot of like the filter using WHERE clause, and also, you know, does a lot of the joints, I know then regularly, kind of like the hitting the complex query gets, you know, slow because the query need, kind of, like the full table scan. So in order to resolve that, you know, the for these frequently in queries, you can kind of create an index, right? So basically, speed up the selected queries for those particular columns. So, you know, I was able to use this query analyzer using and explain kind of a cure, and then see some bottlenecks, and no and then I added the index on the certain columns which improve the performance of the SELECT query, you know? So that is kind of what it is. So it just an object kind of that help and finding faster without doing the entire table scan.

Speaker 3 50:56

One more last question, we say for four minutes. Left, DynamoDB, what is the, you know, pros and cons using Dynamo DB.

Speaker 4 51:07

So, talking about the DynamoDB and pros and cons here, so, you know, so, so basically, you know, DynamoDB is, and, you know, non rational database, right, provided by the AWS. So it's really useful for the, you know, high performance, and, you know, high scalability, right? So it is fully managed, like, the end serverless and, and, you know, it is just can handle millions of requests, you know, like, within a second. And you can like, you can also like the horizon, please, scale based on needs. So it is, you know, being on the kind of relational and it is a flexible schema, like the way you have, like key, like the field value for the partitioning, and it is, like, very, pretty well incorporated, and then by the AWS, right? And so that is kind of like, your major advantage is very faster and, you know, very, you know, partition tolerant, and, you know, available like that. So, but one of the like, the some of the like, the disadvantages here is that you don't have a lot of flexibility on your query. So because, like the you must use a partisan key, and also, you know, like this, some like the sort key, if you want, and like your data, like a more link is just like the whole thing goes into, like your value, right? Rather than the likely some kind of, like the structure data, right? So, if you have a kind of, like, the structure data with a high consistency requirement, then you know, like it's gonna likely, like it's not, does like, not provided that, yes.

Unknown Speaker 52:56

So what about two

Unknown Speaker 53:00

minutes left. Sorry, yeah,

Speaker 1 53:05

just so we don't look over that's quite all right. Quite all right. Certainly we have enough time. We could turn these more questions here, but we are at about two minutes left, so I just wanted to ask a new one more time. If you had any additional clusters for for us here that we can, we can answer, so,

Speaker 4 53:28

yeah, you know, like the I like to know, like the you know what, like the, like the like them, right? So, also know, like, like, what does the you know, like the Optum provides, like, the that, like timeline, like, when they will tell us, like, they got the job.

Speaker 1 53:51

Absolutely, yes. So we are we? This is a fast pace kind of interview and hire process, so we are hoping to make some decisions fairly quickly. We've got the interviews lined up for throughout the rest of this week, so it will be probably closer to the end of the week before we make a decision, but in any event, we'll be making it fairly quickly. So I will be in contact or my my vendor manager will be in contact with your agency, where we will provide updates from there. But if you have any questions, in the meantime, my email is on the invite, and you can reach out and ask any questions, and I'll be happy to field those answers for you.

Unknown Speaker 54:30

Okay, thank you for the answer.

Speaker 1 54:34

You're welcome. All right. Well, we are just about at time, right at two o'clock on my talk here, so we will wrap it up a new thank you for your time today and speaking with us and letting us know you a little bit here.

Unknown Speaker 54:48

Great to know you

Speaker 4 54:49

and as well. Thank you for my side, having me here and giving the opportunity to explain myself and answer some questions here.

Unknown Speaker 55:02

Thank you. Thank You.

Ashim Karki-Eastportanalytics-SRV-Final

Speaker 1 00:00

Some introductions here. My name Brian. I'm a software engineer on the application development team here at youthport. I joined a little over a year ago. I mainly work in the application development portion, which is basically a Java Spring Boot application with an API layer that interacts with you finance, we have a couple in our stack, but we have one big one that we're kind of looking to get help with. Nice, cool,

Speaker 2 00:33

yep, and I'm Jenner. I started a couple months before Brian. I'm also a software engineer. Up until about three weeks ago, I think by this point, I was on same teams Brian, but I recently pivoted, and now I'm on our sandbox team. I'm doing still some Java development, but a lot more SQL heavy and kind of ETL process heavy work. Sure, nice. So

Unknown Speaker 00:59

tell us a little bit about yourself.

Speaker 3 01:00

Sure My full name is ashil Karki, and I've been working as a full stack Java developer since past over five years now almost, and I've been primarily working on, you know, Java based applications, you know, utilizing Spring Boot, spring frameworks, and, you know, using hibernate, rest web services for API development. And being primarily involved in working on the microservices, and been building microservices with designing them, collecting the fundamental functional and non functional requirements as well. And I've also been collecting these requirements, gathering our design doc together, implementing them, and I've also been involved in front end. Work with Angular mainly, and can work with JavaScript and typescripts. And the applications that I've worked with has been primarily deployed on AWS, and I've also worked with services, including AWS, EC two instances, ECS, you know, SMS, SQS, lambdas, you know. So I do have pretty good experience working with, you know, databases as well. And currently I'm the we have a team where we have currently five developers on the team, one scrum master and, you know, other DevOps people, front end people as well. But, but my, mostly my work is on back end side, you know, writing Java back end code and helping the team, you know, with the code reviews and also different kinds of production, you know, support TVs as well. Cool, think awesome?

Speaker 1 02:46

So what? What made you attracted to this position? Was it something that you found or

Unknown Speaker 02:54

a different situation? Or did you like?

Speaker 3 02:57

Yeah, so, like, so I was, I was actually trying to make a switch after my last position, and I'm actually right now in the mood of exploring new opportunities, challenging opportunities, basically, to advance my career, professionally. And honestly, I feel like I found this position. Cindy reached out to me, and after that, I got into deep dive, into the roles, and, you know what the company is, and everything. So I think, you know, it should be really a good opportunity for me to learn new things and, you know, explore different opportunities, basically, advance my career, professionally, and, you know, so it's been very nice, you know, opportunity for me, and I, this is my second round, so I was, you know, first round as well. It was really nice talk with the other people, you know, who was taking range of the first time. And you know, it's just like a new expert, you know, experience for me and for both technical and non technical, so that I can, you know, be enhanced on my skills and expertise. So, yeah, that's the reason Awesome.

Speaker 1 04:05

Awesome. Thank you. Yeah, we're gonna dive into some basically two part to the interview here. We're gonna ask you some kind of framework and technical questions, and then in the end, we're gonna have two little coding exercises. The key things I kind of want to emphasize is that we know the session isn't necessarily supposed to go perfect. Is mainly about seeing how you approach the problem. So like, if you get stuck or you need guidance, just let it know, and we'll kind of like with you to find a path forward. Absolutely, yeah. So we're not looking to, like, catch you in some like, algorithmic challenge or more, so just seeing if you can, you know, get the general framework in place and basically attack the problem in a way that you know, will make you a great fit for the team, sure. So yeah, I guess going on the technical question, we're going to start out with a straightforward Java question here. Could you explain to us what the difference between dot equals and equals equals is in Java?

Speaker 3 05:14

Sure. Yeah. So basically, you know the double, double equals equator is operator is used for checking the reference in memory. And if the two objects point at the same point in the memory, then it returns true as a true or false. But you know your the doc equals method is used to check the contents. Usually you know it's defined on the object itself. And if you have not defined it, you know, on the object, so it takes it from the parent object class. But yeah, I mean double equals is for the for the for the reference, you know, check, and then, you know, equals up is method for, you know, for content. Check, yeah.

Unknown Speaker 06:02

Awesome. So, hash maps.

Speaker 1 06:07

What is the key difference between a hash map and a concurrent task map, and why would you do one over the other?

Unknown Speaker 06:16

Sure, so,

Speaker 3 06:19

so, so, basically, both of them I, you know, use for storing the key values, and, in my opinion, and but the main difference, you know, it boils down to the concurrency, right? So basically, our has map is not a thread shift, meaning that if multiple thread access, you know, is a hash map, or currently, or any kind of convert processing will require external synchronization. But a concurrent hash map is a thread set, and so then this design designed for, actually, for the concurrent system where multiple threads can access this without needing of external synchronization, like I mentioned before, the one major difference is that the hash map allows, you know, one null key and can have, like, multiple null values, but the Corporate hash map doesn't allow null keys or values. I

Speaker 1 07:25

cool, sure. Jenna, do you want to hop in? Yeah, sure.

Speaker 2 07:33

Can you explain the difference between a checked and an unchecked accession with a Java? Sure. So

Speaker 3 07:43

So basically, you know, the check exceptions are your you know, compiled time exceptions, right? So that means, you know, that means that that they are are caught during the compile time, and the you know, the the extensions are caught or declared in the method signature, it usually throws the exception. And basically it's just a representation of a condition that you know, the reasonable application might want to, you know, recover from, or like, you know. So if you have some kind of SQL exception or, you know, input output exception, it's your check exceptions. You know it is, it is typically used for conditions like when the exceptions are external to your program, right? And you know, want to make sure that you are actually handling them, handling them properly. But your unchecked exceptions are, those are the ones that are, you know, that are runtime So, like they are, like your runtime exceptions, like null point exceptions to the exceptions or array index bound, you know, out of bound exception. So they usually represent, you know, bad programming concept, right? So if you have improper practices of writing your code, then if you are trying to run same method on a null object, then it returns, it throws the non point exception. But, you know, no pointer exception, instead of catching, it gives us, it gives a solution, and you can try and modify or refactor your code, you know, to prevent all these exceptions. So that's the major difference, I guess. All right, great.

Speaker 2 09:25

Let's move on to what are some advantages of using Spring Boot over the standard Spring framework.

Speaker 3 09:33

So some of the major differences, basically, would be the Spring Boot is actually, it's a lot about, you know, production ready features, if you need production ready features, and you know the auto configuration as well, right? So it basically makes your configuration, you know, very simple, because of the feature of the auto configuration, which means that the application are based, like, based on dependencies that you can add on into your class path or that are already configured or you don't need to, you know explicitly, you know, like, define your Java based configurations or or like external configurations, right? And also, it provides the embedded Tomcat server where you don't have, like, you don't have to depend on external server to deploy your application, and it provides, like, a lot of server dependencies that you can leverage for production, for any kind of support, or any kind of features, like the actuator endpoints, support for logging, support for all the different kinds of good tools for just making the lives of our developers actually easy in having the robust, or any kind of resilient application.

Speaker 2 10:57

All right, awesome. I want to ask a quick follow up on that you mentioned on your resume that you worked with both spring and quarkus. What have you liked? Kind of, like, the differences between, between the two? What do you prefer? So

Speaker 3 11:09

the main difference would be, so currently working with corpus, so spring is more like, you know, like, like I've worked with spring before, and I just switched to quarkus Whenever I came to D run. So spring has, like a full feature, you know, mature feature. But quarkus is mainly, you know, designed, you know, for, like cloud native or container or, you know, like other kind of environment. So I think, in my opinion, I prefer spring more than just because of the fact that I've worked with it more. And what was it just like, designed, you know, for for for faster applications, to get faster data and things like that.

Unknown Speaker 11:49

All right, awesome, thanks.

Speaker 1 11:53

So with regard to databases, can you explain what an index is and how does it impact performance?

Speaker 3 12:02

So, you mean, I'm sorry I couldn't hear properly. What was that? Oh, I'm

Speaker 1 12:07

sorry. I was asking if you could please explain what an index is in databases and how that would impact performances.

Speaker 3 12:15

Sure, sure. Yeah. So basically, index is just like database object, and it basically improves the speed of the data retrieval. And you know, it functions like index in your like, basically it just helps, you know, like quick improves the speed. And you know, like quickly locating and accessing rows in a table without having having to scan the entire dealership. So if you have like queries that focuses a lot on certain, you know, kinds of filters, like when certain columns equal something, or like, if you have like a query that leverages on join no or you could just want to, you know why you want to have those indexes to those columns. So you that, so that you have better, you know, retrieval speed, great.

Speaker 1 13:07

So, could you please explain to me the difference between a unit test and an integration test? Sure,

Speaker 3 13:14

sure, absolutely. Yeah, I can explain you that, because I'm working with that, right, also. So basically, unit testing is in a testing in isolation, right? So when it what it means that is, if you are, you know, making out for external service or dependencies at your application that you know that uses defining your behavior with some kind of, some kind of tool, like, for example, makito or and then, you know, test your changes, or, you know, unit of a code, you know, in isolation. So if I write a service, then I will, you know, Mark out those, you know, repositories or external services. And then I would just, you know, try to, you know, test what my block of code is actually doing based on different inputs. So, you know, I'm just, you know, like testing that part of the functionality, but the integration testing on the other side is more about how we know these small units of inner code or the services they interact with each of them, and basically it, you know, validates that these different integrated components and systems, they work together as expected, and if like, it's also used to detect any kinds of interfaces and data flow issues that might arise during the communication between the services that is Not easily captured to unit testing. So those are the major differences

Unknown Speaker 14:46

between them. Great. So back to spring.

Speaker 1 14:52

Have you ever had to configure your spring projects so that you had multiple environment? So like you had a Pua, a local production? And if so, how is that done?

Speaker 3 15:03

So, yeah, I mean, actually, it's pretty, you know, standard approach for defining multiple environments, and, you know, as a Spring application. So what you can do, you know, do is, you know, there's like, application to properties file, or an application YAML file, or, you know, you have, like one application yam file. You have, like a multiple of these kinds of properties files, right? So if you have application followed by a hyphen and then the and then the environment name, dot properties file, then you will, you know, have to you have environment specific properties file where so whenever you start your application and a spring application with a with the Spring profiles, that variable set to these this environment name, it will actually pick up those configurations. So that's actually how you can configure multiple environment using Spring application. Do great. I guess this

Unknown Speaker 16:09

is kind of more of a general question.

Speaker 1 16:12

So how do you approach learning new technologies like, do you have some sort of preferred learning platform? Do you tend to like to dive into the code, the references? What do you normally do?

Speaker 3 16:24

So basically, if I, if I had a very good question, you know, like, if I want to flourish and new technologies or newer things, whenever, you know, I'm trying to learn new technologies, I actually prefer hands on experience and hands of practice. So I would actually start with the reading a little bit of documentations, you know, as much as documentations that I can find, and try to get my hands dirty in the project or the or the actual topic that I'm actually interested in. So like, if I like to do something that I'm, you know, following, or I go to Udemy courses where there's an instructor is working and teaching. There I like to work on the same thing side by side, you know, opening up a project like that in my own computer, and to follow up and try to see if it actually, you know, makes sense, whatever I'm doing. And apart from that, you know, I also make sure that I follow the documentations, like I mentioned before, and try to see if it is something that I can follow up with just by documentation. And if you know, then actually work on it. And then the next thing you know I would do after that would be, is always try to think about, you know, side project that I can try to create my own out of all these new, newer things that I'm that I will be learning. And so if it's it's like a simple, you know, just working on, you know, string APIs, try to make sure that I can utilize it on something else to learn it better. So

Unknown Speaker 17:54

that's what I would do.

Speaker 4 18:01

It's a joke. So yeah,

Speaker 1 18:09

have you ever had to do you mentioned containers? Have you ever had to do projects that had to interact with or be built within containers, such as Docker? Yeah, I have

Speaker 3 18:23

had a lot of working with applications where I have worked with, you know, Docker eyes, and then, you know, deployed into the AWS ECS, you know, so, so basically, dock rising with within application is actually very easy. Basically, you just need to define a Docker file, like the dog file, that contains your base images and like your packaging information of the application and how the application needs to be started and the way you deploy it, upload it into the elastic compute, you know, Container Registry, and then you, you know, define tasks on that ECS service that is used for actually orchestrating these containers. So, yeah, actually, I'm familiar with this approach as well. Awesome.

Speaker 1 19:18

Thank you. Um, so have you when you guys normally work with source control? Do you normally use Git? Yes. So could you explain to me, like, what the importance of a pull request or a merge request is?

Speaker 3 19:31

So the You mean the differences, or even the uses,

Unknown Speaker 19:41

the uses the

Speaker 1 19:44

merge request and pull request to the same thing. It's just a different like, if you're using Git lab, they call it merge merge request. If you're using GitHub, they call pull request. But I would just mention in both, mainly because I wanted to make sure it you understood the topic, not the about the name.

Absolutely,

Speaker 3 20:01

I get that now. So, yeah, basically, you know, whenever you open up a pull request, you're actually comparing the changes that you are putting in, right? So whenever you, basically, whenever you, you know, open up a pull request, you are making the chain, making your changes visible to your teammates. And then, you know, the teammates can actually facilitate code reviews on it, right so it allows them to examine the changes before, when we merge them into and, you know, try to help them identify the bug, improve the code quality, you know, at the end and ensure that all these changes do not follow some kind of empty pattern, you know, or it just, it also emphasizes on, you know, like knowledge sharing, so that the teammates can know what are the changes being merged into the code, so it is basically, you know, primarily for collaboration and communication within the team, Right? And again, we don't want to work on one single, you know, Branch, right? So, so, so basically, pull request to make sure you have that multiple branches, you know, maybe, you know, feature branch is going to develop, develop, you know, branches so that you have, like, a proper version control.

Speaker 1 21:18

Okay, do you want to tap in? Jenner, sure.

Speaker 2 21:25

Can you please give me a overview of what is a REST API and why they're important for web applications? Sure. Absolutely.

Speaker 3 21:33

So. REST API is basically, you know, like the they are just a set of rules, right? So, so this is you just use for interacting with the web services by actually sharing the data in different standard format, right? And one of the most important, you know, aspect of REST API is that it is a state case, so, which means that whenever you make one REST request from a client to a server, all the informations that you needed to process that the request is is contained with within that request, either in the headers or in the parameters or somewhere in the Body, right? So this is state place, and has an architecture of client and server, where you, your client makes a request and the server, you know, server hands out and returns a response, and then it uses the actual standard HTTP, you know, bars like for getting a getting a resource post or actually creating a resource put, you know, for operating a resource and deleting any kinds of resource, so that would be binding.

Unknown Speaker 22:54

Okay, awesome.

Speaker 2 22:57

Let's kind of dive into some of the stuff that you've experienced in your past employments. Can you tell me about a challenging project or a bug you encountered and how you went about trying to solve the issue?

Speaker 3 23:09

Yeah, absolutely. So let me give you one of the examples you know, that I've actually faced so. So basically, whenever I this was one of one of the production issue that we actually faced recently, you know, and so we we use a library, you know,

Unknown Speaker 23:29

where like, the

Speaker 3 23:32

like was, basically, so we use the library where, you know, call the the the Netflix, you know, zoom, which basically is used for routing of the request. And this routing actually goes through a thread pool, you know, created by something called historics, right? So, so this, so, so this. So this actually was a tricky circuit breaker for our application, and then, and then this, you know, this tricks basically spins up on new thread, groups on thread pool, and then it executes your interaction, right? So during this, this, this execution of our thread, you know, what was happening was about three or four days later, the resources within the instance would get exhausted without any kind of proper error or latency, and then, you know, everything would go into a time of, you know, state, so no records would actually pass through. So the only way to recover was to actually restart the application. And then, so I was trying to work on this debug what was going wrong, you know. And because we didn't, we didn't, you know, have any recent changes that was pushed out, and we were actually not aware of what the underlying issues might be. And then, whenever I started looking into it, I first, you know, thought, you know, first thing I collected was the the heat dump, and then put it into the profiler tools to see, actually, what was blocking. And then I was, you know, able to see that external downstream call was always on a waiting state, state, and it was accumulated to our, you know, maximum thread count and all, all the, you know, services were stopping to behave. And the reason we did not see any error or logs around this was because the historic internally calls our fallback mechanism, and it actually logs, you know, any error in a debug, you know, level, right? So instead of error level log, it was like a debug level, you know, which it was actually difficult to see without any, you know, turning bug, you know, like debug logs on. So it took me a little time, you know, while to figure out what was the actual issue. And so they, you know, I want to about, you know, recreating that that was running a lot of performance tests by creating new scripts, I know, on JMeter, and then, you know, running the similar kinds of APIs, you know, on the downstreams, on lower environments, and then recording the third counts. And, you know, once I figured out that certain errors, you know, in the codes caused this? I was able to point out the exact location, and it was like due to an improper dependency injection, where spring management of being was not being injected, but rather being created. So, yeah, that was one of most, you know, challenging issues that I've faced recently, you know. And you know, we had to spend considerable amount of time trying to figure out what the solution was. But fingers crossed. Thankfully, we had a disaster recovery environment, which we, you know, switched off traffic for. So that was something that was very challenging,

Unknown Speaker 26:55

that's interesting, that's a gnarly bug. Yeah, it was like that,

Speaker 1 27:01

cool, awesome that doesn't work with the with the thread, the threaded portion of it, yeah,

Unknown Speaker 27:11

um, let's see.

Speaker 2 27:15

Can you give an example at a time that you disagreed with a technical decision on your team, and kind of how you manage that, that relationship among some of your teammates, yeah, so

Speaker 3 27:27

in that scenario, you're like, Let me think about it. So one time, we were trying to incorporate a streaming engine, right? And so this was an, you know, company with the enterprise or insurer, enterprise, you know, tech backlog, right? So, which means that that all the teams you know, have to follow this and then incorporate, you know, this changes to their applications. So our team, we held up the about the micro services we have. We had this, this take, you know, backlog affected, actually, a lot a lot of people. So the decision was to either follow the documentation you know, that they provide, or to come up with some kind of solution that would be, you know, a company wide solution, you know, that our teams could leverage, right? So, whenever I was working on the proof of concept, I came up with an idea of creating a separate microservice and responsible for handling these data aggregation and, you know, streaming to to downstream services. But like my other teammates, they were actually focused into, you know, just getting the thing done and then not managing an external service for our purpose and and because of this, you know, streaming was, was because my push was pretty sure that my approach, that I was proposing it was, you know, not not only going To help to our team, but, but the other teams that could onboard on their dataset, and then we, you know, we could stream it efficiently, you know, rather than having to manage each one of those services with SDK that they provided, and then they configure and like and then configuring them. So this is something that was not approved for further development, but, yeah, I was, I think I was, it was a missed opportunity. But I feel like I kind of found later that our team, that the team didn't want to increase the number of services that they owned and managed as a request response, you know, could sometimes get a little more challenging, and then, you know, increasing load things like that. So it was something that I disagreed with, but with the proper communication within the team and on what they were wanting, I was too quickly able to get out of that mindset of having to go to my solution. And also actually helped with lot of implementations on this approach in some of the services. All right,

Unknown Speaker 30:11

let's see.

Speaker 2 30:14

We live in a remote world now, and like a lot of us, most of us from the company now work from home. How do you like to manage your time and kind of prioritize your own tasks while handling working from home or working remotely from a co working space and also having to handle, like multiple projects and multiple responsibilities?

Unknown Speaker 30:36

Yeah. So for me, like

Speaker 3 30:40

basically, I, basically, I like to divide my task, you know, but it's based on priorities, right? So sometimes there might be, you know, some tax that I'm working on that night that might not be a very high priority and needs to go, you know, like out as soon as possible. So,

Unknown Speaker 30:59

very high priority. So

Speaker 3 31:00

those things I try to, you know, like as soon as possible. So I try to create a small list on my notes and a circle back to the notes on how the priority of each item looks like, and actually work on them through it. But sometimes, whenever there is, there is unforeseen circumstances, like a teammate needing help on something extra crucial at that time. So I might want to jump on that, you know, and then work on it accordingly. But I feel like, you know, with the increase in remote culture, I have learned a lot about the time management, which is the, basically one of the most important thing in the remote world right now, like you mentioned, and you know on how to prioritize certain tasks and how to divide your attention on certain stuff, whenever needed. And yeah, so I think it's, it's really important skill to know have in today's world, and important, you know, remote world, to have all these skills that go, Yeah,

Speaker 2 32:03

all right, um, I think I'll pass it back over to Brian. I think I'm nearing the end of my list here. Sure.

Speaker 1 32:10

Okay, yeah, I think it might be time to move on to the second phase with the programming assignment or programming

Unknown Speaker 32:19

exercise,

Unknown Speaker 32:22

not assignment.

Speaker 1 32:25

So I'm just gonna give you a minute, though, and this is not any fault of your own, but we've had a couple of issues in the past with people kind of doing sneaky thing during the programming assessment. And so that might be Cindy might have kind of warned you that we're kind of trying to prioritize our approach to this, because it's becoming painful. So so what we're going to ask you to do is, if possible, could you please and I'll give you a link. And so you would just basically need to open up a browser window and zoom to share your browser window with us, and we basically look at a little a little SQL problem, and then after that, we'll go into a little bit of a Java pile. Sure. Absolutely,

Unknown Speaker 33:11

absolutely. Yeah, no problem.

Unknown Speaker 33:15

All right. So

Unknown Speaker 33:22

the first one is a SQL problem,

Speaker 1 33:25

and it's weird because normally we would do it where we wouldn't have you share screen, but So the link is going to be weird in that way, where we can technically see what you're doing from our computer while we're watching you do it. But it's just a weird thing. So the first link, so I don't know if I need to give you some sort of permissions. I've never actually had someone share this thing yet. So if you can't share your share your window here, then let me know. Sure.

Unknown Speaker 34:08

You can see my screen, right?

Unknown Speaker 34:11

No, I haven't seen anything pop up in zoom.

Unknown Speaker 34:17

Is there on the button of your on the bottom of your zoom screen? So

Speaker 3 34:21

you want me to share the screen from zoom, yes, please.

Unknown Speaker 34:26

From zoom in there, there's There we go.

Unknown Speaker 34:31

Perfect. Yeah,

Speaker 1 34:34

all right. So yeah, I don't know if you want to move that guy, if it's in your way or not. I

Speaker 2 34:41

don't know if you can. I think I tried to do it before. I don't think you can.

Speaker 1 34:46

Yeah, I don't know if there's a way to maybe if you move, okay, you can scroll. Okay, all right. So the background of this is that there are two SQL databases. I believe this is using the schema for MySQL, but I think it might be quite good. I'm not positive, to be honest, I don't think it can make too much of a difference here. But essentially, there are two tables, there's the employees, and then there's departments. There are five rows in the employees tables, each of them have a department ID in them that will reference the department table. And so if you look at the right side of your screen, that the actual portion that you can execute. And so at the top of your window, there's a button there, I think it says run or execute. And we can do that now just see what's happened through the results. Got it all right? I wish there was a way to is there a way to get rid of that little thing? I

Speaker 2 35:48

don't know. I don't think so. Unless you've got like, a U block extension or something like that, it's really annoying.

Speaker 1 35:56

Yeah, it's weird for me. It's not there, but so you can move, you can't, you can't even move the losing that's known all right. So if you see here, now you're getting the results. Yeah, I see the results. Yeah, yeah. So the first question I have is, can you order those results by the last name.

Unknown Speaker 36:25

Yeah, sure, sure. So I can

Speaker 3 36:28

use the ORDER BY clause in order to actually short, okay. So the way you can do this is select Start from employees.

Speaker 2 36:53

You shouldn't need to use the semi colon. I think it might actually get mad at the semi colon. Okay, do

Unknown Speaker 37:03

okay, I can

Speaker 1 37:07

execute this perfect as a follow up to that. Though, if you look down at the results there, you'll see there's a John Smith and there's a Jane Smith, yep. Could you make Jane come above John. Basically, I want you to sort by last name and first name. Gotcha.

Speaker 3 37:28

So in that case, I would just add a first name after this last name.

Speaker 4 37:42

So there you go. Yeah. Awesome. I'll

Speaker 1 37:47

give you a slight hint on this one. You mentioned the phrase of what I want you to do earlier in the call about this one. So there's a certain type of technique to kind of marry the two tables so that you get the results, I would like to see the employees detail as well as what department they are in. And by department, I don't mean department ID. I mean the actual department name. And

Speaker 2 38:21

if you need to scroll down on the left panel to see the schema, you can Yeah, so you can see

Unknown Speaker 38:27

on the left side, yeah.

Speaker 3 38:31

So basically, where we can do is Just join on department. Let's Do it. You.

Speaker 1 39:22

So just just, I think if you find what you're saying, I don't

Unknown Speaker 39:26

want to make you type out all the columns.

Unknown Speaker 39:30

Just list the fields that we want to view. Did you read? It

Unknown Speaker 39:41

doesn't look like it ran. There we go. Yes, awesome. Thank you.

Speaker 1 39:54

All right, so that's it for the sequel stuff for now, I don't think we'll be going into the crazy dust there, but good job. I'm gonna give you a link for a Java program, very similar format, hopefully this one allows you to look at it easier without,

Unknown Speaker 40:18

you know, things overlapping Absolutely.

Speaker 1 40:25

Okay. There's a couple of things that might happen here. You can close that Google dialog and on the right and you can hit the X on the J droid, AI. Thing would have been added. Okay, alright. So this is the same kind of thing. You basically have a Java program. It. The way the editor is set up is that you can type whatever you want in here on the left side, it won't like, syntactically correct you, or give you tool tips or anything like that, to be honest, if you need any help with that, just they will help you out. And then when you click the orange Execute button, you'll basically get the results. And so right now, the method will return no so you should see no printed out.

Speaker 1 41:25

Is it there? Okay, perfect. So the goal of this method here, and if you want more green sides, you can hit the X on the input output, or you can just leave it there. It doesn't matter. So the goal of this is basically to take a string into this method called get first non repeated character. And so it takes a string, it goes through that string, finds the first letter in it, in that string that isn't repeated elsewhere in that string, and it returns it. And so in the first example, that would be w, and the second one that E, and then the third one, there are no characters that are not repeated, so it returned null. And in the last one, it would turn the capital T, because it's case sensitive.

Speaker 2 42:17

And before you start, I just want to emphasize that we're not looking for the optimal solution. We're not looking for like you to optimize for memory or anything like that. We're looking for to hear your thought process and how you solve the problem and to get something that works. Sure.

Speaker 3 42:41

So, let me think through this. So basically, we have a string and then we need to see if that string exists, like in the character exists or not. I

Speaker 3 43:04

so I think what we can do is we can split the string into the character array and then basically store the account of the occurrence of these characters, right? So it will have something like, let's just say we'll have three.

Unknown Speaker 43:34

Three so three, yes, one and one. I

Speaker 3 43:49

i so I will be creating a map that looks like that, you know. So whenever I'm choosing the map, I need to make sure that I'm presuming the order of these characters as well, right? So for that, I think I would be okay by I know I

Unknown Speaker 44:17

think I have a plan ready to go. So do

Speaker 3 44:26

so what my thought process is is to actually create a link, you know, has map that stores the character and their count right, and then I would loop to the characters. And then I would add the characters count to the map. And then after I populate the map, you know, I return the first time I get the develop one. So

Unknown Speaker 45:03

so that should be the solution for this.

Unknown Speaker 45:07

All right, let's go for it. I

Unknown Speaker 45:42

Yes, slightly, point in here,

Speaker 3 45:43

yeah, so basically I have created a link hash map

Unknown Speaker 45:49

that stores the hacker account.

Speaker 3 45:54

Now, basically I would populate this map by Converting the given input into our hack Drive.

Speaker 3 46:41

BBB, so now after this execution is done, I would have successfully created the map. Now I just need to loop through this map and return the first character that returns value of one i

Speaker 3 47:36

and if The value never becomes one, then we return now.

Unknown Speaker 47:42

Should so this should be it. And where did I get?

Unknown Speaker 47:50

Do you have a typo in the integer?

Unknown Speaker 47:56

Oh, okay, yeah, yeah, I see. Let me fix that real quick. Do

Speaker 3 48:13

so there's one winner, counting for uppercase and lowercase. So which is x, which is expected output, yep.

Speaker 1 48:25

Okay, now I just have to know off the top of my head, as a Java engineer, I would never come up with the entry the map dot entry.

Unknown Speaker 48:40

Have you done this a lot in the pad?

Speaker 3 48:42

So with this, the executive. So whenever you actually want to execute through to a loop, you know, through a map, you need to create, you know, create an engine set. So sometimes, you know, I try to solve no lead code start kind of problems. And so whenever you, you know, whenever I have, like a have to loop through this, you know, this kind of map I use in a map dot entry. So this is a formulas index for me,

Speaker 1 49:13

because I don't know there's also, you could have traversed the key set. You could have also re traversed the string itself and said, Hey, let me go to the string again. Find the one that had the first one that comes up that had one. Okay, but yeah. I mean, great solution. It looks accurate, yeah. So, yeah, I guess really, then it comes down to what question you have for us.

Speaker 3 49:43

So first of all, I would like to say thank you for both of you have been very kind for me, you know, taking this opportunity and I got to speak with you today, and

Unknown Speaker 49:55

but basically, I would like to, you know, ask,

Speaker 3 49:59

like, if I could know a little bit more about, you know, like the day to day, what day to day responsibilities looks like, and, you know, the what kind of developers that you have on the team, things like that, I would really appreciate that.

Speaker 1 50:18

Cool, yeah, currently on the app developer team. We have

Unknown Speaker 50:26

myself

Speaker 1 50:28

a junior engineer, although he's just titled junior engineer, you wouldn't think of him as junior engineer. He's probably the domain expert, because he's been on this the longest. We have someone like myself, who's a senior engineer, and we have a project manager, an engineer manager, and because we're in the consulting world, instead of us actually having like a business, so to speak, we have somebody called a client lead, and that client lead, or consultant is basically the person that's supposed to be the advocate for the customer's responsibilities and what they want to us, and then it gets translated through project management. And yeah, I would say that's it for now. We're looking to fill in another engineer. We also have engineers spread all over the company, and to be honest, it's really interesting, because when I started, I assumed that they were all like the team I was on, like they were all engineered everything. But there's actually a huge portion of the company, if not a bigger portion of the company, that mainly does analytics. They don't develop applications, so they'll do complex data merging, or they'll write Python scripts to pull data from many different data sources and kind of consolidate them for, I guess you would say, setting in criminal investigations of the Treasury Department. So it's very interesting. The team itself is nice. We we meet each other in person one per quarter in Washington DC, which is kind of neat, because up until I joined this company, I've only been to Washington DC once, and that was in like high school. And so now I get to go there as an adult and see everything differently. And it was kind of funny, because the first time I went to DC, where they stopped me, I took a walk around the White House, just the heck of it, and I bumped into the United States senators, which is kind of cool. Oh, nice.

Speaker 3 52:34

Did you meet the senators from from from around here, or some other states?

Unknown Speaker 52:38

No, it was. It was Bernie Sanders,

Speaker 1 52:43

yeah, he was, she was outside taking a picture with a woman getting ready to get into his car. He had security all around them, and I was actually out on a job going around the way up, because a mile away, I went for a drug, worked on my phone, took a picture, even though I wasn't part of that group, got Bernie 10 feet away from me, and then he jumped in the car. And I think I'm and I was like, but it's like, you know,

Speaker 3 53:04

yeah, yeah, barmont is a general place, yeah,

Unknown Speaker 53:08

I always wanted to visit barmont, but that's so that's, I think he's just saying

Unknown Speaker 53:13

different barmont, right?

Speaker 2 53:19

But to get, like, my perspective on your question. I think the day to day here is kind of loose. Esport in general, is very young on the engineering side of things, I was, I think the we did some things with titles, like before I joined, but I think I was the full first full time like engineer hire, and the teams and priorities have kind of shifted around, so people who had been doing like coding work now have engineer titles and things like that, but it's still very young engineering organization. We work mainly with, like single like machine, like scaling, like we don't, we don't really have, like Cloud scaling capabilities right now, so it's a lot more nitty gritty kind of technical details there, rather than letting cloud services handle things for us, and some of that's to do with like working within our clients, space, on the on their machines and things like that. Some of it's that we just don't have the cloud experience or the resources yet to want to try to press into those areas.

Speaker 1 54:34

Yeah, we have, we have our own stuff in AWS, but on the client, we're not so it's like that we do a lot of our development in house, and then what we'll do is we'll have to code and also go to their Source Repositories stuff, and then, technically, the goal is to eventually be able to do the builds there. Currently, can even do that because it's just a security nightmare, but yeah, so yeah, it's a very unique situation. And what's also nice about it is the one of the key project, if not a few of them that we're working on are very new, so they're using very new technology, very they're very high up on best practices. You know, we're testing a lot of it, we're writing code in a way that, you know, we're not all sharing some dev database, you know, we all have everything kind of intelligently created and architected. And at the same time, we have a lot of great resources at this company. We have people like one of the guys who's our architect, was an architect at Pivotal the people who created the brink we have, one of our more executive members is actually somebody who used to be in the criminal investigation field of the Treasury Department. So we have a lot of like, great resources as well as great opportunities. Yeah, absolutely, yep. That makes sense. Yeah. Any

Speaker 2 56:04

other questions we can answer for you? I know we're coming up on time, so I want to be respectful of your busy day. Yeah,

Speaker 3 56:09

absolutely. So one thing I just wanted to know, if you can just give me, like, a short thing explanation or something. So what are the, you know, opportunities that that we can get for, like, continuous growth. You know, what kind of things like the actual company provide? If you could give me a little bit of idea about that,

Speaker 1 56:30

yeah, that's actually something I have been pushing a lot to it. So right now, the way we have things currently is there Junior software engineer, Baltimore engineer, senior software engineer. And and the kind of that kind of the ceiling for the engineer level until you become an architect, and that's if you want to tell an architect. And so we have a couple software engineers like myself who have decades of experience, who are considered senior, whereas in my you know, if you're working for bigger company, would maybe be considered like senior staff or a principal or something like that. So we are, kind of, according to my manager, we are kind of going to shift things up a little bit. We haven't yet, but the growth they are very much working on, because they know that's a craving we have at the engineering we have a great to have progress, and so they're very much working on that. And that's kind of the neat thing about being a small company like this, is that your feedback can have results.

Speaker 2 57:33

And I'll echo that that I think it is a small company symptom where, like, everyone's just used to getting the work done, reaching across the aisles and just helping each other out. And we are starting to grow a little bit more especially looking for for like future contracts we're trying to deliver. You have to become a little more formalized and get our processes down from that side of things.

Speaker 3 57:55

Yep, sounds good. Sounds exciting to me. All right,

Unknown Speaker 58:03

yeah, that should be all but thank you. Thank

Speaker 3 58:05

you so much for taking the time and having me here. And it was really nice talking to you. Learned many things from you guys. So thank you for that opportunity. Thank you. Thanks.

Unknown Speaker 58:15

Ashim, thank you happy rest of your day. Thank you. Thank you. Bye.

Binamra-Final-spectraforce-amazon-srv-final

Speaker 1 00:02

So we're going to start with introductions. You know, I'll talk, I'll talk about myself and be like, we'll talk about himself, talk about yourself for a bit, and we had a technical question for mass amount of time. Do a behavioral question, and then, and with like, five minutes of Q and A, does that work for

Unknown Speaker 00:16

you? Yeah,

Speaker 1 00:19

cool. So let's get into the intros. Well, my name is Darren. I've been at Amazon for about three and a half years. I work on AFS with Vivek here, and before this, I was on another team called PRISM, all within the AWS FinTech organization. But before this, my team automated profit and loss statements, but I, more recently joined AWS, so still learning a bit, and Vivek will tell you more. Okay,

Speaker 2 00:53

hello. This is Vivek. I joined AWS FinTech roughly around four years ago, and that was my duty, the first job at amazon before joining Amazon, I was with the one of the bank, the Bank of America, on the trading investment over there, I was responsible for leading a team for building algorithmic fail. Over here at AWS bed tag, we are pioneering the building of applications which is responsible for financing and accounting, basically, planning, tracking, tracking, cost, tracking, budgets, right for all the assets that line into AWS data centers.

Unknown Speaker 01:35

Yeah, that's

Speaker 3 01:41

about yourself. Okay? So I'd like to introduce your Bucha. So my name is and I've been working as a software engineer for about five years plus now, and working primarily on Java desktop applications utilizing frameworks like spring habit, net and respite services for API developments also are pretty good. Have experience on working on microservices, and I've been involved in development, developing, designing and also deploying microservices on AWS cloud, some of the cloud services that have worked on with is AWS, EC two, SNS, lambdas, right? And talking about the front end side, I do have experience with Angular as well. And for testing side, I've been working on JUnit and mock tube. So to currently give a bit about myself. I'm also also on this company process of five developers. We will be product owner and Scrum Masters, and on daily basis, I've been involved on, mostly working on back end cyber code, helping the team with code reviews and also perhaps

Speaker 1 03:03

Cool, great. Thanks for the job. Really good to get to know you. Let so why don't we dive into a technical question? I'm going to send you a link in this chat. Let me know if you can join it as a

Unknown Speaker 03:20

interviewee or candidate. Sorry.

Unknown Speaker 03:32

Are you able to join it?

Unknown Speaker 03:36

There's a chat panel

Unknown Speaker 03:40

asking for user. It's asking for username.

Speaker 1 03:46

Just say like, are you an interview? Are you a candidate or interview?

Unknown Speaker 03:57

It's asking for signing options.

Unknown Speaker 04:15

I can share my screen

Unknown Speaker 04:22

or I can say it now, you say, now,

Unknown Speaker 04:29

do you see options?

Speaker 1 04:34

Great, cool. So pick whatever language you like

Unknown Speaker 04:39

first, and then I can, I'll paste the problem in the code editor.

Speaker 1 04:51

So for this question, we're not, we're not going to, like, compile your code or anything, but we're just going to talk. We're going to work through it. You're going to still, I mean, obviously, write code and stuff, but yeah, this is the question. Take some time to read through it. Let us know if you Have any

Unknown Speaker 05:07

questions. Yeah, do

Unknown Speaker 05:39

BBB.

Speaker 3 05:52

So it's kind of like a microservice reservation system, which is basically does a thing of reservation, cancelation, view and block.

Unknown Speaker 06:11

Also, like,

Speaker 3 06:14

do you want it or implemented in a springboard cell, rest control or also, symbols.

Speaker 2 06:23

I think whatever works for you, but this is more of an object oriented sentence with a question,

Unknown Speaker 06:30

so not a system. Yeah.

Unknown Speaker 06:56

So what I'm doing is

Speaker 3 07:01

sort of map to the track observations right. And so it will be given a customer name. And P will be the customer name at the valley right. And it could also be just a simple value like a Boolean saying that's a true or false. And so if I have an entry in the yes map, then it will it would mean something, something like someone has a ticket of that pass that handles the ticket information. So we will say something like, we would have a class ticket,

Speaker 3 07:55

and the ticket would have Like a seat number. I

Speaker 3 08:21

i also then you have a constructors, getters, so so that need to be exposing this, this red and so also that it will connect a single ticket class and that will use for our student tickets now we are on actual ticket reservation system. I

Unknown Speaker 09:00

honest.

Speaker 3 09:21

So, yeah, we'll have this ticket reservation service that accuse of reserving so we'll have methods like resolve reserve tickets, cancel reservations, and so basically, we have all the methods that has been asked.

Speaker 1 09:59

So does the ticket reservation service have any entities or just these functions?

Speaker 3 10:11

Yeah, yeah. I'll have the field variables later. I'm just saving up like, the like, right now. I'm just saving up the Enfield because usually it will have be an interface that will be implemented. But, you know, I'm not doing an indication. Just

Unknown Speaker 10:30

think back check it.

Unknown Speaker 10:45

So now we have

Speaker 3 10:49

reserved our cancer. Now We need to be only positive reliability and

Speaker 3 11:50

so this would probably be the methods that we will need to implement, right? So these are the five talks that we need to actually like this would be implemented in reservation service. So the first thing that I'm thinking there is like for the implementation would be to create a hazmat that stores name of the customer as the key and take it as its value.

Speaker 1 12:22

Okay? What if there are people with the same?

Unknown Speaker 12:29

Were there people with the

Speaker 1 12:31

same? Well, I mean, there have to be a mean, or at least more, at least one. John said,

Unknown Speaker 12:38

so, yeah,

Unknown Speaker 12:44

so maybe,

Unknown Speaker 12:47

maybe we can do something else by the key.

Speaker 3 12:51

So like, let's say, if there are, there is a user class, if there's a user class, right, then we can do like, customer ID and Name, So that that would probably Be A Better, better I

Unknown Speaker 13:47

so

Speaker 3 13:50

we'll have a customer to a ticket as a map. So now we probably have,

Unknown Speaker 13:57

we probably have

Unknown Speaker 13:59

lost finding the customer ID

Speaker 3 14:02

dust on certain flow, right? Like, for you to use someone like we have this customer details available and but, but if not like, we might do, like a dummy or something like that, on the customer that is not reliable, so also, and then we will just create it on the

Unknown Speaker 14:28

fly. When

Unknown Speaker 14:31

we are creating a like dummy IDs,

Unknown Speaker 14:38

for example, let's say

Speaker 3 14:41

it's our guest logs in, and let's say our guest tries to book a ticket, right? So in that case, there will be no costume objects. So for that, we'll just create a costume object on the line. But just in case, if the users who's already in the system, then we already have a

Speaker 1 15:05

look. Let's, let's, yeah. I mean, we can Yeah. Why don't we get Yeah? That's a fair summary. Why don't we we can worry about, like, education after we focus on national compensation. I

Speaker 3 15:32

All right, so let's just assume that customer exists already, and then we'll have some service that gets customer test on some other

Unknown Speaker 15:44

panels and maybe, just maybe,

Unknown Speaker 15:48

but other things That would be attached as well

Unknown Speaker 15:53

to the reservation. I

Speaker 3 16:03

so now we'll not be worried about getting the customer materials for now, but you know, that's something that we need to worry about in like this,

Unknown Speaker 16:16

this customer service or something, something like

Unknown Speaker 16:21

that. What about, what about

Unknown Speaker 16:22

like customer service,

Unknown Speaker 16:27

customer service, like

Speaker 3 16:28

customers and like poster, right? So, like object, customer objects, I would say, because that's usually either retrieved from the context applications like security context, anything like that. If user is not sure, yeah, sure, yeah,

Unknown Speaker 16:46

we won't have to worry about customer creation. But

Unknown Speaker 16:52

if you want, I can try and implement that as well.

Unknown Speaker 16:56

Let's talk about base functions first, and then let's see if we can build one on top. I so

Speaker 3 17:10

is the fourth required to have block list.

Unknown Speaker 17:19

Sorry, what's required

Speaker 3 17:22

is the folks required to have a block list to prevent users for resolving a ticket in that requirement, if one understand,

Speaker 1 17:30

yeah, there is a block the reservation required, yes, and normally that just means to, like, for some amount of time. We can set that arbitrarily, but you would hold me in the reservation, but we haven't paid for that ticket yet.

Speaker 3 17:56

Okay, so if I'm understanding that correctly, then this particular customer will not be able to book any ticket right

Unknown Speaker 18:06

for reserve ticket.

Unknown Speaker 18:09

Why wouldn't they?

Speaker 3 18:15

No, I meant, I meant for the block. So the requirement is on the right so, like, you have to clarify that that basically means that

Unknown Speaker 18:28

user would be locked from reserve.

Speaker 1 18:33

No, sorry, that means that, like, you can they they're reserving, like they're holding on to the ticket. They haven't bought it yet. It's sort of, I think, only compared. It's like, you know, you've in ticket master when you're like, it's in the cart like you. It's in your cart for X amount of time, you know, like that. I

Unknown Speaker 19:22

so

Speaker 3 19:26

then we all need to do is like, like, for like for a particular thing, we need to take every City's arrival as well that we can basically get the allowability with regards to the city and also like then,

Unknown Speaker 19:58

like we do, what we do is

Speaker 4 20:02

And so

Speaker 3 20:20

yeah, we'll have the queues here, which is like basically called the sets that are available. And generally, once the classes initialized, we'd populate this set itself. So

Unknown Speaker 20:44

So why AQ

Speaker 3 20:47

like? We'll be holding a sits like because also like we we are managing sits numbers, right? And it is. It's just like having a fear ordering of a seed collections, because when you do a queue, you generally get a first name, first serve.

Unknown Speaker 21:12

Also you can just

Speaker 3 21:18

pull the poll, and then we can have a create a like data, first variable seat that goes to the first portion. And so that's why we are doing a cube. It's instead of a normal set.

Speaker 1 21:34

So you're saying it because you want the ability to just pull the plate at the most like the available seat. So so,

Speaker 4 21:40

yeah,

Speaker 3 21:46

like just, we're just assigning the sets in the reservation itself, reservations orders.

Unknown Speaker 22:02

So if the if the allowable like, let's

Unknown Speaker 22:08

say if the

Unknown Speaker 22:17

like,

Speaker 3 22:19

yeah, so if the allowable seat is empty, then we can just return false because, because there are no return To when there is no allowable tickets, right?

Speaker 3 22:42

So when if the students, if it's if it's available, then we can Pull that zip and then assign an A ticket. I

Speaker 3 24:13

so at this point, we have a functionality that one user or customer can book one ticket, but, but if we want to multiple, then we can just, yeah, so for now, it's just like, we have functions that one customer can build one ticket, but like, let's say, if we want to multiple, then we can just simply convert into an ARD list. And you want to buy multiple tickets, yeah, if you want to multiple tickets, then you can just convert it into an ArrayList. Okay.

Unknown Speaker 25:00

Do we want that one standing?

Unknown Speaker 25:05

I'll just want to I'll just want to single dig

Speaker 1 25:11

it. How about we? Let's do single digit for now, and then we can focus on that after i

Unknown Speaker 25:24

All right, so,

Unknown Speaker 25:27

so in that case, we can,

Speaker 3 25:35

so if that's the case, we can just cancel and we can just get The costumer again, right? Like once we get the customer, and then we can say

Unknown Speaker 25:50

so from the map, we can just remove it.

Speaker 3 26:09

So in your opinion, like, I don't think we need to check if certain key exists, because it goes

Speaker 1 26:22

when did, sorry, what for? Cancel ticket like,

Unknown Speaker 26:29

yeah, we can just cancel that.

Speaker 3 26:37

Because when it does not exist, we can just Return the products. I codes.

Speaker 1 27:01

So just clarify, what if? So, what if in this so, this is this resolution that's that's referring to your map, right? That's referring to map, yes. So if, what happens in the case that the customer does not exist in

Speaker 3 27:20

the map? That's the like, let me change, let me check and change that. This will order under it and remove method I can

Unknown Speaker 27:35

simply, I can just

Unknown Speaker 27:39

simply check. I You

Speaker 3 28:17

BBB, so we are worth doing a Simple multi to see if the customer exists. I

Speaker 3 28:53

now let's say we need to be all the observations so that we can just

Unknown Speaker 29:01

so We do not need to Return

Unknown Speaker 29:05

of the value I

Speaker 3 29:59

BBB, so Now to check the availability, we just need to post to The size of the related zone, instance variable, right. I

Unknown Speaker 30:48

So what does total tickets represent

Speaker 3 30:55

like? So the total tickets would be like some predefined numbers, so generally won't change. So this would be a final because, like in a theater, you do not have additional sets that comes in magically.

Speaker 3 31:18

So for those main tickets, I'm just creating initialization.

Speaker 3 31:32

Yeah, so for the tickets, like, I'm just creating the queue. So to check probability, we just do the difference between the total size and the size of the queue itself.

Unknown Speaker 32:02

Ah, okay, I'm sorry. So it will just be the size of the queue,

Unknown Speaker 32:17

because that's how many tickets are.

Speaker 3 32:24

So now, basically what we need to do is like to do in order to block reservations, I

Unknown Speaker 33:04

so like, do we have a block reservations?

Unknown Speaker 33:08

You do have to

Unknown Speaker 33:13

implement a block reservation. I

Unknown Speaker 33:23

Okay, so let's breathe in.

Speaker 3 33:47

So if the user or user already has something that's booked and they cannot block it. And at first, like we need to do the same thing, right? Like if the city is

Unknown Speaker 34:09

not allowed,

Unknown Speaker 34:19

yes, I need quite

Speaker 3 34:22

passionate. So like, if the user has already booked and cannot block it, like, what I mean to say, if that, if that's the case, then we can just return false. And if the like user has already booked The reservations, then we didn't want To welcome it.

Speaker 3 36:01

So now I'm just thinking if we should just

Unknown Speaker 36:06

block a SIP rather than

Speaker 1 36:30

I think we're getting a little bit close to time for interest time I'm curious to know, sort of like, well, you don't benefit, But like, just will talk through what you would do, or like to finish the block reservations as well. Yeah.

Unknown Speaker 36:46

Okay, so,

Speaker 3 36:49

like, so what I'm what I'm talking here is, like, rather than just doing a simple Ticket object, we'll just block a sick so we'll have a sip result for a costume. So we can just, we can just change the type of the map, and we'll just have the indicator as key and value of the costume object. So because that way we it would be, it would be easy way to take the CDs I can prevent also, like, we can have preventions of duplicate blockings, and with that, we will basically be able to use it everywhere else. So, yeah, I'll just do like, sit as key.

Speaker 1 37:44

Why does using this? Why so? Why? Why are you choosing to use the seat as the key instead of say ticket?

Speaker 3 37:55

So, instead of like ticket, I'm just using like because like, we do not need to create unnecessary objects, like, because the ticket has not been bought yet, and we don't want to quit the actual ticket itself, right? And so the ticket would only be created when you sell the stuff. So it's a game. Like, let's say like, since we have only one, only, only want to block a sip, and it's usually for a certain amount of time, and and you don't, and you don't have a you don't have to, like, you don't have if, like, if you are using process of the system somewhere else, you don't have to go in and remove the object itself.

Unknown Speaker 38:43

And how does that affect

Unknown Speaker 38:47

reserve ticket?

Speaker 1 38:50

Or does it reserve to affect this reserve ticket and office using this map for integer crest,

Unknown Speaker 38:59

so like,

Unknown Speaker 39:04

it just like,

Speaker 3 39:06

goes off the effect, like, since we are using a queue, we are blocking the queue and will keep on getting emptied and wherever, Like, there's a resolutions. Then we'll just basically be just like going to the next set and create a functionality for the block reservations to be converted into an actual ticket itself. But, but, but there is also things like the actual reservations for a new

Unknown Speaker 39:42

user. Will not be affected.

Speaker 1 39:47

So given the code here, let's say, let's just say there's two seats. Let's say customer Bob has blocked. So now the queue so there's two seats, right? Let's just say it's in order.

Unknown Speaker 40:04

So what happens when Bob blocks

Speaker 1 40:09

everybody? Like, what? What Bob blocks seat one? How does that affect? Like, you know the rest of this.

Unknown Speaker 40:20

How does that affect research taking?

Unknown Speaker 40:23

Is this is as good as Is there any addition?

Speaker 3 40:28

I'm sorry, Bob. What does Bob do? Okay, So let's just say Bronx. I

Unknown Speaker 41:10

So, I mean, it's sort of just, let's Why don't we? Like, I just walk through it.

Unknown Speaker 41:22

So, if

Speaker 3 41:25

Bob blocks, it like given that there's only two sig and if they're there, right? So, so basically what happens is that the block method would basically be already pulled. The block have would already have pulled the arrival set. So UQ will only have the two as the arrival signal. So what happens is, whenever there is, there is a reservation, then only arrival seed would be the seat number. Also, whoever does the next booking would be resolving two sets. So now, if the book has said one, then we would maybe need functionality to resolve it, which will then be an happy path for the for like, if, let's say, if we suggest to release it, then we'll just have to offer the same value to the queue again. So, so so that it can be reprocessed. Say, Okay,

Speaker 1 42:41

okay, that makes sense. So outside of this last question, is there anything else that, let's say, like, you know, we talked about some of the functional functions you're supposed to do? Is there any like, something around those lines that we're missing that you would want to implement? I

Speaker 1 43:09

of things. So because, let's, let's say we block it, right, but blocking

Speaker 3 43:15

so apart from, like, what we have here, like, we also have some kind of auto expiry timer. Because we are blocking, we are blocking and so usually theaters are, like, five or 10 minutes time. So after that, after that, tickets get released automatically. We we will need to schedule the block that will, you know, release book tickets also like, we'd also want some options to have a specific selections which we would need to queue with, but, but maybe, like, with, like, another implementation set. So I think there will be the these, those would be the main thing, like, like I mentioned earlier,

Unknown Speaker 44:10

okay,

Speaker 1 44:12

cool, great job. Let's work. Let's talk about some experiences for behavioral questions. Can you tell me about a time when you didn't think you're going to meet the deadline? How'd you figure out that you might miss the deadline? Slash, how'd you handle similar situation? What was the outcome? Well,

Speaker 3 44:35

like, yeah, sometimes, whenever, like, there's a high priority work like, let's say there, there gets like, there is some conflict work while we've been doing then there might be like, we have deadline issues and that I might be missing. So one of the times that when I was in this situation was whenever I was implementing a stream, streaming engine for applications basically like the stream, stream, our application logs, materials like table or for the audit process proposals. And it like it is really complex implementation, and we need to have a leverage on the audit team itself, on on what the requirements was, what was actually, actual metadata supposed to look like, and what the schema of the output was supposed to look like. And this was something that was that was getting too close for comfort itself and and the way that I handled this was basically during stand up and everyone that the task is not getting completed as soon as I have right. So because of the implementation, there's some of the technical difficulties, maybe some some challenges on the way, and also being very clear on what expectations we're going to be having on moving forward. Like, like, I like the proposed method, and like, we have two phase release. Also we were going to show we need to be very have a very simple audit, and then later we can finance as we go forward. So that was something that I proposed, and I handled it basically being very transparent with the community and the team, and also making sure that I asked for help if someone, if someone, has other issues as well with these stamps and everything.

Speaker 1 46:46

So in this experience, when you, when you're working on a funding issue and you brought the applications, what sort of part was your responsibility? Did you were you implementing it? Did you were like, yeah, just sort of want to know a little more about your responsibilities. During this

Speaker 3 47:08

project. Okay, so, yeah, my primary responsibility here was to to go through the work of the process itself, the phase of the process that we have while working along with the implementations, as we were basically doing some design work, like the initial so

Speaker 1 47:28

you wrote the your designs for this project, design application.

Unknown Speaker 47:35

So you Sorry. Unsoft is

Speaker 1 47:37

quite frank city like worked on the designs told that implementation Yes. And then, were there any issues? I mean, you know, things might come up with anything like that, sort of during this project, that could have risked the risk the deadline, like risk missing the deadline.

Unknown Speaker 48:02

Well, yeah,

Speaker 3 48:03

because we have multiple micro services on architecture that we manage, we needed to make sure that things changes. And I was doing a generic enough thing so that we do have, we do work on all of these services itself, so making it generic is it makes sense that some kind of Mayor have dictated how the messages were formatted, how the extreme was, something a bit technical that may would have made it missed the deadline itself, and that's actually how what was, how it cost, and how it was solved a bit slower. So, because we have multiple applications on different domains, right, and they produce different logs, product differently, so capturing them and then sending it to the school calls to the services that was working on, and we made it bit challenging, so that sometimes we as team get affected, but we never.

Speaker 1 49:15

So you said that because we made a generic development was a bit slower, but you still were able to put that on. Okay, I have another question for you. But can you tell me about a time, but you made a mistake, you know, like, sort of

Unknown Speaker 49:36

like, well, can you tell me a bit about it?

Speaker 3 49:40

Well, every time it's a learning place to be honest, right? So everyone make mistakes, and I think it's one of the things that makes you learn through the phases of the mistakes itself. And also, I messed up good time for like, whenever I made a change to enable or filter for one of the applications we have different applications that serve the clients and then our web application itself. During this we had a filter that runs on a mobile itself, and filters just run for every route itself, and there was a filter that was used to decrypt the logics for influence, and we we have to enable it for our web and our facility enabled in a, I'd say, our environment testing, and it looked really good, no excuse. But like that was also observable so but also, like when we moved to prod, everything looked good at initial time and then later on, like towards the end of the day, which was was also little later itself, like, at 9pm I'd say, like, we started seeing our applications not behaving as how we had been expecting to work, as right. And we had about 500 level issues response. So coming back like, basically, we created our severity three production issues, and the reason that this you offer was due to the Logout flow for applications, and it did not like the footer because of the fact that we were missing headers itself. And these logouts flows would be only be triggered if it was a session timeout or anything like that. So if a user does not log out, then it's fine, but it's a session cloud, then that logout would probably be triggered. And this specific scenario maybe that I missed popularly testing that I have worked in rural environments, we were sort of little captured so that it was a mistake that I made and

Unknown Speaker 52:12

like I learned from tell me

Speaker 1 52:14

a bit more about like so, you know, we find out that, okay, it's not working. Can you tell me a bit about sort of all the process to fix it. What was the outcome in the end?

Speaker 3 52:28

Yeah, so for so basically, the first thing that I did whenever I joined the page on the on the Zoom, was trying to revert the release we had. So let's say like, if we have to put it back in single version

Unknown Speaker 52:44

that we add normal

Speaker 3 52:49

we what I did was basically acknowledge that there was no mistake, there was like, there was a mistake, and then people get Right. So first thing, first, like, I didn't check the logs find out what was scaling. And then after that, I just did a I just like, create branch that will fix the issues, which is, which is, doing a simple, simple node, check on the results, and then posting out and say, if the all the workforce was once Italy in the QE environment tools, before pushing it into the production team. Okay? And

Speaker 1 53:36

then sort of, did you have to convey this issue? I know you mentioned that like, okay, not just a mistake, but sort of like, is there someone just talk to to sort of like, alert this, or alert that there has been an issue, or sort of, what was that part like?

Unknown Speaker 53:55

Well, yeah, definitely, when we get,

Unknown Speaker 54:00

when you get production issues,

Speaker 3 54:03

I please that one with the team, and sort of my team also knew about it. Okay,

Unknown Speaker 54:14

okay, cool. Well, that wraps up the interview.

Speaker 1 54:19

I want to leave since we I think we started somewhat late. I wanted to leave five, five minutes for questions

Unknown Speaker 54:26

for us. Sure. Do

Unknown Speaker 54:30

I have any questions?

Unknown Speaker 54:32

Thank you

Speaker 3 54:35

for the opportunity today, and it was really nice talking to you guys. And I'd love to discuss more about the project and the products, right? And how can I, how can I help team to be more dynamic and be part of typical coding environment, typical coding environment, what do

Unknown Speaker 55:01

so? So you want to know

Speaker 1 55:02

what the project is, what coding environments we use. I mean,

Unknown Speaker 55:11

do you want to talk a bit more about, like, what the project entails?

Speaker 1 55:16

Yeah, sure. Let me start my video again.

Speaker 2 55:21

Okay, so basically, as I mentioned earlier, right? So EFS, which is our application in AWS FinTech, it's known as asset financial system. So what the system is responsible is for collecting all the financial events that are happening in order to procure the assets, right? So if we want to run a data center, it won't run there needs to be the rack assets that needs to be power equipments, like generators, that needs to be cabling, like network cable. All this needs to be ordered, and that needs to be a sample on site in order to run the data center. So that's one part of it. Once this equipments are procured, someone also need to monitor exponentially, right? So there needs to be proper accounting that, okay, Amazon is owning X, Y or X number of racks, which the total cost is, let's say $1 billion for example. Right? And now, if we expect that AWS business to grow over next year by 10% then we also need to basically fine tune or increase the budget for that procurement, right? And once that procurement is done, someone also needs to analysis, are we true to our planning estimate or not? This helps our financial leaders, right? For example, CFO to make the financial decisions for the next year, the next planning cycles, right? So that there is one part of it. So now, when we are procuring all these different systems, sorry, different equipments, like cables and everything, our system needs to collect all these related, related signals that like the purchase order was just as part of this purchase order we procured, let's say, X number of racks. And now we need to make a record in the accounting system, right? So slowly over time, what this system is maturing now, but we are still expanding, right? So we started with the high impact assets. Now, basically we are tracking the racks, generators and all those things, but the smaller things, like cables and all those things, these are still pending. So there is not a systematic accounting still yet done. It's still done like manually, by using some kind of lump sum or approximation logic. But so that is the next part where we are seeing an improvement area in our application, and that is what we focus so even though this assets are low by dollar value, it has very high impact, because the number of cables is very right, like tracking a rack is very easy, because you are moving a very big piece of chunk from one place to another, and you know exactly where it's installed, but now imagine that in your house you have 500 screws. Keeping track of those 500 screws is very problematic, right? So over here, we are not keeping track of screws, but I was using an analogy of basically tracking the cables, right? So tracking cables is very hard, so these are like very complex problem that we need help informing that's where this project is heading towards, and this is just the one part of it, right now, if you are also owning this assets, right, for example, cables or racks, right? Which is, which needs to be depreciated as well. So for example, if you own a car today, it's $20,000 for example, after five years, it might be maybe just $2,000 and when you are doing your tax assessments and all those things, you need to take account this depreciation, right? Because that says how your net asset value is decreasing over time. So similar accounting plastic and Amazon also needs to do, as like all the other companies, right? So that's where, basically, AFS sits at a very critical position to make sure that Amazon financial statements are very accurate. These are the statements that are reported in the earnings based on what the stock price moves based on what the performance of companies recorded, right? So without accumulating all the signals and transforming recording it in the right way. All this financial statements or the performance of a company does not matter, right? It is not accurate if we are not doing this accurately. And that's why the EFS system is very critical in the business case, correctly.

Speaker 3 59:39

Thank you for the information. It's really insightful. Yeah,

Speaker 2 59:43

so this was related to project, right? And then what your goal would entail is that, okay, now we have some kind of designs already ready, how exactly we will deliver the enhancement request, but now, right? Some questions still need to polish those designs, and need to take a lead of actually delivering that feature end to end, right? So you have the requirements, but you need to basically lead from the requirements to the actual delivery of the system, where we are able to confidently test the system. We are able to run the UAT and then also verify the accuracy of the system. Right now, if system is dragging 10 signal to the downstream system. What controls are we facing systematically to make sure that, okay, I have the right dollar amount being recorded like it might be recorded right now, because you tested the system. But what about after six months of time? Right is the system having still automated capability to ensure discrepancies are lagging accuracy? acy right so that's where I think people to help us, and

Unknown Speaker 1:00:48

deliver the system with the right control.

Unknown Speaker 1:00:53

Well, I'd be willing to be the team to do.

Binamra-First-paypal-system-design-srv

Speaker 1 00:06

Hello, yeah,

Unknown Speaker 00:12

doing good. Thank you.

Speaker 2 00:16

Okay, so let me introduce myself and the entry process, and then we can go from there. So yeah, this is Srikant ramdani. I'm working for paper past six years. Mostly I work from the merchant side. So where in the paper? You know, right? A lot of consumers will buy the thing from merchants, and when they buy, the funds will go to the merchants paypal account, and we, as a merchant, services we provide, or we send the payout to their banks, right? So every day, it's a scheduled system where we send the payouts to their banks every day, whatever the amount which they got, and based on the some business rules, right? So overall, we work on the maximum services side. So that is background of mine. And mostly we use Java for raptor services, and we have internal frameworks, and we use Java and the services mostly, right. So, so that is high level background of myself, and I don't want to go in depth of that detail. So to start with. And so this is, hope, you know, this is system design interview, right? So, so here, yeah, we'll, we'll talk about the first 510, minutes our introduction, and then I will try to provide some design questions. So that is interactive session where we discuss and we'll try to implement something the better way and the scalable and

Unknown Speaker 02:10

high efficient

Speaker 2 02:13

system. So that is purely the interaction session where you can ask any questions before you start, and I provide, I will provide you the full details of it. So, right, so, but full interest in session. And then last five to 10 minutes, I will give some time to you so that you can ask me any questions, whatever you have, so that will try to address or answer the questions, whatever I know, right? So, so that's entire one hour process, right? So that's it from my end, and you can start introducing yourself. I have gone through our resume and all the details, whatever we have provided, and I just You can provide high level information about yourself, and from there, we can get started,

Speaker 3 03:03

certainly. So thank you so much for your time. So I'd like to introduce with myself. So my name is Bino brunobani, and I've been working as a Java developer for about five years now, working primarily on Java based applications utilizing frameworks like Spring Boot, spring habits and RESTful web services for API developments. And I have a pretty good experience on micro services, and I've been also involved in designing, developing and also deploying AWS services and micro services. So some of the services that I've worked on with includes AWS, EC two, ECS, SQS, SNS and lambda. So talking about my current team. So my team consists of five developers, and we have product owner and Scrum Masters on a daily basis. And I'm mostly involved on working back end our code, helping the team with COVID Reviews and also productions. Okay,

Speaker 2 04:11

got it. Thank you. Thanks for the missions. Full detailed information. Okay, perfect. So I think Do you have the hacker rank

Unknown Speaker 04:22

link, yes, I do have the hacker rank link,

Speaker 2 04:25

okay, can you please try to access it? Enable to

Unknown Speaker 04:37

So, yeah, I'm waiting in Lobby.

Speaker 2 04:39

Yes, you are waiting in Lobby, right? Yes, okay, got it. Give me a minute. I will try to accept it. Perfect. So let me know when you are able to see the question. I think, Bob, it would be good if you can share your screen, sure.

Unknown Speaker 05:12

So can you see my screen? Yeah, I'm able to

Speaker 2 05:19

see. I'm able to see, I think there is some user or something. Arrow is there? Is that the one you just tracked it?

Unknown Speaker 05:29

Okay? So

Speaker 2 05:32

I think maybe we can wipe it out. This one. Did you try it?

Unknown Speaker 05:37

No

Speaker 2 05:41

or no, just you can wipe it out. I don't know how it came up. I'm already there, I guess. Okay, perfect. So, yeah, so I think I have provided you the question with the system design. It's a purely for sub subscription, subscription services, so you can read that question and you can come up with your analysis. And if you have any questions, yeah, you can come up with questions. I can provide you the details, and from there, You can start implementing or designing the system. Okay, so

Unknown Speaker 07:19

So basically we want to

Speaker 3 07:23

implement subscription, subscription based recurring payment,

Unknown Speaker 07:28

which automatically deducts the actual transaction,

Speaker 3 07:33

right? Yes. So, so something like that, I wanted to know was like, do we support like a fixed amount, would it be like something uses best?

Unknown Speaker 07:46

It's a fixed amount,

Speaker 2 07:51

fixed amount, and it might call one of the service which provide you the service fees. It might change later, but it's not a hard coded value, so that you can call one of the service which provide the service fee for the monthly basis or bi weekly basis or yearly basis. You

Unknown Speaker 08:36

so what I'm thinking is like

Speaker 3 08:41

we basically want some kind of services for fixed amount, recurring payment, where you can create a merchant, right, and which can create a recording subscription for user, and then the user provides, Like, a consent for that recording subscription. And then we'll have a system that schedules and processes those auto payments. Okay,

Speaker 3 09:17

how would you structure this? A So, okay, let me start by thinking about how we structure this in a database, right? Let

Speaker 3 09:36

me think of from the data perspective. So, so So let's say we basically

Speaker 3 09:49

use the tables, right? And so like, which contains stuffs, like, ID, name, payment method and like, maybe email as well. So those would be our user datas, metadata and so now I basically all also need to store one details site and I and

Speaker 3 10:27

so we also have, like, ID, URL, name, something like that. Also like, we basically

Unknown Speaker 10:34

want a subscription table. I

Unknown Speaker 10:45

so the

Unknown Speaker 10:47

subscription will have ID, the

Unknown Speaker 10:52

user ID, and

Unknown Speaker 10:57

the user will send ID and it,

Speaker 3 11:05

and then we also have the amount.

Speaker 3 11:17

So we probably would have the amount currency, and we'll also have the interval that then, what will this status be if it's active or paused or canceled, right? And when will our new debt next due will also be, and also, we also have to edit,

Unknown Speaker 11:37

audit it and

Unknown Speaker 11:41

All right, so

Unknown Speaker 11:48

then we also would have a event tables for this one,

Unknown Speaker 11:57

like subscription ID,

Unknown Speaker 12:00

IDs, status, status of any kind.

Speaker 3 12:12

All right, so, yeah, that this would be like, this would be the thing that I'm thinking of, right? So this would be the four primary tables that we might need. And here the main relationship that we can think of is like a user, where user and merchant would be your high level, and each of them would have a relationship with a subscription, and a subscription will have a relationship with subscription events, so that you have a merchant that will be involved in each stages of your actual subscription itself. So yeah, this will be database structures and so, yeah, what I'm thinking of database design perspective of what data will restore initially is like, we can change it later, so as we move forward, but this, like, this would just create The basic fun functionality that we were looking for. Okay? So now what I'm thinking here is,

Speaker 3 13:35

so basically we have a motion portal, right? I

Speaker 3 13:55

so with having this portal here, so from the merchant portal, what we basically do is like, call us to answer

Unknown Speaker 14:03

a subscription API. So let me think.

Speaker 3 14:20

I So, yeah, within the subscription API, it would be Microsoft based, and we'll be having our design. We would also want to make sure that this is behind a load balancer. So this straight line that you see here is a load balancer.

Speaker 3 14:55

Now for the subscription API itself, we would have different endpoints that would handle the subscription. So for the first one, what we do is that we we create a subscription that we want to post,

Speaker 3 15:20

and then we want to get an the subscription detail itself. So we do get an API, yeah.

Speaker 3 15:32

So, yeah, this would be the subscription like we would get the subscription like this. I

Speaker 3 15:43

And then we also have options to maybe pause or cancel a subscription, right? So we can have

Unknown Speaker 15:54

that we'll probably do

Speaker 3 15:58

a full set point for that one. Do

Speaker 3 16:31

so yeah, we'll have these two endpoints that we want, and then

Speaker 3 16:44

so, yeah, that would be our primary subscription API That would be

Unknown Speaker 16:49

helping to manage the subscriptions.

Speaker 1 16:56

Okay, do BBB.

Speaker 3 17:11

So yeah, now we need to incorporate a scheduler process that looks into our database and finds like which subscriptions need to be triggered and they need to process right?

Unknown Speaker 17:28

So, basically, first,

Unknown Speaker 18:31

school. I Think

Unknown Speaker 18:38

screen sharing is gone. I give

Unknown Speaker 18:46

me a second. Let me stop you.

Speaker 3 19:09

So I don't know what happened. Is it now? Is it okay?

Unknown Speaker 19:14

Now? It came up. I

Unknown Speaker 19:23

so now, like,

Speaker 3 19:26

we need to think like, how we can handle the payments and the schedule of those so we feel that one of the things that we might not have too much control is like, of the payment gateway. Okay, go ahead.

Unknown Speaker 19:46

Okay, so, yeah, this would be our payment gateway that we call.

Speaker 3 20:02

So this is basically a simple external payment service that would like call our like payments rights. And here we basically call in some kind of audit service to log the notifications or the payment status itself.

Speaker 3 20:27

So, yeah, this would be the payment flow that like, but, but our scheduler, actual scheduler, will be calling the payment service. So let me do that. I

Unknown Speaker 20:45

So talking about the scheduler, let's see how.

Unknown Speaker 20:53

So, yeah, we would probably need some kind of

Speaker 3 20:58

schedule based, schedule based, and now, like the, the now the general processing of the scheduler. What I'm thinking is like, it would be

Speaker 3 21:15

a dedicated cargo one, which will face the subscription, do, which the subscription?

Speaker 3 21:29

So yeah, now we'll infect the inject the sales. They do subscription. And also we'll have

Unknown Speaker 21:39

also create, we also need to create an event

Speaker 3 21:58

like this will happen in also like, we do not want to re execute the same same code again and again,

Speaker 3 22:13

right? And then it will call the API. And then also we have to, like, we'd also like to add a retry mechanism. And

Speaker 3 22:28

after that is done like we need to update our status. I

Speaker 3 22:54

BBB, so like, if this, like this is basically change the next view of the subscription table, and that generally helps us.

Speaker 3 23:18

And now the after the scheduler is complete, like we basically want to call the notification service to send an autopilot

Unknown Speaker 23:27

sent to the user notifications to the user itself.

Speaker 3 23:38

So, yeah, I think this would be a very high level design of what the payment gateway integration for schedule would look like, but like we'll be, we'd be leveraging our scheduler and absorb itself also.

Speaker 2 24:00

Okay, so, okay, so that's it from high level development perspective, right? So can you please maximize little bit? I think so now the question the subscription API. Who calls the subscription API and what the properties like, high level properties in the subscription API. Okay, so can you, can you give me simple use case, how this subscription behavior is called, and what are all the minimum properties under the subscription?

Unknown Speaker 24:47

Okay? So like,

Speaker 3 24:50

like, we like, we already have defined data, right? We already have defined our data over here, and it looks like we already have, I would say the data about the merchants also like, because we are integrated with that merchant partner. Like, let's say what we call Netflix itself, right? So we use Netflix portal, or the we say put for the management, right? So the Netflix back end, whenever a user makes a makes a call a back end call, like, like, we need to call the subscriptions of the Netflix back end, right? And you answer, your user makes a subscription that that's what we are trying to do. And so whenever user makes a subscription on the Netflix portal, then the and then the back end of the Netflix will be called on the post API itself. Okay? And also, like, what do I say? Also, if like, if we call this API and then the requested also, the request body will generally be containing its. Request body will generally contain these kind of datas that we want to have, and it will contain details such as, like a user ID,

Unknown Speaker 26:25

merchants

Unknown Speaker 26:28

ID, and also like that.

Unknown Speaker 26:36

So yeah, basically we would add in some kind of

Speaker 3 26:40

amount to it, and then we'd like, let's say an interval on the back end, or like interval on the backside, so that whenever, like, we have these major things that occupy for creating a request, then the response of this probably would look like a subscription ID, a simple subscription ID itself.

Unknown Speaker 27:08

So what is the internal? Yeah,

Unknown Speaker 27:12

sorry, what is it? What

Unknown Speaker 27:14

do you mean? What did the internal means?

Speaker 3 27:19

Also like it will be, it will be, likely, a monthly, daily, week, weekly basis. So I think, like that it will be just dictated when, like, when the next deal will be. So if I'm getting in front of you, right,

Unknown Speaker 27:37

okay, so

Unknown Speaker 27:41

who will provide that amount?

Speaker 3 27:44

So with regards to the amount, it will basically be the Netflix so it will provide the amount at the moment, and we might be able to add in a service integration, but, but during subscription time, it will basically be added there

Unknown Speaker 28:04

itself, amount per bot,

Unknown Speaker 28:07

so amount for these,

Speaker 3 28:10

all the subscriptions that we that that's that needs to be added or to be deducted.

Unknown Speaker 28:22

Okay? What kind of subscriptions?

Speaker 3 28:26

So the subscription would basically, we would have it as like, let's say monthly subscription, or the amount fees that might be like, let's say we having daily subscriptions, like Netflix generally have a monthly software rather than the early one. So it might be like something like $200 or something, and we might need to deduct it every month, so we'll just have the amount for the particular period of time, or we just a simple interval of time, and that is that will be used, right? Okay.

Speaker 2 29:16

So, so, okay, so, how do you pay?

Speaker 3 29:24

So, how do I pay? So, okay, with regards to payment, let's say, like here on the

Unknown Speaker 29:34

on the actual, let's say, well,

Unknown Speaker 29:38

well, the subscription API

Speaker 3 29:42

needs the payment details as well, right? And so we need to have the API for the subscriptions separately. Oh, so like, whenever user, like registers within our service, the user has a the user has a payment method, right? So you you can see the user details on the table, on the database that has a payment method. So it's not just a method, it will, like, basically have a data associated with the with the payment processing, like the authorizing ID and so on, so the particular field within the user table would be used for the process. Yes, that

Unknown Speaker 30:42

okay.

Speaker 2 30:49

So, okay, the payment method. Do you mean to say, so when the second happens? So we need to take data from the payment method and we do the process the sort of time to say

Speaker 3 31:04

so, not in the subscription, but in the in the actual payment schedule job itself. So whenever they like, schedule kicks in and it says, like this, payment needs to be executed today. Like, then it gathers, gathers all the required fields, like the amount to be paid, like the merchant who we are paying to, and the payment methods to be used, right? And so it will be doing multiple queries, and then it will just gather all that information to execute the payment. So it is not needed during the subscription, but, like, it will be something that's need to be on, like for the actual schedule.

Speaker 2 31:49

Okay, so how they know which payment method to use? See, for example, let's say I am a users. I have MX card, I have a Visa card, I have a bank account. I have XYZ, right? So how that subscription knows which one to use? Yeah, as a user, as a user, I have 100 cards, five or six, seven bags. Okay, I have added all those for things. So how subscription knows which one to use, okay,

Speaker 3 32:37

so right now, like, I'm just using one payment method, but like, if you want to incorporate multiple then probably we'd want to add up, like payment table itself, which generally contains metadata about the actual payments.

Speaker 2 32:54

Okay, so I want to be little scalable. Okay, that's fine. So you answered my question. Okay, so next, so, so this, how will you scale this? So, what? How will you scale that is first second question. What kind of schedulers will you use? So, schedule,

Speaker 3 33:20

you have two questions, what is like schedule that we are using, and what? Yes,

Speaker 2 33:28

how do you use the shared goal and how it will scale? I have a 10 million merchants, 10 million subscriptions happening every day. So how will you scale that? Okay. So

Speaker 3 33:43

basically, with regards to the stream list, let's first talk about the database itself. So basically, we want to have a primary database that would read on the alternate reasons for like, let's say we have a database based on reasons, and also, like on the actual subscriptions, API agreements and like it would be buying a load balancer, right so that it would be, it would generally be working in a Docker container service, which is employed by The employed on or by AWS ECS that would generally be easily, originally scaled by the spinning up new instances based on the load, right so there is a higher traffic that, let's say, on the CPU and memory when It increases by certain percent, and if the request volume increases, then we have a auto scaling policies attached to the instances, and it will also need scaling as needed so that how we will do the subscription API, since, like we are using a payment gateway, the payment processing will actually scale, since it's going through the gateway, and so we do not need to worry about it. The the actual notification service that we have, it would be like an event driven architecture, so maybe some kind of Kafka consumers and so on. But like so it would probably be good in that way, right for the actual scheduler, and we might need to find trigger millions of subscriptions that we have in power interval itself. So we need to do what is like, avoid the double bleeding strategy. And probably we might also need some kind of temporal engine. So basically, there will be a temporal engine that handles the back and retires for the backup systems, and it basically skills horizontal,

Speaker 2 35:58

yep, just, I'm still having a doubt on the subscribe that scheduling part. How do you scale the scheduling to multiple service servers, right? So for example,

Unknown Speaker 36:15

if I'm having a REST API, okay?

Speaker 2 36:20

If there are 100 millions of merchants hitting that URL, if 100k merchants are hitting the URL, your load balancer is there, and you have your DCS servers based on the load you said, it will scale up, scale on the good you have a scheduling system. How do you scale into ECS

Unknown Speaker 36:51

servers, into multiple servers? Well,

Unknown Speaker 36:55

with regards like,

Speaker 3 36:57

like, we would basically be using what we call that as temporal scale work and on our like, on a spring boost based service like. So it like the temporal would generally help us to automatic retrieve and workflow the creation system itself and right. So what will be is like, it will be something on called a temporal cloud that we can use and like where all the workflows are, so saved on it in the database, temporal database itself, we're talking and then the workflow would be distributed across multiple workers that would that temporal would spin up On with,

Speaker 2 37:40

I didn't get Can you please elaborate on that model, the workflow model, what you said? Okay, just give me a second.

Speaker 3 37:49

Yeah. So, so basically, what will happen is that, like you would have a temporal that would be driving the RB this, let's say, orchestrating it, right? So basically, in the context of this recurring payment system, we would have a workflow code, which would be something that would be distributed up against workers, and then you would have an external actions that would probably be like an API call, right, so that it works that you want the orchestra your layup that you have made would basically have a temporal records for every decisions that you make up and that you can replay the temporal from any point in any point time. So let's say you have like activities like charging the users or retrying or sending successful emails or anything like that, it would be basically be able to resume from each of these flows by basically distributing across different load multiple workers.

Speaker 2 39:01

How can you describe that distribution strategy?

Unknown Speaker 39:05

Okay, so

Unknown Speaker 39:10

take one simple take an example of

Speaker 2 39:14

1 million subscriptions. Okay. So how do you distribute

Speaker 3 39:23

okay? So with regards to the distribution, we are basically saying, with a large number of let's use this right? So if we are talking about 1 million subscriptions, so that it would work, that would be evenly balanced on the load and avoid all kind of odd spots, right? And basically the main goal here would be to enable parallelism. So let's say you have 1 million subscription and so big subscription will maybe need to build monthly so you basically need to define some subscription queue or where each margin will have a its own dedicated queue on the merchant and sometimes there might be a hotspots, because there might be, like, more subscription on certain levels, or so that we might actually they have it against other data, like the subscription ID, so that we will have, like a shared or distributed partisans, where you can run multiple workers, it's assigned to a particular straight so if, if The if, that's what you're asking for, right? Yeah, I

Unknown Speaker 40:43

BBB,

Speaker 2 40:50

so what about the eventing? What kind of events, and how do you send the events? So I want to send a notifications to the customers ahead of time before we charge. How do you design that? And I also want to send a notifications after the payment succeeded.

Unknown Speaker 41:19

Okay? So,

Speaker 3 41:21

so So for the notification service, like it is integrated as part of the scheduled itself, right? And so, so what I'm thinking is like, maybe using like Kafka for event based systems, right? Because it is a critical part that we probably use, like different events created or canceled, or maybe a reminder itself. So all of these events would be great fit for Kafka because it can handle high throughputs, or as May you have millions of events like you mentioned, like we might not have millions, but maybe billions as well. So that way it would work in the scheduler, would send an event to the officer, and then it would basically do two different consumer topics that would be like consumed, maybe through email, or like a merchant worker or anything like that. So like on that subscription filter, we basically do, like, deduct payments due, and then it sends an actual email notification saying that your payment is due then and then after the payment is successful, right away, send success notifications.

Speaker 2 42:47

The payment is due. When are you going to trigger that?

Speaker 3 42:50

So if the payment is due, we're talking about, right? Yeah, okay, so when the payment is due, we trigger it in a couple of days before it is due. So like, we might have another job that would look for the actual due, right? And we might add another job that would actually look for the subscription that are close to due itself, and then, sort of like, send notifications, like two days before it's due, and like, so that the user can also be notified, like, if the payment is due for themselves.

Speaker 2 43:34

Okay, so what kind of events will you send? So, so what's your thoughts?

Speaker 3 43:43

So with regards to my like thoughts on the events that we think like I'm thinking about certain events like creation of subscription canceling of subscription billing reminder, or maybe we can also have a payment success and failure thing.

Speaker 2 44:05

Yeah, I got the what kind of event. That's what my questions. Is it the email, or is it so how? How do you notify customers? Okay,

Speaker 3 44:15

so we can notify customer like differently as possible. So probably an email would be a best option, or maybe an SMS service. Okay,

Speaker 2 44:31

okay, SMS service. So another question, so you said the recurring payments should not happen. How do you manage and how do you ensure that you are not doing the recurring payments?

Speaker 3 44:48

So with regards to the recurring payments, so like, like, do you mean by recording the payments, or like, like, reprocessing already payment processed or something like that, that we are, that is your question. Is it

Speaker 2 45:05

Yes? So how do you ensure that, if the subscription for this month is already paid, and how can you ensure we should not repay again and are maybe due to do internal issues, maybe you got a timeout, but the payment already made, but you got a timeout from the payment gateway. So in those scenarios, how do you ensure that you don't reprocess it again? So

Speaker 3 45:40

with this recurring process we actually have, we might actually like, we, but we say, like, what? What's the exact term like, we might like, we might need to update stable on place of the job itself that we are looking for, and we would basically do some kind of multi layer indepotency defense, where, like, wherever you have some process that might not be successful, right? And you basically need to look up for the key that, right, basically what kind of key that you are checking for in the MN gateway, if certain was successful after time, after a certain period of time, or, like, it would basically change subscription event table. Let's say if it was successful. And once this, like, the successful is done, like, we basically change the next subscription due, and that way, like, whenever it's reprocessing, it does not pick up the same event again and again. So it would be some kind of transaction ID that we can use for this

Speaker 2 46:48

step, okay, in case of any DB failures, how do you ensure that?

Speaker 3 46:51

So in case of, like, even if you make the payments and you have, you have payment failing options, right? Yeah, okay, so we like, we basically have a, what do you say? Potential back off for that one. So we basically do three retries, not more than that, and then retry it, and an exponential fashion, and if it fails, then we basically update J status to fill and then send out the notification. So that would be something like what we would do, so the user may be able to change the frequency that they are doing upon after the update their payment method, or anything that they might do like would require a user verification or maybe authentication, something like that, or maybe user interactions itself.

Unknown Speaker 47:51

Am I? Am I on the right track? Yeah,

Speaker 2 47:54

yeah, I think yes, you are on the right way, but I'm still looking little more concise information, but yeah, that's okay, so we can move on to the next question. So here in the subscription APIs, we said pause and cancel. Pause and cancel. Are these are post APIs or

Speaker 3 48:25

oh so So, yeah, those would be the post APIs. Post

Speaker 2 48:29

APIs, okay, so, when do you consider to have a patch APIs,

Speaker 3 48:35

oh so, with results like for the patch APIs like so basically, for the patch, APIs like a patch is a like is used for the partial update for particular resource that I'm doing on a post API, because it actually creates an event. So let's say, instead of like, having an actual change of the subscription, we basically, we basically might want to make use of that patch, but this like post comms, postcom came into my Man right now. So, okay,

Speaker 2 49:18

good. So what about

Speaker 2 49:28

question related to I would like to ask

Speaker 2 49:38

the payment the gateway? Yeah, so I have a question related to payment gateway. So the payment gateway is always external API to your company, correct?

Unknown Speaker 49:53

My question here

Speaker 2 49:56

is the payments, the we we don't always call payment API, payment gateway, APS, right? So for example, let's take bank. The transfer mode is Ach, right. So we don't deal with the payment gateways always for any kind of payouts, right? So, so there is different system, right? The file based systems are there. So, so, why did you think of having a payment gateway directly? That is one question, second question, audit service. Who implements the audit service and why did you attach it to the payment gateway?

Unknown Speaker 50:47

Okay, so,

Speaker 3 50:49

with regards to why I'm using the why I call the payment gateways, like what I thought about that it's basically since, in it's an external thing, like in a real world scenario, whenever, like you have, we may have, like, stream of cards or wallets or, like, whatever you want to have on payment thing, like they are usually your Normal payment gateway, but in a multi regional system like this payment gateway would be a that layer, abstraction layer, so that would encapsulate the different kinds of payments, right? So, like, which is like, if you want to upload the file, if it's a bank pool or something like that, it be like the gateway would be responsible for handling those payment for routing services. So it is just an what an abstraction. So it would be like an adapter, or something that would be for the payment gateway itself. So the reason that So, the reason of why I'm an audit service is that as to the payment gateway is to record, like, what, what kind of payment activities has been being handled or canceled by payment gateway. And generally it would be that kind of system itself. So managing the auditing, like the subscription and the jobs like, but like, I basically attached it for the payment related data, like attempting and receiving and so on.

Speaker 2 52:29

Yeah, but how do you audit that particular subscription got already paid or not so, so, so payment right? Payment Gateway it doesn't deal with only for subscriptions.

Unknown Speaker 52:42

Okay, so, yeah, okay.

Speaker 2 52:47

You got my question. Payment Gateway not only deals with subscriptions, it will deal with X number of things, exactly, right, but you want to do the auditing for the subscription. Okay? So where do you where do we handle that? So

Speaker 3 53:05

we might actually move to, like, actual service rather than just the payment gateway. If so we, let's say, like we might have an scheduler and subscription API rather than the payment gateway and like, that way it will, like, we can handle the payment requests, thing and stuff, also, like whenever, like, we inside the scheduler and the software API, like, we can just have the like, instead of like the Basically, what I mean to say is that instead of subsequent API, we can just have payment campaign, right?

Speaker 2 53:48

Okay, I think, yeah, we are good. We are on time. It's almost 1:55pm here, so almost 55 minutes we have, we had a good discussion. I think we have six minutes left for you to ask any questions, so that I will try my level best to provide you the detailed information. Sure. So

Speaker 3 54:13

I would like to thank you again for your time today. And it was nice talking with you and discussing about the design and the payment system. So firstly, I'd like to know a little more about how much developers get involved in single design sessions. Like these designs are all they don't do developers generally get involved in similar kind of sessions.

Speaker 2 54:39

Yeah. So in Paypal, right, everything is structured, so we have a developers, tech leads and the ad techs, right, enterprise text, so we have a multiple structures, so the designs mostly will be done by tech leads after specific domain,

Unknown Speaker 55:03

right? So they design

Unknown Speaker 55:07

based on their domain.

Speaker 2 55:10

We have end to end optics or enterprise architects. Then have a bigger program where multiple domains involved, so end to end architect will come into the picture, and he will discuss with tech leads of each domain and end to end. Architect will draw the high level design off with each domains, right? So by interacting with the domain leads, and then each domain lead will go on the design, low level design, respect to their domain, what to do and how to do, and right, so. So here one example, I can say,

Unknown Speaker 55:54

right, you want to involve a payment gateway,

Speaker 2 55:58

and you have a domain class, subscription domain. The payment gateway is not tied up with a subscription. So the payment gateway will deal with X number of things. It will do the bank transaction, it will do credit transactions, or it will do the peer to peer transactions, or into the auto debit so X number of things, right? So I has a subscription domain. I can't go and call the payment gateway directly, so I have to discuss with day domain leads and see which APIs which really fit to us, right? So, yeah, as I said, Yeah, domain leads will go with the technical low level designs and enterprise level architects, or end to end architects will go with the high level design, with interactive with interacting with domain leads with end to end picture. And that are developers. Mostly they don't like junior developer. Mostly they don't involve in designs, but they will be reviewed, right the design review sessions and everybody involved, and everybody can share their own thoughts and opinions about the design, and iteratively, it will be finalized with the end to end Arctic,

Unknown Speaker 57:28

that was very insightful.

Unknown Speaker 57:32

So any other questions? We have two more minutes left.

Speaker 3 57:41

What do you like it most about working on PayPal?

Speaker 2 57:47

Yeah, it's culture, right? So PayPal culture is very good, so it's

Unknown Speaker 57:56

all user friendly. Like, I mean,

Speaker 2 58:01

we can work based on our comfort zones, so there is no pressure. So we can always go and tell if something going beyond. So we can always go and tell to our managers and comments, okay, we initially designed for 10 days, but it will be ongoing 20 days. So we need more time, so, right? This are flexible, so we can't so I have given a 10 days. I have to finish within the 10 days, right? That's not like that. If you go and ask negotiate, and we can negotiate it. Why it is taking more time. Is there any help needed for you to expedite, right? So, so this all sorts of help provided, and it's all mutual discussion. It's not one direction, it's bi directional.

Speaker 3 58:57

It's always good to work on a flexible environment, right? Yeah.

Unknown Speaker 59:03

so thank you for your time.

Speaker 2 59:06

Yeah, we have a long time. Good, thank you. And I still talk to you. Okay, all the tests.

Binamra-Paypal-Algorithms-SRV-Final

Unknown Speaker 02:47

Hello. Nice to meet you. Very

Speaker 1 03:00

good. Are you located or in the west coast or east coast? East Coast, East Coast. Got it. Okay, so today we're gonna so let me introduce myself first. So I'm Walter working and working team, and I'm representing APL to take our the coding round interview for you. And this are this code, this test gonna be our one hour and with two coding questions basically on the digital structures and basic algorithms, okay? And you're gonna have our 30 minutes for each and, yeah, that's it. And do you have any questions to me before we start? Well, thank

Speaker 2 03:47

you for the opportunity. It's like no questions at the moment. Okay,

Speaker 1 03:54

sure, sure. I just want you to feel comfortable. This is that are not gonna be a hard questions, but you can, you can imagine your

Unknown Speaker 04:03

time. Do you have our hacker record link?

Unknown Speaker 04:07

Okay, let's go to hacker ranking.

Unknown Speaker 04:17

I think you need to put me back to

Speaker 2 04:22

do I think you need to put me back to lobby. Okay?

Speaker 1 04:34

And finally, do you want to share your screen? So, yeah, oh, as this is, this is the company policy and

Unknown Speaker 04:51

cool. I don't see you.

Unknown Speaker 05:01

Okay?

Speaker 1 05:15

This this part, this is not your previous questions, right? So I gonna pull the Question to this tab. Okay, Do

Unknown Speaker 08:00

uh. Okay, So I think

Unknown Speaker 08:24

I kind of like get a problem here. So,

Unknown Speaker 08:28

so we basically have a node

Speaker 2 08:32

that that represents a light, right, and then we basically need to create some kind of mapping between them. Do? Yeah. So, basically,

Unknown Speaker 08:47

so, basically, yeah, we need to,

Unknown Speaker 08:49

we need to start with some kind of initial delayed

Unknown Speaker 08:55

flights. And I think

Unknown Speaker 08:56

we need to start with some

Unknown Speaker 08:58

kind of initial things, like maybe Q to

Unknown Speaker 09:04

Prolog it through the map.

Unknown Speaker 09:14

So yeah, for for example, we have a

Unknown Speaker 09:18

we have a flight from

Speaker 2 09:21

Fall three and flight two and two, right? And since

Unknown Speaker 09:26

one depends on four and

Speaker 2 09:30

two depends on three, so initially, like the delayed flights are one and three. So yeah, so the four will get delayed because one is delayed and all the final delayed, we would have that one, three and four as well. So, so I'm thinking like, like a death fork search for the traversal thing, yeah, yeah, we need To contact the leaf deliver sites in the flight. I

Speaker 2 10:23

So, yeah, we have a flight notes we apply from flight to, I think it would be some kind of first search. So, like, we just need to take it to delayed flight and density flight. So we basically take the

Unknown Speaker 10:41

graph itself.

Speaker 2 10:58

So I think, yeah, that would be an approach here, so I I can get started with Coding, if you are if you are ready. All

Unknown Speaker 11:22

right? Alright, so we have right notes, number of notes, and

Unknown Speaker 11:31

then we have a list of slides here from

Speaker 2 11:35

two, then we have the list of delayed so first thing we build is, like, dependency, right? So it would basically be just a map where, like, the key would be five and two, then, then the value of the list of integers as well.

Unknown Speaker 11:58

Okay, so I

Speaker 2 12:32

So, yeah, this dependencies would be some kind of graph that tracks

Unknown Speaker 12:38

the relation between The flights

Unknown Speaker 12:42

to and from so it's so

Speaker 2 13:02

we can now look for flight.

Speaker 2 13:12

So now, if we look through this flight, then

Unknown Speaker 13:18

I think, I think we can get the value drop it off to dropping?

Unknown Speaker 13:35

Now? Basically We just need to add

Speaker 2 14:13

Yeah, so now this will create a basic graph structures of the

Unknown Speaker 14:19

dependencies of the flight

Speaker 2 14:27

once. Now, once we have the dependencies, like, we can basically look for the delayed flight using the death flood charts, because now we have, like, if the starting flight is delayed, then the next one in the node will be delayed as well, right? So what I can do is like we have to just do a simple depth first shot. Suppose,

Unknown Speaker 15:18

so, yeah, probably the floor, but

Speaker 2 15:21

like we need the delayed of the actual graph itself, and like, if we like, if we need, like, the delayed fact itself. Because, like, we need to be sick if it is delayed or not,

Unknown Speaker 15:36

as well. So I

Speaker 3 15:47

but, but for that we like, we basically need the tracking

Speaker 2 15:53

value rather than the origin one, so

Unknown Speaker 15:57

which like we would

Unknown Speaker 16:00

be what we used to be with the

Unknown Speaker 16:08

the What we used to be with, like assets.

Speaker 2 16:22

Okay, so now we have this helper method that is that is going to be used the depth of search now, like, basically, if it is delayed flight, like we want to add that to our total delay.

Unknown Speaker 16:42

So

Speaker 2 16:45

now, like, like, if, like, if the flight is delayed and it is the, it is the part of the key of the dependencies, then that means, like the all the dependencies like might be delayed as well, so we basically Need to do the unfortunates for all the values. I

Speaker 2 17:37

So yeah, this is going to be a recursion, and we don't have any best guess. So let's see if that best case would be well,

Speaker 2 17:49

so now, if the if the flight is already existing and in a delayed set, then we don't need to do the same thing over and over again, right? So like, what

Unknown Speaker 18:19

would be So, yeah, once, once it's done like

Speaker 2 18:23

then my total delay would be populated. And once that is populated, we just want to sort it based on the actual flight number itself.

Unknown Speaker 18:33

So

Unknown Speaker 18:36

for that, we can just return so

Speaker 2 18:51

yeah, so just, we just convert this set into a stream, sort and then return it to a list. I think, I think that's take care of that.

Speaker 1 19:08

Was talking before we rather we run a test, right? So I just want to understand is this, do you think this is massive solution, or most optimal, optimal solution, considering the time complexity and space space complexity well,

Speaker 4 19:30

so with regards to the time the time

Speaker 2 19:35

complexity. So the reason with this approach is because there are nodes and dependencies for for each flight, and then it can come across those nested dependencies, right to talk about some, some, some, some kind of optimality we are doing a depth first search, like and depth of search generally takes two things into your Play that would be considered the size of the flight of the nodes and also the number of dependencies that we have. So it will still be like a linear one, though, but then under the thing like we might most of the time, like, here might be sorting that we have been thinking of. So so the sorting will probably take more than longer anything that we are expecting, to be honest.

Unknown Speaker 20:40

So

Speaker 2 20:42

I think, I think that would be what's causing like but, but we need to like Travis, everything in a graph. And I think for death, for shots like, it would be the best doing, because we need to go through top to the bottom as well. But if we do the breadth first shots, then we might be getting a best optimal solution, because we are visiting node after node, rather than going from top to bottom and depth first. So just skip that part one.

Speaker 1 21:16

Okay, and what does the capacity here? You say it's linear?

Unknown Speaker 21:23

Means so

Unknown Speaker 21:26

well to talk with the it would be, it would be

Speaker 2 21:29

like, if you skip the sorting part, then it would be linear, but, but if there is a sorting it would be analog. N,

Speaker 1 21:40

okay, then it's longer, right? Because you cannot skip the part, skip the sorting part,

Speaker 2 21:45

yeah, exactly. If you skip the sorting part, then it would be linear.

Speaker 1 21:51

Okay, yeah, let's run it and see how it's resolved. Okay?

Unknown Speaker 22:07

Let's drop the test.

Unknown Speaker 22:18

Yeah, looks like everything

Unknown Speaker 22:19

was passed. I Okay,

Speaker 1 22:32

then let's make some this question live a challenge, right? So can we resolve this in the open instead for the entire analog

Unknown Speaker 22:42

pen? Sorry, can come that again? Can result,

Speaker 1 22:44

yeah, can result this question in O N time complexity, instead of instead of n, log n, right?

Speaker 2 22:51

Yeah, it's n, log n, right now. So you want to go into the quantum complexity O N or log n, yes.

Unknown Speaker 23:01

Okay, so,

Unknown Speaker 23:21

so

Unknown Speaker 23:25

with regards to O,

Speaker 2 23:34

I think for that, I don't think we don't want sorting for that one, right? Maybe anywhere we just,

Unknown Speaker 23:47

we just basically

Unknown Speaker 23:50

need to remove the sorting part

Unknown Speaker 23:53

and so look. So let's see what we can do.

Speaker 1 24:03

So there are a couple downsides with with your implementation for what I see right. First one is the time complexity. It is taking O N, log n, log n, and second. Second thing is, our intake are, how much our space complexity it is,

Unknown Speaker 24:30

what's the what's the space complexity and what's the solution?

Unknown Speaker 24:36

Okay with the Express. So the

Speaker 2 24:40

space complexity, I think we basically, we just have to set tracking of the total delays that then we have

Unknown Speaker 24:49

with the map itself. So it would be,

Speaker 2 24:52

it would be a linear on that case, because, because just like, like, we what do I say? How do I fit this one, because we just would be creating

Unknown Speaker 25:03

a set, right?

Speaker 1 25:06

Well, yeah, but let's see Does, does, oh, that's our O N and O 2n O 3n, or 10. And does it? Does it say, oh,

Unknown Speaker 25:19

like you're saying,

Speaker 2 25:23

Well, I don't think it matters, because if it's just a constant multiple, it will still be a linear right, but, but if you want

Speaker 1 25:31

to this capacity, let's see you have our so we have our 1 Million node, right? If you are taking two are you have a copy of our this, you have a copy of dependency. You have a copy of total delayed then the spec, the we have one, one unit records, which take out maybe 10 gig wise, right? But if you say you have our 202, and it's gonna be 20 gig wise memory usage.

Unknown Speaker 26:05

So do you think that matters or not? Well,

Unknown Speaker 26:11

if you, if you put it,

Speaker 2 26:14

if you put it in that perspective, then that matters like, how like, how to Yeah, that would matter simply because yeah we are, because there is a bad memory users of that one as well.

Speaker 1 26:34

Okay, yeah. So then from, from this two points, right? One is kind of complexity, and otherwise the space capacity. Can we improve? Yeah? Can we improve from this two point

Unknown Speaker 26:49

with regards to

Speaker 2 26:52

Yeah, just like I said before, if we just don't Let me see I

Unknown Speaker 27:00

quite well. Let me see I

Unknown Speaker 27:50

well,

Unknown Speaker 27:52

I can't think of anything that I might

Unknown Speaker 27:55

help in the optimization, because you balance. Okay,

Speaker 1 28:05

yeah, then let's move with the second question, and maybe we come, if we have time, we can come back on This one. Yeah,

Unknown Speaker 28:27

okay, I import a second question. I

Unknown Speaker 28:47

So,

Speaker 2 28:49

so the smaller storm. Okay, so we have an array of n elements, and we need to take the we just basically need to find the gate of the small sphere.

Unknown Speaker 29:11

So is that question to me? No, I'm just

Unknown Speaker 29:15

answering the Reading the Part I

Speaker 2 31:17

All right, so, yeah, if it like Basically, I think we need to construct those peers, and we need to turn the sorted product by basically sorting it in the values I

Unknown Speaker 31:40

loop. Let me think of something so if we do multiple so

Unknown Speaker 32:04

we'll be looking to the past product, right? And like,

Unknown Speaker 32:08

if we, if we sort them, just be it

Unknown Speaker 32:14

will just be able to give me the

Speaker 4 32:17

value access as well. I

Unknown Speaker 32:28

so

Speaker 2 32:30

given an integer error, I just, I basically need to create my cross parameters, right? So we it would be a list of integer error. I there.

Unknown Speaker 32:44

Okay, so you're saying you're gonna create our

Speaker 1 32:48

you're just gonna go are very straightforward solution with create a cross product array, then you sort it and then run another for loop to pull another follow up to put case value is that right?

Unknown Speaker 33:09

Right? That's what I'm thinking

Speaker 1 33:12

of. Okay. Then before you start implementation, I will also challenge you with same question, what God? What's going to be the time complexity and a space complexity, and can we have better? You have to always ask yourself, when doing the COVID questions, right? What's the what's the brute force solution? Definitely, I trust you. If you're here interview with me today, you definitely know I definitely trust you. Can use a brute force or our solution, right? But it may not be the answer that I want, right? So you should challenge yourself. They can. Can I think of our our optimal solution?

Unknown Speaker 34:00

So

Speaker 2 34:03

yeah. Usually, like, whenever you are working with, like, most of the elements, you might able to be, like, you might basically be getting it with some minute, which is basically some kind of priority queue. And I think that gives you a sorted on demand values. So, so, yeah,

Speaker 2 34:40

so, yeah, maybe that's something that I can

Speaker 1 34:51

walk on with. But um, do you think is there any difference? Because are you are creating a list and feeding to the particle. And then particles sort of feel then you pull the seventh or 10th value. So then, what is the difference between the prototype, which is discussed,

Unknown Speaker 35:17

so we don't, we don't need to do a

Speaker 2 35:19

nested loop there, because it's because, if you're talking about a priority queue, I guess it's based on the actual values. So whenever, like, I'm inserting on a priority queue, I can just provide what computers I want. So like, instead of doing like M squared,

Speaker 2 35:47

like read. We're also be doing like n log n as well.

Speaker 1 35:54

Okay, but let's see what, what's the, what's the overall contrasting on the solution.

Unknown Speaker 36:06

So, like,

Unknown Speaker 36:08

if we go with the priority queue approach,

Speaker 2 36:15

then like, we probably would just

Unknown Speaker 36:19

get down to

Speaker 2 36:22

like K, like K, log n, something like, That's it.

Speaker 1 36:36

Let's see. Or how about, how about when you will make, when you build that cross product array, what's the complexity of that

Unknown Speaker 36:46

cross product

Speaker 1 36:49

array? Right? We you need saying that you need to build our cross product array, right? No matter if you think you feed the records into the park queue or you feeding the record into array? What's the complexity of that one? So yeah,

Unknown Speaker 37:04

if we work with the PROSPER array, like

Speaker 2 37:08

in the brute force approach, it would be n squared. So like, but if we are going with the priority queue approach, we want to be making that prosper array, but it will

Unknown Speaker 37:21

just be doing on a mini to

Speaker 2 37:23

represent the index, and then we would see the first element in each row. And then if we push it down with its balance as well,

Speaker 1 37:39

okay, um, Okay, then Let's, let's start coding first, and let's see. Okay.

Speaker 4 38:02

Cool. Now, so, so So to ensure the list is in a lexical

Speaker 2 38:11

Copy order like we we give a just started. I

Unknown Speaker 38:51

So,

Unknown Speaker 38:53

so we can do a priority queue

Unknown Speaker 38:55

of English already, and actually,

Unknown Speaker 38:59

like we what like we can do is like we should be on the

Unknown Speaker 39:04

main hip, and

Speaker 2 39:08

then I wanted like this, that, like It could have two elements of A and B. I

Speaker 4 39:36

now, so first I need to Compare the first time that it's itself,

Unknown Speaker 40:11

and If they are not equal, then that's it. I

Speaker 2 40:58

So, yeah, that would be my priority queue definition for the minimum hip

Unknown Speaker 41:05

and now we can just

Unknown Speaker 41:08

get the first elements and then populate it.

Speaker 2 41:44

Yeah, so this would be my initial setup and and now I can just, I can slip it until, like, until we, until the one I, until I exhaust my solution, so I decrease my care every Time on This One You

Unknown Speaker 43:59

BBB,

Speaker 2 44:17

yeah, okay, so, so now what I need to do is like,

Unknown Speaker 44:22

It is possibly I

Speaker 2 44:53

so once we get to the smallest sphere of the hip like we want to continue exploring to the next possible, starting with that element. So in the rule, like we we move forward to next day. So I think

Unknown Speaker 45:09

we need to basically check The condition out here, so that, like addressing,

Unknown Speaker 45:22

I and like, if you just say, if you can listen out here, we like

Unknown Speaker 45:48

it might just be plus one, obviously,

Speaker 2 45:52

can be less than your heart rate and of the size. And we need to offer to the to keep moving forward as well.

Speaker 2 46:19

This will make sure we get the next element that is better than I, and

Unknown Speaker 46:27

it will just be glued I.

Unknown Speaker 46:43

Yeah, so we just initialized the with like

Speaker 2 46:51

zero as the zero index, and basically, like fix the first element and periods with the smallest, smallest

Unknown Speaker 47:03

possible of the secondary

Speaker 1 47:08

length I see, okay, so I see are in your code from line 41 to 45 right? Well, what's the time complexity? Here?

Unknown Speaker 47:18

Is it from line 40 to 4541 to 45 The while loop.

Speaker 2 47:30

So yeah, since we are decreasing the value of K and also making sure that the hip is not empty the

Unknown Speaker 47:40

tank complexity,

Unknown Speaker 47:44

just be lower on the logarithmic scale,

Unknown Speaker 47:49

because you're just pulling

Unknown Speaker 47:52

like and I think the Pulling,

Unknown Speaker 47:56

they just take the most of the times that's

Unknown Speaker 47:59

hmm. And

Speaker 2 48:03

also, since, like, we are, like, looping with the

Unknown Speaker 48:08

K times,

Unknown Speaker 48:12

so like, K, log, n,

Unknown Speaker 48:22

okay, got it,

Unknown Speaker 48:24

yeah, let's move on.

Unknown Speaker 48:40

Yeah, so, What do I Do?

Unknown Speaker 50:21

So, yeah, I think

Unknown Speaker 50:23

my solution is not optimal as a thought, because,

Speaker 2 50:27

like, it's taken too long time, probably I

Unknown Speaker 50:43

Yeah, why does it take too long time?

Speaker 2 50:56

So I'm sorting here, so I'm sort of definitely take some extra time, so let's see If I can just get rid of the salt. So

Unknown Speaker 51:20

it. So

Unknown Speaker 51:28

I think I relied

Speaker 2 51:29

on the sorted error to make sure my like I get the correct values. So that's not something I can get rid of.

Unknown Speaker 51:42

So okay, all right, so yeah,

Unknown Speaker 52:32

I'm not entirely sure what I can optimize here. To be honest,

Speaker 1 52:38

I can give some hints. Okay, so let's see for, for, I'm not sure if I give you a case, you can still result. Okay, don't be too happy, okay, well, not just kidding. So let's see our I'm gonna give you k, right? K equal to seven, okay, k equal to seven. So do you still and what is the sorted rate the salted array is 111213212223, right? Okay, so if I give you a k as seven, or if I give you k as k as two, what will be the first number of the pair.

Unknown Speaker 53:25

So if k is two,

Speaker 2 53:27

so if we give the value as k is two, that would be one, one, comma, one,

Unknown Speaker 53:35

it should be one, come two, right.

Unknown Speaker 53:38

Okay, if k is two, oh,

Speaker 2 53:42

well, like one one would be the output, right? And one comma two would be,

Unknown Speaker 53:48

yeah, that's what you're saying. One comma two is the

Speaker 2 53:51

output, right? Like one comma, one is the first element, right?

Speaker 1 53:56

So if I give you k equal to four, what is the first? What is the first what is the element?

Unknown Speaker 54:05

So if take four,

Speaker 2 54:06

okay, so if k equals four, then, then it should be two. Comma,

Speaker 1 54:17

two, comma, one, right. Two, comma, one, yeah. Okay. So what's when you want to and what is, what if k equals seven, and

Speaker 2 54:25

what if k is equals to seven? Like

Unknown Speaker 54:30

the value, the value of k is

Speaker 2 54:35

seven, let's say like. So k equals that basically like.

Unknown Speaker 54:41

Mean, you need to, like,

Speaker 2 54:46

444, pills, like, three comma, one,

Speaker 1 54:53

yes, three comma, one, right. So, how did you get this? Are, are one comma, two, two comma, one, three column. One, are you counting start from beginning, or you just directly jump to the to that item.

So how did you get? How did you get this? One, one, comma, two, two. I

Speaker 2 55:16

was just like jumping based on the best, on the size

Unknown Speaker 55:22

jumping. How did you jump? Like

Unknown Speaker 55:24

the numbers,

Speaker 1 55:27

right? So if I tell you k equal to seven, right, you skipped our the pair start with one and start, start with two, right? So if I'm saying k equal to seven, you you don't have to look into the items. Start with one. Start with two. You just your direct job to the items. Start with three. True. So in your algorithm, does this happen in your implementation? Did you directly jump into our that item start with three?

Unknown Speaker 56:08

Well, well, no, we are not doing that.

Unknown Speaker 56:11

Then, which is

Speaker 1 56:13

fine, which one is faster. So for your location, can you just directly jump to item start with three? Then, yeah,

Unknown Speaker 56:21

we can

Speaker 1 56:24

you're starting with the very beginning item, or starting with our

Unknown Speaker 56:30

I start with item

Unknown Speaker 56:33

one right.

Speaker 1 56:36

Can just directly jump to item or that start with three.

Unknown Speaker 56:42

Yeah. Yes, I think we can do that.

Speaker 2 56:47

Yeah. So basically, we're just doing this one to get the simple index spacing. Yeah, yeah. So what do I say? Like, if, if, if I think it currently like we can basically get the element based on the sorted

Unknown Speaker 57:07

on so the only time complexity here would be

Unknown Speaker 57:13

the sort itself.

Speaker 1 57:21

Well, what I mean is, or when you add your element into into a ring, sorry, we add an element into the queue, right?

Unknown Speaker 57:35

You want,

Speaker 1 57:37

you want to start from from the first element in the queue, and then keep going and check if I have any any item like left in the queue, right. Because, why don't you just skip? I mean, you already know your output won't be, won't be any added won't be any item that start from one or start from two, your item will always start from at least a three, right? I see I give you k as a 10, okay, if k go to 10, then, you know? Oh, so my element won't be start with one or two or three. It should start with our four or something, right? How did you get that? Anything in anything improved from your algorithm. You can jump. You just need to jump directly to our number of three. You should not start with with one. Okay, that's what I see.

Unknown Speaker 58:38

Yeah, can we can do that.

Speaker 5 58:42

I You're

Speaker 1 58:48

almost there. I see your solution. But imagine this is very large array, 1-234-562-2000, right, and I'm giving you a very large number, and say, 1 million, right? Are you gonna start looping your pair from 1.1 1.2 1.3 No, no, right? You will start from very large number. Hey, I know it cannot be in the number. Can I mean the pair right the product, the product rate cannot the item that I pick. It cannot start from one it should be start from 100 or start from 200 or something.

Speaker 2 59:39

Yeah, okay, so let me, let's do that. Then. So, well, well, if my key is if my key would be less than equals to n, so if my size of the error rate is also less

Unknown Speaker 1:00:00

than the size of k. Then I think

Unknown Speaker 1:00:06

this is fine. So this is

Speaker 2 1:00:09

case we can break out of this loop. So if that's the case, then we just get the value of k by n, right?

Speaker 2 1:00:31

So this will just jump, and we would probably have some kind,

Unknown Speaker 1:00:39

some kind of study. I

Speaker 1 1:00:50

It's okay. We have only two minutes left. I think, I think you can come up with after interview, maybe 15 minutes after interview or 30 minutes after interview. So basically, our so this is I prepared for today. I just want to give you a couple minutes to answer your questions, if you have any

Unknown Speaker 1:01:15

sure so

Speaker 2 1:01:17

I like to ask like, what is, what is the thing that you like most of your job, and what is the team look like.

Speaker 1 1:01:27

Okay, so basically, are the position that you interview is not fine for my team ppl is for just for general interview, and it's like interview pool, and we take rotate on each of the candidates, and for my team, are us working with our Merchant integration. So we are kind of side facing. We are kind of busy with our dealing with the merchants, dealing with our all kinds of integration engineers, and our higher prostate, higher, highest price priority is to serve customer Well, serve the merchant. Well, we're not fit to the general customer. We use a PayPal. We We serve the merchant who are use PayPal for their payments. Okay, so it's kind of our we obviously are not, not kind of busy, are most of time. But we, if we have, like, live issues and with the jumping immediately, so that's something. And we we have a features keep going on every sprint and every month, and we have to are basically focused on the new initiations and new features, but that's all in the plan. It's not kind of we have our some prototype and we had to get done in one month, so that we are not heavy on that part. We are heavy on serving the merchants and reduce the bug, reduce the latency and make service. Make the service available for car, maybe five nights or four nights.

Speaker 2 1:03:10

Thank you. It was, it was nice work timing this

Speaker 1 1:03:15

year, sure, sure. Take some time. Take a break. Thank you. Thank you. You.

Binamra-Paypal-Final-Srv-role-specialization

Unknown Speaker 00:00

E COVID, is that right?

Speaker 1 00:09

It's fine. It's fine. I'm just making sure, just I'm not butchering your name. So hey, thanks for joining. Let me introduce myself. My name is Venkat Bucha, and I've been in PayPal for about one and a half year. I'm primarily work on a Native website on the back end. So I'm part of the consumer onboarding team where users sign up for PayPal app for like, lot of different countries. So we handle that flow in the native, in certain cases, in the web as well. So part of this interview probably, I'll be taking some notes in between, and just if I'm writing something just don't mind, you know. And I'll divide this meeting into, like, you know, three or three sections start off with introduction, and then probably there will be some challenge question I'm going to ask in the hacker ran, and then I'll leave and to 15 minutes up to the questions at the end. Okay, yeah. So just to kick off, you know, please go ahead and, you know, describe your recent experience with the Thrive and like, you know, high level, like detector. So, okay, cool. So like to thank you for the time

Unknown Speaker 01:34

that you are giving me today. So firstly, let me start with

Unknown Speaker 01:42

myself. So my name is bhanda, and

Speaker 2 01:46

I'm a software engineer for like, about six years, working primarily on JavaScript and utilizing frameworks like spring, boot, hibernate and REST API, so Google services at tribent, we are basically managing an investment portfolio system where we basically take care of our accounts and their allocations and then which associated with those allocations

Unknown Speaker 02:20

for that particular week itself.

Speaker 2 02:22

And so talking about my involvement, I'm involved in multiple mental services, writing them in Java, Spring Boot test technologies, and exposing risk influence as well. And overall, that's a bit about the current project that I want to discuss and talking about my current team. We have a team of five members, and we have product owner and the scrub master as well. And on a daily basis, I'm involved mostly working on back end code. I've been working with code reviews as well as on product and support.

Speaker 1 03:03

So you mentioned basically how multiple micro services, right? So, how is this multiple micro services communicated?

Speaker 2 03:13

So with regards to the competition of micro services, it's like we have a RESTful APIs. So basically we expose multiple endpoints, and all these services are exposed to the Eureka servers, and then we just use discovery clients to make use of those calls of the API itself. And there is a couple of flows that is event based. So we basically publish a message to what we call as that SNS topic, and then different SQS queues as well. And pick those messages, and then it triggers certain lambdas functions, and then it just, we just have tunes of commutations

Unknown Speaker 04:01

on that particular

Speaker 1 04:04

path, sounds good. So, how do you resolve the eventual consistency in the micro services? So, basically, so we have, I see that basically, you know, you have multiple systems where trade orders portfolio, and then, you know, investment growth, like all of these different modules, I'm assuming. So probably there might be, is it one database or different database? Probably might have multiple services. There micro services. So let's just say, if somebody ordered a trade, then it goes into multiple systems, like you have to take an order, you have to update the portfolio you have to update the growth so it might be different different services or different database. So how, how would you think that you can handle that eventual consistency to make sure all of these are in sync? Okay, so

Speaker 2 04:59

one of the downsides of having a distributed system with like, investment, trade, trade orders and portfolio like with each individuals microservices have its own database like and it's basically making sure you have proper consistency, because they are distributed transaction across the server, right? So basically that we resolve it using a architecture based patterns itself, and we break those distributed transactions into a series of local transactions. And then there is a trigger right next to that. So if the if the event is successful, we proceed forward, and if any step, like any failures, should I say, then we basically trigger compensating transactions to roll back onto the previous steps as well. So, yeah, we use an orchestra where we have a SATA manager. And, you know, like, the way it works is, like we are having a trade order that will basically create order based events, and we call those as, like, the investment service, and then it will reserve the investment, and then it will go back to The settlement service. So in between, if anything fails, then we just go back and eventually, like the individual report goes using the, I would say, the compensating actions itself.

Speaker 1 06:33

So if it's a different databases, how do you roll back? Because it's orders is in different database and different service, and then your portfolio management will be in a different service and different databases, right? But if you want to roll back, you have to go through the APIs. So how do you roll back?

Speaker 2 06:53

So rolling back with like, if you're talking about rolling back like each of these actions within individual database would have a compensating transactions as a part of a saga pattern itself. And so basically what I mean is like by compensating transaction, it is a manual rollback. So you basically have an action that creates an order that would be compensated by using a cancel order API. So similarly, like, if we are doing the investment service, you would have some endpoint, like a reserve fund, and then you would have a compensating action to release those funds, right? And so that's kind of like a pattern that we have for each of the transaction and we have a compensation added to it, and we basically have it allocated so that whatever there is like if a failure action is occurring, then there is some kind of retry that kicks in.

Speaker 1 08:00

So all these are implemented in same micro service or transaction management with some, you know, framework. Let's bring both. So, oh, yeah. So this saga, how did you, how did you implement, at least at high level, you know, like you have two services, basically, and like order and portfolio. So if you want to manage the, yeah, manage the transaction between these two, and then how would you use a Spring framework using transaction management to solve that problem?

Unknown Speaker 08:35

So the thing is, like,

Speaker 2 08:39

like, each of the saga has its own application. Like it generally orchestrates the whole thing, but it's off the user reserve, so like, if you are compensating, would be a part of the individual macro services that have the actual action itself. So if we just want to have one application, then that's just using the transaction annotation on its method, and it will basically resolve it by itself as well. So since, but like, since we are using a distributed systems, then we basically have to work on with some kind of temporal SDKs, this and where, like, we basically create a workflow for orchestration engine, and then you basically write its logic in those temporal blocks, and you and on, on your actual micro services, you'll have those, I'd say, the individual patterns that has actual actions and its compensation as well.

Unknown Speaker 09:39

I'd say, okay,

Speaker 1 09:46

just so I think you worked on the client side as well, right? So, so it looks like you are familiar with Angular as well. So what type of data binding that you use in Angular.

Unknown Speaker 10:03

So with regards to the bindings,

Speaker 2 10:07

I'd say there are, like, there are different types of data bindings

Unknown Speaker 10:12

in Angular. And like,

Speaker 2 10:15

basically, Angular is a interpolation, which is just one way, where we basically update the property from the service to the template itself. And there is something called as property binding, which we basically bind DOM properties to component class. So you do have the brackets, but the name of the property and the expressions, and also like, then you have event binding, where you can bind events with the name itself to certain handlers. And then you have

Unknown Speaker 10:51

two way binding using answer.

Speaker 1 10:58

How is the two way binding works? Though, like, for example, you have a template which accepts the email and then you have your, you know, controller. How do you map those two way bindings?

Speaker 2 11:10

Okay, so with regards to the two way bindings, we generally use NG Angie model itself and like,

Unknown Speaker 11:18

what we basically

Speaker 2 11:19

need to do is like you have an email property that you need to do is like a create a model, for example, like you have an input tag where you want to store the email, then all the input tags you have on the Ng model decorator, then you also have the value type of the value that also which, which would, which would be your email itself, and then your actual component, you just have a field that would Be an exact match of the particular name itself. Okay?

Speaker 1 12:05

So you have the link for hacker ranking, right? I think in your meeting invite, there will be a link to the hacker rank. Would you able to join there? And,

Unknown Speaker 12:18

you know, share your screen. So

Unknown Speaker 12:38

can you see my screen?

Unknown Speaker 12:41

Now? Yeah, I can see your screen, and

Unknown Speaker 12:47

Let me

Unknown Speaker 12:49

just get The Link myself. I

Speaker 1 13:41

Oh, to join that? Confused? Yeah, it says in the lobby like,

Unknown Speaker 13:53

I think you have to admit it.

Speaker 1 14:02

I Okay, I'm sending the link again in the teams chart. Can you just bring that up?

Speaker 1 14:20

Okay, I think, yeah, did she able to get it? Okay, just go to the last one. Yeah,

Unknown Speaker 14:30

right, yeah, I did that.

Unknown Speaker 14:35

So you see, I

Unknown Speaker 14:46

Yeah, and you go into the coding Tool,

Speaker 1 14:51

and I'm going to the post the question there. I

Speaker 1 15:16

BBB, so, yeah, that's the question. So we have a string with some digits from zero to nine. And so basically, you have to split that string into, you know, the prime numbers, and then count how many prime numbers that you can make out of that string. For example, if you see that 11375, so split that into that string into different ways, like, 11 is another prime number, 37, five, that is one set, and then 11, 375, that is another set. You know, 111375, that is another set, right? So you can make a three out of that string, three different sets of prime number, right? So, yeah, can you write a program that satisfies this condition? And you know, given the set of

Unknown Speaker 16:11

prime numbers that you can make with that string, I

Speaker 1 16:23

yeah, just take Your time and then read it, you know, understand it.

Speaker 1 16:59

Yeah. If you have any questions, let me know. I think,

Unknown Speaker 17:02

yeah, basically,

Speaker 2 17:05

like, I think this is some kind of dynamic programming, right? So we basically have primes in rings, so we will just create a set of numbers, and then we will just contain all the prime numbers. So

Unknown Speaker 17:27

we need to cut the numbers into three digit Max, I suppose. Okay.

Unknown Speaker 17:54

So let's describe it generating the prime numbers from Two to one. So

Unknown Speaker 18:39

BBB,

Unknown Speaker 18:53

so could you ever to explain like,

Speaker 1 18:57

how are you thinking to you know, at least at high level,

Unknown Speaker 19:02

what? What could be done to solve this?

Unknown Speaker 19:04

Okay, so what I'm thinking is like, it's like,

Speaker 2 19:08

I just want to start by generating the panel members from two to one of seven, because it's just thing to do. So like,

Unknown Speaker 19:20

we would be doing something

Unknown Speaker 19:22

like comparing,

Speaker 2 19:26

comparing numbers of the generated prime numbers. And it has nothing to do with the overall algorithm, to be honest, but it just helps to be like helps to have a comparison against the prime numbers that is being generated, right? And we just use 10 programming approach where I think I just need to figure out how many ways I can split a string and that each piece is valid prime. So basically, this is kind of kind of like overlapping solve problem.

Unknown Speaker 20:09

Yeah, okay,

Speaker 1 20:18

I'm going to just Remove the static Add Numbers method, yeah,

Unknown Speaker 21:58

so should I Stop?

Unknown Speaker 22:00

Yeah, Go ahead. I

Unknown Speaker 22:35

So to check if the number is prime or not, right and okay.

Speaker 1 23:10

Yeah, let me know to that stop anything or any questions. You know,

Unknown Speaker 23:14

basically we just like are trying to check if

Speaker 2 23:17

we pro replicated as a little question itself. So, like some kind of division method, I'm just thinking first way to take it. There is a way, if it's prime, because I can just do it, precursor itself. Okay. So so far, any numbers less than two is

Unknown Speaker 23:41

The best guess scenario.

Speaker 2 24:00

So any numbers less than two is one of the base case, and if numbers is exactly two, then

Unknown Speaker 24:10

that's also another one.

Speaker 2 24:21

Checking if the number is divisible by two, and if the number is less than

Unknown Speaker 24:27

two or this divisible value, then we just

Unknown Speaker 24:30

basically return

Unknown Speaker 24:37

false.

Speaker 1 24:46

So why is that number equals to two? Is written stroke that. So you're saying

Speaker 1 24:56

about two, no, not divided by two. I think the first line, number 34 what is the condition? So

Unknown Speaker 25:03

basically, because it's a prime number, depends.

Unknown Speaker 25:16

Now we need to do the

Unknown Speaker 25:18

other cases as well.

Unknown Speaker 25:22

So we look three. To the square root so.

Unknown Speaker 25:56

And it says if

Unknown Speaker 25:57

the number is a prime or not, so if it's

Speaker 2 26:01

time, if it's divisible by two or it's less than then it's not so

Unknown Speaker 26:08

we can, like,

Unknown Speaker 26:12

if any number is divisible, then we can just

Speaker 2 26:16

return false. So this would be by six

Unknown Speaker 26:21

if it's time or not. So

Unknown Speaker 26:22

So one thing

Unknown Speaker 26:27

is like, I basically

Speaker 2 26:29

need to populate a set for one, Two and two, say, I

Speaker 2 27:24

Yeah, okay, so this would give me a list of set of primes from two to any input. Okay,

Speaker 2 27:40

me. I think we are. We actually need, like to do methods That counts fine, strain. Mm, hmm, important.

Speaker 2 28:22

I and also the question, I guess, is, like you need to do a model of the string as well. Do

Speaker 2 28:58

Okay, so now the way I did this is We basically

Speaker 2 29:15

so this would keep my plans right. So

Unknown Speaker 29:27

and So now, once that is

Speaker 2 29:30

that is completed, I think I initialized my dynamic programming already. I

Speaker 2 29:47

and also now this size will be little better than my strings left i

Speaker 2 30:00

i also what I need to make sure of is at the end of my input, I need to make sure that I initialize it, would Be not my best for itself. I

Unknown Speaker 30:58

i also like in this

Speaker 2 30:59

dynamic case, I need to look as well From the back. So I start from the back. I

Speaker 1 31:34

so what are you doing with that loop? Is it for the Yeah,

Unknown Speaker 31:40

I'm thinking like

Unknown Speaker 31:43

looping it from the back. So what

Speaker 2 31:49

that basically does? It takes character and then so I start from the current index that I'm in right and then I will go all the way to the maximum itself.

Speaker 1 32:05

Okay, so you're trying to figure out if the prime number is from the back, like, for example, one to three. I'm

Unknown Speaker 32:11

trying to figure

Speaker 2 32:13

out maximum of the length itself. And since our prime numbers can only be three digit Max, which is here, we also need to make the issue of the point value as

Unknown Speaker 32:26

Well, right? Okay,

Unknown Speaker 32:29

please continue.

Unknown Speaker 33:14

Now that three over here is like

Speaker 2 33:16

we have maximum of three digit of prime numbers. So now, like once, once we have that, we can just

Unknown Speaker 33:30

get the substring from it. Okay, so

Speaker 2 33:50

and as like so, if this number anyhow becomes bigger than 106, we just break it out. Sorry.

Unknown Speaker 34:02

We need to pass it. I

Unknown Speaker 34:23

And if, like, if partial

Unknown Speaker 34:24

is greater than 106 we just pick out.

Unknown Speaker 34:29

So why is 106 now

Speaker 2 34:33

it's, it's because, like, we had a max, like, maximum time number of 106 in the input. So

Speaker 3 34:40

that's the

Unknown Speaker 34:43

reason that I've gone for 906, okay, so that's

Speaker 2 34:53

good. And if my prime set contains that, I

Speaker 2 35:09

and so if that is the case like then I can populate my dynamic programming and I it.

Speaker 2 35:40

So, yeah, it will just like the calculate the sum of these numbers, and then just keep on dividing, and then it just goes all the way to the top. And the last thing that it does, it's calculate the first one. And so the answer that I'm looking for here should be

Unknown Speaker 35:59

the first position. Okay,

Unknown Speaker 36:13

yeah, so do you ever to print something so I can try that.

Speaker 3 36:22

I it,

Speaker 1 36:36

yeah, just put one value. I think it doesn't, yeah, just try, 3175, yeah,

Unknown Speaker 37:07

I need to convert that to number.

Speaker 1 37:15

So your input is probably the string, and then somewhere, your algorithm uses the numbers in The generate frame somewhere you

Unknown Speaker 38:07

Yeah, okay, I think,

Unknown Speaker 38:10

I think I start from

Speaker 2 38:21

the next does not work, because I got A zero back. So let's See what We can do.

Speaker 2 39:25

Well, I think something is wrong here. The sift is pressed. I'm not pressing it. Sorry. What about that? I think like something is wrong. The sift is being pressed. I'll never mind. I

Unknown Speaker 40:27

Alright. So,

Speaker 2 40:31

so yeah, that I think that's Fine. So let's see what's happening. I

Speaker 1 41:00

it. So, why is it so in your first output, I see there's lot of numbers, seven, 917, 19. Where is that 19 or 79 why it's not part of the string itself? Why is it taking all those numbers? So this is

Unknown Speaker 41:33

just the set of times from two to 106,

Speaker 1 41:39

okay, and then you're matching the next dose I

Speaker 1 42:36

so maybe take a small input and say, like, for example, like, you know, Maybe 317, try that, or 3175,

Unknown Speaker 42:47

yeah, 3175, yeah.

Speaker 1 43:08

So you should be able to see at least like three different sets, right? So 317, five is one set, 31 let me Get up. Of Those

Unknown Speaker 46:21

So

Unknown Speaker 46:23

you have the list outside,

Unknown Speaker 46:26

and then in your

Unknown Speaker 46:30

upper loop, that line number 64

Speaker 1 46:36

so yeah, just explain it to me like, what, what your those two loops And what you're trying to do in that loops. So

Unknown Speaker 46:48

in those two loops, I'm like, basically trying to get the

Unknown Speaker 46:53

biggest numbers to the smaller one. So like, I'm keeping, like,

Unknown Speaker 47:01

the back of the loop, right? So I'm just starting,

Speaker 2 47:04

like, starting from the maximum length, and then I'm like, trying to get, like, three characters from,

Unknown Speaker 47:16

okay, do

Speaker 1 47:24

so how does that solve if you take the max length? So the question is, basically, so you have a string, and then within that you're looking at the max length, like 3175, for example, if you're looking at the max length, but how does it relevant to counting the prime number. So, so, for example, you have a string. And then if you take the problem slightly different way, so you have a string, basically, and you can split that string into multiple ways. For example, 3175, and then three, each one three or one or seven or five is a prime number, right? Or you can split that 31 or 117 or 75 right? Or you can split that 3175, or you know, 175 so 1231121231234

Unknown Speaker 48:20

so each one is

Speaker 1 48:23

a prime number or not. So if it is a prime number, you just count the set, right? So if you take a slightly different approach, Would that solve the problem inside your two loops? So,

Speaker 2 48:40

like, what I was thinking of was like, like, let's say like, firstly starting, let's start, starting With the

Speaker 4 49:00

say the first five.

Speaker 2 49:21

Yeah, so that was like I was thinking, like, did you start with five at

Unknown Speaker 49:27

first? Okay,

Unknown Speaker 49:35

sure, okay, so,

Speaker 1 49:38

what do you think the complexity so this approach. So what is the space and time complexity that you're thinking about?

Speaker 2 49:53

So basically, here we have a we have a nested loop, right? And

Unknown Speaker 50:02

then we are looking from i plus

Speaker 2 50:05

to the minimum itself, and then we just also have method that gets square root

Unknown Speaker 50:14

based on the prime numbers.

Unknown Speaker 50:22

So I think

Unknown Speaker 50:26

it will be the

Speaker 2 50:29

total time complexity. So, like, this loop would be, they go off and because the inner loop can, like, look for three times like and so like, it would also go through and, I suppose, and like, but we are doing it like a soft string for like, small substring for the Space, country, city, it will be a

Unknown Speaker 51:01

bigger event itself.

Speaker 2 51:05

Yeah. So both of them would be linear

Unknown Speaker 51:11

about the function that one in the above. So

Unknown Speaker 51:15

you have another loop over there. Okay.

Speaker 2 51:19

So, yeah. So this one would be linear as well, right? Because it's, it's as this going to going from one end to the other. So this will not

Speaker 1 51:33

mean, yeah, the Generate prime number, that's the one.

Unknown Speaker 51:41

How would it be

Speaker 1 51:44

linear? If we have just one per loop here, where you are traversing through and how it is the it's a similar, like you said linear about having to in a loop and in a loop, and again, your there is one more function there. So how did the complexity will be the same if we have different traversing so

Unknown Speaker 52:12

that would be n square root of n.

Unknown Speaker 52:26

Okay?

Speaker 1 52:30

So, yeah, we have seven minutes left. I just wanted to leave some time for you as well, just to ask some questions and all. But if you want to take a couple of more minutes try to try with your program. That's fine, but otherwise we can move on to the question session. Okay,

Speaker 2 52:51

well, to be honest, it's both ways. I'm just thinking of the code as well, like I'm trying

Unknown Speaker 52:57

to figure out What's wrong. To be honest with

Unknown Speaker 53:02

you, I sure I'm

Speaker 1 53:39

Yeah, since anyway, five minutes left, I just want to leave some time for you as well to clarify some questions. Do you have any questions for me or

Speaker 2 53:50

Well, I'd say, like, I would say, not question, but I would like to know a bit more about how much, what are the tasks that developers get involved in, apart from its people coding.

Speaker 1 54:12

So say that again. Sorry, I didn't quite ask you a question

Speaker 2 54:15

to know, like, how much and what are the tasks that developers generally get involved and apart from the typical coding

Speaker 1 54:25

talks, you mean, is it like meetings or something

Speaker 2 54:28

like, I mean, the tasks? So, yeah, oh, the

Speaker 1 54:32

tasks. Okay, sorry, I didn't quite catch that. So it depends on the role. I mean, I cannot exactly tell you, Okay, so like, you know 50% is you do coding and 50% the other tasks. It's not like, I think it's not like a set stone, not just in here, but in my experience, what I have seen is it depends on your role. It might shift your coding tasks versus the other tasks. Sometimes, you know, if you are in a bigger role, for example, like senior engineer role, you might do sometimes the designs as a like, you know, sequence diagrams and stuff as part of your stuff, you know, tasks, and then probably the rest of the time is coding. Sometimes you have to do the develop. So it's not like you have set up stuff that it's defined, but this is all of the responsibilities that's including coding, code reviews and design and, you know, communication with other teams, communication with the product, everything. So sometimes with a sprint, it might split differently, like, you know, depends on what that sprint looks like. So yeah, that is how it will be. But typically, Senior Manager will contribute, at least in here, will contribute the design and HRD high level design as well, as you know, the coding as well. I see for the insights.

Unknown Speaker 56:03

Let us include. For me,

Unknown Speaker 56:06

sure, sure, I think we had a very

Speaker 1 56:13

good brainstorming time. Sure, any other questions? Yeah, it's not, not at the moment,

Unknown Speaker 56:22

but as just like to throw a bit of

Unknown Speaker 56:25

things question to you. Like, what do you like about PayPal,

Speaker 1 56:30

the team, especially, like, you know, I joined this PayPal team, one of the teams that I'm working with. It's, it's really cool. And you know, lot of talented people that, even though I have lot of experience, but still, there is always something new that I'm learning. So that's the that's the thing, you know, I like about people, and also like leadership. The immediate leadership that I work with is very cool. It's very rare to work with, you know, good leadership, especially, I mean, top leadership, not much of a regular we work with, but you know, the loyal leadership that you work every day, if they are cool, if they are, you know, good. You know, it's a job will be much easier, and your job will be, you know, much good. So that's what I like about, yeah, good

Speaker 2 57:20

to work on a flexible and supporting environment, right? One more like, I would also like to know a little bit more about how the onboarding look like. So

Speaker 1 57:35

onboarding in the sense like you're talking about the technology, or you're talking about the job onboarding would specific way that you're talking about job onboarding, yeah, that's it's depends. It's basically the job onboarding. I don't have much insights how it goes. It's like a HR that they will prepare a candidate for, like, at least a week or two. There might be some sessions, like onboarding sessions that they it's like, each one on one working sessions with the HR, basically, or maybe a group of people joining on that day working with HR probably will be assigned to some HR, and then you work with them. So it's not typically, like, you know, it might be the same for everybody. Depends on which team people are joining, but there might be company onboarding, there might be team onboarding. So typically, it changes. But in a nutshell, we'll work with the HR, you know, to get the onboarded. So it depends on how HR works with the team. So it's nothing, you know, specific to the team.

Unknown Speaker 58:45

I think we're just about on time as well. Yeah right.

Unknown Speaker 58:48

Thank you very much for your time.

Bisesh-CVS-SRV-First

Speaker 1 00:00

Minute for decision to join, and then we'll continue.

Unknown Speaker 00:08

How's your day going so far?

Speaker 2 00:09

It's going quite good. Yeah. How's yours well as

Speaker 1 00:13

well. Nice, rainy, little rainy day in the morning, but quite nice outside. Otherwise. We're always surrounded with lot of the problems at the job, and that's what we

Unknown Speaker 00:27

do, right? Yeah.

Speaker 1 00:41

Are you located in Minnesota? Oh, no

Unknown Speaker 00:46

New Jersey right now. Okay, I

Unknown Speaker 01:20

Hello, hello, answers,

Speaker 3 01:25

hello, hi, ba sesh, yeah.

Unknown Speaker 01:30

Are you ready? Yeah, I'm ready.

Unknown Speaker 01:35

Okay. Can you quickly introduce yourself?

Speaker 2 01:39

Yeah, sure. So my name is B says I've been working as a software engineer for about six years now. So I'm working primarily on Java based applications utilizing frameworks like spring, spring, boot, hibernate and RESTful web services for API developments. I have a pretty good at work experience working on microservices, and I've also been developing and designing microservices. So some of the cloud services that I've been using is including AWS, so ECS, SQS, SNS, things like that, you know. So

I've also used our AWS lambda to and also talking about, like, my front end experience, well, I've worked with Angular and talking about testing. I've worked with JUnit and Mockito. So, so, yeah, so I'm, currently, I'm on scrum team. So my team consists of five developers. We have a product owner and a scrum master, and on a daily basis, well, I'm mostly involved in writing a back end Java code so helping our team with the code reviews and also on production support so things like that, basically, yeah,

Speaker 3 03:03

sounds good. Can you talk about your latest projects, like the text tags you used and features you developed? Yeah,

Speaker 2 03:12

so the tech stack that we use, right? So, basically, it's a Spring Cloud gateway. So base application for, like, financial application, yeah, so we work with orchestration, with the API, so we basically have Spring Boot application, which is like containerized using Docker, and also it's deployed to the AWS ECS and well, for the database, we are using AWS, rds, Postgres, SQL and on the front end, side of the application, well, we have angular that is also deployed on s3 bucket, so it's behind a CDN for observability and also monitoring. So, yeah, so we also use from dashboards and also promises.

Speaker 3 04:08

Okay, sounds good. So in this interview, we will question you on both front end at the back end.

Unknown Speaker 04:17

I will take care of the angular questions.

Speaker 3 04:21

My will ask you about a drama questions, and we both will give you a quick exercise to test you out. Okay, so first of all, can you remember? Okay, tell me how, how much percent of the front end work you've done in the past few years,

Speaker 2 04:41

yeah. So, um, well, I would say it's 30% on the front end side, and mostly I'm working on the back end side. So it's like 7030

Speaker 3 04:53

sounds good. Okay, so for the front end side, do you use? You? What version of encoder be used? So

Speaker 2 05:02

in my recent like projects, well, we've shifted to Angular 18, but like that project was originally running on Angular 14.

Speaker 3 05:13

Have you participated in the version updates migration?

Speaker 2 05:19

Well, I haven't really, like directly, been upgrading the versions, but yeah, okay,

Speaker 3 05:28

all right, let me get to, let me ask you some basic questions. What is the data binding?

Speaker 2 05:35

Sorry, I couldn't get you the data binding. Data binding, yes, oh, yeah. So well, data binding is basically a way for you to, like, bound your templates with actual components, right? So, like, you have one way of binding, where you bind the data from a component to like view using like string interpolation, where you just, like, use a double curly brace. And then you have like, property binding, right? So where you bind a specific property to a certain expression. And so, like, yeah, then you have like, how to say this, like, two way binding, where you were, like, it can be achieved with the help of, like, any more. So, yeah, okay.

Speaker 3 06:26

Can you explain how to use HTTP client with an example

Speaker 2 06:33

how we can use HTTP client, right? So, well, basically, HTTP client is used to make any like RESTful API calls to your back end services, right? So, like, firstly, in order to do that, like, you need to, like, import the HTTP client model into your app model. So like, on your imports on the like, NG model operator, and you provide that model, right? So like, once that is done, like you can specifically just like, start by injecting HTTP kinds into your service, and either it's like in your constructor, or you're just like, using like injector. So basically, that once that's done, you can, like, specify different methods where you can make restful ABA calls, right? So on my experience, I was like, recently working on displaying the position for the clients and how their weights are reflected, right? So I I was like creating a position like where I injected the HTTP client, and I define methods to, like, basically call those back into, into, yeah. So, like, by providing, like, the API URLs. So, so yeah, basically, yeah, that's, that's pretty much sums it up. Okay,

Speaker 3 08:03

can you talk about the difference between the guards and the interceptors?

Speaker 2 08:09

Yeah. So difference between, yeah. So basically, I would say, like, guards are something that provides a layer of like security, like when you're working with the routers. So like, whenever, like, you navigate from from all the routes to another route. So like, so like guard controls that right? So make sure that you can view the resource and not like, like you can do, like some some kind of like protection and prompts. Or for like, doing some kind of checks using guards right like, for an example, like you could activate methods to check if user can do that or not. But for like, interceptors, like these are used for modifying your like HTTP request before, like, it reaches the downstream right? So instead, like, for example, like handling the authorization token injection in each of the HTTP client that it uses, you can define an interceptor where you can actually, like, attach an authorization token to request to our central point.

Unknown Speaker 09:24

So

Speaker 2 09:26

I think, yeah, that's the difference. Okay.

Unknown Speaker 09:34

Can you compare obese and the promise

Speaker 2 09:39

compare? Okay? Like, oh, so observables and promises, like they are like, used for synchronous data, right? So, so like fetching promises are, are like different in a sense that promise usually returns one value and like so, like promise results once, but like observable is a stream of like data which emits data over time, right? So, so I think like that would be the primary difference. Okay,

Unknown Speaker 10:16

what is the async pipe?

Speaker 2 10:19

So, async pipe, right? So, um, so, yeah. So, like, basically async pipe is used to, like, asynchronously calculate your data on an observable. So basically, like, you turn the last emitted value for like, let me give an example. So, like, you have an observable or a behavior of a subject on your component, right? So you want to display that on your front end, and you can basically, like, bind with an async pipe in your ng container or an NG if, where you can do like, a check if the user is, like, available for you to display in a certain component, like, using, like, sync pipes, yeah.

Unknown Speaker 11:12

Do you know about Angular given life stack of books?

Speaker 2 11:18

Yeah. So, um, so, like, basically, like, there are, like, multiple lifecycle hooks in Angular. So firstly, the one that gets executed is ng on change. So like, which basically did exchanges on your component. And it's also called like, when the input data changes right. So it is called before the init. The next hook is ng on int. So

Unknown Speaker 11:49

like this, basically gets called on

Speaker 2 11:52

once after, after your ng on change is initialized, right? So it usually we're all initialization logic checks in, and then you have your ng after viewing it. So basically, it's after the Components View has been rendered, right? And then you have ng on destroy, so it's called before the component is destroyed. And so, so yeah, I think that sums up the lifecycle.

Unknown Speaker 12:26

Have you used Angular Material?

Unknown Speaker 12:29

Angular Material, right?

Speaker 2 12:33

So I don't really have a whole lot of Angular Material experience, but, like, we use the internal library within our company that, like, basically creates custom directives, works on like different type of input types, date pickers and so on, right? So it's like pretty similar to our Angular material, but it is like internal So, okay, I

Speaker 3 13:07

What kind of performance improvements you've done in the front end?

Unknown Speaker 13:13

Sorry, I couldn't get you properly performance

Speaker 3 13:15

improvements. Have you done any of that? Oh, yeah. So

Speaker 2 13:20

performance improvement well in the front end, right? So, like, basically, we make sure that you do kind of some kind of, like lazy loading, like whenever it's possible, or whenever, like, it's available and it's it has to be effective, right? So you could also do some way of like optimization, like where you lazy load images, you could do on proper change detection like strategy, so you could, like, push change detection to reduce read renders. And you could also do some kind of like caching techniques by, like, stacking static assets also so. But like, whenever, like it comes to, it comes to like the API call like you need to make sure like they are properly, like subscribed and you could, like, only making like paginated request, so you could do a school based pagination request instead. So, so, yeah, I'd say, I'd say, like these optimization techniques, okay,

Speaker 3 14:33

what's your single page application and its benefits?

Speaker 2 14:38

Single page application, right? So single page application, like, we could also say it's an SBA form, right? So it's special, like, it's kind of like an Angular application, like, where you have a single HTML page, but like, all of the components that is present in the page gets, like, dynamically updated using some kind of like JavaScript code, right? So in this case, like Angular is a single page application, and your like, React is also a single page application. So instead, like, basically, like re rendering the whole HTML page, so certain components gets destroyed and gets replaced based on the routing techniques that they're using. So, so, yeah, that's what a single page application is,

Speaker 3 15:25

great. Okay, let's move on to a code exercise. All right, sure. Can you share your screen and use any IDE or online compilers You you like?

Unknown Speaker 15:38

Okay, let me

Unknown Speaker 15:40

just get that you

Speaker 3 16:10

Yeah, so now I am pasting a API call REST API call

Unknown Speaker 16:19

in the chat. Okay,

Speaker 3 16:23

so this is a API to fetch some information about accounting. So I would like you to pull the data and displaying the follow information on the Angular UI,

Unknown Speaker 16:39

so the name,

Unknown Speaker 16:43

the flag, image,

Unknown Speaker 16:45

the capital

Unknown Speaker 16:49

population.

Speaker 3 16:59

You have around 20 to 30 minutes To complete this task. Okay, Okay, Cool.

Speaker 1 20:54

And you can keep sharing your thought process. What are you thinking about? And okay,

Unknown Speaker 21:00

so, okay,

Speaker 2 21:02

so, well, I'm thinking like, if I can do the work with this template here, so

Unknown Speaker 21:09

do I need, like, define separate one,

Unknown Speaker 21:11

right? So, but I think, like,

Unknown Speaker 21:15

I can just, like, do it,

Unknown Speaker 21:17

because, like, it's very straightforward, right? So,

Speaker 2 21:21

okay, so let me start by, like, defining the HTTP, like science, and then like, we can, like,

Unknown Speaker 21:31

let's just do that. So since,

Speaker 2 21:35

yeah, so we will define countries

Unknown Speaker 21:41

array that holds the Right? So,

Unknown Speaker 21:46

so yeah,

Unknown Speaker 21:51

okay, so now,

Unknown Speaker 21:59

okay, so I do.

Unknown Speaker 22:07

Okay. So basically the

Unknown Speaker 22:10

constructor that index the HTTP client. So

Unknown Speaker 22:19

now that's done.

Speaker 4 22:22

Need to provide it the import so on, the component decorator. So like the added.

Unknown Speaker 22:37

Okay, so now we just,

Speaker 2 22:39

like, need to like, fetch countries which would be done in our initialization logic so

Unknown Speaker 22:48

we would implement ng on it. I

Unknown Speaker 23:05

Okay. So now,

Speaker 4 23:12

okay, so the thing needs to do is, it's need. It needs to, like, give me the country. So

Speaker 2 23:22

we'll just like, make an HTTP call to that API.

Speaker 4 23:34

So once we have the data, we can, like, just do a subscribe on

Unknown Speaker 23:50

it. So this should give me a list of countries. Let's just basically do a console. Dot log. I

Unknown Speaker 24:13

Yeah, so

Speaker 2 24:15

see, we have data here, so now we basically need to, like, update the template.

Unknown Speaker 24:40

Okay, so I'm

Speaker 2 24:44

so, so the way, like we wanted is like we want the name, flag,

Unknown Speaker 24:51

the region,

Unknown Speaker 24:53

and so on, right? So we

Unknown Speaker 25:03

a base so,

Speaker 2 25:07

so, like, since, like, we're returning an array, I'm thinking of like doing an NG for, so we'll do a division that would basically do an NG For, And I

Unknown Speaker 28:27

Okay, so

Unknown Speaker 28:30

this is like property binding example, right? So

Unknown Speaker 28:34

we provide the property source as

Unknown Speaker 28:38

the countries to

Speaker 4 29:10

Alright, so I'm not, I'm not quite sure, like why The flag is not being renewed.

Unknown Speaker 29:19

Sorry, yeah, flags, I You

Speaker 3 29:54

all right, it looks good. Okay, so what if? Now, will I ask you to

Unknown Speaker 29:59

add accessibility

Unknown Speaker 30:03

to this application. What do you do?

Speaker 2 30:07

You want to add accessibility, right? Okay, so Well, if you want to, like, add accessibility, like you do like, have options for like, using the lighthouse plugin on your like develop tools. So, like, usually, like that helps you identify, like, what accessibility issues exist on your app. And well, then you like, you can like process further. Like, for an example, you need to add the area label and so on. So, which is like usually identified by using the lineups tool on your like developer tools.

Unknown Speaker 30:49

So, so, yeah,

Speaker 3 30:53

okay, sounds good. Yeah, excellent.

Unknown Speaker 30:58

It's all yours.

Speaker 1 31:01

Thank you. I it was

Unknown Speaker 31:06

one saying that happening, live,

Unknown Speaker 31:10

nice, switching gears a little bit, you know,

Speaker 1 31:14

I don't know much about Angular, but I know a little bit around the Java Spring Boot and on the back end side. So can you just start explaining, like, what is your recent experience for wherever you are working? What are you proud of and what are you excited about? That's something you did recently on the back end side, okay,

Unknown Speaker 31:42

well, like,

Speaker 2 31:45

okay, so I would like to mention, like, we work with an orchestration platform, right? So since, like, this is a financial system, like, we basically sometimes work on with an orchestrator. So like, like, having to modify the request and response from the client calls. And like, whenever those like client calls come in and there's a modification, so, like, we had a cyber requirement to, like, audit those changes. And like, for making sure like this auditing are proper. I was like tasked to come up with an approach to like, streamline the process, right? So for that, I started with a proof of concept to create a streaming engine so like that would be very generic, so that like could be onboarded into like any application, right? So now, like our applications like that, we hold within our like team, so, like, it was like 15 microservices along with the orchestration layer. So it was like a really vast microservice based system, like and this, like swimming engine that I had thought of like was supposed to be something like that would integrate with all of

Unknown Speaker 33:04

those services, right? So we,

Speaker 2 33:08

like, we do not have to, like, do a lot of like manual work on those services. So, like, instead, like, just make our RESTful call to to this like services, right? So this generate like RESTful service, like, where we could, like, onboard different, different services based on, on the metadata, or like, and then, like, streamline the auditing purpose, right? So, so this was, like, something I'm very proud of. So, like, I, what I did like for that was basically created a Spring Boot based service that, like excludes a couple of endpoints on onboarding and also the origin. So this service, like will take care of, like changing the data into like standard format, and then like aggregating them. So like batching them and like, then, like, streaming it, like streaming into the snowflake. So like, table for like, auditing. So like, this is like, something that I'm very, like, proud of. And this is something like, I recently accomplished. So it was like something that was, like, long overdue. And like, the reason that I'm proud of like, of it like, it's not just like, because the fact that I was able to get it done so and like, it was like, our production like, but it was like, generic, like, such that like, not just our team, but like, also, like, external teams were also able to use the service for the requirements, like, of similar nature, right? So, so, yeah, this is, like, something that I would say, like, I'm proud of

Speaker 1 34:52

good. So, what framework, what tool you use

Speaker 2 34:57

for the connection? Yeah? So, well, it was a mixture like based, like of executor service. So like we had Spring Boot based application that was internally used. So like command to like spin up different like threads for handling data. So like Java screenwood application and also like multi threading enabled. So,

Unknown Speaker 35:27

so, yeah, I

Unknown Speaker 35:48

I think like you're on mute. I'm sorry.

Speaker 1 35:52

Oh, I'm sorry. I was just asking, have you used Kubernetes or Apache Airflow?

Speaker 2 36:00

You said Apache. What? Careful. Okay, so, um, I have, like, used, um, cuberated in my past projects, but like, recent, in my recent projects, like, we're using AWS ECS for the orchestration, um, we basically have Doppel containers that we upload to the elastic Container Registry. And like, then, like, basically around ECS tasks and service creation using like cloud formation templates. So like, ECS does a very similar work for AWS itself. So we went with ECS.

Speaker 1 36:41

And how do you use, or do you need to use scaling or do you. have a steady applications, or you might need a scaling on that. Oh, yeah, so,

Speaker 2 36:57

so, like, I would say for the scaling part, right? So we, we are, like an orchestration platform, right? So, like, we basically need like scaling in place. So we have lot of clients, so, like, so, like, well, basically, like for our policy, right? So, like, for like a peak hours, right? So we need, like, scaling on our work. So the way, like we like schedule policy is like we have New Relic dashboards, and based on the actual like CPU uses and the memory uses, we scale up and accordingly. So I think, like, we have around like 40 instances running which can, like, scale up. So, yeah,

Speaker 1 37:57

when you say scale up, what do you refer to the scaling? Is it vertical scaling or horizontal scaling?

Speaker 2 38:07

Yeah, so it's horizontal scaling. So, like, we basically spin up newer tasks. So it's like new instances that get to spin up. So, like, because I'd say, like, vertical scaling is not really not like an option for like, micro services, right? Because, like, you can not have, like, too many of or like to, like, big size of a CPU and memory, right? So like, because, like, there are like limits and to handle like these large volumes of data, I'd say like causing load scaling is the best way to do it.

Speaker 1 38:45

Okay? And do you see any challenges when you do high traffic orchestration, um, well, challenges, right?

Speaker 2 38:55

Okay, so, I mean, like, it does come with a certain challenges, like so, like, whenever, like, you have higher traffic, because, like, you need to keep track on how much the traffic is increasing, right? So we were, like, able to identify those early on. So like, using, like, J meter, like performance testing and so, like, basically, like, handle it, right. So sorry, so I guess, like, some of the issues that we had. So, like, we were able to, like, handle the services to its fullest, but like, having, like, some memory leak in performances. But like that was that was like, handled separately, but talking about, like, strictly, like scaling, like, I don't see any problem because, like, it was with orchestrated with within the like, ECS itself, okay?

Speaker 1 40:00

Now, you know, spinning our head around the overall scaling that we get from all the cloud providers, you know, when we have high traffic, we have kind of auto scaling automatically, or instance will be spinned up, and then service discovery will happen, whatever, what not, right? Yeah, do you, or do you have a thought around a basic level of working like when we have a Java application, and we are coding Java application. Do you see similar challenges internally into your Java application, and how? How do you think that they you can do something similar inside your application, not at the scale level or at the overall level of the application, of spinning up more instances and all other things individually. Let's say you have one function in your Java application, and that might be, you know, that might get a big traction in the high load, or some kind of situations where you need that function, very often, need to have more performance at that. How would you achieve that?

Speaker 2 41:15

Okay, yeah, so, so, yeah. So I think like, this is where your thread pooling comes in, right? So you might like have dynamics like threadful resizing, right? So that's like, what the actual like, hysterics, commands like. So I like mentioned, like early, earlier, like kings kicks in, right? So hysterics, like commands is basically a framework that sits on top of the executed service, right? So this, like, helps in dynamically creating thread pools with like, minimum and maximum size. So like, like, it's based on the load, and it spins up new threads as needed. So, so like, so like, this is the, like, the main thing here, I'd say, like, it's like, dynamic thread pool. So,

Speaker 1 42:19

all right, so now, since you bring up the thread or something into the conversation, I will just go to the basic like, what do you know about thread in Java and how, how are they created?

Unknown Speaker 42:37

Okay, yeah, so

Unknown Speaker 42:38

threads in Java, okay,

Speaker 2 42:42

well, I would say, like, basically, it's, it's, it's just like a unit of execution, right? So it's just like a path of execution within the program. So, so in Java, like, I think it's something that is, it's like on stack and, like, I would say, like, it's like, used to paralyze CPU intensive work. So, like, because, like, your CPA has processes, and in these like processes, like, there are multiple threads that your box can run on. So utilizing all those CPU efficiently it can, like be done like, basically, like, paralyzing your processes using like threads. So in order to like, create threads in Java, you basically have two major like methods. So like one, one is to like extend the thread class, and the other is to implement Runnable interface, right? So, but like, you know, like, you can also now do it, like using executive service, or, like hysterics, command pool, or stuff like that. So, yeah, so basically, that that's, that's it, yeah.

Unknown Speaker 44:02

Okay, couple more around thread is, do

Unknown Speaker 44:08

you know, what are the different states of threat?

Speaker 2 44:12

The different states of spread? Right, right? Yeah. Okay. So I'd say, like, there are multiple threats, states that the threat goes through so well initially, like, whenever you have our first trade that you create, right? So, it's, it's in new state, so, as, like, it's just created, but as not started anything yet, right? So, and then, like, after it's like, created, you get into a runnable state. And like, that state is, like, whenever it's ready to run, or, or, like, it's running something, right? So, and like, then, like, you have your like block state. So like, when your spread is waiting, like, to acquire, like lock, and then, like, it's waiting like a state. So like, whenever, like your is like, waiting for like, certain kind of signal and like, then it's like, I think, like, those are the primary ones. So, so, yeah, okay,

Speaker 1 45:23

other than Wait, do you know about sleep?

Speaker 2 45:28

So, like this the sleep state, right? Yeah. So, yeah. So, like, the sleep is, like, used to, like, basically put your thread on hold for a certain amount of time. So if you, if you do that, like, the sleep that activates the thread, like, will basically, like, be waiting for so it would like go to, like, a time waiting state. So, so, yeah,

Speaker 1 45:56

so then what is, what is the difference between the weight and sleep?

Unknown Speaker 46:02

Okay, so, like,

Unknown Speaker 46:04

difference between weight and sleep?

Unknown Speaker 46:08

Oh, okay, so

Speaker 2 46:11

I, like, basically, like, whenever, like you, you have your thread in a waiting state, like it means that, like, it's waiting, like, for in for an indefinite time. So like you're like, you actually have to have a thread that notifies it or has a joint method, right? So, but like time waiting like state basically means like that it is like waiting to be like for a fixed time, so when the like time expires, right? So that's the main difference, right? So waiting state can be, like, indefinite, until, like, there is something to notify, but like, like, sleep state is actually, like, there until, like, your time expires. So, yeah, I think, like, that's, that's one of the main differences between,

Unknown Speaker 47:12

okay, do

Unknown Speaker 47:16

Can you explain Java strings to me?

Unknown Speaker 47:20

Java strings? Yes, okay, um,

Speaker 2 47:27

okay, so like stream. It's a feature like that was introduced in Java eight. So basically, it's used to convert your collections into a stream of data. So it can be like, used to process your collections efficiently by like, applying like, terminal and intermediatory operations, right? So, so, yeah, I think I think that would sum it up. Okay.

Speaker 1 48:07

So, do I really need stream so I can do everything with collections. I'm sorry, like so I have collections, right? Why do I need stream? Or, you know, what is, what is the difference? Is there any benefit? Is there any problem with stream, or I can do everything with collections, okay?

Speaker 2 48:37

So like benefits of streams while using for a collection, right? So, yes, um, so basically, like your stream, um, like not only is like useful, like collection operations, it can also basically be advantageous, because your collection can be done in like, more of a like, declarative way, alright. So, so basically, like, you define, like, what you want to do, and like convert it into like, filter or a map, and like, then like, you could, like, chain those operations together, right? So like, for example, if you like, want to, like, basically, like, filter out employees Like, who do not live in a certain area. Or like, then like, convert the employees into into certain author objects. And then, like, you may want to even, like, sort it. So, like, you know, like, so you could, like, basically work in a single stream, right? And like, then keep on, like, changing those operations until, like, you terminate the operation, right? So you could, like, achieve parallel parallelism on like, by using like parallel streams. So, like, because like, basically like, it enhances the performance, right? So and So, like. The other thing is, like stream does not like change the original collection, right? So it makes more safer as well. So I think, like, these primary reasons are what helps in performance by using performance parallel streams so it has, like, some side effects by not changing the original extreme. So, so Yeah, and like, it's very simple.

Speaker 1 50:27

Can you also explain regards to the streams again, what's the what is the map and what is flat map?

Speaker 2 50:37

So in regards to stream, like flat map and flat map, right? So basically, I would say like map is used to convert our data from one form to another form, right? So, like, when you have a stream, right? So for example, like, if you have a list of string, right? So you want to like, convert them into, let's say like uppercase, right? So if you just like, want to like, basically convert it into like account of characters, right? So you would use map so, so like, it would be a one to one conversion. But like, let's say like, the flat map is used to flatten a stream, right? So basically, like, what you can do is, like, for example, you have a list this summer, and like, then you want to, like, flatten that nested list into a single list, right? So this will be done by using block map so which were like converted into like a single list of string,

Speaker 1 51:48

and how does reduce work in Java, space

Unknown Speaker 51:54

in Java, what was it again?

Unknown Speaker 51:57

Sorry, reduce, reduce,

Speaker 2 52:00

right? All right. So in like, reduce, like you provide, reduce the function, right? So it's just an accumulator, right? So, so you provide accumulator function that is, like, used to, like talking the, like taking, sorry, the taking the previous value. And like, you want to work with the next value, and like, it gives the initial value, right? So like, for example, if you have a stream of integers. So like, you would want it to, like, calculate the sum, and like, then you could, like, provide the initial value of like zero, and then you could do like, A, B input parameter. So like, when you can do a plus b as its function definition. So, so, yeah,

Unknown Speaker 53:00

okay, and then

Speaker 1 53:04

we're in year two reduced what, what does collect do, and is there any difference between

Unknown Speaker 53:11

radius and collect?

Speaker 2 53:14

Sorry, so, so collect, right? So, like, generally, like, it's like, used to convert it into, like, some kind of, like, custom result. So collect is used to, like, maybe build a list or set or map or any other collections, right? So it's like, usually used to do the some product or some kind of, like, concatenation, right? So, so the major, I'd say, like, the major difference here. So like reduce is usually used to reduce the value into, like a single value, but like collect, whereas it's like used for more complex like operations, especially like building other collections, right?

Speaker 1 54:09

Okay, and then, if we call reduce on empty string, what do we get? Do we get any exception? Do we get? What should we expect? Um,

Speaker 2 54:25

so, like, if you, if you like, have reduced on an like empty screen, right?

Unknown Speaker 54:30

So it does

Speaker 2 54:34

have optional, I think, like it returns an optional, so you might get an option of empty,

Speaker 1 54:43

okay, having used HashMap and concurrent HashMap, I I'm

Unknown Speaker 54:53

sorry. Could you repeat that again?

Speaker 1 54:54

Have you used hash map and concurrent hash map?

Speaker 2 55:00

Yes, so basically, hash map and concurrent hash map, right? So I have used a hash map, so it's like for single threaded environment and concurrent hash map is like used for multi threaded,

Unknown Speaker 55:21

like multi threaded,

Speaker 2 55:24

like concurrent hash map. So like, basically for like, provides internal like synchronization, so like, whenever you have multiple requests, like write operation, so it does not have any kind of like race condition, right?

Speaker 1 55:45

So let me clarify that, or ask you a similar or kind of question on that, if the if we have multiple threats, they try to modify a hash map simultaneously,

Unknown Speaker 56:03

how do you anticipate that situation? Okay,

Speaker 2 56:11

so if you, if you like, do not like, want to work with a concurrent hash map, and you just like, want to stick with a regular hash map, right? So, if you like, have multiple threads that actually access the that value that and like or update it like it would lead to race condition, right? So you'll have data corruption, right? So if not like, then you basically have to like, synchronize the object and also like, then you have to do synchronize like blocks where you do, like, put the operation on the method and so like that will, like be, basically be the way to go, like, without using concurrent hash map. But it's not like, generally recommended like, because, like, you're actually defining synchronization. So it's not like foolproof guarantee that everything like would work, because it could be a manual effort to like manage that, right?

Unknown Speaker 57:20

So, right now, okay,

Speaker 1 57:24

switching gears a little bit around what you explained earlier and then asking questions similar to situations around the Spring Boot side.

Unknown Speaker 57:40

Have you worked

Unknown Speaker 57:43

or used circuit circuit circuit breakers.

Speaker 2 57:47

Circuit breakers, right? So on our like, orchestration platform, like, so it provides circuit breaking mechanism. So circuit breaker is just a pattern, like, where you prevent, like, cascading failures. So, so, like, two circuit breakers that I've worked with include resolutions for Jn history circuit breaker. So, so, like, the way it works is that you have everything working fine and, like, which is like a happy state in the circuit, right? So it's also, like, closed, right? So it's a closed state. So like, whenever, like, there are downstream, APIs are like failing, so the circuit opens to an open state. So sorry. So like, that is like, similar to your like electrical circuit opening, right? So there will be like threshold decided on your config and when to like, open the circuit, and under what exceptions and what scenarios. So, like, so, sorry, yeah, so, so basically, like circuit will now be open to dedicated amount of time, and that circuit that goes to a half open state. So which is like testing scenario to see if the downstream services have, like been working for fine or not. And in case it is successful, then it like closes the circuit, making it happy again, right? So, if not like it, will go back to the open state. So between, like these four handling the opening open surrogate cases, there is like a fallback mechanism that kicks in. So, like that a client gets a meaningful response back. So, so, yeah,

Unknown Speaker 59:52

yeah, yeah, that was how

Speaker 1 59:54

long? How long the circuit break? Or the circuit breaker mechanism will continue looking for the answer, if, if something is not working. I mean, will it dry? Oh, sorry.

Speaker 2 1:00:08

So, like, so it kind of like, depends, right? So it's, it's configurable. So I think, like, the default timeout is, I think it's like 30 seconds, right? So, but like, from my experience, like it goes to an open state. So I think, like, that's what we have configured. So like, I think after like, 30 seconds of failure of the like of the calls, we go to an open state, and then we have like second timer, and then like it goes to an open state. And so like, basically, like it just does a couple of request tests, and if those fail, it goes back right away. So half open state is kind of like very short, but the open state, it would be like 30 seconds.

Unknown Speaker 1:01:03

Okay, cool.

Unknown Speaker 1:01:07

On this Spring Boot side.

Speaker 1 1:01:12

What is the at Rest Controller annotation

Speaker 2 1:01:17

the other right? Rest Controller, right? So this annotation is on your controller layer, right. So it defines restful entry points, right? So, so what I mean is, like, it's a combination of control annotation with the response body annotation, right? So the main thing here is, like, whenever, like, you have a glass on annotated, like with the app Rest Controller, so like, you are expected to get a response back on, like, like your typical controller, but like you may get a, like, a response or a view, but as a at a like, a Rest Controller, You could, like, get an actual response data back.

Unknown Speaker 1:02:12

Okay, cool enough.

Speaker 1 1:02:15

You may take a little water break or something. You've been speaking blood for a long time, and then we'll jump into a coding exercise for like 25 minutes.

Unknown Speaker 1:02:23

Yeah, no, no, sorry, yeah, just

Unknown Speaker 1:02:25

take a little

Unknown Speaker 1:02:30

Oh, yeah, I'm sorry,

Unknown Speaker 1:02:34

that's fine. You know, it's been going

Speaker 1 1:02:39

well. I want you to simply show a little bit of the coding proficiency you may have and very simple, but if you can bring up any IDE or Spring Boot application development, you know, tool you may want to and then I would want you to have a Spring Boot application. Basically, what I am looking at is any simple database entity, level of the application where you where someone can interact with your endpoints to add and, you know, update data. You can choose anything, you know, very simple to like a user management system or something like that.

Unknown Speaker 1:03:39

Okay, okay, so

Unknown Speaker 1:03:45

I think all right, so let's see. Let's go

Speaker 2 1:03:52

with employee management. Alright. So like, I'll just, like, create two endpoints which would be used to, like, get by ID and then also create an employee so, And I'll be using

Speaker 4 1:04:07

in memory database For This Bye,

Unknown Speaker 1:05:44

okay, so,

Speaker 4 1:05:47

just like our creators, simple spring initiative project for our employee

Unknown Speaker 1:05:51

repository. So

Speaker 2 1:05:54

the only thing that I need to Google Now is the actual configuration for the in memory database. So let me just get that I

Speaker 4 1:06:20

okay? So once we have this configuration added, we should, like get a data source that like spring autoconfigures for us.

Unknown Speaker 1:06:35

Okay, so,

Speaker 2 1:06:37

well, basically, like, in order to create a REST API, like, we basically, like, need four packages. So it would be, like, the entity where the model is stored. It would be repository where, which is like used to be a data access layer, and we would have a service layer where we define the business logic, and we finally

Unknown Speaker 1:07:03

will have a controller layer, right? So

Unknown Speaker 1:07:05

basically, get the request.

Speaker 4 1:07:11

So yeah, so let's start by defining on the entity, okay, so the employee entity.

Unknown Speaker 1:07:27

So

Unknown Speaker 1:07:30

we use

Speaker 4 1:07:31

the Entity annotation. So which basically, like, make sure it's an actual table, right? So we define and identify for the employee, so which would like basically be a long value.

Unknown Speaker 1:07:46

So then we'll just give some something like, feels Like

Unknown Speaker 1:07:49

actual name or email, all right, so I

Unknown Speaker 1:08:11

Okay, so those should be our fields, right? So

Speaker 2 1:08:14

we will, like, use a lot more for exposing

Speaker 4 1:08:18

the getters and sellers, right? So that's going to make it easier. Okay, so once we have that, so, like, we can move to the repository layer. So repository layer is your like data access layer, right? So we should be making use of Spring Data JPA can

Speaker 2 1:08:54

so we should have a lot of like, inbuilt methods available, like find my ID, find my first name, last name, if you like.

Speaker 4 1:09:05

Follow all the foundation, right? So it also provides save method so we don't have to, like, worry about explicitly, like defining it. So now we can go to our Rest Controller, which would be our like entry point for the API. Okay,

Speaker 4 1:09:29

so for the rest control, I'll give a base pack for the employee API.

Unknown Speaker 1:09:41

So like first,

Speaker 4 1:09:43

simply, like would be just doing like two endpoints. So one is to create an employee.

Unknown Speaker 1:10:03

Okay, so, and the other one would be actually to

Unknown Speaker 1:10:07

get an employee by ID, right? So, okay.

Unknown Speaker 1:10:18

So,

Speaker 4 1:10:21

okay. So, like for each of these methods, like we need proper annotations for like creating so like we would

Unknown Speaker 1:10:30

create a post mapping annotation.

Unknown Speaker 1:10:39

Okay, so

Unknown Speaker 1:10:41

post mapping would get

Unknown Speaker 1:10:43

a request body, so

Speaker 4 1:10:49

we'll just like, do our try catch for exception handling.

Unknown Speaker 1:10:55

Like, yeah, so we don't break the application,

Unknown Speaker 1:10:59

and

Unknown Speaker 1:11:05

let's see. So okay, so for

Unknown Speaker 1:11:10

the get by it like we need to get

Unknown Speaker 1:11:13

mapping with a path variable.

Unknown Speaker 1:11:17

So basically,

Unknown Speaker 1:11:22

we should, like, follow similar strategy here also. So I'd say like,

Speaker 4 1:11:29

the main thing here is, like, we need to make a call to our service layer, and for that, like we do injection to to the like service layer. So let

Unknown Speaker 1:11:43

me work on the service okay. So,

Unknown Speaker 1:11:47

so so far like we

Unknown Speaker 1:11:49

have defined interface, right?

Speaker 2 1:12:01

Okay, so the thing like the service needs to do is, like, basically get an employee by ID

Speaker 4 1:12:11

and so, okay, so then also save an employee, right? So, basically, okay, yeah. So, okay. So now we will create an implementation for the service arm, so, like that interacts with our database. So we will implement employee service, and then, like, we'll, yeah, so we'll also annotate the service annotation, so on the employee service, like, we need a repository to talk to the database, right? So we will inject that also. So

Unknown Speaker 1:13:06

now we implement

Unknown Speaker 1:13:09

the two methods here. So let's

Unknown Speaker 1:13:12

just do this.

Unknown Speaker 1:13:16

Do we find my ID and find ID

Unknown Speaker 1:13:20

returns and optional? So we will throw an exception in case,

Unknown Speaker 1:13:24

like there is no employee, right? I

Unknown Speaker 1:13:41

The phone, like, save, we just like, basically,

Unknown Speaker 1:13:46

okay, not safe. Okay. Now we consider to Okay,

Unknown Speaker 1:13:50

now we can use the service in our shoulder. I

Unknown Speaker 1:14:39

Okay, so this should be an end,

Speaker 4 1:14:44

okay, so just have basic interaction for an employee control to save it, right? So, and then get by ID,

Speaker 2 1:14:52

okay, so, um, do you want me to like, I'll run it and we can demo it.

Unknown Speaker 1:14:56

Yeah, go ahead. All right. Cool. Okay.

Unknown Speaker 1:15:43

Let's change the

Speaker 4 1:15:46

method here. Let's, let's save the exception methods. So okay, let's restart. Okay.

Speaker 4 1:16:05

So we should like now get a proper message here. Okay, so, um, let's create an employee for us, Right? Okay, do

Unknown Speaker 1:17:22

Okay, so let's See what happened Here.

Unknown Speaker 1:17:30

I Let me see What happens here. Okay, you

Speaker 4 1:19:00

okay, I think it's issue with Loki. Let me just add so as if that works. Okay, I Okay, this Okay, so, like, this Community Edition does not work with not work really well. So I usually, like, usually get this error. I might just like manually do, like, the getter status and let me just get The status. Okay, like, let's do it properly. Okay, I

Speaker 4 1:20:07

Okay, now we have to data, so it should now be saved, so we should be good. Now, let's see so

Unknown Speaker 1:20:28

So, default

Speaker 4 1:20:43

constructor, so we created one employee. Let me just like add a couple of more and we could test the get

Unknown Speaker 1:21:00

method. So now we have like, two employees.

Speaker 4 1:21:04

We could do a get employed, to get the claim the employee by ID. So, so yeah, both inputs are working. So, yeah, those would be a very simple GET and POST. Aba,

Speaker 1 1:21:26

okay, now on the employees controller, endpoint,

Unknown Speaker 1:21:33

I want to

Unknown Speaker 1:21:35

protect. I can I can

Unknown Speaker 1:21:39

save, I can

Speaker 1 1:21:41

retrieve. But I want to protect the updates like no one should change the data once it's there. So first, if you can add one more to update the data by the ID, and then I want to block that endpoint. Okay,

Unknown Speaker 1:22:04

let's get that then, okay, update. I

Speaker 2 1:23:06

so do you want me to, like, implement the service as well?

Unknown Speaker 1:23:23

So I was starting on mute, yes,

Speaker 1 1:23:27

yeah. Basically, I want a control endpoint, and then, you know, the ability to update the data. Okay,

Speaker 4 1:23:37

so let me do get by ID and Then basically Do

Speaker 4 1:26:06

Okay, okay, so I think that should take care of it.

Speaker 2 1:26:20

So What? What? What was it like?

Speaker 1 1:26:24

Yeah, just run it and show me that. Like, get one employee first, and then update, maybe salary or something of that. All right, address. Just update the address. All right, let's do that

Speaker 4 1:26:47

so there's no employee. So, like, we should create one, right? So

Unknown Speaker 1:26:53

you give me once again. I

Unknown Speaker 1:26:56

think, yeah, that's fine.

Unknown Speaker 1:27:02

So we would create an employee,

Unknown Speaker 1:27:07

and then we'll update the salary

Unknown Speaker 1:27:11

so we update the advice.

Unknown Speaker 1:27:23

So

Unknown Speaker 1:27:25

before we do a put, let's

Unknown Speaker 1:27:28

do a get, make sure we get a valid response, right? So,

Unknown Speaker 1:27:34

okay, so, so now let's do the update, okay?

Speaker 4 1:27:40

And now let's do a get. So, so, yeah, so, um,

Unknown Speaker 1:27:47

so it's now like protected, right? So you have an updated address

Speaker 1 1:27:55

right now. This is a situation that, you know, we created for our organization, but they found out that someone took advantage of that openness. So I would want to, you know I'm in a situation that, for now, I don't know what I do, but I want to stop using, or I want to, you know, stop people using this put. How would I stop that? Everything else should continue working.

Speaker 4 1:28:23

So, so, like, if you, if you just, like, want to disable the put endpoint, right so

Unknown Speaker 1:28:32

you could, okay, let's see, basically,

Speaker 2 1:28:42

so like, you could, like, use, like, pre authorized in here. Like, but like, you probably need to use spring security.

Speaker 1 1:28:55

So let's say, if you have sprint security already, and I don't care about authorization right now, I just want to disable this endpoint not going into the code changes.

Unknown Speaker 1:29:06

Okay? So,

Speaker 2 1:29:10

so I think, like, you just like, do a pre authorized false so, which would basically, like, always deny your access, right? So I think that would do the trick here.

Speaker 1 1:29:24

Yeah, that can be done. But again, I don't want to touch my code of the controller or anything else. Can we do something from the configurations, from

Unknown Speaker 1:29:33

the configurations? Let me think so.

Unknown Speaker 1:29:42

I think like, from, like, the

Speaker 2 1:29:45

security field to change you could, like, always, like, deny access. So, like, basically define a configuration class where you like,

Unknown Speaker 1:29:57

overwrite

Speaker 2 1:29:59

the run method and then on the like HTTP so where you basically provide the API that you want to like, deny the access to, so you'll have filter chain where you'd be, like, playing with the HTTP, like, security object, so And like any requests that matches API, like, slash, employee, slash, anything like, then like to, like, deny everything, right? So you could, like, also define,

Unknown Speaker 1:30:32

like, the PUT method, okay,

Speaker 1 1:30:39

and then a question on moving forward on this project that you build, let's say now,

Unknown Speaker 1:30:52

my employees

Speaker 1 1:30:55

are moving around a lot, and for the compliance purpose, I do need to keep a track of like, what are the addresses? You know, they change the address. They maybe have only one or maybe two current addresses, but I still need to keep a track, like, for 10 years, how far my how, where my employees were, what were, what were their address? How, how would you handle this?

Speaker 2 1:31:23

Okay? So, um, you just, like, basically want to keep track of the addresses, right? So, so like, for that, you like, probably want to work with versioning or maybe create like address history table. So, like directly, so you'd have one to like many relationship in the address history table, right? So instead, like having a single address string, so you basically create something like one to many for address history table, and it would contain, like your active flag, so that we know, like, which address is active, right? And so like and the others like, it would just be the history of

Unknown Speaker 1:32:11

the employees addresses.

Speaker 1 1:32:17

All right, I think we are at two minute over the time, so I will pause here. Thank you very much for all what you have done today for us. Yeah,

Unknown Speaker 1:32:28

I'd like to thank you guys for the opportunity to Yeah, thank you so much for your time. Yeah, thank you. All right, we'll try to get you the responses as soon as possible. Okay, okay, that's, that's Thank you. All right, thank you. you need to.

Dinanka-lowes-first-srv

Speaker 1 00:00

Tell me about your experience. You know what you've done recently, what were your specific inputs? From there, I will move to some programming questions. One of them, you probably don't need to share screen. Just have to walk me through what solution you're thinking. The other one, I just need you to kind of just maybe share your screen, and then, you know, write some code for me. And then from there some groups come set object oriented programming concept. I'm going to touch on a little bit of spring boots. I'll touch on database. If I get some time, I'll touch on CICD and other things as well, too. And then at the end of the interview, I'll give you 10 minutes back to ask me any questions you might have. How does that sound? That sounds good. Perfect. So get me Why don't you get me started by telling me about yourself. Okay,

Speaker 2 00:59

so hello. My name is Dina COVID. I've been working as, like a Java developer for about like five years now, working primarily on, like Java's like web applications by like, utilizing different frameworks, like, for example, Spring, Hibernate, and also different, like desktop services for different like API developments, I have, like, pretty good experience working on, like microservices, and I've recently been involved in, like designing different microservices, like deploying them, and also like deploying them to, like the AWS cloud. Some of the like cloud services that I've worked with includes, like, AWS, EC two, ECS, SQS, SNS, and also, like lambdas. I've also worked with like React and Angular in the front end side and on the testing side. I've mostly been working with J unit and like Mockito. Currently I'm in like a scrum teams. And this team consists of like five developers. We are, like the product owners, and also like the Scrum Masters on a daily basis, I like mostly involved with working on the back end, like tower. Course, I like helping the team with different stuff, like code reviews and also on some like products and like support details. So yeah, that's what I do currently. That's where my day to day to operate.

Speaker 1 02:23

That's good to know. What I'm understanding is you do have five years of experience, close to five years of experience, primarily on Java and display guide by native and everything else around back end technologies. You also have some experience with front end technologies, and then you have some experience with a lot some testing frameworks, I see you're currently working with USA. Is that correct? Yes.

Unknown Speaker 02:47

Why are you looking for a new role at this point?

Speaker 2 02:53

Any reason why you're looking for a new role? Oh, yeah. So USA, this is like a contract job, and my contact is coming to like, like contacts coming to end after, like, the end of this month. So I'm just, like, trying to get everything like, started, okay, okay, immigration status. I'm a USA, okay.

Speaker 1 03:19

I just want to get all these questions out of the way. Now, you are just a junior, which is really not too far, if you look at it, from Charlotte. But you know, we primarily looking at a Charlotte so with relocation concern or something,

Speaker 2 03:34

I'm open to, like, really racing. Okay, yeah. It.

Speaker 1 03:45

Now I have your resume put up in front of me. Can you tell me about, you know, what are you specifically doing at USA? Why a specific country was the project? About, what are your specific contribution to that project?

Speaker 2 03:56

So in like USA, we basically, like migrated on, like, this monolithic app they had this like use app that was basically like responsible for, like, managing, like the whole mortgage insurance and all the other like domain that they have our department was like basically responsible for, like, migrating the whole maintenance of the like user profile and the experiences of the users on the front end, right? So my responsibilities as, like a software engineer on the back end side, I basically developed, like, worked on, like maintaining different APIs. That's like, responsible for making changes to all the different user like, profile preferences, like so whenever, like a user, like mix like a teams to like how they want their user profile like preference to be, we would basically do like this post call, and they can, like, make those changes there based on that. This will also be like getting like multiple calls to basically retrieve like different datas, so that I was actually like responsible for like this migrations, and I was involved in like designing, in like developing, and also like getting approvals, and then also like promoting likely to prod. Or the application that I've been working on is this like experience APIs, which like basically grabs the like entitlements to basically display different like widgets and also different components in the front end size, so, yeah, it's like a Spring Boot based applications that's deployed on the AWS clouds. This application is like Docker eyes, and we basically like reverse, like AWS ECS for authorizing, like, all these like containers. Amazing.

Speaker 1 05:42

Just taking a few notes here, because now I love how you touch on the points where, basically you develop microservices to, you know, manage user data, user profiles, is what I'm understanding for this mortgage application. Now, could you just walk me through how you start? Because you had a monolith application right now. How did you start? By saying, Okay, this monolith application, this is a component I need to migrate, or I need to create a micro service for this user profile. How do you handle those data, the data structuring of each distance. How did you handle the existing data on the monolith to be on the new service? How did you make it consistent to make sure that how was the deployment added? Did you just fully destroy the monolith application at this one of the new microservices, or were they running parallelly together? You know, those I'm trying, I know there's a lot of question in there, but just, you know, help me understand how you basically, I'm trying to get a full picture of, you know, are you did this migration from a monolith to this micro service, and then I'll follow up on that. Okay,

Speaker 2 07:01

so basically, like, the way we got started with the whole like microsimprocess was we had like different, like components that had like a simple business domain, right? So anything to do with like user profiles, it was on like one domain, and then anything that has to do with like different, like user experience side was on like, a separate domains. So anything there was some kind of, like boundary constraints, right, for example, some maybe like order management, or like loan processing, or like profile preferences, or anything that could be like, stripped out, they were basically, like advised by our like architecture to basically, like take in step by steps, and we had to actually started with all the different like profile, like preferences management, right? So main path here was, it was like, responsible for making different changes to like user profile, like specific datas, and like their preferences, and then also like retrieving, like all those datas. So that's how we basically, like, stripped out, like the domains, the design, and also like the business constant, business context. Sorry, once that was done, we basically wanted to, like, capture the same datas with our like models would have, and then try to basically integrate the whole database for that like particular service that will have like, like, the mistake, like, similar datas and that we don't have any like missing data. So the way we design all these like database was for each like services is basically like, taking what existed on, like the monolithic ones, and seeing if we could actually, like, modernize, like, some kind of structure that we had, and then, like, the very sensor between like other like structures, we basically tried to, like, keep these data as like consistent as possible. But in terms of like, whenever like enhancements were needed, we like, basically decided, with architectures and also like database administrations to basically comply with all that one. So basically, in terms of like promotion of these, like micro services, we had like this feature flag options, right? So we basically did like this incremental updates with all the like feature flags, so that we our like context was actually like setups, like 1% of traffic like will go to like the older app, and 1% will actually be routed to, like our microservices. So we did like a rolling updates by doing like 1% forward by like 10% and then like 50% until it was then like 100% complete. So that was the overall like rolling update process. So then, you know, we there was like this whole design dog that we had to like generate for basically loading these like functional and also like this non functional requirements, how like the data integritys would be used, and how all these like services will actually be like independently held, and what actual like business problem like itself.

Speaker 1 10:02

Well, that's a good point. You brought up our follow up question. Thanks for clarifying that. So what I'm understanding you as part of the migration, you also did the data migration, so you had a new database with a new structure in the new world, right? You know, your own, your monthly application was on which database? Sorry, the

Unknown Speaker 10:23

monolith application was running on what

Unknown Speaker 10:25

database? Oh, it was on Oracle database.

Speaker 1 10:29

And then the new application was running the Microsoft was running on which database.

Speaker 2 10:33

So the new microservices, it was on this like, we had to use AWS ideas like Postgres and one of those, like services also included, like Dynamo TV.

Speaker 1 10:45

Okay, so basically, you had to take this structure of Oracle and then adapt it to a new structure in this state, and making sure that you looked at what fits when mandatory workforce could get rid of so you modernized other structure is what I'm understanding in the new world. And now my question is, you didn't touch up on Canary deployments, right? So you had put services running parallelly. So you started with, let's say 10% of the traffic going to your new service, and assuming 90% was still on the existing legacy one. Now, did you have to get that consistency across both of them?

Speaker 2 11:25

The way we handle this was so, like, you mentioned, like, we had some kind of, like Canary, but for like, our like initial release, we had to do like, incremental objects, right? So there's like a row, like, a, like a route, 53 switch, which we can do that we can just like, read out those control. We basically control, like, 1% of the traffic goes into like, this new service, and then this all, like automated through, like Jenkins, right? So whenever we want to actually, like, increase this percentage, slowly, will you basically increase it, and based on all the data we get and all the error rates and all the like response times and so on, will either like increase or like roll back all these like new services?

Speaker 1 12:12

That's not my question. My question is thinking, I'll give you is an example, you know, Salim and mortgage application user, you know some as a mortgage application in the new cell. You know, you know you had migrated Salman mortgage application from the old db to the new DB, so now you have the same snapshots of data as of let's say, yesterday you did the migration in both the new DB and the old dB. Now today, I logged into the profile and I updated, let's say, my birthday date, right? And then you had routed me to the new microservice. Your router routed me to the new service. Now, how do you make sure that the updates on that birthday is also reflecting in Oracle? How did you handle that? How did you maintain that data consistency across both world

Speaker 2 13:01

so we actually had, like a separate repositories that were basically called, like mortgage, Microsoft repository. This is basically like a cell script. And what it does is, right? It turns on this, like periodic basic we did, like a com where we had SLA for, like a date for all the different like data consistency. So we basically run this like magazine script, which is like a list of different sales script, and that basically, like takes in data from our like old database, is then like massage into form of like this new database, and then we like insert all those like database into like our new database, so we did this, like incremental updates using this magazine script, which is part of this, like separate workflow that ran like every day. Basically, we had this agreement that the like the data, data consistency from old to new, while doing the whole Microsoft phase would be like, somewhat like delay.

Speaker 1 14:02

Okay, okay. So there was an alignment that business, I like, aligned as an alignment with businesses. Let's say it could be delayed until this time you run the script to do those incremental updates across ODB got it so it was not like a real time, you know, update, right,

Speaker 2 14:17

right. So everything that was like in the old database, like too late, like today at like 5pm would be basically like reflected, like tomorrow at like 12am

Speaker 1 14:27

something. Now that was the agreement you had with business. I'm not going to get to the reason why I had an agreement, by the way, discussions, I'm asking you, it's a little bit over your level. You know, I know you're interviewing for se level a little bit so, but you are doing good so far. So I want to just keep you know touching on this. But take, for example, you have to do a real time update, right? So, meaning, if you, if I make a change in you know, your AWS, Postgres dB, I wanted to reflect the next second in Oracle. How would you do that? I Yeah.

Speaker 2 15:05

So, I mean, maybe you can do that, like some kind of, like listener on, like your Oracle DB, right? So basically, you can do some kind of, like real time, like migrations by actually using, like a chain data capture tables, and basically whenever, like, there is like a change in any kind of data, you basically run like lambda or like something that basically, like loads these data. So I think that would be like a way to close you like basically have to have this like listener that does this like, real time synchronization. Okay,

Speaker 1 15:48

listen now, which one way, where would you put a listener?

Speaker 2 15:54

Like the listener would be basically like, on, basically on like, maybe some kind of, like triggers on, like, the Oracle side, right, like the Oracle database one, For example, yeah, I think that would be okay. I

Speaker 1 16:32

All right, moving on. So I'm gonna put a question in the chat.

Unknown Speaker 16:40

Then you just,

Speaker 1 16:43

you don't have to saw code, anything. You just walk me through. Are you going to solve for it? Okay, I'm paying right now. Okay. I

Unknown Speaker 17:08

if you have any questions sometime, okay, I can see I

Speaker 2 17:28

so you just want to, like, reverse the values of like two variables, right? Like in Ts, TCS, basically, fine. So I mean, one of the ways, like, the easiest way to do this is to, like, basically use like a temporary variables, to basically store like a value, right? And then like, swap the other values, right? So, like, for example,

Speaker 1 17:56

oh, I forgot to mention it in like, in this case, there was a, there's memory concern, so we don't have even one kilobyte to do to start a temporary variable.

Speaker 2 18:08

So if you don't actually want to use like a temporary variables, then you have to do some kind of like math, like Edison and suppressants. So I think firstly, we will actually add like two values, like 10 and 15. Then I would like assigning, like, I will assign that to just type

Speaker 1 18:29

that for me in the chat, or what you're thinking, you know, you know, type that into chat. Okay, yes, I I was

Unknown Speaker 19:02

COVID too much. BBB, so he basically becomes a b plus b, right?

Unknown Speaker 19:11

A plus B, sorry,

Speaker 2 19:13

basically have like a B assigned as like a minus b, and then a guess, like we sign, we sign back,

Speaker 2 19:32

sign back to, like minus b, and because that would have like This difference, so it would be something like that. Okay,

Unknown Speaker 19:46

okay, perfect,

Unknown Speaker 19:50

yeah, looks good to me.

Unknown Speaker 19:54

Then my next question, I needed to share your screen.

Speaker 1 19:59

Just, you know, open up any ID could be an online ID, could be a local ID. Can

Unknown Speaker 20:15

you see my screen?

Unknown Speaker 20:17

Yes, I can see your screen. I

Unknown Speaker 20:32

just gonna wait for it to load up. I

Speaker 1 20:59

All right, so I'm giving a list of integers. So the input is you have a list of integers. I would like you to iterate through those, through that, through the integer list, square them, and put the result set in a different list. So basically, let's say you have a list of 234, the output of that list is going to be four, nine and 16. And I want you that is a plus if you read this in Java eight or later. And feel free to use whether an object list or a primitive list or a honorary list, anything, feel free to use that as a as your input and output.

Unknown Speaker 21:42

Yeah, so I'll

Speaker 2 21:43

use like tab x print for this. So let me like in a size a list here. Okay,

Speaker 2 22:00

so I have this like NSLs that basically, like contains a value of like one through a file. And then I can, like, generate another, like list of integers. I then I'll let it, like be squared here.

Speaker 2 22:22

And then all I need to do is basically, like, convert, like the initial list into a screen. And then, since we're like, changing the value, we want to do like a map operator. So we'll just have like this value, and then we'll, like, multiply it by, like a cell.

Unknown Speaker 23:01

All right, okay, 149,

Unknown Speaker 23:18

16, okay,

Speaker 1 23:21

right to me, I, now, if you change three to null, what happens?

Speaker 2 23:30

So if you change like three to nos, it will, like, basically be an error, right?

Unknown Speaker 23:39

Yes, sorry. Do that.

Unknown Speaker 23:42

Should we handle that? Oh, yeah. How do you handle?

Speaker 2 23:49

So the way I handle is, the first thing I'll do here is there's like, two ways, right? If you have, like, an old values, what you want to do, right? Do you want to like, skip like that element, or do you want to actually, like, return? No.

Speaker 1 24:10

The requirement is you want to, you know, just square integers, right? So I would assume you want to skip it. Skip it. Okay, got it.

Speaker 2 24:21

So to skip it, we can just use our filter, right?

Unknown Speaker 24:30

So this surgery here,

Speaker 2 24:48

okay, so since I'm like initializing as like a Lister, It doesn't even let me have like a novel here.

Speaker 1 25:40

Okay, what if your list is empty? What if we have an empty list? What happens?

Speaker 2 25:46

So if I have like an empty list, which will it? Will just like return, like an empty list back, because stream operations will still be successful. Here,

Unknown Speaker 26:00

got it. Got it. Okay?

Speaker 1 26:06

Perfect, perfect, awesome. Now you can stop sharing your screen. Now, moving forward. Skinner, what do you not stand by inheritance in that job?

Speaker 2 26:22

Never. Inheritance on Java. Okay, yeah. Can you please, like, repeat the question, please. I

Speaker 1 26:28

didn't forget the inheritance. What was your question? What do you understand by inheritance? What do you understand? Got it? Okay,

Speaker 2 26:36

so inheritance, like in jabus. So yeah, it's all about like, reusability of the code, right? What I mean by this is that you have like, certain, like base classes that your classes can now, like inherited, like their behaviors, or like the fields from. So basically you can, like, use, like what's present there, and that's basically inheritance there, for example, let's say you have like this, like parent class with like certain fields that your like, your style classes can now like use these fields Same things with all the different like methods, right? So if you have like this parent class, for example, like print methods, then you can have, like, a child class that's basically extends from it, and then it can, like, basically use that print methods, or like, even, like overriding.

Speaker 1 27:36

Okay, so how do you achieve inheritance in Java?

Speaker 2 27:41

So in Java's the way you can, like, achieve like inheritance is by like, one of the ways, like, using like abstract class and then extending from all those classes, right? So it doesn't have like FFA class, but you can actually do it like, like a like a parent class. Let's say, for examples, you have like this, like Animal class, and let's say that animal class has like, like a method, like eat, right? Which basically means that this animal will eat. And then let's say you have like this, like concrete implementations, or like this child class called Dog, right? And then you can extend this from animals, and then you can add like different methods from this, like particular dog, like for example, would be like the back methods. So whenever you basically create like a new, like object of dog, you can basically call the Eat methods from like your animal

Speaker 1 28:39

class, the way to inherit this or that. Do you need a super keyword in the dark class as well

Unknown Speaker 28:50

super in the construct? So

Speaker 2 28:52

basically on like the constructor, you would like one to basically initialize like the super class using the super keyword, right?

Speaker 1 29:03

Okay, sure give me one. Um, my, my Patreon, just Give me One Minute. I

Speaker 1 30:46

I'm ready. Sorry about that. No problem to get my charger. Okay, now I'm just gonna ask follow up questions around the same groups. Concept is, what do you understand by polymorphism?

Speaker 2 30:57

So what I understand about like polymorphism is basically like refers to like multiple right? So basically what I mean to like say is that you can have like similar methods that, like will do different things, right? So you can have stuff like a method overloading which is done with like the same class, and you can actually achieve like overloading by actually like changing, like the whole parameter type, or like the number of like parameters, or like the return times, right? So then you have like method overriding, right, which is basically done by like a type class. And for this, basically you need to have like the same parameter and then like the same return types for like method,

Unknown Speaker 31:46

yeah, so this is what I understand.

Speaker 1 31:50

So you give me an example of, you know, you know, being polymorphic, polymorphism, or how do you do that in travel? So okay, an example.

Unknown Speaker 32:02

So, okay, so,

Speaker 2 32:03

yeah. So let's say you have like an add methods like to basically calculate like our like class. Basically what you need to do is you need to handle like additional of different, like data types, right? So for example, you have like this add methods that basically takes in like an integers, like integer A and like integer B, which also like returns and integers. And then you have like another methods that is also called like ADD, but now this one will take like three, like numbers, right? So it will take into like a, like integer A, integer B, and also like integer C, right? So also you can have another, like add methods that will take like double doubles, right? So you have, you have like double A, and also like double B, and then it will like, return, like a doubles. So this is what like method overloading is, or like polymorphic examples, you can also have like, let's say, like, another way of like overriding, which is like, basically, when you have that, like earlier like examples that I gave you about the whole animal class that has an that has the Eat methods here. And let's say you have like a dog class that basically, like extends from this like Animal class, you can basically override the whole eat methods and says, like dog will eat like kibbles. You're just returning like a different implementations of like the base one, like the base eat methods,

Speaker 1 33:47

got it putting the code in the chat, what's going to be the output of that code. Now feel free to copy, put in an ID so it's for readability. Okay. Thank

Unknown Speaker 34:01

you. Posted in like a notepad care you

Speaker 2 34:26

so you have like a core class that basically calls like a method. Okay? So,

Speaker 2 34:49

okay, so I think since you are not like specifying like the type of norm

Unknown Speaker 34:55

you like get an exception on this one

Speaker 2 34:58

might be like a compile time session, like saying that it is like ambiguous.

Speaker 2 35:12

So because, basically, you have like multiple methods that takes up object to strings, which is like one parameters, it cannot decide which one to go to. So

Unknown Speaker 35:31

I want you to take a second look. I

Unknown Speaker 35:43

yeah, this will not compile.

Speaker 1 35:47

Okay? I think it's gonna return a string method. Is what I think was the default value of string. What's

Unknown Speaker 36:01

the default value of string? What's the question? Yeah,

Speaker 2 36:07

the default value of strings will be just like, empty, right? I mean, since you're using like objects or it will like, it will always be nulls, right? Because object, it's also like No, and integer is also like No, string is also null, right?

Unknown Speaker 36:31

So in this case, excuse me,

Speaker 1 36:38

the default value of string null is actually a value string, so that's the default value. So it's going to scan and say, what's the closest, you know? What's the closest implementation here? So in this case, it's going to return, you know. So it's going to return string method is, what is what I know So, but, you know. But anyways, just moving forward, I'm also gonna put another code snippet in the chat, and then you could just let me know, You know what's gonna be The output of that. I

Unknown Speaker 38:12

for some reason I cannot copy

Unknown Speaker 38:16

this. I copy.

Unknown Speaker 38:19

Let me do one thing, let me put out The text. I

Speaker 2 39:19

All right? So you basically have like three classes A, B and like C, right? And then Class B will extend from like Class A, and Class C will like extend from Class B, right? And you are basically initializing the object A as like C. So okay.

Speaker 2 39:49

So the first output here, it should be, I,

Unknown Speaker 40:03

so do you see? Is the spirit, right? Let

Unknown Speaker 40:05

me copy like this one second.

Speaker 2 40:20

So that should be the first one, right? Because it actually like calls the like get methods from passes and then the value of x is what's on like, classy, because it's not like referring to like, super, right?

Unknown Speaker 40:41

Yesterday, will be 30.

Speaker 1 40:46

So you're saying C display will be 30.

Unknown Speaker 40:50

Well, I mean, that's the first

Unknown Speaker 40:57

so you have class A

Speaker 1 41:00

where you initialized x as 10, and then there is a public method to do get x as 10 right. And then there is a in the display that A,

Unknown Speaker 41:17

in this case, is what I understood.

Speaker 1 41:22

And then in Class B extends a, and then you have a private integer which is 20, but you're not overriding the get x. You're only overriding, in this case, the display correct. And then you have class C, which extends P in class C as well. To you know, you're not

Unknown Speaker 41:53

in class C as well. To you're not,

Speaker 1 41:56

what am I saying in class C as well? You initialize as 10, right x 30. Now your display. You already display method in this case to do a get x Okay,

Unknown Speaker 42:12

in this case

Unknown Speaker 42:15

and in this case again.

Unknown Speaker 42:20

So what that means is, and then you also override,

Speaker 1 42:25

you know, get x in this case, you know, provided a different implementation to get x in this case, right?

Unknown Speaker 42:32

Is what I understood now,

Speaker 1 42:35

since this now, you created a method to do object or display does what new C, A, initialize a, b and c right

Unknown Speaker 42:49

in the test will be correct.

Speaker 1 42:53

So now the first a object or display, what does it give to you?

Speaker 2 42:58

So, okay, so it will basically print a C, like, C's, C, like, it would print the C's like display studies, which, because you are actually like referencing object C here, and not like the super class I

Speaker 1 43:23

so in this case, we'll only print. It's a key printers, since display correct, what will be the get x value?

Unknown Speaker 43:32

Yes, so it will, it

Speaker 2 43:33

will do like C's like display like 30.

Speaker 1 43:41

So when you override get X, did they provide any overriding? Because you're creating object A, but you're implementing object C, you're right.

Speaker 2 43:55

What did we call, like, the same methods, right? Because calling the same methods from like, the supermass, right? So you just it will, like, infer, like, the overriding like keywords, even if you, like, don't provide it,

Speaker 1 44:13

no, mister, you're overriding the display, which is why C is displayed, but it gets x value. Now get x really called get x of Class C, or to call get so the first one, right? Will you call get x of Class C? Or to call get x of Class A, is my question.

Unknown Speaker 44:32

It will call it from Class C. Why is that?

Speaker 2 44:37

Because you are like calling in like, a context of like classes that like, get x methods, right? So you are not like calling class A's like, get methods like directly. You are like in context of like the Class C, I,

Unknown Speaker 45:11

point you over, right?

Unknown Speaker 45:22

I'm sorry I'm not really following here.

Unknown Speaker 45:28

Okay. Anyways, moving on, just in the interest of time,

Unknown Speaker 45:32

just gonna touch on DB queries here.

Unknown Speaker 45:36

So

Unknown Speaker 45:39

directly a query you have a table, employee table, okay,

Speaker 1 45:44

employee table has three columns or four columns. You know, column one is ID, column two is name, column three is salary, a column four is Manager ID. So, write me a query, directors, all the employees reporting to Manager. Select manager, you know, I'll just make up a random number here. Manager, 456, so 456, is a single employee ID.

Unknown Speaker 46:17

So, yeah, sure. So basically,

Unknown Speaker 46:20

it will be a SELECT query,

Unknown Speaker 46:24

where, like with like a where class.

Speaker 2 46:35

So, okay, so let me time it out for you. So

Speaker 1 46:51

it will be like a select, okay, perfect, perfect. Now my follow up question on that is, now I want you to write me, you know, optimize that query to also, you know, return all the employees with the high by the highest by the salary. So where the highest salary is first row to be returned, and then down below. So

Unknown Speaker 47:18

for that, we'll just need to add, like order line.

Unknown Speaker 47:32

So yeah, The search right here. I

Speaker 1 48:08

can I see? I see? Okay, perfect, okay, awesome. Danaka, that was all my question for you today. Do you have any questions for me? I you.

Speaker 2 48:23

So first of all, like, thank you for all your time today. Here it was, like, really nice talking to you. The first thing I want to ask is, what are some like activities that the engineers will like get involved in, like, a day to day basic act like loads here.

Speaker 1 48:40

Very good question. So I first of all give up, because to answer your question, I have to let you know what team interviewing for and what we do. So you interview for enterprise cat and checkout. And basically this team is, you know, building functionalities and building experience towards cat and checkout operation for the entire lows. So basically, take off, you know, any card and check out operation, whether it's Add to Cart, view cards to all the weights for you, creating an order in across lows.com. Mobile apps, Contact Center and the stores as well. This team handles that Now, having said that, what that means is, this is a very busy application, that is, I think I see it as the entry point to $100 billion revenue company, you know, for the entire lows now, it's a very, very critical application, you know. And so on a day to day, one year, Debs engineers, there's a day just like you have, you guys are operating agile USA right now. So we do have stand ups in the morning, and then we have teams in the US and India. So we spend a lot of your money after stand up, collaborating with your India counterpart and getting some what was done with any collaboration. What is next for us to do here. And then after that, you also spend, if you're involved in any with cat and checkout, you're probably involved in every feature that that has been run in the entire lows. So most likely, as an engineer, you're gonna be involved in one significant feature the order and there will be programs that are run for these features, so you probably spending your money as well to maybe joining meetings and attending these programs and giving updates on progress on what you've done, if you have any questions with the partner teams and all of them.

Unknown Speaker 50:33

And then from 1030

Speaker 1 50:35

you can start to see a slowdown of all these activities, whether it's, you know, cross geo collaboration, or partner in collaboration. And then from 1030 you can focus on your core development and any learning you want to do, any self thinking or any new process, or anything you want to do as well too. So from 1030 all the way to five, you can add that time to yourself to kind of be very, very productive. And then the cycle kind of repeat itself. Now, in instances where you're on call, you might be pulled into production issues or production incident and say, Okay, we've got an issue. We've got to take a look. We've got to fix that. So those instances will be there when you're on call. And then there are some instances as well, too, where you know, just because you have a lot of knowledge, because I feel you're very brilliant, and then I kind of see you ramping up very fast. So just because you have a lot of knowledge, just because you have a good skill set, you might be put up and say, Okay, I know you're not on call, but we kind of need your help to get a look at these things as well. And then you might see once your peer also start recognizing you as a very, very valuable team member, your peers, who might reach out to you on any questions they might have as well, too. So that's kind of like that day today, in laws for you.

Unknown Speaker 51:51

Thank you for sharing.

Speaker 2 51:53

Yeah, thank you for that information. I'd also like to know what I can expect, like what I was what's going to happen next? What are like the next steps.

Speaker 1 52:01

That's a very good question. I was going to touch upon that for this interview you passed, right? So I'll tell you that right away. I don't you know what the recruiter is going to give you feedback now, the next step for you is going to be a EUA Virginia, so we can bring you on set so after them to do virtual Now, if you're interested, in a coming on site. Is that something you're interested in.

Dinanka-myacuty-srv-first

Speaker 1 00:00

Been quite a day. I don't know how your day has been, but I hope it's been good.

Unknown Speaker 00:04

So

Speaker 1 00:07

I'm Val and guys with me is Matt Sloan, Matt Sloane will be doing most of the talking on our end, after a couple minute kind of overview from me, we're both directors with acuity. Specifically, acuity is national security practice, and Matt happens to be the solution architect for the team that we're considering you for. So in terms of the agenda for today, I'll start there. I'll probably spend two or three minutes giving you a really high level down into a really fast overview of the engagement, and then we'll pass it to you. Take a couple minutes and kind of provide us an overview of your background. Please tailor it to kind of the technologies that I'll briefly describe or briefly mention. And I only say this because I want to save as much time for you and Matt to kind of have a technical discussion, because after that, Matt will ask you a whole slew of technical questions and and that will probably take 20 or so minutes, and we'll leave a couple minutes at the end for any questions You might have said, If all is well and we're both in a good place at the end of this, the phase two interviews, a about a 3045, man coding challenge interview. It's done live via teams using a platform called hacker rank. So that's kind of the process. Brett larned, so you good with all that? I'm good, perfect. So the engagement we're continuing for is within DHS ice, so Department of Homeland Security, Immigration, customs enforcement, and within ice, this particular engagements within the Directorate known as HSI, or Homeland Security Investigations, think of it as like the investigative office within ice and primarily responsible, and frankly, primarily responsible across DHS for investigating A wide range of illegal activity spanning human trafficking, child exploitation, import, export crimes, financial crimes, weapons and narcotics smuggling, etc. So very broad based investigative authority and we are one of a number of integrators supporting HSI in the ongoing development of a cloud based analytics platform that's called Raven. You know, they're, they're a number of integrators working on various technical focus areas. I don't want to say anyone is particularly or totally siloed, but it's, it's a very expensive platform. Across across the enterprise are probably about 180 190 staff working on Raven. Our primary, primary focus, when you kind of divide up the work, is on the UX design, the UI development, and then the majority of the micro services development there. There are other integrators kind of more focused on the platform engineering and the data analytics side. That's but our focus UX design, UI development and microservices architecturally. Well, let me say the platform AWS and Kubernetes based architecturally, it's essentially built around a centralized Knowledge Graph, which open source, Massive Open Source graph database. And it's obviously heavily leverages AI ml capabilities, a lot of data transformation data processing capabilities currently leveraging Databricks and NiFi. I bring up because we do have people who touch NiFi and then data streaming with Kafka, but the majority of our focus is essentially on the on the applications and services that kind of surround that knowledge graph. And you know, at this in this context, probably speaking, we're talking a micro front end, federated, great GraphQL, micro services architecture. There are 25 to 30 applications. We'll just say 30, for the simplicity of math, all but one or web apps. There is a mobile application.

Unknown Speaker 04:41

It's a React shop. So

Speaker 1 04:44

React, React hooks and TypeScript for the web app, React Native for the mobile app. On the back end, we're talking Spring Boot. These are Spring Boot micro services and domain graph, service graph, QL API. Those are kind of core two technologies, along with, I would say, Postgres and Elastic Search, given where most teams are, you know, Helm, Docker, Eks come into play. And then again, like I said, we have other we have people who touch Kafka and NiFi. But in the end, I think for this discussion, the majority of Matt's questions will revolve around your Spring Boot experience. So at a high level, that's kind of the technology landscape, business landscape, all of our staff, some of the platform engineering and some of the analytics staff live in their own kind of distinct teams, but all of our staff are aligned to cross functional agile development teams. Most are between eight and 12, although there are a couple I think are maybe closer to the 14 mark, but we'll be each is led by an ice PM, and the agent, who's a pm and it'll be in the team is supported by, like I said, cross functional skill sets. There'll be a product owner, which is what we call our VA scrum master, UX designer, and a mix of kind of engineering skill sets with staff from multiple vendors, multiple integrator integrators. So on any given team, there may be staff from two or three companies, and that's normal. That's a common operating practice that we're used to. And then the government's from the government's vantage point, they they don't care what company people are from. They just want to see them work together. So that is very high level overview, down and dirty. I went longer than I wanted, but that's that's okay. Why don't you take a couple minutes, and again, we we've reviewed your resume, but why don't you kind of take a couple minutes and give us the Cliff Notes version of it, and then I'll pass it over to Matt.

Speaker 2 06:54

Yeah, sounds good. So hello. My name is bineka wagley. I've been working as, like a software engineer, mainly on like database like web applications, which basically utilizes, like spring, it utilizes hibernate, and also a lot of different like best web services for stuff like API developments being involved in, like microservices development, we like design all these different services. We like deploy that, develop them, and also we like deploy them to, like AWS, clouds, for example. So services, it basically includes, like, easy to CSS, SQL. All of them are like containerized using like Docker and on the front, like front end side of things. I mainly work with Angular, but I can work with like JavaScript and also like TypeScript as well. I am also involved with different like unit testings by using either like JV Nate or like mockers. And currently I'm in like this Scrum teams and this team, it mainly consists of like five developers, where there's like product owners, there's like a scrum master on like a daily basis, and I mostly work on the back end side of projects, basically using like Java code to help, like the teams with stuff like code reviews and also like production on a lot of different like support duties, for examples. So that's what I do, right?

Unknown Speaker 08:24

Perfect, down and dirty.

Speaker 3 08:28

All you. Thank you. Danaka, right, did I pronounce that correct? Okay, hey, I'm Matt song. Thanks for joining us today. So you've been full stack dev for about four years, five years now. Yeah, it's like four and a half. And why are you looking to leave USAA?

Speaker 2 08:51

Oh, so my contract with USAA is set to expire this February. So basically my project as a contractor is ending. Now, that's why I'm trying to, like, change into something else

Speaker 3 09:03

here. All right, yeah, okay. No, understand that just, just, was a little curious. Talk to me about what you understand to be the major concepts of object oriented programming.

Speaker 2 09:16

Okay, so what I understand is, so basically, there's a, there are, like this four like main concept. So in Java, like, you have encapsulations, which is used in, like data hiding. It's basically like hides, like the old class fields. So you have like, complete control over, like your fields or within like the class itself. After that, you now have inheritance, which is mainly about like, reusability of the whole code. So like outside class can basically, like, inherit, like the old behavior and also like fields that's been like, exposed by like a like a parent class. Then next you have extractions, which is about like hiding, like the whole complexity from the outsides. So we can basically achieve it with like the help of different like abstract methods, and also like different abstract classes as well as like interfaces as well. Then last one there is like polymorphizations, which can basically be achieved with the help of like method overriding which is like done in like the same class. And then there's also method like overridings, which is done with, like the side class there. So yeah,

Speaker 3 10:27

what is the difference between abstraction and interfaces? So

Speaker 2 10:33

from what I understand, basically a like an abstract class. It's kind of like this base class, right? It contains some like base functionalities along with different, like abstract methods. So it may contain some methods that has like implementing, and some method that may just be like abstract right, but interfaces they work with, like, like a blue blueprint, right? So it gives you what certain implementation needs to choose without giving any like kind of base fallback. In Java world, you can have only, like one abstract class that you can extend it from, but you can implement like multiple different, like interfaces. And then basically you can have like this constructor in like an abstract class, but interfaces, they will not really have constructed. All right.

Unknown Speaker 11:27

Spring relies heavily on IOC, right?

Speaker 3 11:31

What are the concepts of of IOC? What? How do you? How do you explain that when somebody, or if they were to ask you, or I'm asking,

Speaker 2 11:41

yeah. So, yeah. So, well, IOC is basically inversion of control, right? So basically it's like, responsible for like, delegating your like object creations, and also for all the different like dependency management to like an outside world, right? In case of like a spring it's done with like a spring containers. Basically the main concept here is like the independence injections, which is basically providing the different like objects that a class need through like a spring containers. And you can then do like dependency injections through different like construction, like settler or like fields IOC also take the case of other things, like, for example, like managing the whole scope of different like beans, or, like managing, like the beans in like overall sense. So yeah, these are, like, the primary concepts, yeah.

Speaker 3 12:37

All right. With, with spring JPA, I have my repository I can declare. Obviously, it's an interface, and it's going to pull data or update data, performs my CRUD operations, right? There's a number of ways that I can write my method or define my method to perform the queries. Do you know what the query methods are that are supported by by a JPA interface?

Speaker 2 13:14

So by like the like the query methods? Let me think so basically, you can have like this, like derived methods, which basically means that you can, like, follow a certain conviction, right? So let's say you have like this, Employee objects, which has like first name, last name, salary and etc. So Right? As you can write like a convex sense, like, for example, find by first names, or like find by last names, or like find by salaries greater than or so on. So basically, when it's like, automatically generates like this query. It works for like, very simple queries. But the other way is basically using the whole like query annotations in like a methods to basically, like, define any kind of like complex queries, like on it. So I think these are like the two ways. Okay,

Speaker 3 14:12

so you've got the derived queries, and then you have the query annotation where you provide us, and sometimes there's a second parameter in that query annotation that comes after my SQL statement. You know what that is and the significance of it. Can

Unknown Speaker 14:32

you please repeat the question, please? Yeah, so,

Speaker 3 14:35

so we talked to the derived query, find by ID, find by name. Very good. Right on with the complex query, where I have my query annotation and I write in my my what my query is. A second part of that query annotation is, is another parameter that I need to set. Do you know what that is? So

Speaker 2 15:00

the native it basically tells your SQL to basically, like, execute or draw different like SQL statements, like directly against, like the database.

Speaker 3 15:12

That's what I was looking for. Are you familiar with entity graphs?

Unknown Speaker 15:16

Did we say entity graphs?

Speaker 2 15:20

I don't think I've known too much about it. Yeah,

Speaker 3 15:23

sorry, just curious. It's a it's a different way of annotating a interface to when you're doing multiple joints and you create an entity graph to query back exactly what you want. Okay, I have a a service. I've got a RESTful endpoint, right? So I've got my controller spring, MVC. We're using JSON. So, you know, Jack's fees marshaling the payload and moves it into an object where I've got an employee, right? So it's got the name, it's got phone numbers and it's got email addresses. So so my my entity itself, the employee, has a one to many with phone and has a one to many with email. Now, for whatever reason, I can't leverage the, you know, the set, the the employee, and then do employee repository, dot save, you know, where would normally would save all the entities, right? Legacy system doesn't matter for this question. I have to define so I get my employee. I create my employee, I call employee repository saved, and then I have my phone, and I call phone repository Save, and now I'm going to save my email, calling the email repository, but something fails, right? Don't know what it is. It's runtime exception. Now I've got two records in two separate tables, my employee and my my phone right and and I've got missing data right, because it's not a complete set. How would you handle that? Using spring?

Speaker 2 17:19

Okay, so what I think you are mentioning is about like this transactions, basically spring can handle like transaction by using like transactional methods, by like, just annotating like your methods, or even like services as like transaction loads, you can basically like, specify like, what kind of assumptions You want to handles this transactions for so basically, for example, if you say, like transactional and rollback for like runtime exceptions. So in case of any like runtime exceptions within like this, like particular methods, the transactions will be rolled back so that you don't have any like missing details as part of, like one, like complete commit, right? Because this is one big consistency issue across, like,

Unknown Speaker 18:10

moving on to spring security.

Unknown Speaker 18:13

What is OAuth?

Speaker 2 18:15

So OAuth in, like spring security, especially, what we do is basically like a security measures where you have this, like open standards for basically authorizing like your user, right? So it's not like authentications, but just by using like this third party tool for authorization instead. So basically what you have is like this resource servers, or like the service owners, which basically grants you access, and then you have this like client applications, which is like requesting that access. And then you have this like authorization to basically conserves, which is basically issues like a token. And then on the API that you want the data it basically validates, is it basically validates is against, like, the whole access token.

Speaker 3 19:08

What's the difference between OAuth 1.0, and OAuth 2.0,

Speaker 2 19:15

so from, like, my experience, but I know the basic difference between OAuth one and two is that what one can besides like the version difference being like just better with some fixes and stuff, one basically uses like user like signatures, especially when it comes to like cryptographs, but or two, it actually uses like tokens. And I think what like two, it makes it a lot more simpler, where one was like a much more much like difficult for us. I don't have really much experience with like what one, but I do know of it conceptually.

Speaker 3 19:57

Yeah, no good. I'm glad you're not working well. Okay, so let's, let's talk about JWT. Then, how would, how does that if I were to implement a spring security solution leveraging JWT, how would that work? Can you describe that process? Mm, okay,

Speaker 2 20:24

so, yeah. So basically what you have in like JWT, just like this JSON web token, right? It contains stuff, like your signatures, your headers and like a payload, right? So basically the user, it locks like, locks in with like their credentials, and then you basically get authenticated. And based on that authentication like tokens is like a generated, like a token is generated. So basically this like that JWT tokens, it gets, like, injected into your authorizers and headers and any like substantial calls get like authorized header added to it. So basically it does like this, like signature validations, and then the validators, like validity of this signatures, it's checked with, like the help of different headers. And then once like that is done, your like payload, it gets used to basically extract like the rules and based on all those different rules that you have, like on your payload, a payload certain APIs are like allowed for this user to like view. So, yeah, you know, so you can basically check for all this like validity of these tokens based on like, some kind of like secret key. Yeah.

Speaker 3 21:43

What other spring libraries are you familiar with? Can you use spring batch?

Speaker 2 21:50

The spring batch? So I have not like extensively use spring batch, but I do know that is used for this, like last data processing together, is I've actually worked with spring like acculator for like monitorings. I've used like Spring Cloud for gateway, and I also use like spring oil for like this cross cutting concerns, especially for like login.

Unknown Speaker 22:18

So you've done AOP programming. I

Unknown Speaker 22:20

mean, I've done some part of it, some of

Unknown Speaker 22:23

it. Okay, how about

Unknown Speaker 22:30

Java reflection? Have you done any of that reflection

Speaker 2 22:36

in your program? We are actually, like, currently, like, primarily moving away from Java reflections, because it's not really like a good standard to basically go in and like, try to modify different classes, especially like during like run times. But sometimes when using like DNA for like test reflections to basically set the fields and like stuff like that, we kind of use it like that,

Speaker 3 23:02

yeah. How about reactive programming? Have you done anything like that using web flux or anything? I think springs got another one, spring reactive or something?

Speaker 2 23:15

Yeah, I've worked with like web clients, which basically works with like mono and flux for all the different like reactive programmings. So

Speaker 3 23:22

explain the difference between mono and flux forming? So

Speaker 2 23:26

basically, a mono is like this single data that it may exist or it may not, and then flux is like this, like stream of data.

Speaker 3 23:37

And what is the hardest thing to wrap your head around, because it's a different paradigm of programming and and I see new developers who aren't familiar with this hit, hit this thing almost immediately. Do you know what, what I'm talking about, and how to get around it? So, so if you were to introduce a junior dev to web flux and mono and flux, and they're going to go and start reactive programming, which is a shift from what I would refer to as traditional programming, right? What's like the number one thing that they're going to hit almost immediately when they're trying to break their controller using web flux. Okay,

Speaker 2 24:21

so I think a lot of this, like we said, developers, they will make that calls and then they'll, like, block this calls after like the operations to get like data right away, which is basically like an anti patterns, because, like reactive programming, it's usually for like this subscription models, rather than, like, dropping the call. Yeah,

Unknown Speaker 24:45

those are all the questions I have.

Unknown Speaker 24:50

Do you have any questions for me?

Speaker 2 24:58

So, okay, so thank you. First of all, like, thank you very much for all the questions today. And the first question I would like to ask is, what do the engineers, like, get involved and, like, let that was what like, day to day activities, like, what are like the team dynamics? And, you know, yeah,

Unknown Speaker 25:14

I'm sorry, finished, please. And you

Speaker 2 25:17

know, like, do developers, do they get involved in, like any, like, high level decisions within, like your architects, or is it like purely governmental?

Speaker 3 25:27

So I will say that the team dynamics vary from team to team. So, and I couldn't say which exact team you'd be going to, but we are the day to day. Typically teams in the morning would be a DSU. You know, here's what I did yesterday, here's what I'm going to work on today. Here's what my blockers and then go about their tickets. It's a two week sprint. You know? You'll do the backlog grooming. You'll understand and interact with the pm to understand what the upcoming priorities are. We've got a lot getting thrown at us with recent events. So you know, really, really under the gun. We I strongly encourage my developers to be vocal if you see something that isn't being done right, fix it. I'd rather be done right? Maybe take a little bit longer. So would our customer right? The other thing is, is we all come from different backgrounds, different experiences. If you have encountered something or have an idea, I expect you to speak up and raise it. Everything's on the table, if we can improve performance, if we can improve delivery, if we can bring new functionality that maybe we hadn't considered, if you're holding back and not not raising it up, you're not, you're not doing the job, in my opinion, right? So, so I want, I want people to have the confidence to to be able to do this right, to raise this up, not saying it'll it'll actually happen. You know that there may be other things happening that will say, now we can't really do it, but, thank you. But, but certainly, I look to my developers to to problem solve and and to look for options, and not just sit there and hammer, Hammer out code.

Speaker 2 27:25

So my next question would be, what's the next step that will happen after like this interview.

Speaker 3 27:32

So the next step is Matt and I will compare notes.

Dipesh Shrestha- Infosys- Final- SRV

Speaker 1 00:00

Also, you know, successfully migrated includes, you know, the maintenance of API. You know, it is used to make, you know, changes to the user accounts, right? And entitlements, right? So I have, you know, experience with, you know, grabbing, you know, different entitlements that you know users are, you know, allowed to view on the funding side. So I have been, you know, primarily working with Java in the back end and, you know, also in Angular in the front end side, and talking about the, you know, cloud services, it is primarily AWS. So, you know, our application is deployed on AWS ECS, you know, lambdas, and you know, as three buckets, and you know, Amazon RDS, so and for database, you know, on a daily basis, I have been, you know, mostly working on, you know, Oracle and you know MongoDB. And you know, apart from that, I have been involved in, you know, code reviews and also products and support.

Speaker 2 01:10

Okay, so you mentioned you've deployed it on ECS, correct? So, so you're using account.

Unknown Speaker 01:20

I'm sorry, no. So

Speaker 2 01:22

my question is, are you using some kind of a deployment pipeline for that? Or how is your deployment process?

Speaker 1 01:27

So basically, you know, we have, you know, internal pipeline that, you know, we use Jenkins, you know, that calls our cloud from is on templates, and all we do, basically, is, you know, to create the orchestration, you know, which is the, you know, combination of different properties that you know we need to pass to, you know, provision, the tags and the services for ECS.

Speaker 2 01:55

And is there any scaling, or anything that you've implemented in the current work that you're

Speaker 1 02:00

doing? Yeah, in terms of, you know, scaling. So, yeah, we have, you know, scale based on the you know, incoming request. So we, you know, have properties, say, you know, to look into the, you know, CPU percentage, and also the you know, memory and, you know, we, we scale based on that. So we all, you know, take, you know, consider all this thing. So in case, if the you know Max CPU users exit, you know about, you know 55 percentages, then you know, we scale the number of the instance by, you know five. And the you know the memory Max can be, you know, 40 percentage. And then you know, we scale by, you know five. Again.

Unknown Speaker 02:47

Okay,

Speaker 2 02:50

so anything, maybe we can begin now question two, yeah, it's more basic for

Unknown Speaker 02:59

Java and Java contents.

Speaker 1 03:02

Your course is a little, you know, small. Can you not able to hear me now, yes,

Unknown Speaker 03:09

but it's better. Now. It's better now.

Speaker 3 03:14

Can you explain me the difference between query param versus path para annotations in Spring Boot,

Speaker 1 03:20

yeah, in springboard. So basically, you know, the querying, your query pattern is used to, you know, read the request parameter, right? And basically, you know, it is used to, you know, read any parameter that is followed by the you know question mark, and it is in the you know format of the key values. So if you want to, you know, pass in a query parameter. So you will have to, you know, have a question mark. And then, let's say, you know, if you want to do a query pattern name, you have a you know, question mark, and that name equals to, you know, deepest so, you know, we can read a query parameter name using, you know the query parameters, but you know the path parameter is of, you know, part of the URL, right? So for an example, you know, if you want to get a user by an ID, then it is, you know, usually have a, you know, slash users, and you know that one that you know one is our inner path. In a path, no parameter, right?

Unknown Speaker 04:26

How do you optimize performance when we have multiple calls?

Speaker 1 04:31

Um, so, basically, you know, optimizing, you know, the performance it, you know, takes a, you know, lot of analysis, right? So, if I, you know, if, basically, you know, you may need to, you know, depend on the processing, right? So if you have many multiple requests coming in in a single call, and you know, you'll need to, you know, do some kind of processing, you know, you know, a later time, you know, you can incorporate some kind, you know, kind of batch processing, but also, you know, something like that. You know, if there is a, you know, multiple calls, then you know, you can, you know, compatible with the future, in order to, you know, get concurrency and parallelism in the place. So, basically, you know, the those data which are, you know, frequently, you know, use, which can be, you know, in CAS memory as well. So you may be able to, you know, use the load balancing from the AWS, you know, in order to distribute the load across multiple instances. So you might be, you know, able to live with the pagination as well. So this will help, you know, a lot of you know, I mean, divide the data that comes from the you know request. So it will help to, you know, you know, decrease the load altogether. And also, also, you don't seen, you know, optimizing the performance of APIs

Speaker 3 06:02

and what layer we need to deal with data validation in a Java application.

Speaker 1 06:08

So talking about the, you know, data validation we, you know, basically, would want to, you know, catch it early on. So basically, you know, it is usually handle in the controller layer of the web application. You know that we're building. So this is, you know where, like, you know you are checking the required field, and you know you are formatting the table. You know if it's done well, so you know you can, you know, do that in the controller earlier, so that you know it sometimes there might be a complex business logic, or the business rules, you know, that needs to be validated early. So making sure of all these things, or you know that the data complies with the system requirement or not, this helps us in, you know, processing this business, you know, really related to all this kind of thing. So basically, you know, the validation is done in the service layer, but the other validation for proper input is done in the controller layer as well.

Speaker 3 07:09

Okay. Difference between dynamic binding and static binding,

Speaker 1 07:16

if I have to, you know, talk about the, you know, differences between dynamic binding and static binding, you mentioned static binding, right? Yeah, okay. So if I have to you know, talk about that. Basically. The key difference is that you know the you know the how the matters are resolved, right? So whenever you know you are using the static binding, it is called early binding. And you know, they are resolved during the compile time, right? So basically, whenever you want to, you know, have a private or the fine and you know static matters, you know it can be overridden. And these matters, you know are, you know, are bound to the matter definition based on the type of the reference which you provide you know at the at the beginning, right? So, but you have a you know dynamic binding, which is you know method called, and that are you know result during the runtime, right? So whenever you have you know overwritten method in case of inheritance, these methods you know are called you know, and are you know, are called bound up, you know, and based on the you know, actual object type. So that's what you know. Dynamic binding is

Speaker 3 08:32

okay. Like, how do you perform credit operations in a database with REST API service?

Speaker 1 08:39

I'm sorry, can we, can we do that again? How do you Okay, product operations, okay, yeah, basically, you know, you have, you know, four different requests, right? So in the product operations, basically, you know, I'm using Spring Data, JPA, so each of the operation can be, you know, mapped, you know, directly to our STD people. So you know, there is a you know get request method. This method particularly means, you know, retrieving the resource from our database and you know, or an external source. So whenever you know, I'm making a get request using the you know, gate mapping in the controller, it will, you know, be accessing, accessing the you know data, which is reading the data. And this is, you know, you know, reading from the R, from the you know card. So similarly, you know, when you have a, you know, create operation, which is usually, you know, down to the post operation. So whenever, you know, you want a post controller or a post mapping on the controller, it is, you know, used to, you know, create something you know, in a database or the you know server, and then you have something called, you know, update, which is done to the boot, you know, mapping annotation. And then, you know, we have something called delete, which is, you know, which is done, you know, for the, you know, deleting, deleting something in the database of the server, using delete annotation, delete mapping, annotation. So, you know, if you are working with Spring Data, JPA, then you know, basically you can leverage this, you know, you know, like say, you know, delete, find by you know, Id and you know, also update you know method accordingly. So to perform, you know, all these crowd operations, you know, from the JPA repository that we are, you know, Springboard provides,

Speaker 3 10:42

what is Docker image? How to customize Docker template,

Speaker 1 10:47

yeah. So, you know, talking about the Docker basically. So Docker is like, you know, machine, right? So it's basically a container image, you know, that makes it, you know, lightweight and standalone software. So the you know the advantages it, it eliminates the you know need of the machine dependency, right? So you know if sometimes you know software might run on one computer, but you know it might not work on other computer, right? So you know if, in order to, you know, make things, you know, better. And you know, running the other, you know, all the computer so it needs to run on the software. And, you know, Docker helps to resolve this. So basically, you know, it just includes everything that is, you know, needed to run a piece of, you know, software, or the project that we are doing, right? So including, you know, what code it has. And you know, what are the, you know, libraries it needs. And also, you know, all the environment variables. Also, you know it is right. You know, customize our, you know, Docker emails. So all you need to do is, you know you need to create a Docker file so it will have some, you know, simple text file telling the doctor you know what to do and what to run, right? So you will have to, you know, have a, you know, base emails, you know we are, which is, you know what you are walking directory is, and you know where all the, you know, kind of resources leave, and you know what commands you need to, you know, run in order to run the application. So this is all you know. All about Docker.

Unknown Speaker 12:24

What are generics in

Speaker 1 12:27

Java? Generics, so in, you know, basically generics are, you know, it basically, you know, allows you to create a class or, you know, interface you know that can operate on various, you know, types of object, objects like, you know, while you know, providing a compile time, you know safety also, you know, if you like to have, you know, a list, then you know you have to, you know, you can have a list of integers. You know, you can also, you know, have a list of strings. So basically, you know, or you know, you can have list of employees object, you know, or any kind of object, right? So basically, you know, that's what generate are, you know. So you can create reusable code of you know, components that work with, you know, any object. And you know, this basically reduces the chance of, you know, class cars, you know, exceptions as well.

Unknown Speaker 13:23

What is wild card in Java?

Speaker 1 13:26

I'm talking about the, you know, wild card, basically, it, you know, it allows, you know, to, you know, wild card, like a parameter, it allows to, you know, have a, you know, generic screen or represent on, you know, unknown type of you know, right? So it allows you to, you know, write a little bit of, you know, flexible and reusable code, basically, you know, it indicates you know that the method can work with the range of types. So not only a single, but range

Unknown Speaker 14:00

difference between concurrent dash map and dash map.

Speaker 1 14:05

Um, yeah, so, you know, talking about the differences between, you know, concurrent has map and basically the hazmat is that, you know, concurrent hazmat is, you know, a traitor. So basically, you know, concurrent has maps, you know, provides internal, you know, synchronization. So if you know, you are working on a multi threaded environment, and if you need to, you know, make sure the data is consistent across, you know, multiple threads you will use to, you know, the contrain has map. So basically, you know, if you use a hazmat then, you know, there will be some kind of data, you know, inconsistency.

Speaker 3 14:49

So you have worked on Zookeeper, right? What is Zookeeper, and how did you use it in your project?

Speaker 1 14:56

Um, so, talking about the, you know, Zookeeper, basically, you know, it's a Kafka server, right? So, whenever you know, you you know, have a Kafka and, you know, if you need, you know, need to coordinate, you know, of a service that you know, manages, you know, all the Kafka brokers, right? So it, you know, basically uses some kind of, you know, structure to, you know, manage, you know, data and you know, other related things in the file system. So it is, you know, very essential aspect of that Kafka, until, you know, Kafka 3.0 right? So we have been using zookeeper to, basically, you know, work for our Kafka server. But I think it's, you know, slowly going away is, you know, Kafka is migrating towards, you know, more modern Kafka, 4.0 and that, you know, eliminates the need of Zookeeper. So it is just, you know, used for managing the Kafka brokers.

Speaker 3 16:01

What is rebalancing term referring to when dealing with Kafka, consumer publications?

Unknown Speaker 16:09

So

Speaker 1 16:12

if I have to, you know, talk about the rebalancing term, you know, basically you know it is whenever you know it is used whenever you have a, you know, you know, distributing partition, right? So basically, you might want to redistribute those partitions among various, you know, consumers, or, you know, consumer group, right? So, you know, consumer group is, you know, just a, basically, you know, a group of consumer so rebalancing, just, you know, it redistributes the partition among those available consumers. So, basically, you know, whenever there is a chance of new consumer joining the consumer group, or, you know, consumer exiting the consumer group, or, you know, also, there might be instance, whenever you know, there is a number of partitions, you know, like, like a topic, you know, changing. Then you know, rebalancing occurs. And it ensures you know that each partitions is, you know, assigned to only one consumer in a group, but not particularly at a time, right? So, yeah, this is all about, you know,

Unknown Speaker 17:20

rebalancing,

Unknown Speaker 17:24

okay, so, one

Unknown Speaker 17:27

question, so, so, because you've worked on micro services,

Speaker 1 17:29

right? Yes, micro services, yeah, I have worked on, okay,

Speaker 2 17:33

so if one of the micro service is supposed to handle payment information, okay, how would you design that service. What will be the key aspects of designing payment information passing through the service?

Speaker 1 17:46

Okay? So if I, you know, have to design a, you know, micro service architecture for a, you know, you know, payment processing, basically, I will, you know, you know, like I would like to, you know, whenever I'm dealing with, you know, anything that has a payment right? So it basically, you know, needs to be carefully considered about the, you know, basically, reliability, security, and also, you know, the integration, right? So basically, you know, we might have to, you know, think of the multiple payment method, right? So, you know, I mean our payment method in the first thing, it might, you know, be a, you know, credit, or, you know, a David, or, you know, it might be, you know, like an external payment service, like, you know, PayPal, or other payment apps on the, you know, portal. And you may also, you know, have to, you know, think of the other banks transfer, right? So, basically, you know, we must have that, you know, ability to handle all these kind of payment processing because, you know, user might need it. And, you know, and sometimes you know, as per the requirement, we may also, you know, need to handle the multiple, you know, currencies, right? I mean, there might be, you know, a transaction in different you know, no,

Speaker 2 19:02

that's more around, that's more around the payment, the payment options, okay, what I'm essentially trying to understand is from so you have, suppose I'm saying you're only handling the credit card. You have credit card information that's been sent to your service. Now you want to process it, right? So how do you process it? First of all, how would you expect the data to be flowing? And then how would you process it? The payment, or the payment information as such?

Speaker 1 19:28

Okay, so if you know, basically, thank you for clarifying. You know out that, so I mean questions. So basically, you know, if I have given a credit card and it's the, if it's the only form of, you know, payment, then basically, you know, we will have a you know, payment micro service where, you know, a client initiates, you know, a payment by, you know, sending a request to the payment micro service. And this micro service, you know, will be behind an API gateway. So we will have an API gateway, right? So, basically, so

Speaker 2 20:03

what is the, what is the function of an API gateway? So, I mean, just, just curious, so you said an API gateway, right? So, so micro services are behind an API gateway. Is it right now, this part of time, so exposed it via an API gateway,

Speaker 1 20:16

yeah. So if I, you know, have to talk about the, you know API gateway. It basically manages the you know request handle, you know, rate limiting, you know performance authentication, and you know, also, you know, authorizing the you know request. And it also, you know, handles the load balancing, and you know it basically, you know, make sure that you know request is valid. And also, you know it, you know, basically works on your, you know, orchestrating, you know, at your, you know, various different you know.

Speaker 2 20:49

Okay, so let the API gateway be so we can talk about the payment aspect, right? So you're talking about payment aspect. So through the API gateway, the data would go through next. How would you process that?

Speaker 1 20:58

Yeah, so, no reasoning, whatever, you know, I said you know earlier. So then you know, you will have, you know, both. I mean, basically the payment service, which will be, you know, responsible for, you know, validating the incoming request right, to ensure that, you know, all the required fields are present and are incorrect. You know format. So this basically, you know, helps you validate the fields that you are, you know, getting on your request, like, as I mentioned earlier, like them, you know, you know parameter, you know investors, or you know amounts, or you know. So

Speaker 2 21:38

one question is, when you're saying validating, you say, have a credit card information with you, correct. You are validating that credit card information against some kind of a validation check that you may have. It can be against a rental database or a central list of credit cards, or maybe a credit agency. You can do it anywhere. So you have received that information now you want to process it correct. Your microservice is going to process that information. How would you go about doing that from a security perspective, you just mentioned right? See, reliability is one aspect of it, but I'm talking more around the security aspect, because payment information is a secure information

Speaker 1 22:18

that's correct. You know, I understood, you know, your requirement. I, you know, basically in that case. So whenever you know we are dealing with payments, I, you know, mentioned that reliability and, you know, also security, other, you know, two main factors, right? So, so we will, you know, basically, make sure that you know our data initially, it is, you know, transmitted through the HTTPS protocol and and then basically, you know, we will be using, you know, tokenization process, right? So, you know, instead of, you know, storing our, you know, processing the, you know, actual credit card numbers, we will, you know, use the payment gateway. Basically, you know that gateway provides, you know, some kind of tokenization mechanism for all the you know, credit card details that we are getting, right. So basically, you know, we will be generating a unique token that, you know, uniquely represents this card information, and you know, only store, you know, the tokens in our database, instead of actual credit card numbers. So this basically, you know, minimizes the rates of, you know, exposure. So, you know, we will also, you know, make sure we have proper PCI environment, you know, as as in, you know, payment, you know, card industry. So

Speaker 2 23:36

do you really need to, really need to store the credit card information, even as a tokenization information, do you need to store it, or can you just process it and remove the information? This storage also one of the criteria that you want

Speaker 1 23:49

to do, basically, you know, I guess you know some, you know it might be needed for, you know, for some process, right? But you know, if your application is just, you know, responsible for initiating, you know, the payment, we might not need to, but you know, I would say securing tokens is, you know, usually, generally, basically, a safer, you know, method i

Speaker 2 24:23

Okay, so you mentioned a loosely coupled and tightly coupled monolithic and the loosely coupled architecture, right? That that's the reason. So you mentioned the API service that you have designed, or maybe the micro service that you've designed, correct? Do you know an example where, in maybe a monolithic application is better compared to a micro service architecture or loosely coupled architecture? Or do you feel that all monolithic, it's time that we move to a loosely coupled architecture? What is your opinion?

Speaker 1 25:00

Yeah, that's, you know, that's, you know, basically, you know, micro service has a lot of benefits, you know, over basically the monolithic applications or architecture, right? But you know, whenever you know, you have a very small or, you know, medium sized application, like, you know, you know, like, very simple shopping website with, like, you know, basic functionality, like, you know, a catalog, or, you know, a shopping cart, you know, and also a payment processing then I believe, you know, monolithic architecture is fine because, you know, it is often easier to, you know, deploy, you Know, and maintain the application, because, you know, the overhead of, you know, managing these services is pretty, you know, small. So if you, you know, have this kind of, you know, small to, you know, medium sized applications, then you know, I, you know, I believe, you know, handling it with a as a, you know, monolithic architecture will be more beneficial with, you know, a single code base, and also, you know it simplifies our development. You know also our deployment, and you know communication aspect you know as well, if you know, in the you know micro service world, I believe you know micro service have you know various advantages over you know each services being, you know, independently developed, deployed, you know, and also maintain. So these are only so

Speaker 2 26:26

are you so? Are you essentially saying that the decision for going with the microservice would be more dependent on the size of the application or the functionality, because, see the functionality that you just mentioned, right? Maybe like a end to end shopping, kind of a flow that itself is also a good candidate for service based architecture, because all of the components can be componentized, and then you can have those services built. And it is not necessarily that only that application. When you build micro services, the core reason is you can also extend them onto other applications in your entire hierarchy, or your entire pool that can piggyback onto them, say, for example, a payment service, correct, a payment Microsoft service can be a standalone services or other standalone service and other applications can also use it doesn't necessarily have to be only for the standalone application, Correct, right? So then, I mean, do you think that size is the only criteria, or are there other criteria also?

Speaker 1 27:25

No, I'm not, you know, necessarily you know, saying that there are other criteria as well. I was just, you know, giving an example. So I'm not, you know, necessarily you know, taking, saying that you know only the size, but you know you need to, you know, make sure that you know, you do take into account, like, you know, scalability, you know, in picture. So basically, you know, if you have a, you know, certain components you know that needs a you know, little you know, fine grained scaling on your you know, particular application, basically, in this case, micro service, you know will be a better approach, because you know you can deploy, and you know, scale each of them, you know each of the service independently, right, based on the you know demand. What is, you know, the demand in our application. So, if you like, you know a different workflow that you know may benefit, you know, by different, you know technology stack, and also, you know, have like, different parts of our application deployed on, you know, a various different infrastructure. You know, you can choose micro services, you know, like this. You know properly, you know, handles, you know, isolation of you know, each services and you know, no, basically, you know they may need to be, you know, handle certain aspect of the flow. Then you then also, you know, you will also, again, you know, you will have a micro services, or, you know, better approach, but, but, yeah, you know, like, you know, there's a lot of things that needs to be, you know, taken into consideration. It's not just the size. Because, you know, micro service, you know, are, you know, designed to, you know, accommodate all the you know changes and you know new features more easily because you know new sheet, you know new services. It can be, you know, added, or you know, you know, or you know, edited, or you know, changed without, you know, affecting the existing functionality, right? So, because you know, on some you know, sometimes on, you know, monolithic, you know, application. Whenever you know you want to grow or, you know, add a new feature, it may cause issues. So, you know, basically, with the existing functionality that we have in our application, okay,

Unknown Speaker 29:39

you had something right go on. Right?

Unknown Speaker 29:50

So

Speaker 3 29:53

like, how do you monitor Kafka application for producing and consuming messages?

Speaker 1 30:00

Um, in terms of monitoring, you know, you know, Kafka. So, you know, like, basically, you know, I know, I will try to, you know, use some Kafka metrics. Because, you know, Kafka exposes, you know, a lot of metrics through JMX. So basically, you know, we can do, you know, rate of record, you know, send per second, or you know rate of error when, you know, sending these records. And also, you know, look at, you know, the latency, you know, on the consumer side, right? So we can, basically, you know, look at the record, you know that that is consumed, you know, per second. So, you know, this, this, you know, this is what, you know, we can, like, we can also, you know, look at the log, right. Basically, you know, the logs that are created, and you also the latency. So we can also, you know, set up, you know, protimors in order to, you know, get the monitoring, you know, through, I mean, more visualized format. Basically, you know, we use, you know, Grafana, and it allows, you know, you to create, like you know, various custom dashboards, like you know, for all our team, you know, and also, you know, we can basically restructure various, you know, logging for various our producer as well as you know, consumers. So you know, in order to, you know, log, you know, like, in any important event, you know, such an ears or, you know, like, processing times or so on,

Speaker 3 31:32

okay, how do you enhance, like, the performance of a database while querying data columns From a table, if it contains

Unknown Speaker 31:41

a few distinct records out of several 1000 rows. Okay? So,

Speaker 1 31:49

basically, you know, if, if I have to, you know, think about that. We, you know, I usually, you know, optimize the performance of the database. So there has been an instant, you know, a lot of times it is, you know, using, you know, appropriate, you know, basically, if we use an appropriate indexing, this will help us a lot, because, you know, whenever you implement indexing on you know, of frequently query column. So let's suppose we know, we query a lot of times to a particular column, then we, you know, we can use the indexing. So basically, it is piece of the, you know, data retrieval. So basically, you know, for distinct records, like, you know, there are different records we can, you know, basically choose a, you know, basically a well chosen index. And you know, if you know, we can also, you know, add, like, unique indexes based on, you know, those unit uniqueness that we have in our you know, database we can sometimes also use, like a caching your layered, you know, to store, you know, frequently access data in in our you know, memory. So basically, you know, we have to, you know that data access layer in our application, right? So sometime, let's suppose, you know, if there is a combination of, you know, a lot of tables you know that you need to, you know, do you know, a workout, or, you know, you may be able to create, you know, like, you know, a temporary view to, you know, access some kind of data. So, yeah, you know, those are the, you know, some of the ways, in order to, you know, optimize the database, you know, you know, reachable through the queries.

Speaker 3 33:31

What is Elastic Search, and have you used it in your project?

Unknown Speaker 33:35

I'm sorry. Can you, can you repeat that

Speaker 1 33:39

again? Search, okay, yeah, so, yeah. So, basically, I haven't, you know, worked with elastic shirts directly, but you know, I know what it does so, and I know what it is. So basically, you know, it is a restful search for, you know, an analytic, you know engine. So, basically, you know, it supports, like a, you know, full full text search, right?

Speaker 2 34:19

Okay, thanks a lot for, you know, attending this one. We'll discuss some more ourselves, and we'll get back with the feedback. Okay, sure, sure.

Speaker 1 34:27

Thank you for, you know, giving me this opportunity to, you know, like, attend this interview. I really you know, like, the way we know we interacted. Thank you for you know your time, and I'm looking forward for that. You know your response. Thank you. Okay thank you. Okay.

Unknown Speaker 34:45

Thank you.

Dipesh Shrestha- JPMC- Second- SRV

Unknown Speaker 00:06

Hey, morning,

Unknown Speaker 00:08

morning. How are you sorry? Good afternoon.

Speaker 1 00:13

How's your day shipping? I'm sorry. How's your day shaping up? Can you hear me?

Speaker 2 00:18

Yeah, I can. I can hear you, yeah. It's been good. Thanks for asking. What about you?

Unknown Speaker 00:24

Pretty good. Happy Friday. Just few more hours.

Unknown Speaker 00:26

Yeah, give me one minute.

Speaker 1 00:30

I want to wear the headphones so that you can hear it better. Sure you

Unknown Speaker 00:40

can you hear me now? Yes, that's better. Can

Speaker 1 00:58

you hear me? Yeah, I can hear you. Can you hear me? Okay, perfect, perfect. All right, so yeah, Deepesh, thanks for joining today. Yeah, so I have your profile and I'll go briefly on with us, like I'm from JP Morgan, Chase safe, current performance team. I'm a lead software engineer. We'll have another interviewer with us here today, so we're going to quickly go over your background. First, what is your experience so far, what you have achieved, and what are your key strengths? If you can give a brief of your

Unknown Speaker 01:33

professional career, that would be great.

Speaker 2 01:38

Yeah. So as you know, my name is deepest resta. So I have been, you know, working as a full stack software developer and Java best application for almost around, you know, five years now. And you know, I have been utilizing various frameworks like, you know, spring Hibernate, you know, Spring Data, JPA, and you know, various developer, various RESTful web services for, you know, API development. And recently I have been, you know, involved in a lot of microservices work where I have, you know, worked on, you know, either, you know, creating new microservices from scratch, or, you know, working on the existing microservices. So, yeah, almost, you know, primarily I have used springboard for developing these microservices. And you know, I have been involved in, you know, like design of these microservices, collecting, you know, functional requirements, documenting them, and also, you know, implementing them as well. So all of these, you know, that I have been working on are deployed on AWS cloud, and I have been, you know, working on, you know, to establish services like, you know, AWS, CC, two, s3, ECS, SQS, SNS, and, you know, develop innovators, lambdas function as well. So, you know, I have also worked on Angular and the front end side. So mostly I have worked on Angular, and I can also, you know, work with JavaScript, TypeScript, and for the testing I have, you know, utilized Z unit and monkey tool, you know. And I'm on a, you know, Scrum team. And my team, generally, you know, consists of five developers, product owners, and, you know, Scrum Masters. And on a regular basis, we are mostly, you know, working on back end system using, you know, utilizing Java code. I mostly write Java code so and also, you know, I help team with the code reviews. And also, you know, some products and software reviews,

Speaker 1 03:40

that's good. Yeah, thanks for walking me through. You mentioned you have worked with Java, right, and AWS prime predominantly on the cloud infrastructure. And you mentioned you have used three bucket and lambda serverless architecture as well, right? Okay, tell me one thing. Under what circumstances do you think that lambda circle, serverless architecture, beats any other containerized application?

Speaker 2 04:09

Okay, yeah, so, you know, depending on basically, basically the, you know, lambdas are, you know, like an event based serverless, you know, architecture. So basically, you know, if you know, if you want to perform, you know, some kind of operations, you know, as a, you know, as a result of, you know, some, some kind of events, you know, like, you know, lambda is our top notch, you know, something that we will be utilizing. You don't have to, you know, worry about, you know, provisioning these servers. So, basically, you know, AWS take cares of, you know, running this kind of course somewhere. So maybe, you know, if you want to, you know, do some kind of, you know, real time file processing, or, you know, whenever you want to, you know, upload a file into as three bucket, and, you know, so lambda is, you know, get triggered when all these kind of, you know, events are, you know, awkward, and you know, in order to normalize them. And maybe, you know, there's some process, or, you know, some kind of, you know, use case, or maybe you know if some kind of messages are, you know, in an SQS queue. And then you know, if you know you need to take some messages, or, you know, messages in its way, and then, you know, sending out to some, you know, other, you know, downstream services. Then you know, AWS, lambdas does a really good job. So, you know, apart from this, you know, it's a shortly process that is, you know, a stateless and basically, you know you need to run it on a full scale. And then, basically, you know, lambda will not, you know, be it by any other applications.

Unknown Speaker 05:42

Okay, thank you. Hey, we have Mukta. Yeah.

Unknown Speaker 05:45

Hey, so,

Speaker 3 05:49

yeah, yeah, you guys can continue. Sorry, I tried to install it. Yeah. I don't know if you had introduction or not, but yeah, we went

Speaker 1 05:57

to, yeah the best. Would you mind giving a brief introduction from,

Speaker 2 06:01

yeah, sure. So I can do that again. So here my name, as I know, my name is divest, resta. So I have been working as a, you know, full stack double developer for almost, you know, five to six years now. And you know, I have been primarily working on double dash web application utilizing various, you know, Spring, Spring Boot, Hibernate. And, you know, I have been developing various RESTful web services, you know, for API development. And you know, recently I have been, you know, developing various microservices work where I have been, you know, working on from scratch for microservice architecture. And also, you know, existing microservice architecture. So primarily, you know, I have been involved in, you know, designing the microservices, collecting various functional and non functional requirements, documenting them, and also, you know, implementing them as well. So, you know, all these applications that have been, you know, working on are basically, as I mentioned, deployed on AWS cloud. And few services on AWS that I have been, you know, working on are, you know, EC two, ECS, SQS, SNS, yes, three lambdas. So yeah, pretty much that on the back end. And for the, you know, front end, I have been working primarily with Angular. And for this thing, I have been, you know, utilizing widely with J unit and Mockito. And if I have to, you know, talk about my current team members. There are five members in my team. So we have, you know, product owners, Scrum Masters, developers, and we, you know, basically, you know, jump up on a meeting on a regular basis. And I primarily, you know, as I mentioned, I work on back end using double code. And you know, I also help my team with, you know, production support, and also, you know, with code reviews and all,

Unknown Speaker 07:56

okay, yep, thank you. Thanks for the introduction.

Speaker 1 07:58

Yeah. So, so, deep is going back to your question that I asked, right? I mean, under what circumstances we would think of using leverage labs or serverless? So tell me one thing you mentioned, stateless, right? Is security group a stateless or stateful? Like, if you're, if I'm to leverage its security group, you can tell me, like, what security group, health like, and what is the status of that like? Is it a stateless or stateful? So in terms of lambda,

Speaker 2 08:28

okay, so in terms of lambda, basically, you know, security groups will be basically like a, you know, virtual firewall, right? So it controls, you know, any inbound or outbound traffic. But you know, the security groups are, you know, stateful, right? So, meaning, you know, it allows you to, you know, incoming requests, you know, and it is allowed by an inbound rule, then the you know response is, you know, automatically uploaded to the full bank, even, right? So there is no you know, corresponding out one goal, so in terms of that it is stateful.

Unknown Speaker 09:06

Okay, okay. Muta, any questions, yeah,

Speaker 3 09:10

so you have worked on, you're working on a screen, right? Yes.

Unknown Speaker 09:16

Okay, okay, how,

Speaker 3 09:20

how do you actually integrate the Spring application with the any containerization platform? The patient,

Speaker 2 09:31

okay, in terms of, you know, if I have to talk about the integration, basically, you know, for containerization specific, I have been primarily working with Docker and you know, these Dockers, you know, are deployed on, you know, ECS cluster. And the way you know you do that is basically, you know, you define a Docker file, right? So Docker file, you know, it just like, you know, normal text file, and it, you know, basically defines how you know, your applications will be built and it will be ran. So on the Docker file, yeah, I will, you know, start by first indicating the image. You know, I will be setting my no working directory. I will be basically copying the springboard chart file. And, you know, maybe some of the ports are, you know, given the, you know, entry point to the run the actual application. So, yeah, you know, once you know, you have this Docker file, so you basically, you know, build these Docker emails using, you know, the Docker build command. And this can be, you know, automated using zenkis, right? So you will do that, you know, Docker build, you know command. And once you have that, you know, container ready, you can upload it to the elastic Container Registry. And you know, once you have that elastic Container Registry done, your ECS, you know, will be defining a tax definition. And and then, you know, this tax definition can provision various services based on this container registry. And you know, so this is, you know, your full level, you know, high level, I would say, how containers can be, you know, educated with this penguin application, and then you can, it can be deployed to the ECS application.

Speaker 1 11:12

Have you used any orchestration tool for containers?

Unknown Speaker 11:17

Yeah, so, in terms of, you know,

Speaker 2 11:20

like orchestration, you know, in past, we have levels Kubernetes and I have, you know, very little you know experience on Kubernetes, but you know as we you know, as you know, as I know, the information. It now provides an abstraction for, you know, automatic, this orchestration, you know, this tool, you know, called Boogie, and this Boogie, you know, help us to, you know, orchestrating the ECS, you know, it's the you know, company signature deployment strategy. It is, you know, very similar to, you know, Kubernetes, in the sense that, you know, define a YAML file and you give the you know, configuration.

Unknown Speaker 12:02

I COVID, any question, any follow up question.

Speaker 4 12:06

Okay, so you mentioned that you're deploying your application in ECS, right?

Unknown Speaker 12:11

So are you

Speaker 3 12:13

guys working on maintaining the infrastructure part of ECS as well?

Speaker 2 12:20

Yeah. So, you know, like I mentioned, we, you know, have our orchestration through Boogie, and then, you know, basically, you know, we provide our metrics where, you know, we need to scale up, and sometimes we need to scale down. And you know, all of the, you know, infrastructure that our application, basically, you know, is deployed are owned by our team. But, you know, when we have a, you know, Site Reliability Engineering team, or, you know, SRE team that, you know, basically handles all the internal of it. So, you know, even though we work on the deployment aspect, you know, it is managed by, you know, our external team. And we, you know, we have, like, a dedicated engineer, right, that supports this.

Speaker 5 13:07

So, how do you deploy your so for

Speaker 3 13:12

ECS related you don't maintain any ECS related configurations inside your application. Is there? So, so,

Speaker 2 13:18

as I mentioned, we do it, right? So we, you know, have a, you know, a text, like a boogie file that contains, you know, these containers. And you like this information for every environment. So this, you know, abstraction, you know that does SRE team has, you know, written, and also, you know, they are picked by the boogie file. And

Speaker 3 13:40

so what does that Boogie file holds? I mean what I did. I mean what why do we need to maintain first is what information it carries, and why do you need to maintain this information within your application?

Speaker 2 13:55

Okay, yeah, so. So Boogie file, as you know, basically like, and, you know, text file. Like, we're having all, all kinds of, you know, configuration of an application. So, basically, you know, it's like to have a, you know, fully managed application that we can, you know, some of the configuration can be written in and, you know, this Boogie file can contain, like, some, some tax related definition, like, you know the container name, the images, you know the environment variables, and also, you know, the memory limit, other, other kind of, you know, contents, like, you know how many tracks you want to, you know, run, or what kind of, you know, security groups, and also subnets, information it might need. So, yeah, so you know, what kind of it can, you know, like other information, like what kind of scaling policy it might need, and also, you know, of various, you know, I am policy that might be needed to be, you know, attached to it. So, yeah, these are the few, you know, basic information that you know that, like this particular bookie file will be containing per environment basis.

Speaker 1 15:04

Can a boogie file control your infrastructure or it maintains your auto scaling policy? Who maintains your infrastructure? Actually, I'm talking about like what kind of technology or policy through which you maintains your maintain your infrastructure. Can Boogie, maintain that.

Speaker 2 15:27

So, yeah, so, you know, it is mentioned, you know, it is basically maintained through the boogie file.

Speaker 3 15:33

So is it related to anything? I think where some is going is to use any infrastructure as code, right? Extract,

Unknown Speaker 15:41

not you like Sri team, but which, which are you

Speaker 1 15:45

aware of any, yeah. Are you aware of any technology that maintains your infrastructure? I know if, even if, SRT maintenance that have you ever interacted with any technology?

Speaker 2 15:57

Yeah. Basically, you know, we use, you know, cloud formation templates, and they have some, you know, some infrastructure as a code through, you know, TerraForm and Ansible. But you know, we, primarily, you know, have been liberating this kind of, you know, plan, like cloud formation templates for our deployment tasks.

Unknown Speaker 16:20

So which one you are using out of this?

Speaker 2 16:24

So, yeah, so no, in my current, you know, like team, they only, you know, focuses on, you know, CloudFormation template side. So the SRU team is, you know, responsible for data forming and support. Do

Unknown Speaker 16:40

you use both or

Speaker 2 16:43

so? Is that, you know, as I mentioned, our legacy application, users, Ansible, you know, but you know our more concern, you know, our more modern application are using?

Speaker 1 16:59

Okay? Sounds good, okay, tell me one thing. I have a lambda applications, and I provision, and it is running for 15 minutes, but I have 20,000 records to process when my time limit is there for lambda files, I see that within 15 minutes I was able to execute 10,000 records, if I need to execute rest of the 10,000 records. What kind of

Unknown Speaker 17:23

mechanism Do you think you can apply?

Unknown Speaker 17:26

You can use any

Unknown Speaker 17:29

AWS capabilities

Speaker 2 17:33

so understanding your you know, situation that you just provided in order to, you know, process that you know, basically I have to process, since I have to process these many records, right? And maybe, you know, you can use, like a simple, you know, simple queuing service where, you know, you basically, you know, have some batch process like, and those messages have can, you know, send the records to the SQS queue, and then, you know, have the lambda function trigger those messages. So, yeah, so each, you know, lambda invocation can process, you know, certain amount of record, and then after you know that is processed, you can, you know, get a different approach. I think you know, the advantages here of using, you know, SQS is, you know, if you know our record is successful, then it will, you know, get happily removed. But you know, if it fails, then you know it it goes to the, you know, dead later queue so that you can, you know, evaluate the year, and then, you know, you can reprocess that in a later time. So that's one of the benefit. And the way you know, that way you know, you won't have a hard limit of, you know, either 15 minutes of, you know, lambda so, but you don't have a continuous processing while you know, you're getting the newer messages.

Speaker 1 18:56

Yeah, SQS can be one solution. But what will inform SQS that? Hey, I am done with 7000 let's start with the 7001 How would you because lambda doesn't know SQS you mentioned right? Having once process, it will be removed. But we need to inform lambda what record the next iteration should be picked up? What do you think we can apply?

Speaker 2 19:21

So in that case, basically, in order to, you know, notify, maybe we can leverage, you know, a state function, which, you know, basically, you know, how we can, you know, orchestrate this. So, yeah, I believe, you know, having a, you know, like leverage, like state function, but, you know, I don't have a whole lot of experience working with those step function and firsthand, but I did have this knowledge. So, you know, we can, you know, orchestrate this processing through now, the step functions.

Speaker 3 19:57

So how do you implement security in spring work for your API. The patient,

Speaker 2 20:05

yeah, so in our project, you know, mainly, we have been, you know, like, like, in order to implement the security in our application, we are using, you know, spring, you know, good starter security. So, basically, you know, this dependency provides spring security out of the, you know, out of the box. And also, you know, it also provides spring security in the, you know, handling our authentication, you know, as a role based access. And also, you know, with JWT UConn, or, you know, also we can implement or two. So all you need to do is, you know, basically add that dependency into our you know, dependency management file and override the you know, web security configure adapter. And you know, we can, you know, have our best access using, you know, the pre authorized annotation on various, you know, different controllers that we want to, you know, have it in and, you know, we which can, you know, check if the rule is correct or not. And based on that, they can provide that, you know, it can provide, like, entrance to the application.

Unknown Speaker 21:13

What is a spring controller advice? Do spring controller

Speaker 2 21:19

controller advice? Okay, yeah, spin controller advice, you know, it is used for, you know, global exception handler. So basically, you know if an exception in any point of our application, then you know we can define an exception handler methods that, basically, you know, handles the particular exceptions and then it, you know, returns a meaningful JSON and like whatever JSON response that we want.

Speaker 3 21:47

And if you have to externalize your configurations in Spring Boot, what do you have to do?

Unknown Speaker 21:53

I couldn't quite guess that. Can you can you do that again? If

Speaker 3 21:57

you want to externalize your configuration application, how do you know that,

Speaker 2 22:02

right? So you know, like, if you know, we have an configuration application, you know, properties specific, you know, for each of the profiles. So you can, you know, either do the you know application that's QA, or you know, dot properties, or, you know, in, you can also do it in those product properties. And also, you know, these properties files would be, you know, loaded, you know, on the profile. So basically, you know, one step above that you can, you know, make our, you know, dynamic Config Server by, you know, utilizing our framework called Spring Cloud Config Server. So, you know, basically it takes us, you know, as an, you know, external repository or various storage location that, basically, you know, holds on our properties file. So this helps to, you know, read this you can, either, you know, use a dynamic configure property to, you know, Render, or, you know, simply just, you know, you can use a valid annotation on your you know, spin company, okay.

Unknown Speaker 23:05

Okay. And

Unknown Speaker 23:08

do you know anything about API

Speaker 3 23:11

contract definition? What do you understand by defining a contract for API?

Speaker 2 23:17

Yeah, sure. So you know, basically, whenever you know you are defining an API contract, it, you know, it just basically tells you the you know specification of API, how you know it should APIs will behave, or, you know, how your you know client should interact with it. So you know, we can define the endpoints on the HTTP method, and what kind of you know other request is you know supposed to have, like, you know the request parameters, or you know the body, and you know what the you know response format would look like. And you know this can be documented through our you know tool called Swagger, or you know or open API as well. Option Okay, okay.

Speaker 3 24:03

And from your API, how do you configure the integration with the database downstream?

Speaker 2 24:14

Yeah, so in terms of, you know, configuring, so basically, you know, talking to the database and Spring application is pretty, you know, straightforward. If we, you know, add the dependency for Spring Data JPA, it should, you know, give it to you out of the box. There are, you know, no other configuration required. So all you need to do is you know it. You know it configures the data, source, URL, also, you know passwords, username, and also, you know the driver, and once you do that, so you can define our, you know, a repository layer. So this repository layer, you know, will be, you know, the interface that accidental JP repository. And then you know what you can do is, you know, basic, basic product operations that you know you need to do with the database. So you can also do more advanced operation by, basically, you know you are defining a query using a query annotation, or your application you know should be having an entity class. And basically, you know entity is nothing, but you know R and it's just an object representation of your database,

Speaker 3 25:20

okay? And what is transactional management in springboard, okay,

Speaker 2 25:27

so talking about the transactional management on, you know, springboard. So basically, you know, it just, you know, down to a transactional annotation. So basically, you know, transaction is a you know, sequence of operation that will be, you know, treated as one, right? So if you, you know, want to some portion, or you know, if, like, there is a sequence of operation that will be, you know, treated as you know, one, right? So if you, if you want to, you know, have a portion of your code to run as single unit, then you know, you can annotate. You know that with the help of a method, with, you know, transactional annotation and spring will internally, you know, take care of, making sure that you know the whole operation is done as in one operation, and you know, or it can be no roll back, otherwise, as a whole operation, okay, okay.

Speaker 1 26:24

So in terms of Java, let's say you mentioned that you have used primary Java, right? So what is the functional interface in Java? What does typically do? Can we experience one of them? Yeah,

Speaker 2 26:39

sure. So basically, you know, in Java, functional interface is a special, you know, interface that was, you know, introduced in Java eight. So basically, you know, it just an interface that has one abstract method and, you know, and basically, you know, there are two, there are, you know, ways that you can, you know, define a lambda expression, right? So basically, you know, it's, let's say, you know, you have something like some of the you know, functional interfaces, like, you know, basic, you know, producer, consumer, five functions, and also, you know predicate. So you can define, like the lambda expression, which is, you know, very you know, simplest way of expressive so the way you know of writing your functions can be, you know, different, and you know it can be passed around as an argument. So you know this functional interface, basically, you know, represents the lambda expression, and it can be, you know, passed around as an argument to your method.

Speaker 1 27:41

So say, for example, I have a functional interface. And five years ago, somebody notified functional interface, right? And today I came in, or you came in, and then you want to refactor that file. So functional interface is a abstract, right, in implementation, in nature, and you are bind by the contract over there in order to extend the class and refactor Java provides additional interface mechanism. What is it called?

Speaker 2 28:13

Oh, yeah. So in that case, you know, I think you are talking about a default method in an interface which you know, was also, you know, introduced on Java eight, and basically, you know, this was for, you know, by acquired, you know, backward, you know, compatibility on your interface, right.

Speaker 6 28:34

Okay. Are you familiar with any Kafka? Platform,

Speaker 2 28:45

yeah, so, you know, when talking about Kafka platform, so basically, you know, our application that I have, variously, you know, work with, I have been, you know, part of both consumers as well as producer. But I haven't, you know, had, like, an extensive work on Kafka. I'm familiar with, you know, terminology and also the technology that is, you know, is, it is in the, you know, pops up more, right? So I'm quite familiar with that.

Unknown Speaker 29:15

Okay, what are the various components of Kafka?

Speaker 2 29:20

Very So, basically, you know, whenever you are talking about, so, like, if you want to talk about, you know, Kafka, and primarily, you know, the producer and the consumer are, you know, are the main topics, right? So, basically, producers are, you know, the application that you know are to our Kafka topic, and consumers are, you know, the application that, basically, you know, subscribe to these topics. And basically, you know, consumers processes this field of records. And you know, the, you know, the basically the topic is just a category. And are, basically, you know, maybe a name to which the records are published into. And then you and, you know, you have a broker. And broker is nothing, you know, it just like a Kafka server. So in your you know, Kafka broker will, you know, it will contain the topics. And days, you know, topics will have partitions and and, you know, like, for the more these partitions, it's like a, you know, single log where the records are, you know, stored. And I know, before the Kafka 4.0 Bucha to keep it, you know, used to be coordinated with all kind of this Kafka brokers. But you know, from, you know, Kafka 4.0 it will, you know, internally, you know, consist of this distribution itself,

Speaker 3 30:38

okay, what is, what do you understand by topic and partition in Kafka? Can you give me a little bit more detail?

Speaker 2 30:46

Okay, yeah, sure. So, you know, if I have to, you know, dig down on, you know, topics. So basically, you know, topic is, is nothing, but, you know, it just like a name, right? So, it just, you know, has a category. So basically, you know, you our topic is, you know, like can be divided into several different chunks. So whenever you know like, but basically you know, these chunks are, you know, are known as a partitions, right? So each partition act is for you to, you know, promote parallel processing. So basically, your topic will contain the records on these partitions, and each partitions, you know, each record in a topic you know, can be basically identified by this unique object. So one of the you know data to topic, it will, you know, it will be modified, or, you know, deleted. And you know, which are, you know, most of like, you know, a data in GPT part and within a partition of a topic, records are, you know, stored in the order they are received? And, yeah, so, you know, basically partisan is like a smaller, you know, order segment of these topics. So where, you know, data is distributed across these multiple partitions for, you know, as a part of your parallel processing. So just you know, to give you a small example. So if you have a you know, Kafka cluster, that is, you know, trying to process various, you know, orders from, you know, you create from a Kafka topic. So namely, you know orders, and you know, then you can have partitions like one two, and you know where partitions can contain, you know, order from, you know, East Coast, or, you know, partisan also one can, you know, contain order from West Coast. And you know our partition, you know to can contain order from the middle Midwest and, you know, so on, right? So, like this kind of, you know, like a real example of how it might look like,

Unknown Speaker 32:53

Okay? And what does really mean by

Unknown Speaker 32:57

Can you tell us something about offset management in Kafka?

Speaker 2 33:02

You Yeah. So basically, you know, we have something called, you know, offset, so it just like a, you know, unique identifier, right? And basically, you know, it is critical in Kafka, because it deals with keeping, you know, track of the position of, like, you know where the masses is, right? So you know, if your consumer is processing a masses, then you know it will have an offset. So each you know, record with a partition has that you know, unique identifier number called you know offset. And this makes you know your consumers, you know, read data in that sequential manner. So, you know, basically, Kafka provides automatic offset management through, you know, auto commit. So where you know, the like, there is an offset, like committing features at a you know, particle, you know, configure interval and it is, you know, defined with a consumer configuration by like, you know, auto commit property. But you can, you know, define some, you know, manual offset management, if you are required to,

Speaker 1 34:13

can the offsets be always same for two different clusters, or can they will be different, not another question is, I will they always be same for two different clusters? So,

Speaker 2 34:28

so you mentioned, like, I think, you know, like, it is a unique within a, you know, partition, right? So, as I mentioned, like, since it's a unique object, is always, you know, unique without partition, but not, you know, like upgrade. So basically, you know, if you have multiple Kafka cluster, you know, some offset value can refer to, you know, completely different messages.

Unknown Speaker 34:56

So I want to share my screen now

Unknown Speaker 35:00

show you a coding block.

Unknown Speaker 35:04

So here

Speaker 1 35:07

we have a problem statement. We have a string input, I would let you read through it. The idea of this program that we are trying to solve here is that I need to find the highest occurring word in this string, it can be loaded from the file,

Unknown Speaker 35:28

or can be an input.

Unknown Speaker 35:37

You can write in any of your preferred

Unknown Speaker 35:42

ID. Yeah.

Speaker 1 35:45

So you can ask me question, if you have any clarifying questions,

Speaker 2 35:49

yeah. So, I mean, we just want to, you know, find the highest occurring world, right?

Unknown Speaker 35:57

So the return will be highest occurring world and it's count,

Unknown Speaker 36:04

okay, yeah, just, you know,

Speaker 1 36:07

you can take a screenshot and for your reference, and then you can switch back to your ID.

Speaker 2 36:14

You just, you know, want one entry on the region, like, not the overall five second. You just won the one entry on the return right, not the overall No.

Speaker 1 36:25

So for example. So which one do you think could be a high stocking word here in the input?

Speaker 2 36:32

Give me a second. It looked like it's the okay. Why so? Because you know you have so basically, you know three. You know the Okay,

Speaker 1 36:56

and in that case, yeah, if this is the then we'll, we're going to return the with this count of three. Okay, you can take a screenshot if you want to, and then I can stop share.

Speaker 1 37:14

But you don't have to write the main method or anything. Basically, what I am interested to know is, like the implementation here, how you come to the logic of deriving that is to correct word, okay,

Speaker 2 37:26

yeah, I guess you know. So I have a question here, though, you know. So if you see the line seven and line 15, so here in line seven is expecting or written type of string. But then here the line 15, you know, return some map, right? So I think you just want to, you know, return the word overall rather than the map, right? I will return,

Speaker 1 37:54

you can return a map. I don't worry about this part. So you just return a map that has the highest occurring word and then count.

Unknown Speaker 38:05

Okay, okay, so,

Speaker 2 38:07

yeah, let me share my screen. Sure. Thank you.

Unknown Speaker 38:28

I'm just kidding. We'll

Speaker 1 38:30

spend like quick 15 minutes on that, and then we'll go back to questions with Bucha if she has any questions. Okay, yeah.

Unknown Speaker 38:44

Are you sharing?

Unknown Speaker 38:44

Yeah, just getting my interleaves, you know, ready before I say,

Speaker 2 38:53

Okay, I'm starting my sharing my screen. Just let me know. Just I will let me know if once it's visible to you,

Unknown Speaker 39:12

you can see my screen, right? Okay,

Speaker 1 40:16

that's fine. You don't have to write the boilerplate code. I'm more interested to see the logic, how you define, and find out the istruc report.

Speaker 2 40:28

Okay, yeah, I have a small question before I'm up for watching. So basically, you know, do you care about the case sensitivity? Yeah, we do. Okay. I that.

Speaker 1 40:43

That's why I was asking, like, how do you come to the conclusion that the is occurring three times? Yeah, it is yes, sensitive, so we need to take care of that.

Unknown Speaker 40:53

So in that case, it occurs only two times,

Speaker 1 40:56

right, correct? So you can consider that the and they are all counted three times, but you will have to compare somewhere so that it is in get sensitive.

Speaker 1 41:14

Like, if you don't take care of that, it won't be same, right? I mean, if you just say that the load is the cost two times, but since lowercase the or camel case, they are all same, we can consider, but we'll have to Do some logic.

Unknown Speaker 41:33

Yeah, yeah, okay. Sounds good. Yeah. I

Speaker 2 41:45

up. So initially, I'll start by, you know, splitting my, you know, given input into a string array, I

Speaker 2 42:12

now, to make it easier, you know, I will basically add, you know, just add this lowercase. I

Unknown Speaker 42:24

Okay.

Speaker 2 42:39

So now I define a map, Basically that's sourced out. But I

Speaker 2 43:53

now, basically you know what this will return is, you know it will return the map that, basically, you know, contains each word and it counts right. So now we just, you know, want to Return the one that has the maximum count, Right?

Speaker 4 45:03

Can You move I

Unknown Speaker 45:47

cancel Can you see the cancel failure?

Speaker 4 45:51

Canceled here, and we don't have the cancel failure. We Need to Add I

Unknown Speaker 47:51

That's a Good Question. I

Unknown Speaker 48:26

I think I should take care of it. Okay,

Speaker 1 48:34

sounds good. Any optimization that you think you can do on this and

Unknown Speaker 48:41

what is the runtime complexity of this code, okay, for

Speaker 2 48:47

this particular thing that I've written here, the runtime, you know complexity would be something, You know, like a big O of n,

Unknown Speaker 49:01

okay,

Speaker 2 49:09

and if I have to think about optimization here, you know it would be to do an early return. So you know, if the given string is, you know, empty or null, then you know, we can do an early return.

Unknown Speaker 49:26

What if the string has a special character? So, okay, so if the string contains special character here,

Unknown Speaker 49:41

then basically, you know, we might,

Speaker 2 49:45

I think we might have to, you know, basically, do some kind of check to, you know, replace it to an empty string. Okay,

Unknown Speaker 49:55

sounds good. Thank you. Muta, over to you.

Unknown Speaker 50:00

Hey, thank you. So So

Speaker 3 50:04

okay, so deep. Have you worked on any design side of API or any application? Are you? Are you getting engaged in your current team?

Speaker 2 50:16

Yeah, yeah. So currently, you know, I have been recently, you know, taking some design work as well. So, you know, I rocking two of the APIs recently. And, you know, those were, you know, called maintenance API and the Experience API. So basically, what I did was, you know, I was involved in some architects call. So I was, you know, trying to gather much of my, you know, functional as well as, you know, non functional requirements. And, you know, document all of them, you know, in our Confluence space so, and then you know, this got approval. And then you know, it got implementation as well. So I have been, you know, getting experience on this kind of design aspect of the development as

Speaker 3 51:03

well. Okay, how about monitoring, observability, different kind of testing. Understand with different kind of testing, what do you understand by different testing life cycle for the API to go to production?

Speaker 2 51:21

Yeah. So I think, you know, the different phases of testing that happen, you know, involved in is, you know, unit testing. So basically, you know, unit test is a must, right? So for any changes that you know, I make, so actually depends. So it, you know, doesn't matter if I, know, making one single change, or, you know, adding a new class, or, you know, whole new class, or anything like that. I always, you know, make sure I have unit test cases, you know, added into my, you know, like this, you know, test cases to, you know, chain in isolation. So once I'm, you know, done with that, I have, you know, I'm, you know, I have integration test, basically, you know, making sure that you know, these isolated classes or the methods that you know, where I have implemented or I have tested, you know, work well when it is integrated together with, you know, and you know, that is one other testing that I do. For unit test. I, you know, work with J unit and monkey do. And particularly for, you know, integration testing, you know, I have worked with Spring Boot test. Now, you know, a label above that is, you know, regression testing. So for, you know, regression testing is, you know, like a more detailed in a sense, you know that I test the new changes that you know we are adding, or you know we are, you know, updating into our application that you know does not break our existing flow, and you know it is done, you know, through different frameworks, like cucumber. So I, you know, I have been, you know, involved in a, you know, little bit of on this. And also, I have been involved in a lot of performance testing your name, you know, JMeter scripts. So, yeah, we, you know, have a dedicated environment to, basically, you know, run performance testing. And once, you know, all of these are good, we have it, you know, ready for, you know, for you know, for production,

Speaker 3 53:24

okay, are you familiar with any other kind of testing other than these two

Speaker 2 53:30

um for, specifically for, you know, unit, you know, I have been, you know, doing unit integration acceptance, as also, I mentioned regression testings. And also, you know, our performance

Unknown Speaker 53:47

Okay, Are they familiar with performance testing?

Speaker 2 53:51

Yeah. So, as I you know, basically, you know, mentioned, we work with JMeter for performance testing. So, you know, we basically, you know, test the application under, you know, load and stress. So, yeah, okay, so

Unknown Speaker 54:09

which tool do you use to run the load test? Um,

Speaker 2 54:13

so if I have to, you know, talk about the actually the tool. It's J meter.

Unknown Speaker 54:22

What is the peak load you have carried with J meter?

Speaker 2 54:26

So for the project, basically, you know, we are trying to master, you know, current production load, which is about, you know, 1000 GPS per second.

Unknown Speaker 54:38

Can you run a test with J meter using 1000 GPS? You DPS,

Speaker 2 54:42

yeah, so, you know, there was a plugin, you know, that was called, you know, concurrency thread pool. You can actually add that plugin to our, you know, J meter, and then, you know, you can configure your TPS given, you know, in the base scenario. So you can, you know you can add, like, you know, like a 15, or, you know, 20 users, and after you know 300 you know 100, or something like GPS, but you know, I think there is an, you know, external plugin that you know you can add to your J meter.

Speaker 3 55:15

But how will you connect your J meter instance with multiple servers to distribute the road.

Speaker 2 55:24

So, yeah, so, like I mentioned, you know, we have a various, you know, dedicated environment. So we, you know, we use a tool called Hercules. So basically, you know, the Hercules sticks in our J meter, you know, XML script. And also based on these individual XML script you know this, you know how to, you know, distribute the load across different users. So basically, you never close, but just so you know is a easy, you know instance that you know manages the running of this scripts.

Unknown Speaker 55:57

Okay, thank you.

Speaker 3 56:00

Yeah, I don't have any more question. Dipesh, so do you have anything else to ask?

Speaker 1 56:04

Yeah, one quick question. So you mentioned you have used a various design pattern, right? Can you given me an example of a creational design pattern? How many design patterns you are aware? First of all, like, what are the types of design pattern? But how many?

Speaker 2 56:20

Yeah, okay, so for specifically about my experience with you know, design patterns, I have, you know, work with a lot of design patterns, basically. So if I, you know, have to topic, you know, talk about the specific type of you know, design patterns. I know, you know, there is a, you know, a creation of design pattern, you know, behavior design pattern, structural design pattern, yeah. So, you know, some of the creation of design pattern include, like, you know, Singleton, you know, factory method pattern, basically, you know, just focuses on the creation of the object. And, you know, and then, you know, there is something called structural design pattern, which is, you know, basically deals with the object composition, you know, how the objects are structured. And there is a you know, behavioral design pattern. So, which is, you know, concerned about changing of behavior. So one of the you know, design pattern that I, you know, I have worked on a regular basis, is a singleton pattern. So Singleton is, you know, basically ensuring that our class, you know, has a global point of, you know, entry, and it has only, you know, one instance throughout the application. And then, you know, there is, you know, also a factory method pattern. So, you know, with basically, it is used to define an interface for, you know, creating an object. So those are, you know, you know, to like creation of, you know, like factory pattern that I, you know, work with regularly.

Speaker 1 57:51

Thank you. You have any question for us, Dipesh, we'll be happy to answer those. Yeah, yeah,

Speaker 2 57:56

sure. So, first of all, you know, I would like to, you know, thank you for giving me this opportunity to interview today. So I'll let you know how you know how it is like working with a team at JPMC, you know, like, what would be like a team structure? Can you please give some insights on that? Yeah, sure.

Unknown Speaker 58:17

Would you like to answer that? I can add up?

Speaker 3 58:21

Yeah, sure. So currently we are working under payments line of business in chains. Chain is our line of businesses. We are part of payments and that too. We are working on a new product, which we are trying to build, a horizontal product which will support multiple method of payments across multiple clients. So how our team is placed is we have multi location team. We have parallel to presence teams in New York, teams in Delaware, teams in Texas, and we are expanding in India right now. Yeah, so it's a growing team. It's a full fledged Greenfield application, entirely modern clock tech stack. We work on several AWS services using TerraForm for their infrastructure, and Java is the Spring Boot is the core tech stack.

Unknown Speaker 59:13

And how our team works is

Speaker 3 59:18

everything is being done by the developer like the sprint teams, we are in Agile practice, so starting from content negotiation to co development, to take it through different testing, lifecycle, deployment, taking it to broad, everything is being done by teams. And once it's it turns in broad we hand it over to our support team, and we do get engaged in all any broad issues. So it's like, yeah, jack of all. Ours is master of one. You can think like that, and that is the way teams work across the firm. So, yeah, so you have to know everything for the entire SDLC to contribute to the delivery. That is expectation. Okay, yeah,

Unknown Speaker 1:00:08

thanks. That's good to know. Yeah. Sounds good. Yeah.

Speaker 1 1:00:13

One quick question, is there anything that you would like to typically learn in your next job opportunity? Can you tell me one thing that you don't know on on the track of learning that you would like to typically look forward to in your next job? Yeah, so,

Speaker 2 1:00:29

you know, as a developer, it's always about learning something new, right? So I will say, you know, like I mentioned earlier, I you know, do have a lot of, you know, working experience with Kafka, that is, you know, something that I want to, you know, Master more. And recently, you know, I have been doing some work with, you know, spring webflox, which is, you know, more reactive way for asynchronous processing, and like our Springboard application, right? So these are the two things that you know I will like to, you know, learn technically, but you know, on a more higher level, I want to, you know, learn more like these, you know, leadership abilities that you know in the future I can, you know, take on to some leadership opportunity. So that's what I look into in my new role.

Speaker 1 1:01:16

Sounds good. Thank you. Well, thank you for your time. Dipesh, appreciate your time and talking to us.

Unknown Speaker 1:01:26

Thank you. Yeah,

Speaker 3 1:01:27

you'll hear back from HR. We'll submit our feedback,

Unknown Speaker 1:01:31

sure. Likewise. Thank you. Thank you for

Unknown Speaker 1:01:34

so you have a good question today.

Unknown Speaker 1:01:36

Yeah, bye, bye. nice

Speaker 7 1:01:39

econ daily.

Dipesh Shrestha- mutualofomaha- SRV final

Speaker 1 00:00

Environment. So, you know, with those teams, right? So I, you know, work together with them as a, you know, a single unit, like looking for a single goal. So, yeah, you know, that's how I like to, you know, handle adaptability, because with collaboration, and, you know, some of the effort on the self, I was, you know, able to integrate myself into the, you know, new set of you know, challenges and you know, and then contribute meaningfully, you know, to the overall project.

Unknown Speaker 00:36

Next question, i i Sure

Unknown Speaker 00:49

go.

Speaker 2 00:52

Yeah, just do with time here.

Unknown Speaker 00:58

What motivates you to excel at work,

Speaker 1 01:03

yeah. So, basically, you know, while working as a you know, team, I would say, you know, my biggest motivation is always, you know, is that, you know, profit, you know, how will willing are we to, you know, solve the problems, right? So, I mean, I'm always, you know, you know, working to solve the challenges. I also, you know, like, technically, you know, to challenge someone, or, you know, from myself, working together on the same challenges, and, you know, learning and, you know, growing that comes with, you know, these kind of problems. So, yeah, I'm, you know, motivated in taking, you know, this kind of direct you know problems. And you know, which are, you know, more complex and you know, and basically, you know, coming up with some kind of, you know, contribution that will help the overall system, and, you know, overall product, and it helps me, you know, continuously improve my skill set. So you know, with which, you know, basically excite me as well. You know, excite me to take a more challenging, you know, tax later on. So whenever, you know, I learn new opportunities, and you know, all the new frameworks that coming in and this market, it keeps me motivated to learn, because I know that, you know, I'm growing internally, even though you know I might be sometime, you know, struggling with that issue in hand, but overall, you know, you know I'm, you know, growing, and you know I'm also having that mindset, you know, to continuously learn. And you know, research on the topics that you know motivates me because, you know, at the end, I want to create, you know, be creative and be innovative, and I'm always, you know, working towards that growth mindset.

Unknown Speaker 02:56

Thank you. Thank you. Sure. Can you please

Speaker 3 03:02

describe a specific situation in which you worked with a diverse group of people over a period of time? Based on this experience, what did you learn?

Speaker 1 03:15

Okay, yeah, so I will, you know, like to, you know, think about a situation when I, you know, work with a diverse group of people. So in my you know, experience, I think, you know, we were trying to come up with a, you know, real time, basically, real time integration of financial data. Basically, we were coming up with, you know, to create some Java based microservices. So, you know, in this case, they were, you know, we were basically communicating with people from different time zones. And also, you know, our offshore team along with our, you know, business partners. So basically, you know, during this period of time, we had a mix of, you know, technical and non technical teammates along with, you know, other other people. So it was really, you know, diverse, basically, you know, we are going to, you know, integrate our particular work with other teammates. And, you know, we had different understanding, and different people had different understanding. And you know, it already, you know, depends on how, you know, these were, you know, modeling system that were coming in. So we were trying to understand how to, you know, integrate to work together. Because, you know, developers were, you know, focused more on building the scalable system, but where it is, you know, a non technical people were focused more on the business aspect. And like all the, you know, most of the over overall workflow. So that was the time when, you know, I had basically, you know, be working with different group of people, you know. And now, you know, the way I and my team, or overall, you know, we came about, you know, having a proper collaboration work and, you know, we made a convention, you know, like effort to, you know, rise the kind of a gap that we have between, you know, non technical and technical group of people. So we usually set up, you know, proper meetings that, you know, overlapped with, you know, our working hours and not and not had any, you know, extra time that you know, made our working hour, you know, off, off of, you know, our working hour. And basically, you know, what I learned from this was, you know, it was necessary to have a, you know, cross disciplinary understanding. And you know, we also, you know, have to be very adaptable, because, you know, at the end, you may need to explain something technical, right for non technical group of people. So you know, so that you know, you make sure that everybody understands. And if you you know, do so by simplifying the you know, context, technical details into, you know, something that is comparable to, you know, day to day object. It makes it, you know, easier for everybody to visualize it. So basically, at the end, you know, with this experience, I was, you know, able to, you know, kind of have different, you know, perspective, and also, you know, improve my adaptability.

Speaker 4 06:22

Of course, I of course. So I think you kind of touched on this already a little bit, maybe at the beginning. But so in our field, right? It's very important to stay kind of current on changes in the industry, the marketplace and etc. Can you maybe elaborate a little bit on how you try to keep yourself informed, or how you kind of stay on top of technology. Think I heard you mention a few things at the beginning, but if you could elaborate for us,

Speaker 1 06:48

oh, yeah, sure, sure. Why not? So, alright. So, you know, in order to, you know, always try to make myself better, try to make you know myself, you know, engage with continuous learning, you know, with upcoming new technologies and framework. I, you know, continuously motivate myself. So, you know, I try to keep myself informed by, you know, reading different kind of blogs. That is, you know, coming up. I, you know, my usual activity is, I get up and, you know, read a technical blog. And also, you know, there are some kind of other, you know, resources like online learning. And also, you know, since I'm in a JAVA related field, I mostly, you know, go with the Java related, you know, changes and the libraries and the newer version that comes, you know, sometimes all of a sudden. But you know, apart from that, you know, I'm always invested in, you know, taking online courses. Currently, you know, I'm enrolled in one, one for, you know, software architecture in AWS. So, you know, that's something that I like to do. And sometimes, you know, I do experiment with, you know, this, newer technologies. And also, you know, I spend some time trying to, you know, read the documentation and and then, you know, I try to create some small project out of, you know, my personal, you know, experience and on free, you know, time we use, you know, various cutting new technologies to see, you know, how they perform in real world scenarios. And, you know, I might be able to, you know, work with those technologies in, like, a professional, you know, setting, or, you know, professional setup. So, yeah, I mean, that's what you know, I do, in order to be up to date in the market with the latest technologies and trends. Basically, you know, that's arising into this day.

Unknown Speaker 08:42

Okay, great. Thank you.

Speaker 2 08:48

You give me an example of a time when you influence others to do a job or process in a new or different way.

Unknown Speaker 08:58

How'd you go about doing that? So,

Speaker 1 09:04

I believe, you know, I remember a time, you know where we were, you know, working with, you know, large data sets that might, you know have been, you know, embedded objects with, you know, Jason, so that, you know, we were, you know, integrating the objects within the JSON. And basically, you know, we were having to compare the equality of these objects, right? So these objects that I'm, you know, talking about are basically, you know, they are concerned to the, you know, customer data that might contain sensitive information. And you know, on, the only thing that, you know we were testing, you know, earlier was, you know, converting that decision into an object. And then, you know, doing on equality check on, you know, particular work or the fields, or, you know, different data types, right? So this, you know, converting the, I mean, like, it has a, you know, really time consuming, you know, talks, and we were basically, you know, spending a lot of time, you know, trying to handle the exceptions in our use and a lot of the age cases that, I mean, we need to take care of, because that might lead to some, you know, failure later down the line, and it might, you know, and since it has a lot of, you know, manual process, it was not too efficient. And then, basically, I remember, you know, after doing some research, I was, you know, able to look into the you know pattern of how the, basically, the you know data look. And, you know, I I saw, you know, what we could use in order to, you know, check this comparison. And, you know, I also, you know, check, you know, right? We, you know, we were able to, you know, handle, like, eight, or, you know, different kind of data types. And, you know, types of the data, and looking at those, we could, you know, find some kind of pattern in each of them. So I started, you know, so that's thing using a, you know, Jackson data bank library, so, you know, to basically convert the JSON into, you know, another, you know, stingified object that could be, you know, compared directly, rather than, you know, having to, you know, extract the fields manually. So, yeah, so, you know, we updated the, you know, on the version manually. This was done through, you know, a series of meetings between me and, you know, my T tech lead and, you know, and we were, you know, boiling down, like various different requirements, and then, you know, creating, like an interface to, basically, you know, compare all these objects. And I started with, you know, proof of, you know, concept, and then we got a proof. So that was a, you know, exciting part. And then we followed through the by other developers, you know, by giving some, you know, simple, you know, or simple demo. And, you know, how would it work? So this was the time when I, you know, when I motivated teammates to, you know, basically, try to think out of the box. And, you know, come with, you know, a solution, basically, you know, that was different way, you know. And then we were actually, you know, doing in the current setting, correct, okay, sure. What

Unknown Speaker 12:33

would you say your career goals are,

Speaker 1 12:37

right? Yeah, so you know, as a Java developer right now, I'm actually a senior software engineer, and, you know, I want to, you know, keep on pursuing this field until and unless I feel comfortable mentoring and, you know, leading the team. You know, considering your short term goal, I will, you know, be moving up onto the leadership position where, you know, basically, I want to take in charge of one or a couple of teams. And then, you know, for a long time, goal I want to, you know, move into something, you know, in an architectural position where, you know, I can design different systems and, you know, enterprise products, okay, yes, okay, thank you. Okay.

Speaker 3 13:34

Moving on to our technical questions. The first one is so the first set is kind of around Java. So how would you rate your level of competency in Java?

Speaker 1 13:50

Yeah, sure. So, you know, if I were to rate myself out of range, 10, I will, you know, rate around, you know, seven or half, you know, or eight basically, you know, I give some of those for, you know, continuous improvement.

Speaker 2 14:10

Just a quick question. Can you elaborate a little bit more on what type of Java frameworks you've worked on?

Speaker 1 14:19

Yeah, considering Java frameworks I have, you know, basically, you know, working on primarily with the springboard spring related application. But you know, I have also worked with Apache, Spark, sure.

Speaker 4 14:45

Risk. So could you maybe describe your experience with automated testing as far as Java goes?

Speaker 1 14:55

Oh, yeah, I mean, so as per, you know automated testing is concerned, basically, you know, we have, you know, we have our unit testing. And you know, with each Java, you know, you know, and that we do is, I make sure I, you know, back them up with, you know, meaningful unit test. And you know, unit test is nothing, but just, you know, testing individual unit of code. So basically, you know, when testing everything in isolation, is concerned about, you know, unit testing. So for this, you know, I have been working with J unit, and also, you know, market tool to mock, you know, of the external services. And J unit, you know, is the framework that I use to, you know, run this kind of test. And then, you know, there is something in an integration testing. So integration testing is basically, you know, to test the checks the you know, integration of, you know, different systems that get integrated, right? And it's a little bit higher level than, you know you need testing. So basically, you know this testing can be, you know, different in different India, and you know it can be interdependent, and you know, and you know, while working together and again, you can, you can do use Spring Boot test. And there is something, you know, called functional testing, end to end testings, where you know you can use certain libraries, like cucumber, you know, to run this kind of, you know, test and write these test cases. And also, you know, there is regression testing, which is, you know, basically making sure whenever, you know, add basically new features to your existing features, so existing features doesn't get, you know, broke. And then, you know, there is something about performance testing, which is, you know, to test, you know, how your application performs, you know, under under a very low condition. And I have been, you know, involved in unit testing, integration testing, also, you know, regression testing, as well as the performance testing, as you know, our first can

Speaker 2 17:08

you describe what Dependency injection is with Java?

Speaker 1 17:14

All right? Yeah, sure, so. Dependency, you know, injection in Java, you know, is a, you know, is a pattern, right? So where we know, you give up the control of, you know, creating the object to somebody else, right? So if you are, you know, working with spring, spring container is the one who, you know, take cares of the dependency injection through, you know, inversion of control. So basically, you know, if a class needs, and you know, instance of an another class, you know, which is called dependency, then you know, instead of creating the class by, you know, you know, a developer, it gets, you know, it's, gets it from, then external source, that is, you know what. Basically, you know our dependency injection means, you Alex,

Unknown Speaker 18:12

yeah, let's see here.

Speaker 5 18:21

What would you say MPC is, and can you describe it? Some of the annotations Spring Boot uses?

Speaker 1 18:29

Oh, yeah, all right, for sure. So, you know, MVC, it stands for, you know, a model view and controller. So it is one of the, you know design pattern, and you know it's, you know, inviting our, you know, architecture into Model View, and you know controller. So you know it represents, basically, you know, it depends our data, and you know, business logic of the application. So you know, our view represents a UI component, while the you know user interface, we can call it view, or the you know user interface, and then, you know, your controller access the you know intermediary between you know model and also the view. So basically, you know, it separates the concerns of each component with a, you know, a distinct responsibility. And this particular thing, you know, promotes the, you know, reusability of the core. So basically, you know, just a design pattern for, you know, designing the web application. So, you know, some of the controllers that I, you know, work with Springboard on a reno regular basis include controller annotation, which is, you know, basically, to, you know, denote that a class is a controller for an NBC pattern. So there is also called, you know, Rest Controller annotation, which is also, you know, RESTful API. You know, class is a, you know, like, which is a combination of, you know, a controller, and also, you know, response body annotation, when it compared to the controller and the other, you know, method annotation for, you know, HTTP verbs, like, get, mapping, post mapping, put mapping, delete, mapping, patch, mapping, and, yeah, there, you know, there is an annotation for, you know, reading the data from your request, like, you know, PATH variable annotation. And also, you know, query pattern manipulation and, yeah, you know various service annotations, and, you know, repository annotations. So these are the, you know, different kinds of annotations that you know we have in springboard for, you know, MVC design. Thank you.

Unknown Speaker 20:42

All right. So

Speaker 3 20:45

regarding Spring Data, do you have hibernate or JPA experience, like, Could you describe what a session is or what a transaction is?

Speaker 1 20:56

All right? Yeah, sure. So you know, basically, in terms of Spring Data, you know, I have worked with both hibernate and also, you know, Spring Data, JPA, so basically, you know, session, you know, in Hibernate is basically, I will say it's a short life, you know, object, you know, that kind of, you know, represent the flow, or, you know, conversation between an application and the database, right? So basically, you know it is used to perform the crowd operations on, basically our persistence object. So we know we can create a session when, you know when we begin a transaction, and you know we close a session when the transaction is completed, right? So it's no longer needed after the transaction is completed. So you know, you will want a session when you want to, you know, manage the state of that persistence object. And you are basically, you know, tracking the changes that is, you know, made to them. And you know, you talk about the you know transaction. So you know transaction, in simple terms, is, as you know, sequence of operation performing a, you know, single unit of work. So if you you know, I mean, if we have to, you know, we're doing a payment, or, you know, increasing the balance of one account, and, you know, decreasing it from the other account. So in this case, you know it will be, you know, part of the same transaction to make the data consistent. So you know, if you are working with the spring, you can use a spring, you know, transactional annotation to automatically, you know, manage the all the transactions, basically within that particular method. Okay,

Unknown Speaker 22:44

so maybe for the sake of any

Unknown Speaker 22:48

forest community, just get down to the AWS section. Yeah, sounds good.

Speaker 4 22:55

Yeah. So maybe just generally, at first forest, could you maybe just describe your experience with AWS development? I know you mentioned several services in the beginning of this, but right, maybe elaborate,

Speaker 1 23:06

oh yeah, yeah, for sure. So, you know, so our application, you know, currently they live in, you know, AWS, we have, you know, work with, you know, monolithic app that is deployed on, you know, AWS EC two, but our modernized microservice, they are, you know, deployed on Docker containers to AWS ECS. So basically, we use AWS lambda for a notification service, and even based, you know, processing. So we use, you know, rds for, you know, Postgres SQL database. And we also leverage SQS and SNS, as you know, messaging system to those lambda triggers. So, you know, we use s3 bucket as well to, basically, you know, hold the static content for our front end Angular application, which is, you know, deployed onto the, you know, SG buckets, and you know, are exposed, you know, to a CVN. So, you know, I have been working with AWS pretty extensively. And you know, having to work with COVID Sending and also, you know, basically monitoring the, you know, logs and how do these applications on AWS. Thank you so much.

Unknown Speaker 24:33

So you mentioned lambda, right? Okay,

Speaker 1 24:36

yeah, so, you know, lambda is an event based service. You know, it's, you know, for serverless computing service, right? So, basically, you know, it just enables you to run a code without, you know, managing any kind of server. So it is a serverless and, you know, it allows you to, basically, you know, execute a code you know, you know, to, you know, response to a certain kind of event that is, you know, triggered now, you know, these events may be a messaging in certain queue. It might be a file drop to your in s3 bucket. It might be in, you know, API call through API gateway, or, you know, something else. So basically, you know, it is a serverless computer, which is, you know, very short lived with the, you know, maximum leave duration of 15 minutes. So you, you know you would want to use AWS lambda when you know you want to create, like a serverless web application. And you know you want to process some kind of, you know, data, like an event based, you know, architecture

Speaker 5 25:50

notes. Yeah. The next question, I think, is, what is CDK and how would you use it?

Speaker 1 26:06

See, yeah, so you know, when you You mean CDK,

Unknown Speaker 26:13

yeah, it's CDK.

Speaker 1 26:16

Okay, so, you know, basically, I will say it is an infrastructure as a core, right? And basically, you know, I have been working a lot with Cloud information templates for, basically, you know, deploying this application into our project. So, you know, we use Jenkins and, you know, we use CloudFormation templates to, basically, no provision these kind of applications.

Unknown Speaker 26:46

So

Speaker 5 26:47

instead of using like infrastructure as code, you have been, well, I guess in like Java, you've been just using straight up CloudFormation templates.

Speaker 1 26:57

So yeah, so I have been, you know, primarily, just, you know, directly working with CloudFormation templates, you know, rather than getting it installed? Okay, yeah, that's That's rough.

Unknown Speaker 27:12

It's so much easier. Yeah,

Speaker 3 27:17

are you familiar with what service is used for, seeing real time metrics in AWS,

Speaker 1 27:24

yeah. So, you know, I have been primarily working with CloudWatch metrics and logs, you know, for, you know, AWS monitor, but basically our application also, you know, externalizes, you know, using a tool called New Relic and Splunk, thank you,

Speaker 4 27:50

and then to kind of continue with those lines. So question around step functions, when should a step function be used for AWS?

Speaker 1 28:01

So, yeah, so, you know, basically, state functions are really useful. Basically, you know, whenever you want to, you know, or the state, like, you know, some kind of microservices, or, you know, there is a, you know, long running process, right? So, basically, you know, if your application, you know, comfort, comprises a multiple service that needed to be, you know, work together like a, you know, very particular order, you know, one after the other. Or, you know, need some kind of data. You know they need to pass between them. You know, then, you know, step function, can, you know, orchestrate, you know, instruction more reliably, and you know, if you have some kind of process that run, you know, for extended period of time, then you know, in order to, you know, manage the retries, or, you know, timeouts, or even even the, you know, parallel execution of the step function, can You know, handle, you know, these scenarios are very seamlessly and you know, so basically, you know, it just a, you know, serverless arm, you know, orchestrator service that lets you, you know, coordinate multiple AWS service into, you know, the serverless or workflow. So the only time I have been, you know, exposed to this, you know, step function was, you know, whenever we know we're doing a ETL, you know, all like a, you know, extract, transfer, load, process where, you know we had a flat file, you know, drop into the s3 bucket. And then we trigger, you know, this functions to basically load into a temporary table and then transforming it onto a you know, platform as a service.

Speaker 2 29:54

Thank you. You mentioned some work that you've done with KPIs. Can you give us a little bit of overview of what type of work you've done developing APIs, and what tools could you use for that?

Speaker 1 30:09

All right, yeah, sure. A lot of my you know, API development has been, you know, through Java and springboard, best you know, applications. So I have been, you know, working in, you know, with a lot of RESTful APIs. The most recent ones that I have, you know, completed, that you know, is maintenance API. This API, you know, all this, you know, Project exposes a couple of endpoints that you know is responsible for, you know, changing the user account and the profile preferences. So basically, you know, we expose multiple posts and getting points to, you know, make these kind of changes, and, you know, get to, you know, the status of those updates. So the another API that I remember, I, you know, I have worked with, basically, you know, includes an experience API. This is, you know, an API, which is, you know, used to reach it, you know, the different entitlements, you know, that the user has for, you know, basically, you know, like a wizards on the front end component. So, yeah, I mean, I will say, I have, you know, pretty Good experience working with you know, API specifically, API,

Unknown Speaker 31:37

development, to deal with. What makes an API restful?

Speaker 1 31:47

Oh, yeah, so you know, whenever you know we have an API, so it can be a restful when you know it means that it is, you know, stateless, right? Basically, what it you know, means is, I that whenever there is an API call from you know, client to a server, it contains all the information needed, you know, in order to process that request, right? So, basically, you know each request is treated, you know, independently, and you know, the request contains headers, basically parameters, you know, body that is, you know, responsible for, basically, you know, making sure that the request is valid. And, you know, contains all the informations and, you know, and so on, right? And basically, you know, this is a, you know, I mean, clear, like, separation, you know, between, like, where the client and the service is maintained, right? So there are also, you know, other constraints, like use of proper HTTP burps, like get or, you know, post, or, you know, delete or boot. So forgetting a resource a post is for, you know, basically, you know, creating a resource and put this for, you know, updating and similarly, you know, deleted for deleting, right? So these are the, you know, basic properties that you know, make our API restful. Thank you.

Speaker 3 33:16

I know we're kind of short on time. Is there anywhere you want me to jump to? Or, yeah, I was about

Speaker 2 33:20

to mention when we jumped down to the first part of the JavaScript. Okay.

Speaker 3 33:26

Can you describe your experience with view.js, react, Angular, or another modern JavaScript framework?

Speaker 1 33:34

Yeah, sure, yeah, of course. You know, my front end experience recently has been, you know, a lot of with, basically, you know, Angular. So in my current project as well, you know, we have used Angular in our, you know, a monorepo. And what I mean by that is, you know, you know, we, we have a micro, you know, front end that is, you know, deployed into a bigger application. And, you know, it is all wasted in a single repository. And, you know, I have been involved in creating, and, you know, develop, you know, develop, you know, developing one of the, you know, standalone application, basically, you know, it's called the App Catalog, where users can view, you know, the catalogs, basically, you know, available to them. And also, you know, make changes to them based on those entitlements that are, you know, talk to you, you know, earlier, which I mentioned you earlier, so, and I've also, you know, involved in creating, like, the search services for, you know, sharing the data. And also, you know, reasonable components like Excel sheet exporter and so on.

Speaker 4 34:49

Awesome. So I guess the next one is kind of along those lines. Still, as far as view React or Angular, I guess you're more familiar with Angular. So what are some elements of a good component design in your mind.

Speaker 1 35:05

Oh, yeah. So, basically, you know, for you, you mentioned for a good, you know, component design, right? So, no, okay, okay, so in front end, basically, we, we, we, you know, have various, you know, good component design practices, especially if I have to, you know, talk about Angular, you know, is that you know our component, it should be very single, right, responsible. And also, you know, it must, basically, you know, content, certain components that will, you know, basically have certain responsibility. And if you, you know, have multiple responsibilities. Basically, if you are trying to, you know, put it in together, you still, basically, you know, have, you know, consider splitting into a smaller and a more focused components. And, you know, basically trying to make these components reusable and, you know, accessible across the application. So, you know, so that, you know, basically, you can use the data bindings, like input and, you know, output to, you know, pass the data and the events. So this, you know, allows you to, you know, use the component to work on different contexts, right? So you can, you know, basically, also, you know, avoid any kind of you know logic in the templates. So basically, you know that our template becomes clean. And you know, also, you know, if you are done through the you know transformation with the component class itself, then you can, you know, implement some kind of lazy loading for any kind of you know feature model. So that you know, our application, you know, becomes responsive and you know, has lower initial load time. So along with that, you know, I will say, you should, you know, have a proper either handling, you know, to ensure we have, you know, proper alerting services in place with, I mean, with, you know, a proper notification to the users, and yeah, so, you know, those are some kind of good practices whenever you know you are designing an Angular component, and you know, you just want to make sure your components are clean, maintainable, and, you know, efficient and across, you know, available, You know, across the you know, application so you know, anybody you know joining a new team, you know, you know new like anybody joining our project can jump into it. We just, you know, looking at what's present in the components, and they can understand, well,

Speaker 2 37:40

switch to a really different topic. Can you give us experience? Which type of databases have you worked with?

Speaker 1 37:48

All right, yeah, so I have, you know, worked with relational databases that includes Oracle and Postgres SQL and, you know, non relational databases, you know, with like DynamoDB on AWS, just going to

Speaker 2 38:06

kind of continue with this line of questions in the work that you've done with databases, did you write your What did you write your own SQL? And what tool did you use for that,

Speaker 1 38:22

yeah? So, basically, you know, with, I mean, you know, we have our database visualizer, you know that we use, you know, process provides a busy admin, right? So that, yeah, I can use in order to, you know, write my SQL queries. So, yeah, you know, I have been involved in this, and, you know, logging into this database and writing queries, you know, on the visualizer. So I have, you know, also been involved in writing complex queries using joins in the sub queries, also to, you know, extract the data for, you know, basically our production subordinates, but you know, on a Java application, instead of, you know, directly writing queries. I have, you know, involved primarily with a query annotation in the Spring Data JPA, and you know, also using, you know, criteria builder API for you know, complex queries,

Speaker 2 39:30

maybe talk about the Git or Bitbucket experience,

Unknown Speaker 39:35

sure.

Speaker 5 39:37

Have you worked with Git or Bitbucket or any other code management tool.

Speaker 1 39:44

Yeah, my current project, you know, is stored in GitHub. And, you know, I have been working with Git, so basically, you know, I have been, you know, working with, I would say, get a lot in this experience. Quick question. So, basically, you know, over my experience, you know, a bit bucket is, you know, just your source control instead, you know, of, you know, Bitbucket I have used in GitHub, and you know, GitHub,

Speaker 2 40:24

okay, I'm asking because we're in the process of migrating from Bitbucket to GitHub, so currently we're using bitbuckets GitHub. So oh

Speaker 1 40:34

yeah, thanks for that information. But you know, and then all our, you know, SVN, so yeah, that's that we've

Unknown Speaker 40:42

been getting. That's that years ago. That's, that's old school.

Unknown Speaker 40:52

I wasn't sure where you want me to go. Just keep

Speaker 2 40:56

going. Let's talk about the code management process, and maybe we can go to build soft, okay,

Speaker 3 41:03

um, do you have experience within, like, an SDLC process or a code management process, and how do you guys use it?

Speaker 1 41:12

Ah, yeah. So, you know, basically, to talk about our code management we, you know, we as a developer, will have, you know, work on a feature brand. So you know the you know feature brands, name in the you know brand and the name of the you know, JIRA ticket assigned to us. So we have, you know, that kind of naming convention, which you know feature slash, you know ticket number. And then, you know, once we are done working on that, we push our core. So this coordinates to, you know, have a, you know, have to have a, you know, few reviews before you know, we get, you know, to develop brands. So you know what we have. You know our pull request. You know, up when you know we can choose to, you know, deploy our application to our development environment. So you know our you know, to do our development testing and you know, but once you know, we get proper code reviews and you know, we get it merged to, you know, our developments from, you know, our developer brands, we can basically, you know, deploy it to a QA environment where, you know, QB is where you know, testers can test. And once we feel, you know, comfortable with the changes and the you know, regression cycle is passed, we, you know, stays this out to, you know, release plans. And you know that release brands get, you know, released during the real estate

Unknown Speaker 42:42

and Okay, sounds good.

Speaker 4 42:45

So it sounds like, yeah. So, I guess one, one thing that we're wanting to know too, or getting to know, your experience with MuleSoft, if you've used that or not, and what your experience has been,

Speaker 1 42:58

okay, yeah. So, you know, I haven't, basically, you know, directly work with the mail stuff, but, you know, I have that, you know, knowledge about it. Basically, you know, it's an integration platform. Basically, it allows you to, you know, Connect application through APIs. But I will say, you know, I haven't directly worked on that, but if given any chance, you know, I will be, you know what I mean, I will be getting it, you know, to learn and, you know, understand it better and utilize it in my work.

Speaker 2 43:34

Last question, and then we give you a chance to ask questions, is, what is your experience gathering requirements from a business area. Tell me about a time when you had to be with a business partner and gain requirements for them. How would you go about doing that?

Speaker 1 43:52

Okay, so, so in terms of, you know, gathering the requirements. Basically, you know, I have been, you know, involved in requirement gathering whenever, you know, I'm on a task with, you know, requirements like, you know, we had, you know, and we asked, you know, to develop, you know, risk analysis tool for our existing services. So, you know, we basically, you know, we had a you know, business analyst from the, you know, from Risk Management Department, working, basically, you know, closely with us. So we, you know, whenever you know, I'm working in such scenario, I set up an initial meeting to talk through the you know, project goal and milestone and various expectation so basically, you know, it gives me a brief overview of, you know, what we are walking through and what we want to achieve, and what are the expectation. So, if you know, everything is match, you know, and you know, if I want to, you know, understand, what are the matches? Or, you know, what are the what are, what did it matches? So, you know, I make sure I have, you know, enough preparation before you know these meetings is you know as well as but during the you know discussion in that meeting, I'm actively, you know, understanding, and I'm actively listening to the details that has been, you know, mentioned in those meetings. And basically, you know me myself. I basically encourage the you know, business analyst to elaborate on the you know, specific requirements. And, you know, not keeping anything, you know, hanging, you know, if there are any other you know questions that I might have, so, you know, I will ask it to, you know, make sure I'm communicating properly. Basically, you know, on, you know, how these risk managements will be, I know, stored and displayed, and, you know, manipulated in, basically, in the front end applications. And, you know, I, you know, create a proper documentation for, you know, what we have gathered in these meetings. And, you know, understanding all these things. So we were, you know, we basically, you know, we're able to list, you know, both the functional requirement and the non functional requirement. And, you know, I would say it's a, you know, constant process, basically, you know, a review and, you know, validation from both the developer and the business end. But you know, that's how, you know, I like to go about, you know, gathering requirements from the business. Great,

Unknown Speaker 46:33

other questions that you want to ask.

Speaker 2 46:40

So you have questions for us. Um,

Speaker 1 46:45

so, yeah. So, basically, you know, first of all, I would like to thank you for, you know, giving me this opportunity. You know, I had, you know, wonderful time with you all answering this question. So, you know, my first question will be, you know, I wanted to, you know, ask, what kind of you know, opportunities you know, does you know, our developer Have you know in this team to, you know, grow. So what support does the you know company provide, basically, you know, for the professional development and growth.

Speaker 2 47:19

Can start with, sorry, start with them, and so good thing is a mutual reunion, especially last year, is a lot of emphasis on career growth and development. We have lots of learning paths. We use Udemy and pluralsoft opportunities to you know, everybody has a development plan. We want our analysts to continually learn and grow, and it's encouraged here. We have multiple different training tracks. We have we have heavy emphasis now on certifications, especially with things like AWS and yieldsoft, and we pay out small bonuses for attainment of the certifications. And I think we have you mentioned, we have a career path for the engineer, and as you work up through that career path, you can get more and more responsibilities. We also have an architect career path. So as you get more experience and you want to go down the architect path, we have a career path for that as well. So we've been the last few years, I would say we've been really trying to mature architecture practice here at mutual. It's something that's been a better focus. So

Speaker 1 48:42

yeah, yeah, you did. You did. Thank you for, you know, answering that question to me that you know, I would also, you know, like to know about, you know, some of the you know, challenges the you know, team is facing currently, if it will, you know, give some insights on that. You know, how

Unknown Speaker 49:01

would the other guys like, chime in. I'm

Speaker 3 49:08

trying to think so. I think one challenge that sometimes a lot of teams at mutual face is just like getting scope decisions on time. So that's currently something that we're working with on a particular project that we're working on, that we're we went in through our planning, our quarterly planning, and we still don't really know the scope of the project. So I think that can be a challenge that I, as the product owner, am doing my best to work through. But I feel like that can be a frustration for the team at times, but I try to mitigate that by just being as open and honest as I can about like, where we're at with the decisions and how I'm trying to push them forward. That's, I guess one thing that I see

Unknown Speaker 49:49

Alex or Amy have anything to add,

Speaker 4 49:53

yes, for I guess one thing that you know feel feels like, is the thing on our team is there's always, there's no shortage of work, and there's always new things going on. So I'm sure Alex can attest to that too. It's like, you know, Oh, you don't know Python. Well, guess what we're going to write something in Python? Oh, you don't know glue. Guess what we're going to do something with glue? So, yeah, it's, there's, yeah, there's definitely, if you want to learn and grow, there is definitely a lot of room to grow, learn and grow on our team, a lot of new technologies, getting into services we haven't used before, that kind of stuff. So,

Speaker 1 50:29

yeah, yeah, sounds good. Thank thank you for answering that. Thank you

Unknown Speaker 50:36

other questions here.

Speaker 1 50:38

I think I'm good for now. So So I again, you know, thank you for this opportunity, and you know, I'm looking forward for the next steps. Okay, so

Speaker 2 50:49

kind of on the next steps. We're doing several interviews here this week, so we should have an answer back to, back to the end of this week. So sure, thank you. Yeah, because there's

Unknown Speaker 51:03

this, this is the last step in the review process.

Speaker 1 51:08

That's good to know. Yeah, thank you. Thank you. Thank you. Thank you all. You have a good rest of the day, bye, bye, bye.

Diskshya-Lowes-Srv-first

Unknown Speaker 00:00

You doing

Speaker 1 00:11

in is pretty good experience In developing highly scalable

Unknown Speaker 00:37

retail sector you

Speaker 2 00:44

Sure, sure, sure. Well, let me start with my first name. My name is My full name is Victor, and I have been working as a software developer for six years now, and working on Java web applications. I'm utilizing, you know, framework like Spring, Hibernate and RESTful APIs for API development. And recently, you know, I have been involved in development of, you know, like orchestration platforms that have been involved in, you know, like designing, developing, you know, like microservices, and then deploying the microservices to AWS cloud. And I have also worked on stripper and cloud gateway. And you know, like my primary take a strike is Java packet, I would say I have experience working with AWS solutions like, is it your instance is situ and lambda, you know, like, I have a pretty good experience with Spring Boot test applications. Also on the front end side, I have experience with Angular. And on this inside, I have used journey and mokito. And currently, you know, I'm not, you know, like my team, you know, like my team consists of like five developers. We have product owner, Scrum Master, on a daily basis. You know, I'm involved in, like, mostly working on the back end, especially on Java code, you know, like helping with team, doing a code reviews, and also, like, you know, I also do as well. So, yeah, that's all about my

Unknown Speaker 02:35

introduction.

Unknown Speaker 02:39

Yeah, I am working as a contract. Can

Unknown Speaker 02:43

you talk about the

Unknown Speaker 02:47

project which you're working on right now?

Speaker 2 02:50

Sure, definitely. I'm currently working with jpmg kids. Like project is in like, orchestras platform. And, you know, like our team is has a lot of microservices we use, you know, like we have our own orchestration microservices, and so it's a, you know, like legacy metrics that we are migrating to Spring Cloud gateway. So basically, you know, my responsibilities was, it was, it is primarily involved in like, you know, writing functions and features factories for like, you know, like request response and manipulating those like, you know, then, like, orchestrating that request into like, multiple downstream that, you know, maybe on our own layout team or, you know, like it's, it's a microservices orchestrations like, kind of like, get well there, and it also adds the resiliency and it requests, You know, metadata. You know,

Unknown Speaker 04:01

it responds to the manipulation. Response,

Speaker 1 04:07

the manipulation sector orchestration. Just want to get like business

Unknown Speaker 04:16

context behind it.

Speaker 2 04:19

So, you know, since it is a banking app, we are, you know, like we need to have

Unknown Speaker 04:26

central point of flight for

Speaker 2 04:29

different downstream services that may have, you know, like that. Maybe you know, like our we have partner with like other vendors that deposit our partner. And then even the fur, like money movement, or anything you know, like that has to do with banking. It has a simple point, like simple point of entry, where you know, once you know you want to actually make a call, you know, like, once you actually make a call to like these services, we will first route and then, like, any request to this particular services, you know, like this particular services would have, like, all These services included with some kind of metadata that is attached to it, and based on that metadata, you know, like, before we say, before sending that request to the services, it applies these functions and it filters the factories associated with it. And after that, you know, it basically return the response. And so it's, it's just a simple gateway, but it adds a lot of features to it, like, you know, like, one of the feature is like token, tokenization services, where, you know, other is like removing the request reader, like, are like changing the request for it by adding some fields and so on. So, yeah, that's that's all about, which is

Unknown Speaker 06:14

the consolidation of multiple APs.

Speaker 1 06:20

So that's it. It a two way communication

Unknown Speaker 06:26

at the same time

Speaker 2 06:33

they are passing the information, kind of like API gateway. You can compare it to like API gateway, you know, like that. That is the main part of the applications where, you know, so it's a main part, and it's a spring form gateway that is based on API gateway, and it's, you know, like family, like all restaurant company, it does all the restaurant Application section,

Unknown Speaker 07:02

okay? Entry point for any external consumers, yeah, yeah,

Unknown Speaker 07:13

this is authenticated before

Unknown Speaker 07:16

entering into the banking

Speaker 2 07:18

system. Well, for like, for authentication, basically, you know, like, even before our layer, we have a you know identity gateway, where this identity in, basically, you know identity by I mean, like, it's a particular requester, right? So the way it works is, like, first there is a firewall layer with, you know, like, with the aid of this, and basically that's, that's where the first request gets in, and then it, it goes to the, you know, like, identity provider, which is sign of controller, which, like users can sign in and make sure that the user validate is valid. And then from there, it comes to the layer internally. And so this, this kind of is like how the you know, like security is added within our layer, we have stream security, plus we have best access control. So yeah, that's how we do the authentication

Unknown Speaker 08:36

orchestration is being involved.

Speaker 2 08:41

So it's like 800 API staunch

Unknown Speaker 08:52

orchestration API

Speaker 1 09:01

is not having, is there any errors or

Unknown Speaker 09:06

the failures

Unknown Speaker 09:09

exceptions for this administration?

Speaker 2 09:12

Well, for exceptions, it's where we handle like we have a global exception handler as a default, and, like, any kind of error from downstream gets written back. But for like, actual handling of the scenarios, we have circuit breaker in place, right? So we make sure that when, you know, like, certain services on the downstream fails, then we do not do I mean, like those APIs are like, you know, we have a fallback response which, like, after like certain threshold is made, and for like circuit breaker, it is one of our first resiliency in place to return our fallback response. And apart from that, we also have like, rate limiting in place, where, which, which is, you know, like very simple rate limiting, and it prevents, like, any, any kind of burst of traffic. And we also have, like, you know, like horizontally scaled application so that it can handle, like, large amount of rules. And we do have default error filter that runs in case of failure. So that's, that's kind of like, you know, for distressing where we we we handle large, large volume of data if there is larger error rates. So that's how we handled.

Unknown Speaker 11:00

So you tend to include at a single point in time,

Speaker 2 11:21

like our big transactions per second is like 1400 but, but we do have like 14 stands that is running at a time, and then, you know, basically, will work with, like spring club quickly, which is basically It handles, you know, like while running this request in like multi threaded manner, using, you know, like executor service internally. And so it's like highly resilience framework, which is basically established for, you know, like, for like, multiple requests. So that's how we

Speaker 1 12:07

can use to scale the calls and everything. But just curious, making a call to Orchestration so clearly,

Speaker 2 12:31

800 API, So like one time call is one API, right? So we have like 800 now that we onboard so, so you know, like client can call any of one of those 800 APIs. It's not like one request that gets to like 800 APIs, but it's like ability for clients to use that 800 APIs out of, out of 800 APIs. And like, if you can make, you know, like another API on the downstream, then client cannot use that feature unless it's onboarded, right? So, yeah,

Unknown Speaker 13:20

exposed to the clients each other, interface in a

Speaker 1 13:31

generalized way, exposing to our clients. Well,

Unknown Speaker 13:38

we're exposing, you know, like,

Speaker 2 13:42

like, let me put this in a different way, right? So it's like a layer, right? So, like, main reason that the project adjust is, it's because that it serves to be like, global point of entry that that adds a layer of protection for this endpoints that do critical banking operations, like maybe like money movements, maybe like deposits or anything related with that, car holders data. So, so what our applications does is actual API owner that do not need to worry about is where our clients can make the request from our one device your request is coming from. So you know, like our layer handles that security for making sure it's, it's, it's a valid device, and it's a valid applications are user making, making that call. So let's select today, you you create a new endpoints where you know you want to transfer money from one account to another. Then that way, you know it will be exposed. I mean, it will, it will come to our app, right? So you will onboard your app to us, and by providing that swagger documentations are showing what endpoints you want client to handle. So it will, it will be a one endpoint that will be exposed to the client that you know, that they can use, actually, that they can use. So

Unknown Speaker 15:33

that they can use exposed

Unknown Speaker 15:48

Kafka or

Unknown Speaker 15:50

in this project.

Speaker 2 15:55

So we have used Kafka for like it's we as Kafka for, like, internal notification system where, for, like, some kind of logging transmissions where, you know, so one of the, you know, use cases was for, for our applications was failing because of the notification service, which is going down. And it was a part of, like, part of, like, little older applications. We, we, you know, we basically migrated it to use Kafka topic to send the masses. And this consumer would read that masses and then send the email, which, will, you know, basically decouple that email message, and then you know that on the other part, we also have used big for like streaming, like any cardholder or any consumer related transactions data to the audit table. So for that we use Kafka topic so, so that was the two cases, two users that we use Kafka for in our projects using Kafka

Unknown Speaker 17:16

for normally,

Unknown Speaker 17:18

and we are Using

Unknown Speaker 17:21

some notification right API.

Unknown Speaker 17:39

Messages

Unknown Speaker 17:41

from the API do, well,

Speaker 2 17:46

I mean, like it would be, like it would be a process within our actual orchestrations, where it only gets involved, if you know, if like cardholder data, and it will be a part of like filter. Since we are using orchestral layer, we have a filter that you know, actually take care of, like publishing to the topic. So for like, if certain scenarios meet like, for like, certain filters run based on like, certain metadata or certain process That happens.

Speaker 3 18:23

So, yeah, well,

Speaker 2 18:40

for criteria, you know, it's, it's a kind of like,

Unknown Speaker 18:45

like in kind of manipulations to

Speaker 2 18:49

for like, our consumer data, our consumer transactions. And then, you know, like we, we basically enable our features. So it's, it's a public publish to the topic, and then it usually, you know, like configured during the onboarding of the request or during the processing where, you know, you manually involve the run method of that particular feature. So it's not like publishing based on criteria, but it's actually, enable a filter for, like, after certain time period. So

Unknown Speaker 19:32

how many?

Speaker 1 19:35

So what is the TPS of your API?

Unknown Speaker 19:40

What is that? I'm sorry.

Unknown Speaker 19:48

Well, it's like, like, you know,

Speaker 2 19:53

transaction per second is like, we can go up to like, 1400

Speaker 2 20:02

by always like it, you know, it depends maybe we we do like batch processing. I'm not sure much about it. So Kafka volumes are, you know, I have, I have not worked much on it, but, you know, like, that was one of the use cases I just worked on. So I'm not sure about it.

Unknown Speaker 20:41

Yeah, I've

Unknown Speaker 20:49

series

Unknown Speaker 20:55

application, multiple

Speaker 1 21:02

applications. There are some keys, a single application.

Speaker 2 21:12

You know, it's, it's a single application. So it's one application. But you know, like there are various flavor, like one would be for your Android applications, and one one would be for like iPhone applications, and one for like web applications, actually.

Speaker 2 21:41

So to configure, you know, basically we have three shells, which, I mean, like we have three Springboard applications, and we have like, three main classes, but it uses the same library, which is exactly like same thing. And the reason it is spread out is like, initially it was because of, like, you know, like we we are separating the functions, but everything in it is, you know, like, actually the same, same, except for like, certain application properties, fine. So whenever we have to add features. We just work on

Unknown Speaker 22:23

same company library actually.

Unknown Speaker 22:30

What is the build framework? Which you're using build framework, you mean

Unknown Speaker 22:37

it's a Maven

Speaker 2 22:44

I used, yeah, I have used Gradle, but I'm more into math.

Unknown Speaker 22:52

In the past, I have used Gradle,

Speaker 2 22:58

well in previous projects, I think

Unknown Speaker 23:04

with MasterCard, I think I have used it. I

Unknown Speaker 23:12

went to the APA.

Unknown Speaker 23:15

So,

Speaker 2 23:22

yeah, so give me a second.

Unknown Speaker 23:31

Yeah. Sure. Right to

Unknown Speaker 24:03

get into the logics and

Speaker 1 24:06

everything are the endpoints from top down bottom,

Unknown Speaker 24:13

from the starting to

Unknown Speaker 24:15

getting the data from any database and

Speaker 1 24:19

getting the result back in a very skeletal format.

Unknown Speaker 24:23

Sure, sure.

Unknown Speaker 24:31

So

Speaker 2 24:40

let's just think of like employee endpoints. So

Unknown Speaker 24:45

we'll just do

Unknown Speaker 24:53

another one, whatever you pick it in your profit, right? Okay, so let's say like we have

Speaker 2 25:08

downstream endpoint, right? And then

Speaker 2 25:27

and then we have the consumer money movement account

Unknown Speaker 25:32

downstream.

Unknown Speaker 25:34

And then now

Unknown Speaker 25:39

I can see That should be Fine, Okay, Okay,

Speaker 2 27:00

phone. So I think, yeah, this would be, like, very skilled, late enough Like how

Unknown Speaker 27:22

request works,

Unknown Speaker 27:33

looking for

Unknown Speaker 27:40

API endpoint actually I

Speaker 2 27:45

API so, you know, like, let me, let me, let me. Explain the flow. Okay, so your, you know, like your iPhone app, or all other app would, you know, basically be calling this change and cheers endpoint for like iPhone, wave, like accounts, like I have mentioned here. And so it will, you know, you know it will go through, like your firewall, and to go through your, you know, signing control. And you know, after that, you know, it comes to the Spring Cloud gateway, orchestrations, and after that, it comes to our layer. And basically what it does, I mean, like, what it does is like, it allows this service, which is, you know, like, which is basically ancillary service which leverages our database that stores our route, and then now, like, all that information for that, for that particular endpoint. So it basically, you know, like, grabs the routes associated, and then it does a matching, you know, like matching those informations and then route, and then, you know, like other request mapper parameter, like requestator and a version. And you know, if there is a match, then it extracts that information and but you know, like, there are, there are just most requires, requested ones, which is like the route, host and feature. And once, you know, like, once that is done, it will be ready to, like, execute the orchestrations. And the way it works is like it goes, you know, it goes to our common module, and it checks what filter needs to run, and it run, it drawn, it will run those filters from that common module, which is just a dependency on our shell applications. And then after all that function is wrong, it will run like Client module, which is basically our HTTP client module that makes the downstream calls. And then, you know, like, from there it goes to the like, actual API Zoners, and then it makes the gate calls to like that actual endpoints that that basically provides us for the dogs, after that response is, you know, return, then we do have, like, another, common modules, which will filters around that dependence. And if, you know, like, if there is like, any kind of response or manipulations are requested or not that that's an, you know, like over, that's, that's, you know, like, that's the overall flow. So, you know, like downstream partner is like onwarded to our orchestration. But you know, like client, you know, client only calls our applications directly, but, but you know they, they do not know that what the town is, so it is What it looks like. So, yeah, that's, that's an

Unknown Speaker 31:21

orchestration. API.

Unknown Speaker 31:24

I am giving an item number, you have to

Speaker 1 31:29

call an endpoint to get the entire item information, for example, information from an endpoint and

Unknown Speaker 31:44

information from another one

Speaker 1 31:51

is Which model, what are these things? The request? I mean, information, and this information

Unknown Speaker 32:18

basically the number

Speaker 1 32:22

I aggregate themselves

Unknown Speaker 32:30

back. So basically, for that,

Speaker 2 32:34

I mean, you need a Mars filter, right? So basically, here, come the

Unknown Speaker 32:54

sponsor is no,

Unknown Speaker 33:02

no, no, I'm just

Unknown Speaker 33:06

writing in somebody, someone

Unknown Speaker 33:12

that's fine.

Unknown Speaker 33:19

So looking,

Speaker 1 33:28

we can write the endpoint actually, okay. So,

Speaker 2 33:40

so we want to endpoint right, that that picks an item number, and then you want to merge that response from different endpoints, right? Okay, do

Unknown Speaker 34:18

BBB, so it will be a great item.

Speaker 1 34:22

I item,

Unknown Speaker 34:32

what is that? I'm sorry your voice is which method, right? That way it took

Speaker 2 35:15

so on our, you know, like actual server layer, we would have, like, some kind of mapping that stores

Unknown Speaker 35:25

item number to the API calls needed, and

Speaker 1 35:33

actually in blimination template, not the verbal warnings. So if you can write that, should be really nice. Okay, for example, starting point of this,

Unknown Speaker 35:48

which will be the starting point.

Speaker 2 35:55

So that would be a controller like so This would be a compressed controller. I

Unknown Speaker 36:10

But The

Unknown Speaker 36:27

I BBB, so, yeah, this would be a best controller

Speaker 2 37:58

trigger request, and then, you know, call the orchestration layer to get the item And and now we have The orchestration service, which is

Speaker 2 38:56

it's a service and make that first call a vehicle using

Unknown Speaker 39:07

the reactive Web. Well,

Speaker 2 39:12

yeah, I mean, we'll be using reactive link. I

Unknown Speaker 39:20

What is,

Speaker 2 39:24

well, so basically it is, it is a non blocking code, and it's asynchronous. So instead of, you know, like getting the item, you

Unknown Speaker 39:38

get a reactive stream. And it will be, you know,

Speaker 2 39:43

it will handle like high conference workload with like list number.

Unknown Speaker 39:50

So that's why there is no

Unknown Speaker 39:57

use of using. I

Unknown Speaker 40:05

it like

Speaker 2 40:07

for a simple one we might not but for our applications. So let's say you have like multiple APIs calls for Like, one items, and we will so

Speaker 4 40:35

in Okay, so it.

Speaker 2 40:59

So basically, you know, like, once we get that item details, then we we can combine them together, and then, you know, I think we can just shift in and then get The data. What is that?

Unknown Speaker 41:19

Okay, so for combinations

Unknown Speaker 41:25

again, we can Jay

Speaker 2 41:32

so for reactive Streaming, we can jab right i you?

Unknown Speaker 41:59

Come help me,

Unknown Speaker 42:22

home, get out.

Speaker 2 42:30

It will return so

Speaker 2 42:43

us well, like, I think I forget how you get the item from something like first or second item,

Unknown Speaker 43:02

behind, since,

Speaker 2 43:09

you know, like we have two data, right? So we have item detail one and item detail two, so that's why we're using that.

Speaker 5 43:24

It so for error

Speaker 2 43:33

forever. I mean, like you, you basically have a monop,

Unknown Speaker 43:40

like how you want to handle the errors.

Unknown Speaker 43:47

I mean, like you, you

Speaker 2 43:53

know, like, I mean, like you define, like, how you how you can handle the error.

Unknown Speaker 44:01

You can define. I

Speaker 2 44:08

because, yeah, thank you so much for your time, and I really appreciate talking to you first. You know, like, I would like to know, what are activities developer at low scale, apart from COVID or what?

Unknown Speaker 44:40

Little bit more activities,

Unknown Speaker 44:46

some examples,

Speaker 2 44:49

like like design or making. Like, you

Speaker 1 44:59

know, depends upon well as the stories We

Unknown Speaker 45:21

have a productivity of

Unknown Speaker 45:32

business So I can explain

Unknown Speaker 45:53

that sounds great.

Unknown Speaker 46:00

In.

Unknown Speaker 46:03

So, yeah,

Unknown Speaker 46:26

thank you so much for your time.

Unknown Speaker 46:28

Appreciate it.

Dolma OneNetwork- Srv-Final Offered

Speaker 1 00:05

Pretty good. I've been

Unknown Speaker 00:16

working with them for 18 years now.

Unknown Speaker 00:19

So the company was

Speaker 1 00:23

recently working with them. I lead a logistics business unit for a network responsible for the logistics side of implementation. But I did start my journey with the network as a developer, so I have some background of the development as well. And be talking to you. So how about your search?

Speaker 2 00:47

Yeah, so talking about myself, I've been working as a dhaba group for the past six years now, primarily work on the web application, and then utilizing frameworks like screen mine. And then I've been using hibernate and web services that been primarily involved in developing company services using screenbook, and I have experience on Angular and company based system. I have also worked with AWS and like AWS, EC ECS, SQS, SMS and lambdas, I have been involved in writing like unique test cases. Currently, I am on a scout team. My team consists of five developers, one product owner and a Scrum Masters. And on daily basis, I'm involved mostly working in the back end, data, code developing, and I help with the team, with the code reviews and also with product support as well.

Speaker 1 02:00

Okay, so CEO working with all the medium freight line, okay, so what do you do? What's the project about?

Speaker 2 02:13

So what basically involved is with the micro services migration. So we have a monolithic applications that we are stripping out in different service like in developing these surveys and a couple of surveys that are recently developed in maintenance of API, and then experience with API maintenance. So API primarily used the Change User Profile preferences and then experience. API is used to grab the entitlement for the users that make them available to view different widgets and components on the front end. So I work basically with Spring Boot and Java to develop a microservices and also design them and help develop them and deploy them as well. So

Unknown Speaker 03:15

it's mostly on the back end technology, the

Unknown Speaker 03:20

back end like use Oracle

Speaker 1 03:24

Oracle database at all, or just the microservices APL of litigation.

Speaker 2 03:31

So I have worked with databases in the past, and I have worked with Oracle databases, but in recent project in microservice, I have been involved with the postgres SQL and AWS. So one of the services I worked with, a dyno DV on

Speaker 1 03:54

AWS background, of the supply chain, logistics, basically, and all those different parties, you're mostly on the development side.

Speaker 2 04:11

So I am primarily on the development side, and I have been working with a clients to on how to work the user profile, or how it should look like. So I have sometimes more requirements for adding those components into our application, but I'm more into the development side

Speaker 1 04:41

mine. I was just talking to you and paid for interest on what so do you have a background on this role? What are the roles and responsibility going to be? What's the project all about? Do you have any questions about the company itself on your roles? Um.

Speaker 2 05:02

So I did talk with previous engineers, and then gave a brief overview on what role are pretty comfortable with. So he explained it quite well.

Speaker 1 05:19

Okay, okay, so for my I think let me also give you some few more details about the roles. So one network and I will talk about one network, because as we are working with the other team, we are indicating more. So the processes may change, but I will describe the current processes that we are following as part of the current processes. We have product teams that work on the development of our platform, supply chain applications right then. We have industry teams who are responsible for implementing those solutions and delivering it to the customers. Now, as part of our implementation, some of the other things that we need to develop, we may even enhance the product or extend the product to address the customer requirements. That's where the industry teams come in, and we, just like our product teams are supportive we have within the industry teams, we have subject matter experts, we have developers, Production Support Engineers, and QA, ESR, all those roles so this, and then we have different directors As part of this character we're looking for a senior or mid level developer who can work with guidance and work on those projects, and then the product is actually not fixed. The product can keep on changing. So sometimes you may have to work on a world project, may have to work on a different project. So it requires some kind of context switching. Is not huge, but you may have to do some context switching,

Unknown Speaker 07:10

and then you may

Speaker 1 07:14

historically within the company, the people who are working in multiple hands at a time, sometimes

Unknown Speaker 07:22

developers, sometimes

Speaker 1 07:25

operations to figure out issues with their queries, they might be able to make a production software to deeper the problem, right? So at times, all those different, different roles and responsibilities, if that's something that you are interested in. Do you have any reservations?

Speaker 2 07:46

So no, in my experience, I have worked with development, but I have worked with product support as well, and I have also been the subject matter expert of the applications that we own, we can have applications, so I'm pretty comfortable. And yeah, we can bring the features from different components. So yeah, I'm comfortable with the requirement. Okay.

Speaker 1 08:19

How are the trials actually done sometimes you may have to

Unknown Speaker 08:25

travel, because a few days here

Speaker 2 08:30

and there, I am open to travel. I have worked with AWS cloud and also work with accountability engineer. And I have experience in the past, but I'm pretty experienced working with all the services required for the application development, either on the front end or the back ends. So our legacy application was deployed on AWS es, and then our modern applications on the AWS ECS, we have been working with gateways like

Unknown Speaker 09:15

RGS, okay,

Speaker 1 09:21

we have our own cloud. We use our own cloud where we deploy our application. More and more we are going into the OCI or the cloud, but for instance, we can run it into Azure AWS. There are some customers who run on the interview side, not mostly in Azure, but in the BVR that most of their solution runs on the Azure, so plenty for prosperity to work on those different different areas. I don't have to take me into questions. Is there anything else that I can help you answer or.

Unknown Speaker 10:00

Anything about the company, anything that you could know.

Speaker 2 10:05

So, yeah, so I would like to know, like, how do you measure the success of an engineer that is joining this team right now? So what do you think would be

Unknown Speaker 10:21

the developer really needs to

Speaker 1 10:25

absolutely so there are multiple ways that we measure any mutual performance,

Unknown Speaker 10:31

because in your

Unknown Speaker 10:34

role as a flexible are you in working with

Speaker 1 10:40

your quality of development. If you're developing something, when it goes to the testing, how many bugs that are coming in, or it's ready to look it's no need to worry about it. And there are developers who are like those, right? Everyone understands they are. You will have this as a software idea, the quality of the work that is one of the important things that you measure. The second is means your communication with the peers you're able to mentor them right asking for means problems that you have identified escalating in time at the right point in time, right rather than waiting figuring out that bubble up so those things are measured,

Unknown Speaker 11:29

then at that initial level, I think

Speaker 1 11:33

the limitations, but I think it's more true for the quality of birth.

Speaker 1 11:47

Anything you are local here in Dallas, yes, okay,

Unknown Speaker 11:58

our offices are this is very near to bilayer.

Unknown Speaker 12:03

It's on 635 and you're not required

Speaker 1 12:07

to come to office, because I don't come to this once in a while. I don't have

Unknown Speaker 12:19

much.

Speaker 1 12:21

I understand you have strong technical skills. All those are taken care of. I'm just here to ask if you have any questions for me, because I can help answer.

Unknown Speaker 12:34

I also would like to know that,

Speaker 2 12:39

what are the growth opportunities that the company provides for engineers or anyone that are involved in the team for a long term goal and move into architect for like, a short term goal and competition.

Speaker 1 12:55

Okay, that's a great question, and if you see I'll give you my examples. So I joined as a developer, became a development lead, then the development manager, then I actually went into a different role on the research and development side. So I led the research and development team, and I went back to the pre sales team. I led the pre sales team for a while, into the business unit team, and then I let I'm actually still leading a business unit as vice president. So qualities, depending on we have a lot of people in my team. So for them, let's say their main goal, or the main interest, is on individual contributor. So they want to focus on one project at a time, work with their team and move on. So they're pretty good at what they do. So in viewing under one of the good thing is, it depends on where you want to grow, and you have grown the qualities as a manager role as well as an individual contributor role, right? So you have a positive role if you want to grow into an individual contributor role. And it could be your technical lead, you can become an architect. Then there are different targeted levels, staff, engineers, Principal Engineer and office if you go into the management side, pile on the technical, different opportunities you can grow as a DPM, technical product manager, development manager, Development Director, as a VP. So I think plenty of growth approximation. We have close to 8000 employees now, so pretty large company,

Speaker 1 14:56

right? Don't have anything else,

Speaker 1 15:01

I'll just give that feedback, and someone will reach out to you to gage the next steps. Thank you. Thank you for your time. It was great talking to you. Thank you.

Dolma OneNetwork- Srv-Second Offered

Speaker 1 00:00

For past six years now, and during this time, I have primarily worked on database applications and framework like spring hybrid and Vespa wave services for the API development I have, I have, like, good experience working with micro services like building with the Spring Boot, and have been involved in designing them so collecting and non functional requirement. And recently, I've been working on our discussion Spring Boot, and for the cloud service, I have worked with AWS, and then for the database, I have worked with people. And for the front end, I have worked with Angular and company based system. Currently I am on a scrum team, and team consists of like five developers. We are we have product owners, Scrum Masters, and it is conducted on daily basis. So I have been mostly involved on back end exam and then help with The team with the coke sector as well.

Unknown Speaker 01:29

Okay

Unknown Speaker 01:42

to basically, I was working with a client

Unknown Speaker 02:02

for the for the

Speaker 1 02:05

News. Okay, so?

Unknown Speaker 02:32

Application,

Unknown Speaker 02:36

particularly

Speaker 2 02:42

in particularly services. So can you explain what published food web services used and what was the

Unknown Speaker 02:59

development

Unknown Speaker 03:02

I have been

Speaker 1 03:04

signed with the task to create a microservices with the first API was a microservice called the maintenance API, and this service was finally used to make changes to the profile reference. So to make the additional changes you want to make the profile in a overdraft protection or enroll in the new plan, you would use this micro service. So I was also involved in designing their logging through the Bing boot based application? So I was involved in provisioning the service agent, Bs ECS, for deploying the application, for the database. We use the MySQL as our back end database for this. Apart from this, we have experienced API that second microservice using design a developed basically responsible for grabbing a entitle link for the deploy, basically control the entitlement components and that to be deployed in the front side. So those are the two microservices that I was involved and then worked with the microservices as well, including resource service and also on the network side. I I was pretty comfortable

Unknown Speaker 04:54

with working with

Speaker 2 04:57

you, microservice, designing the microservice like what was the process to build the data streams and then make sure that the process is consistently running.

Speaker 1 05:15

What I basically do is create a data stream, basically have a particular topic, and producers and consumer set up, right? So basically, producers use this to set a send data and then use the and those particular topics, we will basically have a standardized data using JSON, and whenever we pass those records to the topic, producers are configured with the proper serialization and Bootstrap servers and basically sent back to The consumers. So we have consumer groups, where we allow multiple consumers to read from the different different partitions of the same topics for the higher throughput. And these consumers are basically set up decentralization of the group ID, where different data gets decentralized. So whenever there is an error within the data processing consumer, we have mechanism with the exponential lockout. And if there is a failure, even after that, we send it to the basically process it in the later time. So that's how making sure it's

Unknown Speaker 06:49

visible. This particular was more related

Unknown Speaker 07:02

to the videos, right? So

Unknown Speaker 07:05

what was the like normal

Speaker 2 07:08

architecture? Are you using the existing, existing libraries, or you have built the custom libraries for managing the video streaming platform?

Speaker 1 07:21

So also for the so basically we we work through the content delivery network. Basically it was where the delivering the video content basically were used to using the AWS. So it's done through AWS platform. We needed to restart the tickets and it would add any kind of Prop like DNS, right? So whenever a user request a video, the student starts from the nearest locations, and then the video ingestions and process done through the API gateway, which enables the creations of the store and larger photos get uploaded, please, we use and make it to the PostgreSQL and DynamoDB for the related database and needed. So basically, the API on the Gateway and AWS make the best phone calls, which enforces the

Unknown Speaker 08:34

which enforces the

Speaker 1 08:37

basically makes the API call behind and then the streaming before video at once. So if you would load it maybe like 2030 frames, and then it goes like to the 10 times, and it goes to the next 20 or 30 frames. So we have fully utilized the class solution using the platform, dr, AWS, different stores And then API gateway. So basically, for the communication,

Unknown Speaker 09:22

what? For the first

Unknown Speaker 09:26

product which you're working on.

Speaker 1 09:33

So, yeah, first I was working in a high injection process we had at care. First, we had multiple clients based on a different states, and then states would provide a standardized format, as for the RESTful API, but some states would still work with just non standardized file. So we work on loading and market non standardized data into a standardized format. So for these, the first applications was a standalone data applications, so where we would load the file and then create the content containerizations And then second, the application was with the AWS that used the lambda and process the JSON and then convert the Healthcare Center data. And for those, there were two parts, and the outcome of AWS lambda

Unknown Speaker 10:44

will just

Speaker 1 10:46

standardize the JSON type and then expose the microservice endpoints on for the patients, data expectations of the explanations of The benefits, data and coverage, data and network, bheda. So we were in a very high, high level architecture. We use the custom library called in free client. So basically, these were developed, it helps to develop the mind of service. It was based on Spring Boot. And these were the services were deployed using the AWS ECS for containerized a lot of

Unknown Speaker 11:45

which

Unknown Speaker 11:48

render

Speaker 1 12:09

so I have worked post risk And then Oracle, and also have worked with a dynaria. Okay,

Speaker 2 12:26

so from the front end perspective, can you give the example of the front end development you have

Unknown Speaker 12:38

participated front

Unknown Speaker 12:41

end, middle? Simultaneously.

Speaker 1 12:47

So I have, in recent project, I have worked on a catalog component to display the catalogs of the freight and items under the shipments. So this also basically a provided a create a new shipment process. So this catalog of the service that I, that we created, we use the Angular and creator standover applications. Basically, this will initially during papers tracking of the application, and then it would call the back end and catalog that was available for the users to view based on the user entitlements. And based on these entitlements, the right catalog would be included. So to edit this catalog items through the Self Service protocol, we had a self service component, catalog component that was responsible for each of these the service

Unknown Speaker 14:00

component would basically be coming back

Speaker 1 14:05

and make the HTTP like a post request for the HTTP client, so to make the changes, or to get people of The data, the data was from the from a get that would be cached using the session storage. So on the back end side, we would basically get the request from the from these posts and cost and serve them using the Spring Boot based on Spring Boot based app, like the service,

Unknown Speaker 14:45

okay, you

Speaker 2 14:53

normally work on the development side, or you have also worked with the customers for the implementation and support.

Unknown Speaker 15:03

So I

Speaker 1 15:06

more heavily, primarily work on the intergovernment side of the services, the teams with the five other development developers in their group in the multi production so I get assigned products, production support duties.

Speaker 2 15:38

You can get like, different issue, right? Because the production support

Speaker 2 15:46

like very difficult to identify what was the problem and everything what

Unknown Speaker 15:54

approach you able to

Speaker 2 15:58

solve it effectively. So can you explain me like,

Unknown Speaker 16:12

basically,

Unknown Speaker 16:16

in a recent, recent,

Speaker 1 16:23

change, we pushed our SRE team, so they pushed out a changes to update our Amazon Redux to AI. So basically, the Amazon Redux two version had a inbuilt capability to theoretically clean the temporary folder. So our application basically leverage on the Cloud Config Server. So which we use the dynamic configuration? Config solutions. So basically, this Config Server was pulled by our main applications to grab these configs. So I was cased one day to check for the errors right. And then, because the client, we were getting 500 500 error code path, and it seems to be coming from our back end. So doing further analysis on the strong logs and the new valid we were able to pinpoint that the 500 was basically originated from the our applications. And looking at the logs, I saw that a Config Server call was failing to make the order results that was basically did that moment was the restart our application and was

Unknown Speaker 18:01

doing a

Speaker 1 18:04

to reload the resources, and this basically helped in the solve the issue, and for the more, for the more people, I think we found that the update in Amazon Linux to cause a folder to delete after 10 days, and then screencap and keep use that folder. Initially we were to come out with the solutions not to use the temporary folders so that so in my recent project, the issue we faced, and we were severe. It was severely and we were offset. It affected. A lot of customers

Unknown Speaker 18:56

when you're working with different

Speaker 2 19:01

developments for completing the development class, what are the different methods you have used, and what is the pros and cons based on Your experience, on those

Speaker 1 19:20

I have been primarily working with a scrum Agile methodology, where we have two weeks of sprints with a dedicated task in France. So we have also worked with a Kanban board, where we basically follow the continuous delivery approach, where we just pull as capability comes, right? And I have worked with both, but in my opinion, I prefer agile board because, because it is really bounded context and and the task, and then specific time frame where you can complete the certain task. And it makes a makes it more structured, and working on the structures. So also the way Kanban is more flexible depending on what capacity

Speaker 3 20:31

you have, different

Speaker 2 20:38

tools like use, capturing the issues, fixing it and Delivering

Unknown Speaker 20:51

it

Unknown Speaker 20:57

between

Speaker 1 21:00

so but one of my past projects used a GitHub as our source control and for the continuously we Use teaching things,

Unknown Speaker 21:22

consumer, deployment,

Unknown Speaker 21:41

regular, deployment

Unknown Speaker 21:51

execution.

Speaker 1 21:59

So we use the engine files for the standard project, but also use like a groupie based software project where we create a like speed job that sets up the pipeline for us. So I have experience building the pipeline Also using the using other

Unknown Speaker 22:27

pipelines. Okay,

Unknown Speaker 22:49

infrastructure.

Unknown Speaker 22:54

So I have

Speaker 1 22:57

experience with AWS, mostly

Unknown Speaker 23:20

it's breaking Real

Unknown Speaker 23:32

bad, sorry, going to the

Speaker 1 23:45

so I have work with work on clouds in my previous work, but then we also had a open shift container or deploying For our applications. So I have experience in both our application and also.

Unknown Speaker 24:16

You may have

Speaker 1 24:33

sorry, maybe it's my internet or something

Speaker 2 24:40

you must saying.

Speaker 1 24:58

So actually, this is my personal laptop. I don't have any quotes in here. I can solve any coding silences you might have for me. So I don't have a data project in my personal laptop.

Speaker 2 25:20

Anticipate. So

Speaker 1 25:30

I have not actually contributed in any open source project yet, but I have identified some issue and then commented on so not in our like internal projects. So I have this issue on what certain issues exist. So we had, we had an issue where Secretary river library did a matching for the second there was a anti pattern. So I had to, so I had to point out that to create a for creating a new conversions of some library, great.

Speaker 2 26:33

So for the API definitions,

Unknown Speaker 26:40

will API mechanisms.

Speaker 1 26:44

So we use, we use swagger, basically, to define the API properties to be to the exposed endpoints and then make it available for the users. So you can use spring Fox dependencies, so with two it makes available for exposing your endpoints through an endpoint. What are the

Speaker 2 27:20

different data aspects you can use? What are the different data sets you can use

Unknown Speaker 27:30

when you're working with integration

Unknown Speaker 27:34

different data

Unknown Speaker 27:42

types? The

Speaker 1 27:47

time, really, I have worked with patients just because the fact that it is lightweight and easy to read. And then I have also worked with XML pattern once or twice, I would say the XML is more rigid and it is difficult to understand because of the XML test mutation. So yeah, so these are the primary so the I would say JSON and XML. You may

Unknown Speaker 28:31

so basically,

Unknown Speaker 28:33

in the project,

Speaker 1 28:36

we have a first instituted CSV file. We basically had a flat file that contained rows of data in a certain structure. So we have worked with CSV file ingestion and then processing them. So this was an API. It was a standalone application that was used to read the data file we have

Unknown Speaker 29:18

the global

Unknown Speaker 29:22

logistics where logistics

Unknown Speaker 29:30

services infrastructure system

Speaker 2 29:36

integration is very complex system we Have depending on Create remapping to the

Unknown Speaker 29:56

customers

Unknown Speaker 29:58

or the customers

Unknown Speaker 30:03

or the service providers,

Speaker 2 30:06

from The service provider standard, Ada, definitions For

Unknown Speaker 30:13

the different shipments.

Speaker 2 30:20

Business, API,

Unknown Speaker 30:40

yeah,

Unknown Speaker 30:51

we Have the

Unknown Speaker 31:15

customer

Unknown Speaker 31:20

claims,

Unknown Speaker 31:32

triggered

Unknown Speaker 31:43

the complex

Unknown Speaker 31:49

infrastructure

Unknown Speaker 31:56

and

Speaker 2 32:02

get the interesting. So it sounds really

Speaker 1 32:20

good to do. Good to me, like it sounds like a good opportunity. And I feel like if there's something that I could contribute and learn more about it, or be driven to, like solving complex challenges, and then take on my challenges and come up with fishing solutions. So yeah, thank you for the information. It sounds really interesting, and then yeah, the infrastructure is also really laid out.

Speaker 2 33:00

What is different position or different

Unknown Speaker 33:08

opportunity?

Speaker 1 33:10

So my contract is coming to an end. I'm currently in a contracting role. That's why I'm looking for a new opportunity.

Unknown Speaker 33:21

So so

Unknown Speaker 33:31

like expectations,

Unknown Speaker 34:00

perspective,

Speaker 1 34:06

so what I would say, of some of the expectations would be like a collaborative environment, so where I can have a team to lead on at certain times, and also I don't always, I don't want always asking teammate for help or some kind of collaboration work would be really appreciated where team discusses and try to come up with unique solution. And then, apart from that, teamwork I'm looking for somewhere actually grow so in the long term, like I would want some time to learn more about architecture, or, like the role where I will grow my skills and then move up with other position and design systems also provide a solution. So, yeah, these are the primary ones that I would say that I'm looking for primary new things in The next eight.

Unknown Speaker 35:41

So the Other

Unknown Speaker 35:59

potential,

Speaker 4 36:09

so We had so

Speaker 1 36:40

COVID. So, I mean, I have worked on the future. I have, like, I have worked on products and also on the Customer

Unknown Speaker 37:15

Support.

Unknown Speaker 37:31

Again, requirements,

Unknown Speaker 37:38

testing,

Speaker 1 37:57

is actually cheating, But I go back

Unknown Speaker 38:07

Yes

Unknown Speaker 38:13

about the

Unknown Speaker 38:42

ground development

Speaker 1 38:44

side I would Put myself like in the simple sprint.

Speaker 2 39:01

And can

Unknown Speaker 39:07

sit in the jail.

Speaker 2 39:20

From the front end for React, I haven't worked

Unknown Speaker 39:39

with React yet.

Unknown Speaker 39:41

From that side project, but

Speaker 1 39:46

professionally, I haven't done it, but I have only done with JSP template using the JavaScript projects and also Angular. I'm more familiar with Angular recent projects, and to rate it myself, like the JSP template wide, I would say around five or six, because a little bit I need a refresher. And for the angular side, I would rate myself around seven and

Speaker 1 40:47

so I've been primarily employing these services myself, like creating Docker file for the containers and writing the Jenkins files for the developments. Clicks on Jenkins. So I would read myself somewhere around seven, seven and a half, and the deployment, so we leverage the top cloud formation, like templates initially and these deployments. So I'm pretty comfortable with this concept. I

Unknown Speaker 42:00

stuff.

Speaker 1 42:13

So I have been involved with SSH instances, gathering matrix, like logs file. So I have also, I have also worked with creating those logs and file or specific keywords, and then I have connected this remote host using a hurry. So I'm pretty comfortable with it, but I'm not sure all these concepts, and I'm still comfortable so rating it would be around six or seven, but I have been involved myself in studying the

Unknown Speaker 43:06

Splunk,

Unknown Speaker 43:12

like a Splunk

Speaker 1 43:15

for water agents and then like new redications to string those two streamlines or double our metrics to a deep dashboard

Speaker 2 43:39

content in data,

Speaker 1 43:45

so I would date around seven

Speaker 4 43:54

and a half

Speaker 2 44:00

out of 10. So

Speaker 1 44:12

I'm answering based on my experience. So I've been working in this like hands on Amazon experience, so I'm pretty much comfortable on those techniques and technology.

Unknown Speaker 44:50

And

Unknown Speaker 44:54

supporting team members as

Unknown Speaker 45:20

everything Like

Unknown Speaker 45:26

that scenario that

Unknown Speaker 45:55

that.

Speaker 1 46:00

So I've been in a situation where all the requirements were not clear and all the informations were not provided in the study. So just the general idea what needs to be done in this scenario, I would say this kind of situations, I have gathered all the requirements and then come up with solution and then present through it. Then I have worked independently as Well as people,

Speaker 5 46:42

Both so the

Unknown Speaker 47:34

When there are

Speaker 1 47:53

sure, yeah. So whenever I have questions, I would document this question and then basically reach out To the right person to collect the answers in the

Unknown Speaker 48:32

hours. So those kinds

Speaker 1 48:59

I have worked on consultation spaces so where I have to support offshore hours also and then on the weekends, I am comfortable with The rotation

Speaker 2 49:18

and space. Industry to

Unknown Speaker 49:39

drugs.

Speaker 2 49:55

Also provide.

Speaker 1 50:12

This is IV for assessment, and Then I have

Speaker 2 50:20

first second. Thank

Speaker 1 50:25

you.

Unknown Speaker 50:54

Basic content,

Mahim- State Farm-SRV-Final-Offered

Speaker 1 00:00

Working with AWS, AWS engineering teams for the last seven years. Oh, great. Okay,

Speaker 2 00:09

good. And lead software engineer been working on migrating a legacy application onto AWS, can I say farm for seven years now.

Unknown Speaker 00:19

Oh, wow. Okay, sounds good.

Speaker 1 00:22

All right, would you like to tell us a little bit about yourself and your background and maybe why you were interested in this position? Sure, sure. So

Speaker 3 00:29

my name is Mahim, and I've been working as a Java developer for about five years now, and primarily I work on Java based web applications, where I utilize frameworks like Spring, Hibernate and rest web services for mainly for API development. And recently, I've been working a lot with microservices, where I design and develop the whole REST APIs and and then I deploy all those to the cloud services, mainly AWS. So the primary cloud vendor is AWS. And in that I usually work with AWS EC two, with lambdas and then s3 buckets. And currently I am also on my scrum team. And the whole team consists of around five developers, and we have a couple of testers, and we have also, we also have, like, a product owner and a scrum master. So on a daily basis, I am more involved with the back end Java code and where I help my team with with code reviews and also with production support duties. And the role that I'm applying for here, the AWS engineer is, I think it's like a good fit for me, and I think it is very interesting to work at State Farm, and I think it would also help me expand my knowledge on the whole AWS infrastructure. And I also really wish to put my software engineering skills to the next level. And I think so. I think so. I think it's a very interesting role that I would like to be a part of.

Speaker 1 02:29

Sounds good. And I think I saw on your resume, you're in Chantilly, Virginia. Is that right? That's correct. Yes, okay. And the reason we like to ask is the way that State Farm makes our offers is based on your physical location to our buildings or our corporate locations. So obviously that's beyond the 50 miles of any of our locations. So if we do extend an offer, it'll be a, excuse me, it'll be a remote position, which may still mean that you might get requested to travel, but State Farm would reimburse for that travel and things. So

Speaker 3 03:04

okay, sounds, sounds very good. Okay. And just to

Speaker 1 03:07

let you know, so Jose and I are part of a hiring committee, so we're actually aware of multiple needs at State Farm. So as we go through the interview, we are probably going to ask you lots of follow ups. It's to help us better get to know where your current skills are, where you might have preferences. Like, I know you mentioned already having a lot with the back end, so some things like that, but it's really just to help us get a better understanding of where you fit with our openings. So

Speaker 3 03:33

sure, sure, like sounds, sounds good. Okay? And

Speaker 1 03:37

then we'll make sure you've got plenty of time towards the end to ask us anything about State Farm or the position as best we can too.

Unknown Speaker 03:45

Get started. Yes, definitely. All

Speaker 1 03:47

right, so first question for you, solving a problem often necessitates evaluation of alternate solutions. Can you give us an example of a time when you actively defined several solutions to a single problem? Okay, sure, so.

Speaker 3 04:09

So, I think like so. I think one of the times that I can think of was mainly working on a on a proof of concept for solving a tech backlog. So, so it was more on a on a enterprise tech backlog system. So it was like, so it was basically meant that everybody in the organization had to work on this backlog together. So so I, I took charge on, on coming up with a solution for that problem. So, so the backlog was to integrate a streaming engine. So basically the streaming engine,

Unknown Speaker 04:51

it meant that we had to, like, aggregate

Speaker 3 04:55

any and all kind of logs that that we touched with the with a PII, right? So, so in so since, since we're an insurance company, so we work with a lot of health data and with personal identifiable data. So so any kind of interaction with those data needed to be, needed to be logged for, for audited, for auditing purposes. So, so we could not log it into Splunk or any other places, but, but through, uh, through a snowflake table. So for this purpose, while, while we were working on the proof concept, I came up with, mainly, like, two solutions. So one was to directly incorporate the SDK provided by our cyber team in the micro services and configure it according to the needs of the application. So application the writer. So it was like a helper function with clients and and everything was configured to basically make that that particular call to write to the to the table. So, so the alternate solution was that the one I provided was, it was to basically create an abstract service that acted on a on a on a microservice. So that was it took a request from any client, and then, based on the metadata that is configured to the to ingest that data, the messages, the incoming requests were in a, in some sense, like a in a batch. It was sent out to the to the snowflake table. So so I think the first was a little more direct and and the teams, they did not have to worry about adding clients to the to this API, or onboarding their schema. And the second one was more involved on onboarding their data to to our service and to and just making a REST call, right? So so I think so. I think this is one of those time I came across like a multiple, multiple ways to solve a particular problem. So later on, we went with the second approach that I, that I suggested, which was to create the micro service, because it was more it was more robust, and it helped multiple teams interact with the same API, so, which reduced the whole effort for the developers.

Speaker 1 07:26

Did you have any trouble convincing them to switch from the CDK to that solution?

Speaker 3 07:33

So So initially you could say, like, you could say there was some resistance, because, like, some of the teams, did not want to, you know, change the way things already are. And I think, but, but then the changes that were integrated because of the fact that the application was, it was contained, right? They did not want to make those pi is have the related data call all over the all over the network. So, so, so there was some pushbacks, but we just had to come up with a secret, secure way of handling that data. So, but after, I think, but after providing the information of how the application would perform. I think all the team members kind of saw that way of working. So I think in at the end, I think I convinced them to work with the whole setup, and there was, like, a lot of advantages over the system, so I think they also saw the resiliency that it provided. So I think it helped them. But there was a lot, there was a little pushback, for sure,

Speaker 2 08:58

on the on the same vein of multiple so this is a new microservices that's now sitting in between these teams and another API, correct.

Unknown Speaker 09:10

I'm sorry. Would you see that again? Yeah,

Speaker 2 09:12

sorry. Let me make sure these are on the that that new service that you created a microservice that was that's sitting in between multiple other services, right? You're almost like the gate, the

Speaker 3 09:30

service that I were, the one we exactly like you mean that the one you so I think like that was so it was like an orchestration for the for the streaming platform, right? So, rather than the the teams having to integrate the SDK that was provided to make those calls, we came up with the abstraction to to serve this, that same purpose. But yeah, like, you know, it was like in a centralized spot, sure, yeah, gotcha. The

Speaker 2 10:01

reason I ask is because, basically putting your way, putting yourself in the way of that service now you've become that single point of failure for those teams. Potentially,

Speaker 3 10:16

well, um, yeah, potentially, it could lead to us the single point of failure, but, but I think like,

Unknown Speaker 10:26

because this is

Speaker 3 10:29

like, because these were like, log messages that that were being sent, and we would just store it and then process it in batch, and then have those, and have multiple retries, and then with the and with the exponential back, back off in place. So I think even though you know, it was if there would be a single point of failure, it was it was not that the data would affect the customers, right? So, so I think the the internal auditing tool. So, so, even so, even if it add field with the exponential, back off in place. I think the field messages would get retried and then, and then we would know through with the console we had. I think we would be able to figure out and because there would be no customer impact, I think, and the rest response from the API would not, would be affected as much, I think

Unknown Speaker 11:32

it wouldn't. It wouldn't lead to a catastrophic failure as

Speaker 2 11:35

such. Gotcha, yeah, awesome. Thank you. Yeah.

Speaker 2 11:43

It's always interesting when you have that, like, it's nice to have, like, a micro service in the middle, and then next thing you know, you're now just in the middle every once way,

Speaker 3 11:53

exactly, yeah, yeah, that's the other Yeah, exactly. I

Speaker 1 12:03

All right, sounds good. So next question for you, can you walk us through a recent project or assignment that you completed and tell us the process you used to ensure it was complete and accurate?

Speaker 3 12:19

Okay, so, so So the recent, so the most recent project would be the one that I'm that I'm working on right now. So, so it's basically, it's basically the integration of a tokenization function and so, so one of the primary application that it works on. So basically, so, based on on the route that the clients and the the clients on board their APIs with the with the orchestration services that our team provides. So, so basically, we act as a as a gateway layer, right? So, so it's not like a direct gateway, but it's an orchestration gate gateway, and and the way that our system works is it's like based on the incoming request, and we match it with the with the metadata config, and based on that metadata, we apply certain functions and certain certain filters and and one of the requirements that we had was for the secure data, and which was so the bank account number or any any kind of tax IDs, and we had to secure it over the network. So we had to tokenize all of those calls and the way it was available to the to the downstream service, and that would basically be based on the authorization scope and the certain values that that could be tokenized by our direct microservices and so. So the way that I worked with this was from the start of the project to the to the final to finishing the whole project. So it was like first I gathered the requirements, and I went across our entire applications, our client IDs and our client credentials that needed to be to be onboarded, and for these different scopes for authorization, and so how these tokens needed to be generated based on on those scopes that that were defined. So, so basically, I had to write the helper function to, you know, do the tokenization, which required extracting the field that needed to be tokenized, and then make it configurable so that it would support multiple, multiple fields based on on that metadata, config and and so, so it was so I also had to do the graceful it was a failure in case of, in case of errors in tokenization, which is, which is to basically not tokenize it, right? So, so I added multiple unit tests so that we could cover the different situations and scenarios that could arise and and after that, the coding was after the coding was done, my team and the developers, we pushed it for the QA to test it. So, so I think, yeah, that was kind of like one of the so I completely completed that whole tokenization field. And I was, I think it was highly configurable, because of configurable because of the fact that whenever the clients onboarded their APIs. They they can provide whatever feeling they want to tokenize and on the request body. So, so we can pick that field and then tokenize it. So, so I it was like I worked from initial requirements gathering to the whole coding aspect, and then even pushing it for the for the QA, so, so the whole team could be more cohesive. There was more collaboration all around.

Speaker 1 16:30

Can you talk a little more about the pushing it? What? What did that look like? What were the steps involved? There?

Speaker 3 16:37

You mean like after, after my team and the developers push q8 Yes,

Speaker 1 16:43

yeah, that process of pushing the q8 Yeah. Can you describe that a little more

Speaker 3 16:47

Sure, definitely. So. So the way we promote any of our changes is to first, first do a local testing, and then after that, we create a feature branch through which we open up, open up a pull request to develop our our branch on the GitHub right so, so, so my team meets, we give the we provide the code reviews, and then and The other people on the and all the other teammates also provided. And so basically once, once we have all the code reviews addressed, we merge the chains to the to develop branch, and from from the developer branch, we basically deploy it to QE and and the once the deployment is automated through Jenkins, which is our the CICD tool that we use. So so for the for the for the product promotion, we basically got to release the candidates from the from the developed branch, and and all of that is automated through Jenkins, so usually that's how we can deploy it, push it.

Speaker 1 18:11

And I know you mentioned unit testing already, that you guys did a lot of unit testing to cover all the all the different variations of what could or could not happen. Did you have integration testing that was built into like your Jenkins pipeline as well?

Speaker 3 18:27

So there were, so there were a bunch of integrations testing that we that is part of our CICD pipeline. So mainly, we use cucumber for it. We also have performance testings using JMeter. This is not automated using the Jenkin pipelines, but it is used separately before the product promotions. And we also had to, we also do regression testing in place. So, this is done, so this is done before and and the prod from, so that is done before the prod promotion, on the on the release candidate. So, so I think, yeah, there are various kinds of testing we do throughout the whole pipeline,

Unknown Speaker 19:17

including the integration testing.

Speaker 1 19:22

I sounds good. Jose, any other questions on this one? Not this one, all right, sounds good. So next question for you, can you provide us an example where an application you supported was during technical errors and you had to troubleshoot for resolution? Okay?

Unknown Speaker 19:44

So,

Speaker 3 19:46

so, so, yeah, like, like I mentioned, we work with a with a lot of functions, and one of the functions that we that we had enabled for our mobile clients was for for web applications. So we had a feature request to to enable the that particular function factory for the web and our function, and our function was like highly configurable in a sense that it should run for a certain for certain routes, and and that was so. So during this we basically had to enable the log out API to run in this particular function factory. So, so the Q is, we're testing this flow. And so everything worked smoothly there. There was no issue in in the lower environments. But once we promoted this to prod, everything worked initially like, but what we did like, we also did a gradual deployment, and everything was okay, but later on, like at night, we started getting alerts that was triggered because of the failure in the Lola API. So so we had to, so we had to, like, quickly jump in and check what caused the error first and why was it not and why was it not caught earlier by all the all the team? So basically, I got into the zoom session and then checked if our application was, if our application was the point of failure. And I saw that, in fact, it was so. So I was able to look into the into the New Relic dashboards, and then debug, debug the cause on Splunk to check for logs and other sources of errors. So, so it was basically, it was basically our application. So the way it reacted to this was basically rolling back the release to the previous stable version. And then we tried to recreate the error locally first, and then solve in that local environment. So the so the error had So, so the error that we found there was the application was logged out by a session by a session timeout, so the request missed some of the headers that was needed for the particular function factories to run. So So, because of this, the particular scenario or so, because of this particular scenario, our application failed, and the QA had a different version of the app running which did not which, sort of, like skipped the required headers. So, so that was one of the times we had a huge technical challenge, and I think we had to resolve it very quickly. And this resulted in, like, a severity in three different incidents we get to like so you know, so there was a pretty decent amount of customer impact, you could say, because that

Speaker 1 23:17

sounds good. So just thinking back on that, if you wanted to prevent that from happening, what? What could you have done differently? Okay,

Unknown Speaker 23:30

so, so, I think

Speaker 3 23:35

so, to prevent something that as such, I think, I think maybe thorough testing is something that comes to mind immediately, testing all the scenarios where the headers might have not existed, right? So, so that's one thing, and it was, but it was like very trivial headers that usually all the requests have, and we assume that these headers will always be present. But looking back at it, I think the way we handle it was to see if the if the QA and

Unknown Speaker 24:15

the prod had the same version of the API is running,

Speaker 3 24:19

and because it they didn't, we would probably want to, like, go with the testing, with the with the same version, so that we have, like, we have parity across the across the environments, before the promotions. And also, maybe we, we also should have added more integration testing for the for the scenarios with no headers. So I think those were the things that I would have done before pushing it to fraud, right? So, so I think the extra confidence that we had that there would be no issue was and because the issue was because the code was running all the time, I think, I think that that was a bit of a point of confidence on our part, and because it was only for web clients. So I think so maybe like thinking about like different versions of the APIs before promoting it with a would have really helped us in the long run.

Speaker 2 25:30

I have a follow up question, just more so on. So I know you mentioned that you're building parts of your testing suite, and then you send it off to keyway, right? I and QA might have a different environment. Would you? Would you have preferred that your team would take care of those QA tasks? So, like, if you're if you have the application in hand to be able to check it instead of going to QA, you, you would be the ones to check it and then go to production. Um,

Unknown Speaker 26:08

so, so I think

Unknown Speaker 26:13

so, I think it's really

Speaker 3 26:15

it would be a good opportunity, because I think while we develop the code and the various scenarios, I think different parts of the application needs to be tested with with certain changes. And the only thing that helps, I think, when the QA does

Unknown Speaker 26:38

their testing,

Speaker 3 26:41

maybe they are unaware of the implementation, right? So, so I think, like looking at different parts of the requirement and then testing it accordingly, is is more important than maybe, like doing everything ourselves, I think, like a little bit of QA testing would also add an extra layer of validation right on someone else's changes. So, so if I'm doing my own testing in a developer environment, I think having an extra QA QA person might might validate against their understanding of their stories and the features that they understand. And maybe they can always bring us, they can always bring certain point of view to the entire application, right? So I think everything by somebody with doing some, everybody doing their part, and then working cohesively would be more fruitful, rather than us doing everything I think

Unknown Speaker 27:44

got Thank you, yeah.

Unknown Speaker 27:55

All right, how does anyone take the next section of spam?

Speaker 2 27:59

All right, let's look here. So I saw in your resume you have a couple different programming languages, and I hear at your functions factory and a couple of the things. So which one of these is your preferred language? Because I'm hearing a lot of Java, right?

Speaker 3 28:16

So my primary language is Java. My preferred Yeah. Oh, go ahead, sorry. Oh, no. I mean on the on the back end, it's Java on the front end. I mostly work with Angular, but I can. I've also worked with Python and c plus, plus you're in there,

Unknown Speaker 28:36

gotcha. But your preferred one would be Java. Is Java,

Speaker 2 28:39

that's correct, gotcha. And if you had to kind of rank your skill set, let's say one is you've just heard the name of the programming language before, and then five is you could teach this as a college course. How would you rank like your Java, Python and the safer front end, your Angular?

Speaker 3 28:59

Okay, so I think this would be like, I don't want to sound either too incompetent or either too promoting of my own skills, but I'd say like, maybe four and a half on Java, around around three on Python and maybe Angular would be around three and a half. I can also do C++ I'd say, like two and a half. Maybe gotcha awesome.

Speaker 2 29:35

I think, I mean, I work with Python all day, and I will still also just never take a fire that's just me.

Speaker 3 29:41

Oh, I like Python too. Actually, it's, it's very it's like writing pseudo code much, much easier, right? But I think it creates a little bit of a complication here and there. Maybe, like, some prefer, some people prefer, like, object oriented and classes over functional, like Python, maybe just, just, that's how that's the debate you see over stack, like Stack Overflows and Reddit. So

Speaker 2 30:14

I know where you're coming from. As someone who has to use Python every day, I know exactly where you're coming from,

Speaker 3 30:19

right, right? It's a lot of debate you read on Reddit and makes you, makes you wonder, like, why people prefer one language over the other.

Unknown Speaker 30:29

People have their favorites. Is what I'll say, definitely,

Speaker 3 30:31

definitely. And I'd say, like, I think Python is the is definitely the easiest to get into. Like, if you, yeah, if you're just beginning, if I were to tell my niece what to study, I'd say, take life on,

Speaker 2 30:48

just prepping here for the next question. Sure. Okay, so I'm gonna paste in here some code for a pace of code. I want. What I'd like you to do once I paste the code is kind of walk through it, and as you're walking through it, kind of tell us what you like, don't like, if there's anything you would want to change, and if you feel the code would be good for sending out to production. Okay, gotcha

Unknown Speaker 31:14

let me paste here. And

Unknown Speaker 31:18

before getting started,

Speaker 2 31:21

as you're going through. You can highlight, if you look on lines two to 14, you can highlight on the screen to kind of point our attention, okay? And as you're going through, kind of like, talk out loud so we can see, like, what you're thinking about as you're going through as well.

Unknown Speaker 31:36

Definitely, definitely,

Speaker 2 31:38

yeah, feel free to get started. Okay,

Unknown Speaker 31:45

okay, okay, so,

Unknown Speaker 31:49

so, basically,

Speaker 3 31:53

basically what you have your looks like an im policy document. If I'm correct,

Speaker 2 32:02

I should ask before you continue, are you familiar with the language that you see here?

Unknown Speaker 32:07

Right? I am. Yes. Gotcha. Would you

Unknown Speaker 32:10

be able to tell me what language you're saying? It looks like, Give me. Give me one second. Oh, you're good.

Unknown Speaker 32:28

Yeah, it's not meant to be a trick question or anything like that,

Speaker 3 32:31

right, right? No, I'm just trying to, like, I think you can understand, like, certain aspect once you see the Code,

Speaker 2 32:40

yeah, yeah, like, I have policies, right, right? So,

Speaker 3 33:18

so, so I think basically we have our, like, it's a im policy document, right? So, right. So the, it's the, it's the configuration language, right? The HCI,

Unknown Speaker 33:41

it's a um, HCl,

Unknown Speaker 33:44

yes, or HCl, HCl,

Speaker 2 33:48

yeah. So what, what you're looking at here is HCl. Okay,

Speaker 2 34:02

that's guess I should also ask, are you familiar with kind of like infrastructure as code,

Speaker 3 34:10

a little, I've done very little pieces of it through some like cloud formations and some TerraForm, but just a little bit not, not as much

Speaker 2 34:24

Gotcha. And for TerraForm, what have you kind of built in TerraForm?

Speaker 3 34:30

So, on TerraForm, so I've basically done not like build stuff in TerraForm. It's more of like looking at it and then making, like, minor changes here and there. It's like, we have a dedicated DevOps team that handles the TerraForm parts. So, so it's not like I'm pretty much involved on it on a day to day basis. So awesome.

Speaker 1 35:02

Oh, awesome. Oh, I was gonna say, as you look through this, is there anything that pops out as good or bad to you?

Unknown Speaker 35:10

Yeah, just give me. Give me a second.

Unknown Speaker 35:15

So let's see.

Speaker 3 35:19

So it's kind of like, like segregated. I think it's a like, clear structure, which, which basically like, like has, I think the effect that you want and what actions that you want to take on a particular resource, right? So I think like, you have a clear, clearly defined IAM policy document here, which is, which is easy to follow. I can just look at it and see, like, see what you're doing, and then

Unknown Speaker 35:55

you have like,

Speaker 3 35:57

each block which is, which is very descriptive, right? So I think so, I think it's pretty good, like, like, you're having descriptive blocks of different parts you

Speaker 2 36:25

if you wouldn't mind just for me. Oh, thank you. I was about to ask if you could highlight so I can know what you're looking at. Okay.

Unknown Speaker 36:36

So, yeah,

Unknown Speaker 36:38

okay. So basically, like,

Speaker 3 36:41

I think it's like math for managing, like, you have, like detailed effects, action and resources and set and you have set up with what statement you want. So I think it's, I think it's very descriptive, and it has like a clear purpose. And then you also have a very, very granular,

Unknown Speaker 37:10

granular code,

Speaker 3 37:16

like granular permissions that allow the PMs, you're just giving it to this particular resource and so on so. And then you also have a similar thing for for deny. I

Unknown Speaker 37:52

I think that's, that's kind of, I

Speaker 3 37:58

think that's what I kind of like think about this, but I do see, you know, like, there's a lot of wild cards available here, so, so that's Something that's usually not recommended, because it opens up all the rules under that I am account, right? So, so, so, I think it might be good to have a specific, specific I am under the resource Section. I

Speaker 2 38:58

And on that same note, for resource sections. I'm gonna just move around some code a little bit. Give me one moment, I'm gonna move this deny. On top of this allows so lines two to 10, and then my lines 12 to 24 here.

Unknown Speaker 39:19

So I have this detach role policy,

Unknown Speaker 39:23

and I have a similar detach star.

Speaker 2 39:27

So let's say, for just this detach policy, or for this detach action between here and here, when we run this policy, which one would kind of happen? Would we be allowed to detach a role, or would we not be allowed to detach a role for this policy?

Speaker 3 39:55

Yes, so, so once you move that, I think that we have detached rule policy for the

Unknown Speaker 40:07

for the wild card, right?

Unknown Speaker 40:11

So I think it basically

Unknown Speaker 40:21

so. So let's see what happens.

Speaker 3 40:40

So, I think, I think it should deny. I

Unknown Speaker 40:45

think it should deny for,

Unknown Speaker 40:58

yeah, I think it should deny for everything

Speaker 2 41:04

got so for that, deny, role policy, detach, role, policy, action,

Speaker 3 41:12

yeah, yes, I think Yeah. I think so, yes, gotcha. And that's

Speaker 1 41:17

based more on like, Oh, I was gonna say, why is that? Okay? So,

Speaker 3 41:34

I think so, because you're like, restricting the because

Unknown Speaker 41:42

you're restricting the

Unknown Speaker 41:46

flow policy for, I

Speaker 3 41:51

think, because you're restricting the flow policy for all the for all the resources and then, and then You're adding and allow later on, right? So I think because of that, it will deny for all, and then just change the one for the two specific resources to have it allowed. So I think if that's that's how it should work, right? So first it will deny everything, and then and then update that by allowing those specific resources that we mentioned on on 23

Speaker 1 42:30

and is that because the deny is first, the statement for the deny is first and then The allow is second, or Does the order matter? Um, so

Speaker 3 42:53

I think, yeah, I think the order of the statement is it's not how it directly affects how the access is evaluated. I think because each each statement should be evaluated independently,

Unknown Speaker 43:12

I think that's

Speaker 3 43:14

yeah. So I guess to have

Unknown Speaker 43:20

to have a second part of this. I think,

Speaker 3 43:25

I think all the statements are evaluated in the independently, but usually, but for, usually do, for the explicit, deny. I think it always takes precedent, right? Because, because it overrules everything, and because it makes sure that the stricter rules are there. So this explicit deny in any statement will will take precedence over like allow statements. So I think that's why it'll allow. It will deny the detached rule policy for everything

Unknown Speaker 44:14

awesome. Thank you. All righty

Speaker 2 44:19

and I am going to take us to our next question as well. Sure you one second here for this one. I have a Python and a JavaScript example. Which one would you prefer to do? I um,

Speaker 3 44:44

I think, I think, let's go with JavaScript for now, for today.

Unknown Speaker 44:51

Me one moment here. If a

Speaker 2 45:00

little output window pops up, you can minimize it. It's the little arrow to the right of the Run button. If it's not already minimized, make sure. Make sure all the lines are here. So there's 53 lines in here, and it's going to be the same thing as we did before, kind of walking through the code, letting us see, letting us know, like what you're seeing, if there's anything you want to change, want to keep and then in the end, kind of letting us know. I said, Would you want this to go to production or not? Oh, definitely. Just

Speaker 3 45:43

so, so basically, it looks like, like you're doing a,

Unknown Speaker 45:49

like a Dynamo DB interaction.

Speaker 3 45:53

And it's based on, on, like a lambda. I think you just, I think you have a lambda that interacts with the diamo dB. So, so you're doing a parsing of the event body, and then,

Unknown Speaker 46:13

and then, if the body does not exist,

Speaker 3 46:17

then, then you're converting the whole event into a payload. So, so you just basically create a payload for,

Unknown Speaker 46:27

for maybe a

Speaker 3 46:31

or maybe for the interaction. So, so now for the first thing I see here is if you're doing a rule rule validation or a feature validation, like you're you're just doing a basic payload validation, where you're returning an HTTP 400 and so, yeah, I think This is basically, it's a simple validation to check if,

Unknown Speaker 47:04

if any, any of the fields are,

Speaker 3 47:08

are, are working. And then you're also building your DynamoDB ramps, like, you know, the like, the table names on the environment, the rule from the actual payload. And then you're also doing output operation here, which is to update the value from the filter compare here. So and then I think the next is basically making the DynamoDB operation, and then, and then, if it's successful, you're meeting a 200 and with a failure, you're returning a 500 or or whatever the status quo you get from the database. So, so, I think so. I think the good things that are with this part of the application and the code you provided is you have, you have an input validation in place, right? So, so ensure, you ensure that the requirement, required parameters, which are, which are the rule of and and the filter and the key together, they are validated before even being presented to your table. And then you're returning a 400 response back right away. So so in so you don't make unnecessary calls. You're just using the table name from the environment variable, which is usually a good thing, because you don't want to expose it. And then you can deploy across environments. You also give a hint that you have access control allow original error for for a course policy in place, and then you are also, you also have a proper error handling in place. So in case of errors, you return, you log the errors, and then return a response. And that is that is meaningful. So.

Unknown Speaker 49:29

So I think some of the things that I would

Unknown Speaker 49:33

change here would be

Speaker 3 49:36

in case of success, I think in case of processing, if, in case of processing is unsuccessful, then maybe we want to add a log, maybe. And then also, one thing that I noticed is that you're allowing everything on the course, even though, even though you're mentioned for it. So maybe I would. So I'd like maybe make the course setting a little more strict.

Unknown Speaker 50:10

I think,

Speaker 3 50:12

I think, I think those would be the changes that I would like to do

Unknown Speaker 50:18

other than that, yeah, would I be correct here,

Speaker 2 50:22

or I wouldn't say it's a correct or incorrect answer. So to kind of give some context, this code is pulled out from a merch request that we've given like that is from our actual code basis. So we take these samples to kind of show folks who are interviewing a State Farm, like, how would you walk through a pull request of say, code that you've never seen before? Like, if you're if you're on the team, oh, for the first time.

Unknown Speaker 50:54

Oh, interesting. Interesting. Yeah.

Speaker 2 50:57

So I, I want to ask something, actually, because I I saw that you have a lot of Java, and I saw that you

Unknown Speaker 51:04

seem to have familiarity with like lambda.

Unknown Speaker 51:11

In what context Have you kind of used lambda before?

Unknown Speaker 51:15

So because I've worked with

Speaker 3 51:20

on AWS. I've worked with lambdas for like, multiple purposes. So it's so on the so it's just a it's like a fire and forget, kind of architecture, right? So it's, it's event based system, so, so it's just a function that it's, that it runs a piece of code. So a couple of use cases that I've had is it was basically in in response to to a file drop and and not if, and on a notified system, and then in and in response to a message on a on a SNS topic, which triggers like a downstream notification service. So I think these were, these are some of the cases that I've worked on with lambda, specifically

Speaker 2 52:12

gotcha and in terms of, like, what programming languages have you used in in lambda?

Unknown Speaker 52:19

So for lambda, I have

Speaker 3 52:22

primary I work with no JS for that, for lambdas, but I could always like work with other languages too.

Speaker 2 52:35

Catch mostly asking, just like we've had folks at State Farm who use, let's say, like Python for their lambdas. There's folks who use no JS for their lambdas, like, my team's doing Golang. Interesting. Yeah, it's like, it varies. So kind of having, like, because at some point you always going to have a lambda for something, right, right? But, yeah, definitely that helps, at least for the coding questions, that's all I had. Jackie,

Speaker 1 53:05

yeah, let me bring us. I know we're getting close on time, so let me ask a very formal HR question. Can you perform the essential functions of this job with or without a reasonable accommodation?

Unknown Speaker 53:19

Yes, okay,

Speaker 1 53:21

all right, sounds good. What kind of questions you have for us? What can we help with?

Unknown Speaker 53:27

So I think,

Speaker 3 53:30

first of all, just thank you for the opportunity today. It was really nice talking to to both of you, and I'd like to know what like, like, I think what defines success of an engineer on this rule, right? So maybe, so maybe, like, like, six months down the road, what would, what would make me consider considered a successful engineer for this particular role?

Speaker 1 53:56

So I can tell you from a leadership perspective, when I have new hires, new people, to State Farm, what I'm looking for is usually within the first month or two, I want them to be able to take something, it could be something small, but I want them to push something to prod, and go through that whole process of, you know, this is how you test it locally. Like you mentioned, this is how you take a branch, you do the merge, right? You get the feedback from your peers, all of that, and then you go through the prod approval process as well. So to me, that's success is you have a basic understanding of your product. You understand all of the different tech pieces involved, so the different languages, the different AWS services, right? And you should be able to get to the point where you can confidently push code to prod on your own. So that's, that's what I think of when I think in terms of, like, success, of this was a good hire for State Farm

Speaker 3 54:58

and and I think, like, when would I like your back? Maybe.

Speaker 1 55:07

So unfortunately, with the holidays, the timing is a little bit off than what we would usually do. Typically, we would say within a week, with the holiday schedules, it could be closer to a week or two, but you can always reach out the person that reached out to help you get the interview set up. You can always send any questions their way, and they can let you know kind of where it is. So, typically, yeah, typically, within a week or so, you should hear something back, okay,

Speaker 3 55:37

and I think, and maybe, like, what kind of opportunities that I have for growth.

Unknown Speaker 55:47

I'll let Jose talk about that one. So

Speaker 2 55:49

in terms of opportunities, quite a lot. I was not a software engineer. I was actually a database administrator intern. So I was in the data space, and I, I was given the opportunity to basically grow into my role. So, yeah, so like, I got assigned a mentor, I got transferred to a software engineering team. I was given, like State Farm pays for a lot of resources to upskill yourself, so there's an ample opportunity to kind of grow into the position. And also, like, my mentor is still my mentor, and I have other mentors as well. Like, there's principal engineers who are my mentors, other engineers who are my mentors. So like using them to kind of learn how to get to that next step in the roles. So there's a lot of opportunities for product. Well,

Speaker 3 56:44

it's interesting that you made the switch from a DBA to a software

Speaker 2 56:49

engineer. I I wrote a note automation script so we're like, it's gonna probably be a better idea of putting,

Speaker 3 56:56

yeah, it makes sense. Makes sense. Okay, um, I guess, like, one more would be given. Like, you know how AI is kind of like taking, starting to take over. I heard, like, in Google, like, 20% of the code is now being written by AI in some sense. Like, do you in state form use AI? Like, how is AI being integrated in the system?

Speaker 1 57:22

Short answers, yes. So we absolutely are looking at it. And how does it provide value at State Farm, across numerous different places. So we're looking at, how does it provide value within like our systems? How does it help our engineers, especially when it comes to things that are more template, like code or things like that. There's definitely opportunities there. So yes, this is definitely a very big one, and we have many people focused on it. Actually, one of the teams that I have is focused on AI, ml, generative AI work for claims. So we're looking at all kinds of different use cases and more. Can you get those out and deliver to our claim handlers to help them do their jobs better or more efficiently? I should say, so yeah, there's definitely a lot of AI being leveraged.

Speaker 3 58:20

And I guess, like, like I said, because he was trained within State Farm to make the switch from a DBA to a software engineer, I think maybe, like, State Farm would give us opportunities to, like, you know, help us understand ml. And, yeah,

Speaker 1 58:37

yeah. So one of the things as a leader, I'm given an expectation that I need to invest in the growth of everybody on my team. So we actually have a formal goal that everybody, including leaders, get like, 40 hours worth of growth and training and learning in within a year. A lot of people go beyond that. I mean, I'd say Jose, I know I'm well beyond that, with all of my stuff that I've learned. So most people are going beyond that, but yeah, there's lots of opportunities. And like we we have bonuses that you can earn for getting, like AWS certifications, and State Farm will help pay and reimburse you when you do pass the test. So we've got a lot of programs like that to help as well. Don't

Speaker 2 59:21

forget the reimbursement for your masters. Yes,

Unknown Speaker 59:26

that's a big one. That's a big one. That's a, yeah, I

Speaker 2 59:29

have multiple co workers who've actually taken the masters. So I'm in Atlanta, so like, we have Georgia Tech right here. So multiple my co workers have taken their AIML masters at Georgia Tech, and they got it first. Oh,

Unknown Speaker 59:41

come me and come me and

Speaker 3 59:44

like, if it's free, masters, please, yeah, how do I keep staying at State Farm would be the big questions.

Speaker 1 59:55

So I will say State Farm is very invested in their people. They understand that that is a large investment, like, I know Jose mentioned going from like DBA over to programmer, but we also have people that start off in, like our call center areas get, like, their Bachelor's in programming, and then we bring them over, or we bring them over and have them be our product owners or things like that. So we're very good about trying to keep that knowledge of State Farm as long as we can. Yeah,

Speaker 3 1:00:24

I think, I think that's really important. I think, like, that kind of brings a certain love towards the place you work, if you really push your employees to make the jump into higher roles, and then, like, maybe from customer service to soft ranging, that's a huge leap. I think, yeah, and I think it's, it's great, like, State Farm actually does that. That's very, that's very noble in, in some sense, and that's, and, like, really helps the employee in their their whole lives, like making a jump of that sort.

Unknown Speaker 1:00:58

But institutional knowledge is king,

Unknown Speaker 1:01:01

right, right, definitely,

Speaker 3 1:01:05

definitely, any questions. I think that should be it, and I just look forward. I'm just looking forward to hearing back from you.

Speaker 1 1:01:13

All right, sounds good? Well, we appreciate it. Thank you for your time, and have a wonderful afternoon and holidays. You too. Holidays. Happy holidays. Thank you. Bye.

Milan dhakal-jpmc-final-srv-offered

Unknown Speaker 00:24

Hello Oh, hello. Hi. Let's see hi how are you? Alright,

Unknown Speaker 00:36

I'm good, thank you

Speaker 1 00:41

and then everyone who was here, okay, so Melinda is here to talk about the requisition 300 936 And this particular position is for Coachella with AWS specifics and this is for Columbus, Ohio location. Is that anyone else to join us now to scenario good to start?

Unknown Speaker 00:58

No, I think we're good.

Unknown Speaker 01:01

All the way to smoothen and oh, what do you think the renting Thank you? Yeah.

Unknown Speaker 01:05

All right. Thank you.

Unknown Speaker 01:08

All right. Thanks. For joining.

Unknown Speaker 01:12

How's it going, man? Yeah,

Unknown Speaker 01:13

everything going good. I'm excited.

Unknown Speaker 01:16

And now you're in Columbus.

Unknown Speaker 01:18

And now not at Columbus now.

Unknown Speaker 01:25

I relocated to Carlisle Pennsylvania

Speaker 2 01:31

not too far. The weather isn't raining already. Either. There's a pretty big storm coming in. I don't know a bit of Pennsylvania but somewhere there is not so cool.

Speaker 3 01:40

Yeah, but yeah. I also see the news as well.

Speaker 2 01:48

Okay, all right. So you're gonna talk a lot of technical details, some of your experience that you have on a resume. I have a resume on my phone. So I'm gonna keep looking at my phone. So don't worry about it. It's only because I kind of wanted to pick some of your experience and we'll talk in more detail. And John Hardy is my colleague here he is going to kick off the interview and started asking your technical questions. And last 10 minutes, we will basically ask, have you ask any questions for us or anything like that? So there's some time so let's get that started. And

Speaker 4 02:26

Alright, so I just wanted to ask you start with some Java questions just to start.

Unknown Speaker 02:34

So to start, what is a constructor?

Speaker 3 02:37

Well, yeah, sure. You know, so basically, the constructor is you know, kind of like, mainly for the methods. So but basically, it is used to create an object. So, whenever you create an object for a class, so basically you call the constructor so for in constructor, so it can consist you like some of the parameters, which basically we use to set the like, initial values for attributes or also, it also can, you know, like, make

Speaker 4 03:13

sense. That was my little joke by the way to start starting with the constructor. Do you know, do you know the order of initialization? For for an object, there's something called an order initialization in Java can you explain that?

Speaker 3 03:35

Sure. Yeah. So basically, I guess, like, it is like order of initializations right. You know, basically, it is mainly, you know, static fields and also like, static initializer block. So it's basically like, you know, kind of initializing in order. So, they appear in in basically in the class. So, so, basically, you know, static fields and also blocks are like that, that they basically initialize only once so, when the class is like first loaded in the memory and also then or instance fields can also incident the initialization block that are invoked. So, basically, they are invoked, like right before the constructor is called. So, and each time when like an object is created, and all these fields and all blocks are also like initialized and after that you know the instance fields and also initialize the blocks that have been initialized and also constructor is called yeah all

Unknown Speaker 04:46

right. That's good. That's good.

Speaker 4 04:49

All right. I ran into a situation where I had done a commits and then I have a co worker come along and do another commit, which accidentally undid my commit What could I have done to ensure that the commit he had made was discovered quickly? Oh, rather than a couple of weeks later, I found out Oh, he did a commit on the application portion. Is there anything you can think of that would have helped detect quickly that something changed was not meeting the expectations?

Speaker 3 05:32

Yeah, well, basically, you know, I guess we could have like proper code reviews in place like to pull the request to write basically, like, instead of having like working on the same, like brands, so we would create like multiple like brands and also based on the features and they have like the proper commit messages. Basically, the place in that history and also that prevent any kind of like, like fourth that cheered like brands. So, you know, like, like, those are like some of the ways you can make sure that like, you know, also like you do not have to override the like sorting like comments as well.

Speaker 4 06:19

Okay, all right. All right. Then the next thing I want to talk about was test driven development. So where do you start with test driven development? What What's your first step? Well, yeah,

Unknown Speaker 06:32

explain test driven development to me. Sure, yeah.

Unknown Speaker 06:37

Sure. So, basically, you know,

Speaker 3 06:40

test driven development is kind of kind of, you know, writing the test. Also, it is the kind of like, read or like mainly for the green phase, right. You know, basically you can determine the like functionality, so, that you want to implement and also the like behavior of you want to achieve. So, like when you start by writing like when it failing the test, so because you do not have the like source score yet. So you you need to test that you know, defies the like desired behavior, but it also it will fail. So, it is basically like ensured that like test failing is valid. Then when we go basically, when we go ahead like make the test and fast with which is kind of like the green phase, so, where like we write like minimum amount of code possible. So, so, you can write the like simple like simplest possible code just to make the taste fast, tasty to like pass so, when, also like when you focus on the functionality rather than the like optimizations, and then our next phase will come into play. So, which is like, kind of like, you know, refactoring those are implementations So, the like, you know, kind of the step by step test driven development. Yeah,

Speaker 4 08:05

all right. Um, I think you mentioned Maven on the resume. How would you go about getting all the different components and better used in a project? How would you what, could you run a Maven command? And retrieve that list? Yeah,

Unknown Speaker 08:25

yeah. So, well, basically.

Speaker 3 08:28

So, when, like, we can use the you know, which is mbN or dependencies or tree

Unknown Speaker 08:39

Okay. All right. That's all I was looking for.

Unknown Speaker 08:43

The dependency tree.

Speaker 4 08:47

US Spring Boot. What's the big difference between Spring Boot and a normal Java application? Well,

Speaker 3 08:56

yeah, so basically, like, so in the Spring Boot, you say like it's spring work between the difference between the Spring Boot and Java. So it is like, you know, when basically, in a Spring Boot, sorry, so here like we have like dependency injections and also like version control. So basically, Springwood is it is like conventions over the configurations upwards.

Unknown Speaker 09:30

But in like so, I guess

Speaker 3 09:34

in like basic, like Java application, so you have to create an object so that you have to work with them. So but in Unity Spring Boot, it provides like dependencies for like those that object is from the memory from the spring container and then also we can use those objects. So for that, like we don't have to worry about like maybe mainly about like the business logics and also rather than how these objects are, like configured, so yeah, I think they are, like main difference. Is Java and Spring Boot.

Speaker 4 10:13

The next step, what types of authentication have you used in your applications? Yeah,

Speaker 3 10:20

so Well, authentication. So yeah, basically, I have worked with that before with that as well. So, so for that, you know, like I use like, what is called sorry, I forget like is the token sorry, JW token so

Unknown Speaker 10:47

Okay. All right. Do you

Speaker 4 10:52

know the HTTP header that that is used in but it gets passed back and forth, like the when the client makes the requests, it's loading it into a, an HTTP header? Do you know what that is? Oh,

Unknown Speaker 11:07

yeah. authorization header.

Unknown Speaker 11:16

Alright, so that's the JW T.

Unknown Speaker 11:19

When your application talks to the database

Unknown Speaker 11:24

have you used Kerberos?

Unknown Speaker 11:28

So far, Kurvers say I couldn't.

Unknown Speaker 11:33

So there's a

Speaker 4 11:36

there's an authentication mechanism, bad JPMC is really pushing our applications to use and that authentication radicalism in in the Linux world. That's Kerberos. In the Windows world, it's a GMSA account in the group managed service account. Have you used either of those types? of authentication?

Speaker 3 12:00

You know, I have heard about it but you know, I don't have like durian Hansard experience on that

Speaker 4 12:15

I mean, the thing I was wondering about your problem solving skills. So recently, I faced a situation where we have a file transfer team within GPMC. They are moving from their northeast data center to their Northwest data center. And what so they tested connectivity in this new data center. And they they're connecting to a one of our external partners. So we have an external partner that has an SSH host out on the internet. They tried to connect from the northwest data center to be SSH host. They told me that the client might well our our external vendor they are blocking connectivity from JPMC to them from this new data center. I went to the client, the clients. They have everything open. So I'm faced with a situation where JPMC is saying the file transfer team saying it's blocked the client, the vendor is saying it's open. How would you go about trying to solve that problem? Well,

Speaker 3 13:42

so yeah, okay. Let me think. So, so in this scenario, so basically, in this case, so I would basically you know, take the sub net so that correctly, you know, like configurations on both northeast Northwest and also in Northwest mainly Northwest data center and also like external partners. So, basically, basically, you take the like, like Ingress, and also make sure that rules also say if, like, we have any kind of like firewall rules and also like both sides, they are correctly so particularly it takes for the like ports numbers and also like it expose and also I would test the like network or like connectivity, using the trace router or also like pinning to the servers. Or Furthermore, I can also go away to like looking for the like firewall logs and also even in the like, SSS like service logs, if possible. So I will also make sure that like you know, DNS configurations, which is correct or not. So, if there are any connectivity within using the hostname, so, for that, I will also verify the routing tables for the like, data centers, but you know, like, those are the like some of the ways

Unknown Speaker 15:16

I will look at it.

Speaker 4 15:19

Tell you what I did. I tried to connect. It's an SSH service. Do you know what port SSH usually runs over?

Speaker 3 15:30

Well, so, so basically, it is SSH is for 22. Yep.

Speaker 4 15:38

So I was at home one day, and I just opened up an SFTP client application FileZilla did a test connectivity for the our external vendor, and from home it works at that point, I trusted what the what our vendor was providing. And so at that point, I started escalating the issue with the file transfer team. What the file transfer team didn't tell me is that their application it doesn't actually go out of the firewall. It goes out through a proxy, a socks proxy. And what what they didn't know is that the socks proxy team are actually doing their own migration from the northeast to the northwest. They did not have all the rules on the bat were in the Northeast they didn't have all those rules in the northwest. So unbeknownst to the file transfer team, it was a proxy that wasn't allowing the connected so I didn't get to the end. Well, at this point, I'm waiting on the socks proxy team to fix everything. So yes, there was that basically was the firewall issue. That's all the questions I had. Do you want to go into some other issues?

Speaker 2 16:59

Yeah. I mean, I was just saying your conversation. A couple of things interest me a lot. First thing about unit testing. And your resume you didn't mention that you were basically participating in TDD.

Unknown Speaker 17:12

Can you explain like how, what was what was?

Unknown Speaker 17:15

What does he mean

Unknown Speaker 17:18

by second person?

Speaker 2 17:21

I mean, it's very interesting because and I kind of wondered what your experience with TDD like Do you like it? You appreciate it? Do you have any concern about it? Like just share your experience? I don't want to go into like technical detail of how you write it and all just like, like if you asked me, I have a big reservation about TDD and you know, that comes with his own baggage and doesn't make sense. And also, let's just talk a little bit like you say that you participated and you were doing it. So tell me about it. Tell me a little bit more about it.

Speaker 3 17:48

Okay. So, well, you know, basically like we have a fear of programming assistance with TDD. So which is like one of our like a driver and like other one is implemented. So So basically, we were, like adopting it like stress driven development, like process to be adopted within the team. So which is everyone's need to follow? So I think, from my perspective, so, like, initially, hinder like our process because it required different ways of problem solving, and try to come up with like, you know, failure scenarios and also taste right for right for me. Like personally, so I feel like I was like, learning like different ways of thinking through the problems and I was like, appreciating it. But you know, like for the overall team, I did not go out as a plan so for for our, you know, particular team because of the amount of delivers, like we had to complete in a certain amount of time. So, for me, like I personally enjoyed it. And also I would want to like work with a test driven development work.

Speaker 2 19:14

But But how would you do you write unit test first, like what are in defined unit test first, and then you implement a code accordingly? Or you write the code first and then you write unit tests? Oh, yeah.

Unknown Speaker 19:25

What was your approach in your team? So well,

Speaker 3 19:30

for a test driven development, so for basically, we had like around, you know, six months of that. So during that phase, like we were the test first, then we implement this is what like, except for that six month we have like, we would write our code first and then unit test. Yeah.

Speaker 2 19:57

Basic question. According to you can what is unit test, like why it is important?

Unknown Speaker 20:04

Yeah, thank you. For what

Unknown Speaker 20:06

are you looking for in unit tests?

Unknown Speaker 20:09

So well, basically,

Speaker 3 20:13

it is mainly for you, I'm looking for like in in unit test. It is that like, right you know, like you know, when we have like, his block of code, which is working mainly for to check the like, it is working or working as expected or not. So basically, it can have functions methods, so, like, whatever you have so isolating on that, so it is the piece of like functionality and also it validate the like blocks that expects, like, it works as expected. So right on our providing like multiple scenarios that you know, code block that expecting the certain things so for that you know, I give like error data for like, I'm expecting either and like exception or you know, like certain kinds of response back. But, if, but, if that does not go through our like, happy verse, so if I'm providing like proper parameters to that matters, then you know, I'm expecting the like, functionality of going to like certain services and also like, writing the database or something, something like that. So, basically, like, that is what I think is like, unit test is like basically I'm looking for like unit test is like, correct it it has a like a beret or assertion. So also it covers the like, is it mainly for AIDS cases, like you know, like, really for the normal scenarios, so in our conditions and also in other methods. So I look for like how it isolated the test and I'm like, making it more meaningful, like, rather than covering the line, covering the lines and also because you can write a test like oh, you know, like he doesn't have like proper resources and covers everything to pass the like, like, mainly for the like sonar scan. So but like, you know, when we have like, it's not meaningful enough for like someone's who comes in to read through the like unit test for the documentations like, of the source code, or it does not add, like any value on there.

Unknown Speaker 22:43

Yeah, that's about it.

Unknown Speaker 22:47

No, that's great. So

Speaker 2 22:51

you said that you write the code first? Well, you're following the ad. And so you already kind of define the test cases that you would have implemented, then your team write the code and then you write the unit test on it. That's the process, right? So the things you write the code first, is there ever be the time? I mean, did you ever come across a situation where you or your team started writing unit tests and you're like, well, this code is not good. We cannot write unit tests on that. Is that ever happened? Like how did you come across a situation like, what would you do? Like, you know, you're upset you've written like, let's say, 20 lines of code, right? And when you start writing unit tests, you realize that I can cover all the things that are no dependency on there and you have to break down the code. I've just kind of wanted to understand that ever have an experience like that, and you had to take an action and refactor your code because you're in a test. You can write it.

Speaker 3 23:49

Yeah, oh, yeah. That happens. That happens a lot. Because basically, like, if you have like certain code blocks, that's not like testable through the maybe like, oh, he started fields or like having the final classes. So that can be marked. So this is that is the basically you have to make sure that you have like proper implementations also, that is testable code. So right so we have to make sure that like any piece of code that you write is testable. So also, you know, it does not matter. Like if it's like, one line of codes or like, one or like, more than that aligns, so you know, like, it's our whole class that is change. So, for that, you have to make sure that like refractor that course so that you can or test in test in isolation so that I have like conveyed like my teammates like, like whenever you write a certain code, which is not testable, so for that, you know, I asked them to refactor it so that you can use like some kind of mock or like, even like, like other objects to write the unit test in isolation.

Speaker 2 25:14

So that's a good example. You're pretty awesome decent code and it's in production and a lot of bunch of unit test cases written that everything is good. In one of the sprint, somebody you or your team come and change the call for some of their functionality or some defect or something like that. When you make a change, your code works just fine. but bunch of unit tests start breaking. What would you do? what would be your course? of action in that case. Well,

Speaker 3 25:48

so, mainly in that scenario so I guess I would go ahead and also rebuild the. melody or unit test and make sure that changes are like in the ballad history. So basically, also like part of that, you know, I go ahead and see like, if something's related to like, set up the test and also maybe like mock objects is working. Like it's not working like as expected or like certain other scenarios, but I would not want to change my like assertions to mainly for to test the right since it is in embedded assertions. So, basically, on the test block, so I would say that like test the certain methods should return the some things then even read the like code scenes. So it will always written that like it will make sure that basically, rather than like changing the assertion, so I go into the like setups and see if, like, if I need to make changes to set up the blog, so that like additional code changes or sooner like break the having also like existing test but that is like more tests to cover the like additional scenarios. So that like any kinds of like my Microsoft accent that see if like any changes is set up the test. Yeah, that's about did

Speaker 2 27:22

you say you will change some assertion, the unit test that broke because of the code change, you will go and update those unit tests as well.

Unknown Speaker 27:30

So no, no, no, no. Okay.

Speaker 2 27:37

I will would you do like to say fie in a test case failed because of the code update. Okay. And you were part of the team and somebody else changed the code. And you've been asked to fix it. Like, what do we fix it? Because code is working just fine. It's just that unit has broke.

Speaker 3 27:50

Yeah. So I would say for the setup of the class, so So usually our code is working fine. And also unit test is broke, then the code probably has a like defect, right? Like, because assertions that I have would like usually that means like, it it works as like expected like follow up assertion that fails, then there is like defect like somewhere on this the chord changes made by like someone's that that was by a defect. So mainly, I would see like where the issues and like, like I would expect it like it would be in the set of blocks of odd like marks that that used that were not proper. So. So well, like any kind of like my course of action, so I would, mainly I would validate the assertions are like in is still valid, but if it's valid, so I would say for for like my set up for my mock.

Speaker 2 29:00

Conversation and your resume you did mention about the bluegreen deployment. Can you offered a bit more about it, plus, you also say that you've done a bluegreen deployment through elastic brands taught in GA D, get an AWS so just walk me through about the whole process for bluegreen deployment in terms of AWS, what tools you have used and all that stuff. Sure. Well, what was the reason like why did you go with blueprints? Like how did that happen? That that concept came into the picture and your team would have to do it and then how okay.

Unknown Speaker 29:43

So okay, so.

Speaker 3 29:46

So basically, like we have two stacks, which are mainly designed to buy two colors, which is blue and green. So for that, like you have like one environment, which are actively serving the like products and traffic and then also like, green, which is also inactive. So which is basically that's a stack where you deploy like your new or change that you want. So it is basically like blue green, mainly a blue and green deployment is kind of the process you have to follow for the like minimum like downtime. So also you have the like your activity stack then after that you have an inactivity stick so yeah, well basically like when you want to release to happen, so you would have you have that like blue and taking like active traffic gray. So which is like new versions and also then we test that or green or environment or to make sure that it works like as expected. Also, furthermore, you can run like some kind of integration testings, if you have like in like place or like, like auto testing. So once all the testing is validated, you can switch in the traffic from blue to green, which makes a green now active and then blue is on inactive. So for the for the next time. So whenever you want to like make any changes so for that you you will be putting your code to the grain. So sorry, like, sorry, to the mainly for the BlueStacks Yeah, then after that, it flee. It flipped over so you know like by doing this or you can like monitor the like current flip traffic and also if there is any issues we can quickly roll back by switching the traffic back to the previous colors and also after the certain number of hours or like any days so you can clean up the inactivity like mainly that that inactive stack for mainly for the resource management.

Speaker 2 32:14

So how did you do and like you said you have used AWS to your use been Beanstalk and and all those things. Can you talk a little bit more about your own process? Like how have you made a bluegreen deployment happen?

Speaker 3 32:29

Oh, yeah. So well, basically, like or when is this blue green or like deployment in the best of you know, AWS? So you know, mainly for the like auto scaling groups. So mainly, wait, we have to we have like released the soft fix with the blue end which is with the green tech. So we have like active load balancer so that a test like one of those like auto scaling groups. So during that release, so like the off color or which is not taking the like traffic gets deployed and like once we do our like validations so we attach that like active load balancer to the off colors. So that's how our process looks like like we manage through the Jenkins pipelines and also like our source course, mainly that is in the gig. So So through the Jenkins so we pull the light source from the GitHub then we run the run, run basically all these AWS scripts, so.

Speaker 2 33:48

Yes, for questions. I got one more question. For you and then I'll open up where think God was looking at your resume this kind of interest me a lot

Speaker 2 34:15

think we can still kick off the DoD question. I think I'm done. I'm gonna have you talk if you have any question anything for us? Yeah, yeah.

Unknown Speaker 34:28

Definitely. I have a few questions

Speaker 3 34:31

with you guys. So, you know, like, I like to know about like, team size, what the team is look like and like Team force, mainly for the like, like we talked before, like test driven development that we talked about before and me I know also, like the team is structures like, how many developers are there? And also, like, I want to know about like, next steps.

Speaker 2 34:59

or so for today's project, the one that we are hiring candidate they were mainly looking for the skill on the Java side react, and then AWS, the team structure is still creating a team. But we have a board of like a five developers on the backend side, five on the front end, and then we do have an expertise on AWS as well. So typically, is going to be the person who is going to look at your code code review code assessment. He's going to help you build an infrastructure on AWS and all the stuff is going to work with you every day and you

Nimesh Bohara- DHS-First-Saurav otter ai

Speaker 1 0:00

doesn't work out like Robert way calling and shot. It's meant to be nice to meet you

Unknown Speaker 0:12

for being able to do as well so

Speaker 2 0:14

I am the CTO here at mochi rabbit.

Unknown Speaker 0:20

Very smart and lovely people. So

Speaker 2 0:22

it's really nice to meet you this morning. Nice to meet you too easy California.

Unknown Speaker 0:28

So

Unknown Speaker 0:30

things are a little bit

Unknown Speaker 0:33

out of sync right now, but I'm good.

Unknown Speaker 0:34

It may look like he's in outer space, but

Unknown Speaker 0:37

it's just the West Coast.

Unknown Speaker 0:40

Right? That's just the plain of taking on.

Unknown Speaker 0:46

It. Wonderful.

Unknown Speaker 0:49

Tracy, do you want to start with your part?

Speaker 3 0:53

On Tuesday, I heard I heard from Jeff on some call right now but he will join us in about 15 minutes.

Unknown Speaker 1:00

So we will start

Speaker 3 1:03

so oh my gosh, should totally blank. But we all did an introduction Nimesh I would love if you could give maybe a high level. Read out of like your resume. You know a little bit of your background a little bit about you.

Speaker 4 1:19

Sure, yeah. My name is you miss. I'm here located at New Hampshire. So I've been working as a Java developer for like six years now. So primarily, I'm working on this wave application. And I've utilized like, frameworks like spring hibernate, RESTful web services, and I pretty much experience working with micro services. And I'm like design the micro services and develop them. And like even like deploy them to like high environment into the production. I also worked on like AWS Cloud for like deploying these applications utilizing services like AWS ECS lambdas, s3 buckets, and there's an RDS for like databases. So I do have experience working on the front end side. So particularly I use Angular for that. So but I can also work on with like JavaScript and TypeScript. So on the testing side, I pretty much have good experience working with J unit, and Mockito. And I've been working with, like, the integration testing, and currently, like, I'm on like a scrum team. And the team consists of like, five developers. We have like a product owner, Scrum Master, and like we, we do like scrum on a daily basis. So I'm mostly involved in like, working on like backend Java code, helping the team with like code reviews, and also in like production. Support duties, and so on. Yeah, sort of, particularly like, these are like my tools and technologies that I've used throughout my career.

Speaker 3 3:11

what's your what's your favorite part of development? Do you like running back in more kind of wandering?

Speaker 4 3:18

I would say like, I like to get back in a little bit more than the front end. But if there is like a task, I can do like front end, too. So yeah, but particularly, I would say like back Okay, cool. And when you join

Speaker 3 3:34

a new team, I mean this would be a scrum team as well as operating under safe, agile. So how do you go about gaining credibility with a bunch of new team members?

Speaker 4 3:47

Anyway, so certainly getting credit, credit. Credibility, it's like a little bit like challenging thing, you know, like, it's about like, Oh, if it is like a in a new team, we will have new challenges and like if there is like a smaller task and trying to get it into like, more communication as process possible, so like, right, like joining the meetings, providing any kind of insight, like as an outsider, and joining as a new team, provides like any kind of insight with the feedback terms like starting with the communication at all, so yeah, smaller task vulnerability fix or like any kind of backlog that can be done without like whole knowledge of the whole domain. So I like to start like a small and then build my way, like, build my way on top of that, like designing and developing the bigger big features and like, doing like bigger stories. So yeah, I would, I would send again, again, credibility in that way. Okay. Sounds good.

Speaker 3 5:03

And what would you say are the I guess the key ingredients to building and maintaining successful business relationships like with your product owner with your team leads? Sure.

Speaker 4 5:18

Yeah. So yeah, building relationship is like really important. So basically, what I think is like, the main ingredients for like, building a good relationship between like, anyone in the team is like effective like communication, be open with the communication. And basically, we can do like a clear and transparent communication. Like, let's say like, if, if, if I'm like stuck on like something that if I have a blocker, then we can communicate properly. So if I need like, help or like need to communicate that I would like to communicate in both ways, right. So with by listening and by speaking, and I like paying attention to the teammates, and also providing updates like on what like I was working on avec. So the next thing that is important that I would say is like basically be collaborative, and also have like a proper sense of like a teamwork and accountability of what we are doing. Right. So we will need to be like providing help to the teammates and also then asking for help. So, we have to be like a reliable and you know, like when there are like certain responsibilities like placed upon you, like we will need to like provide as much as like, responsibility for that specific task. So that I so that like these kinds of tasks can be done and then we can even be like a creatine ingredient for like the overall development. Okay.

Speaker 3 7:05

You mentioned something about like communicating updates to your team and keeping them informed of what methods do you like to use, but your team members know about your project progress.

Speaker 4 7:21

So for that, what I would typically do is I like to keep it like short and sweet. And then what I also like to do is like, give like a technical detail, and like, like enough product details so that like anyone can understand, like what I what is the issue that I'm facing with like if I'm blocked with let's say, like, external team member, I would say I'm I'm unlocked on this feature. So if I'm able to give like a progressing story that I'm taking, so I'll say like, is much more left but if someone wants to be like on detail, we can like think on like a separate call and get into the depth of that. So that I can I don't take up like much time on like daily standup and can utilize like the time a lot easier. And the standard time would be like, right on point for like all other people. So yeah.

Speaker 3 8:33

Tools are like what ways do you like to communicate? You told me like, you know, what you're communicating. How do you communicate that information? Okay, stand up.

Speaker 4 8:45

Yeah, so, yeah, basically, like, do I get my stories like in JIRA? I can do that with like comments on JIRA. And you know, like, and say that these are the blockers and these are the updates that I have been doing and can follow like sub task of our stories as well. And you know, like basically, like, the block like the sub task or like, or even putting like a complete state I can communicate using like Slack, or like Microsoft Teams, so I, I post my update on the team's channel, so that the methods I use like our verbal communication on a daily standup I post my update on a JIRA on the story status. And also like I communicate on slack if needed. I can also send like an email out to like external team team members, like if I'm stuck because like external team member, so if needed, I can do that. So that will be my means of communication.

Speaker 3 9:52

Well, I like to have your comments on because sometimes people don't put enough effort. And you mentioned you'd like to share some details as much as needed, you know, technical stuff, and I'm kind of wondering about how you like to share complex technical information with your stakeholders that are not technical. Like, I don't know technical stuff. So sometimes those things I can get lost in it. What is your way to like, relay that to people like me.

Unknown Speaker 10:29

So, basically, what I can do is like,

Speaker 4 10:34

I can use like a reference, you know, like, I can make second analogy of like, what the technical technicality is right? So, then what I can do is like, I can make it like more understandable, and like avoid any kind of like technical jargon, or like even acronym and use like simple term to make it like simple for people to understand. So, I will also like pro like to provide like a high level overview, like, so that you know, like, I start with like a purpose and like benefits, and rather than going into like, smaller details, and then also provide like a context and relevance of like what the topic is, if possible. So I can also create like a small task chart like that help people visualize the task of what it is. So that's like how I would like relay my communication for like someone who may have may not have like a technical expertise like in the team are for the external funding stakeholders. Yeah.

Unknown Speaker 11:48

Gotcha. Okay.

Unknown Speaker 11:50

Let's switch it up a little bit.

Speaker 3 11:52

Can you tell you about a time where, if you didn't see eye to eye with a colleague and this was kind of causing her turmoil or disruption in your work relationship, and it was ongoing, how did you approach that? How did you address that? Sure.

Unknown Speaker 12:14

Yeah, so.

Speaker 4 12:17

So yeah, I think like, what do you think is like, conflicts are like kind of, sometimes it's like, commonly like a workplace that you know, like, when we are working to like, the task, there'll be like conflict and how we build like our personnel as well as like person, professional experience. I think one of the time like, when I was having a conflict was during the technical implementation of like, how we are comparing multiple data and had an I had like, a certain point of view, and my teammate had like a, like separate addition, he wanted to go forward with so but you know, it's like, kind of like hamper like one complete holes print, and how we wanted to address that issue. So it was the kind of hampering like just not me, but the other colleague too. But like, what we did was like the whole team because when we were kind of like, blocking on the feature on the on the computer, but by the but the way we did was like we went through like resolving, this was like how we resolved was, we arranged a meeting to discuss the issue so that we could be more focused and uninterrupted with like external factors. And then you know, like what, what I made sure was, I listened listen actively what he had to provide. And you know, like, I listened to his perspective, I didn't mean to interrupt him and like asked him like, any question that I had, like, fully understand what like what he was thinking through. And the next thing that I did was like, I shared my perspective after I listened to him, like I shared my perspective right on my on my side of the story, and then later on, we did like, what we did was like a we went we came up to like a common ground and common ground by listing like the pros, pros and cons, like of like, each of our approaches, and then what we did, like was like a, we came up to the conclusion along with like, you know, our environment of like certain teammates have to have like an external stance on at like, at the point, it is like, so you know, like, I guess the whole point here is like, the way to resolve the conflict, where we are, is like not to see like eye to eye with a colleague, but to make sure like, we have like an effective communication plan. So we have like a focus group to talk through the issue and then come up with like a solution that will not only like resolve the conflict, but also come up with like a better solution, combining both approaches. So in this way, like I handle the consequences.

Unknown Speaker 15:28

Thanks for the example

Unknown Speaker 15:30

because people get sick

Speaker 3 15:35

and they just say, I might do this. So it's good to have like something that you actually went through and experienced. And I'm taking it I don't want to assume but is that your typical way of dealing with conflict is just arrange a meeting and then kind of go from there with the active listening or does it depend on the situation?

Speaker 4 15:53

Oh, yeah, that's what like, I like to do because like, we both can express like our understanding and like, the picture like of how we are like taking the issue in hand. So yeah, a one on one with with the colleague is like one of my best way to handle the issue.

Speaker 3 16:17

Any other questions? Anything? Usually Jeff would like pop in and like start tackling some technical things. And I really wish I knew sort of stuff to ask about that. But if you answered it, I'd be like, sounds great, but I really wouldn't know. So don't be so honest. You guys, though, the only way

Unknown Speaker 16:37

to be would you like

Speaker 3 16:45

so do you ask for forgiveness or permission?

Unknown Speaker 16:51

Interesting questions.

Speaker 4 16:59

So, like, it's an interesting question. So like, so what I think is like it's sometimes like something might have like a lecture at something might have like a major impact. I would definitely ask for permission. Like you know, like, if there is like making a big change, like you know, overall project, but for things like very time sensitive of something, like innovative and I'll try to ask for forgiveness because you know, like this forgiveness for task would be more on like, as ality and adaptability right. So, I would say like how certain like the task maybe like the beneficial. So, you know, like from my nature, I kind of like do more on like the forgiveness side because for like, overall picture I, I strive for, like, better, so better for overall for the team, and also on the application that I'm working on. So if like something works out, it's going to be great for us. But if it doesn't it usually go and I usually like go and make sure like I asked for forgiveness for going out, like going out of the box and I I don't want to go like two out of the boundaries. And like step onto like somebody else to toes. So yeah, I like to ask ready for forgiveness because that's how we innovate and like new ideas come up and we get into new things.

Speaker 3 18:46

To say that, yeah, innovation is something that we do strive for. Prevent I think, for the most part, they'll probably be permission. Yes, based on product requirements and things like that. But yeah, interesting and like, it depends on the situation.

Unknown Speaker 19:06

That was actually good timing because I see that Jeff has joined

Unknown Speaker 19:11

Jen, welcome to the

Speaker 3 19:14

inner department to the party here so I guess I can beat the mashed in my boxes to my left. Hi, this is Chuck.

Unknown Speaker 19:24

Give a quick intro takes a few minutes

Speaker 5 19:31

to get but anyways, sorry, I'm late. First of all, no problem or in any event, a different technique on CPMs with us for three years now. And then developer for 15 years at this point. So

Unknown Speaker 19:51

it's a pleasure to be here. And

Unknown Speaker 19:57

can you recap for me really quickly?

Speaker 3 19:59

If I need to apologize. We did some introductions. We have not really talked about program, but we've covered more like I was just trying to get a sense upon the mesh works and things in regards to like interacting with team members and stuff like that. So I didn't hit on any any technical things

Speaker 5 20:18

that you're like. Yeah.

Unknown Speaker 20:26

From the getting rid of that, so.

Speaker 5 20:29

I think so. My interview is going to be mostly focused on how your thought process is performing. Like I want to know what the question was about the second one, because I do care about wanting to understand how you approach problems how you approach

Unknown Speaker 20:45

like working with others that kind of stuff. So

Unknown Speaker 20:49

when you

Speaker 5 20:57

let's talk about like, what you are doing when you get a set of requirements for a ticket like walk me through your thought process of what the mental checklist of what you're looking for in that ticket and then walk me through recognition and getting that code out into the world

Unknown Speaker 21:19

from getting the ticket

Speaker 4 21:22

from getting the ticket to like, getting the PR approved and all those things. Sure, yeah. So firstly, what I look for us, like, you know, like in JIRA ticket in JIRA, I look into the description of the ticket, and then the acceptance criteria, right. So what are the criterias for the acceptance? So, you know, like, if I'm not like clear on how the story is like described, like during our planning, let's say in our planning or grooming session, I look for like what a problem in hand is, and like, what the requirements are like we are trying to achieve and what we are trying to achieve in the story. So, I do like, what things need to be approved or like, what things need to be approached and what happens like when something is triggered or in this flow person. So and I also like to look for like, what kind of priority of like certain ticket is so if there are like tickets to that as like a high priority, I pick that one up and then and then start the next thing

Unknown Speaker 22:41

and the next thing that I do is, is there is also

Speaker 4 22:46

a ticket right so if I'm waiting for like another team member to implemented implement, like, certain features to them, or like if I'm waiting for my teammate to provide like a like, any like, because to my task, then I make sure that that is properly noted. That basically, then what I do is like, I start with my technical implementation, you know, like and then writing the code, you know, like setting up like certain stuff, based on how I do it. And then what I do is like, I plan my work by breaking down the task into like, smaller, manageable tasks. So that you know, I can break it out into like, small small sub tasks, and work on it accordingly. So if needed I can also try to see like, if there is any kind of like design pattern or like any best practice that I can apply for like easier understanding, and then try to see like, make like if there is like any proper documentation or in that like for that particular ticket. So then I open like a pull request. I guess like before that I do like, once I'm done with like coding, I do unit tests for my code changes and make sure that that test, passes the test. I mean the coding unit test, and then I will make like a pull request for my team members to review my changes. And then once I get like all the approval, I do the deployment and then validate my changes and like lower environment and then you know, like move from there for like, production release or like hiring like

Speaker 5 24:42

let's say, during the process, you get some feedback from one of your team members. walked me through you know, not everything. How do you evaluate good suggestions versus ones that are probably not worth bringing in as part of this

Unknown Speaker 25:07

Yeah.

Speaker 4 25:10

So for that, I like it's like a good suggestion or like something that's like, not really not relevant, I guess like, I would like to understand like, what the committee is trying to say, and like what the teammate is trying to provide a feedback on like, actually, new in the like, in like, actual context of what the problem is. And like the observation is, and basically, I would like to see if like, it provides like a value by making those changes, right. So basically you know, like, sometimes, like, some kind of problem or an opportunity that my that may be on the table for like a later discussion. So if that says, aligns with the goal of the story that I'm trying to achieve, and so like, if if, if, like, I don't implement this, that solution what we want and what could have.

Unknown Speaker 26:13

So that's something I would like to take into account.

Speaker 4 26:19

My team, my team take lead and other like team members to see if like, they have like something that might add like a value. So if I have a doubt on like solution, you know, like I can, I can do these things and station from tech lead or recruit other team members.

Speaker 5 26:43

A little bit here. So some team members will request what is it that you were talking about being critical about this? What were you looking for when looking at someone else's work? Why? Put me back why would you ask what kind of stuff enters your mind when you're reviewing other people's code?

Speaker 4 27:04

Okay, so when I'm giving someone else a pull request, yeah. So I think like, I look for multiple things. The basic first of all, like the basic idea is to provide like a proper code, like with copper code quality, and like has like any it has like a high metal level, right? So I also make sure like it aligns with our requirements. So basically, these are like some of the things I look for like quality in the code. So if, if the code is like, easy to read, and it has like a meaningful variable naming, let me see if there are like comments that that are needed on like certain block. So we can also look for like styles and standards. I also check for like, reputation of like the functionality, if there is any functionality of reputation, so I check on them like the DRI principle, which is like do not repeat yourself. So, I also would like to like, check for the correctness and see that like how it handles like the use cases. So you know, like if there are like any error condition, like there is like if it is not like handled properly. And if the change, meet the requirements and acceptance criteria, I look for like any other vulnerability that might have been exposed, and like how it handles like sensitive data, if it is needed. So And honestly, like, I also take like for the testing, like if there is a unit testing or not, like if proper testing is that like adding the value for the test and take any kind of like integration testing is done or not that that like it interacts with like different components. And like the test covers, like all those tasks, test task or not. So if possible, like I also look for like documentation, like Java Docs, or like even like, with like code comments, so that like when I look for like, so like when I do for good comments. Like it's like being true like the COVID swimmin psychonomic maintained able to stay so that whoever is like joining the team, on the later date, has like easy understanding of like of the over over flow.

Unknown Speaker 30:01

I have a question that kind of a little bit different here.

Unknown Speaker 30:05

Let's say in nonprofit

Speaker 5 30:07

projects, you've found a deficiency in some way. Let's say that, you know, it's not performance, or it's using a pattern that's been kind of depreciated. We use a different federal one. Now I'm moving you know, what to say better, what just what's better, right? But it flies in the face of the existing, like, existing pattern of the system. So there's a case to be made to be committed to be the change that we made here, but you have to get buy in from your teammates. What is the proper way? How do you sell this new approach that they want?

Speaker 4 30:56

Yeah, so I think like, I would follow like the same method, I like usually do. So which is like to list on all the advantages and disadvantages. of the chains. Like with the existing flow, and what would be like the new pattern, which makes it like a little different from like, what we're using, and then like, using a similar pattern, which would make the code like this streamline and use like a sort of similar pattern, and you know, like how the infrastructure is. So my personal approach would be like to list down the advantages and disadvantages and provide the team with that, and see how like, what like how we can approach with that and then how we can like change to like more advantages in like overall scenarios.

Speaker 5 32:01

So what say that one of the things that the value to bring the same changes, like how do you what do you do at that point in time? Do you try to invent some new clients, you know, just abandon the idea entirely. Like what what is it that you like, let's just say that you're really passionate about this, you want to change, but what are you going to do?

Speaker 4 32:28

So in that scenario, what I in that scenario, what I would do is like, I would like to go on, like one on one with him. And like I can think like and think like what we talked about with a client. So but you know, like, we can also go on like one on one to convince him and show him the advantages and the disadvantages with the approach. And you know, like, try to get him like on board with his idea as well. But because like, if, if like, the whole team is on board with the flow, and you know, that like the whole team will would be more like worse and like on the pattern. So because like sometimes the pattern gets like, on like, how we implement it, and if somebody is unknown of the pattern, and it might become difficult for like particular person or even a creator, or even like a particular person might implement, like a different pattern of what's going forward. So I would like to create like more segregation, and then a more streamlined process. So yeah, I guess I would try to make inconvenience so that the using this pattern, and voting for this are really necessary. For like better betterment of the process. So in this way, I would like to convince you.

Speaker 5 34:08

of a typical question. So what are your considerations? Database real quick on this project, that has are largely in charge of creating and maintaining database structure. So you know, we have DBAs consult with but when, you know, from your perspective, if you're creating the relational database model, what is it that you are doing to make sure that you know, you are keeping the databases performance, you know, clean and from just, you know, data consistency, and any other things that would go into the process of you know, maintaining a good data database

Speaker 4 34:53

so, yeah, in order to do that, you know, like, in order to have like a universe that is like copper, data consistency, and like performance, you know, we would need to do like, ensure normalization, that, you know, like the database schema is like normalize and like he's at least on the third level, and so we have like a proper data types in place, right, so we can use like appropriate data type with like in cases, square screens, and like have common with constraint in place. So not all are like unique, so check. We can check like, and make sure we have like a proper primary keys and foreign keys in place to define like relationship between the tables. So we can ensure that database operations like adhere to the properties to promote like the integrity and have like a proper Kriegers in place and like have application level or data validation, or like for for like performance specific, we will need to make sure that the proper indexing is in place, and like more frequent query data. So, we have like a proper schema based on like, requirements, considering like the data access pattern, and so on. So we can like we could like create like a views, which are more like a temporary table for like, for the for the frequently exist data and availability of the database, of course, like multiple reasons, either through like a replica or like any other approaches. So I think these are like some of the things I can take into consideration for for doing that here.

Unknown Speaker 36:57

I'm pretty much at this point. So

Speaker 5 37:03

let's say on CPMs anywhere, it probably feels like we're gonna do a brain surgeon.

Unknown Speaker 37:08

Unless I look what's already been done.

Unknown Speaker 37:10

I didn't I

Speaker 3 37:12

didn't do anything. program related. I didn't share any of that yet, but I didn't know if he would or Johnny wanted to. Yeah, so your

Speaker 1 37:26

backgrounds both JavaScript and Angular. Yes. Okay. That can you tell me about a project you've worked on in Angular and sort of what your approach was to working within that framework?

Unknown Speaker 37:42

And opposing sci fi? So

Speaker 4 37:49

I worked on like, developing like multiple components like in Angular. So one of the components that I was working on was did a graph from the backend back inside and then like, deploying it into like a tabular form in the angular dashboard. So this page was like, supposed to be you know, like, previous table that that we were using, like, across like, different components.

Unknown Speaker 38:21

So we were using an

Speaker 4 38:24

internal component to display this data. And then my task, particularly was to like, populate the table by making a back end call to one of the API's that we will like we had we had like filtering it out, like based on like, certain requirements, and then adding that functionality and to export that data into like, Excel document. So it was like one of the most recent tasks that I did, but you know, like, like, I do work on Bangalore on like, a regular basis. So yeah, we have like mono repo. And we have multiple components, which are deployed on Punisher. So yeah, I've worked on Angular industry like it was like one of my recent projects.

Unknown Speaker 39:17

With them, but if you have a

Speaker 1 39:21

difficulty news and extend across the province, can you tell me how you would make that component available to the components?

Speaker 4 39:33

Can you repeat Sorry, I couldn't hear you bigger conference. Can you? Can you repeat that one more time?

Speaker 1 39:40

So in your projects, you have a component that you need to make it accessible to other components within the project. How would you make that component available?

Unknown Speaker 39:52

One component making it

Speaker 4 39:55

so please, and for that, what I would, what I would do is like I would in order to make that like one component accessible to that other component. Basically what I do is like I make sure like I use like a service to use it. So basically, I would create like an external service and then basically share the certain data and ease it into those service into the component. So so that could be like a receiver component. And we can inject that into the service and then use it you know, like with certain like, decorators like input, output or like, data sharing, you know, if we need like, if we need to save the data, but usually this would be done.

Unknown Speaker 40:48

And this, this is how I would do it. Yeah.

Unknown Speaker 40:53

There's an easier way to do that. Usually

Unknown Speaker 41:06

easier way to

Speaker 4 41:09

do one from one company to another. Yes. So there are two. Yeah.

Speaker 1 41:17

So easier way might be to declare that component and a module and then the module available.

Speaker 4 41:26

Yeah, so we can do that straightforward using the Lipson account component. set during the template itself

Unknown Speaker 41:38

our work

Speaker 6 41:46

they would still like this I'm sorry. Okay. Yeah, he's struggling with that clock here. So sorry, we didn't let Jeff know that piece of information that I hear is struggling with a bit of a cough or some type of cold. But a

Unknown Speaker 42:19

couple of things really quick.

Speaker 2 42:23

I bet the resume Institute does some work in AWS.

Unknown Speaker 42:27

Correct. Yeah.

Unknown Speaker 42:32

Tell me. What's the difference between

Unknown Speaker 42:36

basically

Unknown Speaker 42:42

told me okay.

Speaker 2 42:45

Are you familiar with the term CloudWatch Yeah. What is it?

Speaker 4 42:53

Sorry. Yeah. Blog watch. So basically, cloud watch is like a monitoring tool, which is our like, it's like a service provided by AWS. Right. So basically, you know, like, whenever we want to deploy our service into the AWS, we can use like a cloud watch for log management. And it basically enables us to like store and access the logs maybe like from from like an easy to try instance, is it an instance or like our lambda, and then we can let's set up like a certain Block Groups, and stream those into like, organize, and then organize like these logs. So and then what we can do is like, we can basically filter out these logs with like 10 specific data, and then set up like a certain alarms like based on like, specific metric to trigger the notification. So that you know, if we have like a sudden, let's say like CPU utilization or memory or like something, we can get notified through SNS service, and then we can we can be like aware of of the application if it's in like critical state. So we can also create like a customizable dashboard to visualize all the metrics in the growth in the like in a graph. So yeah, basically, like Cloud watch is, I think, that

Unknown Speaker 44:23

what service train cloud watch and cloud

Unknown Speaker 44:28

watch and cloud create.

Speaker 4 44:33

So Cloud watch and cloud play right? Different so yeah.

Unknown Speaker 44:39

So the

Speaker 4 44:41

discovery, I would say is like a cow crane is like a primarily, like a logging service that records like API calls, and, like even made within an AWS account, right. So basically, it provides like a visibility into like a user activity and resource changes. So basically, if there is like an API call, it automatically records the call. Made in the API is like the AWS service and you know, like, by getting into the by getting into like mowing coastal colonies, the time of the color, and what equates pyrometer art and also maintains like a history of this course. So it is useful, like whenever we want to audit a user activity and like resource changes, and investigate any kind of incident like are like an authorized xs by the euro. Dollars these like most people CloudWatch is like mostly used for monitoring the application and a lot like within the application.

Speaker 6 46:09

Actually, can I just, just to be mindful of the time that we

Unknown Speaker 46:15

I just wanted to share but

Speaker 6 46:18

yeah, I just wanted to see if there was anything we could do. Just a quick a quick summary of the program in our government, even if you could give just to give him some context. We committed

Unknown Speaker 46:43

we didn't work

Speaker 5 46:45

we were going to contract project or contract the acronym but we work for basically a fish management system for Florida integration Bill was massive. A big part of the process that will put the scheduling system is a big, you know, appointment. scheduling system at times lots of people what better to go up to on their phone and provide those employment centers we have details of the reasons they are coming into the center. So once they fill up, there is a tool called CDC. The primary tool is a an application that runs on a desktop, it's a fingerprint to take pictures, and it will also take personal information about the applicant. And it looks at some type of TCP connections and this is where I work in a data store. of personal information for applicants for benefits this guy for people that are looking to become a citizen or green card, people that are applying for doctors. For Nashville, I see situations like that. So we saw lots of data about these people. The providers, the details of their applications, and there

Unknown Speaker 48:04

will be going back on track. Some people

Speaker 5 48:08

already have a football that would affect their application for benefits. So we do interact with a number of services. Lots of agencies, partners within with the ionic that also do similar with different

Unknown Speaker 48:23

applications that that is

Speaker 5 48:30

our business standpoint, our application is a Java back end with a React buttons if we have like 25 micro services work on an Oracle database. We have a Kubernetes cluster that we work on. That way all of our stuff Kubernetes and Docker files out to us as emergent Australian addresses are probably the right choice. And we don't want to wait outside of a few major tenants. There's a lot more data that we use. But you know we do a lot with SQS you know, we have all began to use the Kubernetes layer that Amazon provides and you know we use RDS and all the other kind of like normal stuff. That was a big red version. We didn't get a lot out of it get questions asked him I was

Speaker 4 49:23

sure yeah. But I think you answered all those things. I mentioned all this process, and I couldn't like visualize it. Like, I use all this biometric and all these things. So yeah, I could visualize that that was a good one. Yeah. So I had like one question. So because it's like so sensitive data and all do we require like a top level security clearance for this or

Speaker 5 49:51

required it's it's called public trust, but it's not like a DOD level clearance.

Unknown Speaker 49:58

That's the kind of some of the things

Unknown Speaker 50:02

like it's kind of a surprise, but if your background

Speaker 5 50:05

it's a profit, it's like less restrictive than you do that you do have to pass a background check in order to get productive access. It's okay, yes,

Speaker 4 50:15

that sounds good. And yeah, so yeah, one last thing would be like, is this like a final interview or will there be like any next round on this one?

Speaker 3 50:32

So with our program there is gonna be for an additional interview, which is with our the overall tech leads that sort of goes on CPMs and NAS applications. So that will be like the final step. After this, yeah, go ahead. Probably will be the one to like, update you on that on next steps for sure. Yeah.

Speaker 6 50:56

I just want to do a quick thank you to everyone on the call. Thank you for everyone's time, and focus on this interview really appreciated. And we met any other questions on your end. Are we going to wrap?

Speaker 4 51:10

I think that we're good to read. Thank everyone for your time. And yeah, it was nice speaking with you.

Unknown Speaker 51:18

Nice to meet you as well done. Thanks,

Unknown Speaker 51:19

Jeff. Face to face.

Unknown Speaker 51:24

Take care and thanks everyone. Bye bye.

Nisha-WellsFargo-SRV-final

Speaker 1 00:00

We're doing great here. Give us few seconds here for Swapna to join, and I guess then we can get started. Where are you located? Nisha, trying to find you resume. Sorry.

Unknown Speaker 00:22

Just came out of a meeting and jumped on here.

Speaker 1 00:30

We have Misha on the call. Thank you for being on the video call here with us. Misha, appreciate it.

Unknown Speaker 00:39

So I think

Unknown Speaker 00:43

HR has

Unknown Speaker 00:47

within my space in North Carolina at

Speaker 2 01:01

this time, Charlotte, right.

Speaker 1 01:09

Okay, okay, sounds good. So I think we have your resume in front of us. Nisha, it would be helpful for us if you can take few minutes walk us through your resume, give us a brief overview of what you have been working on, what your experience has been, and just give us overview of your latest project that you have been on. As you know,

Unknown Speaker 01:32

my name is Isha. My full name is

Speaker 1 01:36

and I have been working as like software engineer for about like six years now, and working primarily on Java patients using like framework like Spring, Hibernate services or API development, pretty good experience Working on microprocessors and also like being involved, like,

Speaker 1 02:17

work on includes, like AWS ECS, like s3 and also like that. So I have, like, worked with JUnit, and also like makito and talking about the front end side, basically, I worked with Angular. So to talk about my current team, actually, like my current team consists of the five developments we have, like product owner and all like scrum master, and also on like, like daily basis, and mostly work on backing up the Java code, which is like helping the team members With the port reviews and also I production support. So

Speaker 1 03:09

production support door there for my dog. Keep on working this project, what that project is about. So recently as Carta, I am, like, working on the stock plan administration platform. So there we were, like, managing like investment portfolio. Also like we are basically like, take care of the account, the allocations of the like each of the account investment, and also like with each of this investment held on the particular of actual account, right? So only like we all also like using like Microsoft actually help to the waste management so it like it is a multi asset configuration, and so if like user honors, like multiple assets based on different hierarchy that we have, so we basically, like handle the risk and also allocation of sales. So primarily, like RESTful API development being involved in the creating, like, lot of microsolid points and like, also, like we can say, primarily risk full API development been involved in creating microsolicit importance. And also, like integrating them with the dashboard. So finally, this work, like I'm doing that work, and it is migration for so we do this features already in the legacy application and

Unknown Speaker 04:57

Spring Boot and AWS. So

Speaker 1 05:03

your main engagement has been mainly from the back end or from the front end. How much comfortable are you with the front end application? So for me, it's kind of like 70% on the back end and like 40% on the front end side. It is something that I get more experience on the backing side I'm comfortable with, okay, and how many team members you have in your HR team currently? Currently we have like five team members who are engineers, and there we also have like, product owner and okay, and you are considered as a senior developer, junior developer, lead developer, what kind of role you have right now? Senior Developer, um, so when you guys are doing a development within your sprint, are you guys doing the testing and everything also to consider the story as done, are you performing like functional testing or just the unit testing, or is it just the development. So basically, we follow something called STL. We are like, responsible for unit testing on the code that like we write. And then again, we are also like, responsible for the like riding, automation testing using also we have our regression testing shoes that, like blends, already set up. And sometimes we also need to add certain test cases here as well. And recently we also, like started picking up the into in the playwright, which I'm getting started on. Okay, that sounds good to me. All right, so stop my email. I'm gonna hand it over to you guys for you to get started with your technical questions, and we'll go from there.

Unknown Speaker 07:16

What are some different

Speaker 1 07:31

my previous project, basically, I have worked with a lot of like these, all patterns like singlington, like decorator pada and also like bigger patterns. So it is basically the way you create an object, so you only have, like one instance of objects throughout the whole application. So it's usually useful for the clients, or maybe like even a configuration manager. So you have a single strand for like, for example, we can say, when you use a log for you can give instance of the logger, which is like kind of things. And you also have like a pattern, actually, you can attach like additional behavior dynamically without a single chain structure. Or there is some like SATA pattern, which is for like micro corridor architecture for the distributed transaction, and also like system. And then you also have, like, build a pattern, basically, which is, which is basically to create like objects in a baby, like same by state manner. So these are, like, some of the normal design patterns that I usually work with.

Speaker 2 08:57

Okay, that sounds very good work.

Unknown Speaker 09:10

I

Unknown Speaker 09:22

know you mentioned like

Speaker 1 09:41

ECS and where is our like microprocessor is our diploma. So we also use like lambda use cases where we like basically trigger station flow based on like matches on the actual tube. So we have like, stacked content on the content development on the like AWS ash tree, which is behind the CDN, and we basically work with AWS rds, like database layer, but these are the primary services that I actually work on sounds good,

Speaker 2 10:32

and to deploy those applications to AWS. What deployment tools or pipelines have you used

Speaker 1 10:42

this we work with Jenkins basically, and like we on the Jenkins pipeline. So the way we actually join our Microsoft offices using like Docker file within our application. And this Docker file basically we build like Docker images through environment and to the environment. So all of this is done through the AWS Cloud Foundation campus and integrated like part of Jenkin pipeline. So we actually do similar process for the CO deployments as well. So it is also, like automated through Jenkins, through and like where you basically pass in the view name and set numbers, and it basically creates a new pipeline every time we like, follow like blue green department. So, yeah, basically we do this

Unknown Speaker 11:55

blue green department. Okay?

Speaker 1 12:15

So you asked about the API sites like security and API.

Speaker 2 12:28

How do you secure your API? What kind of authentication, authorization features?

Speaker 1 12:35

Okay, so basically API, we basically have screen security right? The way it works is users, and initially they also get, like, redirect instrument, identify provider. We like, actually, we don't own the service, but we have an identity provider. So whenever our user basically tries to come to our application, they get redirected to the identity provider. And if the user is not authenticated or like the user provider that gets their username and also the like password, once it is like validated, they return to like keydogly tokens, and again, like after that capability token gets authorization later. And we also like validating authorization, like leader, using security and based on tokens, we like to rule The effects and control how the

Unknown Speaker 13:59

security works. Hi,

Speaker 1 14:02

national. Can you walk me through the steps you would take to implement an API, starting with defining the requirements all the way to deploying the API? Like, what steps did you take in your current project? So

Speaker 1 14:22

the like, like, talking about the whole lifecycle. The first thing is, obviously your design session. So I haven't been involved in, like, designing the microphone. You have your requirements, like gathering where you are basically document product owner, or like you are, like stakeholders, or what the API needs to do and how, like, who will use it, and what kind of operation you need to do so, talking about, about the all the functional requirements, right? So, like the basic meat of the application, like what needs to happen and what are the expectations and what are the input and output format. So it's kind of like, like defining your contact of the application, like, how the talk about what specific time I worked on the maintenance and API and responsible for the profile can say, like maintenance. So here I call I gather these functional requirements about, like, what it actually needs to do, and what are all the functionalities that needs to like support. So I put that into like into our design document, and then I also add like, non functional requirements, like, like you can say, like, so that would be basically the first thing that you need to do that you are basically like, design those APIs by using maybe, maybe we can say like, like schemas or entry point of taxation request mapping that matches each resource that you are trying to do. And then the thing you do is, like, rely on the data from other services. So this is actually where you actually make the external services call. And then after that, you have our which is basically it is used to talk to the database, like the work with the same data, GPA, and also like write the different conditions. You will also like add regarding controller advice or global. And once you are done with so after that, you can, like, run it locally and take it but like, higher environments, you basically, like, start with the right next type packages required to do the application. And it is also like, gives you steps to build on the application, and after that, then the application needs to go through to deploy it. And usually like comprises of premium sales and sales and the like deployment after all that process basically

Speaker 1 17:57

like checking to the course, feature band and also like request to like go to the like develop branch. And once we like get through that to like get enough approval, you can get it finally mauled to the developed and that you can launch the kick in five times to basically add reviewed and got the application that would be a very high level flow of everything using basic micro. Okay, so you were involved in all four stages, is what you're saying, starting from defining the requirements, implementation, test and deploy. I like, I'm involved in everything that I mentioned here, like all the COVID stages,

Unknown Speaker 18:54

okay? Or

Unknown Speaker 18:59

did you use an Excel team, like, you

Speaker 1 19:04

know, we did, like, use JSON format for that. So, like, we basically use JSON. Okay, so did you define the JSON, or someone else defined then you just used it. So, talking about Jason, we are, like, migrating for our project. So we basically looked into what are the details? And basically like, make sure in our system, like my product owner, we basically finalized that structure, and we had, like, basically proposed what we initially thought of, and then I did more review guys. So basically we had to design, talk constraints and also get approval from the architects. So that's what I did, initially, provided they're like, for staff, what the response would look like, and what that schema would look like, and then after that, I was like, it was more clarity. After that, okay?

Unknown Speaker 20:15

And for your unit test, what did you use? Okay,

Unknown Speaker 20:22

and how did you test your API?

Speaker 1 20:26

So, like, how is your API called? So basically, like, you test, my test is a manual testing using postman. So basically, I have already different call commands that actually different and different cases. And that is like first step we do, and after that process, we basically integrate the front end application. So these are the like, two different things that I do. And since I, like, I have developed a feature somebody is take care of the writing the automation to basically, like, make sure it is properly covered and all. What did you use to build

Speaker 2 21:18

Your package, the app? So

Unknown Speaker 21:42

locations, annotations?

Speaker 1 21:46

Common Spring Boot annotation? Basically, Spring Boot provides a lot of annotation. So talking about the common annotation that works with the RESTful API, like it will be like risk controller. So risk controller is just for the controller layer. It is basically a combination like offer controller, and is also responsible for like body animation, which basically means like it is responsible to be provided rather than you, right? So it adds like as an entry point, and also like gives you some kind of response that maybe you can say education,

Speaker 1 22:42

which is basically a component annotation of depository, anode annotation, which which is basically like buying your classes with sorting like metadata on like what is talking about, the dependency injections related to annotation you have also, like wired annotation, which is to like inject dB, and then you also have a qualified annotation which is like name of the beam you are trying to inject in. So you would obviously have the configuration annotation with class, like with teams are defined. Then after that you have like been annotation, which is annotation on a method that actually being returned by the game. So basically that all the like common annotation do

Speaker 2 23:45

finally, the difference between an abstract,

Speaker 1 24:00

so extra class is basically a base functionality, right? So whenever you are actually you are actually defining an abstract class. This is where you want to define your condition, like of the plug in methods, maybe. So it is a pure shared code and like shared change between the classes. So another thing about like, about the abstract classes, that it can have the constructors and the way like other classes inhabit it, is using the like, I think, keywords now and then and then. The interfaces are different that they can like pure contract, right? So it usually like defines what your implementation has to do, and it cannot have the constructor. So to talk about the multiple instance, dava actually does not suppose multiple inheritance. So you could have, like, exchange multiple extra classes, but you can definitely implement, like, multiple multiple interfaces. Just

Speaker 2 25:19

a couple more questions. Do you know how to make an object completely encapsulated?

Speaker 1 25:28

So talking about the object being completely encapsulated. So basically, in order to make an object like completely encapsulated identity, actually access to the internal fields, and you only have some kind of, like control way to access them. So firstly, you make all the sales that is within the class private, and after that, that's the very first place. And then after that, if you make public letters and feathers, they cannot be accessed like those files. And you you also have some kind of like collections you want to like, avoid any kind of like mutable field. And if you really want to take a step further, you can also like, make it immutable by like, Will adding final to prevent any kind of sub casting. Or also you can, like, say, I think like, we'll use to do that. So how did you say that you're used?

Speaker 2 26:40

And so how did you say that you use cucumber for your body? You give like examples of like, what framework did you use, and how did you go about executing them,

Speaker 1 26:56

talking about like cucumber, so we have actually used internal tools that is like, orchestrated, that testing like, like you can know that cucumber, it is like Behavior Driven framework.

Unknown Speaker 27:18

So basically, like, upload this

Speaker 1 27:27

syntax. So basically, you just provide the option and the features and the plug inside specific how.

Unknown Speaker 27:40

Continue,

Speaker 2 27:47

specifically, did you write the scripts yourself, or did someone generate scripts yourself for you? Or how did you go? How did you

Unknown Speaker 27:58

go about the testing there,

Speaker 1 28:02

actually, we do have a cucumber test runner with our test classes, right? So we actually do run with cucumber, and we provide all the options within our test classes, and those feedback and something we like, you can

Speaker 2 28:35

marry working on migrating legacy applications, or, I'm guessing, in your favorite

Unknown Speaker 28:46

general idea, like,

Unknown Speaker 28:49

are you switching? Are you operating a Java version? What high

Unknown Speaker 28:55

level? What kind of

Unknown Speaker 28:59

work did you have to do all that.

Speaker 1 29:02

We are basically moving legacy application, like new school based. So we are not like we are now moving into our old micro architecture using like modern Spring Boot and also like Java 17. So that's what we are migrating. Yeah. So application

Speaker 2 29:24

was not deployed in cloud, so you just move it into Cloud for the first time.

Speaker 1 29:32

Basically, yeah, we, we are, like, moving towards the like, Java 17 and Spring Boot.

Unknown Speaker 29:41

What is the legacy application written in Java?

Unknown Speaker 29:45

Legacy application written in Java,

Speaker 1 29:49

basically legacy application. It was in Java, Java. It was in Java, the lower framework. So

Unknown Speaker 30:09

on anything else

Speaker 1 30:19

was in your journey when you have been working here in different projects, knowing that nowadays, with Gen AI and COVID coming into the picture, how much use of that you're Making. Talking about that part, I have found basically AI to be very helpful whenever I'm like working with the testing part. So sometimes when I'm actually working with that is automated testing, a lot of this scenarios. So when we actually blog, like blogging that we add more into our DS code, and also I can ask, like envision, to give me scenarios that I might have missed. So it basically gives me different scenarios that needs to be automated. So that's something that I found really useful, and I think it is a very good to be like Incorporated, but there is not a way that you can totally rely like, on the actual logic. And so because, because, while working on the project at home, physically, I feel there, there is a lot of mistakes and it makes the part, I think it is a very good tool to get into anything. So currently looking for a new

Unknown Speaker 31:53

position here. Can you

Unknown Speaker 31:57

tell me why the student is being paid here?

Speaker 1 32:00

Are you currently working? Or you are out of project? Yeah, I'm currently working. So basically, my work at I'm working at carta right now, and it was a contract role, so I'm looking for, like something more permanent wise. So my role is Tata is coming to an evening next like, middle of next month. So that's why I'm looking for the new opportunity. Currently. Are you in San Francisco there, or are you working from North Carolina?

Unknown Speaker 32:32

So I'm not in North Carolina right now. Okay,

Speaker 1 32:42

sounds good to us. I think we ran out of our questions for you. Thank you so much for being on the label here. Appreciate all your time. We'll get back to the HR and let you know.

Unknown Speaker 32:54

you.

Pintu Jha- Garmin- First- SRV

Speaker 1 00:00

You doing? Yeah, I'm doing great. Thank you for asking. How are you? Thank you.

Speaker 2 00:08

I am Michael Mattingly. I'm a architect for Product Support Team in it Web. I've been at Garmin for a little over 14 years. My team started off with literally just one person, and then I was the second person to join for our team. And I've been on the same team the whole 14 years, and we've grown from just a tiny little team to now. I think we're, I don't even know the exact number, but we're somewhere around 1520 total people, including a team in Romania, a team in Hyderabad, team here, Bas business analysts, product owners, all of that. So our team handles and owns the support.gov.com domain, helping customers. If they have issues or questions about their devices, they can go there. It also, we also support all the applications that the call centers have. So when a customer calls in, the agent gets a pop to their their station, and it says what, who the customer is, what devices they have, etc. We handle all of that. So anything that can get that customer back to doing that love with their device, that's where we are. Patrick,

Speaker 3 01:38

yeah, I'm Patrick. Been here at Garmin for about five years. I am in the pharma data space. I am in the product data and device data space. So we help manage, maintain and distribute data about the devices, serial numbers, things you would need or be required of for registration or for accessing, export those other things, product data, main is really around specs, information about the devices in general. That information is used by a lot of recruiters to determine behaviors of their devices. What they do present a website purchasing is even exported to third parties, for Google Store, Pinterest ads and sorts of things as well. So do a lot of work with high volume changes or bulk changes that people need. We accept and

Unknown Speaker 02:37

we publish those out to consumers.

Speaker 3 02:40

Been here, like I said, about five years, and I've spent that entire time in the space I've worked in other domains, but while I'm here, this has been

Unknown Speaker 02:55

thank you so much for the Introduction.

Unknown Speaker 02:59

Absolutely tell us a little bit about yourself.

Speaker 1 03:01

Yeah, sure. So. So my name is pin two, and I've been working as a Java developer for over six years now. And then, I'm primarily working on a Java based applications where I like utilize frameworks like Spring, Hibernate and rest wave services for API development. And like, I've been, like, involved in microservice development, and also, like, I've been involved in designing them and then also deploying them. So like, you know, implementing also. So my primary experience with, is it AWS cloud, and also, like some of the services that I work with, includes AWS, PCs, SQS, and then also like the lambdas. So what I do basically is also like, do the code reviews on a regular basis. And also, like I work with Java back end coding or work with like product support. And then also like that goes around the rotational basis, so on the front end, like I work with Angular react, and then I do use like JUnit, and then also mokito for back end codes. So I, like, pretty much, like, I have good amount of experience working with database as well, including Postgres and Dynamo. So yeah, I think that's, that's pretty much about me. And then, like, I've been, I've been also the part of agile team for a while now. So like, I follow a lot of ceremonies, and then also, like, I have team of five developer, so I have also, like, product owner, and then scrum master. So yeah, that's that's pretty much about myself.

Speaker 2 04:58

Okay, very good. So solid. You have some front end, some middle back end, and also database, which you have a favorite of those or,

Speaker 1 05:11

well, yeah, like I do, like working on the back end technologies using Java and, like Spring Boot. So mostly on the back end, I would say, Yeah, okay, that's good.

Speaker 2 05:28

And you said you on your current team, or in past you you had agile, and it's involved some analysts and some product owners and yourself, some developers and all right, and then can you tell us a little bit about how your development out of that comes along? Like you have intervals that you fit work into? How do you break up the work? How do you get it from ideation to production?

Speaker 1 06:05

Yeah, so basically it's like two weeks sprint, and we we get assigned stories and then the feature to work on, so during, like the spring sprint planning. So we, basically, what we do is, like, upcoming sprint is how it is going to look like. And then, you know, like, sometime I pull up the stories that I find, like interesting, and then sometime I get assigned to assign from the tech, tech lead. So basically, like, all these stories that are, like prioritized, and based on the priority we work on those stories. So yeah, I mean, I think that's pretty much how the stories are assigned. And then I do, I do get involved in backlog and refinements as well. So what I basically have to give information on, on the like, the and also, like, provide the insights. So like, talking about, like, those backlogs, sometimes it also needs to be worked on, and then what kind of grooming we need to do for the certainty stories that would be like, ready for planning stories in that and then, like, it's a regular cadence. So we have a epic, and then like those epic is basically broken down into multiple features, and then also stories. So yeah, basically, like, stories are worked on the priorities, priority basis. Okay, can

Speaker 2 07:38

you walk us through some of your more recent, I guess, work experiences as far as like something that you're really proud of you've done or achieved,

Speaker 1 07:51

sure, so the one that I recall at the moment, I would say, in my current project. So here, like, basically, we have an enterprise take backlog, right? So, so this basically is a huge initiative within the whole like enterprise to to incorporate some, some kind of change, right? So sometimes, like, I might be like, the Java version upgrade, or sometime it might be resolving our vulnerability, or moving from one, one platform to another. So So recently, like we, we were, we're kind of given a backlog, backlog to incorporate a log, like streamlining so for for auditing purpose, and for for this, like, you know, the path to move forward was to incorporate an SDK and then configure that SDK to send the data to snowflake table. And I think, for for that, I basically, like picked up the story, and initially, like started, started with a proof of concept, and then now the thing that here was, like, we own like, 1515, 20 microservices, and then making these similar changes across all the services seemed actually little bit redundant and also like inefficient. So as a proof of concept, like what I did was basically proposed a separate microservice which was kind of responsible for handling this. And so I actually came up with the design that would like, basically take, take in a metadata to onboard that particular client who would be calling this service. And like, based on the client, the data would be like managed, and then also like, this particular service will handle, so taking the data and, like, fetching it and then pushing it down to the service table, right? So like, I think, like, this is one of the recent project that was, like, I'm kind of very proud of and then also, like, the main reason being is, like, we were able to cut down the working time for all those like microservices. And then, apart from, apart from like, where we are also like, able to offer team onboard the service to, like, push the data as needed. And then because, because also, like, it was highly configurable based on the metadata, so they would provide, like, during those onboarding So, yeah, I mean, I would say, like this, this one was actually the tough one in the recent days. Okay,

Speaker 2 10:53

that was a design that you came up with and for for certain reasons.

Speaker 1 11:00

Yeah. So it was, it was something that, like I proposed, and like I had to discuss with with my tech lead on, like, how, how it would look like, and then he actually liked the idea. And then we basically had a discussion like designing with the architects to move forward with it.

Speaker 2 11:24

Okay, nice, yeah, what do you if you're coming across something new, new tech or new idea or something like that, what is, what are some techniques you use to either learn or figure something out?

Speaker 1 11:42

Well, to to learn, I would say, like, I think one, like, whenever I am in a in a front, in front of, like, new technology, I always kind of try to get hands on on the documentation. And basically, like, look at the documentation, and then little bit of having a small idea and what, what it goes right, apart from like, so like, basically, like, get my hands, hands actually hands on to and then also jump into the technology. So if, if it's a new project then that I'm working on, I would like to, like, go with the unit test, and then also, like, see what different services to do on a unit label, and then try to add like break points in debugger, and then try to get more more info, information on it. So and apart, apart from like, apart from that, I like to basically create a demo, or maybe it could be a small project of my own that encompasses that particularly particular technology so that, like I can, I can little bit play around with it. So I would say those are some of the approaches that I do, like read the documentation and go on with the experiment. So having the hands on experience,

Speaker 2 13:24

what, what led you to apply in that environment. So

Unknown Speaker 13:32

I think like,

Speaker 1 13:34

So, the the opportunities that was like, it was basically the opportunities, I would say, like that put forward, and then it seems like very interesting. The fact is, like it helped me. It helps me, I would say, explore more on what I have been learning. So my experience, and I think, like it would be also good, good experience in overall, so, I think, and opportunities for me to grow myself for my future, and because it's also a lot of interesting thing that you are working on. So And apart from that, like in terms of tech, tech system, I think, like Garmin has a pretty good work also, where it actually values every individual that they work with, and then it has a very collaborative environment, I would assume. So that's the reason. Like, I'm thinking like, those are the primary reason. Okay,

Speaker 2 14:38

and you're familiar with Kansas City. It looks like you were, you're here,

Speaker 1 14:42

yeah. So I actually did my under, sorry, I did my graduation there, like University of Central Missouri. So I spent quite a few time there, like around over, like a year or two so. And then I do have my colleague working there, like we used to play together. So I'm pretty much like, I think I was also the part of some Garbin marathon. So yeah, I know quite a lot. So yeah, I'm always like, active, and then I try to explore. So

Unknown Speaker 15:22

yeah, that's the thing. Okay, cool.

Speaker 2 15:26

All right. So I think just make sure we have enough time to do this. We have a coder pad for you exercise to do. Do you have that link? Or I can paste it in chat here?

Speaker 1 15:38

Yeah, I think we do have, I think I am already logged in, okay,

Speaker 2 15:46

all right, yep. So there are a couple exercises there. One is starting with a simple calculator, okay, as soon as you get logged in, and there you can see it, just let me know, sure.

Unknown Speaker 16:02

So, I'm here. Okay,

Speaker 2 16:08

all right, so yeah, the first one there is the simple calculator, and it's just you have unit tests that are already written. You have a main class is executing it, and you have the one method there add of where you can start your code. You can add classes if you want. You can add methods if you want. But the idea is to to add your code that follows the acceptance criteria and the requirements over there in the description, and please ask any and as many questions as you can or as you need to me. And there's just, there's no one way to do this or anything. It's just

Speaker 4 16:59

seeing how you do here. So that sounds good.

Unknown Speaker 17:13

So

Speaker 4 17:15

we have a couple of acceptance criteria. So went to actually,

Speaker 1 17:23

two positive numbers are provided, so return the sum and

Unknown Speaker 17:37

so basically, I on.

Speaker 1 17:44

So do we have any kind of specific exception like that you want to throw on the exception case,

Unknown Speaker 17:53

you can throw any exception you want.

Speaker 1 18:15

So like, I think we can start by parsing the given integers. So during the parsing, we can catch any exceptions that's thrown, and then we throw it so and I think, like we can basically move forward with it. So I think it's, it's very straightforward. I a

Speaker 1 18:49

little more on extreme so on the second acceptance criteria, when one or more extreme is not a number, then,

Unknown Speaker 19:02

then, like, basically throw an exception. So would that

Unknown Speaker 19:07

that mean is like

Speaker 1 19:10

empty string is also on this case? Yes,

Speaker 1 19:25

so if that's the case, then I think I should have a pretty good path forward so I can write it. I

Unknown Speaker 19:44

do you need me to share my screen before I type? No,

Unknown Speaker 19:48

it's okay. We actually can see

Unknown Speaker 19:52

what's going On. I'm starting

Unknown Speaker 19:57

typing so i

Unknown Speaker 20:26

So,

Speaker 1 20:30

so basically, when I parse and teaser using teaser dot parse method, it actually throws a number format exception. I

Speaker 4 20:47

and I will basically catch that and also throw a generic exception. I

Unknown Speaker 21:19

BBB,

Unknown Speaker 21:28

okay, so

Unknown Speaker 21:31

once I have those two numbers sparsed,

Unknown Speaker 21:35

then I need to actually take

Unknown Speaker 21:38

care of so,

Speaker 4 21:42

yeah. I do need to take care of the third Okay, so, yeah, I do need to take care of the third acceptance criteria, because

Unknown Speaker 21:52

I actually covered The second acceptance criteria. I

Speaker 4 22:56

it now like I need to basically check if the number is greater than 1000. So

Speaker 1 23:09

for for the fourth acceptance criteria, which this can basically be done by using A simple If statement, i

Unknown Speaker 24:35

i Okay so I think I should be able to run this now.

Speaker 4 24:58

I think this is past, right? So, yes, yeah,

Unknown Speaker 25:08

okay, we have, if you see down, there's another

Unknown Speaker 25:15

one below it. Okay,

Unknown Speaker 25:22

this is,

Speaker 2 25:25

this is finding any user has been assigned a role on or after a given date, and the given date is March 15, 2022 okay. Okay,

Unknown Speaker 25:46

so let me actually start by creating, creating the

Unknown Speaker 25:58

Yeah, so starting by creating the object. So

Unknown Speaker 26:21

let me take On this one? Sorry. I

Speaker 4 26:53

so an ideal world will have a getter, setter and constructors. So

Unknown Speaker 27:05

I'll Just do constructor here. I

Unknown Speaker 27:45

Okay, so

Unknown Speaker 27:47

this would be

Unknown Speaker 27:57

my, my audio that that I'll work with. So

Unknown Speaker 28:02

just guess, like it's like a proposal. So

Unknown Speaker 28:10

now let me turn I

Unknown Speaker 28:14

have a method that finds user,

Speaker 4 28:21

okay, so, so, it Seems like I have a method that finds users. I

Speaker 4 28:58

so some basic thing is like We need to parse the given I

Speaker 4 29:33

so I think we have a list of users here. I

Unknown Speaker 29:49

all right now will be given

Unknown Speaker 29:59

our input to string

Speaker 4 30:02

and We need to convert it to a local date. I

Speaker 4 30:29

so I actually just want to test if this parts works or not, and then i'll proceed further forward. I

Speaker 4 31:00

it. So I think I need to use a simple date formatter and provide a pattern for it. So I think that should be the easiest way. Now I

Unknown Speaker 31:54

Okay, so sparse mouth,

Unknown Speaker 31:58

I should be able to use the

Speaker 4 32:18

ah, so now should be able To go stream the given list of users, and then filter it after the date, and then filter non empty rows, and then return The value i

Unknown Speaker 33:50

Okay, so,

Unknown Speaker 33:56

let me create the data now. So I So.

Unknown Speaker 37:31

Basically it didn't work. So see i

Unknown Speaker 38:08

Okay, yeah, I think I was kind of looking further along with it. I think

Speaker 4 38:16

so good work. Yeah. All right.

Speaker 2 38:29

Any questions or anything over what you done comments. So I think, I think no,

Unknown Speaker 38:44

pretty step forward.

Speaker 2 38:46

Yeah, okay, very good. I'm gonna end that part of it. Okay, yeah. So,

Speaker 2 38:55

okay, so we just have a little bit of time. I want to make sure we give you time to ask any questions about Garmin or groups or technology, anything that you can think of

Speaker 1 39:14

like firstly, like, I would like to thank you so much for for these opportunities today. And then, like, I would like to ask a little bit more about the team, like, how it is structured. Like, do everyone join at, I mean, the same huddle, or, like, do you have, like, a dedicated small, smaller hurdles for the for the different groups? Like, how does it work?

Unknown Speaker 39:42

Yeah, so,

Speaker 2 39:46

so Patrick and I are on two different teams in it, web development. And Patrick from a teams, do we have total I don't know that. I can answer that

Unknown Speaker 39:59

ish, there's a few. E

Unknown Speaker 40:01

commerce,

Speaker 3 40:04

partisan, first, Nexus, Eva, product support, CPS, CBS, mes. I mean, there's,

Speaker 2 40:10

we have different teams. We we kind of operate, I would say, in the same framework. I'll just say it that way, each team, then can really operate under their own guise of that framework, but for for different reasons, we all work under like an agile methodology. We all use Atlassian products for story management and Scrum board things like that, Confluence for all kinds of stuff. Right? Confluence, but and then my team is probably different from Patrick's in the terms of we're just a really large team, and we've gone back and forth between operating as a large team or a couple of separate teams broken up right now, whereas one large team, but I think we're actually having, pretty soon, talks of how we would break, break us up into smaller, more dedicated groups. And same with E commerce, I think actually has three teams, doesn't it?

Unknown Speaker 41:27

Yeah, so

Speaker 2 41:29

it just depends. And then, Patrick, you want to describe yours?

Speaker 3 41:33

Yeah, I mean, essentially between product and device, they were historically not the same team, then they combined the teams, and then they're trying to break them back apart again. And so same resources are essentially executing both missions for device and product at the moment. So are at our scrums and our daily ceremonies are executed together. But you know, soon as we get resources, it's our desire to hire, we will be separating the two sets of tasks between them. So generally, ours, you know, we've got the Cluj. We tend to execute very closely with Cluj. They're usually coming online when we leave. So in Romania, six hours different.

Unknown Speaker 42:24

It's eight hours different, eight hours,

Speaker 3 42:25

yeah, so when we're getting online in the morning at eight, it's, you know, like they're wrapping up in their evening. So it's like three four o'clock, something you have to do. So we execute very closely with them. So we do our scrums and our ceremonies with them. Eventually, like I said, we'll, we'll divide it up as soon as the vice does, just because the topics are different. But I would say, you know, our guiding principle is, is, if you're sitting there saying a bunch of things that are irrelevant to the majority of your people, then your scrubs too big, and so I think we try to adjust based on that. So I think what Michael is probably indicating is that when you've got so many things that you're talking about that half or more aren't really interested in because they don't affect them, that is probably too big and it's more efficient. So we make adjustments, which I see as a good thing. If it's getting too out of control, tailor back. We'll try to figure out a different way, try to get, you know, give right information the right people, without flooding everyone with kind of confusing Yeah, and we do operate on like the two weeks. So this is where we are very similar, because it's more of our kind of department. Framework is a two week Scrum, which is coordinated across or two week sprint, sorry, coordinated across our whole board. Everyone sprint, start and stop at the same time, and then most every team will do some form of a stand up, slash scrum ceremony daily. We tend we skip Fridays. I don't know about you, Michael, but we don't do ceremonies on Fridays, just because that's the end of the day for collusion Friday. And so we don't make a payment.

Unknown Speaker 44:07

That's kind of where we differ because flexibility,

Speaker 2 44:09

yeah, and on our teams, I failed to mention this earlier, and probably why, it's also little larger, but we do have developers, QA, Bas, POS, project manager, and then we also have front end developers, back end developers, and then there's even a couple people who want to do both. So they they do both, and it's kind of up to the developer if they want to do the front end and learn it. It's awesome. It's great. But So yeah, that's it's a lot of lot of members, and a lot of you know the roles across the whole development cycle. Yeah, so yeah and

Speaker 1 45:00

yeah, that actually helps. So I guess the the other other question that I remember would be like, What does like Garmin provide, like engineers for in terms of, you know, like the growth or improvements, like any specific

Unknown Speaker 45:20

improvements of what, like training, yeah,

Speaker 2 45:27

well, I can give an example of one. I feel like there's really good opportunities out there. As an example, we just cut a deal, and I can't remember who the company is, but it's a really good training facility, and they have tons of good topics to cover, and it costs, so they're making sure people were really want to use this service, and you had to dedicate eight hours a month to learning from this. But like I said, it's really good stuff, so they open that up to people. And then there's also, I think we have, oh, something with Udemy, or whatever it's called.

Unknown Speaker 46:13

And I

Speaker 3 46:15

mean that to let out a not so hidden secret is, if you have a public library card for whatever your area is, Udemy is almost always free. That's true. It is a bit of a nightmare to get through all of the logins. If you are dedicated and persistent, you can pretty much get into any unity, short of the ones I think that give you like credit. I think you can audit kind of thing, kind of like a college but, yeah, so in library card you get through, but it is encouraged, I guess, from a corporate standpoint, internally, we have, like a soft skills kind of thing. So public speaking, maybe organization, time management, kind of probably the stuff that's not as attractive to most people, but internally, they do have classes that are facilitated. You go down and attend it and someone's facilitating it. But as far as only training, you know, they will accept proposals for conferences. Lot of times. I think if you go and speak at one maybe they'll pay for it. If you go and represent Herman, I think I've heard of

Speaker 2 47:25

that, but some, I know there's some they've paid for in the past, whether whether you spoke or not.

Speaker 3 47:34

We, we do have the local Java developers thing that actually meets at Garmin, I think, monthly. Now there's a chapter of some sort of Java developer

Speaker 2 47:43

user group, yeah, Casey jack, and then there's obviously that, if you want to get a certification, or upper level, type of degree, or something like that. There's always proposed that, and they can give, they love, definitely love continued education. Manager in the end aviation, we also have basically get your private aviation license. for free. Okay. That's nice.

Speaker 1 48:22

Yeah, so that's actually our courses that I have. So, thank you so much for your time today.

Speaker 2 48:29

And thank yous. Great meeting you and you did very well.

Pradip Dhakal- JPMC-Final-SRV

Speaker 1 00:00

Use, how's that go? So, yeah, you know, we follow the Aussie scrum methodology, you know, and our sprints are like two weeks. We follow all those ceremonies, like daily stand up and, you know, sprint planning, and, you know, grooming, backlog refinement, you know, all the like, retrospective iteration, planning and so on,

Speaker 2 00:28

how long you've been working on that same team.

Speaker 1 00:33

So in the team, you know, I have been working, I would say, so for about so over two years now,

Speaker 2 00:48

would you say that, like the product owners are very much involved in your day to day,

Speaker 1 00:55

so on the not on The like day to day product owners, you know, like they,

Speaker 2 01:03

I don't know what you call them, product managers or product owners, but

Speaker 1 01:07

yeah, they usually like, you know, just talk to the business and, you know, they like, grab the requirements. And you know, we work with product owners to, you know, work on those requirements. And so, you know, we call the them, you know, we so they are, like, mainly involved in the grabbing the requirements, right? So, you know, these are the they gathered the requirements, like, from the client. And, you know, we talked with the product owners, like, for refining those into the features, you know. So we take, you know, part in, like, creating these stories, you know, out of those features. So product doesn't product owner does, you know, have a, like, high level of influence, but, you know, we work like closely with this account team, within the team itself, you know, and Eastern master, part of the delegation. Nice.

Unknown Speaker 02:06

How long you been the owners in the corporate world?

Speaker 1 02:10

So are you asking, like, my experience in the corporate world? Yeah. I

Speaker 2 02:15

mean, you started 120, 15, yeah. So,

Speaker 1 02:18

like I said, you know, I've been working like, for over five years now. Okay,

Unknown Speaker 02:27

why are you looking to leave?

Speaker 1 02:30

So actually, at this, you know, client, right? I'm like, you know, my project is, you know, almost about to end. And also, like, I'm exploring the new opportunities, you know, I'm trying to grow my personal, you know, professional experience. And also, you know, at the current role, you know, I do find a lot of improvement, you know, that I made professionally, but you know, I still, like, you know, thing to change because, you know, like, the actual project that the team was handling is completed, right? Like I said, so now we are like, kind of out of work because, like, we migrated all of the microservices that we needed to. So that's pretty much the reason, you

Speaker 2 03:22

pretty much the reason go, um, so I'm just skimming through your CV, any any direct experience like designing for mobile devices. Um,

Speaker 1 03:43

so, you know, basically our application, like, it handles, you know, traffic from like, all the three stacks, you know, including iPhone, Android and web apps. So I didn't, like, actually handle the direct mobile application. But you know, mostly I work through the proxy, you know, I mean, I worked through the the context router.

Speaker 2 04:12

Gotcha, who were you interviewing with? Was it Raul? Was from

Unknown Speaker 04:20

jpmt. Have you interviewed

Speaker 1 04:22

with? So previously, I was interviewed with Mr. Gonzales and Mr. Mahmoud, and second round of interview was with, I say, Mr. L, I don't know his like. Name was paid long.

Speaker 2 04:42

Gotcha, I don't know who that is, so

Unknown Speaker 04:48

any questions for me?

Speaker 2 04:49

Like, I'm not fine. I'm not going to give you the, I don't know the technical answers, but I can give you like, culture, about the company, about the project. Any, any questions for me,

Speaker 1 05:03

sure. So, you know, I would like to, you know, know, like, you know, a little more of the you know, product you know, that the team is handling, like, you know, what is, you know, like in a very high level, right? What is the team like currently handling?

Speaker 2 05:22

Yeah, so give me a little context. JPMC had tons of mobile applications right been sprung up through the different things. Plus, you know, you've got your email, you've got Microsoft stuff that they want on their chat, blah, blah, blah. So about a year ago, the company made a decision to create a employee experience organization. We always had one, but it wasn't really funded very much, and nobody really cared about it. But with the great resignation and all that, it was time to invest a little bit more in our employees. So with this new with this new organ, funny, and the desire to create one application for our employees that will hopefully allow us to, you know, we're never going to be able to deprecate them all. But, you know, get rid of a whole bunch of applications. The direct the application that we're directly replacing is on the app store. I think it's got, like, a 1.1 rating currently, so the bar is really low, but we brought in a lot of folks that have

Unknown Speaker 06:59

what's the word I'm looking for, like

Unknown Speaker 07:02

consumer grade

Unknown Speaker 07:05

experience working on products.

Speaker 2 07:10

The Walmart, like a bunch of other, we brought a whole bunch of people in the work on mobile where, you know, they're very much used to, you know, delivering stuff that makes money for the bank, right? So with that, the goal is doing right now is we're building an application that will be kind of like a daily, what's the word? I'm looking co pilot or daily, you know, assistant that

Unknown Speaker 07:42

you walk around with you all day long. You

Speaker 2 07:43

know, you got your phone and you know, you could use it to get into the building. You could use it to schedule meetings on the go. You could use book a room on the go. Like, oh shoot, we got a meeting. We need a room. Or just book it. Book things like, obviously, see things like news in your area or in events. A lot of personalization was going to go into this, you know, manage all of your HR stuff. Like, Hey, did I get paid this week? Or, you know, whatever it may be that's, that's it. I mean, we basically started with a blank slate.

Unknown Speaker 08:21

We are about

Speaker 2 08:25

eight months into it. We've got, we're just hit doing our like third release, which is kind of like a gamma release, not full full throttle yet, but, but we've just hitting the button on the third release. And you know, the good thing is, mobile is just one surface of many in the bank. Obviously it's the you know, where everybody's going, but you still gotta worry about web and other stuff. So the way our team is working, it's a lot of good buzz around our team and and the rest of the org is noticing and looking to kind of, kind of do what we're doing. The hard part of the job is, you know where to play experience that's only but as you can imagine, like the team that owns the booking system, like the seats, that owns the conference rooms, that owns the guest register like they're all different teams, very silo. So the hardest part of our job is trying to get integration with all of those different teams, getting them to work with us, getting time in their backlogs, you know, all of that stuff. Oh, that becomes the biggest challenge of the job. But I'd say, you know, overall, we're building something really cool. It is.

Unknown Speaker 09:46

It's predicated on,

Speaker 2 09:48

we're building a new headquarters in 70 Park. And so the original thinking was that app was going to be the companion app for that building.

Unknown Speaker 09:57

Now, oh, just make it the companion

Speaker 2 10:00

app for everybody now, in every building, so that where we are, you know, and it's we gotta we have a good, tight team where desperately need resources more than anything,

Unknown Speaker 10:14

lofty ambition.

Speaker 2 10:17

We good people. We need people are going to be accountable. We need people are going to, you know, not sit back and look for work, right? Like, hey, I noticed this. What can we do here? Like, like, people who are very, more proactive, as opposed Hey, I got a JIRA ticket assigned to me, you know now I'll do work, right? We're not like that whole crew, like, where we have a very all of us have similar personalities, where it's just like, well, no matter what it is, we're figuring it out. So I'd like, that's, that's what makes it really good. In that sense, it's, it's definitely harder work than some kind of but it's also and, and and team around us is really good. It's getting better every day. So, you know, I think, I think it's a really good spot to get into it.

Speaker 1 11:16

Thank you for your answer. And also, you know, like I feel like, you know, from my current experience in at Old omenium, like we work as an, you know, orchestrator there. So, you know, being orchestrator, you know, we love, we have, like a lot of, you know, cross functional teams. And like we actually have to, you know, ask the, you know, different teams to incorporate, you know, like, what the changes we are using and, like, you know, what, like, if you are trying to change the way, you know, we are, like, a kind of requested on, right? So, you know that, yeah, you know, like that, that they have to, you know, make the changes and we work on them, yeah? So, you know, it means, you know, I feel like, you know, that's kind of like in the same realm of, you know, like multiple teams working together for the big common goal, right? So, yeah, that's a pretty good experience, you know, I have as well. And I think, like, you know, it would be pretty good to handle the kind of like scenarios as well, you know. So yeah, again, thank you for your like, all the introduction and the answers that you give me about regarding the team structure and how the project is and, you know, yeah, and I like this kind of challenges, and I'm always up for it, you

Speaker 2 12:37

know, I could tell. I could tell. So if you have any other questions, feel free to hit me up on like, LinkedIn. or whatever. But let me know if anything, and good luck. Hope to see you soon yeah

Unknown Speaker 12:48

hope to see you. Thank you.

Pradip Dhakal-JPMC-SRV-First

Speaker 1 00:00

Even the same experience that we give, you know, our consumer right? So everything internally, all the websites, all the mobile app, you know, is up to par with a consumer grade experience, but also, you know, and the reason we don't do that is, you know, employees get happy, but also productive, right? So that's the kind

Unknown Speaker 00:20

of goal, right? So

Speaker 2 00:22

we want to go a little bit to your experience.

Speaker 3 00:28

Yeah, sure. So, you know, my name is Pradeep, and I have been working as a full stack Java worker for over five years. And you know, I have primarily worked on Java, you know, basically, you know, the Java based web application, like utilizing frameworks, you know, using ice, Spring, Hibernate and RESTful web services for the API development, you know. And you know, I'm pretty good working with the microservices. And, you know, I have been involved in, you know, like designing the microservices and like implementing them, you know, and working with the springboard for the development of this, you know, microservices, right? So, you know, also for deploying these applications. You know, I have been, you know, primarily working with the AWS and some services you know, that I have been utilizing from the AWS services, right? For example, s3 bucket, right, ECS, SQS, SNS, lambdas, rds for the database and so on right and on the UI side, I have a pretty good experience working with the angular you know, I also have a good knowledge in the unit and, you know, Mockito for writing the unit test cases currently, you know, I'm a part of this Chrome team. We have like five developers, a couple of testers, you know. And I'm mostly working and I'm back in Java code, helping the team with the code reviews, you know, and I also, like, have been involved in products and support. So, yeah, that's a little bit

Speaker 2 02:14

about Thank you. All right, so let's talk a little bit about latest process, right? So since here that you've been working on RESTful API for like, 40 minute authentication, and you mentioned here a little bit about JWT authentication and RESTful API. So do you want to fully ask, you know how further that entire you know, hand passing information from getting from a username and password to an actual session. You know that we can rely on how that process works,

Unknown Speaker 02:54

okay? So you know,

Speaker 3 02:57

right? So basically, you know what happens is like whenever we have our, you know, the external part portal, right? So, you know, the that we did direct for login, right? So let's say user comes in and they provide their like, you know, username and password for logging in, right? And then the you know, the client sends these credentials to the, you know, the server for authentication endpoint, you know. So it's you know, usually you know, post request, right? Because it's you know, in the you know authentication, and you know, then the solver needs to, you know, verify the identity platform that we use, you know, and you know, where the actual authentication takes place, right? So the server, you know, simply the server verifies the, you know, username and password, you know, using the actual table, you know. And then, like, once that part is done, you know, it generates a JWT token, right? So it's based on our web token, which actually contains the you know, user information, and you know, which are usually known as the claims. And you know, it is signed with a secret, you know, with like public private key combo, you know, to ensure the integrity, and it may contain the errors, right? So now these, you know, three parts are the token. You know the token is generally separated, you know, by the period. And you know the JWT is actually, you know, Return, return to the client you know, which is stored in like either the, let's say, local stores are like the secure cookies, you know. And it is, you know, authentication header like when making calls to the downstream API right now, whenever these JWT tokens, you know, they are passed to our services, right? The server actually, you know, validates this JWT you know, by the signature, you know, taking the signature and the expiry time. So the the signature is, you know, like primarily, you primarily used to make sure that the you know, claims has not been tampered with, right? So you can decode the token, and then, you know, change the date, you know, change the main values. And you know, pass it down, right? But you know that will fail your signature validation. So you know, that's the, you know, main reason we have a signature attached to it. And you know, like, you know, basically because of this, this makes your token validation very important, you know, and the you know. And you know, because, let's say, if we haven't, like incorrect token, then you know, we are not authorized right, to be the research in the RESTful APIs. So you know, all of the these information, they needs to be included because the RESTful APIs, they are meant to be stateless, you know. So the complete information for the request, and the authentication should appear like within the whole request overall. So, you know, the token consists of, you know, like the claims, which consists of, you know, like the role for the role based control. So you know, that's the overflow for validating the claims. And you know, we use spring security for all this.

Unknown Speaker 06:24

Okay, great. And I guess I'm sorry.

Speaker 2 06:30

Okay, so perfect. This, this, this is a very troubled version. Thank you. And I guess what happens when you get to the point where the token has been expire, like, how can we make sure that the user continues having the session?

Speaker 3 06:46

Yeah, so when the token expires, right? So you know, basically, when you know it expires, right, the user usually has to, like, re, authenticate themselves, you know, to get the new JWG token. But also, you know, there is a concept of refresh token, you know, if it is implemented, that can also be used. So, you know, basically what happens is like, whenever the token expires, right, our users will be redirected back to the logging base.

Speaker 2 07:21

Oh, yeah, sorry, refresh token. So you mean, I was taking notes, by the way. So when there is, when we have a concept of refresh token, how is that signing work, in terms of, like, you know, when you're ready to use our fresh token, what are the rules that that we implement for our refresh token, you know, in terms of expiration as well. And what else, I'm sorry, can you please repeat that? Yeah, sure. So absolutely, if you could, if you could divide it deeper on how we handle this refresh token, right? So let's say I am a request, or I cannot make a request because I'm, you know, my channel with this fire break. So what do we do with this refresh token? And what are the rules for the refresh tokens that you will get? Like, are they doing it forever? Can you do anything? I want everything with them? Or there are special rules on how refresh tokens have been managed.

Speaker 3 08:17

Okay? So basically, you know, if you do not have the refresh token, right, you have to, you know, real thing, take it using the login base, right? But if you have a refresh token, then, you know, once the access token expires, then the client sends, you know, a refresh token to, you know, refresh the endpoint on the server. And you know, refresh token is, you know, like, check to see if it's valid and not expired, or, like, you know, hasn't been removed, right? So because you know your server may revoke your refresh token as well, and you know, once that is done, you know that a new access token is issued, and sometimes even a new refresh token is issued as well. So yeah, that's how you know, the overall refresh token flow works. And after the expiration, the refresh token is, you know, actually sent to the authentication server. Yeah.

Speaker 2 09:13

Okay, perfect. And then another question that has to do with, you know, at which, at what point do you, would you like put this validation of the token, right? So let's say that you have, I'm sorry, my mobile engineer, so my termination here might not be

Unknown Speaker 09:32

mobile champion.

Speaker 2 09:35

They have your edge, right? So you have demand connections, and at some point you have an HTTP server or some something that can press an SAP request and route it to an application right and write it to an instance where the application is actually running. So which point will you do validation that the token is incoming, this value, like this sequence of different steps that our request, you know, goes through before actually relating to navigating.

Speaker 3 10:04

Okay, so in this case,

Unknown Speaker 10:12

actually, can you please repeat one more time? I'm sorry,

Speaker 2 10:14

yeah, sure, absolutely. So essentially, you have like, multiple steps or multiple ops that are requesting to do before getting to the iPhone application, where the code that where the business logic code is probably not the one that is going to is going to process the input and, you know, generate an output based on that, and go back right? So there are a number of layers in between that. So I guess what I'm wondering is like, what, what is the layer where we could totally mandate to ensure that request, you know, makes it to the application layer only if it's valid.

Speaker 3 10:50

Okay? So, you know, basically, you know, each layer, right? This will come with its own security protocols, right? Because some security might be, you know, blocked by the firewall, you know. And you know, some might be like they need to be like like when the authentication authorized, this token is usually, you know, validated during the request, you know, right to the to protect the resource, right? So anytime a client makes a request to the endpoint right that requires access to the protected records, that's when the token needs to be validated on every request you know, and whenever a user makes a logout, then you invalidate and revoke these tokens. Did that answer your question? I guess? Yeah, I think it was

Speaker 2 11:45

indeed, but I guess what I was looking for. So maybe we can, we can rephrase the question, or maybe take the question one step back.

Speaker 3 11:56

Whenever a user tries to access any kind of protected resource and a system like, insert Perfect,

Speaker 2 12:03

okay, okay, I think, I think, yeah, but I guess, I guess I wanted to go with more like that is a high level, high level mechanism that happens there, yes, something or someone checks for the validity of of this token. What, I guess, what was wondering more is like, you know, what is the component that takes on that? Or, how will you write, how we write a piece of software that will do evaluation that a token is valuable, you know, where will you place it? How will you make sure that the application receive requests on the you know, if the token is actually valid? Like, what is it? What is the kind of concepts that you can use, or constructs that you can use in order to guarantee that the application does not get the request? Like, where ISO phone network?

Speaker 3 12:53

Oh, yeah. So let's say, like you have a springboard base to the microservice, right? Then, you know, you can use the screen security, you know that is integrated with, you know, utilizes a filter on your request to authenticate and, you know, authorize your request, right? So in that space, right? Your you know, application will use your filters to authenticate. You know, which is like the web security. You know, configure adapter, right? So, you know, if you like, you're using an API gateway or an orchestrator, right? The AWS API gateway comes in and, you know, with an authorizing, authorized lambda, you know that you can use to take the validity of the request. If you are using, like other gateways, you might have to, you know, use the custom implementation of the token validity as well. So, you know, depending on the kind of architecture we are using, you know, we'll have to implement it, implement them accordingly.

Unknown Speaker 13:56

Anything I

Speaker 1 13:58

was going to change the topic, to change the topic, change the topic? Console, yeah, that's

Speaker 2 14:04

okay. I have more questions, but we can see that. Yeah, okay, so

Speaker 1 14:09

pretty obviously, in your resume, one of your strongest skill sets is Java, right? So, in Java, right? You know, can you tell us? Like, we just want to assess your understanding of Java and how we know it, right? So, things like threading, memory management, exception handling, the Java runtime, the Java SDK, like, you know, can you tell me what you know about those kind of those areas? Okay,

Speaker 3 14:38

so, you know, like, basically, right. So I have worked on, you know, like, Java for over five years, and, you know, I have worked with, like, all these concepts that you have mentioned, and in my recent project, you know, I have been, like, primarily focused on a lot of multi threaded, you know, highly concurrent environment, you know, where we are, like, basically using the framework, you know, called the history command, you know, for managing the thread poles, right? So basically, in, you know, on the history, come on, you know, what you do is, like, a bunch of requests that you have, similar in nature, right? So our application handles, you know, a lot of downstream endpoints on the orchestration layer. And basically, you know, based on the configuration that each of the endpoint might have. So this will be grouped on something of the similar command. And you know, these commands, you know, go through the same group of the pool, you know, but like all those through the

Speaker 1 15:45

different Are you trying to explain to us how you implemented a multi threading system? So I wasn't, which part of the job are you trying to explain currently?

Unknown Speaker 15:54

So I was talking about multi threading.

Speaker 1 15:58

Okay, so you How is okay? Cool. So how is multithreading implemented in Java? Or how would you implement multithreading in Java? And I think you're giving us an example of something you did, right? So, yeah, okay, cool. Just tell us a bit more detail on that.

Speaker 3 16:15

So you asked me to, like, continue the multithreading.

Speaker 1 16:18

Yeah, absolutely right. Yeah. So how is multi trading achieved in Java? And you know, what is, what is the exact example you you explain,

Speaker 3 16:25

yeah. So, you know, basically we can use, like multi trading in Java, using the thread class, extension of the trade class, right? And the implementation of the renewable interface, right? So those are the, you know, two primary, main, primary ways. But you know, on top of those, we have, you know, services that can actually handle the management of those trades, right? So one excellent class is, like, exhibitor service, where you can, like, even, you know, use the different method it provided for the trade management, right? So the executor service, basically, you know, it allows you to manage the pool of trails, and, you know, handle the task submissions without, like, directly interacting, you know, with the trade objects. So it has, like the, you know, execute where you have to, you know, submit a task for execution. You know, you have, like, you know, shutdowns, like, where, you know, you have a sort down of the thread once all of these tasks, you know, they are completed, and so on, right? So, yeah, I do have a pretty good work and a pretty good

Speaker 1 17:38

just very basic question then, so, in your mind, what is a threat, right when we say, we talk about threads right? And obviously use them to create concurrent programming, right, concurrent systems. But what is the thread within your mind?

Speaker 3 17:55

Yeah, so you know, like talking about the thread right thread is, you know, is, basically, is, it is an execution, right? So it is, you know, small task that runs independently.

Unknown Speaker 18:11

That's my understanding of it, yeah.

Speaker 1 18:12

But the small task, I mean, that could be a process, right? Yeah. So, so it might have a press, yeah,

Speaker 3 18:21

yeah, it might, you know, share resources from other tasks as well, but it's just an, you know,

Unknown Speaker 18:27

independent unit of execution. Let's say. Why

Speaker 1 18:30

is it independent? What makes it independent? What are the kind of the mechanics that

Speaker 3 18:36

Okay, so, you know, basically, you know, this.

Unknown Speaker 18:42

The Independent is, you know,

Speaker 3 18:47

okay, let me think about it. So, so basically, you know each thread, you know it that has a stack, right? Basically, I think it works. Means, like your tasks, they execute independently. You know, basically,

Speaker 1 19:08

I understand they execute in, you know, separately, right? Not sure, why independently, but, but what does that mean? I'm just trying to see, you know. But what does when you say, execute independently? What do you mean by that? Right? I'll just give you give us an example, right? Give us an example of, you know, even a Hello World example, or something you've done in the past, right, from your experience, right? Say that this is what we use threats for, and this is what it achieved, like, that's the goal, okay?

Speaker 3 19:40

So, you know, basically, right, in, okay, so in my recent one, right, we had a little large scale implementation, you know, basically our application takes in a bunch of requests, right? A lot of you know, those requests, to be precise, you know, and it manipulates the like request header. And then the responsibility, you know that that holds is that, like, it has to pass this request down to the, you know, downstream system. You know, to pass this, you know, request to the downstream system, right? We have to, like, you know, leverage of closable, HTTP cover and this sgtp call, you know, it has to be like, non blocking, asynchronous. So for this purpose, we basically spin up, you know, new trade to take care of this request execution. So

Speaker 1 20:38

let me, let me understand so you have a system where you need to fire off lots of HTTP requests to downstream systems, right? And to achieve parallelism, like you want to send off multiple requests to multiple back end system, right? To achieve parallelism, you're using threads. Is that we so you say, Okay, so let's say you've got 10 downstream systems, right, and concurrently you want to call 10 of them, right? You're using threads for those, right?

Unknown Speaker 21:07

Yes.

Speaker 1 21:09

Sorry. Any questions is that the system design was, am I not completely clear with that one?

Speaker 3 21:20

Like, okay, so let's say, so basically, you know, it's a multiple downstream systems, right? Yeah. I mean, just

Speaker 1 21:27

to simplify the example, let's say you got 10 downstream systems, right? They're all REST interfaces, right? And you got requests coming into your service, and you want to be able to call those 10 systems in as much scalable manner as possible, right? And, you know, concurrently, possibly, right? So, because you want to get the most maximum performance from your from your service, right? So to achieve that, you're saying for each call, for each service, you have a thread. Or what would your concurrency architecture would be for that,

Unknown Speaker 22:02

okay, so for this, right,

Unknown Speaker 22:06

you know, the concurrency architecture so, you know,

Speaker 3 22:10

basically, you know, so, yeah. So when you have this, right, we have like, you know, layers that actually, SARS has an orchestrator, right? So, you know, you know this orchestrator, you know, like, I think,

Speaker 1 22:30

sorry, I don't want to, like, we say, there's lots of frameworks and things like that, right? That would help you achieve and build a system, right? But we're talking about threads. So in the in the world of threads, how would you achieve the concurrency you want, right? Or do you not use thread or do you use something else? Right?

Speaker 3 22:49

Awesome, yeah. So, okay, so you know what happens is, like, you know keys of this command, right? They use as a thread pool, you know, to manage the concurrency. You know, the concurrency of the execution, right? So it, you know, allows you to, like, configure the, you know, isolation strategy, and you know how these requests are made downstream through the parallel stream, right? So, basically, you know, allowing you to make on, you know, current execution of these services, right?

Speaker 1 23:22

Okay, I get it, but I'm just saying, How is thread? How are threads helping you achieve that, right? Rather than these packages you, I think you mentioned you use, right? We started off talking about threads, so I just wanted to understand, like, how would you power thread? What are threads been employed here to achieve the goal you're looking for.

Speaker 3 23:43

Yeah. So regarding the trades, right? So, yeah, you know, basically, you know the trades. You know they actually, you know, help, let's say like, you can spin up like multiple threads, you know, for a system, and you know, program can do, you know, lot of tasks, you know, at the same time, right? One thread can handle one task and process data, and, you know, have the, you know, response same, you

Speaker 1 24:17

know. So I get that right, but I'm saying in the example you gave where you got 10 downstream systems, right? We made 10 up. I think you said, got a number of downstream systems, right? Let's say you've got 10 downstream system and you need to call them all right? They're all HTTP REST interface, right? And you want to be able to call them in parallel, right? And are you going to use threads for that, or are you going to use something else, right? How would you create that parallelism in your system?

Speaker 3 24:42

Okay, so now to obtain the parallelism, right?

Unknown Speaker 24:50

In this case, you know, like,

Unknown Speaker 24:54

right? So when you have, like, multiple, yeah, so, okay,

Speaker 3 25:03

you know, I mean, in this case, right, I would still use the parallel stream, you know, and for the parallel calls, you know, in order to make like, those downstream calls,

Unknown Speaker 25:17

you know, it might not Be like,

Speaker 3 25:20

you know, relapse, the like, you know, multiple trades, but it will be like parallel streams of calls.

Speaker 1 25:30

Okay, so just moving on to other areas in Java. What are the areas in Java? So we talked about threading, but other areas in Java, things like exceptions, memory management, what is the JDK, JRE, JVM, those kind of things, right? So can you just tell us a bit more about, like, each of those areas that you've used? Yeah,

Unknown Speaker 25:50

sure. So,

Unknown Speaker 25:53

right, so talking about the other areas you

Speaker 3 25:58

mentioned. So, you know, basically, okay, so talking about the JDK, right? It is like, you know, complete development kit for, you know, building your Java application. It provides, you know, the necessary tools, you know, to, like, write and compile your server programs, right? So that's the JDK, right? And you know not, you know, you know nothing, but you know runtime environment right in here, like we can run these are applications, right? It provides you with the libraries, and you know, with other components. You know that need execution for the Java program. And you know it also contains the JVM, which is Java virtual machine, and that executes the Java byte code, right? So, yeah, there's a, you know, these are the, like, three primary components, you know, like, when talking about Java

Speaker 1 26:56

and memory management, how's Java does memory mentioned? Yes,

Speaker 3 27:00

you're so talking about the memory management in Java, right? You know, basically the Java, you know, divide its memory into like, different areas, right? It has heap. It has a stack. You know, heap is where, like, you have, like runtime data, where the Java objects and their like instance variables are stored, right? Basically, you know it is dynamic, and you know where it is allocated,

Speaker 1 27:31

what's what's stored in a stack. Could get a, sorry, what is stored in the stack, in the runtime stack,

Speaker 3 27:40

so the, yeah, yeah. So the stack stores, like, the, let's say, our local variables, method calls returns to the objects. You know, it's like, you know, separate stack for, like, it's straight, okay,

Unknown Speaker 27:58

and the inputs in a in

Speaker 3 28:01

the heap, yeah, so in the heap, like, we have, like, you know, kind of shared resource, you know, it has runtime data and, you know, The instance of the variables and objects,

Speaker 1 28:21

okay. And then, and how much heap space does the Java program have? And how is that managed? Okay,

Speaker 3 28:32

sure, so, yeah, basically, I think, you know the heap is managed through, you know, environment variable. You know it is used like, whenever you provide the, you know, the XMS parameter on your Java execution, right? Like you can set it, you know, based on your environment, you know, here, like, you know, you can set the initial hip size using the XMS, and you know you can, like, set the max heap size.

Speaker 1 29:13

And what happens when, when you hit the max heap size, what happens there?

Speaker 3 29:18

Okay, so you know, when we hit the max hit size, then, you know, once we hit it, then basically, you know, there, you know, you get the, you know, out of the memory error. And you know, the, you know, the garbage collector will, you know, actually try to, you know, free the memory, but if it cannot, then you'll get out of memory error, and your application might be like hanging on some processes, or it might sometimes get crashed out or degrade the performance, depending on

Speaker 2 30:02

it. Can I talk something and can you, can you talk a little bit of how the garbage collector works? You know, if you know, like different methods, you actually manage your memory. That's, that's what's been allocated, right here. So I was, you know, what, what was the garbage collector there, how it works, very general, right? It doesn't mean to be very specific conceptually, you know, why is it sad way?

Speaker 3 30:26

Okay, so, you know, basically, you know, like thinking about the garbage collector, right? It is, you know, part of the JVM, right? And it is actually automatic memory management, right? You know it helps in, you know, reclaiming the memory that is, you know, used by the objects that are, you know, no longer reachable, and you know that are like needed by the applications, right? So Java is like, you know, one of the languages you know that automatically, you know, manages the memory through the garbage collector, you know. And you know, basically, you know, like the objects in Javas, you know, they are, you know, created in the heap and and when, like, there are no more references to that object, then it becomes eligible for the garbage collection. So the way it works is that, like, you know, there is a marking phase, you know, where the garbage collector identifies the objects that are still in use, right? So, you know. And like, once marking phase is done, the risk collector, it will, you know, sweep through the hip and collecting the objects that are not active and alive. So that's the basic two phase, you know.

Unknown Speaker 31:53

And I think, yeah, great, perfect. Thank

Speaker 2 31:55

you. And, and, I guess, you know, what is the disadvantage? So, perfect, yeah, Grace connection, and what is the disadvantage that you know, like, manually, manually can assist languages like C for instance, have you know, like, what are the kind of issues that Hawa is providing, preventing by having garbage? Panethal,

Unknown Speaker 32:15

okay, so,

Unknown Speaker 32:18

yeah, so thinking about it, right,

Speaker 3 32:23

you know, the disadvantage, you know? So I don't know, I don't actually know a whole lot about it, you know, but I it might be some kind of, you know, CPU overhead, because, you know, it performs based on, you know, like, because, you know, it consumes, like, some kind of resources on the CPU, right? So,

Speaker 2 32:52

oh, you mean when you have an average, like, correct, yeah, okay, that makes sense. Sorry, what, I guess what I meant. But you know, you're going to a similar point, which is like, I mean, what are the disadvantages on the yellow scheme, like, when you have, like, manually, where you're manually handling the memory allocation, right? So something will happen if you're not, if you don't pay enough attention where you're manually handling that. You know that it doesn't happen in Havana, because you know, right? So where are we talking about? Where is that concept that so

Speaker 3 33:22

you are talking about the disadvantages of, like, manually handling the memory allocation. Of course, yes, so yeah, you know. So basically, you know, like, if you are, you know, allocating the memory, you know, then you fail to fit. After you use, you know, then you are, like, definitely supposed to have some kind of a memory leak, right? And, you know, it might be difficult, you know, to track, because, you know, you have an allocation. And then, you know, I think it makes it a little bit more complex, because, you know, you are trying to manage something, and you are trying to handle the allocation right. So sometimes, you know, like there might be issue with the incorrect sizing, which makes it a little more static, per se, you know. So also, you know, you are increasing the complexity of it, you know. And I think those are the advantages of, like, manually memory management,

Speaker 2 34:36

perfect. Yeah, that's what I am looking forward and, but I guess the Okay, and I guess the other question I have for you, so does that mean that, because there's a garbage collection, I have a garbage collector, memory leaks do not exist, Inshallah, or they do exist. How they can happen, and what do you do to like, you know, like, detect them. I'm sorry. Can

Unknown Speaker 34:57

you please repeat one more time?

Speaker 2 34:59

No absolutely. So does the fact, then that there's a garbage collector means that in Java, there's no way to get to a scenario where we have a memory leak, or, you know, we actually can get in a situation where a memory leak happens. So I guess the question, you know, but not that's, that's the first question, memory leaks exist in Java, I guess. Okay, so,

Speaker 3 35:23

yeah, so thinking about this, right? So the, you know, even though, like, there is a garbage collector, right, memory leak is, you know, it is bound to happen, right? If you have, like, let's say, some kind of, like, improper implementation of your code, right, the garbage collectors, like, just looks for the you know. If your object is, you know, like, no longer in use. But if you have, like, some kind of static fails that remains in the memory for the lifetime, then it holds a reference to an object that is, like, no longer needed, you know, so the garbage collector will not clean it. And like, sometimes, you know, I think the collections also, you know, hold objects that may not be like, no longer needed, and it might lead to the memory legs. So I think, yeah, you know, like, there might be cases, like, when you have some, you know, improper implementations, and that could, you know, lead to, like, memory leaks.

Speaker 2 36:28

And what do you use to detect those? Like, you know, figure out where enforcement, where you're coming from, how to fix situations. How, how do you detect them at all, right,

Unknown Speaker 36:39

okay, so we can, you know,

Speaker 3 36:42

look at the, you know, profiling tools using the Visual VM and also look at the heap DOMs, you know. So there are, you know, other monitoring and telemetric tools as well, you know, to detect them at this moment, you know, we can make sure that we have a proper monitoring in place for that. And you know, apart from that, like, you know, we might want to basically have a, you know, proper, you know, we might have to, like, proper resources, you know, like we should try with the resources, you know, if we you have like, let's say something that needs to be closed, and you need to make sure, like they are closed, you know. So those kind of things are like, what we need to make sure. And, you know, we should have a proper collection.

Speaker 2 37:40

Very cool. And follow up question there just going, going back a little bit to the, you know, to multi processing us, or for the executing multiple tasks parallel, right? So there's multi threading, but there's also some other technique, right, which is like multi processing on the UDL. Instead of having multiple thread, multiple threads, you have multiple processes. How they are different? Like, well, how is a thread different than a process? I guess?

Speaker 3 38:11

Yeah, so, thread and a process, right? So, you know,

Unknown Speaker 38:19

basically, right?

Speaker 3 38:22

I will say, like, the, you know, thread is like the smallest unit of execution within a process, right? And process is like, you know, independent program, you know, that contains, like its own memory, space, code and resource, right? So, you know, like, when, like, you know the process, like, execute independently, and you know the and you know, basically, like they are scheduled, you know, which has its like own context. And you know they can, like, directly interfere with another process, right? But you know you can have multiple trays that you can run concurrently within a process and which allow multiple choice to, you know, execute in parallel.

Speaker 2 39:18

Okay, so you mentioned something that is important that I want to download him on this idea of memory ownership, right? We saw you mentioned that process has its own memory space, and yes, plus one, it is. So how that you know what kind of like different scenario it creates when we're talking about like multiple tasks and multiprocessing, right? What can you do on a thread that you cannot do on a process, I guess, because of these existence of sharp memory space and how do you have you know? What are the techniques that you can use to actually, like, overcome these limitations that here? Okay, so,

Unknown Speaker 40:05

basically, you know,

Speaker 3 40:08

you know, like I mentioned, right trade, you know, eating the same process, they share the same memory and resources, right? So this makes like, you know, whenever, like, you are, like, switching the context between the thread, it becomes, like, less expensive. But also, you know, there's a drawback, you know, like, which is like, you know, say you are using the same resources, you know, this might actually, you know, affect the other resources. And, you know, like, if you are like,

Unknown Speaker 40:47

you know, using the let's say,

Speaker 3 40:51

same resource, right? Like I said, it affects the other resources. But you know, on like, talking about threads, they can share the same memory space and allowing them to, you know, access and modify the share variable. And, you know, processes cannot do that. So, yeah, you know, I think, like, what you can, you know, do for achieving this is maybe you can use some kind of inner process communication, you know, to communicate between the process,

Speaker 2 41:25

perfect. Yeah, that's why I went to get awesome. All right, cool, and I guess switching that topic just a little bit. Can we talk a little bit about the auto scaling we have? We are most time, so we have something 12 more minutes. I want to give you a little bit of time. So if you have questions, but can we try to, you know, summarize in five minutes. I'll just feeding What do you know about it? What are the kind of criteria you use to set it up? So, yeah,

Unknown Speaker 41:54

maybe with AWS, with the context

Unknown Speaker 41:57

of an AWS, yeah.

Speaker 3 41:59

Okay, so, you know, sure, right. So, like, I have been like, you know, working with the auto scaling in AWS. So, you know, I can, like, so, basically, you know, it is like, it helps to, you know, maintain your application availability by automatically, you know, scaling your instances up or down, you know, based on the different criterias, right? The so, you know, the basic criteria is that, you know, let's say you you would like to use, you know, let's say the CPU utilization, right? You see. And then the application, you know, currently, let's say it scales up based on this CPU uses. And if there is a high CPU uses, let's say over 55% you have, like, your memory utilization, you know, maybe there is an exit in the sorted memory, right? So increases your instances. You know, in number of instances you might also, you know, you might want to do, like some kind of schedule, scheduling, right? I mean schedule scaling, you know where, you know, if you have a predictable load. You know, you can scale your instances up during the peak hours, and then, like you scale it down, you know, when you have, like, your normal or less traffic hours. And you know, one of the important aspect is that, right? So this is, you have a predefined, you know, health check endpoint right where you know, if the health check endpoints returns an unhealthy instance, you basically, you know, automatically tear it down, and then, you know, you spin off a new one. So, yeah, you know, the auto scaling, you know, integrates with the load balancers, you know, to ensure that the traffic is very distributed among the help, you know, distributed among the instances, right? And also, you know, I think that's, that's the, you know, key point that, you know, we need to talk about the Arctic scaling.

Unknown Speaker 44:21

Okay, yeah, I think that's of course.

Unknown Speaker 44:25

No, I think we only have 10

Speaker 1 44:28

minutes left, so probably gonna just give Pradeep some opportunity to ask us any questions, any clarifications.

Unknown Speaker 44:38

Okay, definitely. So you know,

Unknown Speaker 44:41

so you know,

Unknown Speaker 44:45

you talked about, like,

Speaker 3 44:49

the platform right for employees, you know, to have the same experience as their clients, right? So, like, what are the primary tools you know that the team is using like, like, is it all like this framework based application, or a new framework or technology like that the team is currently working on? Like, can you please elaborate more on that part?

Speaker 1 45:16

So mobile application with the front end is mobile. You know, for IMS, it's objective, sorry, swift and for Android, it's Kotlin. For the back end system. You know, we're building Spring Boot services, microservices that connect to various other systems that we have in the firm, right? So for things like booking a meeting room for accessing to, you know, people's calendars for, you know, answer cases we're connecting to other systems to allow people to order food, which is more of a client side integration, but there's a lot of back end integration. It's all right. And generally, you know, we want to deploy to public cloud. So we have a bias towards deployed to public cloud, but in but at the same time, we have a lot of legacy systems in place. Some of them are in house build. Some of them are vendor that bring it integrate with

Speaker 3 46:21

so like, these mobile applications and the web applications, they will be doing the same back end service. Or, like, Would it be, like, separate based on the stack?

Speaker 1 46:31

Yeah, ultimately, if we had a completely clean, you know, Greenfield architecture, we'd have mobile and web all hit the same kind of middle tier and hit the back end system, right? But because of, like, where things are in their life cycle, right? So, you know, sometimes web has already built some back end system already, and they're using it, and we want to modernize that. So we don't want to have any new clients for mobile, because mobile is relatively new. We're kind of starting fresh. But, yeah, ideally, like, you know, we want to create a middle tier that is used from, you know, desktop and from mobile. But obviously the final downstream system is whatever the final downstream system is like, then there's only one room booking system. There's, you know, so and so we just use that right from mobile and from Web.

Speaker 3 47:29

Yeah, okay, yeah. I think that's pretty much question from my side. You know, thank you so much for your time and answering my questions.

Unknown Speaker 47:39

Thank you so much so we'll give our

Speaker 1 47:42

feedback to the hiring team, and then we obviously be in touch with you pretty okay. Thanks very much for your time. Yeah, thank you.

Unknown Speaker 47:52

All right. Thank you have a bit. Thank you.

Pradip Ganesh- Infosys-Final-srv

Speaker 1 00:00

We did like, you know, it's like particular says sequel

Unknown Speaker 00:09

about Data Encapsulation

Speaker 1 00:11

over data encapsulations Yeah. So, this I would explain this, like, this is basically our Data Encapsulation is you know, creating a like capsule you know, right. So, you are basically protecting the field of the class. And then you know, protecting the field from the class and then like, exposing the getter and setter methods. So, which is basically data you know, hiding you you have also complete control over the field within the class itself. So, this is what that encryption means.

Speaker 2 00:48

Okay, and how do you achieve that like, how do you expose the data and control how, what is visible to someone using the class and what's not?

Speaker 1 01:03

Oh, yeah, so, basically, the fields of your class need to be private, right? So basically, it's not exposed to the outside world, then you use the private access modifier. And then you can provide public methods which are, you know, getters and setters. And that allows you to control that access over these private filters, and also allows you to retrieve the data, then, you know, use the setter allows you to modify the data and then you know, you will basically on this status, you can make sure that you edit the input. Okay.

Speaker 2 01:42

And what is the difference between protected and default access modifiers

Speaker 1 01:47

So, the difference between the protected and this modifier is you know, yeah, so, basically, you are like, protecting the customer access modifier is like, you know, if you if you if this member is declared as protected, you can access within the same packets, but also by the sub classes, right. So, even if they are in the different packets, but the default is like packets, you know, sub private right, basically, it means that the member can only access within the same paglen.

Speaker 2 02:31

What is the difference between dot equals method and equality operator that have an equal to operator? Okay,

Speaker 1 02:39

so, the difference between these two are like Eli is like, basically, the double equal operator is used to compare the difference or also the memory address, but the equal method is used to compare the content or we can say the value of the object.

Unknown Speaker 02:59

Okay

Unknown Speaker 03:05

have you used collections?

Speaker 1 03:07

How can you make collections? Yeah, I have worked with collections.

Speaker 2 03:12

Okay. Do you know what streams and I'm

Unknown Speaker 03:16

sorry I could miss coming in for that stream. Oh.

Speaker 1 03:20

So yeah, so stream is a you know, a feature of Java, Java eight, like for collections, processing, I have worked with it.

Speaker 2 03:31

Okay, so what what exactly is a stream stream

Speaker 1 03:34

right. So this, I will explain this, like, you know, this stream is basically like, what do you call that? It's basically, it's an abstraction with, you know, with represent like, like a stream of data, right. So it can represent any collections or any kind of stream, which is a sequence of elements and basically, it is for processing you know, the element of a collection rather than like storing any kind of data right. So you are provided with the intensive unary operations like filter map and you know, map map a flat map is there to transform data into another form and and then you have also terminal operators, terminal operators for the for like for each and collect and reduce to and basically like, you know, and extreme by either producing same kind of result or side effect.

Speaker 2 04:40

Okay, so in a Spring Boot application, right, that's the main metric what annotation was that main method? Like the class, the domain method?

Speaker 1 04:53

So yeah, so in Springwood, basically, like, in this application, you can use, you know, a springboard like it's been good, no, Jason's it's been whatever the situation.

Speaker 2 05:04

Okay, yeah. So, in that class, what would be the notation that you would use in the class that you write the main exact annotation

Speaker 1 05:16

notation? application integration, is probably our blog application notation. Okay.

Unknown Speaker 05:22

So what does that notation do?

Speaker 1 05:25

So, basically, in a Spring Boot applications, this basically like, basically Spring Boot, this Spring Boot notation is convenience and of like, what do you call that? You know, these three notations which is like, you know, which is like configuration notations, which is indicating that the class can, you know, can be used as a user spring container source, and which also allows you to check the beans and then basically it also combines the intervals auto configuration as well, which automatically configures your applications based on the dependencies present. And then it also has component is, is can annotations, which allows you to discover and register beans to all your packages right. So, basically, it is the combination of these three integration

Unknown Speaker 06:25

components can What is that unnoticed?

Speaker 1 06:29

Oh yeah, so, this scammy is like, you know, so basically these components can any notation is used to specify like, you know, the packages to scan the components, right. So, basically this spring container looks into the component scan like, into the component scan and looks for the package that is defined within the two within that to look for the annual test and on the class, like, you know, the components service, repository and controllers. So, basically it also it allows the spring to automatically discover those beans into the application context in that packets. So,

Speaker 2 07:16

is it mandatory to mention any common package like the packaging, now incompetent scan

Speaker 1 07:26

Oh, I think this what do you call that? Basically, you know, like, you can use the components scanned by itself, you know, and you can, you do not need to specify any kind of base package. You know, and this spring container should automatically be able to scan the package within the class and also look for any kind of sub packages, right. So, like, if you are in a main method in the package, and then other methods are also our other classes. are in the install some packages, and it automatically scan those packages as well. Okay.

Speaker 2 08:16

So, in a Franken application, right. Why do you require Angular? Why not just use vanilla JavaScript? Well,

Speaker 1 08:28

I think, you know, basically, you can use the benefits of a script as well. But but by using Angular, it is just to you know, like, easy for the structure and organization, right. So basically, Angular provides a very modularized approach to creating your front end applications. So where you can use components and then make the make them reusable. Right. So basically, whenever you create a component, your company, you create a template, you create a TypeScript file. So you create like, you know, you create like our style sheet and then and then package that, that component and then you can separate different components and concerns into different aspects. So it is like it makes your code more maintainable and reusable. So that's, that is why you want to use the frameworks or libraries like you know Angular or react something like that. Okay.

Unknown Speaker 09:38

So in an Angular application,

Speaker 2 09:41

what is the difference between difference between a module and a component

Speaker 1 09:46

in Angular, this component is something like is the difference between some modular component is like you know, is that your module is like I would say, model is like a container right? So, it is a block of code that is dedicated to like a certain kind of workflow or close the relative to capabilities right. So, it is it is like using the nugget modal light decoder decoder, and it contains like components services, and it also it can also contain other modules and derivatives and also so, on. But, the component is law is either a small building block or any user interface, right. So, it controls like a template and, you know, basically contains the logic to manage that template and it is also used to use within the component data integrator okay.

Speaker 2 10:51

How do you how do you communicate data between different components

Speaker 1 10:58

to communicate the data between the different components like basically to communicate data between the components. So, you you can use, like, what do you call that? You can use our input properties where you know where you you can what do you call that

Unknown Speaker 11:22

let me let me think about it and then this.

Speaker 1 11:26

So, so, you will find while communicating the data between the components right, or you can use different input properties where you can pass data from side using the properties right, like a parent component and it can also bind hello to properties in the child component. And using property binding you can use some kind of, you know, event emitter to share the data from side to component. And then you can also use some kind of what do you call that share service to share data between these audited components.

Speaker 2 12:05

Okay, so, you can use it to the input or output data annotations to pass data between parent and child component What about like sibling components? Like if that no, no parent child

Unknown Speaker 12:21

relationship between the two components, right.

Speaker 2 12:25

You mentioned that your application has like the like you're able to communicate data between your companies.

Speaker 1 12:33

So yeah, so in that case, I would be I want to be car service using observables. Right. So basically, our car service can use to present it that data accident

Unknown Speaker 12:55

I'm sorry, what was that in your laptop?

Unknown Speaker 13:01

Yes, that's right. Yeah.

Speaker 2 13:57

Okay, can you write a method that takes two collection of integers as input okay. And returns one collection of integers and method what it does is returns the intersection between the two input collections. Okay. The common elements between the two collections okay.

Unknown Speaker 14:53

So yeah,

Speaker 2 17:52

You want to explain the logic that you're thinking of

Unknown Speaker 17:57

Yeah, sure.

Speaker 1 18:00

So, yeah, so basically, what I am doing is, I'm converting the list, list to one into a stream and you know, and then I am printing out the elements that are, you know, only existing in that list too. And right, so I'm collecting them into another list. Okay.

Unknown Speaker 18:33

Okay, and

Speaker 2 18:35

any reason why you chose to make the metric private?

Speaker 1 18:41

Oh, the reason behind this is like, not really like basically, if, you know, I am working on single class stuff like this. I just make it private so that I have a main method that calls this method directly.

Unknown Speaker 19:01

Okay, and how would you call it in the main method so

Unknown Speaker 19:08

under the Yeah,

Speaker 2 20:58

Okay. Let's say intercession method is in a different class than the main method. Let's say the class is called customer. Okay. How would you call called the intersection method?

Speaker 1 21:17

So, for that, first thing is like, I would need to make that method public.

Speaker 2 21:24

Yeah, that's fine. Let's assume that it's public. Okay. And it's inside a class called customer. Okay. No refactor? Your main method to call that intersection method.

Unknown Speaker 21:39

Alright, so yeah.

Unknown Speaker 21:56

Okay, I think I'm done with my questions, Mark issue. Yeah, sure. Sure. I'll take

Speaker 3 22:06

a break. Hi. Hey, so I'll ask you some other questions about from Java or Java. How, how's your exposure to SQL and database? Oh,

Speaker 1 22:22

yeah. So you know, I have worked on you know, the SQL database. So I'm pretty good about it.

Speaker 3 22:30

Okay, so before we forget that, can you copy your notepad content onto the chart, as well as the one you wrote? So that's there. And while you do this, can you tell me what's DDL statements and DML statements? Yeah.

Speaker 1 22:51

I'm sorry. I'm sorry. Could you please repeat your question? I couldn't hear that was coming in.

Speaker 3 22:55

Yeah. Do you know what a DDL statements and DML statements in our DBMS

Speaker 1 23:00

Okay, so DDL and DML Yeah. So basically these are DDL or data definition language and but this DML or data manipulation language right. So, this DDL statements are used to define and many of the objects in database like tables and schemas, like those are basically create alter and drop and don't get annual Ebert you know, your DML man, you know, you are you are like in the DML or you are like manipulating the data within the existing database object like you know, so they focus on data stored, like you have your SELECT statement, insert this insert statement, already segment and so on.

Speaker 3 23:47

Okay, it's delete a DML statement or DDL statement.

Unknown Speaker 23:52

Delete right

Unknown Speaker 23:55

so it's a DML statement, delete. What?

Speaker 1 24:02

So it's in basically, using a delete, you can remove records from the table, right? So it's focusing on the data right? So not on the structure of the table.

Speaker 3 24:14

I can remove records using truncate to is trumpet also DML.

Speaker 1 24:21

So, in case about truncate saw, truncate would be DDL you know, basically, truncate is used to manage all the objects right so basically not one, right.

Speaker 3 24:38

Okay, but technically I'm on manipulating the data there too, right. What's the difference? Yeah.

Speaker 1 24:50

You are not manipulating a particular data right. So you know, DDL are used to manage all the objects in the database. So it affects you know, the effects and usually, if you are truncating, it is not that reversible, right? So you need to have proper backup for the truncated statement. And basically, there is no roll back. What I mean, you know, yeah, and it cannot be a part of transaction.

Speaker 3 25:21

Okay, and what about drop? What does Trump do?

Speaker 1 25:25

So, in database this drop is basically like it deletes an existing database object, right? So you can delete a table with the help of drop.

Unknown Speaker 25:39

Okay, what's a view

Unknown Speaker 25:42

and when should I create it?

Speaker 1 25:45

So this view is like, basically, you know, it's, it's a temporary you know, it's what we define this is a temporary table, I would say, in a database object, and which is for like, storing the data, but you know, like, you are combining data for from like, maybe multiple tables. And if you have a complex structure of the table, where you know, you need to, you know, you need to construct complex queries for the for creating the data, you can create a virtual table or we can see a view that behaves like a table but represents the data instead of storing the data. And so, you can you know, like typically, like basically create a data abstraction so that whenever you want to access data that may require a large query, it can be divided into a single view.

Speaker 3 26:47

Okay, and when is it? When should we use like, it's basically like, Are you saying it's more about like, reporting stuff is where we should use view.

Speaker 1 27:01

Oh, this like, basically, if you run a complex where is with multiple joints, and conditions, like, like, you know, on a regular basis, you can either create a view which is like, you know, you're a virtual table, and that order represent the data directly. So it can provide a layer of abstracts and by, you know, by hiding the complexity of, of the underlying, what do you call that? My underlying underlying the table is structured so, basically, you want to have to relevant data to be focused on rather than, you know, needing to know the details of the tables right. So that when you want to use that, you know,

Speaker 3 27:48

all right, I will be giving you two tables on the chat window, if you can copy it over to your notes and and then we will create some queries around it. Yeah, absolutely.

Speaker 3 28:04

So we have table salary, we have table employee and you have respective columns like Job IDs, Salva location, etc, right. And similarly for employee, if I want to find out where does the employee who has the second highest salary, recite what will be my query

Unknown Speaker 28:35

main thing about this one

Speaker 1 29:04

so you want the employee with the second highest right salary, right?

Speaker 3 29:09

I want to know the location or the second highest salary Oh,

Unknown Speaker 30:58

me explain this query. If you're done.

Unknown Speaker 31:10

Yeah, so basically,

Unknown Speaker 31:14

in this query

Speaker 1 31:21

so I'm just, I just I have an outer query that returns or locations if the you know, salary, hello, metadata certain value, right? So that's the outer query where you are doing like, select location from salary with sale value is something and then you have an inner query, right? Which is used to retrieve a distinct salary from the salary table in a descending order. And you are restricting the data into one bedroom, and then you are like skipping the first row, which would be the highest salary

Unknown Speaker 32:06

whatever it is to do.

Unknown Speaker 32:11

So did the district right.

Speaker 1 32:14

Those are distinct will usually return the distinct or unique salary right so it removes all the duplicate values.

Speaker 3 32:26

Okay, so you want to get all the duplicate values you want to order them descending Lee, is that descending?

Unknown Speaker 32:34

Yes. And what

Unknown Speaker 32:36

are the other two parameters limit and offset?

Speaker 1 32:41

So, yes, you know, this limit only returns one bedroom and you know, this offset, you know, escapes the roads. So, since, you know, we are providing a one and it gives the highest salary.

Unknown Speaker 33:00

Okay, is there any other way to do it

Unknown Speaker 33:09

you know,

Speaker 1 33:12

you may be able to use like, you know, the row numbering or I think this would be like the best way.

Speaker 3 33:25

Okay. And if we need to display the employee name and you give this address and you can write me another one of getting the employee name of, let's say full time salary.

Unknown Speaker 33:42

Oh, would you write that?

Speaker 1 33:43

So I know if so I want the employee

Unknown Speaker 33:50

name. employee name right. Okay.

Unknown Speaker 33:58

Just give me a second.

Unknown Speaker 34:06

Skim Yes. again.

Unknown Speaker 35:37

Okay.

Speaker 3 35:52

In what cases we shouldn't be using distant

Unknown Speaker 35:56

and then what case we should Oh, so.

Speaker 1 36:02

In which case, you know, like so basically, you know, this distinct keyword is used to eliminate duplicate rows like from the original set, right. And however, you know, right. And so, you know, you can you can know, whenever you are aggregating your data you may want we want to ensure that the only unique values are considered in the calculations. For example, like, like, when you count unique customers who made a purchase, you know, and I So, if you want to get that, you want to use the distinct keyword, but if you have like, you want the exact match, if you want the exact match, and find and find how many of those exact matches exist, then you don't want to use distinct because it escapes a lot of data, right?

Speaker 3 37:08

Yeah, okay. Got it. Do you have any exposure to Unix?

Speaker 1 37:17

So like, I have been, you know, working primarily on Unix and Linux. So

Unknown Speaker 37:26

like, have you been doing on Linux?

Speaker 1 37:29

or so in Linux like, so our easy to you don't like a Linux base actually. So our application is deployed on Linux based person Mason's. So sometimes

Speaker 3 37:45

if I need to figure it out, what task is running on the system? How do I determine that?

Unknown Speaker 37:57

lets me find

Unknown Speaker 37:59

so in that case, like

Speaker 1 38:03

so. So for that you can use like, you know, ps command, PS Condor,

Unknown Speaker 38:11

which command can you write it on the notepad? So

Unknown Speaker 38:19

it's a ps command.

Speaker 3 38:21

Okay. Suppose we want to find out how many instances of it does suppose with start select a product one, product two, product three, right? And you want to figure out if all the instances operativa running what will be my command?

Speaker 1 38:37

Oh, so in that case, you know

Unknown Speaker 38:44

good morning like

Speaker 3 39:00

I wouldn't have you used something similar before. Have you done this kind of thing?

Unknown Speaker 39:04

Yeah. So let me

Unknown Speaker 39:12

let me type it for you. Okay.

Speaker 3 40:25

Yeah, what is the Lucy Hi finale.

Speaker 1 40:32

So, Wi Fi right. So the Wi Fi on like, gives you like word count. Okay, so it gives it like, it gives you the account the number of lines corresponding to like the how many methods are there Okay, okay.

Unknown Speaker 40:56

Suppose I have a file abc dot txt.

Speaker 3 41:01

Okay, how are what are the different ways I can check the content of that file?

Speaker 1 41:09

Okay, so in that case, let me you can use

Speaker 1 41:24

see, you can use scat and you can use this list

Speaker 1 41:36

you can you can also use the like head and tail as well.

Unknown Speaker 41:45

What is the differences between them?

Speaker 1 41:49

Okay, the difference between like you know, these two are is like how to tell this, this this actually hate is used to display the first few lines of the file and like it typically like 10 lines and then you know, you can specify how many you know, number of lines and then date is used to display the last last few lines. So it is like you know, what do you call that which is our default to and to add lines, but you know, like, like, how will you specify how many lines

Unknown Speaker 42:33

Okay, and what's less and what scat

Speaker 1 42:36

so this basically this guide is used to like the what you call the to display the content of file, so it displays everything of that file. And then you know this allows you to build the content of the file on the skin, you know, right. So, so you can use less, you know, less like to what do you call that build content, per like per pages per page, which like, you can what do you call that you can scroll

Speaker 3 43:16

Okay, and if, if I have defined ABC, right, and suppose in which I want to figure out like how many times or where all the pattern for the probates what should be my command

Unknown Speaker 43:33

Okay, so in that case

Unknown Speaker 43:49

this

Speaker 3 44:14

Okay, so Okay, and this will repeat this will give me one output

Speaker 1 44:24

okay. The output, what it will give me is like so, it will give you the line number along with you know, matching lines of like abc dot txt file that you know that contains the water

Speaker 3 44:43

okay, okay. If I have a file and location A and I have to have the same file and location B, would you copy the file and delete it or just plain move the file and be done with it? What would we do? In

Speaker 1 45:04

that case? Let me take so I think my best approach would be to copy and delete because, you know, you want to make sure that the file integrity is safe right. So basically, we would like in you the cp command to copy you know, to copy the location A and B and then use the rm command to delete the file later. Because because you can have a backup of the file until you start using the file and the location be right

Unknown Speaker 45:48

and why should I have a backup of a file

Speaker 1 45:51

so to keep a backup of a file, like so, in case your file is on location B gets corrupted, right? For some reason, and during you know during the transfer purpose, you know like or if you maybe the if maybe the disk is like you know, disk drive system pills, you know, take system fields.

Speaker 3 46:17

Okay, and if I need to figure out what is the disk space in a server, what is the command for that?

Speaker 1 46:24

Oh for that command, you know this what do you call that?

Unknown Speaker 46:30

Let me this

Speaker 1 46:59

this df command shows how much free space is free

Unknown Speaker 47:04

okay and what is hivernage

Speaker 1 47:08

Okay. So, this is you know it does represent in a human readable form, so I think it gives you like tabler form it each unit

Speaker 3 47:24

answer what was that? What was the last part? How does it display the data over

Speaker 1 47:29

the displaying the data? Yeah, so it's like it's it is a triple display in tabular form, right? So it's basically in terms of gigabytes and megabytes.

Speaker 3 47:44

Okay, and if I need to see it in kilobytes, what should I do?

Unknown Speaker 47:49

Oh, for the kilobytes, like, you know,

Unknown Speaker 47:55

for the kilobytes, you

Speaker 3 47:57

because sometimes when you do movement at a good level, it might look like the same, right? But the file size might alter. So I want to know exactly, it's all the file sizes are retained same. So I would need to check on the kilobytes structure right. So what can what should I do?

Speaker 1 48:18

So you should like I know what they call that. So it use you should just be able to you know, use Das, you know,

Unknown Speaker 48:32

I shouldn't just use dash

Speaker 1 48:35

no fashion. Yeah, yeah, you have tears

Speaker 3 48:45

okay. Oh, and if I want to see the memory of the host, what should I do?

Speaker 1 48:50

Okay, so for the memory of the host

Speaker 1 49:02

I think there is a top command like you know, and also a free command.

Speaker 3 49:11

Alright, okay, I pause here. Do you have any questions for us?

Speaker 1 49:16

Oh, yeah. So, I would like to know, like, like, like, you know, what are the rules compromised? You know, comprises of like, what kind of development to work that Odin, you know, developers who is being hired between like

Speaker 3 49:36

so it's a it's a big project, we have multiple roles, right. So there is no one definition role which we are gonna put you in or like, you know, we are looking for the candidates. It's like multiple, you know, full stack roles across multiple projects. So they just cannot find the right fit. But all the projects are run on like the Linux boxes, and they also have connection to database, either Sybase DB two or MongoDB or whatever right. And then on the other platform, so we are just trying to get assessment on all the levels. Okay. So yeah, there is no particular fixed you know, audition, which we are looking for, there are multiple questions.

Speaker 1 50:24

One more question I have like, what would be the next step of this process like?

Speaker 3 50:29

Yeah, so we will be discussing internally and providing our feedback and then somebody from the recruitment team should be reaching out to you

Pradip ganesh-workshield-srv-second

Speaker 1 00:00

We don't like mentoring to, you know, nearest, and yeah, sounds like, and, you know, and

Unknown Speaker 00:13

this, am I audible?

Speaker 1 00:16

What's that? Yes, yes, yeah. So, yeah, it's right. So I have, like, I've been working, like, mentoring to, like, like, two years, I've been mentoring the junior staff.

Unknown Speaker 00:27

Nice, awesome,

Speaker 2 00:31

cool. So let's dive into some technical stuff. Basically, I just have some react questions, and then maybe we'll do some examples where I'll have you pull up an IDE or an editor or something like that, of your choice. We won't turn on AI, and we'll make sure the AI stuff is off. We do use copilot here if you want to, but not for the interview. You know, we do some pseudo code and go through a couple of technical things. We'll have some fun writing some code, all about writing practical examples. I'm all about, you know, seeing how you process and how you work on creating components and creating certain, you know, things that we'll actually use. So we won't be inverting lists or binary trees or anything like that today. So hopefully that's okay,

Unknown Speaker 01:18

yeah. Oh, cool.

Speaker 2 01:22

So, let's just start with some cool react questions. Can you explain what the virtual dog is?

Speaker 1 01:32

So this, yeah, so basically, this mercurial dome is like, kind of like a representation of actual dome. Okay, so it's like abstraction away, like React uses this help to optimize the you know, updates by you know, basically comparing the you know, actual dome against the virtual dome, right? So instead, like, directly manipulating the actual dome, it present and it basically like creates a copy of it. So this is actually, yeah.

Speaker 2 02:04

Have you ever heard of the word reconciliation of the virtual DOM? Yeah? Or how? Yes.

Speaker 1 02:09

So it compares between, like the previous version of, you know, the different person. And then just the calculation is actually a frequently, very cool

Speaker 2 02:21

let's see, can you explain the difference between a controlled and uncontrolled component in React?

Speaker 1 02:30

So talking about this control and uncontrolled this component, so basically, like this, this, this controlled component where, you know, react fully control the form and also input state, right? So basically it holds the input value and any, you know changes trigger or any object, some kind of you know, handlers. And then there is an also on control component, which basically likes DOM manages the state, right? So basically like this uses like riff to access the input value as well.

Speaker 2 03:10

Very cool, exactly. Yeah. Man. Can you name some hooks and react? Hooks that react uses?

Speaker 1 03:18

Yeah, so in this react functional component. There are lots of books we use, right? So basically we use like state hook, like for use state for state management. So basically that gives you an initial value and also function to change that value as well. So there is a huge effect hook as well that basically handles the side effect. So basically, you can call an API and also subscription and any kind of like DOM manipulation as well. You can also use, you use it also. There is also use context books as well, passing the data between components and and, you know, there are like, without, like, having work, like two months prop trading them from parents to decide component, and there are other books as well, like user if and other like use reducer as well. Yeah,

Unknown Speaker 04:11

yeah, man, what's the

Speaker 2 04:15

difference between something like you mentioned, like state management with, you know, having state in a use state hook versus a used context hook.

Unknown Speaker 04:26

What, what would you use each for? And when

Unknown Speaker 04:30

would you use them?

Speaker 1 04:31

So if we compare about the use state hook and use context hook, so basically, like, the main difference is that youth state is used for local components, like the state management, right? So it is also used for managing the state within the single component. So whenever you have a component that needs to handle the own data, right, like basically, basically increasing our counter, something like a button clicks when you want to use the data. But in case of use, context API is like for global restoring, restoring and sharing the data in the various application component right whenever you want to, you know, share the state across the multiple components. It works by basically creating a context provider that holds the CRT statement.

Speaker 2 05:25

Cool. Hello, use reducer. Did you say, I think you mentioned that one a second ago.

Unknown Speaker 05:31

Experience with use reducer and use that one.

Speaker 1 05:37

Talk about that. This. Basically it is like managing the, you know, those state complex logic, right? So, yes, it use this, use reducer, who basically provides some kind of structure away from handling your state transaction, right? So on your like, you do, like, like, use state you like, basically have like your counter and state counter, but on using on use reducer, you will have like the like difference type of actions, like, and then, based on these actions, you can manage the state right. So for example, like, you have an add up card button, and then you can, like, increase the implement that number, and then also item on the card, and also decrease the number of card in also reset as well, so that one particular reducer function will be able to handle that state and next

Speaker 2 06:36

and as well. Yeah, yeah. I remember when VC reducer kind of came out and kind of came available to react. I don't know if you remember when that actually came out, but it replaced a lot of the Redux stuff. Right? Exactly replaced, but it can't substitute it for Redux, right, correct. Are you familiar with Redux too? I assume you are, since they're so similar in terms of stuff. But have you used Redux before? Yeah,

Unknown Speaker 07:02

I have worked with Redux

Speaker 2 07:05

as well. How about asynchronous Redux, or any libraries that helps you to convert Redux into an asynchronous nature?

Unknown Speaker 07:12

If I talk about this synchronous Redux,

Speaker 1 07:16

I mean like, I have worked in like handling, like asynchronous operations in Redux. But, you know, I haven't like done it too much to be honest. And I think this is like middleweight, called a tank that you can use, but I haven't like,

Speaker 2 07:34

right, right the I think there's two popular libraries, thunk and saga Yeah, sagas, and you don't have too much experience with those, or you, you do have experience with those, yeah,

Speaker 1 07:48

I work with as juniors office in Redux, but I don't have much experience. Too much.

Unknown Speaker 07:55

Yeah, that's okay.

Speaker 2 07:58

I think you know, as long as you understand the asynchronous nature and what Redux does, and how Redux is not asynchronous at nature, right? Cool. How would you optimize re renders in React application?

Speaker 1 08:18

So, in React, like, you know, it will like so, in order to, know, optimize, re render in React, I would say like so this basically, firstly, you know, there is, like, some kind of a component memoirs, right? So there is an in React, use memory, react memory as well, right? So, yeah, it helps actually, to re render in the component. And then you can also, like some kind of that callbacks to avoid these unnecessary functions, and it is enters in our component. So, so these are two of the primary ways I basically use use memo and then the callback

Speaker 2 09:09

functions. Awesome. Man, um, there's use memo, and that's a great one for optimizing performance. And what, what exactly does use memo do? I don't want to give just keep talking. I want to let you talk.

Speaker 1 09:25

So if I have to talk about this, use memo in React. This basically it memoizes, okay? And in a simple term, you can say it actually guesses, okay, the result of functions. So it only recomputes whenever you know you have some kind of dependencies in this, right? So basically, you have used memo books, where you pass in some kind of like functions that returns a computed right, and then you provide like dependency array, like, similar to how you lose use effect, okay? And whenever those dependency changes, it will, like, recompute, and it, if the dependency stays the same, it just return the same variable. And also, every in every re render, it only you know, runs so when it like us, there are some certain dependencies, changes else it just, you know, returns the same area.

Speaker 2 10:27

Yeah, that's awesome. How would you quote, unquote, memorize a function?

Speaker 1 10:37

So, so while memorisings of function, you know, this to to memo is actually a function. I would say we can, just like, you know, call a callback hook, right, basically preventing those recreation of the functions, right? So basically you would have something like a handle, handle click functions using the callback,

Speaker 2 11:02

yeah? So they use callback hook to quote, unquote to stop the function from

Unknown Speaker 11:07

RE executing on every vendor, right? Yeah.

Unknown Speaker 11:10

Very nice. Man.

Speaker 2 11:14

Cool. Let's grab our IDE. We can keep coming back to some questions. I got some other questions about GraphQL and maybe some build time questions, just to, you know, vary things up a little bit, but I'd love to see some code. I'd love to, you know, get our hands dirty a little bit, if that's okay. Yeah, all right, let's, let's grab a, let's grab a. IDE, you want to share your screen, and maybe, let's write this. Let's make fun. Let's write a list of items that are fetched from a fake API. You don't have to write the API logic and stuff like that, but let's write the list of items. And let's do an infinite scrolling list, however you want to do that, and work through it one at a time. If we don't get to the end, we don't get to the end, but we'll see if we can get can get some kind of like an infinite scrolling list going on the screen,

Speaker 2 12:10

even if it doesn't compile or anything like that. It's just pseudo code. Let

Unknown Speaker 12:14

me start setting my screen.

Speaker 1 12:20

I actually have a React project set off so I can use it. Yeah, perfect.

Speaker 2 12:27

Are you a fan of copilot and AI and stuff like that?

Speaker 1 12:35

I had co pilot added to my do like it. I don't use it too much, that much to be honest.

Speaker 2 12:46

Are you a fan of I see you have a JavaScript one going on here. How did you set it up? Was it like a create react app? Or what did you choose to get it all set up with? Yeah,

Speaker 1 12:55

the same, the Create react app. Cool.

Unknown Speaker 12:59

You ever used? Viet,

Speaker 1 13:03

no Viet or how about beat? Just like much easier to hit up. Yeah, I heard about that.

Unknown Speaker 13:10

Yeah, of course.

Speaker 2 13:12

Awesome, cool. So we got an app going, probably delete all the stuff in between the div tag and I

Unknown Speaker 13:24

Oh, the classic create react app.

Speaker 2 13:33

All right, so our goal is to create an infinite scrolling list. I how

Speaker 3 13:40

we go about that.

Speaker 2 13:48

So I guess we have to start with maybe getting something from an API or a fake array, if that makes sense, with a certain number of items in it. Yeah, sure. So

Unknown Speaker 14:03

let me create a hook to store that item sure I

Unknown Speaker 14:25

so this is where I'll be storing my items.

Unknown Speaker 14:38

So basically, since I'm

Speaker 1 14:39

doing an API call. I would basically be doing you this checkbook, which was like, basically load an item on a page. I

Speaker 1 15:05

so the use state I will be passing a function and then a dependency.

Unknown Speaker 15:19

So let's see and using usually want to extract that into a whole function, make it easy to work with.

Unknown Speaker 15:27

Okay, so

Unknown Speaker 15:48

So let's see, what

Unknown Speaker 15:57

do we do here?

Unknown Speaker 16:08

Try to call this API Using a phaser right and

Unknown Speaker 16:33

this This. But I would Just keep the

Speaker 1 17:02

keep so basically I have checked if the response of this API is okay, not okay, and then I just throw an error. So now basically what happens is I need to wait for the response back right, and then once I get the item, I would

Unknown Speaker 18:02

just once I have The data,

Unknown Speaker 18:05

I can just save that item

Unknown Speaker 18:12

using it,

Unknown Speaker 18:36

so basically, this One grabs my response, and then

Unknown Speaker 18:40

there's my items that would already had. Now, let's see what I need

Unknown Speaker 19:02

to do. Need to do and the same items, I

Speaker 1 19:08

highly suspect this synchronous call might need to be wrapped around. I think I highly suspect that this synchronous call might need to be wrapped in but,

Unknown Speaker 19:27

yeah, so since the same thing, you could just do

Unknown Speaker 19:31

You the callback. Okay? Do

Unknown Speaker 20:00

Now let's see, once this is done,

Unknown Speaker 20:14

probably need a way to display this. I

Unknown Speaker 20:33

so I think this bit appear

Speaker 1 20:35

needs to be no dependency on other use effect I the user,

Speaker 1 20:45

because, you know, it might change. And for that, we might want to run the you know user framework.

Speaker 1 20:58

Now, basically we just need, you know, have scrutiny List, and let's see what I can do. Sure you

Speaker 1 21:25

so for each of the item, I can create a list item I

Unknown Speaker 21:53

I'm just assuming there is going to

Speaker 1 21:55

be an ID on the item, because actually that is Item requires a key, right? And then we will just split the item tighter. Yeah,

Unknown Speaker 22:04

definitely. Is

Speaker 2 22:10

it ever okay to use the index that

Unknown Speaker 22:13

you're on as the key? Yeah,

Speaker 2 22:19

would you ever consider using that you don't have to change anything. Just kind of curious, what

Speaker 1 22:27

you think about it? Yeah, I think if we use, you know, index as a key, it can sometimes be acceptable, but it might not be, you know, might not correctly update the UI when the list changes, I think it can lead to some kind of project or in future, yeah,

Unknown Speaker 22:48

yeah.

Speaker 2 22:50

I usually find, if it's just a static list, maybe you can get away with it, maybe. But if it's a list that you're changing the order, yeah, index actually represents the item ID, and react tends to not track it too well. So if

Speaker 1 23:07

you read off the, you know, all the list, it can, you know, probably be an issue.

Unknown Speaker 23:13

So you can, you know, you can use for it.

Unknown Speaker 23:25

So this would be our base pipeline. And

Unknown Speaker 23:28

now I need to, basically,

Speaker 1 23:36

I need to let the React know that I am scrolling, and then for each scroll, I need to basically load more data, okay, and for that thing, what I can do,

Unknown Speaker 24:00

that.

Speaker 1 24:15

So we need to store some kind of pagination state, I guess I

Unknown Speaker 24:44

so you will initially be in the first page, okay?

Speaker 1 24:49

And whenever my current page changes, I want to read up the page API, right? I No,

Unknown Speaker 25:12

let's see i

Speaker 1 25:23

i need to basically assure this page when I read the end of the list,

Speaker 1 25:36

so I can probably provide the ref on my list item I

Speaker 1 26:21

BBB, so if I'm at the same index as the length of the item, then I would

Unknown Speaker 26:33

basically call or call back. So

Speaker 1 26:47

I guess it will be better to, you know, load the data as when you know we are the way on the second to last. So I will subtract some some numbers. I

Speaker 1 27:09

now I actually might not know what I need to do on the to handle last item here, but it will probably like just call back and BBB.

Speaker 1 27:38

So the way I handle no one to handle. The last item is, let's see what I need to do.

Unknown Speaker 28:01

So I probably just need to increment the count, I guess I

Speaker 1 29:11

I think this would be some kind of pipeline on how this you know, in finally, scrolling list should work. So do you see anything that I might be heading to the wrong direction.

Speaker 2 29:23

Oh, I like what you're doing so far. And actually had to look up adding a callback into the ref as well, because I've actually never attached a callback to a ref before. I've only ever attached a ref to a ref. So that's really interesting. It seems like you can call that and going to give you, it's going to give you the reference to the to the node correct. And that node is going to execute that function handle last item. It's going to in, it's going to

Unknown Speaker 29:54

increment that page by one. Now,

Speaker 2 29:59

I think our scrolling list and correct me if I'm wrong. You know, this is why I like working on this kind of stuff. I think your scrolling list is going to just continuously execute that. Because no matter what, it's not going to wait for the user to get to the bottom of the scroll correct. It's just going to say, hey, when the item set length is this, boom, do it, and it's always going to have an index that equals items dot

Unknown Speaker 30:24

length minus two. Does

Unknown Speaker 30:27

that make sense what I'm saying?

Speaker 1 30:32

Yeah, actually it does. Basically. What you are saying is that ref would always have the index that actually matches

Unknown Speaker 30:45

the item, like length that the minus two, right?

Speaker 1 30:49

And I guess we need some kind of use Rev hook to basically have an inter interaction with the users, right? Yeah,

Speaker 2 30:58

I think if you use and I mean, correct me if I'm wrong, because I'm not right all the time. And that's the beauty of working together to figure out the code here.

Unknown Speaker 31:09

But it seems like this would make more of an infinite,

Speaker 2 31:11

infinite loop of just calling the API over and over again, right?

Speaker 1 31:18

Right? Yeah, that is exactly right. So

Speaker 2 31:24

is attach a use ref to it. This is which this would work, as long as you're also checking that that ref is in the viewport is observed. If that makes sense. Have you ever used an intersection?

Unknown Speaker 31:42

Sorry, have you

Unknown Speaker 31:44

ever used an intersection observer? No,

Unknown Speaker 31:45

I haven't used that too much. I don't think

Speaker 2 31:50

that's okay. That's cool. Man, yeah. So basically, what it would do is kind of, can you see my camera here? Yeah, cool. If this is the top of the div, and this is the bottom of the div, and hopefully that item lives down below the bottom of the div, right as it scrolls up into the viewport, should be the trigger that actually calls the handle last item, as long as it's not into the viewport or not in as long as that threshold hasn't been met, It shouldn't be calling API. I think that would probably spruce your code up here a little bit and prevent that infinite loop that's possible, I think, in just looking at this. But other than that, this is great. I've never attached a handle last item. Never attached a callback directly to a ref. So I learned something new today. So thank you so much.

Speaker 3 32:42

Yeah, super cool. What else?

Speaker 2 32:54

Handle last item, we're gonna set the current page. The current page changes. It executes your callback function again,

Unknown Speaker 33:01

yeah, I think, I think we're looking pretty decent here.

Speaker 2 33:04

The dependency of the use effect, right? We do have to have that technically, according to our React rules, is the dependency So, but that function shouldn't be changing as long as current page doesn't make the use effect call again, you know? I mean, yeah,

Speaker 1 33:22

yeah, exactly. It's like, just a recommendation, yeah. So yeah.

Speaker 2 33:28

And then let's see, what if there is no more data in the back end of fetch. We would also want to kind of handle that situation to say, hey, oh, there's no more. What our current page is last page. So tracking that variable would be kind of a good thing to do, if that makes sense, then we can prevent that. Most likely, our API would send back that request saying, Hey, you're on the last page. So seriously, awesome job. I love what I'm seeing here. Let me grab my notes and we can keep going. Maybe we can do a couple more questions and write a little bit more code. You don't have to delete any of that. We might use some of that to keep going. Um, so my next question was, what strategies do you use to optimize react applications, and you're showing me that clearly you're memoizing or you're putting a call back on the function. So that's awesome. Have you ever used code splitting?

Speaker 1 34:27

So if I talk about this, code is split, yeah. So exactly I, you know, use code split. It is basically like loading only the nasal JavaScript code, right? So, I mean, like, I've been like, explicitly, like, dedicated for, you know, like for splitting, but, you know, you can just, you know, to react and like lazy loading components, and also, like, when it's needed, and it just like, like, Lazy, Lazy loading, kind of overall sense, like, where users, like users to dynamically import the components and also spends the loading features while the component is loading up. So, but you can also, like, think something, doing something like router, you know, very cool.

Speaker 2 35:26

Say, for example, our infinite scrolling list here has 10,000 items on it. And instead of, well, let's maybe we'll put something where real estate maybe 1000 items, right? Instead of doing an infinite scrolling list like you've implemented here, you fetch all 1000 items at once, right? Have you ever dealt with large data sets on a React application where you fetch all of them at once and dealt with probably some performance bottlenecks, right? Because your React is going to try to render them, and it's going to probably struggle, because we're probably going to be doing a little bit more than just showing the title, I assume, which is just text. What if we're rendering, you know, large amounts of graphics for each one of them. How have you dealt with, or how can you deal with large lists like that? Okay, so

Speaker 1 36:18

for this, like, I guess, I guess, like, instead of, you know, re rendering all the 1000 of data you know, so you basically resonate the data you know, and your face only what we actually needed here, just like your limit and offset. So what is like, very standard way you just like, implement implement pagination, and if your API allows it, and then you just, like, use, like, loading page, loading page, and after and after that, you know, lasting is done and but, you know, apart from that, I think there is also a concept of virtualization as well. So that's one day, you know, that also renders the current items and

Unknown Speaker 37:04

items are using as well. Yeah,

Speaker 2 37:06

so they're not rendered unless they hit the screen. Nice. Cool. How familiar, familiar are you with building and building for production and going down performance, I should say, not performance going down cdci and creating scripts to handle the build processes of a React application.

Speaker 1 37:35

So this, CICD pipelines, this, we do have like, you know, dedicated to you know, personnel who you know, handles deployment. But I have been, like, involved in some kind of deployment strategies. So basically, like, I have built a command which, you know, basically creates, like, your scripts, and then you know, you know, like, also upload that. Like, imagine s3 as well, and also like holding place seven. And then you also, like, you basically direct, you know, cloud platform as well that s3 bucket by enabling standing re entering, and that's how you know our biggest,

Unknown Speaker 38:23

yeah, that's awesome. What kind of tools have you built with?

Speaker 1 38:29

So regarding the tools that I use, like I use, you know that you're, you're talking about like NPM, so

Unknown Speaker 38:40

like build engines, or

Speaker 2 38:41

I'll give you an example. Vite is a tool that we build with here. Have you had experience using vite or webpack or create react app, or any of these tools to actually build the production asset?

Speaker 1 38:58

Yeah. So well, this Webpack is our default, right? But I think we have been, you know, primarily working with the, you know, create a React app. And then there are like, you know, you know this. There are like stocks about moving, you know, weight as well, because the advantages that provides and we never did it, yeah. Oh, cool.

Unknown Speaker 39:27

So how about TypeScript? Are you familiar with TypeScript?

Unknown Speaker 39:30

Yes, I'm familiar with

Unknown Speaker 39:32

TypeScript. Sweet.

Speaker 2 39:35

Let's see. Can you convert this component into TypeScript, just the component the top level, and let's take in maybe, maybe just write an interface for me, just to show me that you can add that interface as the props, and tell me what props we can imagine that it's not the app anymore, but maybe it's one of the components nested down into the tree, and it's going to receive props, and we'll receive UU ID of the list of items, or something like that. So we can create

Unknown Speaker 40:12

a simple interface. I'm

Speaker 1 40:42

I forgot the type of UID, so for now, I just use it as a string. That's completely fine.

Unknown Speaker 40:57

Oh, and let's add maybe

Unknown Speaker 40:59

I'll let You keep going here. You're good. You

Unknown Speaker 41:33

I can call it, yeah, that's fine.

Unknown Speaker 41:38

I'm not too worried about this in text going on below that. That's fine.

Unknown Speaker 41:42

So what are the items as an item array? Or is that props

Unknown Speaker 41:45

that gets

Unknown Speaker 41:48

given to this component?

Speaker 1 41:55

So I thought we are getting the items array, the props, but if we want the want, no, well, we can just, you know, Get the items and from,

Unknown Speaker 42:10

from the props, right? I

Unknown Speaker 42:19

BBB.

Speaker 2 42:27

Cool, and then let's add another property to the item interface. Let's add a user type and user type, let's make that a union of two different types, and you can pick those

Unknown Speaker 42:52

cool and how about

Speaker 2 42:53

the next one going down, we'll do a self reference to another item, Like parent item, and the parent item. Let's just do, let's use, oh, what are they called? Now I'm going to a utility class, a utility helper, the pick helper, to pick out the ID. So it'll be an array of items where they're just their IDs.

Unknown Speaker 43:22

Think utility method, utility class, utility

Unknown Speaker 43:25

sorry, not following,

Unknown Speaker 43:29

sure. So we'll add one more. We'll call it

Speaker 2 43:37

associated items or something like that, and it's going to be a self reference to itself. It's going to use the utility

Unknown Speaker 43:46

method, a utility class called pick

Speaker 2 43:54

and let pick out just the item to ID and the title I'm

Unknown Speaker 44:17

that is the one. Yeah,

Speaker 2 44:20

just a little complex issue there. It's gonna just be Yep, I'll give you a hint. Left Angle, less than sign,

Unknown Speaker 44:32

yeah, my BS code is strong. Oh no.

Unknown Speaker 44:39

No worries.

Unknown Speaker 44:44

Sorry, we'll pick the item,

Unknown Speaker 44:49

and it's going to be The auto complete,

Unknown Speaker 44:55

the ID or the title you

Unknown Speaker 45:22

facili of TypeScript definitions,

Unknown Speaker 45:38

something like that, right? Something like that. Yep, everything

Speaker 2 45:41

the ID, and then you can do the union sign again or title. I'll do it cool. Let's go ahead and I know we're kind of just making some goofy Frankenstein interface, just for the heck of it. Just, you know, seeing how we're doing here, let's go ahead and extend that interface. Can you extend it with the props that would be associated with a React HTML, div,

Speaker 2 46:17

so we'll get all the lovely elements from a div extended on top of this interface.

Unknown Speaker 46:29

So we want to extend this interface

Unknown Speaker 46:33

with the reactive sure i

Unknown Speaker 47:32

Hi, COVID. Last but not least, let's

Speaker 2 47:37

see, can you add on the first interface? Can you add a generic? Yeah, sure, a generic, and then go from there. I

Unknown Speaker 48:09

cool, and how would you use that generic type?

Speaker 2 48:13

And I think example, maybe we'll add another one like I

Unknown Speaker 48:23

Yeah,

Speaker 2 48:29

yeah, super awesome. Love TypeScript, and I hate TypeScript, but I love TypeScript. It's

Unknown Speaker 48:39

one of those things, super sweet,

Unknown Speaker 48:42

you know, I think when I was

Speaker 2 48:43

when I was a junior Dev and I was learning JavaScript, I struggled and struggled, struggled. And when I jumped into TypeScript, it made me realize, you know, really great patterns to use. Now I'm just like, okay, when to use it, when not to use it. Cool. Let's do some GraphQL stuff. Can you write a quick let's see you live a query. I actually just, let's just talk about it for a second. What's the difference between a GraphQL mutation and a query? So this

Speaker 1 49:19

query, so basically, this query is to face data, right? Basically, if you want to retrieve something, you know, query something for your in the back end, and you would use a GraphQL query. But mutation is like it sounds to mutate or modify the data, right? So this mutation is like, like a scientific data

Unknown Speaker 49:45

nice. And

Speaker 2 49:48

have you ever used any GraphQL client side libraries?

Speaker 1 49:55

So this, I have worked with, you know, Apollo client, you know, which, which is a, you know, wrapper above the GraphQL

Unknown Speaker 50:02

i which

Unknown Speaker 50:04

library. I'm sorry

Speaker 2 50:07

I missed it, Apollo client. I do love Apollo client, very cool. And let's see Apollo client. They had some caching right in that they use with Apollo client. Can you describe what, when you would use the Apollo clients, caching versus Redux versus use context?

Speaker 1 50:29

Oh, so that's very nice question, actually. So if we differentiate about this graph given and, you know, context, API and Redux, so this Apollo client, you know, caching has actually, you know, it caused, you know, problem for me in the past, and which when I was, like, very new to the caching, actually. So basically, it is like normalized casing to restore the you know, query result. So whenever you want to have, like, phase and delay, basically, like, ensure the efficient updates. So I think it, you know, crashes based on the idea of the data, but, but that might be wrong as well. But, like, if you have some, you know, UI state models, or anything, you know, it may be a complex right, because tight key, you know, coupled with GraphQL, and you know, like you don't want to do some kind of modifications on it, right? But your Redux, what you your readers, right? But it works with any kind of APIs, types and to reshape each other GraphQL, and I like webs that line like a row socket, and what do you call that? So it is an ideal, ideal for the complex statement manager. But if your application is very small and does not have, like, a lot of complex logic, and adding readers doesn't make any sense, because, right? So, because it's very heavy, if you talk about the Redux, it's very heavy. So in that case, you want to use, like, use context with, you know, basically good for the smaller application.

Speaker 2 52:13

Very cool. Yeah. Have you ever used React query instead of Apollo client to handle GraphQL? Compost? I know it's, it can be used for restfully,

Unknown Speaker 52:27

and I'm just kind of curious.

Speaker 2 52:30

Awesome. I thought he's just double check my list. I'm just going down make sure we talk about everything we can talk about. I

Unknown Speaker 52:37

thought there was one more, but I might be

Speaker 2 52:39

mistaken, and if not, then I have finished my questions for today for you and I really enjoyed our conversation, man. And let me ask you.

Unknown Speaker 52:52

Do you have any questions for me while we're here?

Unknown Speaker 52:56

Yes, I have a question to ask. Sure.

Speaker 1 52:59

So firstly, thank you so much for your time today, and it's really nice talking to you. And firstly, I would like to ask, how is the app and that we are working on it. And so what kind of micro front end is it like, stand alone? And how is it structured?

Speaker 2 53:21

Sure it's, it's a React front end. It's built with feet. So it's a client side rendered app. I guess that's the last thing I didn't, ever, we didn't get to is client side versus server server side applications. I'm sure we can get there another day. We keep talking. Could talk all day about this kind of stuff. So it's a lot of fun. So we are currently on a client side application, react, server, feet. I think it's deployed, just like you were talking about, on an s3 bucket with CloudFront. Let's see. We use two different APIs, so we have a RESTful API that's utilizing Redux and Redux sagas, so I don't think it's going to be an issue for you. If you are familiar with Redux, then I'm sure you can understand sagas and anything like that. And then we also use React query for our other API, which is the first one was built in Node, the second one built in rust, and it's a rust GraphQL API. It features mutations, it features code Gen, not all TypeScript, but we're trying to get to all TypeScript. And it features subscriptions as well that allows us to listen for real time events that come in and react to those events accordingly. We're using an d, U, I library. I think we talked about that last time on the call. Yes, familiar with material, UI, or any of those other typical libraries. As long as you can read documentation, you're good there. It's a fun app. The app's pretty well structured. React. Writer, let's see react. Writer, Dom six and react 18. We're trying to prep for the inevitable merge into 19. We do try to keep up to date. You know, we let things come out and let them settle for a second. So as react 19 comes out and it gets stable and becomes a new platform that actually people use. We do try to keep our our software updated, and we could keep progressing toward that is a goal of the team, to not get left in the dust. If that makes sense, we use some really neat libraries, like, have you ever used React PDF?

Speaker 1 55:39

React PDF, yes. React PDF, no, to be honest,

Unknown Speaker 55:43

no, I haven't used, I guess,

Speaker 2 55:47

online. It allows you to write React code and it renders it into a PDF preview. Yeah. And then we're using high charts libraries that allows us to create dynamic charting and reporting. As you can imagine, being in the industry, of incidents coming into your workplace, you want to be able to display those and give accurate reports to your customers to say, Hey, you're in trouble here. You need to address this. Look at this chart, and you know, here's here's the data in a visualized format, if that makes sense. It

Speaker 1 56:19

sounds interesting. So, so that's all question I have. And so what will be our next step? So

Speaker 2 56:28

again, I'll take it back to my team. We're going to confer and talk about how you answer some of these questions, and go from there. And then I need to go up and talk to Chris Green, the CEO as well. So okay, yeah, have another, most likely, another phone call, because you're still in Pennsylvania. And then let me ask you this, when is your what? How long would you need in order to if you were offer the position, how long would you need in order to come down here? It's

Speaker 1 57:01

probably like one week to 10 days. Yeah, maybe.

Speaker 2 57:07

Okay. And do you require sponsorship or anything like that for your application, hearing.

Unknown Speaker 57:14

No, I don't suppose.

Prajwol Tiwari- AffiniPay-SRV-Final-1

Speaker 1 00:00

Code, you know, helping the team with code reviews, and I also provide some products and support, you know, some products and support duties. So, yeah, that that's what my works look like,

Speaker 2 00:10

nice. So you mentioned having, like, a fairly robust team with Scrum Master. We don't have something with that much like complexity, but we have quite a bit of things going. So do you have any examples or ideas in terms of or not ideas, I guess more in terms of like experience of working within because this role is really going to be kind of working with our web dev team, obviously, but having to kind of almost be working within your own little space, like our head of web dev get direction strategy, but this role really needs to kind of be autonomous in some cases. So like, you'll have your your your task lists, working towards our epics and stories and whatnot, but it'll be able to just be looking at your things and just be banging away. Won't be like be have daily swums, but it's not probably to what year might be used to. How is that? How would that feel?

Speaker 1 01:14

Yeah, I, you know, remember talking about this in my last interview as well. So I would say that, you know, I have good experience, you know, taking on certain project or, you know, you know, self learning and creating, like timeline on how the stories looks like, you know, working independently, I would say, and, you know, trying to complete a feature. I, you know, personally enjoyed challenges, and I like to take a lead on certain aspect of the work that I do. So like, you know, the recent one I remember, was an integration of our streaming engine. So basically, you know, we had to come up with a way to integrate one log streaming service with all of our services. And for that, you know, we had a very, you know, faint description of what needs to happen. And then, you know, I was one of the you know, person to take lead of, you know, taking on that feature, you know, gathering requirements from like person you know, who posted the enterprise, take backlog, you know, dividing stories, and then, you know, working on them. So, you know, as it come through, you know, so even though there is no structure, you know, I'm very mindful of what works needs to be done. What Works need to be, you know, split out and delegated. And, you know, I feel like it's one of the thing that I have excelled on in the past as well. So having a scrum master and fully phrased outside Screen Team is nice, you know, I won't deny it, but even without that, I feel like, you know, I'll just do fine,

Speaker 2 02:50

okay, yeah, because right now, I'm sure you might have gotten some of the lay of the land, but we're trying to get migrate all of our web properties under one stack and but also creating, ideally where I would like to be in, you know, a year or two is having a lot of microservices, having behind the scenes that can be grabbing data, pulling things in, and also helping to streamline most of our backend systems. Because we have both a proxy form that we built out and we're using Marketo. And I think there's ways that we could really be one looking to thin things out and make it much more simplified across the board, and also referencing, like making sure that our CDN is referencing just one single source of truth, because I think when we've got a bunch of different code stacks that we've got to work through, is that something that's like exciting for you, like getting really in the weeds and figuring it out, like making things simple and fast across the board.

Speaker 1 04:02

You know, sorry, were you saying something? Said, No.

Speaker 2 04:05

I said, that's kind of a vague question, but that's kind of how we've been having to operate. We're scaling up, but it's we're just in that, like, kind of in the crawl, walk, run, like, for like, we're just trying to get things right, and then we can start to build out and build out,

Speaker 1 04:24

yeah? So, yeah, absolutely, you know. So I remember talking to Alisa about the integration part of the website. So, yeah, I think, you know, that's like a very good thing to, you know, bring up, right? Because I think, you know, I did talk a little bit about of the product earlier in my previous interview. And I think that there's something that was really interesting, and I think that's, you know, that's how you should, like, you know, build up from what you have to build up, you know, resilient systems. And a resilient system, you know, they usually comes through when each individual, like have a good grasp and understanding, and everyone contributes in a way that they are working toward, I would say a greater picture

Unknown Speaker 05:15

that's good In terms of, say you've written some test cases.

Speaker 2 05:27

Yeah, okay, cool. What? I'm sorry. I'm just reading your resume as I'm looking through it. I

Speaker 2 05:45

work? Yeah. I mean, I think that ideally, within this role, it's really coming in, because right now, probably taking a look at existing systems existing databases that we have, and seeing what we can kind of streamline out. Have you had experience in terms of like having to do looking at legacy and then saying, right, we can build it the better this way. What would that do? Have you ever worked within that kind of framework so it's really looking at like, all, right, we have this big database and this big proxy engine that's been handling all of our forms in various places, migrating into something more streamlined. Is that something that you feel comfortable doing, or is that something you want more

Unknown Speaker 06:33

help with in terms of strategy on it?

Speaker 1 06:37

So yeah, so let me give you, you know, talk about one of a couple of micro services that, you know, I have helped in modernizing. So I would say that, you know, one of our world application is this, you know, monolith was in monolithic architecture. It was a monolithic app where, you know, each request in proxy and transformation is handled, you know, through one point of entry, and it is this big module, right? And the effort to like the whole project is about, you know, modernizing this existing application. There was a couple of aspects that would be split out. And with my knowledge, you know, within the system, I found a couple of areas that would be stripped out, you know, one of like the maintenance aspect of the, you know, profile preference. So whenever, you know, user makes changes to their profile, or anything that they want to do with their profile, if they want to make changes to that, they will, you know, go through the same flow over and over again, but I knew like it had to be modernized, and I proposed that we take this part of the application and convert it to a newer or better or small service that handles these chains. So that was something that, you know, I took on ownership of, and, you know, I did some designing with the architects, you know, implemented them and also deployed them. And there was, you know, also another aspect called the Experience API call for different components to be viewed on the front end side. So, yeah, to summarize, you know, I would say, personally, I've, you know, I feel like I am, I have experienced enough to, you know, call out for any pieces that needs to be converted from, like, a legacy standpoint, to a small, manageable microservice. And, you know, take ownership of these as well. So yeah, like to answer your question, I do feel comfortable working on such settings. Cool.

Speaker 2 08:43

When it comes to what your preferred work environment, what's your, what's your in terms of your experience? Like, where are you copyist? Is it being autonomous? Is it having a good, good group of people to work with? What's or do you like to just go heads down? Like, let me give me my project. Let me just work on it, kind of thing.

Unknown Speaker 09:04

So, you know, personally,

Speaker 1 09:08

from my perspective, you know, I do value a good collaborative team. But you know, I know that, you know, I don't want to be always doing, you know, programming session. You know, I feel like a good team is a really strong need for overall project, for the project to be successful. And, you know, personally, I do want my, you know, you know, downtime hit down time to do, like, a couple of hours or more a day, to just focus on my work, and then the rest, you know, I can do a team collaboration, or, you know, the settings where I can ask questions to the team, or, you know, even architects and so on, you know, I feel like, that's what is the ideal scenario where, you know, I feel like, you know, a time for autonomously working and also having time for, like, continuous feedback And during the iteration, because, you know, I enjoy working on teams where there's clear communication between like everyone, right, each one of us and helping each other to get better. But I, you know, I personally also value like I have my, you know, you know, I feel like I have to have my voices heard when I am conveying the, you know, any kind of suggestions to the team? So, yeah, I would say that, you know, that's what I value personally, you know, to sum it up, a mix of collaboration along with, you know, autonomous time, yeah,

Unknown Speaker 10:33

okay,

Speaker 2 10:39

what's in terms of, like, your own personal places you go and look for for inspiration and ideas where, what's, what's your kind of go to, sites, leadership, thought leaders, kind of things like that.

Speaker 1 10:53

Okay, so, yeah. So I feel like, you know, I currently, I think I have been doing a little bit, you know, watching a little bit more of TED talks, you know, I watched some of the TED Talks video, you know, try to get inspiration on like, you know, listening to their stories, you know. But more on like, the technical sides I do like so I do want to, you know, explore on GitHub, and, you know, Stack Overflow, you know, for trying to figure out how other people are coming up with new solutions, and, you know, how they are integrating different systems. So, yeah, these are the, you know, two places I go for inspiration, on a on my personal side, and, you know, I look into certain work that people put up, and then I'm like, you know, motivated to do something similar on my own. And, you know, I try to take example of their work, yes. And I also like, like to, you know, read books. So basically, you know, I'm inspired by environment that, you know, push creativity in both, not just only in technical or my field, but also on, you know, personal level. So I also, you know, take inspiration from, you know, most of people say, What, uh,

Speaker 2 12:16

when it comes to just your free time, what do you do for your free time? So

Speaker 1 12:20

basically, you know, whenever I get time to enjoy or during my free time, you know, I like to, you know, read book, particularly, I fan of soccer, you know. So I play soccer when I'm free for, you know, having a mental freshness. Sometime I go to go out for a run. And sometime, you know, I feel like a little I, when I feel like I'm a little bit more productive, I try to work on my side projects, you know, build up some technical skills that, you know, I may be logging behind. And if you ask me personally, you know, I like to watch motor sports, particularly motor GP. So these are the things you know that I do on regular basis. So I think you know that I think my refreshment is when I go out, have fun, play soccer with friends, you know, once couple of more GB races. So, yeah, that's what I do. Cool.

Unknown Speaker 13:11

What questions do you have for me?

Speaker 1 13:14

I do have one personal question for you. You know, I check out your LinkedIn, and I really like your bio, like and I want to get an instinct from you, like, how have you been able to convince your mom, like, what you do for like, 20 years cut I'm in my in the same pursuit, like, for five years. We say that again. So I looked up your LinkedIn bio, and, you know, I really like your the tagline that you have been trying to, you know, explain to your mom that what you have been doing for like, 20 years, so I'm in the same pursuit. So, yeah, yeah. So, yeah, that was a silly manner. So apart from that, I would like to know how the team works, right? How is like working on the team in general?

Speaker 2 14:01

Yeah, that's a great question. So we've been through kind of a lot of change within marketing as a whole. I've been with the company for almost a year now, and we've had to kind of rejigger a lot of things and a lot of teams because we've got, we've been also acquiring other companies and having to kind of move them into our stacks, get them, you know, under the fold from our team and modeling. And for the most part, we're we're also going through a change as a company. We just got acquired, so we have a new amount of new round of funding and have a new CMO, new SVP, and the structure of the team has actually gotten a lot better, specific to the web team like so my oversight is on the demand, house, SEO, CRO, design and then Web, and we've got different leaders under each one. Ryan Clayton is leader for the web team. Alex is over design, Esther's over CR and SEO, and Carlos is over demand. With the web team, we've moved into a much we're running on two week sprints, and then everything runs through JIRA. We get a usually task flow comes in from Asana. Those things get pushed to our backlog. We do our groom planning session, and then two week sprints. We're closing out the sprint right today, before it was kind of a very like pseudo ad hoc, where, you know, we had to store the backlog, didn't weren't putting any points to things. It was just like, What can we get done? We've revamped that to like, be looking at velocity, be looking at volume, we're looking at capacity models and things like that. Gotten a lot better in terms of, like, Alright, here's what we can plan to do. Here's what we can do. That's not to say that we haven't it's not a perfect model yet, because the person that was previously in this role was working within Brian's team, but was working kind of in their own court. We didn't know exactly what they had always on their plate, where I'd like to see what's going on in the boards, to then be like, okay, good, good. Yes, here's how we're performing to the tasks we are in the fit mix also of making a big change, of getting all of our web properties. So we've got case peer docket wise, CPA charge my case and law pay, are the five kind of core brands, and we're trying to get those under one roof. And with that, we have a lot of different integrations that have been used in various places that we need to just kind of streamline out and clean up. We use Marketo as our capture engine, or CRM for our or it's more of our automation platform. We're using them as our engine to be capturing leads and then funneling into Salesforce then be putting into different workflows for the sake of your role, it would be really kind of looking at all the back end apps, microservices, databases that we have that are all over the place, and seeing how we can get that really formulated under one good piece of architecture. So it's connecting both our best like, we use BigQuery for kind of handling a lot of our big data sets, but how we can start connecting a little bit more of some of the services to those to make sure that there isn't a lot of failure. Like, whereas we have testing going on, we use datadogs doing most of our daily tests, anything that we're running, just like as a always on tool. But we also have a lot of delineation in terms of who owns what, so that we have we lean into the product of the app size. So this is another thing that you know I would love to be tackling in the eight months, once we've got everything else, is that we reference the product, the actual app, the my case, app, and we use that there's a bunch of handshakes and web hooks going in and out of them to connect the dots for a user, and some of those have been breaking nearly and that's because There's just no single source of truth on this, and this role is really coming in, having working with me and with Ryan to kind of come up where, how are we going to tackle this with, like, what's our six month goal, nine month goal and 12 month stretch? So it's, it's really kind of saying, like, let's just sit down and be like, right? The first phase of this is getting rid of my way I'm looking at is like, we have this proxy form tool, ideally I'd like to be referencing Chester, Marketo capture tool, which is just the engine to grow that being able to handle that is also removing all the excess dead weight that we have from our form side and connecting that to the cleaner database, seeing what other microservices we're bringing in, some validation tool onto it, and how we can connect that as well. So some of it front end, but some of it's also back end, to make sure that our operations team can find the right things to work on, and then this role, you you know, would be helping to connect those dots for it and so, but it's also having some oversight on, you know, some of the other tools that we have out there, like our GTM four or GTM Google Tag Manager Tools, make sure They're great. Is this tracking looking right? Do we have everything hooked up, corrected to that? And then using any of the frameworks that we got going to Contentful, being Astro, and a couple other like, I think they're built on some Angular stuff, and so I think there's definitely crossover of what you're doing again, like, I'm more of oversight on it. My coding days are way back when I was doing that 14 years ago. So I'm not, I don't, I don't really try to text editor or any sort of, any sort of back end tools. I just try and ask the right questions and

Unknown Speaker 20:22

the

Speaker 2 20:25

team itself is very capable. This role is going to be really, really pivotal for us to like, grow and being able to help scale us up, because it's connecting a lot of dots and a lot of services that we know we're going to need like we have various amounts like, Wistia is a hook that we use for our video system. I'd love to be able to build a little micro app inside of our new Contentful stack that basically pulls all the videos in and allows it just to use a reference code to then be pushing it to the site, like it doesn't have to do, like a one to one frame it in. It's more of like, can we pull it in praise microservice that then has it framed into the UI for someone to then say, Oh, here's the video I want to use. Boom, click and drag. It's in there. So it's also working with Ryan's team in terms of, like, right, some functionality we'll want to get in the app, specific the Contentful stack. Like, they'll be, it'll have a lot of different moving parts, so it won't be just specific to one main line of business. It's going to be touching a lot of things. So that's that's a lot that I just explained, but any questions as I explained that in a very

Unknown Speaker 21:41

all at once, kind of thing?

Speaker 1 21:43

No, I mean, absolutely. I mean, it sounds very interesting. And thank you so much for detailed explanation. And you know, I think you know, that sounds really good explanation of what you guys are doing, you know, trying to move forward. And I feel like, you know, I would be able to contribute in a very good way for you, you know, for, like, taking ownership of certain stuff and helping the modernization of what you guys are aiming for. What So, yeah, thanks so much for explanation. And, you know, sounds very interesting. Nice. What else any

Unknown Speaker 22:19

other questions

Speaker 1 22:23

you, I think you mentioned, like you used to, you know, code, or, let's say you were part of development team 14 years back. So, yeah, I want to know, you know, any piece of advice for me or for someone like me, you know, um, I mean, I think

Speaker 2 22:42

in terms of this role, or just in terms of your own career, in

Speaker 1 22:45

terms of career, yeah, what's your personal life perspective?

Unknown Speaker 22:51

Personal perspective is that

Speaker 2 22:54

you always want to be trying new things and always really trying to see one be as you know what we do. Everything moves really fast. Like, not as every like, obviously, with AI, it's changing the game quite a bit for everyone. But I would say it's always being trying to be looking around the corner when people are mostly just looking at what's in front of them. So it's, it's all right, I know that this is going on, and it's more in terms of, like, as they're working with the team, and as you're working with, you know, your senior managers and whatnot, being like, well, there's this. We do have the optionality to maybe try this thing out. If it's saying, like, Oh, if we lean into this, or like, ways that we can connect the dots, because it's more about the business, like, that's the thing that I've taken me a while to learn, is like, it's not just about this little thing you're working on. It's like, what's the impact for the whole business? So from a back end perspective, it's like, how can I give X team 10 more minutes to be working, because I've speed sped up this interface. I've made this thing easier to use, or I've removed some dead weight that it takes for them to get there. Because anytime we can get efficiencies in terms of team, that's going to have good value for impact of the business, meaning we can move faster. We can be thinking better, and we have easier systems to work with. So it's more about like, when I was managing a larger team back at master control, I had 15 developers, half back end, half front end. The thing I was always trying to push them to be thinking about is like, who's using the application that we're working on, and how do we make it easier for them? If it's even just naming convention, if it's a if we can pull in better thumbnails, if we can create a better tagging system, if it's something that we can just remove that's causing extra things for them to get into, because my background is in UX user experience design, and so I'm always thinking in terms of, like, empathy to the person that actually has to touch the thing that we're building. And that's where it really helps, is in terms of, it's just understanding, like being in development, it's the we build things for developers, but developers aren't the ones that are using the final output, unless sometimes they are. But in most cases, what we're trying to build is something that makes it easier for that simpler person, simpler being, maybe just not as advanced an understanding of all the things that we do. How can we make their lives easier? Like we don't. We don't need bloat. We need that's, you know, why Apple runs the show. They've thinned everything out to being super simple. And that's really important, because it's and being concise is almost always harder. You know, it's like, that's the challenge, and it's easy to stand up something if it has 16 layers of things that they have to go through, because that's what the developer thinks about. Because our brains are very like mathematical and a lot of people aren't like that, so you have to think on both sides of it. That's kind of a long winded answer, but you know that's, that's my take.

Speaker 1 26:23

Yeah, sounds, you know, very insightful, and thanks so much for sharing. Yeah, yeah.

Unknown Speaker 26:31

Cool. Any other questions before we wrap up? No,

Unknown Speaker 26:33

I don't have any particular in mind right now, but,

Speaker 2 26:38

yeah, Bram will, um, yeah, look forward to we'll be meeting with Bonnie and Ryan. Super nice. She'll have some good questions for you. And yeah, we'll be talking more soon. I hope so. Looking forward to it. Thank you.

Unknown Speaker 26:55

Thank you so much. Have a good day. Bye.

Prajwol Tiwari-AffiniPay-SRV-Final-2

Speaker 1 00:00

Ryan probably met the other Ryan, the senior front end developer here, so working on all the other front end stuff on the websites. I don't know how much Charlie went into the team structure, but yeah, we work under Ryan Clayton, so he kind of manages the web team. Yep, oh yeah, oh,

Speaker 2 00:28

sorry. No, I was just missing that. I had experience, you know, I had, you know, talked to Ryan the other week, last week. So yeah, he talked me throughout, you know, introduced me about the work. So, yeah, nice.

Speaker 3 00:45

Excuse me. I'm Bonnie. I obviously also on the web team front end things. I've been here for about five years, but this was my first, because it still is my first, like, web dev job. So I started as a little, a little web dev, no regular web dev,

Unknown Speaker 01:06

yeah. I mean, that's what we do, yeah, I can

Unknown Speaker 01:12

go a little more.

Speaker 1 01:14

I've been here about three years at afinipay. Before that, I was another marketing agency, like, like outside agency, doing like websites and things there. And then I've also been in other, um, inside, in, like, marketing team. Yeah, so marketing is a little bit different. I've kind of spent my whole career up to now, like in marketing as a developer, so it's a little bit different than, like, a software side. So, yeah, yeah,

Unknown Speaker 01:55

yeah. It's nice, you know, meeting you guys and yeah, and,

Speaker 2 01:59

you know, hoping to hear more about your journey and stories and yeah, carry on for what we need to do.

Speaker 1 02:06

Cool, yeah. Do you want to go a little bit over kind of your background, things like that? Just, yeah,

Speaker 2 02:11

sure, sure. So, yeah. My full name is praj Tiwari, and talking about my professional journey, or my experience, you know, I have been working as a full stack Java developer for a little over five years now. You know, working primarily on Java based web applications, right, utilizing frameworks like Spring, Hibernate, you know, RESTful web services for API development. And, you know, I have pretty good experience, so working on micro services, you know, build a building then using Spring Boot framework. And apart from that, you know, I was just good knowledge and experience working with AWS cloud services for, you know, hosting these applications, you know, utilizing services like AWS, ECS, SQS, SNS, lambdas, you know, yes, three buckets and so on. And talking about my front end experience, of course, you know, I have worked with Angular, and I have also worked with the JavaScript and TypeScript, and talking about my testing side, or testing experience, you know, I have worked with J unit and Mockito. I am, you know, currently on a scrum team. And the team, you know, it consists of five developers on a daily basis. I have been primarily, you know, working with back in Java code, you know. And I have been, you know, working with code reviews. And I also some, you know, production software duties. I perform them as well. So yeah, and that's what my, you know, day to day looks like, and that helped me my professional journey so far.

Speaker 1 03:50

Oh, and where are you based out of? Where do you live currently?

Speaker 2 03:54

I'm currently in Virginia. Virginia.

Speaker 3 03:59

I work within Texas, yeah, but like different parts of Texas, yeah. So

Speaker 2 04:04

do you guys live in Austin itself, or is it like a remote job

Speaker 1 04:09

I am in Fort Worth Texas outside Dallas. It's like a three hour drive for me to the office if I need to. Yeah. So yeah, and

Unknown Speaker 04:21

I'm in the opposite direction. I'm in Houston.

Speaker 3 04:24

I just moved here from Austin, so very open to obviously, like, remote. I mean, like, clearly you're from Virginia, but

Speaker 1 04:31

yeah, yeah, I would say we go into the office. I do about once a quarter we'll have like an in person meeting like every like three months or so, and then meeting

Speaker 3 04:49

the whole like, demand gym team, yeah.

Speaker 1 04:54

Cool. Can you tell me so kind of that format interview, kind of, you did meet with the other people on the team, and then you did your technical thing with Tanner. So we won't go into like, very like detail, like code stuff, but just have a few questions, kind of open ended, just kind of talk about that. Yeah. So can you tell me about just like a favorite project, something like could be something at work, side project, you know, something interesting does not be your favorite, but anything like, just kind of, you know, what you worked on, What your part was, yeah,

Speaker 2 05:40

yes. So, you know. So basically, you know, I would like to talk about my experience, you know, working on, you know, whole migration effort of, you know, monolithic. So basically, you know, we were going to, you know, massive change in, like, the architecture world, and we were, you know, taking different aspect of the application, you know, converting them into a very service oriented Microsoft architecture, I would say, and I was, you know, kind of, like, excited and all. And then I wanted to, you know, contribute on this aspect. And, you know, I was, thankfully, I was assigned, you know, you know, migration of couple of services, including the maintenance API. So basically, you know, in the past, you know, in a monolithic everything was, you know, handled in one singular module. And you know, whenever a user had to, you know, make changes to their account profile, preference, like where they want to, you know, they say, want their notifications into kind of like protection. Then they would, you know, they want against their account. Then it will be, you know, all set up in one word, basically, you know, took, you know, lead of, you know, like stripping this service out and then converting into a microservice. The reason, you know, I find this, you know, interesting, you know, is, is that this was where I was first introduced to, like, architectural designing, you know, gathering all functional and non functional requirements, documenting them, of course, right? And getting reviews from like the company, you know, from architect to, you know, to see if it is like fit on the standards and so on. And, you know, also, I was involved in, like, implementation of the back end side, you know, getting deployed on AWS, of course. So, yeah, this was, you know, something really interesting for me. And because, you know, I was able to watch it evolve, right, from, like, something that existed to a very, you know, high level picture, along with very low level implementation details as well. So, yeah, so that's something, you know, that was very exciting and interesting for me. And this is something that I did. And you know, the reason this was more interesting was particularly also because, you know, I kind of, you know, laid off their groundwork for other migration effort, right? So with the pattern that I created for this migration, no, we were able to migrate a lot of services in same pattern. And I, you know, quickly followed this by an experience API, which is for entitlements for the front end components to, you know, load different component, and then other team members, they were, you know, they were able to follow the same pattern. So basically, you know, not just learning on my own for, like, a whole, you know, high level architecture, but also, I would say, helping the team to speed up their development process. So, yeah, that's why, you know, I think this is something that was really interesting for me. So, yeah, that's, that's one instance that I can say with you, with you cool.

Speaker 1 08:54

Um, so kind of related to that. Um, so we are kind of going through a like migrating from to a new stack right now. I don't know how much Ryan Clayton went over. We're kind of on like a view based static site generator, or going to Astro, which is a different static site generator, but supposed to be, it's a lot quicker. So like your project that you worked on, like, how, how do you select the tools and like technologies for projects? So with our migration for on our back end, we can have, like, we're kind of open to, you know, either maintaining our current stack, or, you know, you kind of can develop whatever tools you want to use coming up. So how would you determine which tools and which technologies you want to use for the project? Like, if you can use whatever you want, basically, like, how would you determine which ones you know

Speaker 2 10:03

you would want it. So, yeah, I, you know, remember talking about it in my previous interview with Ryan, of course. So if you ask me for my personal perspective, so, you know, basically, you know, determining what tool would be the best fit, you know, the first you know, it depends on different scenarios, right? Actually, it requires a lot of recess. But the major one that I think would be like the project goal, like what the project or the product is supposed to be or supposed to perform, like, if you have something that is, you know, you know, a real fire, or for one time up or, you know, these two operate quickly, something like, you know, AWS, lambda, it can be, you know, which can help you, you know, whole lot, or some kind of functions, right? But depending on these scenarios, you know, some things that I consider are how much scalable the products needs to be, what kind of performance it needs, like, especially when you are migrating from like, existing service to another service. And I, you know, I think performance is a huge factor. And also, you know, we need to make sure that, you know, there is compatibility to be, you know, integrated across the existing system based on toggle switches. You know, we need to look into the tech stacks, like, what kind of tools work with the language that we are working on or we are working with. Like, you know, how AWS lambda would be the best fitted for nauseous or Python kind of language, and not Java, even though it has support for it. But Java, you know, sometimes it's not a good option for lambdas, of course, and you know, apart from that, you know, I kind of like prioritize tools that offer I would stay, say, strong automation, you know, compatibilities, and also monitoring integration for, you know, smooth migration and less manual, you know, if you know a intervention for, like, you know, a deployments and, you know, monitoring, so that there are some of the things, you know, some sometimes cost can be a factor as well, right when choosing the tool. So, yeah, you know, these are some of the things that I take into consideration, right when choosing the new product or new technologies, or to, you know, trying to create a product, or even migrating the products.

Speaker 1 12:32

Cool, yeah. I have any questions, yeah. I guess

Unknown Speaker 12:37

sort of in line with that is,

Speaker 3 12:40

like Ryan said, You like have the ability to pull in any new technologies or uphold the current stack so on and so forth. Something that we were definitely lacking is documentation. So I guess, how comfortable do you feel creating like a new system of documentation?

Speaker 2 13:01

So yeah, you know, basically, in my experience, you know, from my personal experience, I have been, you know, working with creating documentation on things that I have worked on, like, for an instance that I gave you earlier, for the maintenance API, the, you know, whole architecture is documented well, right? So I have, you know, conference space that has a list of functional and non functional requirements, you know, all with all the associated links, of course, and the deployment strategies and everything is, you know, all well documented into a conference space. And basically, you know, other teams that we are trying to, you know, do similar migration would just, you know, clone that documentation and fill in their parts. So I feel like, you know, a very strong documentation can be a very good source for anybody, anyone you know, new coming in and trying to come up to, you know, speed up the system. But for, you know, system for tools, you know, technologies that is lacking documentation internally. You know, I feel like I have, you know, solid experience working on documenting what I know, because our, you know, Legacy system is not well documented, and I had to actually go through the code, you know, read what the code was doing, you know, and I look at the unit test as well to see what the different scenarios it would handle. And then try to, you know, create a small, one stop shop for me to look in for different aspect of the project. So, yeah, I feel, you know, comfortable in like, you know, you know, starting like a new documentation phase, where you have different style pages on different components, and then list out the functionalities. You know, what it is, support, supported, you know, which might be the growing dog, right? So, basically, you know, initially we want to know the whole flow, but once we get started, we can build on top of that, you know, making it a very robust recommendation,

Unknown Speaker 15:04

nice, yeah, like,

Speaker 3 15:10

sorry, my brain is just shooting off in lots of directions, okay, so, yes, going along with documentation. Because obviously that would be for people on our team, people, other people who are coming in to work with that. How comfortable do you feel or like, how would you explain some of these things to non technical people, since we are in the marketing department, and most of the people we work with think that we're wizards, and everything we do is magic, and they ask us to do something, and then it shows up on the site, and they're just like, this is incredible.

Speaker 1 15:44

Or there's also instances. So we need, like, a new service, and it costs money, we'll need to, like, explain that so they can get binds from, like, whoever needs to approve it. Like, you just need to like, hey, we need this. We need the we need help with this. Like, like, basically explaining, like, boil it down. How would you go about that to, like, a non technical stakeholder,

Unknown Speaker 16:09

yeah. I mean, totally Yeah. So

Unknown Speaker 16:13

you know, whenever

Speaker 2 16:15

there are non technical people involved in, you know, explaining something that is technical. What I kind of like to do is, you know, to make it less technical. You know, I try to put less technical jargons as as less as possible. Then also, I try to provide it with examples, right? Because you can, you know, always come up with real world examples or scenarios on explaining certain aspects. So, you know, we can, you know, basically, you know, explain projects, requirement, no goals, to choose those tools for handling the current and future needs of the system. You know, ensuring that you don't, you know, put non technical weight like, you know, for an example, if you are looking for an automation explanation, you would say something like, you know, cleaner service for the office, you know, instead of spending the time doing something, you know, manually, we say, we say things so that the system runs and monitors itself, like, how a cleaning service does it, right? So that's kind of like the way that I explain with example. And also, if there is something like, you know, if you need to take cost in account, like in picture, then I would like to explain that like it is a kind of, like, upfront investment, of course, right? Which will save money in the long run, because you basically be improving a certain aspect of it, which will reduce, like, your manual time working on it. So, yeah, I feel like, you know, some experience, you know, coming up with, you know, examples that can relate to the real world to help non technical people, you know, visualize what the ask is and what the technology that we are working on. Because I like to show them, you know, the bigger picture, and also want the investment in and are, you know, typically working towards, you know, on the long run, you know, providing the picture. So, yeah, that

Unknown Speaker 18:22

totally makes sense, yeah.

Speaker 1 18:25

Um, so for our websites. So what we do on the demand gen team? So our team is responsible for keeping make sure that the site is up, and a lot of the business of afinipay is generating leads through our forums on the website. So we use a service drop Marketo, and then we take those forms and they filter into Salesforce, which generates leads for the salespeople. So that's like, a big part of like, the sales part. Like we need, like, uptime on our websites, and there's an issue on that. It's a big problem, like the site goes down, or anything like that, or the forms break like that's a very big issue. So hypothetically, if our Marketo, like Marketo itself, is down, it's happened in the past before, and then, for instance, they don't have a estimated time until services back up. And then, if our leadership team is asking for a solution, like, what would you do in that situation? What would you suggest? Like, in the meantime, like our forms are down, we need like, something like, what would you like suggest in that situation? Okay, so,

Unknown Speaker 19:44

yeah. So

Speaker 2 19:46

hypothetically, you know, if there is a service is down and there is no, like, estimated time, maybe, you know, we would like to create a simple status space right away, like, you know, it can be, like a separate page that can provide the update about some things going on, right? With the back end team, of course, which can, you know, help manage customer experience, which is, which can, should be the first priority, right? So and so you can, you know, work on resolving the issue. It can be based off of feature flag, right? You display the space whenever that, you know, feature flag is torn on. Maybe, you know, if the form for the lead generation is down, it is, you know, really essential that we maybe, you know, create a temporary alternative, you know, it can be, you know, other kind of like Google form or type form that redirects user to that Backup page to provide their form. And then also work on like a backup system in the meantime. Then, you know, maybe also we can leverage these backup systems for, I would say, simplified landing page, and, you know, have a partial functionality available because, you know, personally, I think communication is the key here. And we keep the customer, you know, we keep involved, I would say, through some kind of channel, and they know that there is downtime, right? And everything might not be functional. People are usually okay with that, like work around for things to work on, you know, similar manner. So that's the approach, you know, that I, I would take

Speaker 1 21:29

cool so say that forms go down. We have alternative forms we just throw up on, like the website. We have a lot of forms. So to mitigate this issue in the future. How would we handle this so we don't lose any submission data? Is there any do you have any like ideas? How would you like the service goes down, but we want to, like, keep tracking like submission data again. Do you have any like suggestions for that? How would you go about that?

Speaker 2 22:05

So for that particular instance or scenarios,

Speaker 1 22:11

it's nothing that, sorry. It was nothing that would have to be done, like immediately, but like a long term backup solution, like in the meantime. So it's nothing like, oh, we need done right now, but something like, we can implement like, so we have these, like, fail safes if service goes down again,

Speaker 2 22:30

yes. So basically, you know, maybe in that scenario, we can use some kind of cloud storage for, you know. So it's being like AWS, CS three for off site backup, right? These services, you know, they would store these backup data. And we can also, you know, schedule an automatic backup of, like, the data that we have. And, you know, maybe income, incorporate some kind of storage fallback, right? If a user feels out, you know, in alternative form on the browser, we can use local source, you know, temporarily save their input. And I would also say that if this form, you know, if submission fails, you can provide them with an option to resubmit the data from the like local stores. When the system is back online, they can, you know, they can be some kind of queuing methods, right? So, like, instead of, you know, directly sending the data to your downstream system, what you can do is, you can, you know, make use of AWS, SQS and SNS pair, and they, you know, these offers, you know, deadly sequence solutions. So whenever the initial, you know, processing fails, so, you know, I will say that, or it will do certain number of retries. Then after that, it goes to, you know, queue, and these dead later queue, to be more precise about it, is something that you can reprocess in the future time so none of the data is lost.

Speaker 1 24:12

Oh, yeah, that's kind of what we have in place right now for that. We kind of have a like a proxy we send to and then it can rewrap depending on the form it could read route to like directly to our like virtual terminal or Marketo itself. So we're kind of doing something similar right now. So

Speaker 3 24:34

cool, and we only have a couple of minutes left. Do you have any questions for us?

Speaker 2 24:41

So, yeah, I do, you know, I would like to ask, like, what would you know, you say is the best thing about working at affnipay? I mean, what? How have been your experience, you know?

Unknown Speaker 24:54

Um, yeah, I can go first.

Speaker 1 24:58

What's great about at least being on the marketing team. And then we have such a small team, we get to kind of decide what we like want to work on. Basically, we get to choose our own like stack, basically, of our project. So we're migrating to this framework like it was already in place. The Grid sum view framework was already here, but we it was time to upgrade, so we looked into like, we want to do Astro like, okay, it's pretty new service, so we're gonna do that. Let's test that out. And then there's like, kind of gives you the flexibility to just kind of like, find your own path of what you want to do. I think we're in a good place now. Like Whoever joins for this senior like back end role, kind of like, can define what they want to do from now, like, as you build out the team, like, we should be getting more back end developers, like another maybe more junior person, you can kind of define roles, like anything that you're excited to work on in the future. You kind of like, decide that for yourself. Basically,

Speaker 3 26:07

yeah, that's a that's all incredibly true. My very first thought, I'm not gonna lie, was healthcare benefits. We have really nice benefits, which is nice, but it is true. We do get to, kind of like, create our own path. We get to sort of figure out what we want to specialize in, because it's a growing thing. Our team within the department, especially now, we have some like higher level people who sort of see more of the value of our team growing and US specializing, which is cool.

Speaker 1 26:42

Yeah, we just, we got another round of investment, so we have a lot of senior people joining, and they're want to invest in this team. So we kind of, yeah, kind of decide your path for the future. What you want to do is, like, one of the great things also, like, the benefits are, they're nice, but good compared to, like my last position,

Speaker 2 27:04

yeah. And on top of that, I think we got paid insurance as well, right? So, yeah, yeah, yeah,

Speaker 3 27:14

we got the pet insurance. We they have, like, donation matching, which is cool, so it's like $1,000 a year. And you could as long as it's like a 401 K charity thing, like, they'll match it to, like any of them, which is nice. There's a $1,500 like development, like personal development thing for, like, conferences or workshops or books or just like subscriptions and things. Basically, it's like event pay likes to have all these like, extra benefits that are just nice.

Speaker 1 27:43

And I would say we're at least our team. We're really flexible with, like, time you need to take off, oh, yeah, like, vacation. Like, I yeah, just anything you need to take off. You're like, Hey, I gotta be out for this. Or just, like, I just need a break, or I want to travel. I went to travel in Asia over Christmas. So it's just, like, took two weeks off. Like, it's, it's unlimited too. So up to, like, they say, like, a month, but I've, you know, a few days here there doesn't matter. So our

Speaker 3 28:23

general team rule is, like, we want people that we trust to get the job done. And so like, that's how I feel about everybody on our team, is that I'm like, I don't care how many days, like, Ryan is taking off, because I know he's gonna get all this work done,

Speaker 1 28:39

yeah, as long as you, like, schedule things ahead, like, Hey, I'm out here. Like, have coverage. Like, it's pretty likely and, like, that's one of the great things about finipay. Like, yeah, you don't have to worry about, like, Oh, I'm using all my vacation days. Like, it's kind of, it's unlimited. Like, so, yeah, that's one other good thing. Yeah,

Unknown Speaker 29:03

any other questions?

Speaker 2 29:05

Thanks so much. Yeah, no, nothing that in particular right now I'm good for now. I just like to say that, you know, I might be a little bit sad later if I joined that in there, because I, you know, you guys won't be coming to the office regularly, but hope to see you in the closing of the quarter. Yeah, of course, by the end of the quarter, And it was, it is very nice talking to you guys, thanks so much for your time.

Unknown Speaker 29:29

Yeah. Have a good one.

Unknown Speaker 29:38

Thank you, third letter.

Prajwol Tiwari-Infosys-srv-Final

Unknown Speaker 00:00

Projects and stuff,

Speaker 1 00:02

sure. So, yeah, my name is, you know, prajal Tiwari. And talking about my professional experience, excuse me, I have been working as a full stack Java developer, and it has been a little over five years now. You know, working primarily on back in Java technologies, you know, utilizing frameworks like spring Hibernate, you know, rest wave, service API development and, yeah, and I do have pretty good experience, you know, working on microservice architecture build using Spring Boot framework. And, you know, I have good experience working with AWS cloud. And, you know, I have been working with services, you know, including AWS, EC two, ECS, SQS, yes, SNS, lambdas and so on. Sorry. You know, I do have experience working with Angular on the front end development, and I have worked with JavaScript and TypeScript. I have also, you know, good experience working with J unit testing. And you know, currently I am on a scrum team. And you know, I have been primarily involved in back end development. And you know, currently the team, it is focused on, you know, function generation for request manipulation on a microservice architecture. So, yeah. So I had been, you know, involved in writing these, you know, functions to manipulate the request to be sent to downstream APIs. And we have a team of five developers here. And, you know, we have product owners and a scrum master. And, like I mentioned, you know, on a daily basis, I have been involved mostly working with these function factories in the backend side and yeah, and I also help team with the code reviews, and I also provide, you know, production support, and I do these duties.

Speaker 2 01:58

Okay, that's nice introduction. So, you know, this position is for Infosys, and we do a lot of, I guess Infosys name is good enough for introduction. We have lot of several, several clients serving across different needs. So before going into the details of this, I would like to have some more. So in your profile, you have mentioned that you have completed your masters.

Unknown Speaker 02:30

When did you complete your masters and

Speaker 2 02:35

and when it comes to the bachelor's degree, when have you actually completed it? So yeah,

Speaker 1 02:39

I you know, completed masters this year in the month of May. And, you know, I did it from University of South Dakota and coming to bachelors, I completed it in, you know, year of 2021,

Speaker 2 02:55

and you said you got five years experience, Yes, where did you have all these five? I see hardly few years,

Speaker 1 03:03

actually, you know, I started working when I was studying back in 2019, and, you know, so in Nepal, like there is no any restriction, like you cannot work while you are in basalt. So I started working while I was in bachelors. And my Bachelor's classes, you know, it used to run in the morning shift, and I used to work in that daytime as a full time employee. Fair enough.

Unknown Speaker 03:40

So let me start some basic questions on

Speaker 2 03:45

have you, are you familiar with the oops concept in Java?

Speaker 1 03:51

Yeah, yes. I have you know, I am familiar with the option chain in Java.

Speaker 2 03:56

Yeah. Can you tell me, what are the basic things of that? Sure.

Speaker 1 04:00

So, yeah, there are, you know, four, four primary concept in Java. So basically, you know, you have encapsulation, which is kind of like, you know, you know, data hiring, right? So what I mean is that, you know, you hide the fields of class, and class has a complete control over the field. And then there is inheritance, which is about like the reusability of the code, so that, you know, child class can inherit the behavior and field exposed to the parent class. And there is also the concept of abstraction, which is about, you know, hiring the complexity from the outside. So, you know, we can achieve this with the help of abstract method and abstract classes as well as interfaces. And there comes polymorphism, right, which can be, you know, achieved with the help of method over loading, which is done within the same class, and then also the method and overriding, which is done on the child class by, you know, inheriting and giving a different implementation. So yeah, these are the four primary oops concept in Java.

Speaker 2 05:10

So can you tell me what exactly the difference between interface and an abstract class? Sure.

Speaker 1 05:18

So yeah, describing the you know difference between the interface and abstract class. I would say that you know, you know interface. I would describe interface as a blueprint, right? So basically, it defines a contract that class can implement, and it is used in a you know, specific way, like what a class must do, but, you know, it doesn't tell how you are supposed to do it, right? So interfaces, they are, you know, ideal for basically defining your capabilities. That is, you know, shared across multiple classes. And when it comes to abstract class, it is kind of like a base class, you know that cannot be initiated, but you know it can have, like abstract methods, you know, concrete methods, which is used to share, like common structure or behavior.

Unknown Speaker 06:17

When it comes to the actual

Speaker 2 06:20

syntax and how do you define that in the interface and in the abstract class?

Speaker 1 06:26

So, yeah, you know, basically an interface is, you know, it is defined using the interface keyword, right? So you can do a public interface. Let's say we are creating vehicle one, then public interface vehicle. And then you can, when it comes to abstract, you know, defining the abstract methods, you know, starting on Java eight, you can, you know, also do default methods in interfaces for backward compatibility. And also, there is this abstract cast, which is also done using the abstract keyword. So, you know, you can do something like public abstract class vehicle, and then you can just, you know, define like the abstract methods by using the abstract keyword itself.

Speaker 2 07:16

So in the abstract class, can I have some methods defined with all the code in it.

Speaker 1 07:24

So in the abstract class, yes, we can have, you know, concrete implementation in abstract class as well,

Unknown Speaker 07:32

complete implementation, concrete, I mean, yeah, okay. Okay.

Speaker 2 07:43

Can I say something about multiple inheritance in Java? Yeah,

Speaker 1 07:47

so, talking about, you know, multiple inheritance, basically, multiple inheritance in Java is not like it is not supported, but I think you know, it is in classes. It is not supported, right? Basically, Java, it doesn't allow a class to be inherited from more than one class, which mean you cannot do something like, let's say, a class vehicle extend to another class, right? Because there is problem called this diamond problem, where classes could, you know, inherited, you know, conflicting implementation. But in Java, you can do inheritance using interfaces. And what I mean by that is, you know, Java class, it allows you to, you know, implement multiple interfaces, and whenever you do this, you can override the method on your concrete implementation.

Speaker 2 08:44

So you're saying, because of the diamond problem, we can't implement it. But in general, if I want to extend the features of a class into a subclass, can still do that. So

Speaker 1 09:00

you can extend one class, but you know you can implement multiple interfaces in Java.

Speaker 2 09:10

Can you can come back that again? You can extend one class, but multiple interfaces, okay, yes, all right, let me ask you some questions on the string and string builder. Can you tell me what exactly the difference?

Speaker 1 09:27

So, yeah, highlighting on the you know difference between string and string builder. So basically, I would say that difference is that you know string it is immutable, right? So once a string object is created, it cannot be changed. And you know, whenever you try to modify that string, such as, you know, try to concatenation or replacement. It basically, you know, forms a new string object. But when it comes to string builder, they are they are mutable, which means that, you know, the object can be modified after creation. So, you know, string being immutable, it is also thread safe. And what I mean by that is the, you know, they can work properly with multiple threads, but string builder, if they are not thread safe, because since they are mutable. So yeah, and there is also a concept of string buffer, which is mutable and thread safe, because it provides, you know, internal synchronization.

Speaker 2 10:29

So let's say I have a string, string a equal to XYZ, and then I say string b equal to a, what will happen to

Unknown Speaker 10:44

string a, if I'm overriding

Speaker 2 10:48

string a, what will happen to the value of that for the object?

Speaker 1 10:53

Okay, so string a is XYZ, and string B is A, right? Yes. So, so, you know, you are, you know, setting I would say string B to string a. So, you know, string a is created with the value of x, y, z, and then basically what in Java is, you know, string literals are stored in a, you know, common pool. So if another string with the same value already exists in that pool or reference, you know that existing object. It is used instead of creating you know new one right away. So whenever you make a reference of string B equals A, then you know that that will make the reference to the same object. So you know, string a and b will now be put into the same memory location where the value x, y, z is, I mean, stored. Okay.

Speaker 2 11:55

So tell me something about the difference between what exactly this try and catch and try the throw and throws differences between these two. Where do we use that? Sure.

Speaker 1 12:12

So, yeah, basically, you know, try cats. It is used for exceptional handling, right? So, you know, it's a block where you handle exceptions, you know. So let's say you have a code that might throw an exception, and you want to make sure that you properly handle that, and for that particular reason, you know. Then you start a code block with the, you know, try and, you know, curly braces, and then you write the code that might, you know, through the exceptions, right? And then after that, you know, you try to catch that block where you would, you know, specify the type of exception, and you'd have a code block to, you know, handle that exception. So basically, you know, these used for exceptional handling, and you know you can have multiple try blocks to handle a different exception separately, but it depends on the hierarchy, right? So you would have the specific exception on top, followed by one more book road or exception now, or through statement it is, you know, used explicitly to, you know, through a particular exception, I would say, you know, whenever there is an error in, I would say, whenever there is an error in processing, and you want to throw, like an, let's say, an illegal argument exception, because the the argument that was passed was incorrect, and you use or throw keyword to throw that exception, and you know, then a throw keyboard it is, you know, using method declaration to indicate that the you know it is using method indicator, indication to, you know, detect that throws this exception and it it needs to be handled by method calling this exception.

Speaker 2 14:05

So from the declaration perspective, can you tell me, like, where exactly you define that throw versus throws?

Speaker 1 14:10

Okay? So, yeah, defining through versus, you know, throws, right? So I would say that you know through is defined within a code block, right? So within the code, code block you would write, you know what you want to throw, and when it comes to throws, it is defined in the methods signature. When

Speaker 2 14:38

you mentioned method signature, just one example. How do you define that? I don't I'm not asking you to write the code and show me, but just tell me some example,

Speaker 1 14:49

like a code so code wise, I would say, let's say, avoid do something. Parenthesis throws, you know, and it's a runtime exception, you know, and then, and then, curly presses. Do you

Speaker 2 15:07

okay? Do you define a hierarchy there, when you're displaying the throws? Can you define more than one exception?

Speaker 1 15:13

So, yeah, you know, you can define multiple exception there, like, you know, if we want, like, maybe illegal argument, exception, you know, input, output, exception, you can define those. You know, it is comma separated list. Okay,

Unknown Speaker 15:36

that's fine. One

Speaker 2 15:38

more question on the Java stuff, so you have something called finally.

Unknown Speaker 15:44

When do we use that

Unknown Speaker 15:47

final, right? So, yeah, basically,

Speaker 1 15:51

not final, finally, yeah, yeah. So finally block, you know, it is, you know, it is used with, you know, try cast block to execute a code and that, you know, the code that must run regardless of the exception being thrown or handled, right? So let's say, you know, whenever we have a try cast block, if you that you want to execute some code block that needs to be, you know, executed regardless, you'll use a finally block. And it is usually used for like, you know, clean up activities, generally,

Speaker 2 16:28

okay. Are there a case? Is there a case where the finally block will not execute at all?

Speaker 1 16:36

So, so, talking about the case, you know where. So I think you know if in particular, like, let me think so, let's say less, let me think about it. So I think you know, whenever we do a force exit, or, let's say, you know, we forcefully dominate our Java virtual machine, the finally block, then in that case, it will not execute. And you know, also, if there is, you know, interruption by the JBM, the finally block in that case will not execute.

Speaker 2 17:19

How do you define that exit in the code.

Speaker 1 17:24

So, yeah, so defining, you know, it's, it's a system dot exit.

Unknown Speaker 17:34

So on the collections,

Speaker 2 17:37

you have tree sort and

Unknown Speaker 17:47

let me come back on that.

Speaker 2 17:50

So let's take a simple list. How do you sort that list? Array list? Let's say array list.

Speaker 1 18:01

So yeah, I mean, you know, sorting and array list, or, let's say, basically just a list, you can basically used collection dot sort method.

Speaker 2 18:13

Without that method, will you be able to sort it? Are there any other Java properties, any other collections, which from the complexity. Can you tell me, like, you know about object or notation and all the stuff from that perspective? Can you tell me, like, what is the complexity of that?

Unknown Speaker 18:37

If you don't mind, can you repeat that once

Speaker 2 18:39

again? So, so you have sorting techniques right in all the same page. You can use Java dot, collection,

Unknown Speaker 18:47

what is the complexity of them? Oh,

Speaker 1 18:49

so collection, dot, sort is, you know,

Unknown Speaker 18:53

big O of, you know, N, log n,

Speaker 2 18:58

and what exactly the sort method used behind the scenes when you use the collections of sort,

Speaker 1 19:04

so, you know, so, yeah, behind the scene, you know, collection dot sort internally, you know it, I think it's, it's Hybrid, you know, between more shirt and, you know, more. So there is it. Is hybrid between, you know, more short and insertion sort, yeah. All right,

Unknown Speaker 19:33

that's fine. So

Speaker 2 19:37

let me come back on some micro services related things. So Srijit, I will hand over to you. Well just I will ask with just one simple question, so what exactly the Spring Boot is? Why do we use it, and when it compared to spring what is the difference? Sure. Okay,

Speaker 1 19:59

so, yeah, basically, you know, talking about a Spring Boot, it is a very popular framework, right, which is built on another top, you know, on top of Spring framework, basically, you know, it extends the functionalities that Spring framework already provides. And rather, it simplifies, you know, and these generation of spring related projects. And it is some of the you know, features that Spring Boot provides on top of spring and it provides, you know, something called Auto Configuration. Basically what it does is, you know, during the bootstrapping, right, it scans your application and it configures the Spring application and the dependencies that you know you have added, and this particular, you know, feature, it reduces the need of any Java configuration or any XML configuration that you have to do when you are working with Spring application, right? And, you know, Spring Boot, it also provides an embedded Tomcat server, right? Which it means? What it means is that you do not have to rely on external solver to deploy your application.

Speaker 2 21:10

Prajna, that's fine. Thank you. Do you want to take over?

Unknown Speaker 21:17

Yeah, yeah. So,

Unknown Speaker 21:21

yeah, hi, hi, prajna. So you, you are

Unknown Speaker 21:24

currently in us. Yes, I'm currently in us.

Unknown Speaker 21:26

How long you been in us?

Unknown Speaker 21:28

It has been a little over one and a half years.

Speaker 3 21:31

I'm happy. So in us, which all projects you worked in? Sorry, can you be there? Which

Unknown Speaker 21:35

one projects you worked in us? So,

Speaker 1 21:37

you know, I have been working currently with Carta, and that's the, you know, only company that I have, which is the one, Carta, Carta,

Unknown Speaker 21:46

what is the technology working in Carta? So

Speaker 1 21:48

I had been using, you know, so Java. And the project that I am working on is, you know, portfolio management dashboard

Speaker 3 21:59

with Carter. Know which technology I'm asking about which technology.

Unknown Speaker 22:02

So, you know, it is using

Unknown Speaker 22:05

Spring Boot framework, particularly Spring Boot three

Speaker 3 22:08

batch, in spring batch. Have you worked in spring batch?

Speaker 1 22:14

So, yeah, you know, we have had some kind of batch processing done when I was working for, you know, hamru patho, back in Nepal, back in home country.

Speaker 3 22:26

So, how can you see? How do you, how do you implement spring batch to to to pull data from database and update in another, some other location. Can you explain the steps? Okay,

Speaker 1 22:40

so, yeah, you know, basically, you know, you can, you know, create a Spring Boot application, and you have this, you know, spring based dependencies added, you know, right. So, basically, that would be the first step, and you can define the domain model that you want to use. And once you have it, then you basically, you know, you would be, basically be defining batch processing job, where you would enable the, you know, batch processing and configuration. And through the annotation, I think it's just the you know, it enables batch processing annotation. And once you have that you would create the beans for jobs, right? Like you can create different beans for different steps, like step one, step two, and so on. And, you know, you can create beans for readers and also, you know, processors and writers. So, yeah, and those, you know, those are the three primary bins that you basically need. Then once you have that, you can, you know, run the batch job by using a job launcher provided by the Spring framework itself. And I would say that, you know, these job Launcher will basically taking the job being and then the parameter of these job by defining all these jobs and steps, and the reader, processor and writer and you will be able to read from the database, you know, process that data. And

Speaker 3 24:13

so I got that. So how do you, what do you in the YAML file? How do you, how do you make sure that say, how do you, where do you give the count of the number of queries to be hit? Like the configuration, for example, I say I need 100, 100 queries or something at a time. How do you, how do you frame that? Where do you defend the configuration?

Speaker 1 24:34

Okay, so you know, if you want to, like, define, or, you know, let's say, if you want to define a, you know, I would say a number of batch query count, batch dot query dot count. So where do you design? So, yeah, we can define these parameter in application, dot properties file, and then we can load this configuration,

Speaker 3 25:02

alright, yeah, so, do you know a concept of deadlock in threads? Yes,

Unknown Speaker 25:08

can I explain the concept of deadlock in threads? So,

Speaker 1 25:10

so, yeah, you know, basically defining deadlock, you know, it is nothing. But you know, just when two or more threads are blocked forever, you know, right? So I would say, you know, each of these threads are, you know, waiting for resources, and they just keep on being blocked forever. So whenever you know there is at least one resource is inhaled, you know, in a non shareable mode, which means that you know, only one thread can have access to that resource at a time. Then if another thread is also, you know, requesting that resource, then it will have to wait until that object is released, right? So, but if a thread is, you know, holding at least that resource is waiting to acquire, you know, additional resources held by another threat. And during that waiting period, you know it doesn't release the resource right away, right? And because it already has that resource, and then you know resource may not be, you know, forcefully taken away from any you know threats, right? Because you know, you know resources allocated and can only be released from a thread that is holding it. And then, you know, if there is also circular chain of two or more threads where each thread also resource, you

Speaker 3 26:32

know, definitely, that's fine. Prajel, I think one last question I want to ask is, like in, have you worked in Java lambda? Yeah, yeah. I

Speaker 1 26:45

have worked with, you know, Lambda expressions in Java. That

Speaker 3 26:49

was just wondering. So you have worked in all these you just have one and a half years experience, right? So, just just wondering, how did you manage to work? Is it? Is it? Can you explain to me, how did you manage to work work on all these things in one and a half years? Like, was this all involved in that? Or you studied yourself separately, outside of projects? So,

Speaker 1 27:13

you know, I have one and a half years of experience in US itself, but like, in total, I have, you know, five years of experience. Oh, okay, you

Unknown Speaker 27:22

mentioned Okay, okay, correct, okay,

Unknown Speaker 27:23

yeah, all right.

Speaker 3 27:25

I think that's all from my side. Said, Yep.

Speaker 2 27:28

Thank you. Thanks. Jit. Subilesh, you want to ask? You have any couple of questions? Anything

Unknown Speaker 27:36

you want to share? Yeah,

Unknown Speaker 27:39

did you work on stream APIs, on on Java, future,

Speaker 1 27:43

yes, yeah, I do have, you know, worked with stream APIs, you know?

Speaker 4 27:51

Yeah, could you please write a short code to print the email numbers in the list by using streams and filters. Streams and filters, okay, yeah,

Unknown Speaker 28:00

my streams, yeah, visually,

Speaker 4 28:03

yeah. You have ArrayList. You have a list which contains 10 numbers. Assume, okay,

Speaker 1 28:07

so like one quick question, do you want me to you know, share

Unknown Speaker 28:11

my screen? Yes, yes.

Unknown Speaker 28:33

That should be Fine.

Unknown Speaker 28:36

Just create A List. I

Speaker 2 29:40

So, yeah, perfect, perfect. Thank you. My pleasure. That should be good. I think we can wrap it up. If anybody has any questions, there's a time we can ask Manoj, before wrapping up, you got any questions for us? Sorry, sorry, I totally took a wrong name,

Unknown Speaker 30:14

Manoj Tiwari, somebody, somebody

Speaker 1 30:19

was sorry. So, yeah, you know, like you mentioned, I do have, you know, few questions, if you don't mind, like, you know about the next step, and also, if you have any like information on the project,

Speaker 2 30:34

I won't be able to disclose all the client details and stuff, but we are considering candidates who can implement microservices. This is for a retail client where you have to write several microservices for several e commerce, just e commerce, just think about whatever the concepts will be there so you got to write down all those cases

Unknown Speaker 31:06

just one

Speaker 2 31:08

second and rest of the questions like, whatever, what about the next steps and stuff? HR team will reach out to you, and they will discuss with you, all right, that's all I got. Anything else? So

Speaker 1 31:24

yeah, I think that should be thank you so much.

Unknown Speaker 31:30

Thanks. Thanks. Timothy,

Speaker 1 31:31

thank you so much for your time. Bye, take care. Bye. Good day. You.

Pramod Wagle- Infosys- SRV-Final-Offered

Speaker 1 00:00

Beers. And, you know, I've been working primarily on Java based applications, so utilizing frameworks like Spring, Hibernate and RESTful web services in order to develop APIs. And, you know, pretty good experience. I've also pretty good experience working on Microsoft basis architecture, built using springboard. So these applications that I have been working, you know, are deployed to AWS cloud. And also I have experience working on AWS lambdas, s3, buckets, rds, ECS, tools and SQS. So being a full stack Java developer, I also have tons of experience on the front end side, so I have experience working on Angular, and is also, I've also worked on Java Script and TypeScript on the testing side. So I have used, you know, JUnit and mokito, and currently, right now, I'm on a scrum team, and the team consists of five developers, and we have pro product owners and Scrum Masters, and, like, on a daily basis, you know, I'm mostly working back on Java code, you know, helping the team with code reviews, and sometimes also, you know, products and support duties. Good. You

Unknown Speaker 01:28

have something I think.

Unknown Speaker 01:34

No, I don't have any question to ask about this.

Unknown Speaker 01:39

Well, this is start from some

Speaker 2 01:43

Java. You work a lot of Java. I saw the resume, right? Yes. Can you tell me something about, oh, you know what? Like before starting this one, there is a procedure, actually. Can you show your ID first? Yeah, of course. Yeah. This is like, can you

Unknown Speaker 02:03

see my ID?

Unknown Speaker 02:07

Can you zoom there? Yeah, of course.

Speaker 2 02:10

Yeah, perfect, yeah, perfect. Got it. Thank you. Yes, no problem. That should be the first step.

Unknown Speaker 02:17

My technical,

Speaker 2 02:21

like, connecting, yeah, what I was asking is, like,

Unknown Speaker 02:28

can you explain something about, like,

Speaker 2 02:33

ops, concept, polymorphism, you know what, whatever you used, you applied in your previous projects, okay, yeah, how can you recall that, if you worked anything related to that?

Speaker 1 02:45

Okay, yeah, so, got it. So, you know, basically I would say polymorphism is a very standard concept, right? So, you know, polymorphism is, you know, basically it is method overloading and method overriding, right? So what it is, you know, you have the certain methods that can be overlaid, it overloaded within the same class. So, right? So you can have one single method that you may you saw with some name, and there are other methods of the same name, so sometimes it may contain different kinds of parameters or even different return types, right? So that's what method overloading is. And there is other concept which is method overriding, right? So whenever you have a parent class that has a certain kind of method, and then you have a child class, so you want to give a different implementation. So it's just the overriding the existing method and giving a separate implementation, right? So these are two kinds of polymorphism in Java. Like, for example, you may have a processor, you know, where you have an interface called processor, and you want to, you know, provide different implementation based on the different type. But there is an abstract class that contains the, you know, same calculator processor, and you have the you know, abstract effort behavior, you know, you know parent abstract class. Or you can, you know, have a dedicated behavior on your child classes. So that's what you know like. So method over training is, I mean polymorphism is,

Speaker 2 04:44

so you said like method overriding and Metro overloading, right? Which one is the runtime, which one is the compile time one, and which one is the runtime one?

Speaker 1 04:54

Oh, runtime polymorphism is a method overloading, right? So basically, what happens is when a subclass provides a specific implementation of the method,

Unknown Speaker 05:10

so it happens in the runtime.

Unknown Speaker 05:13

So, okay, okay, so you want to go

Speaker 3 05:24

um, sure. I guess, um, I guess with with respect to Java eight, could you explain the the difference between functional interfaces and,

Unknown Speaker 05:42

man, I have this written down, sorry, do

Unknown Speaker 05:44

you explain the difference using

Speaker 3 05:47

functional interfaces and lambda expressions?

Speaker 1 05:50

Yeah, of course, you know functional interfaces, right? So basically, your functional interfaces are a special kind of interfaces, right? So it is just an interface that contains, you know, exactly one abstract method, right? So it may have, you know, multiple default or static methods. But the thing here is, it is, it only has one methods that is unimplemented, right? So the main purpose of functional interface is that it is used as a target for your lambda expressions. And now you know lambda expression is nothing but just a simple way. It's, I would say, a concise way of implementing, implementing the abstract method of the function interface. So it basically, you know, allows you to treat the functionality as a method argument or a return type.

Speaker 2 07:02

Okay, next. Okay, so still there in the Java, I want to understand little bit about collection. Did you work in collection? Something? Yeah, of course. How do you understand the collection, and what are the like major components of collection you worked in your previous project.

Speaker 1 07:21

Okay, got it, yeah, so, you know, I have worked with list sales map. So basically, you know, list, it is an ordered collection of objects. So basically, you can, you know, duplicate, you can have the proper order, and you can, you know, access elements by index on array list, right? And there is a set, so which is, and you know, ordered, but unique collection of object. So said, basically, you know, does not allow duplicate and then order is not guaranteed unless, you know, you use specific implementation called, you know, preset. So there is a map, right? So which is like key value pair, where data is stored in a key value and where the value can be, you know, retrieved using a key. So, you know, those are the three types of, you know, collections that I have worked with. And on the regular basis, I would say I also worked with, you know, steam API for, like, processing those kind of collections.

Speaker 2 08:38

So what's the basic difference between, you know, like, asset and preset has

Speaker 1 08:44

trade, okay, yeah, asset and preset, right, yeah. So the basic, you know, the basic difference between a hash set and tree set is that you both, you know, you both. These are specific implementation of the set that you know, maintain some kind of ordering, right? So, or tree set maintains elements in a sorted order, but has it, you know, it's just a random order, right? So, whenever you know, you enter elements in a has said, so it will be stored in just a random order. So then there is a tree set right, so which are automatically arranged in the ascending order. So it also, you know, may also be arranged in regards to the comparator that you know that you provide, okay? Okay,

Speaker 3 09:45

I guess, with regards to tree sets, could you explain, like, how that, like, how that data set is implemented in the background, just, I guess more specifically, why is it called a tree set? You know? Oh, okay,

Unknown Speaker 10:00

yeah. So,

Speaker 1 10:04

so basically, you know, the trees, the tree set is uses, like tree data structure, so, which is a self balancing binary research, right? So there is a, you know, this is a column tree, because it organizes data in a hierarchy, like your tree structure, where, you know, each node has parents and up to, you know, two side. So that's why the diagram is like a tree. All right, thank you.

Speaker 3 10:51

Should we continue asking questions about Java? I had a couple of questions I wanted to ask about a spring or

Unknown Speaker 10:57

Yeah, yeah, please, yeah.

Unknown Speaker 11:02

I guess, well,

Speaker 3 11:04

I don't know what order to ask stuff about, because I guess I have a variety of questions about, like, a bunch of different

Unknown Speaker 11:12

topics. Feel free like anything.

Speaker 3 11:18

This isn't like a requirement or anything. This is just like a thing that my team works with a lot. Do you have any experience with Eclipse? Vertex?

Unknown Speaker 11:27

Yes, you know, so

Speaker 1 11:32

I look I have only, like, used extensively, supreme boot for my, you know, on my professional side. But Eclipse, I have only, you know, use like I have done personal projects. So, yeah, yeah,

Speaker 3 11:52

yeah, it's the team that I'm working on uses vertex a lot, but there's a lot of teams that use Spring Boot, a couple of teams that use vertex. I'm just gaging to see whether people have experience with vertex. With regards to Spring Boot, I guess. Can you just give me, like, a high level overview of, like, what Spring Boot is and why would want to use it in an application, yeah,

Speaker 1 12:21

of course, being a full stack Java developer, right? So I have been using Spring Boot a lot. So Spring Boot, I would say, is a label higher than Spring framework, right? So basically, it is an extension which simplifies your development. So Spring Boot, you know, comes with all the features of Spring framework, but it mainly helps in creating standalone applications by, you know, providing different kinds of starter dependencies. So if you include a Spring Boot starter wave, Spring Boot starter data, spring, boot, starter, security and like so on. So these functionalities that you may want to build different kinds of risk full web services so it will come, you know, out of the box, and all you need to do is simply just make changes in properties file and after that, these dependencies that you need would be provided to you through various, you know, something called auto configuration. So basically, you know what happens is that it will scan through the packages and add those dependencies as needed. So it is very efficient because it promotes, like, you know, convention over configuration approach, and it also, you know, provides embedded server, making you you don't have to use external server. So this build in Tomcat or any other kind of servers for your applications to deploy. So, and, you know, it has a good support for microservice development and so, you know, yeah, so these are the reasons you know why I would, you know anyone, anybody wants to use Springwood. So this

Speaker 2 14:20

expression, like, by the way, which, which is the embedded server in Springwood,

Unknown Speaker 14:25

embedded server, it's Tomcat.

Speaker 1 14:32

But you can also, you know, use JD, sorry. JD, do.

Speaker 3 14:42

I thank you. Um, there was something else I wanted to ask about spring. Oh, um, wanted to ask, could you just give, like, a high level, a high level overview of some of the, I guess, annotations that you would be using in spring, in particular ones that you would be like, editing classes with stuff like that. Like that. Just give, I guess, an explanation of, like, how those differ, what their purpose is, stuff like that.

Speaker 1 15:09

Okay, of course. So annotations, yeah. So you know, in spring, you work with a lot of annotations, right? So that help you give the metadata for you know what a class is. Now your basic core annotations include like include like configuration, so marking a class as a configuration class with the bin definition, and then there is a bin annotation. So which is far method that you know returns or type of a beam. So now for specific kind of classes that you may like use controller or even service you know so data access layers, you have dedicated controllers, like rest controllers, or REST API and or control annotations for any, you know, controller class. Then you have service annotations for service layer, you know, so a business layer, and there is, you know, there is repository layer, so for any repository here, so xx in your database, there is a component annotations, so marking it as a spring managed class. And then there are, you know, other annotations for different kinds of dependency injections, like auto wired and qualifiers. So, you know, I would say these are basic annotations that will work with a regular basis using Spring Boot.

Speaker 3 16:51

Alright, thank you. Did you have any questions about spring around?

Speaker 2 16:57

Ah, yeah, just adding to the same thing like annotations. Do you use quite frequently, like, like, get mapping, post mapping, right? Yeah,

Unknown Speaker 17:10

of course.

Speaker 2 17:11

So can you just like, just like, just give me the scenario, like, what type of scenario you use, like gate post delete, very common one, you know. Did you use that before? Of course, like it, right? Like in cooperations. And just give me the simple example,

Speaker 1 17:35

yeah, of course. So basically, gate post delete, path. So let's say, I'll give you one good example. Let's say just one. Yeah. Okay, yeah. So basically, if you want to, you know, retrieve all the employees, right? So you want to get employee, employees that is stored on your database. So you would have API for getting those using a gate mapping, right? So if you just to get employees, so which will, you know, trigger the gate mapping, annotate method, and after that, it will, you know, retrieve all the employees, right? So it is like, you know, getting the value from your data. So now there is, you know, post mapping annotation. So post means you create, so basically, whenever you have the method that is annotated with post mapping, so it basically means that you want to create a new employee, right? So whenever, you know, make a request with a post, HTTP word on an employee resource, then you will be, you know, providing data to create an employee. So now you know, there is a method called put so it triggers a method that is annotated with put mapping annotations. And now put is for, you know, updating and existing resources, or, you know, creating new resources. Or if you want to update the data of an existing employees, then you can use, you know, put mapping controllers URI, and there is a delete mapping too. So basically, delete mapping is for, like, deleting employee. So, so those are, you know, I would say four primary ones, like paths. So it is mainly for partial update. And also, you know, officers for like, haters and stuff, yeah, pretty good.

Unknown Speaker 19:49

So you mentioned REST API, right? Like,

Unknown Speaker 19:54

can you hear me clearly? Yeah, okay,

Speaker 2 19:58

so why REST API is stateless.

Speaker 1 20:04

RESTful API stateless? Yeah, so one of the main reasons you know REST API is stateless because it doesn't hold any state, right? So basically, each request that you make using these APIs are independent, so, which means that whenever I make that request, when I whenever I make the request, all the information that are needed to, you know, successfully complete the request that is contained within the, you know, whole request, right? So there is a, you know, like a server side, so it doesn't need to, you know, store any kind of session, and the server, you know, does not keep track of the client states or anything like that. So that's why, you know, it's like a stateless design. Let me give you a one example. So if you have that, you know, get employee ID call, right? So you do, so you would, you know, slash employee ID or a gate call, then you would have headers to provide what kind of data that you have. So you have authorization header that is included and so on, right? So if there are, you know, other cookies needed, so it will also be included. So everything you know, needed to get an employee, so will be, you know, present on that request.

Unknown Speaker 21:34

Did he use a postman to test the APIs?

Unknown Speaker 21:38

Yes, of course.

Speaker 2 21:39

Okay, just, just give you the example. Like, how did you, like, write this, whatever you said already, like, that is, everything applies there, right? Yeah, so that's what I want to figure it out. Like, how stateless is this easy to I mean, I mean, just, I'm making simplify on, you know, like,

Unknown Speaker 21:59

okay, greatest apply from postman,

Speaker 1 22:02

yeah, okay. So, so in postman you can select HTTP verb, right? So you can select guest gate, POST, PUT, delete. So first thing is that you simply choose the request verbs. And once you make the selections, then you will put the put it into the URL, right? So if you are, you know, testing it locally, then you'll use localhost and then the path. Otherwise, if you are using servers, then you would use the domain, and you know path for your resource. And once you know, you are done with that. So there is a section for that request parameters, and there are you know, query parameters that you you know you may want to include in it into your request. But if you don't have any, then you know you would have to go with the header section where you know you would be adding like authorization header SF and content type headers and so on. So then there is a, you know, also a body side where you know you can simply pass any kind of request body for like post operations. So once you know all of that is done, though, the whole request is ready, so you can send the request to your server and it responds back. Perfect.

Unknown Speaker 23:27

Thank you. Yes.

Speaker 3 23:32

So I wanted to ask some questions with regards to tests and whatnot. I guess just in a general sense, can you explain the difference between unit testing, regression testing and integration testing?

Speaker 1 23:46

Yeah, of course. So, so basically, you know, your unit testing, I would say, is a testing, basically of the unit right? So basically, you are testing the individual components or even units, of course, in isolation, right? So you do not want to depend upon external source for this kind of unit testing. So as you are focusing on the single unit. So which may you know, which may be a method or even a class, that you validate that each unit, you know, performs as expected. So you would be, you know, testing that method against different data set, like different cases and so on. As you know, it is a part of a development cycle, right? So whenever you know I'm working on any change, so I add unit test to back to that change. So, and now you know there is integration test, so which is a, you know, testing those units combined as a part of applications, right? So, basically, you in integration test verify that, you know, all the pieces work together as expected. And you know, basically it involves multiple methods and units or even modules, even to you know, check their interactions and how the data flows from one to another. So for unit tests, you know you can use tools, various tools such as JUnit and mokito. But for the integration test, you can use you to Spring Boot test, or even like a post pan for API testing. And then there is a regression testing, right? So regression testing is a full of taste. So which, you know, test the existing functionality to ensure that any new code, any new changes that you add, does not, you know, break or adversely affects the existing code, right? So basically, it covers the entire applications and also some specific parts that you might be affected by the change. So you know, it covers you can use a Selenium for regressor testing, and usually these tiers are done as a part of your pipelines. So yes, these are, you know, different kinds of testing. And I would say I have pretty good experience working on, you know, unit test, integration test for the regression testing. So, yeah,

Speaker 3 26:37

fine. Thank you. Um, you mentioned in passing. Matito, um, are you familiar with power mock at all?

Unknown Speaker 26:46

I couldn't quite hate you. Can you repeat that?

Speaker 3 26:49

Um, you mentioned in passing? Matito, I wanted to ask, Do you have any experience with power mock?

Unknown Speaker 26:55

Yes, yes, of course, yeah.

Speaker 3 26:58

Um, can you, I guess, explain a scenario in which you would use power? In which, can you explain a scenario in which makito doesn't work for you and you need to use power mark, you know, you have, like, any example of where it's hard to explain, like, exactly what? Straight up saying, that's fine. What is the, I guess, use case of power mock. Why would you use power mock instead of Mockito in certain scenarios, you know? Okay, yeah.

Speaker 1 27:36

So, basically, you know, so power Mockito is primarily, you know, used whenever you know you need to mock different kinds of static methods, final classes or even construction, right, constructors, right? So these, you know, active, particularly difficult by using Mockito. So I think you know, with Mockito, right? So Mockito is just enough to mock different kinds of static methods now, but like the final classes or final methods or even construction constructors, they cannot be mocked with just Mockito, right? So for mocking those you would want to use power Mockito. So I think, you know, whenever you have a legacy code or which may not, you know, follow any kind of good practices and have used lot of static methods and final classes, so power Mockito, you know, can help you write the test for this kind of code. So it may be, you know, useful, so whenever you have a complex dependencies graph, so power Mockito, you know, can facilitate that testing, you know where you can prepare multiple classes for tests. Thank you.

Speaker 3 29:03

I think that's pretty much all I had to ask. If you have any other questions you want to

Speaker 2 29:09

ask, yeah, this list, this is good things. Okay, yeah, so you're working a lot of microservices, right? Yes. Basically microservices, everybody knows, like working in a development field is like micro application, just in general language, right? So, what are the marriage and demerits and micro services positive and negative sides? Yeah? Okay,

Speaker 1 29:33

yes. So, so, yeah. So, basically, you know, micro services helps in, you know, a lot of areas like scalability and development, right? So basically, you know, some parts of your applications depend a lot upon your scalability, and because, you know, you may be getting getting a lot of traffic on one side application, so, you know, while other applications you know, can be slope at that time. So yeah, so when each you know service can be skilled independently, so it allows different resources to be relocated, sorry, allocated, and you have the proper and faster development, right? So, because you know microservices are small and independent service well, you know you can make changes and have faster and frequent developments where you know a team can deploy changes without waiting for the entire application to change. And you know, you have, like, dedicated service, you know, that is a mainly responsible for certain components, you know, so that makes a little bit easier work to work with, and also, you know, easier to maintain and operate. So these are, you know, like, your basic advantage of using microservices. So when you have, you know, like a dedicated single responsibility, responsibility working for a business school, right? So the backside, or the downside of using microservices that the complexity that it introduced, right? So, because you have a lot of these services that are distributed systems, introducing, you know, lot of complexity in you know how these services interact, so how to monitor these services and so on. So since you are, you know, in a microservices world, microservice world. So when you know, you need to communicate with microservices by using different kinds of RESTful APIs, or other means, you know, through a network. So which may sometimes, you know, introduce some kind of latency. So all the challenge that you know, have is a proper design pattern to, you know, make sure that you have a proper transaction management in place. So these are, you know, some disadvantage using a microservice.

Speaker 2 32:18

And I saw your resume like you were in most of the FinTech companies, right? Credit, great home agencies. Did you guys use third party APIs to validate some datas?

Speaker 1 32:33

So you mean overall, or you mean all the feedback

Speaker 2 32:39

specifically like what I want to ask KV is, like, maybe your project is working or not, I'm not sure, because there may be, like, a lot of projects for the company, right? So what my question is, like, you have to deal with third parties, right? Or to navigate the credits, like you have to work with banks, maybe you work with other, other credit agencies, whatever. I don't know. But how do you guys work with that, that third party APIs like, Do you have any any idea? Did you involve in that type of part,

Speaker 1 33:14

you know? So, yeah. So basically, you know, we have API and API docs, which are provided to us. And, you know, then we have a gateway layer that is responsible for making calls to this, you know, external third parties libraries, right? So, so whenever you need to make any kind of validations through, you know, external services. So we rely on those gate gateway so well, when you know, we call this gateway directly, and you know, then provide our request to them, and then after that, the gateway, you know, get the request down to the respective downstream endpoints. So it has, you know, any kind of sensitive informations, like, you know, PCI, PII, then you make sure those, you know, those are encrypted or tokenized. So if it's not, you know, we just send using in our request format. So which is normal.

Speaker 2 34:23

I'm sorry, just sorry to interrupt because, because of the time limitation. I'm just making this just, just what I want to confirm is like, how do you test that that type of API calls you can directly call to, like, for example, some bank, right? You can directly call to that real API. So how do your agency, I mean, if you worked in that one, yeah, I'm not sure, like, if you're working there or not, but how do you test those type of API calls, like, before going to live, you know, like with the real call, did you guys use some in between mechanism, something

Speaker 1 35:00

so yeah. So, let me gather my thoughts here. So, so, yeah. So, from my, you know, past project, so we had, you know, dedicated your QA environment, you know that they would expose on the lower environment, and then, you know, you would use products and domain for products and environment. So we have, you know, a dedicated environment for testing on the lower requirements, you know, for this, external calls.

Speaker 2 35:35

Anyway, okay, let's, let's talk about something about like database. What type of database you are comfortable

Speaker 1 35:45

with personally, you know, I have worked with SQL database, including, you know, Oracle and Postgres, and I've also, you know, worked with databases like DynamoDB on AWS.

Speaker 2 36:01

Okay, so, did you use like, some query with contains, like, not like condition, like, there is a list of some like, not like condition like, not, no, T, not like, condition, did you? Did you remember you use that, not matching condition? Yes, a query of that one, yeah, yeah.

Speaker 1 36:30

So you know we can, yeah, yes, I have used that.

Speaker 2 36:38

I mean, did I make question? Yeah, yeah, no. So, so there are, like, negative condition, also, we can query, right? You have, you have, you did work on that? Yes, yes, yes. Like, not in the eighth grade. Like, there are list of students, eighth, ninth, 10th, right? And if you have to get the data, like, not on the eighth grade, like, how do you write that?

Unknown Speaker 37:09

Okay, yeah. So

Speaker 2 37:10

this table name, you know, like, this database, this table, right? But not in the earth, like, not type of grade.

Speaker 1 37:18

So you want to get the data from, not the eighth grade you want to get the other day? Yeah,

Speaker 2 37:24

there is a eight, 910, grades, student list, right? Okay, so there is a student like, for example, ABC, school database, name and like, name list. And then from that name list, from the ID, you have to just like, maybe grade, least right on the grade only sort it out. Oh, okay, accept, accept, estimate. Did you write that at a query? Yes,

Speaker 1 37:51

of course. So I just understood your question right now. So you simply, you know, you select asterik from you know students where you know where you put the grade, so that's right, so use it.

Speaker 2 38:07

There should be like keyword, not like, and then whatever you don't want to start that one and the percentile sign, right? Okay. Another thing is, what, what, what do you mean by error and what do you mean by exceptions?

Unknown Speaker 38:26

How do I differentiate this?

Speaker 1 38:28

Yeah, of course. So basically, you know, errors and exceptions are like, you know, what I would say is various scenarios. So when, wherever, you know, whenever there is an issue during the accepts execution of your program, so you you may encounter an exceptions or an error. So either is like a serious problem, right? So you try to resolve it, and you cannot cast this kind of errors. And there are, you know, they are related to environments like, you know. So there is an, you know, in and out of memory error, but because you are using too much error, and there is, you know, stack overflow error, so because the stacks, because of the stat stack size and the limit, right? So there these are on recoverable error, so beyond the application flow. And now there are exceptions, right? So exceptions are whenever you have any error during the execution of your flow. So it is not it is not normal, and the exceptions are you know, handled using try, catch block, and you know these are possibly due to NTC anticipated problems during the execution flow. So like, whenever you know you send a null value for the data that is supposed to be null, so you might encounter, like a North pointer exception, right? So if there is, you know, any error with a database, you might encounter SQL exception, right? So that's what the you know difference between errors and exceptions.

Speaker 2 40:16

Ah, what do you understand by throw and what do you understand by throws

Speaker 1 40:21

through and throughs, yeah, of course. So basically, you know through means that you want to you know through an exception, right? So whenever there is any any error scenarios, so like, let's say you are checking and is, then you want that you know above that it to be above 80, then you can do if you know it is less than 18, then through a new illegal argument, etc. So, right? You know, basically it is written within the method body, and now throws is in a method signature, and it indicates that this method will throw or check exception that you know that must be handled.

Speaker 2 41:15

What about another keyword you know, like a little bit, not confusing, but like when you are using we are mostly used Final and finally. But how do you explain what is final and what is finally?

Speaker 1 41:27

Yes? So, so it is used a lot. So basically, final keyword is used for constants, right? So you can use a final, variable to declare constant. So you can use a final method to declare that the method cannot be overridden by your any of your sub classes. And then, you know, you may use final class like making it Immutable class, so you know it cannot be subclass, right? So these are, you know, there are three primary use cases for final keyword, so making like constant, constant things. And finally, you know, you try and catch block. So once you have a try and clap cats, where, you know you have some kind of code, and then you have an exceptions that you catch. Then you know, let's say you want to execute some piece of code, regardless of the situation. So you know if there is a success scenarios or even exception scenarios, then you can use your final block to write those blocks of code, you know, that needs to be run regardless, regardless of the flow of the code. So finally, you know, is for like, Clean up, clean up code executed, you know, regardless of the exception. Okay. Okay,

Speaker 3 43:05

anything. I have some coding questions to ask, but I don't know if we have the time for it sounds like we only have about 12 minutes of the schedule left. I could ask them anyways, if they want to move on to the like coding portion.

Unknown Speaker 43:25

But other than that, I don't really have any other questions.

Speaker 1 43:29

Sure, you know, I can, you know, share my screen, you know? Yeah, sure. I

Speaker 3 44:06

All right, so we can see your screen just confirm. Imagine you're working on a, I guess, a landlord, property management system. The point is that you have two lists of two different objects. You have landlords which each have a name, and you have apartments which has the name of the landlord that is managing that apartment. Can you write just some code it doesn't have to run or anything like that? I can verify, you know, by just looking at it. Can you write some code that groups the apartments into lists based on the landlord that manages them, and identify any apartments that aren't managed by the land, by any landlords in that you already know. Okay, that

Unknown Speaker 44:54

makes sense.

Speaker 1 44:58

Can you? Can you repeat that, please, if you don't mind, yeah, sure. Actually

Speaker 3 45:02

have it written down already. I can share this in the teams chat, if that would be more helpful. Yes, okay.

Speaker 3 45:22

And why is that not sending? Oh, I see What. There You Go.

Speaker 3 54:49

Sorry. You don't have to add a test data or anything like that. We'll just assume the little sorry to populate it. Okay, okay, got it. Got it. I don't want you to have to, like, I guess, spend a lot of time On that, You know? Yeah,

Unknown Speaker 56:37

uh, so I'm done with recording here.

Unknown Speaker 56:42

All right, thank you.

Speaker 3 56:45

Did we want to ask more questions? Or it's already over one

Speaker 2 56:50

Oh, yeah, time is already over. If you have anything from what you can, you can ask yes.

Speaker 1 56:55

So, like, may I know about the next step. You know,

Speaker 2 57:01

basically, I think EHR will connect with you guys, right,

Speaker 1 57:08

yeah. So, yes, connect to you. So, yeah. Also, you know, can you, you know, give me a little bit more about the project that I'll be working on in the future.

Speaker 2 57:21

Ah, you know, what, like, basically, honestly, I don't have any clue, like, which exact project you gonna be deployed to, as you know, like, this is, like, you know, multinational, like, like, international exposure, IT consulting firm, and then, okay, giving service to different clients, right different domains.

Unknown Speaker 57:48

I hope HR will handle all those things.

Speaker 1 57:52

Got it. Got it. Okay, so like my point of contact is gonna be my recruiter, right?

Unknown Speaker 58:00

Yeah, yeah, sounds good. Okay,

Unknown Speaker 58:04

yep, thank you for your time. Yeah,

Speaker 2 58:06

it was really nice to talk to you guys. Yeah, thank you for talking about something. Good. Day. Bye.

Pramod Wagle-Final Round-Goldman Sachs-SRV

Speaker 1 00:00

And no problems with the flow. If there are any, we attend to it, we come up with workarounds, we get the things done somehow, right? And then go for a strategic fix, okay, yeah. And thereby, like, improve the systems, right, coming up with automation or bulk fixes, or enhancing the systems, making it better, right? So this is all in our scope. There is no end to it. There's no limit, there's no start, there's no end. We are all over. Okay, so that's what we are.

Unknown Speaker 00:34

That's pretty awesome. Now, that's

Unknown Speaker 00:36

pretty awesome, you know,

Speaker 1 00:38

yeah, how about you, buddy? Can you give us an intro? Give me an intro, like, where we start and What? What? What are you?

Speaker 2 00:46

Yeah, of course. First of all, good afternoon. So, yeah, hi. I'm Pramod, and I have been, you know, working as a full stack Java developer for like, five years now. So I'm mainly working on Java based web applications. So I have been primarily working on microservices, and I have been involved in, you know, designing, developing and supporting different kinds of microservices. So the work in mainly, I work in monolithic applications to microservices, or, you know, developing microservices from scratch. So along with that, I've also been working with, you know, AWS cloud. So for, you know, mainly posting these applications, and I have, you know, pretty good experience working on provisioning these services using Cloud, such as AWS, EC easy to ECS, SNS and, you know, lambdas. So yeah, I also have, like, a front end experience of working working with Angular, and currently, you know, I'm on a scrum team, right? So currently I am on a Fed's trading and I have been involved in development of credit risk analysis too. So by using Java and screen related frameworks. So basically, so we take portfolio data from multiple APIs, and then, you know, provide comprehensive data to the users. So it is like, and, you know, automated created worthiness and automated ratings before generations for like, you know, an analytic evaluations of the credit risk. So, yeah, so I work in a team with five developers, and I have been involved in development mainly on the back back end framework, so I have been also involved in production support and so on. Yeah, okay,

Speaker 1 02:57

so this credit risk canal analysis tool, can you elaborate a little bit more on on that, like, what exactly were your users doing, and what was that you developed?

Speaker 2 03:14

Okay, yeah, so you want me to explain the uses and what did I develop? Right? Okay, yeah. So, basically, you know, our team mainly integrates multiple systems, so and you know, multiple down downstream services, right? So, whenever you know a user wants to, you know, whenever user wants to know any kind of borrower borrowing there, there is a credit risk, right? So a check with, we check with, basically the borrower with default to their financial obligations. So to the lender, such as, you know, not paying a loan back, right? So basically, it just takes those kind of risks, right? So we develop our credit risk analysis tool so where we leveraged multiple downstream services to pull in data from final statements, credit histories and market data. And also, you know, what

Speaker 1 04:21

do you mean by leverage downstream systems,

Speaker 2 04:26

leveraging downstream systems? So, yeah, so basically, you know, I mean that we make an API calls to other services, okay,

Unknown Speaker 04:37

but what does the downstream do

Unknown Speaker 04:40

down? What is downstream here?

Speaker 2 04:41

Downstream, yeah, yeah. So, downstream is basically, you know, mainly used, you know. So there are any kind of these are, you know, these are any kind of financial data for the user, or, you know, credit histories. So, so, okay, yeah.

Speaker 1 05:05

Where do you get the data from? Inputs from,

Speaker 2 05:10

or the input data? Yeah, the input is from the front end applications. Like, you know,

Speaker 1 05:17

okay, so how are you calculating the credit risk. How are you calcul, assessing the credit worthiness? So,

Unknown Speaker 05:27

yeah. So basically,

Speaker 2 05:30

what we get data from front end, right? So we get information from the user, that's, you know, that that they are trying to borrow, or anything that comes to our system. So then we use, you know, some of those data, and like make API call to our internal services, you know, which provide the data that is necessary for calculating the worthiness, right? So all these data,

Speaker 1 05:58

exactly, yeah, what? What what do you take to calculate the worthiness, and where do you take it from?

Speaker 2 06:08

So, so you want me to explain, or where do I get the data? Yeah,

Speaker 1 06:13

so let us say I entered some value in the front end, and your your process and your application is getting the data from that front end, yeah, okay, and now you have to process that information that you got from the front end. Okay, yeah. How are you going to process that for that use? You're telling me that you take some data from somewhere. What data are you taking? Where are you taking from, and how are you taking it?

Speaker 2 06:48

Okay, yeah, so basically, the personal data is like that is entered by the user, right? So such as name, address, employment status and income level. So these are entered by the user. So when, whenever they make a request in our call, then we basically do some like kind of international calculations, like the debt to income ratio. So we use the and we also use other, you know, downstream services to pull credit reports and like and it is like a credit score and payment history. And after that, we also try to find if there is already an existing debt obligation. So the information is basically provided by our other services, so within our fixed kids rating system so that that's how you know that team does not own and what all these, you know, micro services we use, like data gets aggregated, right? So, and then we have different kinds of designing engine. So this designing engine consists of, you know, I mean, consists of bonds of rules, right? So, and based on the data that we entered into the decision engine, the credit worthiness is returned

Unknown Speaker 08:20

based on

Unknown Speaker 08:24

No What did you enter?

Unknown Speaker 08:27

You said you entered something.

Unknown Speaker 08:30

Entered personal demographic information,

Speaker 1 08:33

personal demographic and you pull the credit reports, debt obligations, credit rating and all these things from credit bureaus, and you use that to process the application, the credit worthiness, yeah, basically, okay, okay. And what was your role in that? What? What did you develop?

Speaker 2 08:56

So, my main role, yeah, so I developed, you know, using Springboard like microservices. So I developed controllers and other services like and the controllers are took in the data from the front end right, and the services made API calls to you know, and other services to pull in those data so that are needed, and it I also integrated, which I already told you, this is, this is an engine, okay,

Unknown Speaker 09:33

so that's what I did,

Speaker 1 09:36

okay, okay, about the so that was your last assignment with the fixed ratings stuff, right? That was for like, how many so that's like, almost seven months now. Yeah, okay, and before that, what before that? What was it? Where were you working total experience from all so

Speaker 2 10:04

it's five and a half years. So I was working on Northwestern Museum,

Unknown Speaker 10:11

okay, yeah, and that was for how long

Speaker 2 10:14

that was for like, one, one year, one year. Okay? I started working on 2023 of January to February of 2024 Yeah,

Unknown Speaker 10:26

okay, okay, good.

Speaker 1 10:28

And what, what is that? What was that before?

Speaker 2 10:31

So before that, I used to work in Nepal, in gas technologies,

Unknown Speaker 10:38

okay, okay, that was for three years. Is it

Speaker 2 10:40

almost three years? Yeah, that was for three years. Yeah. Okay, great, great. Okay,

Unknown Speaker 10:45

now coming to your support experience.

Speaker 1 10:50

What kind of support experience do you have, and what did you support? Okay,

Speaker 2 10:56

yeah, so my support, right? So, yeah. So, mainly, you know, I supported the applications that we own, right? So, so I do have pretty good experience working on production support. So whenever, you know, there's any kind of incident in the production like, we have internal tool called ion. So this ion tool is used to, you know, pays engineers that are on a call. So our team, you know, does all does on call rotations on a weekly basis. So so I get, you know, assigned for products and support duties every four weeks. And then, you know, when I'm not in a call. So I have like responsibilities, right? So one of the responsibilities is, you know, actually run the releases, but the other, you know responsibilities to make sure that the products on application is healthy, and you know, there is any you know, incidents that are being addressed. So sometime, you know, these issues might be big and it needs to be tackled right away, you know, by either before releasing or, you know, sometimes trying to get a box fixed, right? So, yeah, that's it.

Speaker 1 12:17

So at the beginning of the call, I briefed you on what I do. Okay,

Unknown Speaker 12:25

what do I do?

Speaker 2 12:28

Yeah, you said that you have been working. There's a team basically, you know, and you are maintaining and you know, there's any kind of changes that you know that needs to be fixed. Then you are, you are trying to fix that kind of problem. So, you know, long term solution you so you make, is that any changes to the applications become resilient, right? So you have

Speaker 1 12:58

so I, so basically we, I said, I maintain and support the applications for front office trading. Front Office traders who trade on rates, products, on external products. They have the applications that allows them to trade and the processes to handle that right. I want to understand from this perspective, what, what are your users doing?

Unknown Speaker 13:30

I and what did you support?

Unknown Speaker 13:33

So you want me to explain your role, the

Speaker 1 13:36

support, support as an example of support.

Speaker 2 13:42

Okay, exactly. Support, okay, yeah, got it. Got it. So let's say, you know, basically, you know, in the, you know, last month, so we had a incident, right? So sometimes we require users to upload a large PDF document. So So basically, depending upon the and the kind of the document that they uploaded, right? So, so it might, you know, contain informations that is required for credit worthiness, like the statements and so on. So we have these applications that is supported on multiple platforms, like iOS, Android, desktop, right? So, and you know, the issue recently we have related was Android applications. So these, you know, Android applications whenever a user uploads a document, so that they will, you know, getting who it response score back. So now the poet response code back was from the intermediate layer, so that did not make it back from our system. So we had to support cause up this. So this was, you know, some incident that was, you know, opened. And whenever the user uploaded PDF from their from their iOS devices, they were successful. But whenever, you know, they were doing it on Android, they were not successful. So we, we were on a call. And this call we had to, you know, debug that the response code was coming from and, you know, trying to call back. So any changes that was, you know, pushed out recently to the Android platform,

Unknown Speaker 15:37

and how did you resolve it?

Speaker 2 15:40

So, yeah, basically, so we have a context router before our application, right? So, you know, this context takes router, takes in the request coming in from, like, the front end applications, or, you know, other platform then, and after that, then it routes it to different, different, you know, orchesters. So this contract router is nothing, but, you know, just a pass through, you know, proxy. So on this, you know, proxy, they had set up time up and what the time what time out was reduced from like 32 seconds to 10 seconds. Okay, okay.

Speaker 1 16:39

Are you ready for the Do you know what? Like you understood what we are working, right? Are you ready to be a part of a production support

Unknown Speaker 16:49

role? Yes. Outlook,

Speaker 1 16:52

okay, then okay, because you said you were working in the development assignments, right? Yeah, okay, and this is for New York. You will be able to relocate also, yeah, that would be fine with me. Where are you located? Currently, Virginia, okay, okay. Okay, so you have heard about this coder pad thing,

Unknown Speaker 17:20

right? Can you repeat

Unknown Speaker 17:23

that? Coder coder pad, coder pad, you

Unknown Speaker 17:26

said, Yeah, coder pad.

Speaker 1 17:28

So basically, we, we have you heard about it, yeah, of course, yeah, yeah. So we have a small assignment. Basically, there is a broken program, okay, yeah, okay, you just have to fix it right

Speaker 2 17:45

now. Yeah, okay, yeah.

Speaker 1 17:48

And I see your preferred language is Java, yeah, yeah. So I'm going to give you a Link, okay, a URL, okay, yep, that I

Speaker 2 18:25

so you are like, saving the link on an email,

Speaker 1 18:29

or, yeah, I'm going to send you the link, okay, and I want you to share screen with Me, okay, yeah.

Speaker 1 19:29

Okay, can you share your screen and bring up this URL? A PP,

Unknown Speaker 19:39

you're okay to start your screen. Yeah.

Speaker 2 19:45

So, yeah, can you see my screen? Yeah, okay.

Speaker 1 19:51

A, PP, AP, dot. Coder pad, C, O, D, E, R, coder pad.io/t, T S in Tom, all capital Letters, R,

Unknown Speaker 20:16

R s in Robert,

Unknown Speaker 20:18

n Nancy,

Unknown Speaker 20:22

number 3n,

Speaker 1 20:26

n s in Nancy, yep. T Tom 79

Unknown Speaker 20:34

Yeah, enter, yeah, got it, yeah.

Speaker 1 20:39

Okay, so you can put in your name, okay, and your email ID and sign in, okay.

Speaker 1 21:08

Okay, so what you have here is a

Unknown Speaker 21:15

program,

Unknown Speaker 21:18

yeah, and

Speaker 1 21:21

the purpose of this program is to find the first non repeating character in a given pattern. Okay,

Unknown Speaker 21:34

you got it right?

Speaker 1 21:36

Pramod p is the first non repeating character,

Unknown Speaker 21:41

okay, yeah,

Unknown Speaker 21:44

right to be returned. Okay, I

Speaker 1 21:52

the the method you see fine for client number 22 has not been implemented. That's That's to be implemented. Oh, okay, so I want you to implement a method to get the first non repeating character in a given pattern.

Speaker 1 22:17

You can run this any number of times. Okay, you can introduce system out print lines, basically to see how your program is processing. Okay, yeah, and we want you to implement that find first non repeating character in a given input string, input string,

Unknown Speaker 22:47

you go ahead, yeah, okay. Do

Unknown Speaker 23:09

Where are you typing?

Unknown Speaker 23:10

Sorry,

Unknown Speaker 23:13

I don't see you typing anywhere.

Unknown Speaker 23:14

I'm thinking, actually, no, I

Unknown Speaker 23:17

see you typing buddy somewhere.

Speaker 2 23:18

No, no, I'm not typing. My head is right here in the keyword I'm not typing.

Speaker 1 23:23

I heard you typing, can you type it on the screen? Yeah, you

Speaker 1 23:52

What are you? What's your idea like, what's what? What I'm thinking about.

Speaker 2 23:56

So I'm thinking how to approach this. Yeah,

Speaker 2 24:08

so I think, you know, we can keep a hash map, but then to store how many, you know, times our character occurs in a map, right? So basically, I think if we create a map and store the character so with its occurrence, then we'll have a map that lists, you know, how many time it occurs. Now, the problem with the hash map is that it does not preserve the order. So we need an implementation of a map that preserves the input order now. So I think Link has map, you know, actually preserves the order, you know, in which the items that were entered. So, you know, I think I'm gonna use a link that that has map, you know.

Speaker 2 25:16

So let me start by creating A map first. Then I there

Speaker 2 25:44

so I think basically we need to look through the characters. And, you know, let me import the map as well.

Speaker 2 26:42

So you know this will just basically be a simple loop, right? So that will you know store how many times you know certain character appears. So like, since we have you know this, and since we have been using a hasmap the first time, so we give the value one, so would be the character that actually appeared twice first.

Speaker 1 27:14

Sorry, what can you repeat that if you give character one?

Speaker 2 27:17

Yes. So if we you know. So now what I mean is, so since we are using a linked hash map, right, so it preserves the order so in which, in which we enter the items, right? So all we need to do is we know we need to check for the key whose value is one. So, so with that, you know, so you understand what I'm trying to say, right? Yeah.

Unknown Speaker 27:53

So for that, I think We can use entry set, yeah.

Unknown Speaker 28:50

So, yeah, so, I think that looks like it,

Unknown Speaker 28:54

okay. What? What happened here? So,

Speaker 2 28:56

you mean the whole thing, yeah, so, you know, so all the you know tests that are, you know, passed in the given input.

Unknown Speaker 29:16

Okay,

Speaker 2 29:25

okay, then, oh, so you want me to explain it further? Yeah,

Speaker 1 29:30

I was just waiting for, like, how did it get passed? Is what I'm saying.

Speaker 2 29:33

Oh, okay, yeah, of course. So, so, you know, basically. So let me, you know, just give an example. So how the map looks, right? So let me,

Speaker 1 29:47

I can you explain with respect to this program, this process, where did it start, and how did it get executed?

Speaker 2 29:59

So in the context of this program. Yes, okay, yeah, so let me so yeah. Basically in this Yeah. So in this line, right? So you can see a line 27 so we are simply execution

Speaker 1 30:17

start. Where does the execution start?

Unknown Speaker 30:20

Execution start.

Unknown Speaker 30:24

Where does it start? So, the

Speaker 2 30:26

main method is the where the execution starts, okay. So here in you can see line 67 okay.

Speaker 1 30:37

And then from, from there, where did it go? So,

Speaker 2 30:41

from the main method, so it went to line number 46

Unknown Speaker 30:46

Okay, and what does it do there?

Speaker 2 30:50

So, in this so, yeah, so I think the method 46 is defining our different test cases. Okay, so we have like, three inputs and three outputs, right then it is looping through the inputs, you know, validating the inputs. Uh huh,

Unknown Speaker 31:16

yeah. So,

Speaker 1 31:24

what is it validating? Then, what is it? What is it validating?

Speaker 2 31:27

So on line 56 will, you know, it actually calls the method, uh huh. So, as you can see, it calls, you know, it calls the find first method, so, which is our, you know, actual implementation, okay, and then, you know, it gets the output from the find first method, and Then it takes with the output array, okay, yeah.

And basically, you know, then after it matches, we print success. Otherwise,

Speaker 1 32:08

what is it matching? What? What is it? Can you can explain for one example, can you tell me, like, what happened?

Speaker 2 32:16

So, yeah, so, so, for the input, you know, let's say Apple, so it is comparing it with the output. Aha. So, yeah. So the fine, you know, find first method will, you know, simply return the character a, right. So then, which is same, right? So it will, you know, which will be true. Aha,

Unknown Speaker 32:51

and,

Speaker 1 32:56

okay, what if there were, what happens to P,

Unknown Speaker 33:02

what happens to P, yeah, P,

Unknown Speaker 33:05

will P be

Speaker 1 33:07

How is? How is P not returned here?

Speaker 2 33:13

Okay, yeah, so in our implementation, right? So which is at line 35 so it is returning the first occurrence, right? So whenever the value is one, it just returns it.

Unknown Speaker 33:33

Okay, and what is the value for p?

Speaker 2 33:36

So, yeah, so the value for P is, you know, would be two, right? So it will be not even. So then after that, we proceed inside the conditional on 934

Unknown Speaker 33:53

Okay, and what is the value for L?

Speaker 2 33:56

The value for L, yeah, so the bell for L and E would be one,

Unknown Speaker 34:03

okay. And why are they not returned?

Speaker 2 34:06

So they are not returned because, you know, we are preserving the order then the first time it encounters the value one, so it gets returned. In our case.

Unknown Speaker 34:21

Okay? Now,

Unknown Speaker 34:26

what will happen if there is no

Unknown Speaker 34:33

non repeating character?

Speaker 2 34:37

Oh, okay, yeah, so in that case, if there is

Unknown Speaker 34:42

something like this, something like this.

Speaker 2 34:48

So, yeah, so it will, you know, return line 39

Unknown Speaker 34:55

Okay, and how would you so now,

Unknown Speaker 35:00

can you run this test? Yes, yes, sure.

Unknown Speaker 35:11

What happened now,

Speaker 2 35:13

so, so, basically, so the test field, because we are, you know, validating that, you know, so d is the first non repeating character. So what are on our input D does not even exist. So the test failed

Speaker 1 35:32

input D does not even exist. What do you mean? So, see,

Speaker 2 35:41

we have d, oh, sorry, sorry. D exist. I'm sorry I didn't see D there. So yeah, I didn't see Yeah. So it returns, yeah. It is you

Unknown Speaker 35:55

who is typing this program,

Unknown Speaker 35:56

right? Yeah, yeah.

Unknown Speaker 36:01

So I thought it was, you know, B, not D.

Speaker 1 36:07

Okay. So now, what will happen? How will you make this test pass?

Speaker 2 36:12

So, yeah, so in order to pass this test, so you simply need to, I think, change the output.

Speaker 1 36:25

Output, can you do me a, okay, output to what?

Unknown Speaker 36:31

So, you know, like,

Speaker 2 36:39

hold on. So, yeah. So, so this zero, right? So I don't know what to return zero in the parenthesis. I don't know what, sorry, so, so I don't know what to return zero in the Give me. Give me. Give me one second. Let me figure it out. I

Unknown Speaker 37:14

you understood my question, right? Yes, yes,

Speaker 2 37:20

so give me one more second. Let me check you.

Speaker 2 38:03

So you know the output array it instead of D, it needs to be zero in the A, parenthesis, parenthesis, okay, so this way you know it should pass the test. Why?

Speaker 1 38:21

Why are you return? Why are you putting in parenthesis?

Speaker 2 38:27

So basically, it's a character array. So this way it should pass character array. Yeah, so it needs some character right, so, and that is zero, so, which is also known as a null character array, right in the character array, okay, okay, so basically, that is, you know, null, so I could just do zero, since I'm returning that, or just, okay, yeah,

Speaker 1 39:08

okay, can you test if that works? Yes, sure.

Unknown Speaker 39:19

Okay. I

Speaker 2 39:26

Ah, so basically, that is a small so I could, you know, just do zero, since I'm returning that.

Speaker 1 39:38

Okay, and what if we needed to return the last non repeating character,

Speaker 2 39:47

last non repeating character, yeah, okay. So, so if we know, if, like, we need to get the non repeating character.

Unknown Speaker 40:05

Let me think,

Speaker 2 40:08

I think we should be able to change the line 34

Speaker 2 40:20

Okay, so, yeah. So we would, you know, basically, change that,

Unknown Speaker 40:28

okay, change to what?

Speaker 2 40:31

So, yeah, so we let me think this through first this, this is getting the first occurrence. Do? Hands.

Speaker 2 40:46

So like, if we need to find that last note repeating, then I think we would look for the characters on the given input from the back. And then, like, check if the map value has one.

Speaker 2 41:08

How do you do that? So, yeah, like, I'll give you one example, right? So in Apple, so I would loop it from E and then L, and then p, then air right? So if the value of any of this character is one, then I just return it right away.

Unknown Speaker 41:30

Okay, and can you,

Speaker 2 41:31

I can implement that, yeah, I can implement it for you, If you Want. Okay, I

Speaker 2 43:38

So, yeah, this should be the one From the back. So I'm just competing from the back of the given input.

Speaker 1 43:46

Okay, and for why do you have to change that ddcc to D, DDC, so you change that, right? I had aa, Bb, DD and CC. Why did you have to change it to

Speaker 2 44:04

No, no one, actually, I didn't have to change it. I was just validating that it worked okay, so I can, like, change it back to double, double D, and then give a proper, proper output. Uh huh.

Unknown Speaker 44:24

Okay, this should still work, I guess.

Speaker 1 44:29

Okay, great. There is one more thing I just wanted to check. Yeah, thank you so much. There is one more program I'm going to clean This out. Okay, yeah, that's fine.

Unknown Speaker 44:57

Yep, so this is a program,

Speaker 1 45:08

as you see at line number 19, takes a string and returns the integer value. Yeah,

Speaker 1 45:24

yeah, and it is broken for some value, some tests now,

Speaker 1 45:36

in this case, we Want you to fix it. Hmm,

Unknown Speaker 45:50

so, yeah, okay, I

Speaker 2 46:09

so like, let me delete, let me read it through, Right? I

Speaker 2 52:43

So, yeah, so look looks like, I think I was able to solve it for like, you know, the positive numbers. So I think I need to fix it for the negative numbers. Now, okay, I

Unknown Speaker 53:11

How do you say you fixed it for awesome number,

Speaker 2 53:16

yes, so for the given output, right? So I've simply printed it so like you can see the output the Yeah.

Speaker 2 53:33

So yeah. Now let me try to fix it for the negative numbers as well. So that's what I'm trying to say. Yeah.

Unknown Speaker 54:00

To get online,

Unknown Speaker 54:30

okay, yeah, so all tests are passed,

Speaker 1 54:33

excellent. Can you, can you just explain how it worked? Yes,

Unknown Speaker 54:37

yes, of course. So,

Speaker 2 54:40

so basically, you know, the way this function,

Unknown Speaker 54:46

function, you know.

Speaker 2 54:50

So function at way, right? So, yeah. So though it works is that it was taking its character and then it was basically taking the sky value of the given digit. So now you know some of the issues that it had was it was using a multiplier of two, right? So basically you can see whatever it was. So calculating it to it would be double it. So it was also using an incorrect use of the string length, right? So, because it was subtracting the length by one, and then also on the loop. So it was subtracting it again and making that it smaller loop, right? So instead of, like, looping it two times, so it only looked for one time on it, so your time and one time, yeah, the loop.

Speaker 1 56:02

What is that? What? What? What does that mean two times and one time? Buddy?

Speaker 2 56:06

It means looping two times and one time. So basically, what I mean that is for, you know, any string of length three so it have only looked one time with like, you know the problem that you provided.

Speaker 1 56:30

What does that mean for which number?

Unknown Speaker 56:35

Which number

Unknown Speaker 56:38

one? So you said

Unknown Speaker 56:41

it repeated only two times,

Unknown Speaker 56:43

yeah. So online,

Unknown Speaker 56:47

for which number, online, 25

Unknown Speaker 56:50

Yeah. But for which number,

Speaker 2 56:54

which number? So, yeah, yeah. So for 4042, it only looked once,

Speaker 1 56:59

okay, yeah, and then for the other number,

Speaker 2 57:04

so it should, you know, should actually look to time. So for the other number, it would have been looked one less than that. So okay, I needed to look

Speaker 1 57:13

for, okay, okay, okay, okay. And further, what? What was the special thing that you did at line number 3737 Yeah,

Speaker 2 57:30

yes, 37 so users needed to multiply it by 10 instead of 100, right? Because, if you are trying to accumulate the digits, which is by multiplying by 100. So it would have been added zero after digit, every digit.

Unknown Speaker 57:50

Okay, okay, okay,

Speaker 2 57:58

and yeah. So for 42 it would have been 4020, yeah.

Unknown Speaker 58:09

42 it would have been 4024.

Speaker 1 58:12

Okay, okay, okay, okay. And how did you overcome the negative number case, yeah.

Speaker 2 58:19

Negative numbers, yeah. So basically we have a multiplier, right? So, yeah. So I just checked if the character is negative and if the character is negative, then, then you just multiplied it by the negative one. So you know by changing the multiply to negative one. Okay, okay,

Speaker 1 58:48

okay, excellent, excellent. Just to complete the process, can you once tell me again, like, where did it start?

Unknown Speaker 59:00

How did it go through the test cases?

Speaker 2 59:03

So you mean the whole thing? Yep, okay, yes. So so on line 66 so it calls this method, so to do test pass, right?

Unknown Speaker 59:20

So, yeah, and

Speaker 2 59:24

in in after that, you simply it calls some method on line 47 so you know, on line 47 basically it goes to line 51 and then after that, then it calls the method,

Unknown Speaker 59:49

okay, okay, okay, beautiful.

Unknown Speaker 59:54

You have any questions for me specifically?

Speaker 2 59:56

Yeah, I do have like, a couple of questions, right? Like, what?

Speaker 1 1:00:03

Thanks. Thanks for running through this. I'm going to close this thing, okay,

Unknown Speaker 1:00:09

thank you. I mean, I'll close that

Unknown Speaker 1:00:14

coder pilot exercise, yeah.

Speaker 2 1:00:22

Okay. Yeah, go ahead, yeah. So, like, this is like, more products and support focused, right? So, how much of a role is like? Is it a rotate, rotation basis, or is it always on call. You know, it

Speaker 1 1:00:41

is always on call. Okay, it's a production support work.

Speaker 2 1:00:49

Okay, okay, got it. Got it. So, like, I would also know, like, what kind of ticketing system that you use for the product, products and support, right? So do you usually get paste on your phone or, like, on Slack or something like that? We

Speaker 1 1:01:11

know we have ticketing system, internal ticketing system. I mean, similar to JIRA. Yeah, it's almost equivalent of JIRA. And so tickets are raised, and then we take it from there. And sometimes it's on email too. Sometimes it's on chat, it's it's almost everything, actually, sometimes they may come to your desk. We are co located with our traders, so they can come to your desk at any time.

Speaker 2 1:01:51

Okay, yeah, sounds good. So I do like the role that you mentioned. Is really exciting, honestly, something I could, I feel like I would be able to provide very much. Okay,

Speaker 1 1:02:03

excellent. Thank you so much. And just one last question, do you know anyone in Goldman Sachs, no, you don't know anyone in Goldman Sachs, okay, okay, buddy. Very good. Thank you so much for your time, yeah, I'll communicate through Bala and let you know what next, yeah,

Speaker 2 1:02:24

yeah, yeah, thank you so much. Yeah. Thank you. Thanks.

Unknown Speaker 1:02:29

Bye, bye. YouTube

Speaker 3 1:02:34

was number

Unknown Speaker 1:02:41

again, my latest one.

Speaker 3 1:02:45

Telegradiza there firsteir Lonely ire diving, one other.

Rajeep Karki- Freddie Mac- First-SRV

Speaker 1 00:01

basically, if I understood correctly, your role is kind of singer day, right? Yes, yes.

Unknown Speaker 00:06

So I also helped review the code.

Speaker 1 00:10

Okay, and you work both on front end and back end, yes, so how much you rate yourself on the back end and how much on the front end?

Speaker 2 00:20

So that's a pretty good question. Out of 10, if we

Speaker 1 00:25

say 10 is maximum out of 10, how much you rate yourself?

Speaker 2 00:28

So I rate myself like 70 to 30, so seven out of 10 more for the back end.

Unknown Speaker 00:36

And how about front end? Front

Unknown Speaker 00:38

end, let's say like 6040 Yeah. Okay, 6060.

Speaker 2 00:44

Is like, I'm pretty comfortable with both front end and back end.

Unknown Speaker 00:49

I'm a full stack developer, so I should have the knowledge of that both.

Speaker 2 00:54

Yeah, I would say, like 60. I can do work and pretty confidently, six out of 10. Yeah,

Speaker 1 01:01

good to know. Good to know. And with the version of Angular you're using right

Speaker 2 01:06

now, we are using Angular 15 and Yeah.

Speaker 1 01:14

And how about Spring Boot? Which version of Spring Boot and Java, the

Speaker 2 01:19

Spring Boot and Java, correct? So we are using the Spring Boot three and Java version of 17.

Speaker 1 01:28

Great, great, good to know. So, so let's say you are teaching a school kid. Okay, and you need to explain the kid. What does mean by encapsulation? How can you explain

Speaker 2 01:49

so, encapsulation? Let me think how to explain the kid about encapsulation. So I would say, like, basically, I would start with a very simple terms, because I'm explaining the kid. I would say, you know, like you can imagine of something like what I say, a box so, and even the word itself defines the encapsulate. So, basically, you have a box and which has your favorite toys in it. And we have a toy, like a car or doll or something like that, right? So that the kid can understand, and you have, which is everything inside the box, and you will open the box, and nobody can open it, only you have access. So most of the toys are in there, right? So because of that block, nobody can open, you have to have the right key for it. And once the key has a key, if you I will say, like, if you have a key, you can download that box. And if someone wants to do that, play with the toys, they need to have to ask you first, so it's yours. Yeah, basically this is like encapsulation that how I would explain to kids, I think that's from our my head, so twice inside the box, and the information within it, I would say, let's write it as a classes in it, in the developer world. And then you have like, keys, which is basically like, set up several rules. And then if you have to mess with it twice, you really have the getters and setters, and that's how we relate it, right?

Speaker 1 03:25

How you are going to distinguish, distinguish between private and public access to the to the kid, let's say public access. What is private access? What is private then protected? So how are you going to explain?

Speaker 2 03:43

So I'm as I'm explaining all those to get that, I have to have basically simple rules. How would I start? So I want to start basically saying it to explain to the kid. I can just basically explain it using our, let's say, house, so that the kids are delayed. We have a house, and then everybody can come from the front door, right? So then what you can do, like everybody knocks in the door, and like everybody can access the front door, which is okay. I would define it as a public access so everyone can exit it, and then you have your house, then you have your bedrooms, right? But the bedrooms are private because nobody can access it, because everyone wants a private bedroom. So you are allowed to enter that room only if you have access, or somebody introduced to that room, or when we buy new and we can say that, hey, you know, I want to show you. So in that case, will be like protected, and which is like our family room area, like a living room, right? So everyone can be in that living room once you are inside the house. So you know that that's will be the concept of the protective. And those are the primary concept of, I would say the access modifier, which you're asking me.

Speaker 1 05:06

So what my requirement is, let's say you have one class. Okay, let's say Class A, and there are two methods, method one, method two. So I want a subclass, okay, but I just want to inherit only one property. Let's say method one. Do you have a solution on that, or how you are going to implement you got my requirement? Let's say you have, let's say three methods plus a, method one, method two, method three, but I want to inherit the class A, so the my inherited class is child class is B, but I just want to inherit a and b. I don't want to inherit C. Method, okay, yeah, so is there any way that I can achieve that.

Speaker 2 06:01

So you want to inherited Class A and B, but other C

Speaker 1 06:06

only one class. I have three methods. Method one, method two, methods three, yeah, I have a child, class p, which is going to inherit Class A, but I don't want to inherit all the properties a, method one, method two and method three, just the two method, method one and method two. Is there any way I

Speaker 2 06:25

think I got it so you can just use like, if I'm not wrong, third method as a private method. So the private method, what you can does is it cannot be narrative. So the methods that you want to inhabitate, you can just mark them as a private

Speaker 1 06:43

okay, what are the new features that were added in Java 17?

Unknown Speaker 06:50

So let me think about it. The new features,

Unknown Speaker 06:53

yeah, so as compared to Java 11,

Speaker 2 06:56

so, yeah, basically, yeah. So, basically the new features, Java, 17 eagles are like one out of my head. Now, just is easy. One is a multi line. So basically, they actually have a concept of seed classes, which basically restricts the classes can irritate, right? So you can also, what you can do is you can describe permits keywords, and then also you can have a pattern matching with a switch cases, so you can check if the object is of the certain type in a switch block, it's right. And there is also, I think, multi line.

Speaker 1 07:36

Can you explain a bit on the same classes, how you can achieve that? Can you explain a bit more, let's say, in terms of plus a and plus b, let's say and how you are going to make it the permit. You said something about permit, right? You are using, going to use permit to make it a city class. So how can you, can you explain a bit more on that?

Speaker 2 07:59

So explaining the permits that basically, if you want to work with the like classes, rights, and let's say city classes, then you have to have a parent class. Parent class is necessary, and you define it as public right after that, what you can do is, then the class A permits the Class B, right? And then in that class, what you can do is you can define the methods like, I would say, math one, so whatever, when you are defining a subclass, let's say B, then you can have a public class B which extends a, I would say,

Unknown Speaker 08:39

so, okay,

Speaker 1 08:45

so, can you tell me, what is your Java stream, basically,

Speaker 2 08:49

sorry. Can you repeat that question? Like, Java stream, stream, JavaScript,

Unknown Speaker 08:55

not script. Stream, Java

Speaker 2 08:56

stream, sorry, my bad, because no problem. Yeah, basically, Java stream, I would define it as a collection processing framework. It is introduced in Java eight, as far as I remember. So it is used with working with collections, especially. And you can convert any collection to stream and then do the processing on it, like filter operation, for instance, or to filter your data, you can also use like map operations to convert the one from the other end of the data, and then try things like collecting the different collections and reduce it to the other forms, for instance. So basically, I would say it is a very powerful tool introduced in Java eight,

Speaker 1 09:41

great and you understand by Spring Boot? Yeah,

Speaker 2 09:46

okay. Spring Boot is why he used these days and a full stack Java developer, we use a Spring Boot. Spring Boot is like a framework used for developing a standalone web application. I think that's the most important thing. And it is a framework that is based on top of a spin framework which enables auto configurations of your dependencies, and also it provides lots of products and supports for the label features. So if I have to add something, I would say it is a powerful support tool for developing the services with annotations and auto configurations. Yeah.

Speaker 1 10:25

Okay, so have you? Do you have any experience on upgrading your application from SB two SB in the sense Spring Boot two to Spring Boot three?

Speaker 2 10:38

Yes, I do have that. So as far as not as far as I do have, but not that much. It is a contributor a little bit in the past, which I had done work on the part of operating JDK from 11 to 17, and also Spring Boot two to Spring Boot three. So we had a company wide initiative for supporting, long term support. Yeah. Okay.

Speaker 1 11:03

So, can you tell me what is the changes that new features, not new features, in the sense in new plugins or the features that have been added in three spin goods for you? So, major challenges, so I mean to say the major challenges you faced while moving from Spring Boot two to three.

Speaker 2 11:25

So it's about challenges, correct

Speaker 1 11:29

challenges. Yes, I mean to say, what are the technical difficulties or that you faced? So, yeah,

Speaker 2 11:37

to migrate what from Spring Boot two to Spring Boot three, you have to first get rid of Java X validations, the framework to Jakarta validations, right? So there's a pretty primarily changes, right? So all of these persistence and survey packages, they got to move into new packages. Then they also should have basically main challenges comes in the way when there's a minimum JDK support, and also, like, there's a bunch of changes on how spring security works within it, and we have to remove a lot of deprecated methods, and also try to change the wave security, or let's say, configure adapter, what we have to adjust the secure filter chain and the changes the way application properties was configured by using the Configure profile key rather than Using a spring profile key and so on. So there are other, yes, I would say the major changes which we

Speaker 1 12:47

so what I'm trying to understand is, let's say I want you to develop one endpoint of n, yeah, it's a simple endpoint. So the input will be nothing but user ID, or you can say employee ID, okay, and it will be the input. And what the API is going to do is it will go to the database, fetch the information, let's say employee information, employee name, department, salary, and all of the stuff, personal information or something like that. And responded back to the user, okay. Hello,

Unknown Speaker 13:41

Hello, can you hear me

Speaker 1 13:46

traverse and how it comes back, comes back with

Unknown Speaker 13:53

payload. Can you tell me? Can you explain a bit? Okay,

Speaker 2 13:56

I just had a like, disconnection. Sorry, I missed the part because I was saying, like, I your voice got disconnected.

Speaker 1 14:06

Okay, so basically, I want to understand a REST API where the input will be employee ID, and it will go to the database, fetch all the information, and it will respond me back with the payload. So I want to understand from end to end how it works.

Speaker 2 14:24

So from end to point, end to end point, basically what

Speaker 1 14:29

end to end means, how it will go to the different layer of your REST API, how it will hit the database, how it equal to the database, how it will build the reserve set into a DTO, and how it gives you back to the client.

Speaker 2 14:46

So I would like to answer this question, saying that the first entry point in a Spring Boot based RESTful API would be controller, right? So you have to have the Rest Controller class, which where you would have gate mapping annotation, and in the gate mapping annotation, what we can do is we can have a path like employee ID, and then employee slash ID, where the method will read PATH variable id, and then use that to call the service layer. So your Rest Controller is the engine point, I would say. So now you are on the controller, you would pass the employee ID to the service layer, and now the service layer will work as layer of any kind of the business logic. So you have to be calling external services or databases, then convert the data from the one to another. So first thing, what do we do is service layer. We call your repository layer to fetch the data, and repository layer will be then the extension of JP repository to talk to our databases. So there are set of data you want to fetch from the database. And if data exists, in that case, you will basically convert the entity, which is the object that represents your database object, to a dt. DT, right? So what did you do? Is dQ is data transfer object which would face these details what you need to return to the users. So basically that will conversion is successful. And what you can have is you will just return that. So there are different layers which it can interact with. What we can also do is the Rest Controller being the entry point, the service layer being the business logic point, and repository layer paying the database interaction point.

Speaker 1 16:46

So you are using jPf to connect to the database, right?

Unknown Speaker 16:52

So can you explain a bit how you are

Speaker 1 16:56

executing your query into the database in which saved some service, right? So how you are executing this particular query? So yes,

Speaker 2 17:07

here what I would do is on the repository. I will just simply use extending GPA repositories, which provides inbuilt function called Find by ID that could be one option. So on your service layer, what you can do is you can have a dependency of the employee repository, and then you would just try to find out the ID method and pass the ID on that particular employee. Now you need to some kind of, I would say the customer queries. So you can also provide those within the repository by using query annotations, and then you can use that

Speaker 1 17:52

method. So let's say now my requirement changes. So the thing is, I want to fetch the data from, let's say multiple tables. Let's say four tables. Okay, the data should come from a join, and the join consists of around four tables. So how you are going to define that join? And so they have, yeah, they have primary and secondary key relationship. So how you're going to define on your Java end,

Speaker 2 18:26

so there are primary and secondary keys as well? Okay? What we can do is we can take like multiple tables involved when fetching these cases. So you can do like a native query, right? So basically, on your entity side, what you can happen, what you can do is you would have the many, one to many to one relationship, for instance, with your department or other entities at your employee rated, right? So you basically use one to one, many to one and those kind of a relation. And what is

Speaker 1 19:02

many to one? Can you explain a bit of what is that many to one?

Speaker 2 19:05

So it's a many to one is a relationship. What we have in especially in a database. So mainly, many to one is relationship where many of these entity can have many of the IDs, for instance, can have a relationship attribute, can have a relationship to the one attribute another table. So, so, so, yeah. So basically, you have an object with many entities in one of the table and then get a map to an entity on the other table. So basically you have multiple employees on for practical example, is, for example, you have multiple employees and on the same department, right? So you can have that relationship with one entity of the other team. So

Speaker 1 19:50

what will be the relationship here? Let's say I have employee table, and I have 100 employees, and all are unique employees, obviously, and I have a department level, let's say HR department, technical department, then I have Secretary department, okay, so from employee to department, what would be the relationship? So in terms of many to one, or one to many, or one to one, so it will be

Speaker 2 20:18

like many to one, right? Because you have multiple, multiple or many employees on the one department. So one department feel will you then annotate to as many to one, so

Speaker 1 20:30

many to one on employee side, or a minute to one on the department side, department table.

Speaker 2 20:35

So that's, I would say, many to one on the employee side. That's how we should work,

Unknown Speaker 20:41

okay? And on the department side, so

Speaker 2 20:44

on the department so I side, if I'm not wrong, I would say, basically you don't need to do that as because you have already back mapping. So because, yeah, that will suit. I have a reference column and the name of the employees, but the relation here would be one to me,

Speaker 1 21:01

okay, and how you are securing your API endpoint. I mean to say, I want to understand your security features over there. Okay,

Speaker 2 21:13

so what we can do is we can use a spring security, that's how I do, to implement the security so we have a identity provider, which basically what it does is it takes wherever you want to have the access our applications. So we can then forward an identity provider where you pass into the login details. For instance, your username or password. You have to have the correct username and password, and once you have that, the identity provider injects JWT token. So now the JWT token is a three part token, which that consists of a header signature and a payload of our application. So basically, you know, these, decodes the header and then looks at our claims that can have a role you have, and based on these rules, what we do is we can secure our endpoints, right so you have these role based access control. So I would also like to add that we just do, like a pre authorized annotations, whereby providing a role which has access to those particular endpoints. So,

Speaker 1 22:27

so in which layer you are going to implement on this? Can you tell me whatever you talked about, the JWT and all everything, so in which layer it will go, so where you are going to implement and how it will be injected in your endpoint.

Speaker 2 22:44

So the layer, I would say it will be basically,

Speaker 1 22:48

if I leave about the layer, how it will be injected, I want to understand whatever you said that I understood. It's a JWT token, and it will have some stuffs over there. And okay, but let's say how it will be injected. Let's say when you are hitting on postman, or let's say from the from the from the front end, Angular side, you are hitting this end point. So how it will be authenticated will call this particular code.

Speaker 2 23:19

So yeah, so basically, I would say the tokens are especially would be injected into the request header. So what happens to the bearer tokens comes as a part of authorization header with bearer prefix. So what we can happen is the, basically the identity provide flow would be the identity provider team, or let's say, like the portal where you have username and password provided, right? So that would be part of your login flow, and once the login and the login response would consider the token, you will have that token, and then insect that into your request header as an interceptor part of your request, okay,

Speaker 1 24:07

okay, and let's say I have two endpoint in my application. One is get employee data, let's say get API. Another is save API. And I have two kind of users in my application. One is admin, one is only read only user. So how you are distinguish? Going to distinguish the read only user so that they don't access the Save API,

Unknown Speaker 24:34

so they don't have to access the same API,

Speaker 1 24:38

right? So you have two endpoint, okay? And two kinds of user, one is read only and one is admin. Admin has access to all the endpoints, and read only, user can only access this get endpoint. So how you are going to restrict the group who are read only so that they should not access the Save API?

Unknown Speaker 25:01

What do you do is we can use a role based access control,

Speaker 1 25:07

sorry, firing will define that role and everything. So

Speaker 2 25:11

we can basically, you know, like on the I would say, as roles will be up as part that I will be saying that the decoder token, and you will get the role of the user has a right or not, and if

Speaker 1 25:28

it Yeah, I will interrupt here. But how I will get so let's say I need to create those kind of roles, right? Let's say I'm asking you to develop this endpoint and create two, two kind of users. So how you are going to do that?

Speaker 2 25:48

So let me think about if I understood it basically correct. So that would be, I would like to start that will be stored on the user details table so you can assign,

Speaker 1 25:59

let's say I will. I will. I would rephrase this question in this way, okay, let's say you have one application developed, okay, and there is no security as such as of now you just hit the end button and get the response back. Okay, now I want to, but I want you to design is, let's say I have two kind of users. So let's say I have 10 APIs, and there are three endpoint. Those are kind of safe endpoints. So I want to ask you, I want to I want to do is just create two groups. One is read only groups, another is admin groups, and restrict this read only group to save any kind of data in the database. Okay, okay, so how you're going to design this? You understood my requirement, right? First of all,

Unknown Speaker 26:58

I think yes, I did.

Speaker 2 27:03

So I will try to answer best as possible, which I have understood, right, right? So basically, I would say that what we do is we have to create an admin and read only group for the user, right? So I would also have the user details, for instance, the manager right where I would have the username and a password for our users. Then I would say like I would specify certain role to assign to the particular user, right. Then what we can do is we can use this user details. Is this. User details is identity for us provided by the Spring Boot and spring security, to be precise, and you can also create the spring security's user details object, and the role for that particular user will be loaded into the context once the user authenticates and now, based on that role, gets the role loaded information of the context. So you can basically add a check on on your controller if you want to to see if the user has role that matches to that required role or not.

Speaker 1 28:19

Okay, so let's, let's go back to that end point. Okay, employee, endpoint. So once you're done with the development, you must be doing some kind of unit testing, right? Yeah, are you using Jane it? Okay, so at some point you are getting the database to fetch the data, right? So how you are doing that in the unit testing? I mean how you're testing this particular scenario.

Speaker 2 28:48

So now we are talking about unit testing. So I have done a few unit testing, and what I have a practice of doing is that you can use the, let's say for JV or mokito framework in order to mock on the external service calls, and in my unit test, I just mock out repository layer. But if you want to do some kind of, let's say, integration testing using a Spring Boot test for the local or the test environment, what you can do is you can spin up the memory databases with some pre loaded data. So there are, I guess, from my point of

Unknown Speaker 29:27

view, how you can load the means,

Unknown Speaker 29:30

the what is the static data for testing data?

Speaker 2 29:33

So the question is, how I can load the steady database, test data? Yeah, okay. So basically, if you're working with a Spring Boot, what we I guess, in this case, you will just need to set up the s2 database, and then on the Resources folder, you can just provide the SQL files what you need to run once the application starts. And then what happens is that it loads those database onto SQL file.

Speaker 1 30:06

Okay, so basically, you can have the s2 database and you can have static, secure file, right? Okay, that makes sense, yeah. Okay, so what is your composite key in your database, relational database?

Speaker 2 30:22

So in relational database, I think the composite key, or are very necessary composite key in a database, I would simply out of my head now, would say that it consists of, it is a primary key which consists of one or more than one columns, right? So instead of having a one column as a primary key. What we can do is we can have multiple columns and use them as composite heat,

Speaker 1 30:46

right, right? And let's say in your table you have around half a million data, or 1 million data so and my query is going to patch around 1000s of data, maybe 50k or 100k

Unknown Speaker 31:08

so

Speaker 1 31:11

do you have solution for it? And so it will take a lot of time and then building the results set right and sending it back to the user. So do you have any solution, either on the back end or on the database side, so make it faster.

Speaker 2 31:28

So here we are, as I understood, here we are using with use, like a big data or like certain quality of data. So the fastest way for me, or as I understood, is the data retrieval to make sure that you have a proper indices on the most column, which is where necessary. And these indices, or joins write without indices from database which can scan the entire table. But I would say that index would speed up the data rotable process. So indexing, I think, is very

Unknown Speaker 32:05

important. Okay.

Speaker 1 32:07

And in Angular, what is a two way binding? Can you split the two way two way binding? Okay, okay.

Speaker 2 32:18

So in Angular, I would say two way binding. I would define it. Let's say you have a data flow in both directions in a simple way, and then from one component to view, the to view, and from the other component to do. Let's say for the TypeScript, if the data are bounded together, which we have it is called two way bounding, or, sorry, two way binding. You can use, let's say NZ model directives

Unknown Speaker 32:51

for our two way binding. What

Speaker 1 32:53

is a directive? You talked about directive, right? What is a directive, by the way? So

Speaker 2 32:58

to like to out of my definition, what I would do is basically directive is, I would suggest a kind of like, you know, like what you have is set up for instructions, or instructions, what you need to happen and how you do. Do am needs to process these things so you have something like attribute directives like ng class or ng style, which basically gives, I would say, what kind of a classes and styling needs to happen. And then you can also have a structure and direct tips like Ng, and then if and only if for that some kind of emotional comparisons or grouping we have,

Unknown Speaker 33:44

okay? So

Speaker 1 33:47

let's say I developed one front end application, okay? And I have a feature, small feature, and it's a big organization, and I have multiple team working for different different feature of the website, but I am developing a common feature, okay, yeah, and what I want to do is this common feature should be used by all the teams, okay, yeah. So basically, what I'm trying to do is reusable purpose of my application or my picture or my module. You can say, so do you have anything that you can do on the front end, on Angular? So

Speaker 2 34:37

to solve that problem, what we can do is we can create, I would say, reusable features that will be very useful. So basically, reusable features, yeah, yeah, yeah. So what you can do is you can basically just create a shared library. And when you have a share library that we can use, we can use it for multiple teams. So, yeah.

Speaker 1 35:06

So can you elaborate a bit how you can you create a shared library? Yeah,

Speaker 2 35:12

so I can refer back to my experience, how I did it. So we have a shared library called app shell, which is basically like basic components. Basically you have like ng generated library, and then in this library, what we can do is you will have the component directives and services which you can add, and then on the components you can have, like share button or shared forms, or something like that, and you can use for your whole team.

Speaker 1 35:45

Let's say you have two component in your application, component one and component two, okay. And there is no relationship between them, no child and parent relationship. And in the component one, I have a variable called x. First question is, in component two, can I access the variable x from component one?

Unknown Speaker 36:13

So yes, how?

Speaker 2 36:16

So we have component one and component two. You want to access the component two from the

Speaker 1 36:21

component one. I have component one, and variable x is inside component one. I have another component, component two. I want to access variable x, which is inside component one. Yeah. Can I do it?

Speaker 2 36:34

So if you have, like, basically, if you have no relation between these two components and component to one has a variable x, right? And that you can directly, you cannot, like directly access that variable, right, because there is no connecting point between them. So if you really want to access that, that you need to create a shared services that acts as a bridge between those unrelated components. Otherwise,

Speaker 1 37:04

if I have a parent child relationship, let's say component one is parent, component two is the child of component one. Now the scenario is different. Now, can I access

Speaker 2 37:17

so in this case, I would say yes, if your component one is a parent and then the component two is a child, I think you can use that input and output decorators to pass the data. So in this case, you should be able to okay

Speaker 1 37:32

and how you are unit testing your front end. I'm mostly interested in the Angular app,

Speaker 2 37:41

sure. So what we are basically doing in this part is, regarding the testing is so far what we do, we use, basically use, like Jasmine, what? That's what we are doing right now.

Speaker 1 37:58

Okay, and do you know what is reactive form?

Unknown Speaker 38:05

Can you repeat that again?

Speaker 1 38:07

Sorry, reactive form. Reactive reactive form. Okay. So

Speaker 2 38:10

to my understanding, what I have user experience for, my learning is we have a same kind of for some kind of abstractions above reactive form that we can use to build forms. It is a way to provide, I would say, like schema and a layout to those form so that in order to build like form controls and form groups, right? It just makes controlling the form behavior provides a dynamic ways to manage the data. Yes, so that's, I would

Speaker 1 38:42

say, and on the AWS side, you said you have experience on EC two, then you said the s3 right? Yes. So what is easy to, basically,

Speaker 2 38:51

so in a simple ways, if I have to say, like, what is easy to, easy to is simply the virtual machine, which it, would you, would you can, like, use it for what you're missing, like, aw, easy to AWS servers where you can run anything we want, or you can install like operating systems and then run any applications that are exposed to it. So it is especially, I would say, like physical server and exist somewhere in the cloud.

Speaker 1 39:25

Any any exposure to serverless lambda feature?

Speaker 2 39:30

Yes, serverless lambda features? Yes. I have worked with lambdas for event driven processing. We also use like to listen on SQS queue to process, yeah.

Speaker 1 39:44

And have you have any experience on DynamoDB or something like that? Any database? Yeah,

Speaker 2 39:54

sure. Yeah, because DynamoDB and I, if I'm using an AWS here, yes. So I work with a non regional databases. It is basically the non regional databases.

Speaker 1 40:04

So what is DynamoDB? By the way, how it is different from different database means how it is different from Postgres. I would say so,

Speaker 2 40:12

as I already mentioned, like DynamoDB is non regional database. So basically, is provided by the AWS, and it is basically the stores key and values fields. So it does not any it doesn't do, let's say any kind of like strict it doesn't have any kind of strict schemas. So it just have a key based from documents and then where you can store any kind of data you want to have it. So it is very easy because it is non registerable based non resident databases. It is easy to have stability and availability. Maybe that's very popular and easy to use. So some kind of a less consistency, but you can also achieve a high consistency with it using a high consistency mode of DynamoDB. That's how I would define the DynamoDB, yes.

Speaker 1 41:04

And in AWS, let's say indication of any error or defect, how you're debugging? Can you tell me so

Speaker 2 41:13

AWS debugging, I would say that if we are in within the AWS, let's say

Speaker 1 41:20

you have something deployed on easy to instance, okay, so and you've got some error, deployment error. So how you're debugging? But what? What is going wrong with that

Speaker 2 41:33

answer? Yep. So I would, I would say, basically, we can look at the issues with the deployments, it would have basically error so we can, we can simply refer back to the SSS into the eye to the box and where to go. The deployment folder is, and it usually you're, you know, like OBT OPD folder, where you can define your deployment configurations, but it is not easy to then you can, what you can do is you can use a cloud formation templates,

Speaker 1 42:11

okay, let me replay this so once, let's say you're developing A feature, okay, and in your local it is working. So what is the next step you are doing and how you are deploying for for the testing team to test very fine. Can you tell me

Speaker 2 42:33

Sure? So we can, for instance, we can use simply GitHub as our source control. So basically I would change my brands to a feature brand and then push my changes. So once our team is, let's say, do a code reviews, I can then merge it into a developer branch, and then we can use or run the Jenkins pipeline in order to deploy our changes right. So when we run the Jenkins pipeline and then check it. Pipelines, basically builds our artifacts and then runs, or some scans, like, let's say, vulnerability check, and then it goes to the deployment process. So it's basically then deploys to the Dave environment for the Dave's or, let's say you can be QA to test it in a higher environment.

Speaker 1 43:26

Okay, so what is, what do you understand? Understood by API gateway.

Speaker 2 43:34

So API gateway is basically, from my understanding, it is a central point for, I would say, text as orchestrations layer, where you can have a single point of entry and based on those route mapping, then you restore in some form, also for what's the request to those services. So you can, what you can have is you can have a request routing mechanism, but also have some kind of authorizations with it, using authorized and lambda, so you can implement, let's say, safety features like weight limiting or circuit breaking and so on. So it is a global point of entry. So

Unknown Speaker 44:23

any experience on microservices means,

Unknown Speaker 44:29

yes, tell me what is a microservice.

Speaker 2 44:32

So I have worked on the microservice environment. Basically, microservice, as the name already refers, is a smaller unit of a service and that is independent and can be deployed in the whole unit. So I would also like to say, from my experience, we have worked on multiple microservices, but while I primarily developed and maintained data maintenance, API was responsible for making changes to the users provide preference. So in a nutshell, it is just unit of a basic work deployed as a complete service so that it is a scalable and maintainable within itself. Okay,

Speaker 1 45:17

so are you using Mac or Windows?

Speaker 2 45:20

Well, I use both, so I'm working with especially MAC, which I like to prefer. That my preference,

Unknown Speaker 45:30

but I'm familiar with the both systems.

Speaker 1 45:33

Okay, so I have one question for you, so can you tell me that the recently you faced one technical challenge, that what is the challenge was, and how you resolve that recently that you faced or or whatever you can remember.

Unknown Speaker 45:54

So any technical challenge,

Speaker 1 45:58

yeah, it could be technical error or any challenges you wanted any feature to implement. And yeah. So let's talk about the challenges. One challenge,

Speaker 2 46:14

so one of the most recent challenge that I was when I was incorporating a streamline engine with like matrix commands, right? So these commands were basically using wrapper on the top of agriculture service, which basically spins up to the new trade. So to do some kind of a processing and what do we need? What we did is, or what I did is, we incorporated streaming engine, which collected the applications logs, and then we tried to create a bundle of to push into this API call. So during this our applications, we performed, let's say, around 10, 1015, mm, minutes, and after that it will start performing by out memory error. So that was the problem. And what do we do is, is we use all the memory. And what do we do is, we use this memory, and sometimes we'd also use up the trades, right? So it was pretty inconsistent, but it was really difficult for me. I still remember that to debug it, because debugging is not that easy with huge number of data, so historical commands like for different threads. So this was my technical challenge, and then I worked recently to debug. What I do is so basically do the configuration, like over five to 10 threads Max, and in order to debug it, so I can replicate the issues, running it locally, and then decreasing the size of intelligence, VM, I would also have tried to run out of the memory quickly, which I noticed, and then I had to write quickly a performance test script to hit the instance with a high throughput and then check back what happened. So it was not sort of like doing those kind of back and forth work for further investigation. We also found that there was a command, basically, which swallows the exceptions and then basically it doesn't clear the resources. So the user said, because it does not, like, let's say, like a try with resources. So basically, I had to provide our custom implementations to the particular flow, and by overriding this method, we were able to resolve it, but the following the documentation didn't basically point to what we were using wrong. So that was something I recently solved, and it was an experience.

Speaker 1 48:55

Okay? So I have a last question for you, that if you scale up vertically, and if I ask you to scale up vertically or scale up horizontally, what do you understand by this two method?

Speaker 2 49:11

So to scale up vertically and horizontally, I would say the vertical scaling is providing a more CPU and memory to our applications, right? So, you know, you just increase the hardware of your applications, and that's on the most of our original databases. That what we do. And that's only possible options, right? You can just provide a higher memory power and CPU power to handle, let's say, I would say x, kind of a higher traffic. But horizontal scaling is more about spinning of newer nodes or instances to handle the higher traffic. Basically, you can distribute the traffic across multiple instances of the same application to handle the higher load. Fauci these are two primary scalability concepts, I would

Speaker 1 50:07

say, great. So I'm pretty much done with the discussion, so I will give you the opportunity to ask me any question you may have.

Speaker 2 50:17

So thank you. First of all, I would like to thank you for giving your time and giving me this opportunity. So out of my head, I would say, like, the first question I would like to ask is, what are like, let's say day to day practice of the engineers and the team of our feature project or feature client. And what is like, let's say the requirements are gatherings along the implementations of the features. Yes, overall,

Speaker 1 50:47

yeah. So basically, our team is a small team, but yeah, we have couple of application and it is on Spring Boot Angular, then couple of node applications. We have couple of small application so basically, we are developing the applications and we are supporting them as well. But mostly our main what is that propriety is to develop the application so the thing is, we are currently working on the upgrades and some business stories that enhance the feature. And yeah, we are in an Agile team, two week sprint and Delhi Scrum. Then we have our team comprise of a business analyst, Scrum Master, devs and leads. Then, yeah, we don't have QA as such. We are the developers. We are the what is that called tester? So it's a end to end development. You need to what is that? What is it? Understand the requirement from the business analyst or product owner, develop the feature, test it out, and send it back to the product owners for the acceptance. So this is how it works. Thank you. Any more questions? Yeah,

Speaker 2 52:18

thank you for explaining me, explaining it to me, and it sounds like the practices of workplace is pretty familiar to me, so I would like to ask one more questions, and that would be, what would be the next step for me? But that's the only question, like, what will be the next step we have this? Yeah.

Speaker 1 52:44

Okay, so it was nice talking to you, Raziel, and thanks for your time. Thanks for your time, and it was really good experience. Okay, and I will pass on the feedback to our chat team, hopefully they will get back to you soon. Okay, thanks for your time.

Unknown Speaker 53:02

Thanks for your time, sir. Thank you so much.

Unknown Speaker 53:05

Yeah, perfect.

Rajeep-Barclays-SRV-final

Speaker 1 00:00

About, let's say, five partitions and four consumers.

Unknown Speaker 00:07

How would the partition assignment would be?

Speaker 2 00:11

So basically, what happens in Kafka, in this situation is like we have more numbers of a partition, as you mentioned, as compared to the comparison consumers, then what happens is each partition is assigned to only one consumer with a particular consumer group, right? So then the consumer is in group shares the partition as possible. So let's say if you have five partition then it might what can happen is that the first consumer will do the two partitions and then the rest with the one each. Okay,

Speaker 1 00:51

what important for the other way, if you have less number of partitions and more brokers, let's say, Take example, for for, what is it? For, partitions, right? Yeah. My example could four partitions,

Unknown Speaker 01:15

five brokers, yeah. So

Speaker 2 01:17

four partition we have here, four partitions and five brokers. So yeah, so whenever there are some more consumers, as compared to the partitions, then one of the partition, I'm sorry. I mean, like one of the broker. So each of the partition is assigned to the broker, and basically, Kafka does not require the number of partitions to require to be equal to the broker, right? So each broker would be assigned a partition, but the other broker would just, I would say, like, stay idle in this case,

Speaker 1 01:55

okay. And do you know what real sync replica is?

Speaker 2 02:01

So can you repeat that question again? Async replica? Yes, yes. So yeah, basically, what is talking about this? Let me gather myself. So you know, like when there is replication logic, or when you have the followers replicate the data. But in asynchronous applications, the follower will replicate but do not immediately acknowledge the rights. So basically what happens is the data returned to the leader will not be immediately available to the follower, which basically be like possibility to catch up.

Speaker 1 02:45

So okay, so let me ask you a different question, right when I enable transactions, right on our producer, right? So, how does, or what? What are the significance of insect in sync replicas when it comes to a producer that's in a transaction,

Speaker 2 03:07

okay? So, yeah, basically, whenever you enable, let's say transactions, in this case, on a producer, right? So basically, in sync replica become more critical for your I would say, let's say the guaranteed, to guarantee the ultimate city or consistencies, isolations, what do we have? And also the durability, sa properties. So basically, it's like a set of a replica that always stays on or up to date with a leader of those. And then, whenever there is a transactions of our successful compete, then if there's a crash on the leader, the leader will be chosen from the ISR. That's our so

Speaker 1 03:59

if, let's say a transaction is open and there are no minimum number of ISRs. Then what happens?

Speaker 2 04:08

So you're saying minimum number of IRS, ISR, right? Yes,

Speaker 1 04:13

yes. So let's say the configured ISRs on the on the topic is three, and I'm producing the message, and there are no

Unknown Speaker 04:26

three ISRs available. What happens?

Speaker 2 04:30

So let me gather myself, let me understand this question properly. So what happens is, so you have like, three ISR and you have configure it to three, right? Yeah,

Speaker 1 04:44

yes. So configuration says is three, but runtime there are three, not no three ISRs available when a producer is producing a message, what happens? Okay,

Speaker 2 05:00

so basically, I would say, like, I guess, in my guess, or in my experience, for knowledge, if you have a minimum replicas created, then the Kafka, basically the producers cannot write new messages, because the producer sends the messages to the leader. But since the current number of the IRS is less than three, then the basically your leader will reject the right request by some kind of not enough replicas error. So it will probably be a block to write the data, which will basically makes that durability is not compromised a compromised, yeah, okay,

Speaker 1 05:43

got the answer, yeah, yeah. The producer cannot accept the message because there are no many minimum number of isomers available. Okay, so I

Speaker 1 06:06

multi threading. Core, Java multi threading. Okay, so let's talk about multi threading. Sure. In the

Unknown Speaker 06:17

later versions, after JDK five,

Speaker 1 06:20

there are newer ways to create a thread, but

Unknown Speaker 06:25

so we have a callable thread and a runable thread.

Unknown Speaker 06:29

Can I check in? There's noise here.

Speaker 1 06:34

Okay, so then, so, how have you used threads

Unknown Speaker 06:42

the latest versions of JDK.

Speaker 2 06:44

So how I have used threads correct? So I haven't like directly implemented those, but I have worked with, let's say, the commands, which basically is a wrapper and educator service, which basically is like some idea, right? So instead of manipulating your trade directly, what we can do is we can create an implementations of the stakes commands, which internally does callable interface. So you are callable, you can callable are runnable specifically has two different model tradings, right? So basically, callable allows a trade to return a result or some kind of a check exceptions. I'd

Speaker 1 07:34

say I have, sorry to cut you. Let's say I've implemented callable thread. So, and I need to submit this into the thread pool executer. So,

Unknown Speaker 07:49

how have you created the thread pool executor?

Speaker 2 07:53

So, basically talking about thread pool educator. So the way to create a thread pool educator what I have done, or from my experience, is can be through our educator service, I just study, let's say that using educators dot the thread pool creations strategy, I would say, and then you can just submit a task to those educator service, right? And so for each callable you can just submit attached to the executable service,

Speaker 1 08:23

right? So hold on to that thought, right? So you either create a third pull executer via the third pull executer constructor, or you use the utility method called executors and then call you via the executors, you can create a new fixed thread pool, or a cash thread pool or shuttle thread pool. Is that what you're saying?

Speaker 2 08:53

So if yes, that's That's exactly

Speaker 1 08:58

right. Okay. So let's say, in the first case of creating the thread pool execute or using its constructor.

Unknown Speaker 09:07

Have you used it? Anytime or No,

Speaker 2 09:10

let me see regarding my projects, not that I didn't think that. The reason that I

Speaker 1 09:19

ask is there is something called a thread pool strategy, okay? And I want to know whether you're familiar with that, and if you're familiar with that, then what is meant by thread pool strategy, where call, it's called collar runs, what is meant by that?

Unknown Speaker 09:38

So I have to say,

Speaker 2 09:42

so for my concept in a thread pool strategy, I would say that it is, let's say the fixed rule where you can create like, let's say the fixed number of threads, and you can also basically provide how many threads you want, right then you can have like cache thread pool that you want, or which is basically cache shortly, or workload basically, either threads can be removed from a certain time, and you can basically create a simple threatful By doing, let's say sequencing. Can

Speaker 1 10:23

you focus the answer on the color runs policy,

Unknown Speaker 10:27

okay. So just on the just on the policy.

Unknown Speaker 10:31

Okay. What is meant by color runs, okay.

Speaker 2 10:34

So basically, I would say, like colors policy. Basically, I would think about, or so we think, in a way that it is a way to reject a circular whenever you cannot accept, like a new task. Yeah,

Unknown Speaker 10:58

they work. How does it work? Okay,

Speaker 2 11:00

so the way it works is, like, whenever you your thread pool is some kind of a fool, or maybe there is some kind of a slowdown for the or something like that. Instead of, instead of, like discarding or rejecting masses, what happens is the calling thread executes the task itself. So basically slows down the color, and they're reducing the rate of, I would say, new task. So, yeah, that's the kind of like how it works overall. Okay, okay.

Speaker 1 11:36

And then the second way is to use the executors method, and it has some utility methods called New fixed thread pool, new schedule thread pool. What is the difference between a fixed thread pool and a schedule and a cache thread pool?

Speaker 2 11:50

So the difference between these pools, I would say, like, basically what I would mention is I also mentioned earlier is, was the when it's a fixed pool, is whenever you want to dedicate the number of threads on the thread pool, right? So you just provide how many threads you want on your thread pool, then you basically have some kind of a CPU intensive tasks where you have the evaluated how many threads you might work on certain tasks, right? So if you want to, let's say, dedicated 40 threads for one task, what you can do is you can use a new trick, threads, thread pool with fixed number, and then also provide the scheduled thread pool is kind of a different what when you want to run the tags and the certain intervals, or maybe some kind of a delay, right? So you can create a fixed number of threads, but it has delayed executions. So then cache.

Speaker 1 12:57

So I'm not sure whether you got the question. I'm particularly interested in the cash thread pool, and how, or how do you differentiate it with the fixed thread pool?

Speaker 2 13:06

So, yeah, so like, so my, I would like to say that cash are like a short leave threads whenever you have a short leave or on predictable work, right? So there is no fixed number of threats, Okay, okay,

Unknown Speaker 13:31

go back to your resume. I

Speaker 1 13:46

i Let's talk about few design patterns. Don't see that in your resume, but then I think you would know, let's see. Okay, how do you write a class which is single term, a single term class. You can take an example of a person class. How do you make it single term

Speaker 2 14:09

so? So basically, a single term class, first you need to do, let's say, so you'd have a private constructor for that person, right? So what you do is you define the construction so that the no other classes can initialize personal right? So that's the first method. And the other thing is, I would like to say that you need to have a private statistic, the private static field, or fields variables that holds that person's instances. So let's say I have a person, and whose person is a private static right? So now I would have a public gate instant method, which is like a public static gate instance. And then I think, yeah, sorry, I think, yeah. So that will tell the public so it will return a public person, yeah. So study gets the instance, and within that get instant matter. You would do a null check of that object, and then if the object is long, then you can definitely initiate the object basically just written, okay,

Unknown Speaker 15:28

do you have to worry about,

Unknown Speaker 15:33

what is it in a multi threaded

Speaker 1 15:36

environment? Can it happen that more than one thread can check null you try to create the single term object. Can that happen?

Speaker 2 15:47

So basically, what you do is, if you have, like, some kind of double checked locking with a single time, if you are, like, running in a multi threading environment, right? So what will happen is you can do a synchronized method for the get instance, and basically it acquires a lock. So if you don't do that, what happens is that you may just acquire the log on singleton class and then check if it's no null, and again, check if it's null and

Speaker 1 16:19

then. So how do you modify your public static get instance method

Unknown Speaker 16:26

to take care of the synchronization?

Speaker 2 16:30

So to modify that, I would introduce synchronized keyword on the meta definition, yes, on the method, on the method. Is

Unknown Speaker 16:41

there any other way?

Speaker 2 16:43

Let me think about the other. What the other options could be, the other options from my head now it will be like synchronization of the class itself, rather than the method. How

Unknown Speaker 16:54

would you do that? So

Speaker 2 16:55

basically, what we do is we have this given standard method. So we would check if it's not first, and then after, after the null check, I would get a synchronized block with the person of the class, and then the class itself, yeah, answer

Unknown Speaker 17:13

the question, yes.

Speaker 1 17:19

So I said, Forget about the multi threading and Singleton. You still have the person class, and you have two variables in it, first name and last name, right? And you want to make it immutable. How do you make the person class immutable?

Speaker 2 17:35

So the question is about making it immutable. So in order to make it immutable, I would say that I have to make sure that we have a final QR, right? So first you declare the fields as a private and a final so you don't have that any external access to it. And then you can also define your class definition as a final Okay, yeah. So as private, and you can also expose the constructor, and I would say the getters method, so you cannot change the value after it once it is initialized. So basically no center methods.

Speaker 1 18:20

Okay, so now let's add one more variable to the person class. The person class has first name, last name, and a third one called list of address, right? So it's a collection of addresses. So now, do you have to do something, or how do you what are changes that you want to do on the personal class to make it immutable.

Speaker 2 18:45

So to make it immutable, I would say that you have the list on your object, so you have to take care of making this permanent on addition to that. So whenever you do a gator on that, on that list, what happens is that you know you can either return or the copy of the list, or whenever you initialize, let's say the constructor. I would say that you just do on Modify, unmodifiable, I would say list or something like that, yep, okay,

Unknown Speaker 19:35

there's a line in the original that says,

Speaker 1 19:39

In a world in developing code for obtaining beam references in Spring framework using dependency injection. What is meant by that?

Unknown Speaker 19:52

Do you remember what you wrote on the resume?

Speaker 2 19:57

What is that so talking about if I get the answer question correct, and if I remember everything. So I think what I mean by that is just referencing your dependent, dependence injections pattern, you know, like, so you have something, let's say I think it's one of the project, I think, where we have to configure, like qualifiers and primary means. So I work on that, and I think

Unknown Speaker 20:30

it's what we are talking about there.

Speaker 1 20:34

So what is there to develop? Right? There's nothing to develop now. So,

Speaker 2 20:37

so that's, that's the I would like to add that that's the line I would say refers to, or I would say that that line means that I work on a configuration file that

Unknown Speaker 20:49

define the beans.

Unknown Speaker 20:53

Yeah, the class configuration, okay.

Unknown Speaker 21:04

Let's talk about little bit about it.

Speaker 1 21:07

Let's say I checked out a branch and

Unknown Speaker 21:13

I made a few changes,

Speaker 1 21:16

but meanwhile, somebody comes by my desk and said, Hey, I need some help on this other branch. Now I want to help the person, so I need to kind of check out the other branch. The changes that I've done, I don't want to lose

Unknown Speaker 21:34

what do you do in this scenario?

Speaker 2 21:37

So in this scenario, I would say, like, what you can do is you can do basically a couple of things. So in my perspective, one thing I would do is that get, let's say the enables is basically self for your changes. So by using a get stash command, it basically saves all your modified changes into some repositories, store location, got

Unknown Speaker 22:01

the answer? Yeah. Thank you. Okay,

Speaker 1 22:11

okay, I see that I mentioned a lot for j so I kind of wanting to ask this question. Have you heard about MDC?

Speaker 2 22:20

Did you mentioned MDC, yeah. Okay, so

Unknown Speaker 22:27

MDC regarding, so it's like just logging. Perspective

Unknown Speaker 22:31

of logging, yeah,

Speaker 1 22:35

so there is MDC in SLF 4j, MDC in log 4j, what is that? Where do you use it?

Speaker 2 22:42

So especially in the login contest, I would say it's a kind of diagnostic context with logging where you can, let's say, share like, the request IDs. So it's a kind of, it allows you to store like, let's say there's some kind of request IDs or sessions. You have a transactions ID across the different login sessions. So let's say, you know, in the production, you have lots of traffic. So what you can do is, you know, if you have, like, same log for the different users or the different sections, so what can happen? It just kind of hard to track and the follow with them, with a request, all you have. So you can use you can Yeah,

Unknown Speaker 23:21

let's talk about Maven a bit.

Unknown Speaker 23:25

See, do you have Maven?

Unknown Speaker 23:28

I thought I saw Maven. Yeah,

Unknown Speaker 23:29

I have worked with Yes.

Speaker 1 23:36

It's just that not everybody knows everything. Just want to know.

Unknown Speaker 23:41

Can you recollect something? So in Maven, there's,

Speaker 1 23:47

there's something called snapshots, right? And the releases, so now and then dependencies. So the question that I'm having is that when I say MVM, deploy, how? And it's a snapshot. How does we will know where to upload the artifact? So what specific tag might it infer to see where to upload the artifacts? If you can

Speaker 2 24:21

remember. So basically, again, gathering all the informations to answer these questions and in a right way, so for a different environment. So we can create different releases and snapshot versions. So I think if you use, let's say, the release flag on the Maven deploy commands it, what it does is advise to release and but it will do a snapshot. If you don't do a release flag,

Speaker 1 24:52

you answer a different question. But then let's see. So let's say I have a pawn and

Unknown Speaker 24:58

it says 1.0 snapshot.

Unknown Speaker 25:02

Yeah, and I'm saying Indian deploy.

Speaker 1 25:05

So the Indian starts clean, then it starts building it. Then it creates a jar, let's say, or the war, or whatever, and then it starts uploading. How does Maven know where to upload, is my question. Okay,

Speaker 2 25:19

so how does Maven knows how to

Speaker 1 25:25

upload? Yeah, so I can have multiple repositories. How does it know where it has to upload the article? So

Unknown Speaker 25:33

basically, on your distribution management

Speaker 2 25:36

with the Maven you can, let's say, give the repository name, and you can have the repository, or you can have it like

Unknown Speaker 25:43

a snapshot, snapshot.

Speaker 1 25:45

Sorry, you mentioned. Did you mention the tag? I thought I missed the repository tag. Or is there a specific tag

Speaker 2 25:58

I was I was actually saying, like, either the distribution management or something inside it.

Unknown Speaker 26:02

Yes, that's, that's what I wanted, yeah, okay.

Speaker 1 26:06

Because what are repositories for? I mean, if you see just repositories tag in the settings, dot XML, what do they do? So

Speaker 2 26:16

the purpose of it is, I would say, like, basically, the reports are very tagged kills, or you are a map. He tells you a map and where to look for, let's say, to build or download the artifacts which you have, right? So, like, let's say that I got

Unknown Speaker 26:35

the answer. Thank you. Thank you.

Speaker 1 26:41

Okay, let me ask you a trick question, spring. Okay,

Unknown Speaker 26:46

so there is,

Speaker 1 26:50

what are the different different scopes in spring? That's a very basic question. What are the different scopes?

Speaker 2 26:56

Okay, the three scopes like, let's spring bin is a singleton. About the other scopes, like prototype, scope.

Speaker 1 27:08

Let's stick to I know there are other things but singleton and prototype. So now I have a situation where every time I want to use a beam, I don't want the same instance of the beam, right? I want a different instance of the beam. So the obvious choice is to use Scopus.

Unknown Speaker 27:36

Scopus. So the obvious choice now,

Speaker 1 27:41

the situation is I have a top level class, which is singleton injected within this is a dependent being,

Unknown Speaker 27:53

and it's a prototype,

Speaker 1 27:55

okay? And I don't seem to get it. Why I'm not getting a different instance of the being every time, because I want it that every time I get this Deen, I need a different instance of it, right? Yeah. So Is my understanding correct, or I have to do something else.

Speaker 2 28:21

So to that question, let me think about it.

Speaker 1 28:26

Second repeat, top level beam injected bean, an injected bean, I want a different instance every time I have to work on it, whereas the top level, the outside bean is a single term bean.

Speaker 2 28:42

So these beans are like a creative ones when singleton bean is initialized, right? So for the singleton beans are created once during the startup, and the prototype beans, although that ends, are created only when, let's say he's like explicitly requested. So if the prototyping is directly inside the singleton, it resolves like only once at the startup they

Unknown Speaker 29:09

answer the question,

Speaker 1 29:11

the perfect answer, yes. Now, now let's build up it. But now that I'm in a situation that I wanted to get prototype, but I'm not getting it because, because of the answer what you said, How do I solve this problem?

Speaker 2 29:27

Okay, that's a good question. So to solve, to solve it, let me, let me. Give me some few seconds. So I would, let's say, for my point of view, I would have never like had to deal with some scenarios. Maybe I would say, like, you have to explicitly get this pain from this class, or maybe some kind of, like, different approach, where you attribute the pain only when you need it.

Unknown Speaker 29:57

Okay, so,

Speaker 2 29:59

but I hope that I am going getting the right way in this question, sir, I'm

Speaker 1 30:03

saying you're going the right way. So there is a beam, and I want an inner beam, but straight away, if I just inject it, it wouldn't work. So how else can I get a beam?

Speaker 2 30:18

Yes, so I have to think about it. I'm just sometimes like, I have to make

Unknown Speaker 30:26

my question Okay.

Speaker 2 30:29

Thank you for answering the question. So I would say that, like my guess, my guess is, like, you have to work with lazy loading of the beam using the provider content context.

Unknown Speaker 30:43

So, yes, I think

Speaker 2 30:45

provided context. So because I think like application context is also an options, but it is not generally recommended, I would say to play with application context of like Spring Boot applications. So I would like to say that, like a provider context is just like a lazy loading wrapper for us. Okay,

Speaker 1 31:12

that's, that's the answer to this. For example, you can inject the context aware. Implement context aware, so the context is injected, ask the context to get your bean the container will inspire inspect is, what's the scope, and if it's a prototype, and every time it create a new base,

Unknown Speaker 31:30

right? Yes, you give me a hint. So I was like, I just Okay, sure.

Unknown Speaker 31:39

How good are you in your next Yes,

Unknown Speaker 31:44

but very good test, I

Speaker 2 31:50

would say there's always like

Unknown Speaker 31:55

for the progress everybody has. So let

Speaker 1 31:57

me ask you a stupid question so that you know I know that you're smarter than that, right? So how do you check whether process is running?

Speaker 2 32:09

So how do I check little processes running? Yes,

Unknown Speaker 32:15

okay, let's say

Speaker 2 32:17

I would use a command like L so by mine or all, or of, let's say,

Unknown Speaker 32:26

ps command. Okay.

Speaker 1 32:36

I can ask complicated using cut and grip. Let me ask, how do you

Unknown Speaker 32:44

push

Unknown Speaker 32:47

an application to background processing?

Speaker 1 32:50

But you got an app, and then you want to push it for brand background as a background, background process,

Unknown Speaker 32:57

yeah?

Speaker 2 33:00

So basically to put in the background process. So I guess you can just do an

Unknown Speaker 33:06

app person sign for background processes.

Speaker 1 33:12

Okay, I'll just type PG. Okay, sure. Next. Okay, so ask a few questions. I see Oracle Database mature. Have

Unknown Speaker 33:40

you used union

Speaker 1 33:44

queries? Yeah. Do you know the difference between union and union all?

Unknown Speaker 33:50

Yes. So let's say like, I

Speaker 2 33:51

haven't like, I know about union, but I haven't like, practically had a chance to use it. So basically the main difference, what I can say is like a union, and union all is that, let's eat hands, some hands, like a duplicates, right? Both of them, but the combined results of two or more SELECT statements. But in union, you basically removes, basically removes the duplicates, and the union allocates the all the duplicates, whatever you have, just throw the rows, okay,

Speaker 1 34:29

okay, another simple question. Let's say you have two cables. There's a foreign key relationship between them, and then I want all the rows from both the tables,

Unknown Speaker 34:43

irrespective of

Unknown Speaker 34:46

the joint condition is met or not,

Unknown Speaker 34:49

right? So how do I write such a query?

Speaker 2 34:53

To write such a query, I would say, like, if you want to do such a query, basically for the two tables with, let's

Unknown Speaker 35:04

say with, with all those rows.

Speaker 2 35:07

And you can do, I would say that you can use a Full Outer Join, or, like, basically FULL OUTER JOIN, which matching like rows and on the matching rows from the other table. Okay.

Speaker 1 35:28

And some rest micro services, I will probably leave that. Okay, let's talk about GMs. Yeah,

Unknown Speaker 35:38

so you want me to

Speaker 1 35:40

No, no, I'm just trying to frame a question about JMS.

Unknown Speaker 35:46

I tell JMS you have used Spring. When

Unknown Speaker 35:54

did you configure

Speaker 1 35:57

a JMS listener using Spring? Like if you remember all the SMS rules, but basically, conceptually, how did you configure

Unknown Speaker 36:09

JMS listener using Spring?

Speaker 2 36:13

So I guess, like, basically, you know, some of them are, like, active and queues, basically you have the connection factory where you have, like, the proper URL.

Unknown Speaker 36:24

On top of that, I would say, like,

Speaker 2 36:28

what you can do is you have the username and password, and then you can go to the

Unknown Speaker 36:33

connection factory, next. So, next,

Speaker 1 36:40

so, yeah, you configure the connection factory. Yes, what's next?

Speaker 2 36:45

Okay, so I was saying like, you have, like, your destination for those queries, and now you can, or you need to create the JMS. But maybe, like, I would say that enabling JMS annotations on the JMS applications, you can also just inject the GMs producers and send them the masses,

Speaker 1 37:08

right? Very generic answer. I was looking for a specific answer. So let's talk about,

Unknown Speaker 37:20

how did you configure transaction?

Unknown Speaker 37:24

Right your application is trying to send a message.

Speaker 1 37:28

If you are the developer writing code and just walk me through. How did you get to enabling transaction and also sending a message?

Speaker 2 37:40

So what we do is basically for that purpose. I would say we think, how we do it, we can do like transaction management for active MQs. I think that there should be, like a property for some kind of acknowledgment, and we can, if you want, we want to have the acknowledgement to be done, and then the way you can configure the connection between the transaction is basically defining the transaction manager. So, yeah,

Speaker 1 38:19

okay, so you configure a JV Transaction Manager,

Speaker 2 38:23

right? Yes, so, okay,

Speaker 1 38:27

so you have a GMs connection factory, you have a GMs Transaction Manager, right? And then you need some other beam to send a message. What is that being

Speaker 2 38:39

so you are asked me, so I think so that should be like JMS template.

Speaker 1 38:45

Okay, let's stick with it, right? So there's a connection factory, and there's a GMs Transaction Manager, and then there is a GMs template. Now I'm not sure whether you much documentation. Have you read with JMS template. But what JMS template, what they say is it's totally inefficient that it creates all the JMS artifacts and discards it for every message. So to increase performance, what can you do to make it better?

Speaker 2 39:18

So to make it better, we can use, like, say, a few options. Let me think what could be, what could be done for it to make it better?

Speaker 1 39:33

I repeat the last part because usually in my question, there lies the answer, okay, the question, right? So we have a connection factory, and then you have a JMS Transaction Manager, then you create a GMs template, right?

Unknown Speaker 39:48

And as for documentation, GMs template

Speaker 1 39:52

creates all the GMs artifacts and discards it for every message that is being sent. So what can you do with the JMS template to make it efficient?

Speaker 2 40:07

If I get it right, maybe use connection pooling strategy, where you define some kind of, I would say, like pools, and to retrieve the connections factory, so that, you know, like, GMs and might discard it. But you can also do, like, it's a wild guess, yes,

Speaker 1 40:32

I You're, you're going towards an answer, right? Build on it. Okay, so

Unknown Speaker 40:39

it is. You have to do some kind of, so,

Speaker 2 40:41

yeah, of something, yes. So you have get the, you have this, you have to get generous templates from the cash connections factory, I would say, to reuse the, sorry, from where cash connection, okay, and so from the cache connection factory, you can, like, use the connections so that, let's say,

Unknown Speaker 41:06

even though, from like connection

Speaker 2 41:08

pool, right? And I guess that there might be some kind of configuration settings which you can work on JMS templates, but I don't recall what act configuration you need to Okay,

Speaker 1 41:24

okay, you might think you didn't give me the answer, but you gave me the answer. You have to use a caching connection factory, which caches the connections whereby it doesn't discard the artifacts that are required to send a message. So this is about sending a message. What about receiving the message?

Unknown Speaker 41:41

Okay, yeah,

Speaker 1 41:45

I'm pretty sure you did your GMs template to receive messages, right? So what did you do to receive a message?

Speaker 2 41:53

To receive a message? So in order to receive masses from my experience, what I did is let me think few options for it. So now we are talking here about, basically, I was just, yeah, I'm trying to, I'm trying to get some like options.

Unknown Speaker 42:25

It has been a while, so

Speaker 2 42:27

I have to get it to the point so I don't

Speaker 1 42:33

generally, I've seen from last 45 minutes you already get you always get to the answer. Build up. Build on.

Unknown Speaker 42:38

I'm trying the best.

Unknown Speaker 42:41

So you have, let's say, that

Speaker 2 42:44

connection factory. So maybe that connection factory, from that, what you can do is

Unknown Speaker 42:52

you basically retrieve some kind of,

Speaker 2 42:56

maybe a masses from the sessions which you which you create? Or you can create

Speaker 1 43:04

that's real writing to the JMS API, but I'm say I'm using Spring, so if I'm listening to messages, I wouldn't use JMS template. What is that the spring provides to listen to messages? I was

Speaker 2 43:21

I was, I was thinking in the other way, okay, so, so far, using like spring TMS, you can create, let's say, I would say at GMs Lister. I mean, like GMs, listener and audition, yeah, okay,

Speaker 1 43:44

okay. Do you know what the annotation does behind the scenes?

Unknown Speaker 43:50

Behind the scenes, what I've

Unknown Speaker 43:51

used it, but let me think about it,

Speaker 2 43:55

what it helps or does in the background process. So

Unknown Speaker 43:59

what it generally does I

Speaker 2 44:02

don't have much I didn't have much knowledge. Okay, so

Speaker 1 44:06

the answer I'm looking for is something called a default message listener, container, okay, right? Creates a container, which is a springs MDB,

Unknown Speaker 44:18

okay, I learned something.

Speaker 1 44:24

Okay, I think we come to an end of interview. Do you have any questions for me?

Speaker 2 44:31

I let's say, from basically the project point of view, are we working completely with Java, the spring and so on. So almost like developers get chance or they don't have to involve in a designing and sourcing architectures.

Unknown Speaker 44:50

So

Unknown Speaker 44:53

first of all, what are the things that we work on?

Speaker 1 44:57

From a technology perspective, we are on Kafka and Web. MQ, solace, that's our middleware, Oracle, database, spring, spring, boat,

Unknown Speaker 45:10

all those things. Then,

Speaker 1 45:16

when a particular work is assigned.

Unknown Speaker 45:22

We,

Speaker 1 45:25

as leads, don't generally try to spoon feed. We expect people to come up with their own thought and design and implement and then come up with the solution where we just review the PR and suggest what is right, is wrong, but, but when, when things are not in the right direction, we probably help you with the work right? Thank you so much. So business wise, we are in the business of settlements. So what we do is that we have there are external settlement systems, mostly like, for example, if you heard of gloss from Broadridge or Nomura, is I star, or internally, there are other settlement systems, which are bank settlement systems. So what happens is, when you get a trade, you determine whether the trade you process before or it's a new trade. Based on that, you either correct your previous trade or escaped as a new trade. What you do is that different settlement systems have different formats. So you change from one format to the other format, sometimes XML, sometimes JSON. You also have to enrich the trade with different power you call enrichments. Like, for example, the Counter Party is the legal entity, everything. And then you form the tray the format for the external system to take, and then you send out the message, that's the business that we are in.

Speaker 2 47:10

Okay, thank you. Thank you so much for answering my questions, and yes, I really have like, overview of what we'll be doing in the browsing? Any other questions? Yeah. So what will be like next round? What will be the next step for me? So

Unknown Speaker 47:35

I so I wouldn't know how many interviews are

Speaker 1 47:39

before me and after me, I wouldn't know it's it's kept a secret by HR, right? So I don't even know whether you are in my team. So I am thinking that definitely we have a position open, and that's the rule they try to fill in. And hence, I am interviewing you as part of the lead, but I wouldn't know what's next, because that's HR, who kind of

Unknown Speaker 48:08

streamlines the process.

Unknown Speaker 48:11

But are you here in the east side? Or

Unknown Speaker 48:15

yes, I'm in the East Coast, Virginia,

Unknown Speaker 48:20

so you are up for relocation or no? Yes, okay,

Unknown Speaker 48:25

I haven't asked that question

Speaker 1 48:28

before. Okay, so sorry, I wouldn't know the answer to your question, but yeah, I would say, best of luck and all the best any other questions. Thank you so much.

Speaker 2 48:43

Thank you so much for your time, and I really appreciate it. And thank you also for giving me a chance to share my experience and knowledge with you, and I have, I guess, chance to learn something today,

Speaker 1 48:58

there's always something we always know. Thank you very much. How nice.

Rajeep-K&LA-SRV-Final;

Speaker 1 00:00

Primarily involved in the microservices development, and recently I've been involved in designing and developing also by deploying these applications to the AWS cloud. So some of the AWS cloud services that I've been involved, I would say, like AWS, easy to SQS, SMS and lends. So on the front end side, what I've been doing is, or I have experience with Angular, as our current project is based on Angular for the front end. And on the testing side, I work with Jenkins okito and also some cucumber for automation. So currently I'm in a scrum team. We have five developers, we have a product owner in a Scrum Masters. And on daily basis, what I do is I'm involved, mostly on the back end, Java code, helping the team with the code reviews and also helping the production support DTS. So that's a short introduction.

Speaker 2 01:09

All right, thank you for that. So I can kind of take over from here, and I'll give you a little rundown of what kind of to expect. So right now, I'm going to send some credentials in the chat, it's just going to be pretty much to log in what you already did on your async. So more or less, what we're planning to do is just pick up where you left off on that async and kind of pair together to solve some problems. We'll probably do that for about 20 minutes, maybe more, and then we'll kind of move on to Q and A after that. All in all, it's a relatively short interview, so we'll, we'll see how things go. But the what I will ask you is, once you do log in and put everything in to share your screen, and then we can kind of see if there's any extra little settings we need to adjust, because sometimes this is a VM for us, so sometimes some of the settings need to be tweaked a little, but just get it right where you left off. Okay,

Unknown Speaker 02:16

any questions on any of that? I

Unknown Speaker 02:20

think I'm good to go.

Speaker 2 02:23

Okay, so now that we're in here, if you can start the server, what I'd like to do do first is just, can you actually through through the app demo what problems you were able to complete? You don't really have to show the code. I'm just more interested in seeing the results on the app. Sure. So give me some

Speaker 1 03:01

time saying if I'm not wrong, I was working on that

Unknown Speaker 03:05

calculation part

Speaker 1 03:09

which I clearly remember, and that's left on because also of the time constraint which I had, and also like, yeah, that's what I remember. One is a COVID shoot Park.

Unknown Speaker 03:30

So let me see if I can remember.

Speaker 2 03:33

Yeah, I know it's kind of tough after taking a week or two breaks. So No, no worries there.

Unknown Speaker 03:38

Yeah, I

Unknown Speaker 03:40

remember that I didn't solve everything,

Unknown Speaker 03:44

so the calculation part of

Unknown Speaker 04:02

this completed

Speaker 3 04:10

this. So I think I completed the flame type, but I was not able to get

Unknown Speaker 04:18

invoice total. Yeah, that was I had a

Unknown Speaker 04:21

problem getting the

Unknown Speaker 04:23

total invoice.

Unknown Speaker 04:27

Yeah, that's the next thing. I spent a lot of time with it.

Unknown Speaker 04:37

So what else that

Unknown Speaker 04:41

I also created the invoice from the

Speaker 1 04:46

incendiary screen, and I was able to complete the

Unknown Speaker 04:53

game line uttering defect.

Unknown Speaker 05:00

I think what next?

Unknown Speaker 05:04

The next defect is also done when

Unknown Speaker 05:08

I completed that claim is

Unknown Speaker 05:10

screen or jumping.

Unknown Speaker 05:16

So, so would you be able to Oh,

Unknown Speaker 05:19

yeah, one thing which I

Speaker 1 05:22

also like to mention. Was I was not able to get this Ruby working. I think that was I also mentioned giving the email if I'm not wrong,

Unknown Speaker 05:34

yeah, okay, yeah. I

Unknown Speaker 05:44

yeah, I think.

Speaker 2 05:46

Okay. So, so can you start the server and demo these, the ones that you were able to complete?

Unknown Speaker 05:51

Okay, sure, yeah, yeah. So, let me start. Okay, do?

Unknown Speaker 06:00

Give me some time.

Unknown Speaker 06:19

Okay, so The application started, let me see.

Speaker 2 06:57

Okay, so, so, yeah, we've got the claim type descriptions here. And previously it was showing MP and D right there. And now it's showing dental and medical. So that's definitely one of them, which is great. So yes, right, okay, I think

Unknown Speaker 07:18

the other one was, I

Speaker 3 07:31

so whenever I go to this patient,

Unknown Speaker 07:39

I try to create

Unknown Speaker 07:42

new invoice.

Speaker 3 07:47

I think that's the one right with

Unknown Speaker 07:53

the ID is present.

Speaker 2 08:00

So it looks like this one's partially working. It does get the patient ID in there, but the Begin and End date don't quite prefill. But looks like, hey, that's that's partially working. And then did you? Did you say you have the ordering done? Or

Unknown Speaker 08:20

the claim line ordering. I'm not sure on that.

Speaker 3 08:23

Read me, yeah, I think the ordering was done as well.

Speaker 2 08:28

Okay, so if you hop over the claim screen, I think if we do a show lines, we should be able to see that one. I

Unknown Speaker 08:44

Okay, so those lines are in order, so that's good,

Speaker 2 08:49

right? Maybe a little CSS run off there, but the lines are in order, so that's what we're looking for on this one.

Unknown Speaker 08:57

We've all been there.

Unknown Speaker 09:01

All right.

Speaker 2 09:04

Would you be able to go back to the first problem? So I know you said that you were still kind of working on this one, right? And I think, I think this is the one that I would like to spend some time pairing with you on to see, you know, kind of how you might go about figuring out what's going on here, what you would look at and actually go in and try and fix it.

Speaker 2 09:40

And kind of way, the way that this can work too is, you know, if you do have questions or you do want to ask things, you know, I won't, I won't bug you the whole way through. But if you do have, you know, questions or want a sounding board, you know, we're, I'm available for that too. It's not like totally just blindly close this while we watch, I can kind of pair with you a little at least

Unknown Speaker 10:04

Sure. Thank you. I

Speaker 3 10:49

All right, so we have place where the calculation is done,

Unknown Speaker 10:52

as for The date notes in the

Speaker 2 12:00

I think the other date is 630, 2022, just, you got it.

Unknown Speaker 12:32

Okay, I should have actually started my server in a deeper more I

Unknown Speaker 12:53

so it looks like that final totals $182,397.56

Unknown Speaker 12:59

and I think, based on the read me, we're looking for 182,300

Speaker 2 13:06

and something. I guess, if you pop back to the read me,

Unknown Speaker 13:14

yeah. So it needs to instead be $182,330.28

Speaker 4 13:19

Yeah. So so Tubby,

Unknown Speaker 13:33

okay, I

Unknown Speaker 13:40

so 330, I, again, This One.

Speaker 2 16:22

I Okay, so it seems like for that discount record, the coverage begin and end is, you know, the coverage would seemingly apply for at least this claim, right?

Unknown Speaker 17:11

Yeah, right. So

Speaker 3 17:14

for this claim, I should, I guess, apply because it's on that particular

Unknown Speaker 17:22

Data. Range so

Speaker 2 18:31

so if you don't mind me asking, So, so what are you kind of looking through here? Can you kind of describe what you're looking for? Maybe any thoughts you're having as you go through this.

Speaker 1 18:46

So I'm actually trying to see if there's some discrepancy, discrepancy between those data comparison, or if there is an, let's say, error in calculating that's what I'm trying to use. Or, you know, like if there is an error or in multiplying or something. So I'm just like trying to see if the datas are, let's say, correct or inaccurate. So Well, the line summing, I can see it's inaccurate.

Speaker 3 19:37

So let's save so the discount record is this one thing, right? We

Unknown Speaker 19:45

have a discount record.

Unknown Speaker 19:52

So discount Record Is If

Speaker 2 21:55

so I can probably at least assist a little here and eliminate one One route. So it's not a it's not a routing issue.

Unknown Speaker 22:10

It is, it is more of

Unknown Speaker 22:20

a issue. I guess you could say, I

Speaker 3 22:40

it. So is it a problem with the calculation? So, okay, so based off when the patient has insurance and service details for those claims, insurance care will cover the percentage of the claim, amount of those calculations the patient, right? So basically, based on when the

Speaker 1 23:11

insurance for that particular service ends, or something

Unknown Speaker 23:15

like that, right?

Speaker 2 23:21

So, so what do you add? Can you repeat that? I just want to make sure I understood the question.

Speaker 1 23:24

So I was saying that when the patient insurance and services data for those insurance claim

Unknown Speaker 23:32

insurance

Speaker 1 23:36

of the claim amount after those calculation, patient, right? So it's basically based on when the insurance for that particular service ends, or something like that. If I'm not wrong,

Speaker 2 23:51

yeah, so services, so you need to be having coverage

Unknown Speaker 23:57

during that time span. Yeah? Do?

Unknown Speaker 24:04

So the so it should be the

Speaker 3 24:24

Okay, this country got shoes, if You have covers or not, if you have coverage.

Unknown Speaker 25:02

And now, based On the claims I

Unknown Speaker 25:49

all Right? I

Unknown Speaker 27:00

so for the circle, We have COVID and A's, so is

Unknown Speaker 27:05

the issue that

Speaker 3 27:07

basically, it's reporting all the data instead of the ones that the user has selected.

Unknown Speaker 27:20

Right since we don't have absolutely

Unknown Speaker 27:49

date and ended In

Unknown Speaker 27:50

the Account, right, right?

Unknown Speaker 28:01

So then, how would we

Unknown Speaker 28:04

limit which or

Speaker 2 28:07

how would you change it to limit claims that are within that invoice, begin and end, date range? I

Speaker 3 28:38

so from our controller, we need to make sure that the data we check as the invoice, start as well. I

Speaker 2 29:06

if that team's Microsoft is sharing your screen, I think you can just click that, hide there, rather than I know it's probably been bugging you all interview there. All right, nice. Get that out of your way. Yeah.

Unknown Speaker 29:18

Okay, thank you. So yeah. Okay,

Unknown Speaker 29:26

so basically, I think I need to do a query. I

Speaker 3 29:44

so I think I just need to create a query that basically

Unknown Speaker 29:51

retrieves basically this date

Speaker 5 29:55

between, right, Yep,

Speaker 6 30:14

it needs to be fine By the ID And okay,

Speaker 3 30:31

So I'm I might need to do a custom query here, because I don't remember

Unknown Speaker 31:39

this specification.

Unknown Speaker 31:42

So, so something small that you might have missed.

Speaker 2 31:46

Try, try just adding all after the find, I didn't notice you missed that. See it, see if that'll try that, and see if it, if it still doesn't, then, yeah, you can continue on what you were doing. Just okay. So so that didn't matter. I just wanted you to try that. Just

Unknown Speaker 32:07

make sure that wasn't messing anything up. So Fine. Also,

Unknown Speaker 32:16

between,

Speaker 3 32:20

let me see I do this, if it works. We can do fine, all

Speaker 6 32:38

between, okay, maybe that works. I

Speaker 2 32:52

I think, I think it's only complaining, Just because it's saying it's never used so I

Speaker 2 33:39

if You go to that hammer on the left there. Yeah, go ahead and click the Build and yeah, you can do the hammer there below build or control, if not, I'll also do it for you. I

Unknown Speaker 34:08

so let's see if My claims

Unknown Speaker 34:12

work now. I

Unknown Speaker 34:40

now, okay. Now, okay.

Speaker 3 35:14

I'm just going to do a custom query. Makes it easier, I Guess. Yeah, yeah. I

Unknown Speaker 36:20

so is it looking something like that,

Speaker 2 36:26

right? So you're checking off that claim entity and what to bring in based off of that, and it's still not very happy. I

Speaker 2 37:06

Okay, so maybe we need to do a couple a parameter out optional patient ID not found.

Speaker 2 37:22

So if we need to, just like the async, if we need to hop over, do a quick Google, to pick up some syntax, things we can always do that seems like it's not happy based on some of the parameters you're passed in, passing in, just based on the query. It doesn't seem like anything is too egregious. So it seems like it's just having trouble linking up

Speaker 2 37:51

the patient ID, unless it's spelled differently, which

Unknown Speaker 37:58

seems to match to me. They

Unknown Speaker 38:04

okay, it's Yeah, yeah

Unknown Speaker 38:07

seems to be matching, but the

Unknown Speaker 38:10

console shows like it differently, right?

Unknown Speaker 38:18

Since I

Speaker 3 38:41

I think I can add a param here which should like, okay, okay, it worked on here. I think

Speaker 1 38:48

the bill is lagging some reason, starting now without no change.

Speaker 1 39:01

Okay, so the claim size decrease, it was 11 earlier,

Unknown Speaker 39:07

now is five,

Unknown Speaker 39:16

and we can also see the claims.

Unknown Speaker 39:18

So here it's all on six

Unknown Speaker 39:23

screens. I guess it should work.

Unknown Speaker 39:38

All right, cool. Thank you.

Speaker 2 39:41

So now, now it's lining up. So, so, yeah, we just needed to source the claims from the right data service, and I think now we got it going cool. So, so I think that's the main item I wanted to to work through in the pairing. Since that one's a little tricky, going into the database and doing a few different various things. I did want to throw it over to Elena or Doug if they wanted to do any sort of pairing stuff. If not, we can kind of just move things on to Q and A Sure.

Unknown Speaker 40:16

Yep. Okay, cool.

Speaker 2 40:22

All right, so at this point you can disconnect from the

Unknown Speaker 40:27

VM there,

Unknown Speaker 40:30

I think at the bottom there, there's like,

Speaker 2 40:33

if you hit the start or the Windows icon, there's like, a disconnect, yep, and then, yeah, perfect. Go ahead and disconnect, and then you can stop your share. So we're good on the parent portion here, and really we we have some Q and A here, so we'll, we'll ask you a couple can questions, and go through a few things there, and then we'll, at the end of interview, give you a chance to ask us questions and kind of go from there. Does that? Does that all sound good?

Unknown Speaker 41:08

Yeah, sounds good to me. All right,

Unknown Speaker 41:11

perfect. So

Unknown Speaker 41:15

I did want to ask,

Speaker 2 41:19

Do you have experience with SQL databases, and specifically, which platforms have you worked with?

Speaker 1 41:26

So talking about the SQL databases, in the past, I have worked with the Oracle, but recently, I have worked with Postgres,

Speaker 2 41:39

okay, and on on what level to which have you worked with it? Are we talking like SELECT statements, maybe some more basic updates, or are we talking you've needed to create tables and delete tables and modify and add columns, and then, I think you know, probably beyond that would be like, database ad and stuff. So kind of, what, what sort of operations were you doing?

Speaker 1 42:06

So basically, on the like, I'm primarily involved in writing the complex queries using more like joints, and then also updating the database with the schema using liquid base,

Speaker 2 42:24

cool liquid base. So with your liquid base setup and your joins, what, what sort of would you say your queries, primarily that you've been writing or, like updating things on liquid base. Has it been primarily through like repository methods, or like changing things in the app, or like views in liquid base? Or would it actually even be modifying like the DDL, like changing columns via liquid base and whatnot?

Speaker 1 43:00

So if I understand the question, right, yeah, so just changing the column in liquid base, you know, like you can just do it by updating the database schema, so changing the codes in a liquid based by adding a chain set to where we basically have those records. Yeah. Layers,

Speaker 2 43:23

okay, okay, um, have you had just a few random one off things? Have you ever had to use anything like query hints out or FULL OUTER JOINs or explain plans? Have you had to do any explain plan analysis?

Speaker 1 43:39

Yeah, sure. So I think, like I had to do an,

Unknown Speaker 43:46

like many

Speaker 1 43:49

those queries in that past. And then I also, if I, if I did get your question correctly, we were trying to do an migration from the Hibernate to query as like APA API to the Create a chip APIs. So it is a small effort, but for some of the joints that we are using, it was causing some production performance issues. So basically, then we had to, you know, like, run, explain the commands on our postgrads to basically see what was stopping. So we, basically, what we try to do is we added an index on one of the joints which were missing earlier. So those kind of work.

Unknown Speaker 44:39

Okay, cool. I cool.

Speaker 2 44:45

All right. Kind of branching out from that a little bit. I think you mentioned this earlier in your intro here. So can you describe how your previous projects have approached code reviews? Were you responsible for code reviewing other people's code?

Speaker 1 45:01

Yeah, yeah, that's some that's like interesting fruition. So yeah, I, or we as a team, we regularly get involved in our code reviews to see that if there are any changes that we made, I had to get at least one approval on GitHub to be merged into the our developer branch, which we have. So basically, you know, like the changes you push down, you open to the request, right? And those March requests, you basically ask for the teammates to review. And the main thing was, like, we look for his functionality and the standards that we have set within the team and also for the company, like, you know, the current patterns, like the naming of the variables or users of a constants and error messages and so on. So apart from that, we also like do like test units or unit testings, to see if we it has a proper unit tests and covers like different scenarios and so on. So it usually is a very constructive way to do that. That's how I think that's my approach. So yeah, I've been involved

Speaker 2 46:14

in Have you ever had a circumstance where you've had a difference of opinion with another developer as a result of a code review?

Speaker 1 46:21

Yes, that does happens. Yes, we all know that. So we were arguing specifically, like about if we should do like a simple string comparison or objective mapper comparison, so on one of our dataset. And then, basically, my idea was to wrap an incoming object through the object mapper and then do the another comparison approach. But my teammates just wanted to do like strings significations of the incoming object using the twisting methods. So then compare those, you know, like on the reason we had like contrasting view because of the requirement. So our requirement was like, so supporting a sorting fields, but it was very open, and we could be like, accepting more fields or more discoveries were made. And for the equity check, we didn't want to leverage any fields. So basically, the oxygen mapper would be like configurable to just fit those fields and that we wanted and then compare. So yeah, we did have to sit on the meetings to compare and contrast the different approaches we had in our pros and cons, so basically, come up with a way that we could basically fit the requirements in the long run. So yeah, I feel like whenever we had, like, contradictions, you know, like, in my opinion, like it's opportunity for both both members, or any team members, even for the whole team, as well, I would say, to learn on the different approaches and try to see what best fits for our different scenarios. So open conversation is always a best method, I guess.

Unknown Speaker 48:12

All right, sounds good.

Speaker 2 48:17

Last question I've got before I let Doug or Elaine, ask any questions they might have. So what are the characteristics you look for in a company to help you thrive?

Speaker 1 48:29

So characteristics about the company, I would say like, for me personally, I think like, the most important ones are like, because we work a lot in a team. So team collaboration is a very crucial for me, and where you know, like you have a very supportive team that supports my work together, and we do the work together to achieve our common goals. But apart from that, like providing opportunities for continuous learning, or, let's say, for a group which everybody desires, maybe under some courses, like on a regular basis, or, you know, with some time,

Speaker 2 49:12

yeah, definitely continuous learning, part of part of the gig, right? All right, yeah. Doug or Elena. Did you have any questions you wanted to ask before I kind of open it up to Ruchi,

Speaker 7 49:28

I can I have one so in your past experience, can you talk about, sort of how, how the dev team worked with the business users? Did you have bass on your team? Did you work directly with you know, your business users or clients in any of those situations?

Speaker 1 49:48

Yes. So definitely, we need some math this business analyst on our team who basically, you know, like complete masses, from our date teams to the business people, but I have been involved in meetings like this, involving the business during our UAT phases, where these user asset 10s test things, and we have come with, come up with, like, different scenarios, let's say, with them, and then we try to discuss in a more non technical way. That's how we'd like to put it. And not everybody gets involved in direct communications with the businesses, but I've been involved in those discussions as well, and I try to explain what to do, right? So it is easy to work, I would say, to explain what you are doing to technical teams. However, when it comes to the businesses, I was trying to keep it very, you know, like a handy examples, so they can understand or they can visualize things. Yeah, everyone might not be, you know, like, familiar with technical jargon, which I am very aware of. And, yeah, it's not like much regular meetings with the businesses. But yeah, we do occasionally have times during some of these testing pages or some kind of requirements gap, we have

Unknown Speaker 51:10

awesome thanks.

Speaker 2 51:15

All right, cool. Well, at this point, I want to give you an opportunity to ask us any questions you might have had. I know you've had an opportunity, at least with Rachel, to ask questions, but it got the desk now, so maybe, maybe that'll change some of the questions you ask. So now it's

Speaker 1 51:33

my personality. So yeah, I would say, like, I guess one of the questions that I have for you is, like, what do you like? What do you like most about working here? Like, what makes you excited about working here?

Speaker 2 51:56

I can do this. So I would say, I feel like I can make a difference. I guess there's a lot of a lot more power to do good here. I've had previous employment a large insurance company, and it's very nuts and bolts. Like, you know, you sit in your seat, you have the access to this small slice of the stack, and you do X, Y and Z, and you get it done, and you don't really get that feeling of ownership or understanding of what's actually going on. So feel like I get a much larger slice to be able to focus on things that feel like I can actually make a difference on and kind of contribute in a more effective manner. So I

Speaker 1 52:42

can definitely understand that, and I can definitely feed

Speaker 8 52:48

you. Yeah, I can add to that. Unfortunately, I have to chop off the Colin just a couple minutes for for a two o'clock but, yeah, I think for me, it's sort of that the company treats you with respect, that, you know, it's sort of like they treat you like an adult and a professional. And so, you know, definitely in line with what Sean was saying, that you know, if you think you can improve something, if you think you can make a difference, obviously there's responsibility on you to put forward a sincere effort. But if you're willing to put forward the sincere effort, the company will, respect letting you give it a go and explore.

Speaker 1 53:27

I have one more question, so, like, regarding the opportunities. So what about like, what you guys provide for, like, the growth of engineers on a team?

Speaker 2 53:42

Oh, I'm just within our team, a lot of what we do, at the very least, like I pair with people, just to kind of help them out. And I think most everyone I've never seen someone say help. I'm stuck on something. And, you know, nothing happens. People will help each other out. And I think, you know, ultimately you end up as a developer. You just, you need one on one time with other devs to kind of improve your skills, learn things, and ultimately share knowledge, because not everybody, not one person, is not going to know everything, and it's all about collaborating. So that's that's kind of what we try to do. Remote makes it tougher. But you know, if everybody's online, I just happen to calls with people. It's not really a big deal. It's just kind of what you do. It's part of normal day. So some days, I'll get on calls with people a lot of the day, and I'm sure it's true for others. Other days, people kind of know what they're doing. So it really depends on what level of assistance or help can be provided. Sounds great.

Unknown Speaker 54:48

Sounds great to

Unknown Speaker 54:51

me. So, yeah, I would say that thank you so much.

Unknown Speaker 54:55

Really, knowing the whole

Speaker 1 54:58

like the group members, and I think it's really exciting that to know about the team.

Speaker 9 55:10

Awesome. Well, I can hop back in here, then, if no further questions from anyone else, okay, well, this is the end of the process. So you've made it through all the steps. So what it looks like from here? Of course, we have several candidates, you know, in final stages. So we'll be in touch with you as soon as we can about our decision, but it probably won't be until probably nearing at the end of next week. So I'm thinking, sorry, looking at the calendar, thinking February 7 is probably one. You'll hear more from me. Do you have a question? Sorry. Okay, so you sound looked a little surprised, but you know, of course, you know, just because I, you know, won't be reaching out until probably end of next week. If you have questions that come up after today, you can always feel free to email them to me, and I'm happy to answer them, but hopefully we'll be in touch sooner. But I'm giving myself a little bit of buffer time. So, but we do really appreciate the time that you you took in the process because we know you know you spent some time at home working through things independently and then came on this call as well it's a big commitment, and we appreciate you spending the time doing that.

Unknown Speaker 56:18

And, yeah, just thank you. Yeah, I would also.

Rajeep-paypal-SRV-First

Unknown Speaker 00:00

Ground experience and

Speaker 1 00:02

what excites you to be in this role, definitely. So first of all, thank you so much. Thank you so much for giving your time.

Speaker 2 00:12

My name is Rajiv khaki and I've been working as a software developer for over almost like five years to six so working primarily on Java based web applications, so utilizing a framework like Spring, Hibernate and REST APIs for the microservice development. So I have been also involved in designing, let's say, like the microservices and deploying them by doing our AWS cloud, some of the cloud services that I work on, it goes like AWS, easy to instances or SQS, SNS and lambdas. I do have a pretty good, like, a front end experience with Angular as well as the React. So currently, like, I am in a scrum team. The team simply consists of, like, let's say, five developers. We have, like, product owner and Scrum Masters on our daily basis. Like, let's say I'm involved mostly and working on the back end Java code, helping the team with the code reviews also, and help also in the production support. Yes, this is my like, short introduction. I would say,

Unknown Speaker 01:19

awesome. Got it.

Speaker 1 01:23

Okay then. So the purpose of today's interview is to evaluate like problem solving skills and coding skills. It's designed to be interactive, so I'll try, you know, feel free to ask questions. I'll try to be helpful. I'll present you with a coding problem, and you can use for most of the interview to work through it and feel free to, like, think aloud, so I can try to understand, like, how you approach the problem and stuff. And towards the end, we.

Rajeev Bam-Lowes-First-Srv

Speaker 1 00:00

A basic migration from like, you know, monolithic. So basically, we define a data module on what kind of a signaling filter that the user might, you know, like prefer to view, and any kind of prefer stores or like, you know, reason, or like, anything like that, anything like that. And notification toggles that we had right so it was a Spring Boot based application exposing like those in pointer. We use Postgres to, you know, to pass this preference. And we had, like, multiple gate post talking points for like user preference, and also like a put in point to like, you know, to make update on that if needed. And so that that was the most recent one we did, and where, like it was one that I took charge of, like, you know, designing and also lefty gloving

Unknown Speaker 01:02

when build and REST API

Unknown Speaker 01:05

and and that REST API was,

Unknown Speaker 01:08

like, you were getting some data,

Speaker 2 01:12

like somebody was posting it, like it's it's an, it's an, it's a post API like, which we exposed for someone to call it. Is it

Speaker 1 01:22

so is it user preference, right? So we have a crowd operation,

Speaker 2 01:27

okay, entire crowd the journey. You build it for the profile updates and all. And that was like, what kind of database you said we used?

Unknown Speaker 01:34

We use Postgres for Postgres

Speaker 2 01:37

database, okay? And you build this integration completely, like exposing all the PS needed for the current operation and whatever the database interaction, everything you will, yeah. You were responsible for building this,

Speaker 1 01:53

yeah. So what I did was it, I believe, from like, design document getting approval from, like, our architecture, like working on, who was actually working on the services, and then, yeah, I was able to involve in the development by, like, In doing, okay,

Speaker 2 02:20

okay, so if I have to ask you, can you hear me? Yeah, I can hear you. I think there's some network delay. Actually, I am not able to hear you. Well,

Unknown Speaker 02:34

can you hear me? Now,

Speaker 2 02:38

I can hear you, but somehow I'm not able to see your image like your video.

Speaker 1 02:47

Can you? Is it good? Can you see me? My video is on so can

Unknown Speaker 02:52

you, I can't see you, but it's blurred. Oh,

Speaker 2 02:56

now it's little better. It's getting little better. I don't know what was there any network range or something? Now, it's coming little better. I can, I can see you now, okay, I don't know, some network glitch. Okay, that's, that's good. So you, you, you develop this entire so this was mainly as a back end developer, or did you develop the front end also for this entire thing,

Speaker 1 03:22

for this particular one, it was just a back end part, so front end was integrated already so it was just like, you know, migrating of like existing workflow to like new services. So front end was already done.

Speaker 2 03:40

Front end was already done, okay, but you have experience building front end as well. Yeah, because you mentioned he was a full stack developer, so, so was it like with React front end development?

Speaker 1 03:54

I have worked primarily with Angular, but I do have experience in like, reacting past experience. But it was, it was Angular for this.

Speaker 2 04:08

Okay, so I, if I have to understand your COVID stat, I think you have more experience working on Java, Spring Boot, Kafka, AWS. I

Speaker 1 04:23

did I miss anything? Yeah, most of them, yeah. That is, those are okay.

Speaker 2 04:28

Those are the core things, and, and it's almost like four years of experience, which you have? How many years of experience?

Speaker 1 04:37

Almost five years of experience, okay, almost five

Speaker 2 04:41

years of experience, nice. So we will get into the technology questions, but before that, we'll do some coding exercise. So do you have any I think, do you have access to laptop? I mean, you can, you can write code now, right? Yeah, I can write, okay, that's good. So can you share your screen? I'll post a question in the chat, and you can use any of the ID, like whatever you have installed on your machine. I don't have any preference, and whatever you prefer is, give me a

Unknown Speaker 05:16

second. I have engineers, so I think I can, I can use it. That's

Unknown Speaker 05:19

fine, yeah, let me.

Unknown Speaker 05:32

Give me a second.

Unknown Speaker 05:42

I can you see my screen?

Speaker 2 06:12

Yes, I can see your screen, but I don't see the ID yet. Now I can see the ID. Okay. Can you, yeah, just be on this screen. I will post a question in the chat, and then you can copy the same question and put it in the comment section

Unknown Speaker 06:33

in this okay, sure, sure.

Unknown Speaker 06:38

Can you see the question in the chat? Let me check

Speaker 2 06:43

and I can help you, like you can go through the question. I'll give you a minute. And then if you have any, like, any questions about any help in understanding this problem, We can discuss. Sure. I

Speaker 1 08:18

so I think I got the question, question. So basically, we just want to, like, do the URL sorting,

Unknown Speaker 08:29

coding and decoding,

Speaker 2 08:32

right? So given a long URL, it has to encode it and return your short URL, and other way around, with short URL is passed, then it can decorate and give it back view in a long URL.

Speaker 1 08:58

So what I think is we can just do like hexadecimal character or like given, given characters,

Unknown Speaker 09:08

and then we store like,

Speaker 1 09:12

store that somewhere in like hasmap that takes like, you Know, care of like takes a care for saving it. It.

Speaker 2 09:24

Okay? So you're going to put that in a hash map, okay? And when we have to read to you, you can return from the hash map. Okay. Now let's think like, if, if these are, like, millions of the service gets popular, and, if you like, say, people start using it, and there are millions of user who are there are, like, millions of URLs which needs to be stored. So will you be storing it in hash map?

Unknown Speaker 09:53

If there are millions of user,

Speaker 1 09:58

if there are million then the hash map is just going to, you know, like, going to be very bad because, like, because of the memory uses, so you will have to use it to restoring, like, some kind of, like, external, external sources, like, maybe I would say database, obviously.

Speaker 2 10:18

Okay, what kind of data is like? Do you think

Unknown Speaker 10:25

what would be the good database to use?

Speaker 1 10:29

I think non relational database would be better where we'll just, you know, like, like, do the key and value structure and like, where, where we have short code with a long URL and like we can like we can also do like other way around, but it will just basically be a very good for like, an throughput and for original scaling, because if, like, there is no strict schema here, and there is also No need of consistency, right? So I think just a relational database, like non recent database, like DynamoDB, should be good. Okay,

Speaker 2 11:14

okay, let's, let's not worry about like the millions of user for now, maybe if you can, okay, Like, whatever you're thinking, right, can you start coding it?

Speaker 1 11:59

BBB, so we will have to use like, this has map to actually, like, restore the tiny URLs from given input.

Speaker 1 12:12

And we can also, like, if we want it, and like other way around so we can use it. So we want to, like, have like tiny URL map to the main URL. So we might want to do like no another one that's just opposite of this one. I

Speaker 1 12:43

so we create like two methods that will be used for like encoding and decoding. Okay,

Speaker 1 13:20

our problem here would be, like, it always be this base path here. So this base path would always be on, like, low problems

Unknown Speaker 13:33

that always be, would

Speaker 1 13:37

that always be the base path? Or would you want me to do something different?

Speaker 2 13:41

Yeah, the base path, you can assume that it will remain same. You

Speaker 1 14:42

BBB, so what we need to do here is that we need to extract this part out, like after the problem, and then we need to encode it.

Speaker 2 15:06

Be the output. You can assume the base path will same. The input can have anything.

Unknown Speaker 15:16

Input can be anything, okay, so,

Unknown Speaker 15:25

okay, that makes

Speaker 1 15:31

sense. So basically, we get everything and then encode it and so just do the decoding pass first. I think that would be the better way do better.

Speaker 1 15:51

So here we will have actual decoded code.

Unknown Speaker 15:57

So can discord part can just be a

Speaker 1 16:27

decoding part should be used to forward as that so, right? So you just replace the base part and get it from the hash map and just return the value. I think that's

Unknown Speaker 16:51

and just replace the baseball interior.

Speaker 1 17:02

And so for encoding, I think we need to do this in like. We need to basically, like, generate a random ID associated with the URL, and then we can Put it In hash so

Unknown Speaker 18:45

can just get six character of you like it?

Unknown Speaker 18:54

Okay? I

Speaker 1 20:10

so what I'm trying to do here is that if I can see the key already exists in the HashMap, because if that's the case, then, like, we don't want to replace it, Right? Okay? I

Speaker 2 22:55

Why are you doing this? Do While get key. I'm sorry. Can

Unknown Speaker 23:02

you just get key while

Speaker 1 23:05

I'm doing so that I'm checking key already exists in my lab. So if a key is like, you know already is there, then we don't want to, like, replace replace it. Then that's the primary reason. Okay, if I do that, I don't think we might even need two laps. I

Speaker 1 23:55

So the way I'm thinking about this is what happens is, whenever you do go right, I basically get a key associated to for the path and base path right. And then the key would basically be a random UID from like the first six character of the UID. And then, if my map is like, until it contains the that key or then we can generate another one until, like, we get a unicorn, right? So this might be a little bit more processing, but I think this would be the easiest way for now, like, then we can basically, like, append it with the base path and then add it to the like, other aspect, and then return it so

Unknown Speaker 24:51

I think this should be the solution for For Now, I

Unknown Speaker 26:16

so this is our solution, Alright? So

Speaker 1 26:20

we encoded, then we decoded. So that should be our solution that we want.

Speaker 2 26:28

Okay, now let's see so that decoded URL, right? Go, go up to your solution.

Unknown Speaker 26:38

No, from where you're calling it right?

Speaker 2 26:43

Yeah, here can you put that decode URL one more time?

Unknown Speaker 26:49

Just the decode URL part,

Unknown Speaker 26:55

okay, and print it again. I

Unknown Speaker 27:13

Okay, so

Unknown Speaker 27:18

now there's one thing which

Unknown Speaker 27:24

COVID is down. I

Unknown Speaker 27:50

Okay, why these methods are static?

Speaker 1 27:58

Well, the reason here is we are calling it from static block, so only like one main method. So that's the only reason, because you can only access main method directly, which are like static from static block. So main method is static, that's

Speaker 2 28:21

what. Why? Do you think that they have that restriction

Unknown Speaker 28:28

that you cannot call a non static

Unknown Speaker 28:33

is because

Speaker 1 28:36

static methods like you know, like can be, can do not belong in any instance. So it's basically belongs to that particular class.

Unknown Speaker 28:51

Okay, so when you say,

Unknown Speaker 28:54

Okay, what is the class?

Speaker 2 28:58

What is the difference between class and instance? So,

Speaker 1 29:04

class is a blueprint. So basically, it's

Unknown Speaker 29:10

just a definition, right? So,

Speaker 1 29:13

as I said, like on what certain objects has, it is a blueprint of that object, and what are the attribute and, you know, like, it contains, like, what behavior for that particular object, but actual intent instance is like, you know, like once the class gets realized or get initialized. So it occupies, like, a certain memory, and it is an actual entity that you we can interact

Unknown Speaker 29:47

with, Okay, so where these instances get stored

Unknown Speaker 29:51

and where the classes get stored. So, instance,

Unknown Speaker 29:58

it is stored in

Speaker 1 30:02

class, classes exist in the memory, your memory. Once it is defined, it is stored in the, I believe, instance, in the gib memory, in actual

Unknown Speaker 30:17

memory, in the stack memory, sorry,

Unknown Speaker 30:20

tag memory, what classes

Speaker 1 30:26

objects are in, like a heap Memory, where once they get initialized, and class, classes, hmm,

Speaker 1 30:43

it lives on like, you know, like, I mean, it's just a meta space. I don't think, I don't I think it does not actually, like, have any signals, unless any slice. So I don't think glasses

Unknown Speaker 31:00

gets it stored in any memory area.

Speaker 2 31:04

Okay. So what is so you mentioned, what is stack Do you understand the stack data structure? What a stack data structure is?

Speaker 1 31:15

Yeah. So basically, stack data structure is first. Come,

Unknown Speaker 31:24

first come and

Speaker 1 31:27

first out. So, right? So it's when you push an element, then it adds to on the top, and then when we need to, like, remove it, it removes from the top element.

Unknown Speaker 31:41

You said first in first out? Yeah,

Unknown Speaker 31:45

I sure started, oh, sorry,

Speaker 1 31:47

sorry, my bad is last in first out. My bad, sorry,

Unknown Speaker 31:52

what is first in first out?

Speaker 1 31:55

So for the first in first out is, is a queue?

Unknown Speaker 32:00

Queue? Is data

Unknown Speaker 32:03

Okay, have you used queue data structure?

Speaker 1 32:08

Yes, I have used Not, not that often, but, but I have like, in some cases you might want to work with, but not that often, as a stack,

Unknown Speaker 32:20

okay? Kafka cube.

Unknown Speaker 32:24

Let me think about

Speaker 1 32:28

Yeah. So Kafka is Yeah right, because it poses a particular partition. So you, you put some message in, like, you know, like particular parts, particular partition of that topic, then message get processed in order. So, yeah,

Speaker 2 32:46

in order. Is that always guaranteed in Kafka that messages will be always processed in order?

Speaker 1 32:56

So basically, Kafka guarantees that message order orders in a partition,

Unknown Speaker 33:05

in a partition, yeah,

Unknown Speaker 33:09

okay,

Speaker 2 33:15

okay, and how does Kafka puts the messages to in A particular partition, then, like there are, say, five partitions in a Kafka topic. How who decides that this message should go to zero partition they should go to first, this should Goes to second.

Speaker 1 33:31

So, while talking about Kafka partition, there is some kind of we have a partition key that is, you know, like assigned to like certain messages. So when your, let's say, producer, sends a message, then there will be like hash or like something, some key, and that gets calculated and like based on those partition key, it gets to the like particular partition if, if you do not provide any kind of like partition key, then it will just do like, you know, like a round robin. So it will go from first to the first to the last, and then keep on like, you know, like, certainly, okay,

Speaker 2 34:28

let's, let's. So now you wrote this program. So have you used any of the object oriented principle when writing this program? I hope you understand the object oriented programming. So do you think you've used any of the object oriented programming concept here while writing this program?

Speaker 1 34:47

In this program, I haven't used any like object oriented principles here, because, like, like, you know, it's just declaring everything static and, you know, like calling it directly. But it was like an ideal scenario, or we would create an external solution, service which will basically has an interface, and then we implement, like encoder. And

Speaker 2 35:14

so when you why to create an interface, then like, Why do you think that's like an good approach to have an interface.

Speaker 1 35:23

So basically, interface gives you an like, you know, group blueprint on like, what, what your things are, like, what the services do. So basically, it focuses on like, like, certain services do, rather than how it does it, right? So, for now, we might be using UID, like a random identifier, like to get the key. But in the future, you might really want to go with any encoding approach, or some like, you know, some kind of, like, other hashing algorithm, or like, anything like that. If you want to switch from one to another, you don't like have, you know, you don't want to do it again. You don't want to rework your like, actual project user. So it's then you can, you know, to you can use have a new implementation, and it can help with, like in dependency injection, provide, which is provided by spring, and like, you know, change if we want to change the dependency. So that's the basic, basically extraction

Unknown Speaker 36:38

for the functionality. Okay,

Speaker 2 36:40

so you mentioned about dependency injection. So what are different type of dependency injection?

Speaker 1 36:49

So we have three types of dependency injections. One is a field failed injection, and there is construction injection, and we have also sitter injection. So talking about construction cluster injection, which is like, basically for a required dependency, which where your dependency are injected, like directly on the constructor, then you have your sitter injection, which is like a which is like, depends injection done on, like, a public sector method, and then you failed injection, which you directly inject for the class field.

Speaker 2 37:30

Okay. So what if you don't use dependency injection? Can you still code? Like, let's say, if you're not using any Spring Boot framework as well? Can you still build your program or, like, can you still develop a step here? So

Speaker 1 37:44

in case we are not using an spring, yeah, so we can use it to maintain this. Can in just

Unknown Speaker 37:54

as just a pattern, right? So it

Speaker 2 37:56

may use that, yep, good. Idea. So

Speaker 1 38:02

it makes our, like, working life easier. So, you know, like, you don't have to worry about, like, a lot of external sources. But if you don't want to do the dependency injection, you can initiate it, like those objects, like you yourself, then work with it. So instead of like injecting dependencies, you create like new instance of like, those are on those objects. Like, for example, if you are working with like a wrist rest APIs, where you have like a entry point on a controller. So instead of like, you know, injecting your services, you would like to do a new user service, or like a new employee service, and and on the Employee Services, instead of doing like injection of repository, you need to do like a new data access layer. And then, like an each of those, you're just doing a manual. You will be doing manual so

Speaker 2 39:02

you so you think, like, just to avoid writing new every time you are doing that.

Unknown Speaker 39:08

Basically, the idea is, instead of just

Speaker 2 39:11

to avoid, like, just because, do you think that's like too much of a code to just instantiate an object, like with a new and then use it? Is that too much of a work? I mean, I don't think that's too much of a work, right? You can you just have to write one more additional statement, that's it. Is it?

Speaker 1 39:30

No, it's not like just for that. You know, dependency is actually also like if you do a new object every time, then, like, how would you decouple your like, servicing, right? So today you are using, let's say we're using SQL database, and you might want to go with like, let's say Dynam, right? So if you want to, like, basically, do the do that change you would, but I

Speaker 2 40:00

can use an interface in that case, right? I don't need Spring Boot to use an interface. I can still, as you explained, right? If I have an interface, I can change my implementation correct. So if I'm using some database, I can switch to other database, and my interface won't break any contract you mentioned right in object oriented programming. Now, if I don't, but I don't need spring to do object oriented programming, correct, I can still do it spring. Good. Okay, let's do one thing. Can you copy this code, whatever you written, put in the chat and then stop sharing. Discuss further.

Speaker 2 40:55

So you can stop sharing. Okay, so now we were talking about why Spring Boot is needed. So we were talking about why Spring Boot is needed, like we were discussing, maybe I can still program or still build an application without using Spring Boot, then what is the need of using Spring Boot, right? That's what we were discussing. And we discussed about dependency injection. Okay, let's talk about, let's talk little. Let's say, now, let's, let's say we understand that we need spring good. Okay. Now, what are the different modules available in Spring Boot. So models not in Spring Boot, let's say like in spring itself. Spring, do you understand the difference between Spring framework and springboard? Yeah, yeah.

Unknown Speaker 41:55

Okay, so what is the difference?

Speaker 1 41:58

So spring is just, you know, your regular framework, and Spring Boot is like a little bit more like a which is built on spring, but you you will just get, like a simplification, like and streaming of your it helps you in streaming your project. But and Spring Boot we use is like auto configuration in a sensible fashion where, like, you know, like in every everything is in your class and path is available as a dependencies, and sometimes may require, like, you know, a lot of boiler plate in a spring, but Spring Boot reduces, reduce this, like, with, like, auto configuration based approach. So

Speaker 2 42:45

what is auto configuration? I think you mentioned it couple of times, right? So let's see, understand, what is auto configuration? I mean, give an example, if you know you have used Spring Boot, right? So maybe give an example where auto configuration was used by using string boot.

Unknown Speaker 43:03

So basically,

Speaker 1 43:07

while working with Spring Boot, you can add dependency on your like build tools, right? So let's say you add a dependency of an actuator and you add a string restarter actuator dependency. So what auto computer, auto configuration. Does it like? It sends that particular dependency available like in your class pack, and then it gets is available for like for you. So it automatically, automatically configures those bin and then it gives you an like availability of the actuator in point. And like, no you, all you need to do is just like, go to your application properties and like, say, hey, I want this endpoint to be available, like, under actuator, and then it will give you like metrics or like a health or like info, INFO of like in point available. So similarly, if you are like working with, let's say another Spring Data GP, right? So it will automatically configures your database like provided on your once you provide that in your application, property,

Speaker 2 44:24

file, application, okay, so you mentioned about actuator, right? And you mentioned about the members, the metrics. So what are like? Say you are writing a spring good service you have written, if you expose an API, and that API is doing some like, you build an entire application which can do current operation, let's say for Profile Management. Okay, now you deployed this in production, okay, and then you want to monitor it. Because you deployed it in production, it's running fine, but you have to monitor it correct. So what are the different metrics you will monitor? Like, what are the different metrics you think you should monitor for an application, like a springboard application, which is running in production? I hope you have deployed any service in production and developed some metrics dashboards or something like that, also monitor it.

Speaker 1 45:14

Yeah, I'm experienced, like with this kind of with this, some of the metrics that you should consider monitoring would be actual request like, what's the request volume? So how much request, like, you know, per particular endpoint are being hit, like was the network like? Like, you know, like, what's the network latency like? And so on. And also we can check, like, what is the CPU and like, CPU memory usage and like, what is the disk uses are? We also want like, if possible, like to add like, a dashboard to do, like the connection pool uses and so on. And you also want to You see, like, your API response time as well. Like, wow. What's the data volume? If like, possible. So these are, like, some of the primary, some of

Speaker 2 46:14

the primary. Okay, so now, when are you building your service list? How are you like, how are you promoting it? Like, how are you deploying your services to, like, say, your staging, like, Q environment or production.

Unknown Speaker 46:30

Where are you deploying it?

Unknown Speaker 46:31

So we will use

Speaker 1 46:36

Jenkins for, like, our automation. So you will have a pipeline and that we deployed. And through this Jenkins pipeline, we just use like cloud formation template to like provision the like that stack to AWS. So once our micro services are deployed in AWS ECS of fragment instance, we which are like, you know, containerized application.

Speaker 2 47:06

Okay, containerized, right? So, what is containerization? Do you know any technology that is containerization?

Speaker 1 47:15

Yeah. So basically, I have worked with Docker as well. So, okay, so we use the Docker for containerizing and containerization in just the process of, like, you know, like packaging your application and all this dependency into, like, a small portable unit, which, you know, like, which we always might be like, referred to as a container the portable unit, so it usually contains your applications, your dependencies and the runtime and the configuration that is needed for it. And so, yeah, I work with Docker for containerizing my Springboard microservices. Okay,

Speaker 2 47:58

mainly Docker, okay. Have you? So do you have any experience on working with infra, like, etc, you kind of configured the cluster or anything that sort of work. Have you done that? Yeah. So, like, like, when I say in so let's say I you have been given a task to build an Elastic Search cluster, or maybe build an

Unknown Speaker 48:29

Kubernetes cluster.

Speaker 2 48:32

Yeah. Are you experienced with Kubernetes, like you use Kubernetes? Yeah. So

Speaker 1 48:40

in my past project, it we had a COVID But recently, instead of Kubernetes, we use as AWS ECS, which is, which is also very same thing, you know, like, but you know a different tone, because AWS ECS is like a container orchestration platform which is dedicated for AWS and Kubernetes is more like global, so I work with provisioning AWS ECS services and so, yeah,

Speaker 2 49:12

so you have, you have configured that easier that

Unknown Speaker 49:19

AWS service by your own? Yes.

Speaker 2 49:23

Okay, so, so help me understand one thing, like, when you deploy a service, let's say you look at certain memory to it. How does like, it's a Java program, okay, like a Java process right in like you develop a Spring Boot service, basically, it runs as a Java process inside the container. Now, what really happens is maybe, if you're talking about in coupon distance, you have something called pod. I don't know whether you understand that pod, so you allocate some memory to the pod. So what actually, what amount, like, say, you allocate one gig to the pod. Does that one gig of a memory allocated to your hip? Talking about you understand the question, right? So whatever resources you allocate to the container, whoa, like all those resources are actually given to the the process that is running inside that container.

Speaker 2 50:28

So when you allocate the resources, when you deploy, you must be allocating resources, like memory, CPU. So you allocate it to the container, or maybe, let's say pod, if it is Kubernetes, right? So if you are allocating it to the pod, what do you think like? How much of that memory and CPU gets allocated to your process, but is running inside that pod or inside the content? So

Speaker 1 50:55

the memory allocation within certain pod works in is that on a modern JVM, so it will do is it will detect the container limit and then scale, scale the heap accordingly, right? So if you allocate, like you said, one one GB to a container, then the JVM will will be the container away, and then set the like, you know, like Max XMS to the like half of half a bit. And then maybe use the rest of any kind to, like, you know, overhead. So it's like, actually total memory budget. And he will only get like, a portion of that one one year,

Speaker 2 51:45

got it. So let's start little bit about databases, right? You said you have used poses database, so I hope you have experience writing queries in database. Okay, let's, let's take a see very simple example. You have an employee table in database, okay, employee table has employee ID, employee name, employee department and employee salary, four columns, four fields, okay, name ID,

Unknown Speaker 52:18

department, name and salary.

Speaker 2 52:22

Okay, very simple. Now, I wanted to write a query. You can actually write it to get me the list of employee whose salary is, say more than like 60k per year.

Unknown Speaker 52:40

So list of the employee damage, salaries, 60. You said, right, yeah, 6060, so

Unknown Speaker 52:50

I will do is what I will do.

Speaker 2 52:57

Okay, this is one query, okay, and other query is find out the find out this department was in which department the employee has more than 60k salary. So we want

Unknown Speaker 53:13

to figure out, with the department, bases on the department,

Speaker 1 53:21

what I will do is like, I will select everything. So this will be the my like first carry, we will select, select the star employee where salary is greater than 60,000 and for the and for the second question, did you want the count of employee per department who have more than

Speaker 2 53:47

6000 That's correct. So

Unknown Speaker 53:53

what I will do is to

Unknown Speaker 54:00

so if you want to

Unknown Speaker 54:03

find out the count of

Unknown Speaker 54:22

employee, I

Speaker 1 54:34

for the second question, what I will do is, how will you do the grouping? I think that should be the query, because we already have the employee that has the higher number than the 6000 so we'll just do the grouping on it.

Speaker 2 55:00

Okay? And now I want to list down a department where the count is more than like 10. I mean, just exclude the department where the number of employees are less than 10.

Speaker 1 55:18

So the if number is less than 10, we just want to remove the department, right?

Speaker 2 55:23

But if the count is less than 10, we don't have to list that in the school. So

Unknown Speaker 55:27

does? It has to do anything with the salary. No, right?

Unknown Speaker 55:30

Yes. Salary has to be there an additional party.

Speaker 2 55:35

Well, you we are just building the query. So like from starting then, okay, if I want to do

Unknown Speaker 56:11

Sorry, full time with Home Depot.

Speaker 1 56:13

No, I'm contract with Home Depot. So Home Depot is my client and my employees.

Speaker 2 56:30

What's your employer name? You said K force. Okay.

Unknown Speaker 56:45

So for this query, we'll just need to do like, have a

Speaker 1 56:52

having a cloth, having clothes, like on top of the group go by to do it, to do this. So we'll just,

Speaker 2 57:01

okay, yeah, I think right, you I think we are almost part the time, eight minutes so that I am done with the questions from I am. Do you have any questions for me?

Speaker 1 57:15

Thank you so much for your like, you know, like, your time today, and it was, it was really nice talking to you like, so, but I would like to know more like, a little like, more about like, like, what do developers use? Like, uses allocated, involved on like, apart from, like, like, coding,

Speaker 2 57:36

so, so for probably this role, right? I think, let me tell you a little bit about what we do. So we are from engineering platform and architecture team. So engineering platform and architecture team is responsible for building the platforms for those which are used by the other developers. And with that, we build some services on top of those platform we build some services. Like, that's where the development work is involved. We build some services which will help or ease their work while using those platforms. Kind of what you do, like, when you see easiest to write, you go and interact using an UI, and you get the your work done correct. So you don't know what goes in background. Like, lot of things will happen, but you use interact with an UI. So same, same concept, like you will have, we build a new I will build a service which can do certain automation on the platform side. So that kind of do so and, and for, for a typical developer, right? So most of the time, maybe 66 60% of your time might go in, like coding, then we will be 20, 30% will go in meetings and all and, yeah, that's pretty much it. And the technology stack, whatever, I think we interviewed right, pretty much the same technology stack the development purpose. There are other like platform specific technologies which can be different.

Speaker 1 59:00

Are the endpoints, like mostly internal, internal, or do you also expose them to

Speaker 2 59:07

external? It's It depends. Actually we have certain platform or certain projects where we are exposing outside, also external, and so, but mostly So, mostly, I

Speaker 1 59:20

like, I would like to ask, like, what would be my like, a next step moving forward.

Speaker 2 59:32

So once I submit my feedback, right then, according to the feedback, the recruiter will get back to you with the next step. I Step. Okay, sounds good. Thanks for your time. Raju, thank you so much. Okay.

Rajeev bam-Lowes-SRV-second

Unknown Speaker 00:03

Hello, yes, yes, I'm sorry I couldn't hear you.

Speaker 1 00:23

Okay, can you hear us now? Yeah, I can hear you, yeah. So basically, what I was telling you was the camera in the room where we are working for audience for that. Oh, it's just fine. This one,

Unknown Speaker 00:40

hey, Pradeep, hey, Rajiv,

Unknown Speaker 00:50

oh, you're in the same room. Pradeep, which is

Unknown Speaker 00:56

no worries. Okay, you guys can go ahead and start.

Speaker 1 01:01

Yeah, let's get going. Let's start with that quick introduction about your class background on what is in that latest project? Yeah, sure.

Speaker 2 01:11

My name is Rajiv, and I've been working as a software engineer for about five years. So primarily I use Java applications utilizing frameworks like Spring, Spring Boot, and I've also used hibernate, and I've used just Google Web Services for API deployment. So I have pretty good, you know, like, experience working on micro services, where I've been involving in, like, you know, designing, developing and, like also deploying them in AWS cloud. So some of the cloud services that I have worked with, it includes AWS EC two, ECS, SQS and like, SNS and also lambdas. So I do have a pretty know, like a good experience working on Angular as well for the 14 part and for testing, I mostly work with JUnit and Bucha. So currently I'm going to like scrum team. So my team consists of like, you know, five developers. So we have a product owner, we have scrum master and also, like, we on the daily basis, so I'm involved mostly working on the back end, part with Java code, helping, like, team with, you know, like, code reviews and, you know, like, also with

Unknown Speaker 02:36

Support duties as

Speaker 1 02:39

well. How was your typical like, during production support? Like, were you doing, like, hands on support, or engaging any other teams to support applications?

Speaker 2 02:53

So I have my hands on software engineering, so my like more development book. So, like, within your iPad, development of like, you know, like new micro services, but we have a team, like, as I said, like five developers. So we do product, product supports on like, like, a rotation basis. So that's the support role that I have worked

Speaker 1 03:21

with. So so for the monitoring, can you walk us through, like, what were the tools that you were using to take your applications, to set up dashboards and apps, and how were you like, you know, fixing the issues that you are facing when you walk through the example?

Speaker 2 03:36

Yeah, sure. So basically, we have a synthetic dashboard. So we we, like, basically an integration of promises on graphana. So basically we enable, like, our Prometheus in point by, like, adding the dependencies and like, you know, then based on those Prometheus metrics, we have Grafana dashboard, so, which is basically provided by like, our SI and partners. So we have like a dashboard set up, and we just use that, you know, dashboard like, to look at monitoring for our application. So it usually contains, like, you know, like response, like, status code, like, you know, like, response latency, like, what are the like, traffic, by, like, volume, like, by reason and like, you know, so on. So that's our like, primary monitoring, monitoring app and observability tool. So we, like, also have logs on exploring and which we use, if, like, you know we want to look for any application related logs.

Unknown Speaker 04:48

So you did use Grafana in your last project? In it?

Speaker 2 04:53

Yeah, I have used Grafana before. So it was in in the last project we used Graf

Speaker 1 05:00

so then what other specific monitoring tools

Unknown Speaker 05:05

that have been used? So

Speaker 2 05:07

talking about other monitoring tool I have also experienced with new ready, which is an APM dashboard, which usually, like, gets agent like, installed in like, in an instance, and, like, basically, this agent collects all the, you know, like, logs, and, you know, like, you do not have to do anything, like, apart from just installing those agents. So, yeah, that's, that's the one of the, like, other dashboard that I have used. So you just pass it to, you know, like a JVM argument, and, like, literally, start collecting logs, and you know, metrics, so they have

Unknown Speaker 05:49

used you have a lot

Speaker 1 05:51

of experience from a user standpoint, or do you also like know how to set up logging solution? I'm sorry.

Unknown Speaker 05:58

Can you please repeat the question again?

Speaker 1 06:02

So do you have experience using the logging solution from a user standpoint, or, like, have you ever set up a logging solution? I mean,

Speaker 2 06:18

not directly on, like, a solution, or how login goes, just like from like user perspective. But we just use like agent that is provided by like Splunk team. And so we install like those long forwarder agents within our like our instance, like during the deployment part. And then this is prong agent

Unknown Speaker 06:40

collector. So so

Speaker 1 06:42

basically, like you are from an application team where you will be deploying your micro services to some platform, and as part of that, you are using Splunk. So you said, like you are developing Microsoft. Microservices, like, how do you deploy those and when do you deploy those

Speaker 2 07:08

microservices? So, so in our microservices, most of them are containers using Docker container hands, and then we will, like, deploy it to our AWS ECS, this way is done, like, you know, all automated to, we use Jenkins, so we have like different stage, which basically, you know, like, goes to, like, build, you know, taste and, you know, like deploy phase. Then we use, like, cloud formation, like to like to provision, like the services on PCs

Unknown Speaker 07:46

going back to the

Speaker 1 07:49

COVID. Can you, how can you integrate metrics in a JAVA commission started so

Speaker 2 08:02

we sorry I like, I couldn't follow you. I'm sorry. Can you please repeat

Speaker 1 08:07

that again? So did you mention Prometheus in the beginning? Okay, can you walk me through how can they enable Prometheus metrics in a job application?

Speaker 2 08:20

Yeah, sure. So I did integrate. So in order to like, enable promises, you basically have to add like a Prometheus dependencies on your like Springboard application. So you can just, like, look it, lock it up in like, mapping Google strip, and then you just add the dependency on your build tool. So once that is done, you basically also need to let you know, make sure that the actual you have actuator in point enable. So for that, you will just add the springboard starter actuator. And then you know, like on your application, on your application properties, and you make sure that the properties endpoint is exposed under the management endpoint. And once that is done, you can just access it using like actuator, using a province endpoint.

Speaker 1 09:17

So then once you clicks are exposed. What after that? How can you export those metrics like you also mentioned, right? How about

Unknown Speaker 09:31

so once the

Speaker 2 09:33

so once, the way the Grafana reads it is that is basically like, you know, takes the snapshot of what like promises exposes for like, let's say, for example, your promising point will have stuff like JVM memory, so used in bytes and like, you know, JBM request, like HTTP server request, or like, anything like that. It will have, like, you know, all the metrics exposed, so, like to that particular endpoint, right? So it will be some kind of a plain text key, like a key value pair, and then basically a graph on our queries, and then your promises in your graph on it, you would like, you know, I have a data access source configured so to pull that data from like promises, yeah. So in our case, I would be, I would be like actuary in point. So for that, Prometheus, and then, you know, we provide the host, and then, like, you can have queries like those, like those, using like promises query languages. And it will basically create an like available custom data dashboard that you can

Speaker 1 10:47

primarily, speaking of dashboards, what's your experience with dashboards on Alix?

Unknown Speaker 10:54

I'm sorry, can can please,

Unknown Speaker 10:56

speaking of dashboards, what's your experience?

Speaker 2 11:01

Yeah, I have worked with creating dashboard. So, like, in a new, really good dashboard. So I have worked with, like, Grandma dashboard. I have set up them. And, like, I have also worked with, like, when you, like, I'm doing production support. So I have worked with

Speaker 1 11:22

people. Worked with, what other monitoring or logging tools have you used to have the recent project? Sorry,

Speaker 2 11:33

so those were my primary log modules that I have worked with.

Speaker 1 11:38

So you did not use CloudWatch even so, yeah, we estimate that there is no mention of Grafana, but I see the mention of CloudWatch, but you're saying you use Grafana, but not we did,

Speaker 2 11:55

like a lock some part of, like, application logs to the CloudWatch, but it would like, you know, get forwarded to explore. So in the end, like to like fire like, Firehose like, because it's like, it's easier to like, you know, wait,

Speaker 3 12:12

so, okay, so I know you use the Splunk as well as Kibana, but you are logging, right? So what's the difference? I mean, for what purpose you use to Splunk, and then what other type of logs you used Kibana? Is there any difference with respect to your application logs compared to Splunk versus Kibana? I mean, what's I mean, how you decide, like these logs go to Splunk, and these should be something that you can visible from the Kibana. So talking

Speaker 2 12:43

about difference between Kibana and sprung so basically, Kibana is flexible for like, for Elastic Search. And if you have some kind of like query language that you want to learn, like, if you want to like some kind some kind of, like observability in like, in real time analysis, but Splunk, usually, you know, basically a search engine for like certain logs, right? So for our specific project, every application logs that had a turn on, like your log directory would get captured, and you're like, forwarded to Splunk through like sprung forward agent. So that's that was my primary use case of like Splunk, but Kibana was like in like there for start and application that I didn't work directly. So like, it was just have your it has your like ability to do like more elastic search related queries. Okay,

Speaker 1 13:52

can you agree with like a swap query? Like your application blocks around. How would you query those? How to find that error from your application? How do you populate a table with couple of these on Splunk?

Speaker 2 14:11

So you first basically provide the index, like after Splunk, and so basically you would like no provide your Splunk like a Splunk query, which is like, very important. And then the next thing would be like your source that you like, like you like to use and like. So you can do like, a source of your cluster, or anything, anything like that, you know, like, and then you can also specify, specify, like, which host like, if you have any specific host related queries, then you can also like, provide host so there are kind of like prerequisites that you want. And then after that, you can also use pipe. And then, like, if you want to, like, do any statistic based on like, a certain field, then you can use that keyword. Or then if you want to do some like, say time series, then you want to, like, you know, do the time chart keyword, and then you will provide, like, you know, like you can also to advance regex matching using like a regs command, and like, you know, so on. And so that's the kind of like, how you generate like, different kind of like tables or time series.

Speaker 1 15:40

Okay, okay, so stepping into CICD pipeline, can you offer to like, what's your experience with setting up CICD pipelines? Specifically about topper and Jenkins?

Speaker 2 15:54

Yeah, sure. I have. I have been using CICD pipeline. So the way we have our like Jenkins setup is we have a groupie script which basically works in like a place of Jenkins file. So this groovy script has like different stage, like that actually is like in like, configuring like very in every class like, you can create a class of like a different stage. And then this internally, we get converted into some kind of gentle pipe that will, like run. But as a as we our team, get exposed to that, the groovy script that we use like in each of the groovy script will provide like, what parameters we need, like, and what are the inputs we need. And like, what are the downstream jobs? And like, once the job it kill, how it can kick off, and like, you know, stuff like that. And also, you provide like, a part of like, sales script. And this script is like, you know, like, exactly like the what the command is, and so this certain stage need to be run like, you know, so I have a help like, I'm involved in, like, creating this pipeline from scratch, and also, like, you know, like from for running time during our release. I

Speaker 1 17:23

So you do see there is a type script mentioned in your resume. Can you explain, like, what have you done with the Typescript and predictive unified?

Speaker 2 17:35

So I mean, our Angular is like TypeScript based application. So I work with like our with TypeScript regularly. So so any work that I have to do with fronting is generally and TypeScript

Speaker 1 18:02

for apart from the AWS EC to ECS, did you use any other services?

Speaker 2 18:10

Yeah, sure, yeah, talking about like AWS services, I have worked with lambdas. I also have worked with SMS, and I also have worked with SQS, so I have also worked with rds, so I've used these services as well.

Unknown Speaker 18:32

Can you tell me?

Speaker 1 18:37

Why would you use Docker and ECS target created.

Speaker 2 18:45

So talking about Docker, Docker containerization. So basically, the way it works is, like, Docker will just contain your application, right? So, like, basically, if you're, like, packages your app and, like, all the dependencies into our container. But ECS for runs those container without, like, managing any server, right? So it is a service, service, like a computing engine provided by our AWS. So Docker was are just like, you know, allows you to, like, build your application and run it anywhere. So then how you can just run those in no server management needed. So you will just, you don't need to manage any like instance on your own. You just specify like, the container image and AWS will, like, you know, like, handle the like infrastructures and use that like to, you know, orchestrate the scale, scalability, and also, like other internal that you want, and which is, like, basically a little more secure by design. Yeah,

Speaker 1 20:01

yeah, and going back to the Jenkins. Can you explain your experience with all the different kinds of

Unknown Speaker 20:12

pipelines, and which one did you use?

Unknown Speaker 20:18

Normally, we use

Speaker 2 20:21

Jenkins. But different kind of other pipelines I have used, I would say I use, I have delivered experience with GitHub actions. So I other than that,

Unknown Speaker 20:38

I have, I will work with any other

Speaker 1 20:43

pipelines, monthly stage. I want to use your checklist, right?

Unknown Speaker 20:47

So sorry, the different

Speaker 1 20:50

kinds of Jenkins, pipelines, multi stage pipeline,

Speaker 2 20:55

we work with a multi stage pipeline, so it's not one stage also you have different states, like, like, you know, like, if you have your application in GitHub, then you basically need to make sure that there is a check, or, like, the scan the repository and then run the pipeline for like, you know, certain branches, and then it will have, it handles those. So, yeah, I mean, I have to work with those. What

Speaker 1 21:23

was your deployment pattern like? Why do you think Canary deployments, blue green deployments?

Speaker 2 21:30

So talking about like deployment pattern, we were working with Blue Green deployment. So we can Canary attest so it mixed up with Blue Green Canary. So basically deploy like off corner, and then, you know, like, we switch it right away in like a lower environment, but as we move to the broad blue green, and then we do Canary instance. And then, you know, like that takes like active traffic and where the like validation takes place, and then we switch it.

Speaker 1 22:10

Will you have any experience posting any any kind of quality tools, not in terms of user perspective, but the admin stuff, like any logging database. Have you posted anytime or any matrix database?

Speaker 2 22:30

Sorry, I do get the last part. Can you please?

Unknown Speaker 22:33

So basically, my question is around,

Speaker 1 22:37

think about any monetary monitoring solution, logs or metrics. So were you always an end user of these tools, or managed or, like posting the databases, like, for example, blogs, storage using language, storage using COVID? Have you ever owned any of these databases? Or, like always, I have not

Speaker 2 23:04

owned any block databases, so I I've been primarily using the one. So I'm not never

Unknown Speaker 23:13

they do see Kafka. Can you

Unknown Speaker 23:16

explain that? Did you use Kafka?

Speaker 2 23:21

Yeah, sure. So I have been most of the application process. I have been work used to Kafka. So basically, one of our use cases that we had to like was streaming engine. So this basically was supposed to, like, you know, by collecting any customer related information, so any customer related changes that the application does any log related to any it needed to be like a string or like to an snuffle table, but that we have to integrate our Kafka topics, and, you know, like our application would be, would publish a message, Like to the topic, and then we have our consumer that would like read the message and it message is needed, and then then push it down to the like snowflake table. So that was like our use case of we had for the Kafka. So

Unknown Speaker 24:20

have you encountered any issues, or

Speaker 1 24:24

any issues using Kafka, and how did you solve them?

Speaker 2 24:31

So talking about the issues, why using Kafka? Let me refresh my

Unknown Speaker 24:37

memory on that. I mean,

Unknown Speaker 24:41

we basically sometimes when, like,

Speaker 2 24:45

get a broke broker issues, so, which basically, you know, like disrupt Bros and, you know, like the consumer as well. But we did have some like, in sync, replica in place, which, you know, like always, like a kid, you know, like, whenever there is an failure of like, for those brokers. So that was the something that I remember we had in the past, but not any recent issue, issue that that I don't think I recall.

Speaker 1 25:13

Have you ever faced any lag issues while consuming from kata? Talking about the

Unknown Speaker 25:21

lag issue? Yeah, I mean,

Speaker 2 25:25

there are some time that happened which we had to, like we had in the past, I think, was some kind of a rebalancing delay where we were adding and, you know, like removing the consumer, basically, like that was causing us the like rebellions, and it basically paused the consumption, consumption. And, like, I think we basically fixed that by, like, scaling out the like consumer and then during the consumer pulling so, yeah,

Unknown Speaker 25:57

have you ever reset the lack income?

Unknown Speaker 26:01

Have you ever reset the lag on Kafka a little bit? Yeah,

Unknown Speaker 26:09

how will you do that?

Speaker 2 26:12

So, so we basically so to receive access the lag in Kafka, we we have to basically change the committed offset by the consumer group so that it starts like, you know, like in a different consuming from like, a different point. So it's just changing like, where the committed offset, like, the offset is also like, if you keep, like, you know, like, reprocess all the data, or like you like a want to escape the certain data, you just do that.

Unknown Speaker 26:52

Going back to the CICD problem, right?

Speaker 1 26:56

You have a proper image, which is very big in size. What are the steps that you need to reduce the size of the population so that the deployments or the CSV pipelines will be faster?

Speaker 2 27:12

I'm sorry I'm having a big trouble on hearing. Can you please repeat that question

Unknown Speaker 27:17

again? Can you tell me that now

Unknown Speaker 27:19

this little sound, but can sound,

Unknown Speaker 27:28

yeah, okay,

Speaker 1 27:30

so consider a use case. I problem image size is very big. What are the steps that you would take to reduce the size of proper election?

Speaker 2 27:42

So in that scenario, what I will do is like to reduce the government image.

Unknown Speaker 27:51

Basically what you can do is like

Speaker 2 27:57

creating, like a smaller base image that you can have, right? So, maybe big, specific base

Speaker 1 28:03

image, what is, what is your basic image? Basic

Speaker 2 28:08

so, so base images are usually the first line of like, where, like, where the where on your Docker image, where it pulls your base from right, like, right? What? What? How it is built, like, what everything is so it's basically what tells you the like, it's like a template that contains

Speaker 1 28:33

any base images are you used? I'm sorry, can you name any base images which we used, what number of PCs

Speaker 2 28:44

the base image? Amazon, core to Java 70, COVID two, or like python three. Those are some of the base images.

Speaker 1 28:57

Yeah. So reducing the base image has been formed to explore some files. How without specific dependencies that which you want inside the container, so there are properties.

Unknown Speaker 29:12

Okay, that's fine. So now,

Speaker 1 29:16

can you explain me one use case where you set up an alarm in cloud watch. Let's say, you know, there is an increased usage in some BCS instance. Then how are you setting up the monitoring and based on that alarm, what action were you taking to automatically reserve that alarm? Like, have you done anything like this?

Speaker 2 29:39

Talking about monitoring on CloudWatch. So basically, so you can, you give an example of a cloud but we have something similar. We can like our new reading. So basically, whenever you have ICP usage or like your application is like not responding, so that is, your

Unknown Speaker 30:03

resume is not having any

Unknown Speaker 30:06

numeric I'm sorry. Can you, can you repeat that

Speaker 1 30:09

your resume, there is no energy for numeric at all. Have you used numeric that much? Because I usually like probably 10 times. So far

Speaker 2 30:19

I have, I have been using a new ready, but like, you know, like some, some it was in a recent application. So I didn't put, I haven't updated that.

Unknown Speaker 30:35

I missed out. Maybe I have to update that.

Speaker 1 30:39

No worries. Let's go back to the question, could you walk me through like you set up, and you know how to

Speaker 2 30:48

solve an issue? Sure. So as basically, like you have, like, a large set of whatever, there is a high CPU usage in our instance, so whenever, superior, you know, like utilization matrix goes like, maybe, like, over 60% for about like, maybe, I would say, five minutes, we get an alarm. And then this alarm sometimes can be, you know, like, be a page on your phone, or, like, if there is, like, a production issue, if it's in, like, in a lower environment, we'll just get like a wave of attached to, like our team channel, and then we'll get notified about that

Speaker 1 31:37

linear problem statement. For example, say, like, I have a SaaS application, some application, and then have to consume data using an API call. Say, like I'm receiving a 6 million events every minute. I have to pass data to some filters and drop some data extensive compute operation. So I can only consume, like 1 million messages a minute, but I have 6 million messages to consume. I can consume only 1 million messages with one point. So how do I solutionize this? I have to consume all the data, 6 billion messages. Basically look at the multiple bots. But like, how would you make sure the data is accurate? So,

Unknown Speaker 32:31

so if you have, like,

Speaker 2 32:33

basically, maybe do some, I would say, like, segment like, how, if we segment the request and maybe like, we can use partition it, or first thing you can check is like, if your instant like, if your source can filter the data so in your like query that you are doing. So if you can filter the request like and like, instead of getting everything that will help like you little bit. But if that's

Unknown Speaker 33:08

not the possible, then,

Speaker 1 33:12

so I'll have to consume all 6 million messages, which is coming in from SaaS mission, but I can only consumed with 1 million messages per minute using one bond. I could do scaling like six or 10 ports consumed. But how would I maintain the consistency of data I am running like multiple codes, like I can even get a duplicate data right? How would I avoid that scenario, if we have to

Speaker 2 33:41

do like, we had to do like, maintain the consistency that, in that case, what you can like do is like some kind of a multiple parallel worker for per partition, and you know, you can just make sure that you do not like in order to make sure that you have A proper consistency, you would basically do some kind of like item put in prep processing where you will use the duplication keys, or maybe like, you know, through some kind of like record ID, or like, anything like, anything like that. And you can maybe do some kind of like database transaction right to like and like, avoid partially written rules, like, if there is a failure. So in I think those are like, some ways you can do

Speaker 1 34:38

it, okay, performance, right? So what are the so what are the top, I would say, metrics that you keep in mind to monitor your APIs.

Speaker 2 34:53

Yeah. So API performance is, like, very important. So to some of the top API response performance that I look for is maybe, like, API response code, so that the first thing, like, if everything is successful, or, like, like, you know, like, what kind of error code or it is done, next thing, what would be the like, latency? And then, like, you know, like, what is the time taken by? Like, for the request? Next thing maybe would be like request, like, body size, what size of like, you know, like request that you have, or, like, if there is, like, you know, like, any timeouts, or like retries, or any, like, any kind of, like circuit breaker pattern that, like you can see. So if these are so, I think, like, you know, these are like, top ones that I would see.

Speaker 1 35:48

So this is where we have mentioned, right, to monitor all of these. Is that? Right?

Unknown Speaker 35:56

I'm sorry, can you be there again?

Speaker 1 35:59

This is where I believe you mentioned that we have used COVID History promises. Now, can you walk me through like, how do you how do you secure a REST API? Like, what are the things that you keep in mind to secure REST APIs?

Speaker 2 36:17

So to secure and REST API will,

Unknown Speaker 36:24

like, I will mostly

Speaker 2 36:28

use the JWT token like you basically make sure that you integrate it like spring security. Like,

Unknown Speaker 36:36

what is JWT? JWT is a JSON Web

Unknown Speaker 36:39

Token

Speaker 2 36:42

like an like, so it's consists of like three part, which are header, payload and signature, and then the basically header contains, like the information. So basically it consists of like, three part header, payload and signature. So like, where a signature takes both, like a validity of your claim, and header contains the information about like your signature, and the payload contains like, the data that the user, like the user ID or like, you know, like, what are the rules associated with like with those so you probably use that and use like, a rule based access control on like on your own controller and

Unknown Speaker 37:42

inputs the JWT contain a session cookie.

Unknown Speaker 37:47

I'm sorry, can you please

Speaker 1 37:50

repeat that? Does the JWT contain a session cookie? JWT?

Speaker 2 37:59

I don't think so. Like, I don't think it contains any like, decision cookies, like, all the three things that it contains are like heater and payload and signature. So I don't think it contains anything.

Speaker 1 38:17

Sir, do you use neural link drive? Have you use distributed tracing from New Relic.

Unknown Speaker 38:25

Sorry, can

Unknown Speaker 38:28

you Sorry? Am I audible

Unknown Speaker 38:29

now? Yeah,

Speaker 1 38:32

you said you should use New Relic, right? Have you used any distributed tracing feature from your relic?

Speaker 2 38:39

So talking about distribution feature from New Relic.

Unknown Speaker 38:44

It should be distributed tracing.

Speaker 1 38:48

Do you know the marketplaces? Why do you use cases? So,

Speaker 2 38:55

yeah, basically distributed tracing is used like, like, to track. Like, you know, like, simple request throughout, like the flow. So basically, you request the flow through, like a multiple layer and like multiple services. And so right, like a new gives you the ability, like, to trace the request, like, from the span, like a service name, or like operation or the name of a name the duration, and, you know, like so on. So we, we don't use like a distributed tracing. The only thing that we did like is we inject our request ID, like header within like our application. So it is like it is available from our down downstream partners. And so it will be like a part of our header. We can just like, you know, trace that using is needed, and we do not use, like, New Relic, New Relic for the distributed tracing.

Speaker 1 40:02

So have you used load balancers for any of your applications?

Unknown Speaker 40:07

Yeah, talking about load balancer Yeah,

Unknown Speaker 40:11

yeah, they have a question.

Unknown Speaker 40:14

So, for example, say, like you have users,

Speaker 1 40:19

have users, like, maybe, like 100 applications. You're connecting to your application, and your application is behind a load balancer, and you are an internet company, five or six different instances of your application, so

Unknown Speaker 40:34

all the requests from the

Speaker 1 40:37

user applications are getting connected to only, like one instance of your application, not two instances. It's not getting distributed among your five or six instances. So what could be the issue?

Speaker 2 40:50

So for the issues, basically, the load balancer uses, like, some kind of strategy to, like, you know, basically distribute the load across, like, all the resources. So if you have, like, you know, some kind of st session configure, or like, if you have some kind of a weighted distributed configure, so that request, you know, like, goes to a specific instant for any reason that would be, like, something that would cause this there will be, there could be, like, you know, like some kind of caching that takes place in, like, DNS or NS layer server, which, you know, like might be, might be the issue. So in order to, like, make sure you have that, you need to make sure that the strategy you are using is some kind of, like, round robin, so that, like, it goes from one to

Unknown Speaker 41:50

another. Have you ever heard about sticky session? I'm sorry. Can you please do that?

Unknown Speaker 41:56

Have you ever heard about sticky session? Yeah, like,

Speaker 2 41:59

like, I mentioned, like my first point that I would use, like, sticky

Unknown Speaker 42:04

session, which that rural venture keeps

Unknown Speaker 42:07

routing the same client?

Unknown Speaker 42:13

What's your experience? Kubernetes?

Speaker 2 42:18

Oh, yeah. So talking about Kubernetes, I basically used, I have experience with AWS, AWS ECS, which is, like, you know, like, similar to communities, but in my past, I'm, like, I did some of the community work as well.

Unknown Speaker 42:39

So what are the different components of Kubernetes.

Speaker 2 42:44

So talking about different components of Kubernetes, so components that usually controls your, you know, like your API server, your config map,

Speaker 1 43:01

not, yeah, sorry to interrupt. I'm not asking about the Kubernetes components, but you said you have some exposure to Kubernetes, right? Like, what are the different kinds of Kubernetes that you have used in Kubernetes, within Kubernetes, forget about the Kubernetes architecture. Different levels of Kubernetes, different, so

Speaker 2 43:27

basically that we had in the past was, like, limited, and we have some kind of Helm chart that dedicated, like, know, like, how the services are, like, orchestrated and like, you know. So that's what, that's what I

Speaker 1 43:44

have already deployments, people sets, replica sets. Did you have any experience with any of these? No,

Speaker 1 43:55

okay, so quickly touching upon the CICD, what kind of, essentially, the scanning tools have been used, or any model scanning, or like for scanning, for testing, if you were available to any of the tools which has used.

Speaker 2 44:13

So basically, the first scanning that we do is using, which is, you know, like, just for, like test coverages. But is also, it is also used for the other tool that I like, I think used is called mn, which looks for, like a vulnerability within, like the dependencies. So this does like, you know, like static scans of, like, what the vulnerable libraries are that you have, and then there, like it checks like a marked library that, like you we use for looking for vulnerable code that is, like, checked into our new WordPress. So these are the one that I have, like, work with directly.

Unknown Speaker 45:01

Okay, thank you.

Speaker 1 45:06

So on your resume, I seen like you have this three buckets. Can you tell me, like, what command did you use to write data to s3 and create data from history? Yes, sure.

Speaker 2 45:22

Yesterday. Stories. So you can, like, use, you can use AWS CLI. So basically the way you can write it is like, you can do something like, Copy command. So like, like, No, this will just basically copied. But before that, you need to make sure that you have configured it with like, a proper credential by like, you know, like maybe exporting the Access ID, or like, you know, like a secret ID and to delete you will just do like a AWS s3 demo,

Unknown Speaker 46:12

I'm sorry. Do

Unknown Speaker 46:14

you have any questions for us? So,

Speaker 2 46:20

like, first of all, I would really want to thank you for like, this time today, and it was so whenever we talked me through the interview, you didn't mention about, like, owning the logs, or log edition, does this project, you know, like revolts around, like, the logging Like, so does the team, like, provides, like a solution of log acres or anything like that. Like, we must be focused on

Speaker 1 46:49

the position. What we are looking for is kind of an observability revolves around blocks and metrics.

Unknown Speaker 46:59

Most of our questions we

Unknown Speaker 47:01

have the delivery. So we have the delivery

Unknown Speaker 47:06

team, and we provide tuning reports.

Unknown Speaker 47:10

We have petabytes database owning remote so do you have

Unknown Speaker 47:29

anything else?

Speaker 3 47:33

No, I have time set up with Rajiv tomorrow, you know, anyway, but yeah, just join, you know, just to, you know, reason you know what kind of experience you got from the technical side of things. But yeah, very good. One

Unknown Speaker 47:53

question, do we also work in any kind of, like, streaming

Unknown Speaker 47:59

platform for like? Do we platform

Unknown Speaker 48:06

for Kafka management.

Unknown Speaker 48:09

We do have Kafka depending on when

Unknown Speaker 48:12

we have a very big Kafka

Unknown Speaker 48:15

in terms of loss, and we input

Speaker 1 48:17

using Kafka topics, and we have different consumers

Unknown Speaker 48:26

to push the data.

Speaker 1 48:30

Thank you so much. Thank you. It's got Thank you.

Rajiv Ranjit- BnyMellon-Final-srv-1

Speaker 1 00:00

and I've been working as a full stack developer for about six years. And now I'm working on the Java based on applications and using the platform like streaming hibernate and the RESTful web services for API, development and political experience. Working on the microservices architectures are built using the Spring Boot and have been working on designing the microservices and developing and also deploy deploying them to higher environment. So I have also worked with the AWS Cloud for for hosting this applications and being using services like ECq and SQS. And and so also I have also worked on the lambda and s3 buckets. So on the production side, I have worked with the Angular CLI and work with the Java scripts type scripts, etc. So on the distant side, I have worked with the J unit and Mockito. So currently now, I'm on a scrum team. So the team consists of the five developers and couple of the testers. So we have we have product owner and Scrum Masters and on a daily basis, I have work on the mostly on on the back end, back end code, like Java, and helping the team with the code review and also work on the production support duties.

Unknown Speaker 01:50

So yeah, this is my

Speaker 2 01:55

question so on the current episode, when you can given ideas or ideas something you can share

Unknown Speaker 02:02

Yeah, sure. So I started doing it

Unknown Speaker 02:11

I can share my

Unknown Speaker 02:17

sorry, kick in your video. Once again.

Unknown Speaker 02:23

Okay, just bear with me

Unknown Speaker 02:31

so we can

Unknown Speaker 02:34

talk about it or.

Speaker 2 03:29

Okay, so these are basic modules because they play the main players use the key to this object, okay. So let's call it out. So, we need to arrive at some period as a collection of these Employee objects that are sceptical because Kosha headliner is a collection of service provided we indicate the unique names of that collection

Speaker 1 04:02

Okay, so we just want to return or uniquely viewing after the show.

Speaker 1 04:41

So basically, we convert the employee into a string

Speaker 1 04:53

and then we can do a stream processing

Unknown Speaker 05:07

and

Speaker 1 05:10

yeah, now then we want to first let's get the name and so we can do our map functions now.

Speaker 1 05:27

So, we just saw one the name so we do something like employee dot get name

Speaker 1 05:42

and then since we want the unit name, so we just want artistic functions now, then we can relate this to our list so, since we are returning outside so we just need to stick our functions anymore

Unknown Speaker 06:10

okay, now

Speaker 1 06:13

we just return our fate. So this gave me have a list of employee name

Speaker 2 06:24

it sounds good. Question number the next question number 61. Inequality election ready to return that case?

Unknown Speaker 06:34

And value is an option. Okay.

Unknown Speaker 06:45

Stream operations right so, yeah,

Speaker 1 06:50

so, basically as you said it's the same thing we can do the stream API. So yeah, we can use using the collector and

Speaker 1 07:05

basically, you know, it's it's very simple processing so basically posting is the

Unknown Speaker 07:10

convert into string.

Speaker 1 07:16

And then we can directly call it to map

Speaker 1 07:26

and then collectors sub map and data functions. So the first function is how you want to ID to look like

Unknown Speaker 07:41

so this could still be

Unknown Speaker 07:43

the ID of the employee.

Speaker 1 07:49

And then we want to value the function so which would be the just for the employee?

Speaker 1 08:12

Yeah, yeah, kind of fixed into multiple nonviolence No. So give me one second. Let me teach you this shows.

Unknown Speaker 08:31

Sorry, this is a two map instead of mapping.

Speaker 1 08:37

So I think it was the auto change the value

Speaker 1 08:45

so collected closer to mapping and can be held to convert our accumulate into a key value.

Unknown Speaker 08:52

Well, there's so

Unknown Speaker 09:17

Okay, so does that looks good?

Unknown Speaker 09:25

Okay so,

Speaker 2 09:32

let's get us following on politics, but this is what is called in.

Unknown Speaker 09:41

So we had to do question.

Unknown Speaker 09:46

So basically, those checks

Unknown Speaker 09:50

are the same kind of

Unknown Speaker 09:52

buildings have to do here.

Speaker 1 09:56

Searched interview, I can share it again strictly more louder social.

Unknown Speaker 10:03

Payments problem statement.

Unknown Speaker 10:06

Given the list of strings, they could

Speaker 2 10:11

have right now we can we can make them as case sensitive. And then given that only substrings it will return a list of list of strings where were the things which are which are the same characters appear in each list in there. Okay, so we can take a look at what I just said there and then so we need to know how to implement this into live streams.

Speaker 1 10:39

So I just copied over your questions.

Unknown Speaker 10:43

I can remove that or if

Unknown Speaker 10:46

you don't have Lombok.

Speaker 2 10:50

Forget you can remove that and replace that line 21 Every two minutes system dot out dot print line

Unknown Speaker 11:01

enforcement

Unknown Speaker 12:41

Think the Pol Pot commerce

Speaker 2 12:44

segment explains what and what it's supposed to do so you can look at the example and what output it's generating from.

Unknown Speaker 12:58

So basically, we just need to you know

Speaker 1 13:15

and then we just need to, you know, so what I'm thinking is basically create a map. So

Unknown Speaker 13:28

understood that

Unknown Speaker 13:37

if you can try to stay up as much as possible

Unknown Speaker 13:55

see what I can do is.

Speaker 1 14:25

It's a good to have the same set of economic trips and the same set of the characters.

Unknown Speaker 14:37

Then we just

Speaker 2 14:40

didn't have same character right. So thanks for the output. Go through and pick the one list. That is not really real word but then in and have the same little character. Those are the third item in the list is one up to one, but it's the same as eight and eight because we try to case sensitive because no other an atomic Boolean so it goes off into a separate list which is totally element

Speaker 1 15:17

so yeah, I'm thinking and just creating an app that stores it for us, right. So anything that has a sin characters will be stored in a

Unknown Speaker 15:33

map as a key value pair

Speaker 1 15:37

by control process is you know, for a GE so, first showed that the A D into the upsetting orders and then I'll just make a Brooks So, basically on the horse

Unknown Speaker 16:11

I will convert back into a string

Unknown Speaker 16:17

then cold lake

Unknown Speaker 16:23

I will buy

Speaker 1 16:50

so basically what I'll do is

Speaker 1 17:01

I will convert the board into our directory and sorry

Unknown Speaker 17:14

thank you

Unknown Speaker 17:20

so this again I will solve this error

Speaker 1 17:34

then basically I will choose to return out of strike

Speaker 1 17:45

now this will do is this will create a map

Speaker 1 17:57

this will create a map of string to a lot of strain

Speaker 1 18:10

the way it will looks like would be

Speaker 1 18:30

for a Google Ad

Unknown Speaker 18:44

value of

Unknown Speaker 18:49

e to n.

Unknown Speaker 19:30

So now all we need

Speaker 1 19:33

is to collect the ballots from the beads and then just drop onto that list

Unknown Speaker 19:51

so we'll return our collections

Speaker 1 19:55

so we can do all your errands for this

Speaker 1 20:09

think that's what more do you need me to write?

Unknown Speaker 20:18

Big outputs Not really. And so

Unknown Speaker 20:22

if we can run this on record and see what it does.

Unknown Speaker 20:29

Okay, something we're trying to try and find

Speaker 2 20:35

something if you can tell me something in your projects that was referred was challenging. It was something that nobody else could figure out something new or something that was very hard to do and be able to solve on each page to kind of give me an accurate talk about some of those

Unknown Speaker 20:58

so talking I mean, so yeah.

Speaker 1 21:05

I'm trying to recall my memory you know, so

Unknown Speaker 21:14

so. So basically,

Unknown Speaker 21:18

I tweet,

Unknown Speaker 21:19

I mean.

Speaker 1 21:23

Oh, so, okay. I'm sorry. I'm trying to recall my memory. So basically, I've been reading. So we use a library called histories for our work, working with the last number of requests and you know, we basically use the circuit breaking patterns so provided by the sticks, library, so also something that historic does is provide and abstractions. So in order to

Unknown Speaker 22:00

manipulate or trade goals.

Speaker 1 22:03

So with so you you can define the number of tools that you want and what groups will go to or what polls and so on to basically which we were trying to create a concrete implementation of the history command that will fulfill our requirements. So something that had to be backwards like challenging was defining the trade, say concurrent approach and anterior approach where we, you know, make the fallback mechanism be triggered on certain exceptions blocked and not executed callback, but trigger or circuit breaker matrix on the on the sorting flows. So this was happening overall. And then, when Oh, when we defined the logic we made, making use of the dynamic configured. So dynamic content is spring, Cloud Config based applications that grabs a config config from the config servers. So now, the thing that was challenging here was mapping the multiple conflict servers that could be grabbing the contents and managing the circles and also the SFTP client connections to so we implemented it and then once we pushed it to the higher environment, where all where there are a lot of lows our applications didn't perform, as well. So we wanted to so because I've offered a certain amount of pirated label here, it's called from our downloads in iTunes. So our applications would basically not be performing and all of the requests through the timeout so this was one of the challenging at speed and trying to complete can pick out what the failure was and why it was causing our applications to fail. So yeah, and I took a lead to try others to try to resolve this issue where our calls were being directed and time sessions. So what unhandled exception tables are being followed by their instructor commands. So, so the abstract is recall whenever there is any kind of existence, so sallow or that existence and does not belong the proper existent masters means and this is called, called the fallback mechanisms. So during this scenario, our sdb clients the connections and the amount of the connection that would have open would be the close and because of this, we we will exhaust the resource and so that was the reason for our failure and having to resolve is water. So I think this was something that was challenging and having able to find this issue and then resolve it was something that was preferred to me because it was affecting our active hire department. So yeah, that was a challenging for me.

Unknown Speaker 26:03

Okay, one last programming question.

Speaker 1 30:08

So first thing would be convert the word into an LA and so basically we split out the word into

Unknown Speaker 30:20

an array of the words

Speaker 1 31:50

So here, in order to get the widest base, we can just go do the training function. Of this string

Unknown Speaker 32:06

right, all right. So

Speaker 1 32:10

what between the boards so yeah, I'll do that in a second. So I'll just remove the E

Speaker 1 32:26

Hello, wait a minute by that stage and we can collect it as its as a list.

Speaker 1 33:04

Now, basically what I need to do here is I need to reverse this array to our list so we can do

Unknown Speaker 33:17

elected dot reverse

Speaker 1 33:25

and then at the deep end, if we want to join it so we basically just do

Unknown Speaker 33:36

one further backwards shrink

Unknown Speaker 33:43

and then to collect

Speaker 1 33:52

so we can call it by joining them with space

Speaker 1 34:05

I think that's a detour off for both of these scenarios.

Unknown Speaker 34:21

Yeah, yeah, I think that's it.

Unknown Speaker 34:26

Do you want me to run out

Speaker 1 34:40

this really goes business three stitches.

Speaker 1 34:51

Let me paint this true to stream on the inner warmth so.

Speaker 1 35:13

So basically, whenever we are doing this split, we need to receive that looks for the one is this or the multiple

Speaker 1 35:33

so I can Google the riggotts If that's okay with you

Speaker 1 35:52

Yeah, yeah, yeah, I think in slip can also have better rest to bill not to place not basically, on the street functions, actually takes regards to if I pass release

Speaker 1 36:34

Okay, you want me to read the map again?

Unknown Speaker 37:06

Right.

Speaker 1 37:06

So you just need to do something with those it's on our on the string

Speaker 1 37:25

so basically, I chose to do what the empty space here. Not that this

Unknown Speaker 37:34

also didn't work

Unknown Speaker 37:41

okay, yeah.

Unknown Speaker 37:49

So, so yeah, in the meantime, we, we have

Unknown Speaker 37:56

Yeah, I mean,

Speaker 1 37:57

I like to know about the day to day often the developer on the team, and do like, what would be the time instead mostly by your developer?

Unknown Speaker 38:53

Yeah thank you. So

Unknown Speaker 39:01

much. Know.

Unknown Speaker 39:51

I'm having a lot so can I rejoin in a second?

Unknown Speaker 40:03

Raspbmc Hello, hello COVID Danny

Rajiv Ranjit-Bnymellon-Final-srv-2

Unknown Speaker 00:00

It's so

Speaker 1 00:02

in the others are processed and validations. So I

Speaker 1 00:47

so are we sharing? Oh, I certainly am implementing.

Speaker 2 00:56

Oh, no, we can. I can drive it out and work in Sudoku, but can you give any questions about this? Or is it clear what would need to be

Speaker 1 01:08

done? So I think we're, I mean, we are clear on the requirements, right? So it's basically post all week, saving the database and you only want the validation process.

Unknown Speaker 01:25

So let's say, for example, that

Unknown Speaker 01:30

and I are working on a new repo,

Unknown Speaker 01:35

and

Speaker 2 01:38

we can use pseudocode, but I'm curious about what, what big file structures you might have, what files you would have? Let's assume that you know this code base is meant to be developed on by multiple so we want to establish like good software architecture designs, right? So what files? What folders, etc.

Speaker 1 02:02

So first thing is, we want to make sure we are making the process, I mean proper package structures. So we will create a packets like, maybe com, dot bank, dot order,

Speaker 2 02:27

or Sure so to be a bank, even at the bank of the combat order.

Speaker 1 02:37

So that would be the pockets, rather than the importance

Unknown Speaker 02:43

talking about, like, what, broad, like,

Unknown Speaker 02:44

what the

Speaker 2 02:47

folder would you have in this within the package, I assume that the file extension is already, is already, you know, complete, but I understand, like, within the package, what folders, or what, What types of files you would have.

Speaker 1 03:00

So okay, so if we have all of that ready, then we will first create our reputability folder, yes, and then we will create our entity folder. Then we also service the folder and controller folder.

Speaker 1 03:33

Service, the last one would be the service.

Speaker 2 03:39

Okay, what are some what are some files? Let's just start talking about like, yeah, how would you go about implementing this? What files are are you adding? And we, if you need to, you know, add more photos later on.

Speaker 1 03:52

So, yeah, basically, first, first, I will create the other ETG inside the entity folder. I

Speaker 3 03:59

Okay,

Speaker 1 04:06

this Order Entity represents the Order Entity on the database,

Speaker 1 04:20

so the listing would be on the earthen security and it will create

Unknown Speaker 04:27

the entity first what goes in here.

Speaker 1 04:32

So it will like, private log ID,

Unknown Speaker 04:38

private log phone ID.

Speaker 1 04:42

Prior, private global amount and the private string locations.

Speaker 2 04:55

Would you say this was no

Unknown Speaker 05:01

so that would be along,

Unknown Speaker 05:10

right? And then,

Unknown Speaker 05:12

do we have a repository?

Speaker 2 05:15

Sorry, you said you have a repository. Oh, yeah, yeah,

Speaker 1 05:19

the repository would be name order repository,

Unknown Speaker 05:28

so this would be an extension

Unknown Speaker 05:33

of JP repository.

Speaker 2 05:41

Or what methods would you can be curious, but here, which methods would you inherit

Unknown Speaker 05:51

for sending the order?

Unknown Speaker 05:54

We don't need to explicitly

Unknown Speaker 05:58

define any networks, so

Speaker 2 06:00

we'll just inherit the save. It'll be his orb. Azerbai is up, type Order Entity, and then we should do that. Let's move on. So what's next?

Unknown Speaker 06:16

So we can

Unknown Speaker 06:18

go to the, I mean, the order control, order controller.

Unknown Speaker 06:29

So

Speaker 1 06:32

on the order controller, you can create a method called to create order

Speaker 1 06:41

and the create order would be annotated with the post mapping annotations,

Speaker 1 06:57

so that annotations can have a parameter of I guess, so we don't need anything. So if we define our request entity on the top label,

Unknown Speaker 07:14

so

Speaker 1 07:19

order controller, and we would be your request mapping.

Unknown Speaker 07:32

Sorry,

Speaker 2 07:33

I just assumed what's coming into this function.

Speaker 1 07:38

Oh, sorry, the method would be take our interest value. So

Unknown Speaker 07:43

that would be a test

Unknown Speaker 07:46

body of the order,

Unknown Speaker 07:55

and what next

Speaker 1 08:01

it would look like and request body order, yeah,

Speaker 1 08:13

so I guess I have a depart the order photo. So which can do that? I

Unknown Speaker 08:25

Let's go ahead and assume that this has the same attributes

Unknown Speaker 08:29

as the Order Entity.

Unknown Speaker 08:30

Assume that we'll pass the Order Entity sure

Speaker 1 08:37

on the order controller we want to underwire the order service.

Unknown Speaker 08:50

And what methods are we calling order service?

Speaker 1 08:58

So we will be to order service dot

Unknown Speaker 09:03

and create the order.

Unknown Speaker 09:15

And then we will pass it in the order and

Speaker 1 09:30

okay, and then we'll pass. So this, I mean, we'll pass in the order. So we, we will have that order service, and then we will also have a service

Unknown Speaker 09:49

on service, do

Speaker 1 10:12

so the phone service, We would have a method called, get on details, I

Speaker 1 10:27

and Take up on it. So okay,

Unknown Speaker 10:40

So here,

Speaker 1 10:50

would be using fctp, I,

Unknown Speaker 11:09

basically,

Speaker 1 11:16

the way you can Do is that magazines, you are a certificate. Clients dot

Unknown Speaker 11:28

everything that this works.

Unknown Speaker 11:32

Okay? Do

Speaker 1 11:44

so host and slash party slash on,

Unknown Speaker 12:00

id and id,

Unknown Speaker 12:03

you can do this. You can't really do this.

Speaker 1 12:13

So, yeah, I'm assuming the resource would be so basic Jason's object, so which can conform as a map.

Unknown Speaker 12:31

So on the order,

Speaker 1 12:35

on the order service, the first thing is, we will get the small details. So

Unknown Speaker 12:46

so we

Unknown Speaker 12:50

would on the other

Unknown Speaker 12:54

so basically

Unknown Speaker 12:57

start you.

Unknown Speaker 13:03

So it get the font details,

Speaker 1 13:09

and then you pass it so create the order method. Take order as an argument, right? So

Speaker 1 13:26

so then you can pass order, dot get bond ID, so

Unknown Speaker 13:43

Okay, now we can do the trick, right?

Unknown Speaker 13:52

Okay, so what's the apparatus

Speaker 1 14:01

function so you can do multiple way, right? So you can define a bond, details, offset,

Unknown Speaker 14:10

and so that would be,

Speaker 1 14:13

I mean, that would be inside of another packet. So let's create another folder called Dao.

Speaker 2 14:22

Can see assume that we have things actually. I'm just curious what was happening. You're getting the min and the maps right. You're turning them into Maps. Doesn't seem that we have it like this. I know this is not proper next, but it doesn't matter to me. Okay, so we have this and then what happens.

Speaker 1 14:42

Okay. Now next thing on correct order will do

Speaker 1 14:52

if amount is less than a minimum subscription amount or greater than the meaning of substitution amount is

Unknown Speaker 15:11

what amount less than

Speaker 1 15:14

mean or The amount greater and the greater than the max.

Speaker 1 15:29

No, I mean the or amount greater than max.

Speaker 1 15:39

So here we can throw new Lucas validations exceptions.

Speaker 2 15:51

Let's say we show this to product and they say, hey, it would be nice if we could get not just one exception, but like, we get all the exceptions that the order fails for. The reason is because, like, but you might imagine, like a front end might be calling this right, and it's like a form of users to fill out right? And if users fill out that form, then they might want to see all the things that were wrong, right?

Unknown Speaker 16:31

I'm sorry.

Speaker 1 16:34

I can refer to you, can you? I mean what you said last would be repeating for

Speaker 2 16:43

sure. So let's say, for example, that a user still had a form, right, and the form has these fields, like fun idea, mod location,

Unknown Speaker 16:54

and when the user presses submit,

Speaker 2 16:58

they they want to. They want to see everything that failed, not just one thing. So the way we have it now is you'll throw a new exception, and what that means is that the amount is going to fail immediately, even if the location is invalid. So if that's the case, a product or a friend is coming and saying, Hey, how do we how can we get we want to know like which fields failed and all of them that failed validation, and we want to show like an error message below each field that failed. How would you go about designing this to achieve that,

Unknown Speaker 17:41

little things that

Speaker 1 17:44

so if We want to know all the films that failed, I

Speaker 1 18:17

so We accumulate the error, right? So basically, we can do in order service we will,

Speaker 1 18:35

so we will define here map, right? So our and your list.

Unknown Speaker 18:44

Or is it an Error List?

Unknown Speaker 18:50

Please make a list.

Speaker 1 18:59

So I mean, it would be a new analyst.

Speaker 1 19:13

So if the amount validation builds, we can add a bounces on a list. And instead of showing annexations,

Unknown Speaker 19:22

okay? What are

Speaker 2 19:24

we adding to this list? A message you send, I guess. Yeah,

Unknown Speaker 19:32

okay, whatever it is I forgot. What is the job? So?

Unknown Speaker 19:34

So, yeah, we can do like, Okay,

Speaker 2 19:44

so let's say we do this, and that front end comes back and they say, Hey, how do we know which message persons with which field? This could this could be like one message if it fails the environment validation, it could be one message if it fails the location validation, and it could be if it fails both, right? How do they know which one to

Unknown Speaker 20:16

which field to associate the error with?

Unknown Speaker 20:19

Okay, so we can create a map

Speaker 1 20:23

in that case. So this is the also gear list, list, grid and the air map, okay, and it would be all new password, okay.

Unknown Speaker 20:39

What would be the key?

Unknown Speaker 20:42

So here map

Unknown Speaker 20:53

validations,

Unknown Speaker 21:01

subscription

Unknown Speaker 21:03

amount of validations.

Speaker 4 21:15

Okay? Context.

Speaker 1 21:27

So now we'll do the location validations. Okay?

Speaker 1 21:42

Is if the location is not us and

Speaker 1 22:03

and on ID close to equals to one, or one Id goes to close to our phone ID will still post in three,

Unknown Speaker 22:36

I guess,

Speaker 1 22:39

yeah, I mean, we would wrap this in the

Unknown Speaker 22:44

three hour

Unknown Speaker 22:46

block inside our

Speaker 1 22:49

faces, and then this, yeah, and,

Unknown Speaker 22:58

like, block. Okay,

Speaker 2 23:03

you're checking to make sure, considering to see if the body is not one, and

Unknown Speaker 23:11

it's not one, two or three, basically.

Unknown Speaker 23:21

So, so, yeah, it's not

Unknown Speaker 23:25

so decades.

Unknown Speaker 23:28

I mean, we will also add to an error map.

Unknown Speaker 23:36

We would put location validations

Speaker 1 23:42

now, we can do like similar so location validation failed,

Unknown Speaker 23:51

okay? And what else next?

Unknown Speaker 24:00

Now what is your mass?

Unknown Speaker 24:18

S is energy?

Unknown Speaker 24:23

So if your mouse is empty,

Speaker 1 24:35

then we want to create create orders. So we can use the order reproducing dot c, and

Speaker 1 24:50

then, Yeah, then we can return this value directly.

Unknown Speaker 25:01

This,

Speaker 1 25:02

yeah, so say ordering procedure that's,

Unknown Speaker 25:09

I mean, outside the block.

Speaker 1 25:14

What we will do is we would also revalue.

Speaker 3 25:20

Okay, yeah, okay,

Speaker 1 25:28

and we can pass this secure map does

Unknown Speaker 25:42

everything you do. Front end

Unknown Speaker 25:48

get it done? Are they like

Unknown Speaker 25:51

parsing the exception?

Speaker 1 25:57

No, the controller will handle this part right? So the controller will basically region all 500 response code and mirror map on the rates.

Speaker 2 26:13

So we will catch the exception here and then put that air map in the results market. Yes,

Unknown Speaker 26:23

okay, okay,

Unknown Speaker 26:27

so talking about

Unknown Speaker 26:30

what tests you would write for I

Unknown Speaker 26:56

Fun. I one more time you

Speaker 2 27:25

so that's min, max, not fund ID, location test case it would be Right here you can just give me The values I

Unknown Speaker 28:00

so red one

Unknown Speaker 28:10

and it compares a complaint

Unknown Speaker 29:01

On the okay so?

Speaker 2 29:27

So one way we can view testing is we don't want to just catch the edge cases right. Tests serve as like, kind of like a guardrail to

Unknown Speaker 29:37

make sure that the code

Speaker 2 29:40

is working, right? So theoretically, I should be able to, I should not be able to make a change to this code and have the test sub pass. Does that make sense? But I think I can make some changes to this code and Have these tests to pass.

Unknown Speaker 30:06

Do you see why I

Unknown Speaker 30:26

I think we're missing some test cases. You

Unknown Speaker 30:31

see which ones were missing,

Speaker 2 30:33

these three or four amounts, Right? Use your location. What's

Unknown Speaker 30:58

missing, both

Unknown Speaker 31:02

and the whole. Some

Unknown Speaker 31:14

scenario. Sorry. So this is fail, right,

Unknown Speaker 31:20

this past, right?

Unknown Speaker 31:22

This is a pass,

Unknown Speaker 31:30

right? That

Unknown Speaker 31:33

makes sense.

Speaker 2 31:34

But I think we're missing some test cases, because right now, I can go to this code, change some characters your tests would still pass, which is not great. Do

Unknown Speaker 31:50

you see where those are? I

Unknown Speaker 32:08

You may have to read the violations again. I

Speaker 2 32:38

I don't care. We can assume that we have some listing we run the same test case. I think that's not that.

Speaker 2 33:00

I'll give you a hint. We're missing two test cases under location. We're missing two test cases under amount. And I'll give you a hint. Another hint, this was a bit more funding, but I can modify this expression or this expression, and test cases will still pass in, like a like a bad way.

Unknown Speaker 33:25

What do you think that might be? I

Unknown Speaker 33:40

for example, what if I

Unknown Speaker 33:49

just removed this?

Unknown Speaker 33:53

What would happen to the test cases I

Unknown Speaker 34:15

Oh, yeah, hey. Correct we

Speaker 2 34:40

home. Let's, let's, let's assume that the data is well formed. That's fine, but what I just showed you is I just removed that condition entirely,

Unknown Speaker 34:59

but your tests are still going to pass.

Unknown Speaker 35:02

I That's the problem, right?

Speaker 2 35:13

Some more test cases to Make that not happen, I

Unknown Speaker 35:59

the West.

Unknown Speaker 36:09

Okay, And

Unknown Speaker 36:14

what's the outcome of this? Oh,

Unknown Speaker 36:33

makes sense, okay,

Speaker 2 36:35

and then is that? It do? I

Speaker 2 36:50

think we're in low on time. But did you spot the other one? So with that example, I just had in mind.

Unknown Speaker 36:57

But

Unknown Speaker 36:59

what about this?

Speaker 2 37:02

Or what do you see what we're failing here? We're missing two decis here, right? Because we could do something to the top statement and make these tests pass. You see what that is that will

Speaker 2 37:21

operate with? That's correct, we could add this equal sign and your test was still s. So what test cases would you add here to account for that?

Speaker 3 37:42

Or what are

Unknown Speaker 37:49

the outcomes of these expected outcomes? I

Rajiv Ranjit-Wells fargo-SRV-First

Unknown Speaker 00:00

A small business.

Speaker 1 00:05

So we can get started as you just, you know, like you can start with Professor. Okay, so, yeah, my name is Razi Rajiv and working as a full stack developer over six years of experience and primarily working on these avatars applications and utilizing hibernate and rest for web services, the development of those APIs and I have been also online resources and helping me in designing and implementing the microservices, and also I have been working on the financial and the non financial environment as well. So one of these are the primary primary, and these services have been developed on the AWS and I'm so pretty good with the AWS cloud, and can work with service including the AWS Institute and SG rocket and lambda. So also I can add like CloudWatch and so on. So also I do have the work with the service, including the Chrome team using the Angular and currently I'm working on the scrum team. And my team consists of the spike developers, and we have product owner and Scrum Master and honor David, I was involved in mostly working on the backend Java code and helping the team with the code review and also production support duties my job.

Speaker 1 02:07

So I would like to get started with this like we talked about your role and your various experience, and I was doing good experience, some of the experience you have. So let's talk about one of the recent project or product you worked on and give us, give us detail about it, and then what was your role and how you went about it. Ah, yeah, sure. So, um, so, I mean, so basically, basically at face rating, we are primarily working on the migrating the monolog applications to the microservice architectures. And so I was involved into particular APIs recently for migrate. And these APIs were the maintaining aspect of the user preference and for the Experience API, which was responsible for different entitlement to the user has, I mean, in order to History their front end and the component, right? So basically, in the legacy code systems, so we have all the all these factor into the applications, and then for the modernization approach that the company was moving towards, and we were taking the chunk of the applications, and you know, the modernizing them by the counterizing control and containerizing the services, and the developing micro service from scratch and developing them on the AWS EC on the background, and so, yeah, that's one of the recent one that I have worked with. And the use case of this microservices was completely oriented towards the profile reference and their front end experience. And so this microservices were responsible for the changing the preference of their profile. And while using, I mean while user goes to their settings, so

Unknown Speaker 04:41

so they can

Speaker 1 04:43

opt in there. I mean, for what kind of preference they want to change, and it could, it could be like restful if you guys to call those changes and then string one of the one for the entitlement of for our Angular app to determine what weight range and the component are visible for the user, so so that the brief information. So for these systems, and we use like Java 17 for our JDK, and also we had a screen put there, and those applications were deployed on the AWS ECS country, and we used the strong for the login, and we had New Relic dashboard. So yeah, that's kind of the of like a brief overview of the my the recent project that I have done. So were you involved with our business partners influenced the direction of this conference role. Did you confer better all like, Did you do anything with your business partners to

Unknown Speaker 06:10

implementation? Can you describe?

Speaker 1 06:13

Oh, yes, you're true. Yeah. I did some time have to talk with the implementing partner and the business teams. But lot of my communications was through the through, like, business analytics that we had on the team, and I had, like, direct questions, and we, we would set up a meeting, but one of, I mean, most of the time, I would have my business analysts answer the questions that I had, and also we had the product owner. So, so those were all little more dedicated towards handling the business needs. And so, yeah, but we did. We did have the occasional interactions with the business people?

Speaker 1 07:12

Let's say that talk about the situation like, do we talk about people? Like having

Unknown Speaker 07:18

business people, but

Speaker 1 07:18

there are times when engineers out of orders are typically when it comes to, you know, like talking maybe about the design, architecture, setting something up, technology is brought into that discussion. So talk about, talk about the scenario where you get presented. You have to present something in front of the audience, if you're not technical, but you have to explain technology. How did you pull out that?

Unknown Speaker 07:47

Yes, sure,

Speaker 1 07:50

so. So basically, whenever there is a situation like that, I would try to come up with the example that it's like, comparable to the real world scenarios. And so to talk about one of the situations, you know, we were kind of working on the new platform and where, you know, we do, I mean, we had to present, like, the capabilities of the finance and like minus department and they were responsible for approving the integrations of the new platform, but they did didn't Have any technical background, so basically, I actually have to explain to how to how the decision engines and that our service use the interactions with and like the process and interactions with them. So basically, I made sure that I avoid any technical teams as much as possible, and focus on a lot of business benefit and use for use, like a relative example, and like you know, comparing the microservices architectures to like, highly efficient world and huge on the daily basis. So basically, I emphasize the importance of AWS. Cloud would pay the I mean they would play the important role in these modernization people. That explains how the data of the yields work, like provider for the different services in one shop, and focus on the lot of outcome rather than the technical implementations, like implementation of how the things were structured. So yeah, I would explain in this way you mentioned that we had asked

Unknown Speaker 10:15

you the question, um, the question.

Speaker 1 10:23

So for that, yeah, I mean,

Unknown Speaker 10:30

so let me ask you a question. So, how did you

Speaker 1 10:37

coach your squad with some of yours, like, how did you? Were you in charge of breaking up stories and then maybe delegating for other people? How did you approach that as a senior as a senior development I'm sorry I can barely hear you now, once again, but using one microphone to avoid So can you describe a situation when you were one of your projects, or you can delegate it work to other other junior developers. And how did you go about? Oh, okay, yeah. So, so basically, on our team, we usually have the two engineer developers, and they were responsible for these small features development. And we, we senior, like, as a mentor towards the and also I have, I mean, I remember we had a features where we had to integrate our JDK into our, I mean, into our systems, in order to aggregate the deep red box and then provided into snowflake tables. So I, so I, what I did was the the pictures that was, I mean, available to me. So I was broke down to it and, and I broke it down into the small chunk, and, I mean, into the small shoes that was manageable. And, you know, asked there on the comfort on how they were built on each of these particular stories, and giving them a high, high level, like overview on how things were supposed to work. And also giving them a little bit of the low bail implementations, implementation details on what could be drawn so, so all of those were recorded into the Zilla tickets on the stories and then assign them accordingly, and also in that way, they could work on on something that is manageable to them, and also have some kind of comfort with the reaching, back to Me in the case of the contingents, and also working in short term manner, like actually boosted their confidence to taking out the newer tasks and also raising the questions. So whenever there was, like, sorting confusion. So, so yeah, I do like the country will live in like small group of juniors, and also like a future stable man. So, yeah, that's how I will be acting within. So I'm curious more about the project you worked on. Talked about it a little bit. Now I want you to take into a team on the technology side and pick the project. You can pick BG project for your latest company project, and then talk about what was the architecture of what you were working on. I know you did talk about that you do by experiencing violating to, like, microservices or monolithic so talk about your endpoint. How did you break that monolithic structure to micro services and what kind of micro services you had, what kind of business request you had, and like, how did you go about this? So this is, this is supposed to be a little bit more into technical insight of what you did, okay, okay, so, so, basically, the flow that I broke down was based off on The pop up model. And that was, I mean, are I mean, that came off with it during the one of the portal and the same and that portal whenever the user select all this. And so we had to open our form where the users were able to fill up their informations and select the different opening. And once that the forms were submitted. And this is, this is where a service that I purchased recently came to, came into the play. And so we created a news office called the mention app, maintenance app, and this maintenance app was responsible for the two things. And the first thing is to make the changes to to the user's profile and what like, what they prefer, and also make the changes to any kind of the withdrawal or, like a deposit limit that that they have. And also, yeah, so that could be, like the two features along with one of one of the police artists. And so basically, sometimes the data need to be exported down to like the other services, and it might take a while because of the fact that our applications was not fully migrated. So I liberate some, some of some of the kind of like coaching and so on. This maintenance API's we had like to post in point to make the changes to the database and one, get, one, get, import and to grab the status of the chain based on the IDs and these services, you use the CAC in order to store the current state of of those update requests. And also we we use the stream boot for the project, like, like it was the RESTful APIs for the database, and we use the AWS rds, and the database was Postgres SQL, and it was released on the database that we used and our log were integrated into the cloud watch and with which we use for the buy house to stream the and stream the those logs into this bunk. And we also integrated all for the security and and, yeah, we we use the cloud services AWS and so it will. It was on the Docker eyes container, on the AWS ECS. So so that a little bit of overview, like the project, and if you want to go specify on the endpoint. So it was a post API's application endpoint, and which was responsible for creating our new applications, and it will also get API applications and ID to return the state of the applications. And then so overdraft for any changes and the withdrawal rate there are and I mean, they had to say, and so those were the three input I was responsible for, the for one of the particular services. And, yeah, the

Unknown Speaker 19:19

the application that we have done it. Thank you. Do you have

Unknown Speaker 19:30

any questions

Speaker 1 19:33

for Yeah, so I would like to know a little more about the like, day to day of the developer on the team and like what I mean responsible, and how much subject matters of expert would developer and be a certain expert, and Once he get on board into the properties? So what will be like day to day work? That's what I specifically want to know,

Unknown Speaker 20:12

delivers product

Unknown Speaker 20:19

as a product

Unknown Speaker 20:21

owner, so probably Similar to the day

Speaker 1 20:30

to day. Kind of take a more typical agile framework ahead of time, and then you will just basically take stories from there. I'll be able to get it to yourself. You'll elaborate your traditional product. Do struggling, yeah, besides that, besides your own, you know, like putting a development team, also working with the league engineer in the design center, you will also working with the junior engineers in coaching, mentoring, helping them with the solution as they need. So kind of like the role, which is this one, it is pretty much in between, like the junior

Unknown Speaker 21:32

engineers. Okay,

Unknown Speaker 21:34

yeah,

Speaker 1 21:35

that sounds good. And also, I would like to know the possible any details about the project. It's account open, right small business. So the portal on, well calm when you want to apply for a small business loan per line, you basically just fill in, you know your information, and then it gets submitted, and then it gets mounted over to back end, so it's as customer facing, digital apathy.

Unknown Speaker 22:09

Okay, got it?

Speaker 1 22:11

Okay. So the last question, so what will be the next mistake for me now? So we have just started with the interview. We will be interviewing few more candidates. And basically we are involved. We are training the candidates, and once the ones which we like, you'd like to get the technical interview scheduled, that with Diane, and then you don't like, if you are in that stream, candidate, just interview.

Unknown Speaker 22:42

That sounds good. All right,

Speaker 1 22:47

yep, thank you. We're right at the 25 minute Park, and I know we're back to back so Richie, thank you so much for your time today, and I will be in touch. And I shared with Allison that I will also be in touch, so I'll chat with you later.

Unknown Speaker 23:01

Yeah, sounds good.

Unknown Speaker 23:02

thank you. Thank you.

Rajiv ranjit-bnymellon-first-srv

Unknown Speaker 00:53

Okay, okay, okay, Okay, yes, okay, I.

Speaker 1 01:17

Hi, I'm Ranjit. I've been working as a full stack developer for Over five years. You

Unknown Speaker 03:37

Pacific i

Unknown Speaker 03:56

It's a contract.

Speaker 1 04:00

It was initially one year contract that was the project is for

Speaker 1 04:15

So sure, yeah, my current project is, I'm a full stack developer, Java developer, so I'm part of the team to kind of develop the credit risk assessment platform. Basically, it's a micro service architecture to simplify the credit rating evaluations. So we have implemented Java eight, Java 11, and currently moving to Java 17. So you know to improve like code, right? So, so it's a also includes like Angular changes on the UI. So I have implemented some angular features to improve the responsiveness.

Unknown Speaker 04:52

Also integrated

Speaker 1 04:55

our API gateway for onboarding our microservices API into the AWS infrastructures also work with Elastic Beanstalk and elastic ECS clusters. So my main job is to kind of create new API, if needed, or maintain our API that handles the credit risk assessment basically, basically this tool is for the UI as a client. So you know our team, they view the users credit risk and their ratings and evaluations. Basically,

Speaker 1 05:47

so basically, we provide several APIs for the financial market about the consumers that we have. So my part is to write, you know, API and services for Credit evaluations. And we have used some third party library as well as some requested some third party API for that, you know, for example, like the credit reports and everything we maintain. So, yeah, the whole application is to support our credit risk assessment platform. I

Speaker 1 07:10

Yes, I'm more into back end almost 70% of the time. But I'm also well knowledge on Angular. I have worked on this on my before project also, so I don't have issue working full time, full stack. I

Speaker 1 07:46

Yes, it's all Spring Boot based application,

Unknown Speaker 07:51

3.1 point two,

Unknown Speaker 07:53

I guess, or 3.2. I

Speaker 1 08:21

I'll rate myself around Like 3.5 I guess,

Unknown Speaker 08:32

433

Unknown Speaker 08:54

3.5, point five. 3.5 I

Speaker 1 09:18

so record is in Java is like, like new like classes where you don't have to declare any like, you know, getter setters or some basic like, it was introduced after Java 14, like, so basically they have some auto generated methods, like the constructor equals method, or two string methods. So the field are final, by default in Record.

Speaker 1 10:10

Final field or final means the parameters,

Speaker 1 10:21

yes, we have Lombok annotation, yes, yes,

Unknown Speaker 10:26

you have to use the data annotation. I

Speaker 1 10:38

I mean, you don't need to customize the generated method like in records. There is one limitation, if you want to use you cannot customize any equals or has code method, and it doesn't support the inheritance. Also that is one drawback. But comparison to using Lombok annotation, it provides like concise syntax, right? You don't need to use any annotation, or, You know, like Lombok as a library.

Speaker 1 11:15

I Yes, record does not support inheritance. You mean cannot extend other classes. I mean, they can implement the interfaces.

Unknown Speaker 12:13

Sorry, what

Unknown Speaker 12:19

can you repeat? I

Unknown Speaker 12:37

so they are, I

Speaker 1 12:40

mean, yes, they are introduced. I think Java 15, I mean silk classes restrict like which classes can extend them and which can classes cannot extend them. So they kind of restrict the inheritance, right? It provides like more controllable way of inheriting the properties. So

Speaker 1 13:23

I mean, it's in a more controllable way, so they will restrict like which classes, which other classes, which classes can extend or implement them, you need to define that like you have to define the known set of subclasses.

Unknown Speaker 13:48

So you have to use the keyword permit, do?

Speaker 1 14:02

I mean final class, you cannot do any inheritance, right, you cannot extend, you cannot use the subclasses, right, but in this sealed class, you can still allow some of the classes to extend by using the permit so

Speaker 1 14:40

so it was introduced to control the inheritance.

Unknown Speaker 14:48

Sorry, I think your boys cut off.

Unknown Speaker 14:53

I think your voice,

Unknown Speaker 14:56

your audio, is interrupted.

Unknown Speaker 15:07

Your camera froze also,

Unknown Speaker 15:10

your camera froze and

Speaker 1 15:30

I mean, they provide more concise way, right? They also provide a control way to inheritance, right? So it it supports like clear data structure and it's easy to maintain once the code become concise, because you don't need to write the boilerplate code like constructors or has code methods there also, like sealed classes, will restrict other classes to inherit, so it becomes clear you know which classes can extend them and which classes cannot extend them. That way, it is more easy To maintain the code base. So

Speaker 1 16:46

yes, I use the streams. It was introduced in Java eight.

Unknown Speaker 16:52

It is a way

Unknown Speaker 16:56

I have used filters

Unknown Speaker 16:57

maps. I

Unknown Speaker 17:16

i have used flat map. I Have

Unknown Speaker 17:20

Yes, yes, I have editor so

Unknown Speaker 18:25

let me share my screen. I

Unknown Speaker 18:38

I can use a notepad to write the code.

Speaker 1 18:43

I have a IntelliJ, but it's kind of slow Right now

Unknown Speaker 18:49

it's asking me to activate so I

Unknown Speaker 19:28

hold on, hold on, Hold on I.

Unknown Speaker 19:47

Okay, Do

Unknown Speaker 20:13

you said, Boyd method I

Unknown Speaker 24:03

yeah, let Me Import I

Unknown Speaker 25:43

go back to it.

Unknown Speaker 25:47

Go back to it. I

Speaker 1 27:52

so you want me to Write a separate method, or just here with the stream API?

Unknown Speaker 28:11

Okay, sure, give me one second.

Speaker 1 28:18

Yeah, you need a map right with The ID and The employee object you

Unknown Speaker 29:40

give me one Second, my Screen froze. I

Unknown Speaker 30:26

bro Drop the Call So

Unknown Speaker 33:50

team so

Unknown Speaker 33:57

Except,

Unknown Speaker 34:08

Okay. Sonko,

Unknown Speaker 34:32

suppose.

Unknown Speaker 34:42

Tell us your God, sorry. Ticks, okay,

Unknown Speaker 34:47

but scream golden summer

Unknown Speaker 34:50

team call Cardinal meal Joiner. Okay.

Unknown Speaker 34:56

Lola, Bossa. Bossa boy,

Unknown Speaker 35:21

can you hear me?

Unknown Speaker 35:33

Your voice was not coming.

Unknown Speaker 35:42

No, my screen froze.

Unknown Speaker 35:44

Let me write it. I

Speaker 1 35:59

so here I was trying To use the map collectors map I

Unknown Speaker 36:39

Yeah, I was trying To finish this. I

Speaker 1 37:23

so this will give you a map of ID with the employee. So

Unknown Speaker 38:00

yeah, I'm trying to type. I

Speaker 1 39:30

Yes, in my last project, I worked on multi threading. I multithreading. I mean, I use the executor framework yes to define the thread pool, like default thread pool, and then,

Speaker 1 39:49

and then it was For my Banking project. You

Speaker 1 40:58

so it will create like thread pool of for executor service, which are fixed thread pool of size four. It means that I can run for task concurrently. So

Unknown Speaker 41:22

I have to create more than four. I

Speaker 1 41:42

Oh, that yeah, in that case, I mean, the tasks are queued, so the thread are executed Once the thread becomes available.

Speaker 1 42:13

So dot submit method and then, so let me okay, let me write for loop. So

Speaker 1 43:08

so you have to submit through the framework and create new task and then submit it.

Speaker 1 43:32

So after you submit the task, I mean, it will the thread will run.

Unknown Speaker 43:37

It will start executing immediately.

Speaker 1 43:43

So let's say you can create as many tasks, but since we define only four thread, it will execute the thread pools, so the task the remaining tasks are executed. After the your previous tasks are completed. You

Speaker 1 44:23

Yeah, volatile keyword means you use it will let the state of your thread, uh.

Speaker 1 44:36

It will share the state with other threads when the variable is declared as volatile.

Speaker 1 44:49

So change made by one thread to a volatile variable are immediately like visible to other threads. I

Unknown Speaker 45:17

Sorry, can you repeat that? I

Speaker 1 45:36

I mean it mainly visibility of your variable. So when you declare volatile they are immediately visible to all the threads, right, and volatile variables are not cast locally In the thread specific cast.

Unknown Speaker 45:54

So every like,

Unknown Speaker 46:14

I mean, they

Speaker 1 46:16

can be cast in the local trade thread local caches. So for example, if you are using multi threaded environment, right and then one value updates so that variable is updating, but the other concurrent thread does not know the state of that variable, then it might cause some issues. I

Speaker 1 46:56

a trail local means that it is a special class. Basically, it provides trade local variables. So it's, for example, each thread that is accessing variable, they have their own independent copy of The variable. So basically they are private static field. So

Speaker 1 47:35

So basically, you don't need to do the synchronization, because they have their own copy. They are not using the same copy. So

Unknown Speaker 48:08

you mean in the trade local I

Speaker 1 48:23

I mean, no, not the variable synchronized. You have to define the method as a synchronized if you want to use the synchronized method. So

Speaker 1 48:49

so I'm so, I mean you cannot directly define synchronized variables you want to Use the synchronized as a method.

Unknown Speaker 49:14

I mean static method.

Unknown Speaker 49:17

So that means that

Speaker 1 49:20

static method means they are available. You don't need to create an instance of that class. You can access it. I

Speaker 1 49:43

Yeah, static method means that they basically they are belong to the class itself, rather than any specific instance, right? So they are associated with the class itself. So you don't need to create any instance of the class if you need to call the static method, those are static methods. You need to use the keyword static specifically, whereas non static method, you need to create The instance and call that method.

Unknown Speaker 50:33

So in that case,

Speaker 1 50:43

so, yeah, when the static method is synchronized, I mean, it can put lock on that class object rather than the instance of that class, right? So it will it can be applied. So, to ensure thread safe operation. You can use a synchronized static method.

Unknown Speaker 51:26

So partitioning Kafka, are they are like division, yeah,

Unknown Speaker 51:43

yeah, yeah, I'm fine with that. So partition is basically division in Kafka topic.

Speaker 1 51:50

So each topic is divided into one or more partition

Speaker 1 52:10

so each party partition is like a division, or like so it allows to scale Kafka horizontally. If you increase the number of partition, you can spread the data so

Unknown Speaker 52:55

it does not you.

Speaker 1 53:19

So for that case, I mean, you can increase the number of consumer. Instance, for example, you can do the scaling

Unknown Speaker 53:31

one. With the consumer group, right so?

Speaker 1 53:52

Yeah, you can use, like some thread pool to kind of allow parallel processing in your processing. That way it will increase the processing to optimize the output. Or you can use asynchronous processing, so that way you can use a separate thread, using the thread pool to avoid blocking of the main consumer thread, so parallel processing, you can basically do like use a thread pool.

Speaker 1 54:30

So you can define some executed service thread pools and execute those tasks parallelly, concurrently. I

Speaker 1 54:58

You mean like it's a identifier. It is assigned to each record within the partition, so each partition has the offset as ID.

Speaker 1 55:15

So basically, it's a unique number that is assigned to each message Within the partition. So

Unknown Speaker 56:32

yes, yes,

Unknown Speaker 56:42

Can you Repeat that? I

Speaker 1 57:19

Yeah, so I have even number list or number list and a prime number list. I

Speaker 1 57:31

so and then I have all number list which has the list of list of integers, and you want to get only, like one list of integers, including all the even numbers or numbers and prime numbers, right? I

Unknown Speaker 58:06

it's a list, yeah, yeah, okay, give me one second. I can use the stream API.

Unknown Speaker 58:28

So I'm thinking about using flat map. I

Unknown Speaker 59:27

I guess I can just use the stream function inside that.

Unknown Speaker 59:30

Let me Try it. You

Unknown Speaker 1:01:10

Yeah, Can you tell me about, like, about the team And project itself? I

Unknown Speaker 1:02:04

Okay, sounds good. So what should be my next step? I

Unknown Speaker 1:02:16

Okay, sounds good. Oh no, that's all. Thank you.

Unknown Speaker 1:02:27

Yeah, something happening with the Internet? I think I need to change my internet. I

Rekha Bhandari- Second Round-BlankFactor-SRV-Second

Speaker 1 00:00

Know, pretty good experience working on the micro services architecture, and like designing them and developing them and also deploying them. And I also like to have experience working on the front inside, using the Angular and also on database, including Postgres SQL and DynamoDB on the AWS. And currently I am going to scrum team. And my team consists of like five developers, and on the daily basis, I'm involved mostly on working on the back end Java code, helping the team with code reviews, and also, like on the production, support and reviews.

Unknown Speaker 00:45

Sounds good.

Unknown Speaker 00:49

So, Krishna, you want to take the lead.

Unknown Speaker 00:54

I will take after you, unless you can continue. Okay,

Speaker 2 00:57

so, so, Rika, you said, like you microservices? Can you explain what exactly are microservices?

Speaker 1 01:09

Of course, like the microservices is basically, you know, it is a small, like, independent services that is, like it is a part of larger ecosystem. And like it is designed like in such a way that it is loosely coupled from, you know, your service and like, like it focus on specific business, right? So basically, like it is a, you know, single functionality or capability, and I know it has a good focus on, you know, a particular services. So that's what a micro services is, right? So, like,

Speaker 2 01:44

so how did you, how did you design your micro services? Like, what technologies to use and how

Speaker 1 01:50

for designing micro services? My micro services was like, basically, like, written in Java, so it's been good base application. And primarily we did, like, have component that was AWS lambda return in like, no, like nodes, yes, but all the micro services was a part of JavaScript application deployed on the AWS ECS clusters.

Speaker 2 02:17

So what exactly is your micro service? I mean, I just want to understand. What is the like, starting and end of your goal, of your service. I'm sorry. Can you please repeat that? So what, what is the what is that you're achieving through your microservice? Like, what is your microservice doing? Okay,

Speaker 1 02:37

so let me walk you through a couple of ones that I did. One of the recent one was, you know, maintaining the API. So this particularly like this, you know, this particular service was responsible for, you know, making the changes to your user profile preference, right? So the whole domain frontline you have fronting portal that you know represent, like your dashboard. And then if you want to make any change how, like, the shipping information is updated, it's a part of profile preference. And to make those changes, you call the, you know, those services called the maintenance API, which is I know, responsible for the making changes to those profile preferences. So like that one is the particular microservices that I recently worked on, and the one before was called the Experience API, and like this particular API is, you know, useful for making the changes to the user like entitlement and getting the user entitlement right. So user are entitled to view the different component on the front, inside, and these components are basically, you know, control through the entitlements. And these entitlements can, you know, can be either modified and aided, and then we try using like this experience API service. So does

Speaker 2 04:15

your microservice have database? Yeah, of course. So like, how, like, is it like, how do you share the database across micro services? Is it like one database per micro service, or is it like multiple micro services? Use B, C, M, database?

Speaker 1 04:35

So, like, it is a database for service basis, you know, like, the maintenance API has, you know, the dynamo, like, sorry, the push phrase is a, you know, is the back in database. And then experience like, the API has, you know, dynamic, so,

Speaker 2 04:54

so, so your each micro service has its own, distinguished Microsoft database, right? Each service has its own database, right? Yeah, so. So, like, if you have to design a system in such a way like that, I want to micro services to utilize one database. So what is your approach on that? Or, like, I wanted to share data across multiple services. Like, how do you design it?

Speaker 1 05:28

So? Like, in that case for two microservices? Like, if you want to design a system with two microservices connecting to a singular database, basically, you can, like, you would have to see your database, right so the both services will be, you know, connecting to a CR database and like, like, basically, you might need to have a proper, like, a Proper logical operations, maybe through our table partitioning or so on. So like is micro service maintain. Like, mainly, you know, interact with the specific tables or a set of table, reducing the inter systems dependencies. But if you do not want to separate that you like, you need to proper data access layer, where you know each service data layer are separate and they, you know, communicate through our database connection layers.

Speaker 2 06:29

Okay, so, so you have you mentioned. I mean, I see in your profile like you have good experience with technologies, right?

Unknown Speaker 06:38

Yeah. So,

Speaker 2 06:41

so what is a big factory and application context? And when do you use bean factory or without an application

Unknown Speaker 06:50

context? So like,

Speaker 1 06:54

like, basically you're like, being factory. Like it is a like. It is basically, you I would like to say that it's a container like responsible for, you know, like initializing a configuration and managing a bin, right? So, like, it just basically, you know, provides like capabilities needed for the dependency injection. But you know, your application context is in like extension to this pink factory, making it like, you know, like provide advanced and enhanced like functionality so like it handles like initialization and dependency injection features you use applications context in web application where you need to, like, More than dependency injection, like you need event handling or the spring IOP related stuff, but you use, like those bin factory when you need, like, a very simple, basic dependency injection.

Speaker 2 07:55

Okay, so you're saying, so what do you mean, basic dependency

Speaker 1 08:00

so the basic dependency injection, I mean, like you, you do not need any, any kind of extension initialized approach. And you, you know, have a very light weight application.

Unknown Speaker 08:14

So, so what I mean by dependency injection, right?

Speaker 1 08:17

So, in, in the context of like Java, the dependency injection is just a pattern where the objects are the dependencies that you need. Like for your classes are provided by, you know, external source, right? So, so instead of, you know, creating an object of the service you you know, directly inject the dependency from the external layer. Like, for the spring based applications are the, you know, in version of control take care of dependency injection in the spring containers.

Speaker 2 08:53

So, like, how does spring know? Like, what objects needs to be created?

Unknown Speaker 08:58

So in spring,

Speaker 1 09:01

like, so basically, there are, you know, multiple ways, right? So it basically scan through your packets and then look up for the bin definition, right? So it can be done through the configuration classes, which are the classes annotated with the configuration annotation, right? So within this configuration classes, you have a methods annotated with the bin annotation, and it just keep you like how certain bins need to be initialized. And then, apart from that, the components is scanning, you know, like, take like, take place when it scan your class path and looks for the classes that are annotated with, like the component or the services or repositories or the like controllers annotation, right? So based on these like, it will load those like in as well. So whenever you want those objects, so

Speaker 2 10:03

when you so when you create Spring Boot application, right? So what is that like? Key annotation that that does all this process like, so you're saying it goes through components and everything. So what is that annotation, which takes care of all these like, searching for the components or creating those space, like, what is the directation that that does for you? So,

Speaker 1 10:27

like, overall, in like, spring good, it is spring good application annotation, which is the combination of the multiple annotation, the one that take care of, you know, automation also, what?

Speaker 2 10:41

So what are part of the springboard annotation, Spring Boot application? Like, I'm sorry,

Speaker 2 10:50

so what? So what is the annotation Spring Boot application? Whatever the annotation you give right on top of Spring Boot. And what is that exactly?

Speaker 1 10:58

So, like, he delivers the auto configuration annotation and like that one. So, what

Unknown Speaker 11:03

is so what? What do you mean by auto configuration?

Speaker 1 11:08

The auto configuration is, you know, it basically like automatically configure the beans and the settings based on your application and the class part, right? So, for a for an example, you can add Spring Boot starter dependencies. Any Spring Boot will automatically configure our web server like, like your basic MVC framework.

Unknown Speaker 11:34

Okay, so like,

Speaker 2 11:38

how do I include or exclude? Like some of the classes that that being loaded. So you're saying the auto configuration handles the loading rates, but I wanted to exclude few classes to be loaded. So how do you do that? Like,

Speaker 1 11:57

so on the Spring application annotation, or like even the auto configuration annotation, you have a field called the Exclude where you can provide the class that you want to like exclude, right? So you don't load them you can like, basically provide that.

Speaker 2 12:16

So can you create your own Spring Boot auto configuration annotation?

Speaker 1 12:24

Yeah, like so to create ANR, you know, auto configuration, innovation, I think we should like, Yeah, usually like, we should be possible,

Speaker 2 12:35

yeah, so don't know, like, what, what? That's when. Yeah, questions. So, so you have experience with spring security, right? So, so what version of spring security Did you work like,

Speaker 1 12:54

in case of spring security, I think like we had a recent migration like to, like in my last project, so we have, like, a spring migration to the, like, spring six. So we just had a basic migration from, I don't like, actually, this version, but we

Speaker 2 13:19

so that's fine. Like, how did you implement spring security? Like, so, how did you implement implement security on your end, but is your endpoint secure? Okay.

Unknown Speaker 13:31

So, like, how did you do that? Yes.

Speaker 1 13:32

Like, we have role based access control on our influence. So, like, the way, the way we implement the spring security is, you know, like it is basically at your, you know, very spring multi starter dependency on your form, dot XML file, and then you just configure the spring security configuration adapter, right? So just create a configuration class and then enable the web security. And then like override the Configure method on how you want to, you know, authorization and authentication to take place. So then, based on the rules, like is available from the, you know from the also, like, can restrict the access of the influence using the pre authorized annotation and providing the roles on those inputs so on the particular user can, Like, rewild,

Speaker 2 14:37

okay, so, so, what is worth? O,

Speaker 1 14:42

so, or tool is, you know, like it is, like it is like, simply a like authorization. Each standard, basically you are, you know, defining like it is like, you just provide a way to say for access authorizes or tokens, and then like they need to also provide when to certain resources.

Speaker 2 15:12

So we have experience with spring security, right?

Unknown Speaker 15:19

Spring GPA as well, right?

Unknown Speaker 15:23

Yes, so,

Unknown Speaker 15:26

so spring JPA? So,

Unknown Speaker 15:34

yeah, so, so what is spring JPA? The

Speaker 1 15:38

spring JPA is, basically it is, like, I like, I would say just, it is a simple, like, simplification for, you know, interacting with the database, right? So, like, you just provide the interface that make it working with the database easily. Like, for example, if you want to talk to a relational database, you can just use your repositories annotate interface that like external JBA repository. So once you have done the like, have this done, and you can just, like, issue your basic queries using like, very standard pattern, like find all, or find by ID, and also you can, you know, write.

Speaker 2 16:24

So what's the difference between spring GPA and Hibernate? Like, when do we go for hibernate or when we go for spring GPA?

Speaker 1 16:32

So if you have to differentiate between hibernate and GPA, basically your spring GPA is is like it is a part of the spring project which aims, like, for simplified operation, right? It is, you know, it is very high level of abstraction, and it can use the hybrid internally as a like, as a JPA provider. But, you know, it is just an abstraction API, but your Hibernate is, you know, it is a fully complete ORM framework, right? So it provides a way to map your Java object to a database table.

Speaker 2 17:13

So can I use either JPA or hibernate, or is it like I need to use both of them to talk to my database.

Unknown Speaker 17:20

So I think

Speaker 1 17:25

So in my view. Like, you know, spring JPA, like, internally, can use the Hibernate, right? But

Speaker 2 17:33

you can, I got the answer. So next question is like, how do you do transaction management in JP,

Speaker 1 17:42

okay, for for doing the transaction management. The spring, you know, directly provides a transactional annotation. Who is like, you can use in the method to say, like these, method has to be a part of one transaction.

Unknown Speaker 18:00

Okay, so they're like multiple annotations and multiple

Unknown Speaker 18:05

parameters. So,

Speaker 2 18:16

so you have experience with Cloud also, right? So did you use Spring Cloud by any chance?

Speaker 1 18:25

Like I have worked with Spring Cloud gateway a little bit, and, you know, the Spring Cloud Config Server, server for, you know, dynamic config resolution.

Speaker 2 18:36

Okay, so let me get back to JPS. I have one question. So, do you know persistent context? I'm sorry,

Unknown Speaker 18:44

can you please, persistent context? Yeah.

Speaker 1 18:47

So basically the persistent, yeah, basically our persistent context in GBA is, you know, just like, like, it is a managed entity that, you know, like that are used during our transaction or the session, right? So it just ensures that your entities are read, drive and updated and persisted during the transaction.

Speaker 2 19:14

Okay, so what happens if I so you have entities, and what happens if I make change to the entity, but I don't save it to the database. So what will happen then? Like, so, so, so there is an entity object, an employee. Employee has name, salary or the other it is so. So in employee entity, I made change to his salary, okay, but I didn't commit the changes. So what will happen in spring GPA?

Speaker 1 19:51

So Well, your entity, your GPA, it automatically, like process the you know what, system context at a certain point. So before any transaction processing will like that, if you do not want to commit those transaction right, those changes will not be posted. So basically, it does not automatically trust transmit the database.

Unknown Speaker 20:19

So you mean by default it will

Speaker 1 20:23

commit or it will not commit. No, you need to explicitly commit to it will not comment

Speaker 2 20:31

directly. So anytime I make changes to the entity, I need to explicitly commit it. Yes, okay.

Unknown Speaker 20:42

Yeah, so coming back to your other

Unknown Speaker 20:47

be comfortable. You don't have to

Unknown Speaker 20:56

be okay. So,

Speaker 2 21:00

yeah, so, so MongoDB and MySQL, right? So these two, you have experience. So when do you go for MongoDB and when we go for MySQL,

Speaker 1 21:12

for database, my skill is, I know is for additional database, right? So, like, if we has a, like, very structured and predefined relation between the columns and it has, you know, well defined schema that follows, like certain properties, then you will go with a relational MySQL database. But like, if your data is, you know, a kind of document oriented or does not have a like defined structure or like you go with the MongoDB. So that is the major point that declare whether you go with MongoDB or with the MySQL.

Speaker 2 21:53

Okay, so do you have experience with any other relational databases other than MySQL?

Speaker 1 22:00

Yes, I also have, like, work with Postgres.

Speaker 2 22:06

Okay, so, so again, same question, When do you go for MySQL, or when do you go for Postgres?

Speaker 1 22:14

So, in case of like, Postgres, I say like, the postgres is a little more advanced. So, like, in sense that Postgres allows you to, you know, write more like SQL compliance and data integrity features. But my SQL is, you know, it is a very lightweight and it is for a simpler setup and configuration, right? So if we have very, like, heavy application that need to handle the complex queries you You then want to go with push space because it has a proper support. But MySQL is used for, like, more, like really heavy operation, right? So that's why, like, that's how you pick between these two.

Unknown Speaker 23:03

Okay, so,

Unknown Speaker 23:06

yeah, so,

Speaker 2 23:08

so you worked on micro services. Like, how do these micro services communicate? Like, I have, like, lots of micro services running in my home system. How do these communicate?

Speaker 1 23:22

So, there are, like two ways of communication between micro services. One is the synchronous communication using REST APIs, and the other uses asynchronous communication, either by using the, you know, the Message Broker, or something like Kafka.

Speaker 2 23:41

So when do you go for asynchronous communication?

Speaker 1 23:48

For asynchronous communication, whenever you are, like, do not rely on the data or, you know, or the response back right away, and you have, like, some kind of event that you know triggers on the external work or the progress, then you would go with the N asynchronous communication, right? Because you know you like you do not worry about the response. Okay,

Speaker 2 24:18

so what message service did you use? Or did you use any event,

Unknown Speaker 24:24

like Event Bus, anything during this

Unknown Speaker 24:28

asynchronous microservices communication?

Speaker 1 24:32

So for the like, I was little bit on the AWS like, we use SQS, SNS to trigger the lambda. So the SQS SNS is a messaging service that is provided. So

Speaker 2 24:47

what's the difference between SQS and SNS? Like,

Speaker 1 24:51

if you have to differentiate the SQS and SNS, SNS is basically on notification service, right? It has you know different topics, which has you know public subscribe model, and it allows you to send a message to multiple subscriber at once. So but like in context of SQS, it is a simple, like queue service. It provides like queuing of the data between the components, right? Like you, you can send a message to Q that will be passed down to a like a subscriber.

Unknown Speaker 25:31

So

Speaker 2 25:33

we have Spring Boot actuator, right? So what is Spring Boot actuator?

Speaker 1 25:39

Talking about the Spring Boot actuator, it is like, it is just one of the dependency that is, like, really useful for the products and features. So it, you know, exposes multiple inputs for, like, readiness and liveliness, right? So,

Speaker 2 25:55

so what is, what is read, what is readiness and liveness? Like?

Speaker 1 26:02

It means so basically it means your readiness and liveliest, sorry, liveliness are to expect that defines whether your application is ready to serve traffic or if it is running on LD or so. You know the readiness means the your application can now take the traffic and liveliness means that your applications is still running and like in a healthy state.

Speaker 2 26:36

So we have experience with some of the AWS, right? So, AWS cloud watch, so, so use AWS cloud watch for your monitoring.

Speaker 1 26:52

Yes, like, I have, we have it like, use the AWS cloud was for, like, only a couple of services, but a lot of our monitoring is done through Splunk and new, like, really, okay,

Unknown Speaker 27:06

so what is J meter used for?

Speaker 2 27:10

I'm sorry, J meter, like, I see J meter in your yes,

Speaker 1 27:13

we have a dedicated, like, for water. So the J meter is, I'm sorry, the J meter is, you know, used for the performance script. So, like,

Unknown Speaker 27:30

so did you write any, any performance scripts?

Speaker 1 27:35

So we sometimes have to run performance testing to, you know, load and stress the test our application, we have like, dedicated application called like Hercules, which is basically on EC two stance that, you know, runs like this test and scale them accordingly. So I have written some like, like gem scripts to test the load registers, and you just, you know, provide a UI where you can configure the request and like the you know, the failure and success scenarios and then run them.

Unknown Speaker 28:16

Okay, so how do you scale your applications?

Unknown Speaker 28:20

How do you make sure your application, your microservice, scale?

Speaker 1 28:24

So, like scaling is, you know, primarily based on the you know, current state of the application, right? So we have it on the AWS ECS within the you know, auto scaling. And the auto scaling get trigger when you know the CPU percentage is above the 45% and also the memory uses is above the 50% we have a minimum and maximum policy, or the test with, you know, the Auto Scaling group, so based on The incoming traffic the member uses and the CPU usage may scale the application accordingly.

Unknown Speaker 29:06

Okay, so,

Speaker 2 29:10

so did you heard about how do you make your services resilient, like, like, one microservices communicate with the other micro service. But if micro service B is down and micro service is waiting on the response from me, like, how do you handle

Unknown Speaker 29:30

so like, including a

Speaker 2 29:41

moment like I'm talking about fault tolerance. Like, okay, two microservices are there, like, micro service a and micro service B, like, I'm waiting. So if micro service B is like, is not responding back, like, how does micro service a should handle it out? Like, what is that whole process for like, like fault tolerance, right? So this, how do you handle fault tolerance in these micro services?

Speaker 1 30:11

So for resilience, right? You can implement the certain design pattern. One of them is like circuit breaking, where you know, if you are down, like your downstream service is not performing well, you can just call the fallback method until the downstream service gets, you know, time to recover. So this one and then like, there is like, rate limiting where like, which is basically making sure that you do not, you know, overload the downstream service by, you know, limiting the number of the calls you can make. Other one includes, you know, balancing like, which means you do not always like, hit the same instance, but, you know, distribute the traffic to a downstream service accordingly, yeah, those are some of the resilience pattern you need to use in order to, like, have a proper, scalable application. Okay,

Unknown Speaker 31:13

I'm good. I'm good. Thank you. So

Unknown Speaker 31:16

Krishna will

Speaker 2 31:18

ask you questions in Java. Okay, so I'll pass it to Krishna,

Speaker 3 31:27

yeah, I can see that you have mentioned you use the Java 811 and 17, right? Yes. So when have you migrated from Java 11 to 17?

Unknown Speaker 31:41

So like at Hawaii, it is like

Speaker 1 31:45

11 to 17 was, like, kind of recent, you know, it was about six months ago in my like, recent project.

Speaker 3 31:54

Okay, so can you please explain the differences between Java 811, and 17, please,

Speaker 1 32:02

three of those. Like, if you have to differentiate among those three versions, I say like, the primary difference is between like, like, how the underlying, you know, the structure where the modified, right? So basically, Java eight introduced, you know, like, a lot of enhancement and new features, including, you know, stream API for collection processing and also, like, incorporate a very like standard data API to work with your Java 11 also introduced a new HTTP like client API for making the sttp request, and also like it also introduced new HTTP client API for making HTTP request, and also introduce a local like variable syntax. But the latest double 17 made like silk classes, which is like these classes that restrict, like extension, and also it introduced pattern matching for, you know, switch classes. So yeah, those are the primary differences that like is that can be seen on our day to day use, but internally, they have changed on how the memory are managed and how the garbage collection works.

Speaker 3 33:30

Okay, so, and you also mentioned that you have experience with TypeScript and JavaScript, right? Yeah, can you please expect me the benefits of TypeScript over JavaScript.

Speaker 1 33:45

So, yeah, basically the type, sorry, the type is script, like it is a main suggest that, like it just strongly typed. So, right? So you have a static typing on it, which means that you provide a type, right? So if you are trying or number you like, can give the type number or the stream and so on. So like this is a primary here which makes it, makes it easier to detect error earlier, rather than during the runtime, which reduces the box.

Speaker 3 34:22

I Yes, okay, so I'm going to paste a small program in the chat box. Can you please tell me the output?

Unknown Speaker 34:35

Yes, step by step. Let me see

Speaker 1 34:40

let me see so you are doing a loop from one to 10, and then you are printing a value of i. And after that you are going to another loop from zero to five, and then you are printing up the value

Unknown Speaker 35:19

of j, and when J reaches to

Unknown Speaker 35:23

you are going up doing a break. So,

Speaker 1 35:32

so, yeah, so, basically, the way this code goes is like it will have like,

Unknown Speaker 35:46

I so like,

Unknown Speaker 35:52

so I, zero and

Speaker 1 35:55

J, zero and j1, j2, and test this zero, yeah, go ahead, and then,

Speaker 1 36:17

then it will look for I again, and then

Unknown Speaker 36:23

I, 1j

Speaker 1 36:26

zero, and then J again, one, j2,

Unknown Speaker 36:32

and this one,

Speaker 1 36:37

and then It will continue this pattern until then, and loops until the value of i is nine, and then it is done.

Unknown Speaker 36:49

Yeah, yeah, when will the tongue is printed? I'm sorry.

Speaker 3 36:58

I was asking, When will be done. Is printed? Okay for dawn? Yeah,

Speaker 1 37:06

so the dawn will be printed after, I think, 10 loops, or when the value of nine,

Unknown Speaker 37:16

yeah, after 10 loops or

Unknown Speaker 37:22

so, after this nine is printed, yeah, yeah, after this nine is printed,

Speaker 3 37:32

yeah, so, and you also have mentioned You have experience in Angular two as well, right? So, how data will be transferred from one component to another component in Angular?

Unknown Speaker 37:53

In the context of Angular,

Speaker 1 37:56

like to moving data from one component to another component is a little, you know, complicated, right? So because you can either do, you know, parent style or the relation, and then do input or and the output decoration in order to like pass this data, and you can also like route or resolver, where you can define how you want to resolve this so, like data field, but if you do not, like have those, you know, parent side component relationship, then you need to define a shared Service to share the data across, like those components. Yeah.

Unknown Speaker 38:43

Okay, so

Speaker 3 38:48

can you also tell me a few methods in hash map? Yes, you're

Speaker 1 38:58

so like you have, like your PUT method, which is used to add the data to your his has map. And you have a gate method that is used to get and values based on the key. And you have, I think, a key method to check if the key exists. Like contents, like contents, key method, and then you have a remove method, which is used to remove a specific key. Okay,

Speaker 3 39:34

so can you explain me about the optional in Java eight? Yeah. Like,

Speaker 1 39:45

so, yeah, optional. Basically, it is a wrapper object that was, you know, introduced to prevent, like, non pointed, except some you like, you know, instead of having object return, you would return an optional, which may contain a value or not, right? So basically, using these you like you to check you to see if the object is prevent before, like present before, and doing any like processing, which is like, basically prevents any non pointer exceptions. Okay. How do you

Unknown Speaker 40:28

handle internal server error?

Speaker 1 40:34

So in fact, in case of internal server error, like there are 500 labels for on HTTP or previous APIs, right? So, yeah, this basically means that something, like something went wrong during your request processing, right? So, like there may be some kind of non pointer exception or handle exception. So in order to handle these kind of exceptions, you, like, initially, me may need to configure, like, figure out the root cause behind like these and try to fix the, you know, the block of the code. But like, if, like, if you want to return, you know, the meaningful response back, you can create a controller, advice like which is like global exceptions handler and like that, like that, define the exception handler method that you know the returns are meaningful response back to the user.

Unknown Speaker 41:36

Okay, so how do you

Unknown Speaker 41:38

handle the exception handlers?

Speaker 1 41:43

So like, the way to, like, handle the exceptions in Java is using the try catch block. So in a so in try block, like is where you write the code, where you might have an exceptions, and then there is a catch block like which is a block where you can handle the certain exception, right? So you can have multiple cache blocks that define what exception it needs to handle, but you know, it goes on a hierarchy order, so the smaller exception come first, and then the next bigger exceptions. You can also throw an exceptions on your method signature, but you know it would need, need to be caught by the method calling that, like calling that a method, okay?

Unknown Speaker 42:44

Can you explain me about the concurrent hash map?

Speaker 1 42:50

Yeah, like so, concurrent hash map, it is basically a thrill, safe, fast map, right? So it provides, you know, like internal synchronization. So if your application is running on multiple thread, your component hash map works, you know, properly on those multi threaded environments, right? So it provides like a proper locking mechanism to, you know, reduce the data inconsistency. So

Speaker 3 43:26

so how do you optimize database interactions in your projects? Unlike, what specific techniques did you use to enhance performance?

Speaker 1 43:39

So like some of the technique that we used in the database optimization is, firstly, like, if you have a, you know, a database query that uses, you know, filtering, or where the clauses or the specific columns you need to, then you need to index those columns, right? So, yeah, basically, like, indexing those column enhances the the real operation by a whole lot. And also the other way is, you know, not like, not load a lot of the data, right? So integration integrates some kind of, like pagination that so that you don't have, like, load the hundreds of the records at once.

Unknown Speaker 44:37

The other way,

Speaker 1 44:40

the other way is to, you know, use the casing where, like, which is basically load, frequently access static data in memory and then read it from The memory. And, like, invalid the data them periodically. I

Speaker 4 45:00

Okay, so

Unknown Speaker 45:08

I was looking at your resume,

Speaker 3 45:12

so I can see that at old domino freight line, you have used Ng and x, right. Yes. So can you explain how you handle state management in Angular applications, and why did you choose NGRX?

Unknown Speaker 45:32

So the state management

Speaker 1 45:37

like in Angular, like the state management in Angular is done, you know, through the NGRX framework, where you know, you define the different stores, which is, you know, like a central repository, repositories of your state. And basically, you know it like it has a observer state, like, which can be, you know, access from the components and the services. So basically you can like define the action so that describes any like state change. And basically defining the action, you create an action, and then you define the reducers, which determine how the like transition state are handled. And then you define the like selector that are used to query and those particular stores in specific slice of state, and like you may like need to define the effects along with that right? So basically, NGRX is just a proof powerful tool to manage the state, which costs it off like your core, like excels, reducer and reducers and the selectors,

Unknown Speaker 46:49

okay, how did it depend fit your project?

Speaker 1 46:55

So basically, the NGRX, it just makes sure that our like application state is, you know, like, predictable and like and it will like, it makes easier to manage, right? So like it like it is just a for it just follow our, you know, Redux pattern, so that our state is never like, muted directly, so it was always like on a stable pattern, yeah. How

Unknown Speaker 47:27

did you improve overall application performance, for overall

Speaker 1 47:35

performance on the front end or the back end? Back end. So for the back end we can

Speaker 1 47:53

so for improving the performance of the backing you need for you know, include, like, the pairs processing, parallel processing, and implement the some kind of caching technology so the multiple steps, like, accordingly, depending on what kind like, what kind of error you are having. So, okay, sounds good.

Speaker 3 48:24

So can you like discuss a scenario, like, where you have used Kafka for data processing, and how did you ensure reliability,

Unknown Speaker 48:38

like, unhandled any errors?

Speaker 1 48:42

So if I have to recall a scenario for the Kafka, like, one of the recent use case was like when we were, you know, we were to separate the notification service from our application. So we initially like call this service within our application itself. But you know, whenever this Notification Service went down, our application like would fail, and we would like become, you know, irresponsible, right? So we need to start our application so, like, in order to fix this kind of issue, we like, basically, like, had a Kafka topic where, like, we publish a sorted data, and then like, these messages would be consumed by one of our application which is configured to be like, like a Kafka Consumer, And then this Kafka Consumer like, would be responsible for, you know, making, making this call downstream, right? So only for some what, like, what was responsible for. Our main objective application was like, just send the data to Kafka topics. So this basically improved our overall application in the sense that I like I did not have to worry about the notification service. So now the way we handle the error is, whenever the notification service went down, we had, you know, retry mechanism with like exponential backup. And whenever we exhausted, like this retry number, we would forward our, you know, fill masses flower, you know, our dead later, queue, yeah,

Speaker 4 50:39

so Krishna, just Krishna, just Krishna, I

Speaker 2 50:43

have a question. So, so how do you do the retry process in Kafka? Like, is it something you manually trigger anything? Or, like, how does it work?

Speaker 1 50:52

So, basically, we have a content like portal where, like, we can provide, like, certain configuration, and like, you can just reprocess the data, and so you don't have to worry about doing any like anything else. So all the deleters queue message would be, you know, displayed on the portal. And then you can just, you know, click to the triggers.

Unknown Speaker 51:22

Krishna sorry, Bucha, you can take you can take off,

Unknown Speaker 51:26

yeah, no problem.

Speaker 3 51:31

Can you explain your experience with swagger?

Speaker 1 51:39

Yeah, like for the swagger cities, you know, just a documentation tool that provides the information about, like, what influence you are, like exposing on these application, what are the required parameters and the variables, and what are the, you know, expected outputs, response code And the entity, right? So like it serves as a documentation for the developers to have a reference point like when integrating the entity, right? So like it source is a documentation and like for integrating certain APIs, like to their service. So I have worked in creating these swagger stocks using the spring Fox and I have also worked, like, logging, like, looking at the swaggers and making request integration.

Speaker 4 52:36

Yeah, I'm good. Okay, so, yeah, I

Unknown Speaker 52:42

have, like, we have, like, seven minutes left, so I'll just

Speaker 2 52:46

ask few questions here and there. Sorry, I can be comfortable. So, so you used executor service earlier, right? Do you ever use executor service

Speaker 1 52:57

for, yeah, basically, the executor service like I have used, so it is used for the parallel processing on the multi threaded like, you just handle the thread pooling we have worked with, you know, defining the certain tasks for, you know, processing the certain data, and then execute the service, like, takes care of the thread.

Speaker 2 53:23

So they're, like, multiple ways to create the threads right there. Like, you can create fixed thread pool, multiple ways. So, so do you know, like, what? What are the different ways to create thread pool?

Speaker 1 53:36

So, yeah, like you mentioned, like, we, like, you have a fixed thread pool where you can that's fine,

Unknown Speaker 53:52

if nobody will use it.

Speaker 1 53:57

Yeah, you can create, like normal, fixed, right? So I know about the fixed thread pool, and you know, a single thread work.

Speaker 2 54:07

Do you know, like, When do you go for like, fixed thread pool? Or like, When do you go for single threaded or like, if you go for single fixed thread pool, like, like, how do you come up with that number like you want to have like, 10 threads like, so how do you know like, like, how many threads do I need to use this processing?

Speaker 1 54:36

That case like to talk about the single thread, right? You would want to go with like, single thread when you want to make sure that the tags are executed one at a time and like in order like they were submitted, right? So that's when are you? Like you is you use a single thread? But if we talk about the fixed thread pool, you basically need to, like, do some analysis on, you know, what, like, what are the result of the load testing and the utilization, and comes up, like, with something that you know does not blow up, like, whole process for one operation. So, like, we have used, like

Unknown Speaker 55:24

max of 15 to

Unknown Speaker 55:27

20.

Unknown Speaker 55:28

So, so

Speaker 2 55:29

how did it? How did you get that number in our 20?

Speaker 1 55:37

So, like we were basically looking at how many process we wanted to, like, run in, you know, parallel, which is about the 20, and then we, you know, initial, like, initially started with 15, and then increased after 20, because our application, and, you Know, the instances box could handle, like, more load.

Unknown Speaker 56:05

So

Speaker 2 56:07

I see you have experience with the REST API. So like, When do you go for rest and when do you go for not non rest applications?

Speaker 1 56:19

Okay, so in that case, the REST API are, like, pretty known for, you know, being stateless, right? So basically, what it means is that your single request, you know, contains all the information that is needed, so through the process that particular request, right? So it contains the authentication, like data, it contains the input parameter and so on. So, like, what it mean is that, like, subsequent request does not have to, you know, hit the same server does not like the server does not need to have any management of the state of the data. So like that is like. That is why the REST API is, like, very useful. But so when you use the REST API, but the rest for the non REST API, like the Graph QL, they are, you know, useful whenever you have very, like, very tight coupling operation with complex communication, right? So your RESTful API is, you know, just like it is, just like a standard HTTP method to, you know, communicate across resource.

Unknown Speaker 57:39

So when do you when do you use GraphQL? GraphQL

Speaker 1 57:44

for the GraphQL like it is for querying your database based on like what

Speaker 2 57:51

graph I mean? GraphQL like you're saying like rest and GraphQL, right?

Speaker 1 57:56

Yes, the GraphQL is like it is for querying your data based on like data based on what you need, right? So you can face the complex data and embedded data like within one so if you have embedded objects and you need like you need to grab all those objects certain fields you can, you know, just like you use our like GraphQL query.

Unknown Speaker 58:24

So, did you heard about GPRC?

Unknown Speaker 58:27

You asked for a GPRC Yes.

Speaker 2 58:40

So yeah, we are on time. Krishna, do you have anything else, any other topics, or anything? I'm good.

Unknown Speaker 58:53

I think we

Unknown Speaker 58:55

are good. Okay, so like,

Speaker 1 59:01

thank you so much for your time today, and it was like, really nice to talking with you. Then can I know the like, next mistake?

Unknown Speaker 59:14

Yeah, your recruiter will contact you.

Speaker 3 59:21

We will share our feedback. Then, after your recruiter will contact you.

Unknown Speaker 59:30

Thank you so much.

Unknown Speaker 59:33

We have a great evening.

Rekha Bhandari- WorldPay-SRV-final

Speaker 1 00:00

Which Bhandari and I have been working with Java developer full stack for like, about like five and a half years, working primarily on the back end software engineering using the Java Spring and Hibernate, and I have been building microservices and designing them and implementing them and also deploying them and developing my team using the Spring Boot. So I do have a pretty good experience using like Angular on the front end side. And currently I'm a part of Scrum team, and my team, like consists of five developers and the product owners and Scrum Master. And on the daily, daily basis, I'm mostly working on the back end, Java core and the Java core, helping the team with code reviews and also with the products and support duties. Okay.

Speaker 2 01:05

Thank you, Rekha, for that. Appreciate it. Let me do just a brief overview for you, Rekha, about what we're looking for, why we're talking today. So we here at WorldPay are going through a, I guess, a couple of projects here to modernize and rationalize some of our systems, and as such, we are looking to ramp up our team here in the US, and that's why we're talking to you today, to see if he might be a good fit for to join our team. One of the things we're looking for, again, is obviously senior Java developers, the area we're working in, as I think Lakshmi and raw maybe alluded to in their introductions is the product disputes area at a super high level. If you go to your local merchant, whether it be your grocery store or Best Buy, wherever it might be, and they're a customer of world, pays that transaction when you dip your card to the terminal, that transaction likely is going to come to our systems for a fraud check, and we'll basically do through a third party tool and some other things we have. We'll do some checking on that transaction to try to score it for potential fraud. We'll give that score back to the payment gateway, and the payment gateway will then use that in terms whether or not to approve or deny the transaction. So that's a very, very high speed process. We're talking about milliseconds as far as the process goes. The other side of the coin is the dispute area. If you've ever opened up your bank statement and gotten a charge on your credit card or debit card that you didn't make and didn't belong to you, what you're likely going to do is pick up the phone and call your bank, and the bank will then open up what they call it a dispute case, and we actually handle or service that side of the system as well. Okay, okay, so today we are talking with you. Today's gonna be a technical interview. Brahm and Lakshmi are going to ask you a series of technical questions, and the goal of which is just to understand the depth of your knowledge in certain areas, as well as the breadth of your knowledge in certain areas. Okay? If there's questions you don't know the answer to just say, I don't have you haven't done that before and have experience with that. It's quite alright, okay, but we're just trying to understand how wide and how deep your knowledge goes. Okay, sounds good. Okay, you gave a brief overview of the work you've been doing. Would you mind, just for my benefit, to sort of review the technology stack you're used to working with, like, what versions of Java and so forth you're working with?

Unknown Speaker 03:32

So like, all, like, overall, I mean, like,

Speaker 1 03:36

so old. The our project was, you know, basically at old domain front line. I was, like, involved in using Java eight and 11, and then we recently migrated to Java 11 projects to Java 17, even though, like, we didn't use, like, any features of the Java 17 we, you know, migrated for the Spring Boot. There like three updates, right? So the application is written primarily in Spring Boot, and we utilize the AWS cloud services for the deployment this application. And we like use the services, like easy to ECS, lambdas, and so, yeah, like, that's, that's the, like, take a stick stack in, and in the front inside, like, I've been using Angular, the person is like, 11, and so I think, no, no, it's the 16. I'm sorry. I'm so sorry the whole one, it was the angular 11. So we moved to Angular 16 for the Okay,

Speaker 2 04:48

thank you. And that last question I have, I'll turn it over to ram and Lakshmi for their questions. But if you had to rate yourself in terms of your Java skills, right? So 10 being expert, like, you know everything about Java. There's nothing new to learn. One being a novice, where would you rate yourself with your Java skills? I'd

Speaker 1 05:07

say like seven to five to eight on like on on 10. I rate myself for 7.5 or eight.

Speaker 2 05:17

Okay, all right. Thank you. Lakshmi, dive in first. Yeah,

Speaker 3 05:22

sure. Thanks, Jeff, yeah. So here, aka, like, I reviewed, reviewed your resume, and looks like you have good experience, and that, can you talk about the carta project in detail? Because that was interesting, like CICD and cloud services, cloud native services you've used. So you want to go in depth of that experience,

Speaker 1 05:47

yeah. Like it cartel, yeah. Basically, we were, like, basically, we're developing the multiple micro services for, you know, the equity management. So, like it is, it was a portfolio management dashboard, and it Carta. We were, like, I personally was, you know, working on the couple of the microservices. So one of them was, you know, maintenance of APIs. So, like this, maintenance API was primarily used for making the changes to the user profile preferences, right? So whenever user want to make the changes to the proof preference performance to the user, we would be calling this micro services. And then this micro service use a couple of endpoints. The primary was, you know, the posting points to make a change to a preference. But we had a gate calls to basically get the, you know, status and the state results of the preference. And then we had a like Experience API, which was, you know, basically used to, you know, grab the entitlement that was useful for the user, right? So this entitlement was, like control and like what widget and equity details was visible for that particular user. So, like, there are the, like, two primary microservices that I help in developing, and I designed these. Like together the functional and non functional requirements, and get approval from our architects, and then I develop it. And then for the deployment. Like deploying this, we use the cloud services AWS, using AWS ECS, and the deployment was on, like news done through Jenkins,

Unknown Speaker 07:53

like we internally use cloud formation template.

Unknown Speaker 07:57

So, yeah,

Speaker 3 08:01

that's all, okay. That's good, yeah. So have you built the microservice from scratch, or, like, you added or maintained or enhanced the microservices? So,

Speaker 1 08:09

like, basically, like these two that I talk about, they were from scratch. I was like, from like, gathering the requirements until pushing into the products.

Speaker 3 08:23

Okay, okay, so, so what would you consider like then building a microservice, since you have built something from scratch, right? Like, I mean, what would you say like, if I come with a requirement and say, Hey, I need a microservice for this functionality, for instance, like fraud detection, like we have, we're building a dashboard, and we need to look up some transaction information, something like that. So we are building a service to APIs for like, get and like, update transactions and stuff like that. So how would you design that microservice? What would you think about the considerations when building a microservice from scratch.

Speaker 1 09:00

So in such scenario like basically, when designing microservice site, you basically talk about the functional requirement and the non functional requirement. So the functional requirement are the things that your app need to do, right? So if you are doing a fraud detection microservice, then your endpoints need to be, you know, related to that fraud detection, right? So you would basically define a clear and consistent API for communication between the services, and I would typically consider a RESTful API, just because of the fact that, like is popular and very stateless and has a proper JSON structure for, you know, defining your data. And first thing I would be designing the API structure and how it looks, and then I would identify several boundaries to ensure that your particular micro service that you are deployed has a particular singular, like single responsibility, right? So you do not want to incorporate, like, your fraud detection, your payment and so on, so like in one single micro service, but segregate that into a simple, simple micro service on yourself. Then I also want to ensure like that each micro service would have a like database of like its own to maintain the autonomy and minimize like any kind of coupling with the other micro service. So yeah, those three primary things that I would consider for the functional side, but like, if we have to look, look for the non functional side, I would consider about, you know, the scalability, right? So I would design in like it in such a way that like it will scale independently based on the like load and allowing the better resource optimization. And then I would like also consider introducing the resiliency, like, you know, retry and circuit breakdowns, like breakers and fallback strategy, or some kinds of, you know, rate limits, limiting. So then finally, I also would consider, you know, the monitoring and, you know, logging sorts like, so that if there like, will be any issue, then it will be easier to look into. Like you can like, also, like you can do on dashboard, through our New Relic or the the time press, and also logging through the Splunk.

Speaker 3 11:50

That's, that's fair enough. Yeah, thank you. So like, the RESTful APIs, have you have written, right? So, like, can you explain? Like, like, a rest mapping, like for the kit and port or like post services, well.

Unknown Speaker 12:08

So basically, you

Speaker 1 12:11

know each of this like mapping that you mentioned specific, like, specifically so as a particular purpose on, like, restful environment, right? So your Git mapping is used to, like, retry our resource, and the post mapping is used to create, like a resource, and then the put mapping is used for updating a resource. So, like, the maintenance API that I work with we use the posting points in order to make any change to the profile preference, right? So whenever, like, a user goes to the portal, select or like drop down for maintenance and then submit a form on the front inside and it execute this on the back end POST request, and then, based on this request body that the post request comes with it, it make our database interaction through our service here. But like between those there, there's a lot of like validation and business logic. So don't,

Speaker 3 13:23

that's fine. That's fine. Like, so have you implemented anything related to the security securing the APIs? So like,

Speaker 1 13:34

on the security side, we basically use the spring security then, like, they have a role based access control our, you know, identity provider uses our user and the power like password authentication. And then once you log in, it inject, like our JWT tokens into a session, right? And then, based on the session token, it, like all of our API requests come with, you know, these BRF token and our application basically validate or token by using like signature validation and then looking like it's a claim for like, expiration, and then, you know, the extracts on The rows for that particular user on the claim, and based on those claims, role and certain inputs are exposed.

Speaker 3 14:28

Okay, okay, that's good. Yeah, I'm done. Ram, like, do you want to take over? Yeah, okay,

Speaker 4 14:33

sure. Thanks. Lakshmi, I reckon like you mentioned, like you use the EC to UCs and lambda actually in your introduction better. Like, where did you use and how did you use them?

Speaker 1 14:46

So basically, our EC two is for, you know, legacy code base. Our application was deployed to EC two and, like, this was the legacy code, right? So we had, like about 40 instances that was, you know, deployed behind the load balance and on the Auto Scaling group. So, like there were where our primary legacy code base leaves. So okay,

Speaker 4 15:21

so you say, deploy the easy to write. What does it mean? Like is directly deploying the easy to are you running any other server and easy to deploy?

Speaker 1 15:30

For deployment? We like we run the server, we run our job application

Unknown Speaker 15:39

in which are running through

Unknown Speaker 15:46

the Jenkins, but where is your job application running? Actually

Unknown Speaker 15:48

the Tomcat

Speaker 4 15:51

server. Okay, so may know, like, why you're deploying a tomcat server in the sitter. Like, what is the reason?

Speaker 1 16:02

So, like, ETs are is Springboard based application, right? So, like, it has, like, it's embedded AutoCAD server. So, like,

Speaker 4 16:12

okay, that's fine. So you say you also mentioned ECS Mena. Like, where did you use that run? What is the purpose of it

Speaker 1 16:22

for? Yeah, the GCS is for the microservices. We containerize these services, and this container were created through the Docker and the DCS is used for running those containers.

Unknown Speaker 16:38

What about lambda?

Unknown Speaker 16:41

For lambda, one like,

Speaker 1 16:45

one of our use like, we use the lambda for, you know, the event based system, right? So we publish the message to SNS topic, and then that like get picked up by SQS. And then whatever they, like is a message in SQS, the lambda, like Gears trigger, and then, like runs on process. So like to basically,

Speaker 4 17:12

okay, so you also mentioned TerraForm, not

Unknown Speaker 17:18

you mentioned one

Speaker 4 17:21

cloud permission is mentioned. Cloud formation with Jenkins actually like what you're doing with the cloud permission there

Speaker 1 17:29

for like, cloud formation template is, was like, basically used to provision your AWS resources, right? So basically it is simple yml file that defines you to, you know, provision your infrastructure, right? So it is like, basically your terra homework for AWS specifically, we basically use Jenkins for our CICD, and then I use Jenkins. We basically trigger the cloud formation template, and then we use the like Jenkins to pass the parameters for our

Speaker 4 18:03

Yeah, okay. So, so in Java, like, what is a polymorphism?

Unknown Speaker 18:10

So, like, polymorphism is, like,

Speaker 1 18:14

it is a, you know, multiple form, right? The polymorphism. So it is achieved with the help of overloading and also the overriding. So like, first of all, the overloading just means that you are providing either, like, different parameters or different return type on the same class, but the method overriding is done in a child class by, you know, inheriting and giving or different implementation of the parent method.

Speaker 4 18:47

Okay, so what is the use of wrapper class? Do you know? What are the wrapper classes?

Unknown Speaker 18:53

Yeah.

Speaker 4 18:55

So why do you use it? Like, what is use of the why they are there, actually.

Speaker 1 18:59

So basically the wrapper classes are, like, if I have to define wrapper classes in Java, like, it basically convert your primitive data type into the object, right? So, like for integer you have int is a like primitive, but integer is a wrapper. Okay? I

Speaker 4 19:21

Okay? So when you, when you create the new object, right? So what is invoked from the object class actually, what is called, sorry, like, when you create the new object from the class. So what is called from the class actually, which, which, which part of the code usually is called actually.

Speaker 1 19:46

So, like, whenever you create a new object, then, like, like, it's a constructor,

Unknown Speaker 19:55

okay, can you overload the construction? Yeah.

Speaker 1 19:58

Like, constructor is basically used to, you know, invoke like. You can create a multiple constructor by giving the different parameters depending on like, what you use like, like, what like your use case and like the scheme

Unknown Speaker 20:22

is, yeah, like you can just, just, like,

Speaker 1 20:28

you can just create a multiple constructor with different parameter,

Speaker 4 20:32

okay, so it's like, so how do you Make your class? Read Only. Read

Unknown Speaker 20:39

Only. Okay, yeah.

Speaker 1 20:44

So in order to create your class, you know only it is called immutability, right? So basically, in order to do that, you just make all fields final. Then you also want to make your class final and like you like, you only expose the getters methods, and then you like, do not provide any set of method.

Speaker 4 21:10

Okay. So what is the difference between array and array list? The

Speaker 1 21:17

array and array list basically like difference between the array and Error List is that the array list is, you know, dynamically sized area, right? So the list is a form, like collection framework and like it automatically increase in size, but once you create an array, you cannot change the size of the

Unknown Speaker 21:41

Okay, what is the Java streams?

Unknown Speaker 21:44

JavaScript, right? Java streams?

Speaker 1 21:47

Oh, streams. So basically, Java stream is one of the popular features. Like that is introduced on Java eight. It is basically used for licensing, like functional style, right? So you like it like, I would say, the intermediate operation, like filter and map to basically modify the data on, like, on the collection. And then you have your terminal operator, which is used to provide a result based on like, what operation you provide, right? So some of them includes, like calling and reduce,

Speaker 4 22:26

okay, I got it so you have the link list and Array List option to use it like, in what scenario you prefer the array list, in what scenario prefer the link list,

Speaker 1 22:39

the array list and linked list. So the linked list has a pointer to, you know, the next node. You basically want to use a linked list whenever you have, like, frequent in insertion and deletion, like, towards the beginning or middle of the off, right? So, like in that case, you just basically need to update the pointers rather than moving your elements,

Speaker 4 23:11

yeah. So in Spring Boot application, you need to configure the multiple databases how you can do it.

Speaker 1 23:22

So, like, in order to multi like, in order multiple database, you basically define the bins, right? So, so in that case, like, so in its each bins, you can provide name, and in the configuration classes you like, you would have, like your either a primary bean or named bean. Whenever you like, auto wire them. You can auto add them using the qualifier by providing the name of that beam. Okay. So

Unknown Speaker 23:54

what is advantage of spin data, JPI,

Speaker 1 23:59

the spring. The advantage of Spring Data GPA is that it basically provide an abstraction for working with, you know, the database. And then it also provides, like, introduce a boiler, boiler plate board that helps you to like, in the crude operations, all right? You can you also use the Save pins and like, find all delete method directly. And then you can also, like, support the native query using the query annotations.

Speaker 4 24:37

Okay, so you also study you worked on the Docker too, right? Docker file, yes, may I like, can you tell me saw the Docker commands you can use,

Unknown Speaker 24:48

yeah, like so,

Speaker 1 24:50

some of the Docker command, like

Unknown Speaker 24:56

so, basically, to list all your

Unknown Speaker 25:01

is the docker ps like, docker ps

Unknown Speaker 25:05

like, dash, a that leaves all the emails and,

Speaker 4 25:08

okay, great. So in your in your resume, you're the maintain, the implemented state management with the NGRX, actually, what is that? Actually,

Speaker 1 25:20

the NGRX is like, basically it is like, on the Angular right, you basically don't have an inbuilt function to manage the state. So, like to manage you can use the external tool called ngi x it represent

Speaker 4 25:40

data. Yeah. Okay, so in the Kafka, you also mentioned you worked on the Kafka. So what is the Kafka? Tell me, like, how the Kafka works, just in one, just like, simple way. Like, what is the Kappa? How it works? Okay,

Speaker 1 25:56

talking about Kafka, it is like just a Message Broker. It is like public subscribe messaging model. So basically you have your like consumer that, sorry, you have your producer that produce the masses to walk off like to a Kafka topic, and then you have your consumer that reads the message. And then basically you have your topic like, which is comparative to our database table. This is where your masses are stored. And then you have your Kafka broker, which is like a, you know, server that is used for, you know, mass like, managing the data. And you may have a multiple partisans, like, within a topic for you know, horizontal healing

Unknown Speaker 26:44

and Okay, got it? Okay. Thanks, Erica,

Unknown Speaker 26:49

thank you. Yeah, Frank,

Speaker 2 26:51

I know we're coming up on time, but I want to make sure I save a few minutes for you to ask any questions of us based on what you've heard from all three of us. Do you have any questions for us today.

Unknown Speaker 27:04

First of all, thank you so much.

Unknown Speaker 27:07

The opportunity that

Speaker 1 27:10

you have provided to talk like, it's very nice to talk. And I would also like to know about like, how much of work is cloud based, and does the engineer on the team would be doing it more SRE related duty, or like, Do you have any like dedicated SRE team?

Speaker 2 27:32

So I'll start in. Lakshmi RAM can chip in as well. So it is a cloud based application that you'd be working on. It is deployed in the AWS cloud. It's a microservice application. And then in terms of the SI

Unknown Speaker 27:47

Lakshmi or RAM you would tackle that one,

Speaker 3 27:51

yeah, like, the cloud work you asked, right? Like, mostly, like, you will be working on, hands on, like, Java microservices, like, in the back end system, and probably, like, since you have full stack, like Angular, we have Angular application for onboarding the merchants. So that's, that's our most of the tech stack, right? And then, like, there are, like, cloud native solutions we use to, like DynamoDB and s3 buckets, and we have Kafka queues as well. So all this like You'll get hands on experience, but the majority of the work will be in the microservice code base itself. And we do have a dedicated SRE team which monitors the production, and then they work with developers to review the broad incidents and all that. To your question about the SRE Okay, so

Unknown Speaker 28:40

I Okay. So, so

Speaker 1 28:44

that's the, you know, the production incident, are people on call, on rotation basis, or, like, is it like, always a dedicated product and a support developer?

Speaker 2 28:57

So again, to what lash was sharing a minute ago. So we have a dedicated production support team that monitors and supports our production environments and applications. They have. They have the first line of defense if anything goes wrong in production, if they need support. There is a structure, basically, of tier two, tier three, tier three support that they need to call out to some cases that may require development, but in most cases it does not.

Speaker 1 29:26

Sounds good. Okay, so, yeah, that's all the question that I have, and thank you again for the great opportunity.

Speaker 2 29:34

You're very welcome. I know we're at time, so thank you again. It was a pleasure meeting today, Rekha, I will talk with Ram and Lakshmi afterwards. I know we have some others to consider as well. I'm hoping to make a decision quickly, so I'll get back to Dan with my feedback one way or the other. Okay.

Unknown Speaker 29:53

Yes, nice talking to you. Rica, all the best. Yes, nice

Speaker 2 29:55

meeting you. Thank you for your time today. Have a great weekend. Thank you. Bye. Bye.

Rekha-Fidelity-SRV-final(1)

Speaker 1 00:00

Not be, you know, forced to work on, like, on a multiple interface if you are, like, if you don't need to. So, like, like, you just, you know, separate your interface into a like, smaller chunks. Then you have, you know, only required interface. And then you have your like, dependency in version principle, which means that your class should be, you know, like, depend on the abstraction, and the abstractions should not depend on any details, right? So, like, just the basic guideline for having a proper, modular or the manageable software,

Unknown Speaker 00:41

okay? So what is your

Unknown Speaker 00:46

favorite design pattern in Java,

Speaker 1 00:50

like, like, if I have to compare like, I basically work regularly with single agent design pattern. And, you know, like, like pattern, detector pattern, as well, as, you know, the sagger Like, saga pattern, but yeah, like, I guess those are the some of my favorites. And like, I also, like, enjoy all like, circuit breaker pattern. Also,

Unknown Speaker 01:16

okay, did you use the soccer breaker

Speaker 1 01:18

pattern? Yeah? Like, I have utilized, like, some of them, like, in some case, my, you know, the current, you know, orchestration, like, I work with his strict circuit breaker. Like, we basically working in a like, orchestration, like platform that has, you know, the strict circuit breaker is a part of my, you know, resiliency. And like, they were, like, they are very like, slowly, you know, migrating it to be a resilience for the circuit breaker.

Speaker 2 01:48

Okay, so what are the benefits of advantages we have in Java eight, like, which, which JDK are using today?

Speaker 1 01:57

The JDK version, yeah. Like, I have worked with Java eight, and, like, also with Java 11, and currently I'm into Java 17.

Unknown Speaker 02:09

Okay, so do you know whatever

Unknown Speaker 02:12

the feature started in 17?

Speaker 1 02:17

So like, in 17, basically, some of the features in the Java 17 that includes the CIL class like it basically mean that you have a restriction on the class like that, like you can extend, or, you know, implement the certain interface. So like you can, you know, just permit your classes to, like, extend your abstraction. And then you have a pattern matching, like with the switch classes cases. And, you know, like, where you like, you can just, you know, switch the case based on the object type. Then you have, you know, a multi line string, like with a like triple A course. And then there are also, like, some other like improvements with, like the garbage collector. But you know, like, these are the new features, okay?

Speaker 2 03:11

So you might have familiar with Java eight features, right? Can you describe the streams API, like, how we use a stream, save PA, and then let's, let's go with the beginning,

Speaker 1 03:26

yes, yeah. So, like, basically the steam API is, you know, it's just a call exam, just for colleagues. And, you know, like the colleagues and processing so you can, you know, basically convert any colleagues and that you have into a stream, like, within, you know, intermediate, like, intermediate reform of your collection, right? So when, like, you can, you know, apply the different function that, like, maybe your field filtering, where you can filter, like, filter out the database, or, you know, like certain predicate or or you know, like you can map or data based on, like, some kinds of functions where, you know, you can like transform, or that like data from one form to the another, right? So then also like, like later collection that stream into the another collection by, like, you know, maybe using to list, so like to list function, or, you know, to set, or maybe, you know, even, like, it can be used to, you know, reduce it to something other, alright. So, like, that's what is like, basically stream. So it is basically a very powerful tool in order to work with, you know, colleagues and framework

Speaker 2 04:46

so what kind of data they can supply to stream or type of collections, yeah,

Speaker 1 04:51

like, as I have already mentioned, basically, they are the stream are, like, we can work. We. The collection framework, right? So we can basically pass the supply the colleagues and type like list or write, or, you know, arrays, or you can, you know, convert or list into a stream. And you can convert an array into a stream,

Speaker 2 05:17

okay? So we have different functions we can use in stream site, like intermediate ternary operations, like the No Water intermediate versus ternary.

Speaker 1 05:30

So basically, intermediary operation are those like, which basically, you know, gives you the another string, right? So, like I already mentioned, that the Peter method or the map method, or, you know, the the flat map, or like, like, these are all the you know, intermediary operation, because like, they return the another stream, and then you have your, you know, Turner operator or Like, which basically mean that you cannot perform any other like operation on that stream. So, you know, it is that stream and basically gives you a result.

Unknown Speaker 06:10

Okay, you

Speaker 2 06:13

mentioned that flat map, right? Do you know what is the difference between map versus flat map? Yeah, I don't,

Speaker 1 06:22

basically, you know, your map is, like, it is used to convert, like, one form of data into, you know, the another. So, like, if you want to, you know, basically, like, like, all, like, all the employee ID from a list of the employee you can, like, just do a map operation, and then, like, just get all of the ID. But in case of flat map, the flat, like, you just pattern the multiple stream of the data into a single stream, right? So let's say you have on, like, nested list that is, like, you know, list of list of the string. And you like, when you want to, you know, perform operation in order to, you know, like, just call it, it like quality as a single list of the string, right? So in that case, you can use the fat map in order to, you know, perform that kind of operation. Okay,

Speaker 2 07:21

so we have a static method in right, in Java, right? So can we make a static method abstract?

Speaker 1 07:33

For static method to abstract? I think like in like Java, like we cannot make our static method into abstract because, you know, the static method, I like, they are directly associated with the class itself. So, you know, like, the abstract method. So by definition, the abstract method, like, does not have a body. So that's why I don't think, like, we can make the static method is a F stack, because, like, you cannot combine the F stack and static

Speaker 2 08:08

together. Good. So have we used executor service in one of any one of your earlier implementations with multi

Unknown Speaker 08:14

threading? Yes, for executor service.

Speaker 1 08:18

Oh, yeah. Like, basically, we work with the history command, so which is an abstraction over the executor service, okay?

Speaker 2 08:30

So and have you all the memory management anytime, like Java memory management,

Speaker 1 08:39

yeah. Like, so like, only time I have done that is, you know, bypassing like, for bypassing the like SMS, like keywords on like, on the Java key. You know that for in like Java key startup script.

Speaker 2 08:58

So say, like, if I remember if your application is crashing with memory leaks or high memory usage, right? So if I give you opportunity to analyze and come back with the root cause, like, how do you approach that problem?

Unknown Speaker 09:15

So the basic thing

Speaker 1 09:19

like that I would look into is, you know, I think like first, first is to see like, if I can found out, like, any kind of exception that might lead to like, where the like air is occurring, right? So, but if nothing makes sense to me, the first thing I would do is, you know, I would, you know, log into the instance where, you know, application is running. So, and then like, and then use the heap Dom, right? So I would abstract the, you know, Heap DOM by using the script. And then you can, like, basically visualize the hip dump using the PM, right? So, so, any other proofing tool, and then. You can see, like, what is causing the bottleneck and like, what is holding, like onto the memory. So, so once I figure out I can, you know, basically, see if I can optimize any part of the code or change the way I have, like, you know, processing the certain stuff, then I can make, you know the changes then some of the like performance case, obviously. Okay, so

Unknown Speaker 10:32

let's go with the simple programming

Speaker 2 10:37

test. Like, can you write a small program to reverse a string without using String reverse function? Yeah, of course. So new stream and just try a little bit and explain how you approach the problem. So my screen first, I

Unknown Speaker 11:15

can you see my screen? Yeah, I

Speaker 1 11:27

so in order to reverse the string, I think I will just like, use a call it, like, correct array.

Speaker 2 11:36

Anything you can use, we can take, take that string into a list or a calendar, whatever it's

Speaker 1 12:26

so this can be used to convert it into a character in it, right? So all I need to do is, you know, I need to, like, look from the back and then, like, storage, it into some kind of variable.

Speaker 1 13:02

You so I think I just use the steam builder to store my palette. I

Unknown Speaker 13:19

BBB,

Unknown Speaker 13:41

yeah, I think that should take care of it.

Unknown Speaker 13:44

Okay, so the farm loop,

Speaker 2 13:48

so you started with character, then minus one, and then you are incrementing it. Okay,

Unknown Speaker 13:56

got it, okay, yeah, it should be decreased. Yeah. Do Okay,

Unknown Speaker 14:05

let's move on to the spring.

Speaker 2 14:18

So which portion of spring are you using Spring Boot, of course, for micro

Unknown Speaker 14:22

services? Yes, I have been working with

Unknown Speaker 14:26

Spring Boot. So what is your Spring Boot version?

Speaker 1 14:30

So, like I just recently mandated to Spring Boot three for a long thumbs up.

Speaker 2 14:37

Okay. Are you producing any APIs are just consuming somebody's API of how your architecture look like.

Speaker 1 14:44

So like, I'm doing both. So like, basically, I'm producing an API and a couple of the endpoints and is used for, you know, making the profile preferences things. But you know, I'm also, like, consuming certain API for, you know, the data processing. But. You know? Yeah, I have like experience with both of

Speaker 2 15:03

them. So where do you publish your APIs? Do you are using any tool in your company for publishing it, managing, managing the APIs,

Speaker 1 15:20

like so like, in order to manage the API, we just, like, present a swagger, you know, I think, like, that's what you are talking about,

Speaker 2 15:30

right? Swagger documentation, right? Like, somebody want to know about about our API, still, but say, like, if I am the consumer, right, if I want to consume our API you're exposing, so, right? How can I as a consumer? How can I start consuming your APIs? Where can I locate your APIs, in the in your enterprise level platformer?

Speaker 1 15:56

So basically, you have to go through an like onboarding process, that you have, and you know, orchestration layer that you know, sees like in front of your application. So it is an API gateway, like it is written on, like Spring Cloud gateway. And you know, you basically like onboard, so like your the route that you want to eat onto your application, and like based on your metadata, like, it will forward your request into our application.

Speaker 2 16:27

Okay, so which authentication mechanism you're using? The

Unknown Speaker 16:31

authentication mechanism,

Speaker 1 16:34

yeah, like, we have a, you know, dedicated, like, identity provider that handle the authentication. But, you know, we handle the authorization through spring security.

Speaker 2 16:44

Okay. So do you know how to protect your API with roles like in Spring Boot, we have a few annotations we can use to protect our APIs, right to our to protect from unauthorized users. So how do you know how to do

Speaker 1 17:04

that? So, yeah, like, it is very straightforward. So if you are working with spring security, and you once you have a token that is, you know, present on your like authorization header, you can, you know, basically, extract the claims from the like pillar part of the, you know, authorization token, right? So, like that claims, like you should, like content, the rules that the you know, user has, and, you know, like on the waste, like controller you have, you know, like you can basically use the pre authorized annotation and like that provide the role that should be, you know, able to access that particular endpoint. And like, in this way you can, like, basically prevent your, you know, unauthorized access to a multiple API, right? So, like, if you have, like, you know, delete endpoint that should, you know, accessible. Like that should be only accessible to admin and you like you, you can have a pre authorized annotation with like. With a like has like, it has role for of the admin. Okay, just my desk.

Unknown Speaker 18:19

So, okay,

Speaker 2 18:22

so have you worked on this meta documentation

Unknown Speaker 18:27

when you're developing your APIs?

Speaker 1 18:29

You mean with saga? Yes, I have. Okay,

Speaker 2 18:38

so in spring, we have our different bean scopes, right? Can I explain the bean scopes?

Speaker 1 18:44

Yeah, like, bean scope, like, they are basically, you know, they are singular scope, like, which means that you get a single instance of the object throughout the application, right? Like, like, it's a different scope. So any pin, like, define any spring would, you know, it will be always a single letter. And like, you have a prototype scope in that means, like, our new instance of the bin will, like, always be created, like, every time you inject or or like, or you retrieve it from the spring container. And then you have a, you know, request scope bin, like, which is basically a particular bin for, like, one HTTP request. Then, like, you have also a system scope pin that is, you know, basically it is a pin that is, you know, created for, it's like, HTTP system. So, like, yeah, those are the kinds of Bin, then you can also, you know, change the scope of the particular bin by using the like scope annotation.

Speaker 2 19:47

Okay, so can you tell me what are the different ways of injecting a spring mean

Speaker 1 19:53

in like, in our application, Springboard application, yeah. Like, you can just, like, do a cost. After in like injection, or, you know, the sitter injection and the feeling like fill injection, right? So you usually want to, you know, like, do a constructor injection. That way you get all the pins that you need, like that need in the application available. But you can also use, like, a setter injection for the optional dependencies. So like those are the different ways that you can inject the bins.

Unknown Speaker 20:28

Okay, so can you differentiate

Unknown Speaker 20:32

primary versus

Unknown Speaker 20:36

qualified? Annotation

Unknown Speaker 20:41

primary versus what I'm sorry

Unknown Speaker 20:45

qualifier so

Speaker 1 20:46

basically when you have a like multiple bins of same type like there might be you know like ambiguity on what we need to be used right? So if you have a you know multiple data source then your application will, you know, throw an error on, like, on what to be, you know, inject onto our application, right? So a couple of ways you can get like around it by, you know, giving a default bin, right? So, like, if you can use up, like, primary annotation on certain bin, like bin definition. It means that, this means will be like given preference when you inject that interface and then now the qualifier is like, little bit different. The Qualifier is used, you know, when, like, you are, like, auto wiring, like those bin Orion, you know, you know, a qualified annotation. You can, you know, provide the name of the particular bin that you want to inject. So, like, like, those like, that's the difference, right? Primarily, is basically used for default bin, and then the qualifier is basically used for, you know, selecting the extract bin that you want. You know,

Speaker 2 22:00

okay, so have you used controller advice in our springboard? Applications

Speaker 1 22:05

The controller advice? Yes, I have. I have used the controller advice. So, so it is basically, you know, used for the global accepts on handling, so whenever, if you want to, you know, provide a meaningful, like response back to the user, like, whenever there is, like, some kind of exception, then you can basically use a controller advice, like, for an example, let's say you're like, your application throws, like, all you know, illegal argument exception, then you want To return, let's say, like, 400 bad requests to the user. Then, like, you can just, you know, just to accept an annotation that, like, particularly, particular exception, then like return or response entity, like error, force and stuff and so, like, that's, that's the like primarily uses for the controller APIs. Okay?

Speaker 2 23:05

So if I want to make asynchronous calls right from my Spring Boot, application so we can use make use of Spring Boot provided. Annotations Do you know what annotation I can use for asynchronous

Speaker 1 23:20

programming? So, so in case of synchronous programming, like, spring would just provide the likes in like, annotation Yeah, that's

Speaker 2 23:34

what, yeah, that's what I meant. So if I want to make a call, right? Like, how do you what? What spring provided class I can use, I want to call an external API from my service, right? So how to invoke an API?

Speaker 1 23:54

So, like, in order to call an external API, you can you know, basically, like, just use the Rest Template that the like, spring provide, or also you can, you know, like, close the you know, like, basically you can, like, use all like, closable HTTP client, like, which basically support the connection pooling. And then the only thing you need to do is to set up the URL or the ending the request header and like, like, if like, if like, like, if it is a POST call, or set up a like request payload, then execute that Call.

Speaker 2 24:36

Okay, have you used aspects before

Unknown Speaker 24:41

the experts

Speaker 1 24:43

aspect, yes, I have worked with the expert like, in my case, like it was used for, you know, some kind of, you know, cross cutting console, right? If you want to execute some, like, filters, or, you know, the log the or the method execution time. Or have some kind of, you know, global accepts on handler, like, within the place, and then you can, like, work with the expert. So the spring of like, provide the spring AOP with like, like, we take care of all, like, all kinds of, like, cross cutting console. Okay,

Speaker 2 25:21

so, yeah, let's move on to like, have you what, what framework you're using for database? Like, is it Oracle or anything

Speaker 1 25:29

else? Currently, I'm involved in, like, Postgres, so I have, like, primarily work with push areas, but I have also worked with Dynamo DB for no SQL site

Unknown Speaker 25:45

as well. Have you used? Have you worked an article I

Speaker 1 25:47

have not used, but in my past project, like I have used, but in my current project, I haven't. Okay.

Unknown Speaker 26:00

So you are using,

Speaker 2 26:04

okay, so, not much knowledge in relational databases. What do you that is, what you mean the relational like Oracle, or the SQL like relational databases, yeah,

Speaker 1 26:15

like push space is also a relational database that is provided by, you know, got it? Like it is, like more modern and like it is, bless, okay,

Unknown Speaker 26:33

okay, let's lightly touch upon that

Speaker 2 26:38

if I want to create a composite primary key. So do you know how to do it in our through spring? Is there any PF framework in order

Speaker 1 26:51

to make the composite? So you want to, you know, any spring good, right? You mean, or like,

Speaker 2 26:59

directly, JPA, JPA framework, like, you want to create a composite primary key,

Speaker 1 27:06

like, basically, you can just do an like embedded ID by, you know, providing the embedded, like embeddable class. And for example, if you want to have the employee who's the component, like composite key is based on, based on the employee ID and also the department ID. Then just like, use like an embedded ID annotation for, you know, the embeddable class. Okay,

Speaker 3 27:33

fine. Have you done stored procedures, like, have you had

Unknown Speaker 27:40

an experience?

Speaker 1 27:41

Yes, yes, I have worked with these store procedures as well.

Speaker 2 27:47

So what do you mean by cursor, like, Where will you use that cursor, the cursor?

Speaker 1 27:55

So, basically, a cursor is, like, it is a just, you know, it is used to, you know, transverse and like face the row from the result set, right? So basically, it is just used for, you know, like row by, you know, for the row processing. And like, if you have some kind of, like, complex logic where you need to, you know, perform an operation on like, individual Road, then, like, you can use the cursor. Okay.

Unknown Speaker 28:27

Do you know what is a trigger?

Speaker 1 28:30

Yeah. Like trigger is, like it is, you know, the trigger is,

Unknown Speaker 28:37

basically you can

Speaker 1 28:39

whenever you have any kind of like situation, or, you know, let's say like, some I'd say like is kind of like, like over time, right? So like, the encapsulate the logic within the database, but, but like, it triggers automatically based on the certain function, right? So it is used to, you know, maintain that particular trigger of or, you know, the execution of the procedure.

Unknown Speaker 29:12

Okay, got it so

Speaker 2 29:16

small query, if I want to fetch second highest salary from an employee table. Do you know how to solve the problem through select coding

Unknown Speaker 29:29

to face the second highest?

Speaker 1 29:33

So I think, like, if you want to select the second highest, I think basically you can just limit, like, limit one, or, you know, offset one, right? So basically, you would select a salary from the employee table and then order, order it by salary in descending order, and then you would just limit it by like one value was like, by offsetting the one value i. Okay.

Speaker 2 30:07

So in fiber network or JPA, we have a different levels of caches, right, first level cache, second level cache. Do you know in which scenario we use prime first level cache and then second order cache.

Speaker 1 30:27

Like, I think, I haven't directly worked with different level of cache, but the thing that I want to know is that the first level cache is, like, it is basically associated with the session object, and, you know, the second level cache is like, I think it is associated with the entity manager. And, you know, CRA across the sessions. Okay, okay.

Speaker 2 30:55

So last question, ma'am, the database. If I have any performance issues with my SELECT query. So if you got an opportunity to improve the performance of a query, right, like, how do you approach it, or kind of options you can think of.

Speaker 1 31:18

The first thing you can do is you can, I think you can just print the query execution plan by, you know, using the explained keyword, so like, it should just give you how the query is executed, right? So you can change like, check for all of the indexes that you have on your database, right? Or like, if you have, like, joins on your query, or, you know, a lot of where classes, then you can add the index on the particular column in order to, you know, speed up the data retrieval process, and so that you can, you know, prevent any kind of, you know, full table scan. And now, like, apart from like that, you can also use the, you know, pagination, or, like, limiting how many data like you can return, and also, like you can add some kind of, you know, caching, like, whenever you are, like, retrieving the data from the database, or if like, it does not like, change it, so like they are, like, some of the ways that that can be used to improve them. So, like, you shouldn't, like try to limit or using the like Slack, like Slack all the rows instead, like, you should just select the certain columns that you need. You

Unknown Speaker 32:40

know. Okay, so

Speaker 2 32:45

So did you work? What is your J Unit knowledge? Like, which? How extensive us, alright,

Speaker 1 32:55

yeah. Like, do you need? We have to, like, we have to write the JUnit test cases on like everyday, like day to day changes. So every course change that I make like, I need to write DJ unit test. So

Unknown Speaker 33:13

what is the version we are using for your J units?

Speaker 1 33:19

Currently, like I have been using, I think, Ace of five, like the unit version five.

Speaker 2 33:27

So can you tell me the difference between mock being Versus Spy bean mock?

Speaker 1 33:36

Yeah, like, whenever you provide a mock bean, you have, you know, like, complete control of the bean, right? So it's just a, you know, a mock version of the Spring application, and, like, it just, you know, replace the external service. But, you know, in case of inspired bean, it is a, like, mix of that mock and the actual object, right? So you can, you know, get the original behavior on the method that you want, but, you know, you, like, you can also define the, you know, assorted behavior on the, like, other method that you don't want, right? So, like, it's kind of mix of the real object and the mock object. Okay,

Speaker 2 34:22

so what kind of flavors using for mocking?

Speaker 1 34:28

So, like, we primarily, like, are like working with Mockito.

Speaker 2 34:35

So do you know how to mock a void method which is not threatening? Nothing like a void method is an example,

Speaker 1 34:44

yeah? Like, you can, you know, just, like, do nothing. Like, just do a nothing operation for marking the object, okay,

Speaker 2 34:56

going back to the database, things, right? Like, have you heard of normalization? Yes,

Unknown Speaker 35:02

can you explain what is third normal form?

Unknown Speaker 35:05

Third Normal Form?

Unknown Speaker 35:09

So, like, in normalization?

Speaker 1 35:16

Yeah, like, like normalization is, you know, basically it is used to design the database in order to minimize the redundancy, right? So it means that you have your like database and like with the database table into a multiple tables, so that you do not have a, you know, redundant data. So I think like third normal is like it should like I either I think like it eliminate or like it redundant data, or I think, but I'm not sure. I just, I just forgot. I think, okay,

Speaker 2 35:53

that's fine. Do you know what is the asset principles? What are the asset principles?

Speaker 1 35:58

The acid principle? Yeah. So they are basically, there are four principle the first like, they basically, like, go on your database transaction, right? So you have your atomicity, which basically ensure like each transaction is, like, treated as a single unit of work. Then the second one, you have your consistency like that, ensure that your transaction can, you know, only bring the database from like one valid state to the another valid state. So, like any operation is like consistent so, and then you have your isolation, which means that, like that, concurrent education transaction are, you know, isolated from each other, and then, at last, you have your durability, right? So, which basically guarantees that once the transaction have been omitted, then it will even in committed state, even the system, like fails or anything happens,

Speaker 2 37:02

okay? So we have indexes in database right whenever we need to increase the performance of the table, the column will do the indexing right. So we had different types of indexes available in relational databases. Do you know what are all those? Do so

Speaker 3 37:20

I think

Speaker 1 37:28

for indexes like they are like, I think there is, I can remember the primary index, which is for the primary key, then you have your like unique index which are like content is a unique value. And then I think there is, like a composite index, which is like a, you know, multi column index that I think, like those are three that I'm aware of. Okay,

Unknown Speaker 37:56

so have you created a view

Unknown Speaker 38:00

anytime in database?

Speaker 1 38:02

I haven't like directly created the view, but I have worked with views like they are just, you know, the temporary table which is used to, you know, like they are basically stored the data is like intermediary format, right? So if you have a very complex SQL query that need to be executed or the multiple times in order to, you know, retrieve your data, then you can, like, instead, create a view like, that was that temporary data, and then just use that view like, is your source like, rather than using whole table.

Speaker 2 38:40

Okay, so we have different types of constraints in database, right? Like we can apply any kind of constraints on the table, on the whole columns. Do you know

Unknown Speaker 38:50

few constraints we can use of if

Speaker 1 38:56

you have to talk about the constraints in database, I can remember some of them, like, some of the constants on your database, like, include the primary key constants, like, which basically mean, like, it has a unique and that cannot be, you know, that can contain all the values. And then you have your foreign key constants, which is used to, you know, establish the relational like relationship between the two tables, right? So if you have a cross reference on like one table to another table, like employee to like employee to department, you have a foreign key constant, and then you have a unique constant there as well, which basically guarantees that all the like value in that column is a different from other and also, other than that, you have a like, not normal cost, right? It makes sure that the value is not not, I think, like, that's all in my head right now. Okay,

Speaker 2 39:59

okay. It. So we have different functions in Oracle, right Oracle, or any relational databases? Can you give me the examples of any functions use

Speaker 1 40:13

the functions in database like so, basically, some of the functions in database include, you know, I uh, like, like, one of them is concat function. It is used for, you know, concatting the strings that like you have, like you also have a mean function as well as the max function. And also, like some of the other functions that is used for the numeric processing. So, like these are, like your aggregate function. Then you can also, you know, do your group form, like grouping function by using a group by, okay,

Speaker 3 40:58

yeah. Well, thank you.

Speaker 2 41:02

So you said you have some experience on the cloud, right?

Unknown Speaker 41:07

Is it AWS or anything else? Yes,

Unknown Speaker 41:09

I have been working with AWS. Okay,

Unknown Speaker 41:16

so AWS and Spring Cloud. Is it? Yes?

Speaker 1 41:19

I like primarily, we have worked with AWS,

Speaker 2 41:24

okay, what is your like, where you deploy or EC two or Eks,

Speaker 1 41:30

like our, you know, legacy application, like the we like, we usually deploy them in EC two and in our modern application. We deployed them in ECS cluster, you know. So we do have some of the lambdas that you know, basically get a trigger on the certain emails that you know, like we utilize the AWS RDS for our database.

Speaker 3 41:58

Okay, so I so

Speaker 2 42:04

for code quality, like, do you review your peers code?

Unknown Speaker 42:11

So, like you say, like

Speaker 2 42:15

some if your team member is writing some code, or do you perform any peer reviews? Or peer

Speaker 1 42:20

reviews. Yeah, like we like, sometimes we do, but like, whenever we post our board, you know, to our like feature brand, we all like, we pull, like we do, pull request to the developer brands, and then, you know, we need to, like at least one approval from our peer developer, from the team, you know? So that's, that's how we can review the team, Team code.

Speaker 2 42:46

So if they can, if you get an opportunity to review other code, right? How do you start with it? So what is your minimum criteria?

Speaker 1 42:59

So some of the thing that I look like whenever I doing the code reviews are, you know, the correctness, right? So basically, I, first of all, I, you know, ensure that the functionality is good and like, also, like, see if the is its gasses are, you know, covered with the proper like, proper like, exception handling and so on. So, other than that, I also, you know, look for the clarity, right? So, I look for, you know, the naming standards that is in that is basically used in variables or the methods or the classes, you know, so and also I look for the any kind of, like consistency. Then, apart from that, I also look for, you know, unit test in order to see if the poor, you know, of course, it is meaningful, or, you know, covering the different scenarios, yeah, like this. Like, those are the primary things that I look into, like, whenever I'm doing some reviews.

Unknown Speaker 44:00

So our tool used for code quality,

Speaker 1 44:06

we like, we use, like the sonar scans for, for using

Speaker 2 44:13

the code. So, do you know we have different types of vulnerabilities in sonar support? Right? Like, critical, high, medium. Dana, what comes under critical?

Unknown Speaker 44:27

Like, in critical, I think

Speaker 1 44:35

I can't, I don't think I can remember any time. But what comes under it, but I have worked with like, in dissolve, like I have been working with in disjoining them, but I don't remember, like, what comes in that scenario?

Speaker 2 44:49

Okay, yep, that's fine. So did you create a custom exceptions? Anytime?

Unknown Speaker 44:56

Yes, yes, I have.

Speaker 2 44:59

Can you explain. Me how to create a custom exception. Basically,

Speaker 1 45:02

we create a cost, like the custom exception by, you know, you can just like, extend your your exception class or the runtime exception class, and then also, like, then provide the constructor on how you want to, you know, initialize the exception. So, so like for, so, like that, yeah, that's that that is pretty like, step forward. So you just extend the either the exception class or the runtime exception.

Speaker 2 45:40

So, do you know what is exceptional propagation?

Speaker 1 45:48

Yeah, like, basically the exception, like propagation, they are, like, it is, you know, just a process by, you know how your exception gets, you know, like, forwarded, you know, like call stack, right? So basically, you know, whenever the exception is thrown into a particular method, so it basically, you know, propagate that exception, or like of the call stack, right? Basically, it goes back to the previous method that you know cause it, and also like, check if there is a try, catch block or not, so if the exception is handled or not. So, yeah, that's how the exception propagation occurs. Okay, yeah, thank you.

Unknown Speaker 46:36

So do you have any questions

Speaker 1 46:40

to me? Yes, I do. So first of all, thank you so much for your time today, and it was really nice to talk to you. And I like to know, like, what are all the tools that, like teams, like currently are using right now in the application, like how the application deployed to the cloud, or the in the application live in like today,

Speaker 2 47:03

yeah. So what to see is a Java based application. So we have different applications in our cache management. We have one big application, which is monolithic currently, but which we have, which is running in AWS Eks clusters. So we use Kubernetes, but deployments. And this is a Java based application of Java and diversity 17. We you we are using data as Oracle

Unknown Speaker 47:36

for data processing. We use Informatica also.

Speaker 2 47:41

We use JSP as a front end currently. So we are migrating to our Angular and we use heavily stored processes for all our data operations. And we deal with external banks. We get files from the banks, and we every day, we get multiple batch feeds, and then we load the data to system that is about our main application. And we have sub applications which are micro services running on Spring Boot. So those are just used to we have internal consumers within our portfolio, they need our cache values, so instead of using that UI, they access our app services and get the data what they want. Those are Spring Boot applications. I think that is pretty much like we are running in AWS cloud since two years. It's not blue green deployments currently based on the need and requirements. We go for installs pretty much monthly or quarterly. That's how it is when we follow the Agile Spotify model. So all our teams are following agile, and we follow the scrum, Scrum model.

Unknown Speaker 49:06

That's pretty much okay.

Speaker 1 49:08

So, like, what will be my like, next step for the interview?

Speaker 2 49:14

Now let me check with my manager. He might take interview after this. This is purely technical, and then he needs, like, 15 minutes of your time for managerial round. Let me check with him after that, like we both have a discussion on the shortlist, and we will inform your

Unknown Speaker 49:35

counterpart,

Unknown Speaker 49:38

definitely thing. So I

Unknown Speaker 49:54

they are from. Which are you from?

Speaker 1 49:56

Dallas? No, I'm currently in Virginia. I.

Unknown Speaker 49:59

Under any

Speaker 2 50:05

plans to look relocate or Yes, I can relocate, because fidelity, we follow alternate week work from office, so one week office, one week work from home,

Speaker 1 50:19

that won't be any problem for me. I can relocate,

Speaker 2 50:24

okay, yeah, I'm just being my manager, like waiting for his response.

Unknown Speaker 50:30

All right, thank you so much for the time.

Speaker 2 50:34

Stay in this room and let him respond. Yeah.

Unknown Speaker 50:41

So

Speaker 2 50:43

let me go through the resume one more time. So you're currently working with?

Speaker 1 50:49

Is it financial company? No, it's kind of, you know, logistic of money, free.

Unknown Speaker 50:54

Client, okay, I

Speaker 2 51:04

I see you're from Nepal, right, yes, okay, so, when did you move to us? In 2022

Unknown Speaker 51:12

okay, changed

Unknown Speaker 51:13

masters here. Yes, I did. Okay. Do?

Unknown Speaker 51:23

Let me give you a call

Unknown Speaker 51:27

not to use at the desk.

Speaker 2 51:38

So if we, if we give the hands on coding, like, are you interested in to have full hands on coding, or what is your expectations out of this role?

Speaker 1 51:51

Yeah, like, I'm comfortable with hands on coding because I have been doing hands on coding on my like, previous project as well. Okay, do

Unknown Speaker 52:11

Oh, I don't

Unknown Speaker 52:18

know it's not answering my phone too. I

Speaker 4 52:20

Rohit

Speaker 2 52:29

mentioned anything about, like, how long? He mentioned just one hour, yes,

Speaker 1 52:39

but it was, you know, originally, like, hours, given the slot for one and a half hour. So, like, but it costs,

Speaker 2 52:47

typically, we we take 90 minutes and to use one or 15 minutes, tech team, take up. Take Yes, it was originally for one and 15 minutes.

Unknown Speaker 52:59

Yeah, so let me, if

Unknown Speaker 53:05

you already can take a short break. Okay, present.

Rekha-Fidelity-SRV-final

Unknown Speaker 00:00

UI, we have specialized team.

Speaker 1 00:04

We kind of like looking for someone API development right now. So which one if you're if you need to, like, Gage yourself between like UI and API development, can you tell me the percentage, like, out of 100, how

Unknown Speaker 00:23

much you have, like the UI developer,

Speaker 2 00:30

okay, I would say, you know, I'm, like, more on back inside, I have already explained I'm in percentage. I would say, you know, 70% on back in API development and rest of like, like 25 to 30% on fronting side.

Speaker 1 00:47

Okay, so we'll start with the equal related questions. Then we move on to like, JP, then spring, and then finally Java. So let's start with SQL. Can you please tell the different kinds of join which are available in order to SQL?

Speaker 2 01:12

Yeah, like talking about the joints on SQL, I can remember the like, basically we have the inner join that returns, you know, the matching rows from both table right? So then, like there is a left join, like that returns all of the rows from the left table, and you know the matching row from the right table as well. And like the next one is the right joint. We just, you know, return the all rows from the right table and the matching rows from the left table, and then we have the full joint and the cross join. And, you know, the cell joints and like these are the different types of joints that exist in SQL. Okay,

Speaker 1 02:04

can you tell me what the union is and how different it

Unknown Speaker 02:10

is the unions?

Speaker 2 02:13

Yes, so, yeah, you know, basically the unions is, you know, they basically, you know, like returning everything, right? So, like, we just, you know, the combined combine the result of two or more slack queries. And basically the main difference between union and union all is, you know, the union, basically, they removes the duplicate records from the final result, right? So, like, similarly, the union all like it is used to, you know, combine those slick statement, but you know they, like, Do not remove the any duplicates. Okay,

Unknown Speaker 02:56

can you tell me what's the

Speaker 1 02:57

difference between NBL and the NBL to function? So,

Speaker 2 03:08

so you know, like, so if you have to differentiate between which function, I'm sorry, can you please? You said like and BL and MDL two, right? I Yeah, you know, the NBN is, you know, they basically, like, replace the null with, you know, some kind of, like, replacement manual, right? So, you know, if the expression is null, then you know, it just, you know, like, return the replacement value. But you know, the NBL two it like, basically checks, you know, whether the expression is null or, you know, not, and like, it just return the different values.

Unknown Speaker 03:55

Okay, so

Unknown Speaker 03:56

what is the difference between coalesce

Unknown Speaker 03:59

and NVL? So, so,

Speaker 2 04:03

like, in that case, the basic difference between other qualities and NVL is, you know, that

Speaker 1 04:15

it's commonly Co Le, yes,

Speaker 2 04:18

yes, yes. Call it. So, like, basically, NBL, you know, it replaced the, like, I already explained that it replaced the null values with, you know, a specific So, basically it like it is used for the specific replacement. But you know, call is, you know, it returns the first null value from, you know, like, like, like an, you know, the expression, you know, the first null value, so the first non null value. So, like, if you have a list of expression in a police So, basically, evaluate its argument from, you know, the lift, and then it returns the like, not null value.

Speaker 1 05:05

Can you tell me what a pivot is and what is the use case

Unknown Speaker 05:10

where you will be using a

Unknown Speaker 05:12

pivot? Can you please, like, leave it one more time

Unknown Speaker 05:18

what a pivot

Speaker 2 05:20

is. Basically, I haven't worked with viewers right now, so I have no idea about that. I'm so

Speaker 1 05:31

sorry. Created a view. Can you tell me? Like, what are you

Speaker 2 05:41

so, basically, like, view is a temporary table, right? So, basically, what I mean is, you know, like, by that, if you have a complex query that is used, you know, that is used to, you know, perform the multiple joins and, you know, the combine the data from different slake queries. You know, you can, like, basically compute a view and, you know, basically store that result, you know, temporary table. You can, like, use this view as a, you know, the table in your application, and it basically help us, you know, to speed up the, you know, the process of data retrieval. So, yeah, like, that is what the view is.

Unknown Speaker 06:25

What does materialize?

Unknown Speaker 06:28

I'm sorry, materialized view.

Speaker 2 06:31

So basically, the materialized view is, you know, like the materialized view in the database that it basically stores your result on, you know, on the, you know, the some location also, instead of being a, you know, the virtual table, there will be, you know, specific space in your memory, like that. It will store, okay, so if you're

Speaker 1 06:59

going to be like you like the view is operating on the table, right on the table, we are going to be like modifying a column, like adding or reviewing the column, or reading the column. What is that we need to do to ensure views also up to date?

Speaker 2 07:22

So like, basically, like, in order to, you know, mix your your views, like, stay update, like, update up to date with, like, the data table or modification, then you basically need to, you know, do some kind of, like, extra configuration, right? So, like, you know, the for standard view, it usually, you know, picks the, you know, like those changes up and like, because it is just a virtual table that, you know, fetches those data. But you know, for the materialized view you you might need to, you know, like you need to do, like incremental update, or like over time, so or you know, or do the some kind of like refresh logic.

Speaker 1 08:16

If you have a complex query, how do you

Unknown Speaker 08:21

measure how your query is performing. Okay,

Speaker 1 08:25

what are the remediation which you will be like trying to take to make it more performing.

Speaker 2 08:33

I'm so sorry. I cannot guess the last word. Okay, how do you measure

Speaker 1 08:40

its performance? Its performance? Okay, if that happens to be a query which is like not performing, well, what are the next step or actions you will take to make the query more performing?

Speaker 2 08:53

So, so, yeah, basically, there is a query execution plan that you can, you know, like you can use, or use the, you know, the EXPLAIN command that, basically, you know, give like gives you a list of process that your, you know, SQL will run like, like. It will also, if you are doing the full table scan or the like row scan. So I'm so sorry, the column is scan. So in order to, like, get that this data, so this will help us, you know, to understand how the database engine is performing. And now the thing we need to, like look into is, you know, like, when there are full table scan and you know, the like, store, like, sorting operation, right? So this will help us, you know, to identify like, where we need to like, you know, look for the index, indexes. And you know, if we are like, like, our indexes in right sport, or like, if you like, if you have the indexes in your like clause, like, where like they are filtering, or, you know, like, some kinds of joints. And then you know, you need to, like, like, basically, make sure that we are using right, like, right kind of join. And then we also, you know, need to, like, introduce some kind of like PG nation in order to limit the number of the data that are loading. So, yeah, these are the some kinds of like ways you can analyze your query execution plan. And also, you know, like, also used to, like, improve the performance. Okay, any questions

Unknown Speaker 10:42

from you on SQL?

Speaker 1 10:47

I think you pretty much got it. Let's move on to GP.

Unknown Speaker 10:57

Can you tell me what an entity

Speaker 2 11:02

the entities. So, yeah, basically entities is, you know, the object reference of, like our database, you know, the database table. So basically, whenever you have a database table, you can create an entity on on a Java where, like that represent the table. So, like it is a, basically, it has, you know, all the attributes, and also, like it contains some of the, you know, the extra, like transient attribute. Then you can, you know, represent that table is an object on the Java can you tell

Unknown Speaker 11:40

me the different

Unknown Speaker 11:42

things? Entity relationships?

Speaker 2 11:46

Ah, so basically, the different relationship in Java, like you can have, is the first one is like one to one relation where, like one object, you know, maps to the one relationship. Like, you know, there is one entity on Table A, like it goes, you know, to like entity on Table B and I, you know, like, there is one too many. Like, where a single table relates to a multiple entities on the on on the another table. So there is, like, also many, to many, where you have multiple table entity that relates to multiple table entity on the like, other other one.

Speaker 1 12:35

Okay, so can you just, like, explain the product. So let's assume that you have a product entity, and then you have reviews entity. Okay, let's one too many. One product can

Unknown Speaker 12:49

have, like, many

Speaker 1 12:52

reviews? Can you just explain, like, what are the annotations? Like, it may not be under accurate, but at least like, how you will be approaching it. I just want to understand, you know, both these entities. You want to establish a relationship between them.

Speaker 2 13:10

Okay, so in that case, so basically, the way you do this is, I, you know, you do have your product class, where, you know, you would have your primary key for the product, right? So like for the like Id annotation. And then there is a list of the reviews object field on like on that class like, which would be, you know, annotate with one to many innovation, right? And then on the like, on that, you know, one too many you would, you know, basically like, want to, you know, list out, like, what you are mapping, like, mapping by and you know, what kind of like cost, like cascade strategy you want. But you know, on the review side, you would like have many to one annotation on the product right so you basically, you know, determine like, which is the joint column and like you want to use. You have to for a

Speaker 1 14:19

table you have some for a table, you will have an ID call a primary. So there are different ways in which you can say like you want the ID to be associated with the entity. So can you tell me the different identity?

Speaker 2 14:41

Oh, yeah. So the way to, basically, you know, generate those identity is, you know, using the like generated value annotation. And then, you know, you have a different strategy, right? So you have in the auto, which is, you know, default, so basically, like it, you know, like delegates the you know, database provider to, you know, to set the ID. And you know, there is an identity like, which basically, you know, or to, you know, the increase for MySQL or the postgres kind of database. And like, also, there is some kinds of like sequence where you refer to the database sequence or to give the next value.

Speaker 1 15:27

So if you happen to like, use the Spring Boot application, right? And if you want to like, query a table and

Unknown Speaker 15:35

you create entities,

Speaker 1 15:38

what else you need? Like? How would you be able to like, query the table and get the result? What other items will see for you to be able to like,

Unknown Speaker 15:48

use that entity and get the

Speaker 2 15:50

data? Yes, so, like in Spring Boot, it provide, you know, the Spring Data JB interface are to work with database, right? So basically, for that, you just need to, you know, configure your spring starter data GPA and dependency on your build tool. And then, once that is done, you basically configure your data store by, you know, using the application properties and the like Data Source Configuration once like that is done, let's say you know you have a product entity that you define a product repository, that extend your JPA repository, where you provide your product and you know a type of the identifier On your generics. And then once you know that is done, you can, you know, get the basic functionalities, like, out of the box, right? So by just, you know, using this repository, or, like, if by using, you know, dependency injection, you can, like, do find by ID, or, you know, like simple operation, like, same and so on. So like, but you know, if you want to, like, do more like, kind of say, like the complex queries, then you can, like, define the method on the interface and then use the, you know, the query annotation to write like native query to run those query so the Spring Boot, you know, makes it, I know, like, really easier for like, We using like Spring Data, GPA, can you leave?

Unknown Speaker 17:39

You all can you

Speaker 1 17:45

leave when, like you're creating a table?

Unknown Speaker 17:52

I'm sorry. Oh, I couldn't hear that. I'm

Speaker 1 17:58

sorry. We have audit columns, right? So generally, like many things like, approach it differently. So wherever you're working, right, though, right. So if you create the table, what are the different audit columns that you add whenever you create a table?

Speaker 2 18:18

Oh, so, like some of the audit columns that are, you usually like aid, like, include, you know, include the your created by and like, created it updated by and updated eight. So like, those are the four primary, primary one that you use. And also, you know, sometime you might also like, want to include a person, right? So, but, you know, depending on if you have the version enabled,

Speaker 1 18:52

okay, so, in spring, it gives us, like, easier ways on how we can populate the values for these or like

Unknown Speaker 19:04

the manually

Speaker 1 19:06

updating these values. Are you aware of, like, how to get it done? How screen does it or like, what annotations

Speaker 2 19:15

so screen had, I think, the annotation called the like audited annotation that you basically, you know, enable the JPA audit. And then you can just use the audited annotation and like on the entity, which basically, you know, refers to like, or, you know, enabling, you know, of the like annotation on those. And then you can, you know, just to like, created by annotation. And then the created error annotation that you know, like particular fit that should restore. And you know, you can like, basically like for is define the base entity and then like, take care of that. So like, which other entity X came from?

Unknown Speaker 20:06

Okay, okay,

Unknown Speaker 20:09

any questions from your end? Yeah,

Speaker 3 20:12

can you explain me about if you want to establish connection to two different database from a single Spring Boot application? Now, how do you achieve it?

Speaker 2 20:24

So basically, the Spring Boot allows you, know, you to define multiple means, right? So what all you need to do is like you need to have a collection properties to the data source application.

Unknown Speaker 20:41

So data source, or two data sources here

Speaker 2 20:45

something like, so basically you need to have, you know, multiple configuration that, that can, you know, restore the like data source, URL and username and the password on the application properties. And, you know, you would like basically be defining the bins, and each of those bins should have a specific name, right? So let's say you have a like Postgres, like data source, and then you have a Dynamo DB data source. Then you would basically define it the two data source like bins, and one name the postgres and the other name is Dynamo. And then, like, basically, you can, you know, inject those data source in your service here, and, like, work with those data source,

Speaker 3 21:33

okay, then you know about any connection coding mechanism you want to explain about that, how it works in Spring Boot,

Speaker 2 21:41

the connection coding, mechanism coding. So basically the connection pulling, can, you know, like, they can be done, like, through, you know, like, like it comes, like, embedded when, you know, like, with the Spring Boot. So I think it's called Hikari connection pooling. And then you just, like, you just basically set up how the connection so, like, do you want by, you know, configuring the pool size and, you know, the timeout, so it can be like, taken care of, you know, during the application, properties file, right? We

Unknown Speaker 22:31

can want to Java.

Unknown Speaker 22:38

Can you what is

Unknown Speaker 22:39

the latest version of town you're aware of?

Speaker 2 22:41

I So currently, the latest version that I haven't worked with is 17, Java 17. Okay, can you

Speaker 1 22:51

list the features which were introduced as part of Java

Speaker 2 22:56

17? Ah. So, like, basically, Java 17, like it introduced, you know, still class, which basically is a, you know, permit, like permit secure that allowed, you know, like with child class, can extend our, you know, the parent class, basically, you know, the like boxing, the encapsulation. And there is also a pattern matching, like a matching for the, you know, the switch switch cases, where you can match the type of the object, you know, whenever you are, like, dealing with the switch cases. And there is also, you know, the multi, like multi string using the like triple double port. So, yeah, like, those are the ones I'm aware of.

Speaker 1 23:44

Okay, unlikewise, Java eight introduced lot of features, right? Can you also discuss a few unexplained

Speaker 2 23:56

So in Java eight, some of the features of like Java, it includes, you know, like optional class, which is basically a wrapper class for, you know, handling the non pointer, like exceptions, right? So basically, like, the optional offset of, you know, you like, you do a have the like optional, and you basically have to check if the objects is a present or like not to proceed forward. So, like, there is also, like, stream API, like, which is in like, introduced for, you know, working with the colleagues and easier. So, like, you can convert any collection into a string, and then you can, you know, apply the intermediary operation, like filter and map. And then, you know, like, then also apply the terminal operation to basically in in the stream and get a result. And then there is also, like, a functional interface. And also the lambda expression, which, like, basically is a functional interface as well. Like it is an interface with a one of like abstract method. And then the lambda expression is just the way to represent like that the functional interface, if you You okay,

Speaker 1 25:21

if you did touch base about like strange like, I cannot use trains to do an iteration, and I need to fall back to using a for loop or a for each loop. What is the use case that I not be able to train? Uh.

Unknown Speaker 25:43

So, like, basically, you know,

Speaker 2 25:47

that case, like, when you need some kind of like indexes, like of the object, right? So basically, let's say you want to, you know, get the like object by the position that it appears on the list. And then you, like, you would need to work with a for loop, right? So, and like, whenever there is some kind of exception that might occur. So like, within the processing that you are trying to do you need to, you know, do like traditional formula.

Speaker 1 26:28

Okay, so do you mean to say like exception handling cannot be done when we are doing streams,

Speaker 2 26:38

like it can be done. But you know, it's not that easy,

Speaker 1 26:43

okay, so if you need to do exception handling in stream,

Unknown Speaker 26:47

how would you do

Speaker 2 26:51

it? So that case, like whenever there is exception within these streams? So,

Unknown Speaker 27:00

like you can, you know, basically think,

Speaker 2 27:12

but like you can just, I think you can wrap it with a try catch block, but you know, like it gets little bit tricky. Then whenever I think the best way you can do the exception handling is, you know, some kind of wrapper function that, you know, basically map your exception into other object, or basically, you know, checks the you know exception scenario by using the filter or, you know, map before, like, even processing anything. So we just, you know, need to have a, like, robust check before processing. Okay,

Unknown Speaker 27:56

so I think I'm

Speaker 1 27:57

more or less done with my questions. So what we try to do for the remainder of the time like,

Unknown Speaker 28:06

can you go to, can you

Unknown Speaker 28:07

share your screen

Speaker 1 28:09

and go to trello.com? T r e l, l, o.com,

Unknown Speaker 28:18

t r e l, okay.

Unknown Speaker 28:26

Video link, yeah, watch video.

Unknown Speaker 28:28

Little, little

Unknown Speaker 28:36

on the left hand, you have the watch video, yeah,

Speaker 1 28:40

so what we wanted you to do, I want you to watch this TV, okay? And you assume, like, this is an application which you are going to be building, okay? I want you to, like, break this down into individual tasks, and then, like, you also need to come up with APIs which will be needed to support these. So what are the 80s, which are needed? And also, like, I want you to come up with a data model on which these APs are going to operate. Okay, you're good to watch the video.

Speaker 4 29:21

All right, team, let's use Trello to get this taco truck on the road.

Unknown Speaker 29:26

I'm going to set up a Trello board to plan the project.

Speaker 4 29:31

We got our chef, food buyer, marketing and design all on the board. We can always invite more as we go. Let's all make some lists so we can see the status of everything that needs to get done. Fashion cover it for now, we can all add cards for our individual tasks. Nice. This card is for our most delicious task. Can't wait to see the menu.

Unknown Speaker 30:00

And can you do make a list of produce we'll need?

Speaker 4 30:06

We should aim for all facul Ingredients by Tuesday, for obvious reasons. I'll add a due date and a reminder real quick. Let's move these from to do to doing, so everyone can see where we're at.

Unknown Speaker 30:21

Looks like we have signage ready for the truck. Let's check it out.

Unknown Speaker 30:26

Feel I love these colors.

Speaker 4 30:29

Checking these off. Ooh. Let's move this task to done. In fact, we can automate that now, any cards with completed checklists will automatically move to done. There we go. I got a good feeling about this. I'll add another list to plan where we can go next. This will help us build our business goals as we roll. We're getting there. Let's check in on our Brooklyn launch team. I'm at the truck now, and it looks great.

And not only are we all done, we're ahead of schedule and crushing our goals. First round of horchatas on me, from food trucks to the fortune 500 Trello takes your business wherever it needs to go. Give it a try for free@trella.com nothing

Unknown Speaker 31:22

wrong. Good luck. Look at it one more time. Go for it.

Speaker 2 31:28

Okay, so from like that, what I grab is like thing, like we basically are the platform.

Speaker 1 31:37

Yeah, let's open a notepad and start writing it out. So that way, like we can have the list of tasks, and also, like we can have the list of

Unknown Speaker 31:51

APIs which is associated with those.

Speaker 2 32:01

So basically you have to do right so, basically to content, like your title description,

Speaker 2 32:15

and then, like, basically is food thing. Let me just focus on that

Unknown Speaker 32:22

content, like recipes and so on.

Speaker 2 32:35

So this is what I gather, right? So basically, you have to do the entity that contains the title, description and recipe. And now I also saw that the recipes are the to do. Could be in the To Do state, or, you know, doing the state, or, like, Dawn state, right? So, yeah, these are the three different state that I can introduce.

Speaker 2 33:21

So it will have a string title, it will have a string description, and it will have a list of recipes, and then the enum of the status. Do

Speaker 2 33:53

maybe now, basically, you know, we can also, you know, separate the recipe out

Unknown Speaker 34:04

by what produce in it, so we

Speaker 2 34:09

can have recipe entity. Okay?

Speaker 2 34:25

So on each of on each of the to do. I think we just have one recipe, right? So

Unknown Speaker 34:38

instead of having a list of recipe

Unknown Speaker 34:42

we we would have one recipe.

Speaker 2 35:00

And then so, from Corona recipe, we will have one to map, one to one mapping.

Unknown Speaker 35:13

Okay, and then I mean, we can also

Unknown Speaker 35:21

declare a product class.

Speaker 2 35:30

So it's basically be simple. We should contain like, name, quantity, so

Speaker 2 35:43

BBB, and then it will have a back reference to the recipe.

Speaker 2 36:02

And then it will have one too many relationship between the recipe and the product.

Speaker 2 36:15

Okay, so I think those would be my models and that I can work with. Okay,

Unknown Speaker 36:26

do you think this is sufficient?

Unknown Speaker 36:40

Let's proceed like it's sufficient. Okay,

Unknown Speaker 36:44

what would be your next course of action?

Speaker 2 36:50

So let me basically the status enough. Let me define that as well. I

Speaker 2 37:08

so we have three status, which is like to do in progress and Don, right? So once we are done with these three data source, so basically, we need to handle the multiple operation right? One is like to create or to do, and one is to move in progress, and then the other one is to update, and then the other is to move to complete.

Speaker 2 37:56

So basically, what I like to do is I'll create a post mapping.

Unknown Speaker 38:10

That would be

Speaker 2 38:13

the post mapping would be on our base API of the to do. I

Speaker 2 38:29

so we have this base API, then we have a push mapping where we accept our request. Worry of the to do so. So like this will use to, you know, create a new to do item, and then the Set Status would be on, on like to do so and then we would basically do an update. So like, whenever you want to update anything like within like that recipe, you can, you know, do a like update on that recipe. Okay, so all you need to do is, like, pass in an ID on the put mapping. I

Speaker 2 39:40

it. Yeah, so, like that is one flow, and the other flow is to move the to do into in progress, or the Don, And then we can create, like another, food mapping.

Unknown Speaker 40:12

Okay, BBB.

Unknown Speaker 40:48

Oh, God.

Unknown Speaker 40:50

Yep. I this is good.

Speaker 5 40:56

So pretty much, like we have come to the end of the interview,

Speaker 1 41:01

what I would like to do is like, I would like to, like, take some time for us to take some questions before we wrap

Unknown Speaker 41:12

up. So can you let us know if you happen

Unknown Speaker 41:14

to have any questions?

Unknown Speaker 41:18

So my question,

Speaker 2 41:20

yes. Oh, okay. So thank you so much for the time today, and it was really nice to talking to you. And firstly, I would like to know a little bit more on the product and the tactic, the how the team works on the daily basis.

Unknown Speaker 41:38

We are going to be like working

Unknown Speaker 41:39

on a modernization

Unknown Speaker 41:42

project this application

Speaker 1 41:48

is responsible for, like identifying the prices for equities from all the global markets, not just the domestic market, but across the different market. And it is also responsible for calculating for mutual fund, the heart of fidelity, liquidity. We have lot of mutual fund. And the particular application is responsible for calculating the nav so that is the short window every day within which, like, we should be getting data for like, 20 or 30 and calculating it out to the credit to all of the which should be done within a short span of time. We have more than five application, like within the domain where we are working on, but this is one of the most important application which feel important.

Unknown Speaker 42:49

And we have, like

Speaker 1 42:54

three team onshore and green in India. So all of the creating, we have come working from West Lake office, and I forgot, are you, like, from Texas, or like, where are you from? Currently, I'm in

Unknown Speaker 43:13

Virginia, but I can be located, okay.

Speaker 1 43:20

Okay, yeah. So basically, like, we have 16 and we have some UI focus team who work only on UI and a little bit of maybe API side of things. But we have API focus team to like work just on the API. We have a clear segregation here. We see like that was well for us, so we have gone with that segregation.

Unknown Speaker 43:42

But yeah,

Speaker 1 43:46

that's pretty much the tech that is. Like we Angular UI for the printer. I think we are

Unknown Speaker 43:54

on version 18 for

Speaker 1 43:57

Java. We are using Java 21 one, three.

Speaker 2 44:11

Well, okay, thank you so much for the information. So I think that's all I do have right now. So

Speaker 1 44:23

okay, thank you, thank you for your time. Appreciate it.

Unknown Speaker 44:27

Vijay, anything from

Unknown Speaker 44:30

appreciate it okay

Speaker 3 44:31

from my internship. Thank you, thanks for your time. So what

Unknown Speaker 44:37

will be next? There will

Unknown Speaker 44:40

be HR. We will have to convey our feedback to

Unknown Speaker 44:43

lead within our team.

Speaker 1 44:46

Then we'll make a take addition, and we'll share the addition with HR, and the HR should be able to, like, reach your team.

Unknown Speaker 44:55

So it will be today or the next day.

Speaker 1 45:00

We will be giving our feedback as possible, but when the is going to be reaching out to you that I'm not pretty sure? Yeah,

Unknown Speaker 45:12

thank you so much. Really nice talking to you.

Unknown Speaker 45:16

Thank you. Have a good one. Bye. Bye.

Ronaj Pradhan- Infosys- Srv-final(1)

Speaker 1 00:00

Duties as well. Okay, could you just go about your recent experiences, like, which company recently you work?

Speaker 2 00:12

Yeah, absolutely. So, yeah, talking about my, you know, recent company and project, you know, recently I've been project with, you know, Geico, and we are, like, currently working on developing a kind of a platform as a service where, you know, we, you know, kind of, like, ingest files from, like multiple clients and like multiple states, and then convert those, you know, data into, like, a standardized format. So, like, yeah, we are like, working on, like a microservice based, you know, architecture, and we kind of, like, ingest all of those files and, like, make those data available, like through like REST APIs, so yeah, like, we have the endpoints and that exposed, like, with the, you know, memory information and the benefit Information, the coverage information through like, the Spring Boot, and, yeah, we were like, We also have, you know, front end portal, and we own the experience and, like, the maintenance aspect of the front end portal as well, and where, you know, we have a micro service, you know, that is like, kind of responsible for, like, making any calls for, like, changing, you know, references preferences to the profile and like, also, like the different wizards that is, like, kind of available for users to view. So that is kind of like a very high level overview of what the team manages and what I've been involved mostly in,

Speaker 1 01:38

okay, how many different in gestures you created, and what are the databases back end you interacted with?

Speaker 2 01:47

So, yeah, so talking about, you know, like our back end database, like it is, primarily we are using like, Postgres SQL, like it is, you know, AWS, like RDS and like, we ingest, like the files from like, four different states as they are, like the non standard data, where, like, they provide it, you know, like a, like, a comma separated file. And we have, like, Apache Spark application, you know, that kind of runs on database cluster, and that also, like, you know, loads this data, and that puts them into, like an SNS topic for like, it to be processed by kind of like a lambda and put into the database. And then, like, we use this database to, like, expose this data, you know, through the microservices. I Okay,

Speaker 1 02:46

but you didn't explain me, like, what kind of data is it like you might be supporting a product, right of what kind of product, and what is the back end data you wrote, The ingestors You support the investors,

Speaker 2 03:02

okay, so, right? So, so, like, it is like a member related, and it is kind of like insurance related, related data, right? So, GEICO is an insurance company, and, like, you know, these, like four states, we are using, like a comma separated files, like for storing those data. And you know, like, we get files that hold like information, like the memory information, which is the basic, you know, protected information that is like identifiable. And we get the you know coverage information, which also, like, gives the information about what kind of like coverage they have, and like we and we also have, you know, explanation of benefit information, which is, you know, any claim that the user might put on their insurance claims. And then we also, like have separate files for networks, which is like, like the service provider. And so the like, those are the four different types of, you know, files that we get, and then, like, we created, you know, different types of, you know, tables that store these, you know, in the standardized format.

Speaker 1 04:14

Okay, got it basically your insurance related all the data related content and the documents, basically you are collecting from different data sources and providing the micro services, and you're exposing like you are saving those metrics into post West. QR, okay, got it. So what? What challenges might you face when implementing a micro services?

Speaker 2 04:45

So talking about, you know, the challenges you know, that I faced while, you know, doing these kinds of things, you know. So, yeah, like, basically, you know, microservice are really small units of service, you know, that, like, expose different things and like into certain functionality, like it is bound to, like, certain application context, or like, kind of, like our service that it provides. So, like, whenever, like, there is a collectively, you know, like working of a service platform, like, they need to, know, communicate within each other, right? So we have to, like, fix issues of how like these services, you know, kind of interact with each other. Because, you know, like, whenever they are, like, communicating, like, over a network, like, there might be some kind of, you know, network latency issues, or like, other kinds of issues you know, that might cause issues on, like, a longer period. But you know, whenever you know you are trying to modify this data across, you know, multiple services, since you know they are not like one single unit, like there is going to be, you know, issues with their consistency, right? So, since we are dealing with like the member and covers and like explanation or like benefit information, like, we had to make sure that the data was like, consistent, and because, you know, sometimes there may be, you know, issues where member may change their cover information, and then we are processing a claim on the downstream but The data might not be, you know, like, up to date, so there has to be, like, some kind of, you know, saga pattern involved to, you know, manage the consistency of the data. And like, you know, these, like interconnected services, you know, like, you need to make sure, like, you are not cascading a failure from one service to the other so they are, we have, like, integrated something like a circuit breaker pattern to, kind of, like, make sure we don't, you know, disrupt the whole services.

Speaker 1 06:53

Okay, but how did you handle data leaks? Okay,

Speaker 2 06:58

so, yeah. So talking about, you know, how I handled, like, the data leaks, right? So, so, like, sorry, like, do you mean, like, what kind of data leaks or like, are we just like, talking about, like, the general

Speaker 1 07:15

in terms of micro services security, like, how the assurance of the microservices security. How did you implement and how did you handle the data leaks in case of any sensitive data?

Speaker 2 07:29

Okay? So, yeah. So, yeah, basically, you know, talking about that, you know, whenever, like, we were no kind of dealing with, like, any kind of, like, the sensitive data, you know, we use, like, a common, shared, like encryption service, which, like, encrypts any, you know, types of sensitive data. And, you know, and fast is that as a, like, a payload, and like, we know, provide the, you know, some kind of, you know, double encryption payload that, like, consisted of the like, you know, public key for, like, the prescription. And then we also made sure that we had, you know, you know, input validation in place to make sure, like, there was, like, kind of no script attached to the data that was kind of like coming in from for like might have been triggered some kind of, like a SQL injection or cross site scripting. So, yeah, those were, you know, the two primary ones. But yeah, we also made sure we had, you know, proper exception handling, and, like, we had, you know, secure logging of the, you know, sensitive data, so that, like, the no sensitive data would be kind of logged.

Speaker 1 08:39

Okay, could you explain, like, the common patterns you implemented, like you already mentioned, saga pattern and circuit breaker. So could you explain little bit more detail, like, what are the platforms and how

Unknown Speaker 08:57

you implemented those patterns?

Speaker 2 09:00

So, okay, so, you know, yeah, so, talking about, you know, so, yeah, saga pattern, you know, is for, like, a microservice orchestrator. So, yeah, basically, you need handles the like, the data consistency, like, which comes with a, you know, transaction, commit and rollback. So, like, whenever, like, everything success, like, everything is good, but like, if there is any failure, then, like, the rollback mechanism is triggered, and then, like, it runs so that all the data, like, across the microservices are consistent, right? And the circuit breaker again, you know it's kind

Speaker 1 09:38

of one second, okay, saga pattern you explained right, right now. Okay. I want to understand like, see, basically all our systems are distributed systems, right? Distributed systems. And you want to make sure that a data consistency, but all all the micro services, what state you are maintaining, what kind of state you are maintaining in the microservices.

Unknown Speaker 10:04

Okay, so,

Unknown Speaker 10:07

talking about, you know,

Speaker 2 10:10

talking about what kind of, you know, like, I'm not sure, like, I follow the question, like, Okay, I

Speaker 1 10:17

understood. Um, so I'm just asking, like by default, all the micro services are stateless, right? All are stateless. So how you are like in your micro service, the APS. The APs are asynchronous calls or parallel calls. What kind of APS you implement

Speaker 2 10:41

for those you know, so, so there's a lot of them that are synchronous calls, and those like, like, there are the asynchronous calls I handle, like, through some kind of, you know, like, messaging queues. And we don't have like, lot of those, you know.

Speaker 1 10:57

Okay, so while you implemented the parallel calls, right? How did you handle the data consistency? Because those are parallel calls. So anytime, like a one thread can one API can update the data, one API can read the data. So that is where, like, how did you implement the saga pattern in that scenario?

Speaker 2 11:19

So, okay, so, yeah, like, you know. So, basically, you know, we handle, like a long running, like a transaction with an orchestrator. So we had, like a central, you know, orchestrator service that kind of, like manages a sequence of events that interest that each step, like, it's completed, or, like, roll back and, like, if there is any failure, like, it would basically roll everything back. So that's how, like we had, like we did, we did not have like a choreographic, you know, but we had like a central like, orchestrator that, like, is that, like a data consistency across, like a multiple synchronous operations,

Unknown Speaker 12:05

did you implemented any testing, like,

Unknown Speaker 12:09

between the micro services, during

Speaker 2 12:11

the micro services? Yes, I did implement, you know, we had, you know, like the intervention testing and, like, I have also, like, some regression testing, you know, in place so to test the flow of the data from like one form like like to the other.

Unknown Speaker 12:30

Then they did you hear about the contract testing?

Speaker 2 12:35

Which one did you see? Did you say like contract so for this, you know, we did not use any, you know, contract testing in our team. Like there was, you know, separate team that would, you know, bring it up. And we, like, sometimes our tester would take it up and, like, they would be helping, okay,

Unknown Speaker 13:02

I Oh, sure you want to go ahead.

Speaker 3 13:08

So I'll ask you some Java eight programming problems, right? So can you share your screen? Sure, just a Second. I

Unknown Speaker 13:59

sorry, okay,

Unknown Speaker 14:09

all right, so for the first problem,

Speaker 3 14:15

write a program that finds the maximum number from a list of integers using the stream API

Unknown Speaker 14:25

also explain what you're doing as you're coding it.

Speaker 2 15:18

So, okay, so, yeah, let's like, start by Like, reading, you know, A list of integers. Okay?

Speaker 2 16:03

So, like, in order to use the like, no stream API, like, the first thing is, like, we need to convert the like, individual list, you know, into a stream using the stream method. Okay, I

Speaker 2 16:33

and like, then we know we can, like, use the, you know, building, like, the comparator, like, for integer, So, and

Speaker 2 16:55

like like should return me an optional I know,

Unknown Speaker 17:03

and so, you know,

Speaker 2 17:06

like, I can bring the Like map Max like, if It is like, present, yeah.

Unknown Speaker 17:44

Okay, yeah, that's

Unknown Speaker 17:45

good. All right,

Speaker 3 17:53

right, a program, right? That counts how many strings and lists have a length of three? Okay? Okay,

Speaker 3 18:05

can't just use the same name. It's fine. Start from scratch. I

Speaker 2 18:26

So let's, let's assume like, that's like, this is a list of string.

Speaker 2 18:35

Now, you know, so like, in order to count like, you know, we can use, you know, Count method of the like, a stream API. Okay, so, but like, we like, only want to like, count, you know, count the ones that, you know, have the length, so, length of three. So we can use, like, the filter method to filter those out. Okay, so?

Unknown Speaker 19:38

Okay, do

Unknown Speaker 19:44

now the next problem

Unknown Speaker 19:47

convert a list of strings topper case.

Speaker 2 19:55

So like you know, for this, we can use, you know, math function for this. I

Unknown Speaker 20:44

Okay,

Speaker 3 20:47

all right, so now I'll ask some ask you some Spring Boot questions.

Speaker 3 21:02

Um, create a simple bean in spring.

Speaker 2 21:14

So, like, do you have, like, a specific type of bean that you're looking for, or just a normal

Unknown Speaker 21:23

being normal.

Speaker 3 21:32

So like create the bean and also show me how you will retrieve it. I

Speaker 2 21:44

Okay, so, yeah, let me like, Create, you know, configuration class First i

Speaker 2 22:40

Hi, uh, so, yeah, the you know, being, you know, integration is to like, like, specify that, you know, the method returns a bean, and now Like, we can create, you know, simple controller to, like, retrieve this bean, like, using an auto wire.

Speaker 3 23:46

Okay, so how would you retrieve the bin in your public side? Word made of your Spring application.

Unknown Speaker 23:59

So

Speaker 2 24:14

so like to, like, kind of, like, retrieve it in the no, like, the public static void main, like, So, usually, like, you would retrieve it, you know, like, it isn't like an instance variable, but like, if you need to get it from, like the main method, like you need to get the application in the context,

Speaker 3 24:36

Yeah. So can you show me how that's done? I

Speaker 2 24:51

and so you know, like, so the way you know, do you do that? You know you're basically like whenever, like you do the application, like on like whenever you run like a string application, it returns like our application context.

Speaker 3 25:17

So show me how you would retrieve the B using the application context. So,

Unknown Speaker 25:28

like, Do you know the context you know? So

Unknown Speaker 25:33

context has a method called, you know,

Unknown Speaker 25:42

I like, get bean, okay,

Unknown Speaker 25:50

and then you know, you can like this,

Speaker 2 26:08

so like you can in the past, you know, like

Unknown Speaker 26:15

the Class of the bean that

Unknown Speaker 26:18

we are looking for, I

Unknown Speaker 26:29

Okay,

Unknown Speaker 26:42

um, so for next problem is

Speaker 3 26:46

Create a risk controller with A single get endpoint. I

Unknown Speaker 27:40

So yeah, like, since,

Unknown Speaker 27:46

since, like, we're working, you know,

Speaker 2 27:52

with the customer service, like, we'll basically be creating, you know, customer controller and, like, We can annotate it, like, using

Unknown Speaker 28:01

The rest control annotation, okay, so

Speaker 3 29:00

Okay, so Now, how would I have multiple

Unknown Speaker 29:05

endpoints in a single controller?

Speaker 3 29:11

Let's say I wanted to end points in this inside the stress controller. How I go about doing that? So

Speaker 2 29:16

for this, like, you know, so you can just, like, keep on, like, defining more method and so, basically, no you can create another method and like, in like,

Unknown Speaker 29:28

like an update, Like, we can no gate Mapping,

Unknown Speaker 29:34

okay, demonstrate that I

Unknown Speaker 30:26

Okay,

Speaker 3 30:29

demonstrate how you load a property From application dot properties into this Rest Controller.

Speaker 2 30:40

So, so, yeah, like, we can use, you know, like, a value annotation to, like, to read the property. So,

Unknown Speaker 30:52

okay, yeah.

Speaker 2 30:58

So, yeah, let's say, you know you have, you know

Unknown Speaker 31:04

I limit as your property, then you know

Unknown Speaker 31:08

we can do something like a

Unknown Speaker 31:53

Okay, that's good.

Unknown Speaker 32:02

Create a bean with a prototype scope? Yeah.

Speaker 2 32:11

So, yeah, sure. Like, do you want me to, like, change the existing one?

Speaker 3 32:17

Create a new bean with a product by Avatar the scope, deep prototype, Sure Sounds good. You.

Speaker 3 33:30

Okay, in the spring bean you just yeah, this one demonstrate how the life cycle looks. I

Speaker 2 33:45

So, so you want to know, like the lifecycle hook of this prototyping

Speaker 3 33:49

and just demonstrate how they're used. I

Speaker 2 34:10

so I'm not sure, like I followed that exact thing so you want,

Speaker 3 34:15

so you're familiar with that. You're familiar with the bean life cycle, correct, spring life cycle. Right? So, can you tell me what the spring being like? The spring being my first life cycle is spring,

Speaker 2 34:30

Green Life Cycle, yeah. So, right? So you know, like, basically, you know, like, whenever, you know, like, whenever, like you like, instantiate, you know, a bean like, the first thing you know, like is a list of properties and and then like there is a like, a pre initialization where like it will, you know, try to like a set the application context and like run like The Post construct method, if it is defined. So those are like the like, the first pre initialized, like steps and when, like a bean is, you know, like in ready to use state like a bin is initialized, and like can be used when, like you called, and then, like, there is a, you know, destruction method, like, which is like, which is the application context is closed and like, a beam is like, no longer needed. First thing is like, it calls like, a pre destroy method. And then you know, it may call the destroy method like, if it is a disposable beam.

Speaker 3 35:39

Okay, so can you implement the lifecycle hooks In this aid?

Unknown Speaker 35:57

Yeah, Let me Do it. I

Unknown Speaker 37:53

so I'm Done. Okay? I

Unknown Speaker 38:06

moment i

Speaker 3 38:24

Alright, so I'll ask you some generic Java questions.

Speaker 3 38:32

Okay, let's put me the four access modifiers in Java.

Speaker 2 38:38

So, yeah, so you know, basically, you know, those are, you know, like the public package, you know, default and like the private,

Speaker 3 38:50

okay, what do? What does each of those x modifiers do?

Speaker 2 38:55

So, yes, so talking about, like each of these, you know. So your public is the you know, most you know, broader one. Like that means, like, everything can, you know, access this method or like class. And then you have like a private, which is like, you know, very strict one. And private is kind of like only accessible within the class that you have declared in. And now you have to protect it, like, which means, like, it can be like, access, you know, kind of like within its own package, and also by the subclasses on like different package. And then there is like a default like, which is, you know, only accessible, kind of like within, you know, like its own package, and so it's like, kind of like, like a private package. Okay,

Speaker 3 39:50

what's difference between double equals and the equals method?

Speaker 2 39:56

So, yes, so talking about, you know, though. So basically, you know, double equal, you know, checks for, you know, like a reference on the memory, right? So, like, it is like, used to like, compare the memory address of like, the true objects and the equals method. You know, it kind of like checks the content like of the two values, right? So you can, like, override the two equals method in your object class and then like, define like, how we want, you know the contents to be compared. So basically, you know equals method is like, intended for like, the logical comparison of the equality of like, the two objects. Okay, do

Unknown Speaker 40:56

okay,

Unknown Speaker 41:01

was the final keyword used for Java,

Speaker 2 41:05

okay, so my keyword, right? So, yeah, like, basically, you know, the final keyword, like it is used for, like, defining, like, some kind of a constant, right? So, basically, you know, whenever you define a variable as a final like it just means that like it is a constant and it cannot be like reassigned whenever like you are defining a method to be final, and like it is preventing like your subclasses from like, overriding this method and like, whenever like you define no class to Be final like, it means that, you know, you it cannot be like a like subclass. Then basically like it is used for, like, the security and yeah, like you also want to use, you know, final to, like, declare some kind of constants and like also have, you know, immutability in our objects. Okay,

Unknown Speaker 42:05

was different to throw and throws in Java and

Speaker 2 42:09

throws. So, yeah, so basically, you know, the difference between like the throw and throws is that, you know, you're, you know, like the throw keyword is used to like, explicitly, you know, like, like, kind of like, throw an exception. So let's say like, whenever, like you are facing, you know, some kind of error behavior, then like, you would use the throw new runtime exception to like the throw a dedicated exception. But throws is used on, like, the method signature details, like, like the color, like, what kind of exception, like that this method might be like throw. So,

Speaker 3 42:51

okay, I'll ask you some git questions. Okay, what is the difference between git pull and git fetch.

Speaker 2 43:03

So, yeah, so git pull, you know, like it is kind of, you know, like a mixture of, you know, git fetch and like the MERS. So git fetch is, like, used to, like, update, like you're like, it retrieves, you know, like updates from a remote repository. But you know, it does not kind of like merge the changes in your like local brands. So basically, you know, it gets like, all like, the comments, references and everything but your git pull, right it like, it along like, like, it fetches like, then merges to like the local repository. So, okay,

Unknown Speaker 43:47

how do you stage files for a commit?

Unknown Speaker 43:51

So, yeah, talking about, you know

Speaker 2 43:55

this, you know. So like, in order to stay like, like, as a like stays a file for, you know, a comment like, you can, you know, basically do git add and like add the file names that you want to stage,

Speaker 3 44:10

okay, and if you want to stage all changes, what would you do? Yeah,

Speaker 2 44:16

you can do like the git add, like, dot, you know, period sign,

Unknown Speaker 44:22

yeah, okay.

Speaker 3 44:25

And how do you do the last commit, but keep the changes in your working directory.

Speaker 2 44:34

So for this, you know, you can do like the git reset, you know, soft. Okay. And

Unknown Speaker 44:46

how do you switch to a new branch?

Speaker 2 44:48

Yes, switching from like the new branch is like, so you can just do, you know, like, get check out and like the brand's name. Okay. I

Unknown Speaker 45:07

and

Unknown Speaker 45:09

how do you

Unknown Speaker 45:11

do a checkout and also

Unknown Speaker 45:15

switch to new branch in one step.

Speaker 2 45:18

So, like for this, you know, like, basically, if you do the like, the git checkout, you know, and like, das B with the, you know, branch name, like, it creates a new branch of that name, and it checks out into that branch.

Unknown Speaker 45:36

Okay, all right, that's it.

Speaker 3 45:42

Do you have any other questions? No, I'm good. Okay, all right. So thank you, ranaj,

Speaker 2 45:52

thank you so much, guys and like, what will be the next step after this, you know, interview? Can I know, like, a bit more about it, you know.

Speaker 2 46:15

So I have to reach out to my recruiter, yeah, so, yeah, like, can I, like, do you also, like, know, a little bit more of the you know that the project and like, like, if it's something, like, if, like, this, like, closable or something,

Speaker 3 46:37

I have no idea, honestly. I mean,

Unknown Speaker 46:44

you'd have to reach out to your recruiter,

Unknown Speaker 46:47

okay, yeah, that's good. Sounds good?

Unknown Speaker 46:54

All right, so thank you. Have a good evening,

Speaker 2 46:57

yeah, thank you guys. I really appreciate you guys for taking my interview. I really appreciate it. Thank you. Thank you guys.

Ronaj Pradhan- Infosys- Srv-final

Unknown Speaker 00:00

Person to join the call First

Speaker 1 00:03

two to Three Minutes. You

Speaker 1 02:50

I don't know if the other person is joining, which is kind of annoying,

Speaker 2 02:56

because I wasn't supposed to be the primary panelist, But I guess I can get this. So I guess just tell me a little bit about yourself.

Speaker 3 03:09

Yeah, sure. So hi, good morning, everyone. So yeah, my name is and you know, I've been working as a uniform stack Java depot for over six years now. And primarily, you know, I've been working on Java based, you know, applications utilizing different types of frameworks, like Spring, Hibernate and like web service for, like, API development. And basically, you know, I've also been involved in lot of, you know, microservice development. And, yeah, recently, you know, I've been doing some design, like collecting, like functional and non functional requirements, and also like implementing these requirements. And like I've also liked on some kind of, you know, serverless projects, you know, where I've been kind of, like handling some kind of, you know, AWS lambdas for some even, like basic architecture. And I've also been in part of, like, Scrum teams, and that consists of about, you know, like five developers, and we have no product owners and Scrum Masters. And, yeah, we use, you know, AWS for our cloud platform. And I've also been, you know, working with, you know, AWS s3 buckets. So, yeah,

Speaker 2 04:23

thank you. Um, I was just looking over your resume real quick as well. Um, of course, you know, it has the standard stuff that we're looking for, Java, spring, all that stuff, I guess to, I guess start, get started with Java, I guess. Could you just tell me a little bit about, like, Java streams, what they are, what they use for that sort of thing.

Speaker 3 04:56

So, okay, so, yeah, basically, you know, talking about, you know, Java strings, you know, they are kind of like powerful, you know, features that has been introduced in Java eight, right? So, like, from Java eight, like we've been working with, you know, collections, so which are, like the basic list, assets and maps. And like Java, you know, it introduced concepts of, you know, streams. And these are, like, used for processing our streams, right? So you can, like, filter your streams based on certain conditions. And also, like, we can collect it into certain other kinds of, you know, streams. And we can also, you know, radio, see it and modify it accordingly. And like, let's say, you know, for example, like, we have a list of employees, and we want to like, filter the employees whose salary is, like, greater than a certain number. Then, like, we can convert the list of employees into a stream, and then use a, you know, filter method to like, filter based on the salary. And then, you know, we can collect it into like, a separate list. So that's what stream is, and it's a really powerful tool for, like, working with collections.

Speaker 2 06:07

All right, um, can you, I guess, explain the difference between, like, functional interfaces and lambda expressions?

Speaker 3 06:17

Okay? So, yeah, like, basically, you know, functional interface like these concepts were introduced, you know, in Java eight, right? So functional interface is like a special kind of interface, like it only has one abstract method, right? Like it may have like default methods, but the only key thing here is, like it only has one abstract method, but a lambda expression is a way to represent these functional interfaces. So, like, if we have like a functional interface with like a certain method, you can define a Lambda expressions to kind of represent that functional interface. So some of the function interface, you know, in Java are like, you know, callable, runable, like predicates, consumers and basically like, we can define, you know, Lambda expressions representing like those and like so lambda expressions are also known as like anonymous functions because, like, they do not have a name.

Unknown Speaker 07:25

Okay, thank you.

Speaker 2 07:30

I guess, could you, like, could you explain the difference as well between hash maps and hash tables?

Speaker 3 07:40

So, okay, yeah. So, yeah, basically, you know, hash map. You know, hash map, it is an implementation of, you know, our map interface, right? So the hash map, you know, it is not synchronized, which means, like, like, whenever there are multiple threads accessing a hash map like it won't be thread safe. So, like, sharing data across, like multiple thread like concurrently, will have some kind of like issues on data consistency, and like hash tables, you know, is synchronized, like internally, so you know, it is like a thread safe by default, and like, we can use it like across, like multiple, you know, threads and has map also allows like one null key and like multiple null values. But hash table, like doesn't allow like null keys or null values, so like hash table will send you a null pointer exception when you try to insert any null

Unknown Speaker 08:47

okay, I'm

Speaker 2 08:51

trying to see if I have so I like, I wrote out some questions beforehand. So I'm like, seeming to them to see if I think that pretty much covers everything that's at least necessary for Java. Presumably you're familiar with JUnit, mokito, all that stuff.

Unknown Speaker 09:12

Are you familiar with power mock?

Speaker 3 09:17

You said, Yeah, use a power mock, right? Yeah, I'm pretty familiar. Yeah,

Speaker 2 09:22

can you explain one scenario in which you would need to use power mock over mocking tail?

Speaker 3 09:27

Okay? Sure, so, yeah, like, like, basically, you know, power mock is a tool, right? So that provides, you know, additional features for, like, mocking any special kind of like classes, right? So if you sometimes, you know need to mock a constructor or like a final class or like static methods, then we use, like, the power mock so, so like, recently with the Mockito fight, like, you can use the Mockito to like mock static methods, but Mockito still does not support mock of like Final classes or constructors. So if you want to like define the behavior of how like a constructor is like working, or how the final classes should be mocked, then you prepare that class for like the test using like the prepare for the active annotations, and then you also initialize a power mock, and then you can mock those, you know, final classes. So like, that's the use case of power mock, and power mock has, you know, special runner that you need to run in order to like for the power mock to work. So, yeah, like, that's the use case of like power mock. So

Speaker 2 10:48

okay, thank you. Um, the only thing I can think of for testing, um, can you explain the difference between unit testing, progression testing and integration testing.

Speaker 3 11:05

So, okay, so, yeah, like, like, basically, you know, testing like, like, testing is basically like, like, what means like is an individual part of the application and without, you know, interaction or like after like the external services. So basically, you know, you like, mock out any external parts, and then you just, you know, test specific portions like, by defining the behavior of the like external service by like, using some kind of like box. So you can use like the J unit in Mockito for like the unit testing. And there is this like concept of, you know, integration testing, which involves, you know, these specific units, like how they are integrated and how like they can, you know, like, interact with, you know, each other, right? So basically, you know if you identify the flow of the application, and also like issues that occurs when you know different components and system interact with each other. And you can also, you know, use the JUnit, you know, Spring Boot test for this. But also you can do the manual test using APIs like postman and so on So, but now a regression testing is like a fully automated testing suite, right? So basically, whenever, like, you develop a feature, you will like, write a regression test, and also like a regression test that validates the functionality of our application. Now you'll be like, like, adding newer future. And regression testing is important because this regression testing makes sure that whenever you like add like new future, it's not breaking the existing functional functionalities of the application. So that's what a regression testing is. So you can use that cucumber or Selenium to write the recursion testing.

Speaker 2 13:06

All right, thank you. I guess moving on. This isn't like necessary. But I just wanted to check, do you have any experience with Eclipse? Vertex?

Unknown Speaker 13:18

Vertex, yeah.

Speaker 3 13:22

Eclipse vertex, right? So, no, I cannot, not use like, Eclipse vertex for now, right?

Speaker 2 13:28

All right, that's fine. I just thought I'd ask, because the the team that I'm on, we mostly use vertex, even though, you know, these interviews say often ask for Spring Boot. Just figured I'd check, you know,

Unknown Speaker 13:45

I guess moving on to so like

Speaker 3 13:48

vertex is used for, like a reactive programming, right? So I've used like web flux instead, yeah, okay,

Speaker 2 13:58

yes, good to know. Thank you. Um, with regards to spring, I guess, can you just explain what the what the use case is of Spring Boot, you know? Like, why would someone want to use Spring Boot for their application?

Speaker 3 14:17

So, so, yeah, like, you know, Spring Boot is a really nice framework, you know, that is, that gives, you know, out of a box, you know, production ready features, right? So, um, like, basically, you know, like, if you want to run a, kind of, like, a standalone application, and you do not, you know, want to worry about, you know, a lot of, you know, boilerplate code, like you could, you know, work with Spring Boot. And instead, like, like, it helps you to like, you know, it provides a lot of, you know, standard dependencies. And also it has, like, an embedded, you know, Tomcat server so that you don't have to, like, you know, configure external servers to run like applications. Now, whenever you know, you like, add a certain dependencies on your depend, like, the dependencies, right? So it auto, like configures those dependencies to like, a thing called, you know, auto configuration, where, like, it reads all the available, you know, beans from our glass pass, and then also that it provides it to you. So, yeah, you know, basically it has a lot of, you know, microservice support, and it is also, like, a lot easy to work with, like, whereas, you know, and you can also, like, focus on, like, business logic, rather than, like, configuring, you know, a lot of these components. And, you know, yeah, that's why, you know, you would want to use Spring Boot on, like a lot of the you know, applications you know that I have been using, like, working with, you know, have been Spring Boot and, like we have been working with, you know, RESTful web services using, you know, annotations, like Component Services and so on. And I do not need to worry about, you know, configuring those, you know, just manually through, like an XML but I can just use an annotation, and spring would also like auto configure those dependencies for me to work with. So that's why.

Unknown Speaker 16:12

All right, thank you.

Speaker 2 16:17

That was another question that I intended to ask. Let me I can't find it in like my list of hosts, so let me remember what all it's asking about.

Unknown Speaker 16:31

One second. Sorry. I

Unknown Speaker 16:50

I guess,

Speaker 2 16:53

so I know I remember generally what the what the question was asking about. So I just asked that instead. Don't need that specific question. Could you just explain, I guess, at a high level, some of the critical annotations for spring, you know, like components, configuration, all those.

Speaker 3 17:17

Yeah, absolutely. So, yeah, like, you know, basically the first like annotation for Spring Boot is, you know, Spring Boot application annotation and like, it is done on the class that contains your like, a main method. And like is a combination of a bunch of, you know, annotations internally, like auto configuration and so on. And basically, you know, it's just telling you that this application, like the Spring Boot and just go in and read your class path resources and to look into the dependencies that is available. And then there is, like, a annotation called, you know, configuration. And this configuration annotation, you know, marks a class as a provider for, like, the spring beans. And, yeah, the spring beans are nothing but just objects you know, that are managed by the spring containers, right? So the class you know that is annotated with, you know, configuration you will have to, you know, annotate, like, have an annotation called a bean annotation. And these bean annotations, like, specify, you know, different methods that give like the object of that beam. So that is one annotation, and there is another like annotation that's called, you know, primary annotation on a bean which, like, specify like that if there are, like, multiple beans of the same type, and like, use that as a primary pin. And there are like, different class level annotations, like, like component, service, repositories controllers, like the rest controllers. So like, the service annotation is used for, you know, the service layer the repository annotation is used for, like the data access layer controllers and rest controllers, they are used for, like a controller layers. So the Rest Controller is a combination of like the two control like annotations and like they are called, like the controller and the response body. And usually, you know, typically, we use for the RESTful web services. And there is also, like the auto wire and qualifier used for like the dependency injection. So those are like the primary annotations that, like you can work with whenever, like working with the springboard applications.

Speaker 2 19:39

Thank you. I guess, on the topic of dependency injection,

Unknown Speaker 19:44

can you explain the difference between like,

Speaker 2 19:50

like field injection, versus constructor injection, versus, you know, like getter injection,

Unknown Speaker 19:57

you know, dependency injection? Yes.

Speaker 3 19:58

So alright, so talking about, like these things, you know, so we can do the dependency injection in like the three ways, like you mentioned, right? So which is like, constructor injection, fill injection, and the seven injection, right? So fill injection, like you are, like, using a class field, right? So, on an instance variable and using, like an auto wired annotation on an instance variable to inject, right? So, basically, you know, if I have a like, a kind of, like service class with, you know, dependency or a repository, I can, you know, just define a priority employee repository and then like auto wires, basically, you know, directly injects into a private field without, kind of, like needing any kind of, you know, sellers or constructors. And it is very easy to end quick. And the advantage of that is, it is that easy to use, but sometimes, like, it can become, sorry, become very difficult to test, right? So now constructor injection, like we I prefer constructor injection because I feel like it's the best one to use, because, like, it's used through the class constructor, and, you know, like, it ensures that, like, like, our dependencies are immutable, right? So it cannot be changed one set. And it makes sure and mandatory, like, sure that all of its mandatory dependencies are initialized whenever you are working with that object. And now the setter injection is useful whenever, like, you have an optional dependencies, right? So, yeah, like, you would be doing dependency injection through a setter method. So it makes it kind of like flexible to, like, inject dependencies during runtime.

Unknown Speaker 21:54

Okay, thank you.

Speaker 2 21:57

I believe that's everything that I was looking to cover for those. So if you could share your screen, I have Some some coded questions To share as well.

Speaker 2 22:15

Sure. Yes, sure, so one second, I have a couple of questions, I guess the first question, I'll share this in the WebEx chat once I'm or once I finish reading it. Could you write a Spring Boot configuration class in Java using Spring boots configuration annotation, and in that class define two Beans, beam one, beam two, that are each type, array, list, and use component scan to scan the package or dot example. So probably be, I guess, easier to parse through Once I actually like share it. There You Go. So

Unknown Speaker 28:26

All right,

Speaker 2 28:28

thank you. Let me I'm for these questions I'm asking. I don't expect them to run, of course, I don't kind of just, you know, visually verifying that they are correct. Let me get another one of these questions. This one is specifically, I guess, gaging your competency with with, you know, writing test classes whatnot, making sure you're familiar with JUnit etc, given a class calculator that has methods with a method signature, public int add with, you know, two ints, int a and int b, and public int subtracts with, also two ints to a, int a and B. Can you write a JUnit test class that does or that tests two different test cases for each method and creates a new instance of the calculator object before each test using the appropriate JUnit annotations? Let me share this real quick. Yeah. Sure, sure,

Unknown Speaker 29:43

it's formatted improperly, but That works. I

Unknown Speaker 34:58

All right, thank You. I guess that's

Speaker 2 35:02

everything. That I wanted to ask. Um, did you have any questions for me?

Speaker 3 35:08

Yes, sure, you know, yeah, like, I like to know, you know, like, like a primary, like a tech stack that the team is working with, and, like, a little bit more about, you know, that project, if that's like, available?

Speaker 2 35:25

Um, I'm not so I can't really go into that just because I don't know what team you'd be put on if you were accepted. If I'm understanding this interview correctly, this is just an interview for Infosys, and not for the client that I'm working with, contracted to a client that, you know, that NDA and all that stuff. But I mean, generally speaking, it seems like a lot of the projects that Infosys is working with just uses, you know, Spring Boot, interacting with the SQL database, that sort of thing, you know. Um, as far as that goes, though, I can't, I don't know much more outside of, you know, what my team works on, I can't really go into what my team works on.

Speaker 3 36:14

So, like, like, sounds good. So and like, you know, like, what would be the next steps after this interviews like and I know about that as well.

Speaker 2 36:25

Um, so, I mean, my role is that I'm just, like, a, I mean, my actual role is, I'm a software engineer. I'm not even supposed to be doing these kind of things. My role here is that I'm is that I'm just, you know, the secondary panelist. I'm just running the tech side of the interview, as far as, like, you know, next steps and whatnot, you'll probably have to reach out to the recruiter that you were working with, you know, the person who set up this interview for you, they'd be able to help you with next steps on that. Thank you so much. Yeah, no problem. That's everything. I'll close out the call.

Unknown Speaker 37:04

Thank you for the time and I really appreciate it, thank everyone.

Ronaj Pradhan- Mondo-First-SRV

Speaker 1 00:00

Look into Phil, just to give you a little bit of context to answer, to tell me about yourself and answer some of the questions I have. And as as we go into this conversation, I may interrupt you at different at different points, just asking you to go deeper, asking you specific questions about about things you said. I prefer to have, like a conversational interview, rather than just listen to somebody about themselves, and feel free to interrupt me if you have any questions about about what I'm describing. So I'm the senior engineering manager for the USI division of products at tour price. My teams, I have two sprint teams that are responsible for the logged in Section of the public website where advisors go to do business with your price. In some cases, that's that's running reports, or having our specialists and analysts analyze the advisors book of business and make recommendations to them. The advisor also can can log in and self service and things like upload a new book of business if their holdings have changed, set up a watch list, and different things like that, access old reports. The other team is responsible for an internal application that's basically used, it's like a workflow engine that's used to service those requests that come in from those advisors, do the analysis and go through like a workflow process. Think of like intake comes in. It's a pending state. They work on it. It's a progress, you know, similar to how you know, you probably have worked with tickets in JIRA or rally, or anything like that. So it's very, very similar type of idea. Front end is Angular. The public website version, the login version of the website is runs through Adobe Experience Manager, but it's still mostly an Angular application. Back end is Java, for the most part. We've got an Oracle database. We've got a SQL Server database in the process of migrating over to RDS Postgres instance for some of the data. Yeah, I think, I think that's good kind of background. Any any questions around, any on any of that, or any other information you'd like to know.

Unknown Speaker 02:22

You know, I know, I think that's like, pretty good product right now.

Speaker 1 02:26

Okay, cool. So, so I've got your resume here. If you see me looking over to the right, I'm just, I'm just referencing your resume and looking at things. So tell me a little bit about yourself. I think, are you, you currently with Geico? Is that, right? Yes, I'm currently with Geico. Okay, so tell me a little bit about what you've been working on with Geico. And,

Speaker 2 02:45

yeah, sure, yeah. So basically, you know, like primary at GEICO, so we've been working at data transformation platform, where, like, we taking data from, you know, like a, like, a non standard form, right? So we have, like, a lot of clients in different state, and that, like, stored their data in like, you know, like a different file format, and that we take that, we take that and just like, convert it into, like a, like a canonical, like JSON, to like a standard, standardized data format. And this, like,

Speaker 1 03:21

so sorry, are these, like, Geico reps in different states, or is this, is this like customers in different states?

Speaker 2 03:28

Like, these are like, you know, like the like, these are the Geico like, offices, right? So not

Speaker 1 03:34

the representatives in different states, yeah. Okay, so, so they don't have a standard, a standard way that they're storing data. Everybody kind of rolls their own. And then, and then you've got this, one of your projects was to kind of standardize their their format.

Speaker 2 03:48

So, like, initially, when like GEICO started, like, they did not have, like, have like, a standardized approach. So like, so all our representatives were, like, still on like, Legacy side. So where the like just store into like, a comma separated file. So like, for modernizing this, like, we didn't want, like, the representative to change their way, like, but while, like, ingesting the centralized, you know, system, like, should take like, like, kind like, take care of that kind of format. So,

Speaker 1 04:18

so it's like an ETL process to be able to handle legacy format as well as the current format. Yeah. So, okay, great. I would say,

Speaker 2 04:29

like, like, I would say like, you know, like, 20 to 30% like, of the initial work that I, like, did was, like, the ETL process, but, you know, after that, you know, like it was a lot of, you know, microservice work for like, kind of, like, exposing, like, that flat file data, like micro service based, into a micro service based, you know, architecture

Speaker 1 04:51

and what, who is consuming those micro services

Speaker 2 04:56

for this, you know, like, so these micro services, Like, are available to, like the clients and customers, right? So, like the customers, like, whenever, like, they log into their portal, they can see, like, their information, right? Like, what covers like they have, and, like, what kind of benefits they have and any so

Speaker 1 05:17

the Geico Portal was the consumer those services then,

Speaker 2 05:20

yeah, the guy portal, right, exactly,

Unknown Speaker 05:23

okay. And was it,

Speaker 1 05:26

were there microservices also used by, like, the mobile app, or was it just like the web portal,

Speaker 2 05:33

like for this, like it was like, so we are, we are, like, just a back end team, and before our layer, we have like orchestration layer that, like, takes care of all these platforms. So like, like, we serve like, all, like different devices. But like, before us, like, there is an orchestrator and that takes in the request from like, all of the devices. So yeah,

Speaker 1 06:01

okay, so it's an orchestration layer you were abstracted from, from necessarily which which device or which application was accessing the data. The orchestration layer kind of figured out where to go to get what data it needed. And then we'll call your services to grab the data that it needed and sound like your your services were pulling back, like the client, information about their their policies and things like that. Is that,

Speaker 2 06:25

right? Yes, exactly. Yeah. Awesome, yeah. Sorry. Keep going, yes. So like, we, like, you know, we just, like, work, like, regardless, like, off the device. So like, we just, you know, look at certain like, like, so there is like an orchestrator based on, like, Spring Cloud gateway, you know, that kind of, you know, acts like a reverse, like, a reverse proxy on, like, what kind of like servicing needs to, like, call into and like, whenever we want to, like, kind of Like, expose our endpoints, like, we have to, like, onboard into an orchestrator, and then, you know, I like our microservices, just look at the like the headers and like the correlation, like, like there is like prefixes and so on to determine, like, what kind of like device request it came from. But like, like, regardless of where it came from, like, we serve them as long as the request is valid.

Speaker 1 07:27

So, okay, okay, great, and tell me a little bit about you saying the request is valid. So tell me a little bit about how you handle security,

Speaker 2 07:38

all right. So no talking about how, you know, we handle a security, so basically, so all of our requests are like, back to the spring, spring security, right? So we have, like, a JWT token present in the authorization header, and we like, validate, you know that using the spring security, right? So, like, in your basic JWT token, you have like, the three parts, like, which is your the header claims and the signature. And like we do the signature validation and and then you know, we know to look at the headers to see like, like, if your signature is like, valid or not, so if it is, like, tampered with or not, and like, and they they're within that payroll, we look for the claims or the rules for the user and like, based on that role, like certain endpoints May, like need to be exposed or not, right? So like, that is one phase of the security. But like, before us, there was a, like, a device authenticity check with just like, checking the metadata of the device, like, to kind of see if the Geico app, like version that you're working with is like, kind of valid or not. So like the if the device OS like version is, you know, Jade broken are like, not and like, all like kinds of you know, stuff. So, like, it's done by like, other kind of token, like, which is called, like, authenticity token, but, yeah, apart from that, like, there is some, like, input validations and like so on. So

Speaker 1 09:15

not really thinking about, I mean, to me, that's that it's not really security, but less about like what operating system or what device. But how did you make sure that the request that you were servicing was allowed to be requested by who ever was requesting it? Like, for instance, how did you make sure that I didn't somehow, you know, grab postman or something and try to get your information out of Geico, you know, by changing query parameters or calling, you know, a service with different parameters on it.

Speaker 2 09:54

So, yeah, so, basically, you know, like, like, we just like, you know, like, validated, you know, authorized and header, right? So the author is in the like, in the error, content contains your like claims and so on. And so that's like, what we validated. So the basic like, when you log into your like the Geiger portal, right? So it relates to your like identity provider. And this identity provider is where you, you know, get the information of your like, username and password and like, it validates that, and based on that, it just like index that like into the authorization header, like into each request that you're sending it to. So, like, we just validate that authorization header.

Speaker 1 10:46

Okay, okay, and did you filter it to make sure that any information being sent back was only for that particular user?

Speaker 2 10:55

Like, for the filtering, yeah. Like, I mean, like, all these would, you know, be based on, like, on, like, some certain, like, kind of identifier for the user. So, yeah, like we would do so we would not, like, we do not, like, expose, get all endpoints for, like, just a customer, like a role, and like, it would be like, for, you know, or representative, or, like, some other role that would be like, able to view the progress is like information. So I think it needed, like, it needed to have like, a admin, like access to get all the covers. But like, also, yeah, like customers would only be able to like, view, like the progress information, or like by ID, or like benefits and so on.

Speaker 1 11:42

Okay, so we talked, we talked a bit about microservices, which is great. I've got a good understanding of kind of your background there. And Thanks for, thanks for going into that level of detail. Did you do any front end work as well? I assume you're as me. You've got some angular CSS Ajax, yeah,

Unknown Speaker 12:03

yeah. So I did the angular work.

Speaker 1 12:07

Credit for that same app on the for the portal, for the portal, yeah. So tell me a little bit about about what you worked on on the front end as well,

Speaker 2 12:19

sure. So basically, you know, like, so one of the, like, the recent ones that I worked on was, like, creating, like, like an Excel sheet service, right? So, basically, you know, we have a lot of table that, you know, we can deal with in a portal. And we, like, like, basic task is to, like, basically create a service that would like, take in the table data and, like, Export Excel sheet. So we just, you know, I

Speaker 1 12:48

think what Tell me a little bit more about what that data is like, what is, what is that sheet having it, what kind of type of data and how is it used?

Speaker 2 12:56

So, so this is an, like, like, an abstraction, right? So this is not like, a set table, but like, so like, this service is supposed to, like, take in any like kind of table present in the front end, and, like, convert it into an excel sheet if needed. So, like, that's one shared service that I worked on. But So

Speaker 1 13:20

yeah, I wanted to shift to talk about the front end, run end work,

Unknown Speaker 13:25

as opposed to the services. So

Speaker 2 13:27

no, like, like this is, you know. So, so, yeah, like this is a front end Excel service. So, like it is, you know. So, like, so when you like, whenever you log into your portals, you can see like, a list of like benefits and a list of covers that you have, right? So like, you can, like, view it in like table or format. And I gotcha,

Speaker 1 13:54

when you're saying service, would it? Would it be fair to call it like a front end component, as opposed to a service where you've got this kind of table component that can just take whatever, whatever data is coming back from the back end and show it as a table versus a list, versus something like that.

Speaker 2 14:10

I mean, like, the component is like, specific to a use case, but you know, like, like, this is a shared service that like different component use, right? So, okay,

Speaker 1 14:24

but, but it's still a service, right? It's not, it's not, I guess what language was it written in? Was it Angular?

Speaker 2 14:30

So this was, like, written in, yeah, it was written in Angular

Speaker 1 14:35

service. Okay? So this was kind of a shared service that would do what, convert the data into a HTML format that that was usable by the,

Unknown Speaker 14:50

I guess, just display. So,

Speaker 2 14:52

like, basically, no so like you do, like taking the data that is displayed on your like, an Angular component, right? And then, like you do, like, leverage that Excel service and then convert it into like the Excel sheet. So basically, if I were to, like, give an example of like, one for like, the concrete use cases, so we had a list of like coverage like that a user had. And, you know, like on the table which covers, like the ID and like what the coverage was for the expiration and so on, like in the table format. And then, you know, on the component of that coverage, you know, you would basically be injecting like those Excel service and defining the metadata, which is the header and so on, and and then, you know, the data bound to that component would be like putting as like a data for an Excel sheet. And then, you know, whenever the user clicks on that like export button, like it will go in into the like that Excel service and convert that whole data into like Excel sheet, and so

Speaker 1 16:01

it's like a shared library where, where you could, you could just put a button, basically on any table, and they could click Export, and it would, it would, assuming you did the data mapping correctly, it would give them an Excel sheet with all that data I got you. Okay, very cool. Did you work on any of the, any of the front end, like, any of the pages or or things, or was your front end work mostly like libraries, and, you know, helper services, or shared, shared services, if you want.

Speaker 2 16:31

Also, like, no like, so, you know, like I did, you know, I have worked with like components as well. So, I guess, you know, one of the recent one that I have worked on was, you know, like a, like a component for, like, a self service portal, you know, where the users, like, would, basically, you know, provide, like, the information about, you know, like, sorry, sort of the services like they want to log into. And, like, we just leverage the Angular, like, the reactive form for that data, and then we, you know, we ingest that data and also, like, create a restful call to the back end. So, yeah, like, I worked with the reactive forms with Angular to kind of like, serve, like, the self service portal within, like the within the Geiger portal.

Speaker 1 17:23

Okay, okay, cool. So I'm going to switch, switch gears a little bit here. Tell me what you think is

Unknown Speaker 17:32

difficult about working in an agile environment.

Speaker 2 17:37

So, yeah, so I would say, you know, so to talk about. So the difficulties that, basically, you know, I would face in in kind of, like the azale environment, I would think, like, since Azad, you know, it wants you to, you know, properly, like, kind of like, like, adult changes, sometimes, like, those frequent changes can, like, become, like, pretty difficult. And like, you know, it may be a little bit, you know, hard, like, a hard to plan ahead, and like, like a long term manner, and like meet deadlines at the same time. And like, it like, makes you think on your like feed, but still that frequent changes in the requirements sometimes like it makes it a little bit difficult. So I think that's what one of the major challenges is like while working with azaleah,

Unknown Speaker 18:36

how about, how about,

Speaker 1 18:41

you know, getting your merch required. Requests approved or or figuring out how to, I don't know if there's merge complex, you know, incorporate into the right branch, and keeping track what branch you should be working on, and getting them into, you know, a shared dev or staged environment with with everybody else do. What are some of the difficulties you run into doing that?

Speaker 2 19:04

So, yeah, like, so you know, talking about so this, you know, so, I guess you know, like, if you, like, do not have, like, a, like, a clear, like, set of branching strategies in place. It becomes, like, difficult and like, like, any conflicts, like, with some other developers, like, working for me at the Geico and, like, we had like, preset of branching strategy, you know, that we work with. So we always, you know, ran some, some of the same, like, developed branch and then work on our like, stories. And whenever, you know, some developers, you know, complete their work, we deployed on dev and we handle the validation quickly so that you don't, you know, like, the dev environment is always open for any developers to work on, right? So, like, did

Speaker 1 19:57

the individual developers deploy to the dev environment, or did you as kind of like the primary or the lead,

Unknown Speaker 20:04

were you responsible for deploying into the dev environment

Speaker 2 20:07

for the individual developers? And would

Speaker 1 20:11

you ever have somebody that deployed the wrong code or hadn't merged with the latest branch or it, you know, accidentally, you know, missed some changes somebody else may have made and, and, you know, broke some functionality.

Speaker 2 20:24

Yeah, yes, yeah. And that has happened like whenever, you know, like developers forget to like, like the pool in the pool into our fetch and get the latest code into their like the local so that happens at times, but, but, you know, like whenever that happens like, we usually start to see, like, some error. And then, you know, once we communicate across the team that, like, something is missing, then we usually like, the first thing is, like, we go in and check what was deployed recently, like, on, like, those environments. And then we try to, like, figure out, like, if we can so it was a newly introduced, like a defect or like, if, like something was missed on like that branch. So we can easily do like a Git compared to see, like what changes went in. So, yeah, like that happens, but like, I think, like, once you communicate like properly within the team, like it can be easily solved as well. Okay,

Speaker 1 21:23

and how about, how about keeping track of all the administrative stuff we're using rally or JIRA

Unknown Speaker 21:31

for like we are using like zero.

Speaker 1 21:36

Is that 00? Yeah. Sorry. Is that like a customized version of JIRA or

Speaker 2 21:41

so. Like, it is like, no, it is like, zero. Like, the same thing,

Speaker 1 21:49

oh, JIRA. It is JIRA. I gotcha. Sorry, I got you. So how about the administrative pieces of keeping, keeping JIRA up to date, like, like making sure that stories are in the right status, make sure the milestones are, are related to the right story. Or the epics, I guess, in the case, do we use rally so I think of milestones, but the epics are, are attached to the right stories. Or the stories are attached to the right epics. Or people are putting in actuals, you know, the actuals that they worked and estimation and all. How about, how about that part of the Agile process?

Speaker 2 22:23

So, yeah, no, yeah. So talking about that, you know? So basically, you know, the story is still the story estimation, and everything is done by the developers. So like we are doing, like doing the planning and like the grouping session, like the epic management is done by our like, the product manager and and, you know, like, whenever we have the planning session, like, we basically make sure we have a proper stories in place with proper estimation and acceptance criteria. And during that session, like, we also, like divide soft task to certain stories. So like, it is easier to track like for like, like for like burn down. So like on the charts that like Scrum teams usually use. And like as a developer, you know, whenever we work on certain pieces, we just log in like, how many hours like we work on certain aspect, maybe you know the story or the sub task, and like, based on that, we move forward. So, like, it is a combined effort. And from that, from like, the product manager along with the developers on the team,

Unknown Speaker 23:38

okay, okay, cool.

Speaker 1 23:42

Thanks for not um, I think I've got a good understanding of kind of your background and capabilities. And all I appreciate time. Do you have any questions for me, or is there anything I can answer about to replace the team opportunity?

Speaker 2 23:56

Yes. So like, thank you so much for the opportunity today. And like, it was really nice talking to you. So the first thing I would like to know is, like, what do you think is the like, the most important, you know, skill that a developer, you know, has to have for this like, particular role?

Speaker 1 24:13

Yeah, good, good question. And thanks for asking so. So I'm, I'm an old school developer. I've been, I've been developing since I was in fourth grade, fifth grade, after school, like, on Apple 2e when I cut my teeth, and what I found in my time both, both, you know, being an engineer, lead engineer, but also, you know, as a manager of the team, it's problem solving, like, like, I'll be honest, like, like, a lot of folks are focused on, what, what specific skill, what specific language, what specific thing that the what specific new technology, what specific paradigm you can talk about, which is all, which is all well and good and helps, I think, make things go a little bit smoother and a little bit faster at the beginning, because you're speaking Some of the same language, but, but problem solving, like just, just being able to problem solve. Some of the folks that haven't been successful on a team just haven't been able to had, weren't able to problem solve. I mean, and I say that in a very general sense, like they didn't, you know, they didn't know what to do when they lost power. They didn't know what to do when, when, you know their internet went out and how to get connected, even though they were a couple miles from the office right, just come into the office and work like, it's not, it's not a big deal. Go to Starbucks, you know, if you need to, you know, all of that's totally fine. They just weren't able to problem solve. And it carried over into, here's a user story for you to solve. And just couldn't figure out, like, even where to get started and do that. I'd much rather, you know some, I mean, you know, you're a developer, so, like, you figure it out, and it seems like you've got, you've got that background, right? If you're completely unsure about code, you bring it down locally, you get it to run. Maybe that takes a couple of days to get set up and running, and then you start making changes. You start putting in output lines, you start, you know, putting in break points and understanding what the code does, and you figure it out, and that's what I mean by by problem solving, is that that you just understand how to think logically, and that you know nothing going on in the computer is magic, like it's all being done because of the code. Whether you're fixing a bug or adding a new feature functionality, just just coming up with a methodology of going through and doing those things is, to me, the most important skill.

Speaker 2 26:26

Thank you so much. Thank you for the answer. Yeah, I think, like, that's one of the most important skills, like the developer needs to have, is that, like, eagerness to learn and then solve any kinds of you know, issues that you come across. And I think, like, that's one of the things you know that I enjoy about this career as well. So I guess, like, the next question like that I would like to know is, like, what are the next steps for the hiring process?

Speaker 1 26:54

Yeah, good. Another good question. So, and I was going to cover that anyway, so the next step is to talk to the lead for the team that we're looking to fill this position on. She's technically on vacation for the next three weeks, but she's working a couple of days each week, so I think she's in next Wednesday and the following Thursday, whatever the 17th is. Think it's the following Thursday now the following Tuesday, so I got to coordinate her schedule with some of the folks here, so be talking to

Unknown Speaker 27:29

either two of the leads or a lead, and

Speaker 1 27:33

one of the senior folks on the team, they'll probably do a coding exam with you to get a little bit deeper. I know my questions went a little bit deep, but, but they will go deeper on the technical side of things as well, and probably about an hour. We may, we may throw an extra person in there as well to get there, to get their thoughts to like one of the our lead Bas, or something like that, sure. So basically, it'll be another hour long interview with with two to three people. Okay, so

Speaker 2 28:03

like, I wanted to ask you, like, Would it be, like, like, a virtual interview, or, like, like, Do you know something about it, you know? Yeah, that that's,

Speaker 1 28:13

that's a good question, too. Um, so trying to coordinate. So I'd really like to bring the folks that were we're making it through this interview into the next interview on site to do an interview, and now that you mention it, next Wednesday, likely won't be a good day to come one site. So I don't I still want to get through, like, the interview process. I don't even know if we can coordinate dates, so it might have to be the 17th. But to move and coordinate dates, I'd like them to try to make the 11th work for that, and then maybe, maybe bring you in just for an in person, quick, like, 30 minute conversation with a couple of the folks on the team in person. I

Speaker 2 28:53

say that because, like, I was in my stand up today, and my, like, my manager, he like, set me up into like production, like support, so it might be bit difficult for me to, like, come on, like, all on site, like next 23rd so, like, it would be difficult for me. So

Speaker 1 29:12

we'll figure that out. If we can schedule the interview for next week, it will. The main interview will definitely be virtual, like, like that. That hour long interview with the two folks will definitely be virtual, because one, the one lead I need to talk to you, Will, Will is in India, so she's not coming into the office, but she'll be virtual. But she's she's just there on vacation. She's normally in the office just the end of the year. Everybody's traveling, yeah, so sorry. So my, my thought on that is, we'll, we'll figure things out from there and see how everyone everyone, how comfortable everyone is. We've

Unknown Speaker 29:53

had a couple of folks who, um, we

Speaker 1 29:58

had problems getting into the office at all, so we'll, we'll figure that with you. You are local right I mean, it looks like you're.

Speaker 2 30:05

Yes, I'm, I live in like Roma right now. Okay,

Speaker 1 30:09

just a little bit of a drive but it's not it's not too bad. We're normally two days. In three days work from home. So, so if you get the.

Ronaj Pradhan-Infosys-Final Round-SRV

Speaker 1 00:00

Right now we are doing two days at office. As you see, our environment is around the office and things like that. So that's pretty much about the project. As such, we will learn more as soon as we go can you give us a brief about yourself? How much? Yeah,

Speaker 2 00:16

yes, absolutely. So hi, good morning. So everyone, my name is before

Speaker 1 00:21

you go, and I'm sorry I forgot to introduce he works. He's the lead developer on this project. He's been with this telecom domain for quite some time, well experienced from that time. Yeah, anything you want to be sad? Yeah. It's pretty nice to meet you. Yeah, go ahead. Thank you. Yeah,

Speaker 2 00:46

sure. So, you know, my name is Pradhan, and I've been working as a, you know, Java developer for about six years. And yeah, you know, I work primarily on like, Java based web application, kind of like utilizing like frameworks such as, you know, string hibernate, and like rest for which services for API development. And we primarily working on, you know, a micro service recently. And we've been migrating, you know, monolithic to micro service and bringing them down, you know, designing these micro services and getting, you know, and getting functional and non functional requirements and approvals, implementing them and then deploying them for the company. And I've also been, you know, primarily working with the Jenkins to AWS through, like the cloud formation templates. And some of the services on AWS that I work on, like the ECS, you know, SQS, SNS and lambdas, and on the front end side, you know, I have worked with in Angular, and can work with JavaScript and type scripts as well. And currently I'm like a scrum team. And the team consists of, you know, five developers, and we have, you know, like the product owners and Scrum Masters, and on a daily basis, you know, I'm involved mostly on working on the back end Java code, helping, like the team with the code reviews, and also with, you know, products and support duties.

Speaker 1 02:15

Okay, thanks. So you overall have a five years of software development experience is what I can see from your museum, right? And you have worked on what is your front end? Which you have used on your Angular, or you have used React, yeah,

Unknown Speaker 02:33

primarily I've used

Speaker 1 02:36

Angular. Okay, that's nice. And on the process area. What are the methodologies have you used in software development?

Speaker 2 02:47

Laws? Is So? So, yeah, you know, I've been primarily following like, like, like, a methodologies for this and but, you know, I have had project which follows in a kanban as well?

Speaker 1 03:02

Okay, you are not worked on a safe agile mode.

Speaker 2 03:05

Safe agile mode like no, it's

Speaker 1 03:09

a scale agile framework. We use it primarily in launch complex water projects. Actually, that it's involved the 22 Mike systems has to go and open the same time to use scale agile framework. That's where safe agile view. And just for a brief, can you explain, I think your last project was in Geico, is what I see, right? Yeah. So how do they implement this safe i or agile process? Can you elicit, you know, from from your requirements, to allow you put it back into production. Can you elicit? Can? Can you give us some information on that process, not the technical aspect, but at least, I want to know the process of it.

Speaker 2 03:59

Okay, yes. So, you know, basically, you know, we have a, like, a planning iteration, alright, so basically, there isn't like iteration, you know, planning meeting, whereas there's like no stakeholders, like input, and there's also, like, a lot of requirement gathering in this spot. And, like, it's like a week long event where, you know, teams get together, and then they try to come up with, like, you know, what needs to be done, and all the requirement, what all the requirements are, and they are, all, you know, documented in a very high level. And then basically there is, like, a team structured meeting, which is kind of like more refined and detailed. And like these meetings are like, either like the sprint planning or like backlog refinement, right? So basically, in the sprint planning, it is during at the beginning of the sprint, which happens every, like two weeks, like where we as a team, like plan and work and try to figure out, like, which product backlog we need to work and what can be completed. And we just, you know, estimate the efforts and discuss, discuss what the dependencies are and what the blockers are. And then there's also a backlog refinement meeting where we, you know, prioritize the requirements in the backlog, right? So basically, we discuss the items and also estimate the complexity, and then break larger items into smaller ones. And once that is done within one single sprint, we have like different you know, daily stand ups, where it's developer, report what they work on the previous day and what they would work on and the current day. And there is, you know, any, if there is in blockers or dependencies in the team will be working on, you know, like the features, and once the feature development is done, like the next days, is like testing through like automation, and also through like, testers. So like, basically, we have a couple of, you know, testers in our team who do their manual testing, along with, like, running a full set of regression. And there is, like, if there is any issues during the testing cycle, we also fix them. And once, you know, we have all them, you know, we follow pretty, you know, standard release process. Our code is on like GitHub, so we follow, you know, branding strategies, you know, using like a feature develop and when brands and so, you know, we got to release like candidates whenever we are ready for releases. And we release that branch, right So, and then, you know, there is a sprint review, and also retrospective. A sprint review is at the end of the sprint, and where we review, like, what happened. And then retro, retrospective is just, you know, self reflection, and, yeah, like, those are, like, the kinds of an overall process of the other methodologies

Speaker 1 07:02

Okay, very nice. I mean, that was a complete process. Can you give a sample on what has happened on a retrospective and what was the outcome of that? I know that retrospect is a summary which we generally look back at the Can you give an example of that happened and why it was helpful.

Speaker 2 07:24

Yeah, so, yeah, like, basically, you know, retrospective, we just do, like, like, a, like, a whiteboarding where so, like, it's a meeting of what we what went well, and, like, what could have we done? And, like, what other like, excellent ideas, like items like we take so basically, like, we just reflect on that. And I first like, thank one of my teammates on helping, the helping, on the work that I delegated to him because of my workload, and we were able to complete the tax in hand so that, you know, like, that was what went well section. So, but then what would have like done, basically, you know, like we said, like, like, we need to have a little, like, weekly check ins with the developers for, like, a different, like, complex stories. So, yes, sometimes there are stories that don't always have a clear set of guidelines, and there might be dependencies, and, like I wrote up at a point where we might need, like, a need a weekly check in. So, yeah, like, basically, you know, we have, like, we have to get started of, like, every meeting at Tuesdays at 9am to discuss, like, any kinds of dependencies or other complex techniques. So like, what's happens in a retrospective, and so it's just like a self reflection on how the sprint went and what could have been done better.

Speaker 1 08:55

Okay, what is your story estimation technique? How do you estimate for your story? Specifically, what is the process or methodology you go by for your story estimation, and how accurate was it

Speaker 2 09:14

so? So, yeah, for, you know, basically for stories. You know, story estimation is based on like, like Fibonacci series. And basically, whenever I do like estimation, I do kind of like, you know, relative sizing, right? Basically, you know, we have a collaborative estimation where, you know, lot of developers do estimate together, and then we, like, agree on something. And whenever you know, I am estimating a story, I take you know different several factors into account, like the complexity, the effort and unknowns. And right? So if it is a complex chain like that, might have like, kind of like a very small change needed. I would, you know, estimate a little higher compared to something that is smaller, right? So I, you know, take these three things into consideration when I am kind of like estimating and like I would like, typically, we have been doing 13 story points at as one sprint, World of

Speaker 1 10:22

Work. So for a story, which is as a front end impact, a back end impact, and of course, it interacts with couple of multiple applications, what would be your story point estimation for that? I'm just putting you a random thought. How do you

Speaker 2 10:45

so, yeah, basically, you know, for, for the, whenever we know, like, I have, like, a kind of story that kind of affects, you know, all the components, like, like, the front end, back end, and another integration. So, like, the first thing that would be to, like, the split, you know, each story into different soft tax, right? And the soft tax can be, like, broken down, you know, based on, you know, different types of, you know, layers and which. And this helps to, you know, ensure that every aspect is accounted for. And so we can do like a front end, you know, soft task, like backing, no soft tax API. And then also, also do the database, you know, interaction, soft tax. And we can estimate, like each of these, you know, soft type stocks, like separately. And we can, like, aid with the estimate, making it, you know, easier to see, like, how much effort each component would require. So that would be the way to go,

Speaker 1 11:50

Yeah, but in terms of your Fibonacci points, how much points would you given a ballpark

Speaker 2 12:00

number. So, yeah, like, it would like, I would say, like, it was somewhere around eight, but like, depending on the like experience, if something

Speaker 1 12:09

fine, thanks. I got it. So, one more question I had. So, which company are you? Are you with Geico directly? Are you working for another company? So,

Speaker 2 12:19

yeah, for the Geico one, you know, I've been working, you know, as a contract,

Unknown Speaker 12:26

contract. So basically, you are with another company.

Unknown Speaker 12:30

What's the name of your company? So,

Speaker 2 12:32

yeah, the name of the company is like the do technology. Thank you.

Speaker 3 12:41

Thank you. Thank you for your explanation and everything. So your current project is in Geico, right? Can you please elaborate a little bit about the architecture of your project that you're working on? So what are the components that you involve. Just explain, like, what exactly the business perspective Geico, and then explain, like, how you're connecting different systems, like UI to the backend. Can you please do that?

Speaker 2 13:10

Sure, sure. So, yeah, like, basically, you know, we have a microservice based architecture, and we have different service like in Spring Boot and that are like, kind of deployed on AWS cloud, right? So basically, what our team is responsible to do is we take in data from, you know, multiple clients, like, you know, multiple states worth of data which are like, non standard flat file, and they are like, like, comma separated data, and these data, like needs to be, you know, convert into, like, a standardized format. So our team, you know, basically provides a transformation as a service, and we leverage, you know, something called an Apache Spark to spin up different clusters to load this data inefficiently, and then we process data into like database and for more streamlined processing of our application to expose. So basically, we expose these, you know, claims and COVID related information to like these applications. So basically our application that we expose in a Spring Boot based microservice. Like, we have no services for claims related data, we have services for corpus related data, and then we also have no service for like, memory related data. So we have like, three different, you know, microservices, like primarily, but we also like have the ones that interact directly with the UI for processing, and we have these like deployed on AWS. So we utilize services like AWS, like ECS, AWS, lambda. Our database is rds with Postgres SQL, and we use, you know, AWS storage for storing files, so yeah, like, that's a little bit, you know, like, overall architecture of our application.

Speaker 3 15:11

Thank you. So you mentioned that you will be handling, like, like you have transactions from different states, right for Geico, like, right? There will be, like, many states of data, right? Like, for example, like New Jersey, Florida, you take any, any state, right? So there are 50 states in us, so you will be handling the data right. So are using any services that, like, kind of, like architecture, where you handle, like, these millions of transactions, or maybe 1000s of records. What kind of like framework you're using to handle,

Speaker 2 15:51

yeah, so like, like we are using basically like, you know, we handle the data from, you know, a subset of states, and only, like these states, you know, having, like, a non standardized, like format, and they're like, you know, like eight states, right? And there is a, you know, kind of like a framework from Apache called Apache Spark. And it is basically, you know, a big data framework similar to Hadoop, but for Java, and this basically, is, like, advantageous, you know, you know, sense that it does not, you know, it does. Are like, kind of, like resilience, like driven development, where, like, you can, like, basically, like, spin up a cluster and nodes to process the data. Like, parallel, okay, Oh got it, sure.

Speaker 3 16:40

So you worked on Spring Boot, right? Spring Boot applications, right? So your application to Spring Boot,

Unknown Speaker 16:49

have you used any what is the ORM

Speaker 3 16:52

layer that you're using, like hibernate, JP or hibernate, or you're using JDB template

Speaker 2 16:58

I work with, like hibernate and spring data, like JBA, JPA, can

Unknown Speaker 17:05

you please name a few annotations that you used in JPA?

Speaker 2 17:09

Yeah, sure, absolutely so for this, you know, basically some of the annotations that, you know, I worked with, like a spring data, JPA, include, you know, like the Entity annotation. The Entity annotation is just like a definition of like database object, as you know, Java object, right? So basically, like your table, like you can mark as like a JP entity and like, like this object, like, kind of like represents, there is like a field level annotation, like you know ID for like the primary key and column for you know column. And there is no transient annotation that indicates feel should not like persisted into the you know database. And they are all like a like, just traditional based on the annotations like one to many, or like, many to many as well. Okay,

Unknown Speaker 18:06

so, have you used Kafka in your,

Unknown Speaker 18:09

in your, in your project Kafka?

Speaker 2 18:15

Yes. So, you know, you know, I have, like, not worked extensively with Kafka, you know, I just like, know how it works. Okay, that's okay, got it. So do you know,

Speaker 3 18:30

have you used Java eight in your current project, or any other version of Java that you're working on?

Speaker 2 18:34

So yeah, for no projection I have been using, you know, Java 811, and also 1717,

Unknown Speaker 18:45

so can you

Speaker 3 18:47

maybe elaborate a little bit like, what are static methods and interfaces in Java? Like static methods in interfaces, right? Oh, what are like static methods interfaces? Can you explain what?

Speaker 2 19:01

Yeah, like, like, basically, you know, since you know Java eight, you know you have the ability to include, like, the static methods. So again, you know set like, the static methods are something that belongs to a class, right? So, rather than the object itself, like, what static method in interface, you know they belong to interface directly and rather than the class that it implements, right? So basically, like those static methods do not need to have any implementation, because you cannot inherit those static methods and and they must also, like call them method using the interface name directly. So in order to access, you know, these like static methods, you can, like, you just call the method name through the interface.

Unknown Speaker 19:51

Okay, so

Speaker 3 19:55

got it. So when it comes to like hashing, right? So you know difference between, like, hash map and hash set,

Speaker 2 20:04

can you tell? Yeah. So basically, you know, hash map, you know, each store is eating, like a key value pair, where, whereas, you know, asset is, like just a unique, you know, on ordered collection of offset.

Speaker 3 20:20

Okay, so, have you heard of concurrent hashing, like congruent HashMap?

Speaker 2 20:27

So, concurrent, yeah. So, so, yeah. Basically, you know, your regular unit hash map is not, you know, thread safe, which means that you know, if you work it with a hash map in a multi thread environment, you need to provide, you know, some explicit you know synchronization. And like, you have to do your own synchronization, floating right. But like, if you have your concurrent hash map, which takes you know, care of this internally, making it like a thread set. So if you would want to, like, work on a multi threaded environment, where you want to make sure that multiple threads work with your map properly. So you want to use concurrent hash map,

Speaker 3 21:11

okay, again, when it comes to hash map, right? So let's assume that the hash map has same hash codes. So what's going to happen? How are you going to handle that?

Speaker 2 21:26

So, yeah, like, basically, like, that's called, you know, collision, right? So basically, that's what like collision is. And the hash map has a COVID zone handling, you know, because, as map internally, uses, you know, area of buckets to store entities, right? And like, each bucket can have all, like one or more entities. And when you know, like, like to like, when two keys has has like the same bucket index, like, they collide and they are, like, stored in the same bucket. And so basically, hash map, you know, effectively manage this, like collision, by using a linked list, you know, internally, and depending upon the, you know, like the number of entries in the box, and it basically, you know, stores it in the same bucket as a link.

Speaker 3 22:22

Yeah, have you know anything about Big O notation? What exactly is big O notation?

Speaker 2 22:27

Yes, so, yeah. So basically, you know, big O notation gives you the, you know, the space and time complexity of an application like it just, you know, represents that, like you it is used to describe the complexity of the algorithm, right? So basically, if you have, like, a nested for loop that looks each time for any element, you have a big O notation of n squared, which means that you know the time complexity of this application is n squared. So like, if you increase the input, like, by so many amount, the processing time will increase by the square of that amount, right? So that's what it is. Okay, got it. So when

Speaker 3 23:15

it comes to the design patterns, right? So what exactly are the design patterns in Java, and why we, why we need to use design patterns when we, why we need to consider them

Speaker 2 23:28

patterns. So, so, yeah, like, basically, you know, design patterns, you know they are, you know, like your career, small structure and like, the behavior, you know, design pattern that is used for like, different types of like, specific purposes, and they typically have, like, a standardized solution for our, you know, recording problems, right? So if you know you have, you know us, like, if you have something similar that you want to achieve, like you might want to go with a standardized approach to work with certain kinds of like objects, right? So let's say you know, for an instance, you are like wanting to get a same instance of an sttp client and in order to make a downstream call, right? So, instead of, like, having some kind of helper functionality that, you know, tries to give you the same object back when you ask for that object, you can, you know, use what is known as, like a singleton pattern, where you define an object and where you define a, you know, class that has a no private constructor, and that has like a static method, and that also has a, like a global point of access that returns, you know, you this, you know, stdp client, right? So that's what it is. And it's just a way of, you know, standardized approach for kind of, like, resolving certain problems, right? So, Singleton is one example that is widely used, but there are other patterns, like factory method for object creation, and there's also, you know, like a decorator pattern for adding like new functionalities, and these like a saga pattern for, like, orchestrating, you know, transaction on Angular services.

Speaker 3 25:21

All right, okay, that's good. Ronald, so as you said, like decorator, right? It's on the Java side. So let's right. Let's talk about the class decorators right in Angular. So you work in Angular, right? Rona, you have experience working with Angular, right? Yes. So what exactly like class decorators, and where did you use the class decorators in Angular,

Speaker 2 25:43

yeah. So basically, you know, for your, like, a class decorator is like, I would say, you know, it's basically applies to a class definition, right? So on decorators, they are like your, you know, it allows you to attach metadata to the classes, and it adds, you know, properties and methods. So basically, like, it just informs, like the angular about how to how the process the class, right? So you can use the like a component decorator, and it is the most widely used one, and it marks this as a like an Angular component. And then you can pass like in other like items, like that, selector, template and styles that you want, right? And there's also an insectable, you know, decorator, which you know, like marks this as a service, and that can be kind of like injected into classes. So that's what, you know, decorator is, what is

Unknown Speaker 26:43

the current version of Angular you're using?

Speaker 2 26:47

Yes, so currently, we have been using, you know, like, migrated into, like, the Angular version 15.

Speaker 3 26:56

Okay, so let's take a scenario in Angular. So basically, let's assume that a user trying to log in onto a screen from development, from developer perspective, how we can invalidate that login and redirect them onto the again, onto the login screen? You got it, right. So how would you do that? Like?

Speaker 2 27:21

So, yeah, like, like, basically, if you want to, like, like, invalidate the login, like, the session of the user, and then like, redirect back to the login screen.

Unknown Speaker 27:37

Like, basically, you know,

Speaker 2 27:40

basically what we can do, I guess, like, we need to check if like, the logging is valid or not, right. Or like, do you want to, like, just invalidate it regardless.

Unknown Speaker 27:54

Have you like, interceptors? Maybe

Speaker 3 27:59

interceptors in Angular So,

Speaker 2 28:03

so yeah, like I have worked really like to guard, to like the protected routes and restrict like different accesses, and if the like a user is not like, if like the users are not authenticated.

Speaker 3 28:20

So you said you just wrote Kafka, but you have knowledge of Kafka, right? Can you tell, like, what exactly the architecture of Kafka, like brief, not in not in detail, because you've not worked on it, but just like, what exactly the Kafka contents in

Speaker 2 28:42

it? So, yeah, like talking about, you know, like Kafka, basically, you know, Kafka is like a publisher, subscriber, right? So you have like a basic consumer that puts the like the message into a Kafka topic. And then you have like the like the consumer that gets the message from these, you know, like the Kafka topics, right? And basically, you know, you have a, kind of like a Kafka broker, which is basically, you know, your Kafka server, and like, you have your like Kafka topic, right? So, if you want to like, like, compare topic to something that, you know, we usually, usually use, it's kind of like a database table, right? So, like, it is also like a category, or like a feed name. And then, you know, I talk about, you know, producer and consumer, which is, are the main components that is, like, responsible for, like sending the message, and also, like the processing the messages, and Kafka, I know it runs on, like, on all of the items in Kafka, they are managed by an external server called like a zookeeper. So there is like an external server for managing the brokers, called Zookeeper. And each Kafka topic will have no partition. And this is where, you know, you have, like, have your order records, and like, the records which is processed is like, kind of like, handled through like, the integer call, like the offset. And offset can be, you know, managed by your consumer. Okay,

Speaker 3 30:18

let's assume that the current application that you're working on, right? It's consuming a lot of memory, a lot of memory. At times, it's throwing out of memory exception. So, you know, how do you find that? Like, what is, what are the methodologies that you would use just to find it? I'm not just, I'm not asking it. How to fix it. How do you find, actually, the memory leaks for your application?

Speaker 2 30:42

So for, you know, memory for finding the you know memory leaks, it can be challenging. So usually we have to do like, some kind of you know, analysis. So you whenever, like, there is an issue in order, like in one of our instances on, like the AWS EC tools you can, like, log into that, you know, EC easy to and then try to capture, like a heat term. So you can, like, use like a, like a J map and J stack to get like this, you know, heat dump. And once you have, no this heat pump, you can visualize, like a VM or like J profiler to like, actually, you know, profile these applications and and also see, like, what part of the application is like consuming the most memory. And like, once we have this information, you know, we kind of like sense which part of the application is like retaining a lot of these memories, and we can then try and find out, like, what kind of, you know, data structure, or what kind of like caching, or like, what kind of object is like holding, like this memory. And then we can go on, like, fixing that. Okay,

Unknown Speaker 32:02

so that's good. So when you, when you

Speaker 3 32:06

go into design phase, right? So before you start developing your application. So what would you consider, from your perspective as a developer, when you, when you, you know, make things ready to start working on the API, development a new API, from a developer perspective, what would you consider Raj, yeah.

Speaker 2 32:27

So yeah, like, like, if I am to kind of like, like, work on a new story and the things that, like, I have to consider doing these phases, like, basically understand the purpose and like the functionality requirements of the story, right? Basically, you know, functional requirements are like, the requirements like that has been handed out to you like, you know, these are the things that like, they are must have. They must have, right? So let's say, you know, if a client says, you know, I want to get a list of like users, like back then, then that is your functional requirement, right? So it we must have like for like, a completion of the story. So that is, you know, like the like the first take into consideration, but then I am the next consideration is the like non functional requirement, like usually, which usually like results around the performance and the securities, the like reliability, and also like the monitoring aspects. And I try to see like, what kinds of like non functional requirements needs to be around this area. And then, you know, to basically, like, figure out these documentation as well and some, like, other things that I consider during this development is what kind of, you know, API design like I need, and what kind of modeling I might need for the response and, like, what kind of error handling, like, techniques like I need to utilize. And if like, there is an like on like documentation that I need to add for like the development aspect. And then there, when I move into like the testing aspects, you know, I think about like the unit testing, integration testing, and also like the regression testing. So these are like the things that that I take into consideration, you know, when I'm working on any like feature or API development, okay, that's good.

Speaker 3 34:30

So just, can you give an example, like, where you would be implementing collections in your Java application? So,

Speaker 2 34:42

collections, yeah, well, you know, like the collection like, is used a lot. So whenever, you know we are, we like to get any kind of like data from, like the database. And if we are getting like, like a list of objects, we would, you know, use like the collection framework. So for an instance, we had a database for, like, managing like these, like the cover information, right? So basically, whenever we had to, like, get all of the like cover information from our database, like we would, you know, get, like, a list of these, COVID is back and like, like, and this list is one of the collection framework. And so, you know, yeah, like, we are like, we we are also like, one is want, like to on order and like the unique objects. Like, instead of list, you will be like, kind of using a set, right? So, and if there is, like, some requirement of adding, like, a key value, mapping of those objects and like, you'd be using a map collection. So they're like, different collections with different use cases, you know? Okay, yeah,

Speaker 3 36:01

exactly. Ronit, I can you please explain a scenario that what was the most challenging time when you were able to handle your project like and how were you able to meet the deadline or the requirement? Can you give you an example?

Speaker 2 36:19

Yeah, absolutely. So, you know, yeah, talking about, so, basically, you know, I was working on one of the, you know, feature for, like, the streaming engine, right? So we had a, like, a company wide policy called, like, the like, like an enterprise tech backlog, where, you know, all of the teams within the company, like, they have to, like, incorporate some kind of, like change for making an application, like, kind of resilient, right? So for this purpose, you know, like, we all had a requirement where we had to, like, aggregate our application logs and then also, like, push it into like, the database, like snowflake table for this. You know, since, like, we own like multiple microservices, we have to, like, basically, like, add some kind of, like SDK and like configuration into each service. So instead of that, like, we decided to create a like microservice. When implementing this microservice, you know, the integration of these microservice basically, was a little bit challenging, because we had to take into like account of like different types of, you know, metadata creation of each of the object type and like batch process them, and also make sure that application ran at like, a full scale handling, like, multiple micro service calls, right? And the challenge here was, like, we were using library called his for like, like hysterics, for, like, the thread pooling. And basic thing was, like, if there was any error, like, it, like, swallowed those errors, and then later, after, you know, certain amount of time we were getting, like, a denial of service, basically, like, no calls to the service, like, was being went through unless, like, we restarted the service, Right? So on. This was a technical challenge, like having to figure out, like, where the application failed and without the help of any logs, because internally, like, all the logs were locked at, and we are not, like, getting any locks like, and so it was one of the like challenges, challenges that I had to, like, recently solve. And the way I saw these was, you know, basically, you know, to go step by step, like debugging online, like IntelliJ, trying to match the like, exact use in production. And yeah, like, then by the due date of that enterprise take backlog. We were kind of, like, able to fix it, and we, like, did take, like, a pair programming session, but it was a really nice, you know, learning opportunity for me on, like, how that library worked.

Speaker 3 39:16

What is the framework that you were you you're using for genit writing genits for your code.

Speaker 2 39:23

Okay, so, yeah, we are using JUnit and mock user for this mocking. Okay,

Speaker 3 39:30

so from database perspective, right? So how would you optimize a query when you have what you check on the column, right? Let's be specific, from the column perspective on the table. What is, what are a few things that you would check when you want to optimize the retrieval speed of that query?

Speaker 2 39:52

So, so, like, you know, if you want to, kind of, like, optimize like or retrieval speed of a database based on a column, you can basically, like, create an index on that column, right? So basically, like, if there are columns that that is used in Where are, like joints a lot, you can like, index that query and it will help in, like, speeding up, like the data retribution. Okay, thank you.

Speaker 3 40:24

I will hand off this interview into Vikram.

Unknown Speaker 40:27

Thank you so much.

Speaker 1 40:31

Thanks for taking some time. I think we are done for today. So definitely, hello.

Speaker 2 40:42

You hello, hello, hi, I can't hear you.

Speaker 1 41:00

From echo, I muted my speaker and for starting, sorry about that. Thanks so much that you took some time over here for attending this discussion. We certainly get back to you. Meanwhile, do you have any questions for us?

Speaker 2 41:17

Yeah, like, can I know, like, what the further interview process will be on my regards, you know,

Unknown Speaker 41:25

yeah, so the next step will be notified by

Speaker 1 41:29

our HR, or whatever it is, may supposed to be giving the feedback. And based on the feedback, we have to come to a consensus. And selected you will be notified as so is it is generally, very soon, okay, yeah,

Unknown Speaker 41:48

you are at the Maryland right now, yes, currently Maryland,

Speaker 1 41:54

okay, okay, and you are flexible for moving out of Maryland? Yeah, open

Unknown Speaker 42:00

to relocation. So that's no problem. Okay,

Speaker 1 42:03

you can come down to here in Florida.

Speaker 2 42:10

Yeah, I'm open to anywhere. Actually, no problem there.

Speaker 1 42:14

Yeah. So that's one very vital, because this is in facility work we are doing. So that is the game changes. What time so much. First place in time will see our course. Okay, thank you for your time, you know.

Ronaj Pradhan-Publicis- SRV- Second-Offered

Unknown Speaker 00:00

Which are like a fresh college graduate.

Speaker 1 00:05

Okay, all right, sounds good. So let's structure our conversation like this. I'm gonna make up an imaginary use case, something that your imaginary client might want to ask you to build for them, and we'll use that as the kind of springboard to talk about various technical aspects of it. So the use case would be like this. Imagine your client comes to you and says, we have a an existing relational database. Let's say it's Oracle. It has data in it that we want to give our users access to. But currently we don't really have anything, so you need to build something fairly simple. More specifically, the data is, let's say, is orders something like what you put in on Amazon website when you want to buy something. So order for a something, and your client wants the users to be able to create those orders, to modify them as needed, to delete them and to view their say, active orders and maybe the history of orders that they placed. So something not really that complicated. When you hear that, give me top five, maybe or more, questions that you would ask your client right away to clarify how your solution is going to work.

Unknown Speaker 01:51

Let me just think about that. You know,

Speaker 2 01:56

yes, so like, if we are working on like, kind of, like, auto management system, like that should be, like, able to do your basic operations. So the first thing that I would like ask was about, like, the functional requirement, which you pretty much clarify it should be able to create, read and update, delete the orders. But the other question would be, you know, there would be, like, the kind of roles, role based access or permission needed for users to, you know, allow certain permissions to those users. Like, I'm just trying to ask if, like, is there any kind of authentication or authorization, like, of any kind you know, like that, would you know will be required for the requirement? So that would be my first question. And so something else that would, you know, I would ask that would be, you know, like a volume of the orders, right? So, like, what is it? What like is like your, you know, SLA for, like, a DPS, right? So, how many orders are we expecting to kind of create, and how many concurrent users will be kind of like accessing this system. So yeah, and yeah, something else that I would also like to ask, like, is that if the system needs to be like, kind of like, integrated on some kind of, like external services, like, like a payment gateway, or maybe, kind of like a dedicated notification service. So that would be something like some of the questions that I would ask the client, and yeah, I'd also, would also like to like note, at least like, talk about, like the necessity for like the scale scalability and like the reporting and like the need and dashboard features and so on. And so those would be some of my questions that I would ask them right for this particular use case.

Speaker 1 04:22

All right, good questions. Let me pretend that I am the client, and I'll try to answer those questions as I heard them. You mentioned crowd operations. Yes, exactly. That's what they're asking. Not something super complicated. Is there role based access? The answer is no. The user role hierarchies flat, they will be called any authentication state your client says the simple user ID and password to log into the system for user will suffice volume. It is a low, low volume system. We don't expect more than, say, 500 orders a day, number of concurrent users, the total population of users, since your client is about 100 and they all in us, time zones, three of them, there is no geographical aspect to user population integration with externals. Good question. So there is a existing back end system that effectively is, well, you supposed a payment gateway of some sorts. Let's assume it is a payment gateway of sorts. And yes, it does expect notifications from your system upon events that concern the order, such as order created, order attached, update happened, order deleted. The system doesn't want much detail with this. It just wants to be notified when a particular order with order ID XYZ was touched in a particular way, scalability, based on what I already told you before, low volume, 100 users, the client doesn't expect the system to be super scalable, but they will leave it up to you to decide, based on the numbers that you heard, what is going to happen with scalability and what are the best means to achieve that. We'll talk about this as we progress reporting. Plan was to keep it simple so the ability of users to run effectively the query their own orders is as much reporting as the client wants to see at this point in time.

Unknown Speaker 07:13

So with those answers in hand,

Speaker 1 07:18

give me an idea at the high level, how do you start designing the system? What would be the moving parts of your solution? What kind of technologies you would choose for a client? By the way, says we are big Java shop, so we already have bunch of Java Java related technologies approved for use. So let me know how that affects your choices,

Unknown Speaker 07:47

thinking about all of this. So

Unknown Speaker 07:49

let me recollect my thoughts on this.

Speaker 2 07:53

So, yeah, like, so the first thing that is, like, I like your authentication and authorization since, like, like, we have, like, a simple spring security implement our basic, you know, username and password login mechanism. And since, like, since, like, it is a, like, a flat and the role hierarchy authentication, we just like, verify the user credentials, and once the user logins, there will be, just be, like a session token for like, any particular subsequent request. And yeah, we will then be working with Spring Boot to kind of like, expose the RESTful APIs, right? So these risk APIs would be able to perform your basic CRUD operations. So your end point would be like, post orders, get orders, get orders by ID, put orders by ID, and also like, then delete orders by ID. So that would be like the primary entry points, and, yeah, basically it could be a classic spring with a service layer, data access layer for, you know, interacting with the order, like stable, and also doing the business logic as needed. So that would be our kind of like the core application logic, right? And so now, since, yeah, since we want to integrate it to an kind of, like, external API payment gateway, we would probably be, like, using a REST endpoint from the payment gateway, and then, like, work with the payment gateway through the rescue calls. We can, like, just do the HTTP client service that calls this payment gateway, like, securely. And then again, I need notification to write the service. So we probably want to do this through some kind of, you know, messaging queue. We may use Kafka to send topic messages where a notification service will, kind of like, listen to that messages and then also send email notification like, optimate, this kind of like a synchronous because operation should not be blocked if the email service goes down. So yeah, like that would be our core application. We can create, like a front end application, like using Angular to have a, like a kind of dynamic front end that loads this, like this data from, like, the back end database, or like, you know, back end services now to, kind of, like, talk about the scalability, right? So, like we have, like low volume, so it will be very simple our database, like we do not need anything complex because of the volume that we have for the API. And we basically start with a couple of instances, and then, since this would be behind our board balancer, it would like be spread throughout and then we would also put an Auto Scaling group like above it. And so in that case of increasing, like a CPU percentage, we increase the instances by one like further down the road. And if you basically like encounter issue with that scalability, we might also add a caching layer so that it would be like a very high level design on what like this API solution would look like.

Speaker 1 11:52

Okay, sounds good, so let's imagine that you shared the design plans so far with your client, and plans sounds very excited. This sounds good, and then they bring up some concerns. So they heard you say, I'm going to use spring security or some basic quad. You are going to build a data access layer. You also said that 5050, chance for notifications you might go with either REST API or Kafka front end is going to be built using Angular client is all for it, but they say, as of now, you are the only person on the team. We are in the process of hiring people, but they are not joining the project. From day one. Would you be able to utilize all these technologies by itself, or for some of them, you might need to bring in somebody who you think is more an expert than yourself. So Spring Boot, you said you've used it. So Spring Boot symbol would be only you, your hands all with it, data access layer, Oh, tell me what you're going to use.

Unknown Speaker 13:21

Data access layer, sure.

Speaker 2 13:23

So, like, you know, so we would be basically using, you know, Spring Data, you know, JPA, or like, for the data access layer. And like the spring data, JPA, it kind of provides a lot of abstractions to, kind of like support these basic cloud operation, but we also can, like, use native queries, using, you know, query annotation, and if there's any like need of complex queries, right? So, like, I'm comfortable, like working with Spring Data JPA and so like data access layers should not be a problem for me.

Speaker 1 14:06

Okay, sounds good. So let's move on to notifications. Your client says you're spot on with Kafka. This is what the payment gateway actually happens to use. So would you be able to drive the Kafka integration with payment gateway yourself? Or you're going to need to bring in some experts. Wait for Kafka aware developers to get hired. How hands on you are with Kafka or any messaging for that matter.

Speaker 2 14:46

So yeah, in my like recent projects like I led the effort in decoupling notification service in favor of Kafka based notification service. So our application was directly utilizing and like email service, and this was causing our application to go down during the notification service downtime. So I basically led efforts in sending messages to our Kafka topics, and then also like consuming that through the downstream service, which would like all the notification service indirectly. So I have a pretty good, you know, experience with Kafka as well, you know. So we would basically need like, like order payment initiation, where the order is created, topic will carry the payment initiation necessary that like a kind of like trigger, like the payment gateway, and there would be, like, some kind of status update call to, like, contain the payment result. So, you know, our application would basically produce messages, and then there will be a consumer that will be reading that at the time. So yeah, like, Kafka is something that I have experienced as well.

Unknown Speaker 16:14

Okay, sounds good.

Speaker 1 16:18

Can you tell me more about the choice REST API, as you said, versus Kafka. How do you as a technologist, assume you have a choice, you can go either way. How do you decide which way to go? What are the driving things in your decision making? Or if you think that messaging always is just good, stressed API, fine, but please do explain

Speaker 2 16:47

so. So the main difference between you know your REST API and Kafka is like, address is like, like, it is designed for like, request, response based calls, right? So, like, what that means is that it is synchronous, and whenever you make a call, you expect a response to do some kind of modifications, right? So, so if you want something that is kind of like a fast response time,

Unknown Speaker 17:22

then we use like the

Unknown Speaker 17:26

image froze, okay,

Unknown Speaker 17:32

at some point, but 1015,

Speaker 1 17:35

seconds back, your image froze. And what do you stop coming through. I don't know whose end this happens, but you can repeat what you said in previous 1520 seconds. I lost that

Speaker 2 17:50

so you didn't hear anything from my side.

Speaker 1 17:54

I didn't hear about 20 seconds worth of what you were saying. You were describing the messaging based notification mechanism and how that would work when the model is not request response. So after that, there was a cut off,

Speaker 2 18:16

sure. So, like, I was explaining, like, the difference between, you know, the REST APIs, and also, like a Kafka, and when it is possible to use, like both of the RESTful APIs as well as Kafka. So, yeah, thinking about it. You know, whenever, like, we need some kind of, just give me a second like. So, so the main difference, let me go this to this again. So, main difference between the REST API and Kafka is that the REST API is kind of like designed for request, response based call. So what that means is that it is synchronous, and whenever you make a call, you expect a response to do some kind of modifications. So if you're fetching, you know, some kind of

Unknown Speaker 19:22

some kind of, like,

Speaker 2 19:24

like fast response time. And if you are preaching some kind of, like a user detail, or, like, some meeting, some kind of perform data or retrieving product information, we want to use the REST API. And it would be like a direct response based system, but like the Kafka is more like a streaming service. So what I mean by that is, if you have asynchronous call where the response of this processing does not dictate, like where the user sees so if there is an event that that kind of, like, triggers any cascading effect downstream, then you would want to go

Speaker 1 20:16

with, yeah, I just, I just had the Same kind of break your audio and video. It froze for a moment. Let's see if this keeps happening. Maybe it makes sense for one of us to redial in and see if that goes away. I cannot put my finger on either your end my end of network. Don't know what it is. Okay? Let's imagine what happens next. So what happens next is client eventually hired you developers onto your team, as they promised, and one of the developers after he looked at the details of design that you proposed at the higher level, comes back to you and says, I figure out that, based on requirements, that when order is incoming from UI, there is some extensive processing or computation that needs to happen on the order. And developer says that he proposes to do those computations on the order in parallel, but says he I am not exactly well versed in Java trading, so Can Can you point that developer in the right direction, and just outline what kind of technologies they would be looking at, what they need to do to actually code their intent to use threads, sure. So

Speaker 2 21:57

for this, so, right? So if the developer wants to work on something, you know that is parallel, it can like, basically mean that they want to look into the kind of like the executive services, right? So, like the basic, like the basic, what do you call it like, like the basic, like service that gives you more control over, like the thread management and also provides a kind of way to manage a pool of threads, right? So where you would be passing in tasks, and then it will give you, like, a future response back. So that is something that the developer can look into initially. But like, if we need more fine grained system there, they might also work with, like, the trick commands, like, which is basically a fully managed tool for concurrency management. So these are like, those are like two things that I would look into. But apart from that, I would also encourage the developer to look into the like, the spring integration with parallel processing by using the async annotation and tax executor dependencies that they can use. So those would be some of the things that I would recommend.

Speaker 1 23:36

Okay, sounds good. So let's assume that something has been coded already. So your initial design already took shape of code. This code so far, lives on your machine and on the other developers machine. You kind of divided responsibilities. What needs to happen next?

Speaker 2 24:02

So so, yeah, so, basically, like, we need to create, like a repository on a source control so, you know, maybe, like a, like a big bucket or meetup project where we can work out. So initial setup of the project, it would basically be creating that repository, and each of the developers would kind of like, need to check in their code to a particular branch and then create pull requests to the developed brands, right? So like, where the other developers can actually do the code review. So like, we'll also ask other developers to put in their code review, and we can make changes, like, according to the request. And once that is done, we can integrate some kind of, like CICD tools, maybe use Jenkins to build and deploy this application and something and that I like, like, says, like this services like, like, make it like, to make it content, containerized so like, Someone would be working on, like a Docker file, and this Docker is nothing but just creating like object the emails of the application so it runs on everyone's machine and in regardless of their system, right? Basically, they can just run a Docker build and then docker run to, like, basically set up your applications and run it. So those are the three things that I have next. Okay,

Speaker 1 25:55

what happens next? Client comes in and says, Oh, by the way, I have a little bit of surprise for you. Early on, we said, databases Oracle. Well, we were talking and interacting. The database team decided to migrate all this data into Mongo, so now you have to assess what kind of impact it's going to have on your work.

Unknown Speaker 26:23

Can you talk to me about that?

Speaker 2 26:27

So, right, so, so, okay. So, since we are working like, with a data access layer, right, like, we basically will have very small chain code wise, because we just need to change the like the dependencies and certain queries. But more changes would be on, like how the like the transaction is managed and how the data is represented, because on Mongo, it uses like a JSON like format for storing the data. And Mongo does not provide the same level of consistency as your relational database, right? Because it has high availability and partitions, but not high consistency. So we will, like, need to adjust, like, some kind of consistency across, like the multiple services, even though it has, like, eventually consistency. So that is something that we would want to consider. And, yeah, I think code wise, you know, we just update the dependencies and use the Mongo repository instead of the JP repository.

Unknown Speaker 27:58

Okay, sounds good.

Speaker 1 28:02

Then what happens next is your client says, By the way, we are starting to adopt AWS, so we want your solution to be deployed to AWS. Can you please talk to me about different options for this to happen that you can actually implement.

Speaker 2 28:29

Yeah. So, yeah. So basically,

Unknown Speaker 28:34

we would like deploy our you know,

Speaker 2 28:38

like deploy your Mongo to the AWS, rds, and basically in the relational database service from the AWS, you can do that. And then for our application, we have a choice of running, like a EC two, like which is a complete virtual machine, or an ECS, which is a container service. So also have like Eks for like Kubernetes, but since ECS is more AWS managed solution, we would go with ECS. So the way we would set up the deployment is you would have your Docker file on Jenkins, on your build stage, and you would run the Docker build. And once the build is completed, you would upload the Docker image to Elastic Container Registry, and then ECS, we basically are using a cloud formation template to stand up these services, right? So on the tax definition of the container, we basically pull in that image and then, like, run the start startup commands. So yeah, that would be something that we would look into, and we might want to expose our API behind an API gateway, which, like, acts as a single port of entry for all of these services. So, so yeah, like the those would be the services that I would look into and integrate.

Speaker 1 30:24

Okay, which one of those services or which ones you actually already used on your other projects in real life?

Speaker 2 30:34

So yeah, for for this, I've used EC two, ECS, s3 lambda. Okay,

Speaker 1 30:46

can you tell me, from your perspective, how do you choose? See you have functionality and it's a green field your choice to implement it?

Unknown Speaker 31:02

How? Would you choose between

Speaker 1 31:05

eventual deployment of a Java app into ECS, versus doing it all as lambdas? So

Speaker 2 31:14

so far, like for me, a lambda is just a function, right? So what that means is that, like, it reacts to like certain kind of trigger, so if there is something that is short processing or it needs to happen in the course of an event, right? So if I drop a file into a sq bucket, and I need to do some processing on that file. I would go ahead with lambda, and if there is some misses on like SQS, and then I need to maybe transport that misses from, like one spot to another. I would use lambda, but for an ECS, I would basically be running an application which is always live and and, you know, like it would be some kind of a long running service where I would be serving user continuously and always be, kind of like listening On for, like, a call from external services. So, yeah, like, that's when I would choose between the lambda and the ECS. Okay,

Speaker 1 32:30

which one do you think? Well, you partially answer this, but I'll give you explicit question. Java app in ECS versus lambda? Which one do you think it's better the imaginary use case that we've been discussing?

Speaker 2 32:48

So yeah, for this, you know, for our use case, we would do the ECS because lambda is best suited for kind of short jobs, and it also has a hard limit of, like, 15 minutes. And one of the disadvantages, like, of like lambda, is that it has a, like, a cold start, and with Java, like, it's pretty bad. So, yeah, so ECS is the one that I would choose. Okay,

Speaker 1 33:25

there's another aspect. I'm curious to hear your thoughts about your application. By design, it goes to database, to the back end, and basically manipulates the database data. How would that affect your solution if for whatever reason, you decided to go with lambda instead of Java app in ACs.

Speaker 2 33:51

So, yeah, so thinking about it. So like if my lambda process interacts with,

Unknown Speaker 34:05

with, you know, with the database.

Speaker 2 34:09

Basically the lambda will need to, like, correctly. It should correctly connect to the database. So, like, you know, the connection pooling, it might have some kind of issues, like the calls start and the latency might cause, like, certain issues with it. And, yeah, but so I think for simpler queries, it should be fine, but if the queries, it gets a bit like little complicated, it might actually be a problem. And Okay, fair enough.

Speaker 1 34:57

Now comes the next step. Can you tell me what kind of technologies you're going to rely on to actually monitor what is happening with your Java app that is deployed in a container in ECS, what what other things that you're going to be curious to look at, and what kind of technologies you're gonna use to make it happen?

Speaker 2 35:24

Yeah, so basically, we would install Splunk forwarder agent within our ECS, alright? This, this basically will capture all application logs and forward it to the Splunk index. So that would be the first thing that I would add in order to increase the logging visuals. And after that, you know, we would also integrate the Spring Boot actuator dependencies to have, like, our real life metrics available, like from the actuator endpoints. And the other thing that we would incorporate is the New Relic agent, which captures the system information, like the CPU, uses the memory and like the request response code and latencies, and then provide a dashboard. So we would be integrating these for observability within our application. So in case there is any failure from any region, like the threshold that we have set, we get alerted, and then we can come and, like, fix the application in order to have, like, a minimal impact to the customers. Okay,

Unknown Speaker 36:53

let's imagine that

Speaker 1 36:56

three months down the road, the there is more requirements from the client and your internals. Internals of your implementation started getting more complicated. It used to be simple, crud, perhaps single microservice doing that, and then all sorts of weather processing came into place. Something else. I won't make up what that something else is, but enough to say that design became more complicated. As a result, you have five microservices already, instead of one from observability perspective, in addition to what you already mentioned, Splunk then metrics with actuator, neuralink, what else would you want to capture to figure out more details about how now those communications Between your multiple microservices are happening. So think about it. Uh,

Speaker 2 38:02

let's think about it. So for this, you know, if we have no multiple microservices like running to achieve this task, like we would also look into, into the data flow and also the ownership, so we would have some kind of, like the ID that can track between all of the requests from one service to another. And right now, we would incorporate some kind of service discovery to look into those services that are in play. And we would also have some kind of like a distributed transaction management system, maybe using a saga or something. But apart from that, on like a particular response and request monitoring, we would do the like the throughput, and we do that like the error rates, we do the latency and response time. So some of those would be what I like would think of implementing. Okay,

Speaker 1 39:13

at the beginning, you did mention scaling. That was actually one of your top first questions to ask scalability. So now comes the time Maya comes back and says, all of a sudden, we're expanding our geography of users. It's not going to be 100 it's going to be 1000 they're all over the world. So the volumes passing through the system increase scalability is firmly on the table. So tell me about different ways you can actually utilize scalability, and different kinds of scalability that you might think of, and how that would practically look in the chosen stack that would be Java and operating in ECS in the container.

Speaker 2 40:09

So, so, yeah, basically on ECS, you can specify, like a certain matrix where you want to scale out. And by scaling out, I mean, like horizontal scaling, which is just basically kind of increasing the number of instances, is pretty cost efficient because you're not using Merck CPU, but you're just standing up your instances. So that would be on the like the main service. You would be deploying this service in the multiple regions, so your US east, US West. And if you are doing it like against multiple countries, you can do on other kind of, like geographical locations as well, and be on different availability zones. And we'll also have a load balancer that would go, that would go and use a geo routing to basically like, kind of like route to your nearest location, and also like the database would be like scale, in a sense that we would be kind of like using multiple like, read, like a red replicas of the database, so it is easier to like, kind of like read the database, and we might incorporate some kind of like caching technologies To store the frequently accessed data, right, and the product catalog and other statuses and so on. So yeah, like, those would be primary things to kind of consider, but here we would also also incorporate some kind of, like the rate limiting to basically handle the amount of load that we want, and also have some kind of resilience, using socket breaker patterns to basically make sure that our system doesn't go down.

Unknown Speaker 42:16

Okay, sounds good.

Speaker 1 42:19

Now you mentioned horizontal scaling as opposed to vertical scaling, I guess would be, although I don't think you said the word vertical scaling. Can you tell me if you can think of scenarios where vertical scaling is actually going to do a better job than horizontal scaling, in your opinion, other scenarios like that.

Unknown Speaker 42:50

So, all right, so thinking about

Unknown Speaker 42:55

thinking about it, you know,

Unknown Speaker 42:58

of vertical scaling,

Speaker 2 43:00

it can be better when you have like a load traffic system with like a predictable load, maybe some kind of a legacy system that is initially designed for a single instance and always beyond and always beyond that. And you know, so. But since our system is primarily in our microservice space, whenever we have multiple service it, it might not be a good idea, right? Because we don't want to increase the CPU and memory on like one system, one instance. But yeah, vertical scaling is pretty, pretty used. When you have, you have a resource intensive application, like, on, like a legacy system, and, yeah, I think that's where I would use, like a vertical scale. Okay,

Unknown Speaker 43:58

all right, sounds good.

Speaker 1 44:01

Let's get out of the imaginary use case. I'll take you back to something that you said at the beginning, when you were describing your role as senior engineer. I heard you right, that this, this make you one, just one of the engineers on your team, or you act in a role where you oversee other people's work or tutor them, perhaps, tell me more about that side of your activities.

Speaker 2 44:30

So like our team, like has like five developers, two software senior engineers, one junior, and they may also have like, two TDPs, right? So I'm one of the senior ones, so I'm involved in hands on coding, but I also overlook, you know, like the juniors and TDPs.

Unknown Speaker 44:57

So sorry. Okay,

Speaker 1 45:03

so I'm hearing your hands on technologies at this point, five years from now, if you make plans that are in the future, tell me more about how you see yourself doing what in five years, which direction appeals to you.

Speaker 2 45:23

So talking about where I see myself in, like, about three to five years from now, my next goal is to like to be some kind of leader, maybe lead a team or project. You know, that is my short term goal, but but a little bit longer down the road. You know, I want to be some kind of architect, right, so where I can devise solutions for multiple teams. So that's my short term and long term goal.

Unknown Speaker 45:58

All right, sounds good.

Speaker 1 46:04

We figured out the answer to the question. I'm looking at your CV, and we did talk about your cloud experience. Do you have any certifications from public cloud providers? Any

Unknown Speaker 46:18

certifications currently? Certifications? Okay, certifications.

Speaker 1 46:25

Okay, good. Now I would like to give you an opportunity to ask me questions.

Speaker 2 46:35

So, so, right. So thank you for your time today, and it was really nice talking to you going through, like a discussion for creating the service for our imaginary plan. So the first thing that I would like to ask is, like, what do you like like most to like working at, like a safety

Speaker 1 46:59

end? There are many things in no particular order. It is a collaborative environment. So people are never alone in terms of they also always have access to expertise when they need some help. The way sapient projects are structured, they're always it is always a teamwork that's involved thinking doesn't really do place a single contractor on the project and just have them do work for a client. I don't even know if this happens nowadays. In the past, it happened, but very rarely. So team, you're always part of the team environment. The teams sometimes that combine, they include both a mix of sapien people and client people, and sapien figured out quite well to integrate those teams to the point where, if you don't know specifically whether your colleague, who nowadays a lot of times are virtual, so just hear them on a call, if you don't know that there for certain with the client or sapien, you could not tell the difference in the way communications happen the way collaboration and cooperation happens. Another thing that I knew like about working for sapien, the sapien cares about these people staying on top of the latest and greatest technologies. So there's a lot of internal resources dedicated to giving people access to learning resources that includes all sorts of courses, some of them might even be sponsored or reimbursed if you put some expenses in self education. Another thing that they really enjoy about being with sapien is the nature of the work that we do. It is project based, of course, sapien would love to have projects run forever, but in reality, this is not exactly what's happened projects have, you know, timelines, and those timelines are not measured in years. So in some ways, it is working for SAP. And sometimes feel like you are changing jobs without a change in job. This year, you might be working on a series of projects for one client using the technology stack, and then next year, you go into a different project for a different client, perhaps very often different business domain, quite often different technology stack. So to me, it is a big asset and a big plus, because it keeps me motivated to always learn something new to be on top of things, and things do not feel stale. Or sometimes some people who would be with the same employer for 10 plus years doing the same thing, say, working with the same systems, if you've been supporting it 10 years of that might appear boring to many people, myself included. So we sapien, there is much less chance of that. So that's my short list. Does that answer a question? Yeah,

Speaker 2 50:40

pretty much, yeah. Thank you so much. And I would also like, Yeah, I think it does. And I would also like to ask you, like, like, the next steps in the interview process.

Unknown Speaker 50:55

You know, that would not be me.

Speaker 1 50:58

I suggest you as that of people who sent you the invite, or you've been in touch with I am not on the side of the logistics of the process. Thank

Speaker 2 51:10

you so much. Yeah, thank you so much for your time. Do you

Unknown Speaker 51:14

have any any other questions?

Unknown Speaker 51:15

I think,

Speaker 2 51:20

I think that's it for now. That's all that I have right now, in my mind.

Speaker 1 51:25

All right, sounds good. Thank you for time I enjoyed talking to you, and good luck with the rest of the process.

Speaker 2 51:31

Thanks so much for the time I really appreciate it as well thank you so much. All right, bye. Thank you, bye bye.

Ronaj- AscendingDC-SRV-First

Speaker 1 00:00

So, yeah, you know, talking about the migration project, like, I was in charge of, like, migrating, like, a couple of services, you know, one of the main service that I've migrated, like, migrated include the maintenance API, you know. So this maintenance API, initially, like, used to be, like a more model on, like a legacy code, like base that opened up when the users would, like, make changes to the profile and, like, it would just call this whole like workflow within the model. And now, like, we basically, you know, stripped it out and created a kind of, like a microservice, like and the back end services was, like my part, and the way I worked on it was, like, first, like, I created, like, a design documentation, you know, and like, gathering the functional and non functional requirements and putting it in paper, and also getting approvals from the architect. And so, like, on the you know, functional I listed how the like endpoints would look like, and what we would like support and how the responses would look like, and also so on and on. The non functional requirement, I actually listed, like, the logging, secure, security, and any other scalability kind of aspects that was needed. And once I got the approval from architect, and I basically, like, went ahead with the implementation on, like the Java Spring Boot application. And after that, you know, I went with the deployment to AWS Cloud using Jenkins as our like CSC tool. And we have, like, managed pipeline, so, like, all of it could be done, like, by creating so we had to create, like, a YAML configuration file that helps, like, provisioning the services. So that's kind of like a workflow of one maintenance API. But I was also involved in like Experience API, which is like the entitlements for the widgets and components to be displayed on the front end, right? So these are, like, the two primary services that I like work with the migration project.

Speaker 2 02:07

Cool, that sounds really impressive. So you mentioned you deploy them into AWS. So can you give me more details, like what kind of services you're using to deploy this to ACA endpoints?

Speaker 1 02:20

So yeah, so for our, like the legacy code base, it is in like, easy to write. And these do, like, these new microservices, they are containerized using Docker. And we also, like, we use, like AWS, like ECS on a fargate to deploy it as a, like a container application. So basically, we just like, kind of like provision, like AWS ECS task and a service to deploy these, like containerized application, and we use Jenkins that basically, you know, leverages the cloud formation templates and like to probe like for the provision of these services. And our database is AWS Postgres, SQL, and we also use AWS lambda for one of our use cases to basically, you know, send notification to users. So these are some of the services you know, that I have directly, you know, light dried hands for on, like maintenance API. Okay,

Speaker 2 03:22

cool. So during the process, do you have any challenges or difficulties when trying to, you know, using or deploy ECS, let's say just for level. So

Speaker 1 03:38

talking about the difficulties, right? I guess, like, we have like a, like a kind of, like, very streamlined pipeline, like, where we just provide our, you know, parameters to the configuration like the YAML file. And this configuration, like YAML file is, like, very true, thorough on what like a configuration means to have, like the security groups, like the VPC and IAM roles and so on, and all of all the CPU and all of the information that's like needed for, like, scaling and so on, all right. And I think one of the challenges that we, like, we had, like, basically, was to make a change to a security group to like, add in like, add an ingress rule on one of the services and and this had to go through like, like, ISO approval, which took a little bit longer than expected, but, but apart from that, you know, because of the streamlined process that we had, we didn't have too many problems.

Speaker 2 04:41

Okay, so are you in charge of, you know, this whole AWS deployment process, or someone else has built the jness pipeline, and you just use it

Speaker 1 04:52

so on that. So, like, like, so like, I mentioned, like, we use, like, a managed pipeline, so we own the pipeline that our services use. But, like, but the way it's generated, generated is true, like an automation tool, right? So we just create a YAML file, and this YAML file, like, contains all the configurations that is kind of like needed for the service. Like, what kind of like a build tools it uses, where it needs to be deployed, and if it is using, you know, AWS, ECS, SQS, or anything else like that, based on this, there are like pipelines that is generated, and then our team is responsible for actually running those pipelines.

Speaker 2 05:40

Okay, so basically the AWS resources is provisioned by you. Basically you have confirmation templates to provision like VPC ECS tasks. Okay, cool. Then give me some details on how did you design the VPC architecture for your application?

Unknown Speaker 05:59

The VPC architecture. So

Unknown Speaker 06:03

you mean the VPC architecture, right?

Speaker 2 06:05

So basically, how did you decide which subnet and the security groups for the ESAs or, how do you make sure that your esas containers can be accessible from the internet?

Speaker 1 06:17

So, yeah, so thinking about that one. So, so, yeah, we used a VPC, you know, that was, like, available for our account. And, like, we did not have to, like, kind of like create a new VPC or subnets. We kind of like used a pre existing one. Like, I don't know how it is configured, like, internally, but, but we just like provide the names of the VPC and like the subnet within our service.

Speaker 2 06:46

Okay, so let's say, right, you use the default VPC, and there should be some subnets, but you have to at least decide which subnet you have to use. And also you have to also design the security groups rules. For example, you can put it either in a public subnet or in a private subnet, but behind the load balancer. So what kind of strategy you used for your, you know, application in general, so that it is secure and also it is accessible from the internet.

Speaker 1 07:15

So, yeah, so it was a latter one, like, we had, like, private like, subnets, like, behind the load balancer,

Unknown Speaker 07:23

okay, okay, gotcha.

Unknown Speaker 07:26

And

Speaker 2 07:28

you also mentioned there's that you use to send in Notifications. Is that correct? Yes. Okay, cool. So yeah, I think that's all the questions I have for now, so I'm going to send you a link so we can start a COVID Session. Okay, absolutely.

Speaker 2 08:20

For the light, sure.

Unknown Speaker 08:39

Can you see my screen? Yes,

Speaker 2 09:02

I'll be by the way, you can choose any language you want for this question, but let me paste The description. Be afraid. I

Speaker 3 10:50

Okay. So, okay, so we are basically like creating

Speaker 1 11:01

our like assigning a unique random number to like participants from Like redefine pool to have unique

Unknown Speaker 11:21

and non duplicate so basically, like

Speaker 1 11:26

for this, we need to populate pool of numbers from one to like, a number of participants and then like, have a shuffle. So

Speaker 1 11:53

BBB, so I'm not quite following what we are actually implementing this whole class that takes care of solution, is there a specific case?

Speaker 2 12:04

Yeah, I think you can start by, you know, implementing the core function. Just imagine, you're trying to just implement this function, right? The input is predefined pool, and whenever a user invoke this function, we just simply get a random number from this pool.

Speaker 2 12:26

You know? Yeah, it's like read the list like the element from predefined pool, or at least for array. But the only limitation is that you have to make sure that each element of this array will have the equal possibilities to be chosen. So basically, you can just go through from index zero to index right the last one that you have to think of a way that each time I call this function, I will get a unique element from this predefined pool you

Speaker 1 13:08

Okay, so, yeah. So, let me start by, like, creating a class that is used to, like, drag these store,

Unknown Speaker 13:20

price going. So, okay, so, or price drawing, we would

Speaker 1 13:37

basically use a list to store, like the pool of numbers.

Speaker 2 13:42

It's pretty fine, so you can, we can discuss about this later, but right now, you can regard it as an input parameter that's already provided. So,

Unknown Speaker 13:59

okay, I'm sorry, okay, just let me

Speaker 1 14:14

create. Okay, all right, so there's that that's a bit fine number, and then we want to do

Unknown Speaker 14:27

that. So can we basically have like a setup, assigned numbers?

Speaker 1 14:42

Okay? Okay, now, whenever, like, we sign a number. So

Speaker 1 15:02

can we need to create method that signs a number and returns I

Speaker 1 15:25

so what I'm guessing is, once assigned a number, I remove it from my pool. Okay,

Speaker 1 15:38

so that means, if the pool is empty, like that means everyone is assigned, so I will just return

Unknown Speaker 15:49

empty response back. You.

Speaker 1 16:19

Okay, so now if that's not the case, then we basically, like, Remove it from the pool And then, Like, that's i

Speaker 1 17:52

So, so I get the number, and then I add to the Like assigned number, and then like return that number.

Speaker 2 18:06

Okay, so what question is? So basically, you're just trying to pop this list, right? Because each time your assignment is

Unknown Speaker 18:18

the last element of this list, do?

Speaker 2 18:23

So, right, yes, yeah, so I think we may need to modify this, because if you look at item number 12, there's a rule that's specified, traverse or populate is not allowed. I

Unknown Speaker 18:59

Okay, so let me think. I

Unknown Speaker 19:50

So like,

Unknown Speaker 19:52

if we do not want to pop I

Unknown Speaker 20:03

then we, like,

Unknown Speaker 20:05

we just probably have to, like,

Unknown Speaker 20:08

leverage the assigned numbers.

Speaker 1 20:14

Like, not worry like, about the pre design pool. You

Speaker 2 20:23

Yeah, I'm not quite following, but you can try to implement the code so so that I can, I can try to understand what you're talking about.

Unknown Speaker 20:41

So let's say, like we have this field, like

Unknown Speaker 20:45

that source a number of participants.

Speaker 1 20:52

Like, if my assigned numbers and like the number of participants is same. I then the deck means I have, like exhausted everything

Speaker 1 21:18

that would be the game is over. Case

Speaker 1 21:28

like now in the case the game is like, not over. I just want to like loop until like the sign numbers and to

Speaker 1 21:48

assign numbers exist on. So we will use like a random Number Generator. I

Speaker 1 23:12

so I think you should take care of the assigning like new number. I

Speaker 1 23:28

because we are, like, randomly generating a number from one to like a total participants, and then, you know, like adding it to the list. It like it does not exist,

Speaker 2 23:43

okay, so why the condition for the while loop is assigned number contains aside. So does that mean that if it contains so you have to keep moving like,

Unknown Speaker 23:57

yeah. So it will keep on. Like,

Speaker 1 24:01

right? So, like, if it contains, then that means it will not be distinct, right? So,

Speaker 2 24:09

okay, and in your case, this assigned, is it the index or the actual number or element from the predefined pool

Speaker 1 24:21

for so, like, I think it would kind of like, probably be in like index. So like, if we like, like, if you like, use the like, a predefined pool. Instead of adding the assigned number index, we will use the predefined pool value. So here I'll probably do something like, like a return the value, rather than like the assigned index like itself.

Speaker 2 24:52

Okay, so yeah, let's talk about the performance right now. So what does the case the worst scenario for this few lines of code.

Unknown Speaker 25:04

So talking about

Speaker 1 25:06

performance, I think like for the worst case, it keeps like on retrying until and until, like a signed number is found. Like so it could keep on going, but I would still think it would be like a big off and runtime complexity.

Speaker 2 25:26

Okay, so why is it the worst in a row is also off and can do a detailed analysis.

Unknown Speaker 25:38

Oh, yeah. I and

Speaker 1 25:43

like, basically, like, what happens is, like, whenever you do the random number generator, right? So probably just kind of keep on, like, generating the number over and over until, like, you find the number that does not exist, right? So it will generate a random number from from one to the number of participants and check if the number kind of like already exists, right, and if like it's always true. Then it keeps on looping, because the number have already been assigned. So it will have to probably generate the multiple random numbers over and over again.

Unknown Speaker 26:35

Okay,

Speaker 2 26:37

and so that means the worst scenario, time travel accident is still often,

Speaker 1 26:46

yeah, I think the best is of one, like, like, the best is like, O of one, and the worst is like, O of n.

Unknown Speaker 26:59

Okay, sure, so what about the space?

Unknown Speaker 27:09

So? So, okay, so here my assignment.

Unknown Speaker 27:17

So the sign number is the

Unknown Speaker 27:21

stores that has set. So, I

Speaker 1 27:26

think so. I mean, for the hash, for ant participants, stories would be like the big O of n, and I think that's the only thing, okay,

Speaker 4 27:42

okay, cool. So right now, you know, of course, will there will always be, you know, follow up questions. And from my point of view, right now, we have, like, two possible ways to improve this. The first one is the while loop. I don't think there might be a better way to handle this while because, as you mentioned there, the worst case is that you keep looking through right and another way for direction is that, as I also mentioned, the space complexity is often so I'm wondering if there's a way that we can, you know, avoiding This kind of extra space for this hatch set i

Speaker 1 28:47

Okay, let's see, since the while loop keeps retrying random numbers until An

Unknown Speaker 28:53

assignment is found, I

Speaker 1 29:40

I guess we just have to do some kind of like manipulation to be predefined pool itself. But,

Speaker 4 29:55

yeah, sure, yeah, what kind of manipulations you're allowed, you know, to modify The

Unknown Speaker 30:20

print platform. So I think since, like

Speaker 1 30:24

since the random number like, it gets one through n and two like, from zero to n minus one, then we can probably get the index from like, one through all of those numbers, like, except for the last one and and then once you assign it, if we, like, swap it to that last number, then we probably would have, like, a better approach where we do not need to use like the hash set at

Speaker 4 31:01

all. Yeah. Okay, yeah, that sounds like a good approach. And, okay, okay, so yeah, let's talk about more on the API designs for this problem. So as you know right now, you have implemented the base paths, right? So now I want to discuss more with you know how you tend to deploy the simple applications. So basically, what I want to talk about is, first, what kind of technology step you want to use for deployment, and then is what kind of database, or in general, imagine me as a stakeholder, and I'm asking you, hey, I want this application to be deployed and to be able to use by our users. So just give me some ideas like that discussion as we can start from there.

Unknown Speaker 32:06

Okay, sure.

Speaker 1 32:08

So, yeah. So for the back end technology like we would basically be using like Java Spring Boot application. Like Spring Boot just provides a very easy way to create a REST API, standalone production facing application, and we probably would go with like, a Postgres SQL database and like for the other thing about like for Like, like for the content organization, like, we will Docker and like, kind of like, like, deploy on like, AWS cloud. And let's see like, so the Spring Boot application would basically, like, have a couple of like, the flows, which would be like, like a registration part, where you, like, register participant and you have to assign like a unique number to that participation, like the participant, and then you then, and then you like may want to do like a kind of like, maybe a status check for like numbers, like, depending on like, how like concurrent application kind of like needs to be how it like handles the congression issue. And the other thing we would like we need to store in the table is like the like the participants data, like, like where you have the like the ID of the participants, the name of the participant, like the assigned number, and also like the timestamp in which like they were like, kind of assigned. So in my view, like, I think that would be like a general, you know, like the design of the back end service, like a RESTful web service, to kind of like, register and assign a number and optionally like, like, also check the status of like assignment database probably like a Postgres for like, a good persistence layer. And then, like, we would probably like, containerize this application using like a Docker, and then like, deployed on it, like, AWS ECS, and we, like, like, we probably have, like, proper scaling policies in place, like, based on the like, like, the load that it will increase. And I think like, those would be my like, go to, like, high level design. I do this, okay,

Speaker 4 34:42

okay, cool, that's totally makes sense. But can you give me some reasons why you choose, for example, Java and spring and ECS and Postgres, for example, for this kind of application,

Unknown Speaker 34:58

well, or that

Speaker 1 35:05

so I, you know, so I guess, like Java, you know, because it has been, like a prime be my primary experience in developing, you know, rest for web services, and it has a pretty good, you know, handling of concurrency, and so, like you you can do, like, pretty good multi trading using executor services, services like trade pool and like you know, you can also, like, leverage a lot of the frameworks from Spring, right? Like, because you can get, like, a faster deploy development, less configuration and a lot of startup dependencies available for security and also database and so on, and it is like a microservice ready application. And the reason I chose Postgres is just for having, like, a kind of, like, proper consistency and also good performance. But you can substitute the postgres with, like a DynamoDB, because we do not, like have multiple tables, and using Dynamo table would probably be like, faster, you know, reachable speed. But I guess the postgres just for, like, a constraint that it provides, and like a transaction properties. And it has so like and like it has also, like containerized JSON, because it is portable, you know, and you can also basically, kind of, like, deploy it on any stack, and then like ECS for, like, managing these, like, orchestration of those containers, right? So that's like, those are the reasons I chose each individual, like, frameworks, okay,

Speaker 4 36:55

yeah, that's totally makes sense, and I think I'm done with all the questions so far for today, but I want to give you some time back so you can ask me any questions you have.

Speaker 1 37:09

Yes, sure. So, so, first of all, thank you so much for for the time today. It was really nice talking with you. So, so, first of all, you know, I would like to know, like a bit about the company, and also, you know the team size, and what kind of you know like as an engineer on your team, you know, what does like the developer get involved with, like, apart from coding,

Speaker 4 37:34

sure, so certainly, as you know, it's a startup and specializing in AWS. So we pretty much do 100% AWS related projects. So we have around 50 plus engineers, I think. But like most of them, are in staffing. So staffing means they work for clients. The role that you're interviewing with is the professional service team. It's in office. And so there's around six to eight engineers, I think so the structure is like, we each have to handle the projects individually. And as for the projects, mostly, you know, as I mentioned, it's, it was related, but, you know, there's, like, over 100 services. Of it has we do all kinds of stuff. Sometimes we do a little bit development like but using sometimes serverless, sometimes microservices. But also, there's a lot of projects that's, for example, migration. We help a lot of customers migrate from on prem to AWS. And also we do, like disaster recovery, do data pipelines, data lake, and right now we're working on some machine learning projects. So it's might be a little bit different from a traditional, you know, IT company, but I would say it's quite friendly, active and easygoing atmosphere here, because there's just eight people. We work separately, but we do communicate a lot with each other, and our tech lead or is also just working with us in office. So yeah, I don't know if that's address or question.

Speaker 1 39:25

Sure. Yeah, thank you so much for that. And yeah, it definitely does. And thank you so much for the context as well. And I guess the next question like that would be like, like, what do you like like most about working here.

Unknown Speaker 39:42

Certainly Do I like the work

Speaker 1 39:44

like? What is the like? What do you like the most about like working in SMB? Oh,

Speaker 4 39:50

yeah. So I've been working here for over four years. So I think I started the job back in developing Spring Boot aggregate stuff. But then, during the past four years, I've been working with so many different things, ranging from micro services, serverless, DevOps, CICD to data lake to migration. Right now I'm focused on migration, so I guess there's just so many things you can and you have to learn, but, and most of them are related out. So yeah, that's totally up to you. Some people may think, Okay, I'm just, I just want to develop Java code for the rest of my life. Then it might be a different story.

Speaker 1 40:40

Yeah, okay. Thank you so much for that. And I think the last question is like, can I know, like, what would be the next steps in the interview process? Yeah,

Speaker 4 40:50

so next step will be interfering with our tech lead, also CEO, and also there will be another senior engineer in that round. So basically, yeah, because they did, they're totally have some kind of random questions. So I would say, of course, there will be coding questions, but it's not like you have to code, but honestly, you have to write pseudo code to explain how you're going to solve it. And then there will also be some questions to understand your background in software engineering, like architecture level and also, yeah, I would say a little bit on the cloud, but we're not expecting too much cloud experience for this world.

Speaker 1 41:41

Gotcha. Yeah. Thank you so much for everything. And I think that's all the question I have right now.

Speaker 4 41:49

Okay, cool. So, yeah, I'll talk with Harry about the feedbacks, and he'll answer my question. Sure, sure. Okay, yeah,

Unknown Speaker 41:59

all right, thank you. Thank you so much. Don't thank

Unknown Speaker 42:09

you

Unknown Speaker 42:15

so much. Tom,

Speaker 3 42:21

so Damn,

Unknown Speaker 42:30

sister took Matthew lens.

Ronaj-Adobe-Final-Srv(1)

Speaker 1 00:00

Hey, hi, how are you doing today? I'm good. How are you Yeah, I'm doing good as well. Thank you. Yeah,

Unknown Speaker 00:06

what was your day so far? Yeah, it's been good, busy.

Speaker 2 00:12

Yeah, give me seconds. So I got your resume from my manager, so he asked for to discuss with you for the current position. So currently, are you in Ohio, or

Speaker 3 00:26

Currently I'm in Maryland, right now. Okay, got it.

Unknown Speaker 00:35

Okay. So,

Speaker 2 00:38

so how do you feel about the current role, do you did my goals? Spoke to you on this current role?

Speaker 3 00:46

Yes, good. Let me know about couple of things about the role and what Adobe is trying to achieve right now with the team. So yeah, I do have a little idea. So,

Speaker 2 00:56

okay, nice. So do you have any questions on this role before we go for the further discussion. Or,

Unknown Speaker 01:05

I think, no, we can get started. I don't have any

Unknown Speaker 01:09

problem after, after the

Speaker 2 01:12

interview, Sure, definitely. Yeah, we'll spend some time for the questions at the end of the interview. Okay, so, yeah, can you? Can you explain, like, what is your current role in the project? Like, I think you keypad, right? Is the current client? Yeah. So can you explain what you what is your current role, and, yeah, about your project?

Speaker 3 01:34

Yeah, sure, definitely. So let me start with a brief, brief introduction. My name is Jonas, and I've been working as a software engineer for about six years now. And I've been primarily working on, like, Java based applications, you know, utilizing frameworks such as spring hibernate, and like primary being like developing, you know, back end technologies, developing microservices, also designing them, and also like deploying them in AWS cloud. And currently my role, it is strictly back in engineering role that I've been involved in, like designing microservices, you know, creating different services, gathering the functional and also like the non functional requirement. And then I also like promoting them to, like high environments. So I've been like, involved with, like, designing them, creating, like, pretty stable and like scalable systems. A couple of components that I like worked with like, include the, you know, maintenance and like Experience API for meeting, maintaining the like the user profile preferences and like the experience for, primarily, like the entitlements, you know, for, like the wizards and components on front end. And I'm also, like, involved with, like the microservice orchestration using, like the API gateway. So, yeah, on a daily basis, I'm involved, like, mostly you are working on, like a back end, you know, Java code, helping the team with, like, the code reviews and also on production support as

Speaker 2 03:05

well. Okay, nice. So, can you tell about this? What is your current project is about? Can you explain the architecture the current project?

Speaker 3 03:15

Yes. So, so the main project, like, like, here is like, like, a modernization approach. You know, basically we have, like, a monolithic app that we are, like, modernizing. It's strictly, like a micro service based, you know, modernization. And, yeah, I've been involved in, like, breaking down, like, some of the components. So one of the most recent one that I've, like, worked with includes, like, the Experience API, so, which is like a general microservice that is like, used by our like front end components to basically, you know, call for like, grabbing like, the entitlements to like, check if like, like, starting with it should be like, visible for the user like depending on like, what account the user has access to and like so on. And not all the service that I recently worked on is the maintenance API, right? So whenever like user makes like changes to like the user profile preferences, like overdraft protection or any other like plans that flow is triggered. So it's Spring Boot based micro service. So we use, like PostgreSQL for our database. Everything is deployed on AWS cloud and like containerized through like Docker and on ECS, yes. So that's like, basically the like REST based microservice that I'm working on right now. Okay, nice.

Speaker 2 04:44

So, how about so I see you in your resume, like you have some experience with the Kubernetes, or you have knowledge in Kubernetes. Did you use it in your work?

Speaker 3 04:58

Yeah, so I like, I have worked with like Kubernetes in my project, like, in, like, Key Bank, like, like, we use, like AWS, like ECS for, like, the container orchestration. But like I work with, I have worked with, like, Humanities in the past.

Speaker 2 05:16

So, okay, so how much, what are the areas that you work in the Kubernetes? Can you explain?

Speaker 3 05:28

So, yeah, on the Kubernetes side. So basically, like, Kubernetes just for like, like, deployment orchestration, right? So we basically have, like, Docker containers that, like, we need to deploy to cloud. And my basic job was to create, you know, these YAML configuration files for Kubernetes deployment or, like a deployment descriptors, basically. So there are, like, different properties that I can set based on on the discussion with the architects, and based on like, also, like the scaling policies, and what kind of policies are like, or like metrics would trigger this, like scaling and like, what kind of like load balancers and so on. So I've been involved in like, in writing on those, like Helm charts and Yammer configuration files, I would

Speaker 2 06:22

say, okay, so you you have hands on on the health charts, creating health charts, right? Yes, okay. So when you try to deploy, for example, if I give you a Docker image, so what are the things that you will be deployed. Like, if I, if you want to create a service, I give you a Docker image. So what are the things that you will be looking if, before you want to deploy

Unknown Speaker 06:53

for that?

Speaker 3 06:56

Yes. So basically, if you have like, a Docker, like, like that you want to like report. So first thing like you want to prepare like is like Docker image. It's usually true like elastic Container Registry, where, like you push your Docker images to some kind of like artifact, and then after that you would like, define like your Kubernetes deployment strategy by, like, creating the deployment yam file, right? So it's basically, you know, contents, your like, API, version, the metadata, what kind of replicas, what template you want to use, and also like, what service you want to run, right? So you can specify what selectors you have, like and what port you have and what kind of like load balancer you have. And then you can check like the Kubernetes cluster, you know that is like running and like also verify if the like the cluster is available so and after that like, you can apply that Kubernetes using like, deployment descriptors, or like a district, like a deployment YAML file and like, then you can check, you know, if the bots are, like running or not by just doing, like a cubicle, like get bot commands.

Speaker 2 08:16

Okay, this is good. So now I want to ask a follow up question on this one. So if I want to deploy service, for example, maybe a API service, okay? And if and I have a code and I gave that Docker image for you, okay, so in order to deploy that code and make sure, like, the users able to access that code, APIs. So what are the things that you need? Like, for example, when I say API means like it has to get the users will get a request, like a get request, all right? And your code will be looking for the routes, as you know, right? I think how API works. We can each class server is running, and if based upon the route, it will go, trigger that API, specific API, and get the data from the database and submit back. So this is the the solution, right? I have that API code and assume like database is already there and you have a connectivity from your Kubernetes cluster to the database, right and for now, if I, if I give you a

Docker image, so what are the components that you Need to create in the Kubernetes Okay, let me say Do you understand my question? Like, otherwise I can, I can give you more details, like, if you, if you want,

Speaker 3 09:53

yeah, sure, yeah, sure. I think I'm following on. So, okay,

Unknown Speaker 09:58

definitely. So,

Speaker 3 10:02

you know, so if you already have a Docker image, so for an API service, you already have like Kubernetes clusters and like database exists. So I think the first thing you need to run like the deployment, like which runs the app, API, application pods, right? So, like, like, these are, like, where your application, you know, are, like, kind of posted, and then you like, need to expose your API for internal, like, external access, right? So, so you know, like, if you have, like, like, two sets of service like facing like, internally or externally. Externally, you can basically use a service selector like, within like that service YAML file with the metadata like of that service to kind of like, like expose what ports you want your application to be, kind of like, exposed on, and what kind like, what kind of like load balancer or like cluster IP, like that you want to, like, expose upon, like so if you have like internal, you can do like a cluster IP. If you have like an external, you can use like load balancer. Now, once that is done, you would like, basically need to, like, set up some kind of like Ingress rules for networking to like, kind of make sure that, you know, like certain domains are like, allowed to make call to these APIs. And then, like, you also have like, some kind of like config map that stores like, like, non sensitive values and secrets. And I think like, those are like the basic ones that that you need. And there would be like, like also like network policies, like VPN and like subnets that also, like take care of, like the ingress and like the egress rules. But like, those are like primary ones. So like the like load balancers, the port, the application service itself, the container like which has a like, a reference to like those particular app like you might also want to go with like, the readiness, or like the liveliness, like brought to make sure your like app is up and running. And like, which is like, also, like, ready for like, like, severe traffic. So,

Speaker 2 12:35

yeah, so I'm not sure if you covered that ingress routes also, because that's the one which will be exposing that ingress route. So whenever the request comes in, that will be the your service endpoint, right? So the request comes to that, users will be using that as a your endpoint ingress route, so that will give you the domain and everything, right? And then, once the request comes to the ingress route, then the ingress route will based upon that, whatever the service that you configure, that ingress route will pass to that service, and whatever the service configuration you pass that service will be tied to that parts. So that request will go to from service service to the route, to the ports. That's where your flask app is running, right, and that's where the code is, yeah, I think you covered. You got entry and yeah, we need, like, a config maps. We need a deployment file. We need a service, service file, service dot YAML file. And we need a ingress.in, English, Ingress routes, dot YAML file. So we need all these things, and once you put it in that, helm templates, right? Like you need to templatize these things, because you should be passing all these in the values, dot yaml, right? Yeah, I think good. Like you have. So do you know, like, what kind of services you can be using in the communities or, or just let me know, like, if you use work on that service or different types of services. Or do you just have the knowledge?

Speaker 3 14:17

Yes, for that, you know, I think, like, some of the services that like, like that you can use, include, like, your default cluster, I know IP service, and I think there is also, like a note, Port service or something like that I have Not, like worked with too much.

Speaker 2 14:42

Then, like three to four different services, like the cluster IP, entire port based, right? Like node port, so all those things, yeah, I think that's fine. Yeah. So you do I say, you said, like you worked on the REST APIs in the back end, isn't it? Yes. Okay, so let's discuss little bit on the APIs and the design the REST APIs and design. Let me know if you are comfortable in that topic, right? So for example, like if I ask you to maybe design for APS, REST APIs for online bookstore. So what are the entities? What are the entities that you will be thinking of first, before you deep dive into the APIs, what are the entities, the main entities that you might be thinking of for the online bookstore? Online bookstore. I Yeah? So,

Speaker 3 15:44

for the online bookstore,

Unknown Speaker 15:48

yeah, the first thing that I would want is, like,

Speaker 3 15:52

there has to be a user entity, you know, that basically represents like any customer or admin or like employees within like the bookstore, right? So, like, even though it's not like a physical store and online bookstore, we like, still need some kind of, maybe, admin and manager to, like, like, have access for like, a like a role as, like, a role based and like, then you'll also have your like, customer, or like, who will be basically your primary customers. And then you will have like a book entity, which will like, kind of like store, like the metadata about the book. And this book entity will have like, stuff like, like the title, author, price, and maybe also like, how many stock, also, like, and so on, right? So, so that would be one, and you might want to like, also, like, categorize those books. So you may create like, separate, you know, Category table, you know, that would have some kind of like ID, Name and Description, you know. And then you will have like entities, like relating to, like the customer orders, right? So, like foreign, you know, example, you would have like, like order table that represents like, any orders that like the customer might have, like placed. And you would also have like Order, order item table which, like tracks books in that order, like with like the quantity and price, like you would have like a also, like maybe a card table that stores maybe temporary card for ordering and like you will also have like A payment table for like, tracking all the like payment details. And this order table would primarily, you know, be used for like, viewing, like, the order history and like, like, like, all the item will have, like, many, many like, maybe like, a many to many like relation, like, between like order and books. So, I think, yeah, those would be like, the primary ones that like, should, like, get us started with that.

Speaker 2 18:07

Okay, so you're saying like, books users is the one thing you need. There's an entity, and the orders is the one thing. And do you say like about anything other one payment is the one thing, right? Yeah. And how about the books? Like, the books is other thing. Like, we need to know, what are the books? Are there? So all these things, like, there's

Speaker 3 18:34

also like book which has, like metadata after book and like a category table, like to store, like the book categories as well,

Speaker 2 18:46

yeah, and yeah. So that is one thing, yeah. I think categories, books, users and orders, or something like the very high level of entities that we need to think, right? Okay, so when coming, okay, let's take a one entity of this one, and then we'll design later. We'll take just a couple of entities, and then we'll go from there. Okay, so for example, if I if, let's take a books as an entity, and can you give me what kind of endpoints that what? Testicle endpoint should we have for the books?

Speaker 3 19:25

So, for the books entity, like, like, so far, like, we primarily want like to interact with books, right? So, so yeah, we should be like, able to like, retrieve the book information. And we should be able to like, update, you know, we should be able to maybe, like, add new book. And we should like, maybe able to like, also remove a book. So like, first thing that I would think of is like, get a like a get endpoint of like books, like, which gets, you know, like, a list of all books like. It would be like, like, paginated, and like, have like searching and like filtering options. And this will be like, like, open to everyone, right? So everyone can get all the book information in any order they want. And like can add like filtering on it so that, you know, would be like the like the first book endpoint, similar to the all the books. And we'll also have like a fine, like fine books by ID, like, which retrieves, you know, one particular book by maybe unique identifier. And this would be like open to public as well. And like, similarly, we may, like, expose, like a get books by maybe category endpoint. So like it will be like slash book, slash categories and slash ID. Like we should, like, retrieve, like all the book by sorting categories. So like this will be like primary endpoint that your general customer would be like able to view and for like internal admin users, like we would like to like post endpoints to like, basically add maybe like a new book to the inventory, right? So it would be like, like a post endpoint for like slash books, you know that will take like a request body, then that will create, like a new book entity. And like we'll also like have a put in like point to like update a book by ID. So it would be like slash book, slash ID, and that will be like an admin operation to maybe update the book title, or maybe the price of the book, or maybe also, like quantity available. So and they will also have, have to have, like a delete endpoint for like admins, like just to see if we want to, know, get rid of like certain books within our inventory. So, yeah, that would be, like, some of the endpoints, API, endpoints for books. So,

Speaker 2 22:07

did you mention about the put or patch input? Yeah?

Unknown Speaker 22:11

So, yeah, I did mention about,

Speaker 2 22:15

okay, so amazing. Yeah, I think you covered it like, so basically, like, we need to have, like, a get all the card operations, right? Let the Get, get operate, get for book or list of books, or specific to a book ID, specific book ID, right? And and post, put a delete or something that we needed. So when you say it's a filtering, you mentioned as a filtering is option, right? Like, so, how can you implement that filtering part in the APIs?

Speaker 3 22:48

So, so the filtering, like, it would be like a query parameter, right? So, like, anything that is like optional, like prams would like, usually go as a, like a query parameter. So you will have like, your books question mark, you know, search equals, you know, right? So this would be like, basically be like, searching for books that are named like Harry like, for example, Harry Potter and in series. So you do that particular search item, and then you can, like, also implement it in your like service layer to filter like based on that particular name as

Unknown Speaker 23:26

well. Okay, good, yeah, that's good.

Speaker 2 23:30

And what will be the response format should look like if I asked for get a books of this category, science and fiction category right? Or fantasy category, something like that. What will be the response should look like in your in your API response

Unknown Speaker 23:51

for the response so,

Speaker 3 23:54

so I think the API response for like, get a book by category would be like JSON, so which will primarily have like the two keys, it would be like category, which would like, be like the name of the category, and that we are looking like by ID, right? And it will have book keys that will have like an array of books within that category. So for example, like, if you send like a request by slash books like slash categories, slash one, then the response will say something like, category fiction, and then books with like a list of books within that category.

Speaker 2 24:39

Okay, so basically, it should be the array of the books, right, like the response should be in the array of books, and each book should be having their own. It should include the price of each book, that way users will know what is the price of each this book, right, and the ID author, not all other generic information. But the main thing is, like, price is one thing, and it might require the stock also, like, how many books are available, so that way user can know, like, how many they can order something like that. Yeah, it will, like,

Speaker 3 25:15

contain, like, entire books, DTO, like, which is, like data transfer object, which are the information that we would want the users to be? Basically,

Speaker 2 25:25

yes, nice. So coming back like, I think you already touched this point, like the users and the users will, everybody will have access to the get books and get books by specific category, and only few people should have access to the post the books or put the books, put slash or batch books, right? So how can you, how can you validate it that users privileges when you are trying to do

Unknown Speaker 25:58

that in the API, user privileges, so

Speaker 2 26:04

how can you based on it, based on the API request? How can you identify, Okay, this, this is this user has the admin privileges. This user has a don't have admin privileges before you continue to perform the operations. Yeah?

Speaker 3 26:18

For that, like, we need to get, you know, the spring security to have like a role based like access control, right? So first thing is, like, before you do any of those operation that might restrict like user would like, first you need to like, sign in. So for that, we can just use like an identity provider, like, you know, like, like ID or something like that we might have, and once that the user is authenticated, right? So, like, they will do, like a, like a username and password match, and then from that response, like, it would be like a JWT token, you know, that would be injected, like, to all the request that comes into this, like particular service. So now, like this author, like authorization, like Bear token that is, like, part of your like, a header would be validated by like, a spring security filter. And then, like here, like we would do, like basic validation, like, like, if the token is expired and if the signature is, you know, tampered with. And you know, like, all the haters are okay. And then, like, from the payload of this token, we would like, like, extract the role of that particular user from the risk controller methods. And like, we would like, like, annotated using like, pre authorized annotation to give like a list of, you know, rules that that would be available to interact, like, with the post, or like boot or also, like delete endpoint in that way. So, so yeah, we will have like a control on, like, who can actually like, kind of like interact with, like, those kind of like restricted APIs,

Speaker 2 28:02

okay, yeah, got it. Excellent. So the Okay, moving to the next one, right. Next topic is like, Okay, we covered the books. Is a one entity, right? So can we get similar endpoints? Like, what will be the endpoints for the orders? For the

Unknown Speaker 28:21

orders? So, yeah,

Speaker 3 28:25

let me think. So, yeah, for orders like, we basically, like, have like, permission to make, you know, purchases like, like track items and status, also like payments and so on, right? So, like, the first thing that we like we would want to create is an like order. So, like, whenever you know, like a restful world, world, you know you have, like, something that relates to like, like a creation, we usually go with, like a post endpoint, so foreign, like authenticated user, we would like create a like a post endpoint so that they can kind of like place an order with books and quantities, right? So, like, for example, in a like a post order, like they will create like a request with the request body of the user ID and the items like they want, and so like the payment method they want, and like other small details, like, you know what, what can be, like the shipping addresses, and so so that that would basically be, like the simplest post endpoint to like this new order. And like that is like the primary one, and then the other would be like, get order by ID, right? So, for example, like, like, you have, like, like a get, slash order, slash one, so like, it will, like, retrieve, like, an order with ID one and like, see, like a status, or like, whatever you want, right? So and like this will be like, protected, because no like, not everyone, like, should be able to view like everyone, everyone's order, right? So that is like your other end point. We should, like, also get, like, all users, like, sorry, get all orders for like, a particular user. So, like you do something like, like slash order, like slash user, slash ID, we should, like, retrieve like an order for that user and and for like admin, like we should be, like, able to view like all the like, the orders within like our system. So, yeah, like, we will, like, only expose, you know, like, like, this last order for admin rights and and like, we can cancel an order, like an admin like we should be, like, in, like, a delete entry point, so endpoint, so we can also, like, update the order status, like, of an endpoint, you know, like, which should be, like, just an admin endpoint. So, yeah, those would be like to post time. Like that would be like the, sorry, like the PUT and DELETE operation. So yeah, like, those would be like the like, kind of like different endpoints. So, yeah, we have like, post endpoint for create, get like order ID, to get like an order ID, and I get all order for like end user and the like also like the admin operations.

Speaker 2 31:37

Yeah, that's like previously you mentioned that get orders should be only for admin, but I think later you included the users also, because as a user, they should also know what are the orders they placed. It it's not just admin, right? So, yeah, that's what I was about to correct that one, but you included that one. Good, nice. So then, once we, I think now we came to the orders, right? Like, now the payments. Now, let's go to with the payments, right? Like, I think we don't have enough time to cover the other areas, but what, usually, what will be the challenges that you will see in the payment process, it is processing a payment, what challenges, what challenges, mostly you will see in your design or in the APIs that you should include address those issues. So,

Speaker 3 32:37

like, usually, you know, like payment, like it, like, it goes through, like a, like a payment gateway, right? So, like, it would be like, maybe, like a, like Stripe or other, you know, extra like gateway that we would like leverage for the payment system. So, like, I think the first thing, like, it would be like payment failure and like, error handling, right? So, like, you know, payments can, like, fail due to maybe, like insufficient maybe funds, or also, like expired cards or incurring in the you know, details, like provided. So we should probably, you know, handle these errors, like, properly when we can, like, allow users with different payment method and, like, provide like, clear information on like, like white fields. And we should probably, like, have like, a good transaction around that, right? Because we don't want to, like, succeed like, certain part of the flow, like the when the payment fails. So like, it should be like one or not nothing. So one of the biggest challenges, like, is making sure, like you have, like proper integration with the systems, and also you deal with like proper like frauds and like, also, like chargebacks and other things would also be like a security and also like compliance issue. So as any like, business pursue, like, pushing any payments like, they must have, like, a proper security and also, like secure environment. So, like, because, like, it is like a cardholder data, like, which needs to be, like, true, like, it is like, sensitive medium. So, so yeah, like, those would be like the top in the primary challenges like that, I see,

Speaker 2 34:25

okay, yeah, so, yeah. Good thing. Good point. You touch the security part. Like, we should never store those raw details, the credit card details, or payment details, in our system. We should integrate with the gateway payment gateway methods. So how are you handling? How will you handle the transaction failures? Scenario?

Unknown Speaker 34:53

Transaction failure, okay, so to handle like, the

Unknown Speaker 34:57

transaction failure,

Unknown Speaker 34:58

right? So, basically like,

Speaker 3 35:01

so basically, seems to like, we'll be using like, like, spring based applications, like, like, spring provides, like, like, a really good transaction management so, so like, you can just, you know, Mark like methods that needs to be like, completed, completed as, like, a single transaction with, like, like a single transaction annotation, like, so, you know, like, in the case of insufficient fund, or like, anything like that, like, we need to, like, wrap that around, like a, like a, like a spring based transaction, to handle the transaction, right? So, like, you have, like, a, like, a pretty good control on like, like, how like, the provocation would look like and like, what should happen when you know, like, what errors should like have? Should you have? Like, transactions sometimes, like, there might be temporary failures. So like, depending upon like, on the like, SLA, you can like, add, like, maybe like retry mechanism, like, for example, if there is a gateway, like timeout, or, like, some kind of, you know, API error, you can like, do an like, an exponential, like backup retry. But you know, in the case of other failures, you know, like insufficient fund or expired like card like you should not, like, do retries, right? So those would be, like, primary ways like to handle transactions. And so, like you just, like, roll those transaction back in case, like, some of these error errors happen. So,

Speaker 2 36:41

okay, yeah, good. So, yeah, you should implement retries in few scenarios. For example, there is a network outage from the gateway, payment gateway. You don't get any response from the payment gateway those times, you might need to implement few retry mechanism. But in other scenarios, like, if you don't have enough sufficient funds, you don't need to retry. So there are few scenarios where you can retry, yes, but yeah, I think you covered that. So the other important thing is, like, concurrency, you have to maintain Right? Like, make sure, like, the stock updates happen automatically to prevent overselling, right? Like, whenever there is a transaction happen, you make sure, like, the stock is also updated based upon that. So otherwise, like, you will be showing the same amount of stock, and other users will be over selling it, and later your transaction will be failed. So in order to avoid those things, you have to make sure, like, once the transaction is done, you have to update those things. You have to take care of all these. It's not within that API, but we have to consider all these things to make sure, like, processing, payment processing, is going through without any failures, right? Okay, yeah, this is good. I think I have, like, maybe we'll talk in another, like, three minutes or so, and then I will stop for the questions. But maybe the this will be the last question. Like, let's discuss about the scalability part. So if we want to, if we have millions of users coming in. So how can you scale this? Your APIs?

Speaker 3 38:27

Okay, scaling the APIs. Okay. So, yeah, for scaling. So basically, like, we would have, like, like these application like containerized and like deployed to, like AWS ECS, like ECS like, similar to like Kubernetes, like, is an orchestration platform. So we would need to, like, like, be handling, like millions of user to like scale dynamics and like main like, the main thing here is, like, you'd have Auto Scaling group, right? So you would basically, like, be like, horizontally, like, scale the service, like service or tax, you know, based on like sorting metrics. So if the CPU utilized on like crosses, you know, 50% maybe we could like, spin new instances, similar to if like memory uses maybe goes over 50% we would like expand like the horizontal like instances. And apart from these like two, primary metric is like, we would also look for maybe like a traffic pattern, like so if there is like, a higher traffic, maybe a given point of time, then we will, like, scale up and and also like, like, we'll also make sure that we have like a like load balancer before these services, so that like, all the traffic is, like, spread out, like, properly against all the instances that we have. And, you know, one of the things that, like we need to consider is like adding, like caching, right? So for those like book data, so, you know, like, whenever you are, like, retrieving book details, like, for the same data, you can, like, always, like COVID, like caching. So like, it helps you to, like, improving the performance. And you can also, like, maybe incorporate some kind of, like asynchronous processing to, you know, like, incorporate, like a payment gateway, gateways and like email notification and like the service to like, basically have like, a separate like process, like running and, yeah, I think those would be, like, the primary ways you can, like, scale your service, yes.

Speaker 2 40:48

So yes, scaling at the API level is good. Like you're you covered most of the points at the API level, and the caching will help you at some points to improve the response time. And there are few things that you should consider on the database side, also when you try to look lookups. Maybe, for example, if more queries are keep hitting the database, you should, for example, the lookups right like, at that time, you need to consider, like, how can you improve the performance on the database lookups? Like, but one of the example is, like, you can create indexes on a specific table. That way database will perform response back. I know you taking care at the API and load balancers. Everything is good, but your if the back end database is still having the issues with the response back then your APIs will be nothing to do with your thing, right? Like, it has to wait, yeah,

Speaker 3 41:54

also, like, on the data, yes, like, look up, you know, like, like, we like, we would probably add indexes, like you mentioned like, but like, they also had like option to like, maybe like, shard your database and then also

Speaker 2 42:08

reading. There are multiple, yeah, there are multiple techniques where you can improve the performance on the database side. Yeah, sharding is one more thing to improve the performance. So, yeah, I think performance, API performance, you already mentioned, like the pagination is one thing, right? Like, I think you mentioned like previously at the beginning of the discussion. But pagination is also comes under the improving the performance, because instead of giving like 1000s of records. At one thing, you can pass to Netflix. So that way it will be eight, like 50 books, or 10 books like that. It will help the response, right? Yeah. So that is, yeah. I think these are all, like good points, and I think we don't have much time. I think we have three minutes. But the other thing I just want to, I will let you know, like, what are the other thing is, like, what are the design patterns that you need to include in the API? In the API design like, what are the good design patterns I will, I will tell that one, just for your knowledge and experience, is one thing, but that we can, we need to do like v1 and v2 like that, just in case, if you want to revert back, it is easy for us to pointing to the specific question, right? That is one thing. And the micro services, I think you already discussed, that we have to, instead of going to monolith one, right? Like break down into micro services that way we can the design should be more flexible, right? Instead of monolith and Graph QL and web sockets, are the other design patterns where you can include for real time status updates. Okay? So, yeah, I think those are the design patterns that you should include when you are trying to implement APIs, right? At least in designing the APIs. Okay? So, yeah, I think it's all good. So yeah, now tell me, like, Do you have any questions on the role, or you have any questions for me?

Speaker 3 44:22

Yeah. So first of all, thank you so much for your time today. It was really nice talking to you. You know, my experience and like, talking about this bookstore. So the first thing like that I would like to ask is, like, apart from, like, your coding, like, how much do developers get involved in, like, similar sessions, like, you know, design and, you know, like, architectural discussions.

Speaker 2 44:46

Yeah. So this is how, even at least in Adobe, or at least in our team. So whenever we work with something we designed right like, we will go through present that in the reviews meeting, the design reviews meeting, so we're working on that specific design, they will present that into the meeting. And people can ask, try to clarify their questions on why we need to go with this component. We can use some other component, those kind of things. Can we? People can ask questions and get clarification. Even that way, everybody will get understand, like, why we are using only this technology, and why we are going with this component, like that. So that's how you will you can involve and you can also review the designs, for example, like, once we have a design document is there, we will share across the team, and you can, you can put your questions, you can review the document, and you can put your questions, or you can get your any, any clarifications, just like that. So that's how you can improve your knowledge and also understand, like, based upon the design patterns. What are the good things? And obviously, anyone can contribute back to the design through the reviews.

Speaker 3 46:12

Yeah. Thank you so much for that information and like, something else that I would like to ask is like, what do you like, like most about, like, working for Adobe.

Speaker 2 46:25

So Adobe is, like it said, as you know, Adobe is more of, like, creative company, so it has a lot of stuff, even if you're interested in other areas also, like AI and Mel, is some other area where Adobe is more focusing on so you can learn new things and you can contribute back. Like it's not just like you have to work only in whatever the thing that you got it in your project. You can look around. You can talk to the people. If you can learn new technologies and you can contribute back to those teams, that way, you will understand, you will understand whatever you are learning. It's not just for learning purpose. You can contribute back and also make bring the value.

Unknown Speaker 47:18

That sounds really great, yeah, yeah. So

Unknown Speaker 47:21

yeah. And, okay, yeah,

Speaker 1 47:25

I think yeah. That's all the questions that I have at the moment.

Unknown Speaker 47:29

Yeah. And thank you so much for your time today. Sure,

Speaker 2 47:32

thank you, and nice talking to you as well. And wish you good luck.

Unknown Speaker 47:37

Thank you so much. Take care.

Unknown Speaker 47:39

Bye.

Unknown Speaker 47:48

Listen to me

Unknown Speaker 47:59

also.

Ronaj-Adobe-Final-Srv

Speaker 1 00:00

It for that, right? I think what product you're working on, and then which phase of the project,

Unknown Speaker 00:11

yeah, yeah. Thank you so much for that.

Speaker 2 00:15

And yeah, I guess something else that is like, like, what are the opportunities for like growth that that developers have in like Adobe.

Speaker 1 00:31

It can be from different perspective, right, from the job code, right? There are different levels of job code, right? So you can when you grow up, right? You might be promoted to the next level and also from the work, right? We work on a lot of different things, right? The thing you're working on might not be same as the thing you're working on next year. Next year, you might work on something different and a new right? So you will always have opportunities to learn new things, and also you can work together with different people, right? You interviewed with Chandra and ranch rather than you might also work with other engineers. Yeah.

Unknown Speaker 01:18

Thank you for that.

Speaker 2 01:21

And I guess, like something else is like, like, in, like, in, in a timeline of like, six months or a year down the road. Like, why would be something like that? You said, like, I would have, like, need to, like accomplish in order to like, be like a successful on this team.

Speaker 1 01:47

So I think we measure success, but deliver value right depends on what value you deliver, right. So you work on many different things, like design, implementation, coding, and eventually you deliver something, maybe a new service, a new feature about fixing what demo or some documentation, and those will be used by somebody, right? And so, what will matter? So what value your deliverables have, right? That is the important thing, and especially for new engineers, right? New engineers. And also, there are some other perspectives, right? For example, you can also contribute to the long term of the architecture. Right? We're building new things. That's because we designed this new system, right? We brainstormed and had the requirements, and we did that architecture, and during that process, we made many decisions, right, what requirements to implement, what not to implement, what kind of architecture to use and what technologies to choose, right? So in the future, you can also contribute to that, right? We're also going to make a lot of decisions in the future, right? You can also contribute to that and make sure we make the right decisions.

Speaker 2 03:22

Yeah, that sounds interesting. Thank you so much for that information. And, like, Yeah, I think those are some of the questions that I have. And, yeah, I think I would also like to, like, like, did you have, like, any questions for me?

Speaker 1 03:41

Yeah, I'm not going to interview anymore, right? Because when I've talked, I have, I have some questions. So your partner, location is Maryland, right? Yes, and you're okay for the location, right? If you can get an offer, yes, your company, OPT. Right, right? And is this your first year or second year?

Speaker 2 04:08

Like, off my OBT, yes, yeah, I'm on my extinction right

Unknown Speaker 04:17

now, first year or something. I suppose there are three years, right?

Unknown Speaker 04:20

Yeah, I'm on my like, first year,

Unknown Speaker 04:23

first year, okay,

Speaker 1 04:27

okay, and we have several other candidates to interview. So after we interview all the candidates,

Unknown Speaker 04:36

and we will decide what the next step

Unknown Speaker 04:40

so I will let you know the result, probably

Speaker 1 04:46

early next week, right? We want to complete all the interviews by this week, so probably early next week, I can let you know the result, and you can also ping me on LinkedIn if you have any questions.

Unknown Speaker 04:59

Okay, yeah, sure, definitely, yeah, sure.

Unknown Speaker 05:03

Okay, anything else you want to discuss? I

Speaker 2 05:08

think that's all for right right now. And, yeah, I don't think, I don't, I don't think I have anything else right now in my mind.

Speaker 1 05:16

Okay, okay, that's good. Okay, thank you.

Speaker 2 05:20

Yeah, thank you so much. Yeah, I really appreciate it. Thank you.

Ronaj-Final-Adobe-Srv

Unknown Speaker 00:00

Sure that we

Speaker 1 00:04

we check the permissions before user is allowed to access the APIs, right? And then you need to limit the request for anything like 10 requests per second for Admin API and 100 requests per second for user API, right? So what features will use in Spring Boot to implement this logic?

Unknown Speaker 00:33

So for

Speaker 2 00:40

so like, as far as for the like, the first one like that that you like mentioned, as far as like securing your application, you can basically use like a spring security So, so far like, like, you just add like a Spring security dependency, like, for the authentication and authorization and like we would use like zwt based authentication, right? So like you would have, like, your base JWT, util like, which would, like, extract your like usernames and like it would extract roles, claims, or any other like, anything that is attached to like, maybe to that role. So you can use, basically, use that. And we can also, like, basically, you would like, create once, like, once per request filter, and then, like here, you would basically do like an, like an extraction of like your authentication token. So from the token you would extract your authentication header or username and role. And then basically you would like define like a like a security configure like beam, which basically is like used to authorized like authorized users, right? So from like this security config, you enable your like web security and then on the like being you would like, modify the like HTTP security object to like, pass like, basically make sure like like the user is like authenticated, and then you like authorize the like the request when like the request matches like the slash, slash, admin slash, you know API to have like authority of admins and on the like the other API like the user, to Have like authority of the user. So, like this would like, take care of making like, sure that that your API are like, like, restricted, right? So, so making sure everything is like authenticated, and then only like those roles can can access those particular users. Now, now for red limiting, you probably need to work with, like maybe third party library. I think so. So I think, like, there is, like a Spring Boot, like pocket for like library that you can, like work with, but we have been handling like red limiting primarily through like API gateway, like on AWS. So I'm not primarily sure the how like directly like Spring Boot dependency can handle rate

Speaker 1 03:32

limiting, okay, so when you implement rate limiting, right? So you're saying that there is no feature that you're aware of, right? But if you use API gateway, what would be the user's experience in that case, if the user is rate limited.

Speaker 2 03:51

So if you are like, like, like, using rate limiting, like on the Gateway itself. So I think like, like it depends on like, how like it is configured. Like, if you handle like, if you handle in like, you if you like, if your gateway itself will be like, like, control the and pass the request and like, I think it should still be like, like, a similar, like experience, but yeah, they would be like, like, able to see like that, like, throttle like, instead, instead of like, seeing any or like, like, Like, like another like, like Mrs. Response back, because usually they are like, these are like, Gateway layer handler, those like four to nine responses and like, same, like another response, right?

Speaker 1 04:55

So when the client sees this four to nine, what is it supposed to do? So,

Speaker 2 05:07

like, I mean, it should, like, should be like, like, a wait, like, like, wait for like, certain period of time and then try again later, but, but like, user would not be like, able to like, translate a response code, like to that, right? So like us developers, like, can know the meaning, but I think we should like, within the API gateway, like, we can add a response, like, manipulation that takes in like any like, 249, and like and responsive response to like a meaningful response back

Speaker 1 05:50

when API gateway sends four to nine with retry after clients are intelligent enough to retry after that period, right? So it will wait for that period until and once it is elapsed, it will retry the request. So user won't be transparent in that case, unless it is too much delayed, if it is like five seconds, 10 seconds, user should not be seeing much of the difference in the application response, but the client would be able to handle that. So it's not once the server sends the four to nine with retry option, the clients would retry after that period and renders the web page. Okay,

Speaker 1 06:34

so I say that you worked in Kafka too, right? What is your use case? How did you use Kafka in that project.

Speaker 2 06:44

So for, for that like, we slide, we, like, started incorporating Kafka for, like, some of our workflow, like, so our application was like, kind of getting hammered with, like, notification service calls and like our service was failing, and we basically just, like stripped out this like notification flow by, like publishing like messages. Like to like a Kafka topic, and then like a consumer app would read those like messages, and then, like all these notification service like to send like emails to the users. So, like, already users, they were relying on, like a notification from our service, but like the notification service would like fail, and then that would cause, like a cascading, like a failure in our service. So yeah, basically, like, we incorporated that. So we basically, like stripped out this process of like, like, sending email notification to So,

Speaker 1 07:50

so you presume you're receiving some notifications in Kafka topic, and your application or consumer, you are a consumer who is consuming these notifications and acting on that? Is that correct? If I understood that? Yes, yeah. Okay, so, so let's say multiple users are sending the notification for email. I mean, I presume your application is doing some email notification to those users, right? So those events are coming in from the topic, and you are a consumer who is conserving it. But how do, how does?

Unknown Speaker 08:26

How do you send emails

Speaker 1 08:30

based on the timestamp you received? Right? So there may be hundreds of events that are happening at the same time. How a consumer will handle all these requests and send concurrently to all the users who have

Unknown Speaker 08:47

who are part of the notification system.

Unknown Speaker 08:52

So for the consumer, yeah, so

Unknown Speaker 08:57

basically, like we would define,

Speaker 2 09:02

like email notification group and and like, like, I think we would

Unknown Speaker 09:10

let me see So,

Speaker 2 09:12

yeah, I think there should be like, some kind of, you know, like, aggregation logic like, rather than like, sending it like, right after like like order, because, like, usually the email service are the ones like that typically fails a lot. So we basically consume like a message from like the Kafka topic, and then basically are like like partition based on like certain Einstein groups, and each of those partitioning like messages are like, stored, right? So you have, like, you have a guarantee within a partition, and then it would be processing like, that particular order. In case of delay in processing, we might want to do like, want to do some kind of delay, and in case of failure, we just use a like, a dead letter queue to like, boost this message to like, dead letter queue, and then like, we try it after a later time.

Speaker 1 10:18

So you said you were partitioning by date is the correct

Speaker 2 10:22

part date? Yeah, I think like that, that would be like some kind of, like a time timestamp group,

Speaker 1 10:33

so as the events are coming in, right? So there may be hundreds of users logging into your system, and probably you need to are signing up, for example, right? So they are signing up into the system at the same time. And how will

Unknown Speaker 10:47

the consumer as a consumer, keep the

Speaker 1 10:51

who logged in, or who signed up first? And they should be sent a notification, email notification first, and the second user should be sent the second time. And so how will you maintain this order? So, to maintain the order,

Unknown Speaker 11:11

so, I mean,

Speaker 2 11:15

like the like ordering of like, like processing like the Kafka messages is like, only, like, guaranteed, only, like, on, like, like, the partitions, right? So,

Unknown Speaker 11:29

like, like, we may partition like, differently.

Speaker 2 11:33

So, like, maybe we like, kind of like, like, assign user like, user ID. So, or you can, like you use some kind of like, like, like a hash based partitioning, because, like, you cannot guarantee like ordering, like operas partitions. So I think like either like partitioning by like user ID or some kind of like, like like, hashing mechanism, because, like, even notification some, like, sometimes can be little like, delayed, right? So, like, that was not our particular use case, but like, like, if you have, like, a very strict use case of, like, having, like, ordering them, then I think you need to do like a partitioning by user ID or something.

Speaker 1 12:28

So now we partitioned by user ID and you got the notification consumer is concerned, but consumer has to go through a restart or it crashed for certain reason, right? How will you not process or process at one time, right? So the events that you have processed previously, the consumer should not be processing the same again. How do you keep that maintained?

Speaker 2 12:55

So, yeah, so in that case, in case of maybe the consumer crashes. So basically, we want to, like, make sure, like, there is a like a proper, proper call offset management, like, within the consumer. Because, like, you need to make sure, like, you have like a have a like a comment offset properly and like, then like you should also like, have like, a proper offset management, like using like, like a manual offset, like commit, Kafka. Like Kafka will like, auto commit, like, offset by default. And like we should not like, we should not be like, with like, which is not reliable, like, for crash, right? So instead of that, we would like comment like offset manual after like, the message like is like, fully processed. So like, in that case, offer like a like completion of message processing. You would like commit, which is basically like, like inserts, no duplicate processing. And then there is also like, a concept of like item, like port and processing like, which basically tracks the process message like in the database, which, like, adds a layer of an extra in architecture. But like that might be something like you want to consider it in in your system. You need, it needs to be that highly reliable, and you know, like your system might crash. So I think, like the proper like offset management should, like, take care of, like, the problem that that you're mentioning, right?

Speaker 1 14:39

So you write the commit offset into some database and when, when the new consumer comes up, it will read the data base and sees until what events it has processed, and it will read from there. Is that correct? Yeah.

Speaker 2 15:01

Like, that's something that like that would like it would.

Speaker 1 15:13

So what is your experience on DynamoDB, for how much familiar you are with DynamoDB,

Speaker 2 15:21

on DynamoDB. So, yeah, I worked, like, a little bit with DynamoDB, like we stored, like the entitlement session part for the Experience API.

Speaker 1 15:41

What's the major functionality of DynamoDB? Why does it, why is it popular, but in what scenarios we should be using DynamoDB?

Speaker 2 15:51

So talking about like the major functionality, so like it is like Fauci, like it is non like relational which is, like, really available to it, which is very available, and like, scalable, and it does have, like, a feature of, like, high consistency, if you need it, like, so, you know, basically, like stores, like a key value based, non relational data with, Like, no need for any, like infrastructure management, because, like, AWS handles that, and basically so, yes, features do like, like, restore table at a specific time for like, a month or something, and then, like, then, then you have, like, an option to, like, enable some kind of, like AC transaction, if required, on like, a high consistency mode. So if you want, like a serverless application, you know, like that has like some kind of real time streaming, or like inventory management, you can use, like dynamic TV, so like, basically you have your like partitioning key which supports like range queue queries and like order data based on based on the partition key and sort key, and then, like you have like a flexible schema like to work on top of that, and that's why.

Speaker 1 17:25

So let's say you built your application. Let's go back to the previous example. Our back system is there, and you see there is a need for that, your application to be expanded to multi region, right? Multiple regions, or multiple GEOS, geos. So how will you do that? Yes, so,

Speaker 2 17:51

yeah, like so in order, like so, in order to scale the application to like multiple reasons, like reasons. So I think you probably need to, like, like, firstly, like, consider about the data. Like, usually you'd have like, your primary database, like, on one reason, and then have like a read replicas on like the other one. So, like, the first thing would be like that, right? So making sure your data is available, like on like both region and then you have like a multi like region deployment architecture, so you can have like active, in case of, like a very high traffic and TPS every application, like, like a banking application. Or you can have like an active passive for like a, like a disaster recovery sessions, but for like a user case, like, we'll just consider the like, the active, when, whether, where the both region are active and like, have like a global, like load balancing to, basically, you know, route the traffic to, like a geo routing, like a load balancer. Now basically, like you like you would have like a multi region application on like both regions, like your like US East or like US West. And you would like create your like auto scaling groups like load balancer and your like services. And then you would have your like, like two different, you know, database on like different region. And then you would like have like, like your traffic routing through, like AWS route 53 where you would like, basically have like, a, like, a weighted route on, like, those services. So, like, if you have, like, any kind of, like, static data, you can, like, use, like Cloud front, like, cache those data.

Speaker 1 19:56

So now you handle the jslb is there, routing is there and the application is running in multi region some, sometimes, if the database is not accessible, right? So you have, I hope you went with a standalone database per region, or did you go with a distributed database? In this case, what did you

Speaker 2 20:19

choose? Like, we sorry, so we have, like a, like a distributed database. So

Speaker 1 20:31

distributed database, how many read replicas and how many right right now, so you

Speaker 2 20:38

can, so we, like, we will have like one master and one read replica. So like maybe like West have a master and and then like East have like, like read replicas for that.

Speaker 1 20:53

So let's say the user is coming to East location, alright, so he the application will be talking to Western Region database, or will it? Will it talk to each region database in order to write the entry?

Speaker 2 21:10

So I think like, like, the writing should be like, handled to the like, the waste region.

Speaker 1 21:20

So writing will go to the west region. So don't, don't you see any latency in this case? Let me think,

Speaker 2 21:30

yeah, that would like, probably like, yeah, probably like, problem in latency. So if a user in this region,

Unknown Speaker 21:42

like, if,

Speaker 2 21:45

like, I think that. I think that would be some kind of like, maybe a trade off that we would go with because, because, like, if you have like, multiple like, right replicas, then you have some like issues with like consistency, right? And in order to make sure, like, you have like strong consistency, you would have to, like, maybe create trade off a little bit of latency.

Speaker 1 22:15

So since the requests are let's assume in that distributed cluster, we trade it off, and then we won't eventually consist here, but strong consistency. So all the rights are happening through West region, but the rates are still going to the east region, for example, right? So I'll assume that, but there is a lot of read latency as well in the application. What? How will you handle that?

Speaker 2 22:53

So, so, basically like, basically like, if you like, have latency on like, like, in the reach. You mean like reading for like, like the West, going to east, like for like, reading, yeah,

Unknown Speaker 23:17

yeah, no, like, we would like,

Speaker 2 23:19

like, we would basically do like a local read, but remote, right? So reading would be on, like, would be on the region that they are on, but like, if there is any like, right operation, we would like do that to the like, the West region. So like, like, it's, it's a, like a, like a command, like command, query architecture, right? So, like, anything like that has, like that had that has to do change would like be a command, like APS, like service, and then like query would feel like, would be to like anything that is like, regarding the reads,

Speaker 1 24:02

yeah, so this is a distributed cluster, right? So you have one right node and multiple trade nodes, maybe one trade node for East region, one rate node for West region, for example, right? So, but you your database needs to go through database upgrades, for example, right? So you have to read because Read node in the best region is going through upgrade. You won't be doing the read upgrades are simultaneously on both the nodes, right? You will typically do in one region first, and then give some time once it comes back, then you do the upgrade on the second region, right? So in this case, what's happening is the West region is going through the upgrade, so all the reads are going to the east region, correct? So this is a one time thing, as it go. It's a continuous maintenance thing. It may not be like everyday thing, but it is like every month, once or quarterly, once you your database has to go through some upgrades, security patches and whatnot. How will you handle that scenario

Unknown Speaker 25:13

for that?

Unknown Speaker 25:19

So,

Speaker 2 25:22

so, yeah, in, in that case, so, like, whenever you need to handle like, like, upgrade like, so with like, minimal downtime, I think the best way, like, here, like, it would be to so, like, so we basically, like, modify the application like, the read endpoints to use other like region replica. And I think, I think the best way here would be to like, like, start up a newer instance with the like, upgraded data, right? So like you, like, stand up a like a newer, like an instance of like a database and and this like newer instance will, like, have an upgraded path, like, attached to it. And then once the upgrade is, like, complete on the on the like, the newer version of the database, like, like you would like, basically, like, add, add or incorporate, like, this new cluster, like, in that distributed system. And like, basically, like, register, like, like the other one, right? So you, like, scale that one instance up and like, then break the other one down this way, like, we would like, have the minimal downtime and so, so that, like, you do not need to actually, like, change, like, any routings or anything like you just, like, you need to work on it, like a, like, a separate database. Like, get it, like, get it up in service and and then, like, switch it active one. So I Yeah,

Speaker 1 27:03

that's a good idea to bring up the new node with the upgraded version. But how does the application know that there is a new version or a new node that is it needs to talk to? I

Unknown Speaker 27:20

I think, I think like this would be,

Unknown Speaker 27:35

I think like,

Speaker 2 27:38

like, like, it is like, still be going through like route 53 and like, like, then like you would like, need to make like changes to like the application like, instead.

Speaker 1 27:53

Why do you go to route 53 because you're going into the public network, because your application and database is in the private zone, right? So when you're in the private zone, you directly talk to your database, or where do you want to go to the public and then come back, which is a long delay, isn't it?

Unknown Speaker 28:18

Yes, yeah, that's true.

Speaker 2 28:29

So, I mean, like, you can also use like, like a database, like discovery, like using like the Dynamo like the like that, using DynamoDB, like instance reader. So, so basically, like, you would, like, read like whatever, like instances, like are up after like a refresh endpoint, like, by describing like the like the endpoints of the instances. So, like, it would be some kind of, like a discovery service, like using, like AWS SDK to basically, like read available, like instances, like on the cluster.

Speaker 1 29:17

You can have database load balancer too, right? So you can have a database load balancer which is sitting on top of the actual database node, right? So the application will talk to the database load balancer underneath. You can have multiple read nodes, whatnot, right? So even though one node goes down till load balancer knows which which node to route the traffic and so on. So that's that's abstracted for from the application. So you will not give the actual host name of the reader, but you will have the load balancer sitting on the top, which is listening to multiple readers under it's like a downstream service, okay, okay, so you can keep my maintain that way. That makes sense, yeah. Okay. So now you got everything, but you your system has to go through frequent changes, meaning, like the application has to go through frequent changes. So you need to build a common CICD system, right? So, what kind of so are you familiar with the Jenkins or Argo CD or git hub actions? Which CICD platform Are you familiar with? Yeah? So,

Speaker 2 30:41

yeah. So I work with Jenkins. So like, we own, like our like, like JKS pipeline, and like we like, like create a group and like this script that defines, like the stages that that the Jenkins should go through. And we also have like, like, a bad script which, like, defines how the application, like needs to be deployed. And so like, like the ruby script is for like stages, and then the like, the best script is for like, the actual script that like provisions, the services. So like, we have a pretty good, like set of pipelines, and we incorporate like, like cloud formation templates for like provisioning the services.

Speaker 1 31:40

I think that is all I have. If you have any questions, please do

Speaker 2 31:45

ask, yeah. Thank you so much for your time today. It was really nice talking with you. And like, firstly I would like to ask is, like, how much like offer Site Reliability Engineering duties do like developers get involved in

Speaker 1 32:05

so this is a small team, right? So our team is a small team, so you own entire thing, like from application development to deployment and maintaining it, and as well as handling the customer issues. There is no dedicated ops or as such. But there is a platform team which is helping in terms of building the CICD, which maintains the CICD platforms, but you code, you deploy, you own the entire thing. So here that that is the model. So that is why I'm asking about how much are you familiar with Jenkins pipelines or Kubernetes infrastructure and so on. So I presume I haven't touched upon the Kubernetes side and Dockers, but I presume you have some decent knowledge on that side as well.

Speaker 2 32:50

Yeah, I'm pretty comfortable working with CICD and Jenkins Kubernetes. Yeah, we talked earlier about Kubernetes in the previous interview as well. So yeah, but yeah, that sounds like, like a really good team, like something I would like to know is, so what, what would do? Like, like, like most about working for Adobe.

Speaker 1 33:16

Been with Adobe for almost five years. So one thing is continuous churn, meaning, like, there are a lot of things that you need to deal with in a day. It's not like it's yeah, you will wake up and you know that you will work on these things, but there are times that you need to general up on multiple things and prioritize continuously. There might be a P zero while you are working on some code change, so you need to stop there and then switch gears to take the call, right? So there are those other things, basically, but yeah, so my previous experience all these years is mostly working on different platforms. You get to know about different distributed, large scale systems, what are their current issues and how they are managing it, what type of issues they're handling continuously. So there is always a learning curve. As I mentioned, it's a small team that you need to know about. The entire CICD develops as well. So application development as well. So that's that's a bigger area too, for anybody who joins the team to learn and learn and contribute more on that side. So you may be coming as a full stack engineer, but you should know about other things as well, or you need to pick up on the job. So that's why,

Unknown Speaker 34:42

sure, yeah, thank you so much for that.

Unknown Speaker 34:46

And yeah,

Speaker 2 34:48

I think yeah, that's all the questions that I have at the moment,

Speaker 1 34:54

talking to you as well. Wish you all the best. Thank you. Yeah,

Unknown Speaker 34:59

thank you so much. Yeah, you.

Ronaj-Paypal-Software Engineer Interview-SRV

Unknown Speaker 00:08

Hello,

Unknown Speaker 00:17

hello, hello, I Think I

Unknown Speaker 03:33

So, Hello, hi, I'm

Unknown Speaker 03:44

Hey, hey, hi.

Speaker 1 03:47

Good afternoon. Good afternoon. How are you doing good? How are you? Yeah, I'm doing good as well. Thank you

Speaker 2 03:56

for asking. Let me introduce myself so I am kaitly. I've been with PayPal for close to just four months now, and I am with the credit platform team and MTS one there, and my team is basically responsible for, responsible for maintaining all the credit loans and providing APIs for those kind of stuff. So, yeah, we can talk more about my team, or, like some other questions that you might have towards the end of the interview. So, but let's just Yeah, can you introduce us off? Yeah,

Speaker 1 04:43

absolutely. So hi. My name is Rona Bhan, and I've been working as a software engineer for about six years now. And I've been working primarily on, you know, Java based applications utilizing frameworks like Spring, Hibernate and like rescue with services for, like, API development, and, yeah, I have a pretty good experience working on, like, the back end technologies, and also, like specializing in, like the microservices. And I've been, I've been involved in, like, designing microservices and deploying them into that AWS cloud. And some of the cloud services that I worked on include AWS, ECS, easy to SQS, SNS and lambdas. And currently I'm on a scrum team. And my team consists of like five developers, and we have like, product owners and Scrum Masters. And on a daily basis, I'm mostly involved, like working on back end, you know, Java code, helping the team with like code reviews, and also on like the products and support duties as well.

Unknown Speaker 05:50

Perfect. Cool, yeah, thanks for that. Um,

Unknown Speaker 05:57

yeah, let's, let's

Speaker 2 05:59

probably jump to the question. So I let you know what this round is about. So this round is like, it's a role specialization, role. So what I'm sorry interview, and what I'm looking for in this interview is basically looking for, like, how you write clean code, but how you implement the object oriented principles into your code, and like, how, basically, how you think about object oriented aspects of it, and, yeah, like the design patterns and all that stuff. So I'm not looking for like

Unknown Speaker 06:42

optimization, or like

Speaker 2 06:45

an optimized solution for my question, I'm just looking for like. I might look into like what data structures that you use at the max, but

Unknown Speaker 06:58

that's all so

Speaker 2 07:01

I hope that's helpful in like, what I'm looking for here. So let me,

Unknown Speaker 07:10

yeah, let me open the anchor

Speaker 3 07:12

line grant link that I have Here. I

Unknown Speaker 07:36

Everything, right? So I don't know about this,

Unknown Speaker 07:45

yeah, I think we should go here.

Unknown Speaker 07:49

All right. Do you see my screen?

Speaker 2 07:54

Do you see the screen where it says, solution, app, numbers, yeah, yeah, I see. All right, perfect, so let me post the question Here. I

Speaker 2 08:37

Yeah, feel free to change the Java. Eight or whatever, whichever you're comfortable with that. So let me, let me read the question, and then, like, you can ask me any questions that you have. So basically, what I'm looking for is like designing a library system where a user is, like, allowed to return or borrow books with some limited number of copies, and if a copy is not available, the user is added to the wait list. And like the system should basically output a message every time a book is borrowed or returned, or if, like, a book is not available at any point in time. So you can assume that the user will not return a book that they do not have, nor will they borrow the book more than once at the same time.

Unknown Speaker 09:39

Let me know if you have any questions,

Speaker 1 09:40

okay, so, yes. So basically, okay, so we're designing a library system, like, where we'll have book and then we'll also have a user, and basically, like, the library is the collection of books.

Unknown Speaker 10:01

So okay, where we can like,

Unknown Speaker 10:07

where we can probably

Speaker 1 10:10

basically write a book by a title. So all this, we need to implement functionalities like borrow book and return book,

Speaker 2 10:24

yeah, if you can, like, yeah, as you like, just jot down Whatever your thought is, so we'll have a book and

Unknown Speaker 10:40

so we'll have a book.

Speaker 1 10:48

This book will contain like title, number of copies total, and also every level of copies and so

Speaker 1 11:09

Okay, so we'll then have like user so which tracks The set of books that like he has borrowed. So BBB.

Unknown Speaker 11:26

And then, like, we'll have a library which basically

Unknown Speaker 11:32

like helps in

Speaker 1 11:34

this library will basically contain like the behaviors, like adding book, also like boring book, and then like returning the book.

Speaker 1 11:51

So okay, so those would be our, like primary functionalities and that we work with. I

Speaker 2 12:03

Okay, I think that that's fair, that three main components that you're working with, yeah, what? Yeah, go ahead and start with the classes like I want to see, like some classes and some methods to each of these. I in

Speaker 1 12:23

Sure. Okay, so let's do that for now. So we basically have a book class.

Speaker 1 12:33

So okay, so the book will contain title total copies, and also like every level copies. So

Speaker 1 12:55

is. So that's like the three attributes will Introduce, also like constructors, getters and setters. I

Speaker 1 13:31

so like, whenever we create a new book entry, we basically have the same number of availability as the like the total number, which is initialized with.

Speaker 2 13:49

Now, what do you mean by total numbers as total available?

Speaker 1 13:57

So like the total number is like, just to keep track of, like, how many books like that particular title has overall and the total available is, like, how many books are available to borrow. Okay.

Unknown Speaker 14:16

Okay, you're trying to test it. That's

Unknown Speaker 14:21

fine. Yeah, we can go back from

Speaker 1 14:25

Okay, so now we basically need to do like A couple of things, like getters and

Unknown Speaker 14:34

some behaviors. I

Speaker 1 15:07

All right, so we'll basically be exposing like, these two functionalities,

Speaker 1 15:17

to get the title and also like like to check if it like, if it's available. I might come in and like, add more later on as a progress, but this would be our base for now.

Unknown Speaker 15:42

So

Unknown Speaker 15:44

if number of copies,

Speaker 1 15:47

like no copies are available, the user is adding it to the waiting list, and the system should output a message. Like, every time a book is borrowed or returned,

Unknown Speaker 15:58

are indicating like, if there are no available copies,

Speaker 1 16:02

okay, so we are not handling the primary case of returning and then basically

Unknown Speaker 16:09

getting the reassign you

Speaker 1 16:23

Okay, so here we will now create a user client,

Speaker 1 16:32

and this user class would The name of the user and

Speaker 1 16:42

the set of borrowed books, and We will save the borrowed books in our set. Do

Speaker 1 17:08

so All right, so we can do like a gator setter and

Unknown Speaker 17:48

that's fine, just separate, different Sure.

Speaker 1 17:57

So okay, so we'll be basically be adding a borrow book and return book functionality on The user, because that's what The user can do. So

Unknown Speaker 18:58

So, all right, so So

Speaker 1 19:00

those two will be the primary use cases of users, Like the borrowing and returning of book.

Unknown Speaker 19:21

Let's see So,

Speaker 1 19:23

so now we need the actual library that manages all of this borrowing.

Unknown Speaker 19:31

So before that,

Unknown Speaker 19:34

we need to add some other implementation

Speaker 1 19:39

in the book to basically borrow for A user and and then return for our user. I

Speaker 1 20:03

Okay, so I'll implement them in a while, but let me just start into actual library portion.

Unknown Speaker 20:17

So we will have a class library,

Unknown Speaker 20:21

and we'll just store

Unknown Speaker 20:26

the list of books in a hash map. I

Speaker 1 20:57

so now the first thing is the library. It should be

Unknown Speaker 21:02

able to add a book.

Speaker 1 21:14

So for that, we can just add it to the hash map. So

Unknown Speaker 21:33

okay, so now in the case of borrowing, we can check

Speaker 1 21:41

the first thing I check as The library like the first thing contains the book I

Speaker 1 22:38

so in case the book exists in the library, we'll just basically do a borrow for our user.

Speaker 1 23:02

So So we basically want to return a book for this.

Unknown Speaker 23:12

So the way,

Speaker 1 23:15

so, the way the model should work is, firstly, to check, check if the number of the user already have this book, all right. So,

Speaker 1 23:33

okay, so since it is given that user will not do it, so do we need to add a check for this case? Which case,

Unknown Speaker 23:46

which case? Sorry,

Unknown Speaker 23:50

this portion here, I've highlighted it.

Unknown Speaker 23:55

I don't see any highlights, sorry.

Unknown Speaker 23:57

Oh the oh yeah on the top, yeah, don't worry

Unknown Speaker 24:02

about that,

Speaker 1 24:05

okay, since it's given I want to add a check if the user already has a Book.

Speaker 1 24:19

So first, check if the book is available. So, okay, so if, if the book is available, then I would basically decrease the number of available copies,

Speaker 1 24:40

and then How to get the user to borrow this book i

Unknown Speaker 25:15

So okay, so if the book is not available,

Unknown Speaker 25:20

We would want to add it to the waiting list. Okay,

Speaker 1 25:37

so we need to keep track of the waiting list in the book itself. So we basically do that, since it's a waiting list, we would do some kind of queue,

Unknown Speaker 25:59

which is

Speaker 1 26:02

our strategy. So we'll implement like just like a first in, first out strategy. So we'll implement a queue,

Speaker 1 26:22

and to initialize this, we can simply make it a linked list.

Speaker 1 26:36

All right, so, so if the book is not available, we'll simply add it

Unknown Speaker 26:46

to the list.

Unknown Speaker 26:56

So the user gets added to the limb list. I

Speaker 1 27:08

it else, it is like a borrowable process.

Speaker 1 27:17

So now that should be a whole bar book flow, and now We should basically focus on returning, returning the book. I

Unknown Speaker 27:40

it. So since it is given that

Speaker 1 27:56

they will not return a book that they do not have, we can simply do a check to see If it is a valid book

Unknown Speaker 28:05

within a library. I

Unknown Speaker 28:33

So Alice just returned the book and

Unknown Speaker 28:45

and the way we want to return a book is

Unknown Speaker 28:50

basically,

Speaker 1 28:57

firstly, I use the users return Book function. And then,

Unknown Speaker 29:06

then we increment our account. I

Unknown Speaker 29:35

Are you talking to? Something?

Unknown Speaker 29:47

I'm thinking what you should add next.

Unknown Speaker 29:56

What were they thinking

Speaker 1 30:00

about in this, like, I'm just like thinking about, if you know, like, the return value it would

Unknown Speaker 30:08

resolve or not,

Speaker 1 30:11

because, like, we are basically passing around the reference of the object.

Unknown Speaker 30:17

Sorry, what was the question? Can you repeat that earlier?

Speaker 1 30:19

Yes, so I'm just like thinking out loud about if the like the return value would resolve or not, like since. So basically since we are, like, passing the whole object reference of the book itself, like the whole object here, like on the borrow method and on the return method.

Speaker 1 30:50

So, yeah, I'm thinking if, like, a set would basically be available to check it or not.

Unknown Speaker 30:59

Yeah, but I think it will, so it should be good.

Speaker 2 31:04

Yeah, sorry, I didn't understand your concern at all. It's set where it's set. I don't see the set at all. You mean the borrowed

Speaker 1 31:15

set so on the user, instead of using, like the book. Instead of having having, like a set of books, I think I should have, like a set of strings to store the titles.

Speaker 1 31:36

Okay, yeah, so, like, because like the book so far, like the available count of the book keeps on changing, and like once we borrow so I don't know if the quality would basically work on the set. So

Unknown Speaker 31:59

I'm it might add duplicates,

Speaker 2 32:03

add duplicates to the borrowed books. That's like assuming if the same user borrowed the same book at the same time. But then we said, let's assume that wouldn't be we don't have to worry about it. So I think that's kind of what you

Speaker 1 32:21

have. So, like, since because, like, because of that guarantee, like, we can just, yeah, I think

Unknown Speaker 32:31

we don't have to worry about it.

Speaker 1 32:40

So the total number here doesn't play like any rule that I did on line 12. I think we can get rid of it. Yeah,

Speaker 2 33:05

I think the redone book thing is incomplete. Still. Do you want to finish?

Unknown Speaker 33:21

I'm sorry when the we return a book,

Unknown Speaker 33:25

we need to increment the count.

Unknown Speaker 33:31

And okay, so that's done, and let's see i

Unknown Speaker 33:52

Okay, all right, yeah, I um,

Speaker 3 34:05

yeah, okay, you're adding like print statements. That's fine, yeah,

Speaker 1 34:12

just adding statements, since it's also required,

Speaker 2 34:18

yeah, okay, what do you think would happen to the waiting list, for

Speaker 2 34:30

the waiting list? So you have a waiting list, right? Like you're adding some set of people to waiting list when they borrow a book, and if it's not available, how's that being processed now, like when with that.

Speaker 1 34:46

So I think the waiting list should be handled once we return the book, right? So once the book is returned, we should be able to pull the waiting list and then get that user to borrow it, and I can implement that. If that that's what you

Speaker 1 35:16

so, so if the waiting list is not empty, then we get, then we like, pull for the first element.

Unknown Speaker 35:36

So this will remove it from the queue, and then

Unknown Speaker 35:40

get the reference, and then we can just

Speaker 1 35:44

get this Start to borrow book for that particular next user. I

Speaker 1 36:13

so I think that should take care of it, the waiting list.

Speaker 4 36:20

Don't quickly look at this.

Unknown Speaker 36:39

All right. Why do you think

Speaker 2 36:45

so, seeing that the book is like handling the users aspect of it as well. Why do you think the book should handle that? For that,

Unknown Speaker 37:07

like, line 34 Yeah, like,

Unknown Speaker 37:14

why so,

Speaker 2 37:19

but think of it in a real world scenario, right? Like, there's a book and there's a library and there's a user like, why would the book know about like, what the user side of logic is like, think of book as a separate entity of its own. Like, there's a user could be a user method. Could be anything, right? Like, do you think, like, a book should still handle the user side of logic?

Speaker 1 37:57

So, like, that's an anti pattern that I have added here, we should be only like handling the simplest, like, plain old Java object with like, like a title, available copies and waiting list after the book. Then we, like, handled everything on, like, probably the library, which basically acts as like an intermediary between like those operations I can quickly refactor the code and and make it handle that, if so, like,

Speaker 1 38:42

when in the library, if you borrow a book, right? So,

Speaker 1 38:59

so in order to borrow. We can check if the book is available. You

Unknown Speaker 39:57

sorry, can you talk through what you're trying to do here? I

Speaker 4 40:00

so

Unknown Speaker 40:04

I'm just refactoring my code

Speaker 1 40:07

trying to, like strip out the functionality so like, we have, like a single response to like principle. So first thing

Unknown Speaker 40:16

that I'm just stripping out

Speaker 1 40:19

of this functionality here. So first thing I'm doing on the waiting list, and then I basically need to make sure that the total available is deducted. And for that, we need to do a set of method. But

Unknown Speaker 40:40

where are you doing?

Unknown Speaker 40:45

Online, 27

Speaker 2 40:56

Okay, so, okay, okay, I think you got so you're moving the entire logic now to library, I don't think you really need to do that, like the borrow still, yeah, now you've moved, like the all of borrow to out of book. I don't think you need to do that right, like that, that can still remain where it was. And does that make sense? So I'm talking about like,

Unknown Speaker 41:27

so you are now, like, you're

Speaker 2 41:31

like, whatever you had was fine for like, a book being at the place where you need to maintain the wait list, and like, borrow the book and like let the user know, but I mean just printing out a system dot error for now, and that's all fine. My only concern was about the user. You making the user call it the user logic,

Unknown Speaker 42:01

I don't think you need that.

Unknown Speaker 42:04

That needs to be somewhere else,

Unknown Speaker 42:08

potentially the library system, like I said,

Unknown Speaker 42:12

Does that make sense? Okay.

Unknown Speaker 42:17

Yeah, it does. Yeah. Okay.

Speaker 1 43:32

So like, when we like, go to the library, and then we borrow the book. Firstly, we check if the book exists and or not, and if it exists, then we get the book, and then we borrow the book. So on the borrowing logic, we just like decrease the total data level.

Unknown Speaker 43:58

If it's possible,

Speaker 1 44:01

then on the user call this book. Now this will be a part of like transaction completely so we do not lose the integrity of that data. So similar flow on, like the Like

Unknown Speaker 44:23

the return Book functionality so

Speaker 1 45:53

so once we move the functionality out, I think there

Unknown Speaker 46:00

needs to be some.

Unknown Speaker 46:04

Change for this, like return action.

Speaker 2 46:11

Yeah, so you're now returning a book, and then you're calling on another user, and you're need to update the new user. So yeah, what do you have? How do you think in person?

Unknown Speaker 46:39

So what

Unknown Speaker 46:42

I was thinking, is

Speaker 1 46:49

on on the return flow, I basically need to handle the waiting list outside. So making this waiting list available to use in The getter method.

Unknown Speaker 47:17

So on this waiting list, I

Unknown Speaker 47:35

basically doing the

Speaker 1 48:00

so If the waiting list is not empty, then we basically call a borrow book. Applause.

Unknown Speaker 48:39

So this particular functionality of calling

Unknown Speaker 48:43

and when calling should be stripped out.

Unknown Speaker 48:48

Let me see. Okay, so now we are

Speaker 3 48:50

like borrowing a book, and then you're calling user, borrow book, okay? And then you're returning a book, and

Unknown Speaker 49:02

then you're doing

Unknown Speaker 49:04

user dot return,

Unknown Speaker 49:07

and then you're all Okay, wait, why?

Speaker 3 49:16

So 995, to 99 is a bit confusing for me, because you're doing

Speaker 2 49:23

a book done, and then you're reducing it from the

Unknown Speaker 49:28

so the books waiting lists.

Speaker 3 49:33

Okay, so you are doing the waiting how are you managing the waiting list? Here?

Unknown Speaker 49:40

Do you have a waiting

Speaker 2 49:44

list, so you're saying the waiting list is now the

Speaker 3 49:50

library, okay? And then

Speaker 4 49:58

total available got updated. I there.

Speaker 1 50:07

Like, basically what I'm doing is I'm first getting the like the first read, like getting the like the first return done for the like the older books, and then holding the waiting list and like then borrowing

Unknown Speaker 50:26

it. Okay, yeah, okay, and that might

Speaker 3 50:28

work. So now, what do you do for

Unknown Speaker 50:35

Yeah, I that might

Speaker 3 50:42

work. So you're now doing a borrow for the same title for the user on the waiting list. Okay,

Speaker 2 50:57

I guess that should work. Is there? Yeah, I think they should work. Do you think there's a better way to model this like, do you think I think this is fine,

Speaker 2 51:21

but do you think there's a better way to maintain this as you go, like, all that you all that, like, if you look at the users logic, right, all that you're in the borrow logic of user, All that is doing is, like, it's just maintaining your a list of books, and you're increasing the count or reducing the count, or like, I guess, increasing the number of books or reducing the number of books. Do you need to maintain it in the user side? Or do you think the library can maintain that.

Speaker 2 52:08

So let's say, Okay, let me put it this way. So this is like a library system. Now let's say you have like another system that's very similar to a library system, and you have a it's the same user at this point. Now is, do you want the user to maintain the power logic? Because I want to be able to just use the user for different purposes. It's the same user. He's just now doing two different he's he's being present in two different scenarios. I just want him to be present in two different scenarios.

Speaker 1 52:56

Like, we would not want to maintain, like, the borrow logic on the user, because like, because it's like user, it's just one particular model that holds user data. So for library, we would probably like do a map that like maps a user like to a set of books that he has borrowed. That would probably be like a better approach. That's

Unknown Speaker 53:22

what I was going looking for, right?

Speaker 2 53:27

Yeah, that's exactly what I was looking for. I think if you let's see how Okay, yeah, yeah, we are good on I mean, we are not good on time, so we have just 10 more minutes. But I think, yeah, I think that will work. And I know you know, you'll be able to code that up. That's fine. So

Speaker 3 53:49

I had to just one more question in your logic. So right now, the assumption here is that the book,

Speaker 2 53:59

book is like you have a unique title. What happens if there are two different books of two different titles? Sorry, two different books of the same title, like maybe different author or something? How would you represent

Speaker 1 54:20

that? Okay, so, so, right? So, like we, we would probably want like an identifier on each of like these items, yeah, even if you can have like multiple users with the same name, right? So usually, like these data models like have identifier, like identity element that we will work with. In case of book, you can just have like a like a unique UU ID that is like used to identify that book. Similarly for like a person, you could like incorporate like a unique identifier.

Speaker 2 55:02

Yeah, yeah, I think that we need that. But that's yeah, that's fine

Unknown Speaker 55:13

asking all right, just,

Unknown Speaker 55:19

just one more question, and then I'll give you some time for

Speaker 2 55:24

your questions. Alright, so what if I want, like, a waiting list that's like, not first in, first out, like, let's say I have like, different preferential users, say, like preferred members that I want to give priority over some of this. What do you think we should do, or you should

Speaker 1 55:56

like so like, if you don't have like, a first in, first out order,

Unknown Speaker 56:03

like anyone

Speaker 1 56:05

so, so, like, you would like, want to have like a priority order on what to do. So there is a like a queue called like a priority queue, that you can use. So, like, it basically works like, within like a competitor, like, which will have, like a higher priority based on, like, for the vid members, yeah, yeah,

Unknown Speaker 56:33

yeah, that's

Speaker 2 56:36

right. Um, okay, this is good. Let me just copy this solution of chaos, and then you can ask any question. Do you want?

Speaker 1 56:49

Sure? Yeah, thank you so much for your time today. It was really nice talking to you. And firstly, like I would like to ask is, like, how much of a similar, you know, interactive and design sessions to like developers get involved in, so, in like, like, peer to peer, programming basis, or like, something like that.

Speaker 2 57:13

Sorry, get your question, peer to peer. What like,

Speaker 1 57:16

peer to peer, like programming basis, like, how much like a developer gets like interactive so, like, how much of like your general design sessions and like interactions happen within the team?

Speaker 2 57:33

Um, yeah. So like, I think you're asking like, two different questions here. One is like, peer to peer programming. And then other one is design, right? So peer, so the team is, like, very, I mean, and my team is, and I don't know which team you're interviewing for, but like, I'm pretty sure it's all common across paper. But then, like, the team generally is, like, very collaborative. You you, if you have some question that you have that you're working on, like, you can like, ping anyone. They'll be readily available. And then like, or if you're, like, working from office, you just go say, Hey, can you look at this? Can you help me? And then I'm pretty sure the team would be very helpful to look into it if you have any issues. And beyond that in general, like you have like code reviews, like PRs and all that stuff. So people, like look into that, and like they have comments, and we address that. That's from an implementation standpoint and from like, a design standpoint, like, whenever we are designing some new feature, we, once we have the right kind of information, we set up, like, a design review with the team, and then like, we go over like or like, even when we are like in the process of designing, and then we are, like, not sure what the team preferences. We set up some meeting, and then we can, like, have some discussions on like, Hey, do you think we should go this way? Or, do you think that we should go this way, what the team put is, and all that stuff. So,

Unknown Speaker 59:16

so, yeah, overall, like, pretty collaborative.

Speaker 1 59:21

Yeah. Thank you so much for that. Yeah. And one other question that I'd like to ask is, like, what are like, some of the opportunities you know, that PayPal provides for, like, continuous learning and growth?

Speaker 2 59:37

Yeah? So I have joined. It's been only four months since I joined, but I noticed that PayPal has like, PayPal has LinkedIn learning membership, and then you name your membership where, like, you can like, it's all free. You can like, enroll on any courses that you like. And I, I personally, haven't found the time yet to do it, but then, like, it's readily available for you if you wanted to learn whatever you want to. And other than that, I think I might not be sure about this, but I think they're not like, if you wanted to do something, like, there's like, subsidized versions of it where, like, they can help you pay that off. But, yeah, don't convey on that information. But I think that's also there. But just these two, I know for sure it's there. You have, like, Udemy learning, and then you have the LinkedIn learning access,

Speaker 1 1:00:45

okay, yeah, sure. Thank you for that. Yeah, so much. Thank you so much for that. And, yeah, I think that's all the questions that I have right now. And thank you so much for having me here. And it was really nice talking with

Unknown Speaker 1:01:00

you. Yeah, definitely, thanks.

Unknown Speaker 1:01:03

Ronan, right,

Speaker 2 1:01:06

yeah, thanks. A lot nice talking to you. Yeah, good luck with your other entrance.

Speaker 1 1:01:11

Sure. Thanks so much. Thanks. Bye. Take care. Bye, bye. Bye.

Ronaj-Paypal-System Design Interview-SRV

Speaker 1 00:00

Hi, how are you? Yeah, I'm doing good. What about you doing? Well, thank you.

Unknown Speaker 00:08

Sorry. I'm gonna lay here,

Unknown Speaker 00:11

so I'll

Speaker 2 00:13

introduce myself, and love to hear a little bit more about you. And then I guess we'll do a little system design exercise. But so my name is Jake. I'm part of the shopping core experiences back end team, so mostly working on sort of back end as a front end type, what we call middle tier services for the shopping and offers division. So if you kind of open the PayPal app on the bottom, you'll see a little button that says offers, and there's, you know, you can get your 3% cash back here and there, that kind of thing. So that's sort of our group. It helps drive engagements. It helps, you know, just follow cash back to users, and sort of an additional value added prop for PayPal account holders in general, and for sort of specific features like the PayPal debit card and things like that, you can get extra cash back. So creates value for those, for those offers as well. So I've been here about three and a half years, I would say I have not been on this team. The entire time I've been working on some similar related features. Our division largely came in through the acquisition of a company called Honey. So I had done some work on my joanna.com but, yeah, I've been in my current role for about a year and a half. And let's see questions people often ask. You know, our stack is mix of TypeScript, Java and some kind of like bespoke configuration stores that we work in, mostly our customers, our direct customers are our mobile engineers, but we also support web experiences. And yeah, I think that's a little blessed in a nutshell, but definitely feel free to ask any questions, and I'd love to hear about you,

Speaker 3 02:27

sure, yeah, thank you for that. And sure, it was nice, yeah, it was nice for me to talk to you. And, yeah, talking about myself. You know, my name is Ronan Pradhan, and I've been working as a software engineer for about six years now. And I've been working, you know, primarily on Java based applications utilizing frameworks such as, you know, Spring, Hibernate and like rest web services for API development. And yeah, I have pretty good experience working on microservices. Like I've been involved in, like designing, developing and also, like deploying them. And yeah, some of fun. And I also am, like, involved in some of the services on the cloud that I work on. Like that includes, you know, AWS, you know, ECS, SQS, SNS and lambdas. And currently I'm in the scrum team. And, like, the team consists of like, five developers, and we have, like, product owners and like Scrum Masters and on a daily basis, like I'm working on, you know, back end, like Java code, helping the team with, like the code reviews, and also on like Portugal support duties as well. So that's cool. Great.

Speaker 2 03:39

All right. Well, I will, I will leave, like, more time for questions at the end, which, to me, somewhat, is kind of the best part. But

Unknown Speaker 03:48

let's take a look here.

Unknown Speaker 03:53

So I think we should both have,

Unknown Speaker 03:58

do you have, like, a hacker, a hacker rank? Link,

Unknown Speaker 04:02

hack a wreck, yeah, yeah,

Unknown Speaker 04:03

okay, so let me get in there.

Speaker 2 04:20

Yeah, I'm on the login. All right, sorry. Give me one second here. I just had to update my computer this morning, so I'm now logged out of everything. Yeah, I got you. Yeah. No worries.

Speaker 2 04:43

Okay, and then hold on, I think I Just need to set up the tab correctly. I

Unknown Speaker 05:27

All right. All right, do you see like,

Speaker 2 05:35

should say, the system design, the marketing glitch. You see that? Yeah. All right, yeah, okay, so, so this is not a coding exercise. Idea here is to kind of just read over the description and talk about, you know, kind of more architecturally, how you would design a system like this. And I think there's some some existing little prompts that you can use or not up to you, and yeah, kind of defer to you how you like to do these things, because everybody works differently. If it's helpful for you to kind of take time in your own head and sort it out, and then we can talk about it after. That's fine. If you prefer to chit chat as we go along, that's fine, too. You know, we're obviously not literally building this thing. The goal really is to share just kind of how you think. So why don't I give you, give you a minute to read it over, and then we can proceed however it's best for You. Yeah, sure. Yeah. Yeah,

Speaker 3 07:27

okay, so, all right, so basically, like, we have a client that sends a long URL to be shortened, and then we want to have, like, a shortening flow, and if there is a rejection flow and, and there is maybe some kind of, like tracking analysis going on,

Speaker 3 07:57

okay, so, um, Like, maybe a couple of questions like I have before we get started.

Speaker 3 08:14

So, um, like, basically, like, how much volume are we talking about? Like, right? So, like, do we have, like, anything set in mind, like any kind of, you know, expected read versus write ratios. Great, great question.

Speaker 2 08:32

So I would say I don't, I don't necessarily have a number, but I would assume,

Unknown Speaker 08:40

you know, for

Speaker 2 08:43

for a URL shortener, I mean, I would assume that it's going to be pretty read heavy, right? I mean, otherwise nobody's reading your URL shortener, probably not a very good, probably not a very useful tool. And I think just for purposes of the exercise, let's, let's assume some massive scale, you know, like, like, it's Bitly, or some one of those, you know, very you got a great marketing department behind you, and it's Gonna be, be very well utilized.

Speaker 3 09:22

So I'm, like, basically clients, like, like, treated like, it's like, a request to the server and

Unknown Speaker 09:31

Okay, so we will have so like,

Speaker 3 09:34

so do I have to, like, keep adding on the diagram that's already provided? Or should I, like, create something else from scratch.

Speaker 2 09:43

You can use that stuff or not. I think the only piece that's relevant is, like, we're not talking about how the URL minification actually works. But yes, you can. You can use the squares that are there or delete them. That's that's just kind of starter stuff. Okay? So?

Speaker 3 10:42

So basically here in between the client and server, we want to add a API gateway with load balancer. So okay, so that way we can handle like the high traffic and we have like, proper routing as well.

Speaker 3 11:06

Okay, so this would be the first step to ensure that we have a pretty highly scalable and resilient system, so that it should be like responsible for handling the incoming requests, load balancing and also rate limiting in case of

Unknown Speaker 11:25

birth state requests. So the

Speaker 1 11:34

server is basically responsible for two things. So the server should basically be responsible for

Unknown Speaker 11:46

generating,

Speaker 1 11:50

to generate the short URL, coding the service, it should be responsible like tracking, also the clicks and

Unknown Speaker 12:16

us so So in the database layer, We want to have i

Unknown Speaker 12:54

Okay, so let's See for the database we would want I it

Speaker 3 13:12

so like, one of the questions I need to ask is, like, how long do you want the URLs to be stored for? Like, does it like ever expire? Is it like permanent URL or like should it have? Should it have like a expiry time attached to it?

Speaker 2 13:31

Well, I mean, there it says that for social media posts. So I assume we, you know, want them to be able to live on as long as those social media posts do So

Unknown Speaker 13:55

far, the database will store the URL mapping I

Unknown Speaker 14:06

and then maybe some stacks for that URL. I

Speaker 1 14:44

So, okay, so, and then the server will also be calling it

Unknown Speaker 14:48

tracking service.

Unknown Speaker 15:00

Okay, all right, so, basically,

Speaker 1 15:09

so Okay, so the way our urlification would work, okay, let me think, okay.

Unknown Speaker 15:40

So the way URL modification is

Speaker 3 15:45

we basically generate a hash for this URL, and then, and

Speaker 3 15:56

then we basically then do some kind of encodings, like off the hash. So

Speaker 3 16:13

I think since hash is not always guaranteed to be unique, we need to add some kind of like a unique identifier

Unknown Speaker 16:24

to that as well. So

Speaker 1 16:50

BBB. So I think once we do this,

Speaker 3 16:57

so I think the hashing technique should basically be the best way, right? So that's what I'm thinking, is like we use some kind of like hash function,

Speaker 3 17:14

okay, SSA, 256, and then we say to take, like, the first n characters, and then we just, like, store this, any characters to the long URL mapping in The database. I

Speaker 3 18:01

so I think the problem with hashing is that there might be collisions, right? So I think the way we want to

Unknown Speaker 18:12

sort of handle this collision is

Unknown Speaker 18:16

maybe using

Speaker 1 18:19

this like this particular long hash which is like, SHA, 256,

Speaker 3 18:26

generally a longer hash and like which like lowers the probability, but doesn't make it zero, bad. I

Speaker 3 18:44

so we might want to incorporate another identifier within the hashing algorithm somewhere to make sure like we do not have a class so collision handles the collisions. I

Speaker 3 19:15

to make so like we can just add a counter to like hash to basically handle the collisions.

Speaker 3 19:35

So, okay, so All right, I think this is going to be the overall so because the actual infrastructure is actually given to us, right? Because we have our server which does the unification logic, we would use the cache with that least like, like, not the least frequency use strategy, and if it doesn't exist in cache, we got a database.

Speaker 1 20:12

So on designing the database, thinking,

Speaker 3 20:19

thinking we should be fine with the non relational database, because we do not have like a strict structure, and we can have like a key value pair. And the reason for this is it becomes highly available and partition tolerant, right? And because, like, consistency should not be a problem for you. You are inserting so something like eight, OS, Dynamo, DB or Mongo should do it. I

Speaker 3 21:10

yeah, I think that should basically be handling our you all short or not,

Speaker 2 21:21

so does the tracking service write the same document? I guess, as as we would use when kind of like following the short URL? I

Unknown Speaker 21:52

Yeah, so like

Unknown Speaker 21:54

in the case of tracking service, let's see,

Speaker 3 22:01

so the tracking service, like it, should not update the same document directly, right? Because that might be some kind of like issue there. So I think we should either, for the best practice, we could do a metadata document. So

Speaker 2 22:21

those are like two different collections or documents, right? Okay, got it.

Speaker 3 22:41

So, yeah, you know, like, Yeah, I think those two different collections for one, one for the like, actual mapping, and the other for, like, the metadata. I

Unknown Speaker 23:19

BBB,

Speaker 3 23:33

so yeah, like the user clicks on on the link and it goes to the load balancer to the like a shopping service, basically, like, checks the cache for a fast lookup if it exists, then it returns a direct, like redirect response back, and then on that click, there's an event that triggers this tracking service as well. So this tracking service would probably be an even base or some kind of Hue.

Speaker 3 24:22

Okay, so, yeah, I think that that should be pretty good. I

Unknown Speaker 24:50

would you want me to explain anything specific?

Speaker 2 24:55

Yeah, just wonder if you could talk a little bit more about like the caching strategy. What kind of

Unknown Speaker 25:04

Yeah, what would be your approach there? Okay,

Unknown Speaker 25:18

so I saw like suggested as a j2,

Unknown Speaker 25:22

56, six,

Speaker 2 25:27

oh, sorry, Caching, caching, not hashing. Oh, got it, sorry.

Speaker 3 25:39

I think the way gassing like should work here is

Speaker 1 25:48

okay, let me see I can write something down.

Unknown Speaker 26:01

Okay, so,

Speaker 3 26:07

so for caching, will introduce caching like first on the cache URL to the long URL mapping, right? So

Unknown Speaker 26:17

that's,

Unknown Speaker 26:20

that's the first one.

Unknown Speaker 26:24

So this basically,

Speaker 3 26:27

if there is a value that exists, and it will redirect if there is a mesh on the cast, and it will touch from the database and then store after and then return. Then we should basically be caching the popular URLs, right? So basically, we would use most frequently,

Unknown Speaker 26:56

most frequently accessed.

Speaker 3 27:00

So basically, for URLs with

Unknown Speaker 27:05

like the high traffic,

Speaker 3 27:07

we use, like a longer time so deep, which helps in avoiding the cache and tracing so we can store up to something like least frequently use dash and well, we'll maybe have, like cash for maybe a couple days, or usually, like the social media posts, so for maybe 24 to 48 hours. So I think that to be the best way, and that are like, basically, like, have time to leave off about, like, 48 hours. And then we basically, then, then, like, we'll have something like a least frequently used, like, to store the hot URLs. So yeah, I think that should be the best for caching.

Speaker 2 28:12

Do you have any idea of what kind of device would do the caching for the device, or planets

Unknown Speaker 28:19

like our application.

Speaker 3 28:28

So, yeah, I think we will just be using a reddish cache to store it in. Do

Speaker 2 28:36

you think there's any value in that, in memory, cache, in the server itself, in

Speaker 1 28:42

the so can, yeah,

Unknown Speaker 28:51

let me think.

Speaker 3 28:58

So, like, like, if I do it in memory, cache, in the server itself,

Unknown Speaker 29:08

then you know, like, it

Speaker 3 29:11

basically does, like, I think it will, like, reduce the network overhead for higher availability, and you do Not have to go to the URL minification service. So it definitely will help. It's very simple to implement as well, and so yeah, like, it will, like, definitely help. And I think those servers should maybe be warmed up based on the popular URLs, right? So not everything, but when the server starts, if there is a very popular URL that a lot of people are clicking based on these tracking services like drag data, and we can definitely do like in memory cache on the server itself. So Right?

Unknown Speaker 30:04

So I guess is there, do you

Speaker 2 30:08

think there's sort of like a weakest link here? Like, if somebody called up and they said, Oh, we're getting terrible we're getting terrible performance when people are clicking through, you know what would be the first thing that you want us to go check?

Unknown Speaker 30:33

So, yeah, I think the weakest link,

Speaker 3 30:38

like on a particular design here should be the URL minification part. And I say that because, you know, like, it has a like, a like caching layer, and we might want to think of some kind of, like proper invalidation strategy

Unknown Speaker 31:01

of the cache to

Speaker 3 31:04

basically have a proper cache in place so that we don't, like take up all our process we have, like one database. So you know, if the database goes down, we need to have, like a proper warm replica ready to basically replace the existing database. But I mean, since we are using some kind of, like non relational we should have, like a partitioning strategy and and then like we have, we can have like multiple standbys available to replace it in case of failure. So I think the first place I would go would be the cache layer and then the database layer to see if those are in service.

Speaker 2 31:55

Great. All right, I think this is good, do I guess, with you reflecting on it, do you have any other thoughts that that you want to add about, edge cases you might think about, or, you know, changes you might Need to make over time, or anything like that? I

Unknown Speaker 32:20

let me see.

Speaker 3 32:24

So, yeah, like, I think we might need to, like, have a proper inefficient URL expiry handling. So if you are exposed after a certain time, like, we need, like, a cleanup process to maybe mark as inactive in the database, and also, you know, like deleted from the cache. But we also need to make sure like proper load balancers with like properties, and also like the DNS you're routing as well. It's great. Thank you.

Speaker 2 33:13

Alright, so where's the time left to leave plenty of time for questions that you have about PayPal or me, or or whatever, anything I can answer for you. Because obviously, yeah, I mean, you obviously want to want to know about PayPal, just like we want to know about

Speaker 3 33:33

you. Yeah, sure. So, like the questions I have right now is so like for the new developers who are, like, just going to join the team. What advice can you give, like, new developers who are just going in and joining the team? You know, in Paypal?

Speaker 2 33:54

I mean, I guess it depends a lot, but I would say, you know, if you haven't worked at a very large company before that's sort of its own

Unknown Speaker 34:05

skill set, you know, regardless of industry.

Speaker 2 34:10

So just sort of understating some of that complexity with the different layers of management, different teams, and you know, PayPal in particular. Actually, one of the best things about PayPal is there's people all over the world working here. It's probably my favorite part of the company. But, you know, it creates issues with coordination and things like that. So I think one piece of advice I would have is to sort of accept that, learn that, be patient, and just understand that kind of navigating the size of the company is just sort of part of the job, aside from the technical aspect. But similar to that, it's, it's a pretty old company for consumer tech, so you kind of have different generations of technology here, and being able to navigate that also is not something that you're going to just pick up in one day. That's its own sort of special skill that requires patience. And I think related to both of them, I would just say, ask a lot of questions. You're only going to be able to figure out so much. There's a lot of people here that have been here for a long time, much longer than me, and they have a lot of history, they have a lot of context. They know a lot of different people, and I would definitely leverage the people around you. I think that would probably be my biggest piece of advice.

Speaker 3 35:28

Yeah, thank you so much for that. I really appreciate you saying all of this. And yeah, I think my second question, say, question is, like, like, in a big industry like PayPal, you know, how does the, you know, team size look like, you know? How does it operate? You know, can you tell me something about that as well? I'm really interested in finding out

Speaker 2 35:51

Sure. So, it varies a lot. I mean, my team so, and it also depends on what you call a team. So, you know, the back end engineers that I work with, there's probably about eight of us. But you know, the client engineers that we work with are very important, right? We work very closely with them. And then there's also a lot of contract workers, so I don't even know how many of them are, maybe 10 or 20. And then, you know, we have managers, and then we have peers outside of engineering that we work closely with. So our product manager, we have data analysts that we work with, machine learning engineers that we work with. So there, there are dozens of people that I work with routinely. And again, it's great, you know? I mean, it can be hard to keep track of everything and everyone, and communication is is a challenge sometimes, but there's a lot of different skill sets, a lot of different people. And it's, it's really quite nice.

Speaker 3 36:57

Wow, yeah, thanks for that. And I think my last question would be, you know, like, since I'm interviewing, interviewing for the position of, I don't know, like they said it would be like a Java developer position, but in the, like, the in the email it came by, like, something, like, staff Python developer. So, like, do I have any idea on, like, what I'm like, what position that I'm applying on? Like, because,

Speaker 2 37:30

yeah, I thought that you were applying for a packet in Java position. Yes, having said that, I don't know. So if you have a question about that, I would email whoever from recruiting you're talking to, because I'm the wrong I'm not. I just got a meeting calendar invite and I come. But I don't actually know if they distinguish between the two. So, you know, I my focus is really on TypeScript. JavaScript. I come from a web background, but I contribute to Java projects, so I actually so I would clarify it with them, but maybe it doesn't matter. Maybe you're back end engineer, and you kind of do a little bit either. My impression is that Java is the predominant backend language at PayPal, and I thought that Python is mostly used for kind of like machine learning. But even them, I see they have job applications. So I've actually, personally never seen a Python app at PayPal. So I don't know. You know, other business students may use it more. I don't know, maybe Venmo, I don't know what they use. So that's a good question. I so I thought that you were at a Java specialty, but I don't know. Yes, if you're confused by that, I would definitely ask your recruiter.

Speaker 3 38:57

Yeah, sure. I'll do that. Sure. Thank you so much for like, talking me through it first. Yeah, I really appreciate it. And, yeah, I think that's all the question that I have on my mind right now. All right, very

Speaker 2 39:11

well. Very well, bullets. And I guess we have a little more time, but I don't, if you don't have more questions, I don't want to take more of your time. I'm sure you have plenty of interviews to do today. So, yeah, we appreciate meeting you. Can I wish you the best of

Speaker 3 39:26

luck. Yeah, thank you so much for that and I really appreciate it for you giving me this time of your day to conduct this interview as well and I really enjoy. Yeah, great. Yeah. Take

Unknown Speaker 39:37

care. Thank you.

Ronaj-Paypal-Technical Interview for Engineer-SRV

Unknown Speaker 00:02

Hello.

Speaker 1 00:09

Thank you. Nice. Give me one second. I think she's good. Bye. So yeah, can you specify how to pronounce your name so that I don't get it

Speaker 2 00:25

wrong? Sure, so I pronounce it. Ronas.

Speaker 1 00:28

Ronas, yeah, okay, great. Welcome to your interview round here for data structures and algorithms. I hope you know it's been going well so far. And, yeah, this is going to be, hopefully very smooth one hour. And the way I see it working for us is we'll go over some brief introductions, and then maybe a couple of lines, and then I'll ask you some questions based on your experience before we move on to coding around which I expect to be a collaborative exercise between us on Hacker wreck. And then, ideally, we have 10 to 15 minutes left at the end for you to ask us some questions that you may have. Yeah, that being said, I'll start first, then I'll hand it over to Sophia before we pass it on to you, to you know, share anything you want to share about yourself with us. So hi, I'm shoma Jyoti, but most people call me shoms. That's like shell of homes, shops. I work on the shopping experience team, specifically honey and the honey extension team. I've been here for a bit, specifically on this team for less than a year, and I was there before on a different team and PayPal for a couple of years before that. Sophia,

Speaker 3 02:02

Hi, I'm Sophia. I'm on the same team as Jones. I've been here for a couple years, and I'll just shadow your interview, and I'll be in the back. Good luck. Yeah, sure. Thank you.

Unknown Speaker 02:13

Yeah, go ahead. Please give us

Speaker 2 02:20

a bit. My name is Brunos brudan, and I've been working as a software engineer for, I mean, like, six years now. And, you know, I've been primarily working on database, web application, you know, utilizing frameworks like spring hibernate and like RESTful web services for AI development. And, like, I've been recently involved in, like, a lot of, you know, microservice development, and also been involved in, like designing, developing and also like deploying them on, like AWS cloud and some of the like cloud services, those are, like, ECS, SQS, SNS and lambdas. And, like, I do have pretty great, you know, strength working on like front end as well, particularly using Angular and I have experience also working with JUnit and mokito for unit testing. And currently I'm on like a scrum team. And the team consists of like five developers, like we have, like our product owner, and also like scrum master. And on daily basis. You know, I'm involved, like, mostly like working on like back end Java code, like helping the team with other code reviews, and also on, like production software degrees as well. Awesome.

Speaker 1 03:34

Thank you for sharing that. It's good to always know the experience that people have and the variety experiences that people come with, I have a question, based on everything you shared, so, based on all the experiences worked in different teams and different different kind of projects. So can you share a time when you disagreed with a colleague or a leader, and how you handled that disagreement, what it was on and I proceeded forward with it,

Speaker 2 04:03

sure. So, yeah, like thinking about that time, you know, yeah, I mean, like, working in a team and like, like, disagreements sometimes, you know, it's bound to happen. And yeah, one of the time that, like, I disagreed with my team, it was like, whenever I worked on, like, a feature for, like, comparing a complex object and like, we had to check the quality of those, you know, two particular objects, and the way we like initialized. Initially thought it would happen was like just doing a simple, like string comparison of like, certain fields, right? But, you know, I was thinking of more like, broader aspect, like, basically, like, stringifying the whole payload and then doing like, the comparison using the like object mappers. And, you know, we basically had, like, a pretty good discussion on, like, why, like, some approaches would be good, and why, you know, the other would be, like elsewhere and, and the way I basically handle the disagreement was like suggesting both of us to, like, come up with, like, a list of, you know, pros and cons on like, which one would be like beneficial, and how any and how we would progress, right? So, so once I had done that, we basically had a mediator with us, and yeah, then we came into an agreement on, like, how we would, like, go on with the process and how it would be beneficial. And the outcome of the outcome of this was we, like, we went with the hybrid approach, basically, like both of our sessions, right? So I think the best way to handle the disagreement is, like, being very transparent and also calm, and then also having, like, open communication with someone, like, who might be with, who might be in, like, disagreement with, yeah. So,

Speaker 1 05:54

okay, yeah, thank you. Also, I forgot to mention this, but at any point, if you see me looking on the speed and typing stuff. Don't Don't worry that. Don't think that I'm doing my work and not listening to you. I'm actually taking feedback, which I have to share. Awesome. Thank you for sharing that. So we'll move on to the coding challenge. Do you have the academic link with him?

Unknown Speaker 06:24

Yeah, I do have it, yeah. Okay,

Speaker 1 06:27

then I see you are already patient Institute to Java 70 is that? Is that the language of choice? Yes. Sophia. Do you have the link? I can share it.

Unknown Speaker 06:45

Sorry, I'm working on it. Okay,

Unknown Speaker 06:53

I'll pull in this chat as well.

Speaker 1 07:00

Awesome. So yeah, let me paste question. I

Speaker 1 07:26

Yeah, feel free to read the question. Ask any questions about the question that you may have, and as I mentioned, I want this to be a collaborative exercise. I'm not trying to use this question to God, to you, just try to figure out, you know, if you can solve, how you solve problems? I'm sure you can just once. Yeah, sure. Thank you.

Speaker 2 07:58

So okay, so we have like array of characters, can basically want to reverse this given string without using any extra memory. So

Unknown Speaker 08:24

right. So, okay, yeah, so basically,

Speaker 2 08:28

want to switch this with the last one. So I think there is an like indication of

Unknown Speaker 08:35

two different pointers working together. I

Unknown Speaker 08:45

Okay, all right,

Speaker 2 08:48

so, so, all right, so I think we basically want to add temporary variable that stores a value, and then we like swap those values in place. Do

Speaker 1 09:03

Okay, go ahead and code it out. I recommend doing it like how the example they have done it so that we can run test cases using the main method. Okay,

Unknown Speaker 09:22

okay. Okay, yeah, I think I see

Unknown Speaker 09:27

so I don't see a method definition here.

Speaker 1 09:31

No, feel free to make yours sounds good. I

Speaker 4 09:59

BBB. So basically, we'll initialize two pointers

Unknown Speaker 10:05

which would start at the start and the end, and

Speaker 2 10:22

and then now, like we like, look while Left is less than right.

Unknown Speaker 10:34

So in that case, I

Unknown Speaker 10:45

so I think all we need

Speaker 2 10:50

to, like, store our character in, like a temporary variable and then switch it. So,

Speaker 4 11:01

okay, so basically we will have another string

Unknown Speaker 11:09

that will store

Unknown Speaker 11:12

the left pointer,

Unknown Speaker 11:17

and then we just swap and

Speaker 4 11:48

so once we like, switch it like, we just move the pointer one level inside.

Speaker 2 11:59

Since we are doing it in place. We don't even need to return the stream. It will just change the original array.

Unknown Speaker 12:17

So Okay, so here first, so it's for example, If we have,

Speaker 2 12:25

like a string of array of a

Unknown Speaker 13:15

I think it's dropped.

Unknown Speaker 13:17

Yes, that look as expected. Let

Speaker 4 13:19

me try. That

Speaker 1 13:26

looks as expected, awesome. So well, it would be an ideal world where this would be the only question I ask you. It is not so there are some more sort of paste the second question,

Speaker 1 13:55

absolutely you might want to reuse the stuff you have, So I am going to just paste it here.

Unknown Speaker 14:06

Line in the middle. This is your second question. I

Unknown Speaker 14:20

Okay,

Unknown Speaker 14:27

so we only moved a pointer

Unknown Speaker 14:31

when we encounter

Speaker 2 14:33

a buyer or swap when we encounter a bowl. Right?

Speaker 1 14:42

Wow. I guess, yeah, but yeah, look at the edge cases mentioned and take care of that.

Unknown Speaker 14:56

And as long as you do that, I think it should work. Sure. Okay,

Speaker 4 15:09

okay, so let me create A set of vowels. First I

Speaker 4 16:02

so we can just do uppercase Should be good

Unknown Speaker 16:17

as well. It's up to you. It.

Speaker 4 16:27

So, okay, all right, so that's done now, basically we just need to check like if the pointers are pointing to the vowel or not, and and then we can string simply swap after that. So for example, so So while left is Less than right and

Unknown Speaker 17:02

power, does not

Speaker 4 17:21

so so if my left does not point to a bowel, we'll just increase the left,

Speaker 2 17:30

and similarly with The right, I

Speaker 4 18:02

BBB, so this is like just moving the pointers until, like we reach

Speaker 2 18:10

Val, and then once we reach that, we basically,

Unknown Speaker 18:20

Okay, yeah, I think that should take care of now. Feel free To test it. I

Speaker 1 20:34

Yes, I think we need A string

Speaker 4 20:39

area, another character. I character. Why? I

Unknown Speaker 21:34

it. So why did you make a change? Specifically

Speaker 2 21:41

change? So I'm just like, I'm just trying to, you know, like, get the like, the character string, because we're just doing a character comparison. So that should basically be like and like I'm doing like the uppercase and lowercase on the set, because, like, you cannot convert a character to an Upper uppercase and then, like, preserve the original value. Okay?

Unknown Speaker 22:28

Yeah, I think yeah, right now, I think

Unknown Speaker 22:31

We have it reversed only in the vows. I

Speaker 1 23:07

What is the time complexity of this process?

Speaker 4 23:15

Time complexity, so, so, so here we are basically doing

Unknown Speaker 23:23

an embedded while loop, right? So,

Speaker 2 23:29

so I think the complexity here would be probably go up to like a big O of n square, all right?

Speaker 2 23:48

Oh, sorry, sorry, like we're doing two pointers, right? So let me see I

Speaker 2 24:08

So okay, so we are doing a while left, while loop outside. We'll bring the two pointers, and then there's another while loop inside. I Okay,

Speaker 1 24:24

that's perfectly fine. If you think it's n squared, I just want to know where you're seeing that that's all

Unknown Speaker 24:31

like, where is the square?

Speaker 2 24:34

So I don't think it's n squared, because, like, we like going all the way through the to complete length. So, like, it should, should be, maybe on a, like a logarithmic scale. But since we have, like, a nested while loop,

Unknown Speaker 25:00

might be, I

Speaker 2 25:06

think it will be pico FM, because each cat would only be processed once. And so even though, like there is a nested loop, it will only mean that the outside loop

Unknown Speaker 25:21

will decrease by that so

Unknown Speaker 25:25

it will be like single, linear scale. Okay,

Unknown Speaker 25:30

yeah, that's, that's kind of

Speaker 1 25:33

so now, so far, you have reversed a string which is just one word, and reverse walls in a string which is also just one word, you're just going to move on to something a little more complex,

Unknown Speaker 25:47

and that is this.

Speaker 1 25:51

So instead of reversing just a single word or doing reversal in just a single

Unknown Speaker 26:04

word. How about if I give you a string

Speaker 1 26:08

of words, you know, reverse the place where the words are. Okay?

Unknown Speaker 26:19

So this is also without extra space.

Speaker 1 26:23

There will be extra space, but once you finish the reversal, the extra spaces should go away. There should not be any extra space between words. There should not be an extra space before and after the sentence.

Speaker 1 26:42

Feels feel free to use any pre existing functions that you feel might Don't, don't do string, dot reverse when I say pre existing function, that's not what I mean, but anything might help smooth out some of the smaller processes.

Speaker 2 26:59

So I think that this is just, you know, this is just, you know, spilling it and then converting it into a stream, and like collating it to a reverse order, feel

Unknown Speaker 27:13

free to Try it out,

Speaker 4 27:20

so we might not even need a stream operation. We should just split it and then loop from back. So

Unknown Speaker 27:40

I will have this

Unknown Speaker 27:50

sentence.

Speaker 2 27:54

We can trim it first, but since you mentioned, there will be normal like leading or trading space. There will do,

Speaker 1 28:04

that's what. So there will be leading and Trading Spaces, and there will be, there might be extra spaces between each word, but when you give me the final result, it should be no, it should have no extra spaces and no leading or Trading Spaces. Okay, I can give you another example with letters,

Unknown Speaker 28:31

yeah, I think I'm good. I think we can proceed

Unknown Speaker 28:39

something like this that should also be

Speaker 4 28:44

first, let's streamline leading and dreading spaces by Using functionality, and

Unknown Speaker 28:59

then we can just split it. I it just split it using our space, and then we loop from the back. We're

Speaker 2 29:20

so then we like store the value in string builder and

Speaker 4 29:45

so If the character at that Word is not space, just

Speaker 4 30:31

so it's not planned, then it just appended.

Speaker 4 30:42

I And okay, so then after each word,

Speaker 2 30:49

we want to append it, but we don't have it like we don't want to open it when it's zero. So I

Unknown Speaker 31:15

Okay, all right, that should be

Unknown Speaker 31:20

it. Free to test it out.

Unknown Speaker 31:24

Feel free to test it out. Sure.

Speaker 4 31:58

Can I think it's streamed and doesn't have extra spaces.

Speaker 4 32:11

So it does have extra spaces, Right? So can I Think I

Unknown Speaker 34:12

What do you think is happening right now? So

Unknown Speaker 34:19

so like the empty string is not

Unknown Speaker 34:25

being Like, basically being considered

Unknown Speaker 34:31

something, something's wrong here. I

Speaker 2 35:08

Oh, here, so I'm adding a space every time, right? So,

Speaker 1 35:22

yeah, yeah, as well, wait and see if you for another five minutes if you caught that good catch. Yeah. Sure? Yeah, I think that should work. I mean, that is working as expected. So, yeah, that's all the questions I had. You have many minutes left. So now is now. It's your time. Feel free to ask any questions to either of us, or both of us,

Speaker 2 36:10

sure, yeah. So yeah. Thank you so much for the opportunity. It was really nice talking with you, and I like to know like, a little bit about, like, more to on, like the day to day of the engineers on PayPal?

Speaker 1 36:25

Yeah, I think both of us can give some insight on that. I'll start first. I mean, I, I won't be able to provide details on each and every part of PayPal, because, you know, it's a vast company with a lot of moving pieces. But from what I've seen and what happens with me, for example, we usually have like a two week sprint. So every two weeks we have sprint planning, and the week before that sprint is ending, we kind of have a retro so except for days on that, like that, for the other days, we have tickets assigned to us, so we kind of get check in, have stand ups at some point in the day, probably Monday to the first day. Some teams do it, I know, like three days a week. Sometimes do it four days a week. Fridays are usually days when you don't want to have meetings, unless you really want to, in which case, no one's stopping. But most people wouldn't serve team meetings. And, yeah, if you, if you're going into office, well, fine, but otherwise, just finishing up on tickets. And if you're on call, then there are help channels based on stuff to your team own. So if you're on call, there might be stuff that you have to, you know, attend to there. So here, anything to add to that?

Unknown Speaker 37:46

Okay, yeah, that's what my day to day would be.

Speaker 2 37:50

Yeah, sure. Thank you so much for that. I really appreciate it. And I think the other question that I have is like, what do you like most about working here,

Speaker 1 38:03

me specifically, I would say I love the people I work with, the team that I work on. I've been lucky to say that it's all the best people I've ever worked with, and not just best people. I'm not saying that just because they're nice to work with, but they're also very competent, and they kind of love all the ownership that comes your way. If you're working at PayPal, there's a lot of, as I say, moving pieces. So there's a lot of opportunity for ownership, and for everyone that I've encountered loves taking up ownership of even new fields, new projects, new initiatives, getting an opportunity to implement new designs, come up with new ideas. So being appreciated for that, and being able to kind of not only maintain things that are already there, you know, it's been here for like 30 years now, so there are a lot of things that are already here, but also building new things, like all across PayPal, we had a bunch of new initiatives go out across every brand and product you can think of, just in the last year itself. And for a company as you know, as was been around since, like the.com era, to be still doing that is means that, you know, if anyone's working here, there's always an opportunity to work on new things. So, you know, working with people who are excited for that and love to work with you on a team is always, always a good feeling. Yeah,

Speaker 2 39:35

yeah, thank you for that. Yeah, that sounds really good. Yeah. Thank you so much for that information as well. Okay, yeah, and I think that's all the questions I have for now. Okay, yeah,

Speaker 1 39:50

no worries. Thanks for asking the questions and also attending the interview. I wish you the best of luck for any further rounds that you have. And, yeah, have a wonderful rest of your

Speaker 2 40:03

day. Yeah, thank you so much for that. I really appreciate it and yeah, you have a great day as well. And thank you for like conducting the interview I really had a good time.

SRV-Robinhood- System design- Bishwo

Speaker 1 00:06

Home, we are going to do architecture interview. So let me explain the process a little better. So for the for the first couple minutes, we will introduce each other, then we will do the architecture question for the last couple minutes, for you to ask any questions, yeah, maybe. Let's start with introducing yourself in the b1 let me set up a, b2 Yeah,

Speaker 2 00:31

and I've been working as a primary packet engineer for about seven years. Primarily been working on Java and Spring Boot related technology, but working mostly on microservices, developing them, and even working on orchestrator. Recently, I've been working with an orchestration platform where, basically we orchestrate all the requests after you log in, all the requests that goes through the Capital One Bank flow is orchestrated through our application. We use a framework called Spring file gateway and Netflix Zoo, and basically we do request manipulation, response manipulation, device validation and so on. So it's like an platform that orchestrate those primarily working on the modernization of this using function factories and predicates and so on, being involved mostly on writing, implementing, sometimes designing these, and also involved in a lot of products and support. So that's a brief overview of what I've been doing at couple one.

Speaker 1 01:36

Okay, sounds good, yeah. Let me also introduce myself as well. So how many times you knew from the Career Institute? Yeah? So, so when the clearing team does basically to do the trader clearing and all the trips that happen during the day. So the trader, the trader clearing is like end of day process to make sure all the trips that Robbie wouldn't always match with the trade that exchange of animals, and also the settlement process, basically to make sure all the shears movement. So these are the two major domains,

Unknown Speaker 02:16

and also there are many small domains as well. I've

Speaker 1 02:20

been working here for almost three years. To eight.

Unknown Speaker 02:29

Yeah, so, so

Unknown Speaker 02:30

that's about me. Okay, cool.

Speaker 1 02:32

Let's move on to the architectural question. Are you on the call the signal by the board,

Unknown Speaker 02:41

the mirror? But I see

Unknown Speaker 02:47

okay, cool.

Speaker 1 02:50

Can you see the question already? COVID is the question to the board?

Unknown Speaker 02:57

Yeah, give me one second.

Unknown Speaker 02:59

Okay, that should take the time. I can

Unknown Speaker 03:03

see it. I can see it.

Speaker 1 03:06

Let me quickly go with question and also the requirements. So yeah, today I will go into job schedule assistant. Have you used any job schedule assistant before?

Speaker 2 03:20

Job scheduling, not like directly, but I think we did, there was some kind of downstream that you did.

Speaker 1 03:29

I guess you may have some idea about what the job schedule system and how Job system works. So there are a couple of requirements for this job schedule system. So the basic requirements is like, users should be able to create a job, and the system should be able to schedule and run the jobs based on the schedule that you provided. And also the system should be able to record the job status, like failure, success for the job runs. And also the system should have a strong guarantee about the job runs. So what a strong guarantee means is, basically, let's say you just schedule a job at some specific time, like, let's say 1pm eastern time, so assistant should around this job at that time, and should not escape the job or delay the job a lot. So let's see, even during the failure, then, the system should be able to know the failure and notify the user, and also the user should be able to view the logs and the status of the running job, previous job. So basically, users should be able to view the job history and also job status and a lot associated with each job run requirement is user should be able to define Sara. Then if the job is taking longer than Sara, then the system should be able to notify

Speaker 2 04:58

the user. Are you familiar with what Sara means? Yeah, it's sorry, several agreement, yeah,

Speaker 1 05:05

okay, okay, sounds good, yeah, so, so, yeah. So these are the requirements, and I will

Unknown Speaker 05:10

continue to drive the interview so.

Unknown Speaker 05:24

Yeah, please, feel free to start. I would like to join.

Unknown Speaker 05:32

Okay, okay, okay, please, yeah,

Speaker 2 06:06

so basically we have set of functional requirements like job creation, scheduling and execution, and we'll also basically have like monitoring and logs for users To view the status of running and completed jobs, right? So, like, users should be able to create jobs with specific configuration and systems should support when the job should run and at specific times or any kind of interval. And based on those like, your jobs will run and handle various like, some kind of status, in a sense. So basically, yeah, we'll have that and there, we probably would want to include some kind of time out, because if there is some long running job, we want to manage that and make sure that are responsive based on that. So that's kind of like your functional side of the application, but there will be more non functional as well. Like how reliable system is, so we ensure that the jobs they're run at scheduled time with strong guarantees, and if there is an increase in load, or if there's a large bond jobs like the system should scale accordingly. We should also make sure that any kind of error that happens within the application we are encountering for Fall tolerance, right? So basically, add any kind of retry mechanisms, if possible, with exponential back off or handle those failures gracefully. We need to ensure some kind of status to have the data consistency and have a better performance. So that's kind of like what I can gather from for the functional and non functional side of the requirement. So let me think about some high level of what I what we might be able to do. So basically, I think we will have a manager, right? So this manager, it should basically be handling like the creation of the job, scheduling of the job, and updating up the status of that job. So this job manager would be one component that is responsible for those then maybe we can add a scheduler that manages the scheduling of this job and initiates this execution at correct time, and then we'll have an executor that executes that job and reports the success and failures. We might add some kind of database to store the job details, the schedules and the statuses and yeah, that should be like a high level architecture, along with, like proper monitoring and logging in place. Of course,

Unknown Speaker 09:12

that is what I'm thinking on a very high level. Okay,

Unknown Speaker 09:29

oh,

Speaker 2 09:32

so So to talk about, like, how, like, there's a how the flow works. So basically, he just submits a job through like, some kind of API, and these API will contain request payload with job details which is stored in the database. So on the job creation component, a user will submit a recreation request to an API, and these details would be stored in the database. And then the scheduling component, it is a cron job,

Speaker 1 10:06

or sorry to interrupt Are you? Are you gonna join some

Unknown Speaker 10:11

diagram I just

Speaker 2 10:18

so and then once the job is created, you have that period scheduler whose schedule time has arrived, and basically adds to the executors. So once the schedule runs, it looks for those API if there is, if the time for that scheduled job has arrived, it adds to that executor serves. And this executor just pulls the job from like your queue, and then runs them if

Unknown Speaker 10:46

it's success or failure. That's what I was thinking.

Speaker 2 10:53

So to get started with the diagram, I

Unknown Speaker 11:10

so let's see.

Unknown Speaker 11:16

We will have a

Unknown Speaker 11:25

first.

Speaker 2 11:29

This is the like the clear client user, right? You basically be

Unknown Speaker 11:40

calling this

Speaker 2 11:42

so all these things will be buying a load bouncer, and will users will be calling this load bouncer.

Unknown Speaker 11:58

How do I make this? I

Unknown Speaker 12:13

find the lookout sir.

Unknown Speaker 12:26

You both have a question. I

Speaker 2 12:45

scheduler here find a load balancer, distributed

Unknown Speaker 12:53

scheduler job

Speaker 3 13:00

that's the A basically

Speaker 2 13:22

schedule our sentence message to The queue and the job executor reads that queue. See,

Speaker 2 13:42

indicator, we will have monitoring services

Unknown Speaker 13:50

or monitoring the status of those jobs.

Unknown Speaker 13:56

While we're doing that,

Unknown Speaker 13:58

scheduler

Speaker 2 14:02

will store the information in some repository connected to a database.

Unknown Speaker 14:22

Yeah, so I think,

Speaker 2 14:29

so, yeah. I think we will. That would be an overall diagram. We are basically here using a message in queue, like maybe a Kafka, to decouple the Job Scheduler with executor, which basically makes it scalable for dispatching those jobs, right? And then we will also incorporate a relational database, maybe post principle or transactional consistency, as we need to guarantee that jobs are on schedule it has proper status and proper logs. We'll also incorporate retries and timeouts to handle any kind of fault tolerance error scenario, and we'll use AWS CloudWatch to store and query those logs and real time monitoring, we can integrate something like a probe use dashboard.

Unknown Speaker 15:28

So yeah, that would be

Unknown Speaker 15:32

this diagram is representing.

Speaker 2 15:35

Now we will also need, like the scheduler, APIs, right? So on the scheduler, we will have Job Scheduler APIs and also APIs for monitoring. So we'll have,

Unknown Speaker 15:53

so we'll have

Speaker 2 15:55

API on the scheduler here to basically

Unknown Speaker 16:06

scroll

Speaker 3 16:10

Okay, so user can

Speaker 2 16:17

basically create a post request

Unknown Speaker 16:22

jobs.

Speaker 2 16:25

He can get those jobs and get the job by ID,

Unknown Speaker 16:33

those would be the two exposed endpoints.

Unknown Speaker 16:39

And I guess we can also get our jobs for that

Unknown Speaker 16:43

were created.

Speaker 2 16:49

We'll also add monitoring API. Since that is one byte a piece for monitoring, we will expose a data endpoint for that job based on the job ID that we sent, and will add a status, so status of that job.

Unknown Speaker 17:09

I'm also thinking of a

Unknown Speaker 17:13

job

Speaker 2 17:16

like to get the logs. We can also do a logs endpoint,

Unknown Speaker 17:23

also doing

Speaker 2 17:27

all the status of all the jobs for making sure everything isn't so these are the endpoints that our job scheduler API would expose for an external user. So

Unknown Speaker 17:48

okay so?

Speaker 2 18:39

Now I think to tie it all in, we also need to think about, you know, how many requests we need to handle, right? If you are handling maybe million requests for a day to schedule these jobs or something like that, that means, I mean, we're getting about 213 per second. So for that case, we need to handle proper load balancing. And in the front we need some kind of API gateway so that your services can be scaled up or down based on some kind of metrics that we add in, like any CPU or memory fluctuation. And, you know, for easier retrieval, we can add database indexing on the jobs ID column whichever jobs are getting right, right, so have low latency for our services, and maybe also thinking of implementing Some kind of caching for The job by ID. I

Speaker 1 20:39

Okay, can you? Can you dive deep a little bit for like, each major component?

Unknown Speaker 20:46

Yeah, sure. So,

Unknown Speaker 20:52

basically, we have this

Speaker 2 20:57

so you're a user making a call to our job scheduler to create a job, right? So basically you have this first step load balancer, which is to basically your load balancer, right? It distributes your income load to like it distributes it up evenly across your given instances, you can use any kind of strategy like round robin, and basically it is used for like your request routing and failover and auto scaling support, right? So if anything goes down load balancer, your auto scaling can then get the instances number up directly. In load balancer, you can also have a health check pinging, which means that if your instance is down, and the health check endpoint that you define, it turns back when the status down, then you can build that instance and then get it up. So yeah, it would basically be used as an interaction for the schedule or service. Now our main service, the scheduler. It will be the central point which is used for managing those job creation, scheduling those jobs, and then dispatching those jobs to be executed. So basically, it keeps track of what the request was, what the timing is, and what are the scheduling rules? So basically, it has these, I guess, three major functionalities, right? One is basically to create a job, right? So it allows one who wants to schedule something for a job. It allows them to define a job with necessary details, like aim schedules parameters or any other custom rules that we might that the client might want to add it. And then basically it stores those schedules, which is the scheduling portion it stores into a repository, and period T checks in for that job that are due to run. So that schism services secondary task and then third task it holds is when, like that time for that job arrives, right? So it dispatches it to messaging queue for executing for the job executor. So basically, it all it does. It takes that message jobs that are queued due, it just pushes it to the message queue.

Unknown Speaker 23:29

Basically, what we're considering here is

Speaker 2 23:33

we have multiple instances running, and if there's one failure, then the load balancer drops to the other instances, then that's why we have a load balancer in front of it. And then we also have this uniqueness, because it ensures that jobs are unique and there is no duplicates to be dispatched to the messaging queue. So yeah, the primary interaction this schedule of services it has is with the job repository to store and then the messaging queue to dispatch the job, so that that's like, that's what the primary responsibility for the scheduling component is. Now we will have another one that sits right after the mess scheduling service, which is the messaging queue, and all it does is like a PubSub, right? So basically, it's this decouples this scheduling to it decouples that scheduling apps and execution of the job, right? So it allows your jobs to be queued and basically be processed by the executor at a later time asynchronously. So basically, the basic functionality it holds is queuing, processing and then retries, right? So if it fails, we might add a dead letter queue for storing it, or we might consider retrying it couple of times, so like, maybe three times on exponential backup, and then if it fails, we push it down to $1 queue.

Unknown Speaker 25:09

That's what this component is about.

Speaker 2 25:13

Now we have this executor service which executes the jobs by picking them up from that queue. It basically executes that job, and then it is responsible for handling the time out of that job, right? So, as we mentioned earlier, we might corporate a set time on for the so that our systems not keep on task. So we'll monitor the job time, and if it exceeds then it can either terminate or mark it for a retry. So this executor service would be responsible for execution of the job time or handling and also updating the status for each job into the job repository. So that's the three major components, three major functionality that these job service, executor service takes basically, yeah, I guess, in a nutshell, it just pulls the message from the queue and processes them and updates this job status and logs to the repository, and then we have this monitoring service which tracks the status of the job, which basically is used for real time monitoring and aggregation. Let me ask

Unknown Speaker 26:41

okay, let me

Speaker 1 26:42

ask you a question. So, so let's, let's say users submit a request to the Job Scheduler. So the job scheduler based on what they are diverse, if we're persistent job requesting to dv, right?

Unknown Speaker 27:00

So can you,

Speaker 1 27:01

can you just, can you just tell me what the DB and what the tables, what the table, then what the tables?

Speaker 2 27:11

Yeah, so since we like, since, you know, we're storing, like, whenever job is sent in, and we need to basically be able to support scheduling. So we'll basically have a jobs table which stores the detail about each job, right, and we might include a name of the job, the description of the job, it was created, what the status is and when it was last executed, right? So it's the last

Unknown Speaker 27:47

time. Yeah, sure, we'll

Unknown Speaker 27:52

have this second so this I

Unknown Speaker 28:00

this

Speaker 2 28:05

jobs table. And the jobs table, you would have your job ID, which is like your primary key. You'll have the name of the job. I have its description. You add in a created timestamp, created at you might want to add a status, if it's active or big drive, and you add last execution time, which was like the timestamp that it was executed. So that would be the first table that we have. We'll also need a job schedule table, right? So this table would hold schedule ID of that table, of course, and then we'll have the the actual job ID associated with the schedule, which will have a foreign key to the jobs table, job ID will have some CRON expression to tell how often it should run, and also include this column called next runtime to indicate when it's scheduled for next and maybe, like I mentioned, part time handling will have like a max duration that the job is supposed to also have any kind of retry policy attached to that particular job.

Unknown Speaker 29:54

We will also try,

Speaker 2 30:00

like once you have an execution in place, we'll have a job executions table. And this was primarily it will have its primary key. Then after that, it will have a job ID, foreign key to the jobs table. It will have the start time when the job started and have the end time when the job ended, and it will also have, like a status job. And it will will also include a retry count, if there was a retry that happened on this table, on this particular job. In case of errors, we can add error messages on this table and execution. And in case of failure, we might want to include any error message on the table so

Speaker 2 31:12

and since we want to make sure we have proper monitoring and logs in place, we'll also create a jobs On the table, which would basically consists of the details about the logs, right? So we'll have this primary key, we'll have the execution ID foreign key, which so that you can track a log directly to what it was with. And once that is done, you also include a basic timestamp or that log, and you will also maybe, if it's a success log, it might just be info, or if it's an error log, you might include the log level as info, and you actually have the actual log message that you want. So that will be the job of the logs. Table,

Speaker 1 32:09

okay, so why do we need like, job table, job schedule. Table, job execution. Table, three tables.

Speaker 2 32:20

So what's the difference between considering so jobs table, it just stores your basic metadata about each job, right? And it has one Status column that indicates if the job is currently active. Calls are disabled. So that's it's just a basic primary metadata and job schedules table, it just defines when and how each job should. Instead of like, having those all together in one place, we just,

Unknown Speaker 32:49

we just create two separate tables to like,

Speaker 2 32:54

yeah, your job schedule table contains your pron expression that defines the job schedule, and it also contains, like the retry policy, right? If it's what kind of retry policy you want, if you do want to retry at all, if you want a fixed retry of maybe certain seconds, or if you want exponential backup. So you're just separating those tables out, and you have the job execution table, particularly here to track each individual execution of the job, including like the start time, end time and the outcome of that execution, followed by the so that execution table, you also have this status column, Just so that it provides, like, a real time execution state, right? So it might say something, like, the job is running, it's completed or it's failed. And then finally, you have the logs table for your storing the log information for each job. Or faster, like, if you like, you can retrieve logs based on an expression ID for this.

Speaker 1 34:08

Okay, so let's say users submit a request to run a job every two hours or every one hour. How does the job scheduler table record looks like?

Speaker 2 34:22

So yeah, so if a user submits a request to run job every hour, the request sample. So basically, it will

Speaker 2 34:41

so So you want me to show it on the same sheet, like the whiteboard, right?

Speaker 3 34:50

Okay, so here we let me create an intrigue. Do?

Speaker 2 35:36

So your schedule would be whatever fluid you want. So it would be like either short fluid or non fluid. You'd have your job Id choose again, like a primary key. You can again have it at the fluid. Then it will have your chronic expression telling how long, how often it should work. It would just show, like the numbers on non chronic expression.

Unknown Speaker 36:01

So for that, something like that,

Unknown Speaker 36:07

then you would have

Speaker 2 36:11

next run time. It would be the time for the next run. So for example, the

Speaker 2 36:29

duration, like the time on policy attached to this, which basically is 1800 seconds. And retry policy, so you want to conclude if it's slowly try, if it's exponential backup. So for this example, we'll have an exponential ankle

Unknown Speaker 36:57

schedule table.

Speaker 2 37:18

On the Job table, you have this job ID,

Unknown Speaker 37:23

including earlier would

Unknown Speaker 37:27

be the same one? Yeah,

Speaker 1 37:30

I don't, I don't need an interest for the job table. So, yeah, so, so once the job was the job schedule record is created, and the police gonna create the job record once

Unknown Speaker 37:49

the schedule record is created,

Unknown Speaker 37:53

which conclusion is gonna create the job records once

Unknown Speaker 37:57

you Have the schedule record created,

Unknown Speaker 38:00

for creating the jobs record,

Speaker 2 38:04

we will be have some kind of scheduler, right? So it is this same component that we made earlier for the job scheduler component. It manages this entire jobs table. So basically it provides an API or an interface to basically include the setting of the job, name, description and etc. So when the new job is defined, jobs schedule a first grade to enter in the jobs table with the metadata on the job. It sets the status to active records the other metadata which were initially known they would be added so it will basically be run on the schedule and create that entry on the jobs table

Unknown Speaker 38:56

about the job execution. Who is going to create the job?

Speaker 2 39:00

Basically, once the job, once the job and schedule are created, the Job Scheduler uses the same scheduled data to determine how to dispatch job to the job executor. So each time the job executes, the job executed, the job executed service, right? It updates those job execution table and also the job box table. So yeah,

Speaker 1 39:25

my question is basically the schedule around the job every one hour. So how many, how many job execution records are you going to create? So

Speaker 2 39:46

so every time, right? So whenever a job, so every time a job, when somebody schedules a job for every hour, it will execute an hour, right? So it will create a record every hour. So once somebody did say, for an example, right? So a scheduler is there to execute. You'll pick up that job send it to the messaging queue for execution. So yeah, so a scheduler will basically pick up that job and send it to an execution so it will so every time it executes, it creates a record every hour.

Unknown Speaker 40:24

Let's say, let's say you have, like,

Speaker 1 40:28

100,000 jobs, different schedule. How did you figure out and how do you handle the creative job application?

Speaker 2 40:40

Basically, if you have a huge number of jobs, right, we have the load balancer in front. And if we have enough stores for our enough processing units, it will process it. So basically, by the job creation time, and let's say, whenever the scheduled time, right? It was set up during the definition of job. Once the job is defined, you will have a metadata to start with, right? So contain like that start time and time and so on. But so the job execution, the Job Scheduler, right? It will pick up that job, that next, and we have the next job runtime set up in that job table. So to look at scheduler, and it's basically responsible for picking up the jobs that are next, and then it will send you to the messaging queue for execution at a later time, asynchronously.

Speaker 1 41:35

My question is, let's say your job schedule table has like hundreds of enrollments, has a different schedule. How does Job Scheduler figure out which job needs for the job execution?

Speaker 2 41:50

So So if each so like for each data that is different in scheduled turn, then from that metadata setup, it will know the next runtime, right? So from the initial metadata that we insert, it will know what the next runtime is. And like, like, once we know that which time the job needs to execute, then this it will look up for the next runtime, the Job Scheduler, like it will basically database and look at what's coming next, and to look for that information from the job table and only put those jobs that are up next.

Unknown Speaker 42:33

Okay, so who is going to create the next one?

Unknown Speaker 42:38

So that would be the two. Does

Speaker 2 42:43

be defined during the definition of the job, right? So it will the jobs will maintain how often is, how often it needs to run, and when it needs to run. So when you define the job that you have certain you can define certain interval and when you want to run the job next, right? So if you will only do like a fixed runtime, you can define all those during that job, definition during the scheduler base.

Speaker 1 43:09

My question is, and so let's say the hourly job, but the first round have another like a 12pm that the next one have a certain PM. So who is going to update the next round from 12pm to start your channel.

Unknown Speaker 43:23

So,

Speaker 2 43:26

so basically, like we we actually have a setup for the enroll, right? So if it's hourly, we have that cron job scheduler that is the job of the scheduler to see how often it needs to run, and it will pick that up every hour. That is the scheduler job we already have within the scheduler

Unknown Speaker 43:52

component, okay, so,

Speaker 1 43:56

so let's say the table has, like a one minute job. So you may the Job Scheduler is gonna query the table to find out which job schedule to be wrong. Is that correct?

Speaker 2 44:09

So once you set up, once you define the top you you know it will record. If you do a job record is created on the job table. So basically, your job scheduler will look for its metadata for you know how to set that job up for execution, and it will set up those records for execution. So if you have, like, a million record, yes, it will look up during its job created record.

Unknown Speaker 44:41

So each job can

Speaker 1 44:44

schedule differently. So let's say one job is like everything, one job around every hour, one job around everything. So how often? How frequently the job schedule across the table

Unknown Speaker 44:57

to figure out which job needs to be wrong.

Speaker 2 44:59

So basically, we need to have some kind of great limiting or something set up in our scheduling that if the job, you know, the number of jobs of high volume get keeps getting higher and higher, we have to set up like a batch processing so it will switch to the next batch instead of reading each record every time. So we trying to do batch processing. We define the number of record that needs to be read during the next batch, for example, right? If they're like you said, if there's a million record created yesterday, and we have, we chunk those million records into maybe 100,000 records, so that it will be divided. And then each set of those 100,000 records, we read of them, and then once we finish processing those data, we can look up to the next 1000 records. So it would be some kind of batch processing place.

Unknown Speaker 45:57

My question is about,

Speaker 1 46:02

so you mean the job schedule system. The Job Scheduler service basically gonna do three things together, request from the user, persisting to database and also pulling the job to start and also scan the table to figure out which job needs to be wrong, and the schedule the job. Understand the job basically Correct?

Speaker 2 46:29

Yeah, that your scheduler is your main like main job. Do you see

Speaker 1 46:36

any problem with one service? Does so many things? Do?

Unknown Speaker 46:44

I mean, yeah, one service

Unknown Speaker 46:49

handling,

Unknown Speaker 46:52

basically, you know,

Speaker 2 46:55

we actually have your job schedule, responsible for management, right? But there are other different functionalities. I mean, it can lead to issues with scalability, but for that, we separate the responsibilities. We create like four different services within that component itself. So like, we can create a job creation service, which is separate service, and it will have a scheduling service as a separate service, and we'll just divide that so that you have four different services taking care of each of those jobs steps, right? So creating scheduling, putting in the database and then polling service. So that way we have different services. This way, if you divide it like this, you'll get rid of a slot failure scenario. If you and you can have the schedule service periodically check the job and basically only move the necessary jobs in the necessary queue. So this will basically eliminate your Singaporean s failure in this distributing system.

Speaker 1 48:12

Okay, so how do we handle job failure? Let's say a job is running our executor, but then through exception and the crashes, how do we handle job failure?

Speaker 2 48:22

So for basically handling your job failure, we have, we can basically set up a retry mechanism and a backup policy, right, like I, like we did on the database schema, so we can, basically, whenever a job fails, you have a set number of retries and what kind of backup policy, right? So we can automatically retry on the fields of on your predefined policy, right? Then you can provide a second chance for you to succeed. So for example, there are some backup policies, right, fixed backup and expand exponential backup, right? So we can implement those so that you don't keep on retrying right after or you can retry after a certain interval. And if you use exponential backup, it again progress down the execution of that. And we can also implement some kind of circuit breaker pattern where all the failed jobs prevent a repetitive job failure if it's failing multiple times, right? So it can cause some kind of outage, or something like that, and it will cause all, it will basically cause all the retries to fail. So if we implement a circuit breaker, we define some error threshold for each job so that after failing sometime, it will try something else, rather than keep on retrying the same thing.

Speaker 1 49:43

So can you explain the workflow? Can

Unknown Speaker 49:51

you explain the workflow? Yeah,

Unknown Speaker 49:55

so basically,

Speaker 2 49:59

whenever you have your job that fails, we so in our service, we have set up a mechanism. So we'll have a mechanism whenever there's failure to retry, right? So, for example, you might have a transient issue, for example, if we find a failure, right? The detection of this failure is the first point, and when it fails to some kind of exception or timeout, or we have any other kind of error code, and the job executor will have a flag of failure, right? So then, once you have that flag before executing the retry, the system will check if the job is actually eligible for retry based on the policy that we provide, right so this retry policy can specify how many times I can retry and what kind of backup policy we want, then we will Update the retry count each time the job is retry, and then update the status of the job, and then basically re execute the job. And after certain like, after we the number of retry that we set, like it exhausts those retry count, we will basically move the message into a generator here.

Speaker 1 51:21

Okay. So basically, you are suggesting the job executor will handle the rich man.

Unknown Speaker 51:29

So, yeah, so the basic, okay,

Speaker 1 51:31

so within the job executor, the hardware,

Speaker 2 51:38

so, in that, so you so you Sorry. Said, Could you come again?

Unknown Speaker 51:45

So whether for the job? So

Speaker 2 51:52

if you have a hardware failure on the job, executor service, basically, in that scenario, we have set up the so first thing if once a machine fail, we have a load balancer in front so it detects any kind of malfunctioning of the service and then brings up a newer instance. Or we can also add some kind of API gateway in front, right. And now this API gateway, it will

Unknown Speaker 52:25

switch a traffic someone, but

Speaker 2 52:28

we did have the messaging queue in place, right, so we will not have any loss of data.

Speaker 1 52:36

So, why? Why? Because, I mean Vera function job is already running the executor while we thought we would not lose the data.

Speaker 2 52:52

So your jobs that are already running will have some kind of some sort of persistence for those job status, right? So since we're saving that in our database, you'll have to basically, for example, let's say for an example, if you have a failure, we can have a multi region deployment database where the database snapshots are available in another region. So our other instance, whenever there's a hardware failure, we can read from those and we have the status of those jobs, which are, you know, failed, right? And since it's abrupt execution, we can then continue those jobs to and to the point, like from the point of failure once it fails. So

Unknown Speaker 53:37

let's say the job execute.

Speaker 1 53:41

How do you update the status in the job executing table to

Speaker 2 53:53

fail? In that case, whenever the job executed itself fails, persisting that into the database, I guess, whenever we have that scenario, like the heartbeat failure, like the hardware failure

Unknown Speaker 54:10

we have, we do have those

Speaker 2 54:14

mechanism for health check, right? So, like the heartbeat mechanism that regularly monitors the job interior oil. So that means like you're continuously checking the job executed service, making sure that the job are executed, and sending your signal for monitoring directly to the database, making sure that it is continuously running and healthy. So in case the monitoring system fails to receive a heartbeat, then the executor within the specified timeframe, it will flag the

Unknown Speaker 54:43

jobs field,

Speaker 1 54:51

because we only have two minutes, I just want to leave it the time for

Speaker 2 54:57

you. Thank you. Thank you for having me here. Thank you for the time discussing about this job scheduler. I guess some of the questions that I had was, what, what do you think? What are like your typical day to day that the developer goes through on Robin Hood for our team?

Unknown Speaker 55:17

Yeah, I would say the day to day worker

Speaker 1 55:21

depends on what the levels you are at. So for example, on my level, I need to spend like, 15, 50% of the time to do the meetings, to basically scooting out the work, gathering the requirements and align on the line, on the goals. So yeah, I will say percent of time. the time you spend on the meeting. And maybe, 20, 30% of the time, spend on writing the code and also

Unknown Speaker 55:51

the stand up at the time, with the record review.

Unknown Speaker 55:56

So, this is my digital looks. Thank you,

Unknown Speaker 56:03

thank you for sharing.

SRV-Robinhood-final- Bishwo

Unknown Speaker 00:01

I work on a lot of things.

Unknown Speaker 00:07

Yeah, tell me about yourself.

Speaker 1 00:08

Yeah. I'm Saurav. I've been like a software engineer for about seven years, working on primarily Java based application. I've been working a lot with microservices. Recently, at Capital One, we own a orchestration platform, not sure if you're familiar, but any requests coming into our service, we just make sure, like they are properly authenticated. Like, if any additional manipulation of those requests is needed, we do like, tokenization and encryption before sending me down to like, the dancing partners, right? So like, for an example, if you have login endpoint, we are the first source of checkpoint that, like, validates your device, like it's a proper device and it's not jailbroken or, like, a bad version of that being used. We just work as that platform, sending it downstream. So we use a library called Netflix zoom, and been primarily working on the migration of that app to spring manage Spring Cloud gateway infrastructure. Been working with Cloud, AWS cloud, primarily for hosting these application and working on AWS EC tools, ECS, on bargate SQS, SNS and lambdas. So yeah, that's primarily it. On a daily basis, I'm mostly involved in backend coding using Java and sometimes a language called Lua, writing unit tests, and been primarily leading the team with the developers coming to me for the help, because I've been a subject matter expert of this infrastructure for a while. And yeah, that's pretty much it

Unknown Speaker 01:44

good to hear Whistler

Speaker 1 01:48

names. Yeah, you can call me Saurabh, okay,

Unknown Speaker 01:54

thanks. Okay, yeah,

Speaker 2 01:58

thanks for sharing that. So let's hop straight into I want to give you as much time as possible for the question. So

Unknown Speaker 02:06

do you see your codes? Do you have the code signaling?

Unknown Speaker 02:09

Yeah, it should be my map. Sure.

Unknown Speaker 02:17

Yes. So once you hop in here,

Speaker 2 02:20

there'll be the question on the left hand side, and then yeah, so the question on the left hand side. Give me some time to read over it. Once you're done reading it, let's talk. We can talk over Yeah, your thoughts of the anything, any clarification you need

Unknown Speaker 02:43

and how you plan to tackle it? Yeah,

Unknown Speaker 02:45

sure. Let me go through this prompt.

Unknown Speaker 02:47

Sound Good?

Unknown Speaker 04:44

Next Question. I

Unknown Speaker 05:00

Next question. I

Speaker 2 05:24

It's you can change the language too, if you to like, Yeah, you said earlier to Java, so You entered The job. I

Unknown Speaker 09:02

She's just going Through the final bits.

Speaker 1 09:20

I'm still thinking through some other

Speaker 2 09:28

I also like to hear even if you're thinking, if you're thinking through, like, parts of under either understanding the question, like just how you plan to tackle it.

Unknown Speaker 09:39

I love to hear your thoughts. Okay.

Speaker 1 10:02

First, so thinking of basically creating a map to store the referral accounts and maybe a set to store the users who are referred so we can track who's already in the platform and who's not. And then we'll proceed with processing the referrals and updating the leaderboard as we go right. So if the referral or referee is already like if they are already on the platform, we'll ignore that referral, and if not, then we'll just proceed forward with like the user is now in the platform and will increase the referral account for like, since it's like chain based, so we'll refer Everyone up the chain, right? So upstream of the referral. So that's the thought process that I'm taking so far.

Unknown Speaker 11:49

When you say that, when you say that the

Speaker 2 11:53

when you see that the set is for if they've already been referred. So you're you're saying that you're basically seeing if the new users list is going to have duplicates, right? Yeah.

Speaker 1 12:04

So I'm going to be using a set, and it does not store duplicates, so like, if we do our contains, then it will be faster in terms since set is for uniqueness, yeah. So that would be like the processing method. And then we'll finally try to generate the leaderboard with like most three users, right? So we'll sort it based on the referral count. And if there is like same count, we'll just try to add an alphabetical sort as well, Zoom comparators

Speaker 2 12:41

to simple. Just simplify it. We can assume that the new users list will be unique.

Speaker 1 12:48

Okay, so we don't want to do like, a string comparison based

Unknown Speaker 12:52

on me, yeah, like, that's not, I don't think we need to have the Yeah. It,

Speaker 1 13:04

yeah, and I think that's who that should take care of, like the given process. Yeah. I

Unknown Speaker 13:23

Okay, sounds

Speaker 2 13:27

good. So how are you going to store the referral as a map?

Speaker 1 13:34

So, yeah, I was thinking like the, it will basically be a it will be a hash map with the referral fund, right? So basically, it will use the name like the referral as the key and how many times they referred as the valley, and then I'll use the hash set for like the referred users.

Unknown Speaker 14:00

Okay, gotcha. I Okay, gotcha.

Speaker 2 14:05

And then it's just your hashtag will store the mapping that's been like the direct like, will store the key, will be the the refer, and then what's the, you know, what's the key in the value for the hashtag doesn't have

Speaker 1 14:23

to list unique hash mapping value. How are you going

Unknown Speaker 14:29

to keep track of

Speaker 2 14:33

so in the case of an example that's in the

Unknown Speaker 14:40

in the question side,

Unknown Speaker 14:43

how will you keep track of

Unknown Speaker 14:48

referrals that are

Speaker 2 14:52

downstream? So for example, like, how will you keep track of C's when C refers D? How will you know? How will you update the number count for a So,

Unknown Speaker 15:12

basically,

Speaker 1 15:18

I was thinking of like, basically a referral chain tracking. Because, since I'm using a map to keep track of who referred whom, it will allow me to, like, look on my user referral and then propagate the referral count and update the chain, basically. So whenever you have C refers D, I need to propagate this update back to chain to B, and because B refer to C, then I'll update B, and I need to again update a. So basically I'll need to go like a tracker referral process. Basically, Gotcha, okay, how do

Unknown Speaker 15:58

you, how do you kind of implement that?

Unknown Speaker 16:09

So you said, deploy our

Speaker 2 16:13

implement. How do you know how to buy to implement that, like the tracking of like the tracking, being able to track from what, when you see the C, C to D, referral, how do you plan to

Speaker 1 16:26

Yeah. So for that, we have a map of a string and string as parameters, yeah. So whenever I update the map, I need to update the referral chain map as well, so we will have that map stored in that

Unknown Speaker 16:48

referral. The first bit.

Unknown Speaker 16:57

Yeah, I

Speaker 1 17:21

do I? Do? I need to write the main method I think, or it would automatically call the solution that I'm implementing.

Unknown Speaker 17:28

I also immediately call that solution function. Okay? Thank You. Of course, I

Speaker 2 19:59

so for Your flow chain, it's a string to string mapping. So if, in the case where one referrer refers multiple users, so let's say like we also have, in the example input, we also have a E, like a refers E. How do you plan to store that? Do you want to are you going to store that as two separate entities

Speaker 2 20:35

for that? Yeah, so like, let me, let me also

Unknown Speaker 20:41

just visual.

Unknown Speaker 20:47

State of

Unknown Speaker 20:50

this.

Speaker 2 20:52

So I'm just saying if you also have

Unknown Speaker 21:01

something like that, chain something like that.

Speaker 1 21:08

Yeah. So for basically, when processing the referrals, we update the referral account and propagate the account for entire chain of users, right? For example, we process the referral separately, and I think that should be the way to go, like by processing each referral as a own entity, separately.

Speaker 2 21:35

I'm asking how you're not how does the referral chain map? How do you want the flow chain map after like, I just want to like, Look state of the blockchain map during your process. After you process, let's just say you don't even need to process the B and C. Just process only these, these two entries. I

Speaker 1 22:06

key so the referral map, it's kinda basically, key is the person who refers, and the value is the referral, so the person who made the referral, right? For example, I have, if a refers B refers C, then d, we like, go from B to A, and then C to B, and then d to c. And if a refers multiple, then we have another record for you know, like, let's say, for example, E to A, another record

Unknown Speaker 22:42

there. So,

Speaker 2 22:45

oh, so, okay, so the refer like person that's being referred is the key, and then the the person that referred them is the

Unknown Speaker 22:52

value. Yes, exactly. All right,

Unknown Speaker 22:56

sounds good. So then the math would look Something like this. Yeah.

Unknown Speaker 23:02

I'm Gonna

Speaker 2 23:19

Do I basically, it looks like you're done. Yeah,

Speaker 1 38:16

think I'm done. I was just, you know, doing Final Validation, just Making Sure You

Unknown Speaker 41:20

I do not evaluate the code signal,

Speaker 2 41:28

like to see what the comparison of your of your test results

Speaker 2 41:38

are? Sorry, no, yeah. Do you know how to navigate the code signal to see why, why the tests are wrong. Yeah,

Unknown Speaker 41:51

yeah, I'm running that individual test,

Speaker 2 41:56

yeah. Do you see what? What's the so? So after you just read the test you click into, you click into test one, and then go into, go under the return values to see what the your your return value is and what the expected return values. Okay,

Speaker 1 42:24

so the return value is 3y 2x one, probably, I just think it's a spacing issue. I think we can fix it.

Unknown Speaker 42:37

Okay, nice, yeah, yeah.

Speaker 1 42:42

I think getting all the tests fast. Now it was just the extra space that's causing it to fail,

Unknown Speaker 42:50

like I added in the

Speaker 3 42:52

front. Yep. Sounds good stuff. Okay.

Speaker 3 43:13

Okay, yeah,

Speaker 2 43:18

we are nearing the end of time, but good job. Thank you. Yeah, I want to leave a little bit of time at the end for you to try any questions you have about my work, or prominent in general, or anything that I might be able to answer. Yeah, any questions that you might

Speaker 1 43:45

have? Yeah, no, I Yeah, no. Thank you for that time for today. And I did want to ask about, like, what, what does your day to day look like on as a back end engineer at Robin Hood?

Speaker 2 43:59

Yes, well, my day to day is it's, it's changed a bit over the time. I realized I forgot to tell you how long I've been here. I've been here for like, three years almost.

Speaker 1 44:11

Yeah, I checked your LinkedIn. You were a different position, security, something in the past, and now you're a software engineer.

Speaker 2 44:18

That was just a change of title. They still do very similar things, like a bunch of reorg things. I actually used to work on a lot, like the token Island, like tokens related stuff too.

Unknown Speaker 44:35

But then,

Speaker 2 44:38

yeah, so then during, during those times, I was a lot more focused on the kind of like maintaining our back end, like total Conditioning Service, and then that there was not as much like product interaction. So I think it was very like heads down coding classic stand ups, mostly like that really depends on the team. Some teams have it like weekly. Some teams have it like every other day, and then yeah, and then so that. So during those, like, the first few years, also probably because I was working here, my opportunities was a lot more just heads down, coding and then casual meetings. But then I think in the past year or so, I've got a lot more involved, like, the scope of my work has changed to the product side of, like, of the actual like, just MFA implementations and whatnot, and that kind of allowed me to be a lot more involved with, like the product folks. So I talk with so I hop into like, product review meetings, or I get pulled into meetings or just discussing, like, how we should put tackle certain problems. It's, it's turned from more of, like an 80% coding, 20% meetings, to more of a 5050, that's a and then also, like, a lot more document writing too, because I have to write like, edge design is awesome, and help with the priorities the product design,

Unknown Speaker 46:12

yeah, hopefully that answers, yeah. I totally

Speaker 1 46:15

understand, once you go like, you get more less coding work, and then more talking and then figure things out, designing,

Speaker 2 46:21

yeah, sometimes I worry about, like, coding skills. I'm just, like, spending almost all day just in meetings. Yeah,

Speaker 1 46:29

I had that sometimes, whenever I get like, a sprint word that worked just for researching and, you know, just documenting them rather than actually implementing something. But, yeah,

Speaker 2 46:40

that's good that you have that experience too. Yeah, any other questions that you might have done?

Speaker 1 46:48

Yeah, I guess. I mean, maybe you can tell me what you like most about working here. What do you think is the best

Speaker 2 46:57

thing? So for me, I actually quite enjoy the

Unknown Speaker 47:03

I know some companies like

Speaker 2 47:07

there. I know a lot of companies just blur out a bunch of like values and whatnot. But I there are some values I truly do see in having worked in modern for quite a while, of just like,

Unknown Speaker 47:21

more lesson, like, the, was it

Unknown Speaker 47:25

the, like, the ability to just, kind of,

Speaker 2 47:30

it's like a blameless culture. I think they were really nice because, yeah, I mean, ultimately, you're, we're worth FinTech companies. So when things, when shit, when shit hits the fan, you know, things, things go down, and then it could be bad, right? But the blame was cultural.

Speaker 2 47:55

That wasn't the energy, like, like, Why? Why did you do that? Yes, and it makes it a lot more like, it makes things move a lot faster. It makes things a lot more open. And it just, it's overall like, whenever you talk to people around, I think it's a lot, it's really friendly too, just like, it makes it also really helps you learn faster, of just like the situations and how, how relevant runs too. Because, I mean, people make mistakes, right, right? So, yeah, so I think that's one, like, core thing I really liked about it, regardless of like, which people I've interacted with, yeah, oh, I think for the most part, like, blameless culture has been pretty, like, apparent in the company.

Speaker 1 48:41

Yeah? That? Yeah. That's really nice to know, because that's what working in a team and teamwork looks like. So thank you for sharing that, of course, yeah. Apart from that, I don't have any questions at the moment. Thank you for having me here. Thank

Speaker 2 48:57

you for your time. Well, thank you for your time and Interesting. Yeah, if you don't have anything else that I mean obviously there's always the recruiter, you can reach out to anything. There's also obviously the next interviewer.

Unknown Speaker 49:14

So, I'll give you.

Sabina shrestha-estes-final-srv-offered

Unknown Speaker 01:40

Okay. Okay, you should come to 12345, Okay.

Unknown Speaker 02:01

Do not Use your mouse. I

Speaker 1 08:28

so yeah, I'm genuinely excited about this role for several reasons that align perfectly with my professional and personal interest. Firstly, the opportunity to lead end to end project execution and manage programming strategies alliances perfectly with my experience at USAA and MasterCard,

Unknown Speaker 08:55

and

Speaker 1 08:57

I have successfully overseen multiple projects, from inspection to completion, I have been working on the dynamic environment where I have used my skills in scoping, planning and positing and risk management to drive Project to successful outcomes. I

Speaker 1 10:09

and also I got a chance to work with the you know, cross functional and global teams, which is highly appealing for me, and my experience coordinating with the diverse team, both onshore and offshore has helped me with these skills to effectively manage communication and collaboration across different time zones. For example, at USA, I integrated teamwork daily using Microsoft team which help us the project workflow and achieve the team environment in a good way, despite the geographical distances and yeah. Lastly, the role focused on like continuous improvement and changes management, which is perfectly aligned with my personal achievement. I believe the importance of adapting to the new methodology and process to enhance project outcomes. So, yeah, overall, these roles excite me because it offers the perfect blend of project management, data analysis and cross functional collaborations. Okay?

Speaker 1 14:10

So in my current role as a IT project coordinator at USAA, I manage a range of project that involves reviewing suspicious activities and resolving the fraud risk issues. Is my responsibility includes developing and maintaining the project timelines using Microsoft Project creating the weekly matrices reports with Excel and also designing interactive dashboard with Power BI to visualize the project data. So one of the main aspect of my job is supporting the project manager by developing training materials, coordinating with the department, making sure the project documents were properly tracked and organized in the SharePoint I provides project meetings, track deliverables and assist In creating meeting agendas and documenting meeting My News. I

Speaker 1 16:07

And yeah, looking ahead, I hope to grow professionally by taking more leadership roles in the project management. I'm particularly interested in gaining more experience in strategic planning and execution at the higher level. And I also want to continue enhance my skills in data analysis and visualization, as I believe these are the crucial for making informed decisions and driving the project success. So yeah, I'm eager to continue developing my skills and taking on new challenges that will help me contribute more effectively to the success of my organization and advance my career In our Project Management. I

Speaker 1 18:31

So, yeah, I have, like the years of experience working with the Microsoft Office Suite, particularly Excel, which has been a very important tool in My roles at USAA and also a MasterCard. So in my current role at USA, I regularly use Excel for data analysis and recruiting. For example, I use export list to excel and create pivot tables to produce weekly matrix reports. Though, these reports help track project success, progress, project progress, and also identify bottlenecks and visualize data trends. So by transferring the raw data into the pivot table, I can present complex information in a more manageable format, which allow our team to make informed decision quickly and effectively. And also during my time at MasterCard, Excel was really important for budget tracking, and it infrared instructor project, I created detail spreadsheet to monitor project expenses using formulas to calculate totals and variances automatically. I

Speaker 1 21:01

and also I am proficient in using advanced Excel functions like VLOOKUP and ads lookup and vva macros. For example, I use V League Cup to cross reference data from different seats, and it was essential for you know, providing information from multiple sources. And yeah, overall, yeah, I have been using Excels for build building project scheduling, and I have developed and maintained project schedules using Gantt chart to visualize the timelines and milestone. So yeah, my experience with Excel has been very important for data analysis. Was it tracking and visualization, and these skills have enabled me to contribute significantly to the project management and decision making process You

Speaker 1 24:59

so talking about the working environment and management styles, I usually prefer the collaborative and open work environment where teamwork and clear communication are encouraged, and I like managers who provide guidance and supports, but also trust their team to work independently. And yeah, I am excited for the place that offers opportunities for learning and professional growth. I value diversity and also enjoy working with people from different backgrounds as it leads to more creative solutions. So yeah, overall, I like balanced in management style and supportive work environment. I

Sabina shrestha-estes-final-srv

Speaker 1 00:00

And then same thing and senior Java developer now. Like have my hands everywhere so and I say the official health on a different team

Speaker 2 00:21

so can you share your information, give us a background and

Unknown Speaker 00:25

experience?

Speaker 3 00:28

Sure. So my name is Sabina and I have been working as a Java developer for about six years now, I have been working primarily on Java best web application, utilizing frameworks like spring hibernate, to web services for API development. Pretty good experience, I would say. And I've been working on microservice architecture, architecture, both using like Spring Boot and also have been working with AWS service for deploying applications. And also utilizing ECS, SQS, SNS and lambdas. On the AWS side, I do have like some experience working on the front end, I have used Angular and I can work with JavaScript and TypeScript also. And on the unit testing I have worked with. For that part I have worked with do you need to add Mockito and have also done a little bit of integration testing. I have used home monitor framework for that. Currently, I am on a scrum team. The team consists of like five developers, a couple of testers, and we have our product owner and Scrum Masters. And on the daily basis, I have been involved mostly on working on backend Java code right now. And I've been helping the team with code reviews, and also production support duties. So that's pretty much about it.

Unknown Speaker 02:11

Guess we can get started with data, the questions?

Unknown Speaker 02:15

I'm gonna go ahead with,

Unknown Speaker 02:16

you know,

Speaker 2 02:17

how do you handle errors in your applications? And what mechanisms do you find effective for troubleshooting. So,

Unknown Speaker 02:30

basically,

Speaker 3 02:34

whenever there is an error, right, I have to make sure like, you had proper error handling in place, right. So you know, like, you can like use, like, try catch block to handle those, and exceptions on method level. But on a more global level, you would like use something called centralized or a global error handling, using like, controller, like advice to handle exception across the application, provide a dedicated ID to that exception type, and then log the cause of that exception, so that you can return a graceful response back to the users, right? So sometimes, it might be error processing charter data, like it might be faulty requests into them. So depending on what kind of error it is, like, you kind of return a failure, either the 400 response back and for these purposes, you need to have proper logging level like so like, I would use logger dot error, and if not, for general application for like key steps. So I would use a use the info label, if it's sometime that that is helpful in develop debugging. And I would basically use debug level. Like starting points I'm very comfortable working with as long so using this kind of centralized logging system. It helps in like proper monitoring of the acts as well and also in troubleshooting, I would say. So, these are the steps that would be necessary for you know, logging and monitoring and troubleshooting, I think.

Speaker 2 04:39

Do you have any experience with Kafka? I'm sorry, I just didn't have it. I haven't really taken a look at your resume yet. But do you have any experience with Kafka and

Speaker 3 04:49

Oh, I do have experience in Kafka and I've worked in Kafka a little bit

Speaker 1 05:01

I mean, you get a general overview of like, the the main components their primary use.

Speaker 3 05:10

So, um, basically the, the different components of Kafka include you no producer, which basically send any messages to Kafka topic and you have a consumer that reads from these Kafka topics. So, those are the primary ones, but then there is a concept of Kafka broker, which is like the server that stores the data and server side clients requests a source and there and then so, basically, broker serves the client request then there is a topic which is kind of like a category two, which records our science and then there is like a force part like portion where you know, each topic has like a sub set of topic records, and there and that kind of like kind of like a manager like Zookeeper, there is a consumer group, which are like a group of consumers that work together. And then I guess, those are some of the key components of Kafka like that take place in Cafe ecosystem and say

Unknown Speaker 06:28

have a user and consumer system

Speaker 1 06:31

setting pulling messages from a topic? How can you ensure that you know those messages will be read correctly, you know, that the

Unknown Speaker 06:47

algorithms

Speaker 1 06:50

How can you make sure that the consumer can properly be serialized was being put by the producer.

Speaker 3 06:59

So, basically, in order for like to ensure that consumers are processing the data correctly, and is a DC realizing correctly, basically, I think we have a proper license and serialization in place. So right, so basically, you have a consumer consumer in Kafka that you need to provide a property for. And the property takes like a serializer, class config, and this serializer class config, so and then these properties are set into our Kafka Consumer. And that's kind of where basically, that makes sure you have the PC razor in place, right? And you can also define certain custom designs that you can use. So basically, the way it works is like, your method is serializing to by stream and when sent to Kafka, Kafka stores that messages and consumer retrieves that message as a byte and then and then using this DC riser that you have set in Kafka properties, I think it will convert back into an object and the message is processed. That's basically

Speaker 1 08:29

the most important very well, the answer he gave us was fighting for. I guess I'm more looking for how would you set it up so that if a producer tries to put a message on a topic that doesn't meet certain criteria, that would be rejected? Okay.

Speaker 3 08:55

So, when producer put certain messages, right, that doesn't. So that doesn't meet certain criteria, right? So that's, that's what he meant. Basically, what I would say is, we have some kind of pre validation or message filtering in Kafka streams. So if you do a pre validation, even before a publish, or like publish a record, it will prevent from that but you know, like within the Kafka stream, you can filter out messages that does not meet certain kinds of properties. So that's one way to do it. I think.

Speaker 1 09:46

I'm kind of running towards like a schema registry. Are you familiar with?

Speaker 3 09:52

So schema registry? So I haven't like work directly in that but I have, like heard about it. And like I've seen people discussing about immediate stuff.

Speaker 1 10:08

And the experience that you've done with Kafka and sending messages back and forth, the format

Unknown Speaker 10:17

or the payload.

Speaker 3 10:20

So, um, we, uh, we are using JSON.

Unknown Speaker 10:28

Oh, have you heard of, or use it at all?

Unknown Speaker 10:32

I haven't used that

Unknown Speaker 10:44

yeah, I want to talk about

Speaker 1 10:47

the schema and schema, evolution and stuff. You know, since you said that, you're not too familiar with schema history. So it's just again, so that you have a point where you're using Spring Data JPA. So can you tell me what's the difference between Spring Data JPA and Hibernate?

Unknown Speaker 11:11

Oh, yeah, sure.

Speaker 3 11:14

So, basically, what the difference between Spring Data JPA and Hibernate is that Hibernate is basically object relational mapping framework, just providing you the way to metal Java objects into database tables and other way around. So basically, it abstracts away the biped in code needed for database interactions, but you like your Spring Data JPA is like Java Persistence API, which simplifies the development and JVM TPA base data access object which is built on top of JP and Hibernate right? Basically, if you use hibernate, these are from like these are managing those database interactions and mapping your right entity classes and use hibernate API to perform those CRUD operations and require configuration of like sessions factories transaction management, the maping config and so on. Does that answer your question? Yeah.

Speaker 1 12:24

So which one is which one is the spec? Um, so

Speaker 3 12:37

I think the gap being

Speaker 3 12:46

so, I think so JPA the interface basically Hibernate is the JPA provider.

Unknown Speaker 12:55

What is Spring Data?

Speaker 3 13:01

Basically, it is it is like higher level form of build on top of JPA, which is like persistent API. Okay.

Speaker 1 13:15

You had certain menu set that was provided in that postman I was looking for. And gap is a specification, right? So based on that, I'll give you a scenario. Let's say that I have a, I have a couple of scenarios one, which includes the GPA, the Spring Boot, microservices, and the other one, which is Angular. So do you have a preference?

Speaker 3 13:46

Let's I would say, let's go with the microservices first.

Speaker 1 13:50

Sure. Okay. Sounds good. So let's say that I'm creating a micro service. It's a simple micro service that does CRUD operations, there are these Employee objects, or employees, let's say employees data with the different departments. Let's say that if I want all the employees from the HR department, and then I want that pictures as some of these inputs that prevent pictures, let's say, just for each of these elements. So now we were talking about two things. One is fetching the data from the database is to get the content to get the image from the benefit system. I had built this system, but now I'm seeing that it takes significant amount of time it takes 10 seconds 15 seconds, okay. And I suppose, presuming that you have paid in high throughput, low latency kind of applications. So how would you how would you fix this?

Speaker 3 14:53

Um, so, first, first, I think I would say

Unknown Speaker 15:01

So basically,

Speaker 3 15:05

I would do it, what I would do it in this case is like I would introduce some kind of asynchronous processing using completable futures so that we can fetch data in parallel, instead of fetching employee data and emails, sequence Li, we use, like a synchronous processing to fetch them in parallel, which, which way you can initiate both operations simultaneously and combine the result once it is completed. And, you know, you can implement some kind of caching mechanism for frequently accessed employee data as well. So when this can significantly reduce the time needed to fetch this data.

Speaker 1 15:55

Okay, so that takes care of the APS, right, the call that you're making the content that was about, let's say, even the database when it's pretty, like you have to do, let's say, it's a publisher, you have to do multiple joins, in order to get that information. And when I'm looking at the logs I see for each employee, that are hundreds of ways being, what's the what what should be, like, what should I be think at this point, like, what would be the issue and how can I resolve

Speaker 3 16:35

if the database query itself takes a lot of time, then first thing is like, you check the indexes. And make sure like, the joints that you are using is like a proper and they should tighten them. And then you know that you can significantly increase the data retrieval time after, like, apart from that, you can basically optimize the queries. And if it are optimizable, right, and then apart from that, you can also like, bad match those queries instead of executing a query for each employee. And we can, I think, group multiple queries into a single batch query, which can reduce the number of database, round trips as well like it. If it is something that is like, very complex, there is a concept called database data view, which is kind of like a temporary table, the table that allows you to like pre compute results, which can be more efficient in running certain kinds of queries. Okay.

Speaker 1 17:51

One thing I see is that we can boy class itself has many, it has one to many relationships with, with other classes, right? So it's a parent, and it has a bunch of child. And when I'm looking at fetching the added object, it is making that a set multiple queries. Right. So now squaring is one of the things you can just write some queries. But let's say if I want to know and that's the default behavior of Spring Data. It will go on and you know, like, fetch one very, and then for all the subsequent children, it will try to grab each other, which I think it's very of its own. So there's a lot of queries, right when using Spring Data. Now, there's something like this suddenly come to mind that if there are so many queries, what do we do?

Speaker 3 18:45

So you have to make sure that the fetch type of the entity fit as a rarity so that you do not load all of those type classes on during the employee data.

Speaker 1 18:58

Right, but then if I'm grabbing the data, and you need the trend data, so how would I How would I efficiently grab the data without having so many ways?

Speaker 1 19:15

So if I have even clarify my question, let's say that you have to get one employee that has one too many with another child object, let's say, let's say cars, for example, employ a car that's very rich, they have 10 cars. Okay, now I'm trying to get him by. Okay. And then for each of them, why he's trying to get each of their cars, okay. When I look at the locks, there are 11 ways into I should have gotten it with just two queries. One is getting the main object and the other one is getting into the child, or probably even one query by doing some drugs. So why am I seeing this happening? What's the problem?

Unknown Speaker 20:04

So I'm,

Unknown Speaker 20:08

what I'm thinking is like if

Speaker 3 20:13

so, like, you should be using the joints, like you mentioned in order to have the proper, right. So you, you can do a fetch you if you have the proper What do you call it, which like, if you have, do the like right fetch join, which allows you to retrieve the parent entity and its associated with Chinese entity are to aid in like single query

Speaker 1 20:47

Okay, the answer though, the problem is that listen, and answer, the people was gonna join fetch. If I can just resolve it using Apache embers, Apollo will always be there in any implementation.

Unknown Speaker 21:03

You can you can look that up,

Speaker 1 21:06

though, because that question, and this has recently worked into this issue. I have a website. And I want to do succession management.

Unknown Speaker 21:18

I want to

Speaker 1 21:20

I want to involve users. Oh, wait a second. Is this just happened a meeting? Or is it an hour?

Speaker 3 21:27

I think this was a 30 minute meeting. That's what it is catch up.

Unknown Speaker 21:32

But he was the first to prepare for it. Okay,

Unknown Speaker 21:36

are you available for now? Um, so

Speaker 3 21:42

can we do it a little shorter, because I have a call to take few minutes. But I have a call. Because I can

Unknown Speaker 21:53

go fast.

Unknown Speaker 21:55

We haven't yet.

Speaker 3 22:03

I, I can go Max up to 35 or 40? If that's okay with you guys.

Unknown Speaker 22:09

We'll see if we go.

Speaker 1 22:13

Okay, so in that case, I'll make it faster. So let's say that this is an Angular application. Okay, I have an officer who's taking care of authentication. Okay, we will call you will make your services secure, we will do that. But before that, let's say that I haven't bought service in Angular. It's trying to communicate to all the other services in Angular that, hey, the user is logged in. Okay. So instead of our service falling, let's say that there are 10 different services in that Angular application. Okay, there's a service a week and a session and a service to get the images and service to get employee information. Now, before for that, first, the user has to log in. What would you do? How would you inform all the other services such as I'm going to talk?

Speaker 3 23:24

So so to have some kind of special management, like, like that, in Angular, you probably need to have like, but I think you'll probably need to add some kind of authentication state. So any service or component that needs to, you know, like that, that needs to authentication state will probably be subscribing to that state to have the observation of the phase two that is emitted by the authentication service. So that's what I think. Right? So use the authentication state and service techs need to react to these changes can I can subscribe to that observable, but also there is a concept of Angular router guard that I think we can use for this purpose like, and angular router guard basically, you know, like that redirects the user to certain pages used on the authentications. stays, I think

Unknown Speaker 24:45

you had

Speaker 1 24:48

this experience with Angular, you've seen quite a bit so much.

Unknown Speaker 24:57

And as far as the security of the application itself, on

Unknown Speaker 25:01

how do you make your app secure?

Speaker 3 25:07

So are you talking about like the back index bit or?

Speaker 1 25:14

Yeah, you can, you can have it again, so as to backup services, or you can have some way to the back end service, you can decide for sure.

Speaker 3 25:28

So basically, for security purposes, putting security is one of the most widely used, you know, there is what we have been using some kind of, you know, like, we have been using some kind of role based access control, using the pre authorized annotation and controller layer. So, we do config, like we do config set up a configuration class that has like the web security configuration adapter that we can configure the GW T authentication for services. And we do have some kind of input validation in place for any kind of request to make sure it is not trying to do any kind of injection attacks. But for front end, to back end, we make sure, like, communication are through like a secure protocol using HTTPS. And we have, I think, see our CSRF protection in place. And all of our like calls are getting routed to the backend through AWS. And meanies API gateway does handle a lot of the authentication and authorization within itself. So it makes it a little bit easier for the front end, communicate with the back end.

Speaker 2 26:57

So let's say you have a API that's exposed externally, basically used by outside vendors.

Unknown Speaker 27:03

What how would you make that visual data security?

Unknown Speaker 27:10

may sound like a turtle, let's

Speaker 2 27:11

say we're trying to expose an external API, how would you make that visual that is secure?

Speaker 3 27:20

So can I think it needs to be properly, it needs to be properly authenticating? So you know, one of the concepts is API keys, you can issue a unique API key for each external behavior, like external vendor that these keys are difficulty case and are rotated on a periodic basis. And whenever a request is made to any kind of exposed endpoint, we first check if the API key is valid, or not.

Unknown Speaker 28:00

A rough surface, a lot

Unknown Speaker 28:03

of dedication. So

Unknown Speaker 28:08

uh, yeah, I have like, work with all art as well.

Speaker 1 28:15

Okay, so in that case, like, what, what are the components to be? Okay, if we're talking about the demo minutes, and that just verify last five minutes for your questions. But the last question will be, you know, what are the components of? Like, how do you set up import a service? What are the components? Where does the flow reasonably you can pick anything anymore? And whether the floor or the generic house, the security mechanism is how does spring security?

Speaker 3 28:51

So on the auth, you have a resource owner, which is kind of like the user who owns the resource. You have a server, which hosts those resources, you have a client application that is requesting access to that resource, and then you have authorization server that is responsible for authentication, authenticating that resource, owner and issuer right. So basically, the way it works is that the client makes a call to the authorization server or authenticate an authorized request. And basically, a token is requested when the client sends the authorization code to the server to obtain an access token. You know, it usually has a client credential going type with client id like and client secrets and then with that token response, you send a request to the resource using I think, and you send it request using the authorization agent in it like if it's validated. I think there are certain validation using DWT key. Maybe the resource is like available to that user or not.

Speaker 1 30:19

So, how to authorize sorry, how do you authenticate that case? And in a negative has been good application, or does the authentication. So

Speaker 3 30:38

authentication is one of the first steps like I think the the, like the user is redirected to the client application. And it authorizes and server that authorization server first authenticates the user, typically through like a user name, password, or some kind of client ID. Right? Okay.

Speaker 1 31:04

What's the issuer? Just make it was an issuer versus what's a claim? What's an issuer?

Speaker 3 31:14

So basically, an issuer is in all art is like the entity that basically gives you the token. Right? So the site is usually authorization server, so like the identity provider that basically signs the token. And the claim is kind of like a piece of information about the user. And it can I think it can include information like the user ID roles and permissions. That answer your question. Okay, so, Sabina, I

Unknown Speaker 31:55

guess, you know, we can do you have any

Unknown Speaker 31:59

questions for us?

Unknown Speaker 32:04

So, I would like to like,

Speaker 3 32:08

know a little bit about the team, like, I see you all, but like if I am to be joining the team, like, who would have been working with and what a day to day looks like? And yeah.

Speaker 4 32:23

So right now, the position may involve an individual who might be moving across teams during PII. So right now we have six strengths two weeks HPI session. So you might be involved with the planning area, which right now is to change. You might be helping out on different stories across entities. They have stand ups. One of the things that seems five days a week, I believe Axolotl doesn't have standards, right? And is that correct? Standards, usually 30 minutes or so, typical, typical setups we have with that, that would be your daily activities that they have. Also they have the grooming sessions. Usually they have once a week, maybe particularly at the beginning, they have a retros. At the end of each, they're supposed to have them at the end of each sprint. So we have most of the customers for Agile. And that's what you would expect.

Speaker 3 33:30

Okay. Yeah, this would be a very typical question. But I have one more question like, well, is this role playing? What happened to the person working in this role? Is this a new P?

Speaker 4 33:46

We had multiple roles, one of the individuals wasn't meeting our expectations. So we had to go ahead and with that person go, we had another person who unfortunately, because of their work visa did not align with what the paperwork, so they had to go. Okay.

Speaker 3 34:07

Okay, so what should I expect next, if this interview is good,

Speaker 4 34:12

the next step would be if we want to go the next step, because we push through the interview process, I would probably have that person go ahead and have an interview with these twin masters who report out the things and have more of a personal interview with other team members just to see what kind of synergy or Dynamics I received that interview would be tactical, it'd be less technical in nature,

Speaker 3 34:41

to me know how long they're expecting it to last because today was kind of 130

Unknown Speaker 34:46

events to an hour.

Unknown Speaker 34:49

The three of us join our list.

Speaker 3 34:51

Okay, sounds good. Yeah. Today I had just the invite for 30 minutes. last interview, Gina reached out to me and say if I was okay, staying more then half an hour I said, Okay, for today I wasn't aware of that.

Speaker 4 35:05

I assume based upon our previous conversation that they would go ahead and communicate that with your

Unknown Speaker 35:12

team or problem.

Unknown Speaker 35:15

Do you have any more questions for us?

Speaker 3 35:16

No, not really. Not right now. Like if we talk again, I might probably have more questions because I ask a lot of questions. So, but

Unknown Speaker 35:27

thank you very much. Thank you.

Sakshi-Bhattarai-Lowes-SRV-Final

Unknown Speaker 00:00

Good morning.

Unknown Speaker 00:08

Let's get started.

Unknown Speaker 00:13

Let's start by giving a brief introduction about yourself.

Speaker 1 00:18

Basically, we wanted to understand more on your current project, but that extract you're using along with the project, what is all about that project as a part

Unknown Speaker 00:32

of that introduction?

Speaker 2 00:34

Yeah, sure. So my name is faxiva, and I have been working as a Saba developer for five years now, I work primarily on Saba web application. I also like utilize frameworks like spring hibernate and RESTful web services for API development. I have been like, mainly working on micro service development, and like, recently been involved in designing micro services, developing them and also deploying them like AWS cloud. So yeah, some of the cloud services that I have worked with, like includes like AWS EC, easy to SNS and lambdas. So we also use, like, trainkins for our CICD pipeline. So yeah, that's about me, like, professionally and so talking about, like, the current project that I'm working on, I'm working at Qatar right now, currently. So the project is about and, like, equity management platform. So the current proceed is, like, basically, like, kind of like a migration project. We are basically, like breaking down chance of, like, monolithic acquisition. And then what we do is, like, we basically are developing these micro services, right? So couple of them that I have, like, recently engineered, include the like, maintenance of APIs and micro services. So like each of these services are returning Spring Boot and like Java 17 and they are deployed on AWS ECS clusters, so the maintenance API is used like permitting changes to the user profile preferences and experience. And APIs are used for like the entitlement management. Oh, yeah, that's basically about the current process that I'm working on. Okay,

Speaker 1 03:01

have you used anywhere for circuit breaker? Probably your project.

Unknown Speaker 03:07

Are you talking about circuit breaker? Yes,

Unknown Speaker 03:13

yes, yes, circuit breaker,

Speaker 2 03:20

yes. I have worked with, like, historics and the circuit breaker pattern so our legacy application, like use yeast and for the modernization our like orchestration platform uses, like resilience for G basically, you know, like solving breaker is nothing, but it's just our design pattern that, like, prevents our cascading filler to the downstream systems, right? So, like, we have a group of commands which are like, basically APIs that are, like grouped together. And based on those groups, we have like particular countries on where the solid keeps will be open and where it should like be a pass through.

Speaker 1 04:15

Okay, so what basically is being provide out of the box for implement this circuit breaker. So,

Speaker 2 04:27

talking of so far, like spring in particular, like the circuit breaker, I think, like it should be like, you know, resilience for there soon like be a good cooperation between

Unknown Speaker 04:48

the resilience for

Speaker 2 04:51

travel and springboard. So, like, that's what I have been working with, like for Spring Boot based.

Speaker 1 04:58

No problem. Thanks. So

Speaker 3 05:10

each question of Java and Spring Boot have you used in your current project?

Speaker 2 05:16

So the version of Java and spring so my latest one in zawa is like zawa 17, and for springboard, it's

Unknown Speaker 05:29

like 2.7 Yeah.

Speaker 2 05:37

But I have, like, started, like working from Java eight and 11.

Speaker 3 05:45

Okay, so are you familiar with streams and the functional programming aspect of Java? Okay, so one more question. Do you have your local IDE set up to run the application and to do the coding exercise.

Unknown Speaker 06:04

Now, yeah, I do have okay.

Unknown Speaker 06:12

So which database have you used? Do you

Unknown Speaker 06:17

mean like in the

Speaker 3 06:20

past or in my current process? Yes, what? Whatever databases are you familiar with?

Speaker 2 06:23

So I have worked with like post based database, releaser and DynamoDB for non releaser. I am also like swell server in the past.

Speaker 3 06:38

Okay, what are the key differences between the dynamo and the poster is, and why did you choose? Or why

Unknown Speaker 06:48

was it chosen by regular relation?

Speaker 2 06:54

So basically, posters like is, it is a vs data based writer. So what it means is that it has a structured schema in the form of like rules and columns in a table. So like, but you know your Dynamo TV is like a key value based structure where, like, you do not have a like predefined schema, right? And then post phrase basically provides a very good consistency and follows the transaction properties so like, but Dynamo is like, it's like more. It provides a more like limited transaction and eventual consistency, and has an advantage in, like scalability, where it can be like, horizontally, scale, but post, post as well. If we need, like, higher power, you have to, like, you know, vertically, scale it. Now the reason that, like, we went with Dynamo TV was like because of the entitlements API, which leverage, like JSON, like data structure for different games, so which are liking on, no pieces schema, so anything can like on will be a particular user type, or Like displaying multiple kinds of components on the fund and based on like, using title means and to like event, you know, strictly standard, we just went with like Dynamo TV, and that was the main reason for using Dynamo TV. But our main application, like, where we are storing, like, you know, the primary catalog and context and stuff. It was on post phase because, like, post phase provides a very good schema and structures with, you know, proper consistency and availability. So, yeah,

Unknown Speaker 09:00

I wouldn't like,

Speaker 2 09:07

I like, you know, you go with like Ospreys as well if you want to have like, like, some kind of real standard data model where you have like, you know, multiple joins and mappings between like different tables,

Speaker 3 09:27

okay, so as we are talking about multiple tables and relational database, you must have worked with JBA, right? So consider we have one parent table and we have like multiple childs for that line, right? So what is the relationship you established at a GPA level? What annotations do you use to establish this relationships? So the

Speaker 2 09:51

annotations that I use like to establish relationships. So in order to do that, like in spring GPA, you can, like, use the annotations, like, one to one, one to many and many to one, right? Basically like you have one to many, where you have, like, mapped, like, stating the parent in type entirety, in basically like it is used for one to many relations paper. And then, like, many to one in the child entity that the time, like, defines that many to one relationship, right? And then also, what you can do is, like, you can also provide the joint column, like annotation which specifies the foreign key column, but you can also have a target, like one on one relationship using the one to one annotation.

Speaker 1 10:59

Do Okay, so I'll give you, if you don't mind, I'll give you one hands on problem. We want you to write it down.

Speaker 1 11:28

You can just open one note, maybe in IntelliJ, you can just open a text file, copy and paste this question. Text file

Unknown Speaker 11:42

and paste this question.

Unknown Speaker 11:51

Can you see my screen? Yes, yes,

Unknown Speaker 11:56

just open the text file. Do

Speaker 1 12:17

discussion, so I'll explain you about the question. So think about I have item class, where I have item ID, brand, price, type of product, and I have created eight objects of those item class, added them into a list. For example, I'll explain one object where you have item ID, let's say item one brand. It says LG, price is 1200 and the type of product is appreciator. And similarly, that I have bands, dishwasher, washing machine, all these things I have created. Now I would like to get a map. I would like to get a map where your map contains E as a brand name

Unknown Speaker 13:01

and value would be

Unknown Speaker 13:03

the list of objects correspond to that brand.

Speaker 1 13:08

Okay. How I can achieve this using the Java eight streams API.

Speaker 1 13:17

You can, you can write this method here itself in this notepad so we can see, so

Speaker 2 13:27

basically like it looks like, you just need to like group by the brand. We can basically use like basic stream functionality and collect it by grouping.

Unknown Speaker 13:41

Okay, can you? Can you write the code for that?

Speaker 2 13:46

Yeah, so, like, do you want me to go, like, find page structure, that item cost. I'm

Speaker 1 13:53

good with that. Just you can consider this item this you need not to create a program. You can go to this scratch, dot, txt file, whatever you have, and then you can start writing the what would be the code snippet for streaming the list and get a map out of that.

Speaker 2 14:20

So basically we have, like, that list item, right?

Speaker 2 14:31

So now I can, like, just do a stream on this, which is just a stream, and then I what I can do is I can collect it using the Collect method. And in the collector I just need to pass the grouping where, you know, I want to group by the brand. I

Speaker 2 15:14

yeah, I think it should just be that. It should give me a list. It should just give me, like, you know, a map of string and list.

Speaker 1 15:28

That's it you need to do, because that would become the coupe. Then, is it something you need to collect that into a map somewhere? Do? Somewhere. No,

Speaker 2 15:45

I think like you should be grouping again, like, you know the grouping should create a map. Okay, okay.

Unknown Speaker 16:08

What is the difference between controller and the Rest Controller?

Speaker 2 16:14

When we talk like about the difference between controller and Rest Controller, is that, like, you know, the controller is used to define your class as a controller that handles our view, right? So, like, basically it returns a string that is the like name of the view, right? So, like, if you have a controller that returns the home base, it will return the string that returns, like, name of the view. But your wrist controller, it's like, you know, kind of like a mix of two annotation, which is the controller annotation, the rest. Response body annotation item. So like risk controller is like, the waste controller is like, it is like, primarily used to return our data instead of use. So if you like, have a controller that is inside a wrist controller, and then have a method that it will return either a JSON structure or an XML structure or just like play screen, but it will be the data instead of a view. Okay, so

Unknown Speaker 17:41

okay

Speaker 1 17:46

without using direct Rest Controller? I'm sorry. Can I repeat the question,

Unknown Speaker 17:51

Can I

Unknown Speaker 17:54

use? Can I

Unknown Speaker 17:55

build a REST endpoint

Unknown Speaker 17:58

API without using the Rest Controller,

Speaker 2 18:04

yeah, basically you can do that and like, you can create REST API without using like the wrist controller. You can just do like a controller annotation, and then you know, on the method, you can just provide the response for the annotation. It will like, instead of returning view, it will be done. Have you worked on

Speaker 1 18:38

the SQL part, like Oracle database or maybe postgrad?

Unknown Speaker 18:49

Yeah, I have, like, worked with Aries as well. So yeah,

Speaker 1 18:57

can you copy this code snippet in the same sketch you

Speaker 1 19:29

Okay, okay, that should be fine within the spaces of extra line contacted. So, so there is an employee class, and I have employee name designation in the years of experience, year of joining. I would like to create a sequence statement which will give me year of joining as 2023, something like this, and it will give me the count of employees which joined in that year.

Unknown Speaker 20:03

Can we write the secret status for this?

Speaker 2 20:12

So basically, the year of training that like you have is a full day. So should I consider it like just to be a year?

Speaker 1 20:25

No, it's a dateful, so you have to extract some data from them. So if

Unknown Speaker 20:37

I have a date, I

Speaker 2 20:48

so like, we might need to do processing on that date, right? So I think, like, we can just get the air using, like, inbuilt your function.

Unknown Speaker 21:01

Whichever you wanted to use it, then all we need to do here is

Unknown Speaker 21:06

just like a group.

Speaker 2 21:09

And in life, you just need to group on this like particular use case, like, you know, grouping. So basically, like group based on what parameter we do. I it.

Speaker 2 21:30

So basically it will be like, select the year of training.

Unknown Speaker 21:39

Sorry, are you writing?

Unknown Speaker 21:40

Okay, I didn't see that. Okay,

Speaker 2 21:45

yeah, I'm writing it's on the bottom. Let me scroll up.

Unknown Speaker 21:56

Can you see now? Yes, I

Speaker 2 22:09

so we like extract the air using the air function,

Speaker 2 22:22

and then we'll like build up count of the probably the

Unknown Speaker 22:29

employee ID.

Speaker 1 22:32

Consider, please consider that year of learning is not a date. Concentrator,

Speaker 2 22:49

so I think we can like just to substring on that, right? So next I

Speaker 2 23:07

So basically, we can, like, just do the nib substring.

Unknown Speaker 23:20

So from the year of,

Unknown Speaker 23:35

and we need the first four disease, right? So

Speaker 2 23:52

and then basically we'll count the employee IDs.

Speaker 2 24:02

Okay? BBB, and now this will like be from the employee table. And all we need to do is like, we need to Group by

Unknown Speaker 24:40

and now we are like just ordering it.

Unknown Speaker 25:01

Yeah, I think that's to take care of it. Okay? So are you sure

Speaker 2 25:20

it's gonna work? It should work basically like we are the string left, like substring from the year of joining, and then like, you know, we are doing, like using an inbuilt command on the count, which

Unknown Speaker 25:50

so If I use

Unknown Speaker 25:55

battery, then it should work. Yeah,

Unknown Speaker 26:02

okay, so you have

Speaker 1 26:03

used count of the underscore id shouldn't be the part of the group by class.

Unknown Speaker 26:12

Will it not throw any error?

Unknown Speaker 26:20

See.

Unknown Speaker 26:26

So if I have a

Speaker 2 26:43

then. So if I have a count, then I think it should not be because, like, we are, like, you know, performing like aggregation with count and so, because it, like, you know, operates over the group data rather than the before it does it. So I think it should Like still be okay. I

Unknown Speaker 27:46

Are you waiting for me?

Unknown Speaker 27:51

Yeah, I'm waiting for you. Okay, okay, I

Unknown Speaker 27:54

thought you're still exploring that answer.

Unknown Speaker 28:02

Nothing. Think they have some

Unknown Speaker 28:06

question. Probably, yeah, sure.

Speaker 3 28:08

So, yeah, can open the Spring Boot, create the project. I'll give you the requirement. I will paste it, and I'll let you know. Do BBB,

Speaker 3 28:31

so you can have a car as a model, let's say, and you can ask me follow questions. So we'll have a car, and it has three years model, year of make and price

Unknown Speaker 28:47

and year of make. You can make it a string.

Speaker 3 28:51

Easier thing. If you want model, you can give any strength and price. You can keep it double. That's the format of that model class, and basically expose the basically create a Spring Boot project around it. Have a database. If you have anything in your local you can use it, or you can use an h2 database for your convenience, and expose these three REST API endpoints to basically save a new car with all the required data, fetch all the cars based on the model number and get all the cars within so use appropriate design things and whatever annotations are required, and you can explain why you are using that while you are coding that will help. And if you have any questions with respect to the task,

Unknown Speaker 29:39

you can ask to read it

Speaker 2 29:50

so you want, like me to create our project using Screen initializer, right?

Unknown Speaker 29:57

Yes, you can use Screen initializer.

Speaker 2 30:02

Do, so let me, like start by that first I will, like, just create A spring initializer project. Okay,

Speaker 2 31:09

so firstly, we'll add like the long book for easy caters and sitters.

Unknown Speaker 31:19

And then we'll add like,

Speaker 2 31:23

the way for like, you know, basic risk on API development. And then what we'll do is like, we'll add the GP for the GPA layer. I

Unknown Speaker 31:45

so like, add

Speaker 2 31:50

like, so add each to for memory, database, and what we'll do is I,

Speaker 2 32:04

I think like this will Be like, okay, to get Me started here, okay,

Unknown Speaker 32:16

do BBB, So I'm gonna let my interview say index it first.

Speaker 2 33:19

So like, while that's indexing, I do like look of the configuration for app, for like to database, are you? Are you? Like, you know, okay with me. Go to it. Sure. Okay. Thank you.

Speaker 1 33:48

Sakshi, just I wanted to remind on the time and perspective, it's you're asking your questions to clarify the doubts, but it's Yeah. Three, so we'll have probably 40 minutes of time to complete this exercise. You're free to ask more questions. It's fine to clarify the doubts, but just considering the time timing also, so probably will have time till 115 probably, I would say at max, we can go with this coding exercise.

Unknown Speaker 34:31

Okay, we have 40 minutes, probably for you

Speaker 2 34:33

to complete the success. I think I should vote for that. Okay,

Unknown Speaker 34:37

sure. Thanks. Thanks.

Speaker 2 35:05

Okay, so basically, like the REST API development, like using Spring Boot, like, you know, contains different layers. So, like, the first layer that I want to start it with is an entity layer and entity is like nothing but just a Java representation of A database, offset so,

Speaker 2 35:40

so like, what we do is we annotate this by car.

Speaker 2 35:55

So, like you provided me with data layers which were Like, modern and you are, Like making the fries. I

Speaker 2 36:45

and I'm like, going to use Lombok for gators and sitters. Okay, so that's it, about my entirety layer. And now the next thing, like, I need is repository layer, which is, like, you know, basically used to interact with the database.

Speaker 2 37:15

And since, like, you know, we are using GP from spring, we can, like, just create an interface,

Speaker 2 37:30

and then what we can Do is like we can extend the JPA repository so

Unknown Speaker 37:46

why are you using JPA repository instead of current

Unknown Speaker 37:50

Are you aware of current repository?

Unknown Speaker 37:58

You're extending JPA repository,

Unknown Speaker 38:03

right? Repository, are you aware

Speaker 2 38:06

of that? Yeah, I'm aware of that. So, like, what is like? You know, it is like, more basic, so, like, it provides you, like standard cloud operation, but it has less features, right? So, like, if I use the like, GPA repository, I can do more on it, like, you know, like, more features. And I have been, like, primarily working with JPA repository because of all the advanced, specific operations that I can do. So yeah, that's the reason why I'm using it.

Speaker 3 38:42

You're going up to stop coding in the stock while you're bothering them

Unknown Speaker 38:47

when it's difficult.

Speaker 2 38:57

So that's my GPA, because it really are now I want to go on like, you know, A controller layer, which is like the entry point. I

Speaker 2 39:38

so my controller like it should be annotated with the Rest Controller annotation, and then I'll provide like a predefined path for like using the request having a reason. I

Speaker 2 40:10

so you want to, like, expose, like three in one slide. So we want like, we want to affect all like, we want to save, and then we want to get like within a price range. So like, let me define the first one. And

Speaker 2 40:39

so I think that we need our model in Our URI path, because, like, It's a required field. I

Unknown Speaker 41:19

BBB.

Speaker 2 41:31

So like from the controller layer, we'll be moving to a service layer. So service layer is like where the business logic happens, and usually on the controller you will like, have a dependency of The service layer.

Speaker 2 41:59

So on the service we like, basically, you know what? We usually have best practice of, like defining an interface of the service. So I will do that first.

Speaker 2 42:21

So we need, like, three function team, which will like be get car by model, safe car, and then, like, get Car by prices. So I

Unknown Speaker 43:46

so like

Speaker 2 43:49

our service, like it will have a dependency on the repository to like, you know, talk to the database,

Speaker 2 44:01

and then we can, like, inject using the constructor injection so

Speaker 2 44:24

so like, in order to get a car by model right, what we need to do is, like, we need to define a method within the GPA repository. And like GPA repositories of course, like these basic queries by just defining the method I

Speaker 2 45:12

Okay, so, like, in order to save a car, we can, like, you know, just call the Same method on the repository, which is like, inbuilt So, built.

Speaker 2 45:28

And now, in order to get like in the price range, I can just use this similar kind

Speaker 2 45:40

so what I will do is like, I will define a method here which like returns a list of R, and I think we can like, just to like, buy the price between I

Speaker 2 46:05

but I would like, you know, come back and taste this out. Sure, I think, like, this is a naming convention For in between, but we can check it later. You

Unknown Speaker 46:53

now like

Unknown Speaker 46:55

in my controller.

Speaker 2 46:58

So what I did is like, in my controller on each of them methods, we can, like, just all those, Like, service calls, Right? So,

Speaker 2 48:58

now, similarly, like we can do our same method, which will like be A post mapping to create a resource.

Speaker 2 49:22

Oh, in order to create a resource, we need to, like, provide a request body.

Speaker 3 49:32

While you're writing the code, can you tell me about what is the difference between post and patch?

Speaker 2 49:41

Yeah, sure. So basically, more post is, like, used to create a resource, right? So what that means is that whenever you send a request body in it, like, goes to the database and insert a new resource. So if I like, keep on sending new post requests over and over, I will like, basically to, like, basically be creating, you know, multiple resources in the database right now. So like, put is a replace. So basically on PUT request, I would like, basically be sending the whole data, and it will like replace that data. So it's like, kind of an update. And now, if I seem like, you know, put request over and over again, it does not change like the state after the first update. So if I like do a patch request, it means partial update, right? So I can use a batch request to make like update to sort in fields on your basic resource. So

Unknown Speaker 51:04

okay, thank you. Thank

Unknown Speaker 51:25

you have experience with swagger. Swagger,

Speaker 2 51:31

yeah, so like swagger is like nothing. It's just a documentation tool that you know exposes like on all the inputs that your application has so it provides request body needed and what the response, course like from the API will be

Speaker 3 51:54

okay. If you have time after the success, if you can implement it, that will be great. I

Unknown Speaker 52:03

I'll try at the end, if I have time.

Speaker 2 52:13

So like basically users, like me, to integrate the Spring box or spring talk dependency it should add it

Unknown Speaker 52:46

so for a price raise, I think like

Speaker 2 52:53

a parameter rather than a variable, because, you know, that makes more sense have a Like key value structure. You

Speaker 2 53:58

and I think like this should be my three API, And I should be able To run This application. You

Unknown Speaker 57:45

okay, so

Unknown Speaker 57:47

my application is

Unknown Speaker 57:51

And I'll count this by APIs. I

Speaker 2 58:20

so right now, like I have No data, let me create One. I

Speaker 2 59:33

so there is an error, let me go in and check I

Speaker 2 59:54

basically checking The Hey, you.

Speaker 2 1:00:54

I think I kind of know the cause of the year, because I you know primary key here needs to be a generated value. I

Unknown Speaker 1:01:31

Okay, let me do try again. I

Speaker 2 1:01:57

Okay, so there was no error, but my response is empty now,

Unknown Speaker 1:02:12

let me try to get by that. I

Speaker 2 1:02:40

Okay, so let me try to debunk the application instead. I

Speaker 2 1:03:13

Okay, so it looks like everything is not like For some reason, I

Speaker 2 1:06:51

so let me create A data transfer object First I

Unknown Speaker 1:07:50

so basically,

Speaker 2 1:07:56

design pattern that is Used to be more ideal internal data structure So

Speaker 2 1:12:11

okay, so we save one card now I'll save Another. I

Speaker 2 1:13:04

So here, like we have now total three cars, one as

Unknown Speaker 1:13:12

well. I think we're good with respect

Speaker 3 1:13:19

to that. Can use the price range. One can you get me between 1010 1000 last year? Let

Unknown Speaker 1:13:31

me do that. I

Unknown Speaker 1:14:16

So, yeah, we have, like, cars,

Unknown Speaker 1:14:30

between 5010 1000.

Unknown Speaker 1:14:38

Sorry, 5000 has the wind range.

Speaker 3 1:14:50

Yeah, so working right? Yes, I'm good from my side. Maka, do you have any other questions you

Unknown Speaker 1:15:08

want to take care?

Speaker 4 1:15:11

Thank you. Thanks. How you doing good? How about you

Unknown Speaker 1:15:18

doing well? Okay,

Unknown Speaker 1:15:22

whenever you want to. Thanks. Thank you.

Speaker 4 1:15:31

Okay, so, just to give a little context about you, know, I mean, first of all, a bit introduction about me. I'm a senior

Unknown Speaker 1:15:41

manager at the same talk

Speaker 4 1:15:46

about, little bit about the project, you know, the team, or do you want little bit to know about it? No problem. Completely fine. So we all are basically working in an order management team. So we have a order management team.

Speaker 2 1:16:05

I'm sorry to interrupt, but is it okay if I stop sharing my screen?

Speaker 4 1:16:16

Okay? All right. So, yes. So we are working in the order management phase here. I mean, we have a pretty big team, around 100 plus people, if I'm not wrong. And you know, we work in two locations India, and us, give or take, we have around 3035, or 45% of the folks here. But majority of the folks is in India, okay? Because those has the office in India as well those India. So, so we have onboard the site. This is all completely homegrown product. Earlier, we were using the IBM proprietary tool, and we moved out of it. Now it's completely homegrown. And then we have, you know, it's a lot of, you know, completely, even with microservices, from a volume perspective, we handle a very drastic volume. We handle, you know, approximately 10 to 15 million transactions per day. Okay, so, and given the order management system, the core responsibility of the order management system is to work as good and fulfill the order, you know, irrespective of where you place them, online store, you know, mobile app, or anywhere, all those transactions come to our system. We are Casper with all of the systems in the HR ecosystem, supply chain, you know, last mile, delivery, inventory, fulfillment, pricing, promotions, payment, settlement, there are so many things that goes through the other one, which gets connected to everybody. That's what everything. So right from the order capture till the order delivery, whatever the flow happens, all thing happens within our system. Okay, so maybe even a lot of complexity as well as well as, you know, sort of scalability in the complexities on a very, very high level here. So that's the reason you're looking for a candidate who has a very strong, you know, hands on experience on the, you know, Spring Boot. So Kafka, given we are mostly back end based services, so we are highly reliant on the event based, you know, so Kafka is the backward for us. Now, in terms of databases, we are using a different databases given its microservices. Some of them are post press, some of them are Oracle, some of them are Mongo, some of them are elastic, some of the Cassandra. So we are all all variety in that manner. All right. Hope this helps with you. Yeah, alright. So we'll get started from there, maybe, if you can just give me a bit of introduction about you, and then we will take it from there, sure.

Speaker 2 1:18:55

So my name is faxiva, and I have been working like as a software developer for about five years, I work, like, primarily on Saba based road application utilizing like a framework or like spring hibernate for API development. I have, like, recently, been working on micro services, designing these micro services, and also different them, and like developing them. So I'm primarily, like, involved in RESTful APIs, or like making changes to the user profile differences and also the user experience and APIs for inculturing managers. Currently, like, I'm on a scrum team. My team, like, consists of five developers, where we have, like a product owners and Scrum Masters on our daily basis. I'm mostly like working on back in Java course. So like helping the team with code reviews and also on products and proper duties.

Speaker 4 1:20:06

What I will do is, you know, I guess we have approximately 4050, minutes, we'll quickly, you know, touch base on little bit of the technical side. You know, even my team has already done that. So I'll not go more into it. Maybe 1015, minutes. Then we'll go into a little bit of your collaborations, ownership and all those efforts as well. So we'll start with a little bit on the Kafka first, you know, how does Kafka maintains ordering? I just wanted to hear from you, you, you know, write that Kafka is guarantees the ordering, right? So, how does the test guarantee the order I want to understand?

Speaker 2 1:20:46

So basically, like, you know, you have your topics and partitions in Kafka, right? So basically, those like the partition, it is like a unit of parallelism in Kafka. Basically, like partition is an order and individual sequence of messages. So you know, within the partition and order is guaranteed where, like, each mass is in a partition is assigned to a unique integer, number called an offset, right? So it's like simply, you know, indicates the position in that partition. So Kafka, like ensures that messages are returned and read in the order of square offset. So for first message produced in our partition, we maybe, like, have an offset zero, and then the next one will have one, and so on. So consumers reading like from those partition read messages in the order of offset.

Speaker 4 1:21:53

Okay. Have you heard about the exactly once processing in Kafka? I'm

Unknown Speaker 1:21:57

sorry. Can you repeat that?

Unknown Speaker 1:22:00

Exactly once, processing

Speaker 4 1:22:05

like anything. So see Kafka order. Kafka guarantees the ordering process is very clear, right? Kafka also ensures the guarantee of the message. But exactly once, is there any way can we? Can we achieve that? Is there anything Have you heard about that? Have you leveraged your experience in

Speaker 2 1:22:29

cost? Yeah, so like basically in Kafka, exactly once processing is like it is like, just to guarantee that exactly

Unknown Speaker 1:22:40

once, without duplication. And

Speaker 2 1:22:44

so like, you need to achieve an item like 14 in the producer level, which, like, basically, you know, ensures that even our producer tries, like, sending messages, it will only be returned like once to the customer topic, so like that. So

Speaker 4 1:23:06

that is from the producer side. Is there anything that we should do on the consumer side to also

Unknown Speaker 1:23:15

say, you taking

Speaker 4 1:23:18

care of it while it is replicating? That is ensuring, even during the applications, it is only one copy more than that, right? But what could be done the consumer said, to ensure the consistency. So

Speaker 2 1:23:35

from the consumer side, like, in order to consistency, like, what we can do is, like, basically, I think like this ensures that consumers only read messages that have been committed in the Kafka topic, which like messages in complete transaction. So if you do like READ COMMITTED isolation level, that's

Speaker 4 1:23:59

okay. And can you explain a little bit about it? I mean, if you, let's say you enable the converted isolation level in the consumer, what are the pros and cons of it?

Speaker 2 1:24:13

Pros and cons basically like the pros and cons about like, whether you like, you know? What I'm trying to say is, whenever, like, you enable your read permitted, right? So like it prevents reading from the incomplete transaction, right? So that's your major advantage to the like consumer should not read messages that belongs to an import progress transaction, right? So, like, if a producer is saying for masses as a part of a transaction, and the transaction has not been committed yet, those messages, like, will not be visible. And it basically like it is that basically exactly one semantics feature in a

Speaker 4 1:25:05

Spring Boot when you write a consumer, is this the default isolation level? Or or not? You mentioned right? The consumer need to use the isolation level as we committed when you write a Spring Boot. Consumer, is it the default isolation level there? I think,

Speaker 2 1:25:28

like, not that. So it is like, have

Speaker 4 1:25:40

you say sign so when you say read committed, let me understand from an exam. You know, explain me the difference between the read uncommitted and READ COMMITTED from a real example. Then I'll try to, you know, understand more what exactly you're understanding from an

Speaker 2 1:26:02

exam. Example. So I guess that the rate on communicate and as Okay, let me think about it. So, yeah, let's say that, you know, your producer sends like messages, you know, within the transaction, right? And then your Kafka Producer sends financial transaction like to a Kafka topic. And these messages are, like, rooting to transactions, right? So basically, like your transaction content, like, you know, multiple messages so that the producer may commit or avoid that transaction, depending on, like, what business laws is that you do. So, like your producer, it just sends the message within a transaction, but the consumer, like, reads messages from the topics and processes them now, like, when you have the default read on, communicate like, you know, isolation, your consumers read messages even when they are a part of a transaction that you know have not yet been committed, right? So, like, I think it may have been awarded. So it basically allows consumers to process all messages, including that you know, maybe in an unfinished transaction. So if you have these messages like that are sent from a producer as part of a transaction, then consumers will read like all of those messages, even though, like they are part of an uncommitted transaction. But if you have like, read committed then consumers will not read any of those messages because like they are like part of on COVID transaction.

Speaker 4 1:28:04

Now, one last question on this related to let, have you heard about the rebalancing happening in Kafka?

Unknown Speaker 1:28:15

Yeah, I heard about it. So like

Speaker 2 1:28:20

the rebalancing in cast. It is like, you know, basically, it is a process of redistributing, you know, your conditions when, like, there is a change in consumer groups, right? So maybe, like, you can have, like, more consumers going, or maybe some consumer leaves your consumer group, it's a slight like redistributes the partition among those consumers.

Speaker 4 1:28:50

Okay? And how can we minimize the impact of the debalancing? I mean, because see, most of the time you know you if the consumer is leaving, or consumer is, you know, coming to the consumer group, which means something is really going on, right? Because on a running system, nobody will do that. So how do you ensure and how can we minimize the rebalancing? What are the things that we can what configurations we can do to make it little

Unknown Speaker 1:29:27

better to

Speaker 2 1:29:30

minimize rebalancing. Can do, like some kind of, you know, the consumer safe and timeouts because, like, consumer Kafka, consumer uses like happy mechanism to detect if a consumer has died or, you know, is slowed, you know, so that is one thing that you can do, like a session timeout. You can also do like, you know, some kind of graceful search down so that insuring consumer use the group by, you know, doing some kind of offset commit, so that, like, business or some kind of rebalancing, and, you know, you might have to do, like, some kind of regular offset garment, okay,

Speaker 4 1:30:20

all right, Okay, moving on to the monitoring side of that little bit. So, you know, I mean, have you developed any of the work in the production, I'm sure, as part of your last previous careers, how do you monitor those services? What tools you use there even more so. So,

Speaker 2 1:30:42

basically, I have, like, a monitor, like we have like, and I have monitor them. So, like, one is like strong, which is called logging, and the other one is like, ready purchase for your Task Force on slums. We can like, what we can do is like, we can basically our application index, and then also, like, find different kind of logs and log

Speaker 4 1:31:11

based monitoring. Which Have you done anything based on this thing or anything on that sort

Speaker 2 1:31:19

Yeah, so we have something like new retic, which is, you know, an alternative for Prometheus. So it is, for me, like monitors, the system has, like the CBO, so like that,

Speaker 4 1:31:36

I understand the health, health part of it basically helps that kind of thing. But what I'm looking for is, let's say you have a REST API. Okay, now for that REST API, if you wanted to know what is the latency, how many of the error percentage are happening, how would you do that?

Unknown Speaker 1:31:54

So, like,

Speaker 4 1:31:57

because if you do with the log based mechanic, right? You know you always understand, like, log is always a little bit of lag Coronavirus processing behind the scene, right? Yeah.

Speaker 2 1:32:10

So, like, as far as I understand, the new release agents, like fixed care of that as well. So, like, it is a fully provided platform. It is not like a log based mechanism, but like it's installing your system that basically grabs the system information, rather than based on the log. So

Speaker 4 1:32:29

that's exactly what I'm trying to understand. So neural relic. So in order to enable the New Relic in the Spring Boot, what you will do?

Unknown Speaker 1:32:39

So in order to,

Speaker 2 1:32:41

like, enable you relic in the springboard. What I can do is,

Unknown Speaker 1:32:51

yeah, so, basically

Speaker 2 1:32:56

monitoring, so basically, like it is chart that you know, that needs to be installed into your system instead of a jar directly on your application. You need to provide it as a zvm argument, and then, like, start by pointing this jar as an agent was so it will not, like, you know, directly be added to the application, but as the part of the startup script, this agent will be monitoring your application, like, directly. Okay, all right,

Speaker 4 1:33:35

now let's assume I'll give you a situation I want to understand. How would you handle this? You are deploying, you have deployed something in production. Okay? Now, due to your change, something breaks in production. Obviously there is a customer impact. You do that now, you know the support team or the monitoring team reach out to you. You know something isn't wrong, can you please take a look? So I wanted to know step by step process, what will be your triaging, or what will be the kind of action you will take at that moment on time.

Unknown Speaker 1:34:13

Help me step by step? So for

Unknown Speaker 1:34:18

the first question,

Speaker 2 1:34:21

if this is like, after a change that I pushed into production, right?

Speaker 4 1:34:29

Changes, change is deployed tonight. Let's assume and then, or maybe yesterday, night and now today, morning, we started seeing issues.

Speaker 2 1:34:36

So the first thing always like will be to roll back the release. Right? Basically, what you know, I need by that is, like, if I have customer impact and like, you know, I want to, like, record that change to my previous stable version, right? So, like, I have been working with Blue Green deployment, and basically, on the blue green deployment, we can revert to the previous color. Now. Like, if I do this, I stop, like, you know, customer impact, so that customers will be able to view and use the application in the previous stage. And I will try to like, treat as it on different instance, on, like, you know, lower environments. So I try to like, recreate the same area that they have in production by, like, looking at the parameters and logs that is available in this plan. And you know, I also try to, like, see if there is any exceptions within the application, any kind of irregular behavior, and like, recreate that on lower environments. And then, like, try to choose it once I have a fully functional understanding of like what the leader is. I will try to push the fix and then promote it on like deployment to like protein, so that will be my way to go.

Unknown Speaker 1:36:16

Okay, now, I mean,

Speaker 4 1:36:20

you I mean, you talked about two types of different deployment, right? Blue, Green deployment is your explanation. Okay, all right. But my question was, I mean, I mean, I understand globally, is there, but how did you even realize that your change has made the impact again? I just told that that yesterday was a change. But will you not quantify that fact first, or will you still be working?

Speaker 2 1:36:52

So no, I want to, like, you know, dive into that. I thought the situation like, was like, Like our application, like, cause the email. But, so if the situation is something else, like humans on that there is an event production. First place for me to go is my task was right. So like my new really task person. Now that show me off.

Speaker 2 1:37:44

Will have actually to the pace to our team. So yeah, in case of error within the application, but, like, but if it's that our application is, like, you know, not responsible for that spike either, and I will go ahead with, like, a rolling back, our rolling back. But my like, initial assumption was that our application was the one that was Swazi

Speaker 4 1:38:12

also joined with me here. I'm sorry I did not stop in between Swati, we were just proceeding with swashi on, you know, the second round. She's also seen a manager along with me in this team. Okay? Your

Speaker 5 1:38:40

previous projects, right like situations where there any situations where the person is incident or some issue happened due to code that was introduced, probably checking which was made from you didn't introduce directly, but somehow, like you missed, probably some scenario or anything, but then went into event into production, you suddenly see there is a spike in the issue happening

Unknown Speaker 1:39:16

is,

Speaker 5 1:39:22

is there any In production? Is there any bug which was introduced by your code?

Unknown Speaker 1:39:29

Yeah, so,

Speaker 2 1:39:32

not directly, right? Also, one of the times that I remember was like, migrated to Amazon. Do so basically, our application, like, uses our dynamic Config Server, and this, like dynamic conflict solver is used to fetch in the application properties. And you know, we made our migration like to Amazon. Do like it was, I think, 10 days ago, and then after like, 10 days, our application Acts it like stop processing and not the request like that was coming in from the user because of some like, missing properties. So, like, we're not sure, like, what the call was, and then, like, we are called in on a Sunday evening to look into the issue, because, like, it was impacting, like, a lot of customers, and this was, like, very big thing, because our applications, just like, became, like I responsive, and were in like, software injury. And basically, because of this, we had to go through a whole lot of, like, the dressing process that the way we basically started dressing was, like, you know, trying to see what APIs were like, feeling. And for that, like, you know, particular API and offer a lot of looking through the issue, we started like seeing that the config solver was like returning an empty response back. And we also found that the conflict solver was the culprit. So even though we knew like this was the issue. We are not we're also, like, not sure what the original issue was. And basically the way, like, the status thing really was, like, you know, because of the particular context solvers. So this basically, really relieve the customer impact, but the tracing of this, like, you know, to continue while and later we were like, like, able to find the root cause, which was a change on how the Amazon images are deleted, like on use files in folder which are spring by

Speaker 5 1:42:03

default. Okay, so for your application, right, ensuring the stability in production. Okay, so what are measures you take, actually, to monitor. It could be health. It could be, let's say, the response time, or even the API is actually not giving, like, lot of failures. How do you ensure what are measures you take? So, users

Speaker 2 1:42:28

identity is like. So basically, like, you have a load balancer, like, you know, it constantly like things the checking point, right? So basically, if it becomes like irresponsive, it just like dominates that instance, and then our like auto scaling kicks in, and then skills of like your instances. So that's the first line. How do you

Unknown Speaker 1:42:58

enable the auto scaling? So, how

Unknown Speaker 1:43:02

did you enable the auto

Speaker 2 1:43:06

skill? So, like, in order to give up the auto scaling. So auto scaling is like first authorities that is provided by Amazon white. So, like, you can have like Auto Scaling group where your applications are deployed under and then, then, like, basically give off, like predefined schema, like predefined conflict on each sort in metrics are made you scale up. And if certain metrics are below certain values, then you scale down, right? So like, let's say, if there is a high traffic to use this over like 50% then user skill up to like, higher number of instances. So that's how you do it in AWS. It's something like, you know, similar on AWS, yes, which is the container service. And you just need to, like, specify like similar context for the aspirations. Yeah. Definitions.

Speaker 4 1:44:16

Yes, I mean, so I covered a bit of already, just before you join. And right now we are COVID, so we are completely done on that. I guess we have another 1617, minutes we want to go to, like contactable questions if you're over with that.

Unknown Speaker 1:44:34

Okay, all right, so

Speaker 4 1:44:38

Sanchi just wanted to understand your little bit on this. So

Unknown Speaker 1:44:43

let's say you have written our code,

Speaker 4 1:44:47

obviously right in a collaborative environment, we do a code reviews for each other. So somebody has done a code review on your code, they have basically put a kind of the feedback for you which is not comfortable in a manner, not an append a particularly they are because everywhere has a different way of writing the code, writing the logic, so they have different impressions that this can be solved in a slightly different manner, like a sort of conflict, right? How do you handle such situations? How do you take those conversations, and how do you ensure that you know everybody agrees on and what needs to be done? Have any such instances happen? If so, how did

Unknown Speaker 1:45:29

you happen? Of course, like

Speaker 2 1:45:32

we are working in a team, like everyone has a different way of coding and different way of solving a problem. And so, you know, in our professional environment, it is like important that you take in like everybody's thought process. So basically, whenever I have like, any conflict that is like related to this, then what I like to do is I like to basically taking that feedback that my teammate provides and see that if it makes like more sense, right? So if I'm not comfortable with it, then what I like like to do is like, I try to set up our collaborator. But meeting will discuss about like, about the solution and the approaches that we have. So if it's a big change that the teammate is soliciting, we'll have to, like, you know, probably access the situation and see if there is like advantage of using other approach rather than what I have written, and if it's better than what I have written, obviously I'll be choosing that particular approach because, you know, clear and collaborative environment will benefit, will benefit, not only like me and my teammate, but everyone. So yeah, because I will, and I will also be like learning something new, and then the other team member will also be like learning something if my idea was better. So I think the way to go about this is to, like, basically staying open to that kind of feedback, and then trying to access it and see, like, if there is some kind of trade off in each scenario, and go with the one that is better in the long run. So basically trying to do, like, some kind of pros and cons list of each approach, and then going with it.

Speaker 4 1:47:31

Okay, now have you, for example, let's say, Have you have an instance where you have to, you know, you have been asked to, you know, work on a project where the timeline is very aggressive. Obviously, you know, you know, sometimes the timelines are very aggressive, but at the same time, you also have to maintain the quality. So how do you handle such situation, and how do you take it forward? So

Speaker 2 1:47:57

when I face such situations, and whenever I had an excessive timeline, and like, basically need to what I do is like, I like, basically break down the work and prioritize like, which tasks are important, right? So if your requirements are clear, I can, like, break down the tasks into smaller and manageable chunks, then I prioritize them based on, like, what the business value and the complexity is. I also like, I also like to, like, look into some of the tasks that may act as a road blocker for some of targets, because I don't want to be blocking somebody in case that might want to zero breaking somebody's else work. But I think that working with the team so that the task can be generated effectively by considering like, different strength of each teammate, and then, like, you know, basically adapting to the changes and being flexible with some it is the way to go, right? Because even though they are, like, tight deadline, so we have to, like, focus on what most, what the most important things are, and basically being clear, like on communication and prioritizing the staff that needs to be done is the way to basically go with such process,

Speaker 4 1:49:25

any such thing which you have done, which, which has improved, the, you know, the theme efficiency, or the, you know, the process improvement, or anything basically trying to understand any sort of innovation, thing that you've done at outside of your regular network, outside of assign work. So

Speaker 2 1:49:44

sort of assign work. So I think, like once a time, like it was sort of a suggestion for doing our team sync once a week. So like this will be just developers, like meeting on a weekly basis here. Like, we will likely talk about any kind of technical challenge he will have. You also talk about the new technology. I think, like, this has helped the team a lot, because the team does, the team member does not, like, have to, like, you know, try and find a dedicated time to talk to somebody. We don't have anything to talk about. We just explore like these. I think this is one of the times that that I suggested that has helped, like, in the efficiency of some of the teammates.

Speaker 4 1:50:46

Maybe you can walk with me through with your, you know, the lifecycle of, how do you write your code and take it to the production, like the complete lifecycle, right? Given the requirement you know, and then you have to take it to the production. What are the steps and things that you follow in your practice today? So the steps

Unknown Speaker 1:51:08

basically

Unknown Speaker 1:51:12

have a requirement, right? So basically,

Speaker 2 1:51:17

I make sure that my requirement is met. So whenever requirements, it's like using the functional requirement. Requirements, that is what the application needs to do. But I also like to look into other like non functional aspects, which may be related to, like performance, reliability and security. So, like, I make sure, like, all of these are covered. So that's the first thing. Like, I gather these functional and non functional requirements. And afterwards I like, after, like, gathering those requirements, I do the designing like, of how the applications will be, what the features developments will be, and how the things will be, like, connected and like, that is a technical and part of my job, right? So, like, I finished that in a qualitative manner, so that it is like, strong and resilient. One side, I'm done with that. The next phase that I do is, like unique testing, which is like, basically testing each of these changes in isolation, so that all the units of code work like as expected on the different circumstances. After that, what I do is like, I do an integration testing, which is like, basically not just in the interaction between the modules. So, like, that's some of the testings I do. Also, like, we have a revision testing that we run on these changes to make sure that, like, it doesn't, like, break the existing workflow now. Like, the next thing that to do is, like, deploy it to, let you know the QA environment, right? So QA environment is where, like the external teams use our service to basically integrate into their systems, right? So I would also like open a whole request to the developer brands to get like whole videos from my teammates. So here, like the pull request will be reviewed, and then, based on the review,

Speaker 4 1:53:33

do you complete the unit test? I mean, I'm hoping that's the magic part of it, right? So

Speaker 2 1:53:46

will request, like the CICD pipeline, if there is not,

Speaker 4 1:53:51

I understand. I'm saying you have written, I mean, you understood feature requirement, NFR, all those things you've identified that whether it's going to break anything existing or not, all those things is done. Now you've written the code as well, right? But I'm guessing your first step isn't your writing the unit test, yeah, so you raise a PR after the unit test of the fourth

Unknown Speaker 1:54:15

unit test. So it's like

Unknown Speaker 1:54:18

after the unit test,

Speaker 2 1:54:22

I I need to have, like, UV test along with the code changes. UV test is a part of my coding

Speaker 4 1:54:30

okay, then you raise a PR and do you have any code coverage kind of thing? Code covers?

Speaker 2 1:54:38

Yeah. So we have, like, you know, the scanner integrated with our pipeline, which, like, stands for these, will have like 85% requirement on both sides and

Speaker 4 1:54:54

Okay, all right, now you raise a PR. I got it now. Code Review will happen. Everything goes on, fine, cold. What happens

Speaker 2 1:55:04

after that? So after that, let's say like our reviews is like, sorted like, December 21 right? Then we got then we like, what we basically do is like, we catch a candidate from the developer branch, and this release candidate, like, yes, field and like, deployed to QA again for our life. You know, last run of revision testing. So we don't like a full regression test on this release. And then we like, basically, it's like, basically space for release. What that means. It just deploy the changes. Yeah, okay,

Unknown Speaker 1:55:57

I'm covered from my side.

Unknown Speaker 1:56:00

Swati, if you have any questions,

Speaker 5 1:56:03

I think we are good. We have only five minutes left right. Let's see if he has some.

Speaker 2 1:56:16

Thank you so much for the time today, it was really nice talking to everyone for talking about both technical and non technical experts. So the first question I would like to ask is like, what is the team structure like? So what is your typical work day like for the developer on the team?

Unknown Speaker 1:56:40

Something you want to answer that the

Speaker 5 1:56:45

very big, okay, and our team is actually wide, but between here and impact, I think they might be knowing growth has its own office value, so little bit shape, and we have both teams do an overlap timing so share to ensure that the business service most of the available days, okay, so from the day perspective, like we do express, like you're in office, right? And then we are actually two days a week office, which is Monday and Thursday right now. And like, we'll probably hit nine, nine, even the office, and then they will leave, mostly badly, 330 Okay, so that's what pretty much, it is way during some days we do, like, when the leadership, or somebody will try to cover like, as many days as we can, but ideally two days a week, right? Separate,

Speaker 4 1:57:37

yeah, it's basically our day starts early, just to, you know, ensure that we do overlap with the India team, right? And then we wrap up around 334, unless, perspective

Unknown Speaker 1:57:52

and commitment

Speaker 4 1:57:56

is at least two days. It's hybrid, so two days in office, and as Fauci mentioned, writer, we are a pretty big team, but scrums are very much divided. We have blocked any scrums within the order management. It's not just one with 100 people, obviously not many team. You know, bot size is a standard, around eight to 12 in one team. So that way we have different, different Scrum teams, like, you know, different different components within the OMS order statements. And you know, many more like that, around eight to 10. So you will be out of one of the board having that respect, capability, working

Speaker 2 1:58:41

with these any more question. So one of the things like that I would like to know is, what do you like most about working at so is that something I'd like to hear from you,

Speaker 4 1:59:00

I can at least put my perspective. Can answer from her perspective. Answer from our perspective. So I joined, you know, Lowe's in 2020 number one, you know, right between the pandemic, the most important thing that I have seen so far is the transformational journey that Lowe's has grown over the last few years. You know. You know, it reflects their commitment to the customer. In fact, the stock prices is also going to be high, you know. So we have transformed a lot the kind of the learning that we are getting from here is phenomenal, as I mentioned to you, right, that initially this entire or management was running on the, you know, some licensed product, you know, inbuilt everything in house here. So it is not an easy thing. This transformation comes with a lot of dedication, lot of commitment and lot of learning as well. So we all feel threatened that that we have done something, you know, from the scratch number one, so that, that, you know, inspires me a lot, that I am able to do something from the scratch and build it second is a large, large data volume handling, large scale, you know, data handling. I believe not everybody is lucky enough to work on such corporations where you get to work on large scale applications. So I feel, you know, that is something which tests my engineering excellence to the grade level,

Unknown Speaker 2:00:27

I'm able to

Unknown Speaker 2:00:29

contribute more inspirations.

Unknown Speaker 2:00:33

Thank you so much that it's

Speaker 5 2:00:40

been for me, also, like close to nine years working in here, knowing so many different different domains, okay? And one thing for sure, apart from the technology, you will learn very good function collaborations and lot of new things, actually, which maybe you might have not done in your previous projects, but it will be a very good learning curve, for sure. Thank you so much.

Speaker 2 2:01:05

I guess I have one more question, sure, sure. Like, what are the growth opportunities for? Like, software developer at lowrise?

Speaker 4 2:01:17

So I can take that Swati, I mean, so you know, first of all, you know, and that's that also is the one of the important thing that, you know, that we take care as a, as a, as a leadership, you know, all the leadership is very much committed to the, you know, employee growth. Okay? So we have a very strong, what do you say? You know, commitment towards it. We all take care of that in very manner from a leadership perspective. Now there are a lot of learning opportunities, learning tools available. You know, within the lows ecosystem, we have a Lowe's University as a, you know, one, one concept where you get a lot of learning within that part of it. They have a tie, tie up with, you know, some of the universities as well, so that you can do some of your, you know, learning courses from there. We also have a licensed rural site, app that will be available to you all the time. You know, you can just learn from that. So those are things that comes, you know, as part of your learning tools that have been provided from the low side. In addition to that, you know, our leaders have set up a very nice framework around the performance government perspective, where we ensure that, you know, all the associates are very much committed. We set the right expectations. We ensure that they are right on the track from their career perspective. And, you know, and we also, you know, very much support on the lateral movement as well. You know, you might be seeing, you know, working in such a big team, sometimes the employee feels that I'm too much into one area. If I want to move from, you know, one thing, to another team, we are very open for that. So we take care of that within the team as

Unknown Speaker 2:03:00

well. Does it help anything else? I think that's all the questions.

Speaker 2 2:03:08

I'm looking forward to Nice talking to everyone

Unknown Speaker 2:03:14

same here, and your feedback to the

Unknown Speaker 2:03:19

recruiter. Thank you so much.

Salim Maharjhan- US Bank- SRV- Final

Speaker 1 00:00

You, are you expecting yesterday? I believe,

Unknown Speaker 00:05

yeah, there is actually

Unknown Speaker 00:08

by Monday. Alright,

Speaker 1 00:12

great. Let me just tell you a little bit about the team position. I believe we have already talked to a few folks about this position, right, and then we'll take it from there.

Speaker 1 00:28

We went deep in US Bank, and he managed multiple payment products like IDP, which happens with GCH, number of other things. To person. It's a very widely used feature in the banking apps. So

Unknown Speaker 00:55

we are looking for a technically person who can who

Speaker 1 01:04

can understand how payments work. But we need somebody who can understand it, who understand technology very well, very hands on. Work with our other team members. Can work with other leads, understand the requirement architecture, design, and finally, start creating the solution. The person should also know how to worry about onshore offshore contractors, because we were able a lot of contracting, contracting folks, and the person should be able to guide them, assist them to their code reviews, look into their code and when they have questions, right? Can handle those things? So that is the position that we are looking for. Tell me a little bit about yourself. What is your background and

Unknown Speaker 01:53

what is your last project

Speaker 2 01:55

about? So my name is Sally. So working as a software engineer for about six years now, and mainly like working on like Java, like, basically Java piece like applications, right? And utilizing like spring frameworks, like, you know, there is hibernate and different web services for like an API development and like, really good with experience working on like microservices. So, like, primarily, I'll say like working on spring based for microservices development. So and also, like, I've been working on like AWS Cloud for hosting these applications on services like AWS, like ECS, EC two, and SMS and lambdas, so on. So, yeah, so I've been working on a scrum team. Like, you know, my current team consists of, like, around, like five developers, and we have like, product owners and Scrum Masters, and, you know, on like, daily basis as well. I can involve mostly working on like back end, back end Saba course, and helping the team with like the good reviews, and also on like production, like production support

Speaker 1 03:17

as well. And looks like you're working in Bank of America right now, tell me a little bit about your product, whatever the product or the business functionality is on this project.

Speaker 2 03:30

So at the Bank of America, so like, you know, we have, like this, I'll say basically, like our layer is, like, you know, in the orchestrator, orchestration, like, you know, between the microservices. So we, like cover, like, a lot of like clients and back in API, like, you know, we're like acting like a embrace, like break between, like, you know, between them, like, you know, it's like, kind of act, like, you know, like a rewards proxy, like, but it's more like or authorization, like, you know, authorization check, like, like the device, authenticity check and like, you know, the request and the response, like header manipulations and all the different features and domains like, you know, we have, like, a Walmart, right, like a Walmart domain and data, other domains like that go on, onto our app to, like, you know, validate their like the app requests. And, you know, also, we do like, like, a request manipulation, like, you know, by adding like, a certain request headers, or, like, you know, changing certain, like, the headers, like, you know, for like, the tokenized and things and like decryption of the request. So, like, yeah, like, the main product that that our team holds is, like, a lot of micro services that may help in, like the device authenticity check and, like, you know, the organization scopes, and also, like streaming engines and so on. And that's, that's our, like, primary big stack like, all of these are like the and all of these are like the primary Spring Boot based and deployed on the AWS stock. So pace, that's basically like, you know,

Unknown Speaker 05:39

at Bank of America, we have, this

Speaker 2 05:44

is like, wallet, is wallet management system, right? Like, you know, we have, like, the Apple Pay. We have, like, other water system, right? So is like, to be specific, like, you know, we were developing this thing where we were accepting like, requests coming from different sources, right? So, but like, mainly, like, you know, on the orchestrator, right? I like, worked on, like, multiple systems. So like, for the I would say, like, you know, specifically, like, you know, we're building this like a tool where, like, the users pull on board to make that call to the downstream system, right and then, and based on their request metadata, like, we will, like, you know, will make me, like me, maybe like, add like SSO ID, or like, some kind of other identifier. And the payload, like, you know, the payload by taking the request header from the payroll and from like, our like session tokens, or like, there's something like, you know, we basically, like, make sure the request is going down to those phase wallet system, and there'll be like, validation and before pushing it through the downstream, like the downstream system. So, like, you know, any APIs that had, like, wallets like that, would basically need to go through metadata check to see, like, you know, if any kind of ID was needed, like, you know, to be injected in the payload and so on, things like that.

Speaker 1 07:40

Did you work in the UI team? Or did you work in the back end business? Business team

Speaker 2 07:50

only, mainly like the back end engineer, like, also, like, mostly the 80% was back end system, but sometimes I'll do like a small change on the UI, but

Unknown Speaker 08:10

primarily as part of this project.

Speaker 2 08:15

So mainly, like as as part of this project, like I built an API, like, I know, for like, user experience. Like, you know, basically the API was primarily, like, responsible for, like, checking, you know, like, what kind of components at the Bank of America users can, like, you know, view on their screen, and like, you know, like certain kind of, like car details, or like you know, like only something that can be only viewed by the user with like, certain like entitlements, like you know, So that, and then also, like I built, then API for like preference, preferencing, like, you know, whenever like user they enrolled and over like, you know, different protection time, maybe they can order any changes to their like user profile, Right? Like we then, like we built this, we developed this experience, this API micro services, and any kind of like, you know, experience, but also there, there was, like, my one maintenance API micro services as well to go, like, hand in hand, right and then to make those changes. And yeah, like, we also integrated other like downstream systems as well that was exposed on, on this API gateway, from this on this orchestration services. So, yeah, I'll say that

Unknown Speaker 10:04

will be

Unknown Speaker 10:07

like in Hungary. Okay,

Unknown Speaker 10:09

I was part of the VW Team

Speaker 2 10:13

Services team, which is on the gels and payment department, sales, jail and payment department that was okay.

Unknown Speaker 10:26

Just one other question, right?

Unknown Speaker 10:30

So you said these services are very primarily on

Unknown Speaker 10:36

Java and

Speaker 2 10:38

spring, right? 17 person and like the legacy project,

Unknown Speaker 10:44

they were in like 1.8 but we're migrating

Unknown Speaker 10:54

everything to 17.

Speaker 2 10:59

So we're moving from like the Spring Boot to Spring Boot three.

Speaker 1 11:06

Did you write any batch as spring batch as part of your

Speaker 2 11:13

So regarding the bats, I was not involved in the bats of we had, and there are other people in the team.

Unknown Speaker 11:24

What was the database that you guys used?

Speaker 2 11:28

Sorry, what was the database technology? What was the database technology? Postgres SQL for the relational datas and work with dynamic on the AWS side for the non relational, like dynamic before lighting of store, like the entitlements, like message previously, for different components to be displaced, but push space, I would say, was the primary database.

Speaker 1 11:59

About some of the new, Java, new feature that Java 17 introduced.

Unknown Speaker 12:06

So Java 17

Speaker 2 12:11

also like you know, basically like you know, it introduced like a pattern matching in six cases, which is one of the new feature, and then also you can do like certain, you know, like certain object on the switch case, input, like, you know, in certain class type. And then also, I will say, like, you know, get introduced something called this seed class, like, you know, allow, which is, like, you know, allowing you to restrict, like, what classes can extend your like, a base classes, and also like it introduced like multi, multi line, string, like, you know, using like triple quotes. And I think, I think those were the three primary ones. Give me

Unknown Speaker 13:03

an example of pattern matching

Speaker 1 13:06

in your actual code. Or did you use the pattern matching in actual code?

Unknown Speaker 13:11

In the pattern matching so

Speaker 2 13:15

in the actual code, like we we did not meet like we did not use the pattern matching in our actual code, right? You know, we just migrated what existed to Java, like the Java 70, like, without like, you know, like enhancing it. Okay, so classes

Unknown Speaker 13:31

you said you use sealed glasses. Give

Speaker 1 13:35

me an example of sealed classes from your existing code, if you use it

Speaker 2 13:44

so regarding, like, you know, stuff, right? So, basically, like we, like, you know, we had, like, I said, we have, like, the organization services, right? Like we basically uses like, abstraction layer. So, you know, we basically, like, made this abstract, you know, the abstract cloud that tokenization services as our primary class. And, like, you know, it was a abstract token class that, like, you know, permitted, like the abstracted token, or like the refresh token, or even like a payment token. So from which is from our organization services, like, if the token type was certain, then we will use that like, use that for a particular flow, right? So that is like, so like our like the primary syrup class, which is for the total class where we use it. Okay,

Speaker 1 14:56

spring, right? What is the default scope in the Spring framework being a scope and Spring framework,

Speaker 2 15:04

this is the default, is the single doc, which lightning means that it's only like one instance of the exist. Give me

Speaker 1 15:18

an example of a fenderator and a prototype from your project where you used a being as figurative and a being as a prototype.

Speaker 2 15:30

So in my previous project, so

Unknown Speaker 15:37

basically, like, you know, like,

Speaker 2 15:41

like, all, all the primary bins that we have used, you know, we have, like, primarily been like singleton scope because, like, the default, it's like the default because we don't basically change The scope of the beam. But one of the builders that we use in the orchestration services that I think we change it to a prototype was, like the token token processor, where, you know, we're, we're using, like a field, like a fielding sense of the access token, or, or, like, the casting purposes, right? And, but like, you know, it will, it will be an issue, because there will be, like, different kind of recourse, right? So basically, like, we change the request token, like, you know, the request spoken processor being as a prototype scope. So we so that, like, you know, we we have like different instances for like different kinds of requests. So, so, like, all the other components were Singleton, but just this particular, like a token processing filter was, it was

Unknown Speaker 17:09

okay?

Speaker 2 17:15

Basically, like the single singleton means they're not, they're not thread set, but, like you know, by default, because, like you know, it means that there's, like, only one, one of the instance of the class, like you know, in the in the application context, and like multi threads can access at The same time, like, in a modifying the state, which is, that's

Unknown Speaker 17:45

why I was wondering. How do you define a token as

Speaker 1 17:49

single and deep? Because all the different users will have different tokens, right and related information?

Speaker 2 17:57

Yeah, yes. So, like, you know, like, like I mentioned, right? So, like, that particular tokenization flow, right? Was the prototype scope, but everything other other than the token related, was Singleton, like, you know, because it was an issue. Give

Unknown Speaker 18:17

me an example. What was single term. Singleton,

Unknown Speaker 18:24

any team that you define as a

Unknown Speaker 18:28

singleton, example for the single

Speaker 1 18:32

in your code, right? You may have defined, like so many teams defined, right? You said, all the tokens prototype, everything else has singlet. Give an example. What else is Singleton?

Speaker 2 18:42

So, like, you know, we we have this request logging filter, right? So that's what, like, you know, as a singleton beam. So it's basically, like, you know, this logging filter that is used to, like, accumulate the incoming request, request data, right? Then a lot of that particular So, and that's, that's one of the example of the singleton beam. But another one is, like a filter that, like, you know, basically injects the SSO ID, like, you know, the ID header in the body, like, into, into the body, right? So there's, there's nothing that's like, you know, the header

Unknown Speaker 19:29

is injecting an ID in the body, one

Speaker 1 19:33

instance of the singleton B, and then you said that safe, will it not overwrite all of the multiple all of the multiple requests and have different only one Id going into multiple requests? Yeah, so,

Speaker 2 19:53

no, like, no, not like, during the processing phase, right? Like, you know, it reads the request header. And then also, like, you know, injects it at the same process, right? So it does not, like, basically store that value, like, anywhere it just reaches, and then it just, like, you know, injects it, rather than, like, storing that state, right? So, so, because the issue will arise whenever, like, you know, you're you're storing the value in a shared state. So like singles and beam is like, you know, it's like a single statement, not like processing. So

Speaker 1 20:31

let me quickly come to the postgres side of thing, right? You said use the Postgres, right? So did you have used any breaker database, or all everything was Postgres

Speaker 2 20:43

at BofA. So everything, everything was Postgres, whatever

Unknown Speaker 20:52

index in Postgres,

Speaker 2 20:56

like indexing in the PostgreSQL, right? So basically, I'll say, like indexing, like, you know, it's basically like, like an object, like you know, that is used to, like, speed up, like the query performance, like you know, for like, you know, lookups, right? So if you have, like, frequently queried volume, you can just, like, define, like, an index on upon it, like we space into the process.

Unknown Speaker 21:24

What are the partitioning in Postgres?

Speaker 2 21:27

Partitioning in Postgres like, you know? So it's basically, like, you know, in the database, right? So it's in Postgres, like, it allows like, like partitioning of your like, you know, table into like smaller and like a manageable pieces, you know, like, just like a soft set of the data, like, which, which can be, like, segregated, like, you know, based on like, certain criterias, or like, you know, or like reins, or like, whenever you can. You can also like, create a table, right? Like, you can also like, define like, the partitioning range and things like that. Okay,

Unknown Speaker 22:16

what are the properties

Unknown Speaker 22:19

of a transaction and post Postgres

Speaker 2 22:24

and the Postgres. So for

Unknown Speaker 22:29

this,

Speaker 2 22:31

this problem is not like, just related to Postgres, right, like, but the main properties for properties are like the AC, acig properties, you know, like the automation city, which is like, you know, saying that, like, your transaction should basically be like, one unit of one unit of work completely. So if, if, like, the work fails, it's like, fail completely, and just report everything. And then there's another one, which is like, the which is the consistency, and you know, which is like, you know, the transaction which will have its consistency there, like, so your database will always, like, you know, be on a consistent state from one state to another, and there is isolation, which means, like, you know the transaction, so be isolated from one another, like ensuring that result from one transaction is it's not like, visible to Another, unless you like, commit those data. And then the last one is the like, the durability, like, you know, which means once like, a transaction is committed, the changes will be, like, permanently saved.

Speaker 1 23:55

Okay, I think only a couple of questions, right? Tell me one time when you have made a mistake and later rectified in your career. Yeah, so

Unknown Speaker 24:11

in the previous time, I'll say like, you know,

Unknown Speaker 24:15

like talking regarding my mistakes.

Unknown Speaker 24:19

I'll say in my career,

Unknown Speaker 24:23

like, so have you done

Unknown Speaker 24:28

anything like that in your project or any work?

Unknown Speaker 24:32

Yeah, so, you know,

Speaker 2 24:35

I'll say like, you know, whenever, like, I post on the like, basically, there's so many,

Unknown Speaker 24:44

like, you know, mainly the filter, like, there was

Speaker 2 24:48

being used on, like, you know, one app, but it was like, you know, disabled on the other app. And because there was no use case, there was filter called the decryption filter, like, you know, which basically took, like, a request payload and then decrypted it in the based on the request header right. And it was, it was, like a very common filter that we had, we're using on the mobile request for, like, in a web request that was coming in, but we did not have have it enabled. And whenever we came across this new requirement from, like, you know, from one of our clients, we just thought it will be like a simple change, and we just switched the feature flag on, that filter on, and everything was okay. But like, you know, we only tested the flow that required decryption, and we pushed it out just right, like, you know, being our deployment the time, and everything was working well, but later than, like, you know, at the night time, like, around three, and we just got like, you know. We got like, you know, issue. We got the piece like, you know, log out all the, all the users, and everything was fading like, you know, because the session, because of the session invalidation. So it was like, one of the, one of the biggest mistake, I say like, you know, because we did not test, like, all the, all the flows within the web application, we just were, like, so confident, since it worked in one app and it would work in another. So some, for some reason, like, you know, the web apps, it did not like the send enough headers, and that needed, like, in a proper like processing. And then also, like, you know, we had this error paper due to some kind of the main reason was some kind of null, null pointer exception, like, within the request header, all those things. So that was, like, one of the mistakes which was lacking, you know, testing. And then later, whenever I'm pushing against all these, I added, like, certain acceptance testing to cover different scenarios of like the missing headers, or like the extra headers, or any request bodies and so on. So this was one of the

Unknown Speaker 27:13

missing and I did learn from it as well, one of your

Unknown Speaker 27:18

highest achievement in your career

Speaker 2 27:22

all the highest achievements for my career, I'll say,

Unknown Speaker 27:30

also, like, you know,

Speaker 2 27:32

so, like, recently, like, I was working on one of the microservices we, you know, we named it like the streaming engine service. So this basically, you know, we had an enterprise big backlog to, like, you know, integrate this STP into our project, like, you know, basically, aggregate all, all of our logs and push it to the snowflake tables. So our our applications like, you know, we held like, 1516, like different microservices. And so we had to, like, you know, make this similar things on changes on the all the application, and configure it like, you know, accordingly. And instead of doing that, I devised a proof of like, in a proof of concept to, basically, you know, like, create one external microservices that, like, you know, that will expose, like certain REST APIs, basically achieve, like, you know, this particular change, right, like, so, You know, I created this services, got it deployed, and then, like, you know, we all, all we had to do on all our services was to do, like, a restful call to the service with with the payroll, and then, you know, the service will basically be responsible for, like, mean, like, manipulating the incoming table to the desired format and then push it to the scope, like cable directly. So this was, like a really big thing for me, because I help my team, particular team, but we, like, you know, expose this API without, like, you know, exposing the API to other teams as well. So

Unknown Speaker 29:22

yeah, this was like,

Unknown Speaker 29:30

it was a contract role.

Unknown Speaker 29:36

Contract

Speaker 2 29:40

role today. Yeah, so first of all, I want to thank you so much for the time today. I know we interviewed more time. It was really nice talking to you, but I will, I do have, I would like to know maybe, like, what defines like a successful developer on the team at the time at the US Bank,

Speaker 1 30:02

like, for a developer coming in the real successes and from the environment you're coming in the real successes, it's a very engineering mindset team. The person has to be technically hands on. On multiple tech stack, there is a lot to do on the technology, technology team, by technology team, we are not stirred by some of the barriers. Like, Oh, you can do this is not, not allowed for development like, this is quite a bit of things you can do that you can go to Kubernetes, create your own pods, develop your services on your own performance testing. Number of things the person should be hands on, technically, very smart the person should be understand what exactly the solution. Done. Code very fast. Build those components very fast. What will be the onshore, offshore model? Communicate to them, find the do the code reviews with them, find the issues or define those issues, or how they can do better or define those things for them, right? And finally, take all of this delivery to good production successfully, which I think you're already doing it as part of the project, right? So here's all the things which are help a person to be successful in the role. Thank you very much. Thanks for your time. I wish you a happy holidays and wonderful new year. Take care. Bye. By

Speaker 3 31:44

way.

Salim Maharjhan- USBank- SRV- Final

Unknown Speaker 00:00

And, you know, on

Speaker 1 00:01

the also on the previously, previous side, like, you know, also worked on opposite spark for data injection, like of the healthcare, like the clients on the BCPS, the Blue Cross, Blue Shield, and, yeah. So those were, like the experience in the financial domain, and also, like in on the healthcare side as well. And also worked on JPMC as well. And then for my school, I graduated from Marymount University, that's in Virginia, here in Virginia, us.

Unknown Speaker 00:41

Where is it located? In Virginia.

Unknown Speaker 00:44

Virginia, the exact address it's which,

Unknown Speaker 00:49

which city is,

Unknown Speaker 00:51

Arlington. Arlington. Okay,

Unknown Speaker 00:53

it's in Arlington. Okay, so outside of Washington, DC,

Speaker 1 00:56

okay, it's nearby 25 I think 25 minutes drive in the DC model and from her from my college,

Speaker 2 01:06

okay? And so, so you just said that you worked on a orchestrator, which was like an API gateway in Bank of America. Is that correct?

Speaker 1 01:18

So, like, yeah, it's not like, primarily, like an API gateway, is like a microservice orchestrator, like, you know, sits before an API gateway. So, like, we, like, you know, I'll say, like some kind of reverse proxy, which basically like routes, either to an API gateway or to, like, a micro services directly, like we, you know, like handle requests and like response manipulation. And it's also like a Spring Cloud, a gateway, like, based, like, you know, Spring Cloud gateway, Gateway based application. So, yeah,

Speaker 2 02:00

so, okay, so let's describe the microservice, which you have picked. So you just mentioned it's doing the manipulation is doing the routing. So what did you use to build this application? How have you built this application? Like, what spring components are within it?

Speaker 1 02:18

Yeah. So, you know, basically, like, you know, we use on the Spring Boot for building the micro services, right? Like, the two main like micro services that I developed recently were, like the maintenance API and Experience Experience API. So, like, you know, like, the maintenance API is like, you know, it's used to make changes to like the user accounts, like user accounts preferences, like, whenever, like, you know, make changes to your like user profile, like we have this enrolling on like, you know, certain services and like certain features, it triggers this maintenance API, and it also, like, you know, like, has get getting points, like, you know, to get the status of those updates and so on. And like, you know, the experience regarding the Experience API. It's used to, like, grab the user entitlements for different like front end components, like you know, to load it as a Spring Boot based application. So we use Spring Boot three along with JDK 17, and it's deployed on the AWS ECS. So also we use Splunk for a logging and new really, for the monitoring purpose.

Speaker 2 03:50

Okay, so let's say I have a spring app and I want to make a HTTP request. How do I make a HTTP request in Java app built using Spring,

Unknown Speaker 04:04

yeah, so,

Speaker 1 04:06

you know, so basically, like, if you want to make a HTTP, like, you want to make like the STP HTTP call, right, using the Java, Spring based application, so you can use like two approaches. Like one is to use the rest templates and the other is to use the inbuilt like the SCTP client provided by the Sava. So like, you know, using using like the spring template, like you know, spring template, or like Rest Template. You can just like, you know, get, like your get for the object, or like, you know, post for the object, right where you provide the URL and return type. But if you use the HTTP client you like, basically just work with like the URI, the body and like, you know, the connection pool and so on. So you can either use any of those two approaches for that,

Speaker 2 05:13

okay, and Rest Template. So let's say I end up using Rest Template. What client does arrest template use behind the scenes? Like, does it use the JDK client? Or there are other options to use along, like, there are other options to use when you use Rest Template. The Rest Template is abstraction, right? So if any are using Rest Template behind the scenes, there is going to be a client which is used. One option is to use the JDA client, which you just talked about. What are the other options? Which are there? Are there any other options, like, do we need to give thought like, Okay, I'm making HTTP call. I'm trying to use rest and play. Do I need to give any particular thought that, okay, I want to use the JDK client, or I want to use something else. Like, what are your thoughts about that?

Unknown Speaker 06:06

Yeah, so,

Speaker 1 06:09

like, you know, one of the, like, most popular option that I prefer is the Apache, SCTP, close able client, where, like, you know, you can basically, like, configure, like, pulling connections by using the connection manager. So, yeah, basically you can, they can, you can provide this HTTP connection connection factory on the Rest Template to internally use. So, yeah, I think it's, it uses that. But you can also, like, you know, configure it during your beam, definitions of the rest template itself, and, like, you know, where you like, define the pooling Connection Manager and HTTP client, or any kind of, like, any kind of factory needed with it. So yeah, but behind the scene, like, you know, by default, will use the approachable as a big client provided by the passing, okay,

Speaker 2 07:09

I'm deploying a spring app. Let's say there are rest controllers on it. What like? What would you like? What is your thought process in terms of what server would I use while deploying that spring app, as in, so would you use Apache, Tomcat? Would you use jetty? Or would you like? What are the other options and what is the thought process in terms of what option would be more preferable in what scenario

Unknown Speaker 07:45

Yeah, so,

Speaker 2 07:51

yeah, so spring, I believe Spring Boot, if you deploy an app by default, it comes with Tomcat, yes. So would you? Would you use the default? Would you use something else? Then? Is there any scenario where you need to move to something else? So just want to understand the thought

Speaker 1 08:07

process, okay, yeah, so, yeah, definitely, like you said, default, it's the Tomcat, right? So there are,

Unknown Speaker 08:17

like, in other options

Speaker 1 08:19

that work well on certain scenarios, the Tomcat is, like, well suited for, like, you know, traditional, like, a traditional Java app, web applications, like, you know, with servlets and so on. But, you know, like, if you want a lightweight, like, you know, like fast, like customizable, like components, well, like, you know, with high, like performance concurrency, then JT might be the like a way to go, because it is, like, you know, capable of like, handling, like high, high current requests. And you know, you also, like, have an example for like Niti and honor to like net is, like, you know, it's widely used for like the spring web of like spring web blocks applications where, like, you know, you handle like asynchronous, like event driven networks. And I think, like, you know, under two also like it has, like the light wave light web server for, like, the low, low latency applications. So definitely, there are like, the use cases for like, you know, each of those servers. So, but mainly I have, like, work with tomca and Eddie. Okay.

Speaker 2 09:44

So have you used any messaging middleware like IBM, MQ or Kafka? Kafka can be used for multiple things, but have you used any messaging middleware or maybe rapid MQ, anything? What do you have used, I think, in your resume, you have mentioned Kafka, but I just want to understand from,

Speaker 1 10:03

yeah, so I've worked with Kafka, and like, you know, I've also worked with the AWS provided, like, SDS, SNS as well.

Speaker 2 10:12

Okay, SNS, is it? Is it like a custom implementation

Unknown Speaker 10:19

of Kafka? SNS, um, I don't think

Speaker 2 10:23

so. Okay, okay, no, I'm not aware. That's why I'm asking you, I haven't used SNS,

Unknown Speaker 10:29

okay? I think it's still

Unknown Speaker 10:34

like, you know, like, yeah,

Speaker 2 10:38

okay, so you have used Kafka Correct? Yes. Have you built any application where you have a Kafka Producer or Kafka Consumer or both?

Speaker 1 10:52

Yeah? So yeah, I've worked on a little bit on the like both sides, but primarily on the consumer side.

Speaker 2 11:03

Okay, so what is the most common concept, I think, in Kafka, on the consumer side is a consumer group. So how would you define consumer group? What? What is, what is the consumer group?

Speaker 1 11:18

Yes. So, like, you know, basically, like, when you say consumer group, it's, it's just like a group of like, you know, consumers like you know that like, work together to, like, you know, mainly consume your messages, right? So like, you know, it may work on like, you know, one or more like Kafka topics, and it just allows like, you know, multiple consumers to like, you know, share the responsibility of like, processing, processing the message. Like, you know. You can have like, you know, you can have, like, a Kafka topic, like, divided into like, multiple partisans, right? And like, it's it's partition, like, you know, is consumed by only, like one consumer, like, you know, within that consumer group. So like that, okay?

And

Unknown Speaker 12:20

in Kafka, let's say,

Speaker 2 12:24

is there a way where you can control that? Hey, I want to pull only these many number of records at a time, or I want to pull these many records, then I want to pause. And then after a certain condition is met, then I want to pause any any idea about that.

Unknown Speaker 12:47

Yeah, so, like, you know,

Unknown Speaker 12:51

basically,

Speaker 1 12:54

like, you know, we control the pull right. Then we might have to use the properties like the record pool, right? So I think there is like a max, max record pool property that you want to set for that particular scenario, right? So, yeah, so I

Unknown Speaker 13:23

think that's, that's the way to do it, okay,

Speaker 2 13:30

and what, what experience you, you have had with different databases. I think you mentioned my sequel MongoDB. So let's, let's maybe talk about my sequel. Is that the so in MySQL, right? So let's say from Spring app you need, you want to connect to MySQL. So what are the different options through which you can interact with MySQL in spring or in Java as a whole?

Unknown Speaker 13:57

What is the traditional way one more to connect to MySQL in

Speaker 1 14:01

Java? So, like, you know, basically you can, you can use a plain JDBC, like, you know, you can just use, like, a direct JDBC, where you just add connector, you know, like, use a JDBC template and, like, then you you could do a direct, direct query. But like, you know, widely used, I'll say, is using your spring data, CPA, where, where you define, like, a repository layer and an entity object to, you know, like, map the data into object, and then, like, you work with that repository. And I think you can also use, like, just hibernate, like, directly, but I have like, enough more leave with Spring Data CPA, because it supports, like, wider operations, like many, like, native queries, yeah,

Speaker 2 14:58

okay, okay, I think that's, that's all the questions I had. Did you have any questions for me?

Speaker 1 15:07

Yeah, actually, but I want to first thank you for so much. Thank you so much for for the time today.

Unknown Speaker 15:14

But I would like to know maybe, like, you know,

Speaker 1 15:18

like the date, like, like the day to day work for a developer, look like,

Speaker 2 15:27

okay, so we we work, so our team works in payments, so different kind of money movement which happens in the bank. So it could be Zelle, I'm pretty sure would be aware of Zelle, then real time payment. So that is the clearing house fed now. I think fed now was released in 2023, so last year sometime. And then obviously we have the traditional ech. So we have this onshore, offshore model also. So you will expect to work. You will be expected to work with folks who are offshore, and also folks who are pretty much distributed in the United States and Canada also. And the expectation is you should be able to do coordination. You should be able to do design, some part of design, then obviously development, and you should also be able to help with the performance testing. So the expectation is, it's not something like, Okay, you're only expected to do coordination, like everything is done by offshore. No, no, no, that's not what the expectation is. You need to do some part of the coordination, like you need to be able to break up work, which the offshore should be able to understand. That's one part of the job. Then other is, obviously, you will win. You will be involved in the design of anything new which is built up. You will be, you will have part ownership of that. Then obviously development, that is, as developers, we all love to do. So obviously we'll be doing that, and then the regular code reviews and performance testing, which which we will do before pushing anything in anything in production. So pretty much, I would say, all round effort is expected.

Speaker 1 17:21

And also what will be the next step after after this interview,

Speaker 2 17:27

I'm not sure. Actually I'll need, I think you should ask that to the recruiter.

Unknown Speaker 17:33

Am I the first person you're talking to? No, I

Speaker 1 17:35

spoke with. No, yeah. Okay,

Speaker 2 17:39

yeah. I think so. I think there will be one more interview. After that, I think that should be there will be one more interview.

Unknown Speaker 17:50

I think you should talk to the recruiter.

Speaker 2 17:54

But yeah, I mean it was it was good talking to you.

Shailesh Paudel- BNY- Final-SRV

Speaker 1 00:00

It's okay. No. Need to know answers for all the questions, right? So we can it's it's practically not possible. So yeah, just be comfortable and start with a brief intro about yourselves and your recent, most recent project, and your role in

Speaker 2 00:25

it, sure, definitely. So, yeah, my name is Charis photo, and I have around like five years of experience with like, full stack Java development, and I've been primarily working on like, a Java based web application, like utilizing frameworks like spring hibernate and like web services for like API developments. And recently I've been like working on, like hands on micro service like development as well. So we're basically using, like a Spring Boot, and I've been like, involved in, like designing them, like implementing them and like deploying them. So we're also using like AWS Cloud for like deploying these applications. And so some of the services that I work with includes like AWS, like EC two, ECS and like SQS, SNS and like lambdas on like a front end, and I also work with like angulars. And currently on my like scrum team, my team consists of like five developers, and we have, like a product owners and like Scrum Masters. Also, along with that, I would say I'm involved mostly working on like back end, like Java code, like helping, like a team with code reviews, also with like production support duties. And so at USAA, I've been, like, primarily, like working with, like back in Java development, where I basically work on like modernization of like monolithic existing applications. So yeah, so in my day to day, I mean, mostly I work work, like 60 to 70% on, like the coding, and unlike basically on technology, on like feature development or like microservices work, and also then rest is either like scrums ceremonies, or like helping teammates and like

Speaker 1 02:25

so on. So are you full time with your current employer or consulting? Sorry,

Unknown Speaker 02:33

are you full time with the current employer or Are you consulting?

Speaker 2 02:38

So I'm basically right now. I'm in contract. I smell like w2 Yeah, okay,

Speaker 1 02:44

I'm fair enough. And can you briefly explain about your latest project, what it does and what functionality that you're responsible for and you're working on?

Speaker 2 02:57

Yeah, sure. So my latest project at USAA is, so it's basically going through like, massive, like modernization efforts, so they have, like, like their existing thing, like, which is like a big code basis where, like, they're like, basically, like changing, like the user experiences, along with like modernizing, like, the overall framework like, so we have been like, tasked recently with like, stripping out like, like two of the like, the primary API's like that are related to, like the front and like front end portal and like user preferences sections. So like, the first like modernization that I did was something called, like a maintenance API, which is, like, particularly on, like an API that is, it's like, responsible for, like, taking care of, like a user profile preferences, right? So, like you as a user, you like, log into like your portal, and basically you make like, any changes to your profile or like preferences, like enroll in like, like, a certain kind of like protection plan and like. So basically you remove any kind of overdraft or like, such scope and so basically, like maintaining API that will take care of like particular flow as well. And basically, I created like a design doc for, I would say, like modernizing like effort, using Spring Boot as our like primary framework, then also developing it. And then like, I also work with other like, Experience API as well. So Experience API is like, similar to like, in a sense that you to like, grab user like entitlements and like, like, what components to load on, like the front end. So yeah, basically, those are the primary, primary projects that I worked on

Speaker 1 04:55

recently, since your work is majorly related to API in the recent ones. So what, how do you make the APIs backward compatible?

Speaker 2 05:08

Okay, so, talking about API making backward compatible, I would say so. In order to like, make APIs backward compatible, you basically need to make sure that you add like, any kind of like endpoints by like, like versioning, right? So you can, like, basically create like, new versions of API that are usually like you are, I like preferred, such as like, but you can also have, like, under like, certain strict schema. And I've been using, like some companies, like preferred version based on, like, like a certain application headers. So we also need to make sure, like, whenever we're like, like, updating those resources, we support like, some kind of default values, right? And so instead of, like, stopping your features, like, right away we like, try to, like duplicate those features, like, gradually so like, those are some of the ways you should do it. And all of those should be like, basically maintained is like, like, some kind of like feature flag, because, like, you know you may need to like enable like, a disabled feature based on like flags, which can be tested in like, like, a flexible manner. So I think these are the ways you can make sure the APIs are like, backward compatible,

Speaker 1 06:33

couple of things. So you mentioned about turning on and off the feature, right? So can you explain how you have implemented it,

Speaker 2 06:42

yes, so, like turning off and on and off. Like, the features, I would say, like, in order to like implement

Unknown Speaker 06:54

so you said,

Speaker 2 06:58

I would say maybe like. So basically, you know, like, it's based based on, like, like, a, like, a configure, like properties, like, whenever we're like, talking about like feature toggle or like, it's either like, simple changes to like, our application property file. So you like, if you're like, maybe using some kind of like database, like, it's just an update to that. So we primarily just use it to like, dynamically, like configuration, like configure it and like, like property service, which basically means that our application leverages like, like a Spring Cloud configuration, right? That basically pulls your like property dynamically and then like, it puts like, what do you say? Like, also like, onto like, the application during like, a runtime, right?

Speaker 1 07:50

So you haven't used any any third party APIs or frameworks for the feature flag, so that's fun.

Unknown Speaker 08:00

So how do you secure the APIs?

Unknown Speaker 08:03

So, how do we secure API? You're sure,

Speaker 2 08:06

yep, okay, I would say to secure APIs, we can use like, like token based authentication, right? So basically, like, whenever we use like logins, we basically have like, like identity provider, which has like a, like a front inside, in, like a controller, or like in, like a portal where a user logs, they log in, like, using like a username and their password. So like this is an like external team. So whenever we use like logins, they basically provide like a username and password, and that like team basically like validates their like user authenticity, which means like they're like a user, like, if they're user or not, and then if they are like a valid users, then we identify like, like With the identity, identity provider injection. Act like also, we basically have like a JSON web token, like also consisting of like a headers and like, like a claims and like a signature. So I would say like on all of the like subsequent requests that like a user makes, or like any action that user takes. So basically this access token will be then injected into like an authentication header, and also it adds like a like a beer token. And we also have our application like it basically uses like, I'll say, like spring security, to like validate that like the token, and it adds up to like a role based access control.

Unknown Speaker 09:45

So I see Bishop join. Hey, Michelle, all right.

Speaker 1 09:51

So I mean, please feel free to interrupt or ask any questions. Vishal, so

Unknown Speaker 09:59

let me give you a use case, right?

Unknown Speaker 10:03

Say you have

Speaker 1 10:06

four or five different microservices, right? And everything must be orchestrated properly, right? And the requirement is, we need strong atomicity, right? So we want all the micro services in sync. Each and every micro service is going to have a state change, right? It's going to modify the state of the objects. And the requirement is, we want the entire transaction right, meaning each transaction should consider all the units of work performed by these five microservices. Okay? And if any of these fails, it should roll back the state of all of the microservices. Let's say there is a failure in the fourth service, for example, it should roll back the states modified by service one, two and three, right? So can you explain how you can solve this problem?

Speaker 2 11:13

Let me think I would say, like, basically, this is a like, a classic example of using, like a design pattern. So basically, so there's like a micro service orchestration, right, which is called, like a saga pattern, right? So basically, it is like architecture where, like, we, like manage these, like distributed transactions. And basically saga is it's nothing, but it just like, like a centralized controller. So it basically you have like a saga which is responsible for managing, like a workflow between like these transactions. So you would basically, let's say, let's say, like a user, once you run like a like a certain task, then you would, like, already have, like, a predefined workflow within the saga where you know, like, you have, like a service on and like a two or three with all of them, like, define, like, defining like a transaction. And then you're like, orchestration. Basically it did. It defines, like, what sequence of order it needs to happen. And like, then, like this orchestrator, would like, will be like, responsible for, like, sending a request to like each microservices to perform like a job, and what is like supposed to do so in case of, I would say, in case of like a failure, like a response from like particular microservices, right? So the orchestrator, it basically triggers, like a rollback transaction where is in, like a saga term known as like a compensation action. So this design pattern, I would say this design pattern will help in, like orchestrating your distributed transaction across like multiple services.

Speaker 1 13:02

Okay, all right, and, and you mentioned about your involvement in building the artifacts, deploying it somewhere, right, and take it forward. So can you explain the tools, starting from your source code repository, of how you build the artifacts and how do you deploy to various regions, and then how do you implement to production? Can you explain this cycle? Yes,

Speaker 2 13:33

I would say. So basically, you have your like source code, like source control first, which is generally like your Git, right? So I've been, I've been working primarily with like GitHub, so you would have your like source control that, like developers work out like you're like branches, and you also have your like, like main, like, default branch, and like, some some, like, some organizations and they, work with like, like release branches. So in my personal experience, we have like, like a developed branch where we as an engineer, work out of and and then like a like a main branch where like, like a post release and then from like developed branch we cut like release branches. And so when we need to do like a release branch, we basically have our actual deployment, which goes through, like a production review. And we have like, we also have like a like a development branch, I, as I said before, and we also use like, like a Jenkins pipeline set up where, like, once we build this pipeline, what it does is it uses like a build tool that we have, and which is a Maven to basically like compile like that application. And it helps us with running like a unit test. So during this phase, it just checks if like application can like like it can compile and like all unit tests that we have are proper. And also we may need to add like like a step to do like like a sonar scan to like, make sure that we have like, a proper, like, test coverage and so on, so. But I would say, like, the immense, like, the main step here is, like, packaging your application, like, which can be done using like, a maven package command, right? And then, so what it does is it creates, like, a jar of your application, and then you can, like, upload into like, like, like a Central Art Factory. So from this central Artifactory, you know, like your deployment cycle, which may be just like straight deployment are, yeah, you might be using like, like a Docker right step as well. So I would say, for like a Docker, you would basically have like a Docker file, which is like, we defines like, what kind of step you have to do like in order to run that application. So, so it would be so you would basically run that Docker build command, which will read to read like a Docker file, and you'll like, execute the steps. So usually Docker uploads like, like a docker images to like Container Registry, right? So I would say, in my case, we have like a primary, like, like a primary working with, like AWS ECS on like a four gate that runs like, like a container orchestration. So, you know, like, whenever we are, like, ready, we run the pipeline then. And like this pipeline will like, define like task for like AWS ECS, which will, like, pull in all the container images from like registry, and then it starts like, basically do like, like a docker run on like that image. So I

Speaker 1 17:00

think that's, that's one got it. Yeah, yeah, I got it. Thank you. I think you answered most of what I would expect you to. Thank you. Then unit testing, right? So how do you develop unit tests? What are the frameworks that you're aware of? So So, sorry, unit testing frameworks. Oh, you talk about the unit test, yeah.

Speaker 2 17:28

So unit testing framework, I would say I have primarily worked with like, like a JUnit and mokito, yeah. Okay,

Speaker 1 17:41

got it okay. So the you mentioned about production support also, right. So what are the observability tools that you use to debug or troubleshooting issues?

Speaker 2 17:56

So observability tools, I would say, in like primary like application, we mainly use like Splunk as our like logging framework. So like all of the logs that our like application generates, they're like streamed into like log by like, some like uh Splunk forward agent. But like the actual observe, like observability tools, like, for like matrix, we use like real so our containers have like, like a new real agents already, like, they already installed into the application matrix, and then like, we use it as like a data set for creating, like, like a dashboard, right? So, so we go to our neurotic, like a dashboard where we can see, like a CPU uses that our like memory had, like what percentage of memory, and also, like the like the disk uses also so, so we can, like, also see, like a response time as well, and like a latency and so on. So these are, like, some of the tools that we use. And do you know

Unknown Speaker 19:03

anything about actuator?

Unknown Speaker 19:08

Actuator, yeah,

Speaker 2 19:11

basically, I would say, what you say? Acutator, correct, no.

Speaker 3 19:16

Actuator. Oh. Actuator is,

Speaker 2 19:20

yes, I would say it's, it's like a tool which is like, provided by spring that that is basically used to give like, those like metrics, right?

Unknown Speaker 19:34

Can you talk about any

Speaker 3 19:36

of those? They basically, and how do you configure that in your application.

Unknown Speaker 19:44

How do we configure that?

Speaker 2 19:47

Yeah, yeah, okay. I would say we need to, like, add, like, a Spring Boot acutator dependencies right on on our project. So basically, once we add those, like Spring Boot, likes starter acuture, so it basically it gets activated. But in order to like expose the endpoints, I would say we would need to like, add like, certain configuration on, like the application properties. And

Speaker 3 20:15

can you explain them? What those are? If you remember top of your head, I

Speaker 2 20:24

let me think I would say like info endpoints, like it basically gives like information about like like app health. And also I would say like health endpoints as well, right? It gives it basically tells you, like your app is, like, up and running, yeah, I would say

Unknown Speaker 20:53

that you can go ahead,

Unknown Speaker 20:57

probably pick it up once you're done.

Speaker 1 20:59

Okay, yeah, okay. So I mean some basic Java questions, right? So we need to be just a lot of high level things.

Unknown Speaker 21:12

So let's say

Speaker 1 21:16

name some of the marker interfaces. First of all, what is a marker interface?

Unknown Speaker 21:24

So I would say it's,

Speaker 2 21:27

it's basically an interface that usually, like, does not have any methods or like, like a fields, right? It just like serves as, serves as, like a tag or like a marker, right? So, basically, so one of the, one of the marker interfaces, so one of the like, most common one is like serialization interface, right? So it just like, make sure that it like our class is like, capable, capable of, like, being like serialized. So, you know, like, like converting, like the, like the bike stream, right? There's also, like a, like a cloneables interface, I think,

Speaker 1 22:13

okay, yeah, that's fine. So the, let me give you a use case, right? So in the distributed application, right? So where we mostly have multiple instances of the product running, right? So it's a clustered environment. Do we have experience with the clustered environment?

Unknown Speaker 22:30

Yes, yeah,

Speaker 1 22:32

okay, so probably I'm going to give you a problem statement, which is not really good to have, but hypothetically speaking, let's say you have a block of code right, and you don't want multiple threads to execute that block of code, right at any given point. So how do you how do you make it possible?

Speaker 2 23:09

So whenever I would say, like, whenever you want, like a block, like a, like a certain block of code, you know, I would say it just like making sure that you have, like a proper, like synchronization around, like the blocks. So you have, so you have, like a synchronization block. So it just means that you can, like, specify that like the section of like a code to be executed by only, like a one thread at a time. So that would be like for like a block of code. But you can also do it for like a, like a method, right? So you have like, like a certain methods that you you don't want, like a multiple threads to run. Then you like, just use like synchronized keywords on, on, like, like a method definition. So also, like, one other way to, like, look at some kind of like, like a locking mechanism, so, which is like, basically providing like, like, I would say, like, an like, explicit lock, right? So, so those are the ways you can, like, basically prevent like multi threads from like, running like same methods. But we'll

Speaker 1 24:18

just, but we'll just guarantee synchronization across the cluster members. It will definitely synchronize the code within one instance, right? That's for sure. But will it work across the across multiple instances?

Speaker 2 24:39

I would say normally, like, they work whenever you have, like, a distributed system, like, like, if you how will

Speaker 1 24:47

it know it's peers, right? So let's say you have two different requests, right? That is going to invoke the same block of code, and one request is going to one of your instances and another is going to another instance right now. How will each instance know? Hey, I'm getting two different requests trying to invoke the same block of code, but that block of code lives in each instance also, right the local synchronization methods that you just talked about works within the instance. So let's say you have two requests hitting the same instance, then it would work. But if two requests goes to two different instances, the method that you described won't work, right? So there are multiple ways to deal with this problem, but like there are APIs or frames, there are frameworks available which can help but put aside that framework, as in the framework, right? So you can, we can even write some custom code or logic to deal with this problem. Can you try to come up with a logical solution. Wanted to think about other frameworks, come come up with your own solution to deal with this problem.

Unknown Speaker 26:11

Let me think about it.

Speaker 2 26:14

So, so we want to this, like, prevent, like, multiple clusters, right? So

Speaker 1 26:20

the block of code is sitting in two different instances with the JVMs on their own JVMs, right? So now the problem that we have is we don't want that block of code across two different instances to be executed at the same time. That's the problem statement and synchronization, the way that the solution that you provided won't work across instances. So we need another way. We need to figure out another way to do it.

Speaker 2 26:52

Okay? So, like, basically the way to go, I would say, like, you know, on, like, our, like, a distributed lock table, right? You would basically, before you acquire, like, before you perform like, any kind of operation, you would like, you like, you know, like, hit, I would say, like, basically, you would have like, a log table that keeps track of, like, your certain methods, and if they're like, executed or not. And before you run, like, a block of code, you would like, get that value like, then, like, after you're done, like, executing that, and you would like, update like, like a distributed lock table to that. And if it's like, if it's not locked anymore, right? So I think what would be like the like, the easiest way to go like, without using like any external

Unknown Speaker 27:46

you're almost you're almost there.

Speaker 1 27:49

But I think this point, this solution, also have a challenge, right? What if the two requests right try to get the data from the database at the same time, and then they think that no other instances are doing the work, and then they enter it's a possibility, right? Concurrency issue, right? So, how can you enhance the solution to prevent that concurrency issue from happening, you are going in the right direction, right? So your solution is almost good, but you have to make it more efficient to deal with this concurrency problem, right? See

Speaker 2 28:42

how so I think maybe, like,

Speaker 2 28:57

just think about this. Maybe like, we have to, like, say, like, some kind of, like, isolation level, like, within, like, like, the database.

Speaker 1 29:09

Well, one of the ways is, have you heard about row level locking in the database? Row Level lock? No, okay. One of the ways is to go with that. There are few other ways to do it. Okay, fine. The intention is to try to see your logical thinking. Yeah, I think I'm done with my question. So over to you, Bishop, okay,

Speaker 3 29:41

yeah, so, yeah. I mean, probably just so, so, the same line. I mean, so, so we are using Spring based application, right? Spring, and we want to implement cash, right? But I say that you don't use any third party framework, right? So you have to write your own cache implementation. So how will you implement it?

Speaker 2 30:11

So we have to write our own cache without any frameworks, I would say. So

Speaker 3 30:20

put some implementation details, right, so that, yeah, so that that will help,

Speaker 2 30:26

okay, yeah. So, if I were to design like a cache, I would, I would say, like, will basically, like, incorporate, like a concurrent HashMap as, like a, like, a basic field that holds, like a cache, like a key and like its value, right? So we would have like, like a control

Unknown Speaker 30:47

or a concurrent HashMap.

Speaker 2 30:52

So because, like, you for like, so for the concurrent hashmag, we just like make sure that, basically, we can work with like, like a multiple threads in like environments. Do

Speaker 3 31:05

you think concurrent hash map is totally thread safe for all cash all type of cash operations? Do you see, I mean, you can think about all cash operations, right, what you will end up doing. And do you think concurrent hash map is suitable, or there is another data structure called a sorted, sorry, synchronized map, right? Is that? So, which one is better, and what is the difference between these two amazing

Speaker 2 31:40

let me think I would say so, like, I've only worked with like hash maps and like, like a concurrent hash map, so I'm not sure, like, what's like. Like, a difference between like synchronized map is

Speaker 3 31:58

okay, right? So, yeah, so go ahead right, then probably what else you need to do? Right? You will use a casing concurrent hash map. What's next? What else needs is required.

Speaker 2 32:10

So let's see, I would say, so we need to also like, add, like, add like, our like, like, a normal PUT method, like, I would say, like, also like, Get method as well. And then try to, like, remove methods and also maybe like, add like and like option to like, like, like, a clear those. So I think like, we want to, like, store our cash entry in like,

Unknown Speaker 32:40

like, like, some kind of

Speaker 2 32:45

you think, like, like, internal objects, like that stores like, like the value, along with, along with, like, like the time to live,

Speaker 3 32:56

okay, Anything for the keys, any design consideration for the keys, for the keys,

Speaker 3 33:12

can it be like any class or board?

Unknown Speaker 33:17

Yeah, like anybody, like any custom class,

Speaker 2 33:20

I'm just like, I'm keeping it as like a String class for like more ease.

Speaker 3 33:26

But can I not, okay, can I not use any custom classes keys that was more restrictive, right? Yeah?

Speaker 2 33:37

Like, yeah. But like, once, I would say, like, once you make it like generic, you can like, create, like, like Caskey object to like, basically have like, like a part where you have like a, like a proper hash code and like, like equals implementation and like, then you can just like, like during like a PUT method. Like, also have like a get method, and you generally pass like cache key rather than, rather than like a cast object. But whenever, like you're inserting it, you need to create like a, like a custom cache, like key object.

Speaker 3 34:15

Okay, so moving on. So you, you just, we, just does the multi threading concept, right? So, so I'm just, I'm not talking about clustered environment here, right? I'm just talking about, like a normal, single instance, right? So, so the, so there is a use case, right? And so, wherein a deadlock can happen, right? So, so do you know how to deadlock will

Unknown Speaker 34:41

happen? You know what the deadlocks?

Speaker 2 34:42

Yes, deadlocks, I'm familiar with them, right?

Speaker 3 34:46

How will how does it normally? I mean no, in what situation it will happen. So

Speaker 2 34:51

deadlocks, they basically happens in situation when, like, I would say, like, two or more like, threads are like, like, block, like, indefinitely, right? Yeah, I would say that. So, like, when they're both like, like a waiting on like resources that holds, like your like, other thread is holding that, right? So, like, if you have thread holding like two like resources, and it waits for like resources, and then like a thread, like it does like opposite, and then like both threads are stuck for like, some kind of like release on those operations. Okay,

Speaker 3 35:35

so how will you write a code? I mean, is there a way we can code around it, right? Wherein we still have the situation where the thread, there are multiple threads, they are they will wait for the resources, right? But is there a way we can code around that, right? How will you ensure that, given there are multiple threads and they're accessing those resources, how will you ensure the deadlock will never happen.

Speaker 2 36:07

You think, okay, maybe, like, basically, maybe you can, like, either prevent, like, a deadlock. You can avoid deadlock by what, like, what you're doing and, like, making sure that like, like, all of like locks that are like have with your threads acquires like in like, maybe like a particular order, right? What I mean by that is like, when you have like, like a multiple threads that are like, like, required to acquire like multiple locks. You always like, ensure that each thread acquires like locking. So like, like in like some defined order, so which basically prevents like a, like a circular way condition. But so the other way to avoid like deadlock is just like implementing like a timeout strategy, like within that like thread. So when, like, if a thread cannot, like, acquire, like, a lock within like, specific, like, specified amount of time, it just, like, it backs out of, like, that operation. So those are, like, the two ways,

Speaker 3 37:15

right? Okay, so, so normal spring API, right? Any of the REST API, right? Probably it's trying to do some CRUD operations, okay, and we are creating that spring API, right. So if we ask you to basically create that API, right, what are like the different layers of classes that you will create right within that API, how will you design, basically, right? What are different set of low level, I mean, just basically, give me a low level design, right? Yes,

Unknown Speaker 37:46

sure.

Speaker 2 37:50

Basically, I would say, for like a design, it consists of, I would say, like, like a, like, like a main, three main layers, right? And I would say are you have, like, initially controller, which is like, you're like a, like a risk control annotation classes here you would basically, like define, like a different entry points where you might have, like, your like get mapping, quote, like, right post mapping, or like a put mapping and like design mapping for like, your like, getting resources, like, like, creating resources, updating, like, editing resources. So like, that's where you like, generally have your like entry point of like application here, I would say you would like, basically you have like dependencies of like service layers. And then service layers is, like, generally, you're like a business layer, right? And so in the service there, you would have like, like a different methods that they basically serve, like those for operation that we have. So you have like method for creating, and you would have method for getting, and then like, like updating and deleting. And within the service layer, you would also have like a dependency of repository layers, right? So now, like the now, like the repository layers is in the layer that it basically interacts with, like a database. And if you're, like, working with, like a Spring Boot, I would say like Spring Data, JPA gives like a very like, like, a good like abstraction just for like, like a simple use cases. So you can use like that as well. But, but like, if you want to, if you want to have, like, more complex like queries, you can use, like a native queries, using like query annotations and like repositories that you like need to use just like interface that extends like JP repository, along with like, like the three main layers, and You would need to like, define like an entity class, which is basically like an object representation of like a database object that you are working with, right? So that's one another important thing. And so now you maybe want to like define like a controller, which is used for like a global exception,

Speaker 3 40:24

okay, so what about the entity classes and, right? How? How will you entity classes, or data transfer classes? So, how will you define that, right? Are they like, any consideration about how the I mean, this is like a contract between all these three layers, right? So how will you do that? Yes, do you have, like, a separate DTO or domain classes or how?

Speaker 2 40:57

Yeah, so, like, basically, I would say, like your entity class is just like, like a representation of, like your database table, right for your DQ is like a lightweight, like objects for basically, like transferring data between, like these layers and so usually when, like, I'm working with like crude operation, like my controller, it takes like DTO objects and also like, like a like service takes in like DTO objects and in like service layer, and it's like DTO it gets converted into like, like an entity object. And you can maybe use like, like a beam. YouTube, so if

Speaker 3 41:39

I ask you to not use the DTO object, right? How will you ensure that only the data that is required is getting transmitted?

Unknown Speaker 41:49

Um, so not using a DTO object. I

Speaker 4 42:00

you so,

Speaker 2 42:06

like so if I don't want to use like a GTO object, I would say,

Unknown Speaker 42:12

especially for a JSON based application,

Speaker 2 42:16

I would say like JSON. I

Speaker 2 42:26

think maybe out, maybe like, you can use go by like a key value, which is like a map, right? Or maybe, like, maybe, is there any

Speaker 3 42:39

way without using that key value,

Speaker 2 42:43

maybe, like, some kind of, like projection, like, if

Unknown Speaker 42:46

you want, okay,

Speaker 3 42:50

okay, yes, fine. So, so you so may one, right? I mean, do you know about this, dependencies and vulnerabilities, right? So how I mean, if we have, if we get an application right, where they where we find, figure out there is any vulnerability, how will you basically solve it right? What is your there is like a prefix, predefined way of approaching it and solving it right. So do you normally solve those, resolve those vulnerabilities on the dependencies.

Speaker 2 43:21

I would say like vulnerabilities can be like, they can be based on like dependency, and like it may be based on like, like implementation of like, a certain codes like that you have done, right? So since, like, since you're talking about like the dependent,

Speaker 3 43:37

I mean, sorry to cut you, right? I mean, talking mainly from a Spring Boot based application, right? So, okay, how will you solve vulnerabilities for a Spring Boot based application?

Speaker 2 43:48

Okay, for a Spring Boot based application, I would say maybe, in order to solve that, you can have like, go into like, a Maven repository and like, also, like a Maven like repository, just like a central repository that gives you like information about, like your dependencies and like their versions. So like, if you have like, let's say like, have like a data vulnerability for like a version, right? And then it is like, so

Speaker 3 44:19

if you have to do this without doing all these things, right? How will you do that? Sorry, if you, if you don't want to basically, end up getting into all those right, figuring out, what are the dependencies, what is the Maven repository, right? So, how will you do that? Spring Boot has come up with the latest approach, right? The way of resolving, uh,

Speaker 3 44:50

yes, earlier, we used to do all this kind of analysis, right? What are the transitive dependences and everything and everything right now, Spring Boot has basically made our life very, very simple. Do you know that? How do you do that? I mean,

Speaker 2 45:10

I mean, I'm still in my experiences. I'm using like, same dependency trees, commands and like, like, I try to, like, come up with like version that is fixed.

Speaker 3 45:21

Okay, that's fine, yeah, no. I mean, see, I understand that you can go into dependencies and right, so, and then you figure out what are the latest, right? So that's fine, right? But now the Spring Boot basically gives us a dependency management where you define that in your palm, and it does right? So you upgrade the Spring Boot version and it does rest of the thing, right? So only thing that you need to worry about is using the latest Spring Boot version and rest it takes care of it. Okay, that's fine.

Unknown Speaker 45:52

So,

Unknown Speaker 45:54

right, okay,

Speaker 3 45:56

so going back to the basic core Java, right? So given I have a list right, or an array right, or two dimensional array make. I'll make it more complex. I have a two dimensional array right, and I want to convert that two dimensional array into Maps. Okay, a map, okay, right? Where, where. Basically key is. The key is row right, right, row one, row two, and the value is basically the list in the first row, right. Okay, do you have, like, way. I mean, do you want to give a crack? How will you convert that using Java, two dimensional area, okay, so, and the map, resultant map is like, it's like, the row one is the key and the value is list of all values in row one.

Unknown Speaker 47:00

Okay, let me think.

Speaker 2 47:12

So so what I think maybe you would like, try to, like, convert that like you would like, like, convert that into like array into like, like a stream. And

Unknown Speaker 47:27

can you convert two dimensional array to stream? Do

Unknown Speaker 47:37

you think you can convert that I

Speaker 2 47:45

I know you can like, do, like, like a like, a nested list, but I'm not like 100% sure that

Speaker 3 47:57

can do. But can you convert a single dimension array into stream is that, can you stream over a single dimension? Yeah, that's fine. So, so the question is, now that you can convert, so how will you do this two dimensional array? Okay, I'm just trying to test your stream APIs. Okay, that's all. So its answer is, within all the stream APIs do I

Speaker 2 48:20

think, I think you can, so maybe, like, basically, just like, convert that, like, like that, that particular array into like, like a stream using RAS stream, like, it should, like, still give, give me, like a stream, like streams or an array, so, like, it should be possible. Okay,

Unknown Speaker 48:54

that's fine. Yeah, very close, actually,

Unknown Speaker 48:58

right, right, okay,

Speaker 3 49:01

yeah, treasure, so, the exception handling, right?

Unknown Speaker 49:06

So,

Speaker 3 49:09

how will you So, basically, we have a REST API. How will you basically implement exception handling for a REST API, and how is it implemented in your application that is fine too, even if it is

Speaker 2 49:23

okay. So exception, I would say you basically like a basic to have, like a basic exception handling. You can use like, like a like, try, catch block, right? So basically where you write like a code that, like that may throw an exception within like a try block. And then you would like, like, catch the exception. And then you define like, what you need to do in like a case of certain exceptions. And you would also want to, like, throw.

Speaker 3 49:54

So where we do write the right so you explain me three layers, right, controller, level, service level, right repository like, Where will you write that?

Speaker 2 50:11

Like, if you're talking like, primarily about, like, Spring Boot based application, I would say like, you would want to, like, incorporate like, some kind of like exception handling on like both controller and service link. But I think Spring Boot provides like a new layer called like a controller advice. I think it's like a like a global like exception handler, right which, which can help you to like, handle like each kind of exceptions and return like responses. So

Speaker 3 50:42

if there is an application wherein you have multiple APIs hosted, right, how will this work?

Unknown Speaker 50:49

Application with multiple API

Unknown Speaker 50:56

hosted, so they all will go in the same controller advice, correct?

Unknown Speaker 50:59

I think, yeah, you'll still use like,

Speaker 3 51:01

so how will you differentiate which API is, what three, I mean, which exception is raised. I mean which API has raised exception. I mean at the end, you have to segregate, right? So I

Speaker 2 51:13

think like you have, you want to have, like, like a dedicated controller, like, like a controller, like, advice for like, a particular exception and like, you need to define like, your base pack for like, like a class, so you can, like, define like a like, a particular controller class. So

Unknown Speaker 51:31

you will create an advice for a controller. Is that what you saying

Unknown Speaker 51:36

advice for a controller? Yes.

Speaker 3 51:38

I mean for the API. I mean that controller basically having one API is yes,

Unknown Speaker 51:48

okay, right?

Speaker 3 51:52

And, and, so then, then, if that advice is there, do you still need to have try cash block or No? I

Speaker 2 52:03

I mean, it's, I would say it's like a good practice, but if you're like, trying to do, but

Unknown Speaker 52:08

if you're catching everything right, what is the use of advice there?

Speaker 2 52:15

I would say no, because you would have like, or maybe

Speaker 3 52:20

I'll ask, what will you if you have those try catch block, what will you write in the advice actually,

Speaker 2 52:33

like so like for advice, I would say, if you have like, like, a dedicated try cash block That gives like, a proper exception and like responses back in like advice, you can do like, like, more like generalized exception, right? Like, some kind of like, even like, like a runtime exception, which is like should not be doing. You basically create like, like that resources and like, not in like, an exception and so on, so you can, like, handle that within, like, a controller, advice, like,

Unknown Speaker 53:07

like, like, a check, exceptions. Okay,

Speaker 1 53:12

sorry, sorry to interrupt. I have to go. Please continue. No, that's

Unknown Speaker 53:16

fine, yeah. Nice

Speaker 1 53:18

talking, yeah, okay, it's nice talking to you. Shailesh, yes, another five minutes,

Unknown Speaker 53:24

right? Thank you.

Unknown Speaker 53:26

Thank you very much. Thank you so

Speaker 3 53:28

on that Mockito, right? JUnit thing, right? So the same example that we took, correct? So we are writing now JUnit, right? So, so how, which layer we will you pack, actually?

Speaker 3 53:52

So you're writing a JUnit, right? And for a REST API, do you really write REST API. Is there any other way of testing other than geonet, that you know,

Unknown Speaker 54:07

any other way?

Speaker 2 54:10

I mean, I would say, like, GM is necessary, like, regardless of like testing strategies, right,

Speaker 3 54:15

right? So, I mean, now that you have say, maybe I'll ask you question different way, right? So we have this controller, we have service layer, and then we have repository, obviously, repositories, we use JPA, right? So there's not much code in there, so we, let's talk about right? So you will write how many junior right? One unit for controller, or one unit for and one unit for service fair. Or you will write like one unit for everything.

Unknown Speaker 54:48

How is written in your application? At the moment? Is it

Speaker 2 54:50

like each each class, each class has like a, like a correspondence, like one test class, like so for like, Okay, if I have like five controllers, I would have like five test class, and then, like, if I have like five services, like each like five services will have a unit test. So

Unknown Speaker 55:09

a concept name is function functional testing.

Speaker 2 55:13

Functional Testing. Yeah, it's functional testing, isn't it? Like, basically, like, like testing, like the functionality of an application to basically intro,

Speaker 3 55:24

have you have, do you have, like, any experience writing functional test cases,

Unknown Speaker 55:32

test testing, you know how it is done?

Unknown Speaker 55:34

I think, yeah. I think I have written,

Unknown Speaker 55:39

yeah, so, yeah,

Speaker 3 55:41

how do you mean? What is the tools and technologies that you use to write function on this?

Speaker 2 55:47

I remember I have written like a like integration test piece, like, for like testing using like Spring Boot test,

Unknown Speaker 55:54

or Spring Boot test, okay, so

Unknown Speaker 56:01

for that, like, you basically, like, annotate your class

Speaker 2 56:05

with, like, a spring with chess, and then you can do, like, like, it's like a web like NDC,

Speaker 3 56:12

or, but, but were you connecting to The live database? Or, how did you suppose, right? Yeah, API, right? So, so with the Spring Boot test, you're actually bringing up the instance Right of Spring Boot application, right? So then what upload the interface, right? Was it, if it is connected to a database, if it is making any external REST API call, right? How do you, how was it? Were they mocked, or were they used live? As

Speaker 2 56:48

I would say, basically we had, like, a temporary database, like, in, like, in the memory, like,

Unknown Speaker 56:56

no which database is that?

Speaker 2 56:58

Like, h2, DB like that was basically we had that

Speaker 3 57:03

so in memory database, right? Do you so poop or right? So how do you then initialize that in memory database, it needs to be initialized, right? Yeah, how the initialization is done for that database? Database

Unknown Speaker 57:18

initialization

Speaker 2 57:22

out like, all, like the all the database in, like, like a config, like, they're like, configuration based, right? So, once we have that, I would say, like, like the scope of that, like, define on like dependencies, like, you can define like test, like, their scope, like, whenever

Speaker 3 57:42

I'm not talking about that actually, I'm talking about the database, has to have certain set of tables right, on which you will right. So how will, how is that done, actually? So when, when this instance comes up, right? It needs to have all those tables ready, right? Or maybe triggers, whatever that is ready, right? So how will you basically do that, that initialization? So,

Speaker 2 58:04

like, we have, like, so during that we have like a base class where, like, we extend, like, all of these, like integration tests, like, during this, like, like a startup phase. So like the base class, we run, like SQL scripts. It's just like, like a simple SQL schema file, like it basically contains like certain tables, like some like test data and like we add like those data if we have like certain, like different, like use cases that we need to, like test as well. Okay,

Speaker 3 58:37

hey, grant you here for the second interview, right? Yeah, okay, okay, I didn't get a chance grant to ask give a chance to Shailesh to ask any questions. So is that something you can cover during your interview? Okay, thanks. Okay, yeah, I think I'm done, right? So probably grant will take it from you, right? And, yeah, thank you. Okay, I'll pop up. Thanks.

Unknown Speaker 59:01

Thanks. Thanks.

Unknown Speaker 59:05

How you doing?

Unknown Speaker 59:07

How are you I'm good,

Unknown Speaker 59:08

good.

Speaker 5 59:12

So can we start, I guess, with your Can you tell me a bit about your project? When you work at your right now you're working in.

Shailesh Paudel- JPMC- SRV- Final-Offered

Speaker 1 00:00

All right, sorry about that, Matt. I don't know what really happened there. Okay, so let me just open up your resume. And let me just open up. Just give me one second, because I'm a bit unprepared at this point. Then let me open up. So you will be hearing a lot of typing, firstly, because I have a Clicky Keyboard, and secondly, because we have to take down quite extensive notes. So I prefer to just write everything down and try and keep myself as quiet as possible. But if it's too annoying, just let me know. Okay, all right, so you said, if it's okay, if we go to like, are you in the East Coast time. Okay, so if it goes to like 145 it's fine, right? Shouldn't just sit here. Yeah, all right, cool. So did. What I'll do is we might have given you an overview of what the project is and what we are looking to hire and stuff. So I'll sort of just reiterate it a little bit. We can then go over some of your Java or whatever, your resume, and go some of the Java questions and

Unknown Speaker 01:06

stuff and after that.

Speaker 1 01:10

So in this interview, we'll focus not on coding, but in Java on Spring Boot. Do you have you have AWS, right? So a little bit you have AWS? Yeah, so little bit on AWS, and then some behavioral questions, right? And towards the end, I'll try and leave some time for any questions that you might have for me, sounds good? Make sense? Yep, all right, cool. So I, my name is Amber. I'm the hiring manager for this position, so wing and I are part of, like the same team. This is essentially for cards organization, credit cards. And when I say cards basically under my sort of skip level, nd, the higher org, the product, if you will, is essentially whole of Chase credit cards, like the whole end to end lifecycle processing of Chase credit cards. What exactly does that mean? What we do is everything that is needed for for for like a chase card to work. So by that we mean when you log on and sign up to chase cards, or you sign up by talking to like a banker or something, it goes. It's our system which generates the terms and services. It goes, pulls your credit info, figures on the risk profile, assigns the APR and all that stuff like, to which bucket you fall under, saves that information. Then it issues you the card where you actually get that in your mail. When you swipe it, it goes to the authorization, which is also done by sort of like by us, to when you log on, to see your to see a credit card statement, to the emails that get generated, to the rewards, the benefits, the points, the whole when you pay, make the payment reducing, like The whole thing end to end, is what under my skip MD is, is responsible for, essentially hold of cuts. Okay? So that's to give you a sense of the product and the area that we are working with, right? So that's on the credit card side.

Unknown Speaker 03:15

We

Speaker 1 03:17

mostly, when I say mostly, around 80 to 90% of the current credit card functionality is written on mainframe, okay, but we've been tasked with, we, like the larger team, is on modernization, effectively, like deprecate mainframe. Now, when we say deprecate mainframe, it's much, it's, it's a very loose term, right? We what we are talking about is, what we're talking about is like the roadmap is like a 1020, year roadmap, right? Because mainframe is just too huge. So what we are going to do is we're going to take chunky functionality at a time out of mainframe. We write it in like Java and AWS and microservice, and then hopefully just to shut down that functionality, and then just route all the traffic to the new to the new system. The first such functionality that my team, the one that we are building up over here is going to be working on, is, is on the credit card issue, like the physical card, right? So if you think about what comes in the on in the in the envelope, it has to have your name, your address, it has to have, like, a letter. It has to just figure out what is the text for that letter. There's, like, a bunch of bar codes and things that you can scan. You have your physical, literal card on the card, what's the image that goes, What's the art, whether it is horizontal or vertical, what goes on the magnetic set, your name? Where does the name go? Because can go in like, multiple places. Or the chip that you have, it's actually you have to generate bite by bite. It's not even like, this is the thing that goes, you have to generate, literally on like byte one. You have to put this like bite by byte, which is also encrypted. So there's, like, a lot of things currently, the B frame system, the one that does it, the output of that is like some 30 page document or something that it sends to another system which does a little printing of it. So we kind of have to do a replacement for that. Not only there's a pure replacement, we kind of want to make it that. We want to make it like it like future proof, right? So it's not like just take it and rewrite everything, line by line to Java. We kind of want to, like modernize it to say that, okay, how do we plug and play some other stuff in future such that some requirements will come by then? How do we sort of fit that in the larger context? That

Speaker 2 05:38

makes sense. Yeah. That makes sense, yeah, so

Speaker 1 05:41

that essentially everything with respect to the project, I'll leave, like the last five or 10 minutes for you to should you have any questions? We can jump onto your this thing. But before I we start off right, there is one thing which I do want to double check. Do you require

Speaker 2 05:58

sponsorship? No, I'm a US citizen. Yeah, US

Speaker 1 06:03

citizen. Okay, fine, because there's a lot of complications which come up. So I basically asked everybody before I start, if you had just said yes, the interview would have been much, much shorter, but it's returned. So why don't you tell me a bit about one of your latest project you working in? Are you in Texas right now?

Speaker 2 06:20

Right now? No, I'm working remotely. You're

Speaker 1 06:25

working remote so, but the company is in Texas. Yeah, okay, so tell me a bit about what you have done in usas and Antonia as a full stack developer, and then we can sort of take it from there, like, tell me about, like, the project, something that you worked on, what was your role? And so on and so forth?

Speaker 2 06:41

Yeah, sure. So basically, at USAA, I was mainly responsible for like, like feature based, and it usually fits on like, a like feature of our like, like modernization of like, like, existing portal, so, which is, like, a very old, like Java based application, and we were like, just like modernizing it. And like, some of the piece like that I have modernized includes like, like maintenance API and also like an experience API as well. So like the maintenance API is basically like the workflow that takes care of like, like a changes to like a user profile, its preferences and like, like the like the experience APIs. It's like a basically, like a flow that grabs, like a user's like account entitlements to like display like a different widgets on like a front, like a front in Portal. So it's basically, it's written in like an angular so like, these are like the two of the most recent piece of work that I have been like, working on, like, involved in, and I've been involved in, like, also, like designing these services, like, like implementing them, and, like, also, I would say, like deploying them also, and all of these application like, they're basically returning, like, like, Spring Boot three, and, yeah, so That's That's has been my primary work, I would say,

Speaker 1 08:03

so, were you mostly sort of hands on developer, kind of the thing? Or,

Speaker 2 08:12

yeah, I would say I was, yeah, hands on developer at USAA. Also, I would say, like, I was involved in, like, like designing aspect as well, to basically create, like a design docs and like, like, like a confluence of spaces and get like, like an architectural or like review, and also like, then, like, go with like a development process as well. So, yeah, I would say I have like a hands on developer.

Speaker 1 08:41

Okay, so in terms of design, what have you, like, What was your role, what was the can you give me an example of, what was the design work

Speaker 2 08:55

that you did? So design, I would say, like, basically, I was involved in like, creating like design documents for like, like, both, like, like a mark microservices. So I would say, like a main aspect of design documents, like, it basically consists of, like, like a functional aspects, and then also like non functional aspect as well. So I would say, like the functional aspect is, like, pretty straightforward, like, like a what the app supposed to do, right? So like, for like maintenance API, we basically had, like, we listed like two primary endpoints which were, like, getting the profile, like the preference and also, like, like, basically updating like the profile preferences. So we listed like, like a structures of like a REST APIs, and like the payloads that it took, and like the response it would like give out, and I would say, like the possible responses code out from like, like a particular API, but also on like a, like a non functional sites. So we had to write about, like, a scalability on, like, how, like, a services would, like, scale on, like, AWS ECS, like, like, what kind of like security like we need to add as well, and also, like, I'll say, monitoring and, like logging, like aspect on it as well. So, like, yeah, these are, like, the different things that I had to, like, elaborate on like or like, get approvals from like architects.

Speaker 1 10:25

So can you dig into something that you've that you had done or specified with respect to scalability and monitoring?

Speaker 2 10:34

Yes. So talking about scalability, and I would say basically, like, for like, we start like, like with the monitoring. Basically, let me think I would say like we had to, like, had like an in new agent for like, New Relic like dashboard. And also I would say like, we installed like a New Relic agent onto our like containers, like, whenever we deploy our application, and also like so this New Relic gathers like information from like, basically running like application. So basically like, like provision to like, gather it and like, then based on like, like the gathered matrix, it creates like a dashboard. So yeah, some of, like, the dashboard items that we have to, like use was, like CP uses, like a memory consumption and like the latency, like services, or like, like a thread pool connection, yeah, I would say that. And like the also, like the number of requests, like responses, and like a response is code and like, so on, like, so that so we have in our like new early dashboard. So I would

Speaker 1 11:46

say that okay. And what about scalability?

Speaker 2 11:51

Um, so for scalability, we basically, like, like, we went behind on, like the auto scaling stuff. So, like, we had, like, a auto, like, dynamically scaled for like a services, and we also hit, like, like a like a certain, I would say, like a certain, like a threshold, like, so whenever I like, like a CPU percentage went over, like 45% we would like, scale up and like, I would say, like, similar, like configuration was there for like memory as well. And, you know, like ECS, like automatically launches also to like or like it terminates for like, a new task based on like, our like auto scaling policy of like, happy users,

Speaker 1 12:40

okay, all right. So why don't we jump into some Java questions? We probably will be jumping a bit back and forth with respect to Java, spring and stuff like that, just fin I hope that's okay. Yeah,

Speaker 2 12:54

right. So in Java,

Speaker 1 12:58

I'll start off with something very basic, right? What's How is hash code and equals be used? When you say, when you're using a collection, let's say like, like hash map,

Speaker 2 13:11

okay, so, so hash code and equals, so I would say like, basically like, in hash map, it's implementation of your like a map interface, where, like the keys are there based on like a hash code, right? And hash code is like, it's nothing, but like, just like an integer representation of like, it's like a key based on like,

like a function that you provide on like hash code methods. So, like, whenever you, like, insert, like a key value pair. Of course, in like a hash map, like hash map basically computes like, like an index of bucket that like the like the key points towards it. And it also basically like some kind of like algorithm as well. And like, if you have like a good hash map implementation, then it'll basically, it will like, prevent collision, right? And like the Eco. So the Eco plays like another role, like in like, finding whether, like two keys that we're using are, like, equal or not, right? So basically, I would say you may have like two keys with like, like a same Haskell, but it might be like two different keys. So like, if you have like, multiple keys that map to like, the same buckets, then I would say, like an equal method is used to check if, if, like, he already exists in the bucket or not. So like, that's kind of, that's like the kind of brief info of, like, how so

Speaker 1 14:45

do you know what is the difference between string builder and string buffer?

Speaker 2 14:51

Difference between string builders? Yeah, I would say basically, like, the difference between string builder and string buffer is that like string builder is, it's not like, it's not a thread safe, right? Because it's not like, synchronized and like whenever, whenever you are using against like, multiple threads, they will lead to like, like consistency issue. But I would say, like string buffer is like thread set safe and because, because it provides, like a synchronization so you have, like a multiple thread, like when you modify, like the object at a time.

Unknown Speaker 15:33

Okay, all right. So

Speaker 1 15:38

now, have you heard of functional interface?

Unknown Speaker 15:43

Functional interface? Yes.

Unknown Speaker 15:44

What are functional

Speaker 2 15:47

interfaces? So basically functional interfaces, where they were introduced in Java a right? I think, yeah. And they're like, like, a special kind of interfaces that use, like, abstract methods, right? So they're used for, like, you're like a basic, like, like a lambda expression, or, like, even, like a method references. So like, if you have, like a supplier, like a, like a, like a predicate, or like a consumer, right? So like, like, those are like a common functional interface that are basically provided in Java eight, I would say, okay,

Speaker 1 16:26

what are they? What's the most common use case for a functional interface? Most

Speaker 2 16:33

common use for functional interface, like I would say most common use for functional interfaces, I basically like to like, represent like a Lambda expressions. So like, because you can like, you can store those like lambda expression as like a like a lambda expression, I would say, like it just like it makes those like, working with lambda much easier, like, for like, passing around, like a parameter,

Speaker 1 17:09

okay, all right, cool. What are the different types of garbage collectors that you that you know? What is garbage

Speaker 2 17:20

collection? So garbage collection, I would say, so basically, like garbage collector. It just It helps you, like it processes like,

Unknown Speaker 17:34

with like JVM, or like, no,

Speaker 2 17:38

yeah, like JVM, it claims, like, it helps you, like claims memory, right? So, like, if you, if you have an object and it is no longer used within like application, then like JVM, like it automatically, like triggers, like a, like a process, to manage, like a memory. So, like, so one of, like, one of the like, garbage collection, I know it's like a mark and sweep phase where, like, JVM, basically marks and objects that are like, no longer like, used, and then, like, it's sweet in like a sweep phase, I would say it just, like, reclaims that memory used by, like, those method objects.

Speaker 1 18:19

Okay, do you know, have you heard of or know any other one?

Unknown Speaker 18:24

So any other one?

Unknown Speaker 18:33

Not right now, that's

Speaker 1 18:36

fine. That's fine. Usually, you kind of don't really do that. Can you invoke garbage collection in your code? Or there's no way to do that. So

Unknown Speaker 18:50

invoking garbage collection,

Speaker 1 18:53

yeah, like in my code, I say, You know what, invoke garbage collection right now. Can I glued up? I

Speaker 2 19:00

think so. Yeah, I think garbage collection is, like, is manual and like, like, there's like, a way to invoke garbage collection by I think using like, like a system class, or like, like a runtime class, it should like an option to, like, request it like a garbage collection. Okay,

Unknown Speaker 19:30

okay, you can use, it's called system.gc

Speaker 1 19:35

that's what you would use to do garbage collection. It's not guaranteed in the sense you can, it's more of a request, but JVM may do it at that point, or may just not ignore it, or may do it at a later point. But it's not guaranteed, but it's something that you can but it's generally considered bad practice if you're using if you're doing that, then it means you have, like, some other shitty problems that you are trying to, like, mask. So, yeah,

Speaker 2 19:59

right, because, like, JVM, it automatically collects, like, the garbage, right? Yes, it should, like automated, like, rather than manual, requests, correct, yep, yep.

Unknown Speaker 20:11

So

Speaker 1 20:14

moving to some, like, spring type questions, what? Tell me some of the annotations that you're familiar with and that you work with in spring.

Speaker 2 20:27

Yes, so spring annotations. It doesn't have to be like spring. I would say some of the annotation that I've worked with, like, like a, like a component level adaptations, like you have, like, you're like a root component annotations, and which is, like, very generic, like, like a spring bean, right? And then, like, you have, like, like, your dedicated, like, you're like a specialized component annotation, which is, like, it's like service annotation, which basically indicates that class is like a service layer. And we also have like a repository annotation, which basically represent, like a data access object correct. And you have like your controller, or like a Rest Controller annotation that represents a controller there, when you have like, like a dependency injection related annotation, like things like auto wire annotation which injects like dependencies. And I think there's also like a qualifier annotation which works with like auto wiring annotation, right?

Speaker 1 21:36

So let's take, you know, a few of these, right? So, you mentioned component, service, repository, controller, right? If you mark a class as component versus you mark as mark it as a service. Does spring do anything different, or does it treat them, or does it have any additional bells and whistles around it, or like, what's the difference between like the component and service.

Speaker 2 22:05

So like, the difference between component, I would say, like the main so, like, the main reason why we have like, like a specialized service, or like, like a repository annotation instead of like, like a component annotation, is that like, like, like, like. So basically, it makes you like, like a, I guess, like a spring spring, to like in like a class as, like a, like a proper service layer. So, so there's like, nothing wrong, just like, like a component annotation for your service there. But it just makes it harder to like, like, understand what the class is supposed to do, right? So it's like, all about like, like semantics. So you, let's you just, like, just looking at the annotation, like, says, like, like a service, I don't know, like that like this class will hold some kind of like business logic. And if it's like, it's like a component, they wouldn't know, like, if it had, like an entry point, or like, if it's like, if it's doing, like, some kind of data access, like operation, or like a business logic, right? So I would say, like spring, like, internally does like, it does not like, have any special operation for these adaptations? Okay,

Speaker 1 23:24

okay, so what is like a one word for one word answer for everything that you just mentioned? There's like a one word answer if you if you had said, will be like, Yeah, that makes sense.

Unknown Speaker 23:41

Would it be semantics?

Unknown Speaker 23:43

Readability is what I would go for.

Speaker 1 23:49

Is basically for readability. You mark it as component versus a repository. It doesn't do doesn't do shit. It's the same thing from other things. Just move for readability. You look at it and be like, Okay, this is this. And so on, you mentioned qualifier. So can you tell me what is qualifier? How exactly is that used in what use case? What's a good use case for a qualifier?

Speaker 2 24:11

Yes, so qualifier, I would say basically, it's like a qualifier annotation. It works with like, auto wire annotation, right? So whenever you have, like, a multiple beans or like the same type, right? You might have some kind of like issues, because like spring will not know like, which one to like inject, right? But if like, if you use like qualified annotation, it will like, tell your like spring what's like, specific beam you you want to inject, right? So, for example, like, if you have, like, like two classes, like implementing, like, like a same interface, like both, both are like spring bean. And then if you want to, like, inject that interface, then spring basically, it knows. It knows like, which one to use, right? So you would have to use, like, qualified annotation, right? So let's say, for example, I get it.

Speaker 1 25:11

I think you know it so integrated. So, so we have sprint boot, which has its own bunch of annotations and stuff, but we also have spring. So why do you need Spring Boot, and why can't we just use like plain spring? Why do we need to have all these other specialized annotations and stuff? Okay,

Unknown Speaker 25:36

I would say

Speaker 2 25:41

basically so Spring Boot just it just adds like additional functionalities and like auto, auto configuration, basically, I would say, makes like life of engineers much easier. And like in your like typical, like Spring application, you need to, like configure, like application context, like using XML or like Java based configuration. But in Spring Boot, it is just like a component scan of like, like a package that you have, and then it create those like beans available for you also. So I would say you don't have to worry about like, configuring those like Spring Boot also, because it supports like, like embedded servers as well. So, so basically, you don't have to externally, like, deploy your application to like a Tomcat, but you can't, like, just like leverage, like embedded, like Tomcat server and basically run your application there. And it also provides, like, lot of like starter dependencies, like like like a chair dependencies that you use to, like, like, monitor your application, like security, and it also provides, like spring starter data for like, repository layers and so on. So I would say this makes life much easier for developing your application.

Speaker 1 27:01

Okay, and obviously, if you're not, had your lunch, I my lunch. Just give me. Did you have

Speaker 2 27:05

your lunch? Not yet, not yet. Okay, then

Unknown Speaker 27:09

I'll also skip, I'm not like eat in front

Speaker 1 27:12

of you now. Man, no, that's just basically cruel. I came in late. I'm asking you to stay late, and I'm eating lunch while you understand. So we'll do the same thing to that.

Speaker 1 27:27

Let me see you also have a Spring Boot. Have you done anything on Angular?

Speaker 2 27:32

Angular? Yes, I have done Angular. Let me Yeah, primarily my front end experience was with Angular.

Speaker 1 27:45

When you create a component in Angular, what are all the basic things that come with it? So

Speaker 2 27:55

some of the basic things, let me think I would say, whenever we create like Angular component we have, we have to, like, run, like, like a generated component, like it comes with, like a, I would say, four basic files, which is like template files, like HTML, and it has yours, like TypeScript file as well, like, like your component file. You also have, like your test file, which is for like, like, a specific file, then you have, like, your like style sheets, which is CSS file, right?

Unknown Speaker 28:33

What are directives in Angular?

Unknown Speaker 28:37

Angular

Speaker 2 28:39

directives, I would say they're like, so they basically serve. They serve as just like a class that allows you to, like, add some behavior, right? You can also have, like, your like, a command directives that it will basically like, like check. You'll basically check for like certain things if they're like true, and then they'll also like, display like, if it's like, if it's true, and then have like, like a loop to any arrays, right? Or you may have like, maybe, like, any kind of like attribute directive, right? You can like, bind your elements to and like CSS class based on, like, dynamically, like, like, certain values that is used in your class. And then you may, you may also have, like, like a component directives, basically, like, specializing, like, like a version of directives that is a it's associated with, like, like a viewer template. All right, okay, so, yeah, basically, I would say just like two min plate with like dome and stuff like that. Okay?

Speaker 1 29:50

And what is a component? You mentioned component. So, what is a what

Speaker 2 29:58

is a component? So, component, in Angular component is like, like a building block of like your Angular application, so, so it just helps you with like, it's like combination of like views, which with like UI portions, and it also like behaves as like, like a part of your like interface, right? So, like your Angular component, like I mentioned earlier, it also contains like, three primary files, which, which are like, your templates, your style sheet, and your like logic file, like it just tells you, like, like, how the UI should it should behave, or how it should like look. Okay.

Unknown Speaker 30:41

I Okay, um,

Speaker 1 30:45

okay, enough of Angular. That's as much. So I've done Angular, like three, four

Unknown Speaker 30:52

years ago. Oh, okay.

Speaker 1 30:55

I used to work on in Google. I used to work on ways and the ways UI was written in Angular. I was a TR for that, but I forgot everything. Like, I just Googled some interview questions while, like, completely forgot everything. But Okay,

Unknown Speaker 31:10

anyway, so

Unknown Speaker 31:15

on AWS, right?

Unknown Speaker 31:18

What's the difference between EC two and easiest.

Speaker 2 31:24

So main difference between easy to and, I would say, basically, so it's like, like a virtual machine, right? It just, it's like an instance which is running, and it gives you, like, you're like a full control of like the server. So which one I would say easy to that's easy to basically, you can pick like your operating system, and you can configure like, like all of your like, like a management features that you want to and I would say also, and then, like ECS is like a cloud container service, basically it helps you with, like comparing something like Kubernetes, which just like an orchestrator service. And also, I would say that it like helps you with running Docker container on cluster to basically help you with your instances. And it may also run like something, like special, like a specialized call, like, like instances and so ECS, it doesn't like help you with,

Unknown Speaker 32:36

let me think

Speaker 2 32:38

like it doesn't help you, like, like, it doesn't give you, like, a complete control over everything, so you would basically have to run like against, against, like a Docker and then define like task and services.

Speaker 1 32:53

Okay, all right, what's AWS lambda?

Speaker 2 32:59

Yes. So AWS lambda, basically, it's just, it's like a it's like a function. It basically helps you with a, like a piece of code that you run in, like a server, so which basically reacts to, like a certain kind of event, triggers, all right, like this trigger maybe, like a call from, like, API getaway, like, maybe, like a message on, like, your like Sq, or it might be like, drop on, like, like s3 buckets. So it's just like a, I would say it's like a piece of cord, like, without any probes, like a I got managing your external server, and it helps you with, like, like an event driven, like, architecture,

Speaker 1 33:50

okay, so what's do, You know? What is? EBS?

Speaker 2 33:56

EBS, yeah. EBS, I think I haven't heard of that, but, like, I've never really used it.

Unknown Speaker 34:08

Okay, I normally work with

Speaker 2 34:09

like s3 buckets, like basically storing data.

Speaker 1 34:14

Okay, what have you used? So I see that in one point you have mentioned that using AWS functions to build serverless application integrating s3

Unknown Speaker 34:24

or something.

Speaker 1 34:27

So can you talk a little bit more about that? What exactly did you do in that?

Speaker 2 34:33

Yes, sure. So, yeah, I would say, so So we basically, we would get, like, like an s3 bucket file, and like we, we dropped that, like once a week, and like this bucket will be like a, like a have, like a certain like entitlement for like, certain like user groups. And then like it then, like, with these entitlements. So the s3 drop was basically because it was from like, a legacy system that it would get, like, transported over for, like, for some kind of like File Transfer Protocol set up by our like external team for our three buckets. So, like, once our like, like, like, around, like, five drops into like the bucket, our lambda, like gets triggered, and then like the lambda converts that file into like, like a JSON structure that basically matches those like entitlements with like, just like a key value lookup. And then there is like a match. We update our like a dynamic and then, so what we do, that's what we do with like lambda functions. Okay,

Speaker 1 35:45

could you talk a bit about the CloudWatch stuff that you have done?

Unknown Speaker 35:50

Um, yes, so Cloud watch stuff,

Speaker 1 35:54

did you do it? Or was it like somebody did and you were, like, nearby?

Speaker 2 36:00

No, actually I was, I was one of the like, so one of the we had to do it in like front end application. So I had to, okay, yeah,

Unknown Speaker 36:09

okay. So what exactly did you do? What in cloud watch,

Speaker 2 36:14

yeah, so, so one of the use cases we had to use, like, for our front end application. So we use, like a, like a cloud watch log for like a specific purposes, so which was to like stream logs, right? To like Splunk so like our, like a front end application, we use like Splunk API to have like us, like a Splunk token to make like a log stream. And we had like a Splunk indexes, right? But it was basically for like, a security concern, but we like, because we were, like, exposing like that token into our code base. So, like, the idea was basically to like, write like a helper function that would work as, like an API for like a front end to call like the service, and then this the service would then be within like a logs to, like a cloud watch. And I would say, we basically created like a log group. Then like a log group was they were able to have like a minutes in, like a new log stream. So like a front, like a fun front, front, front end application whenever, like, it uses, like a console log or like, like a login services, it would call these functions. So these function we're basically using, like a log data to, like, like a cloud also, and I would say there's like an external, like a team integrated into like, like a streaming services to basically stream, like those CloudWatch logs to like a Splunk. So, yeah, that is very like a specific to, like a CloudWatch implementation that I did. I would say,

Speaker 1 37:59

Okay, so let's say you have your like us were Spring Boot microservice. You have all your cloud watch and this everything is deployed, and the user is reporting an error, the user is getting like a 404 or we're not even 404 let's say five or two or like, no response, right? How would you go about debugging this issue?

Unknown Speaker 38:33

I would say,

Speaker 2 38:37

for like debugging, it usually like takes for, like an initial logging platform, right? If it's like, I would say, if it's like a lambda or something, I would go into like Cloud watch log to see, like, a report of like, any kind of issues. If not, then I would just go on like a Splunk, which is basically our, like main logging platform. So I would basically have like, a like, bunch of like, a Splunk queries, like, ready to run, and then like, see if I can find like, like, any type of response codes that matches like what the user is reporting. Once I like, once I find that, then I'll be able to see if there's like, any specific case or like, there's any like exceptions being thrown that like did not and like that was not handled like, properly. And then, like, once I'm like, a little bit more like, informed about like that case, then I would go to like, like, recreating like that issue on like, a lower environment, maybe like QA, or like, even, even, like a like, a local then, like, try to like, like resolve it. And then that's how I'll do it. Okay,

Unknown Speaker 39:43

all right.

Speaker 1 39:46

So now let's say you learn from this, and then you're like, oh, there was a bunch of issues. We need to create, like, a brand new service, right? Your requirement is that the service has to be 99.99 like four nines, five nines, percent up, and is going to have like, maybe, like, 500 QPS, right? So relatively high volume QPS that it has to process how, what would, how would you? And let's say it has to query some information from the database and return the info. And the requests that are coming in are, some are like, UI based requests. And some are some like, batch based process. Okay, how would you do this service? Like, you can start Greenfield, right? You come into our team and you're like, there's no shit. Let me tell you how to build a service. This is how you build a service. So what's your best case service that you are looking to build,

Speaker 2 40:46

let's see. Think I would say, so first we need to like, be considered. We should consider about, like, availability and like, like fall turn, since we're talking about like, like five or nine. So we would like first, we would like, think, like, without talking about, like application, like, we need to deploy against, like, multiple availability zone, like, against, like, AWS, like reasons, or, like, behind, like a load balancer, right? So I would say we would have like that initial, like, like, taken care of first, and then like, because we need to have a high availability. So then we would also need, like, like, some kind of, like, good choice of database, right? If you, if you have, like, a data structure, or you would go to, like, a relational database, like, like, a post grace, but then, like, if it's not structured, you would be used something like DynamoDB on like AWS with high consistency turnout, right? So like, I would say, like with, eventually we need like consistency throughout. So also, though, I would say, like, those two would be like primary things that comes to a picture, but like to handle, but to handle like high like transactions, I would say these database firstly, need to be like horizontally scaled, which basically does not like frequently update, or you would want to add like caching layers as well. Also like before database. So, so I would say, instead of hitting like a database, you hit like hash to make it like faster for like a UI, also similar. So like the UI needs to be cached through like a like a content delivery network, right? And it can also basically return like those data more like quickly, rather than hitting like our like s3 buckets, like whenever we deploy it. And for like a batch processing, I would say you would you basically want to create like, like, a bad job to like process through like, like queuing system. So maybe you want to, like, like rack, some kind of like SQS and like, like lambdas to basically have like, a process, like, asynchronously and like for your primary service, and you would need to like, properly, like, monitor that through a relic, or like, even, like, like Amazon, like Cloud watch. So I would say those are some of the design consideration that I would do for like, different layers that you mentioned for, like high availability and like fault tolerance. Okay,

Speaker 1 43:39

so thing we can say we're done with, like, the technical interview, just a few more, and then we can stop. Why are you looking to change?

Speaker 2 43:49

So this is a contract job, and it'll be done, like, by end of this month. So I was just looking for opportunities,

Speaker 1 43:57

okay? And that's, that's good enough. So I Are you looking for something short term or something long term?

Speaker 2 44:09

No, I want something long term because, like, because I want to grow like my career, and I want to gain like experience as well as much as possible, then like also provide like a I want to have like, pretty good contribution to like team, and I want to, like work my way up to some kind of, like leadership position maybe later in the future. So, yeah, I'm looking for like something long term. I would say,

Speaker 1 44:30

Okay, this one would I have to double check? So don't put me on this one. I believe it might be for like a right to hire. So I mean, assuming this all will start to speak with the other interviewer as well, and see if it does, then it'll be like a three to four month I don't know exactly how much three to four month contract, at which point I think we can decide whether to so. Is that something that you are looking for?

Unknown Speaker 44:53

Yes, okay,

Speaker 1 44:56

okay. When did you graduate? Your resume says 2024 May, but you've been working since, since 2020

Speaker 2 45:10

Yeah, basically. So I had an associate degree with Wake Tech Community College. It's North Carolina, so Okay, yeah, when I when I got my Associates, like, through, like, my cousin, my cousin actually had, like a staffing company. I was able to land, like, a contract job at key banks, so, and then after that, so it was like a COVID time. So I just decided to do my Bachelor's, like, study and bachelor together, so, and that's how I was able to complete my Bachelor's. And then, from my contract, got over with key banks, and then I joined USA. And then, yeah, that's how I did it.

Speaker 1 45:50

So you, so you were going to college while you were working.

Speaker 2 45:56

Yes, it was like a remote so it was also like a COVID time also. So,

Unknown Speaker 46:03

so how many years of experience do you have in

Speaker 2 46:06

total? Total in this December, you'll be five.

Unknown Speaker 46:15

Okay,

Unknown Speaker 46:17

okay, um,

Unknown Speaker 46:20

so you mentioned what you're looking for.

Unknown Speaker 46:25

Can you describe,

Speaker 1 46:29

let's say, why we started with licking nice you know. Can you describe a situation where you introduced something to the team, either you introduced something new, or you went above and beyond. And we were like, Hey, let's, let's sort of do some do it in like this other way, because it will help us save by the money or time or make our lives easier.

Unknown Speaker 46:55

Yes,

Speaker 2 47:00

let me think so, I would say so we normally so at USAA, we get like, something like tech enterprise like backlogs where, like, like, all the teams within the company have to, like, implement some kind of like correction, or like, like solutions to like, like, certain existing problems, right? So our requirements was to basically incorporate like SVK, to basically like a stream, like any like data logs, like that change or like basically to like interacting with like private information of like clients, right? So we're like this requirements, and then we we basically didn't know, like, what we're supposed to do. So like, we were in like a, like a proof of concept phase. So like, we basically wanted to have, like, an easy solution. So we wanted to, like, implement across multiple services. So like, basically to, like to easy up, like, like easy to pick up. So I was looking at like these requirements, and then I figured that it would be like beneficial to create, like an external microservice that would basically take in those like requests and then like, like aggregate, like it all together to like a stream, to like, like particular, like, like a snowflake table. So this was my like proposal, right? So to create, like a micro services that would basically, like on board, like a different data sets as a like a customer, so like at particular like services, and then like, like, send those data to like a snowflake table as required. And then this actually was really, like a big proposal, because, like, we were able to just, like leverage, like, like a restful calls, rather than just changing, like, how the application work by incorporating, like, a new SDK. So I think this was one of, like, one of the products moment, and it was really good because we were able to like offer it to like other teams as well, to use this service. So, yeah, this was out. This was the time where, okay, something. How

Speaker 1 49:08

often do you guys push code to production? Right now?

Speaker 2 49:14

Um, right now we we have, like, I would say, two different kind of releases. One is like data fix that we do on like a, like a regular basis. We have at least, like one or two data fix that goes out like every week. But like our regular application releases is like a one sprint, which is just two weeks, all right,

Speaker 1 49:37

okay, I think we can, we can stop at this point.

Unknown Speaker 49:43

Do you have any questions for me?

Speaker 1 49:46

Questions for you? Relax. Now it's done. I'm not going to ask any more questions.

Speaker 2 49:50

Okay, first, I just want to say thank you for your time today. And it was really nice talking with you. And firstly, I would like to ask like, about, like, designing of like, modern nation is so like, basically, is still creating, right, like, what services needs to be, like, migrated or, like, still like, like under discussion.

Speaker 1 50:15

Um, so just give me one second.

Unknown Speaker 50:21

All right. So,

Speaker 1 50:24

right now, the first thing, as I mentioned, right? It's the the issuing of the credit card, like the physical card. So for the last few months, some of the team is I joined this base in like October, right? So I've been here only for like, couple of months as well. The team that I'm hiring is, like, we're hiring a new team from scratch. I say from scratch is like all new people. So it's going to be around like, 15 people, including myself, who will be working on this project right right now it's more of analysis and design phase. So at a sort of, like very high level you kind of like, get inputs, you do some processing, you have the output. So we are going to focus on is more towards, sort of like the tail end. We want to try to write some reconciliation processes, or the overall thing. So some basic implementation will start somewhere in January of next year. Right now, it's mostly like design. So now this whole thing is probably going to be split into, I don't know, one or many micro services. I don't we don't know the details of that. It might be some like lambda functions or like some step functions that we might be using. So it's all that is what we are hoping to nail down, hopefully by before New Year's way, if not then the beginning of next year. Oh,

Speaker 2 51:46

wow. Okay, gotcha, that sounds good. What else? What else I would say? So another question I would like to ask, what defines like success for like, somebody was being, like, hired for this particular

Unknown Speaker 52:06

position. Well,

Speaker 1 52:10

when you say success, do you mean for the interview or for once you have joined, the two completely different things. Which one are you

Speaker 2 52:16

asking? Like the position, once

Speaker 1 52:20

I once you join, yeah. So for, it depends on the so I'll give you the short answer in the well, I'll give you the non answer. There's no short answer for this one. So it kind of depends on the position, right? So for, for, for junior positions. It's going to be like fresh out of college. Type of folks. Is going to be more of are you able to, you know, we'll assign you some work. Do the court make sure things are getting done on time. For somebody like you, you're going to be not like fresh out of college. So you're obviously going to not be like, just the very bare minimum thing. So you're going to be a little bit more. So when you start on, is going to be more of just make sure all the tasks are getting done on time. You're able to finish all these things. But as in, as things grow, and as the team gets bigger, and as even you get more comfortable, it'll be more of okay. The more you are able to take on the more I'll throw at you as much as possible. So for leadership position, well, let's not get into that here. So So for something, for something like you, it's going to be more of starting off with making sure that your work is getting done on time, and maybe at some point, taking like a smaller component within the larger thing. And you know, you in conjunction with somebody else. Let's say, like a junior person, the two of you, or maybe two, three of you sort of work together and be like, Okay, you have like, a one month sort of goal. Let's say, go ahead, go ahead with this one, and then you sort of oversee the work of, like, the couple of folks. So it's kind of like the leadership thing that you were also talking about in terms of, like, long term if you want, like, more, bigger leadership things. So firstly, you need to be at like, a certain level before you can have a, like, a team of your own and stuff like that. That's probably gonna take time that's probably like a few years, and few in the sense only, like two to three years. And again, this is finger in the air ballpark. The thing that I'm saying, right? If you're like a superstar, like a rock star, it might be sooner as well. So it's not like a hard and fast rule, but generally speaking, that's kind of what you can expect. And again, two, three, also, it completely depends. Some people like it. Some people might do it. Some people be like, No, I don't want to do it at all. And some people, you might try and realize they're not good at it. So there's multiple obviously, there's no point talking in details about those things. So that one aside. But other than that, in terms of like, your day to day, it's going to be more of making sure that, first and foremost, your stuff is getting done on time and getting finished. The larger thing is more of the team comes first, right? So whatever is required for the team or whatever is required for the project, we kind of have to do it exactly, so that comes first. After that is fine. Let's take everybody's wishes and thing into into account, right? So should this thing work hard? You all been a great question where you can also do some UI work. Since you've done Angular, we don't do Angular here. Whatever we have done, it's all react. But I don't think it really makes that much of a difference. You can do one, you can do the other. You can pick it up so you can they you can sort of help out. I generally am very gung ho on people who take more initiative. You know, you're assigned X, you do X, you know, you just sort of meet the requirement, so x plus the same, and that shows initiative. That shows the same, so more the better.

Speaker 2 55:50

Gotcha. Gotcha. Okay, so

Unknown Speaker 55:57

I think that's all the questions I

Speaker 1 56:00

haven't I know it's like post lunch time, but if you have, don't look at the time and stuff. If you all have anything, I can stay for another five minutes. If you have any questions, if you don't, that's also fine, but just wanted to throw that out there.

Speaker 2 56:11

No, that's the other questions I have. Okay. Thank you so much for your time

Unknown Speaker 56:18

today. Apologize for the confusion.

Speaker 2 56:22

What? What would be like? The next step after this,

Speaker 1 56:26

I'll speak with wing, who's the other person who interviewed you, okay?

Unknown Speaker 56:32

And then I will,

Speaker 1 56:35

and then we'll get back to you, okay, through the through the system or whatnot. And

Unknown Speaker 56:43

Yeah, sounds good.

Unknown Speaker 56:46

Thanks a lot. Thank you so much. Bye.

Shailesh- BNY- Srv-Final-2

Speaker 1 00:00

2023 okay, got it. So how do you talk a bit about that, like, what you've been doing there? Yeah,

Speaker 2 00:06

sure, at USA, basically, I'm, like, a software engineer. I work as, like, mainly back in, like, Java developer, like, primarily on, like, microservice development, and I've also been involved heavily in, like, microservice, like, like a modernization of like, monolithic applications, where, like, I've been designing them, like collecting requirements. I've also been like documenting them, like implementing like changes, and also like deploying them as well. So yeah, like, that's the like, like, little bit of info about,

Speaker 1 00:42

okay, so to just take one of them is this been one continuous project, or they've been a series of projects.

Speaker 2 00:51

Um, so sorry. So at USA, we do like, like feature work. So okay, I think, like, every six months we bring like new features, and then we work on that. Okay,

Speaker 1 01:05

so you mentioned the so you're, is it organized all around, like an agile method with you have a team that you work with, or how does it How does it work?

Speaker 2 01:16

Yeah, so, like, we basically do, like a scrum team. We have like five developers, like a product runners and like Scrum Masters. So we follow like two week sprints, and then we do all scrum like ceremonies, like planning, grooming, like retrospective and like so on.

Speaker 1 01:35

Okay, and how, actually in this team, what?

Unknown Speaker 01:39

What role do you play these teams today?

Unknown Speaker 01:41

This team today? Yeah,

Speaker 1 01:43

it's you've been working with the same team, yeah, for how long now?

Unknown Speaker 01:49

About like, two years,

Unknown Speaker 01:50

two years.

Unknown Speaker 01:52

How many so what's your role in the current team?

Unknown Speaker 01:54

I'm like, a senior, sophomore, okay, how

Speaker 1 01:57

many five, five developers, is that right? Yes. So

Unknown Speaker 02:03

what's the, what's

Unknown Speaker 02:04

the composition? Like, as far as junior,

Unknown Speaker 02:06

senior,

Unknown Speaker 02:09

how many juniors you have? So

Speaker 2 02:11

like, I would say we have like one TDP, which is like a fresh college grad, and like one Junior. And I think we all, we also have like, like, two, two seniors, okay, okay,

Unknown Speaker 02:26

gotcha. And the,

Speaker 1 02:29

okay, and who runs the scrum? Just out of curiosity,

Unknown Speaker 02:32

is it one of you guys, or is there a dedicated

Unknown Speaker 02:35

scrum master? Like a scrum master,

Unknown Speaker 02:37

Okay, gotcha. So

Unknown Speaker 02:39

in the,

Speaker 1 02:43

so, when you're doing the, I was trying to paint a picture of what the environment's like. So you have, you have a CICD pipeline that you guys work with, yeah, what? What

Unknown Speaker 02:55

is it? What's the tech stack for

Speaker 2 02:58

that? Yeah, we, we actually, we work with, like Jenkins as CICD, Okay,

Unknown Speaker 03:02

gotcha, okay. And

Speaker 1 03:04

the and the deployment is to what to is it to containers, or is it to machines or VMs?

Unknown Speaker 03:14

It's like AWS, okay,

Unknown Speaker 03:16

all right, gotcha. All right.

Unknown Speaker 03:19

I in what to AWS to,

Speaker 2 03:22

we basically deploy like multiple services on AWS. Like we do like EC two for our legacy application and like js for like a modern application. And along with that, we like lambda lambdas as well. So like, like provision, SQS and SNS as well. Okay,

Speaker 1 03:40

gotcha. All right, cool. All right. Um, so in this process, right? The you're, you've been doing this for two years now. When you're, how do you work with the what's the process about dealing with, like the junior developers. Like, what

Unknown Speaker 04:01

do you? What do you find yourself doing?

Speaker 1 04:03

Are you you have to, you know, provide some sort of guidance for these people, right?

Speaker 2 04:10

Yes. So, like, basically, when we, like working with, like, junior developers and like TDPs, it basically like, like, giving guidance on, like, what needs to happen, right? So, like, get assigned, like, a little like, simpler task where, you know, like, like, the requirements are, like, clearly set up with some like, little ambiguity and like, my task is basically to help them guide towards like, like, the correct approach within like, giving them like a proper like, direct, like, hands on answers, right? So like, like, trying to enable them to, like, learn and like identify, like those problems, and then like, work around it, and then also, like, help, like, any kind of, like technical challenges that like they're having and like they might have. So that's been, like, a role that I've been playing for, like, junior developers. Also, how's

Unknown Speaker 05:02

the sorry,

Speaker 2 05:04

I try to have like, Zoom sessions to work with them whenever, like, is

Unknown Speaker 05:09

everything remote? Do you guys all right now?

Unknown Speaker 05:12

Yes, really, okay. The

Speaker 1 05:17

I know all the developers are, they're all over the place where they're in. Who is it Texas?

Unknown Speaker 05:24

And you guys, Texas? You just remote because you

Speaker 2 05:28

Yeah, our team is all over the place, like, based off of Texas, though. Oh, okay.

Unknown Speaker 05:34

So where are you right now?

Unknown Speaker 05:35

Currently, I live in North

Unknown Speaker 05:38

Carolina. Okay,

Speaker 1 05:42

so the that's that must be tough, right?

Unknown Speaker 05:45

I mean, you're dealing with, like

Speaker 1 05:48

you're dealing with junior developers, and so the, I mean, if you're, if you're doing all this stuff like mentoring them, how do you, I mean, how do you balance you know that you have certain deliverables. You got to meet code quality. You had junior developers. How does that get

Unknown Speaker 06:10

sorted out? Yeah,

Speaker 2 06:11

so I have, like, like, basically, I would say, like, about, like, four to five hours a day, I'm usually, like, involved in my work and like where. So I would say maybe, like coding and maybe like gathering requirements, or like designing. So I would say I have like, these hands, hands down time where I basically, like, work on my stuff. But like, that doesn't mean that like, I'm not, like, available. But I just like, I like to focus. I like to have like focus time and but after that, I have like time open for like, for like, someone to ask questions or like, basically, like, set up any kind of one on one calls if necessary, if somebody wants to like, meet or like, if they want to, like, utilize that time, if not, then, like, I'll, I'll still want to work on my time. So it just comes down to like, managing time properly. Because, like, I think, like, it's attainable that you work on, like, multiple stuff, and then like, also, like, manage those tasks properly. Then I think I have, like, I've learned like that skill over the years.

Speaker 1 07:20

Okay, how do they, I mean, do they do people? Do these folks reach out

Unknown Speaker 07:27

to you with questions? Or how do you make sure that that happens?

Unknown Speaker 07:32

Yeah, basically,

Speaker 2 07:35

like, normally, I would say we have, like, very open communication with our team. So, like, when, whenever, like somebody or like anything, so we bring it up, but like, during, like, like a sessions, or like, stand ups, like, if something is, like quick, and we just, like, try to go over it during like that time. But if not, then we'll just set up, like, a call later, or like, but like, sometimes, just like I like listening to updates, you know, I offer like help to like other team members, like, whenever they need some help, like or like, if I have like time on my hand or like, so, yeah, it's just like a mix of both, like coming up with questions and also like offering help when They need it.

Speaker 1 08:19

Okay, okay. So they don't always, you know, you don't find that people wait for the scrum to bring up their questions. They don't like reach out to you during the day, or you usually, right? We

Speaker 2 08:31

don't always wait for, like, the scrum, okay? We have, like, a slack and we just, like, communicate with Slack.

Speaker 1 08:39

Okay, gotcha, gotcha, okay. And the, I mean, what is, what do you find during this time? It's a very dynamic environment. I mean, there must be things pop up all the time, like issues. And can you think of anything or challenging during that,

Unknown Speaker 09:03

like, challenging what

Speaker 1 09:04

have you anyway? What have you found, like, you know, what have you found to be problems, like, things that are, you know, that to kind of get in the way of being productive? Is it technical issues pop up. Like, are you doing production support? Do you have, you know, is it

Unknown Speaker 09:27

requirement problems, like, what have you like?

Unknown Speaker 09:33

Can you tell a story about

Unknown Speaker 09:35

that? Yes, sure.

Speaker 2 09:38

I think, like whenever, like, there's some like changes in like requirements. It's like, it's usually like, little bit hard to, like, meet like, the deadline, because, like, you know, like, usually, like, we plan for like one stuff, and then we might need to like, like switch the context to like a different work. So I think it's like, one of the reasons, like, once I've been, like, involved in so when I was like, developing like a filter to, like, basically manage like, like a request header that was coming into like, basically, like, strip certain headers and then like, modify it into like, different ones. So when we have like, like a passing downstream, but like midway, we had to, like, switch our context to like, like, innovate like a like a streaming engine to basically meet, like, enterprise wide tech like this, like backlogs, right? So, whenever there's like this, changes in like requirements or, which is, like, very important, which may come due to for like planning, some like, you know, like urgencies, like in,

Speaker 1 10:39

like, vulnerabilities, business, business, business priorities very big versus Yeah, like, yeah.

Unknown Speaker 10:46

So how do you deal with that?

Unknown Speaker 10:47

I mean that that's always, that's always a hassle, right?

Unknown Speaker 10:50

So how do you deal with that when that pops up?

Speaker 2 10:54

Yeah? So basically, as always something like, we have, like, uncertainties that might come into work. But whenever like, something like that happen, I always like, I always like, try to keep calm, and then like, try to like, split up the task. So like, in a way that is like attainable, but like, if it's like, something that will not be completed by like, a solo effort, I try to, like, communicate within, like a team, and then maybe, like, split up the task against, like, multiple team members, so we can, like, divide and conquer, like the and then work together, right? And if that's not even possible, then I, like, I try to make sure, like, to always have communication with my like, like the product owners and like Scrum Masters, that it will require, like, more time than we might have in hand. But like so, I guess, like proper communication and like the effective like, try to have, like, a communication with like so you have

Unknown Speaker 11:51

so you have had examples where

Unknown Speaker 11:56

requirements were like, you would not be able to meet the

Speaker 1 12:01

what was being asked within the time frame, right? And, I mean, sometimes is it some, is that because people were asking unreasonable requirements, or because, like, your own estimate was off, and, you know, you had to basically go back to them and say, hey, you know, we made a mistake.

Speaker 2 12:20

Yeah, that was like, because it was like, a change in, like, a requirements that we were in. So, right? One of the example were, like, basically, like, like, leveraging, like a streaming engine to, like, basically stream all, like, pi 18 logs and like, like a, like a snowflake tables, but like, but later they wanted us to, like, add all the logs like within the application to like particular workflow. And basically it increased like our efforts, and because we had to, like now take into like consideration of like, like a performance factors of like, streaming those logs and like so on. So, so you know, like to basically make sure that your application is like resilient against like change and yeah, you need to make sure like proper testing. So not only like, like development process work, but like it would increase. It also like, like a testing effort would increase as well. So I just like try to have communicate and have like

Unknown Speaker 13:20

tea. So what'd you do in this case? I

Unknown Speaker 13:22

mean, it was like, obviously.

Unknown Speaker 13:23

I mean, those are all very good reasons why

Unknown Speaker 13:26

things can't be done on certain times, time,

Speaker 1 13:31

you know, on a certain schedule. The so what did you end up doing in this scenario?

Speaker 2 13:37

Yeah, so basically, we just had one of my teammates work on, like, like a script of like different test scenarios that we would work on. And we also had like this collection of like proper matrix that we wanted to meet. And also all with that, I also work slowly, like, solely on like a development piece as well. So we had like two works going, like, side by side, right? Also had like a base metrics to basically like, compare like, which basically reduces like the time little bit and helping, like completing the task, even though, like, it was longer than our original plan. I we still took like a little bit less time than we thought we would take. And I think, like dividing the work against, like our, like team members and like conquering, like, the whole time plan was, like, probably, like the way to go.

Speaker 1 14:30

So did you say, in this case, did you meet the time, the pop, the, you know, the published timeline, or

Unknown Speaker 14:37

you or did it slip? I

Speaker 2 14:43

Yeah, we had, like, we had to, like, push the release date by, probably, like, Okay,

Unknown Speaker 14:47

Okay, gotcha. So, so

Unknown Speaker 14:49

the So, how did that go? Like, would

Unknown Speaker 14:52

you end up doing just, basically, just sent out? Did

Speaker 1 14:55

you have to explain to anybody, or was there, did the use was there any pushback from users. Like, what

Unknown Speaker 15:01

was that experience? Like,

Unknown Speaker 15:04

yes.

Speaker 2 15:06

So this was, this was like, and like, internal change. So, like, okay, it didn't see it, yeah.

Unknown Speaker 15:13

Gotcha, okay, gotcha.

Speaker 1 15:16

Did you ever have a scenario where you you're making a change that was going to be impacting the business,

Unknown Speaker 15:26

or is it all the work like,

Unknown Speaker 15:28

you know, in for internal?

Unknown Speaker 15:32

Yeah, I don't recall

Speaker 1 15:36

any features. So, the So, the So, the work you mentioned that you mentioned that you get a lot of work you're doing, is refactoring, is that, right? The

Unknown Speaker 15:46

so, were you refactoring?

Speaker 2 15:48

I mean, no, it was just like, like a feature development,

Unknown Speaker 15:52

okay, so these

Speaker 1 15:54

are enhancements to existing applications, or no,

Speaker 2 15:57

it's like a like a feature development, of like a new feature, like, basically, we're adding like, new workflows to, like, like, basically

Unknown Speaker 16:08

aggregate and Gotcha. So

Speaker 1 16:12

this is an application, but there's so does the existing workflow? Is this existing workflow application is that we're working with, dealing with, like, what's the what do you say adding workflows? Like, what is that? So

Speaker 2 16:23

it's like, a micro service, right? Okay, so basically, yeah, what I mean by that is, like, I just like, added, like, new functionalities, like, like a micro service architecture. And so my job has been, like, primarily, like developing these micro services. And I've been like designing these microservices, and also like developing them using Spring Boot. And like this particular example that I talked about earlier, where we mentioned, like, the like, the deadline was to incorporate one particular feature where we like aggregated logs from, like the application, and then we pushed it down to us, like, like a snowflake table, right? But the, I

Speaker 1 17:05

mean, even though it's a microservice architecture, somebody's got to call that service, right? Somebody's going to use that service. So how does that, as you working on that piece as well? Or is that you basically just put it out there and then somebody will just eventually use it? Like, how does that

Speaker 2 17:29

like? It's basically like part of our existing portal. Like, okay, so what happens is, like, like, whenever like our like, a user logs into like USDA website, it goes to, like a user profile Preference Tab, and they can make changes to that, like, like an overdraft protection, or like, any changes to their profiles preferences. So, like, whenever they interact with, like, like a particular piece of like a portal, then so

Speaker 1 17:56

you're integrating into that, basically, so your your thing would would be triggered by some existing piece of some existing application, right? Okay, got it.

Unknown Speaker 18:06

How's that true? How's that trigger

Unknown Speaker 18:08

mechanism? Is it like,

Speaker 1 18:10

is it an API call, or is it like some sort of bus where you send out a

Unknown Speaker 18:13

notification and then, yeah, it's

Unknown Speaker 18:15

an API call.

Speaker 1 18:17

Okay, gotcha. So somebody has to make that API call to tell you this happened, or you or you make the API call. How does that work? Are you pulling or are they notifying you that something said? It's basically

Speaker 2 18:27

like the like, the front end portal. Basically, when you click like, like, submit, it calls like, like, a maintenance API, okay,

Speaker 1 18:36

okay, gotcha. And you just hooked into that, essentially, in this particular case, Okay, gotcha.

Unknown Speaker 18:43

Okay, got it.

Speaker 1 18:46

So you mentioned here that you work with, you work with a bunch of cross functional teams. Like,

Unknown Speaker 18:50

how many? How many teams are we talking about? Like,

Unknown Speaker 18:56

so how many? Like,

Speaker 1 19:00

yeah, you mentioned in here that you work with teams. Is that right in the cross functional teams? Yeah.

Speaker 2 19:06

So we had like our team on like, like a getaway layer team. We had like a dedicated, like microservice team, and then we had like our SRE team, so we usually interact with like four teams,

Speaker 1 19:18

okay? And so these guys, so, what is this? What are these teams responsible for? Other microservices?

Speaker 2 19:28

Yeah, like, basically, like the microservices team is, like they're responsible for like, microservice portion, right? Like certain microservices, like they like the profile preferences, and like the other one, they do, like account management. And then we have like a getaway team which acts on, like an orchestrator between, like these microservices. And then we have our SRE team, which is basically it deals with, like Site Reliability portion. Okay,

Speaker 1 19:57

so how do you guys coordinate? So how do you guys coordinate all your work?

Unknown Speaker 20:01

How does that work? How does that happen?

Speaker 2 20:06

So, like, the coordination between we, like, as I mentioned before, we like a Slack channel that we use for like, any kind of like, like notification or like information. Then we also meet, like, once a week. So, yeah, okay,

Speaker 1 20:21

Gotcha, okay. And when you're doing you're also doing code reviews, right, as far as because you have community developers and architectural discussions, right? Yeah. So when you're doing code reviews, what kind of, what kind of stuff do you look for in the code reviews?

Speaker 2 20:39

Yeah? So, like, I would say, whenever I'm doing code reviews, I basically initially look for like, like functionalities, like, if I can, like, understand what the code is like, if like, the code changing and what it's doing. Like, I look for like a unit test, which are like, backing these changes as well, to make sure that, you know, there is like, a proper, like, testing done. And then I also, like, try to look for like, like the structure of like, the organization of the code. And if the code is like, it's like broken down into like, like, smaller, like, more maintainable, like a modulars. And then I also look for like, a readability of the code. Because, you know, like, everybody needs to be able to understand the code by just like looking at it. It needs to be, like, clear and concise. Then I look for, like, other general standards, like error handling as well, like security concern as well. So, yeah, those are the things I look for, Okay, gotcha

Speaker 1 21:40

and the So what's an example like you mentioned, and yours that you had? There's some when you do microservices, and you know you doing, you know, you have to have some sort of understanding of the architecture of the applications you're

Unknown Speaker 21:58

working with, right?

Unknown Speaker 21:59

So what is a,

Speaker 1 22:02

what are the some of the issues that you have to, you've run into, and discussions you've had around that,

Speaker 2 22:11

some of the issues so well, like, you, like, sorry for, like, for, like, an architectural design, right? Yeah, I guess, like, some of the things would be, like, data consistency across, like, like a like microservice architecture, because, right, like, so that's one of, like, the biggest problem, I would say, because if you want to like work in like microservice environments, you know that, like, your data might like change, might be changed, or, like, across many different services, and you need to, like, ensure that, like, the data is consistent. So the

Speaker 1 22:47

same data, these, all these services, you know, need to use the same data, right? And they might be keeping local, local copies, yeah,

Speaker 2 22:55

I would say like, you need to have, like, some kind of orchestrator that, like we have with like, like, orchestration team, something like that, to take care of like that person of it, okay,

Speaker 1 23:07

the So, when you having all these discussions, how does that? How does that work? Like, what have you done to kind of make that happen?

Speaker 2 23:20

So, like, I would say, so what i So, what I do is out, like, I try to, like, document, like, my findings in, like, like, the conference talks, right? And so whenever there is like, like, let's say like, designing tasks, like, we need to create, like, a draft conference page, right? Then, like, like, then like, a list of like, finding out, and then like, and like, open ended questions on like, that documentation. And then, like, once we are ready with like, like architects, and so we basically do, like, a design review, and then we like, walk through, like, all of the changes that we're proposing. And like, what kind of design like conversation we need to make, and then like, so on that meeting. So we also, like, we get as much, we try to get as much like information as possible. And we try to then like, like, try to revisit that document, and then we try to meet again at later date. So to see, like, if there's any approvals from like, like architects, or like, like straightforward with like implementation. So, yeah, I would say that kind of process

Speaker 1 24:30

I go through, so having these meetings, or even your scrums, right? I mean, they're people sometimes have opinions, right? Did opinions differ?

Unknown Speaker 24:39

How do you How does that get resolved?

Unknown Speaker 24:42

What do you do? But

Speaker 1 24:44

like, you know, imagine, like, you have a you have an idea, and somebody's like, doesn't agree with your solution. What happens?

Speaker 2 24:53

So I would say, Let's whenever we have like, like conflicts between like, let's say, like, a different design or something. So we, I basically, like, try to come up with like a pros and cons list, and then, like, I try to maybe take like one that has like more like advantages, or, like, maybe come up with like a like combination of both approach so that we get like best out of like both work. So yeah, that's, that's like the general approach that I take to move

Speaker 1 25:23

forward with Okay, gotcha. Do you ever have a year? Have you ever been found yourself in a position where you know your idea was not the most popular one, like people did not like it? I

Speaker 2 25:43

I think, yeah, I was like, So, okay, for example, I was like, trying to propose like, a proof of concept, like, for you know, like comparing, like, different data types by basically just trying to, like, like, erase them and like, what the way to go like. But there was, like, not properly taken by my team members because they wanted, like, extensibility and like, then, like, just like a simple string comparison, and then that wouldn't work, and to make it, like, extensive. So I would say, like part of learning and like growing where you your ideas might not be best, but again, I like to take

Speaker 1 26:25

so what did you do at the time, and would you do something different now, you found yourself in this position where people disagree with you, right? So

Unknown Speaker 26:35

what ended up happening at the time?

Speaker 2 26:40

So I would say, like, basically, I took that like, I took that as like opportunity for myself to, like, like a learn. And so, because I was like, I was looking at, like a smaller, like, more like simpler, like, concrete implementation. But, you know, like people want, like, more like, extensible and like they like, if you don't have to, like, work like, more later on, right? So, like, moving forward, I was like, I always like, looking to make like, making sure that like, the application might be, like, extensible, like, where it's needed, right? So, like, like, something that I learned. And so I would say, if I, if I were to, like, do it today, I would probably basically propose, like, other kind of solution where it would be, like, more more generic, rather than like, direct string, because, you know, like, same functionality, like revisiting, like, the same like, basic functionality might be

Unknown Speaker 27:39

more work, like

Unknown Speaker 27:41

effort. Yeah, right.

Unknown Speaker 27:47

But do you think that was the right thing to do? Or, do

Speaker 1 27:50

you think like you just, you just like, do

Unknown Speaker 27:53

you think the original like, you know, I

Speaker 1 27:55

mean, honestly, like, do you think what your original proposal was the right decision? Or do you think the more generic, more expensive implementation is the right decision?

Speaker 2 28:07

I mean, I think, like, for a proof of concept that we're going with, it was, it was like an okay decision, I would say, because it was just, like a proof of concept, right? So we're just trying to, like, prove that this plan that we're making would work like stream, like stream comparison, even though, like the actual implementation of the solution might take like, like a generic one. So I think like the stream it was like, still like, pretty valid approach, but like using like generic would work like less later in like, original

Speaker 1 28:42

implementation. So they so what ended up happening at the end? Did nothing got done or like this PO, because it sounds like nobody wanted this. POC,

Unknown Speaker 28:52

I mean, actually we went with, like, a generic

Speaker 1 28:55

approach. We did okay. So how long did that take? I

Unknown Speaker 29:03

maybe, like we had it done, I would say,

Unknown Speaker 29:08

maybe within,

Speaker 2 29:12

like the we had, like, through, like a proof of concept phase, I would say, like the proof of concept took about, about a sprint, and then, like, the actual implementation, like, was like following through, if we had, like, maybe done, like, like string away, then like, a proof of concept would have taken about like, half a sprint, I would say, and then like implementation might have, like, might have, like, increased, like, a little bit, but yeah,

Speaker 1 29:36

okay, so all in, Not too bad, right? A couple of sprints all in. Yeah. Okay, cool. So I don't have any other questions. I mean anything. What can I? Can I answer anything for you?

Speaker 2 29:55

Yes, I was gonna ask first. I just want to thank you for your time and opportunity. It was really nice talking with everyone. And I would like to know, like, what, like, what the developer is expected, like, what is it expected on the team? Like, like, what defines as like a success for our developer on the team? Okay?

Unknown Speaker 30:20

I mean, the

Unknown Speaker 30:23

we

Unknown Speaker 30:27

wait to success, there's,

Unknown Speaker 30:29

I mean, again, just

Unknown Speaker 30:31

doing your job. Well, I

Unknown Speaker 30:34

guess right,

Unknown Speaker 30:35

you know the

Unknown Speaker 30:40

focus on, I

Speaker 1 30:41

mean, we want to focus on people reducing quality code, right? Like the, we always want people to be doing a good job. And, you know, that's sort of like the happy path, right? But, of course, you're, you have the reality that there are time pressures, right? So there are the you know. So that's always gonna, that's always something that's always there, right in every job,

Unknown Speaker 31:08

the you know, people who are,

Speaker 1 31:12

you know, work well with others that are because you're working in a team, right? There's a lot of, there's a lot of movie parts. It says, you know, like, if you're in a microservice architecture, right, there's a lot of moving parts. I mean, we're not quite microservice focused, but we do have a lot of services that we implement, right? We have, like, code shared code bases, so there's a lot of inter dependencies and communication has to happen, right? The you know. So people who are, you know you need to be responsive to your peers, right? And try to do a good job. And, you know, raise your hand where there's a problem, right? If you think that there's something wrong or you'll agree with something like the expectation is that you know you're gonna Yeah, because you might be wrong. Right might be wrong. You might think, I might think, oh, you know this is not the right thing to do. Raise your hand. Have a conversation. You know, decide what we're gonna do about it. Yeah. So you know the and what do you do, as far as, like, I mean, do you do you do anything to keep yourself it's a very fast moving industry, right? Like things are always changing. Is there anything you do today to kind of keep yourself plugged in?

Speaker 2 32:42

I mean, yeah, I, I try to keep up with, like, some vlogs. And I also, like, look into i Stack Overflows and like, follow, like, a build off, like, keep myself, like, try to keep it updated and do some like, I also do, like, like, Udemy courses.

Speaker 1 32:58

I don't know that's cool, yeah, yeah.

Speaker 2 33:02

I like, I also, I try to learn new stuff, yeah, just try to keep

Speaker 1 33:07

and the Udemy is through USAA or that sometimes you're doing,

Unknown Speaker 33:12

no, it's just like my personal stuff on

Speaker 1 33:15

your own. Gotcha, okay, cool. Any other questions I can ask for you.

Speaker 2 33:21

Any other questions. So, like, I'd also like to know about, like, what's the, like, next step, like, the process, I think,

Speaker 1 33:32

okay, I mean, the, I guess it's going to be the folks, the HR, folks that you've been dealing with, I'm not sure who's been speaking with you. They'll reach out to you, and I think it's supposed to be pretty quick. They'll go back to you in the next day or two, because, if I understand correctly, yeah, I

Unknown Speaker 33:49

think that's what I heard, that they're trying to be quick about it,

Unknown Speaker 33:54

the

Speaker 1 33:56

but, you know, I'm not the I'm not like, the best person. I'm just saying like, basically, if they don't, if you don't hear back from them tomorrow. Right, I would definitely, you know, shoot them an email or call them and ask them what the next steps are. Right, but my understanding is that they're going to get back to you pretty quickly. Okay.

Shreesha Khada-amex-srv-final

Unknown Speaker 00:35

Hi, I'm

Speaker 1 00:54

Hi, hi. Shisha, let me allow me a few minutes to just get settled down. So

Unknown Speaker 01:02

allow me a few minutes.

Unknown Speaker 01:06

I just came out of another call,

Speaker 1 01:13

and I think there should be somebody from the recruitment should also join. I sure.

Unknown Speaker 01:30

Can you hear my voice? Oh, yeah, I can hear you.

Unknown Speaker 01:33

Okay, good, all right,

Unknown Speaker 01:41

okay, let's get started.

Unknown Speaker 01:45

Let me switch on my video as well.

Unknown Speaker 01:52

Okay, hi, hi, Chandra,

Unknown Speaker 01:57

so can you show your ID proof?

Speaker 1 02:03

Chandra, could you just take the snapshots as required? Yeah, just one second. Okay, sister, can you just bring the ID card little mirror?

Unknown Speaker 02:17

Okay, just hold on a minute. I

Unknown Speaker 02:28

yeah, you can continue the session.

Unknown Speaker 02:40

Okay, good morning. Shisha, Hi, good morning.

Speaker 1 02:43

Could you start with your introduction?

Unknown Speaker 02:46

Yes, sure. So hi. Before

Speaker 1 02:49

you, you know, introduce, just focus on your technical skills and the last project that you worked, what kind of role, what kind of, you know, work that we are doing.

Unknown Speaker 03:01

Let's focus more on that.

Speaker 2 03:06

So, yeah, hi. My name is Risa karka, and I'm full stack Java developer with six years of hands on experience in building robust and scalable applications, and my core expertise, it lies in Java, JavaScript and cloud platforms like AWS, where I have led these, where I have successfully led the development of complex project from concept to deployment. And so I'm also working primarily on Spring and Hibernate, and also working on RESTful web services and developing micro services. So I have experienced working in designing them, implementing them, and also like deploying them to a higher environment. And I do have experience working on the front end side with Angular. So let's, let's talk about the AWS side. So I have worked primarily with services, including AWS EC two ECS and SNS and also lambdas. So I have, like, pretty, I have pretty experience of working on that. And currently I'm in a scrum team. And this team, it consists of five developers, so we have a couple of testers on this team too, and our products, product owner or scrum master, and on a daily basis, I'm mostly on the like, you know, working on Java back end code, helping the team with code reviews and also on production support dates and to talk about my current project. The current project is hands on back end developer. I'm also involved in development of different services that interact with each other to improve the orchestration of the overall flow of traffic from clients to the downstream APIs, and also we act as the orchestrator for micro services and the overall flow of the traffic from clients to the downstream APIs. We also act as the orchestrator for micro services, and we on board different scrums like different services provided by the circuit breaker breaking. And we also provide a resiliency framework and and so on. So, yeah, it's all. I also use Java Spring Boot based applications with different services deployed on ECS clusters.

Speaker 1 05:58

Okay, so you should you are a full stack developer? Yes, Chandra, I will not be able to evaluate for the UI competencies. I will be able to evaluate only from the Java perspective. So that should be fine, right? The position is for a Java lead developer,

Unknown Speaker 06:27

yeah, that sounds good,

Speaker 1 06:32

okay, because it would not be fair if I just you know, judged you for only one skill set, whereas you have multiple ones. So that's why I just wanted to make it clear with Chandra is Chandra the caller is dropped off. Let me see, dropped off. Okay, all right. So we'll start off with the basics. First, you have experience on hands on experience on Java coding, right?

Unknown Speaker 07:08

Java version,

Unknown Speaker 07:10

which Java version?

Speaker 2 07:13

So I have experience on, like I have worked with Java eight and 11.

Speaker 1 07:21

Okay, 11 is the highest that you have worked with, no Java 17.

Speaker 2 07:25

I haven't, like, I have worked with Java 11. So recently, like, it's not like I haven't worked with Java 11. Like, we made some of our work to migrate from applications to springboard. So like, there we took our project from Java 17. But, like, I don't have that much experience with Java 17. I haven't done that much of

Speaker 1 07:46

Okay, so what's the biggest difference that you perceived as a developer when you move from Java eight to 11? Okay?

Speaker 2 07:54

Like, at first, like, when moving from Java eight to 11, like, there was different, like, vast changes. So basically, like, as we know, Java eight to 11 does not have a minor errors, like, I mean, it doesn't have a lot of changes. So interesting features from when moving from Java eight to Java 11, it's basically called a local variable, right? So basically, like, you can declare a variable using the bar keyword, and which you know, like, you can declare variables and without explicit in type, type, right? So, like, that's something that you could have used a lot, and I also worked on it and Java, it has, like a string, strongly typed, and not declaring a variable or type or something a new so it was something that like something it was it also provided a very robust HTTP client, and I did not need to engage in external clients. So, and that was also one of the other ones that I think was the major difference between Java eight to Java 11, but the other ones it were, like, minor changes for, like, you know, performance improvements and also enforcement. So, yeah, like, I think these are the methods, our optional classes and so on. Okay,

Speaker 1 09:23

so you said you were using external HTTP clients earlier, right?

Unknown Speaker 09:30

Why would you need an external HTTP client earlier?

Speaker 2 09:34

So, basically, like, why would I need the external client is on that one, like the external HTTP client we have we need to leverage was using, at first, using Apache, and it needed, it needed, like, basically making sure that we need to have proper setup of the pooling and of the proper setup of the connection pooling, and as we know that a synchronous request, right? So we need to get a lot of dependencies for those and also create a configuration class app. Create a configuration class for those as well. And it doesn't like basically, it doesn't provide a wide range of services. So, but for simple RESTful services, where users don't need to worry about a lot of higher connection capability and and also, like we know that all it was like a lot of things to be added, right? So using Java lab, you could use HTTP client API. Okay, so,

Speaker 1 10:51

yeah, I actually, you know, I'm actually filling out a forms, but I should have set one of the ground rules that I usually do while you know, taking any interview is, normally, I ask a question. If I am able to get the answer that I wanted, I would move on to the next one without, you know, allowing to fill up, because we have to cover a lot of areas. So that's something that I do. Please do not consider it as rude or anything else, because, okay, so, okay, let's ask a few hands on questions. First, you know, since you have worked on Java eight and 11,

Unknown Speaker 11:44

what is a functional interface?

Speaker 2 11:49

Okay? So as we know, that basically functional interface, it is a special interface, like introduced from Java eight and that

Unknown Speaker 12:02

that consists of only one abstract method.

Speaker 1 12:08

Okay, it contains only one abstract method. How many methods can a functional interface have in total?

Speaker 2 12:18

Okay, so talking about the methods, like, how many it can have it so it can have, like, multiple interface methods, right? So, but at first, like, it needs to have only one abstract method. Okay,

Speaker 1 12:36

I just have a method signature out there and an abstract method out there in the stellar functional interface as per your definition. So like,

Speaker 2 12:51

let's take example. Like you can have other methods, other methods at first, but like other methods that did either default or else static, but it cannot be abstract, right? So,

Speaker 1 13:03

no, I'm not declaring anything as abstract. I'm saying that, you know, there is one abstract method. So let me give you a functional interface. So functional interface is my interface. Abstract method is my abstract and then another method is there which says my constructor, okay, I'm just naming as my constructor is actually a method out there, but there is no definition. It's only the signature out there.

Unknown Speaker 13:33

So can we have that?

Unknown Speaker 13:38

Only one abstract is there?

Unknown Speaker 13:41

Okay? So, like, basically,

Unknown Speaker 13:46

on that one, like,

Speaker 2 13:48

at first, like, we can have another method with no definition. It's also basically called abstract, right? So I don't think like we can have any function on that one. I

Speaker 2 14:03

it, you can have a function like that, like, I don't think you can have the have that for functioning interface, like we can that we can't have that in function interface, it won't be a functional interface.

Speaker 1 14:14

Okay, it won't be a functional interface, okay, all right. So imagine this piece of code. There's a main method. First line is, int i equal to zero. Next what I'm doing is I'm creating an array, list of integers randomly. I'm setting 10 integers to this ArrayList. After that, I'm using lambda and stream functionalities to iterate over the list and set the value of i equal to the current value of the list after the iteration. Outside of the iteration I am printing the value of i. Okay, that's the end of the program. So what would be the output of this code?

Unknown Speaker 15:06

So, basically,

Speaker 2 15:14

so if the code is within this, if the code is within the stream, then stream you are changing, then the value, value of integer, right?

Unknown Speaker 15:24

The value of Intel.

Speaker 1 15:29

Okay, you want me to explain again? So in I equal to zero, there's an ArrayList, and then I'm setting the value of i equal to current value of the ArrayList, okay, and then I'm printing i at the end of the iteration.

Speaker 2 15:52

So on this one, I think even though I is modified into the stream inside the stream, like the change. I don't think the change doesn't have any effect on the value well outside the stream, but I think it should still be zero or the initial value

Unknown Speaker 16:17

on the code compile.

Unknown Speaker 16:21

Is that, how the code compile?

Unknown Speaker 16:24

Would the code compile?

Unknown Speaker 16:27

Compilation? I think it's all like,

Speaker 1 16:35

okay, fine, you can try it out in your leisure, and you know, just for your learning whether it's going to compile, and if it's not going to compile, what it would take to this. Have you heard about atomic variables? Yeah,

Unknown Speaker 16:50

okay. Why are they used? Okay,

Speaker 2 16:55

talking about the users uses of atomic variables like, we know that it are atomic variables. Like it are basically used whenever we know we know, like whenever you know you want, right and you know that we have to explicit our synchronization. And also it is used by it is used for threat safety because, like it ensures that the operation on shared variables are atomic, and meaning that they are completed in a single step without any interruptions. Um, and I think on the question like that you mentioned, you mentioned earlier, if you are using a atomic integer, right? So at first, I think like, it will combine, but on a normal integer, I don't think it's gonna compile. Compile.

Speaker 1 17:51

Sorry, you said, You know what was that? It was about compile. So,

Speaker 2 17:55

like, the question that you mentioned earlier, like, if you use atomic integer, it's gonna it's gonna come. It will combine. But on a normal integer, I don't think it's gonna come by. So it was just the answer of the previous question, like on the ARL stream. Okay, all right, you

Speaker 1 18:14

have used default methods and interfaces. Can you give me a practical example? I do not want the theoretical definition. I want a practical example where a default method is applicable. Okay,

Speaker 2 18:38

so here, like we can take the example, so basically, talking about a practical example for default method, let's

Unknown Speaker 18:46

take the example like, let's say we have a Vehicle, right,

Unknown Speaker 18:51

and

Unknown Speaker 18:54

and like, we also know that if

Speaker 2 19:03

so let's take the example of the vehicle. So as we know that we have a multiple methods, like get the brand, and like, we can get the branding, or also speeding up or slowing down, and we, but at first, like we and we want to have a special method that explicitly has a user reusability, and it also allows the functionality to be sealed across multiple implementations or the vehicle, right?

Unknown Speaker 19:34

So like, let's say

Speaker 2 19:37

you are adding a feature for threat protection, like an alarm, right? So at first it's gonna implement a default method, and also, like with turn alarm, and we also like plan tone of the alarm off. And

Unknown Speaker 19:54

as we know that

Speaker 2 19:57

the safety protection. So let's talk about the safety protection it measures. It does not have the

Unknown Speaker 20:04

safety product.

Speaker 2 20:07

It doesn't have, like to be explicitly implemented on any of the implementation of this, vehicles, interfaces, so, but like here, the you are providing, like a default implementation for the alarm, and the alarm is gonna look like, look like the it should be a real world scenario. So if you want to

Unknown Speaker 20:31

provide some default browser, then like

Speaker 2 20:35

we can if we want to provide some default default browser,

Speaker 1 20:44

okay, I think a little bit we digressed from the explanation, but it's fine. You have an idea, so that's okay. We'll move on to Spring Boot. Okay, in Spring Boot, we use two annotations at the red bean and other red component, where and why would you use and what would be the different behavior?

Speaker 2 21:15

So on this one, like the annotations a, um,

Unknown Speaker 21:22

so basically, like, let's talk about the,

Speaker 2 21:27

let's talk about the word, like being a annotation. It in a spring. It is used, it is basically used to define a beam rat. So basically, it is an object that is managed by the spring container, and also it is typically used inside the configuration class and configuration class, which means, like, annotated with the configuration. And you can also use a bin, like when you need to create and configure that objects manually. Because here we need like control over the creation of the objects and and also like they are managed by our spring container, and then the and then like component is defined as a class label on services or other classes that you want to spring the energy, like automatically detect, right? So, and automatically detect, and also registered on the spring context as a beam, right? So, basically, like, here you have the components, like services, repositories or controllers that can be registered within the spring context. So I think, like those are, like your primary difference between bean and components.

Unknown Speaker 22:56

Okay,

Unknown Speaker 23:01

if I have

Unknown Speaker 23:04

a JPA call, okay,

Speaker 1 23:08

would you want me to use it as at the red bean or at the red component?

Speaker 2 23:16

So basically, like when calling the API, so your GPA repository,

Unknown Speaker 23:24

I think it would be a component annotation,

Speaker 2 23:28

but I would explicitly annotate it using the repository annotation, which is especially the specialization of component annotation. And so on this one, I would not be Creating a bean object in a method.

Unknown Speaker 23:50

Okay, I

Speaker 1 24:04

do you have used REST APIs, I believe okay as a service provider or as a consumer?

Speaker 2 24:15

So basically, like REST API, I have used a lot in my projects, and I have worked on both ends.

Speaker 1 24:26

My question was whether you were a service provider as a REST API or you were a consumer of the REST API.

Speaker 2 24:34

Yeah, like I mentioned you earlier to like, so I have worked a lot, and basically, like, we have been when it comes to rest APIs, we have been providers. So we also exposed our endpoints and also like, okay,

Speaker 1 24:51

okay. So that means you were providers, you were exposing your endpoints and you were able to but you have not consumed somebody else's REST API.

Speaker 2 25:02

I have worked in consuming this API. Okay,

Unknown Speaker 25:06

great, okay, so I will,

Speaker 1 25:13

okay, let me ask you a question. First, which industry domain do you relate to? Financials, insurance car, practices, energy, telecom. I mean, these are some of the domains. So which domain would you relate to? Industrial domain would you relate to your work?

Unknown Speaker 25:31

Okay, so,

Speaker 2 25:34

as I have primarily worked on financial domain, financial,

Speaker 1 25:37

okay, great. Then I think this question applies, so I have an intranet based application, okay? It exposes a REST endpoint.

Unknown Speaker 25:48

So there is

Speaker 1 25:52

a UI based, you know, Portal, intranet portal, which invokes this rest and point. What this rest and point does is it captures the financial transactions that happened for the given day, and it makes it available to financial auditors only of the organization. The financial auditors are part of the organization. They are part of the intranet, and they access it over an intranet application, okay, what are the details that are being shared in the REST API? The details shared are the last four digits of the card number, the last four digits of the bank account from which the payment was made, and the amount of payment currency, by default, is always US dollars, okay, when you build this REST API, what kind of security would you build around this REST API to make it available to the financial auditors?

Unknown Speaker 26:58

Is the question? Clear.

Speaker 2 27:02

I yeah, like, I think it's clear so limiting on

Unknown Speaker 27:10

that. So basically like,

Speaker 2 27:16

so basically like, as

Speaker 2 27:28

so for, I think, for securing these requirements, basically like I would use OAuth, OAuth two or JWT based authentication, for securing our endpoints to to make sure that you, you know, like only authorized users can access this APS, right? So here we can also add an extra layer of protection, like, if you have a API key validation, where you can use, like you can issue an API key that is a request to like that is required to access API, which adds on additional layer of security on the token and that is provided within the request. So here, like, we will basically be implementing our role based out, like, role based access control for authorization. And

Unknown Speaker 28:25

we're certain one

Speaker 1 28:27

point which you may have missed, I would like to point is this is an intranet based application. The application the portal that is secure within the firewall of the organization, no request, no data is going over the internet, no request, no data is going outside of their firewall. Everything is happening inside their firewall. Do you think oath is a candidate for this kind of authentication?

Unknown Speaker 28:59

Normally work? Is a third party

Unknown Speaker 29:03

authorizer, right?

Speaker 1 29:06

So should we authorize internal applications to a third party where in the service provider and the users are all within the intranet?

Speaker 2 29:18

So I think if you are talking about, about, like, inside a network. And then I don't think you need a OAuth for internal but basic authentication or something like that is enough, right? So, you know, like, I think at first, you can basically leverage a single sign on that the company, like, uses for, you know, like integrating those APIs and also making it easy, but yeah, like, that would be something that you can do for, like, authentication or purpose, or like, anyways, like, So, I think instead of OAuth, you can do single sign on, and I worked with Okta for single sign single sign on as well. And but I don't think, but I think that's how I would adjust if it's within the firewall. I

Unknown Speaker 30:24

Okay, so you have used Okta.

Speaker 1 30:29

You have used you are saying single sign on. So where would the Single Sign On be implemented on your API?

Unknown Speaker 30:40

Okay, so when

Unknown Speaker 30:43

implementing the API,

Speaker 2 30:48

so basically, like, it will be basically, I think it would be a part of identity provider, and that's how, like, we worked with it. So

Speaker 1 30:58

okay, so I get it single sign on as part of the identity provider. Even Okta can do a single sign on. There are LDAP, SSO, all these things are there. Where would you implement the SSO on your API?

Unknown Speaker 31:15

Something? Where would you implement it?

Unknown Speaker 31:21

So when implementing this,

Unknown Speaker 31:25

so basically, like,

Speaker 2 31:28

I think I would implement this in the authentication gateway, right? So there will, like, some kind of gateways, and there will be also some kind of gateway where you want that implementation, right? So,

Speaker 1 31:42

okay, so where would this gateway be there? I know you probably have used gateway. Is it an API gateway?

Speaker 2 31:51

Yeah, like it is a API gateway or AWS now, now here's the question.

Speaker 1 31:56

API gateways are usually used whenever requests are going out of the network through the firewall outside we can use if there are domains within the you know intranet. My question was very simple, where would you put a SSO, if you know, had observed, or, you know, paid attention, it would have been on the portal itself. The Single Sign On would have been on the portal instead of any API.

Because you are logging to a portal, the portal is accessing your APIs, right? You could do a rbse or a jot token from the portal, which authenticates the portal itself, but log into the portal, all roles responsibilities on the portal are managed on the portal, so that way, only the financial auditors see the APIs to access, right? That's fine. I think you have a lot of concepts. It's more just about articulating. That's okay. I believe you are using some kind of code repository for storing your code, right?

Unknown Speaker 33:12

Okay, which code repository are you using?

Unknown Speaker 33:17

I'm using GitHub on that

Speaker 1 33:20

one, like GitHub, one, right? So are you a pull request committer or approval?

Speaker 2 33:30

So on that one, like in GitHub, like I do both. So basically, yeah, I commit my changes and, okay,

Unknown Speaker 33:37

I'll give you a scenario. I'll give you a scenario. So

Unknown Speaker 33:42

there was a product. Production issue.

Speaker 1 33:45

We had to debug it. So what we did is we added a few log statements, then we committed the code, deployed the code into production again, a few more log statements, committed the code. So this went on for like around 100 times. Okay, so now my code commit history shows loggers added for debugging, loggers added for debugging. Ideally, I would like to see my code commit history show me exactly what functionality got changed. I do not want to see these loggers. You know, logger commits showing up. What would you do to help me?

Speaker 2 34:38

So? So like, if you have been working on a separate branch, let's say, like, that's not the main branch, right? So you know that

Speaker 1 34:51

the main branch is the main branch. And because I have to put it into production live environment, I have to do it from the main branch, so

Unknown Speaker 35:06

help me out here. What can be done? So, yeah, like,

Speaker 2 35:14

I guess if you basically, like, if you want to clean up your commit history, basically you can do to interactive rebase, right? I think on this one, like on your local repositories, you can basically run interactive rebase, and then you can go back with how many comments like you want to go. And then you can know either the scores like either discuss the comments or provide the comments.

Unknown Speaker 35:47

Okay, all right, yeah,

Unknown Speaker 35:50

what is the criteria for you to approve a pull request?

Speaker 2 35:57

Okay, so talking about the criteria for me, basically the criteria it would it could involve, like, proper coding standard. So, you know, like, what I mean, it's like, the quality that the organization has provided, right? So, like, I know they, like, we can take, like, the RE readability and of readability of the code, or like the proper variable naming, and I don't look for certain kind of formatting as well. So if I look for the if I do look for the functionality of of the methods, the naming of the methods, or like proper unit testing and proper unit testing around the methods like that have had been implemented. And I would also look for proper comments and documentation when we when needed. But you know, like those,

Speaker 1 36:55

how much is proper? How would you enforce this? Do you have to go into each PR and every time you set a target. Or how would you enforce it?

Unknown Speaker 37:08

So here, like, what would I do is

Speaker 2 37:11

so like proper talking about, like proper functionality, like proper formatting, it can be enforced using check styles. And also, like textile,

Speaker 1 37:21

sorry, we use textile, yeah,

Speaker 2 37:25

like a proper formatting, it needs to be enforced using textiles. So check styles, and also

Unknown Speaker 37:32

textiles are implemented where

Speaker 2 37:36

check styles, like it can cover an enforced

Speaker 1 37:39

where are they? Where do you add the check style?

Speaker 2 37:42

Or check style is it basically can be implemented on the detail, and also it can, I

Speaker 1 37:48

can disable the check style as well, right? So when I committed, there is no check on the check style. Enforce. How do you enforce the Okay, I think there's a lag. Let me finish. How do you enforce that the code committed adheres to standards? Because check style is at the editor. It makes the editor slow. People do disable the code. I mean, disable the check style. How do you enforce the code quality?

Speaker 2 38:21

So, like, after implementing on the editor, we can, like the checks that it can be integrated into a Jenkins Jenkins pipelines,

Speaker 1 38:29

pipeline, Okay, any other tools that you know of code quality that are integrated with the Jenkins pipeline,

Unknown Speaker 38:38

sonar and,

Speaker 1 38:41

okay, what is your sonar approval percentage for a PR to be approved? 80%

Speaker 2 38:49

80% 80% runs and then line cover. Raise,

Unknown Speaker 38:57

line coverage. Yeah, like 80%

Speaker 2 38:59

branch and line coverage on unit test

Speaker 1 39:05

time coverage on unit test is 80% isn't 80% pretty low, considering 20% of the board is not unit tested. It could result in constructors. It could result in creators to be not tested. Which are the most critical ones

Unknown Speaker 39:21

in us. So how do you catch for that?

Speaker 2 39:25

Basically, like we need to look for the quality of the test at first, right? So when looking,

Speaker 1 39:32

I wouldn't know when a J unit is being written. It says J Unit coverage, line coverage and the branch coverage is 80% cyclomatic complexity is covered at 80% How would you know?

Speaker 2 39:46

So basically, like we would look for the quality of the taste rather than lines, right? So

Speaker 1 39:50

how would you know quality? Quality subjective. Right? Quality is subjective. What is quality for you may not be quality for me, my bar might be higher. How would you know what is quality? You cannot do that when we are doing it right, when we are proving if we are going to line by line. Probably yes. Okay, this commenting is not right. There is no proper English, as per your knowledge, my knowledge, my English might be poor. So I might just allow some of the lines of code or the comments that you have written to be allowed some variable naming conventions I met are also quality subjective, unless there is an automated process that does that, like sonar is an automated process. So no critical, no major violations, I can understand that.

Unknown Speaker 40:37

So how would you enforce the quality that's the question.

Speaker 2 40:43

So basically, like I would look for the method naming of the unit test, right, and I already have

Speaker 1 40:50

textile, right? Sorry, you already have textile to do that. So why would you have to manually look at textile? Can you know, pass or fail it? If the naming convention is not right. Plug into it manually. So

Speaker 2 41:04

No, like you at first, like you need peer reviews from not just one developer, but from a couple of developers before the code gets like, code gets more so also, like you want to know that what I mean by what I mean, is like, by looking at the method name, if it's not the naming quality, right? And I suppose, like I'm looking for what the unit taste is actually tasting. And here, like, you know, if you, if one person is reviewing a reviewing it, you know, like you, you said it's a subjective so for me, it might be okay, but for another developer, like with different knowledge bucket,

Unknown Speaker 41:52

how do you do your deployments?

Speaker 2 41:58

Okay, so for the deployments, like we can use Jenkins for deploying our application, Jenkins.

Unknown Speaker 42:07

Jenkins is usually a build tool,

Speaker 1 42:11

but it does not do a deploy unless you do something special with it. So what is the special that you're doing in Jenkins, that it deploys the code. Normally, Jenkins is the build tool code quality it can integrate so you're checking codes, so it will create a final jar, or, er, whatever it is. But deployment, okay, let me put this question, where is your code deployed? So, like

Speaker 2 42:41

our code, it is basically deployed into the cloud platform like AWS,

Unknown Speaker 42:45

right? So, AWS, yep,

Unknown Speaker 42:47

you mentioned and so

Speaker 1 42:51

how you know you are deploying the port of AWS? We have scripts, or do you run commands?

Speaker 2 43:00

So basically, like, we are using deploy script and like it stays on. Deploy stays and like, if it triggers on AWS, the CLI that is, you know, like it is basically responsible for deploying these applications. And also, either like, it is it be deploying Docker or Docker images, or, like, just a jar to Ecto it is basically like done through the AWS CLI, right? So we injected the credentials into the environment and then

Unknown Speaker 43:39

leveraging AWS cloud formation,

Unknown Speaker 43:45

yeah, okay, we have

Unknown Speaker 43:50

a database experience.

Unknown Speaker 43:53

I do have a database experience. What

Unknown Speaker 43:55

kind of database? Yeah,

Speaker 2 43:56

I have worked with Oracle and post GID SQL and on the relational side, I also worked with Dynamo DB, and DynamoDB on AWS, like AWS for non relational data. Okay,

Speaker 1 44:18

so what is the advantage of Postgres? I for Apple.

Speaker 2 44:26

So the basically, like the talking about the main difference, like the, is it advantage or difference

Unknown Speaker 44:32

advantage?

Speaker 2 44:36

So like the advantage of post GRE SQL, in terms of like JSON support here, like it provides a proper flexibility, right and proper flexibility and performance, so you can also specify the custom data and then custom data types and also operators and it supports

Speaker 1 45:04

so have you heard about large object data types? Yeah, I

Unknown Speaker 45:09

have heard about it.

Speaker 1 45:11

So isn't that sufficient for JSON data types? You can store text in large objects, character large objects flops in Oracle. Do? Would that not be sufficient, but we need a specific, decent type, okay?

Speaker 2 45:34

Basically, when we are choosing like, instead of having a large object, you have explicit, the explicit, like JSON, JSON, which is easier to work with, right?

Speaker 1 45:48

I mean, you still get a JSON object as returned. Right to that object would still be the JSON format only.

Speaker 2 45:55

So I think on this one, the JSON format, it would be like, I So here, like, you don't have, like, the querying capability device. So, I think, like, you know, you have built in function for postgre.

Speaker 1 46:11

And that provides different. got it so you have worked on RDBMS. I have owned

Unknown Speaker 46:20

an RDBMS.

Speaker 3 46:32

What? She said, I sorry, marinate, I have to restart, rejoin

Unknown Speaker 49:26

internet. Go.

Unknown Speaker 49:36

Joining in 321,

Unknown Speaker 49:46

no, Hi, I'm so sorry. My internet

Speaker 2 49:55

Hi, I'm so sorry. My internet connection, it went away, and I need to resume it again. Okay, it's fine. I'm so sorry for that.

Unknown Speaker 50:04

Yeah, I think we were discussing on

Speaker 1 50:10

discuss Postgres. Okay, so what is the standard normalization form for any RDBMS?

Unknown Speaker 50:24

So I

Speaker 1 50:35

That's okay. That's I think most of the people don't know the nomination form. That's okay.

Unknown Speaker 50:47

So do you have any design experience?

Speaker 2 50:51

So design experience, yeah, like, I have been using, like, I have been working with designing micro services and REST APIs. So,

Speaker 1 51:00

yeah, I do what's the most common design pattern that you have used,

Unknown Speaker 51:07

the common design pattern?

Speaker 2 51:11

So basically, like some of the design patterns that

Unknown Speaker 51:16

I have worked, what you have used, yeah, yeah,

Unknown Speaker 51:21

singleton pattern and,

Speaker 1 51:25

okay, can you tell me what is the fastest way to implement a singleton?

Speaker 2 51:32

So the fastest way to implement a single pattern it is to create an nm class. And

Unknown Speaker 51:43

that's it like, yeah, it's like to create an nm class. So

Unknown Speaker 51:49

an nm class is a singleton, by default, something

Speaker 2 51:51

else. Basically, if you want to create a fully faced Singleton, then at first, like, you have to know that we need to define a provide constructor, applied private constructor, and also private constructor and private static field, and then also public it, get instance method,

Unknown Speaker 52:15

private static field. Why would we need a private static

Speaker 2 52:21

field? So basically, like, I'm just telling that the the private static field, it is the instance field, right? So if you want to store that instance in a field it should be it should be private, okay, okay, got

Speaker 1 52:41

it. What's the fastest way to get into singleton in Spring Boot? Okay, so

Speaker 2 52:54

when going into the Spring Boot, like, basically, at first, all the bins in the springs, as we know that all the bins in spring, it has singleton by default, right? Okay, so you know, like any bean would be Singleton,

Speaker 1 53:10

so it would be easy. Okay, all right, let me ask you a scenario question. So, where would you use a singleton object, apart from a connection pool, and you know your HTTP connection pool all these things, those are very common answers. Give me an example that, okay, this is a single term. Okay, from day to day example, you have to give me a day to day example that this is a single term.

Speaker 2 53:52

So I on this question, so like, I guess

Unknown Speaker 53:57

the configuration properties, right? So

Speaker 1 53:59

then they said, Please, no database connection, pool, no configuration properties. Those are very common answers. Those are theoretical ones. I wanted a day to day example, which is a single term. I Okay, so Are you a social media user? Yeah, I am. The number of likes,

Speaker 2 54:24

is that a single term? So, like the logging framework, it's basically a single

Speaker 1 54:28

term. No, the number of likes on a page, is that a single term,

Unknown Speaker 54:35

the number of likes?

Unknown Speaker 54:42

I mean, like,

Speaker 1 54:45

okay, so you see, you see your page on Facebook, somebody likes it on your phone, somebody likes it on the website. But when you see you see the total number of likes all accounted for, right? That's a singleton, right? The number of views, number of likes, the number of hits on a page, are all singletons. Okay. Why would I go so much of you know, bother and create a singleton object when I can declare that as static.

Unknown Speaker 55:23

Static is Singleton, right?

Speaker 2 55:30

So, like, basically, when creating a singleton object, instead of declaring static right now, means like that you have like, a control over how you want to create the those instance. So when like, and then when creating this instance, you can also like, implement like thread of six safe access from the across the environment. And then you can also, like, add those behaviors, right? So you can add methods to the singleton for any kind of changing the logics, like, without affecting that

Speaker 1 56:10

the I can have static methods to access static variables, which can give me the same functionality that you want to create as additional functionality for Singleton. Let's say that you want to, you want to update. They can have a static method to update a static variable. That's very much possible. You want to do a minus one. I can have a static method to do that to a static variable,

Speaker 2 56:44

yeah, but like, it won't have a lazy initialization. And also, like, also, why

Speaker 1 56:48

would I need a lazy initialization when we are doing at the red bean, beans are our singleton so beans can be early initialized, and they would be initialized at the start of the boot application. So the initialization is ruled out here, right when we are doing a connection pool. Connection pool is initialized at the startup. Configurations are loaded at startups, so the initialization does not apply, right? Okay?

Unknown Speaker 57:16

Have you had any experience as a team lead?

Speaker 2 57:20

So basically, on the team leader, I don't have overall experience, but I have laid a couple of projects, a couple of junior developers, and yeah, I didn't lead the whole team, but as I mentioned earlier, yeah, I lead couple of junior

Speaker 1 57:37

developers. So let's say that there are two junior developers under you, and they bicker a lot. They fight a lot among themselves.

Unknown Speaker 57:48

What would you do? Okay,

Speaker 2 57:51

so like, in order to solve this, like, solve this problem. So basically, what would I do is, whenever, like, there is a some conflict between two of the developers, like, conflicting situation, I would usually facilitate our joint discussion at first. So what's the main problem? What, like, what the main problem came from? So basically, I would try to arrange a meeting, and then also, like, discuss the issue together and whether is fight right? So like, I would also encourage each other to share their viewpoint, and then have like, a pro and con discussions of each of the method. And then I,

Unknown Speaker 58:32

if they are together, they fight a lot.

Speaker 1 58:36

So how would you have this discussion? How would you have viewpoints if they are together?

Unknown Speaker 58:44

I mean, like,

Speaker 2 58:46

I can also talk to them individually at first. So like,

Speaker 1 58:51

Okay, have you had any experience with estimation sizing?

Unknown Speaker 59:01

Yeah, I have the experience with that too.

Speaker 1 59:04

What what tools have you used, or what kind of methods have you used for estimation?

Speaker 2 59:10

So basically, like, we can use Fibonacci series for estimating. We

Unknown Speaker 59:14

are doing story point estimation.

Speaker 1 59:18

Okay, story point estimation is what you are doing it. Have you had any formal estimation technique, like function point SMC, or any of these? I mean, it's okay if you don't have Fibonacci series is one that we usually use for Sprint story estimation for a given sprint. But that is not sufficient to do it for a feature level or at a PI level, because the size is very high and anything beyond 13

Speaker 2 59:54

is not correctly estimated. So like, I would say like T shirt sizing for epics or like features, yeah, okay,

Unknown Speaker 1:00:04

yeah, that is what I was thinking.

Speaker 1 1:00:07

Fine. I mean, at your level, maybe you're not exposed to those kind of okay.

Unknown Speaker 1:00:14

At think I have asked enough questions. Do you have any question for me?

Speaker 2 1:00:19

So I don't think so. No, like, I just want to know. Like, I don't. I want to know more about the day, like the physician would like. So here, like, what would the developer be involved in?

Speaker 1 1:00:36

Okay, pretty much similar to what we have been involved that will differ from client to client at this point of time. I cannot commit on which client would be working for, because it would all be dependent on the time of year, joining and the open requirement at that point in time. But typically we work on sprints, we work on agile scrums, and work is usually sprint oriented, and it includes development, deployment, debugging, testing, similar to what you've done. A lead is expected to handle a small team, or small to medium team, and we expect, you know, to manage client interest as well, like you know, understand their requirements, give them solutions, POCs, typical, whatever you have been doing, nothing a lot that, yes, the exact nature of work could vary based on the client, based on the nature of project that you get. Okay, so it could be

Unknown Speaker 1:01:46

different in experience based on the

Unknown Speaker 1:01:50

Do you have a location constraint like you know,

Speaker 3 1:01:55

you're currently based out of, sorry, I did not ask. Currently I'm in, based in Virginia, but like I can relocate to any other states.

Shreesha Khadka- Galaxy- Srv- Final

Speaker 1 00:00

Um, do you let me just check something real quick? Yeah, so can you just explain, like, maybe a recent project that you've worked on, and I'm kind of curious to hear about, you know, what problem you're solving, and also how you're using technology to solve it, and kind of what kind of challenges you have faced there,

Speaker 2 00:22

sure. So currently I'm working with, like, you know, all Dominion fetlines. So let me just talk to you about my recent project. It was like, you know, what we did there? It was, it was mainly happening like, you know, we were trying to decouple the notification service and that we had, like, right, you know, like the notification service, it was basically responsible for the users, and like trends and like it right before it was processed, like, you know, we were sending out an email. And even, like, leveraging our downstream API, and that downstream API, like, you know, how email services are, like, they are not always, they are not always up, right? So whenever there was a failure, like, in the email services our application, it would fail, and the transactions, like, you know, it would roll back. So they are, like, I took a lead on and, like, basically, you know, modernizing these. And what I did was, like, I split out the process first, like, process of calling the downstream notification service. And also, like, you know, put it as a AWS lambda function that would like, you know, basically get triggered by a message on Amazon SQS, which is basically queuing message queue, a quick queuing message queue. And also, like, you know, our application, like it was mainly, it was mainly the only responsibility that it would, like, you know, for it would hold that would be able to put some message to this SQS queue. And also, like, you know, it's lambda would trigger, like a trigger to a call on this notification service. So what I did was, like it was, it made our applications more resilient to the failure on the on downstream services. And basically, like, you know, it enabled us to add a layer of resilience, like residency on the notification service by adding like retries and even like, you know, exponent, sorry, exponential backup on the lambda code, as well as on the ability to reprocess this field messages on a later time. So yeah, like, this was the most recent work that I did at my current company at Old Dominion freightline. And like, to complete this task, like we It took about, like, a couple of months.

Unknown Speaker 02:48

Got it perfect. Thanks.

Speaker 1 02:51

Cool. I'm just going to go through a couple of, like, fundamental concepts that we sometimes deal with. So I guess Can you in terms of, like, some basic data structures, you know, can you explain to me the difference between a set and a list? Sure,

Speaker 2 03:06

so talking about the difference between set and list, so let's go with the set first. So basically, you know, like set, it is an unordered collection of uniquely unique objects, right? So basically, you know, like your typical headset, it does not maintain the order. And you know, it does not allow the duplicates, right?

But like you know, there are some implementation of said that may allow you to preserve some kind of order, but you know, like your list, it is an audit collection of objects, right? So it does not maintain, sorry, sorry. So it does not maintain the uniqueness where you can insert the same element over and over again. So you know, like some of the implementation of list, it includes like the array list, and even like the link list and some of the implementation of set. It includes like, has set and tree set.

Speaker 1 04:07

Okay, cool. And can you give me an example of like, when you may want to use a set versus a list?

Speaker 2 04:16

So on that time, like, you know, basically like, when using this set, like, you know, if you are trying to gather, like, some data that you know, and that is basically, like, you know, a unit that is basically unique, right? So if you have, like, if you have lot of duplicate data that you do not need to worry about, that it's like, duplicated or not, and like, you just need a set of those data stored in our data structure. And you would want to go with set, right? So, because it it mainly stores like a unique value. So let's take an example, like, for example, like, you know username and like, those are unique, right? And you can store like, a set of usernames, because you do not want a set of duplicate items, right? So that's like, where you would want a set, but also, like one of the technical use case of set up and that like is that, like, you cannot retrieve an element on a set that is based on an index. But, you know, like, rather, you have to iterate over the whole set. So that's where you would want to use unleashed, because, like, you know, you can retrieve the element based on the index.

Speaker 1 05:33

Okay, got it. And you mentioned something about hash set as an implementation of a set. What is, what is hashing exactly, and how does it work?

Speaker 2 05:45

Well, talking on the like has said, basically like, you know, hashing, it is mainly a basic function, right? That converts like, you know, that mainly converts your like, that mainly converts your data and your object into a hash code. Basically, you know, it just converts your object into an integer based on the hashing function that you provide, right? So, you know, in order to convert your like, convert your object into hash code, you basically override the hash code method, and that even provide the implementation. And, you know, if you do not use that overriding, then it will take that has code like it will take that has code of the object class, and also like, you know, it is really important concept in terms of data structure, because, you know, because that's how it use. That's how it is using. It has maps, and also, like you know, it is used for storing data based on certain type of indexes and so on.

Speaker 1 06:51

Okay, and can you, I mean, you mentioned about unique set of values, so when we're trying to look up for things, right? So let's, let's, you know, one of the classic examples of using hash is also, like, in map. So we can have, like, a hash map, you know, fairly common data structure that we often use, so,

Unknown Speaker 07:15

yeah, like when we're saving the keys,

Unknown Speaker 07:20

what? How do we handle duplicates like, you know, what

Unknown Speaker 07:23

will happen to these duplicate values?

Unknown Speaker 07:27

Basically, like, you know,

Speaker 2 07:30

when handling the sets like, you know, if your keys, they are duplicate or else, like, Are you like? Are you talking about the like, the highest codes that are duplicate? Or,

Speaker 1 07:41

yeah. So just to clarify, so let's say two scenarios, right? So let's say one is you have, like, if you could tell me what will happen where we are, like, you know, trying to store a key when the key has some associated value. But, you know, let's say this key gets hashed to sort of, like, the same numerical COVID as like, there's something existing there. In that case, what will happen

Speaker 2 08:07

so on this case, like, you know, basically like, to handle the headsets and like, you know, when you have the like same key, and that means, like, you know, that you are trying to add a value to, basically your key that gets converted into a hash code, right? And then, like, you know, based on that half score, like the the way that has maps internally works, it's just a collection, that collection of a linked list, right? And basically, you know, like the link list that you will be storing the value on, that would be based on those index and it's called like buckets in terms of has map. So you know, you would get your integer value from the has called method, and then you will get the has value that is determined like that, was determined, like, which array it needs to go to, and then key value, like the key value, it will be a key, right? That that would do an equal check to check, like, if your key is already present in the bucket, right? So let's say, like, you know, let's say it goes to your first bucket, and then it goes to that first element. And if, like, you know, if it, if it equals that, then it replaces, it replaces the odd value on the particular index right. And now, you know, like, if it does not exist, then which means, like it was the highest code, like, has called collision. And you know, like that means it just stored in the same bucket in a trend manner, right? So that's what a link list is. So that's how it handles, like the duplicate, the duplicates on key and has code as well. Got

Unknown Speaker 09:53

it cool. Thank you for explaining

Speaker 1 09:56

that. Great. So I'm going to share a diagram with you, and we can sort of try to walk, walk through this together.

Unknown Speaker 10:06

Cool. Let me know if you can see my screen.

Unknown Speaker 10:10

Yes, I can see your screen, but it's little too small. Is

Speaker 1 10:19

this any better? Yeah, it's better now. Okay, cool. So I'll give you basically the context of what's happening here. So assume, like a very simple architecture where we have some users, the users have access to some sort of a UI, so they're logging in, and we're assuming authentication, and those things are taken care of through the UI service. The UI service is kind of like a gateway to the UI, and that's receiving some request, the user request, and then it is, in turn communicating with this back end service called the ledger service. So think of it as like a almost like a bank transaction type of thing, right? So you are a user, and you're trying to withdraw some money, and the UI service then goes to the ledger service. The UI service is doing some validation of your user data, authentication, all that good stuff. It's a simple service. And then it passes the message along to ledger service, and the ledger service will do all of the accounting, right? So when it is doing the accounting, because it will have to make sure, let's say, if you want to withdraw $10 it will have to make sure that you actually have $10 right. And maybe there are some pending transactions. So there's a list of things that it will do, and it will also have to notify sort of other services. So there's a lot of things happening here before we can say your singular action here that you wanted to do is complete, right? So there are multiple steps here that we're doing, and then we have a relational database, let's say so this is the picture, assuming that this is all fine. And, you know, we launched it, we initially had 10 clients, and now we have 100 clients, and we are getting reports that things are a bit slow and we're seeing some performance degradation. So two questions. One is, in that general statement that I've made that okay, things are looking bad and there's been some degradation. How you how do you as an engineer, go about and sort of investigate, like, Okay, what, what is happening? I mean, we'll come back to the solution in a bit. But how would you identify the problem? Yeah, what would be your approach to that? Okay,

Speaker 2 12:25

so on this one, like, I guess first question, it will be like, are you deploying like individual instances for those services, or is it like horizontally skilled services behind the load balances?

Speaker 1 12:40

Right now, there is literally one instance of this, one instance of this, okay,

Unknown Speaker 12:49

so basically,

Unknown Speaker 12:53

on this one, like, you know,

Unknown Speaker 12:56

whenever you have this kind of scenario,

Speaker 2 13:00

like, you know, I would gather, like, what specific operations are degraded, right? So, so like, if the data entry it is degraded, are some kind of report generation or, or it's like, or the portal view, like, which part is experiencing performance issue, and also, like, you know what part of the day, right? So if it's something like during the peak hours where a lot of users, they are active, or during, like, normal conditions throughout the day, so that's the first point. And of course, of course, like the action that I like to try to get to, and then after that I get that I will try to reproduce the issue on like, control side, and, like, even a staging lower environment, right? So I, if I can do this like, I can confirm, like, I can confirm the problem, and then have a motor group on what I will be looking into now, you know, like, like, you know, for each of these individuals, like individual services, the way I will basically go is about getting, detecting, sorry, detecting the errors. That is to measure the time taken for user to hit a request to that service, and like the time it takes, like the time it takes, right? So, basically, the request processing time within the individual services. And then, you know, I would use a spunk logs, or what kind of logging like, what kind of logging tools we have to detect any kind of issues within that, within those services as well, right? I would provide, I would probably be checking the observability, the tool that we have, and we might have, like, implemented some kind of New Relic dashboard, or else, like, Prometheus and, you know, like, whenever we will have, like, we will be looking at that dashboard. And even like, you know, see how throughput, like, how the CPU and how the memory looks like, right? And you know, like, once those are mainly done, I would have, like, some idea on how the application is performing. Now, apart from that, like, I will also be looking at that data, database, right? I will inspect, like, how queries are running, and I will see, like, if the queries are slow, and if there are any way that I can explain, like, I can run the EXPLAIN query, basically to get the execution step of certain queries. So, yeah, like, that's how I would, like, that's how I would start by debugging these performance issues that we have.

Speaker 1 15:41

Yeah, great. So I think, I think, thank you for explaining. So I think that makes sense. So what we're talking about is there should be some sort of metrics that we need to look at to essentially understand what the systems are doing and how they're performing, right? So that makes sense, some sort of metrics, and observability tracing and sort of, yeah. So that's, that's good. And generally there are three things that we can look into. Let's say there's three, I guess, dimensions of like things that can be an issue. So we have CPU or compute, which is usually not the bottleneck. Then you have IO bound operations, and then you also have, you know, it could be IO or CPU intensive operations, or it could also be memory constraint, right? Like we could have memory issues. So of the three things that I mentioned, io, memory and CPU. Or if you which ones, do you think in this scenario? Right? Obviously, you don't have all of the information about the system, but just basically from what I've explained, which ones you think might be the issue here. Do you think it is the CPU cycles that the compute capacity is filled up? Or do you think it's IO or do you think it's memory usage? So

Speaker 2 16:53

on this one, I think, like since the issue started, like after. So, you know, so, so, since the SEO started after, like, you have increasing users, right? So it was, like, not something that you saw earlier, but like, but you started seeing the issue, right? And when you basically increase the number of user, I think, like, it's something related to, related to the IO issues, because that is, like, very normal, right? Because you have like, frequent database access. And also, like, you know, if you are hearing like your database a lot, then, like, you know, this, this, like, it's the hits, they are expensive, right? And then, like, it causes, like, a lot of latency over time. And if you have like, more and more of requests coming in, then it just means that your responses, but each API call for addressing through one services, it will be more right. And I'm thinking, like, more often, like I have bottleneck here rather than the other ones. But again, like, you know, CPU bottleneck, it can also be, it can also be the next one that would be that we would be looking at.

Speaker 1 18:11

Okay, so let's say, you know, you've mentioned about IO, you've met, you know, briefly about Compute. What about memory? Would there potentially be some memory issues as well, like memory links and stuff like that. And if so, like, you know, yeah, like, how

Unknown Speaker 18:25

do we even check

Speaker 2 18:28

that? So for the memories issue, like, you know, basically memory issue like, it occurs whenever you are holding a large amount of data in memory, like, for you per user, right? So if you are like casing a large data, you have deficiency, like you might have deficiency in the memory allocation, and it may temporarily cause spikes. But again, like if you have proper resource closures, then you probably will not have those memory issues. Like, if your objects, they are not properly released after request. And also, like you know, it will grow. It will grow until, like all the only garbage collector crisis. But like we will see signs of memory bottleneck by looking at it like the out of memory errors. Okay. Cool,

Speaker 1 19:22

great. So let's say now we have identified the problem, and we have found that, okay, this is causing an issue with let's, let's just go with IO that, you know, there's a lot of, like, frequent database access, and in general, there's just, like, a lot of concurrent users here, and that's causing some issue. In general, this instance alone is not as is, is not sufficient, right? It's not this UI service is fine, but then, you know, and we can also replicate the users the UI service, because they're pretty stateless. You know, user comes in, they make a request, but we have, you know, this one now, what are our options with the ledger service? What can we do to cope with this increased load.

Speaker 2 20:04

So, for the increased load, basically, you know, so for the like, for the particular service, like, you know, if you want to, mainly, you know, like, improve the performance here, like, you know, we would scale this, this, like, this instance, as personally, like I mentioned before, like, you know, like my first thing we just add, like, more laser services behind a load like, behind a load balancer, so it will be a better utilization of the system resources. And, you know, like, so, so, like, you know, we, you would have a load balancer sitting on top that will distribute these calls to each of the laser services. So, you know, like, one bottleneck here would be, like increasing the load on database due to more service instances making the request where we can, you know, like, like, we can make changes to the database to be shared. Or maybe, you know, we have, like, maybe even have been, like, read from the read replicas.

And then, like, our last service itself, like, we would introduce our casing to case, our frequently access data to reduce the database hits, and then, you know, we would do like, batch processing of the data to combine our small database into one, like one big batch operations to reduce the IO overhead. So that's the thing that I'm thinking for now through,

Speaker 1 21:38

okay, yeah, those are all very good considerations. I think we'll stop here, since we are kind of short on time. But that's and again, with these things, obviously there's a lot to discuss. There's there's no this. 30 minutes is obviously not sufficient, so. But again, thank you for going through that with me before we end. I think we have about four minutes. Five minutes. Do you have any questions? Anything that I can clarify before we go,

Speaker 2 22:00

Yes, I do have few questions. But before that, like, I would like to thank you so much for your time today. And I really, really appreciate you, like, sitting down and talking with me and like, for going to the course since, like, I would like to know what you feel like it is, what is the best thing for you working at galaxy,

Speaker 1 22:22

right? So for Galaxy, I think the thing that I enjoy the most is it's like when I joined three years ago, it was a fairly small company, 120 people. Now it's 500 plus people. So it has definitely grown a lot. And then it's also, there's just a lot to do. And I think as an engineer, there's a lot to build, and there's a lot of impact you can make. So definitely, if you're somebody who's interested into building large scale solutions and having a you know, because I used to be in a big beer organization before this, and it's a different dynamic over there. It was a lot more specialized and siloed. And in some sense it was good, but in some sense it was also like, I felt a bit constrained if I wanted to do something that was, yeah, it was just like, your role is very, very, very specifically defined, whereas in galaxy, I think it allows you to really grow horizontally and then, of course, grow as a specialist. So

Unknown Speaker 23:22

that's the thing that I enjoy the most. Awesome.

Speaker 2 23:23

That's really nice to hear. And one more thing, like, I want to know, what does the day to day looks like in the developer life? Working at galaxy?

Speaker 1 23:33

Yeah, sure. So I'm currently a dead lead for the custody detect team. Before that, I was a developer here for like, about two years, and generally, you know, we have, like, a scrum in the morning, and then people would have their Yeah, we try to follow the agile methodology, so it's pretty, I would say, pretty standard across the board. So yeah, most of the teams are four, five people in size, so fairly small team. We do have a global footprint. So we have people in Hong Kong and London and a few other locations as well. So yeah, most of the days, I'd say meetings, not that many. We try to minimize meetings. But of course, if it's a new project, there's a lot of discussion to be had. So we wanted to make sure you don't just go away, because most of our developers, we don't look at them as like task papers we look at them as kind of.

Shreesha Khadka- Galaxy- Srv-Final

Unknown Speaker 00:12

The connection is still,

Unknown Speaker 00:16

can you hear me? Because

Speaker 1 00:18

I hear most topic, so you work as a both front end and sometimes back end, right? Like, which part you are more comfortable. So, like, you know,

Unknown Speaker 00:29

I'm working as a full stack Java developer, and

Speaker 2 00:32

I'm working like 70% on back inside and 30%

Unknown Speaker 00:35

on front end side.

Speaker 1 00:38

Okay, so you're very comfortable with Spring Boot and race micro services, right? Yes. So can you please tell me how you will design a system which actually our system gets lots of connectivity exchange data, like we have different crypto exchange Right? Like, like Coinbase and different crypto exchanges, so we get data pulled every minutes from proposed for different account and store it in our database. So how can you develop that kind of system with

Unknown Speaker 01:14

and which technology you want to use that so on this one,

Speaker 2 01:19

like walking about the technologies. Basically, like, you know, if your system, it is designed on a pool based data integration layer, you know, like, I would basically use some kind of messaging queue, right, like, you know, you know, like, to handle the high frequency data pools. And even, like, you know, the couple the producers from the downstream systems. So basically, you know, I would use this like data ingestion layer with a message queue that will be able to handle the high frequency data pools. And, you know, like, there will be a data processing layer, which you may be like, you may using, you may be using a streaming framework, which may be like Apache, Apache stream, and also like, you know, using Apache Spark for our real time data transformations. And, you know, like, we can use like relational database as well. So if we have a structured data or non relational database, that if we need a, like, very high throughput, but like, you know, eventual consistency is fine, right? And you know, like all of these applications that I talked about, it will be like, it will be behind our load balances so that the request, it is distributed across the different layers, with horizontally scaling. And also, like, you know, we just mean, like, standing up instances when there's a lot of requests coming in, but, but, yeah, like, the the way the data flow works is that from the data source, we would have a messaging queue that will be picked up by like, you know, our processing service, which Is the streaming service using like that is using Apache Spark. So, you know, like, from the Apache Spark you would do, like real time data manipulations, which can be, you know, mainly posted on our database, that is by using like Apache Spark as SQL. So, yeah, that's kind of like my throughput like thought process at the moment.

Speaker 1 03:23

So you mentioned about message processing queue. So what kind of queue you are you can use,

Speaker 2 03:32

well, like, on masses process queue, like, you know, we can, like, either use Apache Kafka, or we can also just use like Amazon, like Amazon, SQS.

Speaker 1 03:45

Okay, so how can you ensure that when you are getting data and stored in database, if some reason, some duplicate data comes in, how you can ensure that it is not stored twice?

Unknown Speaker 04:00

So Well, on

Speaker 2 04:01

this one, like, you know, when they duplicate data comes in, so basically, you know, in order to make sure that, like, you don't have duplicate data, you know, like, you can add your data basically like database constraints, right, which is just like the primary, primary keys, and the constraints are like and unique constraints are enforced with the like, you know, database label, but also like Butch spark it. It allows you to add those logic and one of the process like that you can implement. Like that you can implement. So for that. Like, you know, you can do, like, some kind of Mars logic, or even update logic and then a write logic. But yeah, like, you can easily add database constraints for uniqueness and also at that particular, like, particular duplicate logic within the APAC spark process that we have,

Unknown Speaker 05:04

okay, yeah.

Speaker 1 05:07

And so if you're, if you're pulling data and storing in database, and some user is seeing it in the UI, right? And if that complaints to you that, okay, my data is not getting updated. This data, it looks wrong, right? So, how could you approach finding the issue?

Unknown Speaker 05:28

Basically, you know, sorry, like,

Unknown Speaker 05:33

the issue the like, you know,

Speaker 2 05:36

that's the issue, mainly with the consistency, like, which might be a trade off because of the high throughput right, that we are sending. So if you know, like, if you are using like Cassandra database, it works on like a circular node, like circular node based pattern with eventual consistency, right? Which makes that the like serving of the request might give you are still data, but it won't, like, it won't take too long, right? Like, it won't be too long. So, you know, like, it's an eventual consistency that you are seeing. And basically, like, you know, in order to solve it, we might have to consider consistency over like availability and even like partition tolerance. But, you know, usually it does not last that long. But if we think that consistency, it is very important that we will think of something like a relational database with a read only replica. So this way, like, you know, if you are guaranteed, like, if you are, if you are guaranteed consistency, because of the most fullest and database, they do provide you consistency and availability as well.

Speaker 1 06:53

But whatever you do, like, sometimes is still still, can UI can show wrong data right? So what can be the reason? Like, if you give your tone to debug that, most of the time, the system is working fine, but this particular data looks wrong, so how you would try to find the problem?

Speaker 2 07:11

Okay? So basically, on this one, like, you know, like, your resiliency and like, like, like, like, usually for resiliency and better performance. Like, there is a casting layer that is involved, like, which stores data for performance, right? So that would be my first thing to go to, like, check, like, if the cast was evicted and new data it was populated within the test. So, yeah, like, I would see, like, if the casing it needs to be, like, revisited, and like, you know how we are pulling the data into the CAS. But you know, whenever there's, like, some kind of concurrency is used, then you know, whenever multiple users try to write and like concurrently, like, you know, then the older data, it overrides our new updates. So we might have to think about, like, how we handle concurrent So, like concurrent rights, or else, like, you know, if there is, like, any kind of issues within the read replica itself. So sometimes, like, you know, a read replica that is serving your applications, it might not even be updated in time, because, you know, because there will be, like, some issues within the network itself, right? So I look into those and more over on the most like, on the more programming side, I will see that if the query that we are using, like that we are using for extracting the data that is correct. So because, like, you know, you might have wrong, wrong kind of query for data source, like using bad filters. So yeah, like, these are the couple of things that I would look into. Okay,

Speaker 1 09:00

if you notice that your application is like, memory wise, it is growing like,

Unknown Speaker 09:06

so how do you try to find the problem? Well,

Speaker 2 09:13

like, to find the problem like, basically like, you know, I would look into logs for like, for out of memory error, because, basically, like, you know, that means just that, like your application, it cannot perform anymore, like, because your memory it is full. And in order to, like, basically, you know, figure out memory issues, you need, like, some kind of profiling tool, like visual VM, that can, you know, basically trace, like, what kind of memory allocation it is within the applications, right? So, like, we would also look at any kind of observability tool that we like, that we may have in place for monitoring the garbage collection activity, and if possible, like, we can even log into our instances and then extract the head dump, which we can then export, export into like a visual VM or a visual ID to see like what is holding onto the memory. Usually like it is bad to the like it is it is due to like bad practices for a connection management or resources allocation. And the visualizer, like they usually source like they usually shows you where the memory is being used all out, right? And then, like it is mainly based on that. Like we go about debugging. And usually some of the issues that arise whenever, like, we have open connections to your connections pool that you are not even closing, or you are basically storing, like, large amount of data in strings and list and even, like, basically clearing it. So, yeah, like, these are some of the things that you can that you need to look into, right?

Speaker 1 10:57

Yeah. So can you please tell me about one of your recent problem or challenges you faced in your current project.

Speaker 2 11:05

So Well, talking about the recent problem, like recent challenges that I faced, basically like, you know, we were developing our new feature to integrate to a streaming engine. So, you know, like, like the stream and even, like, aggregate the bunch of the logs that is related to user data, and, you know, stream it to a snowflake table. It was like, you know, mainly an enterprise wide backlog item that, you know, we had to work on. And then, in order to work on this like, you know, we use something called a hist command, and now, you know, like, like,

Unknown Speaker 11:48

his tricks command, sorry, and you know, like,

Speaker 2 11:52

the way that his command work, it is that the abstractions on top of executor service so that you can concurrently on, like, you can work concurrently on the multiple process. Now, you know, whenever we integrate this, we are having our issues for like, the first hour, and then after about an hour or two hours, like, based on the load of our applications, like our application, it would start becoming on. It will start becoming unresponsive. I like, like, this was, like, something that I had to debug. So the way that I want, like, I went about debugging now this was, like, we basically reduced the connection pool and all of the sizes. So for the threats to one like that, I did, I did a load test that to try to figure out, like, where the bottleneck was, and then I was able to see the history command, like, that was like, that was being used in, like, incorrectly, without within our applications. Like, you know, when we wait, we are trying to use our string dependency, but like, never injected it properly, and we are like getting, not like we are getting, and even, like not pointing an expectation that was not being handled. And then, like, you know, like, which, like, it's basically held onto the connection, and even drain the resources within our applications, causing resources to be used by our memory. So, like, doing this, it would be, like training. It would be training. So, like, this was, like, you know, one of the most challenging issue that I had to work into, because, like, you know, the no pointer expectation, it did not generate any logs as it was being, like, swallowed internally. And you know, like it was not causing the resources to be closed.

Unknown Speaker 13:46

Okay, yeah, and that's interesting.

Speaker 1 13:51

So what about have you faced any issues working in a team which is like, geographically different places. So what kind of challenges you can face.

Speaker 2 14:06

So, you know, working basically like, whenever you are collaborating within the team members that are, like, located on different parts of the country, or like, let's say even the world, right, like you are connecting with Officer team. So one of the things that usually challenging it is setting up our meeting that, like the time that works for everyone. Like, you know, on my previous project, we had teams from Europe, and then, like, you know, teams from from the like, from the other end of the United States too. And you know, like setting up a meeting for all these, like three teams like to work together. It was, like, quite challenging.

Unknown Speaker 14:47

And you know, like,

Speaker 2 14:48

I had an issue that it that, like entailed both the teams from India and Europe. And also, like, you know, we always had to find out, like, What perfect timing works for all of us, and also like, you know, that I would say, like, that's one of the most challenging, like, most challenging issue that we have to face. And also, like, you know, but I think that, like, you know, through transparency and then consider it, consider it. Like, some of the time differences that we have, like, we can easily overcome this effectively by using the time that we have, and even, like, you know, setting those time apart. Because, like, you know, we we need to utilize the time right that we have, like, properly and like we have properly working together. Like, you know, when we work together with the perfect time that I would say that it becomes more

Unknown Speaker 15:39

easy. Mm, yeah,

Speaker 1 15:45

so have you like, how, like, how do you you don't have much experience in trading or crypto, right?

Speaker 2 15:55

So trading and crypto, like, you know, mainly, like, my domain knowledge, knowledge, it has been more on financial side and then than the investor management. So, yeah. Like, I do have knowledge morally, on financial endowments.

Speaker 1 16:13

So, but, but, like, Why? Why you are interested in crypto development

Speaker 2 16:19

related development, yeah. Like, that's really the nice question. So, like, you know, I think, like the crypto, like the crypto sites, I think like it creates a one of the more it is, like one of the most, fastest growing field, and, you know, like, how it is in train right now, right? And I think, like it is one of the most cut, like, cutting edge technologies, because it offers a, like, lot of innovative solutions that I want to be a part of, and it aligns with the future of financial systems, right? And, like, talking about my experience, like, you know, I have pretty good experience with financial domain, and then crypto, it is the next step for financing, right? So, like, I think, like, I want to get involved in designing this innovative solution, and then also, like, you know, having contribution for like, those cutting edge challenges.

Speaker 1 17:10

Yeah, definitely, yeah. I think I'm good with my side. So can you, like, if you have any questions we get, yeah, like, we have four, five minutes

Speaker 2 17:22

left. Yes, sure. So yeah. Like, I have few questions, but before that, thank

Unknown Speaker 17:26

you so much for your time today.

Speaker 2 17:28

It was really, really nice talking to you. And one of the thing that I would like to know is, like, what do you like about the most working on working in galaxy?

Speaker 1 17:39

So galaxies like, like, still has that startup nature. It's like, it's not a big company, not not talking about financially, financially, it is becoming bigger every day. But like, we don't have that many people like big companies, right? So we still have the startup culture, so very collaborative environment. People always help out each other. When you are stuck, you can approach anybody, and people will help you. So those quality, like, mainly collaborative nature. I like working

Unknown Speaker 18:13

here, okay, okay,

Speaker 2 18:15

and I want to know, like, what does the day to day looks like in the developer life working in galaxy.

Speaker 1 18:22

So day to day It depends. Like, crypto industry is very like dynamic. Every, every day, something is changing. So you need to be a little bit flexible to work in galaxy. Like, sometimes requirement change all of a sudden, because, because the nature is like that a crypto Right? Very dynamic, and day to day life would be. We also follow agile spam, so that work is discussed for the day there and status update, and then you work on your own. And if you're stuck, we can have like meeting with leads or seniors, and we also have a product manager or product owner, so yeah, like for developer, it would be like, it starts with Scrum and then the development, and sometimes we need to do production support, Also, like Prota basis, oh, yeah, I think,

Speaker 2 19:22

okay, that's really nice to hear like, and I want to know like, what will be the next step in the hiring process.

Speaker 1 19:30

So next, I think there will be one final round, but, but our HR will let you know

Unknown Speaker 19:42

our Yeah, the HR manager will let you know.

Speaker 2 19:45

Oh okay, okay, that sounds good. Yeah, that's it from my side, and thank you so much for your time and like for sill yeah thank you

Speaker 1 19:54

guys. Yeah. Okay. Have a good day. Thank you.

Shreesha Khadka- Galaxy- Srv-First

Speaker 1 00:00

Tell me about as far as, like, the kind of role you're looking for, as far as you know what kind of technologies or the demand that you kind of want to work in after that, we can do, like, a little bit of a deep dive into your resume. You can tell me as little or as much as you want about your past projects and positions, and at the end, I will leave it open for you to ask me any questions about galaxy, sure. Okay, that sounds good.

Speaker 2 00:26

Yeah. So hi, my name is srisa Kalka, and I'm working as a Java full stack developer for almost like five years now. So I'm primarily working on a Java based web application utilizing frameworks like Spring, Hibernate and rest web services for API development, I have been primarily working with AWS Cloud for hosting this application. So like some of the services that I have worked with include AWS EC two and ECS, SQS, SNS, lambdas and s3, buckets, I, and I do have experience working on the front end side too, and there I'm using Angular. Currently at Citi Group, I have been working on, you know, like service based architecture, and also been a handling orchestrator for those microservices, and been on a part of our scrum team. So, yeah, the overall architecture of the project and city group has been all like, Spring Boot based application deployed on AWS EC two. And like, you know, it is easy to like, I have been like, you know, kind of some subject matter expert of the expert of the orchestration and later, and also been, you know, like developing different functions and filter factories along with the like, along with adding different endpoints. So like, you know, to talk about what I am looking for in my upcoming role is another software engineering role that would be like, you know, little bit back, like, little bit back and heavy, but, like, no, but I don't mind, like, working on front end work as well. So, yeah, that's something that I'm looking for.

Speaker 1 02:13

Okay, great, yeah, we're definitely, yeah, I think the role that we have kind of open right now, yeah, it's definitely, it's primarily back end focused. There are some, I guess, you might have, every once awhile, maybe do, like, a little bit of front end work. But having someone who might know, like a little could help maybe do a little bit of front end work might actually be pretty useful, just because, like, a lot of the times when we're doing kind of some reconciliation stuff, we have, like, a lot of back end tools, but it would be good to maybe have someone who's like, a little bit of front end experience, just for maybe helping us stand up some small front end UI, just for kind of, maybe handling some back end operations for our operations team.

Speaker 2 02:46

Okay, yeah, that sounds good. Like I have been were about like 7030 split on back end and fronted. So I have been like, I have pretty good knowledge on working with Angular two. So it's not that I'm just an internal engineer.

Speaker 1 03:01

Okay, perfect, I guess. Can you tell me a little bit, I guess, more about your, I think, experience working, I guess, at your current role with Old Dominion, right mine.

Unknown Speaker 03:13

So can you repeat that one more time? Yeah, sure. Sorry.

Speaker 1 03:16

Can you tell me a little bit about, more about, I guess, like your current position that you're at, and kind of, some of the projects and what the positions kind of entailed,

Speaker 2 03:27

yes, sure. So like, currently at Old Dominion freightline, you know, like, I'm working there since May 2024, and, you know, like I'm working there as a full stack, you know, full stack, Java, DPP, so, you know, like, they're like, We are more heavily focused on banking and so basically, like, you know, we are handling, like, the organization, Lea, basically to mainly organize the shipment details. So, you know, like, we have different micro services exposed to like, manage, manage those of the shipments. And, you know, one of the end points that I have recently worked on are one of the services like that I work from scratch was called the maintenance API, which is primarily working for making changes to data on preferences, like you know how you are tracking detail that mainly needs to be applied. And also, like, you know what kind of widgets you want to view like you want to view like some other API that I worked on includes the like, you know, Experience API, which is basically used for entitlements on different components to be mainly loaded on the front end side and, you know, like on the basic orchestrator app that has been primarily based on a Netflix jewel library that, you know, we are modernizing to, like you know that we are modernizing to Spring Cloud gateway. So, like all of these applications, they are containerized through Docker, and then, you know, like, deployed to AWS ECS, and even, like, we use our Splunk for logging and New Relic for monitoring. So it's a very like Java. Is spring like, like, spring heavy applications with a front end component and for self servicing on the Angular side.

Speaker 1 05:23

Okay, okay, so you, so I think for this one, you said you did it from scratch, and it was back end was made mainly spring, and then you even did a little bit of the front end

Speaker 2 05:32

work too, yes. So on that one, like, you know, like, I'm working there as a full stack Java developer, so yeah, primarily in the Spring Boot like in back in we are working on the Spring Boot based applications. And also, like, you know, I design, and also, like you know, it also gathered the functional and non functional requirements then. Also, like, you know, implemented then, and I even deployed them into the AWS ECS. And also, like, you know, I have worked on the front end side too, for self service component here as well.

Speaker 1 06:06

Okay, and so, I guess, for the back end, what sort of database did you guys

Speaker 2 06:10

use? Yes, so mainly for the back end side, like, you know, like, we mainly use, like, Postgres SQL as our relational database, okay, and then, like, you know, we use, we also use, like, DynamoDB on like AWS for one of our experience APIs.

Speaker 1 06:29

Okay, nice. So maybe, I guess, maybe, like, circle back to spring a little bit. So can you tell me a little bit about spring, as far as, like, why it's popular and, like, why, like, a lot of developed, like, specifically Java developers kind of choose to use it, and why it's kind of been adopted in, like, a lot of organizations, kind of throughout the

Speaker 2 06:48

world. So, yeah, like talking about spring, like, basically, you know, like spring, it's just a framework that, you know, makes working with the web applications easy, right? So mainly, like spring, it is a lightweight and also provides a lot of modular features, which basically, like you know, simplifies your Java development. And also, like you know, it provides features like dependency injection, which is for simplifying the object dependencies and even promoting the loose coupling. And then, like it has support for different frameworks like spring MVC and Spring Data and Spring Boot, which makes working with like Microsoft, which makes working with microservices easy. So if you go like a little further with a Spring Boot, like, you know, it provides features of auto configuration, which removes a lot of like, you know, boiler plate code, and also like, you know, you can just use it like we even just use annotation for connecting these services and modules to dependencies injections, basically by the component scanning during, like, you know, during if You have your dependency in the class path, like class path, right? So there are, like, a lot of those primary features that makes our life as a Java developer easier, and that's why we basically choose it.

Speaker 1 08:15

Okay, you mentioned dependency inject. I think you touched on a little bit about what it is, but could you kind of go into a little bit more and kind of, Why, what's, you know, why we kind of use that pattern. And then to follow that up, could you also tell me a couple ways that it's, like, implemented in spring? Yeah, sure. So,

Speaker 2 08:32

like, the dependent injection, like, you know, basically dependential injection. It's just like a patch and right, like you mentioned, like, which is used to create your dependencies from an external source. So, you know, like, instead of creating those of like, creating the objects our services using the like, new keyword, and even like, you know, you let your spring container provide the dependencies of the objects that you need, right and, like, you know, basically there is a concept of inversion of control which handles this in, like, dependency injections through our spring containers. So, you know, like, it is very popular because, like, it makes your code modular, and then it helps in subtracting those concepts right now, so and like, some of the ways that you can inject in like independence, like you can inject our dependencies in spring is through constructor injections, and even field injection and secular injections. So talking personally, like, I personally prefer constructor injection, because, like, you know, it makes your code clear, and then also it also just provides all of the dependencies that you need. And also, like, you know, but for like, more optional dependencies you may want to use, like a settler injection,

Speaker 1 09:58

yep. Okay, great. I think also earlier you mentioned the MVC pattern. Could you tell me a little bit more about that and kind of how it's implemented in spring as well, sure.

Speaker 2 10:07

So basically, like, you know, MVC it is like MVC pattern. It is mainly called as model view controller pattern, right? So theory, like you separate different components into our module, like which represents the applications data and business logic, and also like your view, which is the user interface that you know, displays the data to the user, and then the controller, which is the intermediary between the model and the view, right? So, you know, like, basically you have a separate component for user interaction, or separate controller for doing the back end and business logic, and then also, like, you know, and then a separate component for doing the interactions between those in spring and like, your model is done through, like the through, like your user class and services and also your controller. Here it is annotated with the controller annotation. And then, like, you know, your views are typical templates, or sometimes, you know, like you can also take the JSON, like just gesture responses as the view, mainly like, if you are working with the rest word services.

Speaker 1 11:27

Okay, yep, perfect, nailed it. I think so. I see also, while you were at Old Dominion the very top, you mentioned developed and maintained or undertook significant refactoring effort to transition a legacy model with application into microservice. Can you maybe tell me a little bit about that experience, as far as, like, what the original, like, you know, monolithic application was like, and then how you guys decided to kind of decompose it and to different microservices? Yeah,

Unknown Speaker 11:53

sure. So, you know,

Speaker 2 11:57

mainly, like, basically, you know, like, I have basically on this, like, I have stripped out multiple pieces of this monolithic app to microservices, like, like, you know, like I mentioned, the mental maintenance API and Microsoft and also, like Experience API. So those were like two of the flows that existed on the more like monolithic that more like that, that, you know, helped in creating micro services. So the old monolithic app was a huge code but like use code base where all the services were tightly coupled. And even, like, you know, everything was within the same service, right? So, you know, in order to deploy our applications, we had to, like, tear down the whole services. And like, you know, like, tear tear down the whole services, and then, like, you know, all the changes it would be deployed as a whole, where you didn't have to, like, you know, when you didn't have have control of what services were deployed in what of the manner, right? So in order to separate those services that may be useful for other components and other teams, we were basically like, you know, we basically we're doing a modernization effort. And for this, I I would like, I was tasked some of the services on the user profile view and also like, you know, it even contains like, like, it even contain different components, like the maintenance of the user profile preferences and even the different results that was visible to the user, and to make sure that, like you know, other teams, they were able to use those features and also, like you know, other components, so they're like, you know, we first were like, we first were went with the

Experience API, Which is basically mainly used to grab the different entitlements for users. And like, you know, those entitlements dictate, like, what components is visible to the user, and then later, like, you know, we also went to the maintenance API to make change that like, to make change to that user profile preferences. So those are, like, you know, those are the two primary ones that I migrated recently.

Speaker 1 14:28

Okay, I also shared see here it says you have some experience working with Kafka. Can you tell me a little bit about

Speaker 2 14:34

that? Yeah, sure. So, like talking about Kafka. So, like, one of it is our, like, one of our, one of our workflows. So they're like, you know, we send a notification out to users through an email. And like, you know, on the legacy code, and like, you know, the notification service, it was baked into our single applications. And you know, like, this would cause an issue, because whenever the email service was down, we would be, like, falling the we would be failing the flow. And even, like, you know, the transactions that happen for like, for that call, what we get roll back, right? So, like, in order to prevent that, we basically used a Kafka topic where, you know, the request processing service would send this request, send the request message through a Kafka topic, and then, you know, like, we created our consumer deal, and these consumers, like, they would read that message, and then even, like, you know, call the notification service or the email service. So this like, so this basically it, you know, like, decoupled that notification service to our primary applications. So the transactions are like, the transaction would not get rolled back and the consumers it would, but they would like, basically be processing those that message is to send an email notifications, mainly to out to the users. And in case of notification failures, like, you know, we added a retry mechanism with exponential back off in order to make the more like, make the code more resilient. And in case of like, failure after three of the atoms, we had a dedicated, like, dedicated day later queue that we could reprocess and, like, reprocess these messages at a later time. So yeah, that is what we used for like, that is what we use Kafka for you. Okay, great.

Speaker 1 16:40

And then I guess do you have any other experience? With any other like NoSQL databases,

Speaker 2 16:47

so talking on like a nose or SQL database? So as far the experience like so, mainly for the Experience API, we used Dynamo DB design and model and like that, handle entitlement, suggestion for it,

Speaker 1 17:05

okay, I'd be curious to hear a little bit more about what the what your build pipeline was, I guess, as far as, like, Were you guys using, like, Jenkins and stuff, and I guess, kind of how it went from, you know, COVID being pushed to actually deployed in EC two. So,

Speaker 2 17:25

so, you know, like, basically, for easy to, like, we had an external team for that. And like the external team, they handled the code pipeline through NC bill and, like, we had it through Jenkins too. But like, you know, for more of the modern applications, we handled, like all the deployment and we had to also run our Jenkins script. So, you know, like, we basically had the groovy script that we could write, like, where we would define different stages for the deployment, like we we would also write the build scripts. And even like, you know, deploy the scripts, which were basically sell scripts. So, you know, like these three, it interacted together. So groovy script can define the stages, build, like, build. And even like, you know, deploy the scripts, basically on how to like, like, how the applications would be deployed internally. And, you know, like, we would like, we would even use the cloud formation templates, but this would always be like, you know, this would also be orchestrated through Jenkins pipeline. So, yeah, that is our deployment deployment strategy. So, easy.

Speaker 1 18:43

Okay, great. Um, I guess, could you maybe tell me a little bit about your, I guess, experience at Citigroup, just be kind of curious here more, just because, you know, in the financial sector. So, yes, sure.

Speaker 2 18:56

So at Citigroup, like, you know, like there also I was mainly working as a Java full stack developer. So, you know, like, I have been primarily working there as a, like, I had been primarily working there as a micro services, micro service gateway kind of layer. So, like, you know, there we had multiple routes that were set behind. And, like you know, this, multiple routes were either API or like gateway or micro services. And our app, it was like it was the first line of defense against this, like against this, so you know, like our app, it was Spring Cloud gateway where users could register their rules. And even like you know, based on this, like, based on these rules, we would forward our request. And also, like, we write functions and filter the batteries. Like you know, we wrote filter for filter for validation, like validating the device version and the OS versions. And even, like, you know, we had functions for encryption, decryption and so on. We even had all, like, Spring Boot application and again, like, deployed it through AWS cloud. So yeah, like, that was, like, pretty much experience with the city group.

Speaker 1 20:18

Okay, great. Um, I guess now leave it open for you to ask me any questions about galaxy. Yeah,

Speaker 2 20:24

sure. So I want to know, like, what does the work culture looks like in like a galaxy?

Speaker 1 20:35

So I guess I would say, like, we're not, we're not quite like a like, again, like a full size like financial institution, like, it's like a city, or like a Goldman or something like that. But the same time, we're not, like, we're a little bit bigger than, like a startup. So I'd say, like, we're almost like a mix between, like a startup and a, you know, your traditional financial institutions. But I kind of like it because, I guess previously, before I started, I was at galaxy, I was at do what you think. So it's good because, I guess, you know, because our, like the teams, the tech team smaller, so you you know, things change a lot in the crypto industry. It's nice because we get to directly work with a lot of stakeholders, so traders, operations, team, audit and stuff like that. So it's kind of nice being able to actually talk to them and get feedback, and kind of get, like, a lot, you know, more domain knowledge and kind of understanding of the industry versus, I know sometimes when you're at bigger kind of institutions, just because they're so large, kind of your responsibilities are kind of more focused on maybe doing tech stuff, or you're kind of far removed from the actual people that you're building stuff out for. So I like it in that sense too. I guess also, just because, a little bit like a medium to small size company, it is nice because we do have the kind of the flexibility to kind of try new stuff and try new technologies, or, you know, say you really want, I know, like one of my co workers, he was really interested in trying to learn go so about maybe a year and a half ago, kind of, we kind of came to an agreement with one of our stakeholders to that we needed to build something out. And so we decided to eat, well, he did it primarily in Go. So it's kind of nice being able to not have to, you know, fully stick to, you know, a prescribed set of technologies and framework and stuff like that. I mean, we are primarily, you know, or at least the position, it's mostly Java, Python, SQL, and then maybe a little delay stripping here and there. But it is nice, just because we have, you know, the opportunity to try and do stuff in, you know, you can kind of put yourself to learn. And then, Overall, I'd say, Yeah, we're, I think, kind of, as we've grown, we've gotten definitely, like, a little bit more, trying to think of this way to say this, like, look more stricter, I guess, with our agile practices, which is good, I think, kind of when you I guess the nice thing is, like, when you're a smaller company, you can kind of react to stuff quicker and push out features a lot faster. But obviously, as you're, you know, coordinated grows and you become responsible for stuff, you need to have a little bit of, like, more structure. So it's good that we've kind of developed that, like, a little lot better over time. And, you know, even though, like, it's kind of like, not the most fun thing when you're at a bigger company of having to go through a bunch of red tape and sign off and validation stuff like that, it's good that we put those kind of safeguards in, because I think it's really like, up the quality of our work. So yeah, I guess that that'd be kind of a quick way to describe, like, what the culture is like here. That

Speaker 2 23:26

sounds cool. And like, thank you so much for the opportunity today. Like, I had really a good time talking with you. And lastly, like, I wanted to know more, like, what will be the next step of the process?

Speaker 1 23:38

Um, so I think you have another interview on, I think, Is it Monday or Okay, yeah, I think you're interviewing with my coworker, Praveen. I think he does. He thinks his is longer, so his will probably, it'll be a little more technical focus, kind of like on, like core, Java and stuff like that. So, yeah, just take a chance to brush up over the weekend stuff like that. But I'm sure you have been, if you've been kind of interviewing and stuff. So I think, you would interview with him. I'll submit my feedback to the recruiters after this, and then, you know, I guess, based off of how things go with him, I think there maybe might be one more interview after that, possibly, like, with our like, I guess, or not, the tech like one of the tech work leagues, I guess. And then I think that one will probably be not super technical, probably be more kind of conversational like this. But obviously it'll probably, you know, try and figure a little bit more about your technical expertise and stuff like that. And so I think that would, I think that would probably be the like the last interview, but they they might throw in maybe one or two more after your interview with Praveen, but yeah, I think that's probably what the next steps

Speaker 2 24:46

will be. Okay, that sounds good. Yeah,

Speaker 1 24:49

okay. Well, if there's been great talking with the day, and I'll be sure to get my feedback to the recruiter and stuff,

Speaker 2 24:55

thank you. It was great talking with you as well, and have a safe journey to Hong Kong. Enjoy your trip in the

Shreesha Khadka- Galaxy- Srv-Second

Unknown Speaker 00:00

Working remotely.

Unknown Speaker 00:00

So I can repeat that one more time, I'm

Speaker 1 00:02

not able to hear who is properly based now. I mean, are you working North Carolina?

Speaker 2 00:07

Yeah, no. Like, I'm based in Fairfax, Virginia, and I'm working remotely.

Unknown Speaker 00:12

You're working remotely?

Unknown Speaker 00:13

Okay,

Speaker 1 00:16

let's say our office is in New York, right? Are you

Unknown Speaker 00:19

expected to work from New York, right? You're

Speaker 2 00:26

Yes, like, I can commute to New York. I used to live in New York before, like, my parents, they live in New York, and I just moved to Virginia, like, for this, for two to three months with my friends. But like, if I get opportunity in New York, then it would be best, like, I can stay with my parents. So,

Speaker 1 00:41

okay, okay, so, yeah, that's good. Tell me what is the most recent thing I had with the development experience, and

Unknown Speaker 00:54

I think you were working with

Unknown Speaker 00:55

the whole Dominion cycle, yes. Okay,

Speaker 1 01:02

yes. So can you just tell me about your products? Sure.

Speaker 2 01:05

So right now, I'm working, like at Old luminantrelian as a Java full stack developer, and I'm working on multiple micro services, you know, like I have helped in our development of a couple of recent, like, recent micro services, including the maintenance and experience API. So, you know, basically the whole effort was migration effort from legacy application to micro services. And, you know, like, I have, I have taken in charge of this, like, you know, stripping all of the services, and then, you know, creating microservice, designing them, developing them, and also, you know, helping deploying them into AWS clouds. So, yeah, that has been my primary, primary role. So what

Speaker 1 01:55

was the legacy application originally like, what was it developed with? So, tell me the overall like, what is the tool? What

Unknown Speaker 02:08

was the

Unknown Speaker 02:10

tool responsible for?

Speaker 1 02:13

And what was it the input and output, you know, things like that, and what was the framework used for the world legacy application,

Unknown Speaker 02:22

and how So,

Unknown Speaker 02:27

micro services architecture? Yeah,

Speaker 2 02:30

so it was still, like it was a Java based application, but, you know, it was a very old Spring application with a JSP servlet, and like it was deployed with recently on AWS EC two. And you know, like it was a front end portal for users to log into their account. And even, you know, view with your shipping information and their profile preferences for basically, like, tracking, tracking their shipments, and even like, for the migration, the like, the we recently, like, we migrate to our Spring Boot based services

Unknown Speaker 03:06

before, it was just a GSP, Java Server Pages, oh yes,

Unknown Speaker 03:10

both a database in the back back end, oh yes, it was

Speaker 2 03:16

so, you know, like the data, database on the back end. It was basically the Oracle,

Speaker 1 03:24

Oh, okay. And then you just migrated to just using Springboard with a REST API. And

Speaker 2 03:37

so, you know, like we converted our RESTful way services. So in, like, you know, in micro service, like Microsoft service, it was in Java, Spring Boot, and then the front end it was in Angular. So I did, like some I did front end work as well. No,

Unknown Speaker 03:55

you need a front end work. Like, whatever

Speaker 2 04:00

you springboard. So I'm working as a full stack Java developer than there. And like, you know, I'm working both on back end and front end as well. So like, if I need to give percentage, then I would say 70% on back end and then 30% on front end. Oh, okay.

Speaker 1 04:20

Okay, tell me when you used when you did springboard, right?

Unknown Speaker 04:26

So what was the

Speaker 1 04:28

most important challenges we had when you were converting into springboard, and how did you work came in, and how did you did you split that legacy application into different services, or it just just migrated to the springboard monolith application. Or, how did you still not make the service architecture if you just migrated it, right? So,

Speaker 2 04:57

so you know, we split our different workflows into multiple services. So, you know, the main service that I worked on, it was modification of the user profile, profile preferences. So, you know, like, whenever user make changes to their preferences, like to the preferences in their profile, it would call this maintenance API, which was, you know, mainly responsible for making changes to the user preferences. So, you know, that was one of the primary services, okay? And then, like, there was another service that, like that I worked on, which was the Experience API. So, you know, this API, it was primarily used to grab the entitlements for the users. And like this entitlements, it was responsible for grabbing the components that users could view on the front end Angular pace. So I guess the challenge that we had were like was it was just making sure that you know the data across these components were consistent, and the inter service communication, it was properly managed, because, like, you know, we have like architectural guide on, like, what service needs to be split out with, like, proper service boundaries. So, like, I like I mentioned, for these two, it was like two distinct services taking care of those two distinct workflows.

Speaker 1 06:26

I see that so it took the took that application split into two Springboard applications, each managing its own set of data.

Speaker 2 06:38

Is so you know, like, on, on that one, like, so, so you said, like, spring, good application or,

Speaker 1 06:51

yeah, like, so you the originally, it was a monolith,

Unknown Speaker 06:56

JavaScript pages in the back end. It

Unknown Speaker 06:58

was not data, huge database, right?

Speaker 1 07:01

You are manipulating all the data within one JSP server, I believe, or I'm not getting it wrong. You're not getting it wrong, but you just had one application. You now you split it into multiple Springboard applications. Yes, we are managing different set of data.

Unknown Speaker 07:25

Is that what it is or

Speaker 2 07:26

so, yeah, yeah. Like, you know, it's Yeah, so, yeah, you know, like, like, each back end application, each back application would be managing the data independently, and you know, like, depending on what kind of workflow it had, okay, and then the front end, like, however, it was a, like, a mono repository. And also, like, you know, it was a single Angular repository that hosted, like, different components that display different parts of the applications. So, yeah, like, back in we were also segregating this, like, the databases, into multiple tables. So, but like, you know, the separation of that signal or or else, like, you know, I call database into, like, into multiple micro service, micro service based database. It was, you know, on that one, like it was mainly being taken care by our database team. Like we developed, we were just provided some on site inside, like on what was needed. So,

Speaker 1 08:37

okay, so what was it? Either was there any dependency between these micro services or they were just independent, self managed, no communication between these services. Was there any cross dependency between these services?

Unknown Speaker 08:57

If as and how did you handle it? So,

Speaker 2 08:59

yeah, like, you know, a lot of these were independent on the ones that I worked on, but, you know, there were, like, there were some inter service dependencies, right? So, you know, like, the two services that I worked on with the maintenance, maintenance and API, like, those were independent, but the maintenance API, it did depend on downstream service, basically, like, you know, to check the validity of the user, like user role, or to see that, like, if they can, if they could enroll in that service. So, yeah, like, for making sure that there was, like, communication between these multiple services. We were using RESTful APIs, are using HTTP client and, yeah, like, that's like, so, so, you know, like, the true, the two ones, the

Speaker 1 09:51

microservices, you didn't have any dependency on the other services. That's what you need to say at

Speaker 2 09:59

all. No, like, you know, the like the tour, like the two of the services are the two ones that I worked on. It did not depend on each other, but, you know, the maintenance API did make a downstream call to separate our wrist service to, you know, like some kind of processing, like processing of the preference change that if the user made before it got persisted into the database.

Unknown Speaker 10:26

Okay, so the

Unknown Speaker 10:30

you you said you make a

Speaker 1 10:31

restful call into other service, right?

Unknown Speaker 10:34

What happens if that service is not

Speaker 2 10:39

available? I'm so sorry, sorry, sorry to interrupt you, but I'm not able to hear your voice properly.

Unknown Speaker 10:43

You're not getting it. Okay, let me see.

Speaker 1 10:50

Let me get my headphones. Are you pretty clear

Speaker 2 10:54

now? Not like, it's better than before, but not like, not the best one. Okay, think,

Speaker 1 11:05

let me get back phones. Just give a Minute. Okay, sure. I

Speaker 2 12:07

Yes, I can hear you clearly now, the sound is quite low. I

Speaker 1 12:29

i can hear me now, yes, I can hear you.

Speaker 2 12:39

Oh, yes, I can hear you. It's fine. Yes, it's more clear than before.

Speaker 1 12:46

Okay, so we need a stop.

Unknown Speaker 12:51

You said microsensors. You said

Speaker 1 12:54

he had to make some downstream API call right to other service. So what if the other services not available

Unknown Speaker 13:02

to make that call.

Speaker 2 13:04

So basically they, you know, like we had resiliency pattern that was within our application. So, you know, like we did, we had a circuit breaker in place, like, which would which would basically not be calling the downstream services that are like, you know, repeatedly, in case our failures, and then also we resulted to fall back mechanism giving the downstream and like down time to downstream services, like, like, you know, Giving the time to like downstream services to recover in case of downstream failure. So, you know, this was the main pattern that we had in our microservices. But like, you know, there are other approaches that we can add to make, like, make your application resilient. Like, you know, like the rate limiting on how many calls you can make to the downstream services. Or also, like, you know, adding retry mechanism with exponential back off. So, so like, but our application, it primarily relied on the circuit breakers,

Speaker 1 14:17

but unity could use moral synchronous mechanism like Kafka, send a message and wait for the response. Yes, you don't have any, you didn't have any, any such mechanism in place, like a synchronous calls makes it vulnerable to any failure, right? Because if you the microservices, you have 10 of microservices calling each other using, like, synchronous calls. It's not a good design. What do you think?

Speaker 2 14:47

So, you know, like a synchronous masses are useful, like, whenever you do not rely on the masses to the user, right? But like, we had a requirement to make it synchronous. And like, you know, we did have one of the services that relied on a synchronous message in queue, but this was built like this was for the email notification service. And like, you know, we want to use on synchronous, on like communication when the when the response is necessary to be like, you know, persisted throughout so that you rely on the message that is coming from the downstream. But you know, like asynchronous messaging queues, like Kafka, or like pub server, or else, like, you know, it is like it is,

Speaker 1 15:35

sorry, I see. But if you you have used Kafka in your very set, in your applications, right? How did you present, yes, so

Unknown Speaker 15:44

why did you use Kafka?

Speaker 2 15:45

Yes. So basically, you know, it was like Kafka. It is mainly used as a messaging broker, like messaging broker between our application and email notification service. So, you know, basically what we did there was we decoupled the notification service for like from our applications, which means that, you know our mail, like our main application, it would send messages to this Kafka topics, and based on these messages the consumer application, It would basically be calling our email service to send the notification like to send the email notification to the users. Okay,

Unknown Speaker 16:30

notifying you use Kaka, okay, okay, tell me in

Speaker 1 16:40

like you, your most of the programming experience is

Unknown Speaker 16:46

in Java, right? Just looking, looking at your

Unknown Speaker 16:52

what, you know, what is GC root means, sorry, can

Unknown Speaker 16:55

you repeat

Unknown Speaker 16:57

that one more time?

Unknown Speaker 17:00

You know, what is good, GC root? Oh, yes,

Speaker 2 17:02

I know. So basically, like the GC root in Java, like they are, like, mainly, they are the objects that is used by your garbage collectors to, you know, like to identify what objects are reachable, right? So, you know, basically they just make sure that any objects are like, you know, that is reachable from your garbage collector route. It becomes non garbage collected, like, non garbage collected, right? So,

Unknown Speaker 17:38

so, so you're like, it's just typical.

Unknown Speaker 17:40

What are the DC roots? Sorry,

Unknown Speaker 17:43

what are the GC roots in Java, meaning?

Speaker 2 17:47

So basically, you know, like it consists of your local variables and even, like static fields and active threats. So they are just the start there. There's a starting point of your garbage collection, like garbage collection travels. Driver says, Sorry.

Unknown Speaker 18:13

Okay,

Unknown Speaker 18:14

what about

Unknown Speaker 18:17

the threat? Was that like

Speaker 2 18:22

you? Your voice is breaking down too much. I'm not able to hear the full sentence that you are saying.

Unknown Speaker 18:28

I think

Speaker 1 18:35

you're cutting off. Yeah, I think it's really I can't hear you properly. Also, can you just turn up the video for now? I can turn off the video. Okay, you can give me better now or,

Speaker 2 18:55

yes, like, it's quite better than before, but not that good. Yeah.

Speaker 1 19:03

Maybe you know what I'll release between the interview. Maybe I think later, I think it's not, not at all, like I say, it's breaking up a lot. I

Speaker 2 19:11

think now I can hear you nicely. I think it's good for now. Yeah, okay,

Unknown Speaker 19:16

you can turn off the video at least.

Unknown Speaker 19:22

Okay. Okay. So,

Unknown Speaker 19:28

so, GC, road, so, what about the thread stack? Yes. So, like, yeah, okay, so,

Speaker 2 19:39

like, I was mentioning the thread stack. So usually, like, you know, it contains references to local variables and method parameters in your like, you know, in your active trade, like in your active trade. So thread stacks, they are considered, like, considered as a part of your GC load. Okay,

Speaker 1 20:01

do you know the what do you mean by immutable object versus immutable objects? Yeah.

Unknown Speaker 20:08

So, basically,

Speaker 2 20:12

okay, so basically you know, like immutable objects are, they are mainly like a constraints, like constraints, right? So basically you know, immutable objects are it cannot be changed after they are created. And any modification that you make on immutable objects, it result in creation of a new object, right? And you know, like one of the most common immutable object, it is string. Basically, you know, immutability, it means that they are thread safe, and also, like, you know, they are simple to use, right? And like, all mutable, like mutable object, it just means that it can be changed, like it can be changed, right? So, for like, let's, let's, let's take the example. So if you have a string builder. Like you can change the string builder once you can create it. Okay,

Unknown Speaker 21:11

okay. Why do you think this thing is

Unknown Speaker 21:13

mutable, immutable, right?

Unknown Speaker 21:15

Is it a mutable

Speaker 2 21:18

or immutable? So, like a string, it's a like, like, as I mentioned, it is a immutable, like, immutable because, you know, basically, like, once you create a string, like, any many manipulation on that string, like, it will create a new string. So, yeah, like, basically, it's immutable.

Speaker 1 21:43

Do you know why it is immutable? It is mutable.

Speaker 2 21:50

So basically here, like, you know, I think it

Speaker 1 21:55

can give me the advantages of advantage and disadvantage of creating immutable. Immutable. Immutable objects.

Speaker 2 22:06

So there is so, yeah, like, basically the decider, like, let's talk about, like, advantage of creating an Immutable object. So, you know, like, they are thread safe. Like immutable object, they are thread safe, right, as I mentioned earlier, too. So you can mainly work with multi threaded environment, like easily, like if you use immutable objects, and even like creating immutable object, it means that, like you know, you can, like you cannot, tamper with that object. So if you have like sensitive data or like password on configuration, it becomes more secure and basically, like you know, you prop. Sometimes are like you sometimes it may have use of immutable objects in caching and even like reusability, because, like you know, you can use like same object over and over, right? But that just means that you have increased like memory users, which is mainly a disadvantage, right? Because, like, you know, whenever you want to modify an Immutable object, you can create a new instance, which means, like, you know, you are using more memory. And because of this, like, you know, sometimes it might have some kind of performance overhead as well. So, yeah, like, these are the disadvantages,

Unknown Speaker 23:35

okay, okay.

Unknown Speaker 23:38

Is it the best way to communicate video across threads, actually, I

Speaker 1 23:41

mean creating beautiful objects, or they can the safest way.

Speaker 2 23:52

So, like, in all, like, in order to communication, like, you know, communication between threads, you might want to consider using the like volatile keyword. I think because you know, like, if you mark like micro variable as a volatile it ensures like visibility across like across the threads, so that you know, like one thread, it modifies the value, and then, like you know, other threads, it immediately see the updated value.

Unknown Speaker 24:30

Volatile keyword. Sorry, yes, mentioned about

Speaker 1 24:36

volatile what? I breaking up again. Sorry, I'm going to cut off my video. Okay, okay. You use the video spinning API from the Java, yes,

Speaker 2 24:56

I have. I have used that like,

Unknown Speaker 25:00

okay, let's say I have,

Unknown Speaker 25:03

yeah, let's say I have a list of

Speaker 1 25:06

user object. User object has a first name, last name,

Unknown Speaker 25:12

okay, let's not have that.

Speaker 1 25:15

I have, yeah, I have list of users first name, last name, and city, city name, okay. I need to city name, city name and county within the city, city and the state. Okay, so I need to get a count of users, number of users, number of users, by county and by state. Can you just write a streaming API just to do that?

Unknown Speaker 25:50

Can we count off

Unknown Speaker 25:52

count of number of users by

Speaker 1 25:53

city? How do you do that? Can you write it and you have a notepad?

Unknown Speaker 25:58

Okay, so do you want? We

Speaker 1 26:00

can share your work, yeah, you can share your screen. Okay.

Speaker 2 26:12

Okay, so can you see my screen? Yeah,

Speaker 1 26:21

let's say I have a user class. Okay, the user class has a string, first name,

Unknown Speaker 26:28

First Name and Last Name,

Unknown Speaker 26:31

and state and density. I

Speaker 2 27:00

BBB. So basically, here we want to count the user, like users by stake, right? Yeah, yeah, okay. So here, density, so here, I think, for that, like, we can use the collectors, like collectors grouping, but,

Speaker 2 27:32

But if you want by state and city, then

Unknown Speaker 27:42

so for example,

Speaker 2 27:51

so, so, can you give me an example, like, how you want to answer to it? I

Unknown Speaker 28:05

Okay, no, I

Speaker 1 28:06

just want to like, let's say I have a list of users.

Unknown Speaker 28:11

List of users.

Unknown Speaker 28:19

Yeah, okay, then I want to say.

Speaker 1 28:22

And then I want to get a count of users by city

Unknown Speaker 28:30

which will give me city

Unknown Speaker 28:34

like state, city in con six.

Unknown Speaker 28:42

I wanted that

Speaker 1 28:47

in each state and in city, what is the car number of you

Speaker 2 28:56

so, like, it's two different outputs. Like, the two different outputs are same outputs as like, you know, one combined. So do you want like? You want a user by city, and you want a user by state, or do you want like a user by city, state and commerce, state and city

Speaker 1 29:14

and state? Yeah, let's say we can start off with state. How do you do it? Okay. Do you do it?

Speaker 2 29:24

So let me do by state first. So basically, for that, we can just do,

Unknown Speaker 29:32

so we can just do a stream,

Unknown Speaker 29:36

and then we Can and

Unknown Speaker 29:40

then, like we can collect it so i

Unknown Speaker 30:09

Okay, I

Speaker 1 30:20

is it collectors counting? I am not sure. What is it. Is it correct? I'm not checking out. So basically, looks okay to me. So basically, just using the grouping bar.

Speaker 2 30:37

Sorry, yeah. Is it grouping by right? Yes, so basically, it just counts, like, how many key exists on the list,

Unknown Speaker 30:45

okay, how many keys exist on the list? Okay,

Speaker 1 30:48

let's say, if you want to, just to get the list instead of counting, Give me the users list. Okay.

Speaker 2 31:21

So I think that that's right. So you just want to group the group by the state,

Speaker 1 31:29

yeah, and group by the state, yeah, to start up with, if you can give me the later on, like a state and city, I

Speaker 2 31:47

so do you want a map inside a map, or do you just want, like? Do you just want a map? Like, map with the key, like, with the key as a city, as a city, state, and then continued it, yeah.

Speaker 2 32:12

So the answer should be something like this, right?

Unknown Speaker 32:21

Yeah, okay, so,

Unknown Speaker 32:29

so for this like, we need to make sure

Speaker 2 32:33

the like, we need to make sure the value of my colleague, like, you know, the Collect operation is another list, so we can do another grouping by so we will do users and add stream, And then we will click based on the state. So

Speaker 2 33:06

now for the video like we need to do, okay for the video, like we need to do another collectors do.

Speaker 2 33:31

So I think that should give you right so, like the key, the key is the state and for the venue, like the key is the city, the list of

Speaker 1 33:45

users get counting. Is going to give a list of users, or is it just to give the

Unknown Speaker 33:52

count? Should it be?

Unknown Speaker 33:53

Oh, I'm so sorry. I'm so sorry.

Unknown Speaker 33:55

Let me do it here,

Speaker 2 33:58

so you're getting that, yeah. So, just, it should be that, Right,

Unknown Speaker 34:04

yes, yeah, I

Unknown Speaker 34:41

let's say grouping bar. And then

Unknown Speaker 34:45

does it just give the list?

Speaker 1 34:50

Okay, so, and let's say, if you're going to get the list of speeds from this users, sorry.

Speaker 1 35:02

Oh, just want to get to collect the list of states. Okay.

Unknown Speaker 35:05

So for that, like we can just do math, okay, so,

Speaker 2 35:11

so for that, like it's

Speaker 2 35:20

just so on the map, like we can pass what we want the map it like we want what we want to map it to,

Unknown Speaker 35:29

and then like we can collect it as a list.

Unknown Speaker 35:39

Yeah, so that should give you a list of states.

Speaker 1 35:42

Yeah, yeah. I mean, it will give the step we need a step, right,

Unknown Speaker 35:49

not the list this will give to the case. So you

Unknown Speaker 36:08

I think, yeah, that's it.

Speaker 1 36:13

Yeah, okay, yeah, this

Unknown Speaker 36:19

is good. Yeah, alright.

Unknown Speaker 36:24

You have any questions for me? Yes, yeah, yes,

Speaker 2 36:29

I have few questions. So first of all, like, I would like to thank you so much for the opportunity today. And like, it was nice talking to you, like, even though we had some technical difficulties in the middle, like, it was really nice. So yeah, like, yeah, I would like to know a little more about, like, day to day of the developer on the team. Like, what developer gets involved in? Mostly, no, our

Unknown Speaker 36:56

team mainly focuses on

Speaker 1 36:59

post operation. Like we have a galaxy is like all the trade we have a trading system. So all this trading data flows into our system, right? But that trading data, we actually are responsible to make sure that all the trading data is collected and there it's a data integrity is protected, and also make sure, like, the end of the day, we generate reports, and also we do the reconciliation of the data and that in for the operation team and alerting at any anything problem, any problems in the

Unknown Speaker 37:40

time not being collected.

Unknown Speaker 37:43

So things like,

Speaker 1 37:45

Okay, so our text tag is mainly Java, and we do Python for reports. Okay,

Unknown Speaker 37:51

I think I believe

Speaker 2 37:55

Python, like I had done some of the stream manipulation using python two. And yeah, like, we had to basically grab the existing data and then, you know, like, message it to something, something else. So I have written, like, some scripts. And yeah, like, I did have one separate course Python in my master's too. So yeah, I do have knowledge. Okay,

Speaker 1 38:19

okay, sounds good. thanks.

Unknown Speaker 38:25

Talking to you, it

Unknown Speaker 38:26

was nice talking to you as well.

Speaker 1 38:27

Okay. Okay, thank you, thank you. But.

Shreesha Khadka- Infosys- Srv- Final

Speaker 1 00:00

About five years now. So I'm primarily working on Java based applications utilizing frameworks like Spring, Hibernate and RESTful web services for like API deployment and yeah, like, I do have pretty good experience working on microservice architecture based using Spring Boot. And like, I have pretty good experience working on AWS cloud services. And also, like I have worked on with AWS EC two, ECS, SQS, SNS, lambdas and s3 buckets. I do have, like, good experience working with databases, including Oracle and Postgres. And like on the testing side, I have pretty good experience, like working on J unit and Mockito, and also like geometry for performance testing. And like, I have even a gap experience with Angular on the front end side. And currently I am on a scrum team. And this team, it consists of five developers, so I have been sorry primarily working on back end Java, like a developer with a 7030, split with front end work on a daily basis. And like I'm involved mostly working on back end Java coding, and also like helping team with the code reviews and also on perks and support.

Speaker 2 01:36

So like, what is your current project? Can you use like, little bit details related to current projects in, like, day to day test,

Speaker 1 01:45

yes, sure. So currently I'm working in a city group, and, you know, like we basically focus on, like a customer onboarding and self service portal process, like we are on a big migration effort where, you know, like we are migrating and existing monolithic app to microservices, and like, I took care of two particular services that is, like, basically responsible for making changes to user preferences, like so one of the service that I migrated was the Experience API. So, you know, like this API, it is responsible for grabbing user entitlements to be like, you know, basically entitlements for user to view, like different wizards and also like components that are available on the front end project, like front end aspect and so, like The risk perish API, it is deployed, deployed on AWS ECS and like, similarly, I worked on a maintenance API which took care of like, you know, changing anything on user profile preferences. So like, like, if they have to make like, any changes to their profile, we will use like, we will be using this maintenance API. So, yeah, like, this is our whole main project, and like Project applications, and also, like back in is in springboard, which is, like, mainly deployed on AWS ECS clusters on like, target instances, and like on and the front end, it is basically in Angular and like, Yeah, I'm on our primary back end developer working on this Java spring work based application.

Speaker 2 03:27

So what was the reason like to migrating that application?

Unknown Speaker 03:33

What issues you guys were getting?

Speaker 1 03:36

Basically, yeah, like, you know, when migrating the application, like, you know, the project it was growing. And also, like, you know, there was a addition of new features. And also, like, components were getting out of hand, because, like, you know, there was only one single repository holding so much right. And like, we know that the micro services, it has a lot of advantages in this aspect, right? Basically, like this single, simple services that work together, like, you know, as a cohesive unit. And also like basically different aspects of the component it had to have, like the different flexibility, and also like scalability, and also, like, we would take a lot of time just fixing some small stuff, right? And like, if there was an issue within the some certain components, right? So, yeah, that's the reason we went with the migration aspect. And also, like, we converted this monolith into a small chunks,

Speaker 2 04:35

currently, like region to migrate into, like micro service

Unknown Speaker 04:41

decoupling that code. That's it.

Speaker 1 04:47

Like, no, like, I mean, there were other aspect too, so that, like, you know, like, I think, like, like, a lot of the reasons were, like, basically surrounding the multi reasons, and like, you know, the scalability and like, because like, we couldn't, we could not, like, have properties, scale services as a like consumer, like customers, and also, like, we were going to use different services. And also, like, you know, we could not have like, like, we could not have a continuous delivery, because like, of the different independences of multiple services, right? Because, like, you know

Unknown Speaker 05:26

what?

Speaker 2 05:27

So let me ask you, what was like your role during that migration. So,

Speaker 1 05:34

okay, so basically, like my role in my migration, like I took in charge of, like, migrating couple of components. Like, like I mentioned, like they were true services in that particular that, like, you know, I was a subject matter aspect of it is, like the maintenance API and experience API. So, you know, like these two services, basically, like, I migrated it from the, you know, like existing workflow to the, like Spring Boot application, like Spring Boot based application from scratch. And so, like I did, like architectural design, and also, like, you know, I implemented them, and I also, like, got approvals from the architect as well. So yeah, like, I was a complete hands on developer for these two flows.

Unknown Speaker 06:24

So again, I'm

Speaker 2 06:25

asking same thing, but first, like, your what you did, like, which Spring Boot version you have worked on, like, were used in the dB.

Speaker 1 06:37

So basically, like, you know, like, like, the as you asked the version for the spring, like, we had a Spring Boot three. So basically, like, you know, we planned on doing the Spring Boot 2.72 but, like, due to the end of life as life support and, like, we migrated to Spring Boot three. But, yeah, like, my role as a Java developer, like, it was to migrate the whole services, right? Like I mentioned again, the two services. So basically, like, I took charge of migrating those two services. And also, like we used JDK 17 Spring Boot, like the JDK 17 and Spring Boot three,

Unknown Speaker 07:19

what steps exactly you took it like,

Unknown Speaker 07:22

then you work us through.

Speaker 1 07:24

Yes, sure. So, like, here to do this, like, basically, you know, like, whenever, like, we had the feature to migrate the maintenance API, and, like, the first part was for our design, and also, like, approval. So basically,

Speaker 2 07:43

like, sorry, so, but I'm asking, like, what classes you created, what like annotation you have used, what like tool you have used for migration, not like the design part or the tech part. Just like, from technical perspective,

Speaker 1 08:04

okay, yeah. So, basically, like, you know, like, I took charge of, like, creating a conference space, which is basically listing the functional and non functional requirement. And also, yeah, like, I had pretty good experience of functional requirements too, okay? So basically, like, you know, like I walked through, I talked through a maintenance FDI structure. And also, like, you know, we had a controller, basically, like, annotated with the Rest Controller annotation. And like, in this controller, we had two main APIs exposed, okay? And it was like, one was the gate API to get the status of the maintenance call, which was like, you know, basically to get the request by ID. And also, like, it was annotated with the gate mapping annotation. And like here. And then there was a man update, which was like, called our post mapping annotation. And like this, this, it took our request body on, like, what aspect needed to be updated, and so, like, you know, to what state, why? So, like, you know, like that was a main entry point, which was basically the controller. Now the like controller, it had dependency of a service layer. And also, like, we had multiple services incorporate. So the main services that we incorporate was our maintenance service. Like this maintenance service, it was basically responsible for a couple of actions. And also, you know, like, one has to make a downstream call to a separate service. And like we even used Apache closable HTTP client for making like HTTP call, which was basically a RESTful API call, to downstream these services. And then also, like, you know, it had incorporated our Spring

Data, GPA, and like, we use that repository to make changes to our databases basically like to update the status of the minions. So yeah. Like, that was our service layer, and then the repository layer, it was annotated with the repository annotations. And also, like, you know, we had some custom queries are built using the query annotations. So yeah, like, this is the basic structure of the springboard applications that I developed.

Unknown Speaker 10:25

So what is difference between the gate mapping and

Unknown Speaker 10:29

request mapping? Yes,

Speaker 1 10:34

so talking about the difference between get mapping and request mapping, basically like, you know, get mapping, it is a special, specific kind of like request, right? So, request mapping, right? So, like get mapping, it only handles like HTTP verb called gate. And also, like you know, it is usually used for driving our resources, right? And also, like it is, it provides like a clarity to the user that you know, like you are basically referring to GET HTTP method and also like, but like a request mapping, it is a very broad annotation that can basically like, store any kind of HTTP method, right? So the request mapping and it takes our parameters, like the value which is the basically the path of which the URL will be, and also like, you know the method on what kind of method it would be. So like, if you are like using a request method, then like you have to like if you are using request mapping, and also like you have to specify a particular HTTP verb on that notation,

Unknown Speaker 11:46

what is difference between request,

Unknown Speaker 11:50

like Rest Controller and like normal control?

Speaker 1 11:55

So on this one, like a wrist controller, like it is basically a special kind of controller, right? So, you know, like this, like wrist controller, it is mainly used for RESTful API. And also, like, you know, it is a combination of two other annotation, which is a controller annotation and the response body annotation, right? So, like a controller, like, basically it is a spring MVC, and also it defines your controller, right? And like it is like typically used to return any views or also, like, you know, any response back to the client, which is like your main entry point, right? But like the risk controller, it is a combination of that controller annotation, and also like another annotation called the response body, which is like, basically used to specify that when it is involved okay. And also like it returns all JSON or XML or also like other form of data instead of a view component

Unknown Speaker 13:00

to what will happen like, if we use the

Unknown Speaker 13:03

at the rate controller in rest AP,

Speaker 1 13:10

so like, like, if you use the like, if you use an at the rate controller in a REST API, mainly like, without using the response body annotation, then I think that it will interpret that, like, as a view name, and also, like, you know, it try to resolve to some kind of template, right? So, like, what template is? Like, your HTML template, so, like, if, like, if, it will typically result in, like, for error, because the template it doesn't exist.

Unknown Speaker 13:49

What increase we need in founder,

Unknown Speaker 13:53

generic Spring Boot application. So,

Speaker 1 14:01

yeah, like, in like, so you said, like in this, what did you say? Like, can you repeat the question one more time?

Speaker 2 14:08

I'm so sorry. So what entries we need to have? Like, in pound, road tax amount, like, generic Spring Boot application? Okay, so

Speaker 1 14:23

like, so to have, like, you know, the January Spring Boot application, I think, like, you need the Spring Boot starter dependencies. So basically, like, you know, like you need to provide the parent, and also, like, you know, which is like the Spring Boot parent. And then like, you need, like, any kind of starter dependency, which might be the Spring Spring Boot starter. And also like, and also like, you know, it might be like the like spring wave. And also, like, you know, it might be all those starter dependencies, right? So, like, soon the Spring Boot started parents. And also, like, Spring Boot starter wave and Spring Boot started. So like, that's a bit like, that should be all the thing that you need for, like, a basic Spring Boot application. But like, there are other starter dependences that Spring Boot has, like the Spring Boot starter data, and also like JPA, and also like actuator, and also like many more that is, you know, like required for like and also like integration of different components. I

Speaker 2 15:43

Okay, so let's see you have an instant boot application. We don't want to have like JT server. We want to use like tonkit server. We will do that. So

Unknown Speaker 16:01

basically, like for this one.

Speaker 1 16:05

So basically, like, all you need to do is, like, you know, sorry, Spring Boot, it measures like, advantages that it has, you know, like server in, embedded with it, like, right? So, like, all you need to do is you need to add the spring boots. And also, like, it also, like Tomcat embedded code dependency, right? So basically, like, you just need to add that Tomcat dependency and that, like, suited, right? Like, basically your Spring Boot start your wave, it already contains Tomcat. So like, if you have any kind of JT dependency added, then like, just remove it. And also, like, it will automatically peak dependency. Sorry, how you will remove

Speaker 1 16:57

it. You can do it. So like duty, it needs to be like, you know that this JT, it needs like, it needs to be actually added to your Tomcat, dot examine so, you know, like, you just remove the dependency from like dependency tree. Or also, like, you know, you directly remove the dependencies section that has the springboard. Like, start your JT, I know the

Unknown Speaker 17:34

click tag, which we need to use to remove those.

Speaker 1 17:41

So like to remove this. It is basically the exclusion tag. So

Speaker 2 17:53

in which year you have used, like Spring Boot three.

Speaker 1 17:57

So basically, like I used Spring Boot three in this year. So, like we did the couple, like we did it, we use this couple of months back.

Speaker 2 18:10

Okay, what were the changes with respect to 7.22 point 7.7

Unknown Speaker 18:19

to like three.

Speaker 1 18:22

So talking about the changes, basically, like, you know, Spring Boot three, it basically does not support the JDK which is, like, basically lower than JDK 17. So, yeah, that that was the first thing that we had to upgrade our project to use JDK 17. And you know, like for that, we had to change the certain components right, like the Java X validation, and also, like, you know, it was deprecated for jackal Jakarta validation. So basically, like, we had to change those. And once that was done, we had to make sure our testing libraries. It were migrated to the tree unit version five. And like, that was one other thing like, we had to do. And also, like, one major change that Spring Boot three, it introduced was the Spring profiles they were used and like, like how these Spring profiles were used. And also, like, you know, we could not do the Spring profiles active, but it was like some kind of conflict on active, active log and active flag. And also, like, you know, there were a lot of changes on spring security too, but there was like, separate team that handled the migration of spring security. And also, like, we just had to incorporate our application with it. So like, those were the, basically the primary changes we had to do. And also, like, we had a very detailed steps as we, like, you know, we used a company standard palm, like palm dot XML, XML as a parent. And also, like, we got all our dependences on it, so yeah, like, these were the major changes that I had to make in like, order to migrate our application to JDK 17 and Springboard three.

Speaker 2 20:15

Okay, so let's select you have like customer repository and big customer repository you already have, we need to create, like JP repository. How you will do that? Can you write that code and put into the chat? Okay, Yes, you Have it.

Speaker 2 21:36

I so what is the disadvantage to use the microservice or linguistic niche?

Speaker 1 21:57

Yeah, so basically, the disadvantages of, like, using the microservices. So, like, so basically, like, you know, my microservices are like, it means a lot of small services, right? Like, the interacting with each other. So, like, once you have a huge number of services are managing this in a system. Like, it becomes complicated, because, like, you know, each of these services, it needs to be deployed or developed. And also, like, you know, monitored individually, or like and like. So like, there's lot of thing going on. And also, like, it becomes complex. So yeah, like that is one of the major disadvantages of Microsoft, like using micro service. And also, like, one of the challenge that, like all the micro services face, is how the data is managed, right? So, like, to manage our data consistency, you have to, like, add up, like, Adam, like you have to, you know, like, add up to a proper pattern, like a sage pattern. And also, like, you know, which is sometimes which sometimes it can get, like, tricky, right? So you know that the data consistency is one of the major disadvantages. And, yeah, like, usually, since you are doing like, an inter service communication, there should be some kind of network problem or latency because, like, of services and also, like, you know, interacting with each other through the network. So, yeah, like, basically, these are the disadvantages. Yeah, okay,

Speaker 2 23:40

which worked on?

Unknown Speaker 23:48

You said, Java 17.

Speaker 1 23:50

I Yes, yes, I like. I have worked with Java eight, Java 11, and also Java 17.

Speaker 2 24:01

What is the difference between the thread, dot star and thread, or trunk.

Unknown Speaker 24:10

So, so, basically, like,

Speaker 1 24:15

sorry, the thread dot start. It is, it is, like, you know, it is used to start a new thread execution, right? Basically, like, whenever, like, you call, you call the start method, the JVM ADA internally invokes a run method of that trade, like, in basically a new call stack, right? So, like, it just, like, creates a new thread, and then, like you know, it calls around the method on that new thread. And like you know, which means that the run method, it executes, compute concurrently with code that like that follows the starter start, but like your thread that run is a method that, like contains the code to be executed by the thread, right? So basically, like the run method, it directly does not have like, it does not start a new thread. And also, like, you know, whenever you call the run method directly, like it will execute in on the current thread, rather than, you know, like just starting a new thread. So basically, like, it behaves like a normal method call, and also, like, you know, it does not create, like a separate, like separate thread for execution.

Speaker 2 25:35

What will happen, like, if we do that like thread or start thread to start, like two levels,

Speaker 1 25:40

I so, like, basically, like, if you like, you know, if you do thread that start like two times, then like, it might like, throw you an exception. Because, like, you know, like a thread, it can be in multiple step like, but state, but like, you know, once the threat it is completed, like it is completed execution, it cannot be started again, right? So, like, it should throw some kind of illegal state exception.

Unknown Speaker 26:21

Let's say you have like one REST API,

Speaker 2 26:25

getting like lot of calls to rest a day. You need to scale the click recipe, what step you will do to scale that click? REST API, so,

Speaker 1 26:42

like, um, like, in order to, you know, like, in order to scale risk, API, like you, I think like you might have to consider, like, you know, some common patterns, basically, like, you know, like, there is a concept of horizontal scaling, and like this, what that means is, like, you know, it's you to, like, deploy multiple instances of your API to, you know, handle the increasing load on, basically the load balancing. And also, like, you can have like, a polar, like policy basically attached to it through some Auto Scaling group. And also, like, you know, once you have a certain matrix that is mainly, like, based on that load balancer than the application, it scales to the another number. So, yeah, like, that is one of the ways. But apart from that, like, if your REST API, like it returns a static data, then, like, you can use in memory, placing and also lands like, basically like what it does, and also like it stores frequently accessed data. And also, like you know, it reduces the load on the server by basically, like, retaining the data from the case itself. And also, like, you can use any kind of asynchronous processing if possible. And also, like you know, which allows your API to quickly respond to requests you know, like, while processing them in the background. So yeah, like, these are some of the ways that you actually like help the scalability of your application and also like your application videos and Like it reduce any kind of load related issues. Okay,

Unknown Speaker 28:55

let's say you have like one

Speaker 2 28:58

analyst, and you want to remove like duplicates. Can you write a program to remove the duplicates? Yes, sure.

Speaker 1 29:14

So basically, like to remove a duplicates from an array list, like, like you can I, you can like, mainly using the Head state And Like I

Unknown Speaker 31:19

So, yeah, Like, basically,

Speaker 1 31:23

you know, like, we will be storing our like in unique into our new list by mainly, like calling the calling the constructor of the array list with the headset. But like, if you do not want to use the headset, then also, like, you know, there is also like processing through the stream, and also like we can do, like convert that list into a stream, And also like we can use a distinct keyword, okay,

Unknown Speaker 32:03

have you used Kafka? Oh, yeah, I have,

Unknown Speaker 32:09

like, producer, consumer, or like both.

Speaker 1 32:15

So like, for Kafka, like, I have experience on both side, like, what side of it in a little bit,

Unknown Speaker 32:25

what is the maximum length of one message in Kafka?

Speaker 1 32:34

Maximum length of one message in a Kafka, it is like, you know, like it is a bit like it is usually, like you know, set to one megabyte, but also, like you know, it can be configured to allow the larger messages, if necessary. So basically, like you know, it is a property on your server properties file.

Speaker 2 33:02

Let me ask, in this way, what is like maximum we can set it so,

Speaker 1 33:09

so, so, like, you're gonna so, so so the maximum properties, like, I think like it is up to two like, it is up to two gigabytes, but it is like, way too high, right? So,

Speaker 2 33:37

okay, which I noticed, and we need to use like to produce like message in what is needed, basically, are

Unknown Speaker 33:47

you familiar of the click, how to create topic, like, how to use it

Speaker 1 33:56

for like, basically, like, You know, in order to work with, like producer in Kafka, like you know, you would basically like, you know, create a service for message, and also like producer, and then also like you know, you would like, you would auto write the Kafka template. And so like, you know, you you would like use this, like, like Kafka template to send the messages. And like you know, in order for this, you need to, like, basically have configuration where you provide the bootstrap server, and also they start like sterilizers for these messages. And also, like you know, you will basically create our topic with the bin, right? So, like you can create a Kafka topic with the topic building and like you know, with how many partitions and replicas you want. And also, like your applications, we will, we would be using that bin.

Unknown Speaker 35:00

What is the significance of partitions?

Speaker 1 35:06

Yeah, so the significant, like, the significance of partisans. It is like, it is that basically, like, you know, it is that it will allow your application to be like, horizontally scale, right? Which means like, it allows your messages to be distributed across this partisans, which mainly like, allows you to run multiple brokers and like it's handling the different partisans, which also like, enables the Kafka to handle, you know, like, high numbers of masses, and also, like it also enables parallel, parallel processing, because, like, you know, each partisan it is consumed by a separate consumer.

Speaker 3 35:57

Let's say we have like one consumer, and we have like two partition

Speaker 1 36:05

so, so basically, like, for the like, basically, like, you know, if you have like two partitions with, like, only one consumer, like, It will still work, because, like, you know, a single consumer, like it can read from multiple party sets, right? So, like the consumer, it will fetch message from both partisans in parallel. So like, however, like you know, the processing of those necessary, it will still occur, like, sequentially, since, like, you know, there's only one stance of, like, only one instance of that consumer and but like, you know, like, I think it is not a good practice, because, like, you know, you would Have an under utilization by,

Speaker 2 37:00

like, parallel processing, or like sequential processing. So,

Speaker 1 37:07

so basically, like, you know, like, what I mean by like parallel, parallel processing, and sequential processing is like, you know, if you have, like, multiple instance of consumers, and also like, you know, different partisans, like all the messages, like, you know, all the messages it can be processed at the same time, right by, like, by the different consumers. But like, if you have only one consumer, then you know, like it will basically complete, like it will basically complete one processing of the masses, and also, like it will be followed by the other one, like one after the other. So basically, like sequential it is like a linear fashion, where you know where each task it must complete before the next one begins. And like, while parallel is understood,

Speaker 2 38:00

I understand like, what is sequential and what is like parallel. But I'm asking, when you say, like sequential or parallel processing, with respect to like Kafka Consumer, what does that mean?

Speaker 1 38:18

Okay, so basically, like, you know that. So basically, like the Kafka, it is, like designed to have a parallel processing, like, when multiple consumer the instances it works simultaneously, right? So, like it can consumer, like it can consume mid message from one or more partisans, right? Basically, like, like you if you have, like, multiple consumers that read from different partisans at the same time. And also, like, you know, basically they persist. So like this message, it is independently. So like, that's what it is, right? And also, like parallel processing, it is very beneficial for like, high through throughput and throughput applications. And also, like, you know, where you want your application to be very responsive, and also, like, you want it to be fast, right? So like, there may be, like, sometimes we needed for a sequential processing where the order in which like, which messages appear to be critically managed. So mainly, like, you want to ensure that the message they are processed sequentially. So, like, that's what Sequential Processing is.

Speaker 2 39:35

Have you worked on MongoDB? Sorry,

Unknown Speaker 39:40

have you worked on MongoDB?

Unknown Speaker 39:42

Yes, like I have worked on

Speaker 1 39:45

Aw, like I have worked on like DynamoDB, and also like I have worked on like DynamoDB on AWS.

Unknown Speaker 39:54

Have you worked on like MongoDB,

Speaker 1 39:57

MongoDB, like, not particularly MongoDB on the projects, but like, you know, I have done, like some practices on from my side, but like on my project at work, I have mainly worked on DynamoDB to store, like, you Know, to store the entitlements and also the Jion i

Speaker 2 40:32

Okay, think I'm good. You don't see the trouble. I so

Unknown Speaker 40:49

like, did you ask anything? Like, I'm sorry I couldn't hear that.

Speaker 2 40:55

No, but I was saying that, and could I don't see that other panel joined it. So maybe we can close the call and provide the feedback to a chat if they need to have, like, another call for like, another round,

Unknown Speaker 41:14

I don't see him in that call. That sounds good.

Unknown Speaker 41:20

Do you have, like, any questions to me? Yes,

Unknown Speaker 41:23

I do have few questions.

Speaker 1 41:26

So, like, I want to know, like, more about the projects that I would be working on. Like, if that's

Unknown Speaker 41:36

so Infosys is like,

Unknown Speaker 41:39

having a lot of clients, right? So

Speaker 2 41:45

what you are joining and what projects we have available, it depends on that. So based on that feedback, HR will reach out to you, and then all the like processing and when you join what projects

Unknown Speaker 42:04

availability, but we'll like

Unknown Speaker 42:08

another call with like

Unknown Speaker 42:11

client, whichever client project we have, and then

Speaker 2 42:16

that is How networks you are, like, in which state, sorry,

Unknown Speaker 42:23

are you like, in which state

Unknown Speaker 42:25

I'm in, Virginia right now?

Speaker 2 42:28

Okay, sure you like, okay to relocate. Or, like,

Unknown Speaker 42:35

I guess I'm hoping location I'm open for relocation.

Speaker 2 42:39

Okay, I'm also, like, not sure from which location we have opened this. Anything else that's

Unknown Speaker 42:52

it from my side, thank you. Thank you for your time. Thanks Jason,

Unknown Speaker 42:56

thank you for that.

Shreesha Khadka-MyStudio-SRV-first

Speaker 1 00:00

Code reviews, and also when, you know, like, I have been involved in some products and support duties too. Okay, and so I'm going to dive a little bit in. So talking about some of the back end services that you're doing. So it sounds like you're a lot of it, it's microservices, right? And then says, Each one is microservices. Are they all

Unknown Speaker 00:28

Spring Boot applications? Are they,

Speaker 1 00:31

you know, some other type of job application. So, yeah, basically, like, it is a Spring Boot applications. And like this, by Spring Boot micro services, they are basically deployed on AWS ECS, okay, and then, and then. So I guess for micro services that there's multiple approaches. Are you guys, like, super micro services. So, like the service LD has, like, a few, a few services. Like, get post and then, and then, and then, if they call other microservices. Or is it kind of a, are you guys kind of logically grouping your microservices? Like, saying, like, Okay, here's a microservice that's all payment related items in this one microservice. Like, tell me a little bit about some of the microservices. Like, how are you guys using microservices? So I would say that it is super detailed, and also, like, you know, so it's microservices. It is basically responsible for at most, like five or six APIs. So, yeah, it is, like, very small microservices. Like, it is also like control through or as like an orchestrator deal, which is basically right, written in Spring Cloud gateway. And also this acts like the entry point. So, yeah, like it is very much topic. So do you do you guys normally so for these microservices, and are they primarily, like, touching the database, or are you guys using the microservices, like, kind of talk to, like, a third party, because I see that you that, that you've done some KYC integration to other systems. So I'm assuming you guys part of microservice per per like, I would say, like, sources, the approach. So, you know, like, it depends on the microservices, okay, so it is basically like the primarily orchestrator lay. And also, you know, like it has its own database that source. And also, like, you know, the metadata for all the other microservices. So I would say that this orchestrator here, it has like, its own database, and also it's like internal to it. And, you know, microservices that I have, like that live like behind this orchestrated they may, or like they may not have an external database, so it is basically, like it is connected to it sometimes. And you know, it leverages an external services which is basically the downstream, and, you know, like they grab data for, like the KYC program. But you know, like it also primarily depends on a lot of services, which basically like labels on one database table, but our primary database, it is for the orchestrated RA and also which has limited data for, like the other micro services. Okay? And then are you guys because, because because I see that you've got experience with like hibernate, but they also can see that JPA, so I'm just curious, when you guys talk to the database, how are you interacting with, are using like Spring Data, Spring, Hibernate, like, What? What? What? What's, what's, what are you guys using to to talk to the database, yes. So basically we are using, let's talk on the project. Okay, so some of the projects like that have basically the simple interaction with database. So like, just by finding the ID, or else, like, you know, just by simple clauses, we are basically using Spring Data GPA. But like, you know, some of the projects that we have, and also, like, we use hibernate, right? So, like main user of using hibernate, because, like, we leverage the Hibernate criteria API, and also we use this in order to construct a complex queries that basically includes join, and also the criteria building takes this criteria, this takes care of the adding the root elements, and also, you know, like adding the joins, and then also like predicates. And so in order to prepare this query, right? So like some of the micro services, if they use, like plain Spring Data, GPA and but like some of them, it used the more intricate hibernate criteria API, okay? And then so obviously, you're with taking both approaches, and you're kind of leveraging the NAD frameworks to to kind of a, I would say, Marsh, the data. But D Do you ever find yourself having to go like, really low level and call, like, spring, JDBC and to write SQL queries directly to query the database. Or are you guys more, more high level where, where you focus on, you know, Hibernate entities. So, like, so like, when it comes on the Hibernate criteria API. So I would say that I like, a little more complicated than the intake, like, inter entity interaction, right? But I would say, like, it is little bit a little better API for like, you know, like the low level queries, or as like, like, we physically don't have to write the queries ourselves, but, you know, like criteria API, right? Like, basically it is a low level query, but also an API, right, so that we can use the methods, like the join methods. And in fact, this join methods, it creates a showing between tables, right? And also it adds the predictor predicates. And also, like this predicate, it acts like the word cause, right? So, yeah, I have written this. But like, but you physically, you don't have to write the whole SQL statement. Okay, got it and then, so tell me a little bit about how you're using lambda functions right now. So talking on the uses of lambda functions. So I mean, like we use a lot of Java eight features in lambda functions everywhere, right? So a lot of like collections, and also like processing, and whenever you know, like we are doing any kind of streaming process, we use lambda function. Lambda is right. So, yeah, like, I mean, yeah, I'm pretty comfortable with lambda functions in Java itself.

Unknown Speaker 07:41

How are, how are you invoking the lambda functions?

Speaker 1 07:45

So, invoking the lambda function. So basically, yeah, like, yeah, I would say that if you talk about, like, invoking the lambda functions in Java streams, right? So, like, they are only invoked once, like you do the terminal operation, right? So basically, you know, like, if you are collecting as a list, that when they enter, like intermediary, the lambda functions, it get triggered. So, but like, is that? Like, is that the question you were talking like, a little more on the AWS lambda functions I'm asking, like, so obviously, for lambda functions, then there's multiple ways that that they can be triggered. So one way it would be like, it could be based off of a rest, rest of it. Another one could be based off of SNS message. It could be an event print message. Like, I'm just curious on how you guys are actually invoking your laptop functions. Oh, yeah. So yeah, like, I was talking about the Java lambda expression, right? So for our, like, AWS lambdas, we have, like, we have been behind API gateway, and, you know, like our lambdas, it sits behind, right? So, like, we have two lambdas and one of the lambdas, like it acts like an entitlement based on the front end applications, whereas, like the other lambda, it acts like a notification service, right? Like email notifications to users. So like the first lambda that acts as an entitlement breeze, it is behind on an API gateway. So, like, we trigger it using a restful call to the API gateway on AWS. And, like, the second notification service, it is based off message on the SQS topic, right? So, like, we have a trigger that is basically configured on the lambda. And also, like, once there is a message on this topic that basically, you know, lambda gets executed, right? So, yeah, like, the two ways we have used are, like, we have our lambdas involved, okay, no, that's, that's good. And then so or so your team of five. How are you guys like, dividing up your work, and then I'm just taking it on a little bit through your process, in terms of, how is work scoped? Like, do you guys collectively break down like all, let's say that you have a story. Do you? Do you collectively break down all the subtasks? Or is it like, Okay, I'm going to give you this user story, and then it's, it's your job as a developer, because you're the one that's assigned that's going to bring you to subtasks. I'm just curious about, kind of, some of the, some, some of the experience that you had there. Yes. So, like, when doing the task, like I, primarily, I have been involved in, you know, like just owing the user story, and also, you know, then, like breaking them down into the smaller chunks. Like, as a team, we just collectively create like user stories for features and also like it's the developers who is working on that. Stories responsibility, right? So like the developer, they create this subtask for that. Can you give you one example of this? I

Speaker 1 11:44

a bit sorry I'm not able to hear you like your voice is breaking now, or just, Oh, I'm sorry. Can you hear me? Now? Yes, I can hear you now. Okay, yeah, so if you can just give me an example of, let's something obviously nothing confidential or like that, but, but tell me, like a time or an example of, here's a user story that you were given at a high level, and kind of this. This is your, your approach of how you broke into subtasks. So, like talking about example. So basically, you know, like, I think some of the softwares subtask like that I created for, like, one of the feature, I would say that it was one of the story that I worked on recently. So it was basically an integration of our catalog, right? And this was, it was on the front end ad. So basically, like, we had a feature to create our version history, right? And like, what we did was we had a feature to like, so we users, we could view the version history of a dedicated item, and also, like, they had this it, it was stored in our database with, like, version IDs with an active flag, okay? And so that was basically the user story. So like, just like users, this would be a way to view the version history, right? So this working on the sub task that I created. It was like to create the back end logic, basically to grab the versions for the given ID of the catalog, and when it comes on the next sub task, it was to create the like front end component, basically to display the version history, okay, and then the other sub task come. It was basically to create a front end history, like front end service, to basically grab the version history and also integrate it with the component. And fourth sub task, it was one or like this taste, and also, like the fifth one, it was to the it was deployed to the higher environment. So I think like this are where the five sub tasks that I decided, okay, and then so go back to the user story. Does that come in as a business requirement? Let's say like from the product, like the product owner or, or is that coming directly from like, like a salesperson? So yeah, like the features, it comes in as a business requirement. So like, yeah, what we do, what we do is, like, we break the features down into the user stories and like the features, they are written by meeting from like product owners and also business. So yeah, like, I think there is also a layer before feature, okay, and these are basically called the epics. And like epics, they are a little bit of more higher living, okay, no, so that's, that's good. And then so, I guess super, super easy, easy question, I guess what, what are you looking for in your next role? So, yeah, like in my next role, basically, like, I'm working as a Java full stack developer for like six years, right? So yeah, like in my next role, too, you know, I'm looking for a Java development positions where I can provide, like, basically what I have learned throughout my experience, and also like, I'm looking something that is challenging and exciting at the same point, right? So, yeah, like, talking about me, I'm a very big problem solver, solver. And also like, you know, I enjoy solving complex problems, providing the good solutions. So yeah, on the next role, I basically want to come up with some problems that I can basically solve in an efficient manner. And also, you know, like, be in a culture where I can provide collaborative and also where I can provide like, where I can learn my way throughout, right? Because, like, I want myself to grow more, and also, like, help the team to grow overall. So yeah, like, a good team where I can support, and also where I can get support in continuous learning opportunity. And, yeah, like, that's what I'm looking for in the next role. Okay, that's, that's good. So since you like problem solving, I'm curious what's, what's the, what's the most complex problem that you've encountered? It could be from a personal project or our professional project, and how did you come up with the solution for it? Yeah. So you know, like doing every task like, there always comes the problem, right? So, like, let's talk about one of the most complex product problem that I encountered. It was, like, around the performance optimization and scalability, and, you know, like, what happened in that? Like, in a nutshell, you know, we had a lot of bottlenecks on the services. And also, like that services we were using, okay? So basically, like, we use the AWS s3 for storing, like, a flat file, okay? Like, we used our AWS lambdas basically to process something. And also, then also, like, we use AWS parameter store to, like, basically store the certain parameters. And like, we even used as AWS SQS in order for like, those lambdas to basically have a bridge for connection, right? So, like all of this, it had its hard limit. And also, like, you know, we were trying to learn Apache Spark cluster that process data, okay, so, yeah, like, it even, like, processed Big Data, and, you know, like generated, I mean, like, large number of calls to put data into that SQS queue. And, like, Now, whenever the message is we were getting into the SQS queue, we were, like, throttling the queue. And then in written, you know, like the lambdas, they were getting throttled, okay? And like, you know, in its interactions with other AWS services, they were also, were you throttling the rights into the queue or the reads out of the queue? No, like, so basically, we were getting throttled through from the queue continue, okay? So, yeah,

Unknown Speaker 19:10

so yeah. Basically,

Speaker 1 19:13

in order to resolve the rate limiting issue, because, like, we had to make sure that, you know, our lambdas were not continuous, right? And then also, like, we had, like, an exponential backup, which is mainly, like configured, and also a proper retry mechanism. And, you know, like, we also added a deadlied queue, which helps on, like, processing so with, so, like, whenever the process it did not complete. So, you know, like we could take in the message from the dead later queue, and also, like that process them at a later time. And, you know, like we had a lot of caching integrated into the lambda itself, okay, so that, you know, like it didn't interact with the parameter stores too much. And also, you know, like a lot of those minor tweaks that had to be done, that had to be like, done step by step manner. And also that, you know, like tested through the three major three meter scripts. And also like to make it into an acceptable SLA that that was the way to go, was, did you guys have a post of SLA that you guys had to process messages with it, like, a specific timeframe? So, yeah, like, we basically, like, you know, we had a very linear SLA and, like, we were there, but we were trying to be more efficient, because, you know, like the lambda, it did not have to be always active, right? Because, like, we would get, like, the file drop in, like, you know, in certain interval, right? And even, like the file, it contained large number of data it would, it could, like, have completed overnight, but I would say that our process, it took, like, a way longer than, like, you know that before, and also, like it had a lot of data that was lost because of the weight limiting issue. Okay? Okay, and then, so, how long did it take for you guys to to to start, I guess to I guess. How long did it did it? Did it take you guys to go from

Unknown Speaker 21:40

problem solving to resolution.

Speaker 1 21:44

So talking about, like, the time that we took from problem solving to resolution. So I think, like, it took us about two to three weeks in total. But like, because the first week, it was the discovery phase, right? And in that phase, like we were looking at the logs, and also, like, you know, we were trying to figure out what went wrong, okay? And also that we'd like, we are even interacted with the AWS support team, and like, the when contact interacting with that team, like they were able to provide us with pointers, and also, you know, like, best practices. So yeah, like, that was the first week, and then the next week we were just doing, like, minor tweaks, and also integrating with this proposal. That's, that's, that's good. So, that's, that's a, that's great context. I know we're just about at time, so I'll be respectful of your time. But are there any questions for me before we before we end? Yeah, I have few questions. Like, I want to know, what does the work culture look like? And also, like, you know, I want to know a little more of the project. Like, is there a specific project that you know, like that, the project is working and also hiring for Yeah, so for us, I wouldn't say we're project based. For us, then everyone that that we hire, then they're gonna we hire for the long term. So in terms of projects, we're on engineering side, we're undergoing a very big project right now, and that one is, we are completely revamping our our core product. So our core product is, is, is our, my studio platform. So right now we're actually redoing the entire front end. So we're we're creating a brand new user experience in AngularJS are not Angular from migrating from AngularJS to next js. So that's, that's, that's what we're doing right now. So we have a front end team that's currently working on that. The things that that we're hiring for right now is we're trying to scale up our back end team, because obviously, with us building a new front end, then we have a lot of existing APIs that we currently are using that our AngularJS user interface is calling. So we're kind of leveraging that and the short term. But then what we're doing is we're exposing more data to the front end web services so that we can leverage them for our new, next js application. So that's kind of the near term pass that you'd kind of be working on with the rest of the team, is that we're just enabling more functionality or creating new APIs for our next js application on our existing platform. Then after we get feature parity with with our v2 front end, then that's where we plan and actually redoing our back end. So back in so. So right now we have a monolithic web service. So when we refactor and start rebuilding from scratch, then we'll take the micro service approach to to refactor the back end of one, one service that it has. So so in terms of, I guess you bring them to two projects, right? Maybe you want to think project based. First Project is feature parenting between b1 and b2 so Angular JS and next js. That's, that's the first, first big project. The next big project is, we're going to be refactoring, re, re, architecting, rebuilding the entire back end system, one microservice at a time. And then while we're doing that, then we're obviously going to make workflow changes for that sort because once we actually start refactoring, that's where we can make a lot of business level changes, which we don't want to do right now with our existing system. But yeah, so once we get to the new one, then we're going to be rebuilding and adding new functionality there, so that that's kind of a long term output. And then for us, for OD, we're always trying to add new functionality to our platform. So it's not like it's a one year thing to your thing. It's, it's once, once we get everything migrated over, then we're going to continue to build on top of it. Yep. And then work culture wise, we try to foster everyone to be in office. So here, if you talk about development team, there's four other developers that are in office right now. Our QA team is in office, our design team's in office, our CS teams in office. Our sales team is in office, so pretty much we're, we're all in office. And then I would say that it's a pretty, pretty fun environment to work in.

Unknown Speaker 26:28

Yeah, it sounds good.

Speaker 1 26:32

Yeah. Any other questions I know we're running over time now, yeah. So I want to know, like, what will be the next step for the hiring process? Next step would be, I'll give feedback to Nadine. And then if we move forward, then you'll probably be scheduled to speak with our senior developer in office that's in office here. And then she'll probably ask you a little more technical, some more technical questions there and then. And then after that, if you make it through that that round, then we probably have you come in office, to speak with speak with us in person. And then that one would be a little more hands on technical interview there.

Unknown Speaker 27:15

Okay, so, yeah, okay, yes you, that sounds good. Okay, awesome.

Shreesha Khadka-walmart-final-srv

Speaker 1 00:00

Oh yeah, that's all about me. Can you tell me about yourself? Okay, so, yeah, hi myself, Theresa Khadka, and I'm a full stack Java developer, and it's been like six years I'm working and, yeah, like I'm working on technologies like spring and also hibernate and like RESTful web services for basically API development. And yeah, like, I have pretty good experience working on microservices architecture, too, and this architecture I have built using Spring Boot. And yeah, like, I do have experience working on AWS cloud, basically for hosting the applications. And I can also work with AWS ECS two and like ECS, SQS, SNS and like lambdas. And also as three buckets on buckets, and yeah, like on the front end side, I have worked with Angular. And also I can work with JavaScript and TypeScript too. And when it comes to testing side, I have experience working with JUnit and Mockito. And like Currently, I am basically on our scrum team. And this team, it consists of five developers. And on this team, like, we have product owners and also Scrum Masters and so, yeah, like on a daily basis, I'm mostly involved on, like working on back in Java code, and also, mainly, like helping with the code reviews, and also, like on products and support duties. Okay, got it. And so when you say back in Java with microservices, so are you working on springboard, or any other Java based frameworks? Yeah, so basically, I am working on springboard. Okay, got it and Okay, and just go through your receipt here in a

Speaker 1 02:09

moment. Can you explain me more about the databases that you have worked on, what kind of operations and what level of

Unknown Speaker 02:19

database activities have you worked on?

Speaker 1 02:21

Okay? So, yeah, like, the databases. Basically, I have worked with both like SQL and also non relational database sites. And I have also worked with like, you know, like Oracle and also post GRV SQL and like, on the relational side, right? So I have also worked with DynamoDB on AWS, and that is basically based on the relational database. And yeah, like, I have been my, like primary work, and also I have worked with, like, simple queries, and also these queries, I have worked using Select and also inserts, but also like complex queries, using Like different types of joins. Okay, and one moment.

Speaker 1 03:22

So you said DynamoDB, right? So what kind of I mean? What was the use case that you that made you or your project choose DynamoDB, and how did it benefit your application? Like DynamoDB, basically like, what made me choose it was basically like it was on the front end, side aspect, right? So, as we know it, like that, like there are different type of widgets and also, like components and micro apps, right? So, like all of these micro apps, like it I had these are, like, visibly, like, visible to user based right, and also on their entitlements. And like they have and and like they have their end groups, and also like they are entitled for, right? So basically, what I would say on this, like it are stored on the entitlements, and also, like, they are stored on the form of JSON, right? But like, they are not standardized, and also it can actually be any of any kind of forms, right? So, talking like, so to have like, this type of non standardized data, what we do is like, we basically use the DynamoDB to store the JSON structure. And like, we could actually have different types of key and also like value fields and then the like query based on the need, right? So like, basically we use DynamoDB to restore the entitlements and also the configurations, which is basically related to, like, the different types of environment entitlements for mainly front end applications and like, we use this To mainly display the components for users. Got it? Yeah, and can you also tell me about the CICD practices that you have followed in your team, and also what was the contribution on that CICD pipeline has been okay. So like how I use CICD and like the contributions of like, what I did, so basically, like, we own our own pipelines, right? And like, we also used Jenkins for our CICD tool. And mainly, like our code base, it's Git. So like, we have everything in GitHub at first, and when we are actually ready for like, merging the code to develop like, different branches, then we have, at first like, we have our code reviews. And then also, like, once we merge our code to develop like, we can trigger a pipeline to mainly like, actually deploy the application to basically like the dev environment, and for Jenkins, like, we have a custom groovy file, and this file, it basically gives the parameters to the Jenkins, and also like to like, also Say the sub different stages that's basically needed, right? And and then we have like cell scripts that has an like outline of what we what like command needs, and this needs, it means to, like run in order to deploy the different applications, right? So yeah, and I have been also basically involved in both of these sale and also like the groovy scripts for mainly deployment of the application. And what kind of test cases have you written there? Okay, so what are like talking about the test cases on my code. So, like, all these six years, I always write the test cases on the unit test. And also, as I mentioned earlier, like, I use mainly, like JUnit and mokito for writing unit test for basically, my Java applications. And so like, you know, like I make sure I proper line, and also, like branch covers and usually over 85% so I also have been involved in writing performances test, like using J meter. And also, I have also been involved in writing integrating like, integration test using mocha framework. And like, all these are run in our deployment script. So, like, after the application builds, right? So, like, it takes the unit test and also, after the application is deployed, what it does. It's like, it basically runs those integrations and the Moca taste got it, and what kind of security practices do you have on your pipeline, like there could be multiple checks that we have on security basis, or what kind of security best checks do you have in your CC pipeline? Okay, um, so when it comes to like security practices, like whenever the application starts to build, right? So, like after each build, it basically uses like map and plugins, and also it is called like check style, and also like npmd And this, like, it basically tricks for coding standards, and also like defects are bad, or also naming, and also like unused imports. So like, those are one of the first trip right that is done. And when it comes to like, the next step that what we do is like, we like, so we do like solar scan. And also this scan, it checks for, like, a bad taste cases, and that doesn't cover, like, all the lines, right? And like, also some box. And then also we have, like, a 4545 scan. And also like this scan, it is basically used for vulnerability of the dependencies of different applications.

Speaker 1 09:56

And speaking of security, so I see have worked on both front end and back end range. So how have you secured your application, both from the front end and the back end perspective? Yeah, so like working on both front end and back end so basically, let's talk about the front end application first. So what we do is, like we make sure that our data it is properly validated, right. So like we have the proper input validation and also sanitation in place, and also, like we have proper CS behaviors to rest with, like different sources of scripts and also styling on the applications and like we have also external services that we leveraged for the authentication, and also like authorizations that basically it uses OAuth for authorization, and also, like, we always like valid user permissions on the server side. And like, not just on the front end, right? So what we do is, like, we make sure that we don't have any exposed API keys, and like, we have a proper course settings to control the different domains and like that can access different resources, right? So yeah, like these are our front end security measures, and when it comes on the back end side, we make sure that we have a proper security in places using like spring security, and also, like, we do some kind of API key validation. Okay, when you say, API key validation, can you explain? What about it? Okay. So, like, yeah, API key validation. Like,

Unknown Speaker 12:01

basically, we know that so.

Speaker 1 12:04

So basically, like we know API key validation, it is mainly used to ensure, like we have, we have proper clients, right? So these clients, like they can access your API, right? So, basically, whenever client wants to access, like certain of the APIs, and like the API key is an API key header, then like, as we know that it is included in the request header, right? So basically, like they are on our back end. And also we do an existence check, and if they like, API key exists or not, right? So after that, like, what we do is, like, we validate that key and either it's verified or not. And also, like, if it's not revoked or, like, inactive, then we like, associate those APIs with keys, right? So, like, the resources like that is trying to access, right? So basically, it is useful permission check, like, if the back end it can access the correct resources or not, and also it's given by the particular API key. So you would share this API key with every upstream or every client that would access your application, right. Okay, so have you explored anything like charter or token instead of API keys? So basically, we used what old and like the API it is like the API key. It is basically another layer of security, right? So this key, it basically it helps us in like rate limiting, and also like certain clients, but on normal, like we use, basically use go t token for authentication, because with AP is you have an overhead help rotating it and then shine with the team again, right? But the jaw tokens are the all tokens. They do it on themselves, right? So, yeah, to you, you're saying that you have used both. Yeah, like, we have used, like the OAuth and JWT both, but like the API key. I have used, basically the API key primarily, and like, I have mainly used it for basic resource check along with the rate limiting. But also, like, you know, like the whole rule, it is basically based on the access control, and it is mainly controlled through the API, yeah. Like, so, it is mainly controlled by the API, mainly, like through the payload rate of the JWT token.

Speaker 1 15:15

So, yeah, what kind of design patterns have you worked on in your application? Yeah, so design patterns, basically, I have worked with, mainly, like singleton design pattern and like this pattern, I have used it all out so basically, like, it's just so to cut your singleton concept as a default with Spring Boot, right, anything that is not cooked into the frameworks that you have implemented or used on your own. So basically, yeah, like, I'm also, rather than singleton pattern, I have also worked with a factory method pattern, and this pattern, as we know that it is basically like defining an interface for creating like the objects, right? And rather than that, I have also worked with the decorated pattern to add the standard behavior and like without affecting the objects of this class, and I have also worked with saga pattern for like my micro services orchestration.

Speaker 1 16:38

So I will do a couple of things now. So I'll give you a scenario, an SQL scenario, and then I'll give you a problem that you can solve it. So if you can share your screen, we can talk about it. If you have an ID, it will be really good. Okay,

Unknown Speaker 17:00

let me see, start saving my screen.

Speaker 1 17:11

Okay, so, so you have a table, it could be an employee table with the columns like employee ID, employee name and employee age. So so this table has lots of duplicates in it, so the entire row would be a duplicate, okay, so you need to delete the duplicates from this table without creating a new table or without adding a new column or anything. So in this the same tables, we have to do it in place.

Speaker 1 17:58

So basically, we want to remove, like the duplicates without creating the new table, right?

Speaker 1 18:14

So, like you said, the basically, like the ID will be duplicate as well, right? So wouldn't it be a primary key. Yeah, Id can be a primary key, but right now we think of it like it is currently exported. It some imported from somewhere, and right now it has lots of duplicates. How do you obtain that?

Unknown Speaker 18:42

So, yeah, like

Unknown Speaker 18:45

so, in order to like,

Unknown Speaker 18:49

delete the duplicates at first,

Unknown Speaker 18:51

We can, I,

Unknown Speaker 19:26

so, basically,

Speaker 1 19:41

so, basically, I'm thinking here like doing some kind of join on the table itself and or Something like, maybe, like using A sub query. Okay, do

Unknown Speaker 20:30

so like,

Speaker 1 20:33

maybe we will delete it from employee and so I

Speaker 1 20:52

so since, like, all ID name and then the and then the A's, they are duplicates, right? So I think, like we want to find all these three calls, and then I

Speaker 1 21:25

so here, like in My sub query, what I need to do is I

Speaker 1 22:18

so like, basically, We want to select like all Of the three again And then I

Speaker 1 23:02

so so like after selecting all three again, then I think we need to partition by something On this one. So basically, What happens here is, I

Unknown Speaker 23:44

so basically, like,

Speaker 1 23:47

I think we need to track the rows that has the same ID and the Name, and then the is right. So like, I think there is a window function like, which is called row number that is mainly Like, I think we can use it here. Okay, I

Speaker 1 24:24

so I think, like, the here, the row number, it needs to be over the partition. And, and this part isn't like, we want to partition it by like ID name, and is again, and so, so like, I think We can save this as a row number. Okay, so

Speaker 1 25:03

So I think like we, We just want to make sure that I

Speaker 1 25:52

so we will have to time box this, because you will not have you'll run out of time for The actual problem. So we'll give it another minute for this one. Okay, so here we just want to make sure wherever the row number is greater than one, then I think like we can delete them. But again, you're selecting ID, Name, age, right? And you're doing it with ID, name, Agent again, so that will delete everything again, right? Because it will not delete just a duplicates, delete everything. So no, but like, as we can see here, that we have a clause, right? So that's okay, that's okay, but you will get whatever you have to duplicate right in your SELECT statement. And when you do a delete from employee where, so when you just delete it with ID, Name is anyway, delete everything, but it will not just delete the duplicate.

Speaker 1 27:09

But it's okay. I think we are running out of time. I will move to the actual problem. Okay, okay, so the problem is that you're given an array of coins. You can say coin denomination, and you're also given a number sum. So you have to find the minimum number of coins required to reach sum. And you can take unlimited triples from the array. Each domination has unlimited withdrawals, so you need to find the minimum number of coins but to reach the sum. So the question.

Speaker 1 28:15

So basically, like, we want to reach to a certain some, like from a given array of coins, right? No, you're given an array of coin denominations and also some a number sum, okay, so you need to find the minimum number of coins from the denomination to sub add up to that sum. Okay, and you can do unlimited withdrawals from the denomination. So when I when you say 125, it's not just one to five. So it's a denomination one, denomination two, denomination five. So you can take unlimited numbers of one, two and five denominations to reach to the sum level. So

Speaker 1 29:05

yeah, something like this, if I understand, yes. So like what I wrote, it makes sense, right, okay, so what I wrote here, it makes sense, right? Yeah. I

Speaker 1 29:39

so here like, basically, we have a sum that, mainly, like we need To reach using a set of numbers. And,

Unknown Speaker 29:54

okay, so

Speaker 1 30:38

so we can discuss on the algorithm, and she's stuck, I can help you out. I'll give you some hints, and then, if you're okay, you can start coding. Okay,

Unknown Speaker 30:54

I'm just setting after method. Okay, so

Speaker 1 31:26

All right? So basically, we have a case, and then the like, if the input sum is zero, then also like we can use, we can just written that j, right? So, yeah, basically, that's like the sample case where you Like, you don't need any more coins, right? Yeah. I

Speaker 1 32:29

so I think like, if we have like the given array in descending order, right? So I think we can, like, subtract it from the value, and then also, like, call its minimum cases.

Unknown Speaker 32:47

So sorry,

Unknown Speaker 32:49

can you come again with your approach?

Speaker 1 32:51

Okay, sorry. So like, here, if we have, like, the given error descending order, right? So, like, we can subtract from the value and then also call the minimum coins. And because here, like, I think there's some, some kind of recursion going here, right? But how would you, first of all, you have to sort and how would you iterate over the sorted array to reach your minimum sum? I

Unknown Speaker 33:31

um, so

Unknown Speaker 33:33

basically, let's take an example. So,

Speaker 1 33:37

so like, if I have 11 as my input for the given question, right? And then in the descending order, then the first element will be five, right? So like I would do, I would subtract 11 from five, and which is remaining is six, right? So like, I can call this method recursively again, and but for the next value, like, which would be five, which would be for five then and then, like, it process that way, Right? Okay, but how would you do it efficiently? I

Speaker 1 34:23

i Get your approach, but would it be an efficient approach? If not? How would You do it? Inefficiently? Do

Speaker 1 35:59

so basically what I'm thinking here that I think we will have to come up with some kind of, like, way to store it right. So I know, like, the like, the minimum number of values which is basically needed to store that value right. So let's say, like, if we have an array, then then, like the index of array, then it is A value, Right? So I

Unknown Speaker 37:55

So, yeah, so here like

Speaker 1 37:59

at first, let me start by creating a table, and also the table that stores mainly minimum numbers of coins, which is mainly required for the value of the index, right? Okay, so kind of a hash table. So it would basically just be an array, but as stable it might, I think it might not be required at the moment, but, like, it's just every right so because, like, we don't need the key values, or for like, a specific basic numbers.

Unknown Speaker 38:42

So I think we can just Use the index here. Okay, so,

Speaker 1 39:31

so for the value of zero here we need zero and then also like we can just initialize it like that, right? So, okay, and I think, like the rest of the items here, it can be filled with the maximum value, like as we don't need, right? So basically, we can do a computer later on, right?

Unknown Speaker 40:11

Sorry, I was turning on mute. Can you come again? I'm

Speaker 1 40:14

sorry. So, like, here's we can see the rest of the items. It can be filled with the maximum value, right? So, because like,

Speaker 1 40:28

so here like, I'm feeling this with the maximum value, right? Because we will do a less comparison later, and then like, so, this is, I think like, so this is just an initialization of the array. So basically what I need to do here is that I need to, like, compute minimum coins, which is mainly like required For all the values from One to Some. And I

Speaker 1 42:12

so like, if my denomination at that spot.

Unknown Speaker 42:20

So like,

Speaker 1 42:23

if it's like less than or equal to my current index, I

Speaker 1 42:55

then like, we can grab that value from like, from my minimum points, And Then I

Speaker 1 44:40

uh. So like, after grabbing the values from my minimum points, like the if inner value, it's not the maximum value, then like, if my inner plus it is, uh, the inner plus one, it is less than the table, right? And then here, like, we can set the table value by table value and also increment it by One and I.

Speaker 1 46:40

It. So here, like once it goes through all the loop. So I think, like this should take care of it, and also, like we are just using an inner loop, right? So, like this loop it to it basically it is used to iterate to the denominations. And so, like it runs so that it runs that length out of the denomination time. And also we can actually get The value out here. So i

Speaker 1 48:20

i So think we need to make this the size of some like, rather than of that, rather than the length, right? So because, like, we need to do this, mainly to store all of it. So I Okay, I

Speaker 1 49:03

so I think that's it work. So, yeah, I think, yeah. I think, like, that's work as we wanted to.

Unknown Speaker 49:18

So can you remove 10? I just have five to 100

Unknown Speaker 49:30

Okay,

Unknown Speaker 49:38

and doesn't cover these cases as well. I attachment.

Speaker 1 49:50

I think, like, yeah, it should cover so like, if I have like, let's say something that is unreachable. So let's say like, I like, let's say, If I remove the one here, then I

Speaker 1 50:23

Yeah, so like the first stuff that it cannot be reached to, I think, like it returns a negative one, and Then,

Speaker 1 50:39

like it should also cover the edge cases and and like the in context of like each cases, like, whenever this sum is negative one, like, I haven't handled This sum that is less than the zero, so I So,

Speaker 1 51:13

yeah, like, I think everything, like All the other age cases, it should be handled. So,

Speaker 1 51:31

so can you try to select 2536, I,

Speaker 1 51:47

and sum is 12, no, 2536, okay, so like for this, I need to do a descending Order of array first so I

Speaker 1 52:24

I don't want you to use art. Can you try it without sorting?

Speaker 1 52:33

So like without sorting? I don't think this might work. I Well,

Unknown Speaker 52:50

it worked.

Speaker 1 53:02

So like for 2536, I see months and like to get to 12, like, you just need two sixes. So I think that this is correct, right? Yeah, I'm just thinking, yeah, about 13. I

Unknown Speaker 53:36

Uh, okay, so,

Unknown Speaker 53:41

so for like

Unknown Speaker 53:44

13.

Speaker 1 54:00

So for like 30 now we would need is six, five and two and yeah, like, either way. Like, is there any number that you want to try? Yeah, I'm just thinking about it.

Speaker 1 54:49

Can you replace that to this with two ones or just one? Yeah, and can you try 14?

Speaker 1 55:12

So for 14, like, it should be Two six and two ones, right? Yeah, I

Speaker 1 55:47

Yeah, yeah, like it is three, so it can be six, five and then three.

Unknown Speaker 55:56

So that sounds like I

Speaker 1 56:04

so I think that sums to 14. And

Unknown Speaker 56:15

can I scroll up to the problem? I

Speaker 1 56:25

Okay, can you explain me once, what, what's happening here? What have you how have you solved it? So, yeah, like here in this code, at first, like, what I'm doing is, I'm like, basically creating an array that, like stores the minimum number of coins, and like that is basically required for the index, right? So what I did for the first index is that like it will create how many coins, and for the second index, and like, it will store, like, how many coins? And like, how many coins are the like, all of these are initialized, it with the maximum value and also of the integer. And that, like, the integer that is making it unreachable, okay? And then like, here, like, I'm starting the zero index, zeroth index with the zero. And because like to make zero and you don't need any coins, right? So, so now here I'm starting the iteration, like the, basically the outer loop. Like it just keep tracks of the numbers from one to the required amount, right? So,

Unknown Speaker 57:52

like it just loops for the sum

Speaker 1 57:56

and the like the inner and like it's after, like, looping for this sum, and then the inner loop, like, basically it loops over the given coins, right? So basically here, like, for each amount, iterates over the coins, and also for like each coins, it checks if it like it is less than or also require it also equate to that current amount, right? And then, basically, like we make update to that minimum required coins by basically doing a sub share sub result check. So mainly like this update means that if we use like one or more coin, like the total number of coin needs to like, it's true would be, it would be more than the coin which is needed for the lesser one before the can I explain 20? Um, so basically, like, you know, like that, it just

Speaker 1 59:15

that is like, basically, you know, that is just grabbing by checking the value at that position, and also subtracting the sum, right? So it is just grabbing the sub result of the number of coins that is mainly required. And basically like it is a representation of the minimum of coins like that makes the amount that is like minus the coin. And, no, I didn't get that. So what are you maintaining in the index, and what are you returning in the value? And what's happening in number line, number 20. Yes. So in line number 20, like, the outer index is the sum right, and inner coin index is the coin. And then, like,

Unknown Speaker 1:00:13

I get the outer index, and in the index, right? I mean, inj.

Speaker 1 1:00:20

So like, basically, here we are, like, representing the minimum number of coins, right? So the outer index, I, it is the sum, and then the inner index, like the J, it is the coin, right? So, yeah, like, on here, in on line 20, like, basically what I'm doing is like, I'm just like, doing this sum, like, minus the denomination of that index, and before that, that's what I'm asking. So what are you for the array min coins record? So what is, what do you maintain in the index of main coins taken, the value of main coin check,

Speaker 1 1:01:10

yeah, so the main coin strike, I'm just maintaining it. So like, how many coins it is like, basically required to risk that index, right? So basically, like, on the zeroth index it has zero because, like zero coin, it is required to make the sum of like, make the sum zero at index one, right? So, like, if you have denomination of one, then you would basically require, like, one coin, so it will just have a value of one and so on. So like, the minimum coins, the rake, it just stores the minimum value that is required for the index. So I'm asking again, I didn't get that. So what is present? What are you maintaining in the index of men coins and in the value of mins? So let me do a bit by let me go it by example. Okay, so I

Speaker 1 1:02:25

so here, like the minimum coins rate, it will basically go from zero to one to one here, right? So I

Unknown Speaker 1:02:44

So,

Speaker 1 1:02:50

so basically it will go like this, right? And then also it's just storing like how many points are required to basically reach that amount. So

Speaker 1 1:03:05

so like, Since here we have one and two, right? So let's say, like, both of them, they are one and like, now like to reach to three. Like we need, like, we need one and two. And then also this, it's gonna store two and, yeah, like, the same thing with four. Also you just need, like, two more. And that is two and two. So,

Speaker 1 1:03:40

so for five. So like, you will the new one, right? Because, like, you have the value one, and so it's just like sorting that, right? So, like, the minimum number of coins required it, it is, like, to the least of that amount. And so you're maintaining the sound in the index and the number of coins in the value, right? Raise that sound. Okay,

Unknown Speaker 1:04:10

yeah, like, that's what I wanted to explain.

Speaker 1 1:04:12

Yeah, that's what continue explaining. Okay,

Unknown Speaker 1:04:26

yeah, so,

Speaker 1 1:04:34

so basically, yeah, like, again, I'm gathering all this sub result, right? So I'm doing this mainly to like, represent the number of coins, and also like, you know, like to update the minimum coins at a certain value. Like, which means, I think if we use one or more coins than the total number of coins needed to make, like I it would be more, right? So sorry, before that, I think we haven't finished with I number 28 right? So how are you coming up with that inner So,

Unknown Speaker 1:05:22

this thing,

Speaker 1 1:05:29

so is this the line that you're asking about? Yeah, okay, how does it work? So, so basically, on this line, I'm looking like I'm looking like I'm just grabbing the sub result, right? So it is like the sum that is like minus the coin at that spot. And

Unknown Speaker 1:06:06

okay,

Speaker 1 1:06:20

good. And then, like, after this, I'm comparing that result, like, to see if that is not the minimum or the maximum value. And also, like, you know, like the here, it inner, you know, the inner result, right? So the inner result plus one, it will be basically less than like, what is on the sum? Okay, say you're talking about like, right? You are now, yes, yes,

Unknown Speaker 1:07:06

yeah, so,

Unknown Speaker 1:07:10

so, like, in like, this line 21

Speaker 1 1:07:13

I'm just like, checking our new peer is possible or no, and that will, like, basically make it better period than what it already has stored, right? So on line 21 it would just check like if my newer setup here, it would be a better combination than what I have what I already have. So I

Unknown Speaker 1:08:05

Yeah, yeah, and like,

Speaker 1 1:08:15

if there's better combination, then what I would do is, like, I would just save that on to the minimum, like minimum coins required array, and then also,

Unknown Speaker 1:08:35

and I think,

Speaker 1 1:08:40

And I think, yeah, like that, should that just should give me what I require.

Speaker 1 1:08:55

And how does it work when you online number 22 so you're just comparing red, so if it should not be a max value, and then in a plus one is less than the actual number of coins for some i and if it is lesser, and you're assigning that to the actual number of coins, right? So how does it work? Yeah,

Speaker 1 1:09:29

so basically like, what it means is that, like, if we use one more one more coin at that index of j, then, like the total number of coins that is mainly like needed to make the amount at Iraq and this i index, it would be like more than the coin that is like basically needed To make in the lesser index. Okay. And so you share your eye, trading on the sum first and then on the points, right. What do you think if you swap it? So, what would happen if you swap it?

Unknown Speaker 1:10:15

So if I swap eight,

Unknown Speaker 1:10:20

basically, I think

Speaker 1 1:10:26

so, like, if I swap the loops here, then I think here, like, we will probably run into some kind of index out of a bound issue.

Unknown Speaker 1:10:41

Okay, so

Speaker 1 1:10:47

And like, I don't think there won't be as many comparisons, either.

Speaker 1 1:10:59

Okay, and what is the complexities, the space and the time complexity of this solution?

Unknown Speaker 1:11:11

So the time complexity,

Speaker 1 1:11:20

it is the big O that is mainly End Time sum and then the space complexity. It is like. It is like the big O of many, like some.

Speaker 1 1:11:42

Okay, so are you sure about this time complexity here? I think it will be even more. No, like the time complexity, it should just be a product, right? So because, like, this product, it's a loops over the sum and the denomination, and it will also loop from one to some, which is basically n. And like, there is another denomination which would be another and so, like, it would be big O of n times some Okay, for I think I'm good here. Do you have any questions for me?

Unknown Speaker 1:12:31

Yes, I have few questions. So

Speaker 1 1:12:35

I basically, like, I like to know more about the team and like, what is the setup? And also, like, what do the developers do on their day to day life basis? Okay, so in general, so you, I mean, there are lots of technologies in here, so you can find whatever technology such out there somewhere that's been used here, and so mostly it is data driven, and so you'll have to work with lots of data here, and depending upon the org that you're interviewing for it could be a position for a financial

Unknown Speaker 1:13:25

application, or it could be

Speaker 1 1:13:30

mostly related to financial application. Yeah, that's what I can say. I can't talk a lot about it, yeah, that's it. And yeah. Like, I also want to know about, like, what are the next steps after that, after this? So, yeah, the HR, the hiring manager, should reach out to you and talk about the next steps. Yeah,

Unknown Speaker 1:13:59

yeah, that's it from my side.

Unknown Speaker 1:14:03

Okay, Thank you. Bye. Thank you.

Speaker 1 1:14:36

Storing they are of

Speaker 1 1:14:58

or the answer the way. It's getting done, look at that example.

Shreesha- collabera-First- Srv-first

Speaker 1 00:00

Been working on micro services, and, you know, recently been involved in designing a couple of micro services involving the Experience API and the maintenance EPA. So like I have built this micro services using Spring Boot and also deployed the application into the AWS clouds. So some of the services in AWS that I have worked with, it includes AWS EC two, ECS, SQS, SNS and lambdas. So currently I'm on a scrum team. And my team, it has, there has like five developers. So like, we have product owners and Scrum Masters as well. So my daily responsibility, it usually includes working, like, mostly on the back end Java code, helping the team with the code reviews, and also on production support as well. So yeah, like, that's a little bit about me. When you

Speaker 2 01:00

say production support, what do you do in production support? Cherisha,

Speaker 1 01:03

so basically, like, you know, on production support. So it is mainly on a rotation basis. So basically, like, you know, every week one engineer it they get, like, assigned with the production support role. So the basic responsibility of that engineer. It is to run the religious and like, if there is any release plan or else, like, but, you know, but apart from that, like, we have to answer the pages that might come like across due to some kind of products and incidents. So, you know, like, if there is, like, any issue with within our metrics dashboard that we have to go look into our application to see that if it's in a healthy state, and like, if there is any issues that we have to help in debugging the sessions. So yeah, like, these are my duties as a part of product and support engineer as well.

Speaker 2 02:01

Thanks. So we have rest of team here, so guys go ahead and get started.

Unknown Speaker 02:11

Praveen Harsha srita, I think you guys should start.

Speaker 3 02:14

Yeah, I can go first. So I'll ask you some Java questions. Okay, what's the difference between checked and unchecked exception in Java?

Speaker 1 02:24

So talking about the checked and unchecked exception in Java. So, you know, like, like, checked, check exceptions, they are something that you need to catch, right? So, you know, like, basically, like, these are the something that happens during the compiled time. So, like, you know, you either catch it or use a like, try, catch block or declare it using our method signature, using the state, like, using a throw statement, right? So, you know, like, there is something that your program is expected to recover from like, like, if you have an incorrect SQL syntax, or else, like, if you are trying to read a file that doesn't exist, then like, it will throw something like object exceptions. But you know, like an object exceptions, they are the exceptions that occurs during the runtime, right? So you know, like, what I mean by that is there like, because, like, there is some fault in the logic. So if you are refer, like, if you are reference, sorry, referencing a null object and then performing like, certain operations on it, then you will basically get a no pointer exceptions or or else. Like, if you are trying to access a value in an array that is out of the index, then you get an array index that is out of bound, right? So, yeah, like that is basically an unchecked exception.

Speaker 3 03:55

Yeah, very good. What's the difference between map and flat map in German? So

Speaker 1 04:01

Well, talking on the different difference between map and like. So basically a map, it is a simple operation, right? That is used to translate data, that is used to transform from one object to the like from that is used to transfer like from the object from one form to another. So, you know, like, it's particularly based on an stream API. So you just have, like, let's say, like, if you have a list of objects that, and then, like, you know, so like, sorry, if you have, like, a list of strings, and then you want to capitalize all of those, then you can just use a map to do a upper case of this objects, right? But your flat map, it's just used to flatten your existing, like, existing stream. So you know, like, if you have a list, like, if you have a list of lists, then it like, it just applies a flatten operation to convert into a list of strings. So like, yeah, that's the main major difference between flat map and flat map.

Speaker 3 05:08

Okay, what is the default method in interface in Java?

Speaker 1 05:13

Well, the different method, so, you know, like the So in Java, so the basic default method, it is mainly introduced in Java eight, right? So the default method that is introduced like it is introduced for the backward compatibility, and also, like you know, it also allows you to give an implementation, which is basically giving the method body of an interface.

Unknown Speaker 05:45

What's the purpose of Rest Controller in

Speaker 1 05:51

Java? So, the purpose of Rest Controller in Java? So basically, like wrist controller, it is a combination of two annotation, right? So, which is the controller annotation and a response per unit ocean, right? So what I mean by that is, like, you know, the main entry point of the application. So, you know, here you have, like, defined multiple endpoints and different HTTP request verbs, so that, you know, like that, so that it can be used to interact, right? So your RESTful API, so like, you know, your class that is annotated with the like, annotated with the controller, and you know, like, it returns a response back to the user in form of JSON or XML or just like a plain string. But like you know, it is always a part of a response body. So rather than, you know, like reading a view, than a controller, like it does risk controller, then it returns a response body. Okay, good. Are you

Unknown Speaker 06:56

familiar with struts based application framework?

Unknown Speaker 07:01

Sorry. Are you familiar with struts framework? So

Speaker 1 07:03

on that one, like, I haven't directly worked with struts. So, you know, like, I have been primarily been working as a Spring Boot, like, Spring Boot based application. So, like,

Speaker 3 07:17

so what? What's a MVC design pattern? So,

Speaker 1 07:20

well, MVC design pattern. So basically, like, you know, MVC design pattern, it is mainly a model view controller pattern, right? So, so, like, you know, the MVC it has three components that is basically like segregating these three different aspects of your architecture, that you have your model and which is used to manage the data and the business logic of applications. So like, the main part that is responsible for manipulating your data, and like, you know, setting up the data and so on. And like, you have a view, like, which is the presentation layer, or as the user interface, and also, like, you know, the user it interacts with the view, and then the controller, it handles those user input from the from the front presentation layer, and then It updates the model, right? So, like, basically controller, it connects this view to the model. Okay,

Speaker 3 08:29

I have a hash set of integers, and I want to sort it. How do you do that?

Speaker 1 08:35

So to sort and has set of integers. So, like, you know, if you want to, like, sort of has said that, you know, basically, so basically, like, I think, like, there is an implementation of assets. So, like, you know, so mainly like, like, there is an implementation of has said, and the implementation of a set, it sorts. But I think, like, you know, this is the best idea. Like, here it would be, like, convert that, say, into an a list, and then, like, just use the collections, like, collections dot cert, collections dot cert, to sort it, okay,

Unknown Speaker 09:24

have to print an array. How do you

Speaker 1 09:27

do that? So like to print, to print the array in Java. So basically like, basically like, you just have to look through the so, so, sorry. So basically like, you just have to look through the array and then like access it by index. So mainly, like you know, you can do for each loop, and then like print each elements. And also, like you know, you can also convert the whole arrays into a string for simply printing all that whole area.

Unknown Speaker 09:59

Can't you use two string method? Sorry,

Unknown Speaker 10:03

you can use two string method, right? Yes.

Speaker 1 10:06

So basically, like, two string methods, so, you know, like, you cannot, like, yeah. So, so just to print that error, so you can use to a string, because, like, you know, it will give you the reference of the object rather than the contents, right? So to string, it needs to be over reading, but you cannot override it for that, like, already class. So, you know, like, you would basically have to use our read class. So, like, so sorry, so you basically have to use our edit class specific to the string method.

Speaker 3 10:38

Okay, yeah, I think I'm good. Sweat. You want to go next?

Speaker 4 10:43

Yes, probably so Sri Jag, you said, like you mostly work in Spring Boot application. Could you please let me know what is the difference between Add Component and Add Component scan annotation?

Speaker 1 10:59

So the difference between Add Component, so, you know, basically like component annotation, it is used to determine, like, basically like tail spring, that it is a component, right? And our component, it is nothing, but like, just us, like this string bin. So, basically, like, you know, a company, like a component. It is a very generic type where you can have like, specific types, like service, repository and risk control it. But also, like, it is just used to create a bin, but, like, but a component or scan annotation. It is used to tell your Spring application. Like, what components you need to look into for, like, scanning those bins, right? So, like, so, if you, like, annotate our component and then give a basic like, give a base package of like, you know, simple package path. It looks like more. It looks those like. It looks like those particular classes, they are inside it, and then package it, package and then, like, you know, loads the spring base.

Speaker 4 12:09

Okay, so you work in Spring Boot, micro service application. So how do you protect your micro service end point?

Speaker 1 12:19

Well, to protect the micro service endpoint. So basically, like, you know, you can mainly, like, use the print security and, like, have a role based access control. So though, like this way, like spring security works, is that, like, you know, you have your identity provided that injects an access token, basically to your request, and you know, based on that authorization headed like the BRE token, that you can decrypt it, and like, you know, sorry, like you can decode it. And then also, like, you know, look into those three parts that it has, which would be the header signature and and the claims. So, like you know, once you do the signature validation, you can look into the claims to see, like, what role the user like, what role the user is trying to access, the endpoint. And also, like you know, that is based on those particular leads. You may, like, allow the users to interact with these certain endpoints or deny it. So, yeah, like, that's the major way to, you know, basically, secure your endpoint on a microservice world.

Unknown Speaker 13:39

Okay, what is the difference between 403

Unknown Speaker 13:43

HTTP status code with 401

Speaker 1 13:44

Okay, well, I'm talking on the difference between the difference between 403 and 401 so basically, like, if you get the 401 response code, then you basically like you are not a user, right? So, you know, like, if you are like you do, if you do not have a valid credentials within the systems, but like, 403 response code, it is authorized, right? So that it means, yeah, it means that you do not have, like, sorry, 403 response code, it is, like, unauthorized. So that so that means, like, you know, you do not have a proper role to view that particular resource.

Speaker 4 14:26

Okay? And how do you deploy the microservice application, the springboard application for different environment when you have different variable environment variables

Speaker 1 14:37

to deploy the microservices based applications. So basically, like, you know, there is a concept of application properties, fine, which can you know? Like, which can be specified for different profiles, so you know. So, like, if you have a properties file specific to our Dave or QE or prod or string, sorry, staging and basically like. You know, whenever you start your applications, you can pass your like. You can pass in like, what profile the application needs to run in, and based on it like, it will load those specific properties. Also, you know, if you read like, if you need to read those properties into any like, into any particular component, you can like, use the value annotation to read it. And this way, like, you know, you can like, basically have some applications are deployed on multiple environments, but you know, like, different valuable videos of those properties.

Speaker 4 15:43

Okay, that's good. So you have worked on relational database?

Unknown Speaker 15:48

Yes, I have worked on relational

Speaker 4 15:51

database. What is the difference between union and union hall?

Speaker 1 15:57

So the difference between union and union hall. So basically, like, you know.

Unknown Speaker 16:05

So the difference? So basically, like, you know.

Speaker 1 16:09

So like, difference between union and union hall in database. So like, you know, the union like it removes the duplicates. Like it removes the duplicates. So like, if you use a union operator to combine the result set of two or more queries, that it basically clears like any duplicates that is present on those results. But you know, if you use a union, a union all that it basically like just combines all the results, says and includes the rules, right? So it doesn't care, like if these are duplicates or no.

Speaker 4 16:49

Okay, good. So how familiar you are with Unix command, so sorry.

Unknown Speaker 16:58

How do you rate yourself in Unix.

Speaker 1 17:01

So basically, I have been working with Mac and in my previous project, like I can, like, I'm pretty comfortable working with Unix and Linux as well. Like they are easy tools, and also, like they are deployed on Linux based machine.

Speaker 4 17:18

So do you run any like UNIX command and go to EC two instances and check the logs in your projects, in day to day activity.

Speaker 1 17:29

So I have used SSS command to log into the EC two

Speaker 4 17:35

okay. I think I'm good here. Aishwod, do you want to go next Sure.

Speaker 5 17:51

Let me just share my screen for a minute. Let me know when you can see it. Are you able to see that? I guess there's a table structure. And

Unknown Speaker 18:07

can you write the SQL query to answer that?

Unknown Speaker 18:14

Do you see a chat box, right? Okay,

Speaker 1 18:16

so do you want me to just write it on chat? Yeah, okay.

Speaker 1 18:28

So basically here, like we want to print marks and numbers per student having marks more than the average marks, right? Okay, so basically, like we want to do select marks and and then the count do.

Speaker 1 19:04

And then we want to select it from the students table,

Unknown Speaker 19:11

and we want to check like

Unknown Speaker 19:14

where the marks is, and I

Speaker 1 19:24

so we want to get the average of the marks from the lab. So we need to Do a sub query. I

Unknown Speaker 20:07

can you so, yeah, like, I think that's it with a COVID. I

Speaker 5 20:36

Okay, that's pretty close. Um, is what I had, is this the key being, when you're using group, you need to use a having clause instead of a where clause. So, yeah, that. That

Unknown Speaker 21:00

makes sense. Yeah. Sense.

Speaker 5 21:04

That's fine. Can Do you? Do you have experience with design patterns? So

Speaker 1 21:11

the design patterns, like, I do have experience with that, yeah,

Speaker 5 21:15

so you know, strategy, design pattern, so, you know,

Speaker 1 21:20

like, I have been working with Singleton. So basically, like, you know, strategy design pattern.

Unknown Speaker 21:29

So, you know,

Speaker 1 21:32

strategy design pattern. So basically, like, you know, on that one, like, it is, like, how you align it, and like, you know, I think like you all, like it always allow you to choose, like, what kind of process you want to run during your runtime guide. So I think like you still have some kind of interface that dictates the strategy. And then, you know, like you have, like, concrete implementation of those, like, of those particular strategies that can be picked up from like that can be picked up during the runtime, right? So, you know, like,

Unknown Speaker 22:14

can you think of an example where,

Speaker 5 22:21

let's say you you have a factory producing cars, and depending on the, on the, on certain parameters you want to create, you know, a certain model. So how does the Strategy Pattern help you go there? So,

Speaker 1 22:49

so the strategy pattern, so, if you have, like, a factory producer, and maybe like, you know, you can like design based on like, if you have a SUV or sedan or truck, right? So, like, if you have your car strategy that returns a card object with a simple method called Create car ride and like your car, it is like what you want to create, right? So it's just a plain object that has like types numbers or like numbers of doors and like other parameters, like colors and so on. And then you know, like you do, you can do like concrete strategies, like sedan strategy or a drops type strategy that implements and that implements that create our that create car, methods like, where you can have that like algorithm to create that particular car. And now, like, on your factory, like, whenever there's a create car, now it called it's it called it delegates into that particular strategy. So, you know, like, you have to pass in, like, what kind of strategy you want to use on that particular factory. And then, you know, like, whenever you can create it, it will just basically create that particular car. Okay,

Speaker 5 24:18

so in the chat, I gave you a small programming question. Basically, you have an item object which has three field, ID, Name and Description. So basically using and you have a list of these items. So how do you create, how do you process the list using streams to implement this given condition?

Speaker 1 25:02

So you want to group by ID, like having count greater than pi,

Speaker 5 25:16

so you have a list, right? What is the first operation you do on it. So, so

Speaker 1 25:25

basically you would convert it into a stream, right? So, like, it just, it's just a dot stream method. So

Unknown Speaker 25:35

what is the next operation that comes to your mind?

Speaker 1 25:39

So basically, like you need to do, like some kind of collections or into a map, right? And, you know, like we would basically, you know, we would basically collect the item, where, like, where the key it is, the ID and like, and the value it is the count of that ID.

Unknown Speaker 26:08

And then,

Speaker 1 26:11

like, we have to filter out the values, like, whose, like, which count is greater than five. Guy, so, like, it would be, like, another operation within that particular map that we collected to like, like, you know, like, we would just take, like, if the entries value, it is greater than five, or something like that. So like, I think that would be the overall process.

Speaker 5 26:39

I want to get the value, how many items there are same ID? How are you getting the count on the items with the same ID?

Speaker 1 26:54

So on the first collection, I would be collecting the ID and the value, right? So you can basically use our collectors dot counting to like, to count like, how many I like, how many times the certain key appears?

Speaker 5 27:15

Okay, okay, thanks. I'm done with my questions.

Speaker 6 27:23

Srisha, do you it's Sri SHA, right, not right. Yeah, it's Risa,

Unknown Speaker 27:29

yes. Risha, do you have any questions for us? Yes,

Speaker 1 27:32

I do have few questions. So first of all, like, I would like to thank you all for so much. Like, thank you so much for the time today, and it was nice talking to you all. So like going to the questions, like, I would like to know, what is the formal process like, what will be the next step for the interview process?

Speaker 6 27:51

So we will talk internally this panel, like, who have talked to you, and then if needed, we will probably call you for an in person interview within the next few days that will be on site in McLean. Hopefully you said you are based in Fairfax, right? Hopefully it's not too far for you. Yeah, so then I think you will know that. I mean, what the next steps are after that.

Unknown Speaker 28:22

Any other questions,

Speaker 1 28:24

yes. So I want to know, like, what does the work culture looks like in Freddy Mac,

Unknown Speaker 28:32

Freddie Mac, work culture, again, it depends on,

Speaker 6 28:37

in general, everybody is expected to be answered. You got, you knew that, right? I mean, this is like, not remote. It's going to be a hybrid situation. And as far as work is concerned, this entire team is focused on maintenance of the applications, which includes, like, production support and as well as any enhancements and bug fixes which come with that. Okay? So we are, we will be supporting monthly close activities, financial close activities for the first few days or the last few days in the month. When you do that, the work culture will be like, you will be supporting those applications. So you'd be expected to, like, look at the logs, figure out if there are any issues. I try to, like, resolve those issues. So when you are resolving those issues, sometimes the time might get stretched. Stretched a little bit depends on, like how you get to the resolution, how quickly you get to the resolution. And if there are any dependent deployments which will fix those issues, you'll probably have to do it afterwards.

Unknown Speaker 29:39

Anything else? Nothing from my side, Do.

Unknown Speaker 29:44

All right, so thank you very much. Thanks for your time.

Shreesha-EquipHealth-SRV-First

Speaker 1 00:00

TV with you. See, you know how get a little more information, maybe ask them general questions. Then we'll do an architecture exercise, and at the end, I will leave some time for you to ask me any questions. But you don't have to wait till the end. You want to ask me questions anytime you're free to do so good doesn't have to be just the one way. Sounds good. Last thing, I have my notes over here on this monitor, so you're going to see me looking over here a lot. I'm not zoning out

Unknown Speaker 00:35

just looking at that screen. All right. Cool.

Speaker 1 00:41

So let's see. Let's talk a little bit about Old Dominion. And I know you haven't been there that long, but still you can talk a little bit about it looks like they brought you on to

Unknown Speaker 00:58

these factors

Unknown Speaker 01:01

into microservices. Yes. What

Speaker 1 01:07

was the benefit of doing that? So,

Speaker 2 01:12

like migrating these, like, basically, you know, whenever we move from, like, whenever you go from monolithic application to a micro services architecture, so it basically makes your domain specific to a service, right? So you have a faster development, and also like and deployment, and also like you know, where you have independent services that may be skilled up as needed. So, you know, one of our services that we use a lot, it is the Experience API service, which is useful for, you know, generating the entitlements and grabbing them. So this, it basically dictates that that work is displayed on the front end site by returning this entitlement, right? So this like, it needs to be highly like, highly available. And also, you know, have a higher number of instances running as like, compared to something that is like a profile preference API, which is, you know, used for changing that preferences setting. So I would say like that is one of the main advantage that it provides. But also, like, you know, it also might like migration that also facilitates like the technology, and like technology, like that is based on the advantage, right? So we can generate, you know, some type, like, some lambdas that is written or in node.js and we can write like micro services written in Java, two, right? So basically, like, you have a like, granular control on, on, like different services. So I So, yeah, like, I would say these are some of the ones. Yeah,

Speaker 1 03:07

cool. So did you end up using both, like Java services running and lambdas also?

Speaker 2 03:14

So, you know, basically they're, like, we had the micro services of primarily in Java and Spring Boot. And then we had, like, some events that we needed to react to, basically based on the messages that was posted by this Java application to SNS topics. And then, you know, this would be subscribed by a queue like which would trigger AWS lambda. So, yeah, I do have experience on both.

Speaker 1 03:53

So you have a lot of points on your CV here. One of the things you mentioned is you use the Java 17 features you mentioned, sealed classes, pattern matching and records. Not gonna lie to you, I'm not familiar with all of those things. So can you explain the benefits of sealed classes, pattern matching and record?

Speaker 2 04:15

Sure. So you know, like, basically like, like shield classes. It like silk classes, like, it is a control over polymorphism, right? So, you know, basically like, you can just create like silk classes, you know, which can permit the which can, like, permit certain classes to extend, like, to extend the particular base class. So, if we go on to example, like, you know, if you create a base class, base class like, which is, which is a sealed base class, then you can, like, basically create a permit keyword on that base class, right? Which gives, you know, a list of class that can particularly, like, extend this, extend the vehicle, so, you know, which improves the encapsulation and also like services of those, right? So that's, you know, that's that basically, that's the primary one that I mentioned, and then the next one. What

Speaker 1 05:24

benefit does that give you? Sorry, what benefit does it why would you want to seal a class instead of just making it final, for example, or just doing nothing?

Speaker 2 05:36

So basically, you know, like it helps like compiler to catch like to catch the extensive early and also, you know, it improves like encapsulation and security.

Unknown Speaker 05:50

How does it improve security?

Speaker 2 05:54

So, you know, like, mainly, it restricts, like, unwanted inheritance, right? So avoids, like, it avoids the other classes. So, you know, for example, like, if I want to have that animal class, like, only specifically allowed to animals, then, like, I cannot have like, a penguin, penguin class there, right? So, you know, like, like, like the penguin, it plants like, extend that animal, right? So it just basically prevents that, which basically, you know, prevents this unauthorized extension. Thank you.

Speaker 1 06:38

And pattern matching is that just regular expressions, or is that something else?

Speaker 2 06:45

So, you know, like, it's, it's a switch case, like, where you can basically match based on the type. So, you know, you don't have to check, like, using the instance of operator, okay,

Speaker 1 07:07

cool, let me see. Sorry, I'm still taking notes. So you mentioned inheritance. I noticed you also mentioned that you applied dependency injection and version of control principles. So what are the situations where you'd want to use inheritance versus composition?

Unknown Speaker 07:35

So, you know,

Speaker 2 07:39

well, like, basically, let's say, like, I think, you know, inheritance is like, whenever you have a relationship, like, when you like, when I whenever there is a relationship, right? So, for example, you know what, like, what I gave earlier, like, the animal, example. So a dog, it is also an animal, right? So it's like, it's kind of like a strong relationship. And then, you know, if you want to reuse the behavior, like, without rewriting the code, like, if multiple subclass, they need the same base functionality. And also, like inheritance, like, provided that this is to mainly prevent any kind of duplication of the logic writing. But you know, our composition is whenever like you have like, you know you have like, had you have like, a relationship, right? So something like dog has a tail, or like dog has paws, or something like that, right? So it is like, basically when you want behavior that can be shared across the like on reality class, let

Speaker 1 08:54

me see. I see you have experience developing with Hibernate. What did hibernate ever give you any difficult problems you had to solve to get it to work?

Unknown Speaker 09:07

So thinking on the Hibernate,

Unknown Speaker 09:11

so I would say that, you know,

Speaker 2 09:17

sorry, so basically, like, I don't recall the exact problem right away, because, you know, like my primary experience, it has been with Spring Data, JPA directly, but hibernate like it does have a very common problem, like called the like call the end plus or n plus one query problem. And when you know, when fetching all the entities, it executes one to fetch the parent class, and then, and then, like, you know, additional queries to fetch a such high class.

Speaker 1 09:53

Yeah, did you ever have to figure out how to solve that problem?

Unknown Speaker 10:00

So, you know, like,

Speaker 2 10:02

so, just like, in order to solve, solve the problem, like, I think you know it. So I think it provides the So, mainly, like, you know, so I'm so sorry. Just like to solve the problem, I think it provides like, you know, you can just use joint fetch, where you avoid like, extra queries while fetching the entities, like with fetching the inner like child entities and

Unknown Speaker 10:44

Okay, cool.

Speaker 1 10:49

All right, you mentioned building responsive UI elements with Angular js, which is cool. How do you make those elements responsive?

Speaker 2 11:01

So you know. So basically, in order to make like the responsive UI, so basically, like, you know, you need to use the media queries on your CSS, right? So, just like, adjust the style that is based on your screen width, like, you know, if you have the media screen with certain max width, and also, like, you know, you can, like, basically have, like, have this responsive Ui, ui so I think like, there's also, like, you know, Angular multi material UI framework that you can use, which gives us like, you know, which gives us built in support for it.

Speaker 1 11:46

Okay, so you didn't have to, it sounds like you're optimizing the rendering of components. Did you ever have to optimize like, benching data and displaying in the front end?

Unknown Speaker 12:02

So to display the data. So, you know,

Speaker 2 12:06

so mainly like, to like, like, I mean, like, you know, I have worked with aid like, you know, the HTTP clients and the Apollo clients for like, you know, waste and Graph QL, QL, basically, to face the data. And also, like, display the data where we need to make like, we need to make sure that we have a proper facilitation of large data set. And then also, you know, like, once you reach the certain scrolling point, you load more data,

Unknown Speaker 12:43

infinite scrolling. Okay,

Unknown Speaker 12:51

so you've done a lot of things. You were

Speaker 1 12:56

executing a DDL for Postgres, so I assume that means you were designing tables. So what are, what are some principles you you follow when you're designing table structures, or database relational data.

Speaker 2 13:13

So you know whenever we are designing the data structure. So what we need to do is like, you know, you mainly like, basically like, you know, make at first, we make sure that like, what we keep in mind is like, what kind of operation we would be prioritized, right? If you want like, faster read or like, or like faster than like, if you have, like, you know, if you have like some kind of memory like, memory constraints, then, you know, yeah, like, if you have like, some kinds of memory constraints, then I think you need to basically account for like, you know, what kind of indexes, what kind of indexes need to be addressed, right? And also, you also need to keep in mind that, like, how much data you need and what kind of relationship that exists between the data. And also, you know, you also, sometimes, like, you also need to think of that kept theorem where you can, you know, where you can have a max of like, mix of two, like against the consistency, availability and partisan tolerance, tolerance, right? And you know, like, I think normalization and denormalization is another one.

Speaker 1 14:39

Okay. When would you want to normalize tables and when would you want to denormalize, when talking

Speaker 2 14:46

on normalization and denormalization. So you know, basically like both normalization and denormalization, like they has advanced advantages against different things, right? So if you want like, like, if you want your data to be written, like to be written right away, without, you know, without like, without the issues with inquiry, with your query, then you know, like you would maybe go with denormalized, right? Basically, you know, it improves the read performance, and also it is useful for data warehousing, or also, like, you know, other, like, other kind of data, where you do not need to worry about how you are storing it, right? But, like, it's just the retrial, right? And if you like, you know, if you want a structured data with proper data integrity and consistency around like, you know, working with multiple joints and have like, frequent updates on those data, then you would go with normalized data, where you store, like, you know, different kinds of data in different tables. Cool. Yeah,

Unknown Speaker 16:09

nice. Let me see.

Speaker 1 16:14

So in addition to joining databases, obviously you're writing code, it looks like you have done code reviews, you look for tools to automatically detect code smells. You've resolved issues Facebook. So what are some coding best practices that you try to follow and that you look for in code reviews?

Speaker 2 16:42

So well, you know, code reviews, like, some of the, some of, like, some of the coding best practices that I look for. So, you know, during the code reviews, like, it is the reliability, right? So that is, like, one of the most important thing that I basically need to have a proper understanding of, like, you know how the code looks like, and even if it's following certain conditions that is set up by the company, like the namely structure, or else, like, you know, the formatting structure. So that's one of the basic thing, right? So, but apart from that, if you have a very long nested conditionals or loops, that might be, you know, basically be basically like causing some of the issues, and there's a way to optimize it, okay? So beside that, I also look for our unit test that support the changes that the port, like that the code is doing. And also, you know, like if there's adequate, adequate testing in our unit level, covering, like, all of the scenarios and then zero is also, you know, checking like if there is any improper logging or unnecessary logging, and the code, like it follows the proper security standards, like, you know, preventing the SQL injection and like SQL injection attack, or also, like you know, or any other like scripting attacks, beside that, also, like You know, error handling is, like, is another one that I look for. So, yeah, I think, like, those are the ones that I would primarily look for in the code reviews. Nice,

Unknown Speaker 18:32

talking about getting,

Unknown Speaker 18:36

getting work.

Speaker 1 18:38

So obviously the engineering team works with a product team or some team that's defining your priorities. Have you ever had to push back or work with a product team who wanted you to do something you didn't think was the right thing to do?

Speaker 2 18:55

So, you know, basically, like, those are, like, the difficult situation, right? Because, you know, you like, we might get involved in something that has a very high technical requirement, but sometimes, you know, there are product teams like, you know that needs to go out as as soon as possible, right? So I think, like one of the time when this happened was when we were working on enhancement of, you know, of one of our filters to basically, you know, to basically, like, stream the data into a table. But during that time, you know, like our product team, they wanted to post hard on the self servicing portal that we were working on. So basically, you know, like we had to communicate within the product team and also explain the importance of the changes that we were doing, and also, you know, like we were doing, basically for stability of our production code. But you know, like later, we came into an agreement that, like some people, they would be allocated for, like, for like this future work and the other one, like other they would like still be working on the self service tool, but it's just like, you know, I would say, like, a give and take, right? So against, like, what needs to go and like, like, one day needs to prioritize wants, like, we have everything set properly, but like, you know, but it's like, something that is high priority. Like, I like to jump in and, like, I like to jump in and fix those issues at first as well. So, yeah,

Speaker 1 20:50

was there ever a time when you had a very tight, very hard deadline that you had to meet?

Speaker 2 20:58

So, like, like, tight deadlines, so, and if the we left, and if the project is given, like the deadlines, it's one of the most important part, right? So, yeah, like, I did have a couple of those encounters. So, you know, we were working on our file upload process, sorry, like a file load process, where we would go, like, we would get a file, and then basically, you know, like, extract the data from the file for processing. And then, you know, we would extract the data and then process it into, like, into a canonical JSON and then send it to SNS topic. So, you know, like this. It basically seemed like it basically seemed like simple at first, right? But there was, like, a lot of complications because, because, like, how they generate, how generic the data was, and how it was supposed to handle the data based on the metadata of the file. So during that time, like, I had some extra work, and I had a, like, very tight deadline. So the way I handled the handle that deadline, like that time, it was firstly being like transparent, that is taking, like, transparent, why it is taking longer than expected. And after that, like, after that was said during my standard I then asked for help, whoever had, like bandwidth to help me with the change. And I think, like, you know, one of my team, one of my teammates, he, like, they helped me to, like the jump, to help me so that we could get it out in time. So whenever I'm in current scenario, I always make it transparent and then try to ask, like, ask for help, if available. Cool, nice. All right,

Speaker 1 22:53

so obviously you're looking for a new position. I think you're currently on a contract that's ending, right? But still, can you? Can you give me an example of something, either at your current position or at Citigroup, something that works really well, that you would like to continue doing, or you'd like to see also happening at your next job?

Unknown Speaker 23:22

I would say that, you know,

Unknown Speaker 23:25

like, basically, I think

Speaker 2 23:29

we do a weekly tech sync. So, like one, there's like, always something, always like someone presenting a new change or a new way of new framework, or, you know, anything that is like, anything that is new that someone is working on, basically, it helps us broaden our knowledge, right? So, you know, it happens like every Tuesday morning at 10, and there's, like, a walkthrough of how the code works, or basically, like, you know, how certain technologies, they are integrated, or something like that. I think, like, this is something that is really interesting. And like, it helps everyone basically get better. And from this, like, I learned a couple of brick best practices in this meeting. And also, like, you know, I also presented how our like, how our history, how our streaming engine worked for the log aggregation. So I think, like, that's something, what I would like to continue,

Speaker 1 24:38

let me ask you the opposite, what's something that didn't work? Well, okay, one of these jobs that you would definitely not like to see in that position.

Speaker 2 24:48

So what did not work? Well, so I would say, you know,

Unknown Speaker 24:59

I need to think of that

Unknown Speaker 25:01

as, so, you know, like,

Speaker 2 25:07

I used to enjoy my work when working. So, like, when working, I used to enjoy my work. So

Unknown Speaker 25:13

I would say, like, you know,

Unknown Speaker 25:16

in one of my work. So we dedicated,

Speaker 2 25:20

like we dedicated for four hours of heads like, four hours of heads down. So, you know, like, time for working. So we basically had four hours blocked and like blocked, like blocked during the day. So, you know, we basically, like, like, we had to like, block during the day. Like, we don't have the meeting across the people. And also, you know, like, we had the focus hours for like, you know, like, for like, four hours. And this one like, like, which didn't work well for me so, and also, like, you know, that was something that I don't like, and it's like, like, it did work that much well. So I think, yeah, that's something that I don't want.

Unknown Speaker 26:09

Why didn't it work well?

Speaker 2 26:13

So, you know, like, the main propose of focus, like, like, for our it was to work on on your dedicated task. So, you know, like, if you have that time frame are just allocated for you, but working on a team like you get pulled into some issues, right? And you get pulled into helping someone like you get pulled into looking at an issue in a QA or production, or like, anything, right? So, like, that's like a four hour. So it was like it, it never was for our rather than, rather than an hour. So, you know, if it was to like, if it was like to come back again, we would need to dedicate certain people or certain teammates to basically, you know, have a focus, like focus hour at a different time, and then the like others they have like others to have that at a different time. So, you know, whenever there are issues like people, they are not in a focused time, then we can address it, and then we converted it office value, right? So I think like, that's how it should be.

Unknown Speaker 27:27

That makes that makes sense.

Speaker 1 27:31

Couple more questions. Can you tell me about a time when you failed

Speaker 2 27:39

so well, you know, I think I would like to basically, you know, I would like to basically remember a point where we were migrating from hibernate, criteria API To JPA, criteria API, okay? Because of the, you know, deprecated library, and also the vulnerability of the Hibernate criteria, the failure that I had there, it was like it was not meeting the deadline, okay? So, like, you know, I had it extended for a sprint because of the amount of work it had. And I was also basically like, you know, basically I was also able to do all the coding challenges, and also like, the coding changes that needed to do. And also like, you know, basically we have a good project after the upgrade, after the upgrade. And also like, you know, but towards the end of the second sprint where I was working on this, like, you know, we had a lot of API call calls like that, we are performing poorly. And even though, you know, this upgrades, it was supposed to improve the performance. So I was not able to complete that in time. And I would say that I took that like, I took that as a failure. I mean, like, you know, after that, like after the sprint closed, we basically created a second part to improve the performance, but still, like, you know, it was a failure for me, in my for in my for me in my book, you know, because I underestimated the effort. And then, you know, it kind of was a learning experience for me.

Speaker 1 29:30

What if you had to do it again, all over with what you know now, what would you have done better at the beginning of that project? Or you couldn't know that? What you didn't know, right? He didn't know how long it was going to take. So how could you improve that situation?

Speaker 2 29:46

So I would say I would probably start with, like, you know, some kind of proof of concept, like, for this kind of work that involves, you know, changing all the database interactions and how it interacted with the newer update frameworks. And also, you know, I would say how it differs, and then basically, I try to have a better story by breaking, you know, different services and chunks into the individual tasks. So if you have like, individual units that gets migrated part by part, rather than everything at once. So yeah, I will do that way.

Speaker 1 30:35

All right, my last question before we talk architecture is, what should I ask you? What do you wish I would ask you? So

Unknown Speaker 30:53

I would say,

Speaker 2 30:59

so that's really interesting question. So I think

Unknown Speaker 31:05

you should ask me,

Unknown Speaker 31:10

What do you want me to know that I haven't asked about yet,

Speaker 2 31:15

thinking about that. So I think you should ask me, like, how I work within a team, or, like, what I look for within a team. So I would say, like, you know, I'm very collaborative and a team player that I enjoy working with. And I think trip team strength, it is very important for working within the team. And, you know, I like to say, like, I like to say what I know to my teammates and what I want to learn from my teammates as well. So I think that's one of my strengths, that I promote a collaborative environment. I think, I think that's something I wanted to

Speaker 1 31:59

express, cool, excellent, all right, let's talk architecture. So the way I like to do this is I will give you, like a just a high level description of some of the things that equip does, and then I'll ask you, if you had to design this from scratch, what would you do? Okay, so I'm just kind of a lot here. You might want to take notes, but all right, whenever you're ready, I will give you the scenario. Okay, I would okay. So, so here, here in equip, we, we treat people with eating disorders, and these people are called patients. We also, of course, the people who are treating the patients are called providers. These are, you know, people that work at equip, their therapists, their dietitians. They're psychologists, doctors, peer mentors, family mentors. These, these people are all working to treat the patients. Some people ask, Are they contractors? Know they work for equip, so they're working here. The patients also have someone in their life who's helping them through their treatment. Usually it'll be a mother or a father, especially for underage patients, but sometimes it's a friend or a sibling or somebody, and those people are called supports. So we have the patients, the providers, and then the support. The main vehicle for treating patients is through one on one appointments with their providers. So we have to have a system where people can schedule appointments with their with their providers. They should be able to schedule recurring appointments, they should be able to, you know, see when a provider's calendar is blocked and not schedule appointments there the patients or the providers need the ability to, you know, change an appointment or cancel appointment if they need to. And all this is done with a with a calendar view that they have so they can see their calendar. After the appointments, the provider is required to write a note. This is a clinical note. It's an official medical document. So once it is written, it is immutable. You can't change it. You can add addendums to it later. You can go back and say, Well, I said yellow, but I really meant blue, but you can't change the yellow to blue within the original note that's immutable. Outside of appointments, we try to get information on our patients through surveys. There are about 40 different kinds of surveys. And these aren't like, how are we doing surveys? These are, how are you doing surveys? How much you eating, how much you exercising, you have suicidal thoughts, all those sort of things. And all these 40 surveys have a set list of questions for each survey. Sometimes they decide they need to change the questions, they'll remove a question, they'll add a question, they'll change a question, and we have to make sure that the answers we store from the person who took the survey. We have to make sure that those answers are associated with the correct list of questions. So if we send out a survey on Monday and it changes on Wednesday, and they answer it on Friday, then those Friday answers have to go to the Monday version of the survey, not the Wednesday version. And finally, we have notifications. These are SMS or email. They'll go out based on user preferences, whether they want SMS or email. And these notifications are things like you have an appointment in 10 minutes. It could also be like you have a new survey you need to go fill out. And there are just absolute tons of different kinds of notifications. So realizing we only have like, 20 minutes, and you can't design every single thing. If I told you we needed to build the system from scratch. What would you do? Where would you start?

Unknown Speaker 36:41

Okay, so,

Unknown Speaker 36:44

let me just recreate which I can.

Speaker 2 36:59

So you have like a healthcare system where you have patients, and then you have like providers, and then you have supports, like three different types of people, three different kinds of entities, patients and their support, and then providers, and then, like the functionality that we should be looking for, for our applications, it needs to provide the CL east of providers, right? So, so that, like, you know, patients, they can choose a provider and book an appointment. These

Speaker 1 37:46

patients do not choose their provider team. We choose their provider team. But So like, if you have a therapist, that's the therapist for this patient, the patient won't have another therapist, but they do get to choose, like when they're making an appointment, they can choose which of their providers they're Making that for. That's true.

Speaker 2 38:20

So the only thing that we can choose is the, you know, like, only thing that patient can choose from patients perspective, it is the, it is the appointment, like, appointment stand, which is, you know, basically going to be an appointment service, which, which is like, you know, which is like a managed calendar and scheduling service and other service that they have control over is a service, right? So, you know, like, so which, basically, like we said, which, which, basically it would be, you know, like, if you choose, the only thing that patient can choose, like, from person's perspective, it is the appointment stand, and which is basically going to be an appointment service, by the like, by the by the patient service and The responses, right? So some of the databases design that I would start with would be like patient entity. So, you know, like the patient entity, it will contain the details of the patient like, like their identifier, their name and email, and also like, a list of appointments and and also like, you know? And then we will, like, we will have a provider entity which basically has some kind of name like, especially in their like, especially in their list of appointment, appointment, right? So the appointment will be related to, like, What kind like, what appointments the provider has. And then, you know, like, the appointment entity, which basically has an appointment date and time with like, you know, with like, with like, a recurring flag. And then, like this appointment entity, it will basically have a mapping of two patient and also provided, right? So a patient, like a patient, it will have, like one to many relation with appointments provider will have one to many relation with appointment as well. And then there are some other smaller entities, like the server response and patient provide, like as patient and also like, you know, provided, provide a map that you said, right? And then the provider note that is immutable, right? And once read like only note which does not expose any kind of like, any kind of functionalities. So, you know, so those would be my database entities out that I'm thinking right now, and

Speaker 1 41:26

I follow you on the notes. Could you explain the notes again? Sorry. Explain what you were talking about with the notes in the database. I wasn't 100% following you.

Speaker 2 41:37

Okay? So basically, you know, we would have a provider note where you should have a functionality of like, functionality of write once, and then also like read write, and then read write. So once a note is saved, it cannot be edited or updated. And you can basically do this by using the update table, field of your column, annotation to false right, which is, you know, which is basically like non nullable or and non operable fields. So,

Unknown Speaker 42:17

so you know.

Speaker 2 42:20

So this patient note, it will have like, you know, it will have like, a map deal with a patient and provider as well. So,

Unknown Speaker 42:35

so basically, you know,

Speaker 2 42:38

so basically here I'm thinking of like, I think it should be many to one. So, like, so like, there are many providers. So sorry. Let me think again. So let me think again of this. And

Speaker 2 43:01

I think, yeah, it's only one provided to many notes, and then one patient to many notes, because, you know, provided, they can have like, many notes, right? And then patient, they can have many notes associated with them, right? So,

Unknown Speaker 43:21

so I so

Speaker 2 43:23

I think, like those would be my database entities to start from,

Speaker 2 43:35

and also, so, so now let's talk about the appointment functionality, okay, so, you know,

Unknown Speaker 43:45

so for the appointment functionality. So basically,

Unknown Speaker 43:58

so basically, like it should be,

Speaker 2 44:02

like, it should be able to view appointment list available times and create an appointment, right? So I think this would, this will, like, basically be a REST API with, you know, users like, where users are the patients, and then first browse like, what time frame is available, right? And then, you know, like this would be able to create an appointment. Like this would be able to schedule a recurring appointment. This would be also able to cancel the appointment. And then also view the appointment, right? So if I were to schedule an appointment like those are the options that I think like that. I think I would want. So I think, you know, some of the end points that I would look for was get appointments for patient and get appointment for providers. So, you know, so this, like, so this, you know, like this, would basically give you the list of appointment the patient has, and also the list of appointment of provider has, and if provided, they want to look at their appointment, basically to create the dashboard, you know, where you have available slots. We should have, like, you know, we should have another end point on to get the available slots for the provider. And, you know, we will have like, we will have a post endpoint, which is used to create an appointment. It would be like, you know, it would be like a slash book like, it would be like a slash book appointment endpoint, which would we should, you know, we should be basically authorized for patients that will, that will take in, like your like, like your other appointment request, like the patient name, patient data, and also the time slot. And then after that, like, basically create that, like, you know, there would be, like, a boot mapping option to cancel the appointment, which basically changes the appointment that is based on that appointment ID to basically cancel the appointment. And so what?

Speaker 1 46:41

Mean with the appointments? Well, first of all, back to the database. Were you you mentioned that you would need an appointment service? Sorry, not the database. You need an appointment service. What other services would you envision here.

Speaker 2 47:05

So you know, here, what I am thinking of is survey service, which, you know, which basically keeps track of the survey questions and answers. Another one would be a notification service, you know, which basically saves user preferences, and then, you know, send a notification that is basically based on those preferences. And I also think, I think the I think the notes,

Unknown Speaker 47:51

I think the notes, I mean, let me think,

Speaker 2 47:56

I think it might be okay to put in the provider service itself. And I think you know notes for the providers should be the provider service and

Unknown Speaker 48:14

and I think those are,

Speaker 2 48:18

those are the services like we would have your patient service, and also you would have your provider service appointment, and like, talking on the service, notification service and, and also, like the, you know, like the survey notification service, And, yeah,

Speaker 1 48:41

I don't want to cut you off, but we are nearing the end of our time, and I do want to make sure that you have some time. If you have any questions for me, what can I tell you about? Anything?

Speaker 2 48:54

Sure? So like I do have few questions, but before that, thank you so much for your time today. And it was, like, really nice talking to you. And I think you know, like, I would like to know, like, what do you most enjoy? Like, what do you enjoy about, mostly, about working here.

Speaker 1 49:15

Well, one of the things that I enjoy is that it's kind of a smaller place, so every individual person has the ability to make

Unknown Speaker 49:26

a big contribution to everything we do.

Unknown Speaker 49:30

So I think we have,

Speaker 1 49:34

we've been hiring more developers, so we have more than we used to, but we don't, I don't think we have 20 developers in the company. So each developer is 5% or more of our entire development. So everyone has the ability to really make an outsized difference if they if they choose to step up, right? If they choose to step up and

Speaker 2 49:54

contribute that way, that sounds shaded, because yeah, and I would also like to know, like, what does the like, since you like, since you mentioned that you are so, like, the team, it has small group of people, right? So, like, what like? How does this ceremonies? Like, you know, like the places, like, sorry, the play like the developers, they gather in a small like, in a single meeting, or, like a daily hurdle, or it's like spread out against certain use cases or domains. So

Speaker 1 50:32

that's a good question. So we're, we do Agile development, so there is a daily meet for 15 to 30 minutes for the team. So we all meet together for that long also regular, just agile ceremonies. So we have a, I think three or four backlog refinements per sprint. It's a two week sprint. We have official sprint kicking off ceremony. We have retros at the end of every sprint. So yeah, the team's together quite a lot, virtually, obviously. But yeah, every day at least we meet. We do try to have no official meetings on Tuesdays and Fridays. But that doesn't mean you're not allowed to talk to anyone. It just means we're not going to put anything on the schedule. Then people still get together and huddle if they have questions or they want to do some pairing or whatever. But yeah, so we try to keep meetings to not every day of the week. That's really

Unknown Speaker 51:39

cool. I think

Speaker 2 51:42

last thing I would like to know, like, what will be the next step after this? So

Speaker 1 51:50

after this, if you move to the next round, then the next step would be a couple of more technical coding interviews. So an inside and a back end side, and that would be it, basically. So, yeah, we would schedule that the back end will be Java, which it sounds like you're familiar with. The front end is usually more react, but I think they can accommodate Angular also, so that's more comfortable. Maybe they can walk you through. I'm

Unknown Speaker 52:23

fine with react as well.

Unknown Speaker 52:29

Yeah, so that's what happens next. Okay, great.

Speaker 2 52:33

Okay, so that sounds really cool, and that's all the questions that I have. Thank you so much for your time.

Speaker 1 52:39

Yeah, thank you for your time. And you know, have a pretty fantastic weekend. And thank you, you too.

Unknown Speaker 52:45

Yeah, thank you for that.

Shreesha-Galaxy-SRv-first

Unknown Speaker 00:00

Hiring Manager for this role.

Speaker 1 00:04

So let me just give you a quick rundown as to how the interview will proceed. We'll start talking a bit about your background and any relevant work experience. Then we'll jump into a few technical questions, and then I'll leave some time at the end for you to ask any questions you

Unknown Speaker 00:20

may have. Sound good.

Unknown Speaker 00:21

It's good. Sounds good. Awesome.

Unknown Speaker 00:23

With that, please

Unknown Speaker 00:24

take it away. Tell me a

Speaker 2 00:27

bit about yourself. So yeah, like Hi, my name is Risa kartka, and I am working as a Java full stack developer for like five years now. So talking about my expertise like I'm working primarily on Java based web applications, and also, like, you know, utilizing frameworks like spring hibernate, and also like API for development. And like, I have been primarily working with microservices architecture too. And like, you know, I have been working with designing and, like implementing these APIs. And also, like, you know, been involved in implementing this with, like this Spring Boot. So like these services, it has been basically deployed on AWS cloud. And like, I can work with services including AWS and like ECS two, ECS and SQS, SNS, cloud Watch and like s3 buckets. And I do have experience working on front end too. Like, there I am using Angular technologies. And can also, like, you know, I can work with Java scripts and tribe type scripts. So yeah, like, I do have pretty good experience working with the cherry unit and Mockito for unit testing. And currently, like, I'm on a scrum team. And like this team, it consists of five developers. So like in this team, we have, like, product owners and then also Scrum Masters and talking on the daily basis. I mostly involved working on back in Java code, basically like helping the team with the keyboard reviews, and also, like, you know, on products, product support duties.

Speaker 1 02:16

But I see that you're currently working for Citigroup. Yes, I am. Okay, can you tell me a bit about some of the code that you've written for them? Like, let's take one of functions or enhancements or workflows that you've created, and let's talk a bit about

Speaker 2 02:38

that. Yes, sure. So like at City group, currently, I'm mainly working on the customer onboarding team. So, you know, like on this team, I'm, I'm mainly like working. Sorry, I

Speaker 2 03:04

so basically, like, sorry, like, are you able to hear me? Or no, yeah, sorry, I can hear you. Okay, I'm so sorry. So like at City group, I'm mainly working on the customer onboarding team. Okay, so you know, like on this team, I'm mainly responsible for customer onboarding and self service portal project. So, you know, like, one of the most recent flow that I'm working up, like, right now, it's basically, and, like, I'm so sorry. So I'm basically working on the customer related data request. So, like, you know, anything that test any customer related data. And also, like, you know,

Unknown Speaker 03:51

I'm basically like,

Speaker 2 03:54

I'm sorry. So recently, like our like in our team, our there are, like, cyber and also, like, you know, auditing people. So, like, with where they, basically, they wanted to log and, you know, like this auditing events are, they are on our, like, basically snowflake table, so that, you know, like auditing and, like every other log gathering and it will be proper. So, like, even, like, they created a big enterprise take backlog. And also, like, you know, we had a kind of, like, incorporate these JDKs that, like, basically they provided. And also, like, you know, they created a workflow where you could, like, basically gather these logs on any data processing which is mainly related to customers. And, you know, like, basically, we create our wrist call to a service that is, like, mainly responsible for sending this data out. And so, basically, like, what we did was, like, we created an external so, like, this executed service. Like it was mainly, you know, like it was like this executor service that would mainly, like, collect these logs in a batch. And also, like, you know, after it reaches a certain threshold, we could make this race call to the external APIs, and that could that would like log this stream. So, basically, you know, like, I was also, like, it was a service responsible for, mainly, like, gathering logs from, you know, like, various different workflows. And then also it was mainly into the into, like, pushing it into the snowflake table.

Speaker 1 05:41

All right, um, anything else you'd like to highlight about your skill set or experience?

Speaker 2 05:48

So, like skill set and some like experience. So for this, like, at City group, like, we are mainly working on like, you know, highlights of like, how, like, I mean, highly, I'm so sorry. Like, we are mainly working a highly concurrent workflow. So basically, like, I would say that we were mainly, like, using a thread pool service. And they're like, you know, our main point of the executor was, like, something called the like, historics, and also, like, basically, you know, we had to make sure that we were highly available. And also, you know, our application, it was mainly like performing on, like, highly skilled and also, like concurrent Manning. So, yeah, like, I do have experience working in, you know, like working with a multi threaded environment in a like, basically high concurrency. And, like, I would say that, like defining this configurable services where, you know, like, you have a certain set of APIs that mainly work on thread pool and the other services too. So like, Yeah, that might be like working in a separate pool, so that, you know, like you have separation of these services that, like that might be needing this certain configuration. So, yeah, like, that is something I have been working on recently. All

Unknown Speaker 07:16

right, all right, let's,

Speaker 1 07:18

let's discuss some technical questions. You are designing an algorithm for an elevator. So this algorithm controls the elevators movement, so whether it goes up or down, what floor it stops at, what order in which it moves right, your input to your system is going to be a floor number and a direction, meaning fourth floor up or fifth floor down, right, and once you've picked the person up on the fourth floor, the fifth floor, they will then select an actual floor in the direction that they indicated. So if someone on the fourth floor hit down when you go and you pick up the person on the fourth floor, when they get in the elevator, they will press either three, two or one. Makes sense.

Speaker 3 08:25

All right, so you want to design an efficient algorithm

Speaker 1 08:31

for this elevator. How would you go about doing so? This is more to get ideas to how you think so, feel free to ask any questions. Make sure you think out loud.

Unknown Speaker 08:45

Yes, sure so let me think about it.

Speaker 2 09:00

Wow. So basically, like, I would say here that if I am designing this, like, elevator service, so we will, like, we are, like, basically scoped into only, like, only one design on elevator, right? Or, like, will there be, like, multiple elevators involved?

Speaker 1 09:22

It will be one elevator for the building. So just one elevator.

Unknown Speaker 09:31

Okay? So

Speaker 2 09:34

like, the you said, like, What about like the floor? Or like constraint would like, it will always be on through the certain number or like, would there be, you know, like special force, like basement, or like parking,

Speaker 1 09:48

no, no, no. Special floors, just floors, let's say floors one through 10. You have an equal distribution, meaning you can expect a request from any floor in any direction, equally.

Speaker 2 10:10

So basically, in this case, what I would do is,

Speaker 2 10:22

I uh, like, from my side, like, I would assume that the request they are like, made with a floor number and the direction. And like, you know, once the user, they are peaked and like the user, they select for the destination floor. So basically, like, what I would do is I would like, set the current floor, which is mainly like a default method of the starting floor, and then like the start it would basically be one, right? So like, let's say, like, that's the first floor, and then maybe like the current status of the elevated or like, the current direction would be, like, the idle, right? Because, like, it's basically not moving, so yeah, like, that's like, my initial state of the elevator. So like, there I will store that. And then like, and then now, like, let's say like, the like user, they make a request, right? So, like, whenever, like the user, they make a request with a certain floor. Or else, like, let's say, like the duty action, then I would basically, like, have two queues to store their rights. So like, if it's like, basically a upward elevator, or else, like, I would store it in, it in Upper queue. And like, basically, like, if it's a downward motion, then like, I would mainly store it into a down queue

Unknown Speaker 11:52

and,

Unknown Speaker 11:54

and now, like,

Speaker 2 11:56

like, if the like, now, like, let's say, but all, let's talk about the way of the elevator. Okay, so, like the way, like the elevator, they make a design on it will check, like both the opq and also like, you know, like the down queue. And so basically, like, if like the both up queue and like the down queue, like they had, they have a request, then maybe, like it will prioritize the direction with the fewer requests, right? So that you know you have that equal distribution in place. But like, if it only has one queue, then I think like it will just go to a particular direction. That's basically like an ideal straight state, right? So like, whenever, like, your elevator it is resting, and then, you know, like, there is request on both the queues. Then, like, I think, like, it will go to one that has fewer requests. And now, like, like, there is a case, like, when the elevator can be moving, right. So what it means is, like, it's already going into a like, certain directions, and then, like, what happens is, like, you know, when it's moving the elevator, like it should continue into the direction, um, it is moving until, like, the queue is empty, right? So basically, like, you know what it means. It's just like, it keeps on going to that floor. And like, if the elevator, it reaches to the last request in the current direction. Then I think, like, it will check the opposite direction for the pending request. And then, like, you know, once all the requests are fulfilled at the elevator, it go back to the like, back to that idle direction. Okay,

Unknown Speaker 13:47

do you think that's efficient? Sorry,

Unknown Speaker 13:50

do you think that that algorithm is efficient?

Speaker 2 13:53

Yeah, I think, like, it sort of balance the efficiency, like, with equal distribution, because, like, the equal distribution of the service, because, basically, you know, like, it handles the upward and downward direction both right. Like, separately, like, it's basically means that, like, no direction, or, you know, like, set of floors, they are, like, basically overly prioritized, right? Basically like, like, we want to, like you want to make sure that you have the constraint distribution. And also, like balancing these two separate cues. And also, like, you know, like directing, like direction the switches, and also like the elevators, it serves, like all floors fairly right, like, while maintaining and minimizing the wait times,

Speaker 1 14:41

okay, what is the worst case scenario for your algorithm, meaning, or building that has n floors? What is the longest that someone would have to wait before they get onto the elevator? So

Unknown Speaker 14:58

the longest one

Speaker 2 15:02

is? Um, so for the worst case scenario in this algorithm, I think that it would be like,

Unknown Speaker 15:15

like, let's say like

Speaker 2 15:19

you have like, basically like, you have suppose, like, let's suppose that you have n number of floors, okay? And like, in, like, talking about the worst case scenario, like the elevator that could potentially travel the, you know, like the full height of the building, which is, like, basically multiple times, right? So basically, like, I would say, on one full trip the elevators from, let's say, from floor one to floor, and it covers like n minus one floors. Okay, so that if it has such like it, if it has k and then like, such strips, then, like the worst case, distance, it would be like k times n minus one. Okay, so it would be something like a big like the be a big O of like k times and like, you know, the time that the worst case it would take trigger,

Speaker 1 16:17

no, I don't, I don't need the asymptotic limit. I mean in in literal sense, if I am a person who's waiting for the elevator, what is the number of floors like, how long would I have to wait like, what is the longest trip?

Unknown Speaker 16:35

Is there n floors in the building?

Speaker 1 16:39

And the worst case scenario, how many floors has the elevator traveled before it stops to pick me up.

Speaker 2 16:50

So the worst case scenario, like the longest time you have to wait. So you know, like, if the elevator, it is basically on the opposite end, and, like the worst case scenario, like waiting for the elevator, it would occur, like, when it's moving right. So mainly, like, if you are on floor and and then the elevator is on the floor one, okay, then the elevator, it must travel the maximum distance to reach you, right? So which is the worst case scenario on our end, right, and even on the other so, like worst case, it would be like the elevator it is already on. So I think that the like, you know, worst case it is because, on the algorithm, because, like, I mean, here that it checks a queue on one direction, right? So, like, if it is on basically like, n minus one floor, and like, mainly it's moving down. And for like, it checks for that up queue. So like, it will go and then it will try to go up in the algorithm. Because, like, you know, that's the closer one in the UP queue. So,

Unknown Speaker 18:10

so, basically, like,

Speaker 2 18:14

so, so, yeah, I think, like the i

Speaker 1 18:29

Okay, all right, that's all the questions I had. I want to leave some time for you to ask any questions you may have about the position, about galaxy, or about myself. So is there anything that I can answer to you? Yes,

Speaker 2 18:44

I have, I do have few questions. So like, yeah, I want to know, like, more about the company, like more about the galaxy, and also, like more about the projects that I will be working on. And, like, I want to know more about the developers on the train, like, how they are structured.

Speaker 1 19:03

Yeah, so this would be a position on the lending team. I am a lead developer for the lending applications, and currently there are 12 developers under me, and so you would be joining that group. We work on lending. So that's the, you know, lending and borrowing of goods internally and externally to other counterparties. And we build tools to help facilitate that lending, the model the risk associated with that lending and to monitor our positions to ensure that we are protected and that our clients are protected.

Unknown Speaker 19:54

And like,

Speaker 2 19:57

and I want to know like more about you know, like, what the project looks like, like the project structures and like, you know, like, what would be the developers be taking on, like, the new workflows, or, like, basically it is some kind of migration effort. Or also, like, you know, on other existing apps.

Unknown Speaker 20:18

Oh, this is a greenfield

Speaker 1 20:21

this is all Greenfield work. We will be building out tools to facilitate the lending and borrowing of branding products, and the developers will be helping to facilitate that. As new product types come in, they would develop workflows that would help to model those types and then to track their performance and their risk.

Speaker 2 20:45

And I want to know, like, what would be the next step on the hiring process, I

Speaker 4 20:51

will give my feedback to Joe and Joe and reach out to you. Okay, yeah, then that's That's all from my side,

Speaker 1 20:59

Alrighty, well, thank you so much for taking the time to speak with us today. Thank

Unknown Speaker 21:02

you, thank you for your time, too.

Unknown Speaker 21:04

Have a great day. You too. Thank you.

Shreesha-ICE-SRV-First

Speaker 1 00:00

Yeah, hi, myself. Srisa karta and I have been working as a full step, full step Java developer for about like five years now. So I'm working like primarily on database web applications utilizing frameworks like spring hibernate and web services, and I do have front end experience as well. So mainly on the front end side, I'm working with Angular and also with React. So like I do have pretty good experience working on databases like SQL and also DynamoDB and talking about my cloud experience, it is primarily with AWS and some of the services that I have like I have included with AWS. It is Ec two, CS, SQS, SNS and lambdas. So talking on the testing perspective, I have worked with J unique and mokito, and also, like, you know, and on our day to day basis, like I've mostly, I have mostly worked on back in Java code, helping the COVID team with code reviews, and also on production support

Speaker 1 01:37

as well. So yeah, currently I'm working at oil Dominion flight line, and here we are working on the migration project. So you know, we have used monolithic applications, and we are breaking down in chunks, and then also, you know, basically developing multiple micro services, talking on the couple of services that I helped in design, and it includes design, and also it includes the Experience API. So, you know, this service, it is primarily used for, like, you know, grabbing the entitlements that basically dictates, like what ways is and components, they are visible to the user on the front end side, and then also, like, you know, we also have maintenance API, which is basically used for making the changes to the user profile preferences. Also we use Java, Java and Spring Boot for the back end with PostgreSQL and DynamoDB for our database, and on the front end side, we are using annually.

Speaker 2 02:53

Okay, okay, so you put in this recent project, right? So let's talk about that first. So let's discuss about that, you know, benefits in Spring Boot, which we are missing, is things well,

Speaker 1 03:12

talking on the benefits like, you know, basically Spring Boot, it is like, you know, it is specifically designed for creating strength and applications, right? So that is basically products and products and readiness. So basically here, what I mean by that is it mainly comes with an auto configuration. So you know, any dependences, like any dependencies that is available in your class path, like, it just gets available, available for your dependency injections. So, you know, like, you don't need to write the boilerplate code to actually manage, like, actually manage those bits right? So you can just use, like, an auto wire that is based on, like, what you have on your class path. And then also, it also provides, like, a lot of starter dependences, like a springboard starter wave, and also, like, you know, Spring Boot starter security and a Twitter, which helps in a lot, like, you know, like a lot of manners, like the web, for like web application security, and also for liberating the spring security and actuator for, like, you know, our products and monitoring and metrics. And other than that, like, you know, like the Spring Boot, it comes with an embedded server, so you do not have to, like, you know, deploy your applications to external servers. So yeah, like that, are some of the major advantages that Springwood provides you. Okay,

Unknown Speaker 04:59

injection in Springwood,

Speaker 1 05:02

so dependency injection. So, like, you know, basically, basically like dependency injection, it is a pattern, right? So with the objects that you need within, you know, certain classes or objects, they are provided by an external source, right? So, for example, you know, like, if the so, like, so, if you have a service layer and you need to interact with other services or repository layers, then you know you can basically inject those dependencies by using the auto wired annotation, right? And then, if you like, if you bring in version of control, they basically take care of providing that dependencies to your services. And then also, you know, you can have like, three kinds of dependencies, injections along like those are constructed constructor injections that is done through the constructor of the class. And then you have the field injections with the which is directly on the field type. And then you have the sitter injections, which is done on the sitter method. Okay, okay.

Speaker 2 06:21

In the we implement the exception in Spring Boot?

Speaker 1 06:28

Well, so to implement the so to implement that, like you know, like apart from try, catch method, like it should block, like it should block, and also, like, you know, it should through the keywords on the method, right? So like Spring Boot, it collect, like Spring Boot controller, the advice like it is our global exception handle. So what you need to do is like, you know, you need to define a class that is annotated with the controller advice, right? And basically, like, you know, on each methods you can use handle exceptions annotation, and then also, you know, like, give the type of exceptions like it needs to handle. So here, like you you will, you will, you will be able to return a custom response to your client with a response code and messages and like so on. So like basically, this is a global exception exception handling that the Spring Boot provides.

Unknown Speaker 07:37

Okay, okay, we'll do the

Unknown Speaker 07:41

exercise later. Have you

Speaker 2 07:44

worked on AOP, Aspect Oriented Programming? So

Speaker 1 07:48

except oriented programming. So like, you know AOP, it basically helps in like, cross court, cross COVID concerns, like logging and logging security on and so on, right? So, like, I have done a little bit of AOP on the logging side. So, you know, we will basically intercept our method invocation and log like and like log the actual, actual parameter. And also, you know, like the timing that, the timing that the method took for executions. So yeah, like, that's what an AOP is, and also, like it is like a cross cutting logic.

Speaker 2 08:30

Okay, let me your name. You can do some exercise, sure? I should just,

Unknown Speaker 08:54

okay, can you see my screen?

Speaker 2 08:57

Yes, I see the browser, teams, microsoft.com, Microsoft,

Unknown Speaker 09:03

you have any idea

Unknown Speaker 09:07

we can use

Unknown Speaker 09:15

integrity? Okay?

Speaker 2 09:19

Do we were discussing about AOPA. Do one example of AOPA,

Unknown Speaker 09:27

which you have already done. Okay,

Speaker 2 09:32

yeah, let's create one class where we can handle

Speaker 1 09:36

the AOP. Okay, so at first, like, did you want me to start our Spring Boot applications, or just do it here, because if I do, like spring initializing, like, I should get the dependences right.

Unknown Speaker 09:50

Yeah, let's Do that. Okay?

Speaker 1 10:51

I don't know where it's not finding AOP, but I Can edit later. But

Unknown Speaker 13:15

I'm just letting letting it Load. Sorry if I forgot something. I Excellent.

Speaker 1 13:27

Okay, so I think it's loaded. So I'm just going to create a dummy controller that Just takes in certain value And then just written it.

Speaker 1 15:25

So I have two endpoints, which is basically gate and post and get it is to retrieve an employee ID, and then the post it is to create an employee here. And I'm also just doing a static response, like this response back, and then I will use AOP to basically, so Basically I have method execution time that is printed.

Speaker 1 16:02

So for here, I will basically create a new new aspect. And

Unknown Speaker 16:16

so I

Speaker 1 16:24

so we can annotate this class as an aspect, and then I

Speaker 1 16:46

and I think for aspect, we need a different package as well. So let me find

Unknown Speaker 17:00

that you can use the third one started

Unknown Speaker 17:04

AOP because,

Speaker 1 17:09

but, like it did not give me the required dependencies. So I think I know the aspect I

Unknown Speaker 17:47

Okay, it came.

Speaker 1 17:58

Now I noted this as a spring component, and now, like here, we would basically have the point guard.

Speaker 1 18:11

So I think for that, we just need our point cut adaptation. I

Unknown Speaker 18:31

Sorry. Can you hear me? I just heard

Unknown Speaker 18:35

I just heard things. Okay. I

Speaker 1 18:44

So point card I should provide The actual part of the controller. So

Speaker 1 19:43

so this would be my reference on what kind of aspect I want. So basically I can do around the methods that I execute and And then I would provide that point guard here. So

Speaker 1 20:27

so I think we need to pass in our joint point for this. So we'll do that first. I

Speaker 1 20:45

So firstly, I'll start the time and in The time and then log the execution time. I

Speaker 1 21:50

Okay, so this, it will give me the method execution time, but I need to actually, but from doing point and from Doing point, I can get the method signature and I

Speaker 1 22:23

so from joint point, I can't get the method signature, but I think it should be A different kind of joint point.

Speaker 1 22:40

It is like a proceeding joint point and I

Speaker 1 23:05

so once I get the method signature, I can get the name of The method, and also I can get the arguments I

Unknown Speaker 23:23

So,

Speaker 1 23:32

so I can here, I can basically log it and I

Speaker 1 24:27

and then basically, here I just get the result, which would be, We should be by proceeding With the joint point, I

Speaker 1 25:20

so It would be something like this, and

Unknown Speaker 25:30

I can paste it out if you want me

Unknown Speaker 25:37

to go back. I aspect.

Speaker 2 25:50

Can you explain this? How the language will call this function, and you added it on the controller level, how this is around allocation you used. Can you

Unknown Speaker 26:10

explain the whole code? Okay,

Speaker 1 26:17

so basically here in this code, Spring Boot, Like it automates.

Unknown Speaker 26:26

Go to the top

Unknown Speaker 26:34

location. Okay,

Unknown Speaker 26:42

so here basically,

Unknown Speaker 26:49

so basically, the

Unknown Speaker 26:53

Spring Boot

Speaker 2 26:57

will enable this Aspect Oriented programming for us,

Unknown Speaker 27:04

you have your code now that

Speaker 2 27:07

let's see how Spring Boot will enable

Unknown Speaker 27:12

this component for us.

Speaker 2 27:15

Is there any specific functionality we need to add to enable this

Speaker 1 27:21

so basically, like, I think you need to annotate your starting class with, like, enable aspect, auto proxy or something like that. So there's a specific annotation, like, needed to enable spring AOP, right? So that is the first thing you need to do. And then once that is done, it will come back and look into the components that is like that Max, that is like, created, sorry, that is marked and like spring aspect for AOP, because the because expert annotation marks this class as an AOP Advice logic, and then component annotation like it registers the class as Spring management.

Speaker 2 28:19

Okay, let's go down to line number 23,

Speaker 2 28:29

location also right with the difference between these two, like the invocation part will be different when we call the function. So like, when this annotation will be happening,

Speaker 1 28:52

so the along the annotation, it allows complete control over the method execution, right? Because here you can proceed with the join point, like join point, right, but before it executes, like before it executes before a method. So like, if you use before annotation, it just executes before the object method is executed. So basically, like around like it can be like an interceptor that follows our method execution, so it can help us, like something, sometimes in modifying the method arguments before execution, and then also like, and then also execute any kind of Extra logic, like before and after the method runs, or else, like you can just, you can even, like, stop the method execution on certain cases. So basically, like around it helps in measuring the metrics. Like I have, like, the execution time that I have here on line number 38

Speaker 2 30:02

can we have more than one around annotation around the same controller side? Let's say you have one around annotation right for the controller class, right if I want to have one more around annotation for the same class along with some different classes, right? Multiple classes? Can I have

Unknown Speaker 30:29

more than one annotation?

Speaker 1 30:33

Yeah, like, I mean, you can have like multiple like around aspects to be applied to the same controller. I think you know the order of execution. It depends on like how precedence it is defined, and also you can use the order annotation to control the execution flow.

Speaker 2 30:57

Okay, so line number 32 right where you are using the Proceed function. Let's say multiple annotation. Do I have to have this proceed in all the annotations? Or only one is enough to have the Proceed?

Speaker 1 31:21

I think here all of the around annotation, it must have the process like, because, you know, it needs to come to this method and then go back to the original education flow, right? Because, like, you do not want to stop on that particular aspect, right? So it needs to be proceeding in all of the aspect. Okay.

Unknown Speaker 31:52

Have you worked on reactive programming? Sorry,

Speaker 1 31:59

reactive so, like, I do have some experience working with mono and flux and even, like spring wave plugs.

Unknown Speaker 32:08

Okay, so can we do a small example,

Speaker 2 32:15

like within service file, service class, we can use the same employee object which you have created.

Speaker 1 32:36

Okay, do you have, like, any, anything specific in mind.

Speaker 2 32:44

Let's create one class in place and add few parameters here. Let's say for same class, name, ID, I

Speaker 1 33:38

Okay, I had the employed class, so,

Speaker 2 33:46

okay, so what you have to do, let's assume you need to get you need to do the API calls to get the data of the employees based on list of ids. Okay, you will be getting a list of ids in the get KPI, and so the database need to filter out those ID and then return the

Unknown Speaker 34:13

employee data back

Unknown Speaker 34:16

to the controller. We

Speaker 2 34:18

don't have to write controller. We can simply write the service

Unknown Speaker 34:25

layer. Let's make sure

Unknown Speaker 34:29

to write, okay.

Speaker 1 34:33

So for that, let me just Add this thing With talks dependencies. I

Speaker 1 35:54

so we will be using our plant to get the actual direct to credential employee data. So to

Speaker 2 36:04

the actual employee data zoom, you can just do the you have the argument of a function, which is the list of ids. Okay, based on the ideas, just write the logic to fetch the data, filter out the employee table based on the idea, and data response back to the controller. Okay,

Speaker 1 36:33

I'll just use this as a sample. I'll not be setting it up then.

Speaker 1 36:41

So I'll create fetch by employee ID, which would basically,

Unknown Speaker 36:50

which would basically make

Speaker 1 36:53

HTTP call to grab the ID right. And since it's it's by ID, we would use a single employee bank, and for single employee, It is a mono And

Speaker 1 38:59

ah, yeah, I think this would take care of it right

Unknown Speaker 39:11

by ID, and then You are calling it through

Unknown Speaker 39:16

The exception heading, also,

Unknown Speaker 39:22

if I

Speaker 1 39:55

so basically here, what I'm doing is, whenever There's an error, we just return an empty model?

Speaker 1 40:16

We can also do an on status check and define some kind of response back if you want

Unknown Speaker 40:31

that. Okay?

Speaker 1 40:49

So it takes a predicate, so which basically means, like, I can check if it's a client error. So,

Speaker 1 41:04

so basically, I want to check, check it's a current error that for that response, I want To check if it's a 401, or not. I Hi,

Speaker 1 41:57

so if it's a photo one, I just written a mono off earlier.

Unknown Speaker 42:13

Okay,

Unknown Speaker 43:02

closer

Speaker 1 43:07

you, yeah, I think there will be something like that. Or for like, I would do a specific exception, and then for other, I would do something like that. Okay,

Unknown Speaker 43:30

UI project, sorry.

Unknown Speaker 43:39

UI project running on a

Speaker 1 43:41

local I just think I have a RV World project running here. Okay,

Unknown Speaker 43:48

so you said UI project, right?

Unknown Speaker 43:56

That one, I don't

Speaker 3 44:00

think. Okay. Applause,

Unknown Speaker 44:33

She's in order so Shari, sir, you are in North Carolina.

Unknown Speaker 45:16

No, I'm in Fairfax, Virginia.

Unknown Speaker 45:20

Oh, Virginia, okay,

Speaker 1 45:21

it's my remote role, so I'm not in North Carolina. So are you sending the link in The chat? Yeah,

Speaker 3 46:56

it's key. Can you? Can you type something? Is it working for me? Okay,

Unknown Speaker 47:17

yeah, yeah, I get that.

Speaker 3 47:22

So you see there is the API posted over there.

Unknown Speaker 47:26

What I want you to do is

Speaker 3 47:30

to this endpoint whatever data you get, stored it as state or whatever format you like, and then show it in the UI. So table phone.

Speaker 1 47:50

So what does this? APM in? Like, can I check it? Yeah, good. Okay.

Speaker 1 48:07

Okay, and this is a React application, right? Because it's a TSX

Unknown Speaker 48:19

react application.

Unknown Speaker 48:30

So basically,

Speaker 1 48:33

for React apps, the most efficient way, or like, even just for node.js app, I think this is just like using the sage API, right? So I can just use our node and node dot fridge API to face the data, and once for React, we need to use Cloud books, because API calls like they are something that creates a side effect. So firstly, I would create a youth state hook To have a placeholder

Unknown Speaker 49:16

for the data. I

Speaker 1 49:40

I'm so I would need to make an API call using use

Unknown Speaker 49:51

using use effect. And

Speaker 1 49:59

so use effect like it basically takes a function and then dependencies as its parameter. Dependencies like a list of parameters that will cause like this. Uses like this. Use effect to be executed for now we can just like, leave it empty. So to make a Fetch API, we just use the fetch method and then pass in the URL.

Speaker 3 50:40

Once you have an awesome problem Once you have An output, Okay, Do

Unknown Speaker 56:30

Can you can you expect go to

Unknown Speaker 56:39

the console. I

Speaker 3 56:47

role just before the API URL, there's a two return, right?

Unknown Speaker 57:05

I think

Unknown Speaker 57:11

so, let me think of that.

Speaker 1 57:16

So I think react like it has double parsing because, like, you know, it calls the use Effect Noise, and I think, like, it's a strict mode or something, right?

Unknown Speaker 57:37

What's the reason we have ideas

Unknown Speaker 57:42

so I think

Speaker 1 57:45

if you use it in strict mode, it intentionally calls the use effect two times. Okay, yeah, yes, so see here on the main TSX it has a strict mode.

Speaker 3 58:02

Yeah. I

Unknown Speaker 58:09

think so, yeah, okay,

Unknown Speaker 58:23

you need to save the file.

Unknown Speaker 58:31

Yeah, yeah. Is this? It's just calling it once, that's good.

Speaker 3 58:38

I'll share one more link with you. So

Unknown Speaker 58:52

context API,

Unknown Speaker 58:55

context API, so,

Speaker 1 59:00

so it's been a while I haven't worked with React. So I don't react. I don't recall it, but I did work in it in the past.

Speaker 3 59:12

Okay, so this is a simple context store on line

Unknown Speaker 59:21

number six.

Speaker 3 59:25

Now what's happening is, example, one component which is starting at line number eight is using the pretext number two, which is online. Variety was not using the pretext inside the default score, which is on line 24 the vertex provider is the parent, for example, one and two, both

Unknown Speaker 59:51

on the right hand side is white space

Unknown Speaker 59:54

at the bottom to click on it. That later,

Speaker 3 1:00:09

example two is not using the context. For example one is using the context, even though that's the case when you click on the click two button, it is always rendering example two. Now one way to solve this is wrapping Example two with the reactor memo function. But can you do it without using reactor memo? I?

Speaker 1 1:00:45

Yeah, I think, let Me see on this. I

Speaker 1 1:01:32

so I think since we are inside the same context provider, we basically need to move it out of The context provided. So,

Unknown Speaker 1:01:51

so, Okay, that Still doesn't flow.

Unknown Speaker 1:03:07

Can you Hear your screen

Unknown Speaker 1:03:10

again? Do

Speaker 3 1:03:40

so now if I want to use Example two inside the context, how can I use it?

Speaker 1 1:03:57

So if you want to use example to inside the context, inside.

Speaker 3 1:04:06

Example two, how can I do that now?

Unknown Speaker 1:04:12

Example two, how will I able

Unknown Speaker 1:04:15

to do that?

Speaker 1 1:04:20

So it will be like that will be separate wrap up context inside the example, so it will be another context provided wrapper.

Unknown Speaker 1:04:36

Yeah, I do have few questions.

Speaker 1 1:04:41

So first of all, thank you so much for the time today. And like, I would like to know a little bit more of the role. Like, you know how much developers they get involved in the backing part and how much on the front end part,

Unknown Speaker 1:04:59

application you'll be working

Unknown Speaker 1:05:02

on both the sides right Now,

Unknown Speaker 1:05:10

different

Unknown Speaker 1:05:14

teams. So that's only

Unknown Speaker 1:05:20

three other people. So

Speaker 1 1:05:21

theaters, okay? And I would like to know, like, second one. I would also like to know, like, you know, what does the like, you know, what the team gets involved like in a regular basis, like, you guys to follow, like the methodologies, or what kind of ceremonies do they have? This

Unknown Speaker 1:05:47

is the platform development team, right? So

Unknown Speaker 1:05:53

components for the whole company,

Speaker 3 1:05:58

you will develop something that is going to be like, very exciting. Very small application will be company wide, so for example, component input the whole organization. So application developer would be able to access that using your code. This way, we can have cohesive ecosystem across the

Unknown Speaker 1:06:30

whole company working

Unknown Speaker 1:06:33

on both the

Unknown Speaker 1:06:37

platform and

Speaker 3 1:06:40

Oh, sounds good. This role is four

Unknown Speaker 1:06:54

days

Unknown Speaker 1:06:58

in office, whenever

Unknown Speaker 1:07:02

you like you and also learning from you at the

Unknown Speaker 1:07:06

same time. That sounds good, like I'm comfortable with it.

Unknown Speaker 1:07:12

Any other questions?

Speaker 1 1:07:13

So I would like to know, like, what will be the next step after this?

Unknown Speaker 1:07:19

I think she will have one more

Unknown Speaker 1:07:22

with our manager.

Speaker 3 1:07:25

If that goes well, you might have one more round with a director.

Speaker 1 1:07:31

Oh, okay, okay, that's great. Yeah, that's it from my side. All right, thank you so much. Thank you for your time. Thank you for that. Thank you.

Sujina-Citi-Final-Srv

Speaker 1 00:00

String. So how can I do a custom query? If that is not possible by the actual limitation we have, yes. So you have

Unknown Speaker 00:16

to write the custom query

Speaker 2 00:19

for this problem. So yeah, in spring query, it only works with like jvql and Native queries, but you can write, yeah. You can write the cost queries, yes. So what if you want something you know that relates to your document from MongoDB. What do you need to do? Like you can do the MongoDB repository rather than like the GP repository. So, yeah. So talking like, yeah, so that's what I can think.

Speaker 1 00:59

Okay, but we still have the same problem where it is not, it's not standard all over, right? Because, like, if I want to drop off the, like the MongoDB and switch to another, another database, so this is going to be a problem for me,

Unknown Speaker 01:23

yes, so yeah, for

Speaker 2 01:28

So, for that, I guess that's the you know, sorry, advantage of using like interfaces, right? So you can quickly switch between the implementations, because you are not, like, strictly, like, tied to it. Okay.

Unknown Speaker 01:49

Are you familiar with something called liquid base,

Speaker 2 01:53

liquid rice, liquid base, liquid base. Oh, yeah. I liquid base, yeah. I worked with it for, for like generations of this simple initial data

Speaker 1 02:08

model, so, so you used it to do the job like the for generating the Java model or generating that as a base model,

Speaker 2 02:21

like generation of, like the simple, it's for the database. I've used it for the database. It's for the database model, right? So, so it's basically changed. But I'm sorry if there is like a change to your schema on the database, it keeps like tracks of a certain like this change log. And then if the change log you know does not exist, then it applies those you know changes, and also migrates the data so you can, like, can have, like, exempt file with some kind of, like liquid base, Change ID or change Save, and then it applies to it based on the change log that you have, based on the tracking table.

Speaker 3 03:05

So yeah, so I have a question, how can I enable Avro format in liquid base? Because liquid base, I believe what format file you used. How can I enable Avro like I want to write all my scripts in the Avro file format. How can I do that in liquid base? Because liquid base, more or less, we all you have used XMLs or yamls. How can I use arrow

Unknown Speaker 03:33

to write my TV scripts? So I

Speaker 2 03:40

haven't like worked with Avro based Kayla, so like, I would, like, I don't know what it's like, should probably be, like, a configuration, you know, and the configuration and for the properties and applications, or, like, like garbages to have that enabled. I Okay,

Speaker 1 04:05

okay. So back a little bit to liquid base. We you said you used the liquid base to sync

Unknown Speaker 04:15

the databases changes, right?

Speaker 1 04:19

Okay, so how about how what we do the other way around. We actually, the user, like the developers that are working on the back end, doesn't, don't have access directly to the database, so they don't actually change the database, but they can actually changed the models on the back end code, and they want these changes propagated to the database. This is a very valid scenario, and I have it all the time. So okay?

Speaker 2 05:03

So, so if we okay for that scenario, so if we want to, like sync your like model changes into the database, I think you would, you know, just do an like automatic schema update in the hybrid network. So where you can just do is, like, use like a hibernate, like, oh, for the so for the update schema in I can do is use the Hibernate, and where you can just do is like DDL, you know, auto settings as update on your applications property. So that's you can do for that

Speaker 1 05:58

scenario. Okay, so how far along did you use Angular? Angular I've been using for five years, okay, but you're using it on your current job, okay, so let's talk a little bit about Angular. Oh, you know how, when we start a new page and we want to root data for the like the UI to present it, so where do you actually put that request? So

Speaker 2 06:42

for that. Where do you put? So where do you put? What do you want me

Speaker 1 06:48

to I'm sorry. Let's say you have a page which is loading a table of data. Let's say you use like you're loading a list of users. So that page, which is called users. When we get to that page, we want to load the list of users so that exact location where we call that HP request, where do you put that? Usually, put that it's not, it's not at the question I want to know, where do you usually put the HP code, the HTTP code, yes, so on the browser address

Unknown Speaker 07:35

bar. I guess it's not

Speaker 2 07:39

related like any piece. If you want to like navigate to then you want to like directly go there, but you just put it in the address bar of the browser. I think so,

Speaker 1 07:53

not the actual route. So I want to figure that component to load the user list. Okay, so where do you actually call the back end?

Speaker 2 08:10

So, so for calling the back end, you just use it in the service layer. So you can just create, like, a different services, like within like Angular that interacts with your back end calls using the angular like HTTP clients, or clients for our request,

Speaker 1 08:33

you can edit a method inside the service that calls like uses the HTTP client to call The back end. Where do you call that method in the component to constructor in it, or where, where did you want to call this method? So, yes, so you can,

Speaker 2 09:00

like, basically do it on ng like, lifecycle. Hook. So basically, once the component initialize, you would load that data so it would be on the NG, on it on the lifecycle hook. So I would say that, Okay,

Speaker 1 09:18

what if that that HP request has a failure. So I actually don't want to show this page if the if the HP request is failing. So that means that I, I don't want to go inside the components at all, inside the engine to call the HP request, I have to do it somewhere else before we get to that component. So where do we put the HP request? Okay, so,

Speaker 2 09:57

so in that case, you know the way can handle this is what I can think of, is like you would like route resolver, where you would make, you know, the API call and then resolve it to a component. You know, by passing that was like the lead response. In case of success, like you can put like input data to the component, and then, in case of failure, you just don't do it all, right? So it would be like a part of your routing module,

Unknown Speaker 10:33

okay, so I have it

Unknown Speaker 10:38

follow up question. Mr. Fauci, it's okay.

Unknown Speaker 10:41

Go ahead. Good, okay. I have

Speaker 3 10:43

a very, very simple question, right? I'm sharing my desktop on a simple post request. Sorry,

Unknown Speaker 10:53

I couldn't get the front first part.

Unknown Speaker 10:57

Don't worry about it. This is my code. Oh, okay,

Speaker 3 11:04

what is what is the problem in this class? Please observe this carefully. What is the problem in This class and where is the

Speaker 2 11:50

Can we have like two request body on a single request? I don't. I've never seen that like two body request On a single request,

Unknown Speaker 12:04

we can you're not.

Unknown Speaker 12:13

Okay.

Speaker 2 12:26

So if you know, if that's the case, then you have, like, a demo controller. You are, like, ingest, like injecting, like the object mapper, and then I'm

Speaker 3 12:37

not asking about the code. You don't need to read me the code, just there is an error in this book. Oh, there is error, or there couldn't be an error. You just tell me,

Unknown Speaker 12:49

okay, just give me a few minutes.

Unknown Speaker 12:50

I'll just check i

Speaker 2 13:09

i would say, like I see as having to request body. The only error I can see is having the request body

Unknown Speaker 13:19

is not the problem.

Speaker 3 13:21

I can I can, I can have two objects, right, being passed in a post request, right? Why do you think that is a problem? I can pass in two requests within a post body.

Unknown Speaker 13:34

We do it all the time. Like,

Speaker 2 13:35

how do you, like, pass like, two objects and a post object, like, it takes, like a single, I remember, like a single JSON object, right? Or a single request body of, like a certain

Unknown Speaker 13:49

entity. We can do it right.

Unknown Speaker 13:53

There's a bitter we can do it.

Speaker 2 14:02

But I feel like, I think it should still become like as a part of like single request body, this one.

Unknown Speaker 14:18

But why do you think this is, I mean,

Unknown Speaker 14:22

like I said, there couldn't be any error too.

Unknown Speaker 14:28

I'm just saying that there could be an error. There could not be an error.

Speaker 2 14:42

It's okay. But I would still like to stick with my so I would, you know, it's like, I would, you know, you cannot, like, have, like, multiple request body and a single request. So I would like to stick to my, okay.

Unknown Speaker 15:02

Got it got it makes sense.

Unknown Speaker 15:07

Actually, that will work.

Speaker 1 15:16

Okay, so we are back into the H request, you said in the route resolver. So the route resolver you either you have to send, you have to return data. So in case of failure, what will happen? Because I don't want to go to that page at all. If there is a

Unknown Speaker 15:47

failure, what would you do?

Speaker 1 15:53

Or better yet, let's do another scenario. Is not in case of failure, but it is case of valid data but empty list. So let's say you were doing the same scenario, returning a list of users, but the list is empty, so I don't want to go to that page at all. The response is valid, but I don't want to go to the page. Photo, what should I do with the empty request? Yeah, okay,

Speaker 2 16:36

so, I mean in that case, right? You would make that API call. Yeah. But you, instead of, like, catching the error, you would be, you know, you would do the catch error and then also catch the empty scenario. And you would like, basically show, like, a message that pass and stay only in the case of, like, successful with the data, so you would activate the router in

Speaker 1 17:08

that case. Okay, so let's go a little like one more level up. So I want to do HP request before all routes, and it doesn't get sent to the actual components or anything.

Unknown Speaker 17:30

So how can I get Okay? In

Speaker 2 17:42

that case, I So, if you are like, you know, navigating from one company to other, right? So you I guess

Speaker 1 17:56

I'm not navigating. I want to make an HP requested, before, before everything before all routes.

Unknown Speaker 18:05

Before all routes.

Unknown Speaker 18:07

Yeah, okay, so for over that, oh,

Unknown Speaker 18:14

yeah, oh sorry, I need to put together some

Speaker 2 18:21

thoughts. So in that case of, you know, the HTTP request, like before all rounds, you would want to have, like, some kind of like app, you know, like the app initializer logic to call the API before, and then anything before it loads, like anything so you would like do, like, it's a kind of like loading config, like, kind of like a scenario where you need to, like, load conflicts, conflicts, and then a typical use Case for initial, like, API call. So in that app initializer, you make this API the API call in that app module, the app module,

Unknown Speaker 19:19

okay, but yes,

Speaker 2 19:23

it would be as a part of like initializer you would provide. So in that sense, I think,

Speaker 1 19:32

Okay, I think we are passing 45 minutes

Speaker 3 19:38

because I believe you also worked on or in, like hibernate, right? Yes, just a quick question before we close. How can I use sprints? I'm sure you're aware of a transactional annotation, right? Automatically and create data fields within my entity, you have an entity object, and there are, of course, sensitive fields, yeah, how can, how can I make spring transactional annotation, using transactional annotation to encrypt these sensitive fields within the entity before it is written to database. So, yeah, so

Unknown Speaker 20:25

talking about it before.

Speaker 2 20:31

So you know to automatically, you know the Encrypt, like the sensitive data. What I can recall like right now is maybe just like. So I don't think there is like a direct support with the like within the transaction annotation, but you would like have to utilize string aspects for that, like before, like transactions. How can hibernate do it?

Unknown Speaker 21:08

Oh, so for the hibernator,

Speaker 3 21:13

we can, we can do with the transaction annotation, what? What kind of

Unknown Speaker 21:18

do I need to modify within transit?

Speaker 2 21:20

Oh, so you have a converter attribute on the entities that can do it. So, I mean, within the entities, you know, what you can do is you can just apply converters, you know, to encrypt, like the passwords, or something like, something like that. So you you know, so you should have, like, an attribute, like converter that you can, like, work with. Okay, all right, thank you. Thank you.

Speaker 1 21:55

Okay, what questions do you have for us? I'm gonna, like, I want to give you some time, yeah, okay, yes.

Speaker 2 22:07

First of all, I'd like to thank you for the time today, you guys. It was really nice talking to you. I'd like to know about how the team works, because, like, what developers get involved in a part, apart from, like, coding activities and everything

Unknown Speaker 22:32

about the team like, so take that, sir.

Speaker 3 22:36

Yeah. So, so, like I said, right, we are, we are doing a lot of modernization within financial sanctions space within city. So we are looking at individuals who can actually come in and bring in their expertise on in our modernization effort, and walk along with us right so there could be efforts that we got to work on, taking legacy applications, redesigning them, I wouldn't say redesign that we're looking at, but, but reshaping them to fit in the modern world with the best practices that are being applied within microservices, and also making them cloud ready. So it's like we can apply these 12 factors and move along these integrations along the line, right? So, so we are looking more for the modernization of work right now.

Speaker 2 23:32

Okay, sorry, so how much of like like cloud applications do you use like cloud?

Speaker 3 23:39

We are moving into Cloud. I don't think there is anything on the sanctions that actually run on cloud today, but in the next in one of these applications, and then following up, we are actually migrating a lot. Okay,

Speaker 2 23:54

thank you. I thank you so much for that information. So yeah, I think that's all the questions that

Unknown Speaker 24:03

I have to know about the next steps as well.

Unknown Speaker 24:08

Sure. All right, thank you. I gotta jump out for the call,

Speaker 1 24:11

yeah. So for the next steps, the recruiters also will reach out. We will. I Will.

Sujina-Paypal-final-Srv

Speaker 1 00:00

If you see the second example, and I want you to if you understood the question correctly, I want you to tell me the answer for second one. Okay.

Unknown Speaker 00:23

It should be negative one, right? Because

Unknown Speaker 00:31

that's correct. So I think

Unknown Speaker 00:35

question is very simple.

Speaker 1 00:38

I wanted to keep it simple first, and if, based on this should be solved in 15 minutes or so,

Unknown Speaker 00:48

then we can have another question, if we have time left.

Unknown Speaker 00:56

Okay, before that, I'll put an example in the

Unknown Speaker 01:01

another example, and you need to tell me the so that

Speaker 1 01:07

we know that you understand, understood the question correctly in the input I'll put,

Unknown Speaker 01:16

if you see, I have put another example. Here

Speaker 1 01:20

you see this input? Okay, tell me what is the answer to

Speaker 2 01:28

this? We basically need to, like, keep track of the maximum sum, right? Yeah, so it would just be 15. Sorry,

Unknown Speaker 01:36

15. I mean, okay,

Speaker 1 01:43

I mean, so if you see that i plus four is nine, when you do minus one, you still have eight left. Is not zero or less than zero. So you should keep that minus one also in the sub array. So here the sub array is whole area. Oh, you just need to add everything together. Yeah, in this case, previous example that we had, the first example, you need to talk because you, as soon as you go negative, why do you want to continue? Right? I mean, make no sense.

Unknown Speaker 02:17

Yeah, it makes sense now.

Speaker 1 02:22

Okay? So let me change that to the basic

Unknown Speaker 02:29

they have. Then let's

Speaker 1 02:34

you can so couple of things you can look up for syntax on Google or chat GPT, if you I mean, we all do not remember all the syntax of every data structure and everything right? So it's okay to look that up. That's allowed. The only thing that is not allowed to switch over another monitor

Unknown Speaker 03:01

and do not search for

Unknown Speaker 03:04

the answer, because this is very,

Speaker 1 03:08

very kind of question. So I would say that, okay, Your time starts now here I think this time. Please think it loud so that I don't want you to start writing the code, and then I'm kind of lost. It would be great if you can think it loud so that I would know what you're doing. And if there is any point where you're going in wrong direction, I can bring you back to the track. Or I would know sometime you might have a different approach than what I have. So I should know that otherwise, at the end, it is very chaotic to understand 20 lines of code. So, yeah, okay, go ahead. Please do

Speaker 2 04:04

so while I'm thinking, here is, I need to, you know, keep track of the maximum sum in some of, some way that you should, you know, start from the actual total of the sum, and then, like, we need to, like, keep track of the sum from, like one value where we don't drop, right? So, so I think I would just need, you know, like, like a, kind of, like a two pointers, you know, to keep track of like, two different, like sums, so, so, yeah, so here, I think in in this case, there are, like, negative numbers too. So I'll just, you know, start my pointers. And I think I would just want to start, you know, I would just like start my pointers at the minimum value.

Speaker 2 05:15

So, yeah, so, so this is my like, first pointer where, which keeps tracks of the total sum that is the maximum, right? So it will be compared with the sum

Unknown Speaker 05:32

that currently like calculating. I

Speaker 2 05:44

and then, you know, I will just start with my current sum, which is like, like continuous, like sub array that I will be summing do.

Speaker 2 06:04

So once this like pointers are like, you know, set up. I just need to move through my area. I

Speaker 2 06:26

so what happens is like if my current like sum is greater than my maximum sum, then I would just set like

Unknown Speaker 06:38

maximum sum to that.

Speaker 2 06:54

The way I'll calculate, you know, the current is like, every time I, you know, some like becomes less than zero, then I need to, like, reset like, I just need to reset My color and

Unknown Speaker 07:12

then add the value

Unknown Speaker 07:22

something you said

Unknown Speaker 07:26

that you will have two pointers.

Speaker 2 07:34

So, yeah, so these are, you know, the two pointers right the max is used to calculate the pointers for the maximum, like the maximum value across my sub array and then the current this one is used for, like the count pointers for the sum of my sub array.

Unknown Speaker 07:56

You said pointers? Did I hear correctly?

Unknown Speaker 08:00

Two pointers or two counters. So

Speaker 2 08:05

it might be just called like values, rather than like pointers. It's just, it would just be like, like two stories, like locations, two counters,

Unknown Speaker 08:16

Max counter or current counter.

Unknown Speaker 08:20

I got confused because you said pointers

Unknown Speaker 08:24

you are moving,

Speaker 1 08:26

pointers are basically indexing, right? So you are on which index you are maintaining, two indexes, but here you are not. I just wanted to understand, did I hear it correctly or not? Yeah, the

Speaker 2 08:36

pointer will be like indexes, but like, we don't want indexes right

Speaker 2 08:48

now. So, yeah, so once we have that maximum value set up, like, we just like, return the max do?

Speaker 2 09:12

So, yeah, so I think this should be right, so they just takes in the value, calculates the sum of the sub array, and if it's greater than the maximum value, then it just returns the maximum

Unknown Speaker 09:30

value. Do you think this gonna work?

Speaker 1 09:34

This would calculate so you're not doing any sub array here. You're basically going ahead with doing total of whole array. But anyways, let's run it and see.

Speaker 2 09:55

No, I'm just like, you know, resetting my counter like, so every time I go below zero, it just resets And then create another sub array down The line, 29

Speaker 3 10:41

we have exception there, line number three seven, which is from the main function. Line number 67, but

Unknown Speaker 10:59

that's coming

Unknown Speaker 11:01

on. Line number 50, main, but bars,

Speaker 1 11:12

oh, make, make those variable you need to change it to right the INT. Integer to heat mapping into integer mapping is this

Unknown Speaker 11:35

is your output

Unknown Speaker 11:38

which is not right? No.

Unknown Speaker 11:43

Collapsed. I think,

Unknown Speaker 11:45

did you change anything there?

Speaker 2 11:51

I should say, like, I put it in like a new line, No,

Unknown Speaker 11:56

don't. Don't change the main method.

Unknown Speaker 11:59

Does you because

Speaker 1 12:01

what will happen if you change the main method? When we run the test cases, the test cases are expecting input into particular format, then only it can validate

Unknown Speaker 12:10

the output. You know, yeah, I'm not

Unknown Speaker 12:15

just not changing that. I'm not changing I

Speaker 2 12:29

BBB, so it wants to like, read each line, right? So I think it should be

Speaker 4 12:42

like In the in a New Line,

Unknown Speaker 13:11

why i i think it was just A way

Unknown Speaker 14:01

we are Passing we pass it

Unknown Speaker 14:27

Bishop, when

Unknown Speaker 14:40

I do me bring

Speaker 1 14:46

the list. Maybe I need to change the question. This

Unknown Speaker 14:53

question is not, oh, we are getting empty

Unknown Speaker 14:57

array, so list is coming empty,

Speaker 3 15:00

which is surprising. So check where the method is called large, so large substitute or large subset array, dot sum array is built over zero to array, count, map object I offer red line to place all request, all substring plus a okay,

Unknown Speaker 15:30

I would say, let's I chose One question.

Unknown Speaker 15:36

But basically at this new logic,

Speaker 1 15:43

currently, you're just iterating from the beginning to end, but you need to move between. So this will only tell you the maximum sub total of whole array, but there is a possibility that you need to move your

Unknown Speaker 16:01

left index. So

Speaker 1 16:04

in the sense left and right or beginning or right, you have to keep moving again, because you might have to draw some some items, some integers, because that might be adding up to negative value. But yeah, so yeah, the first example you see, if you dry around on that, the example that we have you will get you will iterate, current is zero, okay, then you put current equal to number,

Unknown Speaker 16:40

and then current is greater than

Speaker 1 16:44

mix. Then you assign to the mix, and then next you come, you know, yeah,

Speaker 2 16:49

yeah. I think it should still work. I mean, like, we can test it by just passing the valid directly. And if you want, like, you can just test it by passing a custom area. So I think you should still work. Okay?

Unknown Speaker 17:22

Do do you see what is happening?

Unknown Speaker 17:53

It is doing total of whole

Unknown Speaker 17:57

but 15 is not an answer at all.

Unknown Speaker 18:00

If you see you have to put minus five.

Speaker 1 18:13

No five, so answer six, correct. Now, take the

Unknown Speaker 18:18

next example. Okay.

Speaker 1 18:26

Okay, I'll give more example from the test cases. Give me a second, because

Unknown Speaker 18:33

I don't Want to. I

Unknown Speaker 19:26

I'll put Another example in the chat. I

Unknown Speaker 20:02

The game Just one more example i

Unknown Speaker 20:37

right? So those are all great,

Speaker 3 20:39

okay, what is the last output? They can see that

Unknown Speaker 20:47

for the last outputs, for

Unknown Speaker 20:50

the example I just posted in the chat,

Speaker 3 21:00

yeah, This room. So

Unknown Speaker 21:10

let me Just put that in. I

Unknown Speaker 21:56

15. That's

Unknown Speaker 22:02

correct. Moving the

Unknown Speaker 22:05

pointer. Can you do that? Your solution?

Speaker 2 22:14

So, yeah, so this piece here, you know, does some area come so it does not, like, actually matter, you know, if we are, like, just creating a separate sub area or not. So every time we just get a sum which is negative, it just basically changed the pointer. So just like, start from a position, and then we do a max, like Max of those values.

Unknown Speaker 22:50

I'm just

Unknown Speaker 22:56

thinking more examples. I

Unknown Speaker 23:06

The tomb Okay,

Speaker 1 23:23

okay, yeah, I think I was, I just took another interview in which the problem was, you have to go back. And it was very similar question. But yeah, I think this is correct. This is, I think I chose very simple in that one. It was, you have to go back and recalculate. You remember when I asked the question I got I myself got confused. Okay, okay, no problem. I think this was the simple science. Give me a minute and pull in another question. Okay.

Speaker 1 24:04

How confident you are on the Java language.

Speaker 2 24:13

So I would give myself like, I'm pretty, like, comfortable with Java, so I've been working with Java for like, almost six years now.

Speaker 1 24:26

And have you worked with any other programming language?

Speaker 2 24:31

I primarily like work with Java, but I would say I have also worked with Java, like JavaScript, you know, a little bit more like little bit more like maybe I can as

Unknown Speaker 24:46

well. So then you, if I see you,

Unknown Speaker 24:53

if I see you,

Speaker 1 25:00

screen. It got everything got closed. I need to log in again to my okay. So I was able to restore so I see your profile. You

Unknown Speaker 25:21

have studied computer science in

Speaker 1 25:25

bachelor's and as well as data analytics in masters, then, what was the reason that you chose to continue as a programmer rather than data analytics in the data analyst? You

Speaker 2 25:40

know? So I was just like exploring for like, a newer field. And when I like, chose, like, data analysis. I have always been, like, interested in the computers and computer science, but I really enjoy the problem solving, you know, that came with the software engineering. So I always like, I always like, decided to, like, continue this fashion that I have, you know, for building the softwares. So yeah. So I just Yeah, and then I decided to continue with the software engineering.

Speaker 1 26:13

Okay, so if I may ask you some computer science questions.

Unknown Speaker 26:22

So Java

Speaker 1 26:24

is object oriented programming. Can you tell me? I mean, it also supports functional programming also. But what is the, what are the pillar of object oriented programming? Yeah.

Speaker 2 26:42

So, yeah. Yeah, yeah. Java is like, so yeah, we do have, like, basically we have, like, I would say, for primary pillars for object oriented programming. So you firstly have like, a encapsulation, which is like data, like hiding, so you just hide the classes of fields, and you just have, like, complete control of the fields within the class itself. And then there's like inheritance, which is about the reusing of the like, usability of the code. So like, trial the classes and can inherited and feels like exposed by the parent class. And then there is like, you know, abstractions, which is also about about the hiding of the complexity of the code to the outside. And we can, like, achieve this with the help of like, abstract methods and abstract classes as well, and as well as the interface. And then, you know, and then, as far as, and then is the polymorphism, yes, which can be like achieved with the help of, you know, methods like overloading. And then we also have the methods like overriding and overloading, which is like done within the same class, and method overriding is done with the child class. So these are the four pillars.

Unknown Speaker 28:16

I mean, you answered it correctly, and now what

Speaker 1 28:21

total technologies are always misleading, encapsulation and abstraction, you said, same definition for encapsulation. What are the differences? If they have two pillars? Should be something could be different. Both would go hand by hand. Programming language cannot be just supporting one if they want to be objective. Programming would have called it one pillar. Why did they put into what are the

Unknown Speaker 28:53

differences? And it's okay if,

Speaker 2 28:59

yeah, so I would say, like, in fact, encapsulation is basically for bundling the data, right? So is, you know, is just for variables and methods that operates on the data, usually within the class, right? So, for example, like, if you have a employee, it might have, like, name and, you know, email and so on, which is like basically just hiding the internal state with the help of the Gators and setters method, right? So encapsulation is just for the data and for the state of the objects method. So encapsulation, you know? And so if we have like, employee, like service, then that's like, how you want to create an employee. So it is like, create employee, or update employing. Or like something is like, it just like, hides the complex implementation details and just shows the features it has. So that's what I would say of encapsulation. And for the encapsulations, I would say for detail of the data, and abstraction is for like implementing of the details, I would say, like implementations of the details

Unknown Speaker 30:27

also, can you

Speaker 1 30:32

explain how the polymorphism works in case of

Unknown Speaker 30:38

So, give me one example where polymorphism will come,

Unknown Speaker 30:43

okay, so just giving it in

Speaker 2 30:49

an example. So let's say, you know, you know for let's say just you want to have like an animal, so, right? So animal would be your like bass class, and you, you know, so, so just like animal can make generic sound. So it would be a method called, like, make sound, and then you can just have, like a dog, you know, which extends the animal. So here we can just override the actual mix on the method and give a separate implementation, right? So the sound that a dog would make would be like a bark, and then you have like another like cat, which, you know, extends an animal. And then the mix on the method would be like something like a meow, right? So, yeah, it would be that. And that could be one example talking about the polymorphism.

Speaker 1 31:52

And how do we achieve abstraction in leguma,

Unknown Speaker 32:02

yeah. So to achieve

Speaker 2 32:06

abstraction in Java, you know, you can, you know, with the help of abstract classes, I would say, which are just like the base classes, like which contains, maybe like abstract method, or the concrete like a concrete method, then you can just also use like interface, like interfaces, which contains, normally just abstract methods. But since Java eight is like us as I can remember, you know, you can just have like default like methods. Yeah. So, yeah.

Unknown Speaker 32:44

Yes, you talked about abstract class, right?

Speaker 1 32:49

So then what is the difference between abstract class and interfaces? Okay,

Speaker 2 32:56

so for the major difference that I can remember between them is abstract class and interface is that abstract class is the base class you know, which, basically you know bounds, the relationship you know between two classes. So basically it will have, like a base functionalities that you know that other classes can work with or has, like some shared code or like contract users, right? So, if you So, if you are using an abstract class, Java can only extend one particular like abstract class, but interface is just used for like contract and behavior and like definitions of your implementation classes, right? So interfaces cannot be like, instantiated, like by itself, but in Java, you can, like, you know, achieve like multiple inheritance by using like interfaces, because you can like implement as many number of the interfaces like you want.

Speaker 1 34:09

Okay, that is very

Unknown Speaker 34:13

own spot answer. Very Bucha, I like it

Speaker 1 34:18

now. So let's say, if you are in an API development team, okay, you have to define and you work with clients. Could be internal or external clients. First thing you are asked to

Unknown Speaker 34:36

provide the API.

Speaker 1 34:39

So when we say, Can you provide the API documentation? What all details will you put into API documentation?

Speaker 2 34:47

So, yeah, so documentation, yeah, it's very important to so for the documentation change, you just like, so if you have, like, an already existing API

Speaker 1 35:00

that is, you're asked to define, I mean, I'm not putting any use case or any scenario that, like, you know, dog or animal like, not like that, just what all information you will put it into API documentation. So,

Speaker 2 35:16

for the API documentations, you know, I would say that you would have like your base information, like, what does API stand for? Like, the name of the API, the endpoints, that what

Unknown Speaker 35:29

an API stands for.

Speaker 2 35:33

So, yeah, so API stands for, like, applications, programming, like interface. So

Speaker 1 35:43

the last word in the API interface is something that you just explained to me, right? That is what we are talking about. So what? Let's say you have an API

Unknown Speaker 36:00

get customer data.

Speaker 1 36:04

Okay, and it returns object of customer data. You know, object and input is, let's say it takes customer ID. Now, can you write here on the screen? How would you define the interface for This.

Unknown Speaker 36:21

Okay,

Speaker 2 36:40

so you will have like an eight customer like, get customer data, like ID. So it would be like, get mapping ending. So since you are you want an ID, you would just do like a slash.

Speaker 1 36:58

The get mapping is a notation you're using. Yes, okay, I've been good that you know that, but I was like, you can go simple.

Unknown Speaker 37:13

It's nothing to do with Java, nothing to do with

Speaker 1 37:19

it could be in C Plus Plus, or it could be a Python doesn't matter. You have to just define, okay, so method is gap. What is get in the gaps? Stands for gaps. Just request for that. Okay? Just request that, GET HTTP method, yeah, it just used to retrieve a resource. What all methods are available in HTTP

Speaker 2 37:57

so you have get post

Speaker 1 38:06

daily badge. Those are the most. There are tons more. Like, I think 10 more would be there, but not in regular day to day. Okay, so now you say customers less ID.

Unknown Speaker 38:20

What would be

Speaker 1 38:23

written type for the ID.

Speaker 2 38:28

So the return type would be adjacent JSON. I'm sorry, Jason,

Speaker 1 38:38

okay, so basically response, that's your Tp response would be the different type, but you are now. This is at the API. Here at the implementation layer. You need to this is what you're showing you. But

Unknown Speaker 38:57

have you worked on defining the API?

Speaker 1 39:01

How do you like? This is what you're putting, like an example. But API, interface documents contains, like, if you go to, let's say, developer.paper.com you will see, and I would say, if you open, I'll show you what I'm okay. Can you open? Developer dot?

Unknown Speaker 39:25

Okay, and you see the API

Unknown Speaker 39:31

on top? Yeah, these get started

Speaker 1 39:37

and look for, let's say, go down orders, API

Unknown Speaker 39:43

mail, so, whatever.

Speaker 1 39:47

So, this is what I was you put the this is I was asking that how so the quest had a response. So, but it's okay if you have, have you worked on defining the API spec for

Unknown Speaker 40:04

specifications for any

Speaker 3 40:07

product in past? Yeah, I have worked with swagger

Unknown Speaker 40:12

basically. Yeah, okay.

Speaker 1 40:17

What tool do you use to convert swagger into

Unknown Speaker 40:22

programming? The class.

Speaker 2 40:25

So we here, we just, like, we have, like, you should, like, be open, like, API specifications. Okay, good to know.

Unknown Speaker 40:38

Yeah. Do you have any question, like, so, yeah, it was definitely very good.

Speaker 2 40:50

Talking to you. You did talk about a lot of like, wasteful APIs at the end. So how much do the team gets involved in, like, microservices and risk for APIs, along with, like, maybe even driven

Speaker 1 41:06

I see, I see you that this interview is for back end engineer. I would say that, like, you know, you will be highly involved in, like, you know, this interviews for staff engineers expected to you to define the API, the basic API, the contract between two parties, right? So at the staffing engineer level, you will be responsible for, like, you know, defining these interfaces to that. To do that, you need to understand, first of all, the overall product, what you're working on, and what client expects and what like, you know. So implementation is something that maybe back in a genius staff engineer may work or may pass it on to the junior engineers for the implementation part. But this is something is expected from the staff engineer to define all these contracts, works with the peer teams the client, and then, obviously you might not be the hand service, so you might be calling another micro service to get some data they cannot. So all this is something is expected, and that's why I asked this question, and I saw that you showed

Unknown Speaker 42:24

that, you know, obviously

Speaker 1 42:29

problem solving is one thing, right, which is expected from job engineer, but these

Unknown Speaker 42:36

things are important to

Speaker 1 42:39

at this at least on the staffing email that is effective, the person to at least know all this question,

Speaker 1 42:53

any other questions, then I can give you a couple of minutes of your time. Yeah, yes,

Unknown Speaker 42:57

sure, that sounds good. But

Speaker 2 43:00

like, what I just want to know, like, what do you like? Mostly working at PayPal.

Speaker 1 43:07

Okay, so I've been in PayPal for little over 10 years now I like one thing I like most about is work life balance, given most of us have family to support. And one thing like, you know, I want to say that pretty cool. And you can every team has different structures. And when there is, you know, P zeros are going on, things go, but, but overall, if you ask anyone, most of the people will say, the work life balance is good because the PayPal, or wherever you work, that is you do to support the other life you have when ligand, when your work started hampering with your real life. That's where you know. So that is one thing that I like most about it. Second thing that I like that like you know, management has gives you lots of opportunities to prove you you're like yourself and PayPal always works with the latest technologies, latest technology stacks, and allows you to change, like, if you want to move within PayPal from one team to another, you are open to like, you know, like, if you are in team for 12 months, you can change the team, you know, so things like that have changed a couple of times, which and then you build a network. Another thing that I would say I like about the that management is supportive of your ideas. Sometimes idea may come from product, which is they are supposed to bring the new ideas into the system, but it may come from engineering as well, like you know, so they it is always appreciated by company that you put yourself as a customer, and company like PayPal, where I don't know if you use PayPal or not, but I use a lot of people products. I use Venmo wild use PayPal, and there are a lot of places where underneath we don't know PayPal is being used. Like Braintree is our company, linking subsidiary and Braintree provides payment

Unknown Speaker 45:42

gateway for lot of

Speaker 1 45:44

merchant where you are paying using your punching in your card, debit card or credit card. Basically the payment is being processed by brain three, which is pay balance, actually. So all these to know all this, because then you are in a company where products are customer facing.

Unknown Speaker 46:05

You can relate that okay,

Speaker 1 46:09

that how my how my product is being used, because you can use yourself. So it's a business to customer relationship. We also have business to business relationship. So basically, that is, these are the things that I liked about Bucha. And like I said, in between, the work life balance is very important for me. It is, should be important for everyone. But there are quality people. So I don't want to deny that for everyone, but for most of the people, it is, and that is what you get working

Unknown Speaker 46:49

with Baker. That sounds

Unknown Speaker 46:53

really great. Thank you so much for

Speaker 3 46:56

the time. That's all the questions today.

Speaker 1 47:00

Thank you. Thank you. So if I pronunciating in the your name correct is Susanna.

Unknown Speaker 47:04

Yeah, thanks.

Unknown Speaker 47:10

Have a good day. Thank you so much.

Sujina-navyfederal-SRV-Final

Speaker 1 00:00

Us, one of the manager of the savings engineering services team here at Navy, federal. And again, we're, we posted this developer three position. I'll go around. I kind of probably walk you through a little bit how we do these interviews. Scott and I, I'll let I'll give him a minute to introduce himself, but we'll go ahead and have a nice chat back and forth. Feel free to ask questions. We'll do a round of different situational questions. Get to know you a little more, and so forth. And in the probably a few minutes, maybe at the half mark of the hour, we'll have one of our current developers in the team come in and then chat with you as well, more from a technical perspective, and feel free to ask him questions, like, you know, kind of, how does your day look? He's, you know, again, he works directly within the team with that. As I said, Mayor, I'm Mariela Naomi, and I'll pass it to Scott just to introduce yourself, and then we can pass it on to

Speaker 2 01:09

you. Scott Gina, my name is Scott Miller. Nice to meet you virtually. I'm not on camera, so bear with me for that. But yeah, as Mary Ellen said, I'm the manager for the safety engineering services team, which Mary Ellen i are partners in supporting that team, and we're here to interview you today for this position. I've been with Navy Federal now for almost 15 years, and you know, been through a lot, and like certain, times are ahead of us. We're going through a major modernization effort, which Mary Ellen might get into in her brief overview of maybe federal and what we do. But anyway, just looking forward to our conversation for the next hour. So nice to meet you. Nice to

Unknown Speaker 01:59

meet you, too. I'd like to thank you guys for

Unknown Speaker 02:04

this opportunity. Too great. So, as Scott said,

Speaker 1 02:09

exciting times for Navy Federal, as I mentioned, our team mainly focuses on different products. Our team specifically looks for or works on savings products. But again, Navy Federal as a whole, is going through what we call a core modernization, taking our applications, legacy applications, out to the cloud, different but of course, and that meaning of time, the current applications are also left there to be one, maintained, updated, any changes or, you know, so there's, there's a lot of opportunities here at Navy Federal. A little bit about myself. I've been here almost this close right on the heels of Scott 14 years at Navy, fed, loving the team. And again, the team is made up of several roles. We have some developer roles we were doing the transformation to Agile. So we do have some Scrum Masters, some what we call TPAs analysts, that help with the requirements. We currently use Azure DevOps as our tool to manage our backlog, and that those things. And I know I'm mentioning these because I have seen your resume, and I know it looks like you have work through that. So with that, if you can tell us a little bit about yourself, and I'll probably kick it off with your interests in this position and why. So yeah, first of all, my

Speaker 3 03:40

so again, first of all, my My name is Susan, and I have been working as a software engineer. And I have been working as a software engineer for about six years now, pretty good experience working on microservices, Java based web applications, and I have worked on modernization legacy to microservices. Work with cloud, and I primarily with AWS, but I do have some ajar experience as well, and I guess my primary motivation towards the open position is that it feels like a really good opportunity to explore the technology, to give back on what I have gained, you know, from my experience along in the under newer opportunities, right? So because the position does offer a lot of, you know, exciting technologies or work related to it, so that's my primary like reason to because

Speaker 1 04:47

I'm interested in this position. Interesting, could you dive in a little bit of what currently you're working with? USA? I'm sorry. USAA, yeah, USA. So talking more

Speaker 3 05:00

about my experience with USAA, I'm primarily, you know, working on, you know, migration of the project. So, yeah, we have a legacy portal that is, like, hosted on, like a firm, and we basically breaking down like chunks of this, like applications into microservices based architecture, and then, like, basically deploying them to, like, eight of this lab. So I have been involved in designing this microservices, like basically coming with the design of the documents, getting architectural improvements, and, you know, like gathering functionals and non functional requirements. So once that is done. I developed this basically write the code and then also like work in the deployments aspects to AWS using our CICD tool. So the couple of services that I recently completed, like migration, include the Experience API, which is used for grabbing entitlements for the front end basically dictates what, you know, widgets and components users can view. And then you know, maintenance of the API, which basically is responsible, you know, for making changes to our user profile preferences and so, yeah, so Java Spring Boot is a primary tech stack, and we use services like AWS ECS for containerized architecture. And you know that AWS lambda for event based system, so AWS s3 for storage, AWS RDS for database. So that's the kind of overall overview of what I have been involved in

Speaker 1 06:55

this template. Thank you. Any questions that you have, Scott,

Speaker 2 07:02

yeah, just one quick question. It looks like you've been at USAA for just a little bit over a year now. I'm just curious. You know, why are you looking for a new position here? You know, a new federal now? Or just, I know you talked a little bit about why you you are interested in this position, but I guess tell a little bit more about why you're looking to transition from us day to because

Speaker 3 07:29

my contract is coming to an end. I just decided that it will be in March 15 ish, that our project is coming to an end. So that is the reason why, because of, because

Unknown Speaker 07:41

it's a contract position,

Speaker 2 07:48

and are you taking? There's no opportunity for conversion to full time. There is that another reason for not pursuit? It was already like decided before. So,

Unknown Speaker 08:02

yeah, okay, so didn't have like

Speaker 1 08:12

conversation like factor yet, so basically, got it so I so I think I'd like one of these questions I'm looking here, obviously, Navy Federal, kind of, what, what? What interested you about Navy Federal, obviously, USAA, I see the other companies that you worked on. But So what drew drew you to Navy Federal, or, again, those qualities in an organization, because this would not be a contracting position, this would be a full time position.

Speaker 3 08:42

So one of, one of the main reasons also, like, I'm looking for a full stack position now that I think the main thing here at Navy Federal, it has a mission like driven work. So basically, you know, it provides a very tailored and unique solution so probably not only helps in like technological aspects, but also have some like societal like impacts. So, you know, the military members, and I think Navy Federal is basically invading, you know, the financial technology to stay competitive, and which helps me and basically broaden my experience as well. And the work culture here is very good, as I learned from you in this interview. So what I have researched about, like, very strong views and

Unknown Speaker 09:41

yes and a pretty good opportunity for the growth of my career.

Speaker 3 09:47

And that's the main reason. And, you know, trans and also transitioning from, like engineer to architecture and so on. So I think these are some of the primary reasons that makes maybe federal outstanding as a place to work,

Speaker 1 10:07

in my opinion. Thank you. Yeah, we are mission driven, that's for sure.

Speaker 2 10:13

Well, again, some of the work that you've been doing for USAA is very consistent with a lot of the same work that we're doing at the federal as we spoke to earlier that, you know, we are kind of going through a similar phase of application modernization, moving, moving a lot of our our existing on prem, you know, applications and infrastructure to the cloud as part of this modernization work that we're doing. So, yeah, so I think you know, your background and your work history is very applicable to the kind of work that we're doing here at Navy Federal.

Speaker 1 10:56

Could you tell us a little more, or if you ever had, either this, the current projects that you were working on with USAA, or the others, what has been one of the most interesting projects that you have worked on, and what made it interesting? So I

Unknown Speaker 11:18

think one of the most interesting project

Speaker 3 11:22

that I was having a part of was the orchestration aspects of the USA, and how did this, like, different API are basically handled. So basically, what happened is there, there is orchestration for, like microservices that dictates whether you know service calls from your front end or your cloud. So WF roots to write this basically takes you know. So there is a Spring Cloud gateway based application which basically acts as your API gateway. And this cloud this spring, cloud gateways application is pretty basic, but just the definition, but it does a lot of work. So it basically accepts the request, it does manipulations on those requests, basically transitioning from those requests into like other format or adding payload, changing the payload, you know, and all of those are processing as it needs so it needs to do, and then calculate the host that Is needs to, you know, go to and then, basically proxies is to to the host. The this was kind of interesting, because at first it seemed like a very simple gateway applications, but later I came to know that there was a version for each of the device type, like the iPhone device has its own app. Android had its own, and then web applications had its own, and each of those app basically handles more than, like five, like 600 API call so these huge orchestrations layers handle the layer of this, micro services, calls, and then, like doing request, go ends and responses, manipulations and a high in a very high skill keep and being available at all time was something that was, you know, very interesting to me, because it needed to be that a very highly functional, and, you know, available and making changes to, you know, any one app would, you know, affect like other in some manner. So it had to be, like, very careful and forward looking. So I think this was one of the interesting ones that I worked on.

Speaker 2 14:00

Wow, Were you part of a team doing that work, or is this something that you were primarily responsible for doing? And tell us a little bit about that kind of setup you had you

Speaker 3 14:12

at USA. We basically pull in like features, like for six months, so we were involved in this feature work for that like API gateway for like six months,

Unknown Speaker 14:28

and it was that,

Speaker 2 14:30

were you operating as an Agile team as part of that? So, yeah, I was a

Speaker 3 14:37

part of the team. Yeah, that basically handled it for like, so it was like for, yeah, we were operating on an

Unknown Speaker 14:50

Azure team. Sorry,

Speaker 2 14:53

I see you. You currently are based in Gainesville, Virginia. Is that correct? Yes, and this contract with USAA was out of their St Louis, Missouri location. So were you working full time remote for them?

Unknown Speaker 15:11

Okay, okay,

Unknown Speaker 15:14

when we take the next question? Mariela,

Unknown Speaker 15:16

yeah, if you want SCOTT Sure,

Speaker 2 15:21

maybe kind of related to to the response you just gave about the most interesting project. But the next question I have is, you know, did you have a time where you kind of were able to take ownership of a project, or at least a part of a project, and if so, tell us about you know, what that was, especially if it's something different than we do, but also tell us, you know, how things have, what was the outcome of that, of that project or that task,

Unknown Speaker 15:54

and just to help a little bit about that, Yeah.

Speaker 3 15:59

So, yeah. So, yeah. I, recently, I took the ownership of the maintenance and then experience like API modernization. So, yeah, I was like, you know, mentioning earlier, USA is going through the modernization, right? So our team is we basically handle the user account and profile portion. So for the moderate, you know, the modernization of the account preferences and chains, I took charts of the first migration report and which was the maintenance of the API. So for that, basically I had to, you know, first figure out what the domain I had to define and where the responsibilities had to be, you know, segregated. And then I basically had to gather, you know, functional and non functional requirements. And then basically took charge of the whole modernization report, you know, by documenting, by finding, and then getting approvals and reviews from the architects, and then basically, and putting that into, like, your actual deployable. So I think that's, I think that's the after that I proceeded with the Experience API as well, the, you know, the main outcome of the first mean charge, or the, you know, lead on the modernization was, I was able to, you know, create, like a path for others, you know, to follow through, like a modernization report, right? So, because I had a very good, like, I made a very good documentations of what I had worked on that particular report, so the others could, you know, basically, view and change it the part of the project, you know, into more modern frameworks as needed. So, yeah, that that is one of the instance.

Speaker 2 17:59

Okay, thank you for that. Mary Ellen, do you have any follow up questions out of

Speaker 1 18:06

that? I wanted to, I think you touched on a little bit on you had created this pod so others, I assume, could learn. So obviously, you know, I wanted to maybe talk about your team collaboration, if you could kind of talk how that worked out, how you were able to share that, and was it well, you know, was it Welcome? Did people learn? Was it well received? Or if you had any, you know, folks that had any other kind of questions, or how did that work? Can you talk to me a little bit on that team, collaboration with that, with that effort?

Speaker 3 18:51

So, yeah, so basically, we worked in a giant team, and we had like five engineers, the product owner and the scrum master, right? So, yeah, we basically meet, like, daily on huddles, but as a part of this modern day, or like, modernization report, after, sorry, after, I like, gave a little demo to the teammates, they were excited to see the outcome. So even before I had the competition, I sometimes, like peered with a couple of my teammates to talk through, you know, some of the solutions of these supports, right? So because we, you know, we and I personally believe that the collaboration is really good. It has to be really good within the teammates, if there is a enough time and bandwidth for everyone, I think collaborative work promotes like better and robust applications. So we do follow pretty good collaborative efforts. And apart from this work, we do like we also do the code reviews and peer programming sessions as needed. So, yeah, so in my personal opinion, I think the team collaboration is very important, you know, for team building, as long as for the measurement of the development that someone is doing.

Speaker 1 20:15

So a follow up on that though, have you had any situation where another team member was not doing their share of the work, and if so, how did you handle it? Oh, yeah.

Speaker 3 20:30

So talking on that. I mean, yeah, there are times when, like teammates, they might be like feeling like there might be like a falling behind on, like the talks there might have been assigned, you know, and whenever like those kind of situations occur, I usually offer help if they are stuck on like certain like manner. But if they are like locked on like certain manner, then if something is not clear. I try to see if I could help them giving like technical details on the issues, and if it's like non technical, then I think what I have done in the past is I basically created like a smaller sub talks on their talks in hand so they can, like, easily complete like the smaller talks to work towards the, you know, the bigger goal. So, yes, rather than, you know, demotivating them, I try to, like, motivate them to complete the smaller talks for the smaller ways, you know, which basically accumulates, like, towards the bigger goal.

Speaker 1 21:40

So it's easier to break it down. So all of you guys do the work. Got it, any follow ups on that? Scott, or if you want to go to the next question, I'll

Unknown Speaker 21:52

go to the next question.

Speaker 2 21:55

Have you ever had a situation where you had to explain a very technical, you know, complex problem to a non technical person, and or not necessarily a problem, but even a solution, but to a non person. And so tell us. Tell us about that situation. How you how you convince or explain that situation to the other person, based on your experience. So,

Speaker 3 22:27

yeah. So yeah, we do sometimes it happens while working in the workplace, but I do have experience with

Unknown Speaker 22:36

that. So basically,

Speaker 3 22:38

when I was working on the Spring Cloud gateway, the Spring Cloud gateway. The whole architecture is, you know, the kind of like complex. It's very complex. But the way, I kind of had to explain how the Spring Cloud gateway solution, you know, the solutioning was, was to able to help our routing issues that we had was basically given on, for example, for an analogy to something that was, like, very tangible, right? So I explained the concept of, like, the Spring Cloud gateway framework as your Frontex of a hotel. So you know that the main purpose of the spring club gateway was, you know, taking the request and then direct it to, like, where the person wanted to go, right? So, similar to a hotel dogs, right? So they take in the user information and then check the registration and so on, and then direct them to a room or the restaurant or the gym or whatever they want to go. So whenever I had to explain a very, you know, take heavy frameworks and ideas to someone who might not be, you know, non technical, like a business executive or someone, then I had to, you know, come up with analogy beforehand, with some prep, like work done, so that, you know, I help in visualization what the whole architecture looks like. So that is the kind of what I do in the past.

Speaker 2 24:18

Okay, thank you for that, Maryland. Back to you.

Speaker 1 24:23

Um, I was wondering. I know Rob joined us as well, so I if that's okay, could we move on to more of a technicals? I know, again, he could pass more of those deepers. But I'd like to introduce you to rob Kelleher. He's one of our lead developers in the team. Lots of experience. Great guy. I'll let you introduce yourself, Rob.

Speaker 4 24:48

Hey. Are you? My name is Rob Kelleher. I'm on the I'm on Mary Ellis team as a developer, and asked me to sit in here to help with some of the technical questions.

Unknown Speaker 25:07

I'm sorry.

Unknown Speaker 25:09

Gina, nice to meet you.

Speaker 1 25:15

Um, I don't have anything, I mean right now that we can talk through. But Rob, would you like to take on with some technical questions?

Speaker 4 25:28

Um, sure I was going through looking at your resume and had a couple questions. Let me bring up my notes here. Sorry,

Unknown Speaker 25:40

it disappeared behind my camera,

Speaker 4 25:49

so I noticed the a lot of spring and spring boot experience, just exciting. What about legacy software or legacy technologies? What have you dealt with those and either conversion to new or maintenance of them, yeah, just talking

Speaker 3 26:09

about the legacy of technology. So, yeah, I have been, you know, involved in the modernization of the legacy applications and USAA. So I do have experience like maintaining the legacy role

Unknown Speaker 26:26

code and then also like migrating them.

Speaker 4 26:31

Are you familiar with web sphere, IBM, WebSphere? Have you used that, or you have experience with it? So, yeah, I think it

Speaker 3 26:41

was in web sphere. So, yeah, I think it was in Webster, web sphere. I

Unknown Speaker 26:49

don't have, like, a direct

Unknown Speaker 26:51

work experience

Speaker 4 26:54

with it, but I think, I think our application views it Okay, is it you worked on moving it over to a modern platform? Yes, okay, that's exciting. And then the cloud technologies you've worked with looks like, primarily Amazon Web Services.

Unknown Speaker 27:11

And have you done anything with Azure? Yeah, I have done Azure in the past, where

Unknown Speaker 27:19

I have created

Speaker 3 27:22

a function and also use, you know, their buckets, your service, like bus, for training, triggering those functions, okay,

Unknown Speaker 27:35

and the DevOps pipelines.

Speaker 4 27:40

Use Jenkins. We used to use that here. I assume you're familiar with GitHub.

Unknown Speaker 27:49

Look at

Speaker 4 27:58

maybe you can just walk me through, like what sort of what you did to develop, or what was your role in developing those that CICD pipeline, or what your ideal CICD pipeline would look like. I have used

Speaker 3 28:19

CICD pipeline in my recent project as well, so you USA, we basically, you know, have we? We have our own pipeline. So then we get all what we need to do, basically, is to set those pipeline up. So firstly, we create something called a seed job that sees so other Jenkins job in this seed job, they basically provide how this pipeline needs to be generated, right? So for that, we basically create a ruby script which has, like your own parameters and configurations attached to it, and then for the actual deployment job, you need to, like, create, like the shell scripts that is used to, you know, provisions of those services. So we have an abstractions of all, you know, cloud formation templates to provisions those services. So we just use this, those abstractions. So once we run into our speed shops is up with our applications pipelines. And these pipelines, they basically have, like, different stages, and each stage basically triggers a shell project means shell script, so I have experience in building these pipelines and then deploying

Unknown Speaker 29:49

it to the cloud. AWS, great.

Speaker 4 29:53

What about let's talk about automated testing. What is your experience with automated testing? Yeah, I have done a

Unknown Speaker 30:01

lot of a lot of automatic testing. So

Unknown Speaker 30:07

we, yeah, we recently, you know,

Speaker 3 30:10

incorporated the cucumber testing for an end to end testing, and that is something that I have done recently. So apart from that, we do like J unit testing as a part of, you know, a regular development, right? So we do like our own regular unit testing, and then do like integration testing using like Spring Boot test within the application. And then after that, as a part of testing stage of our pipeline, we incorporate

Unknown Speaker 30:47

the cucumber testing. Okay,

Unknown Speaker 30:50

go ahead. Do you use cucumber much?

Speaker 4 30:54

Have you used it a lot? Or rather, I started, I recently started using

Unknown Speaker 31:00

it. Okay, for the last couple

Speaker 4 31:01

six months, yeah, that's, that's something we're one of the technologies we're looking at implementing here, so that would be interesting. And then you talked about at USAA, the front end work you did with Angular and

Unknown Speaker 31:25

RxJS, what

Speaker 4 31:28

was it was that your main focus was front end or back end, primarily Back

Unknown Speaker 31:37

end engineer had to like, make like,

Speaker 3 31:41

certain change. Like, certain changes on the front end that I did do a full feature work for developing a task, catalog, read, update view. So it was a standalone component. And, you know, we basically had to provide a view where users would be able to view different catalogs and then edit it based on the tags. So, yeah, so this is the work that

Unknown Speaker 32:12

I did recently, about 70%

Unknown Speaker 32:19

okay, see each see,

Speaker 4 32:30

spring makes it really easy to develop API clients. What kind of API management tools have you? Are you familiar with, like, meal, soft, that kind of thing? So,

Speaker 3 32:50

API clients for the API management tool, you know, we don't use soft, but we basically

Unknown Speaker 33:01

use spring, good. So list APIs.

Unknown Speaker 33:05

So talking more on that, I would say

Speaker 3 33:09

I think you were talking like, I think you were talking probably about swagger, you know, for the documentation, yeah,

Speaker 4 33:17

yeah, and where we, yeah, that, that whole interface, yeah, but I'm not sure,

Unknown Speaker 33:21

like what Director you were

Speaker 4 33:25

referring to. It was sort of an open ended question, but yeah, that's okay. So you your your application reaches out to an internal API to get data right.

Unknown Speaker 33:39

Those endpoints are managed by

Speaker 4 33:45

usually something like, well, we use MuleSoft, but we've used other technologies as well. Just wondering, if you're familiar with those technologies, you could describe what you used in the past. So, yeah, so so far. So

Speaker 3 34:02

basically, you know, to make a down like downstream call, we reached out to our Spring Cloud gateway, managed like API, so we get a dedicated, like host of those gateway, and based on the URI that like URL that we provide, we get routed to the

Speaker 4 34:30

dedicated downstream services as we want. How did you guys, how do how have you handled client security,

Unknown Speaker 34:39

like the interface

Speaker 4 34:41

security between your application and the REST API. So client

Speaker 3 34:52

security is one of the top requirements. So we use spring security

Unknown Speaker 34:58

for our application security.

Speaker 3 35:02

So we basically have the identity provider that validates the user's identity, basically the username and password, and then, based on those, say, user, username and password, token gets returned as a part of the identity check, and when then we use that token for the role based access controls on the downstream. Some of the some of the clients do have an API key attached to them, so we do not like we do have API key based security as well.

Unknown Speaker 35:41

Okay. Excellent.

Unknown Speaker 35:50

Sure is TLC. I

Unknown Speaker 35:59

think that's all I've got.

Unknown Speaker 36:05

Thank you. That was very interesting.

Speaker 1 36:08

Thanks. Thanks, Rob. Well, now I guess we'd like to turn the tables a little bit. What questions do you have for us? Any one of us? Scott, Rob, myself. I

Speaker 3 36:23

Yeah, firstly, I would like to thank you so much for the opportunity today. It was very nice talking to you all, sharing my experience and getting to know a little bit more about you guys. So first, when I do have a question that I'd like to ask is, what do engineers usually get involved on, apart from their coding work?

Unknown Speaker 36:52

I wanted to know regarding that Rob day to day, sure,

Speaker 4 36:59

I you know, aside from the the actual coding, right, there are

Unknown Speaker 37:08

some of the,

Speaker 4 37:13

the management type things where you have, you know, we have to go and go through the process of getting approvals and access, you know, putting in tickets to access certain services and those types of things. And then currently, we're doing quite a bit of analysis on some vendor products and how they're going to integrate into our own software. And so that's a really big push right now is that analysis of that sort of thing. I can't really go into a lot of detail about it, but, and then, you know, we have, we're sort of, we're sort of in the midst of an Agile transformation, if you will. So moving through learning the the ceremonies, if you will, you know, and how that for each team, yeah, and you know this the teams are. There's a lot of some growing pains there with the teams, but we're that. So that's a lot of what we do on a day to day basis is, you know, aside from coding, which there's, there's, there's a lot of coding to do too. But, you know, right now, the big push is analysis for the some of the vendor products we're integrating

Speaker 2 38:33

to add to, you know, the kind of kind of work that you do day to day is obviously going to vary, you know, from time to time, depending on the nature of the project that you're you're on some sometimes you might be on a project where you're the only developer doing doing some work. Maybe that an enhancement, small enhancement, to an application, or you're working on, you know, a larger project where, like Rob was explaining that there's some analysis, a lot of analysis work going on, analyzing vendor, you know, interfaces and things like that, to see how a solution might integrate with with our current infrastructure, or other other applications. So it's kind of very, quite a bit, you know, we try to, you know, limit to the best we can. You know, the role of a developer to be, you know, a developer to do the technical things. But, you know, not, not everything a developer does is write code. They they do a variety of things. And as Rob kind of alluded to it a little bit, you know, we have a pretty formal change management process. There's other things that are very formalized, from the standpoint of, you know, requiring access that isn't automatically granted to you as a developer who newly comes to Navy Federal, there are things that you have to kind of discover as you go about, you know, doing the work that you're assigned, that Aoi need access to this asset or that asset, you know, and you have to go through the process to find out what kind of request you have to submit to get that access or work on. You know, the change management process, you know, to get approvals and so forth, to move your code to the different environments and things. So there's, you know, I think Rob used the word management. I guess that's a kind of a management responsibility to some extent, but it's, you

Unknown Speaker 40:30

know, exactly what I was referring to,

Speaker 2 40:33

yeah, more than administration of the work that you do, right? You know, overseeing work that you're doing. You know, it's not just a sense of matter, of writing a bunch of software, but that's the core of the job. But there's lots of different parts to it, and it varies, you know, greatly from one effort or project to another. So it really depends.

Speaker 1 40:57

And then, if I can add, obviously, this, you know, we posted as a developer three, Rob's a developer four. So I think depending on the levels as well, but by any means, you know, there's this team has always been supportive to each other. So I think to Scott's point, sometimes we you get into a project that would have a couple of developers. So I've seen how they work together. They collaborate, right? They share ideas, they they test each other's code, things like that. Obviously quality and you, you touched on this a little bit in your introduction of the, you know, the opportunities and security of the financial, you know, industry that we're in is very important. So, So collaboration is important in the team as well. You know, like Rob said, the change management, there's all these processes, but obviously they're there, you know. But again, the team would help as well, if you don't know, they're always collaborating and helping each other?

Speaker 4 42:01

Yeah, I can, I can say, from my experience, I'm constantly reaching out to team members, you know, for somebody's an expert in this, you know, part of this process, and they can help me configure this thing really quickly, versus me. You know, instead of spending a whole day trying to learn it myself, yeah, and then we all learn faster, and we help each other out quite a bit. So it's very collaborative team. Thank you.

Unknown Speaker 42:31

Anything else I would say? I do have the

Unknown Speaker 42:41

next question, and I do. I have, like, what

Speaker 3 42:47

would like most about, like, working at, like, maybe theaters. Like, what do you guys like the most about

Speaker 1 42:56

working with, okay, who would like to take that from? Oh,

Unknown Speaker 43:03

I came to say that Thanks,

Speaker 2 43:07

Rob Dana, you did little investigation and research on the federal and you may mention, you specifically mentioned that you heard the culture, the work culture here at Native federal is very good, which is very true. You know, there's as evident by both myself and Mary Ellen and Rob, to an extent, because I don't think Rob's too far behind us in terms of in your

Unknown Speaker 43:30

five years, right, Rob, you turned

Speaker 2 43:35

we, you know, people come to Navy Federal, and they say, because they like working burning drill. They like the mission that we, we work to, which is the support our members, which are very much similar to the USAA. You know, customer profile, you know, military members, you know, and supporting them and their families and doing with it, you know, providing the support we can to them for their financial needs. And it's something to take a lot of pride in. And so it's, it's something that that helps us, you know, come in each state and do our work and make us feel proud of the work that we're doing. You know, some of the work we do might not have a direct effect on members, but the bigger picture is, is that a lot of the work we do does have an effect in one way or another, onto our membership. And you're knowing that the things you're doing are in support of, you know, their their financial well being as best we can so, you know. So it's a lot, it's a lot to that that really, I think, speaks to the culture and why each of us come in and do our job, you know, each day. So it's a big part of that.

Speaker 4 44:52

That's a big one for me. I'm a 20 year Navy veteran, currently a navy reservist, and I see young sailors in my unit, you know, benefiting from Navy, federal and their services, and what I do so it's really rewarding, because I get to see it directly. And it's one of the things that keeps me excited about coming to work is,

Speaker 1 45:15

I mean, it's been a big company. I didn't realize how big it was when I joined, to be honest with you, you think of a credit union. But again, I think the culture is definitely what kept me here, versus all the other companies that I've been with. But one of the things I'd like to share with you, we are a hybrid environment, so that's kind of new, obviously, that came out of the the pandemic, before we were, you know, on site five days. But I think that changed. And I think that also to the testament of our culture, we are staying hybrid. So a couple of days on site, and then, of course, you have the, you know, the remote option. So those are a couple things that I do, like currently with Navy Federal.

Unknown Speaker 46:09

Thank you so much for

Speaker 5 46:12

the information. Yeah. So the last question, I guess, is, sure, sure, what would be

Speaker 1 46:19

the next steps? I just wanted to know that we actually just kicked off your first interview, so we're still meeting with other candidates the rest of the week. We are looking to fill it up quick, so maybe make a decision in the next couple of weeks. If there's any other questions that you come you can think of after, you know, our conversation. Few you know, feel free to reach out to Kristen Klein. She's been talking to you about the position. So if anything, just reach out to her. She'll walk you through the process after that, you know the and after we finish this election process. So a couple more weeks give us, because again, we've just started interviewing for it.

Speaker 1 47:11

Oh, thank you. We want to, I want to thank you for your time. Thanks Rob for jumping on and Scott again, and nice to meet you. And have a great day. Have a great

Speaker 2 47:24

day, guys, thank you so much. Nice to meet you and talk with you. Have a good day. Take care. Okay, bye, bye. Thank

Unknown Speaker 47:32

you.

Suman Astani- Paypal- SRV- final- System Design

Speaker 1 00:01

Yeah, sorry that I'm a minute late. I was lost track of time.

Unknown Speaker 00:05

I just recently joined

Speaker 1 00:09

Perfect, yeah. Okay, yeah. So you know, my name is AJ. We're going to be going through a system design interview today. I would like to you know, I mean, you've probably talked about this a lot already, but let's just touch on your work experience a bit, and then we can move on from there. Yeah,

Speaker 2 00:26

sure, sure, sure, yeah. So, you know, I have been working as a full set Java developer. It's been around six years now, and like my primary, like, I have been working on Java based web applications right, using frameworks like spring hibernate for data services right, and for API services for like airspeed, APIs for API development as well. And I have, you know, like pretty good experience working on microservices, and I have been involved in, you know, like designing, developing and also deploying them on AWS cloud as well. So I have work on AWS cloud services, like easy to ECS, you know, like SQS, SNS and lambdas as well. So currently I am in at zero price. So my, you know, like, team consists of five developers. So we have, you know, like product owner and Scrum Masters, and on a daily basis, you know, like I have, I'm mostly working on the back in Java code, so I have been with the team with, you know, like, code reviews, right? And also on products and support duties as well. Sometimes,

Speaker 1 01:36

great, yeah. So as far as, like, you know, any projects that you've worked on, not even recently, but in your like career in general. What would you say is your like proudest work?

Unknown Speaker 01:47

My proudest work?

Unknown Speaker 01:51

That's a good one. Yeah, so to think of it

Speaker 2 01:58

like, it's a tricky one. So my proudest, I would say that working at the zero price company. So, you know, like, while I was here, you know, I was working on one of the project where, you know, like, we have to kind of basically generate very, kind of, you know, like generic solutions for our streaming of our logs, right? And the initial recommendation was to indicate that the SDK and kind of, you know, like, make this configure, configuring all the applications that we want, right? So which was basically, mean, a lot of work with, you know, like each of the services, right? But what I did was basically, you know, proposed, kind of create a separate micro service that handle that, right? So we'll basically create a, let's say, a dedicated micro service sitting in the post in point, and at that specific post in point, like it will be able to, you know, like, do what this SDK is supposed to do, right? So we created onboarding logic for mapper of the, you know, like non standard logs to standard format. And then we basically created, you know, this. We call it streaming engine that you will transfer those RESTful APIs calls into, like, streaming of those, you know, logging in the in the slow fit table, right? So that was one of the, you know, like, most, proudest project that I took into Completion. I would say, Yeah, that sounds good. Thank you.

Speaker 1 03:31

I'm sorry. I am. I want to try and get my headphones working. My previous ones died. We're gonna see if this one works. I it.

Speaker 1 03:46

Okay, hopefully you can hear me. Yeah, I can hear you, excellent. I can hear you, great. Now, okay, yeah, so you know, are you familiar with the like, system design format, like using a whiteboard to, you know, create a solution.

Speaker 2 04:06

Um, yeah, so I, I am,

Speaker 1 04:10

okay, cool, yeah. So we're just going to be doing that today in hacker rank. They've got a neat little interface for that. I will send you. I'm going to send you a link in the team's chat to hack your rank right now I'm

Unknown Speaker 04:46

okay,

Unknown Speaker 04:52

okay, There we go. Should be loaded in any second.

Speaker 1 05:08

Yeah, cool. So, you know, here we've got a systems I'm probably going to be working on today, suddenly realizing I forgot to talk about myself. Yeah, but yeah. So I've been at PayPal for five years. I've been working at Venmo the entire time, you know, primarily on their credit card, their debit card, doing all back end engineering and, you know, some system design so and you know, if, yeah, if you're familiar with Venmo, you know, much like PayPal, a lot of you know, peer to peer transactions. And so today, look at that. We're designing a peer to peer payment service this whiteboard. Yeah, so this is primarily going to stay focused on the like the send and receive aspects of Venmo or PayPal. But yeah, and you can see there's been some services set up for you, which you can use or even drop if you think it makes sense to the service. I'll give you a few minutes to read the description on the left, because it does a better job at it than I can and then We can get going. Yeah,

Unknown Speaker 07:04

okay, so

Speaker 2 07:06

it's just a tactical payment system, right? So let me first, you know, like, get rid of what's here, and maybe we can kind of do everything, right?

Unknown Speaker 07:16

Yeah, it sounds good. I

Speaker 2 07:30

so maybe some of the questions that I wanted to add was, does the user that will be using this particular systems be, let's say, authenticated, or will you just, like, assume it as a part of our systems, which is outside of the scope of this design. So,

Speaker 1 07:51

yeah, you know, I mean, users are, like, a really important concept. So, you know, if you want to put in like reference to that, it can be really useful, but don't go too deep into the actual mechanisms of authentication. That's outside the scope. Sounds good. Okay.

Speaker 2 08:11

And so what about the currency conversion? So will maybe just be a one single currency, or, like, will it be a multiple one?

Speaker 1 08:23

That is actually a really good question. Let's say a single currency for now, but I'm going to put a pin in that, and if we have time, we can circle back on it. I

Speaker 2 08:44

Okay, so let's see. I think we can get started with very like a minimal design for now and how I am picturing this.

Unknown Speaker 09:01

So basically, like, we

Speaker 2 09:05

will have a client, right? So I will do is, I will start by, you know, throwing in some diagram first.

Speaker 2 09:20

So this will be our clients, and we don't care about what the clients are. So these are just clients, right? So it could be, let's say, a mobile apps or web apps.

Speaker 2 09:38

So this will be an like identity provider that we will leverage. So this will be, you know, like external where we will, you know, like, basically, kind of make sure that user is logging, and it will basically have a like token injected in our place in the response, right? I it. So now that's our, you know, like, where our primary flow comes into play. So I kind of, you know, like, want to back everything behind an API gateway, right? So this box here will be an, let's say API gateway. And if I get, like, if I get API gateway service as a, you know, like green orchestrator, right? So, you know, for connecting, let's say multiple services for now, for here, yeah. So this will be an API gateway

Unknown Speaker 10:46

that, like that clients will call now,

Speaker 2 10:51

and we have our, you know, like primary user service as well, so which manages user profile, email data, Right so that I want to work with first i

Speaker 2 11:23

So behind the, you know, like user service, like we will have all of the accounts, right? Like service, contact service, like transaction service and so on, right? So this, you know, like Four services that we want you.

Speaker 2 12:03

So we want the of four primary service will be like this for I just mentioned.

Speaker 2 12:22

I so and basically, like, we will maybe, like a payment gateway here that's getting invoked here, so we can come back to this later, right? But I want to, you know, like, talk more about the database first.

Speaker 1 13:02

Yeah, before you get into the database, can you just give me sort of a high level overview on so like account service and contact service. I think those are concepts familiar to this. But then when there's balance management service and audit service, what do you see those two used for sure, I sure. So first

Speaker 2 13:32

what I'm thinking about these two extra service I it. So, like, each year balance management service will probably, you know, like, act as a external service, right? That maintains the, let's say, real time user balances and basically, like, enforce, you know, like, any kind of limits, right? So let's say this would be like an extra validation six here. So that will be like a, you know, like real time validations and updates, you know, and based on, like, you know, like David's or credits as needed, it just, you know, something that makes your reinforcement and management, like, a little bit better, right? So instead of, you know, like encapsulating that within your account, right? So the you know, like, the audit service is basically for keeping, you know, like, track of all the actions that, let's say, the user has, so and you have external service leaders as well. So who's, you know, like, let's say sole responsibility is to, you know, like, keep track of those, let's say transactions and logs and like, you know, like, everything that the users, let's say, try to make, right? So it's, it's kind of like an append only service. So which is, you know, like, stored in some kind of databases, or, like, let's say a log stores as well.

Unknown Speaker 15:07

Yeah, love it.

Unknown Speaker 15:09

Yeah, thanks. We can keep going.

Speaker 2 15:20

So let's think about the database over here, database side first. So what we want to include, and, you know, like, what tables we want to include. So I will make, I make a bigger box for that. So I think, you know relational databases here is easy because, you know, we, you, we need, like, essay, like automation, consistency and like ability to have an audit, right? So we want that, even though, like, it's a little bit difficult to scale, because it's, it's vertically scaling, right? But we have, like, a master database that replicas for performance, right? So we need, we basically need to have that right. So we will do something like, let's say AWS RDS PostgreSQL. RDS PostgreSQL of AWS. So there will be, there will be, probably some tables. So the first table I'm thinking of is like user table, right? So user will have, let's say ID, let's say a name, and let's say an email or phone number as well,

Speaker 2 16:51

and we'll have a contacts, so this will probably have an ID,

Unknown Speaker 16:59

so this will be

Unknown Speaker 17:02

user ID.

Unknown Speaker 17:04

It will have, you know, like,

Unknown Speaker 17:07

context ID as well, and

Speaker 2 17:13

maybe create it like, like. So the metadata, metadata will be here, right, but I am not writing it. Metadata, like creating by and updating it as well.

Unknown Speaker 17:26

Oh yeah, yeah for sure. So

Speaker 2 17:28

this will be our context table,

Unknown Speaker 17:33

and we will have

Speaker 2 17:36

balance table as well. And the balance, you know, like contains user ID,

Unknown Speaker 17:53

and then we have

Unknown Speaker 17:57

our open only transaction, right?

Unknown Speaker 18:02

So we have open only transaction,

Unknown Speaker 18:06

so transaction table,

Speaker 2 18:11

so we'll have, you know, like kind of form to amount. So it will be these fields. So we'll have an, you know, like ID will have, you know, like form to amount status, let's say amount type as well, and so on.

Speaker 2 18:39

Yeah, I think this table will be the tables that we initially need for the design to be initially work on.

Speaker 2 18:52

No, yeah. Now let's go back and let's see how this, you know, like service, interconnect with each other.

Speaker 2 19:16

So basically, you know, like amount service, like it's responsible for account management, right?

Speaker 2 19:28

So let me create, you know, like each of those services here.

Speaker 2 19:42

So we'll basically have a, you know, like post request to create,

Unknown Speaker 19:50

to create a, let's say a request first.

Speaker 2 19:56

So this will be a you know, like a request to certain amount. Will have.

Speaker 2 20:13

So so you have to ID. You have, you know, like amount descriptions. You so

Speaker 2 20:27

this will be a request, you know, like a money and then you can, you Need, actually, in points to send and fulfill, right? I

Speaker 2 21:00

it. So you know, like, basically whenever you request, you create a log into the transaction table, right? And that transaction will be fulfilled using the fulfill in points. So, you know, like that will be responsibility. That will be your responsibility of an account first here, account service, let's say, yeah, so let's go to our contact service here.

Speaker 2 21:53

So this is, you know, like just for searching and adding. So we will have to get in points to look for.

Speaker 2 22:07

So we'll have a posting query here as well. So basically, to create a contact right, and then we will have a gate call contact as well for that, get con get call contact by ID, let's say,

Speaker 2 22:26

so that would be our contact service, very simple now, And the balance services will also probably be simple. I

Speaker 2 22:43

so we'll have a, you know, like kind of a get balance for user ID. So

Speaker 2 23:03

it, and we and we'll have a like force balance, and then we will have a, we call it like a post graded as well over here.

Unknown Speaker 23:15

So this, you know, like doing points

Speaker 2 23:18

will be used to increase and let's decrease the balance as needed, right? So this will be a part of our transaction controller within the account service. So whenever, you know, like there is a request being sent from the fulfillment or sending the money, this will kick into the play, right? And then the transaction log service will basically just be, you know, like saving some kind of, you know, like transaction record.

Unknown Speaker 24:02

I so those

Unknown Speaker 24:12

primary inputs that we are working

Speaker 2 24:18

so I'm drawing a line here, you know, because that will be a, let's say, a Load Balancers before each of those services, right? Yeah. I

Unknown Speaker 24:46

BBB. So now let's see.

Speaker 2 24:58

We do need other services in place here, like, let's say actual notification service or the notification demand. So I think those will basically be part of the API gateway, right?

Speaker 1 25:15

Yes. So I mean, however, we do want to see, like, emails and maybe even push notifications to users. So however that ends up happening is up to you.

Speaker 2 25:29

So I think this will be a trigger from an API gateway as part of the you know, like actual processing, right? So now, basically this is what I'm thinking right now. So we have a user service, so this basically acts as a, you know, central point, right for managing all those

Unknown Speaker 26:00

so it's a user profile, right? So it's,

Unknown Speaker 26:06

it's a user profile access, right? So

Speaker 2 26:11

it's kind of a like a gateway or service, right? So instead of, let's say, duplicating users logic in the service, you centralize it in the user service here. So from here, you know like you will basically be calling each of those individual services,

Speaker 2 26:38

and then for actual so we need the payment gateway because we are handling a balance ourselves, right? So let's get rid of that for now.

Speaker 2 26:56

So all of these, you know, like, basically, will be going to the PostgreSQL, right? So we'll have some kind of caching between the user and the context table, let's say

Speaker 2 27:14

so for the higher availability and, you know, like kind of reducing the database.

Speaker 2 27:29

So let's walk through a flow where, you know, like it's a user requested money and then someone fulfills, right?

Speaker 2 27:47

So basically, you know, they do their regular identity providers like, right? So check where, where you check if the user is authenticated or not. So that flows, goes there. Now let's say user a request $20 from user B, right? So for that, it will, basically will go to the API gateway, so it will call that to the user service, right? It will call the account service and contact service. So for, you know, like basic validation. And then it will call the audit service to, basically, you know, like insert a transaction, or, let's say, request as well. Now, basically, you know, like B now wants to fulfill that money request. So we will, you know, like, initiate that particular fulfillment, right? So they it will call to do a fulfillment request endpoint for now, so from the account service, right? So it will be, it will basically, you know, like fits, like the data from the object service, making sure that the you know, status is, let's say, pending, and the from users and to the users, it's actually what it says. So basically, make sure that the balance management service return, or, you know, sufficient balance, right then it will call.

Speaker 1 29:29

So I am, I am taking notes. Whenever I'm silent, I'm like, legit, just typing, typing, typing, just say no,

Speaker 2 29:38

continue. Yeah. So this balance, balance management is the, you know, like, actual money, money service. So basically it will call the, you know, post David and post credit endpoints on, you know, like automatically, let's say, to make sure that the transactions take place at a

Unknown Speaker 29:58

single unit. I BBB,

Speaker 2 30:15

so that if everything is successful, you know, like, then we go to the notification daemon from the API gateway right, and send our, let's say, even, notification, and that will be end of our, like, whole flow. So, but of course, like, you know, like you will have, you know, like individual residency and all those sort in place to make sure that we don't have failure that breaking so that break down the entire system here. Great.

Speaker 1 30:54

Okay, yeah, so then I think as far as the notification Damon goes, you know, seeing, like, you know, things be triggered from the API gateway as the request is returned, right? That's not something that I do see that often, actually. So can you explain more about what guided that decision?

Speaker 2 31:15

Um, yeah. So for that, you know, like I am thinking, what I'm thinking is, you know, API gateway will be, you know, like, orchestrating those calls, right, and waiting for the response back. Because,

Unknown Speaker 31:35

you know, like, but like you mentioned it,

Speaker 2 31:38

it should not be, let's say as it should be. It shouldn't be like a like client facing service, right? So it should basically send, you know, notifications once there is some event, right? So maybe whenever there is, let's say, any kind of transactions, lock chains, right? And any change in the queue or like a process, and it should be part of that, right? So I think it's a better decision to move it here.

Speaker 2 32:16

So let's move that. So whenever there is an, let's say, entry on the transaction, stable, you know, like to the object service, so we trigger a notification demon, let's say, so some so some process there. Now, like we do that the API, with API gateway, becomes the, you know, like it's it becomes the user service, Right? So we don't need This extra service here. You

Unknown Speaker 33:40

Yeah. So, yeah, I think

Unknown Speaker 33:50

those will be a

Speaker 2 33:52

little bit refined system. So you know that you don't rely on the API gateway or the time facing app, right? So wait for the response back. So in case of, let's say, any event, or on the audit service, so it will trigger maybe a queue based or some kind of, like, pops up model where, you know, like, you will send an email notification up, right? So

Speaker 1 34:23

great. Okay, yeah. So let's talk more about, like, scaling techniques. You know, at a company like PayPal, obviously, scaling is, like, insanely important, yeah. So why don't we run through, just like, for example, the database, you know, how, what kind of techniques would you use in designing the database or even choosing, you know, the type of database that you chose?

Speaker 2 34:52

Yes, so I, you know, like, choose SQL database, because, like, there's, you know, like high need of consistency here is because it balances transfers, right? So basically, we need to make sure all the transactions are atomic and, you know, like it's reflected in all the instances right away. Let's say so, that's why we need relational database here now for the scaling, basic scaling part, like we will have particle scaling here, you know, because that's what relational database has to be. It cannot be horizontally scaled, right? But I think that what I think we need to do is like, basically, we need to make sure we have, let's say, a proper indexes in the database, right? So we'll have a kind of, you know, like proper query optimizations for, let's say, a proper performance, right? But apart from that, like, we'll have a read write replica as well. So we will have a primary, which is about a Read and Write, write, and then we'll, we'll have a replica, which is just a read only, and then our core database will handle the writes. But our replicas will, you know, like read for the it will be used for, let's say, some kind of read operations, right? So we might do partitioning as well. So whenever, let's say there are lots of rights, right? So we probably want to do some kind, some kind of, let's say, or horizontal partisan based on, maybe, let's say user base or other strategy, right? But that might come into play once we start, you know, like hitting a large number of users, right? Let's say so that is, you know, like something we can add up, like, you know, like trading and like partitioning the database for, like, higher scalability,

Speaker 1 36:52

awesome, cool. And then as far as, like the services as well, even like, what technologies would you use to, you know, deploy and manage these services. So

Speaker 2 37:07

I think of like, you know, all those services will be simple, restful based, micro service, like returning, you know, that was returned in, you know, Java, spring, boot, your API gateway can be just AWS API gateway, right? So that's what I'm thinking of right now, because I don't see, you know, like lambdas, except for the notification daemon over here. So I think the notification daemons will be SNS, SQS, with the, you know, like lambda trigger.

Speaker 1 37:47

That sounds great. I'm trying to think what other questions I could have.

Speaker 1 37:57

Actually, I would, you know, this is a good time actually, to circle back to the multiple currencies question that you have to start. Let's say we're ready to go global and we want to start supporting other countries MONEY. How would you update this design to handle that?

Speaker 2 38:21

So let me think about the currency conversion here. So basically, you know, like, we have this cross region, right complexity, then, like, we have a currency conversion rate, which might be just an, you know, like extra service, right? That will be

Unknown Speaker 38:48

incorporated, you know, like, for

Speaker 2 38:52

proper conversion rate, let's say so we will, you know, like, probably use a currency code in the balance table, right? So to make sure that, you know, like, we are storing the currency in the, let's say, correct value, right? So we will, then, you know, like, incorporate that currency exchange service within our system to, basically, you know, do a conversations, right? Say, then we will, probably will just store it like a global sense, amount, like, let's say dollar one, amount in our dollar tables. Let's say, but everything else will be done on, so, everything else other, let's say it will be done on the basis of, let's say, calculation basis. So it will be, you know, like adding a flag on the, you know, like balance table, like, what currency, let's say what currency it is, what will be the currency exchange services. So then, you know, like, we will be kind of using our currency exchange service, right, to keep track of the, you know, conversion rate. Then we will basically do a conversion on the fly testing.

Speaker 1 40:09

Okay, all right, let's discuss a little bit about security. Now, I do remember seeing you know you talked a bit about OWASP in your resume, at the very least. In general, when it comes to this design that we put forward, where do you think are,

Unknown Speaker 40:26

you know, the important points to have good security controls. And

Unknown Speaker 40:35

so what

Unknown Speaker 40:41

I'm thinking of here is I think

Speaker 2 40:44

all the you know, services will be individually secured, right? So all the services need to have a spring security enabled on it. So you would have a, you know, like spring security dependency injected. So you will probably will have a, you know, kind of, let's say, identity providers, right, that will take care of all the client authentications. But on the API gateway, you know, like, you can add a like, authorizer lambda, let's say, to make sure that the user request is valid. And once, you know, like that is done. You basically on the individual, on your individual services, you know, validate that token. Where is it coming from, all the requests. And apart from these, like authentications, authorizations, like, we basically need to make sure that data is protected, right. So we have, you know, like, input validation in place as well. So we need to make sure there is a, you know, like, proper transactional integrity, right, and so on. So for each of the services,

Unknown Speaker 41:55

yeah, that sounds great,

Speaker 1 41:59

okay, um, yeah. I mean, I don't think I have any other direct questions here. We do have more time than usual, so if you've got other things you'd like to add to this design and clean up, then now's a good time to do it. And then, otherwise, we'll just move on to the next section, where we can talk about PayPal. Um,

Speaker 2 42:25

so let me go through it like one last time, and then maybe, like, we can talk for the next Yeah, we can talk about next sections. So let's see here. I

Unknown Speaker 42:53

Oh, yeah, like,

Unknown Speaker 42:55

I don't think like any changes that I want to add here for now. I

Speaker 2 43:08

But here, I think, yeah, like, I think we can over here. I

Unknown Speaker 43:26

Yeah. So you think you're all set there,

Speaker 2 43:27

yeah? Like, I, I think that's all. I don't think that. Like, I have any

Speaker 1 43:34

cool yeah, then, you know, let's go ahead and move on. Yeah, I'm happy. So, so let's do it, all right, I'm gonna lock it shifted to production, okay, yeah, so, you know, I did realize I forgot to introduce myself at the start. I will apologize. I came back from a vacation in Mexico less than 24 hours ago, still got a bit of beach brain going on, right? Yeah, but just, you know, getting back to the swing of things, like I mentioned, I've been working at Venmo for five years now. Before Venmo, I did a ton of work across various fields, mostly in back end or some full stack. So I was working in like, DevOps solutions, cybersecurity, company, sales, training, software, cryptocurrency for a minute, which I'm not the proudest of, but and then Venmo, which has been pretty amazing. Yeah, this section is for you to ask any questions about me or my job PayPal in general, as you have only question I can't answer is, how did I do on the interview? Not allowed, but everything else is fair game.

Speaker 2 44:52

Yeah. So like, thank you so much for your time. I'd like to say I would like to know, like, what do you like most about working with PayPal?

Speaker 1 45:05

Yeah, so, I mean, it's a ton of like, I don't difficult problems going on that have been really interesting to solve. I gotta say, some of it is difficult in the wrong way, because PayPal is huge. So with a huge company, becomes a huge code base, and then things just start to, you know, venmos over a decade old, and you're starting to see its age just in various parts of the system. But even then, it is satisfying to sometimes crack open code that nobody's touched in like, four years and understands anymore, and did, like, get something new out of it?

Unknown Speaker 45:40

Yeah, my current project, I

Speaker 1 45:45

I'm not sure how much I can say about it, but we basically, we've got an important feature that we already offer. But then this company that we work with for it came to us said we are shutting down in four months. So good luck. So I've actually, yeah, as me and my teammates, we got together, and we basically, we had to build the entire thing from scratch. We're going with an in house solution, and we've got like six to eight other teams, like, all involved together, working on just cranking this thing out together. It's been one of the, like, most interesting, like, you know, oftentimes stressful but ultimately, like, satisfying projects that I've ever worked on, we're actually, we're trying to do our prod shake out, like, right now, like during this interview, we've got other people giving it its first like, run some productions. So, you know, we're made some good progress on it.

Speaker 2 46:40

Okay? So like, thank you so much for that information. So it was already insightful. So like, what do you what do you like to say something that, like, new joining at PayPal should be careful about, or, let's say, know about, be prepared about. So if you have any suggestions, yeah,

Speaker 1 46:58

yeah, yeah. Just, you know, just take a breath. Onboarding is extremely difficult. I will say, I always feel so bad for all the new employees, because it's for the stupidest things. It's like you have to use this one tool and put in a bunch of like, random strings into a form field to get the right access to stuff. It's maybe documented properly across like, four different documents, you know, it's you know, it's rough, and it's, you know, become this way because PayPal is also going through its own growing pains. We're constantly, like, updating our systems or migrating to new things. But then, you know, documentation starts to fall out of date, like, knowledge becomes a little more tribal. So, yeah, I've seen, especially, like with really junior employees, like fresh out of college, they come in and they start to, like, panic a bit, because it's completely overwhelming. But everybody knows, we've all gone through it. We see how it, you know, has grown over time. So my advice is, just be aware of that up front, take a breath and you'll get through it.

Speaker 2 48:11

Yeah, that's a good advice, yeah. So I think that's all the voices I have for now.

Speaker 1 48:25

Okay, yeah, I mean I'm good to go on my end. Do you have more interviews scheduled?

Speaker 2 48:32

Um, with PayPal, yeah, yeah. So I have two more.

Speaker 1 48:39

Okay, cool, yes. Then I will give the, you know, the whole spiel about follow up stuff. I'll just wish you luck at your next interviews. And yeah, and again, thank you so much. It's been a pleasure meeting you today's month, yeah, for me

Speaker 2 48:57

as well. Like, it was really nice talking to you. Like, thank you so much for your time.

Unknown Speaker 49:01

Yeah, absolutely, all right, you have a good day.

Unknown Speaker 49:04

You guys will take care. Bye,

Unknown Speaker 49:18

That's our last cast around him.

Unknown Speaker 51:00

On This particular nightmare,

Unknown Speaker 57:05

Nice, You. I'm Talking

Unknown Speaker 1:03:02

About what's up So

Unknown Speaker 1:05:20

what Is This?

Unknown Speaker 1:06:00

I have A Question.

Unknown Speaker 1:06:04

Thank High.

Suman Astani- Vertexinc-Second-SRV

Speaker 1 00:00

Are, you know, like new challenges. So I want to read the way to success. So help different team members work in teams, so as I like, improve myself, making myself better version of myself, right? So that's kind of what I'm looking for. If that's answered your question, yeah, absolutely. You mentioned kind of helping team members and stuff, and that, it's a great segue into my next question of, how would you manage it, or how do you approach that? So, how do i So, basically, like, what I like to do is, like, I like making myself very clear, and like, I do believe in, you know, like, clear communications, right? So that's the key. I think. So like communicating, you know, like, properly with teammates, and make sure, like, it's you know, like it's efficient. So if there is, you know, like, any you know, like, confusions, like, I want to make sure, like, it's cleared up and like, are you in the process, you know, like, rather than, rather than later, realizing that something was wrong. So communication is the key. You know, like, I do get, like, involved in a lot of conversations with them, and make sure, you know that I come the active listener. And while I'm communicating, I listen a lot like I speak less. So that's kind of my approach, is, like making sure I listen to my teammates, along with, like, providing, you know, feedbacks and also keeping myself open for, you know, any kind of, any constructive criticism. So that's that's it awesome. Is there a time in recent memory, maybe, where you helped out a teammate that was struggling on something, and could you kind of walk through what that was? Yeah, if I remember correctly, in my previous project, like, I'm sorry. Like, so what's your what's the I couldn't hear, like, what's the same? What's the poison, yeah, so just maybe an example of a time that you helped someone with something and what, what they were struggling with. Okay? So basically, while I was working in my previous project, right? So I had a scenario like, I do, I do have to request help from my teammates. And, you know, like we have to, we have to let courses and we have to have a there was a team members who was freshly hired in college, like he was a fresh graduate, right? So he started as a role as an internship in my recent previous project. So we used to get actively involved in a project, like, right, like that the team is currently working on. So basically, you know, the team was like, kind of, you know, struggling to, you know, like understanding and understand the basic of how, like, spring profiling work and how, you know, like, you can set off, like, different configuration on application property file, like, see what it was being No. So it was a little difficult, in a sense that, like, it was not straight off, like, changing in like application properties. So I do get involved in, like, the dynamic configuration server that you know, like phases a conflict from the GitHub repository during runtime, and like, replaced what's backing into your applications, right? So I was helping him and like, how to properly set up the dynamic configuration bin. And, you know, then, like, wearing those properties in a failed way. Save that if that like in the property did not exit like, return, default in a certain value. So that was, you know, something that I helped him. So I remember, like, I remember setting up a call I walked into how things work on and like, overflow of how the different mechanism works over there, how things are working. And like, setting up initially, and like, we're able to resolve all these misunderstanding of, like, how it works. So that's, that's how I resolve in my recent project. Very cool, awesome. What does typical day look like in your previous role? Like, day to day, what would it look like? Okay, so day to day, I will say, like, I will mostly be involved in about, like, you know, four to five hours of, like, hands down development time, right? So I'll be, like, actually coding, right? So for about four to five hours, right? And I'll be actually coding four to five hours a day, and the rest of the my times are, like, usually involved in meetings, you know, like some kind of, like, very strong incisions, or, like peer programming sessions as well. And you know, like, I do get involved in like releases as well. So, like, apart from coding like, I'm either doing like different code reviews for my team members, like, you know, like pull requests, like, I'm either involved in any kind of like, as other ceremonies, like spring planning, like grooming retrospective as well, or like, you know, like, like the black, black love requirement, right? So I'm involved in that kind of releases process, like, whether running the releases or, like, validating the releases as well. So that's what, how my day to day goes as a software awesome. So you mentioned validating, go for it real quick before there. That's a good segue there, too. But you actually just mentioned you sort of review code, right? You're reviewing pull requests and all the other things. Just curious. What are you looking for when you do a PR review? Okay, that's kind of interesting. Like, okay, so the so basically, like, whenever, you know, like, I am really reviewing a pull request, right? So, like, I look for a couple of things, right? So you know, like, I need sure the code is, like, readable and, like, it's consistent, right? So basically, like, if there are different methods and there are different variables that are named in like, in a confusing way, or like, it's a non readable way, like, I asked the teammate to change it so that like, it's like, more understandable and like, it's consistent, right? So if like, even the functionality is correct, it might like have like, different certain kind of approaches that the company, or let's say the team, takes for managing the code base, like how much the, let's say, methods are supposed to be and like, how complex it can be, right? So I'll also, in addition to this, like, I will also look at for any kind of validation, right, that might not have been done or that have not been checked. And I if that's if I need to check it personally, I'll check it as well. So I'll also look any kind of like. I also look, make sure, you know, there are different happy path like. I'll also look there are different kinds of documentation, if the developers is writing or README file documentation as needed and so on, right? So that's the pretty much so to add something more on this, like, you have to, like, make a static sense of code coverage and different box using, like, sonar, right? But, you know, like, I'll also look at these extra things, you know, to make they are really proper. They are written readily, properly. So that's what I look into the request of teaching. Awesome. Thank you, awesome. So you mentioned that you look for, you know, some readability stuff. In addition to, you know, even if the code's correct, it might not be formatted correctly. How do you know if it's correct? Though, okay, so I look at the like unit testing, right, that the developers might have written, right? So if they have unit testing, I look at it that. So basically, your unit is, you know, so as a documentation of a change, or, let's say, the features as well. So I'll be looking at the different test cases the developer have region, so how, in, how it's named and, like, what's the assertions are like? So if you know that's, that's my go to the that's my go to like, that's what I look after, right? So I will look at the unit test right for the functionality, and I will also try to make sure that's what the expected result is. Awesome, cool. So earlier you mentioned, you know some of the release validation stuff that you are also a part of. So how do you know production or software is ready for production or it's ready for release? Okay, so I so basically, like, we do follow different, like approaches, or different, you know, like religious plan, right? So basically, firstly, what I like to do is like, I would like, I would like to deploy the changes on the deep environment where we as a developers, can do so our like validations and different sanity tests. Then once you know, all the developers are done with their validation. So we moved to the QA environment, right where our, you know, like testers can run their test cases, and they can even run their regression testings as well. And also, you know, like integrated different backgrounds to run their tests if needed. So if it's effective, we need to make sure that you know that the new changes do not you know the big existing changes to that different recession cycles. And once that is done, we stage these religious brands, we boost these release brands, and we push the release there as well. So like, we do different kinds of like, like, we do different kinds of like products and traffic, you know, like, we do different kinds of things on this as well. So yeah, so actually, that's our kind of release strategy that as a developer, we can approach, and that's kind of how we know the score is ready. Ready for, you know, productions and like, for production services as well. Very cool. You mentioned, when you push to dev, you know, developers can come in and run their tests and any sanity tests that they might have. What do those look like? And how do you run those? Okay, so, so, basically, like the said, just for the dev artist, like, just for the dev are usually, you know, how it works is, like you, you have a manual test, right? So they just, if you are writing some manual so this business should be postman request. So, like, if these, this is integrated, you know, with the front end, then you can do their you can do their front end validations, and if the, you know, requests are good, and so on. But for the developers, you know, usually the testing is like manual testing. However, in the pipeline, if you can different pipeline, right, you do have different accepting, acceptance testing, where the first is, like unit test that gets run during the state build stage right. But there is also like acceptance right, where it's, you know, like just the functionality and the integrations to the environment so it asset and like test into the pipelines that runs, you know, against the, you know, like the data environment. Gotcha. So those, those pipelines, I see you said you mentioned, let's see you implemented some end to end automation using Jenkins, and it significantly enhanced, you know, develop delivery velocity and maintained the quality of software. Can you talk about, you know, what are some of the strategies you used for those Jenkins pipelines, and what some of the quality gates might have been? Yes, sure. So, yeah. So basically, like, what in my previous project? So we have, you know, the integrated, if I remember correctly, what integrated text login in like, in each developers IntelliJ, but also, you know, as a part of a build stage. So whenever you know, like, you are checking your code, and it, you know, like it's, let's say it runs that check. It's the check starts that's comply successfully. And if, like, if that's if it's not, let's say a property formatted right, then it's failed. The pipeline fails, right? So that's the first step. And there is also a scenario gate where, like it does, a static scan of your different artist and code, so sonar can you know flag for any bugs in your code. And also, like the different improv improper code that you have returned for, you know, import for the discoveries, and also too many, if there are too many state if statement, let's say so. These are, these are checked to Sonas as well. So our, you know, like dress strategy is like 80% cocketing, 50% coverage on the both line and branch, you know. And that's one. And then there is other check. Is like 45 scan, you know. So it's a scan for our vulnerability, right on your dependency that you might have included in your project. So these are three checks that we have in our pipeline, here, in our imagination project, very cool. And how did that increase velocity? Okay, so, so, how did I increase

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so, increase velocity, right? So you mean increased velocity,

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right? Yeah, you said in the in your resume that that significantly hence the delivery license. Yeah, so how it works is, like, I have, let me recall, like, how I have did that. So basically, you know, like, without these quality gates, our without this quality gates, our applications, you know what really prone to box and defect, right? So, coming out, you know, even the setup of the pipeline and you are writing the brain test actually takes a little a bit longer, right? But once it has been integrated into the delivery of the proper code and a proper application become very fast, right? Because you can catch those failures as well. So are you like already, as in the process, rather than have like those defective products out of the box, and then, you know, having to come up with a box fix, or a box fix, or a hot fix. So that's how I actually walk out that company. Sounds good. Very cool. So I know we kind of not to suddenly shift gears here real quick. But you know, we talked about your front end and some of the back end experience here too. Just curious, which one you prefer? Do you prefer front end development or back end development? Or I'll do like being full stack Okay, so as a full stack developer, like, I have to be good and good, but I prefer backing. Guys I have been primarily working on as a back end developer, so I like writing back on the back end logic. So I say I prefer backing, yeah, yeah. So same, I would say the same thing, but I started as a front end developer, but there's just a different set of problems that you're solving, right? And speaking of solving problems, I kind of have a, not a again, it's a little bit of a jarring segue here, but could you talk to me about, like, the most difficult technical problem in recent memory that you have, that you've encountered, and then tell me how you solved it. Okay, sure. So the recent problem that I have and how I solved it, let me Okay, so, yeah. So to talk about that in my project previously, while I was working on the, you know, most difficult problem that I saw recently, I will have to say, like, it's like optimizing the performance of our Spring Cloud gateway based application. So that's the main problem that I have solved. So the way that I solve is, like Spring Cloud gateway works is, you know, you have to route that. You have to design and you and it works as an orchestrator for those routes as well, right? So, you know, like, it does not work like your typical microservice API, but it acts as a gateway, like it manipulates your, you know, require request, and then it manipulates your, you know, like responses based on the meta that you provide in so, you know, like, it is a really reciting framework, but, you know, it relies on resilience for a sarca like breaking and has, you know, a lot of support for extensive as well, you know, but the fact that you have to be treat safe and also, you know, integrate with the overall route mechanisms, while, you know, something that can usually have an issue on with the main performance, because, you know, there could be different leagues, let's say of your client connection pool, right? So any kind of not pointers that get, like a sellout. So that's, I remember, you know, last year's we had, last year's, we had an issue in proximity with one of our deployment where, for about you a week, like for about a week, our applications will be good, but all of a sudden all the requests will start failing, and the whole containers became unreducible. So, right? So we'll have a huge number of 500 being collected on the orchestrator, and, like, after it hit at a sorted point, you know, like, it will always be, always been failing, right? So that was a really difficult challenge for me, I would say, because, you know, like, we couldn't do anything off the logs. We couldn't, like, replicate it into our, you know, lower environment. But the one thing that we found out later was that there are different things that I found later was a bit of, I would say the debugging was that that we were basically, like shallowing, and there was like, exception being thrown when we were grabbing that, you know, dynamic configuration that I talked about earlier. And there was also a knob pointer exceptions as well, right? So during one of the, you know, like the functions that was being salad, which are held onto the HTTP client connection from the, you know, connection pool. And then the connection pool was, you know, getting exhausted, right? So that was, you know, one of the most challenging ones that I had faced and spent. Or I think, I think, if I remember it correctly, like I have spent around a week to figure out this, there was no information, information on the logs and, like, also nothing on the like, three terms as well, and keep terms as well. So that helped me either. So yeah, that's one of the most challenging problems that I figure and like, resolve it as well. So I did help, take help from my some of from my teammates, someone from my mentors as well, to help me, like, clear some of the confusions and that I have, and also write some performance script as well, you know, so, but yeah, so it was a really eye opening scenario for me that the framework, you know, lost some of the Euros lost in like derog labels, rather than like euro dollar. So that's how I solve the problems and the challenges that I have faced. Like, if it sounds clear, I it? Yeah, I mean, that's kind of like a worst case there, right? You have 500 small exceptions. You can't find problems, and then you end up finding npes and some obviously the connection pool exhaustion is going to be a big one for for you there. So definitely, for a week, I was struggling on that problem, especially when you start dealing with the web flux stuff and like, you know, springs taking you down like six different classes. And just like, I don't know what just happened, so the solution, remind me again, kind of summarizing for me, it was increasing connection pool counts better error handling, making sure that the dynamic configuration was not breaking on startup, right? Because you were just kind of getting overrun by exceptions, right? Awesome. Looking back at that now, though a couple months or however long it's been, would you do anything differently? Would you change how you tried to identify the problem? Would you change what you ended up doing to fix it? How would, what would you do differently if you could, okay, so, so, basically, you know, I think the only thing that I would have done differently was to try to get access to the, you know, instance, and try to SSL into the box so throughout, like tunnels, and then, you know, see if I could find, you know, like anything within the instant itself. Because, you know, you can run some of your niche staff and like and processes commands as well on denox box to, you know, find what the real connection coding will have look like. So that's what I think I have done differently at the end, and what the connections are enclosed like, you know, like what the what the connections are enclosed in a waiting stage. So that's something I would say there are something preventive measures, or the preventive approach that I should have actually, you know, included into the connection pool, like informations on our health checks, influence that we use. So I think these are the two things that I will have changed if I look back into Yeah, spend less time spinning the wheels, right? I'm just curious as well, you kind of said something in passing, in sort of your previous answer, but you were only encountering listening to lower environments, right? Not in production, yeah? So that, that is a profound effect that we, you know, found you Okay, cool. And then so there was no you had the exceptions for being swallowed, yeah, no meaningful logging. And then you have the SSD would like to have SSH into the box running it. So I wonder what was actually running this application. I'm just curious, like the infrastructure that you were deploying on. Okay, so, yeah, that was easy to, like, easy base. Like it was easy to it was on easy to

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legacy. Yeah, so that was like it

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was this an existing, established application. Why did the team choose to deploy on the EC two versus something like fargate, Kubernetes or whatever else, right? Deploying container versus okay? So we choose that. So our new modernization processes was going through like, you know, AWS, CCS on like Target as well, but this was our legacy applications, right? That was still taking different traffic. So we only have moved like sorted component of it into like this, new Microsoft offices. So do curious. It's always fun. You see, easy to use living for a long time, we have a lot of EC twos here, and a lot of fargate and a lot of Kubernetes. So it's, it's always a fun one to, you know, get to the answer why? So I think you kind of got to a little bit, but I want to explore a little further. There's a bug randomly in the application, could whether it's been deployed to dev or somewhere else, but there's a bug. Let's say I'm getting a 500 every time I send a certain request. How do you with this lesson learned all the other stuff, but how do you go about troubleshooting it now? Okay, so, debugging, right? Okay, so, so I would say that first, say that first, I will say that first you go to, always, to the logs, right? So it's, it's an like application log, I will try to see what's happening and if it's throwing an exception that I can catch in order to find the root cause of the issue. So that's my that will be, like, my first approach, like, if possible, we can use spring AOP to actually intercept the method calls and logs as well. So the method parameters, you know, which helps a lot and unlocks the method parameters as well. So, because, like, but if it's not, it's if it's not, it's not a big deal. That's the something, that's something that I would like to do, like, look into the logs and, you know, try to follow the through the flow of people. That's how it's going. And the next step will be, like, actually, you know, to recreate it on my machine. So if I can, like, recreate it locally, like, it's great, right? If not, like, I will just be using, you know, like postman to hit the environment, right? And try to see what flows its triggers and like, if you know, like, I'm able to actually recreate this. Like, it becomes a lot of easier, right? So that I can, you know, step through its blocks of code, you know, and try to figure out what, what the point of failure is. So that's kind of my approach on finding defects and like, resulting, oh, it always goes perfect, like seeing that, you know, especially spring op and receptor adding method call, then seeing the entry points and being able to follow that stack trace, which kind of brings me to something that you mentioned on your resume, where you talked about log correlation. I'm curious, how did you correlate your logs, and what sort of observability tools or pipeline did you have in place for your applications? Okay, so I'm being white. I've been a little costly today. Sorry about that. No worries. So basically, you know, like our applications logs are, you know, all delivered to Splunk through. And like, Splunk forwarding, as in, so the spawn sprung forward is, like, installed on our, you know, easy to distances, right then, like, it basically listens to our, you know, log files and then stream it down to a certain index, right? And what our, you know, application does is like it adds our message ID to the logs, right? So once our you know, request enters our services, like it's, it's the orchestration, like that acid that add messages ID letters, right? So that that this message ID becomes like, unique, you know, to the particular request, and like, it gets easy to track the flow of, you know, like, how the request is being right, so how the request is flowing from one part to the end, right? So that's what I did. And you know, like, there are unique identifiers, like the client correlation ID, you know, like, which is, like, unique session, so there is a unique, unique to the users. But you know, the message ID headers is, you know, like, actually, what drives the uniqueness of one particular request going from, you know, one systems to others. So, awesome. Very cool. All right, so let's talk a little bit about how you approach an issue when you find it. But how do you find an issue? How do you know there's something wrong when something wrong with a service? Okay, so, um, the No, the way, I know how it's wrong with the service. Like, usually during the, you know, like sanity testing, like if something is off right, or like something is, let's say, taking longer than a normal time, and then that is a little bit of indication that something is wrong, right? So sometimes I do look at the others that we have in place for, you know, like increased numbers of 500 or, like 400 levels code, but, you know, like, but you know, like, usually a lot of integrated partners are the one, you know, like postings on our channels saying that they are experiencing and, you know, anomaly on their as expected behaviors and it's not working correctly. So yeah, then, like, these are the three things that I look into for, like, finding out any issues. Okay, what kind of you mentioned alarms, curious, what kind of metrics and stuff were you monitoring for? Alarms? Like, what were you curious about? So, so, basically, you know, like, basically our dashboard, you know, captures the dashboard captures a lot of things, so it tracks the response code, right? So there are certain amount of, you know, 500 500 500 500 500 and like, 400 euro codes, right? That are acceptable, right? But once it deviates from that initial matrix. You know, we find it alarming, and we have to find, you have to, you have to, like, kind of, look into what's failing, right? So we have to do track of the latency of the cause, like how long the request takes, and, like, how long it deviates, right? It again, there is another reason to devote the systems, like we also, you know, keep track of, you know, memory uses, like CPU uses, you know, like we recently added the trade collections, like collections to see how many trips are in use. Like, how many trips are, you know, waiting, and with how many trips are in state and so on. So, yeah, basically, these are the matrices, you know, that we have been incorporating on our dashboard. Dashboard, nice. And were you responsible for making those alarms and making the dashboards? Um, I would say that. I yeah, I am responsible for making the dashboard, you know, but I do add something, but the alarms were set up by someone else. Gotcha. Okay, so in terms of, like, maybe monitoring, you know, Heap or memory usage, or CPU usage, how do you set the thresholds? Like, how do you know how much CPU or memory is too much? Um, so basically, like, you know, it did take a lot of trial and error. And then we had to do like, you know, like we have to extract data from our live productions, you know, to see what it's look like. And we also also look at the AWS matrices for like, what their CPU usage was on, like an average, roughly. So we have about like 55% to 60% that we should scale up, or we should actually do not worry about, but yeah, usually, usually go about like 55 to 60% uses. And the trade is, you know, based on trade pool configurations that we have on our application properties, right? So we have our, you know, dedicated allocations, and, like, dedicated allocations of trades, and it exists like certain, let's say 80% of that allocation, then you can actually look into the issues. Okay? So you kind of get a baseline of, you know, what's typically expected, and then kind of go outside of those paths. That sounds good, yeah, sort of addition to that, you know, we're talking about baselines, we're talking about alarms and scaling and things like that. I'm curious what tooling and sort of concepts or whatever did you use to load test your applications? Because I think at this point you've talked a little bit about it, but I don't know how you're doing it. Curious. Okay, so,

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yeah, so we, you know, like we have our internal tool called Hercules, so that's the internal tool that we were using so that we run our performance test on right. So we use two kinds of languages for writing this test. So one is like, one is like, one is KC, k6 and like. Other is like, JMeter, J meter. So that's the, you know, like, two, two test cases that we, we work on. So I would say that, you know, whichever developer, so the developers, you know, can family can get familiar with, you know, different tools, and you know, they can do different J meters, because, like, J meters provides a user interface, right? So which is, you know, like, really easy to work on, which is really easy to work with, right? But the key six, like provides you with different proper matrices and like traceable for ease as well. So, very awesome. Cool you mentioned on your resume using Docker for packaging applications in Kubernetes for orchestration. So could you walk me through, you know what that looked like in in a particular instance? Sure. So, so, yeah. So, basically, you know, like Docker So, basically, you know, like Dockers basically our applications, while that was more modernized, you know, was more nice application that we work on was, you know, what was like work on, run on the Docker containers, right? So it was on Docker containers. So each of our project contain, you know, Docker file containing, like what the applications startup will look like, right? So you know, you, you know what Docker is. Docker is the like you can package, say, or package your application into containers, right? And then running for the running it and deploying your applications, right? So running for portability as well. So we had, you know, our standardized pipeline, so we had to set up a file on a call, you know, like, where you can actually define the Kubernetes properties on, like, it's our yml format. So it was, you know, an abstractions on top of what Kubernetes already uses. So basically, you know, instead of the, instead of the deployment yml file we will be using, you know, like different bogey file, and you will be providing, you know, the API versions, the metadata, the specifications and so on. So what I'll say that what kind of template you know, you want to use for that containers and what port of the applications in whatever the board the applications drawn on, right? So that's what our title strategy of deploying applications into, like a coordinate orchestrator. You didn't have to worry about things scalability for the service, you know, making things highly available in realty rates, right? That was abstracted by some DevOps team. You just provided some data, and they went up there, and you'd have to worry about it, right? Yes, yeah, we just provided our like technical data of the application itself. And the like abstraction linears that the DevOps teams obeyed were, like, responsible for a different you know, like scaling policy and, like, how many because we had in you, like, we just provided them, like a basic trace. Also, your concerns were more on the application side then. So you you mentioned it was your modern application, or you modernized it. What were some considerations you made when you were preparing the application for containerization. Okay? So, um, basically, you know, the main things that you can work on. What? Okay. So, so, basically, I think, like whenever, you know, like you, you said that you want quantizer applications, right? So maybe I misunderstood, but I thought you said you you modernized your application in order to containerize it. So I'm curious, you know, how you did that, what your approach was, okay? So, basically, you know, like, we just stripped out, like, we just like, we just, like, take out the certain functionality, right? So if we, if I were to talk about these two services that that helps in, like, modernizing this, where those were the experience and, like, maintenance of the fields, like API, so the Experience API, you know, these are, like, a particular services right, that is responsible for grabbing the user entitlements, right, to see what a user is entitled, entitled to, right? So, and look at the funding side, right? So, basically, they will be grabbing, you know, a list of JSON structure that listed, like, what, you know, like entitlements he had, or what kind of components, and since, you know, this was a separate concerns, we could, like, strip out it, and when it tied to the services you know, that is needed for so, you know, like that, that's how this needed for So, and there's other things that We made it as a standard external applications, right that? Then we deploy it into AWS, like easy to and then there was this maintenance apps, which it was, let's say, responsible for, like, you know, office won't, like maintain this to the its own profile, user profile preferences. So basically, what I say the consideration that we had to take, where the ease of like, you know, decoupling from the one of the and like, and how independently it can run on. So how you can, you know, basically make it as you know, like standalone, standalone application, and, you know, deploy it in a separate services. So how these apps were being incorporated into and like overall systems, right through different service discovery and like registry and so on. So I think, I think these are some of the way, and also to add on, like, one of the things that we had to keep in mind was the entitlements APIs that was, you know, supposed to be highly available and, like, highly performant, because every time you know, a user logs it, it triggers the flow. That flow, right? So it's kind of like the login traffic. So scalability of scalability of the service, was, you know, something that we had to consider as well, yeah, but these are the things that we could consider to be so you mentioned some high availability and some scaling. What were your considerations there? How did you ensure that the application was was scalable with the service? So, rather, I think what makes it highly available? Sort of curious about that too, just to tack that on to Austin's questions. So okay, sure, so I got it. So basically, you know, high availability, like it means that you know your applications remains accessible for, you know, higher percentages of the time, right? So basically, you know you can deploy multiple instances right into the server across, you know, multiple reasons and across multiple availability zones, right? So our applications was a deploy on like US east, right? And us waste two reasons of the AWS. So where we had to, like, deploy on like, multiple zones, like one A, like 1b like, like, that's, that's it. That's, that's what I mean, right? So, so we as a multi region deployment with a DNS based routing. So that's awesome. Very cool. All right, Zach, you want to jump in on any questions on that before we shift topics? Yeah, no, no, I'm ready to shift. Sorry. We only have 15 minutes left, and there's still more questions. I jump over to design patterns. What's maybe? What's your favorite design pattern, or maybe one you've used recently, and why?

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My favorite design pattern, like I do know, work a lot with design patterns on a regular basis. So I work on, you know, like single agent pattern. So basically, you know, it just ensures that your class has one instance right, and provides, you know, like global point of access. So you can use, you know, double synchronizations to make sure only, like one you only have one instance, or like other than other than that, like I have also a work on, which includes the factory method, where you define, you know, where you define an interface, right? So, where you define an interface and to for creating the object, right? So, but you also have your subclasses to other the type of objects. And there is also, you know, like degraded patterns as well. So the degradator patterns, like, it's very useful, right? So in, you know, changing the structures that allows our behavior to add to the independent objects dynamically, like creating, you know, without affecting the behavior of the other objects of the same class, right? So, you know, these are the three design patterns I will. I think I would like to say like, single agent is my favorite, even though, like, it's not too complex, but I am, I like working with a single design patterns helps with all your spring beans are singleton so but kind of keeping with that, one of the points that you mentioned in your resume is extensive error handling and retry mechanisms within Kafka producers and consumers. And I'm just curious, were there any design patterns to handle errors or to handle retries? How did you actually implement this stuff, and did you apply any established sort of, I guess, microservices patterns to solve it?

Unknown Speaker 42:08

So, okay, so, yeah, so the

Speaker 1 42:13

Can you repeat the question, like it was really hard for me to process all those things? Yeah, no worries. Yeah. I'm just wondering, you know, in the case of Kafka error handling and retries and other things like that, what sort of patterns or, you know, approaches that you take to implement that? Okay, sure. So, yeah, I got it. So, yeah, like, we basically use the, you know, we basically use the dead letter queue patterns, right? So on the Kafka to make sure that any kind of error was sent to the, you know, different topics called the dead letter queue, so that, you know, you have these topics that are, you know, stores of messages that are failed, right? And then there is a retry with except, like, exceptional backup patterns as well, where we, like, do not reprocess the messages, like, near our process, but we have, like, let's say, exponential, tiny backup or, you know, handling these failures, right? So we retry three times with our, you know, like with a backup of the different exponent of the exponent that we use, and then, if it was a fail, you know, after three times, then it goes to the, you know, date later skew. So where we could, you know, work on. So to add on, to add more on this like, we process like between, we process it on later times as well. So, yeah, that's it. Yeah, awesome. I would say, sort of in the application code. Let's say you're streaming messages to Kafka, or doing something similar to that, and you are getting failures, and you are placing them for the dead letter Q. How do you stop the application from sending failed messages? Okay, so the app process, the things that we can do is, you know, like, so, yeah, we did have a circuit breaker, you know, you know place to make sure that if you know the consumption is fairly right, then, like, it would not let any cascading effect to the downstream, right? Because, like, you know you want to make sure that you let the application start to, like, recover before sending any more messages, right? Perfect, yeah, you actually said the magic word more than anything else, because I was just, I'm with both sides of it. So thank you. I kind of want to go up a little bit higher. It sounds like, so you have Kafka in place, you have a bunch of other things in place. Sounds like there's a lot of different microservices. How did they find out about each other? How did they communicate with one another? That? Yeah, that's the two part. There was a third part, but we don't need it, okay? So, like, we, you know, like we had a service discovery deployed, right? So, you know, like, it's on this service where, like, registering themselves to, you know, like Netflix, like euroca server, right? So, like we have that services and like and communication, you know, between these services were mostly done using like restful way services, right? So through creating different RESTful API, so talk a little bit about those RESTful APIs. So your Team A, and you have service a, Team B has service B, and they want to talk to service a, right? How would you give sort of API documentation or something like that? What would you give to, you know, Team B to be able to integrate with your service? Okay, that's kind of interesting questions, okay, sure. So basically, you know, like you use our swagger docs, right? That has a sack swagger, a Swagger UI, that list, you know, that list, all those things, right? So our team, like our teams, and then what the other parameters with certain API lead, right? So that's, you know, what we provide to the other teams, you know, to in order to like, more like, integrate their like API calls to hours, right? Yeah, I'm curious about how you use the Swagger doc. So it's open API documents. Whatever was this, something you did code first, or was this contract first? Right? If you need me to explain the difference to I'm happy to, but I'm just curious what your team's approach was and what your opinion on it is, sure, so we

Unknown Speaker 46:56

call it correctly. So I think we had a

Speaker 1 47:00

code first, like just before the ease of the developers, right? So they develop a code and then later, like, adapted to the, let's say, swagger talks, right? So later, which was generated, I think I like it more than the contract first, because, you know, there could be a first thing changes, you know, that could come in and like, update on those contracts that were, like, already no generated could, like, sometimes be very difficult. Yeah, makes sense. I always like to see where people's opinions are on that, because I am a contract first kind of guy. So it's kind of funny. I want to quickly switch context here again. Sorry, not the force to go further here. But you also mentioned on one of your bullet points, a little bit about caching, right? You talked about improving performance by implementing some cache mechanisms, etc. Talk to me a little bit about that. What caching strategy did you use? What cat I mean, what was the cache? Is it a symbol in memory, stores or something else? I'm curious about that. Okay, sure, so let me, let me implement, like, how can I put this, all those things out? So, like we, you know, like we, so we just use our, you know, simple in memory caching. The reason, like, we needed caching. What our, you know, services was, like, I did some, like, depending on the right routes, right? Because, like, it was a Spring Cloud gateway, and these routes are like, statics, right? So unless, you know, like, it changed by some like, production, really. So basically, you know, like, we are set up our casing with a lazy loading and basically, you know, like we, whenever we grab the route, we save it into our memory. And, yeah, that's so that's the improved on performance, because, like, any subsequent calls, like, you know, you can you like, you know, like, how kids work, right? So you don't have to make, you know, subsequent calls to like downstream of or database programming, like those data because, like, it is already stored, right? So that's kind of our casings diagrams, where, like, we store data in memory. Whenever there's a change in, you know, in the database, for these routes, we have to, you know, like, rephrase endpoints, set in place like that. You know, the phrases the death cases, so, you know. So that's what I have, you know, work with. So we use, like, spring casing, spring casing, and like for the memory casing, awesome. Just this is my vanity here. Sorry, but you're caching the roots, or you're catching caching the responses to common requests for those roots. I would say that like casing the roots, casing the roots and what version, I'm just curious, what version of spring were you using doing this, um, if I recall it correctly, like I think we're using so that so our application, you know, like, more recently moved to the Spring Boot, right? So I think the spring versions was, spring version was, like Spring Boot three, and like spring versions was, it was maybe a six. Okay, I just wondered, because, you know, if you were coming from an older version of spring, for instance, and I, like you can tell me to shut up here, because this is just me rambling, but spring actually has a handling mapper, interceptor that will hash all of your roots by default, especially if it's, you know, going through. And I know you're doing like, you know, using ant matchers, or you're using, you know, normal path pattern parsers, or whatever else it is. So they build a cache for you, and they actually do the lookup. And I was, I didn't know if that was something that's either a missing in spring gate and Spring Cloud gateway, or if it was an older version, because this only came in in Spring Boot three, but specifically spring security, six, dot 10. 6.0, dot 10. Don't ask me. I literally just went on the journey about this today. That's why I was curious. Anyway, what's that? Though? Sorry, go ahead. Sorry. So when I mean by route, it's like, it's not a normal route, like, because it's, you know, like, JSON structures that have, you know, metadata and like replacement part of, like, where it needs to go, the domain, like whole stops and any kind of, like functions, right, that it needs to be processed. So that's why we have to, you know, like, store the whole object, rather than just a simple route, right? So our route is like an internal object that we had, we have consist of this data. So, perfect. Austin, I don't know if you had any other questions now we're getting, we're getting close to time, so I want to make sure we have time for Suman. So any questions you have for us? Yeah, about anything. Yeah. Like, I like to, I'd like to know about the day to day after developers on the team, like what is involved and like, how was it the work is done on the daily basis on the development, like focuses versus the meetings and team engagement. So, yeah, I want to know that Sure. So we're a kanban team. The rest of the org is kind of more Scrum, so we're a little bit insulated from that. But generally we, you know, we have a daily stand up. We have not, not too many meetings. We have, like, a Thursday workshop, you know. So we're pretty light on meetings on our team, at least. So a lot of work just getting done, thankfully. And then, in terms of the work itself, we're constantly learning new things and, you know, switching directions and thinking up new projects that are totally different than the last so I think if I had to describe it, I'd say it's about, you know, staying on the edge and learning new things all the time. It's been a lot of fun. We practice chaos on a daily basis. I'm sure do. Yeah, that's kind of interesting, you know. And I like, I'll also, I'll also like to know, like, what will be the, you know, like, next step in the process, if I remember it correctly, like I thought that the next step there will be an interview solution architect, and bunch of other interviews, like, how it works, or what would be the next stage of the process after this one? Yep. So so far, you've talked to Ryan, who's the manager for this team. He probably also had an HR screening before that. But yeah, we're around two, and so I think we'll submit our feedback after we're done here, and then I think HR will let you know the next steps. Yeah, I think Yeah, sounds good. I think that's all questions that I have right now. So and like, Thank you for your time. Like, it was really good to talk to you know about you, and like, express myself. So, yeah, that was a pleasure. Exciting with you as well. I hope you enjoyed the interview. I mean, I enjoyed it. We had some really good conversations here. And I hope you have a wonderful rest of your day, and you'll hear back from vertex very soon, yeah, I hope you guys have a good weekend, long weekend, looking forward to it. All right. Bye, bye. You.

Suman Astani- vertexinc-first-srv

Speaker 1 00:00

I'm gonna ask me anything. Yeah, that's completely fine. Like I was just here, so I wasn't waiting too long. Awesome. Boom. All right, so first order business, did I say your name correctly? Suma,

Unknown Speaker 00:16

yeah, that's correct. You pronounce it.

Speaker 1 00:18

Awesome. So always important to me to just like, get that right and not forget it. So I need my glasses to write. Here we go. Alright, and today is the 27th Yes, 27th August. Oh, so really great to meet you. I appreciate you taking the time to chat with me today, so I have a little structure. We'll just kind of talk through those logistics real quick. So I like to have our first conversation here be more of a conversation than just like a interrogation. Right about your resume. I see all that. I'm really interested, like, what motivates you, how you approach things like, I'm looking to find more about you, divorce from the things done, right? So I'm going to do real quick lightning intro. Me the teams, what the team is, this specific team is doing. Then I'll turn it over to you chat, and then at the end, something like 510 minutes, I'll leave for questions, but feel free to, you know, ask questions along the way. Okay, awesome. All right, so I'm Ryan. I have been at vertex 12 years now. I actually started out in testing. I spent I went through test like testing, test automation, test Architecture, Testing leadership came over to engineering. Spent a long time with engineering excellence, best practices, kind of pseudo build architecture stuff. I'm currently running two teams, so one team off, C off, and mostly our on prem product, but kind of working on working our way out of that whole and then at a team, which is the team you're interviewing for, have a kind of interesting history. So they started as a test infrastructure team in VMware Jenkins. And over the years, they picked up the our Maven build. They picked up moving to AWS. They picked up a lot of that, like back of house, improve everything. You know. Give me 50% improvement on bills and things like that. Yeah. So it's summer. The two that are left anyway, two took an opportunity on the other side of this project, on a different team, so we're still working closely. But the two that are left this summer have been working on

Unknown Speaker 03:03

new production capabilities.

Speaker 1 03:06

And that's a Kubernetes with Argo Helm, you know, standard, whatever stack there that's got some native AWS stuff in the back, moving data and prepping it, but basically taking something that we have, that we built 20 years ago, and containerizing it, repurposing it, moving the data, and making new, new features, new markets, etc, right? So they're working on that. They're working through that, and then what that begins to lead into is more of a system architecture team, development team role. So as you can imagine, like Kubernetes re platforming all that kind of system architecture, and the worlds are starting to blur, right? So the goal, really, for the team long term, is to work ahead of all of the tax domain developers, etc. And we're going to basically be doing the runway work, the work of the runway, not the design of the runway, ahead of teams addressing tech debt, preparing for tech debt, best practices, all that stuff on what is called a system architecture team, but it's still engineering, and so my goal really is to supplement their back of house experience with somebody that has some of the Java, Some of the production runtime, some of the user requirements, etc, you know, that kind of standard, more standard development, and then hoping there's a nice, you know, knowledge transfer both ways. And everybody makes out with, you know, cool, new abilities and stuff. That's me team and our vision. Tell me about you. Yeah, yes. So first of all, thank you for the bright introduction for the team. So to start myself like my new Suman and you pronounce it correctly, like I have been working as a full stack Java developer for about five years now. So to talk about predicts, like, I was primarily working on, on Java based web applications, you know, like utilizing our framework, like spring, like hibernate, and, like building REST APIs for development. So I have, like, you know, like, pretty much good experience of working with Microsoft, microservice architectures, you know, like, I have been like, involved in designing different microservices, using and implementing, like, you know, these services and, like, deploying them on like AWS as well. So, you know, like AWS is my cloud experience. So I have been, like, primarily working on AWS, like, easy to like, like lambdas. So that's the text tag that I have been using on AWS. And like, on the front end, I have experience working with, you know, like Angular, so it's basically a JavaScript and like, type is basically JavaScript, and type TypeScript as well. So, you know, like for testing, like I have been working on, mainly JUnit and like Mockito as well. And I have also, you know, like, some experience with like Mocha testing for like integration. And currently, like, I have been, you know, like working on, I have been working on, like, different, like screen teams. Like the team consist of Scrum teams. The team consist of, you know, like, five develop developers. So it's a like, we have a couple of tasters as well. So we have our, like product owners and like Scrum Masters and to talk about on the daily basis. Like I have, I'm involved mostly working on the back end Java code. So, you know, like helping with the teams, with code reviews, and also, you know, like on the products and support as well. So that's pretty much my right description. So what, what takes that I use and like how I work on So, awesome. Thank you. So, um,

Unknown Speaker 07:16

something about what motivates you?

Speaker 1 07:19

Okay, so that's kind of hard questions, like the what motivates me, right? So to talk about my motivations, I always, you know, like, what motivates me, right? Like, I would say, solving a complex problem, you know, and seeing the like impact of my war, like the impact of the world, impact of my work, that is that made like, that's one of the things that motivates me, like, because, you know, like, for me, it's rewarding, like being to able to take different, you know, like challenging issues, you know, like breaking it down and then build, you know, like solutions that will work And that add values to the teams, or, like, even the whole organizations. So, right? Because, you know, like, it's something that makes our lives of everyone easier, right? So to talk about in my previous job, like, I work on a microservices, you know, that includes, like, processing systems. And it was motivating, I will say because, like, you know, I could actually see how it directly benefits the, you know, like, overall infrastructure. So my by making things more robust and more resilient. So I would say that, like, I'm motivated by a continuous learning. So that's the things that motivate me, and, you know, like growing and like evolving. So, yeah, cool. Thank you. Um, so, all right. So, continuous learning, um, how often do you find yourself learning for a challenge that you are currently facing versus learning and finding out that it helps you later. Obviously, there's a third category of I just learned and it didn't become relevant. But I'm curious how often, like, you're learning path you feel like, percentage rough, I don't know, nothing specific, but like, how often does your own exploration turn out relevant, pretty quickly, okay, that's like an example, yeah, yeah. That's kind of interesting reason. So, you know, like, I'm in learning and, you know, like, finding out something is new, something that is new learning. I believe that learning is finding out something that is, you know, that is new, right? So that helps me a lot, you know, like, it's kind of difficult, because sometimes I might learn something that it's not relevant to my work, you know, that I'm doing, but sometimes, like, I might learn whatever I'm actually like, learning that's related. So in my previous process, you know, like we were, I was working on Apache Spark, right? So I was learning something new, you know, like something like, I was learning something resilience driven. It's a resilience driven deployment. So I was learning something new. So we were, you know, actually not implementing that yet. But like, it helped me out quickly, because, like, it was some kind of pattern that spark leverages and, like, we could process the data faster, right? So I will say that, like, on the, you know, like regular basis, like, I am learning more for solving a challenge that I am currently in, right? So I focus on, you know, like the priority to try and like, you know, solve the challenges that it's on hand and when I and when on my on my free time, you know, like, I often, you know, try to find similar topics. And like, if I can be better on myself by, you know, like understanding, like those topics as well and, and if you know, like it doesn't get used to, if I doesn't get used to soon, then I mean, like it's on knowledge for me, right? So I don't think never knowledge never get wasted. Like, cool, um, switching gears a little bit technic, more technical. What experience do you have with, like, legacy code and like, how old, roughly would you say that code was okay? So my experience talking about my experience on that so, yeah, basically, you know, at my current project, you know, like we were actually working on, like, mitigating the legacy code to the microservice architectures, right? So that's one of my responsibility in my recent project. So I'm, like, pretty much comfortable working with both legacy system and, you know, like the modern system. So I actually do not know, like, how the old code is, you know, but like, it's, it's been there for around 10 to 15 years, right? But like, I know, like, I would like, love to work, you know, like on the like the legacy, modernizing them, like making it more more better, like rewriting them, for making it like, you know, more microservice driven. So, more service driven. So I would say that, okay, what were, what were some of the things you learned? You know, after you started, I'm kind of interested in the adapting versus the, you know, reading the book on legacy code or something, right? What was something you learned along the way in working on that legacy code that was impactful? Okay, I can say that. So, yeah, you know, like, like, basically. So I think some of the things that you know, legacy code has taught me was, I will say that the importance of persons and, like, you know, attention to the details, right? So, because our, you know, like, legacy code base was really huge, massive, right? So, you know, one class consisting of, like, 1000 lines of code, let's say, and one method, like being too long for us to follow through, right? But I think it was very like, important for me, like, to be our passions, and, you know, like, then play already close attentions to detail, like, you know, like, try to see, you know, like, I try to see if I could, you know, create the rough graph, graph, or, you know, like flow chart or, or the complexity on that as well. So and, you know, like trying to understand the full pictures before rushing to make any changes, you know, like adding, let's say, new functionalities to it. So I had to, I would say that I had to learn to take time and understanding it rightly. So that's something that I have learned, and to add something onto it as like, I think something that I also learned was to create, you know, documentations for anything that you can do, right? So, because if I get hired into some project that is, like, very legacy, and it doesn't have any, like documentations, it becomes, you know, like, very difficult for me, right? So, but once I go through the code and I have, like, understanding, you know, like, if I create a, you know, like rough writing, you know, on what I understand, or, let's say, somebody else who is coming into the project again. Can, you know, like, take my draft as a pointer, so even, you know, like, though it might not be accurate, but it will definitely help them, right? So I think that's something that I took, was to actually create a rough draft on what I understand. And you know, like, even like, it might not be accurate. So for newer product, like, I do make sure I have, like proper documentations as Java Docs and like different readme files as well. But you know, like, for legacy, I think that's what I need to learn. Cool. Appreciate it. Thank you for that time for Well, one more here. Yes, you mentioned flowcharts, kind of kind of charting out your discovery. Might actually be the word. How does that exercise differ from you know, for you, like, how do you approach that differently than documenting before you do the work? Right? So if you were writing the same, like if I gave you a set of requirements and you drew them, okay, versus I gave you a system and you sort of mapped it out. But what? What do you do differently if there's anything that you can describe, okay? So

Speaker 1 16:17

I would say that. I would say the flow chart will be, you know, like something that will be helpful, you know, for me to understand the, like whole systems, like whole set of flow, because the discovery, because the discovery phase, it's like more about understanding, you know, the business needs And, like translation, translating into like technical solutions, you know, but if I have an existing system, you know, the discovery phase becomes more about, like understanding what's already in there, right? So I will say that I will start by exploring the code, and then, you know, like exploring a different modules and different methods and trigger line as well. And like figuring out, you know, like how to how these interactions work with each others. So I would say that flow chart are like different creating a full flows are like out of this will be like creating a plugging and playing, how things are there, and like, how you know, like logical operating operators are interacting with different modules and so on. So basically, like, it's about creating, you know, like from scratch versus like plugging in. Like, how different you know, like modules interact with each other, right? So, basically, like, I will say, for a new project, the focus will be on designing a solution, you know, like that meets the requirements from the ground up. So it's something, you know, that is like, a little proactive. Oh, yeah, so that is one of the pieces where I think we're going to be bouncing back and forth consistently, right? So that's kind of where that question is coming from. And there's no right answer, you know, I'm just kind of curious your process. So we, I expect us to receive some piece of functionality that needs to be moved or re platform. And, you know, we may actually be both diagrams at the same time and etc, so. And it's good example of the kind of challenges I'm expecting us to be running after. And by the way, this whole, you know, I say again, I'm expecting because we are in the process of rolling it out today, you know. So, yeah, shifts a little bit here, but okay, as promised. 10 minutes. What do you have for me? Um, for you. So, yeah, I have few questions. So first, I like to know about, like, how the team works in overall, right? So how the developer interacts with each other, and, you know, like, what kind of, how much of, like, peering will there will be between the teams? And, you know, like, how much developers works together, so how the team is compromised. So I want to know that first, absolutely okay. So today there is a dev two and a dev three, and so that would make you know four of us with this position, the team, I have not been told I have to change this yet. So the team is Kanban instead of Scrum. We're kind of sitting next to a scale agile, with the programming kermits and all that stuff. We basically operate on a weekly cycle. So every week, every Monday, we discuss new work that was defined. Point, any work that's not estimated,

Unknown Speaker 20:08

define it, refine it. Kind of plan

Unknown Speaker 20:12

daily stand ups, that kind of standard thing.

Speaker 1 20:16

We try to retro every other Friday. And then all the alternate Fridays we actually have a coffee meeting. It is a no work discussion, just trying to emulate getting coffee in the office, right? We the team. I'll go this way first so we don't have a formal Po. There's a chance we will need a formal po which will kind of work it out as a team who's going to hold that responsibility. Kanban equivalent of a scrum master. We call it the Kanban leader is rotating. So we do monthly rotations of board, hygiene, meetings, facilitation, just keeping stuff going. So the team as a whole are a lot closer to the academic definition of a team versus a work group, so they are kind of all accountable to each other. And I think rotating that board and like getting all those jobs in there help a lot. So they, in fact, are often telling me what needs to be done, because they're they're standing out in front of that. But what that ends up meaning is, you know, I have this expectation, everybody becomes a master in something, something we do, but we're all going to work on whatever we're working on together. So lot of different kinds of work, lot of collaboration. You know, when stuff's difficult, rough, urgent, sometimes there's just like, hand phone calls throughout the day, just informal. There is a weekly formal, we call it workshop, where they it's optional, but basically where it just calls in, goes on mute, and then unmutes when there's a question, as if you were at different desks, right? There's tons of collaboration. So we kind of, we take that, that internal kind of team behavior and culture that. So then what ends up happening is we're supporting like eight to 10 teams that are in the tax domains themselves, like writing the new product features, not architecture the green platform, right? So we're supporting them. We are part of the body that is creating standards and processes and things to make engineering efficient and scalable, safer,

Unknown Speaker 22:51

whatever.

Speaker 1 22:53

And then on the other side is essentially the Cloud Platform group and we serve we end up surfing as a wedge between the two. Cloud Platform group is two of our members that were switched over for this new project, so we're also collaborating along multiple teams. Interesting. Yeah, it's definitely an interesting spot. And then ultimately the team is going to be reporting into a team of principals and architects that I'm the, I'm the formal member of. But, like, I don't know how that's going to actually end up, you know, working and, but, but that's the idea, is that reporting into the principals and architects to try and be six months out. Yeah, that's answered my questions. That answers my questions. And I like, I'd like to also know, you know, like, what kind of opportunities will there for a developer, you know, like to grow their to grow myself, you know, like learning different opportunities, like, you know, like, anything like that,

Unknown Speaker 24:13

sure, yeah. I mean, so

Speaker 1 24:17

easiest way to share is by examples. You know, I my entire career here, my boss, actually, when I started, was in product support. She's now director of engineering. The two engineers that moved over to that other group went from writing one product to, you know, kind of the whole back of house to now developing the future platform features and whatever vision, trying to think of one more, there's one like, off the tip of my tongue, but so we are looking to become, build, run teams. So that's it. That's something you haven't done formally, like, that's hard if we're going to be learning it. We give it, given that most of what we're working on is directly with principals and architects and our principal architect, who's always kind of looking to grow people mentor, he's actually been mentoring me like there's a lot of informal there's a lot of informal opportunity within, the opportunity to be learning, coaching, mentoring, when you Try to grow leads and leaders and principals from within. So, yeah, that's always like something we're looking out for. This who within the teams can increase their impact, sphere of impact, and, you know, grow and all that. Yeah, training, yearly trainings.

Unknown Speaker 26:02

There's I had put us through

Speaker 1 26:04

continuous delivery. The Dave Farley, course, we've got, like, training stipends and books and other stuff too. There you can kind of branch off a little bit. That's good to know. Yeah, that's good to know.

Unknown Speaker 26:19

We do career development plans. So we actually

Speaker 1 26:23

discuss through, like, the 70% on the, I forget the breakdown on the job network based and then formal training. So we actually formalize some of how that looks and set goals for the year based on it and all that that's interesting. A perfect place to grow, right? Yeah, so I think that's all the questions that I have. And like, I'll also like to know, like, what will be the next steps after this interview? So how many rounds, how many rounds of interview. There will be. So how soon I would like to hear from you? So, sure, yep. So I'm going to use both with Roger, right? Yeah, yep. Okay, I'm going to get back with Roger. If we, you know, if we move forward, I've got, we're doing two. It's really three interviews. One's a panel of two. It's a the gentleman who moved across the other project and our team lead. And then we've got one on one with principal architect and one on one with either principal engineer or or dev for depending on availability and all that. And then that's, that's the interview. And they'll, you know, they each kind of have some technical direction that they're they're really targeting. Okay, sounds good. So I know Roger's turnaround time is, but we try to stay quick so you'll hear something soon. Yeah, sounds good. Thank you. That's all my questions, and like, I look forward to hearing from you soon. So awesome. Sounds good, hey. Well, I appreciate the time. It's really great meeting you. Yeah, it was really good to meet you as well. I hope you have a good rest of your day. You too. Thank you. Thank you.

Suman Astani-Bank Of America- Srv-Final-Offered

Speaker 1 00:19

Okay, it looks like it almost There. Just get his audio. I

Unknown Speaker 00:56

Hello, can you hear me?

Speaker 1 00:59

No problem, yes, I can hear Ramesh. Adam and Suman, can you guys hear Ramesh? Yeah, I can hear you. Yep, I can hear okay, cool. So good luck to you. I'm gonna make you the co host. Okay,

Unknown Speaker 01:15

thanks, Jason, I'm dropping Okay. Thank you guys. Okay, thank you.

Unknown Speaker 01:21

Any one minute sure, take care. No

Unknown Speaker 01:23

worries. Take your time. You

Unknown Speaker 02:11

okay,

Unknown Speaker 02:18

let's start. Why don't we

Speaker 2 02:20

start with your introduction. Okay, sure,

Speaker 3 02:26

yes, sir. So my name is Suman. So you know, like I have been working as a Java developer for about five years now. And you know, like I have been primary, how many years?

Speaker 1 02:37

Five years now. Five years, five years

Speaker 3 02:41

now. So I primarily, be, primarily been working on a Java based web applications using, you know, frameworks like spring and like hibernate and like creating web services for like API development. So I have been, you know, like, primarily working on, like, microservice architectures, you know, and been helping in like designing and implementing microservices and like deploying them to AWS, like some of the AWS services that I have worked on, and that includes like, you know, like Ed like AWS, like that include easy to like ECS, right? So lambda, so, like, I do have experience working on the front end side with Angular. So to talk about the testing framework, I have worked with JUnit and like a Mockito for writing different unit test cases, right? So I am currently in a scrum team, and my team consist of, you know, five developers, so we have a product owners and Masters on a daily basis. So on the daily basis, you know, like I am involved mostly working on the back end Java code, I would say I'm helping the team with, you know, our code reviews and also production support duties. So that's my overall experience.

Speaker 4 04:02

Okay? How do you rate yourself in Java in the scale of one to 10?

Unknown Speaker 04:08

How do I rate myself? That's kind of hard questions.

Speaker 3 04:12

I would say like around 7.5 to eight. Okay. Okay. Okay,

Speaker 4 04:29

what was your last project? What was, I mean, the current project, or what or the last project?

Speaker 3 04:35

So currently, you know, like, basically, like my current project is like financial good planner. So that's the main features that I have been working on, right? So, like, we have been primarily working on, like orchestration thereof, like microservices. So using like Springboard framework, like building Spring Cloud applications like this is really bringing you know, like, building Spring Cloud applications is really helpful for, like, orchestrating requests, like requests from the client side to the server side. So like use for request to like manipulations. So,

Unknown Speaker 05:20

okay, so what exactly

Speaker 4 05:22

tell me about what was the last task or the current task that you're doing?

Speaker 3 05:29

So my recent tasks, my current tasks, I will say that like Currently, I have been, you know, creating a filter. So it's called, like tokenization filter. So basically what it is like we have, you know, downstream partners called like queuing. And you know this during is used for changing your secure data, like tax ID or like account number or so on into the like hash value. So, and this is our school APIs that we can use. So basically, whenever you know, the client onboards your route, like, right, based on the you know, that metadata that they provide, you know, like, we can actually, like, convert this sensitive information into a token, right, by calling that services. So, you know, like, basically these filters, like, it's supposed to traverse into the like, JSON payload, you know, that is provided and like, look for the key that is configured as the metadata and their you know, value for the key is converted into the token right before, like, it is passed along to the downstream services. So that was the most recent filter, I think that that I have work on. It's a new requirement so that you know any kind of safety address is prevented

Unknown Speaker 06:58

with respect to the

Speaker 4 06:59

with respect to the actual task, writing the code with respect to what you did with the task. I know you explained the task. Can you guide me through a little bit of code to see what you have done for that particular

Unknown Speaker 07:16

task?

Speaker 3 07:17

Okay, so code basis, right, okay. So basically, you know, like, like, we have interface that all, like, all of the filters, you know, should use our patterns on, right, right. So the do this interface has like, three abstract method so that my, like, on, create a tokenizer. Filters implement on, so on the first one was the order. So basically, this determines, of like, what a particular order. So this filters, you know, it needs to be like, executed, like, right? Because there are, there should be like, 30 or 40 filters that needs to be executed, and it needs to be on the different orders, right? So first thing was that implementing the order, and the second thing that I implemented was the like, filter on tape. So that's the second thing I have the second thing I have implemented so it's so it's called the, it's called the filter method. And here, like, basically,

Speaker 4 08:23

again, right? I don't need, I don't need you to go through the explanation kind of thing. Take one, pick one last task, what? Okay, I have written this class, and whatever you said, filters, right? What did you do in that class to achieve that thing, that particular thing? I want to know what exactly was your task and how you approached it, and what you did in that you don't have to tell me anything that is confidential, or anything of that nature, but I just want to know the process and how you went through the code. What is the code that you have written? I mean, you don't have to extend anything confidential to me. Okay,

Unknown Speaker 09:12

okay, okay, so I tried to explain it.

Speaker 3 09:16

Okay. So basically, you know, like I was tasked with, like, implementing the filter, right? So the main thing that I did on this class was the actual running of the filter on using this. So I basically, you know, use the connections from our like, HTTP, client connections, pool and on this, like, we basically made a, you know, REST call to our downstream partner to, like, tokenize, you know, that to tokenize that particular field. And like, after that was done, like, I basically, you know, replaced the value of the key that was, you know, onboarded with the tokenized value. So that was my overall task. So I was in New I was in charge of implementing the

Unknown Speaker 10:09

spring filters right, Springboard

Unknown Speaker 10:13

filters correct. So it's

Speaker 3 10:18

like function factory. So there's a, you know, a concept of function factory on our Spring Cloud gateway. So it is not a typical filter, but it's a function view that you can run on our routing so

Unknown Speaker 10:43

in spring, right in frame.

Unknown Speaker 10:47

I have a an interface. Okay,

Unknown Speaker 10:51

will build an interface. Okay?

Speaker 4 10:56

That interface, let me call it as a vehicle. Okay?

Speaker 4 11:05

We we have several implementations of those, cutting vehicle. Okay, one is car, one is truck, one is motorcycle, let's say and you have a bus, something like that. Okay, we built our we built our services classes,

Unknown Speaker 11:27

and then we have injected,

Unknown Speaker 11:31

because we have to

Speaker 4 11:34

go to an interface. Are you aware of the principal, object oriented principal, called code to interface? Okay, so we should go to an interface, right? So all our services will say, I know a vehicle, okay, but it does not know what is the type of implementation. So we have an implementation of several implementations correct inside so then Spring Boot, it was complaining that there is more than one so it is not able to associate an implementation to the interface. How do we resolve this problem?

Speaker 3 12:20

So the way to resolve like, I understand your like scenario. So basically, you are talking about like interface, you know, which has multiple implementation, and it's a service, right? So basically, there is a like, a very easy way to tackle this. So there is an like annotations in Spring Boot call the like qualifier annotations, right? So basically, what qualifier does is like it basically specifies, you know, what implementation you want to like inject for that interface, right? So you can simply use like qualify annotations, qualified annotations, wherever you are injecting that, right? So there's also, like another concept right, called the primary right. Basically, if you do not provide you know, any like qualifier, then it can take, you know, one of those pins as like primary beam and like, inject that on this specified right? So those are the two approaches that you can take, but I will say like qualifier is like more useful and handy.

Speaker 4 13:36

So what do you understand by controller advice, okay,

Speaker 3 13:46

like, so, basically, controller advice. It's like concept in springboard, you know, that is used for, like, global exception handling, right? So, basically, you know, like, it allows you to specify, you know, what response you want to send out to the clients for the particular exceptions, right? So you, let's say you have, like, a non form exception, then you can return our, you know, response entity with the response status of 404, or, like, if you have, you know, some, let's say, some kind of genetic exception that you can, you know, send out, like, HTTP 500 with a like, meaningful message and like so on. So that is a controller advice in springboard. Okay?

Speaker 4 14:48

Quality do we use? Is it possible to use? So we have two annotations, right? One is at the right component. Another is at the rate service, right?

Unknown Speaker 15:08

Can we interchangeably use them?

Speaker 3 15:12

So we have two components, so can we interested in the use them?

Unknown Speaker 15:18

Right? No, I'm talking about

Speaker 4 15:21

the annotations called at the rate component and at the rate service cut.

Unknown Speaker 15:30

We didn't know the annotations. Yeah. So basically,

Speaker 4 15:32

why do we use them? Why do we use them, and can we use them interchangeably?

Speaker 3 15:37

Okay, okay. So, basically, you know, component is, you know, a component just tells you, it tells your spring that every component annotations, it tells the steals your screen, that it's a component that your spring bin can manage, right? So, basically, you know, it is a species specifying that, like, it can be used as a spring bin, and service is like a more specific or like a tailor towards like bin that provides some, you know, business logic, or, let's say, some like service functionality, right? So service is used to annotate the service layer. So now, now to answer your second question, you can use a component annotations as our service class, but like it is just, let's say, make it confusing, right? So it is not wrong, but like, you don't, you don't, don't provide like, any specific like tasks for like for better, I'll say for better readability, and like understanding for better understanding, right? So you will use, you will have to use a proper annotations, like, maybe a controller or a repository or like, or a service in C Sharp component, right? So, but if it's a generic you can use a component annotation. So I will say, using the more detail or specific annotations, like it is a better practice, you know, and like it should always be followed.

Unknown Speaker 17:12

That's more to JPA. Can you use JPA?

Speaker 3 17:18

Yes, yes, I have, like I have, I have working on Spring Data JPA.

Speaker 4 17:30

What do you understand by the word projections? In JPA

Speaker 3 17:36

means the town projections, right? So basically, you know, the projections like they are basically used to, you know, like retry only specific columns or parts of the entity rather than fetching the entire entity, right? So, basically, you know, like you can like you basically you can define the like, different projects and interface. And basically defining an interface, you know, with like gators will help you get the properties that you want to retract and like, that's what opposition feedback.

Unknown Speaker 18:20

So I'll give you one sorting problem.

Speaker 4 18:33

You have a list of 11 numbers. Okay,

Unknown Speaker 18:39

can you write a piece of code to sort it,

Speaker 4 18:44

so you can type in the chat message here,

Speaker 3 18:50

so there's 11 numbers, so we have to be an array or list whatever, right? Okay?

Unknown Speaker 19:00

A or list, whatever, right?

Speaker 4 19:03

So it should take your function should take on the list. Also take whether it is ascending or descending.

Unknown Speaker 19:15

It should return the sorted, sorted list. What's your requirements?

Unknown Speaker 19:33

So you just want me to type on the chat, or do I need to

Speaker 4 19:38

just start keeping typing the chart or type anywhere else

Unknown Speaker 19:45

as you finish one line, just Press Enter C,

Unknown Speaker 19:51

so that way I know that You're

Unknown Speaker 20:16

don't use,

Unknown Speaker 20:18

okay, don't use any, any

Unknown Speaker 20:23

already existing sort methods, okay, default method.

Speaker 5 20:38

So you don't want me to use Java stream, sort method,

Unknown Speaker 20:44

no, no, use Your.

Speaker 4 20:50

Got to use your, your your own sorting. You

Speaker 5 21:09

okay, so to write my own sorting, I think I should be able to Do that. Okay?

Speaker 6 24:18

So I think like that should take care of it, right? So, like, but it's a very insufficient sorting, using ball sort, like, you know, where you just go through these numbers and, like, look through it. So okay, what

Speaker 4 24:41

do you call this sorting, the one that you put. What algorithm is this?

Unknown Speaker 24:47

It's called bubble sort. It's called

Unknown Speaker 24:52

bubble sort. Why do you use j plus one?

Unknown Speaker 24:56

So what?

Speaker 4 24:58

So what happens at the end of the array. So,

Unknown Speaker 25:02

basically, you know, like,

Speaker 6 25:06

basically, you know, I am, you know, comparing the current element with the next element until the second last. I don't have to run out of the index. So that's because of that. Okay,

Speaker 4 25:29

I'll give you one thing. Just do this one. Okay, I have a list of address okay,

Unknown Speaker 25:42

I know this problem, desktop

Unknown Speaker 25:50

address list Just on typing, okay.

Unknown Speaker 25:54

Address as

Speaker 4 25:58

name, comma, city, okay,

Unknown Speaker 26:05

you're seeing my chart. Okay. There are address

Speaker 4 26:11

list as 100 elements

Speaker 4 26:22

that has 100 elements. So what we need is we need an output. We need to write a function that will return something like this stuff using streams API, okay. Return the map so you see this one. So once you so string is city and in a key, right key, string is city and the list of addresses. Let's say you and I belong to the same city. If I get a key of our city name, then you and I should be present in the list of addresses. If someone else is there with a different city, it will be different key, therefore he will be in a different list. Can you do that using streams API, convert a address list into a map of key and list of addresses. The key is city in the address.

Unknown Speaker 27:38

Follow me. Yeah, I'm

Speaker 3 27:39

getting, I'm following the requirements. So basically it just all, you know, collection, collection grouping on our like, stream API and like, it's straightforward. I can return, you know, let me, Let me type it for you. I

Unknown Speaker 28:23

So, yeah, that should be okay.

Unknown Speaker 28:30

What is a lambda?

Speaker 6 28:34

Lambda over here. Okay, so basically,

Unknown Speaker 28:40

I will say

Speaker 3 28:42

it like, it's just a like, clear and lambda. It is just a clear and like, concise way, you know, to represent, like, one interface, right? So which basically, you know, it just and like expression that that is, like, really useful in expressing a syntax using like arrow function.

Speaker 4 29:06

How do we store a lambda? What into what? Let's say, store a lambda into a variable.

Unknown Speaker 29:15

Okay, what would be the variable type?

Speaker 6 29:18

So the type of the variable after restoring the lambda and the variable, right? I think, yeah, I think, like it should be a Runnable

Speaker 3 29:27

interface, right, but like it can be any functional interface, right? So basically, like there are running running balls, there are functions, there are predicates and by function. So it's a functional interface.

Speaker 4 29:47

It's a functional interface. That's our answer, right? Something is happening to my computer. Just hold on. What's going on?

Unknown Speaker 30:01

Can you see me?

Unknown Speaker 30:06

Oh, yeah, you are you?

Speaker 6 30:07

Are you are on the call? Like, I can do

Speaker 4 30:13

a video, like, if you're a video, I lost my power. Looks like I lost power. Just hold on one second. I'll try to use another one. Correct another one. Okay.

Unknown Speaker 30:41

Okay, fine. So,

Unknown Speaker 30:59

what's going On? Teacher, tell you what. Boys.

Unknown Speaker 31:36

Functional interface. What's a y function? Sorry,

Speaker 2 31:41

I Sorry, what is the by function? Oh, do

Unknown Speaker 31:45

you ever use it? Yeah, you

Speaker 3 31:46

mean by function? Okay, so basically, like, by function is a like, you know, it's a kind of lambda expression, you know, that takes two values and like return some values, right?

Unknown Speaker 32:19

Okay, okay. Going on?

Speaker 4 32:33

Taking time. Okay, fine. Okay,

Unknown Speaker 32:51

taking time,

Unknown Speaker 32:54

okay, let's come to some multiple

Speaker 4 32:56

team. Okay, yeah,

Unknown Speaker 33:11

why do we need multi threading,

Speaker 3 33:16

multi trading, you know, basically the concept. So basically, you know, monitoring it like this. You know, like splitting your like tasks into different trades, right? So, basically, you know, you can break down your single process and, you know, around it concurrently, which basically, you know, improves your like application performance, right? So, like, concurrent execution allows, you know, like, multiple operations to run, like, simultaneously, you know, which makes the use of those CPU resources that you already have, right? And like, it can help you, you know, like, improve the performance. That's

Speaker 4 34:01

okay. I say you, you wrote a multi threading application. Multi threading is there right now, and server threads are running right one of the thread, one of the thread. I mean, it's misbehaving. It's not it's not being as it's supposed should be. Some problems are happening. Okay, how would you stop that threat?

Speaker 3 34:33

Okay, so, how can I stop that threat? So, basically, you know, like, so you have some kinds of misbehaving trade you can do. So there are, like, trade interruptions, like setting a flag, you know that like indicates whether a trace will stop running or not, right, and check if the interruption take about the interruption status, so periodically, right? So based on that, you can, like, interrupt, stop the trade, and like you can interrupt the trade, right?

Speaker 4 35:19

We wrote a multi credit application, and then we are seeing there are a lot of deadlocks happening all the time. How would you resolve that deadlocks?

Speaker 3 35:33

So basically, to resolve the deadlock in that situation, so are you basically have to, you know, make sure that your you typically have, like, proper deadlocks, like in particular order, right? So, basically, you want to use, like, consistent ordering, right where you enter all the traits you know, that, use that, use that resources, and you have, a, you know, proper locking, and, like, constants order, right? So, like, you have, let's say two locks, like a and b, and then, you know, you always acquire that lock, a before b, like, across all the trails. Then you can, you can also try a lock with a timeout, you know, so that you know you have a timeout after acquiring logs and like, if the locks you know cannot be acquired within that timeout, then the tree can, like, back off and like, retry again, right? So like, whenever you have like collections, right? You make sure you use a proper concurrent collections, like like concurrent has map or like copy on right array, right to make it is a trade safe, right? So, yeah, these are some of the ways you can ensure you have a proper control over deadlocks, right? Okay?

Speaker 4 37:07

Okay, let's say you have a function, okay, you have a method. So what you're doing is in that method, it is making some calls, okay, some service call. Another is a database call, and all that it is doing. One of the database calls. It, if it takes more than five minutes, it should come out. Exit, Exit. Your function. How do you write that? Achieve that way we can answer that. So you understood my problem, right? So you have a method in that method, something else you are calling, okay? But when it is given you called a database function, let's say you calling your database some

Unknown Speaker 38:07

selecting data from okay,

Speaker 4 38:13

what is happening is it is taking a lot of time. Lot of time is more than five minutes. So if it is taking more than five minutes, what I want to do is, I want to come out of that method. See, you made a call to a method. So it is a it is like a hanging that method is hanging. You cannot you. There is no way that you know what is? What you call that word? I can't remember, so you make a call and it is stuck there inside your method. No, it is not coming and executing the next statement. Okay, so if it's going to take more than five minutes, I want to exit from there, exit from the totally out of that method. So how do we achieve this thing? How do you write the code such that you will be able to time out if it is more than I understand

Speaker 3 39:10

your requirements? So basically, what you want to make sure that you wrap your like long running tasks into the into a callable interface, right? So once you have that, you can, you know, submit that task into the executor services, right? And the executor service, service basically returns, uh, like future, like if future is working, but a promise that it will complete in the futures as soon as you right, so, and whenever, like, you do a future like, like, dot gate, you provide a timeout, time that you can, like, specify, like, how you how long you know you can wait the result for, right? And then you know, like, whenever that threshold is right, so it exists, like, out of that processing, so that your resources do not hang up. And you know, like you do not, like exhaust your credit. So that is an approval you can use, right? So, based on the requirement, okay, let's

Unknown Speaker 40:15

go to some database queries.

Speaker 4 40:20

Okay, I have a database, and in the database I have a, let's say

Unknown Speaker 40:29

yes or no column,

Speaker 4 40:34

okay, my mistake, you know, when they did from the UI, when they were supposed to write it as a yes, send it back to the database or the service to as yes, they could like no, but then for the yes, they would know, for the no, they put Yes. So when you now it is after a few months, everybody looked into it, it looks like everything is like yes is NO and NO is yes. In the database, okay, they can fix the code in the front end, but in the database, it is already there, like, yes is a million rows or S 2 million rows are no. But then what's happening? We need to switch them also in the database, correct, I want you to write a query that by running one single query, it will switch both. So you

Unknown Speaker 41:30

want me to write a query?

Unknown Speaker 41:33

Yeah, write a query.

Speaker 4 41:36

How do you update in one single shot, query both? Rows should be updated, okay, yes will be updated to No. No will be

Unknown Speaker 41:50

updated to Yes. So

Speaker 3 41:54

basically, like, what we can do is like, you can update your cable right? And you can, like, set that a flag to be not the flag right? Basically, if there your is or yes and no are Boolean. So, like, it can be stored in their strength. There are strings, okay, okay, so,

Unknown Speaker 42:23

even with how we will do

Speaker 3 42:27

it. So basically, in that case, if it's a string, like, we will want to like, basically like, switch, you know, switch something into a temporary value, right? So basically, I will update my table, and I will, you know, set the quorum based on our case statement. So whenever the whenever that column is s, right, I will change it to a temporary value. And then when the column is, you know, I will change it to s, and then after that it is done. So I will do that,

Unknown Speaker 43:06

so that will take care of, like updating the values, okay,

Speaker 4 43:17

you have a trillion records in a table. Okay, records, we exported all of it into one file. We have a file. Okay, records. Now let's say you create one more table. We want to insert all the tape, all the data into the table. Okay? So, because we have lot of amount of data, we want to finish the job very fast, okay? And also, right, we don't want to do the task A lot of times. I mean, let's say with breaks, I don't want to whatever has been already there. I don't want to redo it again, something like that. But I created 100 billion records have been already inserted, and I don't want to redo the same thing again, right? In the interest of time. What is the best approach that you will follow to do this? Okay,

Speaker 3 44:20

in that scenario. So you know, if that's your requirement, you know where you want to efficiently insert, like I will like probably do like our batch insert, you know, batch?

Unknown Speaker 44:40

Insert. What is a batch? Insert? Yeah. So

Speaker 3 44:42

basically, what I mean by that is, like, what I mean by that, what I mean by that is like, you just mesh the data and, you know, you know, like, you copy that data into a new column, right? And then you do, like, bulk insert into that table.

Unknown Speaker 45:06

You do bulk insert into that table,

Unknown Speaker 45:11

bulk insert.

Unknown Speaker 45:15

What is bulk insert?

Speaker 3 45:18

So, I mean, like, basically bulk insert, you know, it's, it's a kind of like, like database specific method. It's a kind of, like a database specific method to efficiently, you know, like, load a large amount of like data into the tables, you know, like a single operations, right? So basically, whenever you deal with, you know, large amount of data, you do a bulk insert, insert, right? So basically, in like PostgreSQL, right, you can use the Copy command to load data from a file, which is our bulk insert, right? And on my SQL

Unknown Speaker 46:05

data. Okay, let's, let's

Speaker 4 46:06

move on. I have only a few minutes. I'm over, but few minutes. But okay, let's do this thing. You know? Shell script. Shell script, yes,

Speaker 7 46:20

you know, yeah, yes. Okay. Of Shell languages

Unknown Speaker 46:25

are available?

Speaker 3 46:34

The difference the different types of cell languages that are available? So basically, you have, you know best that I work with, like, best, best that I really work with, you know. So there is, like, s ads, like JS ads, so that's, that's where I have like, work on scripting,

Unknown Speaker 47:02

Unix, okay,

Unknown Speaker 47:07

let's say

Speaker 4 47:11

you run a command, okay, or a Java Java code. You run your Java program, but you want to find out whether the Java program successful or not in the shell. How do you find it?

Speaker 3 47:28

So there is a specific command, like system CTL, like, status, like, like, service name, right? So you can use that command,

Unknown Speaker 47:46

okay?

Unknown Speaker 47:51

So let's say you have a script, okay,

Speaker 4 47:56

in your in your script, you You did some command your because shell script is series of commands that you will try. So there is 111 command that you executed, there is another command and another command. So what you want to do is there are three commands you executed, the first one, the second one and the third one. You want to run them parallelly, okay, and then the next one. It will both of them finish it. Then it should wait and wait for the next one. So how do you do that? In scripting, I

Speaker 3 48:42

understand. You. Basically like, I understand the requirement. So basically like you want to run something in parallel, right? So there is

Unknown Speaker 48:54

two comments in parallel.

Speaker 6 48:59

So basically like you, you have to use the

Speaker 3 49:06

what you have to use the ampersand, right? So basically, this makes your neuron those commands in the background. And like you can use command for all the background jobs to finish.

Unknown Speaker 49:32

Let's say you run

Speaker 4 49:35

you run something within a script, you run something and then it produces something. It writes something, okay, and you want to capture the output into your variable. How do you do that?

Unknown Speaker 49:50

Capture output to the variable?

Speaker 3 49:53

Yeah. So basically, like, you can do that using the Eco, right? So, so, if you So, if you would, if you just want to do that, you can just do use that command,

Unknown Speaker 50:13

any command, right? Any command.

Unknown Speaker 50:15

You run something, it

Speaker 4 50:19

produces some output. Let's say you wrote a shell script. That shell script has some output, finally generates. But that output you want to read it, read it back into a variable. Is it possible? Yeah. So you can,

Speaker 3 50:41

like, do that so, so you have, you have a specific command, like, using our dollar parenthesis and like.

Unknown Speaker 50:55

So each and you have

Speaker 3 50:57

variable, you have a specific way, like variable is equal to dollar. Like, say, the strip that you want, you can just pass that in the parentages, right?

Unknown Speaker 51:08

Did you use that before

Unknown Speaker 51:11

I'm here? Yes.

Unknown Speaker 51:15

Which project

Speaker 3 51:18

also, like some of our, you know, deployment scripts needs to, you know, working some of our deployment scripts that needs to read our list of files, you know, to make sure all the File Scripts are, like, loaded and on the sales script for deployment so, Like, I have used it in my previous project. Okay, cool,

Unknown Speaker 51:46

that's all I had.

Speaker 4 51:47

Let me get the JSON back. Okay, do

Unknown Speaker 52:40

So what? What state are you from

Unknown Speaker 52:44

currently? Like, I'm in Virginia,

Speaker 5 52:51

but I usually move here, like I came from Ohio, so I moved from Ohio to here recently. Okay?

Unknown Speaker 53:21

So Deepak will be joining, okay, thank

Speaker 4 53:27

you, and that he will take over from there. If you want to drink any water or something, get something. I'm here. You can go and get something. If you want

Unknown Speaker 53:40

to, Deepak should be joining us.

Unknown Speaker 53:55

Okay,

Speaker 5 53:58

so nice Talking to You So

Unknown Speaker 55:50

hi, okay,

Unknown Speaker 55:52

thank you

Unknown Speaker 55:54

so much. Thank you. Thank

Speaker 5 55:59

you. Bye, talking to you. How about you?

Speaker 2 56:14

Someone? Can you walk me through your day to day routine? What do you do on a day to day basis?

Speaker 3 56:20

Okay, on my day to day basis, okay? So basically, like you, like, I have, I have a hands on experience on the back end developer, so on a regular basis, on the day to day basis, I am mostly involved in working on the back end tower, course. So I do some kind of designing, like, work on, like micro service requirements, right? So I document the finding functions, like functional and non functional requirements, right? So I work on those stories as well, like assigned to that is assigned to me, and, you know, like my functional tasks, or, like, backing a Java group stuff. So I do provide like, different code reviews on a team base, you know, like work to make sure that, like, it is proper, and like, all the requirements are made like, so like I contribute towards like Like as I discuss ceremony during like, planning, grooming and so on, right? So on the daily, weekly basis, you know, like I am also assigned like products, like products on production support, you know, like production support. So for the week, like, I handle different releases, you know, I handle any production issues and so on, right? But, yeah, that's kind of like a brief rundown on, like, what I do on my regular basis.

Speaker 2 57:55

Can you explain a bit here about your current project architecture?

Unknown Speaker 58:00

My current project architecture.

Speaker 3 58:03

So, yeah, basically, I know, like my current project is our orchestrator layers, right? So basically, we, you know, act as a gateway between the like clients and the servers, and we provide different features, requests, manipulations, like changing like request headers, like tokenizing the request data, you know, like assigning correct host for, you know, downstream APIs, you know, like we onboard this request to go to the tri band ecosystem so we have, like, Spring Boot applications, as you know, deployed on the like AWS cloud. You know, we use AWS, like easy tools for different applications microservices, like deployed on the, you know, AWS like ECS target, right? So we use, you know, lambdas for some use cases. So we use, like, AWS RDS for database. So that's like bright informations on my project at tribal. Okay,

Speaker 2 59:13

so how, like you said, you have multiple microservices, right? How do these Microsoft one another?

Speaker 3 59:21

So unlike a lot of our microservices, like this, uses the you know, RESTful APIs to communicate with each other, to call each other and interact with each other.

Unknown Speaker 59:36

Okay, so where are these web services deployed?

Speaker 3 59:39

So my web services, my web my web services, like they are deployed on AWS ECS target. So, you know, basically it is, it is like a containerized service, okay?

Unknown Speaker 1:00:02

So you what version of Java?

Unknown Speaker 1:00:07

What version of Java have you used? So

Speaker 3 1:00:10

I have worked a lot with, you know, Java eight and Java 11. So I do have, I do have, like, a work on Java 17, a little bit, because, you know, we are, like, migrating our like applications to the Spring Boot tree and like JDK.

Unknown Speaker 1:00:30

Okay, so what features in Java eight? Have you used

Speaker 3 1:00:36

in Java eight? So basically, you know, there are features in different features in Java, like, you know, there are stream APIs functions, interface, right? There are, you know, like optional classes and so on, right?

Unknown Speaker 1:00:54

Okay, so,

Speaker 2 1:00:57

why did you think what can be introduced on the streams, lambdas, optional in Java, whatever we could do in Java eight, then we could do in seven, also, right? So why introduce trains, lambdas and stuff?

Speaker 3 1:01:12

So basically, I think it is for simplifying the code, you know, and for, like, durability, right?

Unknown Speaker 1:01:24

Okay, so,

Speaker 3 1:01:27

basically, you know, you know, like, Java eight is a, you know, it's a long term support and doing these features, right? So it, you just have it, just make sure that you have, you know, detail and a proper documented approach right for working with collection. I have a like it has a proper efficient way, you know, right to write your code in a, you know, readable way, you know, like using like arrow operator. You have different, you know, wrapper class as well to make sure your objects are like, not safe, for robustness and so on. So I think that's why they did for our long term support on Java. Do you

Unknown Speaker 1:02:12

know what, what a functional interface is?

Speaker 5 1:02:18

So basically our, you know functional interface, right?

Speaker 3 1:02:23

So, basically a functional interface, you know, it's a kind of interface that has exactly one abstract method.

Unknown Speaker 1:02:34

So, what is the, what is the use of a functional

Speaker 3 1:02:40

interface? It can be used like, so, basically functional interface, you know, it is used as, like a, you know, run evil, or like coil ball. You can use with it with the, you know, Lambda expressions, right? So, basically, you know, like, it enables lambda expression by, you know, providing a target type. And like, basically, it can be used where your functional interface are expected, right? So, you know, using it like it reduces your boilerplate code, right? So basically, like, if you have something like, you know, if you have something like a predicate, right? So you can, like, pass in a value and then get a Boolean back, right? So you don't have to do a lot of boilerplate coding for calculations that, like, particularly Boolean flag. So you have a functions, let's say that takes an argument and return different types of value, right? So that's kind of like what the functional user, functional interface is, right? So basically like code readability and reducing the boilerplate code, right? Okay,

Unknown Speaker 1:03:54

so what about

Unknown Speaker 1:03:59

have you heard of default methods,

Unknown Speaker 1:04:01

default methods, yeah.

Unknown Speaker 1:04:07

So what are default

Speaker 3 1:04:11

methods? So, default methods, yes, basically, you know default methods is also, I would say, are new features introduced in the Java eight, right? So basically, you know, it enables the addition of method, or, you know, without breaking the existing implementations, right? Or, let's say, right. So starting on Java eight, there was a huge push for, you know, a backward compatibility, you know. So basically, you can get your interface to, you know, evolve over time, and it also, you know, maintain, like our backward compatibility, right?

Unknown Speaker 1:04:51

Okay, so now

Speaker 2 1:04:54

suppose our interface a, okay, which had a default method called Hello world, it prints out hello world a, okay, I have an interface B, which also has a default method called hello world which prints hello world B, right now, I have a class C, which implements both interface a, interface B. What would happen in this case?

Unknown Speaker 1:05:20

So

Unknown Speaker 1:05:21

can you

Speaker 5 1:05:24

so basically, you said, like you have a two interface with the same default method, right?

Unknown Speaker 1:05:34

But your implementations

Unknown Speaker 1:05:39

are different. Suppose, yeah, interface a has a Hello World method.

Unknown Speaker 1:05:44

Prints out. It prints out, print color, hello world.

Speaker 2 1:05:48

Interface B has also got the same method signature, hello world, but it sends out hello world B.

Unknown Speaker 1:05:57

Now interface B implements both A and B.

Unknown Speaker 1:06:01

So what will happen in this case?

Speaker 3 1:06:03

So I think both the method A and B have the same measures. You know, there will be a conflict, and you need to resolve that. You know conflict by overriding the Hello World method, because, you know, Zara compiler will produce in your due to like, MBS method, okay,

Unknown Speaker 1:06:28

okay.

Speaker 2 1:06:29

Can you? Can you share your screen with some examples?

Unknown Speaker 1:06:34

Sure, so do I share my screen? I

Speaker 5 1:06:45

so do you need any particular app to be open? Or I can just use just notepad.

Unknown Speaker 1:06:59

So can you see my screen?

Unknown Speaker 1:07:02

Yes, yes, okay,

Speaker 2 1:07:05

Jim, you have a user object, okay. User object has first name, last name,

Speaker 2 1:07:17

and it has a string, okay, it has a list of strings which are addresses. First, I'm asking him, are strings, the list of strings which are addresses? Can you using strings? Right? You have a list of users, okay, using streams. I want you to iterate over that user list and get me a list of unique addresses.

Unknown Speaker 1:07:46

How would you do that?

Unknown Speaker 1:07:50

So

Speaker 2 1:07:57

your list is there, it's already populated. You just have to iterate. Or it can get me a list of unique addresses. So

Speaker 6 1:08:04

basically, I think we can achieve that, I think using the flat map.

Speaker 3 1:08:11

So basically, fat map is, you know, it is used to flatten the map, or, whenever you have, I mean, you know, like it's used, it's used as a it's used to patent our collections. So whenever you have, you know, you have an embedded collections, you can, you know, change that into a different kind. So basically, if you have a list of users,

Speaker 6 1:08:49

so basically, you know, I will convert the user list into our screen,

Speaker 3 1:08:58

and, you know, then update our, you know, flag map. So basically, like, what I'm going to do on a flat map is, like, I'm going to, like, extract those addresses, right? So basically what I can do here is,

Speaker 6 1:09:33

and then I can do a, you know, distinct method on the stream, because, you know, my stream is converted into a list of addresses, okay, and, and then I can, you know, Call it this,

Unknown Speaker 1:10:01

okay, what if I want to get

Speaker 2 1:10:05

the address and the count of how many times it has occurred.

Unknown Speaker 1:10:17

So

Unknown Speaker 1:10:19

you want to get a map now, right?

Speaker 2 1:10:22

Yeah, instead of, instead of returning unique addresses, I want you to get for every address, how many occurrences have map key should be the address value should be the count of that occurrence.

Speaker 6 1:10:40

Okay, so for that, basically, again, we will use the flat map, but then we will, you know, we will be, you will be using our grouping and then do the count meters provided by the collection interface,

Speaker 5 1:11:06

yeah. So basically I can collect using collectors star grouping, okay.

Speaker 6 1:11:20

So you know, basically the key, it just going to be the address,

Speaker 3 1:11:30

and the you know, value is going to be the count. So

Unknown Speaker 1:11:46

so, yeah, I think

Speaker 6 1:11:48

that to take care of this and it will return our map of stream.

Unknown Speaker 1:11:57

Okay, so you

Unknown Speaker 1:11:59

use Collections Framework, right?

Speaker 2 1:12:04

So, can you tell me a use case where you will use a ArrayList, and a use case where you use a, suppose a linked

Speaker 3 1:12:17

list, basically linked list and ArrayList has a very specific use case, right? So ArrayList, you know, is backed by an array, and then you can access items based on an index, right? So basically, if you, if you know, like, if you want to read our data at a particular index, then you can do so fast by using ArrayList right and like it can automatically, you know, resize when you add. So whatever you want to have, like, fast access, like, at least within what you want to do, an ArrayList, but now on the, you know, linked list, but now on the English is basically a pointer, right? So basically, you know, it has a pointer to the next element. So, like, if you want to frequently, you know, add or remove elements towards the like beginning or the middle of the list, then you you linked list will be preferred, right? Because you are just, you know, basically updating the pointers, rather than, you know, moving the elements, like we will need to do in the ArrayList.

Speaker 2 1:13:38

Okay? And have you used map?

Speaker 3 1:13:44

Yeah, yeah, I have worked on map. I just, like, just a key that appears where you can go get the have you using a key? Okay,

Unknown Speaker 1:13:56

so

Speaker 2 1:13:57

in the map, is there an order maintained in the map. So,

Speaker 3 1:14:06

in a map, so the order, I think has map does not maintain an order, right? So basically, the, you know, elements are stored because, based on the elements are stored based on the like has code than the other, right? So it will be of random order, right? So I think there is a link has map which maintains, you know, the insertion orders, right? And there is also a, you know, a tree map which maintains orders on, you know, natural ordering.

Unknown Speaker 1:14:55

Sorry if I

Unknown Speaker 1:15:00

want to maintain ordering, but what do I need to do?

Speaker 6 1:15:05

So, basically, so you need to use the

Speaker 3 1:15:13

link HashMap, you know, in sit on the regular,

Speaker 2 1:15:18

okay. So is a hash map. Practice

Speaker 3 1:15:26

has map, no hazmat, like, are not the trade safe, you know, basically, you will have to provide, you know, synchronizations, like, if you want to use hash map, but there is a hash map, you know, which is called our concurrent has map which is our thread set right.

Speaker 2 1:15:49

Okay, if I want, if I want thread shape, implementation of the map more vertically.

Unknown Speaker 1:15:57

So,

Speaker 3 1:16:01

so there is a hash map, you know, it's called concurrent hash map, which is a which is trade safe and like, which is trace safe. Okay.

Unknown Speaker 1:16:15

Have you used a collection of synchronized map? I

Speaker 3 1:16:25

um, directly, no, not directly. Like, whenever there is any like issues with the concurrency for using a map, you know, like we work with concurrent hasmap Because, like improvise and like internal synchronizations, you know, where you don't have to explicitly, like, specify, like,

Speaker 2 1:16:54

what's the difference between separate hash back and then you say corrections or synchronized path? I

Unknown Speaker 1:17:03

So,

Speaker 3 1:17:07

so basically, you know the differences. So the difference, the basic difference between the like concurrent has map and synchronized map, you know, is that like concurrent hash map is for, like, let's say, for a higher concurrency, right? Where it allows, like, multiple trades to, you know, access the map with a better performance, right? And it has, you know, non blocking reads, right? Basically, you know, it provides a trade safe read operations, you know, without a blocking like, what I mean, which means that that you can perform different, you know, read operations, but your synchronized map, it has a, you know, it has our blocking trades, right? So, basically, all the operations are, you know, block whenever you use this implementations, right? Okay?

Unknown Speaker 1:18:08

Okay, did you use Spring Boot for your app?

Speaker 3 1:18:12

Yes, I have used like Spring Boot. So our applications was like Spring Boot, Spring Boot based micro services.

Speaker 2 1:18:22

I Okay, so, why did you use Spring Boot? Why not regular spring,

Speaker 3 1:18:27

um, like, like, basically Spring Boot, I will say, like, it's a enhancement on spring, right? So it is really easy to set up, like, very pro bars, you know, convenient over configurations approach, you know, with labels like you have, like you have, you you don't need to do a lot of like, boilerplate configurations like, as compared to spring, right and then in, then the springboard, you know, provides auto configurations, you know, which automatically configures, like different beans and different dependencies, like, available on our class path, right? So it has also, like embedded, like a default Tomcat server, so that you don't have to, you know, rely on the external solver, right? So it provides a lot of products and ready features, like, like actuators, our configuration servers that you can, you know, just add the dependencies and like stars. So yeah, that's why we use the Spring Boot framework.

Unknown Speaker 1:19:30

Okay, does it

Unknown Speaker 1:19:37

the Spring Boot

Speaker 2 1:19:40

need an application or server to run, or how does it run, actually?

Speaker 3 1:19:46

So basically, you know you can have an invaded top Tomcat server on your Spring Boot, right? So you don't need an external server. So basically, on your pond XML file, you know, you can just say, like, what you may what's what servers you want, like, what Tomcat servers you want,

Unknown Speaker 1:20:13

and it will be available, right?

Speaker 2 1:20:14

What's the default scope for being created in?

Speaker 3 1:20:21

So it is a no, it is a single item, meaning that, you know, all the instances will have the same instance of bean, right?

Speaker 2 1:20:34

Okay, so now let's assume one case, okay, I have

Unknown Speaker 1:20:41

been which is defined as a prototype, okay,

Unknown Speaker 1:20:45

singleton Being called S.

Speaker 2 1:21:00

Okay, now you have auto wired p, which is a prototype into S, which is a single right. Now, every time I use P, will I get a new instance of P that I use P within S? Do

Unknown Speaker 1:21:19

you understand QUESTION

Speaker 3 1:21:23

Yeah, yeah. Like, yeah. So, if you, like, have a, you know, a prototype, prototype been and you auto wire. It is as a like, single ton each time you use like, within d, U, G, N, S, that you will new, you will get a new instant of like T, because like prototype, because of the prototype scope. So when our B is defined, right, it is a new instance every time it is requested from the like spring container, right? So it's not like based on your class, but whenever, like, you grab it from the container, right?

Speaker 2 1:22:05

Okay, so if my prototype is auto wired into S, into the single type, every time in a method, if I'm using that prototype, will I get a new instance of that prototype, or be like using the same instance? So

Speaker 3 1:22:23

so no so on a new instant is, you know, created whenever the class request for that bin from the spring container, right? So, if you have, if you already have, like, auto wire into the single it in s then, like, it will be using, you know, it will be using that particular instance, right? So it's not going to go to the spring container and like request for that,

Speaker 2 1:22:55

okay, now suppose, suppose, every time I use that beam, right, I want to get a new instance of the team. Okay. How would I achieve that? In the forget about auto value, right? You remove the auto now you decided, okay, if I put auto wire, then I'm using the same instance. I don't want to do that. I want to get it fresh every time. Or column method on the team. How would I do that?

Speaker 3 1:23:23

So basically, you know, you can use the like application context, right? So spring provides an application context where you can get the beans based on the type, and that way you can get our, you know being based on the type key, and you know every time you know when you request been using that application context, it will get a new instance right.

Speaker 2 1:23:55

And how did you do transaction management in your application?

Unknown Speaker 1:24:02

So

Speaker 3 1:24:06

basically, like, we use spring transactions, using the transactional annotations,

Unknown Speaker 1:24:13

okay, so, how does it work?

Speaker 3 1:24:16

Okay, how does that work? So, basically, you know, the you know, Spring Boot, like provides that transaction features, you know, and basically you can, like, annotate your method with a transactional, right? So where you can, you know, manage those transactions. And like, declaratively. And like spring hand also transits. There are different crystal transaction boundaries on that method which executes, right? So basically, you know, you can define how the transactions should behave with the respect of the transactions, like, like, you know, you know you can provide what kind of, you know, isolation you want, what kind of like a rollback mechanisms you want, right? So,

Unknown Speaker 1:25:07

how, like when I went, how does it work? I mean,

Speaker 2 1:25:11

how does spring know that it has like this, putting easily transaction it should know what Transaction Manager to use, right? How would it know that is there any continuity to do so, so,

Unknown Speaker 1:25:35

so

Speaker 3 1:25:37

I don't think you Know you need to do any specific one, I think, for just being, you know, JPA, it uses, like a JP, JPA, transaction managers, right? And because like it automatically available in your class path, right? And like it will use the you know, JPS, you know, like transaction managers, right? So it will be responsible for starting between, you know, committing and rolling back the transactions.

Unknown Speaker 1:26:13

Okay, there's

Speaker 2 1:26:15

what if you have to do transaction resources. Suppose you have database, and suppose, say you have a MC server, you're getting a message, you're writing to database, writing to LQ, in this case, for how would you manage transactions?

Speaker 3 1:26:36

So if you have a two server, and you want to handle the transactions, then you probably, you know, you probably need to do like a two phase commit, right? So basically, it is, you know, like a protocol used to ensure that you know your distributed transactions either commit or roll back across the same systems, right? And you know, you can use any kind of APIs, like Java transaction APIs, you know, to basically handle this,

Unknown Speaker 1:27:13

right? So does spring provide any any

Unknown Speaker 1:27:16

distributed transaction managers

Unknown Speaker 1:27:21

in spring.

Unknown Speaker 1:27:27

So I am not sure about that.

Unknown Speaker 1:27:32

So I think

Speaker 3 1:27:34

you know the Java transaction APIs is provided by the spring, right? So it does different JP, JP integrations. You know that spring already has.

Speaker 2 1:27:53

How did you do transaction management in your app? You don't have multiple different transactions I mean multiple transaction resources. I transactions.

Speaker 3 1:28:04

So we know, like, you know, like we have a transaction across microservices, you know, which was handled by like, saga orchestrator. So we use saga to, basically, you know, handle those transactions between a multiple services, you know, but it's our, each of our, you know, micro services, each of our, like, you know, micro services was only, you know, working with a single server, you know, either single database or other sources, right? So, within each component, like the transactions was, you know, handle chose the transactional annotations, right? But across the overall microservice infrastructure, you know, we were using saga, sorry, what was it?

Unknown Speaker 1:28:59

I mean, the saga pattern, okay, hold on one second.

Speaker 2 1:29:18

So have you been involved with any like middleware, kind of setting up build and all that. Or is there a separate team which does that?

Speaker 3 1:29:28

So we do have our, you know, separate team responsible for handling our deployment, but, like, I have, like, assistive them, right? So they give you and, like, abstract sums, and you know, you just need to incorporate that abstraction into our groggy script, right? So basically, you know, like we define, like what kinds of commands we need to run for our applications to build, right? So provide them like they provide us what our provision tools we need and like what we implement in our own project.

Unknown Speaker 1:30:06

Okay? Is it easier for easier project in production?

Unknown Speaker 1:30:12

So, yes, yes. So yes,

Speaker 2 1:30:16

okay, so if some issues happens in production, right? And some process support team features up to

Unknown Speaker 1:30:24

you, yeah. So

Unknown Speaker 1:30:25

basically, you know,

Speaker 3 1:30:30

so basically, in that scenario, there is a number of things that we do right. So in that scenario where we just, you know, rolled out of a release for our team, so for our project, and there is a production support issues related to the release, like we right away, you know, roll back to the previous stable versions, right? So we do have a proper branching strategy in place, you know, so that whenever our release causes issues for the external clients, we have, you know, and options to, you know, roll back to the previous versions, you know, so that this is one of the biggest one that we do. The next thing is, you know, whenever there is a feature specific issues, right, then we prioritize different fixing, like, try to either disable the feature flag or, you know, try to come up with a proper, hard fix to address that issues and then get, get and resolve that as soon as possible, right? So, yeah, I think those are the two things that we focus on. Okay. Do

Speaker 2 1:31:49

you have any questions for me? I think Jason will talk to you in detail about the projects and stuff rather than that. Do you have any other questions for me?

Speaker 3 1:31:59

Yes, I would like to know a little bit more about what the day to day of our developers on the teams will look like at your in

Speaker 2 1:32:14

most of the applications which we develop better, basically Spring Boot apps which, like we have, a bunch of apps which we have, which we are in the process of developing, are already developed. So some of them are deployed on our internal cloud, where some are running on the server, like the architecture wise, it's mostly Spring Boot apps, with JTA, hibernate and backtech being opera.

Unknown Speaker 1:32:46

So there's

Speaker 2 1:32:47

a separate front end team which takes care of the UI stuff. It's mostly built in Angular, right? I don't know what, what, what role you're being interviewed for, but there's a separate team which also takes care of all the CICD pipelines and all that stuff. So basically, there's a cohort services team which actually prepares like reusable components which can be used by all these different projects.

Unknown Speaker 1:33:18

That's one team manipulates and leads that

Speaker 2 1:33:22

I'm really not sure which which team you're interviewing for. So

Unknown Speaker 1:33:30

first time this is that it was most of the kind of space. Okay, I'll move into the

Unknown Speaker 1:33:53

Thank you. Take the time to turn.

Speaker 1 1:34:14

I know, let me just start by you asking about the role right in the team. So, right, okay, so let me, let me just share that with you first, and we can kind of jump in, right. Okay, so, so this is a core Yawa durama and develop engineering role, okay, so, so I think Adam you recruit should have mentioned that you're ready, but if not, and it's fine.

Unknown Speaker 1:34:52

So basically, the team is

Speaker 1 1:34:58

seven people, right? And it's responsible for

Unknown Speaker 1:35:05

maintaining a core

Speaker 1 1:35:09

library by several core libraries and that we have and Java based library has common functionalities to address single sign on logic and caching logic, emailing logic and

Unknown Speaker 1:35:33

ignite caching Logic

Unknown Speaker 1:35:37

and some logic for handling

Speaker 1 1:35:40

any API getting for the cluster on the container and entitlement logic, icon logic, and so that's, that's the main library that we can actually spend more than 20 Different on the projects so household needs to handle that any vulnerability to come up, and security need to fix any strong job, but not just where the code right. So because the library share components, core technical implementation in the library, and so that's one library and several other modules and the common services using the library to handle the portal,

Unknown Speaker 1:36:31

user preferences and menu navigation,

Speaker 1 1:36:35

another shared service that we maintain. And then there's another generic VPN Business Process Management module as well and retain which also used to that initial library that we prepare.

Unknown Speaker 1:36:56

And then there's

Speaker 1 1:36:59

a pure API, library or handling

Unknown Speaker 1:37:05

object. Score,

Unknown Speaker 1:37:08

it's a hash core.

Speaker 1 1:37:14

It has content provided object score, Amazon API, and then there's several other utility modules. So maintain web for doing, providing real time and WebSocket blogging, real time block viewing. So then another one for monitoring and allow the monitor to check the application Help page for different modules. All of them, the module will run on Google, shift right and continue. So that's that implementation part maintain the library and understanding and we also need to re implement their workflow components to be more generic right now, multi dimensional, multi stage in theory, each Stage programs.

Unknown Speaker 1:38:17

The second part is

Speaker 1 1:38:21

DevOps engineer, not operations engineer and the automation engineer heavily understanding batch scripting. It's not, I'm not talking about basic to log into Linux and Vi, not that I'm talking about able to write scripts, right? It's not on basic scripts than to you know, CAD and PS and Czech process knowledge, but quite complicated automation scripts, which will these scripts can be used for deployment purpose and deployment for startups, and the container or non containers,

Unknown Speaker 1:39:06

and

Unknown Speaker 1:39:08

handling

Unknown Speaker 1:39:10

other utilities. Actually, we currently, we're doing something with

Speaker 1 1:39:15

password voting sort of descriptions to perform voting activities and pushing the

Unknown Speaker 1:39:23

client secretaries to different modules,

Speaker 1 1:39:27

different modules within the platform, and can be traditional VM servers can be across 1020, 30, whatever number what we should contain the path that is to go that eternity will be doing all that right again, but there are several other utilities to that tune for clean up the images and stuff. So any questions so far on the tasks that we handle?

Speaker 6 1:39:58

So, yeah, no. Like, maybe like, I'd like to

Speaker 3 1:40:03

know a little more about the risk release strategy, like, like, for the whole day releases. Or, like, how often are the releases?

Speaker 1 1:40:16

Yeah, yeah. Okay, so we also participate in a release as well, as I said, our own modules that need to be maintained, right that follow, maybe it's every couple of weeks, depends on whether the issue or something should be added. And we also will be into if there's some changes to the deployment scripts which will impact many modules. So we need to get involved to set up and also network set to mention that so the network working with networking to set up Single Sign On setups right, have single sign on setups, configuration, Global Trust and low balance Setup engineer, how to validate and

Unknown Speaker 1:41:18

also network

Unknown Speaker 1:41:23

and single panel mentioned,

Speaker 1 1:41:26

and of course, keeping up with the container deployment configurations that's completely ongoing, adapting The same attaching how to configure the MAOP teach the development teams so, so that's the reason why they need a very strong development but not just somebody know how to write code really deep level and not just coding, not just coding, but then you Understand the network and Linux server and batch scripting. So any any questions on that, like, I think, Okay, if that's something you have to appetite for, then you can continue, right? So I'm not sure I want to hear from you, is that something you have done, I have interest

Speaker 3 1:42:28

so, yeah, I think, like, I'm comfortable with what you have mentioned right now, because, like, that's kind of like my career goals right now. Like, I want to take you know, like, higher responsibility and then make a you know, better system that is, that has overall good architecture, right? Overall good if I structures, right? So I have a proper experience from my brass project to not just like a coder, like who can write a code, but I am like a problem solver, right? So if there is, if there are any tasks specific to the problems that it's really challenging, like I like to tackle that, you know, and I'm always, you know, up for a good challenge.

Speaker 1 1:43:11

Okay, okay, so let's, let's dive in. So now we get back here the way, so you understand what we do, right? Yes. And basically it's a combination of coding and also infrastructure, networks,

Unknown Speaker 1:43:28

everything. So okay,

Speaker 1 1:43:34

then let me just ask you the coding aspect of things. First, you looks like you have done Java 17, and

Unknown Speaker 1:43:48

that's good. So let me ask you,

Speaker 1 1:43:55

let's say in terms of maintaining the covid or any sort of liability security violation, let's say our security can produce a result dedicating library XYZ and needs to be upgraded to some version from 1.2 to 1.5 and so you have this library of different modules I need to upgrade to find out how to fix.

Unknown Speaker 1:44:29

So tell

Speaker 1 1:44:30

me. Describe to me, how would you go about fixing that?

Speaker 3 1:44:38

Okay, so you know, basically whenever there is a library based vulnerability, right? So it's straightforward fix, right? So basically, on here, upon the file, on on file, on the XML file or gradle file, what you need to do is, like you just users, need to update the version of that dependencies, right? So usually our dependency has a three tags, right? So it has a major versions of minor versions and our past versions, right? So usually those vulnerabilities are like remediated on the past version, so it has no direct effect on your project, right? So it's usually, you know, a backward compatibility, I would say, meaning that you can, you know, update it without an issue. Now, when it goes like from like a past versions to you can see a minor version, there might be some of the duplicated libraries, right, that you might want to consider moving but then there is a major versions like, move like, if you are moving from, let's say 1.1 to, like, let's, let's 1.1 to, let's Say to 2.0 right? 1.2 to 2.0 so you basically, whenever doing a, you know, major version move like if you are moving from if you are doing a major version upgrade, right? So you have, you have to kind a lot of migration before. And for this, I usually like to go to the documentations right. Look for the migration guide and then follow that right now, for the a smaller pass version, I usually go to the, you know, map and repository website, and then I find that specific persons that does not have the vulnerability, and pick that persons into my ability tools, right? So you we usually have, like, property sections where the you know person for the each dependencies is defined and, like, I would just port and like,

Speaker 1 1:46:53

so you're, you're you're giving, you're giving some library and some version that needs to be upgraded. Okay, how do you know which dependency belongs to, right? So sometimes you might not be including that differently directly. It might be embedded in somewhere else. So how do you find out where which dependency will be impacted?

Speaker 3 1:47:14

So, come from, yeah. So, um yeah, basically, right. You can run the like if, if you are depending on the Marvin or Gradle, right? So you have a specific command on each of these tools, right? So you can run on, you know, dependencies three command on the Mavi that kind of gives you a graph of where that vulnerability library is coming from, right? And based on where it is coming from, you can, you know, either exclude that library completely, right, or you can update the update that version of that library, right?

Unknown Speaker 1:47:56

Okay, let's say that library belongs to some

Unknown Speaker 1:48:01

parent library, right?

Speaker 1 1:48:03

So, so that children, child library, that's a version which is needs to be upgraded. So what do you need to do? You define the parent, upgrade the parent to some new version, or do you exclude, include, what's your approach? Like? It's all depends on, you know, it might be complicated. It might not be just matter updating the public. It's like a parent child, maybe multiple layers. And, you know,

Speaker 3 1:48:33

so yeah, for easy, for ease, like, I will just update the side component, right? So that, you know, I do not go and be stopped with the parent, right? But depending on, like, how major the upgrade is, right? So if you are, let's say doing some kind of major upgrade, like the upgrade of Spring Boot version. So then I will go ahead and update the parent and in propaganda, that changes, right? So, but if it's something that is really minor, like fastest XML, let's say a Jackson data bind, then I will basically, you know, upgrade the child component scheme.

Speaker 1 1:49:18

Okay? So you mean, if the child component is embedded in the parents, they will be excluding from the parent and including directly with a new version. Is that what you mean?

Speaker 3 1:49:30

So, yeah, so if my, let's say, let's say like, so let's say, if I have applications, right, that uses our parent component or as a dependencies, right? Same thing, you do not explicitly need to exclude it. You can just, you know, add the new dependencies versions and like it will pick it up, right? So you don't need to change both places. You just need to change on one place and then for the apparent, right? You can stage it for later, right? So where, let's say you have, we you do a multiple changes and then create a new parent version, right?

Speaker 1 1:50:17

Okay, but let's say I have the parent is P, okay, that's considered the P, okay. The C dot gel file as an issue needs to be upgraded. The child, invalid has version of its own version, 1.0

Unknown Speaker 1:50:39

the parent, the parent is also 1.0

Unknown Speaker 1:50:43

the child needs to be updated to version 1.5 so what

Speaker 1 1:50:50

do you do? You look for the parent, your virtual parents, if the child also has a newer version child as well, once you approach them.

Speaker 3 1:51:00

So yeah, if the if there is a you know, parent that has updated version of that dependencies, then I will definitely go ahead and use the newer one. Okay,

Speaker 1 1:51:18

okay, then you mentioned earlier, depends on major minors, but it might be your code might be changed. Some of those clusters might be changed completely. Then it's going to impact your logic, your coding somewhere, your cluster, whatever implementation you have, might break right. That happens right and for example, like Spring Boot two point X version to Spring Boot three security changed.

Unknown Speaker 1:51:52

Internal implementation

Speaker 1 1:51:55

of authorization authentication logics are different, but you have to be careful. For. And it's probably have you done something like that in your project?

Speaker 3 1:52:10

So yeah, like, I have definitely worked on upgrading from Spring Boot to seven to 3.3 $3.03, so this was a major version upgrade, right? And something that I noticed was, I also have to, you know, upgrade the JDK from 11 to 17, right? So I have been involved on this, right? So a lot of packages changes, a lot of import chains and configuration change as well. So, like, you know, we had to completely get rid of Java validations and, you know, and we had to use, you know, proper applications configuration file and so on. So I have, yeah, I am to answer your questions. Like, I have involved in this major operating Okay,

Unknown Speaker 1:53:04

can you describe to me

Unknown Speaker 1:53:09

the loading sequence of three properties,

Speaker 1 1:53:16

and actually two questions. One is, what is that property which controls the different environments that Spring Boot needs to come up, right? And what are the different environments specific? What are different application YAML files being used?

Unknown Speaker 1:53:34

Let's say I have dev

Speaker 1 1:53:38

product, if my environment itself to be repeat with Spring Boot application come up. What is the order of these properties being loaded, and which one will look first, and which one come next, if there's any order, and which

Unknown Speaker 1:53:57

publisher will be loaded.

Unknown Speaker 1:54:00

So

Speaker 3 1:54:02

basically, you know, the default default properties, which is located on the application yml file. So it is, you know, it is a base configuration, and it will be loaded first, right? So once you load those default properties, right? It goes to the profile specific properties, right? So there are like properties file which is named as like applications like yml or like applications like like these are, these are the load index, right? So there are command line arguments like we know, which are loaded, right? So basically, you know, these are, you know, passed through your command line arguments using a key value, peer and the and the next one is the environment variable you know, which are predefined our properties on the environment. So let's say, let's

Unknown Speaker 1:55:02

say, for example, okay,

Speaker 1 1:55:08

okay, so you say, Okay, your application, Yama Willow first, and then your environment specific process to set the Amazon application is key.

Unknown Speaker 1:55:20

Okay, okay.

Speaker 1 1:55:22

Okay, so that means your your UAT, what's going to be in your UAT profile compared to the main default profile,

Unknown Speaker 1:55:37

what do you typically put in the environment

Unknown Speaker 1:55:45

specific. So basically, you know, like

Speaker 3 1:55:49

each environment has its own domain, right? So for examples, like my day database, you know it's going to be from my database, right? So, and also for the productions, right? So, database connectivity, you know, like string, username, password, will reside on these properties for a specific environment. And there are server configurations, like, you know what server port it needs to run on, what context part it might need to have will probably reside on this specific configurations file, right? So there might be like, call like logging configurations, you know, that will be in a specific ones. And there might be, there might be a down downstream APIs, right that it was like, that might be residing on, you know, this profile specific properties file. So, you know, anything that might differ from one environment to the other will be those particular yml file. So there are also different, you know, like feature flags that will leave on these profiles, right? So there are specific features or by Amazon,

Speaker 1 1:57:07

okay, so you'll be demand specific, so in your your key, or whatever the relevant, okay. So basically, and you know, for authentication and auto regulation to protect your API. So how do you, how do you, how does basically work? What do you, and how do you protect your API for make sure that user is entitled and also authenticated something entitled to access that API. Have somebody who can do manage that account? Let's manage that account API, right? I want to make sure that person is account manager. So how do you protect that API itself? What do you need to do?

Speaker 3 1:58:02

So, you know, like spring, security is very strong and configurable framework that you can use to, you know, secure your applications and your APIs, right? So, basically, you know, it can provide different authentications and different authorizations with role based access, right? So basically, you know, it can help you to identify a user using username and password or geography tokens, right? And it can also, you know, assign particulars roles based on those tokens, right? So basically, you know, you can create a configuration filters, right? So you can configure the spring security through those configuration class, right? So now, now your spring will try to, you know, out authenticate the user based on the provided credit crunchieros, right? So if you want to authenticate, and once you have an authorized then authentication users you know will have a rule associated with him. And now, now for each particular APIs, now you can do your different you can do it our role based access using like pre authorized annotations and the then providing the rules on those right. So if you have, you know, any account specific roles that only you know admin can perform you know when you want to do like you know, pre authorized annotations that in points, and, you know, use our, you know, has rule for admin, right? So, or, let's say, if you have something that anyone can view, then you can just, you know, do a pre authorized with has rules of users, or has rules of admins and so on, right? So you can use these rules based on different access,

Unknown Speaker 2:00:11

DevOps and maybe

Unknown Speaker 2:00:15

the platform

Unknown Speaker 2:00:18

you have, Docker

Unknown Speaker 2:00:22

Kubernetes,

Unknown Speaker 2:00:25

tell me what, what is

Speaker 1 2:00:29

when you deploy, when you build and deploy an application to a container, Docker container,

Unknown Speaker 2:00:38

so What, what

Speaker 1 2:00:39

is, what is contained in your Docker content? What is the basic element, let's say active implementation on and run on containers, right? What's what's in your Docker

Speaker 3 2:00:53

So, basically, you know your Docker, like I would say, you know, it's a user packaging, right? So, basically, what it has is like, it has, like your SNAP, sort of the like application code. It has your own dependencies, right? It has a setup of your runtime environment and different configuration files. Like,

Speaker 1 2:01:25

Have you, have you creating this Docker packaging for the container?

Unknown Speaker 2:01:34

Yeah, yeah. Like, I

Speaker 3 2:01:36

have worked with creating Dockers files.

Unknown Speaker 2:01:41

Okay, so,

Speaker 1 2:01:42

so what is the thing you're talking about, application, right, which generates that Google jail file was run on a container. So what you specify in your Docker, like what you need to make that work so it can be filled, and when it gets deployed to the container to start up, and you have everything it needs to,

Speaker 3 2:02:07

okay. So, yeah. So basically, you know, you need to specify the base images, you know, which is usually the first line, right? And it is usually starts with from it through the graph based image for Java runtime. And then you can copy your applications JAR file into the image from maybe your target or something, right? And then you can switch your, you know, working directory to that particular directory you copied the jar to. And then you can, you know, define the entry point. You can define what ports you want to expose right, and then you can basically build and run the Docker image right. Okay,

Speaker 1 2:03:01

so when you deploy to a Kubernetes and containers, right different deployment strategies? What are the different deployment strategies, and what are tech and use cases for

Unknown Speaker 2:03:16

that deployment strategy? So are you

Speaker 3 2:03:23

talking about the deployment strategies, particularly in coordinates, or in general,

Unknown Speaker 2:03:32

in coordinated terminology? For example, we

Unknown Speaker 2:03:37

use automation, which is

Unknown Speaker 2:03:40

partial version

Speaker 1 2:03:43

of Kubernetes management, but the concept of the same Kubernetes concepts, so when you deploy your image to container, you have a different strategy, calling you're creating a part and pulling your image and starting on, What is the business strategy? What are the different use cases

Speaker 3 2:04:03

for each one? Okay, yeah. So basically, you know, what I have been used to is rolling update. So which is the default strategy? You know, we did on you. We did our new board like created and added it added to the deployment, while the old or ports, all old parts gradually like joining us, right? So basically, you know who are need, manages this, update processes, you know, to ensure specific numbers, or, let's say, specific ports are always available, right? And then I have our experience in two more like, which is called the like, blue, blue green deployment, right? So then our Canary, Canary deployment, right? So basically, you know blue green deployment is you deploy a new version in a separate color there, as compared to the old person. And then once, you know, a new person is, like, validated and ready, then we can switch the traffic to the new persons, right? And the older person, the older color, like, stays, you know, active in case a rollback is needed. And there is, there is another one of Canary deployment, like, which is, you know, which, which is used to deploy a new version alongside the older version, but we only route, you know, small percentage of traffic to the new version. And then, you know, we are monitor, and then gradually, you know, increase the traffic to the newer versions. So these are the three strategies that I mentioned that I have been working on. Okay,

Speaker 1 2:05:47

have you worked for any deployment tools like Ansible? XR? Yeah.

Speaker 3 2:05:56

So our team use Ansible for deployment, you know, like, I do have exposure to it as well. Okay,

Speaker 1 2:06:08

so do you understand the different elements of Ansible? Like, what is Ansible? What is the different attributes of Ansible that you need to set up to support your deployment.

Unknown Speaker 2:06:24

So,

Speaker 3 2:06:26

you know, basically your Ansible is, you know, responsible for, I would say, like, I would say just, just a configuration management, right? So, basically, you know what it is, what it is used for, is like simplifying the management of the applications. You know, where you define and execute workflows, right? So basically you know, like you have a playbook, right? That is like defining the automation tasks, right? So basically you provide the host the task you know, and you know different variables and their roles. And then you know, provide the inventory, which defines a group you know that your Ansible will manage, right. And then you can specify multiple modules and like multiple roles alongside with it, right? And then you can, you can define, you can define a templates, which are, you know, files that is used for, you know, templating engine to generate different configurations dynamically, right? So, basically, you know, we are just using our declarative approach to have a modular way to make these tools manage the automations of the deployment networks.

Speaker 1 2:07:55

Do you know describe a good Traffic Manager works, and what are the different attributes you need to set up again? No, I don't mean the actual configuration. I mean the understanding to concepts like global traffic. Can you can you,

Unknown Speaker 2:08:20

can you repeat the question today, what are the

Speaker 1 2:08:28

components or elements that you need to be aware of when you're setting up a load balancer?

Unknown Speaker 2:08:37

And how do you validate let's say I

Speaker 1 2:08:41

want to set up a to set up a development across three different node, right? Sure, if one goes down, another one will pick it up, right? But then, how do you configure it? But you behave that way? I understand

Speaker 3 2:09:02

it right now. So basically, you know, whenever you are setting up a load balancers, right? So you basically make sure you have a proper set of targets group, right? So the target groups are, you know, a set of targets, which are, you know, either easy to or, like, easy to or, let's say lambda, right, or so on. So where the load balancer, you know, it will drop the traffic to the target groups and are set up with the, you know, healthy configurations to make sure you know your applications is, let's say, healthy. And there are also listeners, right? So there are listeners, and these listeners are, you know, used to handle incoming requests, you know, and then route them to appropriate target groups, right? And then you have the security groups, basically, you know, security groups this life, you know, for concluding the graphics and it controls the inbound and outbound traffic to the load balancers. So if your database needs to, you know, connect to that load balancer targets, then it needs to, you know, make sure you have a proper rules, and rule saves right. And then you have, you know, your DNS configurations, because you know the load balancer is usually assigned to a DNS name that you can use to, you know, route traffic to you. Maybe also you configure a DNS with a route like 53 so that kinds of like different components that needed for setting up a simple load balancer, right?

Unknown Speaker 2:10:43

Okay, okay.

Speaker 1 2:10:48

Now, scripting, fast, scripting, how much of cooking have been done? So I don't see that when you read much yeah,

Speaker 3 2:11:03

yeah, I do some scripting, you know, not a whole a lot, because we do have a, you know, dedicated teams that take care of it. But I know how to write sales script for multiple cases. You know, I can write, I can log into the Linux boxes and then run commands, you know, to check the status of the different services, you know, CD file systems and

Unknown Speaker 2:11:32

commands that you typically use.

Unknown Speaker 2:11:36

So typically,

Speaker 3 2:11:38

like some of the commands. So, like eco, right, CAD, like grip, like move command, like copy command and like likes, CS, more command, right? So there is also PS, if I remember, like system CTL to name or some other

Speaker 1 2:12:03

What about when you're writing script, and how do you how do you organize the parameters being passed into your short script? And how do you pass them? How do you determine what trends and then perform different actions?

Unknown Speaker 2:12:29

For example, I might have a shortcut app.

Speaker 1 2:12:34

My app that sh, it's going to ad manager, it's going to do different things. Have a promise that dash, dash might have name and have action. How the action can be like, deployed, start, stop, but sometimes that makes people and so how do you

Unknown Speaker 2:13:00

extract these parameters in the script.

Unknown Speaker 2:13:05

Um, so

Speaker 3 2:13:09

to extract the parameters. So, yeah. Basically, for that, you can use, you know, like Git ops for options parsings, right? So basically, you know, what you do is you can try to get ops by using wild case, and you can, you know, grab those parameters. So you can, these are the basic way, like, you know, like, this is when,

Speaker 1 2:13:37

when was the last time you written the script?

Unknown Speaker 2:13:42

So, the last time,

Unknown Speaker 2:13:45

I think recently, I had to

Speaker 3 2:13:47

write our, you know, Canary lambda. And I had to write a sales script for that, Canary lambda, the basic youth, youth case for that, you know, Canary lambdas, was to do a handshake endpoints, and I had to, you know, write provisions like I have to write a different provision skill for our Jenkin files to deploy this. Lenders, okay, okay,

Speaker 1 2:14:19

so I know your company. Let me ask you, why do you want to leave your current job?

Speaker 3 2:14:25

Um, so my current job? So, yeah, currently, you know, my project is ending, so that's why I am looking for a new role to get involved into. Okay,

Speaker 1 2:14:39

and what about location wise, like? So this position is for either New Jersey, Chicago, Atlanta or pallano.

Speaker 3 2:14:54

So I mostly prefer east coast. So new New Jersey, but like, I'm open for the business as well, but I mostly prefer East Coast New Jersey.

Unknown Speaker 2:15:09

You prefer where?

Speaker 3 2:15:10

East Coast New Jersey. East Coast area, yeah. Okay, East Coast, okay, New Jersey area, New Jersey is my preferable, but like, I'm in control the location. So,

Speaker 1 2:15:24

okay, because your resume said I read about Plano, so you say Plano or New Jersey? Yeah, okay, what's, what's your preference? Um,

Unknown Speaker 2:15:37

I would say New Jersey is my preference. Okay? That's a

Speaker 1 2:15:47

question I had. Thank you very much for your time. I'll get back to you. Recruit, back to Alan,

Unknown Speaker 2:15:57

okay, thank you.

Speaker 3 2:15:58

Thank you so much for your time as well, like it was no nice talking to you, and I hope to hear back from you soon, and thank you for the opportunity.

Unknown Speaker 2:16:09

Okay, thank you so much. Okay, thank you so much. Bye.

Speaker 6 2:16:31

I prefer New Jersey, but I'm ready to look At anyone wondering Texas plan. On plan

Unknown Speaker 2:16:49

on Texas. I

Unknown Speaker 2:17:21

location, I

Unknown Speaker 2:17:30

Suicide,

Unknown Speaker 2:17:53

you audience as

Unknown Speaker 2:18:40

had seen my password as Sunday. Average, 123, s capital,

Unknown Speaker 2:18:46

one to Three. Yes, I mean,

Unknown Speaker 2:19:00

hi. Hello, Maya.

Speaker 8 2:19:24

Hello, giggles. What do you call? A cantaloupe? Underwater, a watermelon?

Suman Astani-Bank Of America- Srv-Final-Offered otter ai (1)

Speaker 1 0:19

Okay, it looks like it almost There. Just get his audio. I

Unknown Speaker 0:56

Hello, can you hear me?

Speaker 1 0:59

No problem, yes, I can hear Ramesh. Adam and Suman, can you guys hear Ramesh? Yeah, I can hear you. Yep, I can hear okay, cool. So good luck to you. I'm gonna make you the co host. Okay,

Unknown Speaker 1:15

thanks, Jason, I'm dropping Okay. Thank you guys. Okay, thank you.

Unknown Speaker 1:21

Any one minute sure, take care. No

Unknown Speaker 1:23

worries. Take your time. You

Unknown Speaker 2:11

okay,

Unknown Speaker 2:18

let's start. Why don't we

Speaker 2 2:20

start with your introduction. Okay, sure,

Speaker 3 2:26

yes, sir. So my name is Suman. So you know, like I have been working as a Java developer for about five years now. And you know, like I have been primary, how many years?

Speaker 1 2:37

Five years now. Five years, five years

Speaker 3 2:41

now. So I primarily, be, primarily been working on a Java based web applications using, you know, frameworks like spring and like hibernate and like creating web services for like API development. So I have been, you know, like, primarily working on, like, microservice architectures, you know, and been helping in like designing and implementing microservices and like deploying them to AWS, like some of the AWS services that I have worked on, and that includes like, you know, like Ed like AWS, like that include easy to like ECS, right? So lambda, so, like, I do have experience working on the front end side with Angular. So to talk about the testing framework, I have worked with JUnit and like a Mockito for writing different unit test cases, right? So I am currently in a scrum team, and my team consist of, you know, five developers, so we have a product owners and Masters on a daily basis. So on the daily basis, you know, like I am involved mostly working on the back end Java code, I would say I'm helping the team with, you know, our code reviews and also production support duties. So that's my overall experience.

Speaker 4 4:02

Okay? How do you rate yourself in Java in the scale of one to 10?

Unknown Speaker 4:08

How do I rate myself? That's kind of hard questions.

Speaker 3 4:12

I would say like around 7.5 to eight. Okay. Okay. Okay,

Speaker 4 4:29

what was your last project? What was, I mean, the current project, or what or the last project?

Speaker 3 4:35

So currently, you know, like, basically, like my current project is like financial good planner. So that's the main features that I have been working on, right? So, like, we have been primarily working on, like orchestration thereof, like microservices. So using like Springboard framework, like building Spring Cloud applications like this is really bringing you know, like, building Spring Cloud applications is really helpful for, like, orchestrating requests, like requests from the client side to the server side. So like use for request to like manipulations. So,

Unknown Speaker 5:20

okay, so what exactly

Speaker 4 5:22

tell me about what was the last task or the current task that you're doing?

Speaker 3 5:29

So my recent tasks, my current tasks, I will say that like Currently, I have been, you know, creating a filter. So it's called, like tokenization filter. So basically what it is like we have, you know, downstream partners called like queuing. And you know this during is used for changing your secure data, like tax ID or like account number or so on into the like hash value. So, and this is our school APIs that we can use. So basically, whenever you know, the client onboards your route, like, right, based on the you know, that metadata that they provide, you know, like, we can actually, like, convert this sensitive information into a token, right, by calling that services. So, you know, like, basically these filters, like, it's supposed to traverse into the like, JSON payload, you know, that is provided and like, look for the key that is configured as the metadata and their you know, value for the key is converted into the token right before, like, it is passed along to the downstream services. So that was the most recent filter, I think that that I have work on. It's a new requirement so that you know any kind of safety address is prevented

Unknown Speaker 6:58

with respect to the

Speaker 4 6:59

with respect to the actual task, writing the code with respect to what you did with the task. I know you explained the task. Can you guide me through a little bit of code to see what you have done for that particular

Unknown Speaker 7:16

task?

Speaker 3 7:17

Okay, so code basis, right, okay. So basically, you know, like, like, we have interface that all, like, all of the filters, you know, should use our patterns on, right, right. So the do this interface has like, three abstract method so that my, like, on, create a tokenizer. Filters implement on, so on the first one was the order. So basically, this determines, of like, what a particular order. So this filters, you know, it needs to be like, executed, like, right? Because there are, there should be like, 30 or 40 filters that needs to be executed, and it needs to be on the different orders, right? So first thing was that implementing the order, and the second thing that I implemented was the like, filter on tape. So that's the second thing I have the second thing I have implemented so it's so it's called the, it's called the filter method. And here, like, basically,

Speaker 4 8:23

again, right? I don't need, I don't need you to go through the explanation kind of thing. Take one, pick one last task, what? Okay, I have written this class, and whatever you said, filters, right? What did you do in that class to achieve that thing, that particular thing? I want to know what exactly was your task and how you approached it, and what you did in that you don't have to tell me anything that is confidential, or anything of that nature, but I just want to know the process and how you went through the code. What is the code that you have written? I mean, you don't have to extend anything confidential to me. Okay,

Unknown Speaker 9:12

okay, okay, so I tried to explain it.

Speaker 3 9:16

Okay. So basically, you know, like I was tasked with, like, implementing the filter, right? So the main thing that I did on this class was the actual running of the filter on using this. So I basically, you know, use the connections from our like, HTTP, client connections, pool and on this, like, we basically made a, you know, REST call to our downstream partner to, like, tokenize, you know, that to tokenize that particular field. And like, after that was done, like, I basically, you know, replaced the value of the key that was, you know, onboarded with the tokenized value. So that was my overall task. So I was in New I was in charge of implementing the

Unknown Speaker 10:09

spring filters right, Springboard

Unknown Speaker 10:13

filters correct. So it's

Speaker 3 10:18

like function factory. So there's a, you know, a concept of function factory on our Spring Cloud gateway. So it is not a typical filter, but it's a function view that you can run on our routing so

Unknown Speaker 10:43

in spring, right in frame.

Unknown Speaker 10:47

I have a an interface. Okay,

Unknown Speaker 10:51

will build an interface. Okay?

Speaker 4 10:56

That interface, let me call it as a vehicle. Okay?

Speaker 4 11:05

We we have several implementations of those, cutting vehicle. Okay, one is car, one is truck, one is motorcycle, let's say and you have a bus, something like that. Okay, we built our we built our services classes,

Unknown Speaker 11:27

and then we have injected,

Unknown Speaker 11:31

because we have to

Speaker 4 11:34

go to an interface. Are you aware of the principal, object oriented principal, called code to interface? Okay, so we should go to an interface, right? So all our services will say, I know a vehicle, okay, but it does not know what is the type of implementation. So we have an implementation of several implementations correct inside so then Spring Boot, it was complaining that there is more than one so it is not able to associate an implementation to the interface. How do we resolve this problem?

Speaker 3 12:20

So the way to resolve like, I understand your like scenario. So basically, you are talking about like interface, you know, which has multiple implementation, and it's a service, right? So basically, there is a like, a very easy way to tackle this. So there is an like annotations in Spring Boot call the like qualifier annotations, right? So basically, what qualifier does is like it basically specifies, you know, what implementation you want to like inject for that interface, right? So you can simply use like qualify annotations, qualified annotations, wherever you are injecting that, right? So there's also, like another concept right, called the primary right. Basically, if you do not provide you know, any like qualifier, then it can take, you know, one of those pins as like primary beam and like, inject that on this specified right? So those are the two approaches that you can take, but I will say like qualifier is like more useful and handy.

Speaker 4 13:36

So what do you understand by controller advice, okay,

Speaker 3 13:46

like, so, basically, controller advice. It's like concept in springboard, you know, that is used for, like, global exception handling, right? So, basically, you know, like, it allows you to specify, you know, what response you want to send out to the clients for the particular exceptions, right? So you, let's say you have, like, a non form exception, then you can return our, you know, response entity with the response status of 404, or, like, if you have, you know, some, let's say, some kind of genetic exception that you can, you know, send out, like, HTTP 500 with a like, meaningful message and like so on. So that is a controller advice in springboard. Okay?

Speaker 4 14:48

Quality do we use? Is it possible to use? So we have two annotations, right? One is at the right component. Another is at the rate service, right?

Unknown Speaker 15:08

Can we interchangeably use them?

Speaker 3 15:12

So we have two components, so can we interested in the use them?

Unknown Speaker 15:18

Right? No, I'm talking about

Speaker 4 15:21

the annotations called at the rate component and at the rate service cut.

Unknown Speaker 15:30

We didn't know the annotations. Yeah. So basically,

Speaker 4 15:32

why do we use them? Why do we use them, and can we use them interchangeably?

Speaker 3 15:37

Okay, okay. So, basically, you know, component is, you know, a component just tells you, it tells your spring that every component annotations, it tells the steals your screen, that it's a component that your spring bin can manage, right? So, basically, you know, it is a species specifying that, like, it can be used as a spring bin, and service is like a more specific or like a tailor towards like bin that provides some, you know, business logic, or, let's say, some like service functionality, right? So service is used to annotate the service layer. So now, now to answer your second question, you can use a component annotations as our service class, but like it is just, let's say, make it confusing, right? So it is not wrong, but like, you don't, you don't, don't provide like, any specific like tasks for like for better, I'll say for better readability, and like understanding for better understanding, right? So you will use, you will have to use a proper annotations, like, maybe a controller or a repository or like, or a service in C Sharp component, right? So, but if it's a generic you can use a component annotation. So I will say, using the more detail or specific annotations, like it is a better practice, you know, and like it should always be followed.

Unknown Speaker 17:12

That's more to JPA. Can you use JPA?

Speaker 3 17:18

Yes, yes, I have, like I have, I have working on Spring Data JPA.

Speaker 4 17:30

What do you understand by the word projections? In JPA

Speaker 3 17:36

means the town projections, right? So basically, you know, the projections like they are basically used to, you know, like retry only specific columns or parts of the entity rather than fetching the entire entity, right? So, basically, you know, like you can like you basically you can define the like, different projects and interface. And basically defining an interface, you know, with like gators will help you get the properties that you want to retract and like, that's what opposition feedback.

Unknown Speaker 18:20

So I'll give you one sorting problem.

Speaker 4 18:33

You have a list of 11 numbers. Okay,

Unknown Speaker 18:39

can you write a piece of code to sort it,

Speaker 4 18:44

so you can type in the chat message here,

Speaker 3 18:50

so there's 11 numbers, so we have to be an array or list whatever, right? Okay?

Unknown Speaker 19:00

A or list, whatever, right?

Speaker 4 19:03

So it should take your function should take on the list. Also take whether it is ascending or descending.

Unknown Speaker 19:15

It should return the sorted, sorted list. What's your requirements?

Unknown Speaker 19:33

So you just want me to type on the chat, or do I need to

Speaker 4 19:38

just start keeping typing the chart or type anywhere else

Unknown Speaker 19:45

as you finish one line, just Press Enter C,

Unknown Speaker 19:51

so that way I know that You're

Unknown Speaker 20:16

don't use,

Unknown Speaker 20:18

okay, don't use any, any

Unknown Speaker 20:23

already existing sort methods, okay, default method.

Speaker 5 20:38

So you don't want me to use Java stream, sort method,

Unknown Speaker 20:44

no, no, use Your.

Speaker 4 20:50

Got to use your, your your own sorting. You

Speaker 5 21:09

okay, so to write my own sorting, I think I should be able to Do that. Okay?

Speaker 6 24:18

So I think like that should take care of it, right? So, like, but it's a very insufficient sorting, using ball sort, like, you know, where you just go through these numbers and, like, look through it. So okay, what

Speaker 4 24:41

do you call this sorting, the one that you put. What algorithm is this?

Unknown Speaker 24:47

It's called bubble sort. It's called

Unknown Speaker 24:52

bubble sort. Why do you use j plus one?

Unknown Speaker 24:56

So what?

Speaker 4 24:58

So what happens at the end of the array. So,

Unknown Speaker 25:02

basically, you know, like,

Speaker 6 25:06

basically, you know, I am, you know, comparing the current element with the next element until the second last. I don't have to run out of the index. So that's because of that. Okay,

Speaker 4 25:29

I'll give you one thing. Just do this one. Okay, I have a list of address okay,

Unknown Speaker 25:42

I know this problem, desktop

Unknown Speaker 25:50

address list Just on typing, okay.

Unknown Speaker 25:54

Address as

Speaker 4 25:58

name, comma, city, okay,

Unknown Speaker 26:05

you're seeing my chart. Okay. There are address

Speaker 4 26:11

list as 100 elements

Speaker 4 26:22

that has 100 elements. So what we need is we need an output. We need to write a function that will return something like this stuff using streams API, okay. Return the map so you see this one. So once you so string is city and in a key, right key, string is city and the list of addresses. Let's say you and I belong to the same city. If I get a key of our city name, then you and I should be present in the list of addresses. If someone else is there with a different city, it will be different key, therefore he will be in a different list. Can you do that using streams API, convert a address list into a map of key and list of addresses. The key is city in the address.

Unknown Speaker 27:38

Follow me. Yeah, I'm

Speaker 3 27:39

getting, I'm following the requirements. So basically it just all, you know, collection, collection grouping on our like, stream API and like, it's straightforward. I can return, you know, let me, Let me type it for you. I

Unknown Speaker 28:23

So, yeah, that should be okay.

Unknown Speaker 28:30

What is a lambda?

Speaker 6 28:34

Lambda over here. Okay, so basically,

Unknown Speaker 28:40

I will say

Speaker 3 28:42

it like, it's just a like, clear and lambda. It is just a clear and like, concise way, you know, to represent, like, one interface, right? So which basically, you know, it just and like expression that that is, like, really useful in expressing a syntax using like arrow function.

Speaker 4 29:06

How do we store a lambda? What into what? Let's say, store a lambda into a variable.

Unknown Speaker 29:15

Okay, what would be the variable type?

Speaker 6 29:18

So the type of the variable after restoring the lambda and the variable, right? I think, yeah, I think, like it should be a Runnable

Speaker 3 29:27

interface, right, but like it can be any functional interface, right? So basically, like there are running running balls, there are functions, there are predicates and by function. So it's a functional interface.

Speaker 4 29:47

It's a functional interface. That's our answer, right? Something is happening to my computer. Just hold on. What's going on?

Unknown Speaker 30:01

Can you see me?

Unknown Speaker 30:06

Oh, yeah, you are you?

Speaker 6 30:07

Are you are on the call? Like, I can do

Speaker 4 30:13

a video, like, if you're a video, I lost my power. Looks like I lost power. Just hold on one second. I'll try to use another one. Correct another one. Okay.

Unknown Speaker 30:41

Okay, fine. So,

Unknown Speaker 30:59

what's going On? Teacher, tell you what. Boys.

Unknown Speaker 31:36

Functional interface. What's a y function? Sorry,

Speaker 2 31:41

I Sorry, what is the by function? Oh, do

Unknown Speaker 31:45

you ever use it? Yeah, you

Speaker 3 31:46

mean by function? Okay, so basically, like, by function is a like, you know, it's a kind of lambda expression, you know, that takes two values and like return some values, right?

Unknown Speaker 32:19

Okay, okay. Going on?

Speaker 4 32:33

Taking time. Okay, fine. Okay,

Unknown Speaker 32:51

taking time,

Unknown Speaker 32:54

okay, let's come to some multiple

Speaker 4 32:56

team. Okay, yeah,

Unknown Speaker 33:11

why do we need multi threading,

Speaker 3 33:16

multi trading, you know, basically the concept. So basically, you know, monitoring it like this. You know, like splitting your like tasks into different trades, right? So, basically, you know, you can break down your single process and, you know, around it concurrently, which basically, you know, improves your like application performance, right? So, like, concurrent execution allows, you know, like, multiple operations to run, like, simultaneously, you know, which makes the use of those CPU resources that you already have, right? And like, it can help you, you know, like, improve the performance. That's

Speaker 4 34:01

okay. I say you, you wrote a multi threading application. Multi threading is there right now, and server threads are running right one of the thread, one of the thread. I mean, it's misbehaving. It's not it's not being as it's supposed should be. Some problems are happening. Okay, how would you stop that threat?

Speaker 3 34:33

Okay, so, how can I stop that threat? So, basically, you know, like, so you have some kinds of misbehaving trade you can do. So there are, like, trade interruptions, like setting a flag, you know that like indicates whether a trace will stop running or not, right, and check if the interruption take about the interruption status, so periodically, right? So based on that, you can, like, interrupt, stop the trade, and like you can interrupt the trade, right?

Speaker 4 35:19

We wrote a multi credit application, and then we are seeing there are a lot of deadlocks happening all the time. How would you resolve that deadlocks?

Speaker 3 35:33

So basically, to resolve the deadlock in that situation, so are you basically have to, you know, make sure that your you typically have, like, proper deadlocks, like in particular order, right? So, basically, you want to use, like, consistent ordering, right where you enter all the traits you know, that, use that, use that resources, and you have, a, you know, proper locking, and, like, constants order, right? So, like, you have, let's say two locks, like a and b, and then, you know, you always acquire that lock, a before b, like, across all the trails. Then you can, you can also try a lock with a timeout, you know, so that you know you have a timeout after acquiring logs and like, if the locks you know cannot be acquired within that timeout, then the tree can, like, back off and like, retry again, right? So like, whenever you have like collections, right? You make sure you use a proper concurrent collections, like like concurrent has map or like copy on right array, right to make it is a trade safe, right? So, yeah, these are some of the ways you can ensure you have a proper control over deadlocks, right? Okay?

Speaker 4 37:07

Okay, let's say you have a function, okay, you have a method. So what you're doing is in that method, it is making some calls, okay, some service call. Another is a database call, and all that it is doing. One of the database calls. It, if it takes more than five minutes, it should come out. Exit, Exit. Your function. How do you write that? Achieve that way we can answer that. So you understood my problem, right? So you have a method in that method, something else you are calling, okay? But when it is given you called a database function, let's say you calling your database some

Unknown Speaker 38:07

selecting data from okay,

Speaker 4 38:13

what is happening is it is taking a lot of time. Lot of time is more than five minutes. So if it is taking more than five minutes, what I want to do is, I want to come out of that method. See, you made a call to a method. So it is a it is like a hanging that method is hanging. You cannot you. There is no way that you know what is? What you call that word? I can't remember, so you make a call and it is stuck there inside your method. No, it is not coming and executing the next statement. Okay, so if it's going to take more than five minutes, I want to exit from there, exit from the totally out of that method. So how do we achieve this thing? How do you write the code such that you will be able to time out if it is more than I understand

Speaker 3 39:10

your requirements? So basically, what you want to make sure that you wrap your like long running tasks into the into a callable interface, right? So once you have that, you can, you know, submit that task into the executor services, right? And the executor service, service basically returns, uh, like future, like if future is working, but a promise that it will complete in the futures as soon as you right, so, and whenever, like, you do a future like, like, dot gate, you provide a timeout, time that you can, like, specify, like, how you how long you know you can wait the result for, right? And then you know, like, whenever that threshold is right, so it exists, like, out of that processing, so that your resources do not hang up. And you know, like you do not, like exhaust your credit. So that is an approval you can use, right? So, based on the requirement, okay, let's

Unknown Speaker 40:15

go to some database queries.

Speaker 4 40:20

Okay, I have a database, and in the database I have a, let's say

Unknown Speaker 40:29

yes or no column,

Speaker 4 40:34

okay, my mistake, you know, when they did from the UI, when they were supposed to write it as a yes, send it back to the database or the service to as yes, they could like no, but then for the yes, they would know, for the no, they put Yes. So when you now it is after a few months, everybody looked into it, it looks like everything is like yes is NO and NO is yes. In the database, okay, they can fix the code in the front end, but in the database, it is already there, like, yes is a million rows or S 2 million rows are no. But then what's happening? We need to switch them also in the database, correct, I want you to write a query that by running one single query, it will switch both. So you

Unknown Speaker 41:30

want me to write a query?

Unknown Speaker 41:33

Yeah, write a query.

Speaker 4 41:36

How do you update in one single shot, query both? Rows should be updated, okay, yes will be updated to No. No will be

Unknown Speaker 41:50

updated to Yes. So

Speaker 3 41:54

basically, like, what we can do is like, you can update your cable right? And you can, like, set that a flag to be not the flag right? Basically, if there your is or yes and no are Boolean. So, like, it can be stored in their strength. There are strings, okay, okay, so,

Unknown Speaker 42:23

even with how we will do

Speaker 3 42:27

it. So basically, in that case, if it's a string, like, we will want to like, basically like, switch, you know, switch something into a temporary value, right? So basically, I will update my table, and I will, you know, set the quorum based on our case statement. So whenever the whenever that column is s, right, I will change it to a temporary value. And then when the column is, you know, I will change it to s, and then after that it is done. So I will do that,

Unknown Speaker 43:06

so that will take care of, like updating the values, okay,

Speaker 4 43:17

you have a trillion records in a table. Okay, records, we exported all of it into one file. We have a file. Okay, records. Now let's say you create one more table. We want to insert all the tape, all the data into the table. Okay? So, because we have lot of amount of data, we want to finish the job very fast, okay? And also, right, we don't want to do the task A lot of times. I mean, let's say with breaks, I don't want to whatever has been already there. I don't want to redo it again, something like that. But I created 100 billion records have been already inserted, and I don't want to redo the same thing again, right? In the interest of time. What is the best approach that you will follow to do this? Okay,

Speaker 3 44:20

in that scenario. So you know, if that's your requirement, you know where you want to efficiently insert, like I will like probably do like our batch insert, you know, batch?

Unknown Speaker 44:40

Insert. What is a batch? Insert? Yeah. So

Speaker 3 44:42

basically, what I mean by that is, like, what I mean by that, what I mean by that is like, you just mesh the data and, you know, you know, like, you copy that data into a new column, right? And then you do, like, bulk insert into that table.

Unknown Speaker 45:06

You do bulk insert into that table,

Unknown Speaker 45:11

bulk insert.

Unknown Speaker 45:15

What is bulk insert?

Speaker 3 45:18

So, I mean, like, basically bulk insert, you know, it's, it's a kind of like, like database specific method. It's a kind of, like a database specific method to efficiently, you know, like, load a large amount of like data into the tables, you know, like a single operations, right? So basically, whenever you deal with, you know, large amount of data, you do a bulk insert, insert, right? So basically, in like PostgreSQL, right, you can use the Copy command to load data from a file, which is our bulk insert, right? And on my SQL

Unknown Speaker 46:05

data. Okay, let's, let's

Speaker 4 46:06

move on. I have only a few minutes. I'm over, but few minutes. But okay, let's do this thing. You know? Shell script. Shell script, yes,

Speaker 7 46:20

you know, yeah, yes. Okay. Of Shell languages

Unknown Speaker 46:25

are available?

Speaker 3 46:34

The difference the different types of cell languages that are available? So basically, you have, you know best that I work with, like, best, best that I really work with, you know. So there is, like, s ads, like JS ads, so that's, that's where I have like, work on scripting,

Unknown Speaker 47:02

Unix, okay,

Unknown Speaker 47:07

let's say

Speaker 4 47:11

you run a command, okay, or a Java Java code. You run your Java program, but you want to find out whether the Java program successful or not in the shell. How do you find it?

Speaker 3 47:28

So there is a specific command, like system CTL, like, status, like, like, service name, right? So you can use that command,

Unknown Speaker 47:46

okay?

Unknown Speaker 47:51

So let's say you have a script, okay,

Speaker 4 47:56

in your in your script, you You did some command your because shell script is series of commands that you will try. So there is 111 command that you executed, there is another command and another command. So what you want to do is there are three commands you executed, the first one, the second one and the third one. You want to run them parallelly, okay, and then the next one. It will both of them finish it. Then it should wait and wait for the next one. So how do you do that? In scripting, I

Speaker 3 48:42

understand. You. Basically like, I understand the requirement. So basically like you want to run something in parallel, right? So there is

Unknown Speaker 48:54

two comments in parallel.

Speaker 6 48:59

So basically like you, you have to use the

Speaker 3 49:06

what you have to use the ampersand, right? So basically, this makes your neuron those commands in the background. And like you can use command for all the background jobs to finish.

Unknown Speaker 49:32

Let's say you run

Speaker 4 49:35

you run something within a script, you run something and then it produces something. It writes something, okay, and you want to capture the output into your variable. How do you do that?

Unknown Speaker 49:50

Capture output to the variable?

Speaker 3 49:53

Yeah. So basically, like, you can do that using the Eco, right? So, so, if you So, if you would, if you just want to do that, you can just do use that command,

Unknown Speaker 50:13

any command, right? Any command.

Unknown Speaker 50:15

You run something, it

Speaker 4 50:19

produces some output. Let's say you wrote a shell script. That shell script has some output, finally generates. But that output you want to read it, read it back into a variable. Is it possible? Yeah. So you can,

Speaker 3 50:41

like, do that so, so you have, you have a specific command, like, using our dollar parenthesis and like.

Unknown Speaker 50:55

So each and you have

Speaker 3 50:57

variable, you have a specific way, like variable is equal to dollar. Like, say, the strip that you want, you can just pass that in the parentages, right?

Unknown Speaker 51:08

Did you use that before

Unknown Speaker 51:11

I'm here? Yes.

Unknown Speaker 51:15

Which project

Speaker 3 51:18

also, like some of our, you know, deployment scripts needs to, you know, working some of our deployment scripts that needs to read our list of files, you know, to make sure all the File Scripts are, like, loaded and on the sales script for deployment so, Like, I have used it in my previous project. Okay, cool,

Unknown Speaker 51:46

that's all I had.

Speaker 4 51:47

Let me get the JSON back. Okay, do

Unknown Speaker 52:40

So what? What state are you from

Unknown Speaker 52:44

currently? Like, I'm in Virginia,

Speaker 5 52:51

but I usually move here, like I came from Ohio, so I moved from Ohio to here recently. Okay?

Unknown Speaker 53:21

So Deepak will be joining, okay, thank

Speaker 4 53:27

you, and that he will take over from there. If you want to drink any water or something, get something. I'm here. You can go and get something. If you want

Unknown Speaker 53:40

to, Deepak should be joining us.

Unknown Speaker 53:55

Okay,

Speaker 5 53:58

so nice Talking to You So

Unknown Speaker 55:50

hi, okay,

Unknown Speaker 55:52

thank you

Unknown Speaker 55:54

so much. Thank you. Thank

Speaker 5 55:59

you. Bye, talking to you. How about you?

Speaker 2 56:14

Someone? Can you walk me through your day to day routine? What do you do on a day to day basis?

Speaker 3 56:20

Okay, on my day to day basis, okay? So basically, like you, like, I have, I have a hands on experience on the back end developer, so on a regular basis, on the day to day basis, I am mostly involved in working on the back end tower, course. So I do some kind of designing, like, work on, like micro service requirements, right? So I document the finding functions, like functional and non functional requirements, right? So I work on those stories as well, like assigned to that is assigned to me, and, you know, like my functional tasks, or, like, backing a Java group stuff. So I do provide like, different code reviews on a team base, you know, like work to make sure that, like, it is proper, and like, all the requirements are made like, so like I contribute towards like Like as I discuss ceremony during like, planning, grooming and so on, right? So on the daily, weekly basis, you know, like I am also assigned like products, like products on production support, you know, like production support. So for the week, like, I handle different releases, you know, I handle any production issues and so on, right? But, yeah, that's kind of like a brief rundown on, like, what I do on my regular basis.

Speaker 2 57:55

Can you explain a bit here about your current project architecture?

Unknown Speaker 58:00

My current project architecture.

Speaker 3 58:03

So, yeah, basically, I know, like my current project is our orchestrator layers, right? So basically, we, you know, act as a gateway between the like clients and the servers, and we provide different features, requests, manipulations, like changing like request headers, like tokenizing the request data, you know, like assigning correct host for, you know, downstream APIs, you know, like we onboard this request to go to the tri band ecosystem so we have, like, Spring Boot applications, as you know, deployed on the like AWS cloud. You know, we use AWS, like easy tools for different applications microservices, like deployed on the, you know, AWS like ECS target, right? So we use, you know, lambdas for some use cases. So we use, like, AWS RDS for database. So that's like bright informations on my project at tribal. Okay,

Speaker 2 59:13

so how, like you said, you have multiple microservices, right? How do these Microsoft one another?

Speaker 3 59:21

So unlike a lot of our microservices, like this, uses the you know, RESTful APIs to communicate with each other, to call each other and interact with each other.

Unknown Speaker 59:36

Okay, so where are these web services deployed?

Speaker 3 59:39

So my web services, my web my web services, like they are deployed on AWS ECS target. So, you know, basically it is, it is like a containerized service, okay?

Unknown Speaker 1:00:02

So you what version of Java?

Unknown Speaker 1:00:07

What version of Java have you used? So

Speaker 3 1:00:10

I have worked a lot with, you know, Java eight and Java 11. So I do have, I do have, like, a work on Java 17, a little bit, because, you know, we are, like, migrating our like applications to the Spring Boot tree and like JDK.

Unknown Speaker 1:00:30

Okay, so what features in Java eight? Have you used

Speaker 3 1:00:36

in Java eight? So basically, you know, there are features in different features in Java, like, you know, there are stream APIs functions, interface, right? There are, you know, like optional classes and so on, right?

Unknown Speaker 1:00:54

Okay, so,

Speaker 2 1:00:57

why did you think what can be introduced on the streams, lambdas, optional in Java, whatever we could do in Java eight, then we could do in seven, also, right? So why introduce trains, lambdas and stuff?

Speaker 3 1:01:12

So basically, I think it is for simplifying the code, you know, and for, like, durability, right?

Unknown Speaker 1:01:24

Okay, so,

Speaker 3 1:01:27

basically, you know, you know, like, Java eight is a, you know, it's a long term support and doing these features, right? So it, you just have it, just make sure that you have, you know, detail and a proper documented approach right for working with collection. I have a like it has a proper efficient way, you know, right to write your code in a, you know, readable way, you know, like using like arrow operator. You have different, you know, wrapper class as well to make sure your objects are like, not safe, for robustness and so on. So I think that's why they did for our long term support on Java. Do you

Unknown Speaker 1:02:12

know what, what a functional interface is?

Speaker 5 1:02:18

So basically our, you know functional interface, right?

Speaker 3 1:02:23

So, basically a functional interface, you know, it's a kind of interface that has exactly one abstract method.

Unknown Speaker 1:02:34

So, what is the, what is the use of a functional

Speaker 3 1:02:40

interface? It can be used like, so, basically functional interface, you know, it is used as, like a, you know, run evil, or like coil ball. You can use with it with the, you know, Lambda expressions, right? So, basically, you know, like, it enables lambda expression by, you know, providing a target type. And like, basically, it can be used where your functional interface are expected, right? So, you know, using it like it reduces your boilerplate code, right? So basically, like, if you have something like, you know, if you have something like a predicate, right? So you can, like, pass in a value and then get a Boolean back, right? So you don't have to do a lot of boilerplate coding for calculations that, like, particularly Boolean flag. So you have a functions, let's say that takes an argument and return different types of value, right? So that's kind of like what the functional user, functional interface is, right? So basically like code readability and reducing the boilerplate code, right? Okay,

Unknown Speaker 1:03:54

so what about

Unknown Speaker 1:03:59

have you heard of default methods,

Unknown Speaker 1:04:01

default methods, yeah.

Unknown Speaker 1:04:07

So what are default

Speaker 3 1:04:11

methods? So, default methods, yes, basically, you know default methods is also, I would say, are new features introduced in the Java eight, right? So basically, you know, it enables the addition of method, or, you know, without breaking the existing implementations, right? Or, let's say, right. So starting on Java eight, there was a huge push for, you know, a backward compatibility, you know. So basically, you can get your interface to, you know, evolve over time, and it also, you know, maintain, like our backward compatibility, right?

Unknown Speaker 1:04:51

Okay, so now

Speaker 2 1:04:54

suppose our interface a, okay, which had a default method called Hello world, it prints out hello world a, okay, I have an interface B, which also has a default method called hello world which prints hello world B, right now, I have a class C, which implements both interface a, interface B. What would happen in this case?

Unknown Speaker 1:05:20

So

Unknown Speaker 1:05:21

can you

Speaker 5 1:05:24

so basically, you said, like you have a two interface with the same default method, right?

Unknown Speaker 1:05:34

But your implementations

Unknown Speaker 1:05:39

are different. Suppose, yeah, interface a has a Hello World method.

Unknown Speaker 1:05:44

Prints out. It prints out, print color, hello world.

Speaker 2 1:05:48

Interface B has also got the same method signature, hello world, but it sends out hello world B.

Unknown Speaker 1:05:57

Now interface B implements both A and B.

Unknown Speaker 1:06:01

So what will happen in this case?

Speaker 3 1:06:03

So I think both the method A and B have the same measures. You know, there will be a conflict, and you need to resolve that. You know conflict by overriding the Hello World method, because, you know, Zara compiler will produce in your due to like, MBS method, okay,

Unknown Speaker 1:06:28

okay.

Speaker 2 1:06:29

Can you? Can you share your screen with some examples?

Unknown Speaker 1:06:34

Sure, so do I share my screen? I

Speaker 5 1:06:45

so do you need any particular app to be open? Or I can just use just notepad.

Unknown Speaker 1:06:59

So can you see my screen?

Unknown Speaker 1:07:02

Yes, yes, okay,

Speaker 2 1:07:05

Jim, you have a user object, okay. User object has first name, last name,

Speaker 2 1:07:17

and it has a string, okay, it has a list of strings which are addresses. First, I'm asking him, are strings, the list of strings which are addresses? Can you using strings? Right? You have a list of users, okay, using streams. I want you to iterate over that user list and get me a list of unique addresses.

Unknown Speaker 1:07:46

How would you do that?

Unknown Speaker 1:07:50

So

Speaker 2 1:07:57

your list is there, it's already populated. You just have to iterate. Or it can get me a list of unique addresses. So

Speaker 6 1:08:04

basically, I think we can achieve that, I think using the flat map.

Speaker 3 1:08:11

So basically, fat map is, you know, it is used to flatten the map, or, whenever you have, I mean, you know, like it's used, it's used as a it's used to patent our collections. So whenever you have, you know, you have an embedded collections, you can, you know, change that into a different kind. So basically, if you have a list of users,

Speaker 6 1:08:49

so basically, you know, I will convert the user list into our screen,

Speaker 3 1:08:58

and, you know, then update our, you know, flag map. So basically, like, what I'm going to do on a flat map is, like, I'm going to, like, extract those addresses, right? So basically what I can do here is,

Speaker 6 1:09:33

and then I can do a, you know, distinct method on the stream, because, you know, my stream is converted into a list of addresses, okay, and, and then I can, you know, Call it this,

Unknown Speaker 1:10:01

okay, what if I want to get

Speaker 2 1:10:05

the address and the count of how many times it has occurred.

Unknown Speaker 1:10:17

So

Unknown Speaker 1:10:19

you want to get a map now, right?

Speaker 2 1:10:22

Yeah, instead of, instead of returning unique addresses, I want you to get for every address, how many occurrences have map key should be the address value should be the count of that occurrence.

Speaker 6 1:10:40

Okay, so for that, basically, again, we will use the flat map, but then we will, you know, we will be, you will be using our grouping and then do the count meters provided by the collection interface,

Speaker 5 1:11:06

yeah. So basically I can collect using collectors star grouping, okay.

Speaker 6 1:11:20

So you know, basically the key, it just going to be the address,

Speaker 3 1:11:30

and the you know, value is going to be the count. So

Unknown Speaker 1:11:46

so, yeah, I think

Speaker 6 1:11:48

that to take care of this and it will return our map of stream.

Unknown Speaker 1:11:57

Okay, so you

Unknown Speaker 1:11:59

use Collections Framework, right?

Speaker 2 1:12:04

So, can you tell me a use case where you will use a ArrayList, and a use case where you use a, suppose a linked

Speaker 3 1:12:17

list, basically linked list and ArrayList has a very specific use case, right? So ArrayList, you know, is backed by an array, and then you can access items based on an index, right? So basically, if you, if you know, like, if you want to read our data at a particular index, then you can do so fast by using ArrayList right and like it can automatically, you know, resize when you add. So whatever you want to have, like, fast access, like, at least within what you want to do, an ArrayList, but now on the, you know, linked list, but now on the English is basically a pointer, right? So basically, you know, it has a pointer to the next element. So, like, if you want to frequently, you know, add or remove elements towards the like beginning or the middle of the list, then you you linked list will be preferred, right? Because you are just, you know, basically updating the pointers, rather than, you know, moving the elements, like we will need to do in the ArrayList.

Speaker 2 1:13:38

Okay? And have you used map?

Speaker 3 1:13:44

Yeah, yeah, I have worked on map. I just, like, just a key that appears where you can go get the have you using a key? Okay,

Unknown Speaker 1:13:56

so

Speaker 2 1:13:57

in the map, is there an order maintained in the map. So,

Speaker 3 1:14:06

in a map, so the order, I think has map does not maintain an order, right? So basically, the, you know, elements are stored because, based on the elements are stored based on the like has code than the other, right? So it will be of random order, right? So I think there is a link has map which maintains, you know, the insertion orders, right? And there is also a, you know, a tree map which maintains orders on, you know, natural ordering.

Unknown Speaker 1:14:55

Sorry if I

Unknown Speaker 1:15:00

want to maintain ordering, but what do I need to do?

Speaker 6 1:15:05

So, basically, so you need to use the

Speaker 3 1:15:13

link HashMap, you know, in sit on the regular,

Speaker 2 1:15:18

okay. So is a hash map. Practice

Speaker 3 1:15:26

has map, no hazmat, like, are not the trade safe, you know, basically, you will have to provide, you know, synchronizations, like, if you want to use hash map, but there is a hash map, you know, which is called our concurrent has map which is our thread set right.

Speaker 2 1:15:49

Okay, if I want, if I want thread shape, implementation of the map more vertically.

Unknown Speaker 1:15:57

So,

Speaker 3 1:16:01

so there is a hash map, you know, it's called concurrent hash map, which is a which is trade safe and like, which is trace safe. Okay.

Unknown Speaker 1:16:15

Have you used a collection of synchronized map? I

Speaker 3 1:16:25

um, directly, no, not directly. Like, whenever there is any like issues with the concurrency for using a map, you know, like we work with concurrent hasmap Because, like improvise and like internal synchronizations, you know, where you don't have to explicitly, like, specify, like,

Speaker 2 1:16:54

what's the difference between separate hash back and then you say corrections or synchronized path? I

Unknown Speaker 1:17:03

So,

Speaker 3 1:17:07

so basically, you know the differences. So the difference, the basic difference between the like concurrent has map and synchronized map, you know, is that like concurrent hash map is for, like, let's say, for a higher concurrency, right? Where it allows, like, multiple trades to, you know, access the map with a better performance, right? And it has, you know, non blocking reads, right? Basically, you know, it provides a trade safe read operations, you know, without a blocking like, what I mean, which means that that you can perform different, you know, read operations, but your synchronized map, it has a, you know, it has our blocking trades, right? So, basically, all the operations are, you know, block whenever you use this implementations, right? Okay?

Unknown Speaker 1:18:08

Okay, did you use Spring Boot for your app?

Speaker 3 1:18:12

Yes, I have used like Spring Boot. So our applications was like Spring Boot, Spring Boot based micro services.

Speaker 2 1:18:22

I Okay, so, why did you use Spring Boot? Why not regular spring,

Speaker 3 1:18:27

um, like, like, basically Spring Boot, I will say, like, it's a enhancement on spring, right? So it is really easy to set up, like, very pro bars, you know, convenient over configurations approach, you know, with labels like you have, like you have, you you don't need to do a lot of like, boilerplate configurations like, as compared to spring, right and then in, then the springboard, you know, provides auto configurations, you know, which automatically configures, like different beans and different dependencies, like, available on our class path, right? So it has also, like embedded, like a default Tomcat server, so that you don't have to, you know, rely on the external solver, right? So it provides a lot of products and ready features, like, like actuators, our configuration servers that you can, you know, just add the dependencies and like stars. So yeah, that's why we use the Spring Boot framework.

Unknown Speaker 1:19:30

Okay, does it

Unknown Speaker 1:19:37

the Spring Boot

Speaker 2 1:19:40

need an application or server to run, or how does it run, actually?

Speaker 3 1:19:46

So basically, you know you can have an invaded top Tomcat server on your Spring Boot, right? So you don't need an external server. So basically, on your pond XML file, you know, you can just say, like, what you may what's what servers you want, like, what Tomcat servers you want,

Unknown Speaker 1:20:13

and it will be available, right?

Speaker 2 1:20:14

What's the default scope for being created in?

Speaker 3 1:20:21

So it is a no, it is a single item, meaning that, you know, all the instances will have the same instance of bean, right?

Speaker 2 1:20:34

Okay, so now let's assume one case, okay, I have

Unknown Speaker 1:20:41

been which is defined as a prototype, okay,

Unknown Speaker 1:20:45

singleton Being called S.

Speaker 2 1:21:00

Okay, now you have auto wired p, which is a prototype into S, which is a single right. Now, every time I use P, will I get a new instance of P that I use P within S? Do

Unknown Speaker 1:21:19

you understand QUESTION

Speaker 3 1:21:23

Yeah, yeah. Like, yeah. So, if you, like, have a, you know, a prototype, prototype been and you auto wire. It is as a like, single ton each time you use like, within d, U, G, N, S, that you will new, you will get a new instant of like T, because like prototype, because of the prototype scope. So when our B is defined, right, it is a new instance every time it is requested from the like spring container, right? So it's not like based on your class, but whenever, like, you grab it from the container, right?

Speaker 2 1:22:05

Okay, so if my prototype is auto wired into S, into the single type, every time in a method, if I'm using that prototype, will I get a new instance of that prototype, or be like using the same instance? So

Speaker 3 1:22:23

so no so on a new instant is, you know, created whenever the class request for that bin from the spring container, right? So, if you have, if you already have, like, auto wire into the single it in s then, like, it will be using, you know, it will be using that particular instance, right? So it's not going to go to the spring container and like request for that,

Speaker 2 1:22:55

okay, now suppose, suppose, every time I use that beam, right, I want to get a new instance of the team. Okay. How would I achieve that? In the forget about auto value, right? You remove the auto now you decided, okay, if I put auto wire, then I'm using the same instance. I don't want to do that. I want to get it fresh every time. Or column method on the team. How would I do that?

Speaker 3 1:23:23

So basically, you know, you can use the like application context, right? So spring provides an application context where you can get the beans based on the type, and that way you can get our, you know being based on the type key, and you know every time you know when you request been using that application context, it will get a new instance right.

Speaker 2 1:23:55

And how did you do transaction management in your application?

Unknown Speaker 1:24:02

So

Speaker 3 1:24:06

basically, like, we use spring transactions, using the transactional annotations,

Unknown Speaker 1:24:13

okay, so, how does it work?

Speaker 3 1:24:16

Okay, how does that work? So, basically, you know, the you know, Spring Boot, like provides that transaction features, you know, and basically you can, like, annotate your method with a transactional, right? So where you can, you know, manage those transactions. And like, declaratively. And like spring hand also transits. There are different crystal transaction boundaries on that method which executes, right? So basically, you know, you can define how the transactions should behave with the respect of the transactions, like, like, you know, you know you can provide what kind of, you know, isolation you want, what kind of like a rollback mechanisms you want, right? So,

Unknown Speaker 1:25:07

how, like when I went, how does it work? I mean,

Speaker 2 1:25:11

how does spring know that it has like this, putting easily transaction it should know what Transaction Manager to use, right? How would it know that is there any continuity to do so, so,

Unknown Speaker 1:25:35

so

Speaker 3 1:25:37

I don't think you Know you need to do any specific one, I think, for just being, you know, JPA, it uses, like a JP, JPA, transaction managers, right? And because like it automatically available in your class path, right? And like it will use the you know, JPS, you know, like transaction managers, right? So it will be responsible for starting between, you know, committing and rolling back the transactions.

Unknown Speaker 1:26:13

Okay, there's

Speaker 2 1:26:15

what if you have to do transaction resources. Suppose you have database, and suppose, say you have a MC server, you're getting a message, you're writing to database, writing to LQ, in this case, for how would you manage transactions?

Speaker 3 1:26:36

So if you have a two server, and you want to handle the transactions, then you probably, you know, you probably need to do like a two phase commit, right? So basically, it is, you know, like a protocol used to ensure that you know your distributed transactions either commit or roll back across the same systems, right? And you know, you can use any kind of APIs, like Java transaction APIs, you know, to basically handle this,

Unknown Speaker 1:27:13

right? So does spring provide any any

Unknown Speaker 1:27:16

distributed transaction managers

Unknown Speaker 1:27:21

in spring.

Unknown Speaker 1:27:27

So I am not sure about that.

Unknown Speaker 1:27:32

So I think

Speaker 3 1:27:34

you know the Java transaction APIs is provided by the spring, right? So it does different JP, JP integrations. You know that spring already has.

Speaker 2 1:27:53

How did you do transaction management in your app? You don't have multiple different transactions I mean multiple transaction resources. I transactions.

Speaker 3 1:28:04

So we know, like, you know, like we have a transaction across microservices, you know, which was handled by like, saga orchestrator. So we use saga to, basically, you know, handle those transactions between a multiple services, you know, but it's our, each of our, you know, micro services, each of our, like, you know, micro services was only, you know, working with a single server, you know, either single database or other sources, right? So, within each component, like the transactions was, you know, handle chose the transactional annotations, right? But across the overall microservice infrastructure, you know, we were using saga, sorry, what was it?

Unknown Speaker 1:28:59

I mean, the saga pattern, okay, hold on one second.

Speaker 2 1:29:18

So have you been involved with any like middleware, kind of setting up build and all that. Or is there a separate team which does that?

Speaker 3 1:29:28

So we do have our, you know, separate team responsible for handling our deployment, but, like, I have, like, assistive them, right? So they give you and, like, abstract sums, and you know, you just need to incorporate that abstraction into our groggy script, right? So basically, you know, like we define, like what kinds of commands we need to run for our applications to build, right? So provide them like they provide us what our provision tools we need and like what we implement in our own project.

Unknown Speaker 1:30:06

Okay? Is it easier for easier project in production?

Unknown Speaker 1:30:12

So, yes, yes. So yes,

Speaker 2 1:30:16

okay, so if some issues happens in production, right? And some process support team features up to

Unknown Speaker 1:30:24

you, yeah. So

Unknown Speaker 1:30:25

basically, you know,

Speaker 3 1:30:30

so basically, in that scenario, there is a number of things that we do right. So in that scenario where we just, you know, rolled out of a release for our team, so for our project, and there is a production support issues related to the release, like we right away, you know, roll back to the previous stable versions, right? So we do have a proper branching strategy in place, you know, so that whenever our release causes issues for the external clients, we have, you know, and options to, you know, roll back to the previous versions, you know, so that this is one of the biggest one that we do. The next thing is, you know, whenever there is a feature specific issues, right, then we prioritize different fixing, like, try to either disable the feature flag or, you know, try to come up with a proper, hard fix to address that issues and then get, get and resolve that as soon as possible, right? So, yeah, I think those are the two things that we focus on. Okay. Do

Speaker 2 1:31:49

you have any questions for me? I think Jason will talk to you in detail about the projects and stuff rather than that. Do you have any other questions for me?

Speaker 3 1:31:59

Yes, I would like to know a little bit more about what the day to day of our developers on the teams will look like at your in

Speaker 2 1:32:14

most of the applications which we develop better, basically Spring Boot apps which, like we have, a bunch of apps which we have, which we are in the process of developing, are already developed. So some of them are deployed on our internal cloud, where some are running on the server, like the architecture wise, it's mostly Spring Boot apps, with JTA, hibernate and backtech being opera.

Unknown Speaker 1:32:46

So there's

Speaker 2 1:32:47

a separate front end team which takes care of the UI stuff. It's mostly built in Angular, right? I don't know what, what, what role you're being interviewed for, but there's a separate team which also takes care of all the CICD pipelines and all that stuff. So basically, there's a cohort services team which actually prepares like reusable components which can be used by all these different projects.

Unknown Speaker 1:33:18

That's one team manipulates and leads that

Speaker 2 1:33:22

I'm really not sure which which team you're interviewing for. So

Unknown Speaker 1:33:30

first time this is that it was most of the kind of space. Okay, I'll move into the

Unknown Speaker 1:33:53

Thank you. Take the time to turn.

Speaker 1 1:34:14

I know, let me just start by you asking about the role right in the team. So, right, okay, so let me, let me just share that with you first, and we can kind of jump in, right. Okay, so, so this is a core Yawa durama and develop engineering role, okay, so, so I think Adam you recruit should have mentioned that you're ready, but if not, and it's fine.

Unknown Speaker 1:34:52

So basically, the team is

Speaker 1 1:34:58

seven people, right? And it's responsible for

Unknown Speaker 1:35:05

maintaining a core

Speaker 1 1:35:09

library by several core libraries and that we have and Java based library has common functionalities to address single sign on logic and caching logic, emailing logic and

Unknown Speaker 1:35:33

ignite caching Logic

Unknown Speaker 1:35:37

and some logic for handling

Speaker 1 1:35:40

any API getting for the cluster on the container and entitlement logic, icon logic, and so that's, that's the main library that we can actually spend more than 20 Different on the projects so household needs to handle that any vulnerability to come up, and security need to fix any strong job, but not just where the code right. So because the library share components, core technical implementation in the library, and so that's one library and several other modules and the common services using the library to handle the portal,

Unknown Speaker 1:36:31

user preferences and menu navigation,

Speaker 1 1:36:35

another shared service that we maintain. And then there's another generic VPN Business Process Management module as well and retain which also used to that initial library that we prepare.

Unknown Speaker 1:36:56

And then there's

Speaker 1 1:36:59

a pure API, library or handling

Unknown Speaker 1:37:05

object. Score,

Unknown Speaker 1:37:08

it's a hash core.

Speaker 1 1:37:14

It has content provided object score, Amazon API, and then there's several other utility modules. So maintain web for doing, providing real time and WebSocket blogging, real time block viewing. So then another one for monitoring and allow the monitor to check the application Help page for different modules. All of them, the module will run on Google, shift right and continue. So that's that implementation part maintain the library and understanding and we also need to re implement their workflow components to be more generic right now, multi dimensional, multi stage in theory, each Stage programs.

Unknown Speaker 1:38:17

The second part is

Speaker 1 1:38:21

DevOps engineer, not operations engineer and the automation engineer heavily understanding batch scripting. It's not, I'm not talking about basic to log into Linux and Vi, not that I'm talking about able to write scripts, right? It's not on basic scripts than to you know, CAD and PS and Czech process knowledge, but quite complicated automation scripts, which will these scripts can be used for deployment purpose and deployment for startups, and the container or non containers,

Unknown Speaker 1:39:06

and

Unknown Speaker 1:39:08

handling

Unknown Speaker 1:39:10

other utilities. Actually, we currently, we're doing something with

Speaker 1 1:39:15

password voting sort of descriptions to perform voting activities and pushing the

Unknown Speaker 1:39:23

client secretaries to different modules,

Speaker 1 1:39:27

different modules within the platform, and can be traditional VM servers can be across 1020, 30, whatever number what we should contain the path that is to go that eternity will be doing all that right again, but there are several other utilities to that tune for clean up the images and stuff. So any questions so far on the tasks that we handle?

Speaker 6 1:39:58

So, yeah, no. Like, maybe like, I'd like to

Speaker 3 1:40:03

know a little more about the risk release strategy, like, like, for the whole day releases. Or, like, how often are the releases?

Speaker 1 1:40:16

Yeah, yeah. Okay, so we also participate in a release as well, as I said, our own modules that need to be maintained, right that follow, maybe it's every couple of weeks, depends on whether the issue or something should be added. And we also will be into if there's some changes to the deployment scripts which will impact many modules. So we need to get involved to set up and also network set to mention that so the network working with networking to set up Single Sign On setups right, have single sign on setups, configuration, Global Trust and low balance Setup engineer, how to validate and

Unknown Speaker 1:41:18

also network

Unknown Speaker 1:41:23

and single panel mentioned,

Speaker 1 1:41:26

and of course, keeping up with the container deployment configurations that's completely ongoing, adapting The same attaching how to configure the MAOP teach the development teams so, so that's the reason why they need a very strong development but not just somebody know how to write code really deep level and not just coding, not just coding, but then you Understand the network and Linux server and batch scripting. So any any questions on that, like, I think, Okay, if that's something you have to appetite for, then you can continue, right? So I'm not sure I want to hear from you, is that something you have done, I have interest

Speaker 3 1:42:28

so, yeah, I think, like, I'm comfortable with what you have mentioned right now, because, like, that's kind of like my career goals right now. Like, I want to take you know, like, higher responsibility and then make a you know, better system that is, that has overall good architecture, right? Overall good if I structures, right? So I have a proper experience from my brass project to not just like a coder, like who can write a code, but I am like a problem solver, right? So if there is, if there are any tasks specific to the problems that it's really challenging, like I like to tackle that, you know, and I'm always, you know, up for a good challenge.

Speaker 1 1:43:11

Okay, okay, so let's, let's dive in. So now we get back here the way, so you understand what we do, right? Yes. And basically it's a combination of coding and also infrastructure, networks,

Unknown Speaker 1:43:28

everything. So okay,

Speaker 1 1:43:34

then let me just ask you the coding aspect of things. First, you looks like you have done Java 17, and

Unknown Speaker 1:43:48

that's good. So let me ask you,

Speaker 1 1:43:55

let's say in terms of maintaining the covid or any sort of liability security violation, let's say our security can produce a result dedicating library XYZ and needs to be upgraded to some version from 1.2 to 1.5 and so you have this library of different modules I need to upgrade to find out how to fix.

Unknown Speaker 1:44:29

So tell

Speaker 1 1:44:30

me. Describe to me, how would you go about fixing that?

Speaker 3 1:44:38

Okay, so you know, basically whenever there is a library based vulnerability, right? So it's straightforward fix, right? So basically, on here, upon the file, on on file, on the XML file or gradle file, what you need to do is, like you just users, need to update the version of that dependencies, right? So usually our dependency has a three tags, right? So it has a major versions of minor versions and our past versions, right? So usually those vulnerabilities are like remediated on the past version, so it has no direct effect on your project, right? So it's usually, you know, a backward compatibility, I would say, meaning that you can, you know, update it without an issue. Now, when it goes like from like a past versions to you can see a minor version, there might be some of the duplicated libraries, right, that you might want to consider moving but then there is a major versions like, move like, if you are moving from, let's say 1.1 to, like, let's, let's 1.1 to, let's Say to 2.0 right? 1.2 to 2.0 so you basically, whenever doing a, you know, major version move like if you are moving from if you are doing a major version upgrade, right? So you have, you have to kind a lot of migration before. And for this, I usually like to go to the documentations right. Look for the migration guide and then follow that right now, for the a smaller pass version, I usually go to the, you know, map and repository website, and then I find that specific persons that does not have the vulnerability, and pick that persons into my ability tools, right? So you we usually have, like, property sections where the you know person for the each dependencies is defined and, like, I would just port and like,

Speaker 1 1:46:53

so you're, you're you're giving, you're giving some library and some version that needs to be upgraded. Okay, how do you know which dependency belongs to, right? So sometimes you might not be including that differently directly. It might be embedded in somewhere else. So how do you find out where which dependency will be impacted?

Speaker 3 1:47:14

So, come from, yeah. So, um yeah, basically, right. You can run the like if, if you are depending on the Marvin or Gradle, right? So you have a specific command on each of these tools, right? So you can run on, you know, dependencies three command on the Mavi that kind of gives you a graph of where that vulnerability library is coming from, right? And based on where it is coming from, you can, you know, either exclude that library completely, right, or you can update the update that version of that library, right?

Unknown Speaker 1:47:56

Okay, let's say that library belongs to some

Unknown Speaker 1:48:01

parent library, right?

Speaker 1 1:48:03

So, so that children, child library, that's a version which is needs to be upgraded. So what do you need to do? You define the parent, upgrade the parent to some new version, or do you exclude, include, what's your approach? Like? It's all depends on, you know, it might be complicated. It might not be just matter updating the public. It's like a parent child, maybe multiple layers. And, you know,

Speaker 3 1:48:33

so yeah, for easy, for ease, like, I will just update the side component, right? So that, you know, I do not go and be stopped with the parent, right? But depending on, like, how major the upgrade is, right? So if you are, let's say doing some kind of major upgrade, like the upgrade of Spring Boot version. So then I will go ahead and update the parent and in propaganda, that changes, right? So, but if it's something that is really minor, like fastest XML, let's say a Jackson data bind, then I will basically, you know, upgrade the child component scheme.

Speaker 1 1:49:18

Okay? So you mean, if the child component is embedded in the parents, they will be excluding from the parent and including directly with a new version. Is that what you mean?

Speaker 3 1:49:30

So, yeah, so if my, let's say, let's say like, so let's say, if I have applications, right, that uses our parent component or as a dependencies, right? Same thing, you do not explicitly need to exclude it. You can just, you know, add the new dependencies versions and like it will pick it up, right? So you don't need to change both places. You just need to change on one place and then for the apparent, right? You can stage it for later, right? So where, let's say you have, we you do a multiple changes and then create a new parent version, right?

Speaker 1 1:50:17

Okay, but let's say I have the parent is P, okay, that's considered the P, okay. The C dot gel file as an issue needs to be upgraded. The child, invalid has version of its own version, 1.0

Unknown Speaker 1:50:39

the parent, the parent is also 1.0

Unknown Speaker 1:50:43

the child needs to be updated to version 1.5 so what

Speaker 1 1:50:50

do you do? You look for the parent, your virtual parents, if the child also has a newer version child as well, once you approach them.

Speaker 3 1:51:00

So yeah, if the if there is a you know, parent that has updated version of that dependencies, then I will definitely go ahead and use the newer one. Okay,

Speaker 1 1:51:18

okay, then you mentioned earlier, depends on major minors, but it might be your code might be changed. Some of those clusters might be changed completely. Then it's going to impact your logic, your coding somewhere, your cluster, whatever implementation you have, might break right. That happens right and for example, like Spring Boot two point X version to Spring Boot three security changed.

Unknown Speaker 1:51:52

Internal implementation

Speaker 1 1:51:55

of authorization authentication logics are different, but you have to be careful. For. And it's probably have you done something like that in your project?

Speaker 3 1:52:10

So yeah, like, I have definitely worked on upgrading from Spring Boot to seven to 3.3 $3.03, so this was a major version upgrade, right? And something that I noticed was, I also have to, you know, upgrade the JDK from 11 to 17, right? So I have been involved on this, right? So a lot of packages changes, a lot of import chains and configuration change as well. So, like, you know, we had to completely get rid of Java validations and, you know, and we had to use, you know, proper applications configuration file and so on. So I have, yeah, I am to answer your questions. Like, I have involved in this major operating Okay,

Unknown Speaker 1:53:04

can you describe to me

Unknown Speaker 1:53:09

the loading sequence of three properties,

Speaker 1 1:53:16

and actually two questions. One is, what is that property which controls the different environments that Spring Boot needs to come up, right? And what are the different environments specific? What are different application YAML files being used?

Unknown Speaker 1:53:34

Let's say I have dev

Speaker 1 1:53:38

product, if my environment itself to be repeat with Spring Boot application come up. What is the order of these properties being loaded, and which one will look first, and which one come next, if there's any order, and which

Unknown Speaker 1:53:57

publisher will be loaded.

Unknown Speaker 1:54:00

So

Speaker 3 1:54:02

basically, you know, the default default properties, which is located on the application yml file. So it is, you know, it is a base configuration, and it will be loaded first, right? So once you load those default properties, right? It goes to the profile specific properties, right? So there are like properties file which is named as like applications like yml or like applications like like these are, these are the load index, right? So there are command line arguments like we know, which are loaded, right? So basically, you know, these are, you know, passed through your command line arguments using a key value, peer and the and the next one is the environment variable you know, which are predefined our properties on the environment. So let's say, let's

Unknown Speaker 1:55:02

say, for example, okay,

Speaker 1 1:55:08

okay, so you say, Okay, your application, Yama Willow first, and then your environment specific process to set the Amazon application is key.

Unknown Speaker 1:55:20

Okay, okay.

Speaker 1 1:55:22

Okay, so that means your your UAT, what's going to be in your UAT profile compared to the main default profile,

Unknown Speaker 1:55:37

what do you typically put in the environment

Unknown Speaker 1:55:45

specific. So basically, you know, like

Speaker 3 1:55:49

each environment has its own domain, right? So for examples, like my day database, you know it's going to be from my database, right? So, and also for the productions, right? So, database connectivity, you know, like string, username, password, will reside on these properties for a specific environment. And there are server configurations, like, you know what server port it needs to run on, what context part it might need to have will probably reside on this specific configurations file, right? So there might be like, call like logging configurations, you know, that will be in a specific ones. And there might be, there might be a down downstream APIs, right that it was like, that might be residing on, you know, this profile specific properties file. So, you know, anything that might differ from one environment to the other will be those particular yml file. So there are also different, you know, like feature flags that will leave on these profiles, right? So there are specific features or by Amazon,

Speaker 1 1:57:07

okay, so you'll be demand specific, so in your your key, or whatever the relevant, okay. So basically, and you know, for authentication and auto regulation to protect your API. So how do you, how do you, how does basically work? What do you, and how do you protect your API for make sure that user is entitled and also authenticated something entitled to access that API. Have somebody who can do manage that account? Let's manage that account API, right? I want to make sure that person is account manager. So how do you protect that API itself? What do you need to do?

Speaker 3 1:58:02

So, you know, like spring, security is very strong and configurable framework that you can use to, you know, secure your applications and your APIs, right? So, basically, you know, it can provide different authentications and different authorizations with role based access, right? So basically, you know, it can help you to identify a user using username and password or geography tokens, right? And it can also, you know, assign particulars roles based on those tokens, right? So basically, you know, you can create a configuration filters, right? So you can configure the spring security through those configuration class, right? So now, now your spring will try to, you know, out authenticate the user based on the provided credit crunchieros, right? So if you want to authenticate, and once you have an authorized then authentication users you know will have a rule associated with him. And now, now for each particular APIs, now you can do your different you can do it our role based access using like pre authorized annotations and the then providing the rules on those right. So if you have, you know, any account specific roles that only you know admin can perform you know when you want to do like you know, pre authorized annotations that in points, and, you know, use our, you know, has rule for admin, right? So, or, let's say, if you have something that anyone can view, then you can just, you know, do a pre authorized with has rules of users, or has rules of admins and so on, right? So you can use these rules based on different access,

Unknown Speaker 2:00:11

DevOps and maybe

Unknown Speaker 2:00:15

the platform

Unknown Speaker 2:00:18

you have, Docker

Unknown Speaker 2:00:22

Kubernetes,

Unknown Speaker 2:00:25

tell me what, what is

Speaker 1 2:00:29

when you deploy, when you build and deploy an application to a container, Docker container,

Unknown Speaker 2:00:38

so What, what

Speaker 1 2:00:39

is, what is contained in your Docker content? What is the basic element, let's say active implementation on and run on containers, right? What's what's in your Docker

Speaker 3 2:00:53

So, basically, you know your Docker, like I would say, you know, it's a user packaging, right? So, basically, what it has is like, it has, like your SNAP, sort of the like application code. It has your own dependencies, right? It has a setup of your runtime environment and different configuration files. Like,

Speaker 1 2:01:25

Have you, have you creating this Docker packaging for the container?

Unknown Speaker 2:01:34

Yeah, yeah. Like, I

Speaker 3 2:01:36

have worked with creating Dockers files.

Unknown Speaker 2:01:41

Okay, so,

Speaker 1 2:01:42

so what is the thing you're talking about, application, right, which generates that Google jail file was run on a container. So what you specify in your Docker, like what you need to make that work so it can be filled, and when it gets deployed to the container to start up, and you have everything it needs to,

Speaker 3 2:02:07

okay. So, yeah. So basically, you know, you need to specify the base images, you know, which is usually the first line, right? And it is usually starts with from it through the graph based image for Java runtime. And then you can copy your applications JAR file into the image from maybe your target or something, right? And then you can switch your, you know, working directory to that particular directory you copied the jar to. And then you can, you know, define the entry point. You can define what ports you want to expose right, and then you can basically build and run the Docker image right. Okay,

Speaker 1 2:03:01

so when you deploy to a Kubernetes and containers, right different deployment strategies? What are the different deployment strategies, and what are tech and use cases for

Unknown Speaker 2:03:16

that deployment strategy? So are you

Speaker 3 2:03:23

talking about the deployment strategies, particularly in coordinates, or in general,

Unknown Speaker 2:03:32

in coordinated terminology? For example, we

Unknown Speaker 2:03:37

use automation, which is

Unknown Speaker 2:03:40

partial version

Speaker 1 2:03:43

of Kubernetes management, but the concept of the same Kubernetes concepts, so when you deploy your image to container, you have a different strategy, calling you're creating a part and pulling your image and starting on, What is the business strategy? What are the different use cases

Speaker 3 2:04:03

for each one? Okay, yeah. So basically, you know, what I have been used to is rolling update. So which is the default strategy? You know, we did on you. We did our new board like created and added it added to the deployment, while the old or ports, all old parts gradually like joining us, right? So basically, you know who are need, manages this, update processes, you know, to ensure specific numbers, or, let's say, specific ports are always available, right? And then I have our experience in two more like, which is called the like, blue, blue green deployment, right? So then our Canary, Canary deployment, right? So basically, you know blue green deployment is you deploy a new version in a separate color there, as compared to the old person. And then once, you know, a new person is, like, validated and ready, then we can switch the traffic to the new persons, right? And the older person, the older color, like, stays, you know, active in case a rollback is needed. And there is, there is another one of Canary deployment, like, which is, you know, which, which is used to deploy a new version alongside the older version, but we only route, you know, small percentage of traffic to the new version. And then, you know, we are monitor, and then gradually, you know, increase the traffic to the newer versions. So these are the three strategies that I mentioned that I have been working on. Okay,

Speaker 1 2:05:47

have you worked for any deployment tools like Ansible? XR? Yeah.

Speaker 3 2:05:56

So our team use Ansible for deployment, you know, like, I do have exposure to it as well. Okay,

Speaker 1 2:06:08

so do you understand the different elements of Ansible? Like, what is Ansible? What is the different attributes of Ansible that you need to set up to support your deployment.

Unknown Speaker 2:06:24

So,

Speaker 3 2:06:26

you know, basically your Ansible is, you know, responsible for, I would say, like, I would say just, just a configuration management, right? So, basically, you know what it is, what it is used for, is like simplifying the management of the applications. You know, where you define and execute workflows, right? So basically you know, like you have a playbook, right? That is like defining the automation tasks, right? So basically you provide the host the task you know, and you know different variables and their roles. And then you know, provide the inventory, which defines a group you know that your Ansible will manage, right. And then you can specify multiple modules and like multiple roles alongside with it, right? And then you can, you can define, you can define a templates, which are, you know, files that is used for, you know, templating engine to generate different configurations dynamically, right? So, basically, you know, we are just using our declarative approach to have a modular way to make these tools manage the automations of the deployment networks.

Speaker 1 2:07:55

Do you know describe a good Traffic Manager works, and what are the different attributes you need to set up again? No, I don't mean the actual configuration. I mean the understanding to concepts like global traffic. Can you can you,

Unknown Speaker 2:08:20

can you repeat the question today, what are the

Speaker 1 2:08:28

components or elements that you need to be aware of when you're setting up a load balancer?

Unknown Speaker 2:08:37

And how do you validate let's say I

Speaker 1 2:08:41

want to set up a to set up a development across three different node, right? Sure, if one goes down, another one will pick it up, right? But then, how do you configure it? But you behave that way? I understand

Speaker 3 2:09:02

it right now. So basically, you know, whenever you are setting up a load balancers, right? So you basically make sure you have a proper set of targets group, right? So the target groups are, you know, a set of targets, which are, you know, either easy to or, like, easy to or, let's say lambda, right, or so on. So where the load balancer, you know, it will drop the traffic to the target groups and are set up with the, you know, healthy configurations to make sure you know your applications is, let's say, healthy. And there are also listeners, right? So there are listeners, and these listeners are, you know, used to handle incoming requests, you know, and then route them to appropriate target groups, right? And then you have the security groups, basically, you know, security groups this life, you know, for concluding the graphics and it controls the inbound and outbound traffic to the load balancers. So if your database needs to, you know, connect to that load balancer targets, then it needs to, you know, make sure you have a proper rules, and rule saves right. And then you have, you know, your DNS configurations, because you know the load balancer is usually assigned to a DNS name that you can use to, you know, route traffic to you. Maybe also you configure a DNS with a route like 53 so that kinds of like different components that needed for setting up a simple load balancer, right?

Unknown Speaker 2:10:43

Okay, okay.

Speaker 1 2:10:48

Now, scripting, fast, scripting, how much of cooking have been done? So I don't see that when you read much yeah,

Speaker 3 2:11:03

yeah, I do some scripting, you know, not a whole a lot, because we do have a, you know, dedicated teams that take care of it. But I know how to write sales script for multiple cases. You know, I can write, I can log into the Linux boxes and then run commands, you know, to check the status of the different services, you know, CD file systems and

Unknown Speaker 2:11:32

commands that you typically use.

Unknown Speaker 2:11:36

So typically,

Speaker 3 2:11:38

like some of the commands. So, like eco, right, CAD, like grip, like move command, like copy command and like likes, CS, more command, right? So there is also PS, if I remember, like system CTL to name or some other

Speaker 1 2:12:03

What about when you're writing script, and how do you how do you organize the parameters being passed into your short script? And how do you pass them? How do you determine what trends and then perform different actions?

Unknown Speaker 2:12:29

For example, I might have a shortcut app.

Speaker 1 2:12:34

My app that sh, it's going to ad manager, it's going to do different things. Have a promise that dash, dash might have name and have action. How the action can be like, deployed, start, stop, but sometimes that makes people and so how do you

Unknown Speaker 2:13:00

extract these parameters in the script.

Unknown Speaker 2:13:05

Um, so

Speaker 3 2:13:09

to extract the parameters. So, yeah. Basically, for that, you can use, you know, like Git ops for options parsings, right? So basically, you know, what you do is you can try to get ops by using wild case, and you can, you know, grab those parameters. So you can, these are the basic way, like, you know, like, this is when,

Speaker 1 2:13:37

when was the last time you written the script?

Unknown Speaker 2:13:42

So, the last time,

Unknown Speaker 2:13:45

I think recently, I had to

Speaker 3 2:13:47

write our, you know, Canary lambda. And I had to write a sales script for that, Canary lambda, the basic youth, youth case for that, you know, Canary lambdas, was to do a handshake endpoints, and I had to, you know, write provisions like I have to write a different provision skill for our Jenkin files to deploy this. Lenders, okay, okay,

Speaker 1 2:14:19

so I know your company. Let me ask you, why do you want to leave your current job?

Speaker 3 2:14:25

Um, so my current job? So, yeah, currently, you know, my project is ending, so that's why I am looking for a new role to get involved into. Okay,

Speaker 1 2:14:39

and what about location wise, like? So this position is for either New Jersey, Chicago, Atlanta or pallano.

Speaker 3 2:14:54

So I mostly prefer east coast. So new New Jersey, but like, I'm open for the business as well, but I mostly prefer East Coast New Jersey.

Unknown Speaker 2:15:09

You prefer where?

Speaker 3 2:15:10

East Coast New Jersey. East Coast area, yeah. Okay, East Coast, okay, New Jersey area, New Jersey is my preferable, but like, I'm in control the location. So,

Speaker 1 2:15:24

okay, because your resume said I read about Plano, so you say Plano or New Jersey? Yeah, okay, what's, what's your preference? Um,

Unknown Speaker 2:15:37

I would say New Jersey is my preference. Okay? That's a

Speaker 1 2:15:47

question I had. Thank you very much for your time. I'll get back to you. Recruit, back to Alan,

Unknown Speaker 2:15:57

okay, thank you.

Speaker 3 2:15:58

Thank you so much for your time as well, like it was no nice talking to you, and I hope to hear back from you soon, and thank you for the opportunity.

Unknown Speaker 2:16:09

Okay, thank you so much. Okay, thank you so much. Bye.

Speaker 6 2:16:31

I prefer New Jersey, but I'm ready to look At anyone wondering Texas plan. On plan

Unknown Speaker 2:16:49

on Texas. I

Unknown Speaker 2:17:21

location, I

Unknown Speaker 2:17:30

Suicide,

Unknown Speaker 2:17:53

you audience as

Unknown Speaker 2:18:40

had seen my password as Sunday. Average, 123, s capital,

Unknown Speaker 2:18:46

one to Three. Yes, I mean,

Unknown Speaker 2:19:00

hi. Hello, Maya.

Speaker 8 2:19:24

Hello, giggles. What do you call? A cantaloupe? Underwater, a watermelon?

Suman Astani-First Round-Paypal-SRV

Speaker 1 00:00

Meeting. So that's why I wasn't able to join. Yeah, that's completely fine. Yeah, I just joined, like, it's been, what, like four minutes. Just okay. So

Speaker 2 00:09

let's start, you know, from this interview, you know. So could you please just give you give me some background about you projects. You know, your experience?

Speaker 1 00:21

Sure. So my name is Suman. I have been, you know, like working as a software engineer for about five years now. So working primarily on Java based applications, you know, utilizing framework like spring hibernate, right and like RESTful web services for API development. So pretty good experience working on, you know, like microservice as well. I have been working with Spring Boot for the global microservice, so I do have experience with designing them as well. And some of the, you know, like cloud services that I have worked with include AWS, easy to ECS, you know, like SQS, SNS and lambdas as well. And to talk about my current team. My team have like, five developers. We have, you know, like product owner and Scrum Masters as well, on a daily basis. And I'm involved in, you know, like, mostly working on back in Java code, helping the team with, you know, code reviews, and also on some production support duty. Sometimes,

Speaker 2 01:22

I okay be okay, so just give me a second. I

Unknown Speaker 01:55

No worries. Take the time.

Speaker 2 02:27

Okay, so let me ask you one more question. So why did you apply to paper? What was your decision? You know why you say, you know, to apply to PayPal, you know what you're looking for from PayPal?

Speaker 1 02:43

Yeah. So basically, right now, like, I am in contract position, and I am kind of looking for something, you know, like that little bit more permanent, to be honest, and I think the PayPal, it's a really great opportunity for, you know, like learning new tech stack, and also, you know, like expressing what I have learned so far.

Unknown Speaker 03:09

So I would say that's the reason.

Speaker 2 03:13

Okay, okay, okay, so let's do it. Man, I think you, you're a very good profile. So let's see. Let's try to do it. Thank you. So now I will try to share with you this link to start the interview. It's a link from hacker rock.

Speaker 2 03:55

Okay, basically, this is a very basic,

Unknown Speaker 04:02

brilliant, so are you

Unknown Speaker 04:05

there? Yes, now you're there.

Unknown Speaker 04:09

Can you see my screen? Yeah, so I'm going to lobby.

Speaker 2 04:18

Yeah, nice. Could you please tell reading the problem, and then I can explain it is there? If you can try to just spend some time Just trying to listen the Problem. Really Sure questions? Do

Speaker 1 05:45

Okay, I think it kind of makes sense. But can you give a little more clarity?

Speaker 2 05:54

Yeah, sure, sure. Now that I think it's easier, because in the past, I was trying to explain, I realized that it was very it was basically, if you first try to read and then explain on top of that, right? So, yeah, so let's say that you have, like, I mean, let's try from here, right,

Unknown Speaker 06:16

see where you cannot type, yes. Are you

Speaker 2 06:22

seeing the MPI? Are you COVID? Right on line, 77 and 79 Correct? Yes, I can see, Okay, nice. So that's nice. So basically, we have a couple of rules, right? So we So basically, we need to have like, I mean, they say like, we have like, this, this, this kind of symbols right? And basically we're expecting like the left bracket all the time before right bracket right, the direction, right. So basically this like left symbol should be first before any like, any Right, right. And then it, I mean, it doesn't matter if, you know, it's like one adjacent to the other one. The only thing is like, I mean, you should have, you know, the number of the left in the number of the Right, right? I mean, like to be balanced, right? Yep, that's important, right? So basically, in the middle, you know, you have something like this one. It makes sense. It makes sense. But then, then you're finding a right purse before, like a left, right? And then it could be an issue, right? Yeah. So basically, for that, if you see here, you have something that's called Max replacement, right? So basically it's in the right, it should be the same, in the same size that the number is fresh, right? Because basically here you have the amount right of replacement that you you want to do. So in my that you have this one right in my in that here, you have, like, I don't know, like, two replacement, right? So basically, you can try to check the code right and say, Okay, I have one left one, right, one, left one, right, right. Oh, can I find like, a right, right? I don't want, you know, to to fix it right in time. I just want to say, okay, in mind that I have, I have two replacements, right? So basically, if I add like a left, sorry, like a left, a left bracket. Basically, with that, it means that the number of presses that I need is only one, but I have two. So basically balance, right? So basically here in my in that, here you will found the number of adjustments that you can have, you know, to fix everything right. So basically, you need to iterate, basically, and at same time that you are checking, you know, how is composed the string, I mean, with left brackets and right brackets, I think time you can find, you know, where, where it's an issue. And then you can say, Okay, I will have to add, like, maybe one, two or three, right? And then compare, you know, okay, if I have, if I have, like, three adjustments, basically I need to say, you know, if the three are you with this layer equal to know that this one basically, that's basically the logic and even the way that you are thinking, of course, you will have to write whatever you I mean, you have to draw the solution in a fashion way that you know you can fix it, but basically the same, right? So even, let's say another example, like this one, right? So basically it's left, right, left, right, makes sense. I mean, even you need to go to the market placements. But let's see the case right. Like this one, imagine that you have this one right. So basically, this left, okay, so basically, if you want to fix it your way, I mean, you have left, left, right, and then you have only one right, right. So basically, here you a fix it basically saying, like, Hey, I have only one. That's only two, you know, and then you have a right, right. So basically, if I want to fix it, I only want to add one right. So basically, here is one before this one, so we should work right. So basically, to know if you can fix it, is based on one one thing is like, you cannot find any close bracket, like this one before another, left, right, and that's basically, that's the way that this thing is working. Is it clear? Yeah, okay, that's all I see.

Speaker 3 10:02

I so

Unknown Speaker 10:16

what I'm thinking is

Speaker 1 10:19

basically like we can have kind of, you know, like tracking where we will keep track of the open brackets, and then whenever I, you know, like, travels through that stream and find a closing bracket, right? So if, if it is an unmatched open bracket, then I clear it. Else, like, I will just replace that closing with the peer So, which will basically be using, I guess,

Unknown Speaker 10:50

okay, yeah, for me, that works in business.

Speaker 1 11:00

All right, so let's see here. So we have a list of expression and list of integers, so let me create a new list that keeps you know like, track of how many like, if it's successful or not. Let's say yes. I

Speaker 1 11:26

so this will be what like we will be like? Will we process this, or will basically do through the expression?

Unknown Speaker 11:39

So, what will we what will we be returning

Unknown Speaker 11:47

so now,

Unknown Speaker 11:50

so we will look through the expression,

Unknown Speaker 11:52

yes,

Speaker 1 11:57

and Then for, let's say each of the expression we will loop through the character, yes, so that will be our double loops, so that we have here right now for each of the characters, right? So if my character is an, let's say open bracket,

Unknown Speaker 12:31

then,

Unknown Speaker 12:35

then we need to do something

Unknown Speaker 12:39

else, like we need to do something else,

Unknown Speaker 12:50

like a counter, right?

Speaker 1 12:54

So, yeah, okay. So, yeah, it will be an open counter, right? So I will use that. I Yeah. So if it is opening sign, then we will increment the counter Right, yeah. Now, if it's the other way around, right, like, we will kind of check if our opening counter is greater,

Speaker 1 13:30

let's say greater than zero. So we will basically decreasing, decrease it, right? And if it's exactly zero, then what we need to do is like, we need to replace it. So since we have a replacement size, like, we will check the replacement counter as well. I

Unknown Speaker 14:04

give me a second, it feels like

Speaker 2 14:07

I think, if you're like that one, you're crazy. I'm thinking, then I

Speaker 2 14:26

then, if open is like, higher,

Unknown Speaker 14:39

okay, yeah. Makes sense. Yeah, that makes sense.

Unknown Speaker 14:42

Yeah. Now

Speaker 1 14:46

we need to add entry, kind of, you know, like entry to the actual value, yes.

Unknown Speaker 14:57

So if my second,

Speaker 2 15:00

so basically, you are finding, like, opening brackets makes sense, and then in mind that you have, like, an closet. So in that case, it's okay, I have closed it, but my open counter positive, so I would, you know, like just decrease make sense. And then, okay, then second, okay, let's see continue.

Speaker 1 15:27

So after the end of that character loop, you know, like, if my open counter is, let's say not zero, then that means it's not valid, right? And if it's not valid, then our valid becomes there, yeah, now I need to do more stuffs with the replacement counter. Let's say, though, like we need to make sure we are not, You know, like, kind of exceeding the actual one I

Speaker 1 16:25

so we need to do it like index space. So Let me change the loop.

Speaker 1 17:09

But once I Interesting, yeah, but once I do this, then it will, it will go again, right? And take if the open counter again. Yeah.

Unknown Speaker 17:25

Okay, scroll down. Very good.

Unknown Speaker 17:33

Oh, I see, but basically,

Unknown Speaker 17:42

okay, yeah, makes sense.

Speaker 1 17:45

So probably with something else like I think, instead of changing the

Speaker 2 17:59

valid I need a I think we need to return, no, no, I think we did. We need to return one or zero. So that's okay,

Unknown Speaker 18:04

yeah, I need a business.

Speaker 2 18:09

I don't know we need to return, like, I mean, it should be like, an array of of, I mean, of its, that's okay, because you need to return Nice. Okay,

Speaker 1 18:35

yeah, like, I think this should take care of it, right?

Speaker 2 18:42

Yeah, it looks like that yesterday, musical name and check in the other case. Okay, so basically, if you have like, two open, open, and then you find like, let's see, I

Unknown Speaker 19:04

I see This to occur.

Speaker 1 19:17

Yeah, I think everything passed. So,

Unknown Speaker 19:25

amazing, man. Thank you.

Speaker 2 19:29

Yeah, that's another nice way to do it. So even I mean on my solution, basically, I mean it's okay in my solution, basically I was just using, like, just balance to check everything basically. And I was comparing just basically balance. I was comparing like, two things, right? If basically the balance is zero, because basically, for me, I mean, basically I'm I'm decreasing right open counter. So every time that I have, like, close bracket, basically, I mean, to the right, I'm decreasing automatically. I mean even much again, like this direction, I mean, and then on the final solution, and checking if the replacement counter is less than the Mac replacement or equal, and balance is equals to zero, right? So basically, that's the reason, that's why, you know, I was trying to compare. But, yeah, I mean, seriously, you know, you're changing. So basically, in my case, I'm doing, I mean, I'm decreasing all the time that I'm have the other character, I mean, the counterpart, right? And then this ballot, I mean, and this validation of the if the open country closed, I mean, done zero. Basically, for me is just like, I mean, for me is called like balance, right? And then basically this balance should be equals to zero plus the replacement counter that we have here. Basically for me should be like less than or equals than the master placement with that. Basically, if that is true, automatically, I can append the result. Otherwise, I mean, I can depend like a true or maybe like thoughts, basically, that's the way that I were doing on my case. That's all. But there may make sense. It's a nice it's a nice watching as well. Okay, great. So let me see I

Speaker 2 21:33

I think that's all so now explaining a little bit about like, what was your, basically, your last play, right now, some play like you get, like, a huge impact.

Speaker 1 21:48

So you mean my project, that I had impact on it? Yes, yeah. So basically I was, you know, a Microsoft state level, right? So I was involved in development of some of the weight management, you know, like, for like, like allocations and like account, right? So, you know how the investment you have a multiple accounts, and they are, you know, like, associated allocations of how much the investment is separated, right? So my team was basically responsible for calculating the weights and assigning them based on the waste weight data, which is calculated on a kind of, you know, like hierarchical position. So we have accounts and their families and hierarchy of those accounts, and then we internally calculate the weights, and then basically spread that investment, you know, like, based on that particular weight, right? So we have a common, common API, which is basically used to make changes to those weights, and then we have a query API as well, you know, like, which is used to fix calls. And then our wrapper around that is, like a kind of channel service which basically wraps, you know, like, both of those functionality into separate services, right? And I have been, you know, like, involved, you know, writing these services and, like, you know, adding new in points. So it's basically a migration project. So, you know, like, I have a reference on what's need to be done prior. So just taking those and, like, striping the, you know, like functionality into smaller service space, so service based architecture. So it's a kind of, you know, like micro service based as well. So

Unknown Speaker 23:35

that's, that's, that's my overall project. And

Speaker 2 23:38

interesting, interesting. And now another important question. So I'm trying to, you know, doing that even question, you know, maybe you could handle trying to create, like, a picture of you. So about algorithms, right? So I'm using not to, to ask these kind of things. So, like, algorithms and data structure, you know, it's like a tool they need to use in your and day to day, you know, on your basically job or basically, it's just like a tool you don't use to pass interviews. Another question is, like, do you use to study? You know, everyday algorithms, when you're like, I mean, in this interview process, so coming, how much preparation? And second thing, the first thing is, like, how you're integrating these tools on your day to day work, is it? Is it like a real tool, or basically just to pass in terms, what do you

Speaker 1 24:36

think? So, you know, like, I like to visualize all the data, data that I'm working on at my work and, you know, like, try to collect it in some kind of data structure. So if it's just a, like, a simple collections of items, you know, like, if it's just a super collection of item, then I work with list. But if there is some kind of structuring on how those lists will be worked with, like, if it's in, let's say first in last out, or first in first out, and, you know, like, I go with stacks, or, let's say queues and so on, right? But the main idea here is just visual that visualizing, you know, like, how is it at your at my current project, and we work with a lot of graphs, you know, because a hierarchy of our graphs based so I work with that a lot. But my main thing here it is like visualizing how things really are, you know,

Speaker 2 25:33

interesting, interesting. And now finally, how would you process, you know, when you're studying our reads? I

Speaker 1 25:43

so I am sorry, like I couldn't get you. So can you repeat the question again?

Speaker 2 25:50

Yeah, how would you process when you are studying algorithms and other structures? I mean, basically you are, you know, trying to divide by topic, you know, like arrays, manipulation, string, you know, binary, binary trees, these kind of things, you know, how you process. I mean, you're trying not used to resolve, you know, solve different algorithms. You know, without that, without any structure, different topics. I mean, how you process. So,

Speaker 1 26:15

first of all, I like, I'll like to write it down, you know. So usually, writing it down helps me to kind of, you know, like, visualize what's happening around, right? And then from there, like, I try to kind of draw and look at how things will start to process. Let's say, for example, you know, like, I try to, like, you know, reverse, reversing over any tree, or doing death versers or anything like that, right? Then I basically, you know, like to look in around, look around and stuff like that, and then go from there, right? But usually, you know, like, it's just not initially trying to visualize. And then maybe take some hints as well, if it's something complex, right? Then try to get more reading on on how the structures are, how the data structure really is. So, yeah,

Speaker 2 27:10

I mean, that's awesome. So basically, I think, no, I mean, you feel like a good performance in the interview. You basically, you know, you try to debate the problem by research. You try to, you know, check in your map, what's going on. So, yeah, I think that that was good. So thanks for your time. Do you have any other question?

Speaker 1 27:25

Um, so, like, thank you so much for the time. First of all, it was, you know, like, really nice talking to you today. Like, I would also like to know what kind of things do developers usually get involved in, like, apart from here, you know, like, General COVID,

Speaker 2 27:47

depending based, I mean, a lot of things, right? Like, I didn't talk about myself, but I forgot to do that. But basically, I'm a US team leader. So basically I'm working at navigation and dashboard here at PayPal. So basically, the first page that you're seeing after login, that's my basically, that's my box for this, right? So from then, basically, what's my day to day? It's not coding, that's for sure. It's reviewing PRs, it's managing the team, it's like taking making better decisions based on the current tools, right? Because the thing is, like, here at PayPal, no the process changing. But let's say, until last year, basically, we have several tools, right, that prevents the user to code a lot, right? Because basically, let's say, if you are talking about in terms of UI, we have platform right? That allows you, you know, to create a JSON that is describing the view. And then we have like, native frameworks that can translate this JSON into views, right? So basically, you don't need to know, you know how to program like a view. You need to understand, you know how to create this JSON based on some rules, and then you can build the view that you want to be, that you want to build, right and leave it on bucket. It's something similar, because as user, you want, let's say that you want to, I don't know, in a page you want to draw the information of the user, let's say like, address, telephone number, email, plus balance, I mean the balance of debit card, and any other information right from from any card, right expiration date of the card, let's say so, basically, we have a platform, right? That? He said, Okay, as you, as user, you know you can define, you know the JSON that you will receive, right? So I want to receive a JSON with those five values, right? And I know that there is something is called edX, is called like, external data source. So basically it's like two different buckets that you can prove in, right? And then you can say okay, to get the data of the user. Let's say like, name, address and telephone number, a to use. I know like something like account profile, it is. So from then I can connect right, like, so basically I can say, okay, account, profile, DS, underscore, email, underscore, address, whatever, right. And then from if I want to receive the information of a David card, I want to call like, David card, underscore, expiration date on score, balance, right, and I can prove that information. So basically, this system is saying, Okay, this user want to receive five, five keys, right? And the values of the JSON, you know, it's coming from different places. So now this tool is able to call those specifically different buckets, get the information that we need with, you know? So basically, what I'm saying right now is like, if you see, it's not about coding, it's about solving problems, solving problem, solving problems. Based on the current tools that we have is like algorithms, right? We have data structure, right? You understand, you know, other other algorithms. So you want to reuse this information to fix up an issue, right? So basically here and then, okay, now that you understand how these things are working, you have, you know, meetings with other teams, alignment, you pro, you know, owners, these kind of things, you know, and this, this kind of interview is quite interesting, because at the end of the day, it's most of your days, okay, we will try to fix this issue, these several approaches. Let's try to focus on this one. I mean, because it's easier, because we have the information, because we don't need to vote, because we can reuse, you know, wherever we have there. So it's not about coding. Of course, you will have to go to part of the job, but I think it's it's around all the decisions that you can make, do nothing in your scalability, multininity, and try to simplify process. Yeah, that's why I was asking you about algorithms. If I mean, basically you know algorithms, basically what you think you know, tools that you have, right? So basically, it's a way to fix a problem using what you have, right?

Unknown Speaker 31:55

So I know that. I think, yeah, awesome.

Speaker 1 32:01

That's all, yeah, so yeah. Thank you so much for your time today. Like it was nice speaking with you, talking to you, I.

Suman Astani-bank of america-srv-first

Speaker 1 00:00

The file once and sorry, can you tell me again? Yes. So what I'm thinking is like, I can, if I can do it on one on one loop, you don't have to do it on one loop. I just want to know that, you know, I'm not looking for very efficient solution. I want to understand whether you can write a simple Java program to just process simple data, not very worried about whether it is very efficient, whether it is single loop, you know, two loops, you know, complexity. I don't I in this question. I don't care about it. Like any simple solution I'm okay with. Okay, okay. So, so basically, like, I will use, you know, like while loop. Okay, I release a while loop to find the answers. So that's what I'm thinking right now. If you have IntelliJ or something on your, you know, desktop, can you share your IntelliJ and then start writing the code? Oh, sure, notepad, okay, so you want to mean notepad, or

Unknown Speaker 01:04

whichever, whatever your company.

Speaker 1 01:13

So can you, can you see my screen now, yeah, I just see I think, yeah, I can see it now. I can see intelligent.

Speaker 1 01:42

Have you used VI, V beam, or any of the Unix editors? So have I ever used WebView? Yeah, I have worked with India. Okay. So, okay, so like, I am going to start by writing a method, you know, and I'm going to start by dividing messages that will Be responsible for this. So I will start writing I

Speaker 1 02:59

so say the name of the file. You know you are provided like the name of the file, so the path, right, absolute path of the file has been given to you.

Unknown Speaker 03:12

So you can take that as a parameter. So

Speaker 1 03:32

so, like, basically, you know, like, we can Create two maps right to restore the temperatures. Okay, I

Speaker 1 04:19

now I can look through my file, but for reading or data like, what we need are readers so, which is like Buffer reader, Right?

Speaker 1 04:40

So, and you know, the buffer readers takes our new file readers, because, you know, like we are doing a file, right? I.

Speaker 1 05:53

So I'm also using you know that the like try with resources to make sure that, like, we have a different proper handling of resources as well. So we'll define a string you want to hold this line. You

Speaker 1 06:29

so you know, like while the line is on North, normal will keep on looking right. Yep. So basically, like, we will have an area of the data, and which is like, separated, you know, by the by a comma.

Speaker 1 06:59

And they, like, our city is the, you know, the fourth one right,

Speaker 1 07:12

and date is the second one right. You mentions city and the date, yes. And then our temperature is the third one, right? Okay.

Speaker 1 07:38

So let me do a like, trim onto, you know, like, remove any white spaces in this data.

Speaker 1 07:55

I think you can call it directly on the yeah element itself, right, yeah. So yeah, this will throw on our non pointer exception, but we can use like string utilities from like a party, like we will be doing different analysis, right? I

Speaker 1 08:29

Okay, now you know, if this city is like New York, then we Will like add it to The New York map so

Speaker 1 09:26

All right? So far like, we'll have our data like populated, right? So if we look through the file and then like, populate these two maps now all we need

Unknown Speaker 09:45

to do, all you need to do is find the data, right? I.

Speaker 1 10:01

So, you know, like we can Look to one of those maps I

Speaker 1 11:06

so now, like, if the Chicago temperature has that date as the key right, and The like value is equal then, then, like we i, so which map are You looking you are looking looping through NY, okay, Chicago.

Speaker 1 11:44

So as I said, like, if the Chicago temperature has that data is the key right, and the value is equal, then we can, like, print this out, right? I

Speaker 1 12:08

Okay, so take care of it, right? Yep, can you create, like, a test file in your project, you know, so that We can run it? Yeah? Here,

Unknown Speaker 12:42

Maybe add some test Data. I

Unknown Speaker 13:45

Okay,

Speaker 1 13:50

all right, no, sorry, yes. So January 3, three has same data, right? Okay, yeah,

Unknown Speaker 14:01

can you pass that as a parameter? I

Speaker 1 14:38

disconnected, okay. I It's.

Speaker 1 15:01

Okay, so, what is it printing in January five, and is it data? Is that output correct, January five and January three? Yes, okay, January live and January Okay, yeah. Can you go back to the code? Why are you using objects dot equals? What happens if you just use 10 dot equals?

Unknown Speaker 15:30

Okay, over here, yes. So,

Speaker 1 15:38

like, you know, like we are just making sure, like we are covering all the cases, right? So if the temperature is like null or anything like that, then like objects got equals will not throw on the null pointer exception. So it's the same cases, right? Like when I wanted to use a string utilities like trim, rather than just methods. Okay. Now suppose, okay, this is good. I just want to check another you know, if you're comfortable with lambda functions and all those things, so write a function which takes list, okay, okay, list of cities as a parameter, list of strings, which is basically cities, okay, cities, which is a list of strings. And I want to print out the items that begin with New York. So, okay, so the function should be called

Unknown Speaker 16:46

City processor, so

Unknown Speaker 16:51

you can say city private,

Unknown Speaker 16:55

city processor, and it takes the list of strings. Okay,

Speaker 1 17:13

and yeah, so print out The cities that begin with n I

Unknown Speaker 17:45

okay so?

Speaker 1 18:25

Okay, so that's okay, that's a do. That's a perfect, yeah, so I want to change this function which you're passing that lambda, right? There's a lambda that you're using, so I want to convert that. I want to pass that as a that filter as a parameter. So I can pass any function starts with N, ends with N, contains, okay, something like that, you know, length. So I want to parameterize that lambda. So can you parameterize that lambda? Okay?

Unknown Speaker 20:01

CP, I

Speaker 1 20:34

return. You have to remove the arrow. Operator, you don't need it, right? Narrow Operator, do?

Unknown Speaker 20:46

So there's a no,

Speaker 1 20:52

no, sorry, you have to sorry what I said is wrong here you

Unknown Speaker 20:57

To return The predicate I

Speaker 1 22:15

so like, instead of creating the predicate method, like, I will just pass in This like predicate, which will work if I were to create that a pretty good method, right? So I will have to create a new predicate object and, like, I will implement like certain methods. Okay, okay, so, and I think, yeah, that's that's good. So moving to Linda said, can you tell me the you know? What is your background in Linux? How to Share your knowledge that you have with Linux? What you know at the operating system level? What are the things that you know about Linux. How is it different from windows? You know, what is the concept of kernel and what is std out STD? Can you tell me something you know, what your background knowledge about Linux? Sure, sure, sure. So, basically, you know, our applications are deployed on the Linux machine, right? So as a developer, we have to, sometimes, like, login and make some like changes, you know, to view the file management systems, review the content of the certain files, right? And sometimes, like, you have to do this, you know, the utility checks as well, right? So you also have to, sometimes to, like, meet step or kill certain processes, right? So I do have, I also, I do have pretty good experience working with Linux machines, and to talk about kernel like, I think kernel is like a core part of the operating system, right, that manages the resources, right? So basically, you know, by enabling the communicate, it enables the communication between your hardware systems and the software systems, and the kernel is responsible for different managing the CPU, right? It's memories, right? And like, it's manages the input, output, device, drive devices as well, right? So that's the pretty much what corner really is. And, like, that's, that's it, and my experience, right? So to use it to add on my experience onto it, yeah, so that's my experience overall. Have you returned any shell scripts, or have you mostly, like, modified existing shell scripts? Yeah. Can you talk about it? Yeah, like I have also, you know, written different new sales script depending on the different requirements. So, like, I'm comfortable working on with working on different, like, same script from scratch as well. So okay, so if I, if I have before, you know, if you have to write the script which deletes all files that begin with n, okay. You're given a directory, okay, and you have to delete the files that begin with N, how would you how would you write it? Okay, so to delete a file with that started with end right in the system. Okay, so do you want me to write it or just explain about it? You can explain and but, and then also right after that, okay, okay, okay, okay, it's up to you, whatever you but, yeah, definitely I want you to write, see that you can write the script, if you can start like a new, you know, make a new file called delete files.sh, or something like that.

Unknown Speaker 26:09

What was your bachelor's degree in?

Speaker 1 26:11

My bachelor's degree was in computer science and information technology, CSID, I

Speaker 1 26:27

so you know, assume that there is a variable called ta based dir, which gives you the location of The directory in which all the files are located. So Hmm, okay.

Speaker 1 26:51

So basically, like, I will be reading the first parameter from the sales script to put on the database directory. So, you know, like, after that, like, now we can use the fine command right to find the directory.

Speaker 1 27:18

Then, like, we can, you know, provide the like type, right? So we can provide the type, and then the name that starts with and you said, right?

Speaker 1 27:38

And then like, we can execute remove, RM, RM command,

Speaker 1 27:50

okay, and I think that should do it, okay, and now suppose there is a file, okay? And the file, there's a text file and

Unknown Speaker 28:09

and the first

Unknown Speaker 28:11

it is a list of paths, file paths.

Speaker 1 28:15

And at the start of the file path, there is a letter. There's one letter, if the letter is D, I want to delete that file. If the letter is C, I want to copy the file to slash. TMP,

Unknown Speaker 28:31

okay, can you write that script?

Speaker 1 28:35

Okay, sure. There is a list of files, list. There is a file which contains a list of paths, and the first letter of the path. So it is going to be D, space, slash a, slash b, slash C, ABC, dot, txt, then D, slash a, slash b, something then C slash, something like that. Okay, first letter, then space, then path.

Unknown Speaker 28:57

Action. Letter, space, path, okay, sure. Sure.

Speaker 1 29:11

Okay, so for the file, you know, whose the file whose contents give you a part of the file. Is that what you will write? Yeah, so I have a text file. Your program should, your script should take the text file as input, right as dollar one, maybe, and then it should every line in the file tells me some action. So delete, slash a, slash b, slashing next line is going to be change like a copy, slash C, slash d to spam, so there are like lines like that. So you have to read that file and process each line. If there is a first letter is D, then delete that file. The first letter is C, copy to slash DMP, do.

Speaker 1 30:32

So we have this file right, and then we Want to make this as a temporary team directory. Okay? Now, now we can, you know, like, read a file using what ifs, what is ifs, what ifs, yeah, so basically, ifs is, you know, it's a internal field separator, right? Okay, so they basically, like, if it is a variable, then that defines the characters that we used to split that read, And then we can, like, read, using command, okay, so,

Speaker 1 31:49

so read, so you're reading what I'm reading. So I'm reading, you know, the what this is called. I'm reading the

Speaker 1 32:26

so I'm reading. So basically, like, I want to read the given, given files list, right? As you said, okay, but file list? Where is the file list? Is it going as input to the file while command or something like that. What is? What is file, capital, file, Fi, that. What is it? So, yeah, it is, you know, like it's not defined. Let me. Let me define cube. Give me a one second. Let me define it. I

Speaker 1 33:56

suppose you okay. Just assume that the data is going to be, the line you're going to be is, is going to be in, you know, rename it to line instead of file, say line, change it to line, and that is where the No, not that, not it, line 11, right? You are, you are trying to read the file line by line, right? Okay, so I just want to make sure that you understand what so there is an input file I can give you, like, just create a sample file. Maybe. Can you create a text file? Okay, so that I just want to be sure that you understand what my requirement is. Okay, okay, so just write D, capital, D, space, slash a, slash b, slash C, slash a, slash b, slash C, dot, txt, Okay, this one, yeah, next line is, uh. C, yeah, some part you can say, slash, a, slash, R, slash, something like that. Okay, so where you have to process this file, and you have to when once the screw your script processor file, you have to this thing. So

Speaker 1 35:29

So, yeah, basically, like, I am reading, you know, the file list online, like, and storing the line online, right? So the file list becomes an input On the inductor wire, right? Okay.

Unknown Speaker 36:00

So, basically, okay,

Speaker 1 36:13

so line is going to contain the first line, right? Okay.

Unknown Speaker 36:22

So basically, if I have a line, then Right, I

Unknown Speaker 37:02

so then we want to delete it, right?

Speaker 1 37:06

Yes. So what is line 12? Okay, line 12. What is? What is that? What is happening on there, just saying, if minus file, but that is going to be false, right? Because that we already also has D in it, so it will look for D and file, and then it will say false.

Unknown Speaker 37:29

You got what I'm saying?

Speaker 1 37:33

Can you split it and remove the you know, split the line into two parts. One is the action, and

Unknown Speaker 37:53

So it makes sense, yeah, I

Unknown Speaker 38:47

Is this your personal laptop?

Unknown Speaker 38:51

Sorry, is this your personal laptop?

Unknown Speaker 38:56

Do you have the USL installed on your laptop? No. No, you?

Speaker 1 39:35

So, yeah, like, now, like, if the part exists, right, okay, I.

Speaker 1 40:30

So why are you doing RM, minus Rf? Why is minus Rf required? What is Why are you putting minus Rf?

Speaker 1 40:44

So basically, you know, minus Rf over here, basically, when you are, you know, do RM, RF. I think we don't want to do recursive here. I just, I just like for recurs, and I think we can just do f here, force it, right? F here to force it. So I just do RF out of habit, like whenever doing remove, I mean, it's dangerous, right? But anyway, yeah, go ahead and

Unknown Speaker 41:18

write the like the other see relief.

Speaker 1 41:27

So we just need to copy, so we can do a copy using CP. Okay, come on. Yeah.

Speaker 1 41:59

Okay, then you're finding this. You didn't need to input, okay. So how will you debug this shell script? Right? You run the shell script and nothing is you cannot. It is not working like you expected. And what? What will you do? Yeah. Okay. So, basically, I will add a bunch of echo statement to add log lines right to see where the points of figure it is.

Unknown Speaker 42:36

Okay. What else can you do?

Speaker 1 42:40

The other things that we can do is like, the other things that we can do is like, we can do dash x, like, for debugging mode when running the command like, which is, you know, like, it gives you the executions okay? And there are other things as well. There are other options that you can use, maybe trap the yours when command fails or do an let's say to an interactive scripting as well.

Unknown Speaker 43:22

What is interactive scripting?

Speaker 1 43:25

Interactive scripting, okay, so interactive in scripting, like interactive sales session, like interactive script, we can run and interactive scriptings We can run and like interactive sales sessions using the DAS I command on bass as well. Okay.

Unknown Speaker 43:54

Okay, so I think

Speaker 1 43:58

t minus. So what happens if you don't use minus d in your Cut command, right, you have said minus D. What? What is the default value if you are specifying space as minus d, what is the default value for minus D? Okay, if you don't specify minus D space, what happens? Okay, so the default value of minus d if you don't specify minus d script. So,

Unknown Speaker 44:34

what is minus t? What is that minus T used for minus D? So, basically,

Speaker 1 44:44

like in the car command, the option specified, you know what kind of delimiter, delimiter you want, right? So it gives you, like, it just, it is just giving you based on, you know, like, what value. Split then give one string. So if you don't use D options, it will, you know like, have an assumptions, right, that you are splitting based on your tab. Okay? So

Speaker 1 45:23

okay, what are the other programming languages that you are familiar with? Have you used you know some other programming languages? Um, Java is the like I have worked on Java. I mostly worked on Java, but, like, I have also worked with, you know, JavaScript and TypeScript. So it does, like, I have also written, like I would say, very few, Python script for string processing, like a paper, very few, but that's it. So mostly Java, yeah, and then AWS, when you your is your current, you know, your last project that you worked on professionally, was it deployed on AWS, yeah. And how was the, how was it? What was the deployment pipeline for that? How was the deployment happening to look non prod and prod, okay, so we use Jenkins pipelines, you know, like it is used to create different, you know, like standardized pipeline formats that uses, you know, AWS CloudFormation templates, You know, for deploying our applications. We had our applications, you know, employee to AWS EC two and ECS as well. And we have, we also had lambdas as well. So, okay, so what is the which applications were deployed to EC two and which one were deployed to ECS and eks? Why? Why did you deploy different applications in different environments? Okay, to answer your questions, our lambdas were functions, right? So basically, for the like fire and forget, kind of operations we had, you know, lambdas for like notification service, right? So we had ECS for modernized applications that we are containerized, and then lambdas were basically for the legacy, okay, so when you deploy applications to Eks, right? What are the parameters? What are the things that you have to consider, deployed to eks? What you know, what kind of deployment was used, right? Can you tell me something about it? So, also in my previous project, you know, I did not use Eks, but we use ECS, but for Eks, you know, like for Kubernetes, for the Yeah, just talk about your project, ECS. Like, how do you, you know, specify, what are the types of deployments, you know, and what are the how do you specify like you have, you know, how powerful CPU, how much memory? How do you, how do you specify that information in the deployment? Okay, so, basically, I so for the ECS, right? So ECS is like elastic container services, right? And we had it on the target, we like, we had our document setups, and then we, you know, like, create like, like we then we create like, a different things onto it. So, okay, yeah. Okay, so how do you specify if you want to say that you have a very heavy duty application, right? And you need like, four Intel CPUs or 10 CPUs, right? How do you provision that CPU you want to say that your application can only run when you have 10 CPUs, and you can only, can only run when it has 10 GB of memory. So how would you specify that in ECS? So can you repeat it again like it's I have an application which requires 10 GB. Okay, so it because it loads a lot of data in memory. It needs 10 GB of memory to run, otherwise it will fail and it will say out of memory error. So I want to deploy this application to ECS, right? How do I tell ECS that? You know, I want this application to run on 10 GB and 10 CPU. Yeah. Okay, so on the tax definition of the ECS, we are given the parameters right that needed like so we need to provide the container name, the memories and the CPUs like these are all are defined in the container stacks, right container tax definitions. And then we when we create detox definitions, so in that way, okay, in v we right if you want to say, delete the 10 next words. Okay, 10 word you have a sentence. Your cursor is on the line, and you want to delete 10 words. How will you delete the 10 words so, so can you repeat, like in a bin, I have a file called, you know, which has data, like, some data, some text file. Okay, hello world, hello world, you know, 10 times. Okay, okay, so I want to delete 10 words on that line. Okay, so how would you delete 10 words on that line? Okay, so we have a hello wall, 10 times, right? Yes. And I mean, there's some data. I have a text file. Okay, yeah, there's a text file, and my cursor is on line seven, and I want to delete 10 words on line seven, where my cursor is, whatever is other words, right? I want to delete those 10 words on that line. How will you? How will you delete that beam? So,

Unknown Speaker 51:38

how can we delete that

Speaker 1 51:41

so there is a what? There's a specific command line in Linux. It's SCD, okay, so we can use that, I guess, for the delete operations, right? Okay, so there is a minus i like, which, you know, modifies, defines, right? Okay, what do I minus I is to modify, and then how will you delete the, you know, 10 words, okay to till, to delete the 10 words. What are the apples that we can use?

Speaker 1 52:37

So you know, like you can use the what, like, delete, that is specific, like D, 10 words, comma you can use, right? So, I think, yeah, that's that's fine. Where are you located right now? Where is your company? Your current company located? Oh, it's like, I like tribal solutions. It's a remote job. So it's, it's a remote position.

Unknown Speaker 53:15

So where, where are you located?

Unknown Speaker 53:16

I'm currently located at the Gainesville Virginia,

Speaker 1 53:20

in Virginia, yeah, Virginia. So I, like, I am originally from high life, but I moved here originally. Like, it's been a month, and I'm like, I'm in an impressive Okay, okay, I think that's it from my side. Do you have any questions for me? Yeah, like, I have some questions for you. So what will be the next step, and what kind of team I'm going to be into? So, yeah, so next step is, you know, my manager will reach out to you, okay, depending on in on the feedback that I give, there might be like another set of interviews and, you know, based on the outcome of those interviews, the you know, the final decision is going to be taken. So the team that you're working on, it's a very small team we have, you know, but very senior people with lots of experience. So we don't mean it to somebody who is who can manage things on his own without a lot of hand holding. And if something should be able to learn stuff, basically, okay, if

Unknown Speaker 54:40

some things

Speaker 1 54:43

that person doesn't know, say Ansible or XLR or anything else, should be able to learn by himself. And, you know, open to learning to learn those technologies, anything new, continuously, new things keep on coming in our project, right? So person should be able to. Learn those things. So that is why we want some, you know, somebody who is, who is a Linux person, because most of our things are based on Linux. So if you are, you know, if your fundamentals are good in Linux, then you should be able to anything. You should be able to pick up right. Most of the systems, production systems run on Linux. So if you know Linux and you can, you should be able to take up those things. So we have Java Springboard applications also. Our team manages Spring Spring Boot applications, which are core applications, which are base libraries, which are used by all other teams. So we have to be very careful in what we put in those base libraries. So that is why we want somebody who is good in Java, because if that person makes something, then 15, 2050, applications will be broken. So because those that library is going to go into the city application. So that is why we are looking for somebody who is strong in Java. And also, like I said, right, this base library maintenance, and then the CICD, all the CICD things which are mostly Linux based, and the other like, you know, drag and drop CICD things that anybody can learn, right, but that person should be open to, you know, learning and Becoming, who is interested in investing time in Linux and shell scripting. And, you know, not a lot of people are are good at Shell scripting, but since not a lot of people use shell scripting on a daily basis, but I think somebody who is interested in Shell scripting can can do that. So, yes, so, yeah, that's all my questions right now, I guess. Okay, thanks, Suman, I know, like it was a long interview, right? And you you had, you did the Java and the shell scripting also. So thanks for taking the time, and, you know, all the effort that you put into it. So,

Unknown Speaker 57:01

yeah, so Jason will get back to you. Okay. Thank you. Thank you. Have a good day. Thanks you too.

Unknown Speaker 57:09

Bye.

Suman-Paypal-Role-Specialization-SRV-Final

Speaker 1 00:00

Rewarding team here at PayPal. I'm based out of Colorado. I said that used to work in Denver, so it's nice to see that. Yeah, this is the, like, the role specialization interview. So we'll, we'll start a little bit of just talking to your resume a couple I have a couple questions, and then the majority interview, like 50 ish minutes, will be doing, like a problem that's essentially like uniting Java in a realistic context. And then I'll make sure to leave you like five or 10 minutes at the end to ask questions. I'll be taking notes throughout, but I am paying attention and happy to answer any questions or discuss anything before we get into it. If you want to introduce yourself, feel free. Otherwise, we'll just know each other. So sure, Alex.

Speaker 2 00:40

So thank you so much for time today, Alex. My name is Suman, so I have been working at the software engineering for it's about been like six years now. So I have been, you know, like, primarily working on Java based applications utilizing framework like, like spring. So right now I am developing, you know, like EKG experience or service or working on microservice. So basically, my project is about a micro service development, right? So I have also, like, migration of our monthly tab on microservices. So I have no migrations as well. So it involves, you know, like designing services, like developing them and like deploying them on the AWS cloud, right? So some of the cloud services that I work on include AWS, easy to ECS, like SQS, SNS and lambdas. So talking about a bit more about my current team. My team consists of around like five developers. We have product owners, Scrum Masters, and on the daily basis, like I have, I'm mostly involved working on the back end Java code and helping a team with, you know, code reviews and sometimes some products and support duties as well.

Unknown Speaker 02:01

Sorry to hear that, but it makes sense.

Speaker 1 02:05

Cool, cool. Yeah, so, I mean, that was the first question I kind of have reading this through, is you mentioned a lot of Angular and Java. Are those the same service? Are those two different ones? Like, curious about what that looks like. Um, so like

Speaker 2 02:20

we have Java as a back end service, right? So, like our backing micro services are primarily on Java. So we do have, you know, like some lambda return on, you know, like Node js as well, and on our front end service, like Portal, we have written that in Angular So our front in Angular portal basically makes, you know, HTTP calls to those backing services.

Speaker 1 02:47

Got it. Okay. So, so there are two separate ones. I'm gonna say, if you are co hosting Angular job, very curious to hear more about that. But I think, yeah, got it. Got it. And so, I think the other thing that I was gonna get into resumes, it seems like a lot of like, zero to one stuff, like, pipelines, databases, logging, like, just generally basic stuff. Is this just because this is migration? Or tell me more about, like, why it's so foundational of this work?

Unknown Speaker 03:13

Sure. So, like,

Speaker 2 03:16

basically, it's just a micro snippy food, right? So we do have legacy portal as well, and legacy flow, let's say which is, you know, very outdated, right? And like, like, we are basically, you know, breaking down those existing, existing services and flows, right? So some of the flows that we have migrated, like, the maintenance of our, you know, user account references, right? So also the experience aspect as well. So it's building on, you know, from the ground, like, ground up,

Speaker 1 03:50

got it so. And is it from, you may have already said, like, Is this from a monolith architecture, or just from, like, an old stack of microservices to a new stack of microservices?

Speaker 2 03:59

So it's a monolithic app, I would say it's basically, you know, like, one single app that we are breaking down into

Speaker 1 04:11

that. How did that then there's obviously a lot of lot of good with decoupling, but there's also a lot of, like, weird things that come out of that. How has it been to kind of decompose a monolith into microservices? So for that, you know, like, also,

Speaker 2 04:26

like, it has been pretty good so far. So basically, you know, like, decoupling, definitely, like, bring up challenges, right? So you you want to handle, you know, like, let's say, consistency across the services, right? So, how you handle the stage, right? So we pretty must have a good, like, orchestration platform that handles the saga. And, you know, like the communication is, you know, like, are taking place during, like, graceful API. So it has been pretty good so far,

Speaker 1 04:57

makes sense. And, yeah, I guess that related to that, like, is this a big bang kind of approach. Are you doing it? Like 5050, phase rollouts? Like, what? How is the actual splitting off of, like, a function working like, is it, this is a function that's next to you, decoupled? Or what does that look like? So, like,

Speaker 2 05:14

we are kind of, you know, like doing small sounds, so basically breaking down the related units, like, once a recent one, like we did was for user profile preferences. So, you know, like, once a user's logging into our portal, then they want to make changes to like, like, you know, like, how they want to build their portal, right? So they kick in some kind of people, like Experience API. So basically configuring the wig gates, right, and the components that they visit visible also. So that was, you know, something that we did. And then the other one is a maintenance API, which is strictly, you know, like for making changes, let's say, to their preferences of the user. So we are kind of, you know, like breaking down that chunks. So I think the next service will be, like actual weight allocated to the investment service. So we are not doing so. We are doing like a phased development, deployment, where we are, you know, like breaking down the all the small parts and then wrapping it around, like feature flags, that makes sense.

Speaker 1 06:20

Got it, as you can do, I think all of it, I think D costs are very interesting. But the other thing I was curious about is just you had some mentions about Kafka event driven systems and that. So I'm just curious more about what, what kind of use cases you're using for, cop you're using Kafka for,

Speaker 2 06:38

how you feel about it also, yeah. So one of the, you know, like you use case that we are working on with Kafka is for streaming engine right now. So we basically had a, you know, like, audit requirement, you know, where, let's say any customers related data. So if we use it, then we have to log those messages into our audit table. So for that, like, we basically publish it to the Kafka topics, right? And then basically streams it into a, we call it, like, snow snowflakes table, right? So I feel like it's a really good way, because you have to fire and, like, forget and like, you know, like, you don't have to worry about a whole a lot of things. So yeah, that has been my experience with Kafka. So most recently, you know, but like, in the past, we also have, like, I also have experience with like, you know, notification system driven and so Kafka topics as well.

Speaker 1 07:37

Fair enough. Fairness is mainly, and I'm guessing Kafka's partially picked that because, like, you need to audit, but there's no 100% consistency guarantee, so if it drops on that's okay, or at least acceptable,

Speaker 2 07:49

Yeah, makes sense, yeah. So it's That's right. So we, you know, like, just had around 90% of, you know, like, SLA, yeah, I

Unknown Speaker 08:00

guess that makes sense. Interesting.

Speaker 1 08:03

Well, the other thing, I guess so did the old stack like, have no CR like, mentioned that, you know, setting up automated builds that went from two weeks to two days. And I think that's another passion. Point of mine is that we aren't always best released about, like, what the state of that was before, and how you how you migrated

Speaker 2 08:24

that. So the older stack, you know, think of the older stack deployment was kind of really complicated, so I was not involved in deployment of the older stack, but for the newer stack, you know, like we basically integrated with Jenkins. So I have been involved in basically, you know, like creating the Docker file for our services and then integrating those with Jenkins. So basically, you know, like you build and you test and you you do, like a container in ECR, upload of the image of your Docker file, and then you use, like, you know, cloud formation templates using deployment. So that's what the lower environment would be for an upper environment, like, let's say a broader stage environment, right? You just need to open a change request with, you know, like, service now and then, you know, like, once it is approved, right? You do the deployment within, we call it a blue beam deployment strategy,

Unknown Speaker 09:24

and with some Canary as well. So

Speaker 1 09:27

interesting, but so so with the is the two weeks reference in your resume, like that, that's what the month was. Was deploying out, and these deploying to two days and that, that that's the improvement is just like the old stack did a really slow and the new stack uses better stuff, so it goes faster, because you mentioned specifically, like cutting release cycles from two weeks to two days. I'm just curious about what, yeah, yeah.

Speaker 2 09:47

So our microservices are, you know, like, deployed in two weeks and in the two days deployments are kind of, you know, like data fixes. So we use you we use the onboard on bound routes, right for our services. And every other day we kind of, you we know, like, we do some SQL scripts data as well, and fix the like data in short. So these do not, you know, like, require a change order, right? But our micro service are kind of like two week deployment

Speaker 1 10:20

you're doing, like a man, is that? So it's just like, someone kicks the build every two weeks, and that's the release structure, like, what's the sprint? Essentially?

Speaker 2 10:31

Yeah. So basically, we do, like a blue, green deployment. So it's kind of, it's kind of like a, you know, like couple of manual steps, but it's all about, like integrating as a pipeline, right? First it gets deployed into the off color. We do the, you know, like validations, and we proceed with the next step, you know, like we need attaches, let's say, active stack, and then we do validation again, and then we just flip off the color. So that's kind of, you know, like how

Speaker 1 11:02

it is interesting, cool. Well, let's pivot into the main part of the interview now. So do you have the hack Frank link, or do you want me to drop it in the chat? Happy to do either. Um, so yeah, let me check here. I have it up so I can also put in the chat, if it's faster Marina, I don't know if there's been other ones I

Speaker 1 11:44

Okay, perfect. So happy to read the question, if that's helpful, also happy just to let you read it through and ask any questions. Either way it works, you may

Unknown Speaker 11:59

Sure. So First, let me read through it first. I

Speaker 2 14:16

so it's basically a library management system, right? So where we kind of want to integrate the borrowing and returning functionality for a book for users, right? Yep, exactly.

Speaker 2 14:42

Okay, let me just follow through the prompt, and then, you know, like, try to get an integrated here. So it's ask me start with the users. So let's see if we can get here first. So

Speaker 2 15:02

so the users will be able to borrow a book and then return A book, right? I

Speaker 2 15:39

so on the borrow method, when the users borrow, what happens is, we can check if the users had already borrowed the loop or not, and if it's not, then, like we can Just, you know, like, borrow that particular group, Right, right?

Speaker 2 17:25

So if the users borrowed a book does not contain that particular book, book ID, let's say if the user's borrowed book doesn't contain that particular Book ID so

Speaker 2 18:46

so since, since this, you know, like particular uses, user is called, right? So, basically, borrowing the book so we can pass in, in the instance of these users, okay? And same thing, you know, like for returning, like, if it's actually so if it contains, you know, that book, then we just return it.

Speaker 2 19:22

So, that's take care of the user class. So if it doesn't contain, you know, that book, we borrow if it contains book or not, and we remove those. So this will be your primary functionality of the users class now, yeah, following that, we have a book class. Now, let's go to the unbook class. So we have borrow copy for users which have which kind of, you know, like allows, let's say users to be available, to be made available, right? So if it is successful, then we should print something and if it's not successful, then we need to print something else. So for that, we can basically check The number of available copies over there. I

Unknown Speaker 21:39

BBB.

Speaker 2 21:50

So then we kind of want to store, you know, like the instance of this particular class adds a value in that map, right? So since that is successful, we'll just print it out now For now, let's say

Speaker 2 22:56

now so it's available, right? So COVID is less or not, let's say so if it's not available, right? So then we need to Put that into Our queue. I

Speaker 2 24:22

so that's right. So we have a waiting list to offer, right? So then we kind of, you know, like, print a message now for to return a copy, right? So in the return the copy of the book and gives, give that, give it to the next person on The waiting list, if, let's Say, if that exists, I

Speaker 2 26:09

so this is returning part now, so it does say it give to the next person on the waiting list, right? So we kind of can, you know, check if the waiting list is empty or not, and like, if it's not empty, then we just get The next users, right? Yep,

Speaker 2 27:13

so I think that will take care of the boring logic as well. Here. Thanks. So

Speaker 1 27:21

yeah, thanks for now, let's, let's keep this, keep progressing. We can see if anything comes out as we go forward.

Unknown Speaker 27:29

So the book class now we can move. So

Unknown Speaker 27:33

let's go to the library. So if you, let's

Speaker 2 27:41

say, want to add a book into into it, like you just simply just add there, right?

Speaker 2 28:00

Um, similarly, you know, like for adding a users like added to the users map here.

Speaker 2 28:15

So that becomes the first like, easy one. Now we need to talk more about like borrowing a book for our user ID and our book ID, right? So firstly, we need to get users by the user ID, and let's Say book by the book ID. I

Speaker 2 29:01

so if, let's say both of them are not now, then we kind of can proceed. I

Speaker 2 29:25

and same thing here, like, if you want to get the you know users and a book, yep,

Unknown Speaker 29:55

so I think that's kind of like the process here.

Speaker 1 30:02

Yeah, do we want to run the test first and see if anything shakes out and talk about any first things,

Unknown Speaker 30:14

and this test cases should work. So I can say I

Unknown Speaker 30:33

show i

Speaker 1 30:43

i currently having a problem. Can you see the standard in this coming

Speaker 2 30:50

in goodbye, let's say, like, if it's it, just say too large. So let me Yeah, I see that now. Like formatting message first. Yep. I

Speaker 1 31:20

i think it's possible I really just have any issues, because at least the example one looks great

Unknown Speaker 31:40

to me. I do, yeah, if you can't get the standard work, I can't, yeah,

Speaker 2 31:46

let me download and I can see if I can do a single, single comparison.

Unknown Speaker 31:52

Yeah, I think that's good. Giving me a

Speaker 1 31:56

quest expired so I can't even see that input. But I

Unknown Speaker 32:03

yeah, I can see the

Speaker 1 32:08

account, yeah, okay, well, I have a suspicion about one way that perhaps it's not happy. How do we feel about the fact that the book class knows implementation details the user class.

Speaker 2 32:25

So that's one console that I have so that it's something that I don't like on like what I have done over here.

Unknown Speaker 32:39

Any thoughts about how we can fix

Speaker 2 32:47

it. So let me think so. Let me think how I would Want to handle that. Here. I

Unknown Speaker 33:50

We use the class hierarchy to our advantage point.

Speaker 2 33:55

So, you know, like, I'm just trying to see if I can use the, you know, like, available copies on the book class, you know, and

Unknown Speaker 34:05

to expose that from the book instead,

Speaker 1 34:10

without hearing or what do you mean by use the double copies?

Speaker 2 34:13

So I, you know, like, I'm just trying to get a read out these borrow book on top of the, you know, like on the on top of the hazmat itself. So basically put it in the users, rather than over there, over here. Yeah, I agree.

Unknown Speaker 34:37

So if I

Speaker 2 34:40

like do that, then I need a way, you know, to basically add it to the waste where it is, right. So let me expose like two sitters here, sitters and getters here. But it

Speaker 1 34:59

would be helpful if you just change the return type on the borrow copy method so slightly to give you that indicator about whether you succeeded in returning a book or not, or borrowing a book or not.

Speaker 2 35:14

So if we make it up, yeah, that will help. But the theme here is, you know, like, we still need a way to add to the queue, right,

Speaker 1 35:24

right? And I think, but I think that's fine, right? The book is orchestrating the waiting list queue. That's the book's responsibility. What about a user's responsibility? The user's responsibility is keeping track of what books they have. Okay, yeah,

Unknown Speaker 35:41

it makes sense then, yeah, I

Speaker 1 36:36

Why did you swap the order of the if and the house? I think it's fine. I'm just curious as to what the rationale was.

Speaker 2 36:41

Yeah, so I am doing like like earlier, so I usually like to return if the success scenario doesn't does that. So the if conditional does not get too complicated.

Speaker 2 37:06

So now, when I use users, borrow it right?

Unknown Speaker 37:19

Let me check one more thing here.

Speaker 2 37:26

So this check is not required, because the requirement does say, like, you will not borrow the same book more than once, right? So we can get up, rid of this one up.

Speaker 1 37:37

Yeah, I think it's nice to have and scale with, but it's not required. To run.

Speaker 2 38:36

It so that will be the borrow logic. So you know, like, only user has a user flow, right, and book has the book data.

Speaker 1 38:50

We want to refactor the can remove the get borrowed books, function, map.

Unknown Speaker 38:56

So, yeah, let me draw it. I think that should be.

Speaker 2 39:16

So are we kind of, you know, like seeing the get borrowed books, right? Because that is being used in the library class, being

Unknown Speaker 39:26

used in the library class.

Speaker 2 39:30

So sorry, like I missed the book. Let me remove it. Do.

Unknown Speaker 39:52

So we need the same thing here.

Unknown Speaker 39:57

So we did the same thing here.

Unknown Speaker 40:10

Oh, actually, okay, yeah, so we don't need that, right?

Speaker 2 40:18

Why? So once it is removed, like, because, you know, like, it's always going to be successful from our given scenario, excellent. So we just add there, so we just do.

Speaker 1 40:51

I don't know if that fixes the test, but it might. It might have actually caught some performance weirdness, but you see if they run better this time. If not, I think hack ranks just a little bit broken,

Speaker 2 41:04

that's fine. Let's see that. So if we had a side by side extreme comparison, then maybe,

Speaker 1 41:13

yeah, I think the hacker rank just can't talk to us three right now. Happens this case

Unknown Speaker 41:21

like it's past, right? Zero,

Unknown Speaker 41:22

Yep, yeah, yeah, I agree.

Speaker 1 41:26

How do we feel? Like, what? What if these were orchestrated calls? Like, for some reason the book like, what if the book service was, this was a service. Is there anything you would change, or anything that you would want to add to make it more robust in that case?

Speaker 2 41:50

So you are talking more about, like, a concurrency standpoint, or you are like, kind of talking about something

Speaker 1 42:01

else more referring to, like, what if that was a network call? What would it make you feel like you should change or wrap anything differently if it was a network call than if it was just an in memory call?

Speaker 2 42:17

So I'm, I mean, like, if it was a, let's say, a network call. I mean, first thing, you basically want to make sure that you know those were transactional, right? So, since it's in a memory, it's not a problem, but you need to make sure you don't, kind of, you know, like, have any shear state, right? Because those and like it, whenever a users borrow, it only makes sure that we have a proper response back, right? So maybe, as per our requirement, like, we might add a retry for, you know, like borrowing and like returning pause, depending on like how we, let's say, define the criteria, maybe retry with, you know, putting exponential backup address, you know, like some log on each of these method right to make sure that we have, let's say, proper visibility.

Unknown Speaker 43:13

So I think those will be the some things

Unknown Speaker 43:17

that I can

Speaker 1 43:20

add on. Would that make you want to return a boolean in the return copy, or do you still feel like a void is sufficient in that case? No,

Unknown Speaker 43:29

I know. I think that return will probably be

Speaker 2 43:31

something else, like pose object. Let's say so though that so that you don't have a proper response, like our standards are borrowed and a message of like a book ID, right? So rather than having a Boolean or void, right? You will, you probably want a proper return time, let's say

Speaker 1 43:51

makes sense. See, you would have a return type, just not necessarily a Boolean, because the booleans maybe not verbose enough, right?

Speaker 2 43:59

Yeah. So, yeah. So we will want, like, you know, details, like maybe some event status are, like borrowed or not, and like a Mirada, right, like a metadata attached to it.

Speaker 1 44:20

Excellent. And I know it's not in the requirements, but what would you propose we do in this borrow book and return bookcase where the user or the book is null?

Speaker 2 44:37

So we, like, kind of haven't handled that case. So in those cases, you know, like, we will just basically return false here, so it will be, it will be a failure, like, right? So the first time here, if the user, if the first sign here, like, if the user is not, then you know, like, we kind of return. So we will, we will do like input the validation as well. So maybe using a valid annotations within a parameter on the, you know, like rest controllers

Speaker 1 45:14

and you would, you would be more of a fan of returning a Boolean than throwing an exception. Or do you think this is an exceptional case.

Speaker 2 45:23

I'm sorry. So, like, we will basically, you know, try an exception or

Speaker 1 45:29

unique, yeah, the magic of the hand, valid, got it? Yeah, they

Speaker 2 45:33

need to be handled through, you know, like, you know, global exception handler for, you know, like, proper for

Unknown Speaker 45:38

handling. So it

Speaker 1 45:41

is an exception. Otherwise, it's not just, it's, it is truly exceptional. If you give us something, not just return falls and fail silently,

Speaker 2 45:52

right, right? Yeah. So it will be an, you know, like actual exceptions that you can see the, you know, like stack trace and so on, right? Yeah. Got it.

Speaker 1 46:04

I mean, for fun, do you want to put in what, what you would do on borrow book for an exception, not just on the library layer? I think because it is that if Noel is not equal to user case that's sitting there, so

Speaker 2 46:27

in the you know, like use on the book, if the you know, like user is not, will throw if there is no user, right? So it will draw new illegal argument exceptions, right? So

Unknown Speaker 46:49

so it would be something like that. You know,

Speaker 1 46:59

the only other thing, I guess, to talk through, stylistically serious about, how do you feel about the fact that these are nulls versus optionals any stance on either way?

Unknown Speaker 47:13

So I think I prefer

Speaker 2 47:16

optional more, because, you know, it gives a little more safety on those objects, right? Because you can do not optional, like easy, MT or not, like you check before proceeding. And I think that's an excellent features provided with Java eight, they have added, right? Yeah,

Speaker 1 47:39

for expedience, we use null, but if you were writing this reproduction, you would use optional as nullable on the users.gov

Unknown Speaker 47:50

Okay, I think it's all the questions I have for the

Speaker 1 47:57

this function. I'll give you some time to ask me questions now, so we're pretty much the time. So yeah,

Speaker 2 48:05

thank you so much for the time. You know, like, it was really nice talking to you. It was fortunate that, you know, like, we can get a validation on the, you know, like, actual test cases. But some of the questions that I have is, you know, like, what do use it, engineers, users, usually get involved in apart from coding,

Unknown Speaker 48:28

say, more like, like science apart

Speaker 1 48:32

from coding, in terms of, like, what, what process or like outside groups, like, what exactly about outside of coding? I mean, there's lots, but I just want to make sure I can target it,

Speaker 2 48:44

yeah, so, like, I mean, like designs and, you know, like requirement gatherings and so on. But

Speaker 1 48:52

I think it's, I think it's worth having. I mean, it's somewhat of like a based upon seniority within the team, right? Like there is the more leads and like, the engineering managers have the earlier conversations about the feasibility, and then it kind of funnel funneled down into and it's not entirely based on singular it's based on context. Obviously, like, if someone's really has high context on a specific feature, like our team specifically does, like single threaded ownership things, and so like, the owner of that feature is the one that is the like contact person for external stakeholders to talk through feasibility, low level designs, what have you. And so it's definitely not just like being assigned code. There is, there is ownership. And I think it's pretty standard. I think it's relatively standard. But I think, you know, I suppose that's maybe an overstatement. Some companies may are just, just right COVID. It is, it is a design process. Is a collaboration with outside product stakeholders, and what have you. Sounds

Speaker 2 49:46

good. So you know, like so much that information, so something else I like to know about it? What do you like most about working here at PayPal? So maybe, like, any suggestions about, like, work life balance as well?

Speaker 1 50:02

Yeah, I mean, I think that first off, like PayPal, is PayPal, like every everyone knows that there's therefore a lot of scale, a lot of opportunity to, like, work on things that aren't just a small scale thing, like I was at a start before this, and it's just a different scale, obviously, to work with 1000s of transactions a second, and just have to design around that. So I think that's definitely one of the coolest things with this kind of opportunity, and I think also work life balance, I think is pretty good. I think the nice thing I've said is that PayPal is the first one I've ever worked at where they actually give additional pay, like there's a small, a small stipend, but stipend for being on call. And that's not something that I've ever had before. I'm used to just doing on call as part of the job duties. So that is a nice little like, kind of recognition of that. And so I think that there is definitely like that, that cultural support for it. I think that, as with any company, like, when things get closer to their lines, I think there is more stress and such. And so there may be less of a balance, but I do think that there is, like, there's an element of PTO, things like that. So like, I think there is a culture of, like, longing to recognize that it's not your sole motivation, in a way that, like, maybe some startups are, we're like, hit the mission. If you're not lying with the mission, then get out. So anyone's got any pretenses, large corporation, and there are other things that you're here for besides just loving PayPal.

Speaker 2 51:17

Yeah? So, like, that's all questions I have. So,

Unknown Speaker 51:24

like, I it was really nice talking to you.

Speaker 1 51:28

Yeah, I'll give you, I'll give you a nice back end. I'll get what I need. So thanks. Best of luck with any future interviews. Yeah,

Unknown Speaker 51:33

thank you, Alex, for your time. Awesome. Bye. Bye.

Suman-Paypal-Role-Specialization-SRV-Final otter ai

Speaker 1 0:00

Rewarding team here at PayPal. I'm based out of Colorado. I said that used to work in Denver, so it's nice to see that. Yeah, this is the, like, the role specialization interview. So we'll, we'll start a little bit of just talking to your resume a couple I have a couple questions, and then the majority interview, like 50 ish minutes, will be doing, like a problem that's essentially like uniting Java in a realistic context. And then I'll make sure to leave you like five or 10 minutes at the end to ask questions. I'll be taking notes throughout, but I am paying attention and happy to answer any questions or discuss anything before we get into it. If you want to introduce yourself, feel free. Otherwise, we'll just know each other. So sure, Alex.

Speaker 2 0:40

So thank you so much for time today, Alex. My name is Suman, so I have been working at the software engineering for it's about been like six years now. So I have been, you know, like, primarily working on Java based applications utilizing framework like, like spring. So right now I am developing, you know, like EKG experience or service or working on microservice. So basically, my project is about a micro service development, right? So I have also, like, migration of our monthly tab on microservices. So I have no migrations as well. So it involves, you know, like designing services, like developing them and like deploying them on the AWS cloud, right? So some of the cloud services that I work on include AWS, easy to ECS, like SQS, SNS and lambdas. So talking about a bit more about my current team. My team consists of around like five developers. We have product owners, Scrum Masters, and on the daily basis, like I have, I'm mostly involved working on the back end Java code and helping a team with, you know, code reviews and sometimes some products and support duties as well.

Unknown Speaker 2:01

Sorry to hear that, but it makes sense.

Speaker 1 2:05

Cool, cool. Yeah, so, I mean, that was the first question I kind of have reading this through, is you mentioned a lot of Angular and Java. Are those the same service? Are those two different ones? Like, curious about what that looks like. Um, so like

Speaker 2 2:20

we have Java as a back end service, right? So, like our backing micro services are primarily on Java. So we do have, you know, like some lambda return on, you know, like Node js as well, and on our front end service, like Portal, we have written that in Angular So our front in Angular portal basically makes, you know, HTTP calls to those backing services.

Speaker 1 2:47

Got it. Okay. So, so there are two separate ones. I'm gonna say, if you are co hosting Angular job, very curious to hear more about that. But I think, yeah, got it. Got it. And so, I think the other thing that I was gonna get into resumes, it seems like a lot of like, zero to one stuff, like, pipelines, databases, logging, like, just generally basic stuff. Is this just because this is migration? Or tell me more about, like, why it's so foundational of this work?

Unknown Speaker 3:13

Sure. So, like,

Speaker 2 3:16

basically, it's just a micro snippy food, right? So we do have legacy portal as well, and legacy flow, let's say which is, you know, very outdated, right? And like, like, we are basically, you know, breaking down those existing, existing services and flows, right? So some of the flows that we have migrated, like, the maintenance of our, you know, user account references, right? So also the experience aspect as well. So it's building on, you know, from the ground, like, ground up,

Speaker 1 3:50

got it so. And is it from, you may have already said, like, Is this from a monolith architecture, or just from, like, an old stack of microservices to a new stack of microservices?

Speaker 2 3:59

So it's a monolithic app, I would say it's basically, you know, like, one single app that we are breaking down into

Speaker 1 4:11

that. How did that then there's obviously a lot of lot of good with decoupling, but there's also a lot of, like, weird things that come out of that. How has it been to kind of decompose a monolith into microservices? So for that, you know, like, also,

Speaker 2 4:26

like, it has been pretty good so far. So basically, you know, like, decoupling, definitely, like, bring up challenges, right? So you you want to handle, you know, like, let's say, consistency across the services, right? So, how you handle the stage, right? So we pretty must have a good, like, orchestration platform that handles the saga. And, you know, like the communication is, you know, like, are taking place during, like, graceful API. So it has been pretty good so far,

Speaker 1 4:57

makes sense. And, yeah, I guess that related to that, like, is this a big bang kind of approach. Are you doing it? Like 5050, phase rollouts? Like, what? How is the actual splitting off of, like, a function working like, is it, this is a function that's next to you, decoupled? Or what does that look like? So, like,

Speaker 2 5:14

we are kind of, you know, like doing small sounds, so basically breaking down the related units, like, once a recent one, like we did was for user profile preferences. So, you know, like, once a user's logging into our portal, then they want to make changes to like, like, you know, like, how they want to build their portal, right? So they kick in some kind of people, like Experience API. So basically configuring the wig gates, right, and the components that they visit visible also. So that was, you know, something that we did. And then the other one is a maintenance API, which is strictly, you know, like for making changes, let's say, to their preferences of the user. So we are kind of, you know, like breaking down that chunks. So I think the next service will be, like actual weight allocated to the investment service. So we are not doing so. We are doing like a phased development, deployment, where we are, you know, like breaking down the all the small parts and then wrapping it around, like feature flags, that makes sense.

Speaker 1 6:20

Got it, as you can do, I think all of it, I think D costs are very interesting. But the other thing I was curious about is just you had some mentions about Kafka event driven systems and that. So I'm just curious more about what, what kind of use cases you're using for, cop you're using Kafka for,

Speaker 2 6:38

how you feel about it also, yeah. So one of the, you know, like you use case that we are working on with Kafka is for streaming engine right now. So we basically had a, you know, like, audit requirement, you know, where, let's say any customers related data. So if we use it, then we have to log those messages into our audit table. So for that, like, we basically publish it to the Kafka topics, right? And then basically streams it into a, we call it, like, snow snowflakes table, right? So I feel like it's a really good way, because you have to fire and, like, forget and like, you know, like, you don't have to worry about a whole a lot of things. So yeah, that has been my experience with Kafka. So most recently, you know, but like, in the past, we also have, like, I also have experience with like, you know, notification system driven and so Kafka topics as well.

Speaker 1 7:37

Fair enough. Fairness is mainly, and I'm guessing Kafka's partially picked that because, like, you need to audit, but there's no 100% consistency guarantee, so if it drops on that's okay, or at least acceptable,

Speaker 2 7:49

Yeah, makes sense, yeah. So it's That's right. So we, you know, like, just had around 90% of, you know, like, SLA, yeah, I

Unknown Speaker 8:00

guess that makes sense. Interesting.

Speaker 1 8:03

Well, the other thing, I guess so did the old stack like, have no CR like, mentioned that, you know, setting up automated builds that went from two weeks to two days. And I think that's another passion. Point of mine is that we aren't always best released about, like, what the state of that was before, and how you how you migrated

Speaker 2 8:24

that. So the older stack, you know, think of the older stack deployment was kind of really complicated, so I was not involved in deployment of the older stack, but for the newer stack, you know, like we basically integrated with Jenkins. So I have been involved in basically, you know, like creating the Docker file for our services and then integrating those with Jenkins. So basically, you know, like you build and you test and you you do, like a container in ECR, upload of the image of your Docker file, and then you use, like, you know, cloud formation templates using deployment. So that's what the lower environment would be for an upper environment, like, let's say a broader stage environment, right? You just need to open a change request with, you know, like, service now and then, you know, like, once it is approved, right? You do the deployment within, we call it a blue beam deployment strategy,

Unknown Speaker 9:24

and with some Canary as well. So

Speaker 1 9:27

interesting, but so so with the is the two weeks reference in your resume, like that, that's what the month was. Was deploying out, and these deploying to two days and that, that that's the improvement is just like the old stack did a really slow and the new stack uses better stuff, so it goes faster, because you mentioned specifically, like cutting release cycles from two weeks to two days. I'm just curious about what, yeah, yeah.

Speaker 2 9:47

So our microservices are, you know, like, deployed in two weeks and in the two days deployments are kind of, you know, like data fixes. So we use you we use the onboard on bound routes, right for our services. And every other day we kind of, you we know, like, we do some SQL scripts data as well, and fix the like data in short. So these do not, you know, like, require a change order, right? But our micro service are kind of like two week deployment

Speaker 1 10:20

you're doing, like a man, is that? So it's just like, someone kicks the build every two weeks, and that's the release structure, like, what's the sprint? Essentially?

Speaker 2 10:31

Yeah. So basically, we do, like a blue, green deployment. So it's kind of, it's kind of like a, you know, like couple of manual steps, but it's all about, like integrating as a pipeline, right? First it gets deployed into the off color. We do the, you know, like validations, and we proceed with the next step, you know, like we need attaches, let's say, active stack, and then we do validation again, and then we just flip off the color. So that's kind of, you know, like how

Speaker 1 11:02

it is interesting, cool. Well, let's pivot into the main part of the interview now. So do you have the hack Frank link, or do you want me to drop it in the chat? Happy to do either. Um, so yeah, let me check here. I have it up so I can also put in the chat, if it's faster Marina, I don't know if there's been other ones I

Speaker 1 11:44

Okay, perfect. So happy to read the question, if that's helpful, also happy just to let you read it through and ask any questions. Either way it works, you may

Unknown Speaker 11:59

Sure. So First, let me read through it first. I

Speaker 2 14:16

so it's basically a library management system, right? So where we kind of want to integrate the borrowing and returning functionality for a book for users, right? Yep, exactly.

Speaker 2 14:42

Okay, let me just follow through the prompt, and then, you know, like, try to get an integrated here. So it's ask me start with the users. So let's see if we can get here first. So

Speaker 2 15:02

so the users will be able to borrow a book and then return A book, right? I

Speaker 2 15:39

so on the borrow method, when the users borrow, what happens is, we can check if the users had already borrowed the loop or not, and if it's not, then, like we can Just, you know, like, borrow that particular group, Right, right?

Speaker 2 17:25

So if the users borrowed a book does not contain that particular book, book ID, let's say if the user's borrowed book doesn't contain that particular Book ID so

Speaker 2 18:46

so since, since this, you know, like particular uses, user is called, right? So, basically, borrowing the book so we can pass in, in the instance of these users, okay? And same thing, you know, like for returning, like, if it's actually so if it contains, you know, that book, then we just return it.

Speaker 2 19:22

So, that's take care of the user class. So if it doesn't contain, you know, that book, we borrow if it contains book or not, and we remove those. So this will be your primary functionality of the users class now, yeah, following that, we have a book class. Now, let's go to the unbook class. So we have borrow copy for users which have which kind of, you know, like allows, let's say users to be available, to be made available, right? So if it is successful, then we should print something and if it's not successful, then we need to print something else. So for that, we can basically check The number of available copies over there. I

Unknown Speaker 21:39

BBB.

Speaker 2 21:50

So then we kind of want to store, you know, like the instance of this particular class adds a value in that map, right? So since that is successful, we'll just print it out now For now, let's say

Speaker 2 22:56

now so it's available, right? So COVID is less or not, let's say so if it's not available, right? So then we need to Put that into Our queue. I

Speaker 2 24:22

so that's right. So we have a waiting list to offer, right? So then we kind of, you know, like, print a message now for to return a copy, right? So in the return the copy of the book and gives, give that, give it to the next person on The waiting list, if, let's Say, if that exists, I

Speaker 2 26:09

so this is returning part now, so it does say it give to the next person on the waiting list, right? So we kind of can, you know, check if the waiting list is empty or not, and like, if it's not empty, then we just get The next users, right? Yep,

Speaker 2 27:13

so I think that will take care of the boring logic as well. Here. Thanks. So

Speaker 1 27:21

yeah, thanks for now, let's, let's keep this, keep progressing. We can see if anything comes out as we go forward.

Unknown Speaker 27:29

So the book class now we can move. So

Unknown Speaker 27:33

let's go to the library. So if you, let's

Speaker 2 27:41

say, want to add a book into into it, like you just simply just add there, right?

Speaker 2 28:00

Um, similarly, you know, like for adding a users like added to the users map here.

Speaker 2 28:15

So that becomes the first like, easy one. Now we need to talk more about like borrowing a book for our user ID and our book ID, right? So firstly, we need to get users by the user ID, and let's Say book by the book ID. I

Speaker 2 29:01

so if, let's say both of them are not now, then we kind of can proceed. I

Speaker 2 29:25

and same thing here, like, if you want to get the you know users and a book, yep,

Unknown Speaker 29:55

so I think that's kind of like the process here.

Speaker 1 30:02

Yeah, do we want to run the test first and see if anything shakes out and talk about any first things,

Unknown Speaker 30:14

and this test cases should work. So I can say I

Unknown Speaker 30:33

show i

Speaker 1 30:43

i currently having a problem. Can you see the standard in this coming

Speaker 2 30:50

in goodbye, let's say, like, if it's it, just say too large. So let me Yeah, I see that now. Like formatting message first. Yep. I

Speaker 1 31:20

i think it's possible I really just have any issues, because at least the example one looks great

Unknown Speaker 31:40

to me. I do, yeah, if you can't get the standard work, I can't, yeah,

Speaker 2 31:46

let me download and I can see if I can do a single, single comparison.

Unknown Speaker 31:52

Yeah, I think that's good. Giving me a

Speaker 1 31:56

quest expired so I can't even see that input. But I

Unknown Speaker 32:03

yeah, I can see the

Speaker 1 32:08

account, yeah, okay, well, I have a suspicion about one way that perhaps it's not happy. How do we feel about the fact that the book class knows implementation details the user class.

Speaker 2 32:25

So that's one console that I have so that it's something that I don't like on like what I have done over here.

Unknown Speaker 32:39

Any thoughts about how we can fix

Speaker 2 32:47

it. So let me think so. Let me think how I would Want to handle that. Here. I

Unknown Speaker 33:50

We use the class hierarchy to our advantage point.

Speaker 2 33:55

So, you know, like, I'm just trying to see if I can use the, you know, like, available copies on the book class, you know, and

Unknown Speaker 34:05

to expose that from the book instead,

Speaker 1 34:10

without hearing or what do you mean by use the double copies?

Speaker 2 34:13

So I, you know, like, I'm just trying to get a read out these borrow book on top of the, you know, like on the on top of the hazmat itself. So basically put it in the users, rather than over there, over here. Yeah, I agree.

Unknown Speaker 34:37

So if I

Speaker 2 34:40

like do that, then I need a way, you know, to basically add it to the waste where it is, right. So let me expose like two sitters here, sitters and getters here. But it

Speaker 1 34:59

would be helpful if you just change the return type on the borrow copy method so slightly to give you that indicator about whether you succeeded in returning a book or not, or borrowing a book or not.

Speaker 2 35:14

So if we make it up, yeah, that will help. But the theme here is, you know, like, we still need a way to add to the queue, right,

Speaker 1 35:24

right? And I think, but I think that's fine, right? The book is orchestrating the waiting list queue. That's the book's responsibility. What about a user's responsibility? The user's responsibility is keeping track of what books they have. Okay, yeah,

Unknown Speaker 35:41

it makes sense then, yeah, I

Speaker 1 36:36

Why did you swap the order of the if and the house? I think it's fine. I'm just curious as to what the rationale was.

Speaker 2 36:41

Yeah, so I am doing like like earlier, so I usually like to return if the success scenario doesn't does that. So the if conditional does not get too complicated.

Speaker 2 37:06

So now, when I use users, borrow it right?

Unknown Speaker 37:19

Let me check one more thing here.

Speaker 2 37:26

So this check is not required, because the requirement does say, like, you will not borrow the same book more than once, right? So we can get up, rid of this one up.

Speaker 1 37:37

Yeah, I think it's nice to have and scale with, but it's not required. To run.

Speaker 2 38:36

It so that will be the borrow logic. So you know, like, only user has a user flow, right, and book has the book data.

Speaker 1 38:50

We want to refactor the can remove the get borrowed books, function, map.

Unknown Speaker 38:56

So, yeah, let me draw it. I think that should be.

Speaker 2 39:16

So are we kind of, you know, like seeing the get borrowed books, right? Because that is being used in the library class, being

Unknown Speaker 39:26

used in the library class.

Speaker 2 39:30

So sorry, like I missed the book. Let me remove it. Do.

Unknown Speaker 39:52

So we need the same thing here.

Unknown Speaker 39:57

So we did the same thing here.

Unknown Speaker 40:10

Oh, actually, okay, yeah, so we don't need that, right?

Speaker 2 40:18

Why? So once it is removed, like, because, you know, like, it's always going to be successful from our given scenario, excellent. So we just add there, so we just do.

Speaker 1 40:51

I don't know if that fixes the test, but it might. It might have actually caught some performance weirdness, but you see if they run better this time. If not, I think hack ranks just a little bit broken,

Speaker 2 41:04

that's fine. Let's see that. So if we had a side by side extreme comparison, then maybe,

Speaker 1 41:13

yeah, I think the hacker rank just can't talk to us three right now. Happens this case

Unknown Speaker 41:21

like it's past, right? Zero,

Unknown Speaker 41:22

Yep, yeah, yeah, I agree.

Speaker 1 41:26

How do we feel? Like, what? What if these were orchestrated calls? Like, for some reason the book like, what if the book service was, this was a service. Is there anything you would change, or anything that you would want to add to make it more robust in that case?

Speaker 2 41:50

So you are talking more about, like, a concurrency standpoint, or you are like, kind of talking about something

Speaker 1 42:01

else more referring to, like, what if that was a network call? What would it make you feel like you should change or wrap anything differently if it was a network call than if it was just an in memory call?

Speaker 2 42:17

So I'm, I mean, like, if it was a, let's say, a network call. I mean, first thing, you basically want to make sure that you know those were transactional, right? So, since it's in a memory, it's not a problem, but you need to make sure you don't, kind of, you know, like, have any shear state, right? Because those and like it, whenever a users borrow, it only makes sure that we have a proper response back, right? So maybe, as per our requirement, like, we might add a retry for, you know, like borrowing and like returning pause, depending on like how we, let's say, define the criteria, maybe retry with, you know, putting exponential backup address, you know, like some log on each of these method right to make sure that we have, let's say, proper visibility.

Unknown Speaker 43:13

So I think those will be the some things

Unknown Speaker 43:17

that I can

Speaker 1 43:20

add on. Would that make you want to return a boolean in the return copy, or do you still feel like a void is sufficient in that case? No,

Unknown Speaker 43:29

I know. I think that return will probably be

Speaker 2 43:31

something else, like pose object. Let's say so though that so that you don't have a proper response, like our standards are borrowed and a message of like a book ID, right? So rather than having a Boolean or void, right? You will, you probably want a proper return time, let's say

Speaker 1 43:51

makes sense. See, you would have a return type, just not necessarily a Boolean, because the booleans maybe not verbose enough, right?

Speaker 2 43:59

Yeah. So, yeah. So we will want, like, you know, details, like maybe some event status are, like borrowed or not, and like a Mirada, right, like a metadata attached to it.

Speaker 1 44:20

Excellent. And I know it's not in the requirements, but what would you propose we do in this borrow book and return bookcase where the user or the book is null?

Speaker 2 44:37

So we, like, kind of haven't handled that case. So in those cases, you know, like, we will just basically return false here, so it will be, it will be a failure, like, right? So the first time here, if the user, if the first sign here, like, if the user is not, then you know, like, we kind of return. So we will, we will do like input the validation as well. So maybe using a valid annotations within a parameter on the, you know, like rest controllers

Speaker 1 45:14

and you would, you would be more of a fan of returning a Boolean than throwing an exception. Or do you think this is an exceptional case.

Speaker 2 45:23

I'm sorry. So, like, we will basically, you know, try an exception or

Speaker 1 45:29

unique, yeah, the magic of the hand, valid, got it? Yeah, they

Speaker 2 45:33

need to be handled through, you know, like, you know, global exception handler for, you know, like, proper for

Unknown Speaker 45:38

handling. So it

Speaker 1 45:41

is an exception. Otherwise, it's not just, it's, it is truly exceptional. If you give us something, not just return falls and fail silently,

Speaker 2 45:52

right, right? Yeah. So it will be an, you know, like actual exceptions that you can see the, you know, like stack trace and so on, right? Yeah. Got it.

Speaker 1 46:04

I mean, for fun, do you want to put in what, what you would do on borrow book for an exception, not just on the library layer? I think because it is that if Noel is not equal to user case that's sitting there, so

Speaker 2 46:27

in the you know, like use on the book, if the you know, like user is not, will throw if there is no user, right? So it will draw new illegal argument exceptions, right? So

Unknown Speaker 46:49

so it would be something like that. You know,

Speaker 1 46:59

the only other thing, I guess, to talk through, stylistically serious about, how do you feel about the fact that these are nulls versus optionals any stance on either way?

Unknown Speaker 47:13

So I think I prefer

Speaker 2 47:16

optional more, because, you know, it gives a little more safety on those objects, right? Because you can do not optional, like easy, MT or not, like you check before proceeding. And I think that's an excellent features provided with Java eight, they have added, right? Yeah,

Speaker 1 47:39

for expedience, we use null, but if you were writing this reproduction, you would use optional as nullable on the users.gov

Unknown Speaker 47:50

Okay, I think it's all the questions I have for the

Speaker 1 47:57

this function. I'll give you some time to ask me questions now, so we're pretty much the time. So yeah,

Speaker 2 48:05

thank you so much for the time. You know, like, it was really nice talking to you. It was fortunate that, you know, like, we can get a validation on the, you know, like, actual test cases. But some of the questions that I have is, you know, like, what do use it, engineers, users, usually get involved in apart from coding,

Unknown Speaker 48:28

say, more like, like science apart

Speaker 1 48:32

from coding, in terms of, like, what, what process or like outside groups, like, what exactly about outside of coding? I mean, there's lots, but I just want to make sure I can target it,

Speaker 2 48:44

yeah, so, like, I mean, like designs and, you know, like requirement gatherings and so on. But

Speaker 1 48:52

I think it's, I think it's worth having. I mean, it's somewhat of like a based upon seniority within the team, right? Like there is the more leads and like, the engineering managers have the earlier conversations about the feasibility, and then it kind of funnel funneled down into and it's not entirely based on singular it's based on context. Obviously, like, if someone's really has high context on a specific feature, like our team specifically does, like single threaded ownership things, and so like, the owner of that feature is the one that is the like contact person for external stakeholders to talk through feasibility, low level designs, what have you. And so it's definitely not just like being assigned code. There is, there is ownership. And I think it's pretty standard. I think it's relatively standard. But I think, you know, I suppose that's maybe an overstatement. Some companies may are just, just right COVID. It is, it is a design process. Is a collaboration with outside product stakeholders, and what have you. Sounds

Speaker 2 49:46

good. So you know, like so much that information, so something else I like to know about it? What do you like most about working here at PayPal? So maybe, like, any suggestions about, like, work life balance as well?

Speaker 1 50:02

Yeah, I mean, I think that first off, like PayPal, is PayPal, like every everyone knows that there's therefore a lot of scale, a lot of opportunity to, like, work on things that aren't just a small scale thing, like I was at a start before this, and it's just a different scale, obviously, to work with 1000s of transactions a second, and just have to design around that. So I think that's definitely one of the coolest things with this kind of opportunity, and I think also work life balance, I think is pretty good. I think the nice thing I've said is that PayPal is the first one I've ever worked at where they actually give additional pay, like there's a small, a small stipend, but stipend for being on call. And that's not something that I've ever had before. I'm used to just doing on call as part of the job duties. So that is a nice little like, kind of recognition of that. And so I think that there is definitely like that, that cultural support for it. I think that, as with any company, like, when things get closer to their lines, I think there is more stress and such. And so there may be less of a balance, but I do think that there is, like, there's an element of PTO, things like that. So like, I think there is a culture of, like, longing to recognize that it's not your sole motivation, in a way that, like, maybe some startups are, we're like, hit the mission. If you're not lying with the mission, then get out. So anyone's got any pretenses, large corporation, and there are other things that you're here for besides just loving PayPal.

Speaker 2 51:17

Yeah? So, like, that's all questions I have. So,

Unknown Speaker 51:24

like, I it was really nice talking to you.

Speaker 1 51:28

Yeah, I'll give you, I'll give you a nice back end. I'll get what I need. So thanks. Best of luck with any future interviews. Yeah,

Unknown Speaker 51:33

thank you, Alex, for your time. Awesome. Bye. Bye.

vishnu giri 3 30 interview otter ai

Speaker 1 0:00

Bishwo, yeah, I'm doing great. How are

Unknown Speaker 0:14

you sure?

Unknown Speaker 0:22

Yes, perfect. I'll let the quorum between you both. Just

Unknown Speaker 0:36

give it a couple more months. Okay, I'm just ready

Unknown Speaker 0:41

to join this. That's fine. Okay? You

Speaker 2 1:03

so to a wall, you're not using a background. Sorry, you're back to a wall, you're not using any backgrounds. Oh, no, no.

Speaker 3 1:12

I don't like using backgrounds my desk

Unknown Speaker 1:18

chops a little bit, here and there, right? So, sorry. It chops a little bit here and there. Oh yeah, The backgrounds, Yes. I

Speaker 4 2:20

Bishwo Bear with me. I'm just looking on another screen. Yeah, Totally fine. I Bishwo

Unknown Speaker 4:00

Bishwo Okay,

Unknown Speaker 4:36

this would call You, so I guess he's looking for You.

Unknown Speaker 4:49

Perfect. Thank you. Hey, Bishwo.

Speaker 4 4:58

Hey, bishwoman, myself, Kumar Kumar, we have Shashi Khan. He goes by the Shashi Bishwo for us to get the combination started. And it just gave me like overview of your like, bring good experience on your Java code. And fact, your recent project, area of your class project, I have given day to day contribution to the project, okay, and any other details that you feel that you want

Speaker 3 5:33

to add? Oh, sure, definitely. So myself, bishwnu Giri, and I've been working as a senior software, you know, like engineer for over, like six years, and working primarily, you know, like on Java based applications utilizing the frameworks like spring, have a need RESTful web services. You know, like for web development, a pretty good, you know, like experience working on microservices architectures, and I primarily, you know, like working in designing and developing and also deploying the micro services using Java and Spring Boot based technologies. And pretty good experience working with AWS cloud, you know, like for hosting these, you know, like this applications and utilizing the services, like easy to ECS, SQS and SNS and lambdas. And I also, you know, like, have good experience with Jenkins and the cloud formation tablets for CICD. And currently I'm on on a scrum team. My team consists of, you know, like five developers. We have a product owner and a Scrum Masters. And on daily basis, I involve, you know, like mostly working on back end in Java code and helping the team with code reviews, and also, you know, like a production support device,

Speaker 4 6:50

Okay, what about the data? Database, rdbfs databases?

Speaker 3 6:57

Oh, yes, so database we currently use the post Postgres, SQL on AWS, rds, and we also, you know, like, use the dyno dB. And some of our project that you know, like stores, like schemas, less entitlements, you know, like for loading dashboards. You

Speaker 4 7:26

I'm just give you a quick glance.

Speaker 4 7:39

You have mentioned like, both, like Postgres SQL and like MongoDB,

Unknown Speaker 7:46

how it is used in the application.

Speaker 3 7:51

Yes, so the postgres is, you know, like, it's our, like, final, like, primary database, and this contains the data that is, you know, like specific to the user, right? So if the user has any preference, or if there is, you know, like any claims, are, you know, like a benefit information that is pretending to a user, these are a standard, you know, like a data that's get, you know, like a persisted to, you know, like a user, and they're like a into the postgres database. So that's actually usually, you know, like what our microservice usually interact with, but the dynamic DB that we have it is, basically, is for, you know, like a specific use case. And this use case is for, you know, like the entitlements for different widgets and components for the front end, right? So our back end, you know, like a back end system handles entitlement which contain what different type of users and user group can access in the front end application, and this Dynamo DB stores like indictments for each of the you know, like micro apps on our like mono app and different groups that users, you know, like types are associated with. So the Dynamo DB can be, you know, like, taken as a place for storing, like the config for dashboard metadata.

Speaker 4 9:22

You have any questions before we kind of get to the details.

Speaker 2 9:29

So how do you like differentiate between Postgres and MongoDB? What do you think are the advantages or disadvantages?

Speaker 3 9:39

Yes, definitely. So there are, you know, like, multiple differences between the postgres and the, you know, like Mongo, so, right, basically, you know, like Postgres is a relational database that has a schema, and you have different between, you know, like different tables with them, and you basically have a high, strong consistency between the data, right? And basically, like the Mongo is used for storing, you know, like the documents, or, like a, you know, like a key value pair, where you don't have a strict schema in place, and it's usually more on, like, the availability and partition like tolerance, but might have not as strong consistency as compared to, like a relational database. So, so you know, like your post K Postgres standard has, you know, like AC transactions, and it's really useful for like, like, making sure like, everything is in a consistent state. But Mongo is a base transactions with maybe like and eventual consistency in place. And Mongo is usually preferred if you you know, if you need, like, a high availability and articulation tolerance and horizontal scaling between those.

Unknown Speaker 11:06

And what was your use case for Redis Database?

Speaker 3 11:11

Use Case for Redis. Okay, so use case for ready? So, yeah, so Redis is, is a cache, right? So, basically, we store our, like, our static data we might have for certain, you know, like routes or other configurations that that does not change a whole lot. We basically make sure we have it, you know, like wandering cache so that you don't have to, you know, like make database entry for each of these static data reads, right? Because database operations can sometimes be slow. So some, some like stale data, you know, like, like the explanation of manifest routes and explanation of benefits informations can be, you know, like, stored in our red space, so it's easily available.

Unknown Speaker 12:05

We can go in details. Now.

Speaker 2 12:11

You want me to proceed? Okay, so Bishwo, can you like walk us through the architecture of your project? Current Project?

Speaker 3 12:21

Oh, yes, definitely. So basically, we are working with, you know, like the claims management for user in our in our current project, right? So we have two folds of this application. The first four is, you know, like the data ingestion process, where we try to, you know, like, ingest the data from the multiple clients and states. And this is of transformations as a service that we provide. And for this, we basically use, you know, like a spark platform. And the main application that our team own is, is, is a RESTful API for exposing these claims and benefits data. So we have, you know, like a lot of spring boost, Spring Boot based microservices deployed on AWS ECS for, you know, like fargate instance, and these are tied together on by API gateway. So these services use the postgres SQL as it's like a primary database. And, yeah, I would later, like say that's a very high level or so whenever we want to, you know, like view their exponential behavior for particular claim that they are making, the users send requests to the API gateway, and the API gateway, you know, like, forwards their request to the exponential benefit layer, where It you know, like might call other services and aggregate the data to send back to the user. So, yeah, that's the high level.

Speaker 2 13:48

So I never you have two parts. One is the spark for ingestion, bulk data ingestion, and second is your micro service architecture. Currently, micro service architecture, you mentioned you have API gateway, then you have some rest models in between dash micro services, and then you have databases, right? Yeah. So as you said, you have MongoDB, DynamoDB and Redis. These are the three different types of databases here. Let's say that is not database, but a cache. Okay, so in between this API gateway and this databases, how many macro services do you have? How many different application, Spring Boot applications, or whatever you're using.

Speaker 3 14:26

Oh, yeah. So our team, we are responsible, you know, like for, you know, like for distinct microservices, but they come in different, you know, like flavors, right? So we have, like, a member microservices. We have explanation of benefit service. We have, you know, like a coverage service, and then we have a network service. So these are the four distinct micro services, and it can be, you know, like deploy against different flavors based on on what the state is trying to access this data.

Unknown Speaker 15:00

What do you mean by flavor side? And get that part,

Unknown Speaker 15:03

what is flavor in your comes,

Speaker 3 15:06

Oh, yeah. So it will be using, you know, like the same, you know, like a backbone for the same service. But it will be using a different profile, even, like a different profile, even in, you know, like, even in fraud, right? So instead of just having an application, prod, Yammer. So we have an application, prod, like,

like a YAML, applications, prod, UTF, yaml and stuff like that. So these, these are the state data is distinct flavor, because they have a distinct rule associated with it, and because we have basically deals with like a states, you know, like, worth of data. We need to make sure like these datas are not interacting, you know, like, like each other, to basically separate the concerns. So each state is, it is like own flavor.

Speaker 2 15:57

I don't understand that. But anyways, moving on. Can you, like, walk us through one flow, end to end flow, like, when you get the data to the end of the life cycle, like one entire life cycle. Can you walk us through?

Speaker 3 16:10

Yeah, so, so let's just go through, you know, like the explanation of benefits, right? So experience and, you know, like explanations of when this are, you know, like a process daily. So at the end of the day, you know, like their clients, usually, they basically drop their files that they, you know, like that they process, you know, like the benefit from, you know, like into, into a FTP, and it's handled by a separate team. Our basic team is responsible for reading that file that they dropped into, you know, like the s3 bucket. So the s3 bucket contains the file and the file they are posted. And our ingestion process basically kicks up around, like, 3am in the morning, and it basically loads the data into a Spark cluster. And then, you know, like, the data transform into a standard form comes to in our database. So make it available to be viewed following on the you know, like the on the day when a user goes into the portal and they go into the explanation of benefit page, you know, like they can click on each of the claims that they have submitted, and if the explanation of benefits ready for that claim, they are able to view that explanation of benefits, and they will, like, make a request to our backend service. And the backend service, you know, like, will basically look at the data in the table that has been inserted, and it will basically, you know, like, aggregate into a proper form and then return it back to the user.

Speaker 2 17:41

Okay, and have you worked on this ingestion part, the spark jobs, mapping, grading the files and then transforming them into the database? Oh, yeah, database. Have worked on that part? Also, right?

Speaker 3 17:55

Yes. We have used the spark and the data frames for it so it deployed.

Speaker 2 18:00

So you have worked on The Spark job part also, and the REST services part also, both parts Correct? Oh, yes, okay. Kumar, do you have any high level questions? Otherwise we can, I can start with Java and spring

Speaker 4 18:21

like you. That, like, you have used, like both more with the JWT BLR token. Can you just walk us through, like, how it has been used in your service?

Speaker 3 18:34

Yeah, so basically, we use the JWT, you know, like, for the validating users that are, you know, like that are making the request to our, you know, like particular service. So JWT is, you know, like a web token, which basically, like a component signed authentication, so we use it for role based access control and with our service. And basically, like, done, done using a, you know, like a spring security, where the identity provider module that we have the validates the username and password for, you know, like injected JWT token that we, you know, like we use within our service now. So all these basically used in our machine to, you know, like a machine flow, so, like, if the explanation benefit needs to talk to another service and and we it might, you know, like, need to do some kind of calculation based or a certain on a certain network data. It basically tried to get a token and then it pass it to the machine to machine flow using the service accounts.

Speaker 4 19:39

Okay, so you, you have mentioned, like, pretty much there about, like, the user authorization part to come into user, user function to check in if they can access the data. What about for the AP integration, the micro services, right? If I have a service that needs to talk to NP service because having any authentication in between, I just found that part is complicated.

Speaker 3 20:05

So if you have, you know, like multiple services interacting, you know, like with each other. And those are, you know, like usually done through OAuth, alright? So these are, you know, like, those are the machine to machine calls, and for that, you use, you know, like OAuth tokens. And over tokens basically, like, on behalf of the flow where the authentication is already, you know, like, handled by only the thing that is, you know, like, carries the authorizations, making sure, you know, like, the that the caller who's trying to, you know, like, access this particular service has access to. It's like, it's kind of like logging with like Google flow, and because you are already a Google user, that means Google authentications, you can authorize yourself for certain application, right? And that's exactly what we do. So our explanations or benefits, like basically talks to data in order to order to retrieve the network rules. Like and networks basically the service providers, and that's the, that's the, you know, like a USA has its like in their insurance, insurance name. And we basically use this oath to token to make that request to the server service to apply, you know, like those rules to a particular experience and the benefits, this

Speaker 4 21:24

is something like that you're not using in your application, that you just know the concept. Oh, sorry, yeah, I didn't get your question, can you? Can say, No, Harvey, have you implemented the AP integrations? Oh, no,

Speaker 3 21:42

yes, actually, like, so, like I mentioned, whenever we we call our networks, we use the OAuth tokens.

Speaker 2 21:57

Okay, let's start with Java. Can you tell me, like, what, what are the various access levels in Java? Oh, yes, modifiers and the access levels.

Speaker 3 22:12

Oh, access level modifiers, yeah. So there are, there are probably, like, four access levels, right? So you basically, you have a public which is basically, like, available for everyone, right? So you can, you know, like, access, sorry, the public classes or methods from anywhere, right? So it can be, you know, like it might be the same class, same package is subclass or other packages, and then oppose it to public is private, which is very limited. So you can access private matters within the same class, and then you have protected. Basically a protected is an accessible form from the same class and the same package. And also, like subclasses, even if you know, like the subclass lives in a different package and and there is a default, which is like a packet, you know, like private, which is means it's accessible from the same class and same package, but it is not accessible outside.

Unknown Speaker 23:13

Difference between abstract class and interface.

Speaker 3 23:17

Oh, yes. So abstract class in differences. So the main difference, you know, like the app is like the the class and interface is the interface basically defines, like a contract, right? So basically it defines everything that your class must implement and are there to in, you know, like in terms of rules and but you're, you know, like abstract class is like, kind of like a base class, and abstract class represent is a relationship, right? So it basically is a, like a class that defines a base functionality and act as a, you know, like a parent super class with some, some kind of partial implementations of a base functionality, but interfaces. So they are generally considered like, you know, like a contract that define the behavior in Java. You can implement multiple interfaces, but you can, you know, like extra and one abstract class.

Speaker 2 24:14

Okay, so let's say, if you have a try, catch and finally block, and you don't get any exception. So how will be the exception flow in this case? Like, let's say, if you get exception and if you don't get exception in both

Unknown Speaker 24:32

cases, okay, in both cases. So try catch and finally, right?

Speaker 3 24:37

Yeah. So basically, like, try catch and finally, the finally part here plays a specific, like, a special role, right? So finally is, is something that will execute it regardless of your outcome. So if there is no exception, it will basically go, you know, like, go to the final block and do, what's what's on the finally block is there, right? But if there is like an exception within the trying cache block, right? So the try will throw an exception, it will caught, and in the cache block, it will execute what's on the like cache block, and after that, it will go to the final Can

Speaker 2 25:19

you tell me what is the difference between String, String Builder and string buffer,

Speaker 3 25:24

okay, string, string so, yeah. Basically, the the difference, differences, you know, like, between the string and string builder and the string buffer, he says, right, is that the string is, is. String is an Immutable object like, but a string builder and string buffers are the mutable like implementation to work with the string and the main differences between the string builder and string buffer is that the string buffer is, is, is a is, is, is a thread safe. Is a thread safe? Yeah, so yeah, that's the that's the main difference.

Speaker 2 26:04

Okay, so let's say, if you, if you want a performance, high performance, in their application. So what will you use? Will you use string or string buffer or string builder?

Speaker 3 26:18

Okay, so, so basically, if you want a high perform, if you want a higher performance, and I think it might be depend on the you know, like the use case per se, because the string builder is a good way, because it's mutable, and there is no, you know, like overhead of any you know, like synchronizations, and you know, like you don't keep and creating newer objects every time. So you do, you know, like any operation. So I would usually prefer string builder, okay, like on a simple application, just just to be and

Speaker 2 26:59

where do these live on the Java memory model,

Unknown Speaker 27:05

or where they where they live

Speaker 2 27:06

in the Java Virtual Machine. Where do they leave? String, String Builder, string buffer,

Unknown Speaker 27:13

which area they Okay, okay,

Speaker 3 27:15

yes. So basically, like, the strings are very specific, right? So basically, like a spring literal, right? So maybe like a double code with string values, so they leave on a string pool, which is, you know, like which is inside the heap memory. And the string builder is basically like an object with the heap and then string buffer is also like buffer. So within the heap memory,

Speaker 2 27:45

string, you said, heap memory, or I didn't get the string part, is it heap memory or something else?

Speaker 3 27:50

So strings, so a string is like a bit different. So if you have a literal string,

Unknown Speaker 27:58

then it will go, you know, like it will

Unknown Speaker 28:01

go to your spring pool.

Speaker 2 28:06

And so since we are in the Java memory model, right, so where do class level members live in the memory in the JVM? Okay, so variable in class level, right? Where do they live? And let's say you have local variables, method level variables. So where do they sleep in the memory?

Speaker 3 28:26

Model, okay, so, so basically, like a static variables, these static method and kind of, you know, like a metadata that is specific, you know, like a specific to a class they leaves on, like, like a perm, like a Gen memory.

Unknown Speaker 28:50

And what about the method local variables?

Speaker 3 28:55

So method and local variables, so local variables, each like specified method, get its own stack. So local variables should be on us, on a stack memory.

Unknown Speaker 29:11

So what are the various collections you have worked on for

Unknown Speaker 29:16

which you frequently use in your project?

Speaker 3 29:18

So yeah, so I have worked with the list sets and maps. So the most common that I work with is the ArrayList and the hash map.

Unknown Speaker 29:30

Okay, ArrayList and hash map, okay.

Speaker 2 29:34

So do you know when, in which case you should use analyst and in what case you need to use link list?

Unknown Speaker 29:42

Oh, yes. So,

Speaker 3 29:46

basically, like, so a release, you know, like a simple extension of an Error List, right? So it's just in like a order collections of objects, and you use this mostly, right? So if, if a list gives you, know, like a good iterative solution, you you also get the option to, you know, like, retrieve elements by indexes. So you can use the, you know, like the R list most of the time, like the Error List most of the time. And they are, you know, like only specific class that when you when you would, you know, like a used link list. Because in a link list, you have a pointer to the next row, right? So if you need some kind of operations where you want to insert elements to the middle, or you can just update the pointer and will be, you know, like a performance, right? So I think, like, that's the main thing that distinct versus between a link list and the analyst,

Speaker 2 30:44

okay, you said link list is order, right? So, what is the order of link list? What is the default order of link list?

Unknown Speaker 30:54

So for the linked list, right? Yeah, linked list.

Speaker 3 30:59

Okay, so link, so link list order should basically like, so what's the instruction orders? Right? So whatever you insert it on it, like, follow the order, because you are basically inserting on the node of the point, like the pointer,

Speaker 2 31:15

instruction order. Okay, and let's see if I want a sorted list. Then what can I use?

Speaker 3 31:24

Okay, so for the sorted list, if you you know, like, I have simply a one sorted list based, like, sorry, so yeah. So basically, like, if you simply want a sorted list on a natural sort thing you might, you know, like, want to do something like a tree set maybe, or maybe, like, there is an implementation, you know, like that does natural ordering. So you might have to, you know, like, build something customs, or just use collection frameworks to sort it. Okay?

Speaker 2 32:00

So you said something custom, right? So let's say I have a list of employee and it has employee ID and it is an employee name. So my first use case, I want to sort it by the employee name and second, second is by the employee ID. So how can I implement this?

Speaker 3 32:15

Okay, so employee ID and that you have. So in order to do that, basically you have, like a, as we call a custom sorting so that you want to do, right? So you use a competitor, like computer on the collections, you know, like dot sort method, and then you provide it like the, you know, like the competitor. So on the list of employees, you do a custom, you know, like competitor so and where it compares the name first and then compares the ID,

Speaker 2 32:58

and you're saying compare it, right? So, how do like? What is the way of implementing this? How do you exactly implement like? What do you do? You create a class, or do you do? What are the steps, exact steps to create this custom sorting thing, right? I want to create something which will give you a custom sorting like, two different ways, by ID or by name. So how do you implement that?

Speaker 3 33:21

Okay, so, so the comparator would basically give a custom comparator, right? So that you can, you know, like, pass into the collection frameworks, and you can simply use, like, the method chains on, you know, like the collection sort method, so you will be able to, you know, like, add a natural ordering. So by using the, you know, like Comparable interface as well, and if you go with the natural ordering, you would basically implement a computer interface, and, and he would, you know, like Comparable interface and it they would, you know, like override the, you know, like The compare to method, and, and you will get

Unknown Speaker 34:06

it and map. So

Unknown Speaker 34:11

how do you iterate over a map? Let's say you have a

Speaker 2 34:15

map, key value pair, and how you want to iterate over it, right? How do you usually do that

Speaker 3 34:21

over the key value. So basically, you you have to get the entry set out of that. So you cannot, you know, like, directly iterate over, you know, like a hash map. You would basically convert the map into an entry set. And then, you know, you know, like, you can iterate over the entry set.

Speaker 2 34:38

I Okay. Have you used thread self connections or concurrent connections?

Speaker 3 34:49

Yeah, so I have worked with the, you know, like concurrent and hash map, and also like copy and write our you know, like list,

Unknown Speaker 35:01

okay. Have you used any priority queues?

Speaker 3 35:04

Priority queues, you said, yeah, oh yes. Priority queues, like, I have only like a word in a, you know, like Link, lead, like lead, course selections, not like a reason, so real exam example, in projects,

Speaker 2 35:20

okay, yeah, that's how usual it is. Okay, let's go to concurrency. Now. Can you tell me, like, what is a lifecycle of a thread?

Speaker 3 35:33

Yes. So basically, like, lifecycle of a thread goes through a different, you know, like states. So basically, you know, like, you have a new state where the thread object is just created. Like, you can, you know, like, like, you can just do a new thread, and then you have, you know, like, runnable state. And so once you you do a thread, or dot start the the thread is eligible to run. So it goes to, you know, like, like, goes to random, runnable state wherever, you know, like, whenever you thread tries to, you know, like, enter some kind of synchronizations, but another thread holds a lock against that synchronizations. And then the thread goes, you know, like, to the block state, you know, and because it's waiting to acquire the log. And then there is also a waiting state where the thread is waiting for another thread to return, some kind of, you know, like a signal. And then you have the Terminator state where the thread has, you know, like, finished the executing.

Speaker 2 36:38

Okay, let's say the thread finished execution, and I want to run it again. So how can I do that?

Speaker 3 36:45

Okay, so when the thread is finished the execution, okay, yeah. So for that, if a thread, you know, like is, is already, you know, like completed and finished the executions you cannot, you know, like, restart the same thread again. I think, because that's, that's the because thread has specific states,

Speaker 2 37:12

is the correct answer. And how do you have you implemented, like, multi threading in your

Unknown Speaker 37:19

project? Yes.

Unknown Speaker 37:21

Okay, did you use any library, any framework for that?

Unknown Speaker 37:26

Okay? On the

Unknown Speaker 37:32

Yes, so in our project,

Speaker 3 37:36

it's, it's like enabling, you know, like use thread pools and for, basically, you know, like 11 enabling the multi threaded environment. So it's our, you know, like wrapper around the executor service.

Speaker 2 37:52

Okay, good. And can you tell me, like, how I can make a object thread safe, or a method thread safe, or variable thread, Chef

Speaker 3 38:05

or object? Yes. So basically, there are, you know, like multiple things you like, that you like mentioned, like we talk about the methods and variables and so on. So to make a object, thread self, you basically make that object immutable, right? So that's the simplest way. But if you want to make some methods thread safe, you have to synchronize it, right? So if you do a synchronized method, then it basically, you know, like a be a thread safe method where only, only one thread can access, access it. Or alternatively, you can basically create a synchronized block where, which only you know, like, only blocks that certain part and part of that that you can also you know, like make some different variables thread safe. So the first one is by using the volatile keyword, which basically makes it more like, makes update visible. But also, you know, like, have some atomic variable, which are, you know, like, which is like a third step implementation of those variable, right? So, depending on which one you want, I would, you know, like, I would say those are the ways, okay,

Speaker 2 39:22

ah. I think we are going to the last section in Java. Can you tell me, like, where? What are the new I mean, not new as of now, but what are the additions introduced in Java eight? What are the new features in Java?

Unknown Speaker 39:35

New features in Java eight?

Speaker 2 39:37

Yeah, Java eight. Like, what are the changes from JDK 1.7 to jar to JDK eight.

Speaker 3 39:41

Okay, so Java, and I think, like, we, I think was, like, really huge in terms of features, right? Because it introduced the stream API. So it is a way to, you know, like a way of working with you, with the collections, right? So you can convert collections into a stream and then apply processing in this, you know, like a process. You can implement the lambda interface, Lambda expressions, right? So, which is like a way of representing functional interface. And functional interface is like a simple interface and containing of one method, you know, like of method, of abstract method, and the Lambda expressions are just, you know, like arrow functions. Other features includes, you know, like optional classes. These are which are, like, you know, like a wrapper around the null objects. And another big one is, you know, like a date, like an daytime API, which makes working with local dates and easy to easy. So, yeah, okay.

Speaker 2 40:47

Have you implemented map, reduce, flat map, all these methods of the stream API?

Speaker 3 40:54

Oh, yes, I have worked for all those like intermediary operations that that you mentions. So basically, map converts one form to another. Like, if you have taken an example of the employees that you mentioned earlier, and if you want to, you know, like, take a list of employees and then get a name of those employees back, so you can simply use the map operations on the list stream. And also, like, if you want to, you know, like a flatten, like a nested list, or you can use a flat map reduce so you can, you know, basically, provide an aggregator functions on how different, you know, like fields can be, you know, like aggregated like, maybe calculating the sum of, you know, like salaries of particular department, of particular department, Like, anything like that. So, yeah,

Unknown Speaker 41:41

okay, readymade methods. Okay,

Unknown Speaker 41:44

okay, good, so I'm good with Java.

Speaker 2 41:48

Do you have any questions with Java, or should we avoid Okay, so what spring modules have you used?

Speaker 4 41:59

One question you one question, our mission, or like in the so you're using the open JDK. Sorry. You are used to like using the open JDK,

Unknown Speaker 42:14

open JDK and and the Amazon

Unknown Speaker 42:18

would be called, like the corrector, corrector.

Speaker 4 42:21

Okay. So in the latest version of JDKs, do you know? Like how they want they introduced, like, manage the memory, like what features they have introduced to manage the memory.

Speaker 3 42:36

So are you talking about the Java 17? Bhandari, okay, yeah, so we have, so I would say that they have been, you know, like introducing ways of changing how the garbage collection works and also, like, basically what we have so they have, you know, like, some kind of, you know, like API's that, you know, like, that provides some, you know, like spaces with, like, with metaspace to be, you know, Like, adjusted dynamically, I think, but I don't know, like, a whole lot, because I usually, you know, like, a lot of garbage collections.

Speaker 2 43:33

So Bishwo, can you, like, walk me through, like, various layers in your application, in your Springboard application, right? Can you tell me? Tell me, like, what are the various layers? How do you segregate the application into?

Speaker 3 43:46

Oh, yeah. So we have, you know, like, a well structured, you know, like parts with our applications. So we have a controller and, like, controller, and then advice, which is, like, you know, like a entry point of the RESTful API's. So they enter through the controller layer, and then we have a global exception handler using the controller advice, right? So these are separated from where the actual processing and the transformations take place. So which is a service layer and so, and also like, so we have like an like an explanation of benefit service and the claim service that lives on the service layer. The service layer also have a dependency of other services, like maybe, if you want to call a network service with the with the explanation of service, explanation of benefits, sorry, and we have, you know, like we have a client that that we use as a dependency and that we make a call to the network through the network client. And then we have a repository layer, which is used for interacting with the database. We have an entity layer where it is like, you know, like different database table mapped as as Java object. So we have, we have a transfer transformation layer or or a mapping layer with where the DTO and the entity map, you know, like between each other, so that you know like they don't expose entity back to the clients.

Speaker 2 45:19

And what library or framework they use to communicate with a different micro service. Press Service,

Speaker 3 45:29

yes, so we simply use the web client now,

Speaker 2 45:35

web client, but web client is for asYn asynchronous calls, right?

Speaker 3 45:43

Oh, yeah. So we are working with the, you know, like a more and, you know, like a flux more. So now, because it says, like a reactive, you know, like a reactive programming easy, so it's a lot of, you know, like non blocking operations before that. So we simply work with, you know, like Apache crossable, like HTTP clients.

Speaker 2 46:05

You're using web client only now. Are using like flux monos in your application? Yes.

Speaker 3 46:12

So we are starting to, you know, like use the more of you know, like the flux and mono.

Unknown Speaker 46:18

So your project is completely

Unknown Speaker 46:21

non blocking you're using flux and models.

Speaker 2 46:24

You don't use regular, like, blocking calls, Rest Template calls, Oh, yeah.

Speaker 3 46:29

So, like, you know, like I mentioned, it's like a migration, you know, like a project, right? So we are starting more and more to make, you know, like a more reactive non blocking calls, yeah, okay.

Speaker 2 46:41

But I think that is also outdated. Now there is a new library, if you are aware, REST client. So one Rest Template web client, and now there is a new one REST client. But anyways, good to know. I haven't seen anyone implementing that, but Okay, good to know. So, yeah,

Speaker 3 46:56

yes, I just, you know, like, heard about it, like, yeah.

Speaker 2 47:02

So can you tell us the various annotations you use? Let's say, let's say, how you when you create a new project, right? So, can you walk us through the annotations you're using the new project annotations. Then you're the controller layer, service layer, or down layer. So, like, can you just walk through us through the different annotations you're using in the project, right from the setup to the database layer,

Speaker 3 47:26

okay, so from the setup to the database layer. So basically, on the on the entity layer, which is like a database table representation, you have the Entity annotation. So we, so we have a claims POJO, which is annotated with the, you know, like entity annotations. And also have the table annotations given the table name of the table. With this, we have the ID annotations in the primary key, and then we have a different column annotations, or, you know, like the columns on the like a repository layer, we have the repository annotation, which is basically a JPA repository extensions on the service layer, which we have the service annotations on the controller. We had the Rest Controller annotations, we also have the request, maybe, annotations that provides the, you know, like, the base path of that controller, which is like, you know, like, like one claims, and then each of them methods, like, represent to distinct HTTP. And we have, you know, like, we have a get mapping or a post mapping annotations, depending on what kind of request we are making in the service layer and the and the controller layer, we have the dependencies of like, maybe controller that has a dependency on a service layer, and the service layer has a dependency on the repository layer. We use the auto wire annotations to inject those dependencies on, you know, like on a global exception handler. We use the controller controller advice, you know, like annotations on a beam definitions. We have the configurations annotations and the methods that have been, you know, like bean annotations,

Unknown Speaker 49:10

okay, and what type of dependence injection you use.

Speaker 3 49:15

So we simply use the, you know, like the constructor injections and ingestions, and that's the most preferred and safely because all of our dependencies, you know, like dependencies are required,

Speaker 2 49:30

let's say, a service interface, and you have two different implementations of that, okay? And you need to automate it somewhere. So how do you do that? How does the receiving end know that which mean to refer to, for

Unknown Speaker 49:46

which implementation to refer to?

Speaker 3 49:48

Okay, so you can simply make the use of, you know, like the qualified qualifier annotations, the qualifier notation is used to give the name of that beam that you want to inject. So if you have, if you have multiple beans of the same type, there will be no unique or distinct beam, beam of that particular type. So you can, you know, like, inject by name. So you know, like, you can just use the, use the qualifier you

Speaker 2 50:26

and so what different models you use in your Spring Boot application? I mean, let's cover the let's say you create a new application, right? So what are the conditions we you use for that? Let's forget about the controller layer, all the layers. Let's say you are just want to print a Hello World. So what are the adaptations you will use in that case?

Speaker 3 50:47

Oh, just to print the Hello World. Yeah, I want it to be a springboard. Oh, on the springboard,

Speaker 2 50:54

yeah. Okay, the setup annotations, which we need for the setup.

Speaker 3 50:58

So, so the main thing that you want to, you want for Spring Boot would be a starter dependence. So it would be the Spring Boot application dependency you may want to, you know, like, use a component scan and enable auto configuration dependencies annotations that you may, you know, like you may have, but that should be, you know, like about it, right? Because you don't need anything apart from that. So that's, that's, that's the main thing.

Speaker 2 51:30

Okay, so there is annotation, which is widely used, right, which is the combination of component scan and all those right, primary annotations for setup, Spring

Unknown Speaker 51:42

Boot, application, annotation. Have you used that?

Speaker 3 51:47

Oh, yeah. So I did mention that. So the Spring Boot application, you know, like is the main annotations that you need to make, make sure that is, you know, like that. It's a Spring Boot application so, and it contains, you know, like the combinations of multiple specific annotations. The first thing is the you know, like the configuration annotation, which makes start of a class, or, you know, like a place for beam and like beam definitions. So the next thing is the you know, like enable auto configuration, which means that it should know, like, a scan the class path and find the config classes and configure those beams automatically. And then finally, it's a component scan. And you know, like, which basically scan all of those classes. And like the component service, service repositories annotations

Speaker 2 52:45

going to the DAO layer. So you mentioned about the repositories, and it is right. So let's say you want to write a custom query which is not supported out of the box by the JP repositories. So can you tell me how you can write custom queries

Unknown Speaker 53:00

for the custom query. So, yeah, on the what

Speaker 3 53:04

do you have, repository interface, you can, you know, like a, create a method, and then, you know, you know, like you can annotate that by that method, by using the query annotations, write your own query in, you know, like in a plain string. And then you can use a native query flag by just, you know, like setting native equal true, and then it just, we know, like, should run just like a native SQL query.

Speaker 2 53:32

Can you tell us, like, how you test your REST endpoints or your overall project? What are the various testing methods you use for J units and everything.

Speaker 3 53:43

Oh, yeah. So for each of those, like, individual classes that I have, you know, like I write, I have unit test, unit tests, right? So basically using the JUnit and the mokito to basically, you know, like mock out the dependencies and basically the testing framework. So that's the primary test that I add as a part of my work. And I also work on, you know, like integration testings, using the Spring Boot test, and which basically makes user more beans and containers using the Docker so, you know, like using Docker to set up a local database for, you know, like testings, the integrate, like the interaction between each of these services. Like, that's a that's a two primary like testings, and then I add into the repository itself. And but apart from that, that I do, like, mention, do manual testings using the Bruno, like, basically sending the request into the controller and making sure everything is work aggressive as expected.

Speaker 2 54:45

What do you use for manual testing? Any tools you use?

Speaker 3 54:50

Oh, yes, so yeah. Basically, it's called Bruno. It's like a substitution saying, you know, like for postman.

Speaker 2 54:57

It's like, post postman, okay, okay. And do you do you do any load testing or any bulk kind of testing?

Speaker 3 55:07

Yeah, I have, you know, like we do have a performance testing frameworks in set. So we have a mix of JMeter scripts and k6 scripts, depending on what kind of applications we are in,

Unknown Speaker 55:21

okay, okay, good.

Speaker 2 55:24

Kumar, I think I have covered everything. Maybe you can ask more questions high

Unknown Speaker 55:31

level or areas.

Speaker 4 55:36

So Bishwo like another question from us, and PBR like any questions

Speaker 3 55:41

to us. Yes. So first of all, I thank you so much for you know, like taking your time to talk to me today, and I would like to know a little bit more about what's new, like on what the developers get involved in, like, a day to day basis, like in terms of designing sessions and and requirement analysis and so,

Speaker 4 56:02

um, so the team, the question where we are looking for is, it is to my team. Okay, so we are kind of like a interface engineering team, so we can apply for like across the line of businesses. So this team, like we mainly supports the content management system, so where we are the services exposed to kind of like for other applications to integrate, and then, like, we have the services and modules to kind of do a bulk condition of the documents, like bulk extracting the documents. Right now. Like, we are not on like cloud. We are still like on prem, but we are like, looking to like motor cloud, cloud, which is going to be the top priority for us the next year. Okay, the technology stack wise, like, we are still, we are using the data direct servers. We have the GitLab with full pipeline setup for the scanning world boundaries and stuff, some of that. And we are like, heavy, like, MS, SQL database servers. When we go through cloud, like, we are looking for, like, different database servers. But that's not this kind of data, okay? And then for the development, it depends on the developer's preference, what we are into the shape.

Unknown Speaker 57:09

And for the payment request phase, we use at JFR,

Speaker 3 57:16

okay, that sounds good, yeah. So thank you so much for all those you know, like, like, those are the all the tools that I'm familiar with. Yeah. So thank you for that information. And yeah. So I would like to you know, like, know, like, what will be the next step of the process?

Unknown Speaker 57:34

You just will be in touch with you.

Unknown Speaker 57:37

Okay, sounds good?

Speaker 4 57:40

Okay, yeah. Thanks, Bishwo, thanks for your time. Thank you so much. Thanks, Vishnu, thank you. Yeah.